

Randomness, Quasirandomness, Complexity, and Cryptography
Foundations through Arithmetic and Dynamical Pseudorandomness

QRCC Book-Series Project

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Publication, ownership, and reading guide

What this edition certifies

This edition separates four questions that are easy to conflate: whether a statement is part of the QRCC architecture, whether its proof is reconstructed locally, whether it is imported from a verified source or companion, and whether it is a dated research frontier. Stable identifiers of the form `QRCC-...` are the permanent handles for these distinctions. Page numbers may move between editions; stable identifiers are intended not to.

Ownership classes

Every registered result carries one of four ownership classes.

- A** Local QRCC ownership. The proof or derivation is written in the QRCC source and has passed the local proof audit when its status says *PROVED_AUDITED*.
- B** Companion ownership. The result is deliberately proved in another maintained book/project; QRCC supplies the exact interface, hypotheses, consequences, and a stable companion pointer instead of duplicating the long proof.
- C** External source ownership. QRCC states and uses a theorem from the scholarly literature, with a verified primary or canonical source gate. Reader-facing text must distinguish theorem statement, local consequences, and imported proof burden.
- F** Frontier or open-problem ownership. The item is a dated status ledger, conjecture, or frontier map. A frontier entry is evidence about the state of knowledge, never a claim that QRCC proves the underlying open statement.

How to read a source-owned theorem

For a class B or C theorem, reconstruct the interface in four passes: identify the exact hypotheses; identify the exact conclusion used later; trace all local dependency edges; then consult the recorded source only for the proof burden that QRCC explicitly leaves external. The reconstruction guides at the end of every volume are designed to enforce this discipline.

Citation and locator policy

The backmatter contains a deduplicated catalogue of scholarly sources that are actually live in the 24-volume result registry. Companion and internal provenance are listed separately. Source-gate verification dates certify the bibliographic/theorem interface checked by the project; they are not claims that a paper itself was published on that date. DOI and stable web locators are preferred over transient search URLs.

Edition-local versus dated information

Complexity conjectures, cryptographic standards, security estimates, and research-frontier entries can change after the edition date. Such material is marked by dated frontier metadata. Mathematical theorems are not silently upgraded merely because a stronger later theorem exists; a later edition must reopen the relevant source gate.

Global notation and collision policy

Scope rule

Notation is local to the smallest declared scope. A chapter-level definition overrides a volume convention, and a volume convention overrides this global guide. This rule is essential in a 24-volume work because the same symbols carry standard but incompatible meanings in probability, complexity, cryptography, lattices, and arithmetic dynamics.

Global defaults

Unless a chapter says otherwise, $\log$ is the natural logarithm, finite averages are normalized, $\mathbb{E}$ and $\mathbb{P}$ denote expectation and probability, and asymptotic symbols O, o, Ω, Θ hide constants independent of the asymptotic variable. Entropy units are stated locally; when a formula depends on base, the base is shown explicitly. For finite fields, $\mathbb{F}_q$ denotes the field with q elements once existence has been established or imported. Norm subscripts are written when more than one norm is active.

High-risk collisions

Symbol	Common QRCC meanings and required disambiguation
λ	Security parameter in cryptography; Liouville function in arithmetic pseudorandomness; eigenvalue or lattice-minimum notation in local linear-algebra/lattice contexts. The chapter declaration always controls.
q	Prime power/field size, cryptographic modulus, query count, or alphabet size. Never transfer a bound across chapters without re-reading the local definition.
n	Input length, ambient dimension, number of vertices, code length, or sample size. Complexity-theoretic polynomial bounds use the locally declared input-length parameter.
ε	Approximation error, statistical distance, spectral tolerance, soundness error, or analytic small parameter. Quantifiers and dependency of hidden constants must be read from the theorem statement.
μ	Probability measure/distribution in analysis and dynamics; Möbius function in arithmetic dynamics.
G	Graph, group, generator, or game depending on volume. Structural type is part of the definition, not inferred from the glyph.

Fourier and probability normalization

Finite-group Fourier transforms and inner products are normalized exactly where introduced; the book does not impose one transform convention across Boolean analysis, additive combinatorics, noncommutative groups, and quantum information. Total-variation/statistical distance conventions are likewise restated where constants matter. A reader should never import a factor of $|G|$, 2, or 2π across volume boundaries without checking the local definition.

Complexity and cryptography conventions

Complexity classes refer to the machine/model specified in the local definition; advice, promise, randomized, quantum, and nonuniform variants are not interchangeable. Cryptographic security statements quantify over the declared adversary model (PPT, QPT, information-theoretic, or concrete) and over the security parameter. “Negligible” always means negligible in that declared security parameter.

Source-owned notation

When QRCC imports a theorem, the reader-facing statement is normalized to QRCC notation only when this preserves the source theorem exactly. If a source uses materially different normalizations, the interface records the translation rather than silently identifying the two statements.

Cross-volume navigation and subject map

The table records only formal dependency edges that cross volume boundaries. Absence of a formal edge does not mean that a topic has no pedagogical prerequisites; consult the reconstruction guide at the end of the volume.

Vol.	Subject	Formal dependencies on earlier/later volumes
I	Probability, Information, Fourier Analysis, and Randomness	None registered
II	Computability and Classical Complexity	None registered
III	Finite Fields, Codes, Lattices, and Algebraic Algorithms	None registered
IV	Quasirandom Graphs, Hypergraphs, and Limits	Vol. I (2 edges)
V	Higher-Order Additive Quasirandomness	Vol. I (1 edge)
VI	Quasirandom Groups and Noncommutative Mixing	None registered
VII	Expander Graphs, Random Walks, and Graph Products	Vol. I (5 edges), Vol. III (1 edge), Vol. IV (1 edge), Vol. XI (2 edges)
VIII	Weak Sources, Extractors, and Pseudorandomness	Vol. I (6 edges), Vol. III (5 edges)
IX	Randomized Complexity and Derandomization	Vol. I (1 edge), Vol. II (4 edges), Vol. III (2 edges)
X	Boolean Circuit Complexity and Barriers	Vol. I (10 edges), Vol. II (8 edges), Vol. III (4 edges), Vol. IX (1 edge), Vol. XII (1 edge)
XI	Arithmetic Complexity, Polynomial Identity Testing, and Algebraic Structure	Vol. I (2 edges), Vol. III (2 edges), Vol. IX (2 edges), Vol. X (1 edge)
XII	Communication, Query, Streaming, and Data Complexity	Vol. I (13 edges), Vol. III (1 edge), Vol. IV (3 edges), Vol. VIII (2 edges)
XIII	Interactive Proofs, PCP, and Proof Complexity	Vol. I (2 edges), Vol. II (4 edges), Vol. III (7 edges), Vol. IV (1 edge), Vol. VII (2 edges), Vol. X (1 edge), Vol. XI (2 edges)
XIV	Average-Case, Fine-Grained, and Meta-Complexity	Vol. I (1 edge), Vol. II (3 edges), Vol. IX (4 edges), Vol. X (5 edges)

Vol.	Subject	Formal dependencies on earlier/later volumes
XV	Quantum Complexity and Nonlocal Computation	Vol. I (2 edges), Vol. II (7 edges), Vol. III (3 edges), Vol. IX (1 edge), Vol. XII (2 edges), Vol. XIII (3 edges)
XVI	Foundations of Modern Cryptography	Vol. I (2 edges), Vol. IX (4 edges), Vol. XV (1 edge)
XVII	Number-Theoretic and Elliptic-Curve Cryptography	Vol. III (1 edge), Vol. XVI (10 edges)
XVIII	Hash Functions and the Wang Cryptanalytic Revolution	Vol. I (2 edges), Vol. XV (3 edges), Vol. XVI (5 edges)
XIX	Symmetric Cryptanalysis and Automated Attacks	Vol. I (1 edge), Vol. XV (4 edges), Vol. XVI (1 edge), Vol. XVIII (6 edges)
XX	Lattices and Post-Quantum Cryptography	Vol. II (1 edge), Vol. III (9 edges), Vol. XIV (1 edge), Vol. XVI (15 edges), Vol. XVIII (2 edges)
XXI	Zero Knowledge, Secure Computation, and Succinct Proofs	Vol. I (1 edge), Vol. II (5 edges), Vol. III (4 edges), Vol. XIII (1 edge), Vol. XV (1 edge), Vol. XVI (12 edges), Vol. XVII (6 edges), Vol. XX (3 edges)
XXII	Homomorphic, Functional, and Program Cryptography	Vol. II (3 edges), Vol. III (1 edge), Vol. X (1 edge), Vol. XVI (4 edges), Vol. XVII (4 edges), Vol. XX (6 edges), Vol. XXI (5 edges)
XXIII	Finite-Field Arithmetic, Zeta Functions, Coding, and the Wan Program	Vol. I (2 edges), Vol. II (2 edges), Vol. III (9 edges), Vol. V (1 edge), Vol. XIV (1 edge), Vol. XVII (2 edges), Vol. XX (1 edge)
XXIV	Arithmetic and Dynamical Pseudorandomness	Vol. I (5 edges), Vol. V (7 edges), Vol. VII (1 edge)

Navigation principle

The series has three interlocking spines: randomness/quasirandomness (Volumes I, IV–IX, XXIV), complexity/proof systems (Volumes II–III, IX–XV), and cryptography/arithmetic (Volumes XVI–XXIII). Cross-volume stable identifiers should be preferred to chapter numbers when tracing a dependency.

Part I

Probability, Information, Fourier Analysis, and Randomness

Chapter 1

Finite Probability and Random Experiments

This chapter fixes the finite probability model used throughout the first proof unit and proves the elementary tail inequalities on which later concentration arguments depend.

This chapter is the first proof-bearing unit of the series. Its stable identifiers are inherited from the pre-proof registry. Throughout, logarithms are natural unless another base is displayed explicitly. Every probability space in this unit is finite; this removes measure-theoretic distractions without weakening any of the finite combinatorial, information-theoretic, or cryptographic uses.

Definition 1.1 (Finite probability space and random variable). A *finite probability space* is a pair (Ω, p) , where Ω is a nonempty finite set, $p : \Omega \rightarrow [0, 1]$, and

$$\sum_{\omega \in \Omega} p(\omega) = 1.$$

An *event* is a subset $A \subseteq \Omega$, with $\Pr(A) = \sum_{\omega \in A} p(\omega)$. A random variable with values in a set S is a map $X : \Omega \rightarrow S$. For real-valued X we write

$$\mathbb{E}X = \sum_{\omega} p(\omega)X(\omega), \quad \text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2].$$

If B is an event of positive probability, then $\Pr(A \mid B) = \Pr(A \cap B)/\Pr(B)$ and $\mathbb{E}[X \mid B] = \sum_{\omega \in B} X(\omega)p(\omega)/\Pr(B)$. Random variables $X_1, \dots, X_m$ are independent when, for every choice of values x_i in their images,

$$\Pr(X_1 = x_1, \dots, X_m = x_m) = \prod_{i=1}^m \Pr(X_i = x_i).$$

The product of finite probability spaces carries the product weights.

Positive example. For a fair six-sided die, $\Omega = \{1, \dots, 6\}$ and $p(\omega) = 1/6$. The identity map is a random variable with mean $7/2$.

Nonexample. Weights $(1/2, 1/2, 1/4)$ do not define a probability space because they sum to $5/4$. Allowing a negative weight also violates the definition even if the weights sum to one.

Definition-before-use discipline. All later expectations, conditional probabilities, couplings, and product experiments refer to Definition 1.1 until a later volume introduces general measure spaces.

Theorem 1.2 (Union bound). *For events $A_1, \dots, A_m$ in a finite probability space,*

$$\Pr\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m \Pr(A_i).$$

Equality holds if and only if, outside a set of probability zero, no outcome belongs to two distinct events.

Proof. For every $\omega \in \Omega$,

$$\mathbf{1}_{\bigcup_{i=1}^m A_i}(\omega) \leq \sum_{i=1}^m \mathbf{1}_{A_i}(\omega).$$

Multiplying by $p(\omega)$ and summing over Ω gives the asserted inequality. The pointwise inequality is strict exactly when ω lies in at least two of the events. Therefore equality after averaging occurs exactly when the total probability of such outcomes is zero. $\square$

Worked check. For a fair die, let $A_1 = \{2, 4, 6\}$ and $A_2 = \{4, 5, 6\}$. The union has probability $5/6$, whereas $\Pr(A_1) + \Pr(A_2) = 1$.

Theorem 1.3 (Markov inequality). *Let $X \geq 0$ and let $a > 0$. Then*

$$\Pr(X \geq a) \leq \frac{\mathbb{E}X}{a}.$$

If B is an event of positive probability, then also

$$\Pr(X \geq a \mid B) \leq \frac{\mathbb{E}[X \mid B]}{a}.$$

In the unconditional inequality, equality holds precisely when, up to probability-zero outcomes, $X = 0$ on $\{X < a\}$ and $X = a$ on $\{X \geq a\}$.

Proof. Pointwise,

$$X \geq a \mathbf{1}_{\{X \geq a\}}.$$

Taking expectations and dividing by a proves the first assertion. The same argument applied to the conditional probability space on B proves the conditional version. Equality in the expectation inequality holds exactly when the nonnegative difference $X - a \mathbf{1}_{\{X \geq a\}}$ vanishes almost surely, which is the stated condition. $\square$

Worked check. If X equals 10 with probability 0.1 and 0 otherwise, then $\mathbb{E}X = 1$ and Markov at $a = 10$ gives equality.

Theorem 1.4 (Chebyshev and Cantelli inequalities). *Let X be real-valued, with mean μ and variance σ^2 . For $t > 0$,*

$$\Pr(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Moreover,

$$\Pr(X - \mu \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}.$$

The second estimate is Cantelli's one-sided inequality. Chebyshev is nontrivial precisely when $t > \sigma$.

Proof. Apply Theorem 1.3 to the nonnegative random variable $(X - \mu)^2$ with threshold t^2 :

$$\Pr(|X - \mu| \geq t) = \Pr((X - \mu)^2 \geq t^2) \leq \frac{\mathbb{E}(X - \mu)^2}{t^2} = \frac{\sigma^2}{t^2}.$$

For the one-sided estimate put $Y = X - \mu$. For every $s \geq 0$, the event $\{Y \geq t\}$ implies $(Y + s)^2 \geq (t + s)^2$. Markov's inequality therefore gives

$$\Pr(Y \geq t) \leq \frac{\mathbb{E}(Y + s)^2}{(t + s)^2} = \frac{\sigma^2 + s^2}{(t + s)^2},$$

because $\mathbb{E}Y = 0$. If $\sigma^2 = 0$, then $Y = 0$ and the result is immediate. Otherwise choose $s = \sigma^2/t$; direct simplification yields $\sigma^2/(\sigma^2 + t^2)$. Finally, the Chebyshev right-hand side is less than one exactly when $t > \sigma$. $\square$

Worked check. For a fair die, $\mu = 7/2$ and $\sigma^2 = 35/12$. At $t = 5/2$, Chebyshev gives $\Pr(|X - \mu| \geq 5/2) \leq 7/15$, while the actual probability is $1/3$.

Chapter 2

Concentration and Tail Inequalities

This chapter develops the exponential-moment method in a form that records all constants, endpoints, and independence hypotheses.

Definition 2.1 (Moment generating function and subgaussian proxy). For a real random variable X , its moment generating function is

$$M_X(\theta) = \mathbb{E}e^{\theta X} \quad (\theta \in \mathbb{R}).$$

A centered random variable X is σ -subgaussian when

$$\mathbb{E}e^{\theta X} \leq e^{\sigma^2\theta^2/2} \quad \text{for every } \theta \in \mathbb{R}.$$

A one-sided subgaussian estimate must explicitly restrict the sign of θ .

Positive example. A Rademacher random variable $\varepsilon \in \{-1, 1\}$ with equal probabilities satisfies $\mathbb{E}e^{\theta\varepsilon} = \cosh \theta \leq e^{\theta^2/2}$, so it is 1-subgaussian.

Nonexample. The statement $\mathbb{E}e^{\theta X} \leq e^{\sigma^2\theta^2/2}$ only for $\theta \geq 0$ does not, by itself, define two-sided subgaussianity.

Lemma 2.2 (Exponential-moment tail bound). *For every real random variable X , every $a \in \mathbb{R}$, and every $\theta > 0$,*

$$\Pr(X \geq a) \leq e^{-\theta a} \mathbb{E}e^{\theta X}.$$

Consequently,

$$\Pr(X \geq a) \leq \inf_{\theta > 0} \exp(-\theta a + \log M_X(\theta)).$$

Proof. The map $x \mapsto e^{\theta x}$ is increasing when $\theta > 0$, so $\{X \geq a\} = \{e^{\theta X} \geq e^{\theta a}\}$. Apply Markov's inequality to $e^{\theta X}$ and then take the infimum over $\theta > 0$. $\square$

Worked check. For a Rademacher variable, the lemma gives $\Pr(\varepsilon \geq 1) \leq e^{-\theta} \cosh \theta$; the right-hand side tends to $1/2$, the exact probability, as $\theta \rightarrow \infty$.

Lemma 2.3 (Hoeffding's lemma). *If $\mathbb{E}X = 0$ and $a \leq X \leq b$, then for every $\theta \in \mathbb{R}$,*

$$\mathbb{E}e^{\theta X} \leq \exp\left(\frac{\theta^2(b-a)^2}{8}\right).$$

The constant $1/8$ is the best universal constant in this normalization.

Proof. The case $a = b$ is immediate, so suppose $a < b$. Convexity of the exponential places its graph below the chord joining $(a, e^{\theta a})$ and $(b, e^{\theta b})$:

$$e^{\theta x} \leq \frac{b-x}{b-a} e^{\theta a} + \frac{x-a}{b-a} e^{\theta b} \quad (a \leq x \leq b).$$

Taking expectations and using $\mathbb{E}X = 0$ gives

$$\mathbb{E}e^{\theta X} \leq (1-p)e^{-pu} + pe^{(1-p)u}, \quad p = \frac{-a}{b-a}, \quad u = \theta(b-a).$$

Because a centered variable taking values in $[a, b]$ forces $a \leq 0 \leq b$, we have $p \in [0, 1]$. Define

$$\phi(u) = \log((1-p)e^{-pu} + pe^{(1-p)u}).$$

One has $\phi(0) = \phi'(0) = 0$. Differentiating twice shows that $\phi''(u) = q(u)(1-q(u))$ for a number $q(u) \in [0, 1]$, namely the exponentially tilted probability of the second support point. Hence $\phi''(u) \leq 1/4$. Taylor's formula with integral remainder, from 0 to u and also for $u < 0$, yields $\phi(u) \leq u^2/8$. Exponentiating proves the claim.

To see that the universal constant cannot be reduced, consider a symmetric interval of length $b-a$ and the two-point variable $X = \pm(b-a)/2$. The second derivative at $\theta = 0$ of $\log \mathbb{E}e^{\theta X}$ is $(b-a)^2/4$, so any bound $\log M_X(\theta) \leq C\theta^2(b-a)^2$ valid near zero requires $C \geq 1/8$. $\square$

Worked check. For $X = \pm 1$ with equal probabilities, Hoeffding's lemma reads $\cosh \theta \leq e^{\theta^2/2}$.

Theorem 2.4 (Multiplicative Chernoff bounds). *Let $X = \sum_{i=1}^n X_i$, where the X_i are independent Bernoulli random variables, and let $\mu = \mathbb{E}X$. For $\delta \geq 0$,*

$$\Pr(X \geq (1+\delta)\mu) \leq \left(\frac{e^\delta}{(1+\delta)^{1+\delta}} \right)^\mu \leq \exp\left(-\frac{\mu\delta^2}{2+\delta}\right).$$

In particular, for $0 \leq \delta \leq 1$ the final bound is at most $e^{-\mu\delta^2/3}$. For $0 \leq \delta \leq 1$,

$$\Pr(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}.$$

Proof. Write $p_i = \Pr(X_i = 1)$. For every real θ ,

$$\mathbb{E}e^{\theta X} = \prod_{i=1}^n (1 - p_i + p_i e^{\theta}) \leq \prod_{i=1}^n \exp(p_i(e^\theta - 1)) = \exp(\mu(e^\theta - 1)),$$

where $1 + u \leq e^u$ was used in each factor.

For the upper tail, Lemma 2.2 gives, for $\theta > 0$,

$$\Pr(X \geq (1+\delta)\mu) \leq \exp(\mu(e^\theta - 1) - \theta(1+\delta)\mu).$$

The minimizing choice is $\theta = \log(1+\delta)$, including the limiting choice $\theta = 0$ when $\delta = 0$. Substitution gives the first displayed bound.

Put $h(\delta) = (1+\delta)\log(1+\delta) - \delta$. Since

$$\log(1+s) \geq \frac{2s}{2+s} \geq \frac{2s}{2+\delta} \quad (0 \leq s \leq \delta),$$

where the first inequality follows by differentiating the difference, we have

$$h(\delta) = \int_0^\delta \log(1+s) ds \geq \frac{\delta^2}{2+\delta}.$$

This proves the simplified upper tail; if $\delta \leq 1$, then $2+\delta \leq 3$.

For the lower tail take $\lambda > 0$ and apply Markov to $e^{-\lambda X}$:

$$\Pr(X \leq (1-\delta)\mu) \leq \exp(\mu(e^{-\lambda} - 1) + \lambda(1-\delta)\mu).$$

For $0 < \delta < 1$, choose $e^{-\lambda} = 1 - \delta$. The exponent becomes

$$-\mu(\delta + (1-\delta)\log(1-\delta)).$$

Because $-\log(1-s) \geq s$ on $[0, 1)$,

$$\delta + (1-\delta)\log(1-\delta) = \int_0^\delta -\log(1-s) ds \geq \frac{\delta^2}{2}.$$

The endpoint cases follow by continuity. If $\mu = 0$, then all $X_i = 0$ almost surely and every displayed inequality remains valid. $\square$

Worked check. For $X \sim \text{Bin}(100, 1/2)$ and $\delta = 1/2$, the upper-tail bound for $X \geq 75$ is at most e^{-5} by the simplified exponent.

Theorem 2.5 (Hoeffding inequality). *Let $X_1, \dots, X_n$ be independent and suppose $a_i \leq X_i \leq b_i$ almost surely. Put $S = \sum_{i=1}^n (b_i - a_i)^2$. If $S > 0$, then for $t > 0$,*

$$\Pr\left(\sum_i (X_i - \mathbb{E}X_i) \geq t\right) \leq e^{-2t^2/S},$$

and the same estimate holds for the lower tail. Consequently,

$$\Pr\left(\left|\sum_i (X_i - \mathbb{E}X_i)\right| \geq t\right) \leq 2e^{-2t^2/S}.$$

If $S = 0$, the centered sum is identically zero.

Proof. Let $Y_i = X_i - \mathbb{E}X_i$. Then $\mathbb{E}Y_i = 0$ and Y_i lies in an interval of length $b_i - a_i$. By independence and Lemma 2.3, for every $\theta > 0$,

$$\mathbb{E} \exp\left(\theta \sum_i Y_i\right) = \prod_i \mathbb{E} e^{\theta Y_i} \leq \exp\left(\frac{\theta^2 S}{8}\right).$$

Lemma 2.2 therefore gives

$$\Pr\left(\sum_i Y_i \geq t\right) \leq \exp\left(-\theta t + \frac{\theta^2 S}{8}\right).$$

The minimizing choice is $\theta = 4t/S$, which yields $e^{-2t^2/S}$. Apply the same argument to $-Y_i$ for the lower tail, and combine the two events by Theorem 1.2. If $S = 0$, every X_i is constant on the support, so the centered sum vanishes. $\square$

Worked check. For independent fair bits $X_i \in \{0, 1\}$, Hoeffding gives $\Pr(|\sum_i X_i - n/2| \geq t) \leq 2e^{-2t^2/n}$.

Example 2.6 (Three concentration bounds on one binomial experiment). Let $X \sim \text{Bin}(100, 1/2)$, so $\mathbb{E}X = 50$ and $\text{Var } X = 25$. For the common event $X \geq 75$, three previously proved inequalities give genuinely different information:

$$\Pr(X \geq 75) \leq \frac{50}{75} = \frac{2}{3} \quad (\text{Markov}),$$

$$\Pr(X \geq 75) \leq \Pr(|X - 50| \geq 25) \leq \frac{25}{25^2} = \frac{1}{25} \quad (\text{Chebyshev}),$$

and, with $\delta = 1/2$, Theorem 2.4 gives

$$\Pr(X \geq 75) \leq \left(\frac{e^{1/2}}{(3/2)^{3/2}} \right)^{50} \leq e^{-5}.$$

Thus the point of concentration theory is not merely that every probability is at most one: using independence and an exponential moment changes a constant or polynomial-quality estimate into an exponentially small one. The comparison is finite and exact; no asymptotic claim is inferred from this single experiment.

Chapter 3

Martingales and Bounded Differences

This chapter fixes the martingale and Doob conventions and proves the registered Azuma–Hoeffding, bounded-difference, and Efron–Stein inequalities at the finite normalization used throughout the volume.

Definition 3.1 (Filtration, martingale, and bounded increments). A filtration on a finite probability space is an increasing sequence $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_n$ of sigma-algebras. The *atoms* of a finite sigma-algebra $\mathcal{F}$ are its minimal nonempty members. For a real random variable X , the conditional expectation $\mathbb{E}[X \mid \mathcal{F}]$ is the $\mathcal{F}$ -measurable function which, on each atom A with $\Pr(A) > 0$, has the constant value

$$\frac{1}{\Pr(A)} \sum_{\omega \in A} p(\omega) X(\omega).$$

Its value on a probability-zero atom is immaterial; we set it to zero. This formula is the unique definition up to probability-zero outcomes and implies $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}]] = \mathbb{E}X$.

A real process $(M_i)_{i=0}^n$ is adapted when M_i is $\mathcal{F}_i$ -measurable, and is a martingale when $\mathbb{E}[M_i \mid \mathcal{F}_{i-1}] = M_{i-1}$ almost surely for every $i \geq 1$. It has bounded increments c_i when $|M_i - M_{i-1}| \leq c_i$ almost surely. For independent coordinates $X_1, \dots, X_n$ and a real function f , the Doob martingale is

$$M_i = \mathbb{E}[f(X_1, \dots, X_n) \mid X_1, \dots, X_i],$$

where the conditioning means the sigma-algebra generated by the displayed coordinates.

Positive example. For independent mean-zero variables Y_i , the partial sums $M_i = \sum_{j \leq i} Y_j$ form a martingale in the filtration generated by the first i variables.

Nonexample. The process $M_i = i$ is adapted to every filtration but is not a martingale.

Theorem 3.2 (Azuma–Hoeffding inequality). *Let $(M_i, \mathcal{F}_i)_{i=0}^n$ be a real martingale and suppose $|M_i - M_{i-1}| \leq c_i$ almost surely. Put $V = \sum_{i=1}^n c_i^2$. If $V > 0$, then for every $t > 0$,*

$$\Pr(M_n - M_0 \geq t) \leq \exp\left(-\frac{t^2}{2V}\right), \quad \Pr(|M_n - M_0| \geq t) \leq 2 \exp\left(-\frac{t^2}{2V}\right).$$

If $V = 0$, then $M_n = M_0$ almost surely.

Proof. Write $D_i = M_i - M_{i-1}$. Conditional on every positive-probability atom of $\mathcal{F}_{i-1}$, D_i has mean zero and lies in $[-c_i, c_i]$. Hoeffding’s lemma, applied on each such atom, therefore gives

$$\mathbb{E}[e^{\theta D_i} \mid \mathcal{F}_{i-1}] \leq e^{\theta^2 c_i^2 / 2} \quad (\theta \in \mathbb{R}).$$

Iterating conditional expectation yields

$$\mathbb{E}e^{\theta(M_n - M_0)} \leq e^{\theta^2 V/2}.$$

For $\theta > 0$, Markov's inequality gives

$$\Pr(M_n - M_0 \geq t) \leq e^{-\theta t + \theta^2 V/2}.$$

The choice $\theta = t/V$ minimizes the exponent and proves the one-sided bound. Apply the same argument to the martingale $-M_i$ and use the union bound for the two-sided form. If $V = 0$, every $c_i = 0$, so every increment vanishes almost surely. $\square$

Theorem 3.3 (Bounded differences / McDiarmid inequality). *Let $X_1, \dots, X_n$ be independent finite-valued random variables and let $f(X_1, \dots, X_n)$ be real-valued. Suppose that whenever two inputs differ only in coordinate i , their f -values differ by at most c_i . Put $V = \sum_i c_i^2$. If $V > 0$, then for every $t > 0$,*

$$\Pr(f - \mathbb{E}f \geq t) \leq e^{-2t^2/V}, \quad \Pr(|f - \mathbb{E}f| \geq t) \leq 2e^{-2t^2/V}.$$

If $V = 0$, then $f(X_1, \dots, X_n)$ is almost surely constant.

Proof. Let $M_i = \mathbb{E}[f \mid X_1, \dots, X_i]$ be the Doob martingale. Fix a positive-probability prefix $X_1 = x_1, \dots, X_{i-1} = x_{i-1}$. For a possible value a of X_i , let

$$g(a) = \mathbb{E}[f(x_1, \dots, x_{i-1}, a, X_{i+1}, \dots, X_n)].$$

Coupling the future coordinates identically for two values a, b shows $|g(a) - g(b)| \leq c_i$. Hence, conditional on the prefix, the increment $D_i = M_i - M_{i-1}$ has mean zero and lies in an interval of length at most c_i . The conditional form of Hoeffding's lemma gives

$$\mathbb{E}[e^{\theta D_i} \mid \mathcal{F}_{i-1}] \leq e^{\theta^2 c_i^2/8}.$$

Iteration gives $\mathbb{E}e^{\theta(f - \mathbb{E}f)} \leq e^{\theta^2 V/8}$. Markov and the minimizing choice $\theta = 4t/V$ yield the upper tail $e^{-2t^2/V}$; applying the same argument to $-f$ and taking a union gives the two-sided bound. When $V = 0$, changing any coordinate leaves f unchanged on the product support, so f is constant there. $\square$

Theorem 3.4 (Efron–Stein variance inequality). *Let $X = (X_1, \dots, X_n)$ have independent coordinates. Let X'_i be an independent copy of X_i , independent of every other variable, and let $X^{(i)}$ be obtained from X by replacing X_i with X'_i . For every real function f on the finite product space,*

$$\text{Var } f(X) \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E}[(f(X) - f(X^{(i)}))^2].$$

Proof. We first use the tensorization inequality

$$\text{Var } f(X) \leq \sum_{i=1}^n \mathbb{E}[\text{Var}_{X_i}(f(X) \mid X_{-i})], \quad (3.1)$$

where X_{-i} denotes all coordinates except i . For completeness, this follows by induction from the law of total variance. Splitting off X_n ,

$$\text{Var } f = \mathbb{E} \text{Var}(f \mid X_1, \dots, X_{n-1}) + \text{Var } \mathbb{E}(f \mid X_1, \dots, X_{n-1}).$$

Apply the induction hypothesis to the conditional mean in the second term; Jensen's inequality for conditional expectation bounds each of its coordinate variance terms by the corresponding coordinate variance of f . This proves (3.1).

For fixed X_{-i} , if $Z = f(X_i, X_{-i})$ and Z' is obtained from an independent copy X'_i , then

$$\mathbb{E}[(Z - Z')^2 \mid X_{-i}] = 2 \operatorname{Var}(Z \mid X_{-i}).$$

Average this identity and substitute it into (3.1), giving precisely the stated inequality. □

Chapter 4

Entropy and Information

All random variables are finite-valued. Logarithms are natural; changing the logarithm base multiplies every entropy and divergence by one positive constant.

Definition 4.1 (Entropy, relative entropy, and mutual information). For a finite random variable X with distribution p_X ,

$$H(X) = - \sum_x p_X(x) \log p_X(x),$$

where $0 \log 0 = 0$. For distributions P, Q on the same finite alphabet,

$$D(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)},$$

with $D(P\|Q) = +\infty$ if $P(x) > 0 = Q(x)$ for some x . Conditional entropy is

$$H(X | Y) = \sum_{y: p_Y(y) > 0} p_Y(y) H(P_{X|Y=y}).$$

Mutual information and conditional mutual information are

$$I(X; Y) = D(P_{XY} \| P_X P_Y), \quad I(X; Y | Z) = H(X | Z) - H(X | Y, Z).$$

Random variables X and Z are *conditionally independent given Y* when, for every y of positive probability,

$$P_{XZ|Y=y} = P_{X|Y=y} P_{Z|Y=y}.$$

Equivalently, for every such y and every z with $\Pr(Z = z | Y = y) > 0$, one has $P_{X|Y=y, Z=z} = P_{X|Y=y}$.

Positive example. A fair bit has entropy $\log 2$. If $Y = X$, then $H(X | Y) = 0$ and $I(X; Y) = H(X)$.

Nonexample. Relative entropy is not symmetric. For example, a point mass is absolutely continuous with respect to a full-support distribution, but the reverse divergence is infinite.

Theorem 4.2 (Entropy chain rule). For finite random variables $X_1, \dots, X_n$,

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1}).$$

For two variables,

$$I(X; Y) = H(X) + H(Y) - H(X, Y) = H(X) - H(X | Y) = H(Y) - H(Y | X).$$

Proof. For two variables, terms with $p(x, y) = 0$ contribute zero, while for $p(x, y) > 0$ we have $p(x, y) = p_Y(y)p_{X|Y}(x | y)$. Hence

$$\begin{aligned} H(X, Y) &= - \sum_{x,y} p(x, y) \log p(x, y) \\ &= - \sum_{x,y} p(x, y) \log p_Y(y) - \sum_{x,y} p(x, y) \log p_{X|Y}(x | y) \\ &= H(Y) + H(X | Y). \end{aligned}$$

Applying this identity repeatedly proves the n -variable formula.

Expanding the definition of relative entropy gives

$$\begin{aligned} I(X; Y) &= \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p_X(x)p_Y(y)} \\ &= -H(X, Y) + H(X) + H(Y). \end{aligned}$$

Combining this with the two-variable chain rule yields the remaining identities. Zero-probability fibers cause no ambiguity because they carry total weight zero. $\square$

Worked check. If X and Y are independent fair bits, then $H(X, Y) = 2 \log 2 = H(X) + H(Y)$ and $I(X; Y) = 0$.

Lemma 4.3 (Strict concavity of finite entropy). *Let $r^{(1)}, \dots, r^{(m)}$ be probability distributions on the same finite alphabet, and let $\lambda_1, \dots, \lambda_m > 0$ satisfy $\sum_{j=1}^m \lambda_j = 1$. Then*

$$H\left(\sum_{j=1}^m \lambda_j r^{(j)}\right) \geq \sum_{j=1}^m \lambda_j H(r^{(j)}).$$

Equality holds if and only if $r^{(1)} = \dots = r^{(m)}$.

Proof. Set $\varphi(t) = -t \log t$ for $t > 0$ and $\varphi(0) = 0$. We first verify strict concavity on the closed interval $[0, 1]$. Let $a \neq b$ and $0 < \theta < 1$. If $a, b > 0$, then for

$$F(\theta) = \varphi(\theta a + (1 - \theta)b) - \theta \varphi(a) - (1 - \theta) \varphi(b)$$

we have

$$F''(\theta) = -\frac{(a - b)^2}{\theta a + (1 - \theta)b} < 0, \quad F(0) = F(1) = 0.$$

Hence $F(\theta) > 0$ for $0 < \theta < 1$. If, say, $b = 0 < a$, then

$$\varphi(\theta a) - \theta \varphi(a) = -\theta a \log \theta > 0;$$

the case $a = 0 < b$ is symmetric. Thus φ is strictly concave on $[0, 1]$.

The two-point inequality, iterated finitely many times, gives for every coordinate x

$$\varphi\left(\sum_j \lambda_j r_x^{(j)}\right) \geq \sum_j \lambda_j \varphi(r_x^{(j)}),$$

and equality holds exactly when the numbers $r_x^{(1)}, \dots, r_x^{(m)}$ are all equal. Summing over x proves the entropy inequality. Since every coordinatewise Jensen gap is nonnegative, equality in the summed inequality forces equality at every coordinate. Therefore all the probability vectors coincide. The converse is immediate. $\square$

Worked check. For $r^{(1)} = (1, 0)$, $r^{(2)} = (0, 1)$, and equal weights, the mixture is $(1/2, 1/2)$, so its entropy is $\log 2$, whereas the weighted average of the two endpoint entropies is zero.

Theorem 4.4 (Conditioning reduces entropy). *For finite X, Y, Z ,*

$$H(X | Y, Z) \leq H(X | Y).$$

Equality holds if and only if X and Z are conditionally independent given Y : for every y of positive probability and every z with $\Pr(Z = z | Y = y) > 0$,

$$P_{X|Y=y, Z=z} = P_{X|Y=y}.$$

Proof. Fix y with $\Pr(Y = y) > 0$. The conditional law of X given $Y = y$ is the finite mixture

$$P_{X|y} = \sum_{z: P(z|y) > 0} P(z | y) P_{X|y, z}.$$

Lemma 4.3 gives

$$H(X | Y = y) \geq \sum_{z: P(z|y) > 0} P(z | y) H(X | Y = y, Z = z).$$

Multiplying by $P(y)$ and summing over all positive-probability y proves $H(X | Y, Z) \leq H(X | Y)$.

The equality criterion in Lemma 4.3 shows that equality holds exactly when, for each such y , all laws $P_{X|y, z}$ with $P(z | y) > 0$ coincide. Their weighted mixture is $P_{X|y}$, so the common law must equal $P_{X|y}$. This is precisely conditional independence of X and Z given Y under Definition 4.1. $\square$

Worked check. Let X be a fair bit and let $Z = X$, with Y constant. Then $H(X | Y) = \log 2$ but $H(X | Y, Z) = 0$. If Z is independent, equality holds.

Theorem 4.5 (Gibbs inequality). *Let P, Q be distributions on a finite alphabet and suppose P is absolutely continuous with respect to Q . Then*

$$D(P || Q) \geq 0,$$

with equality if and only if $P = Q$.

Proof. For $u > 0$, $-\log u \geq 1 - u$, with equality exactly at $u = 1$. Let $S = \{x : P(x) > 0\}$. Since $Q(x) > 0$ on S ,

$$\begin{aligned} D(P || Q) &= - \sum_{x \in S} P(x) \log \frac{Q(x)}{P(x)} \\ &\geq \sum_{x \in S} P(x) \left(1 - \frac{Q(x)}{P(x)} \right) \\ &= 1 - Q(S) \geq 0. \end{aligned}$$

If $P = Q$, equality is clear. Conversely, equality forces both $Q(S) = 1$ and $Q(x)/P(x) = 1$ for every $x \in S$, so $P = Q$. $\square$

Worked check. For $P = (1, 0)$ and $Q = (1/2, 1/2)$, $D(P\|Q) = \log 2 > 0$.

Theorem 4.6 (Mutual-information chain rule). *For finite random variables $X_1, \dots, X_n, Y, Z$,*

$$I(X_1, \dots, X_n; Y \mid Z) = \sum_{i=1}^n I(X_i; Y \mid Z, X_1, \dots, X_{i-1}).$$

In particular, taking Z constant gives the unconditional rule. The symmetric rule obtained by expanding the Y variables holds as well.

Proof. By Definition 4.1 and the conditional form of the entropy chain rule,

$$\begin{aligned} I(X_1, \dots, X_n; Y \mid Z) &= H(X_1, \dots, X_n \mid Z) - H(X_1, \dots, X_n \mid Y, Z) \\ &= \sum_{i=1}^n H(X_i \mid Z, X_{<i}) - \sum_{i=1}^n H(X_i \mid Y, Z, X_{<i}) \\ &= \sum_{i=1}^n I(X_i; Y \mid Z, X_{<i}), \end{aligned}$$

where $X_{<i} = (X_1, \dots, X_{i-1})$. The conditional entropy chain rule is obtained by applying Theorem 4.2 separately on each positive-probability fiber of Z and averaging. On the same fibers, the two-variable identities in Theorem 4.2 give

$$I(A; B \mid Z) = H(B \mid Z) - H(B \mid A, Z),$$

so conditional mutual information is symmetric in A, B . Applying the first part of the proof after interchanging the two blocks therefore gives the symmetric chain rule. $\square$

Worked check. If X_1, X_2 are independent fair bits and $Y = X_1 \oplus X_2$, then $I(X_1; Y) = 0$ but $I(X_2; Y \mid X_1) = \log 2$, so the total is $\log 2$.

Theorem 4.7 (Fano inequality). *Let X take values in a finite set of size $M \geq 2$, and let $\hat{X} = \hat{X}(Y)$ be any deterministic estimator based on Y . Put $p_e = \Pr(\hat{X} \neq X)$ and $h_2(p) = -p \log p - (1-p) \log(1-p)$. Then*

$$H(X \mid Y) \leq h_2(p_e) + p_e \log(M-1).$$

Consequently

$$p_e \geq \frac{H(X \mid Y) - \log 2}{\log M},$$

and, if X is uniform,

$$p_e \geq 1 - \frac{I(X; Y) + \log 2}{\log M}.$$

Proof. Let $E = \mathbf{1}_{\{\hat{X} \neq X\}}$. Because E is determined by (X, Y) ,

$$H(X \mid Y) = H(E, X \mid Y) - H(E \mid X, Y) = H(E \mid Y) + H(X \mid E, Y).$$

Conditioning reduces entropy, so $H(E \mid Y) \leq H(E) = h_2(p_e)$. On $E = 0$, $X = \hat{X}(Y)$ is determined by Y , whereas on $E = 1$ and for fixed Y there are at most $M-1$ possible values of X . Hence

$$H(X \mid E, Y) \leq p_e \log(M-1),$$

which proves Fano's inequality. Since $h_2(p_e) \leq \log 2$ and $\log(M-1) \leq \log M$, rearrangement gives the first lower bound. If X is uniform then $H(X \mid Y) = \log M - I(X; Y)$ by Theorem 4.2, giving the second. $\square$

Theorem 4.8 (Data-processing inequality). *A finite Markov chain $X \rightarrow Y \rightarrow Z$ means that X and Z are conditionally independent given Y in the sense of Definition 4.1. For every such chain,*

$$I(X; Z) \leq I(X; Y).$$

Equivalently, if $K(z | y)$ is any finite stochastic map and PK, QK are the output distributions obtained from input laws P, Q , then

$$D(PK \| QK) \leq D(P \| Q).$$

Proof. We prove contraction of relative entropy first. Define joint laws $R(y, z) = P(y)K(z | y)$ and $S(y, z) = Q(y)K(z | y)$. Directly from the definition,

$$D(R \| S) = D(P \| Q),$$

because the common channel factors cancel wherever $R > 0$. The entropy chain rule for relative entropy, obtained by grouping the defining finite sum by z , is

$$D(R \| S) = D(PK \| QK) + \sum_z (PK)(z) D(R_{Y|z} \| S_{Y|z}).$$

Every conditional divergence is nonnegative by Theorem 4.5, proving contraction.

For the Markov chain, regard the channel $Y \rightarrow Z$ as acting on the Y coordinate of the two distributions P_{XY} and $P_X P_Y$. Their images are P_{XZ} and $P_X P_Z$, respectively, because the same channel is used and the Markov property supplies $P(z | x, y) = P(z | y)$. Therefore

$$I(X; Z) = D(P_{XZ} \| P_X P_Z) \leq D(P_{XY} \| P_X P_Y) = I(X; Y).$$

□

Chapter 5

Statistical Distance and Distinguishability

The chapter uses the normalization $d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$ and makes the optimizing coupling and optimizing test explicit.

Definition 5.1 (Total variation and statistical tests). For distributions P, Q on a finite set S ,

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_{x \in S} |P(x) - Q(x)|.$$

A randomized statistical test is a function $T : S \rightarrow [0, 1]$; on input x it accepts with probability $T(x)$. Its signed distinguishing gap is $\mathbb{E}_P T - \mathbb{E}_Q T$, and its advantage is the absolute value of this gap.

Positive example. For $P = (3/4, 1/4)$ and $Q = (1/2, 1/2)$, one has $d_{\text{TV}}(P, Q) = 1/4$. The test accepting only the first point achieves advantage $1/4$.

Nonexample. A function taking the value 2 is not an acceptance-probability test.

Definition 5.2 (Maximal coupling and likelihood-sign set). For distributions P, Q on a finite set S , their overlap mass is

$$\alpha(P, Q) = \sum_{x \in S} \min\{P(x), Q(x)\}.$$

A coupling is a joint law of (X, Y) with marginals P and Q . It is *maximal* when $\Pr(X = Y) = \alpha(P, Q)$. The likelihood-sign set is

$$A_+ = \{x \in S : P(x) \geq Q(x)\}.$$

Positive example. For $P = (3/4, 1/4)$ and $Q = (1/2, 1/2)$, the overlap mass is $3/4$; a maximal coupling agrees with probability $3/4$.

Nonexample. Independent sampling need not be maximal. In the preceding example it agrees with probability $1/2$, strictly below the available overlap.

Theorem 5.3 (Coupling characterization of total variation). *For finite distributions P, Q ,*

$$d_{\text{TV}}(P, Q) = \min \Pr(X \neq Y),$$

where the minimum is over all couplings of P and Q . A maximal coupling attains the minimum.

Proof. Write $p_x = P(x)$, $q_x = Q(x)$, and $r_x = \min(p_x, q_x)$. Since $|p_x - q_x| = p_x + q_x - 2r_x$,

$$d_{\text{TV}}(P, Q) = 1 - \sum_x r_x = 1 - \alpha(P, Q).$$

For any coupling,

$$\Pr(X = Y) = \sum_x \Pr(X = Y = x) \leq \sum_x \min\{\Pr(X = x), \Pr(Y = x)\} = \alpha(P, Q),$$

so $\Pr(X \neq Y) \geq d_{\text{TV}}(P, Q)$.

If $\alpha = 1$, then $P = Q$ and we may set $X = Y$ with their common law. If $\alpha = 0$, the supports of P and Q are disjoint, so independent samples from P and Q mismatch surely and already give the desired coupling. Suppose now that $0 < \alpha < 1$. With probability α , sample Z from the distribution r_x/α and set $X = Y = Z$. With the remaining probability $1 - \alpha$, sample X from $(p_x - r_x)/(1 - \alpha)$ and Y from $(q_x - r_x)/(1 - \alpha)$, independently. For every x , at least one residual mass is zero, so the two residual distributions have disjoint supports and $X \neq Y$ on this branch. The marginals are P, Q , and the mismatch probability is $1 - \alpha = d_{\text{TV}}(P, Q)$. $\square$

Worked check. For point masses at two different points, the overlap mass is zero and every coupling mismatches with probability one.

Theorem 5.4 (Variational characterization of total variation). *For finite distributions P, Q ,*

$$d_{\text{TV}}(P, Q) = \max_{A \subseteq S} |P(A) - Q(A)| = \max_{0 \leq T \leq 1} |\mathbb{E}_P T - \mathbb{E}_Q T|.$$

The indicator of the likelihood-sign set A_+ is optimal.

Proof. Let $d_x = P(x) - Q(x)$ and $A_+ = \{x : d_x \geq 0\}$. Since $\sum_x d_x = 0$,

$$\sum_{x \in A_+} d_x = \frac{1}{2} \sum_x |d_x| = d_{\text{TV}}(P, Q).$$

For any event A ,

$$P(A) - Q(A) = \sum_{x \in A} d_x \leq \sum_{x \in A_+} d_x.$$

Applying the same argument to A^c bounds the negative direction, proving the event formula.

For a test $0 \leq T \leq 1$,

$$\mathbb{E}_P T - \mathbb{E}_Q T = \sum_x T(x) d_x \leq \sum_{x \in A_+} d_x,$$

because positive d_x are multiplied by at most one and negative d_x by a nonnegative number. Apply the argument to $-d_x$ for the opposite sign. The test $T = \mathbf{1}_{A_+}$ attains equality. $\square$

Worked check. For $P = (3/4, 1/4)$ and $Q = (1/2, 1/2)$, the optimal event is the first point and the maximum advantage is $1/4$.

Theorem 5.5 (Pinsker inequality). *For finite distributions P, Q , with natural logarithms and total variation as in Definition 5.1,*

$$d_{\text{TV}}(P, Q) \leq \sqrt{\frac{1}{2} D(P \| Q)}.$$

If relative entropy is instead measured with logarithms to base 2, then

$$d_{\text{TV}}(P, Q) \leq \sqrt{\frac{\log 2}{2} D_2(P \| Q)}.$$

Proof. If $D(P\|Q) = +\infty$ there is nothing to prove. Otherwise let $A = \{x : P(x) \geq Q(x)\}$ and put $p = P(A)$, $q = Q(A)$. The variational characterization of total variation gives $p - q = d_{\text{TV}}(P, Q)$. Apply the log-sum inequality separately on A and its complement (zero-mass terms being omitted). Here the needed finite log-sum inequality is itself an immediate consequence of Gibbs: for positive totals $a = \sum_i a_i$ and $b = \sum_i b_i$, apply Theorem 4.5 to the normalized vectors a_i/a and b_i/b to obtain

$$\sum_i a_i \log \frac{a_i}{b_i} \geq a \log \frac{a}{b}.$$

Applying this once on each of the two blocks gives the binary coarse-graining

$$D(P\|Q) \geq d(p\|q) := p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

For fixed $q \in (0, 1)$, the function of p on the right has value and first derivative zero at $p = q$, while

$$\frac{\partial^2}{\partial p^2} d(p\|q) = \frac{1}{p(1-p)} \geq 4.$$

Integrating this lower bound twice gives $d(p\|q) \geq 2(p-q)^2$. The boundary cases follow by continuity (or are infinite). Thus $D(P\|Q) \geq 2d_{\text{TV}}(P, Q)^2$, proving the natural-log form. Since $D = (\log 2)D_2$, the base-two constant follows. $\square$

Chapter 6

Fourier Analysis on Finite Abelian Groups

The normalized counting measure convention is used throughout: $\mathbb{E}_x = |G|^{-1} \sum_{x \in G}$. This convention fixes every Fourier and convolution constant.

Definition 6.1 (Normalized Fourier transform). Let G be a finite abelian group written additively. A character is a homomorphism $\chi : G \rightarrow \{z \in \mathbb{C} : |z| = 1\}$, and $\widehat{G}$ denotes the character group. For $f : G \rightarrow \mathbb{C}$ define

$$\mathbb{E}_x f(x) = \frac{1}{|G|} \sum_{x \in G} f(x), \quad \widehat{f}(\chi) = \mathbb{E}_x f(x) \overline{\chi(x)}.$$

Normalized convolution, translation, correlation, and L^p norms are

$$(f * g)(x) = \mathbb{E}_y f(y) g(x - y), \quad (\tau_a f)(x) = f(x - a),$$

$$C_{f,g}(a) = \mathbb{E}_x f(x + a) \overline{g(x)}, \quad \|f\|_p = (\mathbb{E} |f|^p)^{1/p}, \quad \|f\|_\infty = \max_x |f(x)|.$$

Positive example. On $G = \mathbb{Z}/N\mathbb{Z}$, the characters are $\chi_k(x) = e^{2\pi i k x / N}$ for $k \in \mathbb{Z}/N\mathbb{Z}$.

Nonexample. The function $x \mapsto e^{2\pi i x / (N+1)}$ on $\mathbb{Z}/N\mathbb{Z}$ is not a character because it is not compatible with addition modulo N .

Lemma 6.2 (Character extension and dual cardinality). *Let G be a finite abelian group.*

1. *Every character of a subgroup $H \leq G$ extends to a character of G .*
2. *Characters separate points of G .*
3. *Distinct characters are orthogonal under normalized counting measure.*
4. *The dual group satisfies $|\widehat{G}| = |G|$.*

Proof. Let φ be a character of $H \leq G$. If $H \neq G$, choose $x \in G \setminus H$, and let m be the order of $x + H$ in G/H ; equivalently, m is the least positive integer with $mx \in H$. Choose a unit complex number z satisfying $z^m = \varphi(mx)$. On $H' = H + \langle x \rangle$ define

$$\varphi'(h + kx) = \varphi(h) z^k.$$

Suppose $h + kx = h' + k'x$. Then $(k' - k)x = h - h' \in H$, so m divides $k' - k$; write $k' - k = qm$. Consequently

$$\frac{\varphi(h)}{\varphi(h')} = \varphi(h - h') = \varphi(qmx) = z^{qm} = z^{k' - k},$$

which proves $\varphi(h)z^k = \varphi(h')z^{k'}$. Thus φ' is well defined, is a character, and extends φ . Repeating this enlargement finitely many times extends φ to all of G .

If $g \neq 0$ has order m , the character $kg \mapsto e^{2\pi i k/m}$ on $\langle g \rangle$ extends to G and is nontrivial at g . Hence characters separate points.

For a nontrivial character η , choose a with $\eta(a) \neq 1$. Translation invariance gives

$$\mathbb{E}_x \eta(x) = \mathbb{E}_x \eta(x + a) = \eta(a) \mathbb{E}_x \eta(x),$$

so $\mathbb{E}_x \eta(x) = 0$. Applying this to $\eta = \chi \bar{\psi}$ proves orthogonality of distinct characters. They are therefore linearly independent in the $|G|$ -dimensional space $\mathbb{C}^G$, so $|\widehat{G}| \leq |G|$.

Every $x \in G$ defines an evaluation character $\text{ev}_x \in \widehat{\widehat{G}}$ by $\text{ev}_x(\chi) = \chi(x)$. Point separation makes $x \mapsto \text{ev}_x$ injective. The already proved upper bound, applied to the finite abelian group $\widehat{G}$, yields

$$|G| \leq |\widehat{\widehat{G}}| \leq |\widehat{G}| \leq |G|.$$

All inequalities are equalities, and in particular $|\widehat{G}| = |G|$. □

Worked check. For $G = \mathbb{Z}/4\mathbb{Z}$ and $H = \{0, 2\}$, the character with $\varphi(2) = -1$ extends by choosing $z = i$ and setting $\chi(k) = i^k$.

Theorem 6.3 (Fourier inversion). *For every finite abelian group G , one has $|\widehat{G}| = |G|$, the characters form an orthonormal basis of $\mathbb{C}^G$ under $\langle f, g \rangle = \mathbb{E}_x f(x) \overline{g(x)}$, and*

$$f(x) = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(x).$$

Proof. Lemma 6.2 gives $|\widehat{G}| = |G|$ and shows that the characters form an orthonormal basis of $\mathbb{C}^G$. It remains to identify the coefficient reconstruction explicitly.

Let $K(t) = \sum_{\chi \in \widehat{G}} \chi(t)$. Then $K(0) = |G|$. If $t \neq 0$, Lemma 6.2 supplies a character ψ with $\psi(t) \neq 1$. Multiplication by ψ permutes $\widehat{G}$, and hence

$$K(t) = \sum_{\chi} (\psi \chi)(t) = \psi(t) K(t).$$

Thus $K(t) = 0$ for $t \neq 0$. Finite Fubini now gives

$$\begin{aligned} \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \chi(x) &= \sum_{\chi} \left(\mathbb{E}_y f(y) \overline{\chi(y)} \right) \chi(x) \\ &= \mathbb{E}_y f(y) K(x - y) = f(x), \end{aligned}$$

because exactly the term $y = x$ survives and normalized averaging cancels the factor $K(0) = |G|$. This is the asserted inversion formula. □

Worked check. On $\mathbb{Z}/2\mathbb{Z}$, inversion states $f(x) = \frac{f(0)+f(1)}{2} + \frac{f(0)-f(1)}{2}(-1)^x$.

Theorem 6.4 (Parseval–Plancherel). *For $f, g : G \rightarrow \mathbb{C}$,*

$$\mathbb{E}_x f(x) \overline{g(x)} = \sum_{\chi \in \widehat{G}} \widehat{f}(\chi) \overline{\widehat{g}(\chi)}.$$

In particular, $\|f\|_2^2 = \sum_{\chi} |\widehat{f}(\chi)|^2$.

Proof. Insert the Fourier expansions from Theorem 6.3 into the inner product. Orthonormality eliminates all terms with $\chi \neq \psi$ and leaves the stated sum. Taking $g = f$ gives Plancherel. $\square$

Worked check. For $f = \mathbf{1}_{\{0\}}$ on $\mathbb{Z}/N\mathbb{Z}$, every Fourier coefficient is $1/N$, so both sides of Plancherel equal $1/N$.

Theorem 6.5 (Convolution and correlation identities). *For normalized convolution,*

$$\widehat{f * g}(\chi) = \widehat{f}(\chi) \widehat{g}(\chi).$$

Furthermore,

$$\widehat{\tau_a f}(\chi) = \overline{\chi(a)} \widehat{f}(\chi), \quad C_{f,g}(a) = \sum_{\chi} \widehat{f}(\chi) \overline{\widehat{g}(\chi)} \chi(a).$$

Proof. Using finite Fubini and the homomorphism law,

$$\begin{aligned} \widehat{f * g}(\chi) &= \mathbb{E}_x \mathbb{E}_y f(y) g(x - y) \overline{\chi(x)} \\ &= \mathbb{E}_y f(y) \overline{\chi(y)} \mathbb{E}_z g(z) \overline{\chi(z)} = \widehat{f}(\chi) \widehat{g}(\chi), \end{aligned}$$

where $z = x - y$. For translation, substitute $y = x - a$:

$$\widehat{\tau_a f}(\chi) = \mathbb{E}_y f(y) \overline{\chi(y + a)} = \overline{\chi(a)} \widehat{f}(\chi).$$

Finally, expand $f(x + a)$ and $g(x)$ by Fourier inversion and average. Character orthogonality leaves only equal frequencies, giving the correlation formula. $\square$

Worked check. Convolution with the constant function 1 produces the constant $\mathbb{E}f$; on the Fourier side it deletes every nontrivial frequency.

Lemma 6.6 (Finite weighted Holder inequality). *Let $U_1, \dots, U_m$ be nonnegative functions on a finite probability space, and let $\alpha_1, \dots, \alpha_m \geq 0$ satisfy $\sum_j \alpha_j = 1$. Indices with $\alpha_j = 0$ are omitted. Then*

$$\mathbb{E} \prod_{j=1}^m U_j^{\alpha_j} \leq \prod_{j=1}^m (\mathbb{E} U_j)^{\alpha_j}.$$

Consequently, whenever $1 \leq p, q \leq \infty$ and $1/p + 1/q = 1$,

$$\mathbb{E}|FG| \leq \|F\|_p \|G\|_q,$$

including the endpoint pairs $(1, \infty)$ and $(\infty, 1)$.

Proof. If $\mathbb{E}U_j = 0$ for some j with $\alpha_j > 0$, then $U_j = 0$ almost surely, so both sides of the weighted inequality are zero. Otherwise set $V_j = U_j/\mathbb{E}U_j$. If all V_j are positive at a point, concavity of $\log$ gives

$$\sum_j \alpha_j \log V_j \leq \log \left(\sum_j \alpha_j V_j \right).$$

After exponentiation, $\prod_j V_j^{\alpha_j} \leq \sum_j \alpha_j V_j$. The same inequality is immediate when a positively weighted V_j vanishes. Averaging and using $\mathbb{E}V_j = 1$ yields

$$\mathbb{E} \prod_j V_j^{\alpha_j} \leq \sum_j \alpha_j = 1,$$

which is the weighted assertion after restoring the means.

For $1 < p, q < \infty$, apply the weighted inequality to $U_1 = |F|^p$, $U_2 = |G|^q$ with weights $1/p$ and $1/q$. This gives Hölder's inequality. If $p = 1, q = \infty$, then $\mathbb{E}|FG| \leq \|G\|_\infty \mathbb{E}|F|$; the other endpoint is symmetric. The specialization $p = q = 2$ is the Cauchy–Schwarz inequality. $\square$

Worked check. Taking $F = \mathbf{1}_A$ and $G = \mathbf{1}_B$ with $p = q = 2$ gives $\Pr(A \cap B) \leq \sqrt{\Pr(A)\Pr(B)}$.

Theorem 6.7 (Young's inequality for convolution). *Let $1 \leq p, q, r \leq \infty$ satisfy*

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

Then

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

Proof. If $f \equiv 0$ or $g \equiv 0$, then $f * g \equiv 0$ and the claim is immediate. Hence assume both relevant norms are positive. First suppose $r < \infty$. The exponent relation implies $p, q \leq r$. For fixed x , write

$$|f(y)g(x-y)| = (|f(y)|^p |g(x-y)|^q)^{1/r} (|f(y)|^p)^{1/p-1/r} (|g(x-y)|^q)^{1/q-1/r}.$$

The three exponents

$$\alpha_1 = \frac{1}{r}, \quad \alpha_2 = \frac{1}{p} - \frac{1}{r}, \quad \alpha_3 = \frac{1}{q} - \frac{1}{r}$$

are nonnegative and sum to one. Lemma 6.6, with any zero-weight factor omitted, therefore yields

$$|f * g(x)|^r \leq \mathbb{E}_y |f(y)|^p |g(x-y)|^q \|f\|_p^{r-p} \|g\|_q^{r-q}.$$

Averaging over x and using translation invariance in the g variable gives

$$\|f * g\|_r^r \leq \|f\|_p^p \|g\|_q^q \|f\|_p^{r-p} \|g\|_q^{r-q} = \|f\|_p^r \|g\|_q^r.$$

Taking r th roots proves the finite- r case.

If $r = \infty$, then $1/p + 1/q = 1$, and the conjugate-exponent consequence of Lemma 6.6 gives, for every x ,

$$|f * g(x)| \leq \mathbb{E}_y |f(y)| |g(x-y)| \leq \|f\|_p \|g\|_q.$$

This includes $(p, q) = (1, \infty)$ and $(\infty, 1)$. $\square$

Worked check. Taking $p = 1$ and $q = r = 2$ gives $\|f * g\|_2 \leq \|f\|_1 \|g\|_2$.

Chapter 7

Boolean Fourier Analysis

The Boolean cube is treated as the finite abelian group $\{-1, 1\}^n$ with uniform probability measure.

Definition 7.1 (Boolean characters, degree, and influence). On the cube $\{-1, 1\}^n$ with uniform measure and coordinatewise multiplication, define

$$\chi_S(x) = \prod_{i \in S} x_i \quad (S \subseteq [n]).$$

For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$f = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S.$$

When $f \neq 0$, define

$$\deg f = \max\{|S| : \hat{f}(S) \neq 0\};$$

for the zero function set $\deg 0 = -\infty$. The L^2 influence of coordinate i is

$$\text{Inf}_i(f) = \sum_{S \ni i} \hat{f}(S)^2 = \frac{1}{4} \mathbb{E}(f(X) - f(X^{\oplus i}))^2,$$

where $X^{\oplus i}$ is obtained by flipping coordinate i . If $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, this equals $\Pr(f(X) \neq f(X^{\oplus i}))$.

Positive example. For the dictator $f(x) = x_1$, only $\hat{f}(\{1\})$ is nonzero; $\text{Inf}_1(f) = 1$ and every other influence is zero.

Nonexample. The algebraic degree of a non-multilinear polynomial expression is not the Fourier degree until the expression is reduced using $x_i^2 = 1$.

Theorem 7.2 (Boolean Parseval identity). For $f, g : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$\mathbb{E}[f(X)g(X)] = \sum_{S \subseteq [n]} \hat{f}(S)\hat{g}(S).$$

Consequently,

$$\text{Var}(f) = \sum_{S \neq \emptyset} \hat{f}(S)^2, \quad \text{Cov}(f, g) = \sum_{S \neq \emptyset} \hat{f}(S)\hat{g}(S).$$

Proof. The cube is a finite abelian group and its characters are exactly the χ_S . The Parseval identity is therefore Theorem 6.4 in this notation. Since $\hat{f}(\emptyset) = \mathbb{E}f$, subtracting the empty-set term from $\mathbb{E}f^2$ gives the variance formula. Applying the same argument to $\mathbb{E}[(f - \mathbb{E}f)(g - \mathbb{E}g)]$ gives covariance. $\square$

Worked check. For $f(x_1, x_2) = x_1 + x_1x_2$, Parseval gives $\mathbb{E}f^2 = 1^2 + 1^2 = 2$, which agrees with direct averaging.

Theorem 7.3 (Boolean Poincare inequality). *For every real $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,*

$$\text{Var}(f) \leq \sum_{i=1}^n \text{Inf}_i(f).$$

Equality holds exactly when $\hat{f}(S) = 0$ for every $|S| \geq 2$. For a nonconstant Boolean-valued f , equality occurs exactly for a dictator or a negated dictator.

Proof. By Definition 7.1 and Theorem 7.2,

$$\sum_{i=1}^n \text{Inf}_i(f) = \sum_{S \subseteq [n]} |S| \hat{f}(S)^2 \geq \sum_{S \neq \emptyset} \hat{f}(S)^2 = \text{Var}(f).$$

Equality holds exactly when no Fourier coefficient with $|S| \geq 2$ is nonzero. Thus an equality-case function has the affine form $f(x) = a_0 + \sum_i a_i x_i$. If it is $\{-1, 1\}$ -valued and at least two of the a_i are nonzero, varying the corresponding two signs produces at least three distinct values, a contradiction. Hence a nonconstant Boolean equality case has one nonzero linear coefficient, and the two-value condition forces $f = \pm x_i$. $\square$

Worked check. For parity $f(x) = x_1 \cdots x_n$, the variance is one and the total influence is n , so the inequality is strict when $n > 1$.

Chapter 8

Noise, Influences, and Hypercontractivity

This chapter fixes the noise coupling and L^p conventions, proves the elementary Fourier consequences locally, and keeps the full hypercontractive and KKL inputs explicitly source-owned.

Definition 8.1 (Noise operator and cube norms). For $\rho \in [-1, 1]$, let X be uniform on the cube and let $\xi_1, \dots, \xi_n$ be independent signs, independent of X , with

$$\Pr(\xi_i = 1) = \frac{1+\rho}{2}, \quad \Pr(\xi_i = -1) = \frac{1-\rho}{2}.$$

Set $Y_i = X_i \xi_i$. Then Y is uniform and, independently in every coordinate conditional on X ,

$$\Pr(Y_i = X_i \mid X_i) = \frac{1+\rho}{2}, \quad \Pr(Y_i = -X_i \mid X_i) = \frac{1-\rho}{2}.$$

In particular, $\mathbb{E}[Y_i \mid X_i] = \rho X_i$. The noise operator is

$$(T_\rho f)(x) = \mathbb{E}[f(Y) \mid X = x].$$

Equivalently, $\widehat{T_\rho f}(S) = \rho^{|S|} \widehat{f}(S)$. For $1 \leq p < \infty$, $\|f\|_p = (\mathbb{E}|f|^p)^{1/p}$, and $\|f\|_\infty = \max |f|$.

Positive example. $T_\rho \chi_S = \rho^{|S|} \chi_S$.

Nonexample. Replacing each coordinate with an independent fair sign with probability $1 - \rho$ has correlation ρ only for $0 \leq \rho \leq 1$; negative correlations require a different coupling.

Theorem 8.2 (Noise-stability Fourier formula). *For real functions f, g on $\{-1, 1\}^n$ and the ρ -correlated pair (X, Y) of Definition 8.1,*

$$\mathbb{E}[f(X)g(Y)] = \sum_{S \subseteq [n]} \rho^{|S|} \widehat{f}(S) \widehat{g}(S).$$

In particular,

$$\text{Stab}_\rho(f) := \mathbb{E}[f(X)f(Y)] = \sum_S \rho^{|S|} \widehat{f}(S)^2.$$

Proof. Expand both functions in the Walsh characters χ_S . Independence across coordinates and $\mathbb{E}[X_i Y_i] = \rho$ give

$$\mathbb{E}[\chi_S(X)\chi_T(Y)] = \mathbf{1}_{\{S=T\}}\rho^{|S|}.$$

Indeed, if $S \triangle T$ is nonempty, one coordinate appears to the first power alone and has expectation zero; if $S = T$, every selected coordinate contributes $\mathbb{E}[X_i Y_i] = \rho$. Substituting into the two Fourier expansions leaves exactly the stated diagonal sum. $\square$

Theorem 8.3 (Bonami–Beckner hypercontractive inequality — source-owned). *Let $1 < p \leq q < \infty$ and $0 \leq \rho \leq \sqrt{(p-1)/(q-1)}$. For every real function on the uniform Boolean cube,*

$$\|T_\rho f\|_q \leq \|f\|_p.$$

The two-point inequality and tensorization are retained as a source-owned deep input in this closure pass; no stronger range of parameters is inferred.

Theorem 8.4 (Low-degree hypercontractive moment and tail estimate). *Suppose $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ has Fourier degree at most d . For every $q \geq 2$,*

$$\|f\|_q \leq (q-1)^{d/2} \|f\|_2.$$

Consequently, for $t > 0$ and every $q \geq 2$,

$$\Pr(|f| \geq t\|f\|_2) \leq \left(\frac{(q-1)^{d/2}}{t} \right)^q,$$

and the best bound obtainable from this displayed family is their infimum over $q \geq 2$.

Proof. Fix $q \geq 2$ and put $\rho = (q-1)^{-1/2}$. Define g by $\widehat{g}(S) = \rho^{-|S|}\widehat{f}(S)$ on the Fourier support of f . Then $f = T_\rho g$. The source-owned Theorem 8.3, with $p = 2$, gives $\|f\|_q \leq \|g\|_2$. Parseval and $|S| \leq d$ imply

$$\|g\|_2^2 = \sum_S \rho^{-2|S|} \widehat{f}(S)^2 \leq \rho^{-2d} \|f\|_2^2 = (q-1)^d \|f\|_2^2.$$

This is the moment estimate. Markov's inequality applied to $|f|^q$ gives the tail bound. $\square$

Theorem 8.5 (KKL influence theorem — source-owned). *There is an absolute constant $c > 0$ such that every nonconstant Boolean $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ with $n \geq 2$ has a coordinate i satisfying*

$$\text{Inf}_i(f) \geq c \text{Var}(f) \frac{\log n}{n},$$

where influence and variance use the conventions fixed earlier in this volume. The hypercontractive influence argument is source-owned to the Bonami–Beckner–Kahn–Kalai–Linial chain.

Example 8.6 (Finite Fourier and concentration laboratory — completed artifact). The executable file `code/volume_01/fourier_concentration_lab.py` computes exact Walsh coefficients on a small cube, verifies Parseval, convolution and the noise-stability identity, and compares the Markov, Chebyshev and Chernoff bounds from Example 2.6 with the exact binomial tail. It uses no nondeterministic seed and asserts every normalization before printing its PASS token.

Chapter 9

Randomized Algorithms and Probabilistic Existence

The probabilistic method is converted into an explicit deterministic selection rule whenever conditional expectations or pessimistic estimators are computable.

Definition 9.1 (Randomized algorithm and seed complexity). A randomized algorithm is a deterministic map $A(x; r)$ together with a uniformly sampled finite seed $r \in \{0, 1\}^{s(|x|)}$. Its resource cost includes both running time and seed length. A Monte Carlo algorithm has a prescribed running-time bound and may err; a Las Vegas algorithm is always correct but has random running time. One-sided and two-sided error guarantees are stated separately.

For a finite product probability space, a pessimistic estimator for a cost C is a sequence $\Phi_i(x_1, \dots, x_i)$ such that the initial value is controlled, for every positive-probability prefix the average over positive-probability next-coordinate values does not exceed the current value, and the terminal estimator dominates the realized cost.

Positive example. Randomly selecting a hash function from a finite pairwise-independent family is a randomized algorithm; its seed specifies the selected family member.

Nonexample. A deterministic algorithm with an unspecified source of “random-looking” constants is not a randomized algorithm until the seed distribution is fixed.

Theorem 9.2 (Probabilistic-method expectation principle). *Let Z be a nonnegative integer-valued random variable. If $\mathbb{E}Z < 1$, then some outcome in the support of the probability distribution satisfies $Z = 0$. More generally, every real random variable C on a finite probability space has an outcome in the support with $C \leq \mathbb{E}C$. On a finite product space, after any positive-probability partial assignment there is a value in the support of the next coordinate that does not increase the conditional expectation of C .*

Proof. If $Z \geq 1$ at every outcome of positive probability, then $\mathbb{E}Z \geq 1$, contradicting the hypothesis. Thus $Z = 0$ at an outcome in the support.

If $C(\omega) > \mathbb{E}C$ at every outcome in the support, averaging would give $\mathbb{E}C > \mathbb{E}C$, impossible. Hence some outcome has $C \leq \mathbb{E}C$.

For the product-space assertion, fix a positive-probability prefix $x_1, \dots, x_i$. The conditional expectation before revealing coordinate $i + 1$ is the weighted average of the conditional expectations

after each value a in the support of that coordinate:

$$\mathbb{E}[C \mid x_1, \dots, x_i] = \sum_{a: \Pr(X_{i+1}=a) > 0} \Pr(X_{i+1} = a) \mathbb{E}[C \mid x_1, \dots, x_i, a].$$

At least one positively weighted term is no larger than their average. $\square$

Worked check. If Z counts monochromatic edges in a random two-coloring and $\mathbb{E}Z < 1$, then some coloring has no monochromatic edge.

Theorem 9.3 (Method of conditional expectations). *Let $\Omega = \Omega_1 \times \dots \times \Omega_n$ carry a product distribution, and let $C : \Omega \rightarrow \mathbb{R}$. Suppose functions $\Phi_i : \Omega_1 \times \dots \times \Omega_i \rightarrow \mathbb{R}$ satisfy*

1. $\Phi_0 \leq B$;
2. for each $i = 0, \dots, n-1$ and every prefix $x_{\leq i}$ in the support of the first i coordinates,

$$\sum_{a: \Pr(X_{i+1}=a) > 0} \Pr(X_{i+1} = a) \Phi_{i+1}(x_{\leq i}, a) \leq \Phi_i(x_{\leq i});$$

3. $C(x_1, \dots, x_n) \leq \Phi_n(x_1, \dots, x_n)$ on the support of the product distribution.

Then one can choose supported coordinate values greedily so that the resulting outcome satisfies $C \leq B$. If every Φ_i is efficiently computable and each supported finite choice can be searched efficiently, this is a deterministic algorithm. Exact conditional expectations are a special case.

Proof. Start with the empty prefix. Given a supported prefix $x_{\leq i}$, condition (2) says that the weighted average of the positively weighted candidate values $\Phi_{i+1}(x_{\leq i}, a)$ is at most $\Phi_i(x_{\leq i})$. Hence at least one supported candidate a satisfies

$$\Phi_{i+1}(x_{\leq i}, a) \leq \Phi_i(x_{\leq i}).$$

Choose such an a and continue. Inductively,

$$\Phi_n(x_1, \dots, x_n) \leq \Phi_0 \leq B.$$

Condition (3) gives $C(x_1, \dots, x_n) \leq B$. The stated computational claim follows because every greedy step is then executable within the specified resource bound. Taking $\Phi_i = \mathbb{E}[C \mid X_1 = x_1, \dots, X_i = x_i]$ makes (2) an equality and (3) an equality at $i = n$. $\square$

Worked check. If C is an integer-valued number of bad events and $\Phi_0 < 1$, the greedy outcome has $C < 1$, hence $C = 0$.

Chapter 10

Mathematical Notions of Randomness and Pseudorandomness

Randomness is always relative to a specified family of tests. Information-theoretic, algebraic, and computational notions are kept distinct.

Definition 10.1 (Randomness relative to a test class). Let U be a reference distribution on a finite set S and let $\mathcal{T} \subseteq [0, 1]^S$. A distribution P ε -fools $\mathcal{T}$ relative to U when

$$|\mathbb{E}_P T - \mathbb{E}_U T| \leq \varepsilon \quad \text{for every } T \in \mathcal{T}.$$

A function $\nu : \mathbb{N} \rightarrow [0, \infty)$ is *negligible* when, for every constant $c > 0$, one has $\nu(n) < n^{-c}$ for all sufficiently large n . For ensembles $(P_n), (Q_n)$ on common finite domains, computational indistinguishability means that the advantage of every uniform probabilistic polynomial-time test, given the unary parameter 1^n and a sample, is negligible in n . An ensemble is efficiently samplable when a uniform polynomial-time randomized algorithm outputs a sample with exactly that law. On a finite abelian group, Fourier bias refers to $|\mathbb{E}_P \chi|$ for nontrivial characters χ .

Positive example. The uniform distribution U zero-fools every test class relative to itself.

Nonexample. Efficient samplability alone is not pseudorandomness: a deterministic point mass is efficiently samplable and is immediately distinguished from uniform.

Proposition 10.2 (Hierarchy of randomness notions). *Let U be uniform on a finite set S .*

1. $P = U$ if and only if $d_{\text{TV}}(P, U) = 0$, if and only if P zero-fools all $[0, 1]$ -valued tests.
2. If $d_{\text{TV}}(P, U) \leq \varepsilon$, then P ε -fools every test class. Conversely, fooling the class of all tests to error ε implies $d_{\text{TV}}(P, U) \leq \varepsilon$.
3. On a finite abelian group, $P = U$ if and only if $\mathbb{E}_P \chi = 0$ for every nontrivial character. Statistical distance at most ε implies $|\mathbb{E}_P \varphi - \mathbb{E}_U \varphi| \leq 2\varepsilon$ for every complex observable $\varphi : S \rightarrow \mathbb{C}$ satisfying $|\varphi(x)| \leq 1$; in particular, this holds for every character.
4. Negligible statistical distance between ensembles implies computational indistinguishability.
5. A distribution may fool a fixed restricted test class while remaining far from uniform in total variation. Assuming secure length-expanding pseudorandom generators exist, computational indistinguishability need not imply statistical closeness. Efficient samplability alone implies no pseudorandomness guarantee.

Proof. Parts (1) and (2) follow directly from Theorem 5.4. In particular, the class of all tests contains an optimal likelihood-sign indicator.

For part (3), let G be the underlying group and define the density $f(x) = |G|P(x)$ relative to normalized counting measure. Then

$$\hat{f}(\chi) = \sum_x P(x) \overline{\chi(x)}.$$

If every nontrivial character expectation vanishes, Fourier inversion (Theorem 6.3) gives $f \equiv 1$, hence $P = U$. The converse is character orthogonality. For any complex observable ϕ with $|\phi| \leq 1$,

$$|\mathbb{E}_P \phi - \mathbb{E}_U \phi| \leq \sum_x |P(x) - U(x)| = 2d_{\text{TV}}(P, U),$$

which applies to characters.

For part (4), every probabilistic algorithm induces a $[0, 1]$ -valued acceptance probability test. Its advantage is at most total variation by part (2), so a negligible statistical bound is also a negligible computational bound.

We now record the failed converses.

- A restricted class need not detect statistical distance. On $S = \{-1, 1\}^2$, let U be uniform, let P be uniform on $\{(1, 1), (-1, 1)\}$, and let the test class contain only constants and tests depending on the first coordinate. Then P fools the class exactly, but $d_{\text{TV}}(P, U) = 1/2$.
- Assuming a secure length-expanding pseudorandom generator $G_n : \{0, 1\}^{s(n)} \rightarrow \{0, 1\}^{m(n)}$, the output distribution $G_n(U_{s(n)})$ is computationally indistinguishable from $U_{m(n)}$ by assumption. Yet its support has size at most $2^{s(n)}$, so

$$d_{\text{TV}}(G_n(U_s), U_m) \geq 1 - 2^{s(n)-m(n)},$$

which is close to one when the stretch grows.

- Efficient generation alone gives no test-fooling guarantee: the point mass at 0^n is efficiently samplable and is distinguished from uniform by the test $T(x) = \mathbf{1}_{\{0^n\}}(x)$.

These examples establish precisely the claimed failures, without asserting any unconditional separation that would require unresolved complexity assumptions. $\square$

Worked check. The restricted-test example has identical first-coordinate marginals under P and U , but the second coordinate is fixed under P and uniform under U .

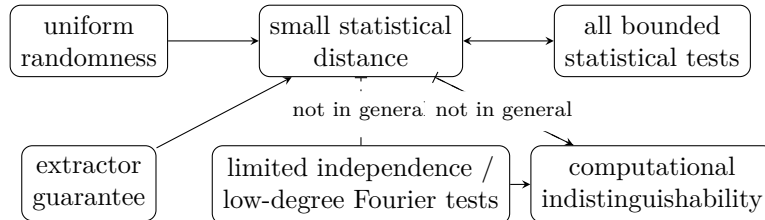


Figure 10.1: Hierarchy of randomness and tests. Solid arrows record implications under the conventions of this volume; dashed barred arrows mark converses that fail without additional hypotheses. Limited independence controls selected tests, whereas statistical distance controls every bounded test.

Internalization and Reconstruction Guide: Volume I

Purpose. Learn to move fluently among probabilistic tails, information measures, Fourier structure, and explicit test classes; the point is not four separate toolkits but a common language for saying when a distribution or function behaves randomly.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

This is a foundation volume. Its prerequisite ladder is internal: begin with the first definitions and reconstruct each later family from those conventions rather than importing unstated external notation. Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-I-DEF-001 (Finite probability space and random variable); QRCC-I-DEF-002 (Moment generating function and subgaussian proxy); QRCC-I-DEF-003 (Filtration, martingale, and bounded increments); QRCC-I-DEF-004 (Entropy, relative entropy, and mutual information); QRCC-I-DEF-005 (Total variation and statistical tests).

Chapter retrieval map

- Ch. 1: Finite Probability and Random Experiments:** enter through QRCC-I-DEF-001 (Finite probability space and random variable); exit by being able to state QRCC-I-THM-001 (Union bound) with its ownership boundary.
- Ch. 2: Concentration and Tail Inequalities:** enter through QRCC-I-DEF-002 (Moment generating function and subgaussian proxy); exit by being able to state QRCC-I-LEM-023 (Exponential-moment tail lemma) with its ownership boundary.
- Ch. 3: Martingales and Bounded Differences:** enter through QRCC-I-DEF-003 (Filtration, martingale, and bounded increments); exit by being able to state QRCC-I-THM-006 (Azuma–Hoeffding inequality) with its ownership boundary.
- Ch. 4: Entropy and Information:** enter through QRCC-I-DEF-004 (Entropy, relative entropy, and mutual information); exit by being able to state QRCC-I-LEM-033 (Strict concavity of finite entropy) with its ownership boundary.

- Ch. 5: Statistical Distance and Distinguishability:** enter through QRCC-I-DEF-005 (Total variation and statistical tests); exit by being able to state QRCC-I-THM-011 (Pinsker inequality) with its ownership boundary.
- Ch. 6: Fourier Analysis on Finite Abelian Groups:** enter through QRCC-I-DEF-006 (Normalized Fourier transform on a finite abelian group); exit by being able to state QRCC-I-LEM-034 (Character extension and dual cardinality) with its ownership boundary.
- Ch. 7: Boolean Fourier Analysis:** enter through QRCC-I-DEF-007 (Boolean characters, degree, and influence); exit by being able to state QRCC-I-THM-018 (Boolean Parseval identity) with its ownership boundary.
- Ch. 8: Noise, Influences, and Hypercontractivity:** enter through QRCC-I-DEF-008 (Noise operator and L_p norms on the cube); exit by being able to state QRCC-I-BB-031 (KKL influence theorem) with its ownership boundary.
- Ch. 9: Randomized Algorithms and Probabilistic Existence:** enter through QRCC-I-DEF-009 (Randomized algorithm and seed complexity); exit by being able to state QRCC-I-THM-021 (Probabilistic-method expectation principle) with its ownership boundary.
- Ch. 10: Mathematical Notions of Randomness and Pseudorandomness:** enter through QRCC-I-DEF-010 (Randomness relative to a test class); exit by being able to state QRCC-I-PROP-022 (Hierarchy of randomness notions) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-I-THM-001 — Union bound. Dependencies: QRCC-I-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-I-THM-006 — Azuma–Hoeffding inequality. Dependencies: QRCC-I-DEF-003, QRCC-I-LEM-024. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-I-THM-011 — Pinsker inequality. Dependencies: QRCC-I-THM-010, QRCC-I-DEF-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-I-THM-018 — Boolean Parseval identity. Dependencies: QRCC-I-DEF-007, QRCC-I-THM-016. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-I-THM-021 — Probabilistic-method expectation principle. Dependencies: QRCC-I-THM-001, QRCC-I-THM-002. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-I-PROP-022 — Hierarchy of randomness notions. Dependencies: QRCC-I-THM-014, QRCC-I-THM-015. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

Common failure modes

- changing between normalized and unnormalized Fourier transforms without changing constants;
- using independence where only martingale or bounded-difference hypotheses are available;
- treating small statistical distance as computational indistinguishability without specifying a test class;
- forgetting support-zero conventions in entropy or relative entropy;

Boundary and degenerate cases

- zero-probability events and degenerate random variables;
- support zeros in KL divergence and equality cases of Pinsker/coupling;
- the trivial character and normalization at group size one;
- noise parameters at 0 and 1 and degree-zero Boolean functions;

Retrieval and reconstruction drill

1. State **QRCC-I-DEF-001** from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at **QRCC-I-THM-001**. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For **QRCC-I-THM-006**, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from **QRCC-I-PROP-022**, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume II turns the abstract notion of a test class into computationally bounded tests and resource classes.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part II

Computability and Classical
Complexity

Chapter 1

Machines, Encodings, and Universal Simulation

Algorithms will be studied through a concrete machine model, but the eventual complexity classes should not depend on irrelevant engineering details. We therefore begin by fixing a sufficiently explicit multitape model and then prove two simulation theorems. The first says that one fixed machine can interpret descriptions of all others. The second shows that several tapes can be compressed to one tape at only quadratic time cost.

Definition 1.1 (Deterministic and nondeterministic Turing machines). A *k-tape deterministic Turing machine* is a tuple

$$M = (Q, \Sigma, \Gamma, \square, q_0, q_{\text{acc}}, q_{\text{rej}}, \delta),$$

where $k \geq 1$; Q is finite; $\Sigma \subseteq \Gamma$ are finite alphabets; $\square \in \Gamma \setminus \Sigma$ is the blank symbol; the three displayed states are distinct; and

$$\delta : (Q \setminus \{q_{\text{acc}}, q_{\text{rej}}\}) \times \Gamma^k \longrightarrow Q \times \Gamma^k \times \{-1, 0, +1\}^k$$

is a partial function. On input $x = x_1 \cdots x_n \in \Sigma^*$, tape 1 contains x beginning at its distinguished origin and every other cell is blank; the remaining tapes are blank; every head begins at its origin. A transition reads the k scanned symbols, writes k new symbols, changes state, and moves each head by at most one cell. Entering q_{acc} or q_{rej} halts the computation. If no transition is defined and the machine is not in a halting state, we regard the computation as divergent.

The *running time* on a halting input is the number of transitions before halting. The *space used* is the number of distinct tape cells visited, with the input cells included unless a later read-only-input convention is stated explicitly.

A *nondeterministic k-tape Turing machine* has instead a finite transition relation

$$\Delta \subseteq (Q \setminus \{q_{\text{acc}}, q_{\text{rej}}\}) \times \Gamma^k \times Q \times \Gamma^k \times \{-1, 0, +1\}^k.$$

Its computation is a finitely branching tree. It *accepts* an input when at least one branch reaches q_{acc} and rejects when every branch halts and reaches q_{rej} . Whenever a time bound is imposed later, every branch is required to halt within that bound.

Fix once and for all an effective binary encoding $\langle M \rangle$ of finite machine descriptions and an effective pairing function $\langle u, v \rangle$. Invalid binary strings are interpreted as one fixed machine that immediately rejects; hence every binary string denotes a machine.

Positive example. A two-tape machine can copy its input from tape 1 to tape 2 by repeatedly reading the current input symbol, writing the same symbol on the second tape, and moving both heads right until the first blank is reached.

Nonexample. A device whose transition rule asks whether the untouched suffix of the input contains a prime number of 1's is not a Turing-machine transition rule: a single transition is allowed to inspect only the finitely many symbols under the heads.

Definition-before-use discipline. All later machine descriptions, universal simulations, reductions, and complexity classes use this encoding and halting convention unless a theorem explicitly changes the model.

Theorem 1.2 (Universal simulation). *There exists a fixed deterministic multitape Turing machine U with the following property. For every deterministic Turing machine M and input x , U on input $\langle M, x \rangle$ has the same halting outcome as M on x ; if $M(x)$ diverges, then $U(\langle M, x \rangle)$ diverges. Moreover, if $M(x)$ halts after T steps and*

$$N = |\langle M \rangle| + |x| + T + 1,$$

then U can be chosen to halt within $O(N^3)$ steps and to use $O(N^2)$ tape cells. In particular, simulation by a universal machine incurs only polynomial overhead.

Proof. Because a machine description is finite, there is an effective procedure that parses $\langle M \rangle$ into the number of tapes, the alphabet, the finite state set, and the transition table. The universal machine first performs this parsing. If the description is invalid it simulates the fixed rejecting machine specified in Definition 1.1.

For a valid description, U stores a complete *instantaneous description* of the simulated computation. A convenient representation is a finite string for each simulated tape, delimited from the next tape, with one marked version of every tape symbol to indicate the simulated head position. The current state of M is kept on a separate work tape. Only cells lying between the leftmost and rightmost locations visited so far need to be stored; blank cells immediately outside this interval are implicit.

Suppose M has so far taken $t \leq T$ steps. Each simulated head can have moved at most t cells from its origin. Consequently the total encoded tape region has length

$$O(|x| + k(t + 1)),$$

where k is the number of tapes of M . Since $k \leq |\langle M \rangle| + 1$, the total represented region is $O(|x| + k(T + 1)) = O(N^2)$. This deliberately coarse bound is uniform even when the simulated machine itself, and hence its number of tapes, varies with the input to U .

To simulate one transition, U sweeps the stored configuration to collect the k marked symbols, scans the finite transition table in $\langle M \rangle$ to find the unique applicable rule, and sweeps again to perform the indicated writes and head moves. If a head moves onto a previously implicit blank cell, U inserts one encoded blank at the relevant boundary. At simulated time t , every such sweep has length $O(N^2)$, while scanning the program costs $O(N)$. Thus one simulated transition costs $O(N^2)$ actual transitions. Parsing costs at most $O(N^2)$ in this simple implementation, and the T simulated steps cost $O(TN^2) \leq O(N^3)$. The stored description, state, and visited tape regions occupy $O(N^2)$ cells.

If the simulated state is q_{acc} or q_{rej} , U enters the corresponding halting state. If the simulated computation never halts, the simulation loop never halts. Therefore halting behavior is preserved exactly. $\square$

Worked check. Take the two-tape copying machine from the positive example and input 101. The universal simulator successively stores configurations in which the marked heads move across the three input symbols, and after three copying transitions the second simulated tape contains 101. The program of the copying machine is data for U ; U itself never changes.

Theorem 1.3 (Multitape-to-single-tape simulation). *Let M be a deterministic k -tape Turing machine that, on every input of length n , halts within $T(n)$ steps, where $T(n) \geq n + 1$. Then there is a deterministic one-tape machine S deciding the same language and satisfying*

$$\text{time}_S(n) = O(T(n)^2), \quad \text{space}_S(n) = O(T(n)).$$

The constants may depend on M and k , but not on n .

Proof. Fix M . The one tape of S is divided logically into k tracks, separated by delimiter symbols. On track i , every simulated tape symbol has an unmarked and a marked version; exactly one symbol is marked, indicating the position of the i th simulated head. The finite control of S remembers the current state of M and, during a sweep, the finitely many scanned symbols.

Initially S writes the k track representation of the start configuration. After t simulated transitions, no simulated head can lie farther than t positions from its origin. Hence every track needs at most $n + 2t + O(1)$ represented cells. Since k is fixed and $t \leq T(n)$ with $n \leq T(n)$, the whole encoding has length $O(T(n))$.

To simulate one step of M , S makes a left-to-right sweep, records in its finite control the k symbols carrying head marks, and thereby determines the unique transition of M . A second sweep performs the k writes, removes the old marks, and places the new marks one position left, stationary, or right as prescribed. If a head crosses a represented boundary, S shifts a suffix by one cell and inserts a blank. Such a shift is another $O(T(n))$ sweep. Thus a single simulated transition costs $O(T(n))$ one-tape steps.

There are at most $T(n)$ simulated transitions, so their total cost is $O(T(n)^2)$. Initialization costs $O(T(n))$ and is absorbed. At every moment only the $O(T(n))$ -cell track representation and constant-size finite control are used, giving the asserted space bound. Because each simulated transition implements exactly the transition of M , acceptance and rejection are preserved. $\square$

Worked check. For a two-tape machine running for at most n transitions, each encoded track has length $O(n)$. A complete sweep therefore costs $O(n)$, and at most n simulated steps give the expected $O(n^2)$ one-tape bound.

Chapter 2

Computability, Reductions, and Undecidability

The point of a reduction is not merely to transform instances: it transfers a solution procedure from one problem to another. Once that logical direction is fixed precisely, one undecidable language becomes a reusable source of undecidability results.

Definition 2.1 (Decidability, recognizability, computability, and many-one reduction). A language $A \subseteq \{0,1\}^*$ is *decidable* if some deterministic Turing machine halts on every input and accepts exactly the strings in A . It is *recognizable* (recursively enumerable) if some deterministic machine accepts every string in A and never accepts a string outside A ; on strings outside A the machine may reject or diverge.

A total function $f : \{0,1\}^* \rightarrow \{0,1\}^*$ is *computable* if a Turing machine halts on every input x with $f(x)$ written in a designated output region. For languages A, B we write

$$A \leq_m B$$

and say that A *many-one reduces* to B if there is a total computable f satisfying

$$x \in A \iff f(x) \in B \quad \text{for every } x.$$

When f is computable in polynomial time we write $A \leq_m^p B$.

For a machine M , let φ_M denote the partial function computed by M : $\varphi_M(x)$ is its output when it halts with an output, and is undefined when it diverges. A property $\mathcal{P}$ of machines is *extensional* when it depends only on φ_M rather than on the syntax of $\langle M \rangle$.

Positive example. Let A be the set of binary strings of even length and B the set of strings ending in 0. The computable map $f(x) = 0$ for even $|x|$ and $f(x) = 1$ for odd $|x|$ witnesses $A \leq_m B$.

Nonexample. A transformation satisfying only $x \in A \Rightarrow f(x) \in B$ is not a many-one reduction: without the reverse implication, a decider for B need not distinguish no-instances of A .

Definition-before-use discipline. All reductions in the classical complexity volumes preserve both yes- and no-instances as above. Later randomized, promise, search, and approximation reductions will be defined separately rather than silently reusing this symbol.

Theorem 2.2 (Transfer along many-one reductions). *Suppose $A \leq_m B$.*

(i) If B is decidable, then A is decidable.

(ii) If B is recognizable, then A is recognizable.

Consequently, if A is undecidable then B is undecidable, and if A is not recognizable then B is not recognizable.

Proof. Let f be a total computable reduction from A to B . If D_B decides B , construct D_A as follows. On input x , compute $f(x)$, run D_B on $f(x)$, and return the same yes/no answer. Both stages halt, and

$$D_A \text{ accepts } x \iff f(x) \in B \iff x \in A.$$

Thus D_A decides A .

If R_B merely recognizes B , perform the same composition with R_B . When $x \in A$, the reduced instance lies in B , so R_B eventually accepts. When $x \notin A$, the reduced instance lies outside B , so R_B never accepts; it may reject or diverge. Hence the composition recognizes A .

The two final assertions are contrapositives of (i) and (ii), respectively. The fixed effective machine encoding from Definition 1.1 ensures that the composed machines are effectively constructible rather than merely described abstractly. $\square$

Worked check. If a known undecidable language A reduces to B , an alleged decider for B immediately yields a decider for A by composition. Notice that a reduction in the opposite direction, $B \leq_m A$, gives no such contradiction.

Theorem 2.3 (Undecidability of the halting problem). *Let*

$$\text{HALT} = \{\langle M, x \rangle : M \text{ eventually halts on input } x\}.$$

Then HALT is recognizable but not decidable.

Proof. Recognition follows from the universal simulator of Theorem 1.2. On input $\langle M, x \rangle$, simulate $M(x)$. If the simulation ever halts, accept. If it never halts, the recognizer runs forever. Thus exactly the members of HALT are eventually accepted.

Assume for contradiction that a machine H decides HALT. Construct a deterministic machine D that, on input y , interprets y as a machine description and behaves as follows:

1. Run H on $\langle y, y \rangle$.
2. If H says that the machine encoded by y halts on y , enter an infinite loop.
3. If H says that it does not halt, halt immediately.

Every binary string denotes a machine by our encoding convention, so this program is total up to the deliberate loop in step 2. Let $d = \langle D \rangle$ be its own finite description.

If $H(\langle d, d \rangle)$ answers *halts*, the definition of D makes $D(d)$ loop forever. If $H(\langle d, d \rangle)$ answers *does not halt*, the definition of D makes $D(d)$ halt immediately. Each answer is therefore wrong, contradicting the assumption that H decides HALT. Hence HALT is undecidable. $\square$

Worked check. The contradiction does not require a machine literally to inspect its own hardware. Its finite description d is an ordinary binary string, and the hypothetical decider H is asked about the ordinary pair $\langle d, d \rangle$. The self-reference is therefore implemented entirely through effective encoding.

Theorem 2.4 (Rice’s theorem). *Let $\mathcal{P}$ be a nontrivial extensional property of partial computable functions: some partial computable function has $\mathcal{P}$ and some partial computable function does not. Then the index language*

$$I_{\mathcal{P}} = \{\langle M \rangle : \varphi_M \text{ has property } \mathcal{P}\}$$

is undecidable.

Proof. Let $\perp$ denote the nowhere-defined partial function, computed by a machine that loops on every input.

First suppose $\perp$ does *not* have property $\mathcal{P}$. By nontriviality choose a machine G whose partial function $g = \varphi_G$ does have $\mathcal{P}$. We reduce HALT to $I_{\mathcal{P}}$. Given an instance $\langle M, x \rangle$, effectively construct a machine $N_{M,x}$ which, on input y , first simulates $M(x)$. If that simulation ever halts, $N_{M,x}$ then simulates $G(y)$ and returns whatever $G(y)$ returns. If $M(x)$ never halts, $N_{M,x}(y)$ never reaches the second stage for any y . The description of $N_{M,x}$ is obtained effectively by inserting the fixed strings $\langle M \rangle$ and x into this finite program template, so the map $\langle M, x \rangle \mapsto \langle N_{M,x} \rangle$ is total computable. Moreover,

$$\varphi_{N_{M,x}} = \begin{cases} g, & M(x) \text{ halts,} \\ \perp, & M(x) \text{ does not halt.} \end{cases}$$

Hence

$$\langle M, x \rangle \in \text{HALT} \iff \langle N_{M,x} \rangle \in I_{\mathcal{P}}.$$

If $I_{\mathcal{P}}$ were decidable, Theorem 2.2 would make HALT decidable, contradicting Theorem 2.3.

Now suppose $\perp$ *does* have property $\mathcal{P}$. The complementary property $\mathcal{P}^c$ is also nontrivial and extensional, and $\perp$ does not have $\mathcal{P}^c$. The first case shows that $I_{\mathcal{P}^c}$ is undecidable. But $I_{\mathcal{P}^c}$ is the complement of $I_{\mathcal{P}}$ under our convention that every binary string denotes a machine. A decider for one would give a decider for the other by reversing its answer. Therefore $I_{\mathcal{P}}$ is undecidable as well. $\square$

Worked check. The property “the computed partial function is defined on input 0” is nontrivial and extensional, so Rice’s theorem makes its index set undecidable. By contrast, “the machine description has an even number of states” is a syntactic property and is decidable by parsing the description; Rice’s theorem does not apply to it.

Chapter 3

Time Bounds and Model Robustness

Complexity theory asks not only whether a computation terminates, but how the resources required to terminate grow with the input length. This requires a small amount of bookkeeping before diagonal hierarchy arguments can be stated without ambiguity. In particular, the time bound itself must be available to a machine at roughly the cost advertised by that bound.

Definition 3.1 (Time constructibility and time classes). A function $t : \mathbb{N} \rightarrow \mathbb{N}$ is *time-constructible* if $t(n) \geq n + 1$ and there is a deterministic multitape Turing machine which, on input 1^n , outputs $1^{t(n)}$ and halts within $O(t(n))$ steps. Equivalently for all standard bounds used below, a machine can maintain a clock for $t(n)$ steps with only constant-factor overhead; whenever the exact-step convention is needed, we state it explicitly.

For a function $t(n) \geq n + 1$, $\text{DTIME}(t)$ is the collection of languages decided by a deterministic multitape machine in at most $ct(n)$ steps for some constant c and all sufficiently large n . Likewise $\text{NTIME}(t)$ is defined using nondeterministic machines, with every computation branch required to halt within the stated bound and an input accepted when at least one branch accepts. We write

$$\text{P} = \bigcup_{d \geq 1} \text{DTIME}(n^d), \quad \text{NP} = \bigcup_{d \geq 1} \text{NTIME}(n^d).$$

Changing finitely many input lengths does not change any of these classes.

Positive example. The linear bound $t(n) = n + 1$ is time-constructible: on unary input the machine scans the n input symbols while copying them to an output tape and then writes one final 1. Standard integer polynomials are time-constructible as well; their routine counter constructions will be used only when needed later.

Nonexample. An arbitrary function $t : \mathbb{N} \rightarrow \mathbb{N}$ is not automatically an admissible clock merely because it grows quickly. If t is not computable, no Turing machine can in general know when exactly $t(n)$ steps have elapsed.

Proposition 3.2 (Polynomial-time robustness under tape compression). *Suppose a language L is decided by a fixed k -tape deterministic Turing machine in time $T(n) \geq n + 1$. Then L is decided by a one-tape deterministic Turing machine in time $O(T(n)^2)$. Consequently the class P is the same whether it is defined using one-tape machines or a fixed finite number of tapes.*

Proof. The first assertion is exactly the simulation constructed in Theorem 1.3: one simulated transition costs $O(T(n))$ one-tape steps because the complete track representation has length $O(T(n))$, and there are at most $T(n)$ simulated transitions.

For the class statement, let $L \in \mathsf{P}$ under the multitape convention. Then some fixed k -tape machine decides L in time $O(n^d)$ for an integer $d \geq 1$. Increasing the implicit constant if necessary gives a bound $T(n) \geq n + 1$ of order n^d . The simulation therefore decides L on one tape in time $O(n^{2d})$, still polynomial. Conversely every one-tape machine is already a multitape machine with $k = 1$. Thus the two definitions of P coincide. $\square$

Worked check. A three-tape algorithm running in time $O(n^4)$ may become an $O(n^8)$ one-tape algorithm under this deliberately simple simulation. The exponent changes, but membership in P does not.

Theorem 3.3 (Deterministic time hierarchy — source-owned sharp form). *Let $t(n) \geq n \log n$ be time-constructible. Under the multitape convention of Definition 3.1,*

$$\bigcup_{s=o(t/\log t)} \text{DTIME}(s) \subsetneq \text{DTIME}(t).$$

The near-linear universal simulation and diagonal language giving this sharp normalization remain source-owned to the classical time-hierarchy lineage.

Theorem 3.4 (Nondeterministic time hierarchy — source-owned sharp form). *Let t_1, t_2 be time-constructible bounds satisfying $t_1(n + 1) = o(t_2(n))$. Then*

$$\text{NTIME}(t_1) \subsetneq \text{NTIME}(t_2).$$

This is the strong unary-padding/diagonalization form; its exact simulation and diagonal proof are source-owned to the Zak–Seiferas–Fischer–Meyer lineage rather than reconstructed from the polynomial-time robustness result above.

Remark 3.5 (Why the source boundary matters). A quadratic simulator is enough to prove robustness of P but not the sharp hierarchy between nearby time bounds. The two hierarchy statements therefore import the exact simulation/diagonalization theorem and do not infer it from Proposition 3.2.

Chapter 4

Nondeterministic Polynomial Time and Verifiers

Nondeterministic computation is most useful when translated into the language of efficiently checkable certificates. The translation is exact: a polynomially bounded accepting computation can be written down as a polynomial certificate, and a polynomial certificate can be guessed nondeterministically.

Definition 4.1 (Polynomially balanced verifier and NP). A *polynomial-time verifier* is a deterministic Turing machine V and a polynomial p such that, for every pair (x, w) with $|w| \leq p(|x|)$, $V(\langle x, w \rangle)$ halts in time polynomial in $|x|$ and returns either accept or reject. A language L has a *polynomially balanced verifier* if

$$x \in L \iff \exists w \in \{0, 1\}^{\leq p(|x|)} V(\langle x, w \rangle) = 1.$$

The string w is called a *witness* or *certificate*.

A language belongs to NP if some nondeterministic Turing machine decides it in polynomial time in the sense of Definition 3.1. A language A is *NP-hard* under polynomial-time many-one reductions if every $L \in \text{NP}$ satisfies $L \leq_m^p A$, and it is *NP-complete* if in addition $A \in \text{NP}$.

Positive example. For CLIQUE, a witness for the assertion that a graph G contains a clique of size at least k is a list of k vertices. A verifier checks in polynomial time that the vertices are distinct and that every pair is joined by an edge.

Nonexample. The assertion “there exists a proof eventually found by some terminating search” is not by itself an NP verification condition: the witness length and the verifier running time must both be bounded by polynomials in the input length.

Theorem 4.2 (Verifier characterization of NP). *For a language $L \subseteq \{0, 1\}^*$, the following are equivalent.*

(i) $L \in \text{NP}$.

(ii) *There are a deterministic polynomial-time verifier V and a polynomial p such that*

$$x \in L \iff \exists w, |w| \leq p(|x|), V(\langle x, w \rangle) = 1.$$

Moreover the transformations between the two descriptions are effective.

Proof. Assume first that $L \in \text{NP}$. Let N be a nondeterministic machine which, on every input x of length n , halts on every branch within $q(n)$ steps for a polynomial q . Because the transition relation of N is finite, there is a constant b such that every configuration has at most 2^b successors. Number the possible choices at each step by b -bit strings, padding with rejecting choices when fewer successors exist.

A witness w consists of $bq(n)$ bits, divided into $q(n)$ blocks. The verifier V simulates $N(x)$ for at most $q(n)$ steps, using the next block of w to choose the transition at each nondeterministic step. It accepts exactly when the selected branch reaches q_{acc} . The simulation is polynomial time by the universal-simulation machinery of Chapter 1; because N is fixed, a direct simulator would also suffice. Thus every accepting branch gives a polynomial witness. Conversely, if V accepts a witness, the corresponding sequence of choices describes an accepting branch of N . Hence (ii) holds.

Now assume (ii). Construct a nondeterministic machine N_V as follows. On input x , it nondeterministically guesses a string w of length at most $p(|x|)$ and then runs the deterministic verifier $V(\langle x, w \rangle)$. To make the guessing stage have a fixed polynomial clock, N_V may guess exactly $p(|x|)$ symbols from the alphabet $\{0, 1, \#\}$. If a $\#$ occurs, the first one terminates the witness and the machine rejects if a later symbol is not $\#$; if no $\#$ occurs, all $p(|x|)$ guessed bits form the witness. Thus every branch has polynomial length. The machine accepts iff one guessed witness makes V accept, which by assumption occurs iff $x \in L$. Therefore $L \in \text{NP}$.

Both constructions are mechanical transformations of finite machine descriptions, so the equivalence is effective. $\square$

Worked check. For **CLIQUE**, nondeterministically guessing k vertex names and then checking all $\binom{k}{2}$ edges is exactly the nondeterministic version of the verifier in the positive example.

Chapter 5

Cook–Levin and the First NP-Complete Problems

The Cook–Levin theorem turns an entire bounded computation into a Boolean formula. The construction is valuable beyond NP-completeness: it is the basic prototype for representing a global computation by local consistency constraints.

Definition 5.1 (CNF, 3CNF, tableaux, and parsimonious reductions). A *literal* is a Boolean variable or its negation. A *clause* is a disjunction of literals, and a formula is in *conjunctive normal form* (CNF) if it is a conjunction of clauses. A *3CNF* formula is a CNF formula in which each clause has at most three literals. SAT and 3SAT are the languages of satisfiable CNF and 3CNF formulas, respectively.

For a time-bounded Turing-machine computation, a *tableau* is the finite array whose row t encodes the complete configuration after t transitions. We use an *augmented tape alphabet*: an unmarked tape symbol a records a cell not carrying the head, while a marked symbol $[q, a]$ records that the head is at that cell, the machine is in state q , and the underlying tape symbol is a . Thus every legal configuration contains exactly one marked symbol.

A polynomial-time many-one reduction f between search/counting problems is *parsimonious* when it induces a bijection between the solution set of an instance and the solution set of f of that instance. Ordinary NP-completeness requires only preservation of existence of a solution, not parsimony.

Positive example. The formula

$$(x \vee y) \wedge (\neg x \vee z \vee w)$$

is a 3CNF formula. A three-row tableau of a machine contains an initial row, one intermediate configuration, and one final configuration; local constraints specify which adjacent rows can occur consecutively.

Nonexample. The formula $(x \wedge y) \vee z$ is not in CNF as written. Likewise, an array of arbitrary strings is not a computation tableau unless every row is a legal configuration and consecutive rows obey one allowed transition.

Theorem 5.2 (Cook–Levin theorem). *For every language $L \in \text{NP}$ there is a polynomial-time computable map $x \mapsto \Phi_x$ to CNF formulas such that*

$$x \in L \iff \Phi_x \text{ is satisfiable.}$$

If a nondeterministic machine for L runs in time $T = p(|x|)$, where p is a fixed polynomial, the construction may be chosen to use $O(T^2)$ tableau cells and $O(T^2)$ Boolean variables and clauses, with constants depending only on the machine. Consequently SAT is NP-complete.

Proof. Let N be a fixed nondeterministic Turing machine deciding L in time at most $T = p(n)$ on inputs of length n . The track construction in the proof of Theorem 1.3 applies branch-by-branch to a fixed nondeterministic multitape machine: the simulator nondeterministically chooses one of the finitely many enabled transitions and then performs exactly the same track update. Hence, after replacing p by a larger polynomial if necessary, we may assume N is a one-tape nondeterministic machine. Modify N so that accepting and rejecting states have self-loops which leave the tape unchanged. Then x is accepted iff there is an accepting branch whose configuration at the common final time T is in q_{acc} .

1. A finite tableau window. In T transitions the head moves by at most T cells. Hence it is enough to record positions

$$I = \{-T, -T + 1, \dots, n + T\},$$

whose cardinality is $O(T)$ because $T \geq n + 1$. Let $\mathcal{A}$ be the finite augmented alphabet consisting of all unmarked symbols $a \in \Gamma$ and all marked symbols $[q, a]$ with $q \in Q$ and $a \in \Gamma$. Its size is a constant depending only on N .

For each time $0 \leq t \leq T$, cell $i \in I$, and symbol $s \in \mathcal{A}$, introduce a Boolean variable

$$X_{t,i,s},$$

intended to mean that cell i of row t contains augmented symbol s . There are $(T + 1)|I||\mathcal{A}| = O(T^2)$ variables.

2. Cell well-formedness. For every (t, i) add the clause

$$\bigvee_{s \in \mathcal{A}} X_{t,i,s}$$

and, for every distinct $s, s' \in \mathcal{A}$, add

$$\neg X_{t,i,s} \vee \neg X_{t,i,s'}.$$

Because $\mathcal{A}$ is fixed, this contributes only $O(T^2)$ clauses and forces each tableau cell to contain exactly one augmented symbol.

3. Initial and accepting rows. The complete initial configuration on x is fixed. For every $i \in I$ add a unit clause selecting the unique initial augmented symbol at $(0, i)$: the head and start state occur at the origin, the input occupies its prescribed cells, and every other recorded cell is blank. At time T , require some cell to carry the accepting state by the clause

$$\bigvee_{i \in I} \bigvee_{a \in \Gamma} X_{T,i,[q_{\text{acc}},a]}.$$

This clause has $O(T)$ literals and is still a CNF clause.

4. Local transition constraints. A Turing-machine transition changes only the scanned cell, the state/head marker, and possibly which adjacent cell receives the new marker. The safest local test therefore compares a complete 2×3 window: three consecutive cells in row t and the same three cells in row $t + 1$.

Call a six-tuple

$$(a, b, c; d, e, f) \in \mathcal{A}^6$$

locally legal if it occurs as such a 2×3 window inside two consecutive genuine configurations related by one allowed transition of N ; windows near the recorded boundary use the permanently blank outside cells. Because N and $\mathcal{A}$ are fixed, the set of legal windows is finite and can be tabulated once from the transition relation. For every $t < T$, every center i , and every six-tuple which is not locally legal, add the clause

$$\neg X_{t,i-1,a} \vee \neg X_{t,i,b} \vee \neg X_{t,i+1,c} \vee \neg X_{t+1,i-1,d} \vee \neg X_{t+1,i,e} \vee \neg X_{t+1,i+1,f}.$$

There are only constantly many six-tuples for each (t, i) , so the total number of transition clauses is still $O(T^2)$. At the two ends of I we use the analogous finite list with the outside symbol fixed to blank.

We now justify the local-to-global step rather than hide it in the phrase “standard tableau argument.” Suppose row t is a genuine configuration, with its unique marked symbol at position h . Every 2×3 window whose top row is disjoint from h occurs only in a transition that leaves those three underlying tape cells unchanged and unmarked, so row $t + 1$ is forced to agree with row t away from $h - 1, h, h + 1$. The overlapping windows centered at $h - 1, h, h + 1$ share two bottom cells at a time. Since each entire window must occur in one genuine one-step transition from the same top configuration, those overlaps force one common transition rule: its written symbol, next state, and head direction agree on all shared cells. Hence the bottom row has exactly the tape contents and the single marked state symbol produced by that one transition. Thus row $t + 1$ is a genuine successor configuration. Conversely, every genuine transition plainly makes all of its 2×3 windows locally legal. Starting from the fixed legal initial row, induction therefore shows that every satisfying tableau consists entirely of genuine successive configurations.

5. Correctness. If $x \in L$, choose an accepting branch of N and write its configurations from time 0 through T into the tableau, extending the accepting configuration by its self-loop when the branch accepts early. Set precisely the corresponding $X_{t,i,s}$ variables to true. Cell uniqueness, the initial row, the local transition clauses, and the final accepting clause are all satisfied, so Φ_x is satisfiable.

Conversely suppose Φ_x is satisfiable. The exactly-one clauses decode a unique symbol in every tableau cell. The initial clauses give the start configuration. By the local-to-global argument, row $t + 1$ is a legal nondeterministic successor of row t for every $t < T$. Hence the rows describe one computation branch of N . The final clause places that branch in q_{acc} at time T , so N accepts x and therefore $x \in L$.

6. Polynomial construction. The tableau has $O(T^2)$ cells. Each cell contributes constantly many well-formedness and local clauses, except for the one $O(T)$ -literal accepting clause. Since $T = p(|x|)$ and N is fixed, a deterministic machine can list all variables and clauses in time polynomial in $|x|$.

Finally, $\text{SAT} \in \text{NP}$ because a truth assignment is a polynomial-size witness and the formula can be evaluated in linear time in its encoding length. The reduction above proves that every language in NP reduces to SAT; hence SAT is NP-complete. $\square$

Worked tableau check. Consider a toy machine whose only nondeterministic decision on input 1 is to either accept immediately or reject immediately. At time 0 the unique marked symbol is $[q_0, 1]$. The transition clauses allow exactly the two legal successor patterns corresponding to the two choices. The final acceptance clause eliminates the rejecting successor, leaving a satisfying assignment precisely for the accepting branch. The formula therefore records the existence of an accepting computation rather than asking a Boolean solver to simulate arbitrary machine semantics from scratch.

Theorem 5.3 (3SAT is NP-complete). *There is a polynomial-time many-one reduction from SAT to 3SAT. Therefore 3SAT is NP-complete.*

Proof. Membership in NP is immediate: a truth assignment is a witness, and each clause can be checked in polynomial time.

Let $\Phi = C_1 \wedge \cdots \wedge C_m$ be a CNF formula. Clauses of length at most three are left unchanged. Consider a clause of length $r \geq 4$,

$$C = (\ell_1 \vee \ell_2 \vee \cdots \vee \ell_r).$$

Introduce fresh variables $y_1, \dots, y_{r-3}$ used nowhere else and replace C by

$$\begin{aligned} &(\ell_1 \vee \ell_2 \vee y_1) \\ &\wedge (\neg y_1 \vee \ell_3 \vee y_2) \\ &\wedge \cdots \\ &\wedge (\neg y_{r-4} \vee \ell_{r-2} \vee y_{r-3}) \\ &\wedge (\neg y_{r-3} \vee \ell_{r-1} \vee \ell_r). \end{aligned}$$

Every new clause has three literals.

We verify equisatisfiability. Suppose first that the original clause C is true. Define

$$y_i = 1 \iff \ell_1, \dots, \ell_{i+1} \text{ are all false} \quad (1 \leq i \leq r-3).$$

If either ℓ_1 or ℓ_2 is true, the first new clause is true; otherwise $y_1 = 1$. For an intermediate clause $\neg y_i \vee \ell_{i+2} \vee y_{i+1}$, if $y_i = 0$ it is already true, while if $y_i = 1$ then either ℓ_{i+2} is true or the first $i+2$ literals are all false, in which case $y_{i+1} = 1$. For the final clause, if $y_{r-3} = 0$ it is true through $\neg y_{r-3}$; if $y_{r-3} = 1$, the first $r-2$ literals are false, so the truth of C forces $\ell_{r-1} \vee \ell_r$ to be true. Thus the new chain is satisfiable.

Conversely, suppose all new clauses are true while all original literals $\ell_1, \dots, \ell_r$ are false. The first clause forces $y_1 = 1$. The second then forces $y_2 = 1$, and induction forces $y_1 = \cdots = y_{r-3} = 1$. But the final clause becomes $\neg y_{r-3}$ and is false, a contradiction. Thus satisfaction of the new clauses implies that at least one original literal is true.

Apply this replacement independently to every long clause. Fresh variables for different clauses are disjoint, so the whole transformed formula is satisfiable iff Φ is. A clause of length r creates $r-3$ new variables and $r-2$ clauses, hence the output length is linear in the total number of literal occurrences of Φ . The transformation is therefore computable in polynomial time. Since SAT is NP-complete by the Cook–Levin theorem, 3SAT is NP-complete. $\square$

Worked reduction. The five-literal clause

$$a \vee b \vee c \vee d \vee e$$

becomes

$$(a \vee b \vee y_1) \wedge (\neg y_1 \vee c \vee y_2) \wedge (\neg y_2 \vee d \vee e).$$

If, for example, only d is true, the choice $y_1 = y_2 = 1$ satisfies all three clauses. If all five original literals are false, the first two clauses force $y_1 = y_2 = 1$ and the last clause fails.

5.1 A first reduction network beyond satisfiability

The point of NP-completeness is not that every difficult-looking problem needs a fresh machine simulation. Once one complete problem is available, one can move between concrete combinatorial formulations by explicit encodings. The next result records three reductions that will be reused repeatedly.

Theorem 5.4 (Selected classical NP-completeness reductions from 3SAT). *The decision problems CLIQUE, VERTEX-COVER, and SUBSET-SUM are NP-complete. More precisely, there are polynomial-time many-one reductions*

$$3\text{SAT} \leq_m^p \text{CLIQUE} \leq_m^p \text{VERTEX-COVER} \quad \text{and} \quad 3\text{SAT} \leq_m^p \text{SUBSET-SUM}.$$

Here CLIQUE asks whether a graph has a clique of size at least k , VERTEX-COVER asks whether a graph has a vertex cover of size at most k , and SUBSET-SUM asks whether a submultiset of a finite list of nonnegative integers sums exactly to a given target.

Proof. All three problems are in NP: the proposed clique, vertex cover, or subset of the input list is a polynomial-size certificate whose defining property is checked in polynomial time. It remains to prove NP-hardness.

1. From 3SAT to CLIQUE. Let

$$\Phi = C_1 \wedge \cdots \wedge C_m$$

be a 3CNF formula. Delete tautological clauses, delete repeated literals inside a clause, and handle an empty clause by outputting a fixed no-instance. These operations preserve satisfiability and take linear time.

Construct a graph G_Φ with one vertex (i, ℓ) for every occurrence of literal ℓ in clause C_i . Two vertices (i, ℓ) and (j, ℓ') are adjacent exactly when

$$i \neq j \quad \text{and} \quad \ell' \neq \neg\ell.$$

Set the target clique size to m . The graph has at most $3m$ vertices, and its adjacency relation can be listed in polynomial time.

If Φ has a satisfying assignment, choose in each clause one literal made true by the assignment. No two chosen literals are complementary, so the m chosen vertices form a clique.

Conversely, suppose G_Φ has a clique K of size m . There are no edges between two vertices belonging to the same clause, hence K contains exactly one vertex from each of the m clauses. Because every pair of vertices in K is adjacent, the chosen literals contain no complementary pair. Assign every variable appearing positively among the chosen literals the value true, every variable appearing negatively the value false, and assign all remaining variables arbitrarily. This is consistent and makes the chosen literal in every clause true. Thus Φ is satisfiable.

2. From CLIQUE to VERTEX-COVER. For an n -vertex graph G and integer k , first handle the boundary cases. If $k \leq 0$, the clique instance is trivially yes, so output a fixed yes-instance of VERTEX-COVER. If $k > n$, the clique instance is no, so output a fixed no-instance. We may therefore assume $1 \leq k \leq n$ and output

$$(\overline{G}, n - k),$$

where $\overline{G}$ is the complement graph. For every $C \subseteq V(G)$,

$$C \text{ is a clique of } G \iff C \text{ is an independent set of } \overline{G} \iff V(G) \setminus C \text{ is a vertex cover of } \overline{G}.$$

The last equivalence follows because a set D is a vertex cover exactly when no edge has both endpoints in $V \setminus D$. Therefore G has a clique of size at least k iff $\overline{G}$ has a vertex cover of size at most $n - k$. Complementing an adjacency matrix or edge list takes polynomial time.

3. From 3SAT to SUBSET-SUM. Again let Φ have variables $x_1, \dots, x_n$ and clauses $C_1, \dots, C_m$, after deleting tautological clauses and repeated literals. Choose the base

$$B = 2n + 4.$$

We use $n + m$ base- B digit positions. The first n are variable columns and the last m are clause columns. For each variable x_i construct two numbers T_i, F_i . Both have digit 1 in variable column i and digit 0 in every other variable column. In clause column j , the number T_i has digit 1 exactly when x_i occurs in C_j , while F_i has digit 1 exactly when $\neg x_i$ occurs in C_j .

For every clause C_j construct two slack numbers A_j, D_j . They are zero in every column except clause column j , where their digits are 1 and 2, respectively. Finally set the target Z to have digit 1 in every variable column and digit 4 in every clause column.

No subset of the constructed list can create a carry: in a clause column the sum of all possible variable digits is at most $2n$ and the two slack digits sum to 3, while $B = 2n + 4$; in a variable column the sum is at most 2. Hence equality of integer sums is equivalent to column-by-column equality of base- B digits.

Suppose first that Φ is satisfiable. For each i , choose T_i if x_i is true and F_i if x_i is false. Thus every variable column sums to 1. If clause C_j has $r_j \in \{1, 2, 3\}$ satisfied literals, its selected variable numbers contribute r_j in clause column j . The deficit $4 - r_j \in \{1, 2, 3\}$ is supplied uniquely as one of

$$1, \quad 2, \quad 1 + 2$$

using A_j, D_j . The chosen numbers therefore sum to Z .

Conversely, suppose a submultiset sums to Z . Since there are no carries, each variable column must sum exactly to 1. The only nonzero entries in column i belong to T_i and F_i , so exactly one of those two numbers was chosen. Interpret that choice as a truth assignment. In clause column j , the chosen variable numbers contribute precisely the number r_j of literals of C_j made true by this assignment. The two slack numbers can contribute only 0, 1, 2, or 3. Since the target digit is 4, the equality is impossible when $r_j = 0$. Hence every clause has at least one true literal and Φ is satisfiable.

There are $2n + 2m$ constructed numbers, each with $n + m$ base- B digits and therefore $O((n + m) \log(n + 2))$ binary bits. The entire reduction is polynomial in the input size. This proves all three NP-hardness statements and completes the proof. $\square$

Worked reduction. For

$$(x \vee y) \wedge (\neg x \vee z),$$

the clique graph has two clause layers. Choosing the vertices labelled y and z gives a 2-clique and corresponds to the satisfying assignment $y = z = 1$. In the subset-sum construction, choosing exactly one of T_i, F_i for each variable makes the variable columns equal to the target; the slack columns then succeed exactly when both clauses receive at least one satisfied literal.

Remark 5.5 (Historical attribution for Hamiltonian cycle). The undirected Hamiltonian-cycle reduction is intentionally not folded into the preceding theorem. In his later retrospective attached to the reprint of his 1972 paper, Karp explicitly wrote that he had failed to prove NP-completeness of the undirected Hamiltonian circuit problem and that the missing reduction was supplied independently

by Lawler and Tarjan. The series therefore gives that construction its own source and theorem record.

Example 5.6 (A concrete Cook–Levin tableau). Take a toy nondeterministic machine whose accepting computation on input 01 uses at most three time steps and three tape cells. Introduce variables $C_{t,j,a}$ meaning “at time t , cell j carries symbol/state label a ”. The tableau formula contains: exactly-one clauses for every (t, j) ; unit clauses fixing the initial row; local transition clauses saying that every length-three window in row $t + 1$ is compatible with the corresponding window in row t ; and a clause requiring an accepting state in the last row. An accepting three-step branch determines a satisfying assignment by setting exactly its displayed labels true. Conversely, the exactly-one and local transition clauses force any satisfying assignment to decode row by row to a legal accepting branch. This finite instance is the small-scale form of the construction in the Cook–Levin theorem; the polynomial bound comes from the $O(T^2)$ number of tableau locations and a constant-size local clause family.

Theorem 5.7 (Undirected Hamiltonian cycle is NP-complete — source-owned). *A Hamiltonian cycle in a finite undirected graph is a simple cycle that visits every vertex exactly once. The language*

$$\text{UHC} = \{\langle G \rangle : G \text{ has a Hamiltonian cycle}\}$$

is NP-complete under polynomial-time many-one reductions. Membership in NP follows by using the cyclic ordering of the vertices as a certificate. NP-hardness is source-owned to the classical Hamiltonian-circuit lineage; in particular the published Garey–Johnson–Tarjan planar cubic Hamiltonian-circuit theorem is a stronger restriction and therefore implies the general undirected statement. The historical Lawler/Tarjan attribution is recorded separately from this published theorem route.

Chapter 6

Space Complexity and Recursive Reachability

Space behaves differently from time because storage can be reused. A polynomial-time computation has only polynomially many steps, but a polynomial-space computation may run for exponentially many distinct configurations before repetition becomes unavoidable. Savitch's theorem exploits this reuse: nondeterministic reachability can be checked deterministically by recursively guessing a midpoint, but enumerating those midpoints one at a time rather than storing an entire path.

Definition 6.1 (Space classes, constructibility, and configuration graphs). For space complexity we use a read-only input tape whose cells are not charged to the space bound, together with finitely many read-write work tapes. The *work space* of a computation branch is the total number of distinct work tape cells visited on that branch. For $s(n) \geq \log_2(n+2)$, $\text{DSPACE}(s)$ and $\text{NSPACE}(s)$ consist of languages decided by deterministic or nondeterministic machines using at most $O(s(n))$ work cells on every input of length n ; in the nondeterministic case every branch halts. We set

$$\text{PSPACE} = \bigcup_{d \geq 1} \text{DSPACE}(n^d).$$

A function s is *space-constructible* if, on input 1^n , a deterministic machine can mark exactly $s(n)$ work cells while using $O(s(n))$ work space.

Fix a machine M and input x . A *configuration* records the current state, input-head position, work-tape contents on the permitted cells, and all work-head positions. The *configuration graph* $G_{M,x,s}$ has one vertex for every well-formed configuration respecting the $O(s(|x|))$ space bound and a directed edge $C \rightarrow D$ when one legal transition of M carries C to D .

Positive example. A depth-first search of an explicitly stored graph on N vertices can reuse a stack and visited-set representation. Savitch's algorithm will do something more extreme: for an implicit graph with exponentially many configurations, it stores only $O(\log N)$ configurations along one recursive call stack.

Nonexample. Writing the entire input onto a fresh work tape and then declaring the input “free” is not legitimate space accounting. Only the designated read-only input tape is uncharged; copies made on work tapes count toward the space bound.

Theorem 6.2 (Savitch's theorem). *Let $s(n) \geq \log_2(n+2)$ be space-constructible. Then*

$$\text{NSPACE}(s(n)) \subseteq \text{DSPACE}(s(n)^2).$$

More precisely, reachability in the configuration graph of an $O(s(n))$ -space nondeterministic machine can be decided deterministically in $O(s(n)^2)$ work space.

Proof. Let M be a fixed nondeterministic machine using at most $c_0 s(n)$ work cells on inputs of length n , and fix an input x of length n . A configuration can be encoded using $O(s(n))$ bits. Indeed, the finite control costs a constant number of bits; the input-head position costs $O(\log n)$ bits; work symbols occupy $O(s(n))$ cells; and each work-head position costs $O(\log s(n))$, which is absorbed because the number of work tapes is fixed and $s(n) \geq \log(n+2)$. Hence for some constant c_1 the total number N of well-formed space-bounded configurations satisfies

$$N \leq 2^{c_1 s(n)}.$$

If an accepting configuration is reachable, then one is reachable by a simple path of length less than N , since cycles may be deleted.

For configurations C, D and an integer $j \geq 0$, define the predicate $\text{PATH}(C, D, j)$ to mean that D is reachable from C by at most 2^j directed edges. For $j = 0$, this means either $C = D$ or $C \rightarrow D$, and can be checked in $O(s(n))$ space by inspecting the two configuration encodings and the finite transition relation.

For $j \geq 1$ we have the exact recurrence

$$\text{PATH}(C, D, j) \iff \exists Z (\text{PATH}(C, Z, j-1) \wedge \text{PATH}(Z, D, j-1)),$$

where Z ranges over all well-formed space-bounded configurations. The forward implication chooses the configuration reached after the first $\min(2^{j-1}, r)$ edges of a path of length $r \leq 2^j$; the reverse implication concatenates the two paths.

A deterministic algorithm evaluates this recurrence by enumerating all possible encodings Z one at a time. It recursively tests the first half; if that test succeeds, it tests the second half. It never stores the list of all midpoints or an entire path. Let

$$J = \lceil c_1 s(n) \rceil.$$

Since $2^J \geq N$, every simple accepting path has length at most 2^J . The outer algorithm enumerates all accepting configurations D and tests $\text{PATH}(C_{\text{start}}, D, J)$.

It remains to count space carefully. At one recursion level we store the two endpoint configurations, the current midpoint under enumeration, the integer j , and constant control information, for a total of $O(s(n))$ bits. The recursion depth is $J = O(s(n))$. Once the first recursive call returns, its workspace is reused for the second; sibling calls are never stored simultaneously. Thus the total work space is

$$O(s(n)) \cdot O(s(n)) = O(s(n)^2).$$

The enumeration may take enormous time, but space complexity imposes no polynomial-time requirement. The deterministic algorithm accepts exactly when some accepting configuration of M is reachable, so it decides the same language. $\square$

Worked reachability check. If a toy configuration graph has at most 16 vertices, then $J = 4$ suffices. To test a path of length at most 16, the algorithm tries every possible midpoint Z and recursively asks for two paths of length at most 8, then 4, then 2, then 1. Only one chain of four recursive frames is resident at a time.

Theorem 6.3 ($\text{NP} \subseteq \text{PSPACE}$). *Every language in NP is decidable in deterministic polynomial space. Equivalently,*

$$\text{NP} \subseteq \text{PSPACE}.$$

Proof. Let $L \in \text{NP}$ and choose a nondeterministic machine N deciding L in time at most $p(n)$ for a polynomial p . On every computation branch, at most one new tape cell can be visited per transition on each fixed work tape. After converting the input to the read-only convention if necessary, N uses at most $O(p(n) + n) = O(p(n))$ work space. Thus

$$L \in \text{NSPACE}(q(n))$$

for some polynomial $q(n) \geq \log(n + 2)$ which may be chosen space-constructible. Savitch's theorem gives

$$L \in \text{DSPACE}(q(n)^2) \subseteq \text{PSPACE}.$$

Since L was arbitrary, the containment follows. $\square$

Worked check. If an NP machine runs in time n^5 , its branches use at most polynomial work space. Savitch's simulation may square that space bound, but the result is still polynomial; no claim of polynomial running time for the deterministic simulation is made.

6.1 Hierarchy, complement closure, and PSPACE completeness

Theorem 6.4 (Deterministic space hierarchy). *Let $s(n) \geq \log_2(n + 2)$ be space-constructible and write*

$$\text{DSPACE}(o(s)) := \bigcup_{g=o(s)} \text{DSPACE}(g).$$

Then

$$\text{DSPACE}(o(s)) \subsetneq \text{DSPACE}(s).$$

The statement uses the read-only-input convention of Definition 6.1.

Proof. The containment is immediate: if $g(n) = o(s(n))$, then $g(n) \leq s(n)$ for all sufficiently large n , and changing a decider on finitely many input lengths does not change its asymptotic space class.

We construct a diagonal language D . It is enough to diagonalize against one-work-tape deterministic deciders. Indeed, a fixed finite-multitape machine using $g(n)$ work cells can place its finitely many tape tracks in one delimited work-tape segment, mark each simulated head on its track, and rescan that segment to perform one transition. This may increase time arbitrarily, but it uses only $O(g(n))$ work space.

Fix the effective machine encoding used earlier. A string of the form

$$x = \langle M \rangle \# 1^r$$

is called a *self-description input*; write $n = |x|$. Inputs not having this form are rejected. From the read-only description $d = \langle M \rangle$ compute a positive integer c_d large enough that one simulated work cell of M , including its tape symbol and head-marker information, can be represented using at most c_d bits. We may choose c_d with binary length $O(\log(|d| + 2))$, so c_d and indices into d use $O(\log n) \subseteq O(s(n))$ space.

On a self-description input, the diagonal machine H first uses the space-constructibility machine for s to delimit $s(n)$ work positions. It allows the simulated machine to visit at most

$$h = \left\lfloor \frac{s(n)}{8(c_d + 1)} \right\rfloor$$

work cells. The transition table of M remains on the read-only input and is rescanned whenever a transition must be decoded; it is never copied wholesale to work space. The simulator stores only the currently used simulated cells, the simulated head position, a state index, and a constant number of counters and description indices.

Crucially, H simulates M on the *entire input* x , not merely on the padding suffix. If a simulated transition would exceed the h -cell cap, H halts and rejects. If $M(x)$ halts within the cap, H outputs the opposite answer. To make H halt even when the capped simulation loops, observe that a capped configuration has an encoding of length

$$c_d h + O(\log n + \log(|d| + 2)) = O(s(n)).$$

Hence there are at most $2^{Cs(n)}$ capped configurations for an absolute constant C . An $O(s(n))$ -bit counter therefore lets H stop and reject after that many simulated steps. If a deterministic capped computation has not halted by then, some configuration has repeated and it will loop forever. All mutable data used by H occupy $O(s(n))$ work cells, so $D = L(H) \in \text{DSPACE}(s)$.

Suppose toward a contradiction that a deterministic machine M_0 decides D in space $g(n) = o(s(n))$. Let $d_0 = \langle M_0 \rangle$ and let c_{d_0} be its fixed simulation constant. Choose r so large that, for

$$x_0 = d_0 \# 1^r, \quad (c_{d_0} + 1)g(|x_0|) < \frac{s(|x_0|)}{8}.$$

This is possible because d_0 and c_{d_0} are fixed, $g = o(s)$, and $s(n) \geq \log_2(n + 2)$. On x_0 the universal simulation never reaches its space cap. Since M_0 is a decider, it halts before a capped configuration repeats. Consequently H reaches the branch where it outputs the opposite answer to $M_0(x_0)$. But M_0 was assumed to decide $D = L(H)$, so

$$M_0(x_0) = H(x_0) = 1 - M_0(x_0),$$

a contradiction. Therefore $D \notin \text{DSPACE}(o(s))$. □

Worked separation. Taking $s(n) = n$ produces a language in linear deterministic space that is not in $\text{DSPACE}(g)$ for any $g(n) = o(n)$. The argument is a resource hierarchy theorem; it does not identify a natural language having this separation.

Lemma 6.5 (Exact reachable-count certificate). *Let $G = (V, E)$ be a directed graph whose vertices have m -bit encodings, let $a \in V$, fix $0 \leq j < 2^m$, and put*

$$R_j = \{v \in V : v \text{ is reachable from } a \text{ by at most } j \text{ edges}\}.$$

Assume the exact integer $c_j = |R_j|$ is available in binary. There is a nondeterministic $O(m)$ -space procedure which, on input a vertex v , certifies exactly one of the alternatives

$$v \in R_{j+1} \quad \text{or} \quad v \notin R_{j+1}.$$

Moreover, scanning all vertex encodings with this classifier computes the exact value $c_{j+1} = |R_{j+1}|$ on every accepting computation branch.

Proof. Membership has the direct certificate: guess a path of length at most $j + 1$ from a to v , storing only the current vertex and an m -bit step counter.

For nonmembership, first check $v \neq a$. Then nondeterministically guess, in strict lexicographic order, exactly c_j distinct vertices

$$u_1 < u_2 < \cdots < u_{c_j}.$$

For every u_i , give a guessed path proving $u_i \in R_j$ and check that $(u_i, v) \notin E$. Since the list contains c_j distinct members of R_j and $|R_j| = c_j$, any successful list is necessarily the complete set R_j . Therefore no vertex reachable within j steps has an edge to v , and together with $v \neq a$ this proves $v \notin R_{j+1}$. Conversely, when $v \notin R_{j+1}$, the lexicographically ordered list of all vertices in R_j is a valid certificate.

Thus, provided c_j is correct, exactly one of the two certification modes can succeed for each legal vertex. Scan all m -bit strings, ignore malformed encodings, and for every legal vertex nondeterministically choose one of the two modes. Increment a counter precisely when the membership mode succeeds. If all certifications succeed, the final counter equals $|R_{j+1}|$; no accepting branch can produce a different value. Only a constant number of vertex encodings and m -bit counters are stored at once, so the space is $O(m)$. $\square$

Theorem 6.6 (Immerman–Szelepcsényi). *For every space-constructible $s(n) \geq \log_2(n+2)$,*

$$\text{NSPACE}(s(n)) = \text{coNSPACE}(s(n)).$$

Proof. It is enough to prove closure under complement. Let a nondeterministic machine M decide L in $O(s(n))$ space, with every branch halting, and fix an input x . Its configuration graph has vertices encoded by $m = O(s(|x|))$ bits. Add one distinguished sink configuration t and an edge to t from every accepting configuration. Then

$$x \in L \iff t \text{ is reachable from the start configuration } a.$$

The augmented graph has at most $N = 2^m$ legal encodings.

Starting from $R_0 = \{a\}$ and $c_0 = 1$, apply the preceding lemma successively for $j = 0, 1, \dots, N-2$. Stage j takes the already certified value c_j and nondeterministically produces the uniquely forced next value $c_{j+1} = |R_{j+1}|$. Thus after the loop we have certified $c_1, \dots, c_{N-1}$. The loop index $j < N$, each count is at most N , and every vertex encoding uses only $O(m) = O(s(|x|))$ bits. After $N-1$ steps every reachable vertex has appeared, because any reachable vertex has a simple path of length at most $N-1$.

To recognize the complement of L , after computing c_{N-1} invoke the nonmembership certificate of the lemma for the distinguished sink t (or, equivalently, use the exact reachable list for R_{N-1} to certify that t is absent). If $x \notin L$, the true reachability sets provide an accepting branch through every counting stage and the final nonmembership certificate. If $x \in L$, then $t \in R_{N-1}$, so soundness of the exact-count certificates prevents every branch from falsely accepting. Hence $\bar{L}$ is in $\text{NSPACE}(s)$ up to constant factors. Applying the same inclusion to $\bar{L}$ gives equality of the two classes. $\square$

Theorem 6.7 (TQBF is PSPACE-complete). *The language of true fully quantified Boolean formulas is PSPACE-complete under polynomial-time many-one reductions.*

Proof. Membership is immediate by depth-first evaluation: recursively assign the outer quantified variable, evaluate the two children sequentially, and reuse workspace. The recursion depth and the stored partial assignment are linear in the formula length.

For hardness let M decide a language in polynomial space $s(n)$ and let G_x be its configuration graph. After the usual polynomial-time encoding, a configuration is an $m = O(s(n))$ bit string. Equality of two configurations is a Boolean formula of size $O(m)$, and the one-step relation $\text{STEP}(C, D)$ is a Boolean formula of size $\text{poly}(m)$: it checks the finite control/head update and requires every tape cell outside the head neighborhood to be unchanged. Put

$$R_0(C, D) := \text{EQ}(C, D) \vee \text{STEP}(C, D).$$

For $i \geq 1$ we build a formula saying that a midpoint Z works for both halves while keeping only one syntactic copy of R_{i-1} . Introduce fresh m -bit strings Z, U, V and a fresh selector bit b , and let

$$P_b(C, D, Z, U, V) := (\neg b \rightarrow (U = C \wedge V = Z)) \\ \wedge (b \rightarrow (U = Z \wedge V = D)),$$

where vector equalities abbreviate conjunctions of the m coordinate biimplications. Define

$$R_i(C, D) := \exists Z \forall b \exists U, V [P_b(C, D, Z, U, V) \wedge R_{i-1}(U, V)]. \quad (6.8)$$

The variables at level i are freshly bound, so prenexing after alpha-renaming causes no capture.

We prove by induction that $R_i(C, D)$ holds exactly when D is reachable from C in at most 2^i steps, allowing equality steps for padding. The base is the definition of R_0 . For the induction step, if a path has length at most 2^i , choose Z at the midpoint (or at the endpoint when the first half already contains the whole path). The cases $b = 0$ and $b = 1$ then ask for the two subpaths, each of length at most 2^{i-1} . Conversely, truth of (6.8) for both selector values gives $R_{i-1}(C, Z)$ and $R_{i-1}(Z, D)$, whose paths concatenate to length at most 2^i .

Let S_i be the formula size. The selector constraints have size $O(m)$ and only one copy of R_{i-1} occurs, so $S_i \leq S_{i-1} + O(m)$ apart from the fixed polynomial-size base relation. Thus $S_m = \text{poly}(m)$. A deterministic machine that halts while using m configuration bits cannot visit more than 2^m distinct configurations without repetition. Hence, with $C_{\text{start}}(x)$ the initial configuration,

$$\exists D [\text{ACCEPT}(D) \wedge R_m(C_{\text{start}}(x), D)]$$

is a polynomial-size QBF true exactly when M accepts x . This formula is computable from x in polynomial time, completing the reduction. $\square$

Example 6.8 (Configuration graph and Savitch laboratory — completed artifact). The executable file `code/volume_02/configuration_savitch_lab.py` constructs a finite directed configuration graph, checks nondeterministic reachability by breadth-first search, and independently evaluates the bounded recursive predicate used in Savitch's theorem. It records the recursion height and verifies the expected logarithmic-depth midpoint decomposition before printing a deterministic PASS token.

Chapter 7

The Polynomial Hierarchy

Definition 7.1 (Polynomial hierarchy). Put $\Sigma_0^P = \Pi_0^P = P$ and recursively $\Sigma_{k+1}^P = NP^{\Sigma_k^P}$ and $\Pi_{k+1}^P = coNP^{\Sigma_k^P}$. The polynomial hierarchy is $PH = \bigcup_{k \geq 0} \Sigma_k^P$.

Positive example. $NP = \Sigma_1^P$ and $coNP = \Pi_1^P$.

Nonexample. The hierarchy level records a bounded number of alternations; unrestricted polynomial-space alternation is a different model.

Theorem 7.2 (Quantifier characterization). *For $k \geq 1$, a language L is in Σ_k^P iff there are a polynomial p and a polynomial-time predicate R such that*

$$x \in L \iff \exists y_1 \forall y_2 \exists y_3 \cdots Q_k y_k R(x, y_1, \dots, y_k), \quad |y_i| \leq p(|x|),$$

with alternating quantifiers beginning existentially. The complementary form characterizes Π_k^P .

Proof. We prove the statement simultaneously for Σ_k^P and Π_k^P by induction on k . For $k = 1$, the existential form is exactly the polynomially balanced verifier characterization of NP , and complementing it gives the universal form for $coNP$.

Assume the two characterizations at level k . Let $L \in \Sigma_{k+1}^P = NP^{\Sigma_k^P}$. Fix a polynomial-time nondeterministic oracle machine M deciding L , and on an accepting branch record all nondeterministic choices together with the entire polynomial-length transcript of oracle queries and claimed yes/no answers. For each claimed yes answer to a query q , substitute the inductive Σ_k^P formula for $q \in A$, where A is a fixed Σ_k^P -complete oracle. For each claimed no answer, substitute the inductive Π_k^P formula for $q \notin A$. The transcript also contains a polynomial-time check that the listed queries really are exactly the adaptive queries issued by M when fed those claimed answers and that the recorded branch accepts.

There are only polynomially many queries and every quantified witness has polynomial length. To see the grouping explicitly, write the inductive oracle predicate as

$$q \in A \iff \exists a_1 \forall a_2 \cdots Q_k a_k S(q, a_1, \dots, a_k).$$

For a transcript position claiming “yes”, put its a_1 variables into the outer existential block, its a_2 variables into the next universal block, and so on. For a position claiming “no”, use the negated formula

$$q \notin A \iff \forall a_1 \exists a_2 \cdots \bar{Q}_k a_k \neg S(q, a_1, \dots, a_k),$$

so its a_1 variables enter the next universal block, its a_2 variables the following existential block, and so forth. Pad witnesses at a common level and concatenate all variables assigned to that block. Dummy variables fill empty contributions. The outer block also contains the nondeterministic choices, query transcript, and all first existential witnesses for positive answers. The final polynomial-time matrix simultaneously checks transcript consistency and the appropriate S or $\neg S$ predicate for every query. Thus exactly the alternating pattern beginning existentially, with at most $k + 1$ blocks, suffices. Hence L has the required Σ_{k+1}^P quantifier form. Complementing gives the corresponding Π_{k+1}^P form.

Conversely suppose

$$x \in L \iff \exists y_1 \forall y_2 \cdots Q_{k+1} y_{k+1} R(x, y_1, \dots, y_{k+1}).$$

An $NP^{\Sigma_k^P}$ machine first guesses y_1 . The remaining k -block statement begins universally, so it is a Π_k^P predicate. Its negation is in Σ_k^P and can be submitted to the oracle; the machine accepts exactly when that oracle answers no. Thus the existential form lies in Σ_{k+1}^P . Complementing proves the universal form. This completes the induction. $\square$

Theorem 7.3 (Polynomial-hierarchy collapse criterion). *If $\Sigma_k^P = \Pi_k^P$ for some $k \geq 1$, then $PH = \Sigma_k^P$. It suffices that a Σ_k^P -complete language belong to Π_k^P .*

Proof. Under the equality hypothesis, a Σ_{k+1}^P formula begins with an existential block followed by a Π_k^P predicate. Replace that predicate by its equivalent Σ_k^P form; its leading existential block merges with the outer existential block, leaving only k alternating blocks. Thus $\Sigma_{k+1}^P = \Sigma_k^P$. Induction collapses every higher level. For the complete-language criterion, closure under polynomial-time many-one reductions places all of Σ_k^P in Π_k^P ; complementing gives equality and the first argument applies. $\square$

Chapter 8

Circuits and Nonuniform Computation

Definition 8.1 (Circuit families and advice). A Boolean circuit is a finite acyclic directed graph of constant-fanin Boolean gates. A family (C_n) has polynomial size if $|C_n| \leq n^{O(1)}$. $P/poly$ is the class of languages decided by polynomial-size circuit families. An advice machine is a polynomial-time machine $M(x, a_n)$ supplied an arbitrary string a_n depending only on the input length, with $|a_n| \leq n^{O(1)}$.

Positive example. Hardwiring a fixed lookup table for one input length gives a nonuniform circuit.

Nonexample. Advice depending on the individual input is not length advice.

Theorem 8.2 (Uniform computation has polynomial-size circuits). *If a fixed deterministic multitape machine halts within $T(n) \geq n + 1$ steps, then its length- n computation is computed by Boolean circuits of size $O(T(n)^2)$ under the elementary tableau construction.*

Proof. In T steps only $O(T)$ tape cells on each fixed tape can be visited. Encode at each time the state, head positions, and symbols in these cells. For every cell the next symbol and head/state flags are Boolean functions of a constant local neighborhood together with the current finite control. A layer computing the next complete configuration therefore uses $O(T)$ constant-size gadgets. Compose T layers and hardwire the machine description; the total size is $O(T^2)$. The input bits feed the initial tape cells and the final accept-state bit is the output. Every layer implements exactly one transition, so induction on time proves correctness. $\square$

Theorem 8.3 (Circuit characterization of $P/poly$). *A language has polynomial-size circuits iff it is decided in polynomial time by a deterministic machine with polynomial-length advice.*

Proof. Given circuits C_n , use an encoding of C_n as the advice string. A uniform interpreter evaluates the acyclic circuit in time polynomial in its size. Conversely, let $M(x, a_n)$ run in polynomial time. Hardwire the advice bits a_n into the circuit obtained from the preceding theorem for the computation of M on n input bits. Its size remains polynomial. The two constructions preserve the accepted language. $\square$

Lemma 8.4 (Universally checkable SAT-circuit certificate). *Fix a length bound m . Suppose circuits $C_1, \dots, C_m$ are given, where C_ℓ is intended to decide satisfiability of formulas whose canonical encoding has length ℓ . There is a polynomial-time predicate $\text{Consistent}(C_1, \dots, C_m; \psi)$ such that, if it holds for every formula ψ of length at most m , then every C_ℓ is correct on every formula of*

length ℓ . The predicate uses only terminal evaluation and the self-reduction identity obtained by fixing one canonically selected variable.

Proof. Use a canonical formula encoding together with a polynomial-time simplification/padding convention whose restrictions never exceed the original length bound. If ψ contains no variables, compare the appropriate circuit output with the truth value obtained by direct evaluation. Otherwise let z be the first variable under a fixed parsing rule and form

$$\psi_0 = \psi[z \leftarrow 0], \quad \psi_1 = \psi[z \leftarrow 1].$$

Check

$$C_{|\psi|}(\psi) = C_{|\psi_0|}(\psi_0) \vee C_{|\psi_1|}(\psi_1).$$

All parsing, restriction, padding, and circuit evaluations are polynomial time in the total input size.

If every such test passes, induction on the number of variables proves that the bundle decides SAT correctly. The base case is the terminal check. For the induction step, both restricted formulas contain fewer variables, so their circuit outputs are correct; the displayed equality is exactly the self-reducibility identity for satisfiability. Conversely, the genuine SAT circuits satisfy every test. Thus one universally quantified encoded formula is enough to challenge the correctness of the entire guessed bundle. $\square$

Theorem 8.5 (Karp–Lipton). *If*

$$NP \subseteq P/poly,$$

then

$$PH = \Sigma_2^P.$$

Proof. By the collapse criterion it is enough to show $\Pi_2^P \subseteq \Sigma_2^P$. Let $L \in \Pi_2^P$. By the quantifier characterization, there are a polynomial p and a polynomial-time predicate R such that, for $n = |x|$,

$$x \in L \iff \forall y \in \{0, 1\}^{p(n)} \exists z \in \{0, 1\}^{p(n)} R(x, y, z) = 1.$$

For fixed (x, y) the predicate “there exists z with $R(x, y, z) = 1$ ” is in NP . By Cook–Levin, in polynomial time construct a CNF formula $\varphi_{x,y}$ of length at most $m = q(n)$, for one polynomial q , such that

$$\varphi_{x,y} \text{ is satisfiable} \iff \exists z R(x, y, z) = 1.$$

The hypothesis implies that SAT has polynomial-size circuits. A Σ_2^P machine for L existentially guesses a polynomial-size bundle $C_1, \dots, C_m$ of circuits of the sizes promised by that hypothesis. It then uses one universally quantified polynomial-length string u to perform two kinds of challenge:

1. if u encodes a Boolean formula ψ of length at most m , verify $\text{Consistent}(C_1, \dots, C_m; \psi)$ from the preceding lemma;
2. if u encodes a string $y \in \{0, 1\}^{p(n)}$, construct $\varphi_{x,y}$ and verify

$$C_{|\varphi_{x,y}|}(\varphi_{x,y}) = 1.$$

Malformed universal strings are assigned a vacuously successful challenge. The matrix predicate is polynomial time because the guessed circuit bundle has polynomial total length.

If $x \in L$, choose the genuine SAT circuits promised by the hypothesis. Every consistency challenge succeeds and every formula $\varphi_{x,y}$ is satisfiable, so all universal challenges pass. Conversely, suppose a guessed bundle passes every challenge. The first class of challenges and the lemma force

the bundle to decide SAT correctly on every formula of length at most m . The second class then implies that every $\varphi_{x,y}$ is satisfiable. Hence for every y there exists z with $R(x, y, z) = 1$, so $x \in L$.

We have expressed membership in L by an existential polynomial-length guess, followed by one universal polynomial-length challenge, followed by a polynomial-time predicate. Thus $L \in \Sigma_2^P$. Therefore $\Pi_2^P \subseteq \Sigma_2^P$; taking complements gives equality, and the collapse criterion yields $PH = \Sigma_2^P$. $\square$

Worked logical shape. The guessed SAT circuits are never trusted. Their correctness is itself exposed to one universal challenge. This converts the nonuniform assumption into the quantifier pattern

$\exists$ circuit bundle $\forall$ challenge polynomial-time check,

which is precisely a Σ_2^P computation.

Chapter 9

Counting Complexity

Definition 9.1 (The classes $\#P$ and PP). A function $f : \{0, 1\}^* \rightarrow \mathbb{N}$ is in $\#P$ if some polynomial-time nondeterministic machine has exactly $f(x)$ accepting branches on input x . A language is in PP if such a machine accepts on strictly more than half of its computation branches exactly for the yes-instances, after padding to a fixed polynomial number of binary choices.

Positive example. The number of satisfying assignments of a CNF is a $\#P$ function.

Nonexample. Closure under subtraction does not follow by deleting accepting paths, because the result need not be nonnegative and branch cancellations have no nondeterministic interpretation.

Theorem 9.2 (Basic closure properties of $\#P$). *$\#P$ is closed under pointwise addition, pointwise multiplication, and sums $h(x) = \sum_{y \in \{0,1\}^{p(|x|)}} f(x, y)$ for polynomial p .*

Proof. For addition, prepend one nondeterministic bit and simulate the machine for f on branch zero and that for g on branch one. For multiplication, simulate the first machine and, on each accepting branch, independently simulate the second; accepting paths are in bijection with ordered pairs of accepting paths. For bounded summation first guess y and then simulate the machine for $f(x, y)$. In each case the construction is polynomial time and the stated branch-count bijection proves the identity. $\square$

Theorem 9.3 ($\#SAT$ is parsimoniously $\#P$ -complete). *Every $f \in \#P$ admits a polynomial-time map $x \mapsto \varphi_x$ such that the number of satisfying assignments of φ_x is exactly $f(x)$.*

Proof. Let $f \in \#P$ be witnessed by a polynomial-time nondeterministic Turing machine M . First put M into a binary-choice normal form without changing its computation tree: each genuinely nondeterministic transition has two labelled successors, while deterministic transitions have one successor. Fix a polynomial time bound $p(n)$ and pad every halting state by a deterministic self-loop until time $p(n)$. For tableau purposes introduce one auxiliary choice bit at each time step, but force that bit to 0 whenever the current transition is deterministic (including the padded halting self-loop). Thus an accepting computation branch has exactly one extension to the time-indexed choice variables; no branch is duplicated merely to obtain a rectangular tableau. Once the actual branch choices are fixed, the entire computation is deterministic.

Apply the Cook–Levin tableau construction with these time-indexed choice bits as explicit variables. The local alphabet and transition relation are fixed, so every forbidden local 2×3 window is excluded by a constant-width CNF clause directly on the tableau/choice variables; no Tseitin extension variables are needed. Add the usual cell-uniqueness, initial-row, and accepting-row clauses,

and add clauses enforcing the dummy choice convention after halting. The resulting formula φ_x has polynomial size.

For each accepting branch of $M(x)$, its normalized choice string determines one and only one sequence of configurations, hence one complete assignment to all tableau and choice variables satisfying φ_x . Conversely, the legal-window soundness argument shows that every satisfying assignment is a single legal accepting computation tableau; reading its explicit choice bits gives one normalized accepting branch of $M(x)$. These two maps are inverse. Therefore

$$\#\text{SAT}(\varphi_x) = \#\text{acc}_M(x) = f(x),$$

and $x \mapsto \varphi_x$ is a polynomial-time parsimonious reduction. □

Theorem 9.4 (Toda's theorem — source-owned). *Let PH denote the polynomial hierarchy, PP the class whose probabilistic polynomial-time machines accept yes-instances on more than half of their computation paths, and #P the corresponding counting-function class as defined earlier in this volume. Then*

$$\text{PH} \subseteq \text{P}^{\#\text{P}},$$

and equivalently $\text{PH} \subseteq \text{P}^{\text{PP}}$ under the standard polynomial-time Turing-oracle formulation. The arithmetization, randomized reduction and counting simulation are source-owned to Toda's primary theorem.

Chapter 10

Relativization and the Limits of Diagonalization

Definition 10.1 (Oracle machine and relativization). An oracle Turing machine may, in one distinguished query step, ask whether a written string belongs to a fixed language A . Classes such as P^A and NP^A are defined by polynomial-time deterministic and nondeterministic oracle machines. A proof method relativizes when the same argument remains valid uniformly after adjoining an arbitrary oracle.

Positive example. The inclusion $P^A \subseteq NP^A$ holds for every oracle A .

Nonexample. A statement proved by exploiting the internal structure of SAT may fail to relativize.

Theorem 10.2 (Baker–Gill–Solovay relativization phenomenon). *There are recursive oracles A, B such that*

$$P^A = NP^A, \quad P^B \neq NP^B.$$

Consequently a completely relativizing argument cannot by itself settle P versus NP .

Proof. For the equality oracle let $A = TQBF$. PSPACE-completeness gives $PSPACE \subseteq P^A$, while a deterministic polynomial-time oracle machine making TQBF queries can evaluate those queries in polynomial space and reuse that workspace, so $P^A \subseteq PSPACE$. Hence $P^A = PSPACE$. Similarly, an NP^A computation uses only polynomial space along each branch; replacing its TQBF queries by polynomial-space evaluations gives $NP^A \subseteq NPSPACE = PSPACE$ by Savitch's theorem. Since $P^A \subseteq NP^A$, we obtain

$$P^A = NP^A = PSPACE.$$

For separation enumerate all clocked deterministic polynomial-time oracle machines $M_1, M_2, \dots$, repeating machines if necessary so every polynomial clock occurs. We build a recursive oracle B by finite extensions. After stage $i - 1$, membership has been fixed for every string of length at most some bound b_{i-1} . Choose $n_i > b_{i-1}$ so large that the clock of M_i on length n_i is strictly smaller than 2^{n_i} . During a simulation of $M_i(1^{n_i})$, answer every still-unfixed oracle query by no, while preserving all earlier fixed answers. Let b_i be at least the maximum of n_i and the lengths of all queries made in this simulation.

If the simulated machine accepts, set every still-unfixed string of length at most b_i to no. Then no string of length n_i belongs to B . If it rejects, fewer than 2^{n_i} strings of length n_i were queried, so

choose an unqueried $y_i \in \{0, 1\}^{n_i}$, put y_i into B , and set every other still-unfixed string of length at most b_i to no. In either case the oracle answers observed during the simulation are never changed later.

Define

$$L_B = \{1^n : \exists y \in \{0, 1\}^n \text{ with } y \in B\}.$$

It lies in NP^B by guessing y and making one query. At stage i , if $M_i^B(1^{n_i})$ accepts then no length- n_i string is put into B , so the machine errs on a no-instance of L_B ; if it rejects, the unqueried y_i is inserted, leaving the observed computation unchanged while making $1^{n_i} \in L_B$. Thus every deterministic polynomial-time oracle machine fails somewhere and $L_B \notin P^B$.

Finally, B is recursive. On input z , simulate the finite-extension stages until some $b_i \geq |z|$; by construction membership of every string of that length has then been permanently fixed. The stage construction itself is effective, although not efficient. Hence B is a recursive oracle with $P^B \neq NP^B$. The existence of the equality and separation oracles proves that a completely relativizing proof cannot settle unrelativized P versus NP . $\square$

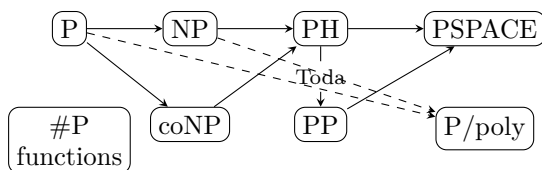


Figure 10.1: A complexity landscape. Solid arrows are proved containments in the book’s stated models. The Toda arrow is source-owned. Dashed arrows involving nonuniformity are reminders that oracle/barrier phenomena and conjectured separations must not be read as proved equalities or strict containments.

Internalization and Reconstruction Guide: Volume II

Purpose. Build the hierarchy from machine semantics and reductions to time, space, nondeterminism, counting, nonuniformity, and barriers, while keeping decidability questions distinct from quantitative resource bounds.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

This is a foundation volume. Its prerequisite ladder is internal: begin with the first definitions and reconstruct each later family from those conventions rather than importing unstated external notation. Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-II-DEF-001 (Deterministic and nondeterministic Turing machines); QRCC-II-DEF-002 (Computability and many-one reduction); QRCC-II-DEF-003 (Time-constructibility and complexity classes); QRCC-II-DEF-004 (NP verifier and polynomial balance); QRCC-II-DEF-005 (Tableau formula and parsimonious reduction).

Chapter retrieval map

- Ch. 1: Machines, Encodings, and Universal Simulation:** enter through QRCC-II-DEF-001 (Deterministic and nondeterministic Turing machines); exit by being able to state QRCC-II-THM-001 (Universal simulation theorem) with its ownership boundary.
- Ch. 2: Computability, Reductions, and Undecidability:** enter through QRCC-II-DEF-002 (Computability and many-one reduction); exit by being able to state QRCC-II-THM-017 (Computable reductions and undecidability transfer) with its ownership boundary.
- Ch. 3: Time Bounds and Model Robustness:** enter through QRCC-II-DEF-003 (Time-constructibility and complexity classes); exit by being able to state QRCC-II-THM-002 (Deterministic time hierarchy theorem) with its ownership boundary.
- Ch. 4: Nondeterministic Polynomial Time and Verifiers:** enter through QRCC-II-DEF-004 (NP verifier and polynomial balance); exit by being able to state QRCC-II-THM-018 (Verifier characterization of NP) with its ownership boundary.

- Ch. 5: Cook–Levin and the First NP-Complete Problems:** enter through QRCC-II-DEF-005 (Tableau formula and parsimonious reduction); exit by being able to state QRCC-II-THM-005 (Cook–Levin theorem) with its ownership boundary.
- Ch. 6: Space Complexity and Recursive Reachability:** enter through QRCC-II-DEF-006 (Space complexity and configuration graph); exit by being able to state QRCC-II-THM-004 (Space hierarchy theorem) with its ownership boundary.
- Ch. 7: The Polynomial Hierarchy:** enter through QRCC-II-DEF-007 (Polynomial hierarchy); exit by being able to state QRCC-II-THM-011 (Polynomial-hierarchy quantifier characterization) with its ownership boundary.
- Ch. 8: Circuits and Nonuniform Computation:** enter through QRCC-II-DEF-008 (Boolean circuit family and advice); exit by being able to state QRCC-II-THM-012 (Karp–Lipton theorem) with its ownership boundary.
- Ch. 9: Counting Complexity:** enter through QRCC-II-DEF-009 (Counting classes #P and PP); exit by being able to state QRCC-II-BB-015 (Toda theorem) with its ownership boundary.
- Ch. 10: Relativization and the Limits of Diagonalization:** enter through QRCC-II-DEF-010 (Oracle machine and relativization); exit by being able to state QRCC-II-THM-016 (Baker–Gill–Solovay relativization phenomenon) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-II-THM-001 — Universal simulation theorem. Dependencies: QRCC-II-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-II-THM-002 — Deterministic time hierarchy theorem. Dependencies: QRCC-II-THM-001, QRCC-II-DEF-003. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-II-THM-005 — Cook–Levin theorem. Dependencies: QRCC-II-DEF-004, QRCC-II-DEF-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-II-THM-011 — Polynomial-hierarchy quantifier characterization. Dependencies: QRCC-II-DEF-007, QRCC-II-DEF-004. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-II-BB-015 — Toda theorem. Dependencies: QRCC-II-THM-011, QRCC-II-THM-014, QRCC-II-DEF-009. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-II-THM-016 — Baker–Gill–Solovay relativization phenomenon. Dependencies: QRCC-II-DEF-010, QRCC-II-THM-022, QRCC-II-THM-008. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

Common failure modes

- confusing undecidability with superpolynomial complexity;
- dropping uniformity when passing from algorithms to circuit families;
- reading a class containment or relativized separation as an unconditional separation;
- using hierarchy theorems without their constructibility or resource hypotheses;

Boundary and degenerate cases

- empty languages and machines with no accepting branch;
- time/space bounds below the model overhead and constructibility threshold;
- promise-free versus promise problems in completeness reductions;
- oracle statements whose conclusion changes when the oracle changes;

Retrieval and reconstruction drill

1. State QRCC-II-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-II-THM-001. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-II-THM-002, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-II-THM-016, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume III supplies the algebraic and geometric algorithmic objects that later complexity reductions and cryptosystems manipulate.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part III

Finite Fields, Codes, Lattices, and Algebraic Algorithms

Chapter 1

Finite Fields and Frobenius

Definition 1.1 (Finite field, characteristic, and Frobenius). A finite field is a field with finitely many elements. In characteristic $p > 0$, Frobenius is $\text{Fr}_p(x) = x^p$; its fixed field under an iterate consists of the elements fixed by that iterate.

Positive example. $\mathbb{F}_5$ has characteristic 5 and $x^5 = x$ for every element.

Nonexample. $\mathbb{Z}/6\mathbb{Z}$ is finite but not a field.

Lemma 1.2 (Existence and uniqueness of splitting fields). *Every nonzero polynomial over a field has a splitting field generated by its roots, unique up to an isomorphism fixing the base field.*

Proof. Existence follows by induction on degree: adjoin a root of an irreducible factor using $F[X]/(g)$, divide off the corresponding linear factor, and repeat. For uniqueness, extend an embedding root by root. If α has minimal polynomial m over the current embedded field, the transported polynomial has a root in the target splitting field (it divides the transported residual factor of the original polynomial); sending α to such a root extends the embedding. After all roots have been adjoined, the embedding sends the finite set of roots injectively into the same root set, hence permutes that set. Its image therefore contains every root; because the target splitting field is generated by those roots, the embedding is surjective. $\square$

Theorem 1.3 (Existence and uniqueness of $\mathbb{F}_{p^n}$). *For every prime p and $n \geq 1$ there is, up to $\mathbb{F}_p$ -isomorphism, a unique field with p^n elements; it is the splitting field of $X^{p^n} - X$.*

Proof. Let $q = p^n$ and let E split $X^q - X$. Its derivative is -1 , so it has q distinct roots. In characteristic p , $(a + b)^q = a^q + b^q$ and products and inverses of nonzero roots remain roots; the root set is a field and generates E , so $|E| = q$. Conversely any field F of size q has characteristic p and dimension n over $\mathbb{F}_p$, and Lagrange's theorem in $F^\times$ gives $a^q = a$ for all a . Hence F is also the splitting field, and splitting field uniqueness applies. $\square$

Theorem 1.4 (Classification of finite fields). *Every finite field has cardinality p^n for a unique prime p and $n \geq 1$; fields of the same cardinality are isomorphic over their prime field.*

Proof. The characteristic is prime, since a composite characteristic would create zero divisors. Thus the field is a finite-dimensional vector space of dimension n over $\mathbb{F}_p$ and has p^n elements. The preceding theorem gives the uniqueness statement. $\square$

Theorem 1.5 (Cyclicity of the multiplicative group). $\mathbb{F}_q^\times$ is cyclic of order $q - 1$, with $\varphi(q - 1)$ generators.

Proof. Let m be the lcm of all element orders. In a finite abelian group there is an element of order m : for each prime power dividing m , take an element whose order contains that maximal power, raise away the coprime part, and multiply the resulting commuting elements of pairwise coprime orders. Every group element is a root of $X^m - 1$, so the field root bound gives $q - 1 \leq m$. Lagrange gives $m \mid q - 1$, hence $m = q - 1$. A cyclic group of order N has exactly $\varphi(N)$ generators. $\square$

Theorem 1.6 (Subfields and Frobenius). If $d \mid n$, $\mathbb{F}_{p^n}$ has the unique subfield $\{x : x^{p^d} = x\}$ of size p^d ; every subfield arises this way.

Proof. The fixed set is a field. In cyclic $\mathbb{F}_{p^n}^\times$, the equation $x^{p^d-1} = 1$ has $\gcd(p^d - 1, p^n - 1) = p^{\gcd(d, n)} - 1 = p^d - 1$ solutions, so the fixed field has p^d elements. Any subfield of size p^e is fixed by $x \mapsto x^{p^e}$; the tower law gives $e \mid n$, and equality of cardinalities yields uniqueness. $\square$

Chapter 2

Polynomial Arithmetic over Finite Fields

Definition 2.1 (Polynomial arithmetic and computational model). Polynomials are densely encoded. Field operations are unit cost unless a bit model is stated. The monic gcd and Bezout coefficients use the usual Euclidean normalization; modular arithmetic always reduces to degree below the modulus.

Positive example. $(2, 4, 0, 1)$ represents $2 + 4X + X^3$ over $\mathbb{F}_5$.

Nonexample. A complexity claim without specifying field representation is incomplete.

Algorithm 2.2 (Euclidean polynomial arithmetic). *Division with remainder, gcd, and extended gcd are correct and terminate; for degree at most N , classical dense arithmetic gives safe bounds $O(N^2)$ for one division and $O(N^3)$ for gcd/extended gcd.*

Proof. Cancel leading terms until the degree drops below the divisor. Uniqueness follows because a nonzero multiple of the divisor cannot have degree below it. Repeated division strictly decreases nonzero remainder degree; common divisors are invariant under $r_{i-1} \mapsto r_{i-1} - q_i r_i$, giving the gcd. Maintaining $r_i = s_i a + t_i b$ gives Bezout coefficients. At most $N + 1$ divisions occur, and the stated classical bounds follow by coefficient counting. $\square$

Algorithm 2.3 (Interpolation and polynomial CRT). *At distinct α_i , the unique degree- $< n$ interpolant is*

$$f(X) = \sum_i y_i \prod_{j \neq i} \frac{X - \alpha_j}{\alpha_i - \alpha_j}.$$

For pairwise coprime moduli m_i , residues a_i determine a unique class modulo $M = \prod_i m_i$, constructible by extended gcd.

Proof. The Lagrange basis has $L_i(\alpha_j) = \delta_{ij}$; uniqueness follows from the root bound. For CRT put $M_i = M/m_i$ and choose $u_i M_i \equiv 1 \pmod{m_i}$. Then $\sum_i a_i u_i M_i$ has all desired residues; uniqueness follows because all pairwise coprime m_i divide the difference. $\square$

Lemma 2.4 (Frobenius factorization polynomial). *$X^{q^m} - X$ is square-free and equals the product of all monic irreducibles over $\mathbb{F}_q$ whose degrees divide m .*

Proof. The derivative is -1 . An irreducible polynomial of degree d divides the polynomial iff one (equivalently every) root lies in $\mathbb{F}_{q^m}$, which by the finite-field subfield theorem is equivalent to $d \mid m$. Square-freeness gives multiplicity one. $\square$

Algorithm 2.5 (Finite-field irreducibility test). *A monic degree- n polynomial f is irreducible iff $X^{q^n} \equiv X \pmod{f}$ and for every prime $r \mid n$, $\gcd(f, X^{q^{n/r}} - X) = 1$. Modular powering and gcd give a deterministic polynomial-time test in the classical field-operation model.*

Proof. For irreducible f the Frobenius-factorization lemma gives both conditions. Conversely the first condition makes every irreducible factor degree divide n . If a proper factor had degree $d < n$, choose a prime $r \mid n/d$; then $d \mid n/r$, contradicting the corresponding gcd test. The algorithm follows by successive modular Frobenius powering and polynomial gcds. $\square$

Chapter 3

Polynomial Factorization over Finite Fields

Definition 3.1 (Irreducibility and factorization output). An irreducible factorization is $f = c \prod_i f_i^{e_i}$ with distinct monic irreducibles f_i . Square-free means every $e_i = 1$. A Las Vegas algorithm always returns a correct result, with random running time of finite stated expectation.

Positive example. $X^4 + X = X(X + 1)(X^2 + X + 1)$ over $\mathbb{F}_2$.

Nonexample. Repeated irreducible factors are not square-free.

Lemma 3.2 (Square-free decomposition). *Given a monic nonzero $f \in \mathbb{F}_q[X]$ in characteristic p , its square-free decomposition*

$$f = \prod_{j \geq 1} s_j^j,$$

where the nonconstant s_j are square-free and pairwise coprime, can be computed deterministically using polynomial gcds, derivatives, and recursive p th-root extraction.

Proof. Write formally $f = \prod_h h^{e_h}$ over its distinct monic irreducible factors. Suppose first that $f' \neq 0$ and put

$$c = \gcd(f, f'), \quad w = f/c.$$

If $p \nmid e_h$, then the h -adic valuation of f' is exactly $e_h - 1$; if $p \mid e_h$, it is at least e_h . Consequently w contains exactly those h for which $p \nmid e_h$, each with exponent one.

Set $i = 1$. While $w \neq 1$, compute

$$y = \gcd(w, c), \quad z = w/y.$$

Then z is exactly the product of the irreducible factors whose multiplicity in f equals i and is not divisible by p . Record this contribution to s_i , replace

$$w \leftarrow y, \quad c \leftarrow c/y,$$

and increment i . To see the assertion, fix one irreducible h with $p \nmid e_h$: its single copy in w survives exactly until the iteration $i = e_h$, while the exponent of h in c drops by one at every preceding iteration.

After the loop, every irreducible exponent remaining in c is divisible by p . Hence $c' = 0$ and c is a p th power. Indeed, writing $c(X) = \sum_j a_j X^j$, the equation $c' = 0$ forces $a_j = 0$ unless $p \mid j$.

Frobenius is an automorphism of the finite coefficient field, so every coefficient a_{pi} has a unique p th root. Thus

$$c(X) = \left(\sum_i a_{pi}^{1/p} X^i \right)^p.$$

Apply the same procedure recursively to this p th root and multiply every multiplicity returned by the recursion by p . If $f' = 0$ at the start, perform this root-extraction step immediately. Each such recursion divides the degree by p , while every nonzero-derivative stage makes progress through gcds, so the algorithm terminates and recovers every multiplicity exactly. $\square$

Lemma 3.3 (Berlekamp fixed algebra). *If square-free $f = f_1 \cdots f_r$, then $B = \{a \in \mathbb{F}_q[X]/(f) : a^q = a\} \cong \mathbb{F}_q^r$ and $\dim B = r$.*

Proof. CRT identifies the quotient with $\prod_i \mathbb{F}_{q^{\deg f_i}}$. In each factor the fixed field of q -Frobenius is exactly $\mathbb{F}_q$, giving the claim componentwise. $\square$

Algorithm 3.4 (Berlekamp factorization). *Compute the kernel of $a \mapsto a^q - a$ modulo a square-free f . Kernel dimension one means irreducible; otherwise a nonconstant kernel element b and gcds $\gcd(f, b - c)$ for $c \in \mathbb{F}_q$ yield a proper split. Recursion, preceded by square-free decomposition, deterministically factors f in time polynomial in the degree and q field operations.*

Proof. Under the preceding CRT description a fixed residue is a tuple of constants. If dimension exceeds one, some computed fixed element has different constants on two components. The gcd with $b - c$ is exactly the product of components on which that constant equals c , so some gcd is a proper factor. Degrees decrease under recursion; kernel computation and gcds are polynomial-time linear and polynomial arithmetic. $\square$

Lemma 3.5 (Constant-probability equal-degree split). *Let f be a product of $r \geq 2$ distinct degree- d irreducibles and put $Q = q^d$. For odd q , a random residue and the quadratic character split with probability at least $1/8$; for $q = 2^e$, the absolute-trace split succeeds with probability at least $1/2$.*

Proof. CRT makes the residue components independent uniform elements of $\mathbb{F}_Q$. The r irreducible factors have disjoint sets of d roots in $\mathbb{F}_Q$, so $rd \leq Q$ and in particular $r \leq Q$. For odd q , conditioned on all components being nonzero, their quadratic characters are independent fair signs; the character-based gcd fails to separate components only when all signs agree, an event of probability $2^{1-r} \leq 1/2$. Moreover $(1 - 1/Q)^r \geq (1 - 1/Q)^Q \geq 8/27$ for $Q \geq 3$. Thus the event “all components are nonzero and the signs are not all equal” already has probability at least $4/27 > 1/8$. If some but not all components are zero, the preliminary gcd with the sampled residue itself supplies a split, so ignoring such outcomes only weakens the lower bound. In even characteristic use $T(z) = z + z^2 + \cdots + z^{2^{ed-1}}$. This is a nonzero polynomial of degree $2^{ed-1} < Q$, hence a nonzero $\mathbb{F}_2$ -linear functional; its two fibers have equal size. Independent trace bits are fair, and failure probability $2^{1-r} \leq 1/2$. $\square$

Algorithm 3.6 (Cantor–Zassenhaus factorization). *After square-free decomposition, distinct-degree factorization uses $\gcd(f, X^{q^d} - X)$ and equal-degree blocks are recursively split by the preceding lemma. This is Las Vegas polynomial time; a conservative dense bound is $O(n^5(1 + \log q))$ expected field operations.*

Proof. The Frobenius-factorization lemma shows the gcd stage isolates the factors whose degrees divide d ; after earlier degrees are removed it isolates exactly degree d . Each nontrivial equal-degree block splits with constant probability, so the expected trials per successful split are bounded and at most $n - 1$ successful splits occur. Every factor is an exact gcd, so correctness is unconditional.

Classical modular powering, multiplication, and gcd bounds give the stated deliberately loose polynomial cost. $\square$

Chapter 4

Linear Codes, Duality, and Syndromes

Definition 4.1 (Linear code, dual code, and syndrome). For $1 \leq k \leq n$, an $[n, k, d]_q$ linear code is a k -dimensional subspace of $\mathbb{F}_q^n$ with minimum nonzero Hamming weight d . Its dual is $C^\perp = \{y : \langle x, y \rangle = 0 \ \forall x \in C\}$. A parity-check matrix has row space $C^\perp$, and the syndrome of r is Hr^T .

Positive example. The binary repetition code is $[3, 1, 3]_2$.

Nonexample. A set of words not closed under addition is not linear.

Theorem 4.2 (Dimension, duality, syndromes, and parity-check columns). *If $\dim C = k$, then $\dim C^\perp = n - k$, $(C^\perp)^\perp = C$, syndromes label cosets, and the minimum distance equals the least number of parity-check columns with a nontrivial linear dependence.*

Proof. The map $T : y \mapsto (x \mapsto \langle x, y \rangle)$ from $\mathbb{F}_q^n$ to C^* has kernel $C^\perp$ and is onto: extend a basis of C to an ambient basis, extend any functional by zero on the new basis vectors, and use nondegeneracy of the dot product. Rank-nullity gives the dimension and then the double-dual equality. Equality of syndromes is equivalent to the difference lying in C . Finally $c \in C$ iff $\sum_i c_i h_i = 0$ for the columns h_i of H , so nonzero codeword supports are exactly nontrivial dependent column sets. $\square$

Chapter 5

Coding Bounds and Asymptotic Rate

Definition 5.1 (Rate, relative distance, and asymptotic families). For a q -ary length- n code C , put $R = \log_q |C|/n$ and $\delta = d(C)/n$. For a family $n \rightarrow \infty$, asymptotic lower bounds mean $\liminf$ bounds. For linear $[n, k, d]_q$ codes, $R = k/n$.

Positive example. Repetition codes have distance rate 1 but information rate tending to 0.

Nonexample. One fixed code is not an asymptotic family.

Theorem 5.2 (Singleton bound). *Every q -ary code of size M and distance d satisfies $M \leq q^{n-d+1}$; hence linear codes satisfy $d \leq n - k + 1$.*

Proof. Puncture any $d - 1$ coordinates. Two codewords cannot coalesce, so the map is injective into q^{n-d+1} words. Substitute $M = q^k$ for a linear code. $\square$

Theorem 5.3 (Hamming sphere-packing bound). *For $t = \lfloor (d - 1)/2 \rfloor$,*

$$M \sum_{i=0}^t \binom{n}{i} (q - 1)^i \leq q^n.$$

Proof. Radius- t balls around distinct codewords are disjoint by the triangle inequality, and each has the displayed volume. Their union lies in the ambient q^n words. $\square$

Lemma 5.4 (Asymptotic Hamming-ball volume). *For $0 \leq \delta < 1 - 1/q$,*

$$\frac{1}{n} \log_q \sum_{i \leq \lfloor \delta n \rfloor} \binom{n}{i} (q - 1)^i = H_q(\delta) + o(1),$$

where $H_q(\delta) = \delta \log_q(q - 1) - \delta \log_q \delta - (1 - \delta) \log_q(1 - \delta)$, with the usual convention $0 \log 0 = 0$.

Proof. First we prove the only factorial estimate needed here. Since $\log x$ is increasing, for every integer $m \geq 1$,

$$\int_1^m \log x \, dx \leq \sum_{j=2}^m \log j \leq \int_1^m \log x \, dx + \log m.$$

As $\int_1^m \log x \, dx = m \log m - m + 1$, this gives

$$\log(m!) = m \log m - m + O(\log(m + 1)). \quad (5.1)$$

Let $r = \lfloor \delta n \rfloor$ and $\alpha = r/n$. Applying (5.1) to $n!, r!, (n-r)!$ yields

$$\log \binom{n}{r} = n[-\alpha \log \alpha - (1-\alpha) \log(1-\alpha)] + O(\log(n+1)).$$

After adding $r \log(q-1)$, dividing by $n \log q$, and using $\alpha = \delta + O(1/n)$, the last summand has logarithmic rate $H_q(\delta) + o(1)$ (the endpoint $\delta = 0$ is immediate).

For $0 \leq i < r$ the ratio of consecutive summands is

$$\frac{\binom{n}{i+1}(q-1)^{i+1}}{\binom{n}{i}(q-1)^i} = \frac{(n-i)(q-1)}{i+1}.$$

Because $\delta < 1 - 1/q$, this ratio exceeds 1 throughout $i \leq r-1$ for all sufficiently large n . Hence the Hamming-ball sum lies between its last term and $(n+1)$ times that term. The extra factor contributes only $O(\log n)/n = o(1)$ to the base- q logarithm, proving the claim. $\square$

Theorem 5.5 (Gilbert–Varshamov bounds). *There is a q -ary code with $M \geq q^n / \sum_{i=0}^{d-1} \binom{n}{i} (q-1)^i$. If $\sum_{i=0}^{d-2} \binom{n-1}{i} (q-1)^i < q^{n-k}$, there is a linear $[n, k, d']_q$ code with $d' \geq d$. Consequently, for every $0 < \delta < 1 - 1/q$ and $\varepsilon > 0$, there is an asymptotic family of linear q -ary codes with*

$$\liminf R \geq 1 - H_q(\delta) - \varepsilon, \quad \liminf \delta(C) \geq \delta.$$

Proof. Greedily choose codewords and delete radius- $(d-1)$ balls. Maximality makes those balls cover the ambient space, yielding the finite bound. For the linear version choose parity-check columns inductively, forbidding all combinations of at most $d-2$ previous columns with nonzero coefficients; the hypothesis leaves a permissible vector. Then every $d-1$ columns are independent and the kernel has distance at least d and dimension at least k ; pass to a k -dimensional subcode. If $1 - H_q(\delta) - \varepsilon \leq 0$, the asserted rate lower bound is nonpositive and a one-dimensional repetition family already has relative distance 1 and asymptotic rate 0, so the claim is immediate. Otherwise take $d = \lceil \delta n \rceil$ and $k = \lfloor n(1 - H_q(\delta) - \varepsilon) \rfloor$. The preceding volume lemma verifies the finite parity-check hypothesis for all sufficiently large n , giving the stated liminf bounds. $\square$

Chapter 6

Reed–Solomon Codes and Unique Decoding

Definition 6.1 (Reed–Solomon evaluation code). For distinct $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ and $1 \leq k \leq n$,

$$RS(\alpha, k) = \{(f(\alpha_1), \dots, f(\alpha_n)) : \deg f < k\}.$$

Positive example. Over $\mathbb{F}_7$, $f = 2X + 1$ at $0, 1, 2, 3, 4$ gives $(1, 3, 5, 0, 2)$.

Nonexample. Repeated evaluation points destroy the root-counting distance argument.

Theorem 6.2 (Reed–Solomon distance theorem). $RS(\alpha, k)$ has parameters $[n, k, n - k + 1]_q$ and is MDS.

Proof. Evaluation is injective because a degree- $< k$ polynomial vanishing at $n \geq k$ distinct points is zero. A nonzero difference of two message polynomials has at most $k - 1$ roots, so distance is at least $n - k + 1$. The polynomial $\prod_{j=1}^{k-1} (X - \alpha_j)$ attains that weight, proving equality; Singleton gives MDS. $\square$

Algorithm 6.3 (Berlekamp–Welch unique decoding). If y differs from an RS codeword of f , $\deg f < k$, in at most t positions and $2t < n - k + 1$, solve $Q(\alpha_i) = y_i E(\alpha_i)$ with $\deg E \leq t$, $\deg Q < k + t$. Every nonzero solution has $E \neq 0$ and $Q = Ef$, so division recovers f in polynomial time.

Proof. For the actual error set S , $E_0 = \prod_{i \in S} (X - \alpha_i)$ and $Q_0 = E_0 f$ form a nonzero solution; this explicit pair, not an equation-count argument, guarantees existence. If a solution had $E = 0$, the degree- $< n$ polynomial Q would vanish at all n evaluation points, hence be zero. At each correct position $P = Q - Ef$ vanishes. It has at least $n - t$ roots but degree at most $k + t - 1 < n - t$, so $P = 0$. Thus $Q = Ef$. Linear algebra finds a solution and polynomial division recovers the unique message; uniqueness also follows from minimum distance $> 2t$. $\square$

Example 6.4 (Complete Reed–Solomon encode/decode example). For the preceding $[5, 2, 4]_7$ code, $f = 2X + 1$ at $0, 1, 2, 3, 4$ gives $(1, 3, 5, 0, 2)$. Corrupt the value at $\alpha = 2$ from 5 to 6. With $E = X - 2$ and

$$Q = E(2X + 1) = 2X^2 + 4X + 5 \pmod{7},$$

the equations $Q(\alpha_i) = y_i E(\alpha_i)$ hold at all five evaluation points: at the corrupted point both sides vanish, and elsewhere cancellation recovers $y_i = f(\alpha_i)$. Polynomial division gives $Q/E = 2X + 1$

and hence the original codeword. Since the minimum distance is 4, one error lies strictly inside the unique-decoding radius $2t < 4$.

Chapter 7

List Decoding and Polynomial Interpolation

Definition 7.1 (List decoding, agreement, and weighted degree). A code is (ρ, L) -list decodable if every radius- ρn Hamming ball contains at most L codewords. For Reed–Solomon decoding in the nontrivial range $k \geq 2$, agreement counts indices with $f(\alpha_i) = y_i$. For $Q = \sum q_{ij} X^i Y^j$, define $\text{wdeg}_{1,k-1} Q = \max(i + (k-1)j)$ and let $M_k(D)$ count monomials of weighted degree at most D .

Positive example. Beyond half the minimum distance a bounded list can still be algorithmically useful.

Nonexample. “Few nearby codewords” without radius and list-size quantifiers is not a theorem.

Theorem 7.2 (q -ary Johnson bound). Let $C \subseteq [q]^n$ have minimum distance d and put $\delta = d/n$. Assume

$$0 \leq \delta \leq 1 - \frac{1}{q}$$

and define

$$J_q(\delta) = \left(1 - \frac{1}{q}\right) \left(1 - \sqrt{1 - \frac{q\delta}{q-1}}\right).$$

If $0 \leq \rho < J_q(\delta)$, every Hamming ball of radius ρn contains at most

$$L \leq \frac{\delta}{\delta - 2\rho + \frac{q}{q-1}\rho^2}$$

codewords. In particular, for fixed q, δ, ρ with strict inequality, the list size is bounded independently of n .

Proof. Fix a received word r and distinct codewords $c^{(1)}, \dots, c^{(L)}$ within distance at most ρn of r . At coordinate j , let $m_{j,a}$ be the number of listed codewords carrying symbol a , and put

$$e_j = L - m_{j,r_j}, \quad E = \sum_{j=1}^n e_j.$$

Since each codeword is within distance ρn of r ,

$$E = \sum_{\ell=1}^L d_H(c^{(\ell)}, r) \leq L\rho n.$$

Count the total pairwise Hamming distance $S = \sum_{\ell < \ell'} d_H(c^{(\ell)}, c^{(\ell')})$. At coordinate j , the number of disagreeing unordered pairs is

$$\frac{1}{2} \left(L^2 - \sum_a m_{j,a}^2 \right).$$

The $q - 1$ counts with $a \neq r_j$ sum to e_j , so Cauchy–Schwarz gives

$$\sum_{a \neq r_j} m_{j,a}^2 \geq \frac{e_j^2}{q - 1}.$$

Using $m_{j,r_j} = L - e_j$, the contribution of coordinate j is therefore at most

$$Le_j - \frac{q}{2(q - 1)} e_j^2.$$

Summing and applying $\sum_j e_j^2 \geq E^2/n$ yields

$$S \leq LE - \frac{qE^2}{2(q - 1)n}.$$

The right-hand side, as a function of E , is nondecreasing on $0 \leq E \leq L(q - 1)n/q$. Since $\rho < J_q(\delta) \leq (q - 1)/q$, we may insert $E \leq L\rho n$ to obtain

$$S \leq L^2\rho n - \frac{qL^2\rho^2 n}{2(q - 1)}.$$

On the other hand, minimum distance gives

$$S \geq \binom{L}{2} \delta n.$$

Combining the two estimates and simplifying gives

$$L \left(\delta - 2\rho + \frac{q}{q - 1} \rho^2 \right) \leq \delta.$$

The denominator is positive precisely for ρ below the smaller root of that quadratic, namely $J_q(\delta)$. Division proves the stated list-size bound. $\square$

Algorithm 7.3 (Bivariate polynomial root extraction). *Given nonzero $Q \in \mathbb{F}_q[X, Y]$, all f with $\deg f < k$ and $Q(X, f(X)) = 0$ are found by recursively choosing coefficients of $f = c_0 + Xf_1$: normalize by removing common powers of X , factor $Q(0, Y)$, substitute $Y = c_0 + XY$, and recurse. The procedure is Las Vegas polynomial time using finite-field factorization.*

Proof. Any solution must have $Q(0, c_0) = 0$. After choosing such a root, $Q(X, c_0 + Xf_1) = 0$ is equivalent, after removing a common power of X , to the child relation for f_1 . At depth k , direct substitution verifies the candidate. If c has multiplicity m_c in $Q(0, Y)$, the normalized child has Y -degree at most m_c , so the sum of child degrees never exceeds the parent Y -degree. Thus at every depth there are at most $\deg_Y Q$ active branches and only polynomially many nodes, each using polynomial-time univariate factorization and substitution. $\square$

Algorithm 7.4 (Sudan list decoding). *If $M_k(D) > n$, one can list every degree- $< k$ polynomial agreeing with the received points in more than D positions; the list has size at most $\lfloor D/(k-1) \rfloor$. The choice $D = \lceil \sqrt{2n(k-1)} \rceil$ suffices.*

Proof. Use the $M_k(D) > n$ monomial coefficients as unknowns in the n homogeneous conditions $Q(\alpha_i, y_i) = 0$ to obtain nonzero Q of weighted degree at most D . For a message f , $Q(X, f(X))$ has degree at most D ; more than D agreement points force it to vanish identically, so $Y - f(X)$ divides Q . The preceding root-extraction algorithm finds every candidate, and distinct candidates give distinct linear factors in Y , bounding their number by $\deg_Y Q \leq D/(k-1)$. Finally $M_k(D) > D^2/(2(k-1))$, so the displayed D gives more than n monomials. $\square$

Algorithm 7.5 (Guruswami–Sudan multiplicity decoding). *If $M_k(D) > n \binom{s+1}{2}$, one can list every degree- $< k$ polynomial agreeing in more than D/s positions. Hence fixed-rate Reed–Solomon families of rate R —with block lengths $n_j \leq q_j$ and hence allowing the field sizes q_j to grow—are polynomial-time list decodable from every error fraction below $1 - \sqrt{R} - \varepsilon$ for sufficiently large block length.*

Proof. Impose Hasse-derivative conditions of total order $< s$ at every received point; there are $\binom{s+1}{2}$ linear constraints per point, so a nonzero weighted degree- D interpolant exists. At an agreement point, substituting $f(\alpha + U) = f(\alpha) + UV(U)$ shows that every surviving Hasse term is divisible by U^s . Thus $Q(X, f(X))$ has at least sA roots with multiplicity when there are A agreements. If $sA > D$, it vanishes identically and root extraction applies. Choose $D = \lceil \sqrt{(k-1)ns(s+1)} \rceil$; since $M_k(D) > D^2/(2(k-1))$, interpolation is possible. Dividing $D/(sn)$ and first choosing large constant s , then large n , makes the agreement threshold arbitrarily close to $\sqrt{R}$. $\square$

Chapter 8

Geometry of Numbers and Successive Minima

Definition 8.1 (Lattice, determinant, dual, and successive minima). A full-rank lattice is $L = B\mathbb{Z}^n$ with B invertible and $\det L = |\det B|$. Its dual is $L^* = \{y : \langle x, y \rangle \in \mathbb{Z} \ \forall x \in L\}$. A *convex body* is a compact convex set with nonempty interior. For a centrally symmetric convex body K , $\lambda_i(K, L)$ is the least scale at which λK contains i linearly independent lattice vectors.

Positive example. For $\mathbb{Z}^2$ and $[-1, 1]^2$, both minima are 1.

Nonexample. A rank-one subgroup of $\mathbb{R}^2$ is not full rank under this convention.

Theorem 8.2 (Blichfeldt principle). *If a Lebesgue-measurable set S of finite volume satisfies $\text{vol } S > \det L$, two distinct points of S differ by a lattice vector. Moreover some translate contains at least $\text{vol } S / \det L$ lattice points.*

Proof. Translate the pieces $S \cap (P + \ell)$ of a half-open fundamental parallelepiped P back into P . If no two points differed by a lattice vector these pieces would be disjoint, contradicting total volume greater than $\text{vol } P$. For the counting statement integrate $N(t) = \sum_{\ell \in L} 1_S(\ell - t)$ over P ; Tonelli and the lattice tiling give average $\text{vol } S / \det L$, so some translate attains at least the average. $\square$

Theorem 8.3 (Minkowski convex-body theorem). *If centrally symmetric convex K satisfies $\text{vol } K > 2^n \det L$, it contains a nonzero lattice point; for closed K the same holds with $\geq$.*

Proof. Apply Blichfeldt to $K/2$. Distinct $x, y \in K/2$ with $x - y \in L$ give a nonzero lattice vector $x - y \in K$ by symmetry and convexity. For equality, apply the strict case to $(1 + \varepsilon)K$ and let $\varepsilon \downarrow 0$; only finitely many lattice points lie in $2K$, so one nonzero point recurs and closedness puts it in K . $\square$

Lemma 8.4 (Triangular slice-centroid volume map). *Let $K \subset \mathbb{R}^n$ be a bounded open convex body and let $b_1, \dots, b_n$ be a real basis. Write $x = \sum_i x_i b_i$. For $j = 1, \dots, n$, let $\phi_j(x)$ be the centroid of the slice of K through x parallel to $\text{span}(b_1, \dots, b_{j-1})$, with $\phi_1(x) = x$. For positive numbers $0 < \lambda_1 \leq \dots \leq \lambda_n$, put $\lambda_0 = 0$ and*

$$\Phi(x) = \sum_{j=1}^n (\lambda_j - \lambda_{j-1}) \phi_j(x).$$

Then Φ is measurable and injective and

$$\text{vol } \Phi(K) = \lambda_1 \cdots \lambda_n \text{vol } K.$$

Proof. Fix j and write u for the first $j - 1$ coordinates and z for the remaining coordinates. The projection P_j of the open convex set K to the z -coordinates is open and convex. For $z \in P_j$ its fiber

$$K_z = \{u : (u, z) \in K\}$$

is a nonempty bounded open convex set. Put

$$A(z) = \int 1_K(u, z) du, \quad M_i(z) = \int u_i 1_K(u, z) du.$$

We first record that A and M_i are continuous on P_j . If $z_r \rightarrow z$, then $1_{K_{z_r}}(u) \rightarrow 1_{K_z}(u)$ for every u away from the boundary of the convex fiber K_z : membership is stable at interior points by openness, and nonmembership is stable away from the closed fiber. The boundary of a bounded convex set has Lebesgue measure zero. Indeed, after translating an interior point to the origin, the scaled sets $t\overline{K_z}$ with $t < 1$ lie inside K_z , while the volume of the shell $\overline{K_z} \setminus t\overline{K_z}$ tends to zero as $t \uparrow 1$. Because all nearby fibers lie in one fixed bounded box, dominated convergence applies both to 1_K and to $u_i 1_K$. Hence $A(z)$ and $M_i(z)$ are continuous. Since $A(z) > 0$, the centroid coordinates $M_i(z)/A(z)$ are continuous functions of the later coordinates. The centroid belongs to the open convex fiber itself (it is a barycenter of a full-dimensional probability measure supported there).

Consequently each ϕ_j has the triangular form: its coordinates $j, \dots, n$ equal those of x , and its earlier coordinates are continuous functions of $x_j, \dots, x_n$. Summing the contributions in the definition of Φ gives

$$x'_i = \lambda_i x_i + \psi_i(x_{i+1}, \dots, x_n), \quad 1 \leq i \leq n,$$

with continuous ψ_i on the relevant projections of K . Since $\lambda_i > 0$, equality of two images determines x_n from the last coordinate, then x_{n-1} , and so on; thus Φ is injective. The same backward formulas give a continuous local inverse, so $\Phi(K)$ is open and in particular measurable.

It remains to compute volume without assuming differentiability of the centroid functions. Realize Φ as the composition, for $i = 1, 2, \dots, n$, of elementary triangular maps

$$S_i(x_1, \dots, x_n) = (x_1, \dots, x_{i-1}, \lambda_i x_i + \psi_i(x_{i+1}, \dots, x_n), x_{i+1}, \dots, x_n).$$

At the moment S_i is applied, the coordinates $i + 1, \dots, n$ still have their original values, so this composition is exactly Φ . For fixed later coordinates, S_i merely translates the i th one-dimensional fiber and scales it by λ_i ; earlier coordinates play the role of additional Fubini variables. Therefore Fubini's theorem gives

$$\text{vol}(S_i(E)) = \lambda_i \text{vol}(E)$$

for each open finite-volume set E arising at that stage. Iterating over $i = 1, \dots, n$ yields

$$\text{vol } \Phi(K) = \lambda_1 \cdots \lambda_n \text{vol } K,$$

as claimed. □

Theorem 8.5 (Minkowski's second theorem). *Let $K \subset \mathbb{R}^n$ be a centrally symmetric convex body, let L be a full-rank lattice, and let $\lambda_i = \lambda_i(K, L)$ be the successive minima. Then*

$$\frac{2^n}{n!} \det L \leq \lambda_1 \cdots \lambda_n \text{vol } K \leq 2^n \det L.$$

Proof. Lower bound. Compactness of K and discreteness of L ensure that the successive minima are attained. Choose linearly independent $u_i \in L \cap \lambda_i K$. Then $u_i/\lambda_i \in K$, so convexity and central symmetry imply that K contains the cross-polytope

$$P = \text{conv} \left\{ \pm \frac{u_1}{\lambda_1}, \dots, \pm \frac{u_n}{\lambda_n} \right\}.$$

Its volume is

$$\text{vol } P = \frac{2^n |\det(u_1, \dots, u_n)|}{n! \lambda_1 \cdots \lambda_n}.$$

The lattice generated by the u_i is a finite-index sublattice of L , so $|\det(u_1, \dots, u_n)|$ is a positive integer multiple of $\det L$ and is at least $\det L$. Since $P \subseteq K$, rearrangement gives the lower bound.

Upper bound. Work with the interior K° ; it has the same volume as K , and the infima defining the successive minima are unchanged. For $1 \leq k \leq n$ put

$$W_k = \text{span}_{\mathbb{R}}(L \cap \lambda_k K^\circ).$$

Then $\dim W_k \leq k - 1$. Otherwise k independent lattice vectors lying in the open set $\lambda_k K^\circ$ could all be shrunk by a common factor slightly smaller than one, contradicting minimality of λ_k . The W_k form a nested filtration. Since each is spanned by lattice vectors, choose a real basis $b_1, \dots, b_n$ of $\mathbb{R}^n$ consisting of lattice vectors and adapted to this filtration so that

$$W_k \subseteq \text{span}(b_1, \dots, b_{k-1}) \quad (1 \leq k \leq n).$$

Such a basis is obtained by extending a lattice-vector basis whenever the filtration dimension grows and finally extending the resulting independent lattice vectors to an ambient real basis using further vectors of the full-rank lattice. The b_i need not form an integral basis of L .

Apply the preceding triangular-map lemma to K° and this coordinate basis. Suppose for contradiction that

$$\lambda_1 \cdots \lambda_n \text{vol } K > 2^n \det L.$$

Then

$$\text{vol } \Phi(K^\circ) > \det(2L) = 2^n \det L.$$

Blichfeldt's principle applied to the lattice $2L$ yields distinct $x, y \in K^\circ$ such that

$$v = \frac{\Phi(x) - \Phi(y)}{2} \in L \setminus \{0\}.$$

Write $x = \sum_i x_i b_i$ and $y = \sum_i y_i b_i$, and let k be the largest index with $x_k \neq y_k$. For every $j > k$, the slice centroid ϕ_j depends only on coordinates $j, \dots, n$, which agree for x and y , so $\phi_j(x) = \phi_j(y)$. Hence

$$v = \sum_{j=1}^k (\lambda_j - \lambda_{j-1}) \frac{\phi_j(x) - \phi_j(y)}{2}.$$

Each half-difference belongs to K° , because K° is convex and centrally symmetric. The nonnegative coefficients sum to λ_k , so convexity gives $v \in \lambda_k K^\circ$.

On the other hand, for every $j \leq k$ the b_k -coordinate of $\phi_j(x)$ is x_k , and similarly for y . Thus the b_k -coordinate of v is

$$\frac{\lambda_k(x_k - y_k)}{2} \neq 0.$$

Therefore $v \notin \text{span}(b_1, \dots, b_{k-1})$. But $v \in L \cap \lambda_k K^\circ \subseteq W_k$, contradicting the adapted-basis property. This proves the upper bound. $\square$

Chapter 9

LLL Reduction and Approximate Lattice Problems

Definition 9.1 (Gram–Schmidt data and LLL reduction). For a basis b_i , write $b_i^* = b_i - \sum_{j < i} \mu_{i,j} b_j^*$ and $\mu_{i,j} = \langle b_i, b_j^* \rangle / \|b_j^*\|^2$. For $1/4 < \delta < 1$, a basis is δ -LLL-reduced if $|\mu_{i,j}| \leq 1/2$ and $\delta \|b_{i-1}^*\|^2 \leq \|b_i^*\|^2 + \mu_{i,i-1}^2 \|b_{i-1}^*\|^2$. Put $\alpha = (\delta - 1/4)^{-1}$.

Positive example. An appropriately ordered orthogonal basis is LLL-reduced.

Nonexample. $(1, 0), (N, 1)$ is not size-reduced for large N .

Algorithm 9.2 (LLL basis reduction). For an integer basis $b_1, \dots, b_n \in \mathbb{Z}^m$ and fixed rational $1/4 < \delta < 1$, repeatedly perform nearest-integer size reductions

$$b_i \leftarrow b_i - \lfloor \mu_{i,j} \rfloor b_j \quad (j = i-1, i-2, \dots, 1),$$

with a fixed tie-breaking rule, and swap b_i, b_{i-1} whenever the Lovász condition at i fails. The algorithm terminates with a δ -LLL-reduced basis of the same lattice. With exact rational arithmetic its bit complexity is polynomial in the input bit length for fixed δ . Rational input bases reduce to the integer case by clearing one common denominator.

Proof. Subtracting an integer multiple of an earlier basis vector and swapping two basis vectors are unimodular column operations, so every intermediate basis generates the same lattice. Performing size reduction from $j = i-1$ downward makes every resulting $|\mu_{i,j}| \leq 1/2$ without spoiling coefficients already reduced at larger j .

For termination, put

$$d_i = \det(\langle b_r, b_s \rangle_{1 \leq r, s \leq i}) = \prod_{j=1}^i \|b_j^*\|^2, \quad \mathcal{D} = \prod_{i=1}^n d_i.$$

Because the b_i are integer vectors, every d_i is a positive integer. Size reduction preserves the lattice spanned by every prefix and therefore preserves all d_i and $\mathcal{D}$. Suppose a Lovász failure causes a swap at positions $i-1, i$. Only d_{i-1} changes. The new $(i-1)$ st Gram–Schmidt vector is the projection of b_i orthogonal to the first $i-2$ vectors, namely

$$b_i^* + \mu_{i,i-1} b_{i-1}^*.$$

Consequently

$$\frac{d_{i-1}^{\text{new}}}{d_{i-1}^{\text{old}}} = \frac{\|b_i^*\|^2 + \mu_{i,i-1}^2 \|b_{i-1}^*\|^2}{\|b_{i-1}^*\|^2} < \delta.$$

Thus each swap multiplies the positive integer $\mathcal{D}$ by a factor below δ . Since $\mathcal{D} \geq 1$, there are at most $\log \mathcal{D}_0 / \log(1/\delta)$ swaps. Between two swaps the scan index can move forward through at most $O(n)$ positions, and each position performs at most $O(n)$ elementary size reductions, so at most $O(n^2)$ such reductions occur between swaps. This polynomial factor is sufficient for the complexity bound.

It remains to prove that exact arithmetic uses only polynomially many bits. We need no external determinant inequality. Gram–Schmidt gives

$$d_i = \det(\langle b_r, b_s \rangle_{r,s \leq i}) = \prod_{j=1}^i \|b_j^*\|^2.$$

Each b_j^* is an orthogonal projection of b_j , so $\|b_j^*\| \leq \|b_j\|$. If the input coordinates have absolute value at most B , then $\|b_j\|^2 \leq mB^2$, and therefore

$$d_i \leq \prod_{j=1}^i \|b_j\|^2 \leq (mB^2)^i, \quad \log \mathcal{D}_0 = O(n^2(\log B + \log m)).$$

At every stage $\mathcal{D} \leq \mathcal{D}_0$. Since each d_i is a positive integer factor of the product $\mathcal{D}$, $1 \leq d_i \leq \mathcal{D}_0$. With $d_0 = 1$,

$$\mathcal{D}_0^{-1} \leq \|b_i^*\|^2 = \frac{d_i}{d_{i-1}} \leq \mathcal{D}_0.$$

After a vector has been fully size-reduced against a size-reduced prefix,

$$\|b_i\|^2 = \|b_i^*\|^2 + \sum_{j < i} \mu_{i,j}^2 \|b_j^*\|^2 \leq \left(1 + \frac{n-1}{4}\right) \mathcal{D}_0.$$

Hence every vector that has completed a reduction pass has polynomial-bit integer coordinates. At the start of any later pass, each current basis vector is either an untouched input vector or the output of an earlier completed pass; swaps merely permute these existing vectors. Thus all basis coordinates at the start of a pass have polynomial bit length.

For temporary values inside one backward size-reduction pass, let M bound the Euclidean norms of the vector being reduced and its size-reduced prefix at the start of the pass. The lower Gram–Schmidt bound gives

$$|\mu_{i,j}| \leq \frac{\|b_i\|}{\|b_j^*\|} \leq M\sqrt{\mathcal{D}_0} =: A.$$

Because the prefix is size-reduced, $|\mu_{\ell,j}| \leq 1/2$ for $j < \ell < i$. If the already chosen backward rounding integers are q_ℓ for $\ell > j$, then just before reducing against b_j ,

$$|\mu_{i,j}^{\text{cur}}| \leq A + \frac{1}{2} \sum_{\ell=j+1}^{i-1} |q_\ell|,$$

so

$$|q_j| \leq A + \frac{1}{2} + \frac{1}{2} \sum_{\ell=j+1}^{i-1} |q_\ell|.$$

A backward induction bounds every $|q_j|$ by $(3/2)^n(A + 1)$. Its bit length is therefore $O(n + \log A)$, polynomial in the input size. Every temporary vector is an input-or-reduced vector plus at most

n such polynomial-bit integer multiples of polynomial-bit prefix vectors, so its coordinates also have polynomial bit length. Gram determinants, inner products, and rational Gram–Schmidt coefficients consequently have polynomial-size numerators and denominators. Exact rational operations, comparisons, and nearest-integer rounding therefore take polynomial bit time.

Together with the polynomial number of swaps and reductions, this proves polynomial bit complexity. When the algorithm stops, all size-reduction and Lovász conditions hold, exactly as required. Clearing a common denominator scales all basis vectors by the same positive scalar and leaves every $\mu_{i,j}$ and every homogeneous LLL inequality unchanged, so rational inputs reduce to the integer case. $\square$

Theorem 9.3 (LLL approximation guarantees). *For a δ -LLL basis,*

$$\|b_1\| \leq \alpha^{(n-1)/4} (\det L)^{1/n}, \quad \|b_1\| \leq \alpha^{(n-1)/2} \lambda_1(L).$$

At $\delta = 3/4$, the shortest-vector factor is $2^{(n-1)/2}$.

Proof. Size reduction and Lovasz give $\|b_{i-1}^*\|^2 \leq \alpha \|b_i^*\|^2$, hence $\|b_i^*\|^2 \geq \alpha^{-(i-1)} \|b_1\|^2$. Multiplying and using $\det L = \prod_i \|b_i^*\|$ yields the determinant bound. For a shortest nonzero $v = \sum z_i b_i$, let j be its largest nonzero coefficient index. Orthogonal projection gives $\|v\| \geq \|b_j^*\| \geq \alpha^{-(n-1)/2} \|b_1\|$. $\square$

Algorithm 9.4 (Babai nearest plane). *Nearest-plane rounding on a δ -LLL basis returns $v \in L$ satisfying*

$$\|t - v\| \leq \sqrt{\frac{\alpha^n - 1}{\alpha - 1}} \text{dist}(t, L).$$

For $\delta = 3/4$ this is $< 2^{n/2}$.

Proof. We prove the guarantee by induction on the rank n . Let $L_{n-1} = \text{span}_{\mathbb{Z}}(b_1, \dots, b_{n-1})$ and project onto the one-dimensional orthogonal quotient generated by b_n^* . Write the quotient coordinate of t as $c_n b_n^*$. Nearest plane chooses $z_n \in \mathbb{Z}$ nearest to c_n , so $|c_n - z_n| \leq 1/2$, then recursively applies the algorithm to the projected target in the affine plane $z_n b_n + \text{span}(b_1, \dots, b_{n-1})$.

Let w be a closest lattice point to t . If w lies in that same affine plane, numerator and denominator have the same orthogonal residual $(c_n - z_n) b_n^*$. On the parallel component, the induction hypothesis applies. Since the induction factor is at least one, Pythagoras gives the same factor in rank n .

Otherwise the quotient coordinate of w differs from z_n by a nonzero integer. Since z_n is a nearest integer to c_n , every other integer is at least $1/2$ away from c_n . Hence

$$\text{dist}(t, L) \geq \frac{1}{2} \|b_n^*\|.$$

At each recursive stage the rounding error in the corresponding orthogonal Gram–Schmidt direction is at most half that Gram–Schmidt vector. These errors are mutually orthogonal, so the returned point v satisfies

$$\|t - v\|^2 \leq \frac{1}{4} \sum_{j=1}^n \|b_j^*\|^2.$$

The LLL inequalities imply $\|b_j^*\|^2 \leq \alpha^{n-j} \|b_n^*\|^2$. Therefore

$$\frac{\|t - v\|^2}{\text{dist}(t, L)^2} \leq \sum_{r=0}^{n-1} \alpha^r = \frac{\alpha^n - 1}{\alpha - 1}.$$

Taking square roots proves the claimed approximation factor. At $\delta = 3/4$ we have $\alpha = 2$, so the factor is $\sqrt{2^n - 1} < 2^{n/2}$. $\square$

Example 9.5 (Finite-field, coding and lattice laboratory — completed artifact). The executable file `code/volume_03/finitefield_coding_lattice_lab.py` checks finite-field interpolation, the explicit Reed–Solomon decoding example of Chapter 6, and exact-rational two-dimensional LLL size reduction. Every calculation is deterministic and uses integer/rational arithmetic whenever an exact identity is being audited.

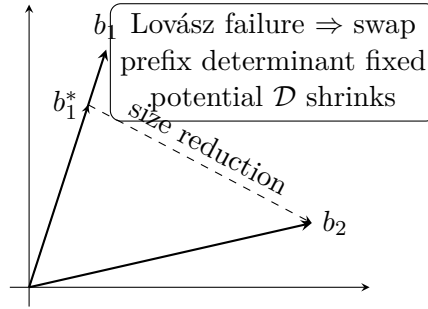


Figure 9.1: LLL geometry: Gram–Schmidt separates orthogonal scale from integral shear. Size reduction changes the shear coefficient without changing the lattice; a Lovász-violating swap decreases the determinant potential used in the termination proof.

Chapter 10

Discrete Logarithms and Generic Algorithms

Definition 10.1 (Cyclic-group discrete logarithm and generic cost). Let $G = \langle g \rangle$ be cyclic of known order N . The discrete logarithm problem asks, given $h \in G$, for $x \in \mathbb{Z}/N\mathbb{Z}$ with $g^x = h$. A generic algorithm accesses group elements only through equality, multiplication, inversion, and supplied encodings; group operations and stored elements are counted explicitly.

Positive example. In $\mathbb{F}_7^\times = \langle 3 \rangle$, the log of 2 base 3 is 2.

Nonexample. An index-calculus algorithm is not generic because it exploits a representation of field elements.

Algorithm 10.2 (Baby-step giant-step). With $m = \lceil \sqrt{N} \rceil$, store g^j for $0 \leq j < m$ and search for a collision with $h(g^{-m})^i$, $0 \leq i < m$. The algorithm deterministically finds the discrete logarithm using $O(\sqrt{N})$ group operations and storage.

Proof. Write the desired exponent $x = im + j$ with $0 \leq j < m$ and $0 \leq i \leq \lfloor (N-1)/m \rfloor < m$. Then $g^j = h(g^{-m})^i$. Conversely any table collision yields $h = g^{im+j}$ and therefore a valid discrete logarithm modulo N . The two lists have length $O(\sqrt{N})$; incremental multiplication generates them with the claimed resources. $\square$

Lemma 10.3 (Collision time for an ideal random mapping). Let F be a uniformly random function from an N -element set to itself and start an orbit $X_{t+1} = F(X_t)$ from a fixed point. If T is the first time a state repeats, then for $0 \leq t < N$,

$$\Pr(T > t) = \prod_{j=1}^t \left(1 - \frac{j}{N}\right) \leq \exp\left(-\frac{t(t+1)}{2N}\right),$$

and consequently $\mathbb{E}T = O(\sqrt{N})$.

Proof. Conditioned on $X_0, \dots, X_t$ being distinct, the values of the random function revealed at the distinct visited domain points are independent uniform images. At the j th opportunity for a collision, the new image must avoid j previously seen states, which has probability $1 - j/N$. Multiplication gives the exact survival probability.

Using $1 - u \leq e^{-u}$ gives the displayed exponential bound. For a nonnegative integer-valued random variable,

$$\mathbb{E}T = \sum_{t \geq 0} \Pr(T > t).$$

Split this sum at $\lceil \sqrt{N} \rceil$ and compare the remaining Gaussian tail with an integral after writing $t = u\sqrt{N}$. The result is at most $C\sqrt{N}$ for an absolute constant C . $\square$

Algorithm 10.4 (Pollard rho for discrete logarithms). *Maintain states $X_i = g^{a_i}h^{b_i}$ under a deterministic pseudorandom iteration, for example a three-set partition with updates*

$$X \mapsto hX, \quad X \mapsto X^2, \quad X \mapsto gX,$$

and corresponding exponent updates

$$(a, b+1), \quad (2a, 2b), \quad (a+1, b) \pmod{N}.$$

Floyd or Brent cycle finding detects a collision with $O(1)$ stored group elements. From $X_i = X_j$ the discrete logarithm x satisfies

$$(b_i - b_j)x \equiv a_j - a_i \pmod{N}.$$

If the coefficient is invertible modulo N , this determines x uniquely; otherwise solve the linear congruence and verify its candidates or restart. For prime N , under the explicit heuristic that the deterministic iteration behaves like a random mapping and that the collision coefficients are correspondingly random, the expected group-operation count is $O(\sqrt{N})$ with constant memory. No unconditional square-root runtime claim is made for the fixed partition iteration.

Proof. The exponent invariant follows by induction from the update rules. For example, squaring sends $g^a h^b$ to $g^{2a} h^{2b}$. Thus every stored pair (a_i, b_i) certifies the associated group element. Cycle finding evaluates the same deterministic map at different speeds; once the orbit enters its cycle, the pointers meet, so a repeated group state is detected without storing the orbit.

At a collision, substituting $h = g^x$ gives

$$a_i + b_i x \equiv a_j + b_j x \pmod{N},$$

which rearranges to the displayed congruence. If $b_i - b_j$ is invertible, multiplication by its inverse yields the unique correct residue x , which can be verified by exponentiation. For general N , elementary linear-congruence theory gives either no solution or $d = \gcd(b_i - b_j, N)$ residue classes when the right side is divisible by d ; the true logarithm guarantees consistency, and all candidate residues can be checked, or the walk can be restarted.

The running-time statement is deliberately conditional. In the ideal random-mapping model, the preceding lemma gives a collision after $O(\sqrt{N})$ expected evaluations. In a prime-order group, a random nonzero coefficient difference is invertible; the exceptional zero difference has probability $1/N$ under the corresponding coefficient heuristic. Restarts therefore preserve the square-root scale in that model. These probabilistic assumptions explain the heuristic complexity but are not used in the correctness proof. $\square$

Internalization and Reconstruction Guide: Volume III

Purpose. Use exact algebraic structure as an algorithmic resource: finite fields support coding and interpolation, lattices support geometric reduction, and both become computational primitives used later in complexity and cryptography.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

This is a foundation volume. Its prerequisite ladder is internal: begin with the first definitions and reconstruct each later family from those conventions rather than importing unstated external notation. Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-III-DEF-001 (Finite field, characteristic, and Frobenius); QRCC-III-DEF-002 (Polynomial arithmetic and computational model); QRCC-III-DEF-003 (Irreducibility and factorization output); QRCC-III-DEF-004 (Linear code, dual code, and syndrome); QRCC-III-DEF-005 (Rate, relative distance, and asymptotic code family).

Chapter retrieval map

- Ch. 1: Finite Fields and Frobenius:** enter through QRCC-III-DEF-001 (Finite field, characteristic, and Frobenius); exit by being able to state QRCC-III-THM-001 (Classification of finite fields) with its ownership boundary.
- Ch. 2: Polynomial Arithmetic over Finite Fields:** enter through QRCC-III-DEF-002 (Polynomial arithmetic and computational model); exit by being able to state QRCC-III-ALG-004 (Euclidean polynomial arithmetic) with its ownership boundary.
- Ch. 3: Polynomial Factorization over Finite Fields:** enter through QRCC-III-DEF-003 (Irreducibility and factorization output); exit by being able to state QRCC-III-ALG-005 (Berlekamp factorization) with its ownership boundary.
- Ch. 4: Linear Codes, Duality, and Syndromes:** enter through QRCC-III-DEF-004 (Linear code, dual code, and syndrome); exit by being able to state QRCC-III-THM-021 (Linear-code dimension, duality, and syndrome theorem) with its ownership boundary.

- Ch. 5: Coding Bounds and Asymptotic Rate:** enter through QRCC-III-DEF-005 (Rate, relative distance, and asymptotic code family); exit by being able to state QRCC-III-THM-007 (Singleton bound) with its ownership boundary.
- Ch. 6: Reed–Solomon Codes and Unique Decoding:** enter through QRCC-III-DEF-006 (Reed–Solomon evaluation code); exit by being able to state QRCC-III-ALG-011 (Berlekamp–Welch decoding) with its ownership boundary.
- Ch. 7: List Decoding and Polynomial Interpolation:** enter through QRCC-III-DEF-007 (List decoding and agreement); exit by being able to state QRCC-III-ALG-013 (Sudan list-decoding theorem) with its ownership boundary.
- Ch. 8: Geometry of Numbers and Successive Minima:** enter through QRCC-III-DEF-008 (Lattice, determinant, dual, and successive minima); exit by being able to state QRCC-III-THM-015 (Minkowski convex-body theorem) with its ownership boundary.
- Ch. 9: LLL Reduction and Approximate Lattice Problems:** enter through QRCC-III-DEF-009 (Gram–Schmidt data and LLL reduction); exit by being able to state QRCC-III-ALG-017 (LLL basis-reduction algorithm) with its ownership boundary.
- Ch. 10: Discrete Logarithms and Generic Algorithms:** enter through QRCC-III-DEF-010 (Cyclic-group discrete logarithm and generic algorithms); exit by being able to state QRCC-III-ALG-019 (Pollard rho for discrete logarithms) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-III-THM-001 — Classification of finite fields. Dependencies: QRCC-III-DEF-001, QRCC-III-LEM-030. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-III-ALG-005 — Berlekamp factorization. Dependencies: QRCC-III-LEM-034, QRCC-III-LEM-035, QRCC-III-ALG-004. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-III-THM-007 — Singleton bound. Dependencies: QRCC-III-DEF-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-III-ALG-013 — Sudan list-decoding theorem. Dependencies: QRCC-III-ALG-037, QRCC-III-ALG-023, QRCC-III-DEF-007. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-III-ALG-017 — LLL basis-reduction algorithm. Dependencies: QRCC-III-DEF-009. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-III-ALG-019 — Pollard rho for discrete logarithms. Dependencies: QRCC-III-LEM-039, QRCC-III-DEF-010. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

Common failure modes

- performing field algebra in a ring where division is not valid;
- counting arithmetic operations as bit operations without a cost model;
- confusing unique decoding radius, list-decoding radius, and covering radius;
- treating an LLL-reduced basis as a canonical invariant of the lattice;

Boundary and degenerate cases

- zero and constant polynomials;
- small characteristic and repeated factors;
- decoding exactly at half the minimum distance;
- LLL parameters at the admissible endpoint and rank-one lattices;

Retrieval and reconstruction drill

1. State QRCC-III-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-III-THM-001. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-III-ALG-005, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-III-ALG-019, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume IV shifts from algebraic objects to large combinatorial structures and asks when many local statistics force global random-like behavior.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part IV

Quasirandom Graphs, Hypergraphs, and Limits

Chapter 1

Random and Quasirandom Dense Graphs

Randomness in a graph can be tested in many inequivalent-looking ways: edge counts inside subsets, spectra, four-cycle counts, and counts of arbitrary fixed patterns. The central phenomenon of dense quasirandomness is that, at fixed edge density, these tests become asymptotically equivalent.

Definition 1.1 (Finite dense graphs and p -quasirandom sequences). Graphs are finite, simple, and labelled. For $G = (V, E)$, write $n = |V|$. For $S, T \subseteq V$, define the *ordered edge count*

$$e_G(S, T) = |\{(x, y) \in S \times T : \{x, y\} \in E\}|.$$

Thus $e_G(S, S) = 2e(G[S])$.

Fix $p \in (0, 1)$. A graph sequence (G_n) with $|V(G_n)| \rightarrow \infty$ is called *p -quasirandom* when it satisfies any—and, after Theorem 2.7, equivalently every—one of the dense asymptotic tests listed there. Whenever we write

$$R_n(S) = o(n^2) \quad \text{uniformly in } S,$$

we mean

$$\sup_{S \subseteq V(G_n)} \frac{|R_n(S)|}{n^2} \rightarrow 0.$$

Analogous uniform conventions are used for two subsets S, T .

Positive example. Theorem 1.2 shows that for fixed p , a graph sampled from $G(n, p)$ satisfies the uniform subset-edge test with probability tending to one. Hence one may select a deterministic outcome for every sufficiently large n to obtain a p -quasirandom sequence.

Nonexample. Let n be even and let G_n be the disjoint union of two cliques of size $n/2$. Its global edge density tends to $1/2$, but taking S to be one clique gives

$$e(G_n[S]) = \binom{n/2}{2}$$

instead of $(1/2)\binom{n/2}{2} + o(n^2)$. Global density alone therefore does not imply quasirandomness.

Theorem 1.2 (Erdős–Rényi edge concentration). *Fix $p \in (0, 1)$ and let $G \sim G(n, p)$. Put $N = \binom{n}{2}$. For $0 < \delta \leq 1$,*

$$\Pr(|e(G) - pN| \geq \delta pN) \leq 2 \exp\left(-\frac{\delta^2 pN}{3}\right).$$

Moreover, with probability $1 - o(1)$,

$$\sup_{S \subseteq V(G)} \left| e(G[S]) - p \binom{|S|}{2} \right| = o(n^2).$$

In particular, $G(n, p)$ satisfies the subset-edge quasirandomness test with probability tending to one.

Proof. The random variable $e(G)$ is the sum of the N independent edge indicators, so $e(G) \sim \text{Bin}(N, p)$ and has mean $\mu = pN$. The upper-tail part of Theorem 2.4 gives

$$\Pr(e(G) \geq (1 + \delta)\mu) \leq e^{-\delta^2 \mu/3},$$

while its lower-tail part gives the stronger $e^{-\delta^2 \mu/2}$. The union bound proves the first assertion.

For the uniform assertion, put $t_n = n^{7/4}$. For a fixed set S with $|S| \geq 2$, the variable $e(G[S])$ is a sum of $M_S = \binom{|S|}{2}$ independent $[0, 1]$ variables. Hoeffding's inequality, Theorem 2.5, gives

$$\Pr(|e(G[S]) - pM_S| \geq t_n) \leq 2 \exp\left(-\frac{2t_n^2}{M_S}\right).$$

Since $M_S \leq n^2/2$,

$$\frac{2t_n^2}{M_S} \geq 4n^{3/2}.$$

For $|S| \leq 1$ the deviation is zero. Taking a union bound over the at most 2^n subsets,

$$\Pr\left(\exists S : \left| e(G[S]) - p \binom{|S|}{2} \right| > n^{7/4}\right) \leq 2^{n+1} e^{-4n^{3/2}} = o(1).$$

Because $n^{7/4} = o(n^2)$, the claimed uniform estimate follows. $\square$

Worked check. For $p = 1/2$, the first estimate gives an exponentially small probability for a fixed relative error in the total number of edges. The second argument is stronger in a different direction: it pays a factor 2^n to control every induced subgraph simultaneously, but the $e^{-\Theta(n^{3/2})}$ tail still wins.

Chapter 2

The Chung–Graham–Wilson Equivalences

We now make four notions of dense randomness precise and prove that they are equivalent. The proof is arranged so that every analytic bridge is visible: four-cycles control fourth spectral moments, cut discrepancy controls bounded tests, and a delocalization argument converts cut discrepancy into operator norm.

Definition 2.1 (Homomorphism counts and C_4 density). For a fixed finite simple graph H and a graph G , let $\text{hom}(H, G)$ be the number of maps $\phi : V(H) \rightarrow V(G)$ such that

$$uv \in E(H) \implies \phi(u)\phi(v) \in E(G).$$

Set

$$t(H, G) = \frac{\text{hom}(H, G)}{|V(G)|^{|V(H)|}}.$$

The number of injective such maps is denoted $\text{inj}(H, G)$. Homomorphism counts allow collisions of nonadjacent vertices. If A is the adjacency matrix of G , then

$$\text{hom}(C_4, G) = \text{tr}(A^4),$$

because both sides count closed walks of length four with a specified starting vertex and orientation.

Positive example. For $H = K_2$, every unoriented edge of G produces two homomorphisms, so $\text{hom}(K_2, G) = 2e(G)$. For C_4 , repeated opposite vertices are allowed; this is exactly why the trace identity has no lower-order correction.

Nonexample. The number of unlabelled embedded copies of C_4 is not $\text{tr}(A^4)$: the latter remembers a starting point and direction and also includes closed walks in which opposite vertices coincide.

Definition 2.2 (Adjacency spectrum, centered spectrum, and jumbledness). For an n -vertex graph G , let A be its adjacency matrix, J the all-ones matrix, and

$$B_p = A - pJ.$$

The *centered spectral discrepancy* is $\rho_p(G) = \|B_p\|_{2 \rightarrow 2}$. A sequence is spectrally p -quasirandom when $\rho_p(G_n) = o(n)$.

Separately, a d -regular graph is an (n, d, λ) -graph when every adjacency eigenvalue on the orthogonal complement of the all-ones vector has absolute value at most λ . Under the ordered-edge convention of Definition 1.1, a graph is (p, β) -jumbled when

$$|e_G(S, T) - p|S||T|| \leq \beta \sqrt{|S||T|}$$

for every $S, T \subseteq V(G)$.

Positive example. For a d -regular graph, $n^{-1/2}\mathbf{1}$ is an adjacency eigenvector with eigenvalue d . If all remaining eigenvalues have magnitude much smaller than d , the graph has strong spectral mixing.

Nonexample. The disjoint union of two cliques of size $n/2$ has an adjacency eigenvector which is $+1$ on one clique and -1 on the other, with eigenvalue of order n . Thus it fails centered spectral quasirandomness despite having global edge density asymptotic to $1/2$.

Definition 2.3 (Cut norm and dense graph discrepancy). For a real $n \times n$ matrix M , define the unnormalized cut norm

$$\|M\|_{\square} = \max_{S, T \subseteq [n]} \left| \sum_{i \in S} \sum_{j \in T} M_{ij} \right|.$$

For a graph G and $p \in [0, 1]$, put

$$\Delta_p(G) = \|A - pJ\|_{\square}.$$

The diagonal entries of $A - pJ$ equal $-p$. Replacing pJ by $p(J - I)$ changes every cut sum by at most $pn = O(n)$, so the two conventions are equivalent for assertions of size $o(n^2)$.

Positive example. If $\|A - pJ\|_{2 \rightarrow 2} \leq \eta n$, then

$$\Delta_p(G) \leq \eta n^2$$

by Cauchy–Schwarz, since $|\mathbf{1}_S^T (A - pJ) \mathbf{1}_T| \leq \eta n \sqrt{|S||T|}$.

Nonexample. A small error in the total number of edges does not control the cut norm: the two-clique example has the correct global density $1/2 + o(1)$ but a cut of quadratic discrepancy.

Lemma 2.4 (Layer-cake extension of the cut norm). *Let M be a real $n \times n$ matrix. If $0 \leq f_i, g_j \leq 1$, then*

$$\left| \sum_{i,j} M_{ij} f_i g_j \right| \leq \|M\|_{\square}.$$

If instead $|f_i|, |g_j| \leq 1$, then

$$\left| \sum_{i,j} M_{ij} f_i g_j \right| \leq 4\|M\|_{\square}.$$

Proof. For $0 \leq f_i \leq 1$ we have the finite layer-cake identity

$$f_i = \int_0^1 \mathbf{1}_{\{f_i \geq s\}} ds,$$

and similarly for g_j . Therefore

$$\sum_{i,j} M_{ij} f_i g_j = \int_0^1 \int_0^1 \sum_{i:f_i \geq s} \sum_{j:g_j \geq t} M_{ij} ds dt.$$

Every inner sum is a cut sum and has absolute value at most $\|M\|_{\square}$; integrating proves the first estimate.

For signed functions write $f = f_+ - f_-$ and $g = g_+ - g_-$, where all four parts take values in $[0, 1]$. Expanding gives four terms, each bounded by the first part, and the triangle inequality gives the factor 4. $\square$

Worked check. Taking $f = \mathbf{1}_S$ and $g = \mathbf{1}_T$ recovers the definition of the cut norm. The lemma says that allowing fractional vertex weights does not increase the nonnegative test strength at all.

Lemma 2.5 (Dense cut discrepancy forces small operator norm). *Let B_n be real symmetric $n \times n$ matrices with $|(B_n)_{ij}| \leq 1$. If*

$$\|B_n\|_{\square} = o(n^2),$$

then

$$\|B_n\|_{2 \rightarrow 2} = o(n).$$

Proof. Suppose not. Then for some $\varepsilon > 0$ and an infinite subsequence, $B = B_n$ has an eigenvalue λ with $|\lambda| \geq \varepsilon n$. Choose a real unit eigenvector x . For every coordinate,

$$|\lambda| |x_i| = \left| \sum_j B_{ij} x_j \right| \leq \sum_j |x_j| \leq \sqrt{n} \|x\|_2 = \sqrt{n},$$

so

$$|x_i| \leq \frac{1}{\varepsilon \sqrt{n}}.$$

Put $a_i = \sqrt{n} x_i$. Then $|a_i| \leq 1/\varepsilon$, and

$$\lambda = x^T B x = \frac{1}{n} a^T B a.$$

Apply Lemma 2.4 to εa , whose coordinates lie in $[-1, 1]$. It gives

$$|a^T B a| \leq \frac{4}{\varepsilon^2} \|B\|_{\square} = o(n^2).$$

Hence $|\lambda| = o(n)$, contradicting $|\lambda| \geq \varepsilon n$. Therefore the operator norm is $o(n)$. $\square$

Theorem 2.6 (C_4 count and centered spectrum). *Fix $p \in (0, 1)$, and let $G = G_n$ be an n -vertex graph sequence satisfying*

$$e(G) = p \binom{n}{2} + o(n^2).$$

Then the following are equivalent:

- (i) $\text{hom}(C_4, G) = p^4 n^4 + o(n^4)$;
- (ii) $\|A - pJ\|_{2 \rightarrow 2} = o(n)$.

Proof. Let the adjacency eigenvalues be $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, and put $u = n^{-1/2}\mathbf{1}$.

Assume first (i). The edge-density hypothesis gives

$$u^T Au = \frac{2e(G)}{n} = pn + o(n).$$

By the Rayleigh principle, $\lambda_1 \geq pn + o(n)$. On the other hand, Definition 2.1 gives

$$\sum_{i=1}^n \lambda_i^4 = \text{tr}(A^4) = p^4 n^4 + o(n^4).$$

Hence $\lambda_1 \leq pn + o(n)$, so

$$\lambda_1 = pn + o(n), \quad \sum_{i=2}^n \lambda_i^4 = o(n^4).$$

In particular

$$\mu := \max_{i \geq 2} |\lambda_i| = o(n).$$

Choose a unit eigenvector v_1 for λ_1 with $\alpha = \langle u, v_1 \rangle \geq 0$, and write $u = \alpha v_1 + w$, $w \perp v_1$. Then

$$u^T Au = \alpha^2 \lambda_1 + w^T Aw, \quad |w^T Aw| \leq \mu \|w\|_2^2.$$

After division by n , the left side tends to p , the coefficient λ_1/n tends to p , and $\mu/n \rightarrow 0$. Since $p > 0$, it follows that $\alpha^2 \rightarrow 1$. Thus $\|v_1 - u\|_2 \rightarrow 0$.

The spectral decomposition yields

$$\|A - \lambda_1 v_1 v_1^T\|_{2 \rightarrow 2} = \mu = o(n).$$

Moreover

$$\|v_1 v_1^T - uu^T\|_{2 \rightarrow 2} \leq 2\|v_1 - u\|_2 = o(1).$$

Since $pJ = pn uu^T$,

$$\|A - pJ\|_{2 \rightarrow 2} \leq \mu + |\lambda_1 - pn| + pn\|v_1 v_1^T - uu^T\|_{2 \rightarrow 2} = o(n).$$

This proves (ii).

Conversely assume (ii), and write $B = A - pJ$ with $\|B\| = o(n)$. Since $u^T Ju = n$,

$$\frac{2e(G)}{n} = u^T Au = pn + u^T Bu = pn + o(n),$$

which is consistent with the stated edge density. Also

$$\lambda_1(A) \geq u^T Au = pn + o(n)$$

and

$$\lambda_1(A) \leq \|pJ\| + \|B\| = pn + o(n),$$

so $\lambda_1 = pn + o(n)$.

For any unit vector $x \perp u$, $x^T Ax = x^T Bx$, hence $|x^T Ax| = o(n)$. The min–max principle therefore gives $\lambda_2 \leq o(n)$, while pJ is positive semidefinite and $A = pJ + B$ gives $\lambda_n(A) \geq -\|B\| = -o(n)$, so $\lambda_n \geq -o(n)$. Consequently

$$\max_{i \geq 2} |\lambda_i| = o(n).$$

Finally,

$$\sum_i \lambda_i^2 = \text{tr}(A^2) = 2e(G) = O(n^2).$$

Therefore

$$\sum_{i=2}^n \lambda_i^4 \leq \left(\max_{i \geq 2} \lambda_i^2 \right) \sum_{i=2}^n \lambda_i^2 = o(n^4).$$

Together with $\lambda_1^4 = p^4 n^4 + o(n^4)$ this gives

$$\text{hom}(C_4, G) = \text{tr}(A^4) = p^4 n^4 + o(n^4),$$

proving (i). □

Worked check. The two-clique graph at density $1/2$ fails the theorem's C_4 condition: its block structure creates many more four-step closed walks than a $G(n, 1/2)$ graph. Spectrally the same defect appears as a second eigenvalue of order n .

Theorem 2.7 (Chung–Graham–Wilson quasirandom equivalences). *Fix $p \in (0, 1)$ and let G_n be graphs on $n \rightarrow \infty$ vertices. The following properties are equivalent:*

(Q1) *uniformly for every $S \subseteq V(G_n)$,*

$$e(G_n[S]) = p \binom{|S|}{2} + o(n^2);$$

(Q2)

$$\|A_n - pJ\|_{2 \rightarrow 2} = o(n);$$

(Q3)

$$e(G_n) = p \binom{n}{2} + o(n^2) \quad \text{and} \quad \text{hom}(C_4, G_n) = p^4 n^4 + o(n^4);$$

(Q4) *for every fixed finite simple graph H ,*

$$\text{hom}(H, G_n) = p^{e(H)} n^{v(H)} + o(n^{v(H)}).$$

The $o(n^2)$ term in (Q1) is uniform in S ; H is fixed before $n \rightarrow \infty$ in (Q4).

Proof. We prove a cycle of implications and keep each interface explicit.

(Q2) $\Rightarrow$ (Q1). Let $B = A - pJ$. For every S ,

$$2e(G[S]) - p|S|^2 = \mathbf{1}_S^T B \mathbf{1}_S.$$

Hence

$$|2e(G[S]) - p|S|^2| \leq \|B\| |S| = o(n^2)$$

uniformly in S . Replacing $p|S|^2/2$ by $p \binom{|S|}{2}$ changes the expression by only $p|S|/2 = O(n)$, proving (Q1).

(Q1) $\Rightarrow$ (Q2). Again put $B = A - pJ$ and define $q(S) = \mathbf{1}_S^T B \mathbf{1}_S$. By (Q1),

$$q(S) = 2e(G[S]) - p|S|^2 = o(n^2)$$

uniformly in S . If $S \cap T = \emptyset$, symmetry gives

$$2\mathbf{1}_S^T B \mathbf{1}_T = q(S \cup T) - q(S) - q(T) = o(n^2).$$

For arbitrary S, T , partition them into $S \setminus T$, $S \cap T$, and $T \setminus S$. The bilinear cut sum $\mathbf{1}_S^T B \mathbf{1}_T$ is a sum of three disjoint-pair terms of the form just controlled plus $q(S \cap T)$, and is therefore $o(n^2)$ uniformly. Thus $\|B\|_{\square} = o(n^2)$. Lemma 2.5 gives $\|B\|_{2 \rightarrow 2} = o(n)$, proving (Q2).

(Q2) $\iff$ (Q3). Property (Q2) implies the edge-density part of (Q3) by taking the quadratic form on $n^{-1/2}\mathbf{1}$, and Theorem 2.6 gives the C_4 count. Conversely, the two hypotheses in (Q3) are exactly the hypotheses of that theorem and imply (Q2).

(Q2) $\Rightarrow$ (Q4). Property (Q2) implies

$$\Delta_p(G) = \|A - pJ\|_{\square} \leq n\|A - pJ\|_{2 \rightarrow 2} = o(n^2),$$

because $\sqrt{|S||T|} \leq n$. We now give the counting argument rather than hide it behind a later theorem.

Let H have vertices $1, \dots, r$ and edges $e_1, \dots, e_m$. For a map $\phi : V(H) \rightarrow V(G)$, write $A_e(\phi) = A_{\phi(a), \phi(b)}$ when $e = ab$. The telescoping identity

$$\prod_{j=1}^m A_{e_j}(\phi) - p^m = \sum_{k=1}^m \left(\prod_{j < k} A_{e_j}(\phi) \right) (A_{e_k}(\phi) - p) \left(\prod_{j > k} p \right)$$

is exact. Sum it over all n^r maps ϕ . For a fixed k with $e_k = ab$, fix the images of all vertices other than a, b . The remaining factors split as

$$c f(x) g(y) (A_{xy} - p),$$

where $x = \phi(a)$, $y = \phi(b)$, $c \in [0, 1]$, and $f, g : V(G) \rightarrow [0, 1]$. Lemma 2.4 bounds the sum over x, y by $\Delta_p(G)$. There are n^{r-2} choices for the other images, so the k th telescoping contribution is at most $n^{r-2} \Delta_p(G) = o(n^r)$. Summing over the fixed number m of edges,

$$\text{hom}(H, G) = p^m n^r + o(n^r).$$

This is (Q4).

(Q4) $\Rightarrow$ (Q3). Take $H = K_2$. Since $\text{hom}(K_2, G) = 2e(G)$, (Q4) gives $e(G) = pn^2/2 + o(n^2) = p\binom{n}{2} + o(n^2)$. Taking $H = C_4$ gives the required four-cycle count. Thus (Q3) follows, completing the equivalence. $\square$

Worked comparison. For a p -quasirandom sequence, testing every induced subset, computing one operator norm, counting C_4 homomorphisms together with edges, or checking all fixed subgraph densities are asymptotically equivalent tasks. Their algorithmic costs can differ drastically; the theorem concerns mathematical information, not computational efficiency.

Example 2.8 (A small Paley-type diagnostic). On $\mathbb{F}_5$, join x and y when $x - y$ is a nonzero quadratic residue. Since the nonzero squares are $\{1, 4\}$, the resulting graph is the cycle C_5 . It is 2-regular with adjacency spectrum

$$2, \quad \frac{\sqrt{5}-1}{2}, \quad \frac{\sqrt{5}-1}{2}, \quad -\frac{\sqrt{5}+1}{2}, \quad -\frac{\sqrt{5}+1}{2}.$$

Moreover $\text{hom}(C_4, C_5) = \text{tr}(A^4) = 30$. This is a finite consistency check, not evidence that five vertices already exhibit asymptotic quasirandom statistics: the Paley family becomes informative only as the field size grows, and finite normalizations such as $d/(n-1)$ versus $2e/n^2$ must be kept distinct.

Chapter 3

Spectral Discrepancy and Expander Mixing

Dense quasirandomness uses a centered adjacency matrix because a general dense graph need not be regular. For a regular graph the all-ones vector is already an exact eigenvector, and the basic mixing estimate becomes especially clean.

Theorem 3.1 (Spectral discrepancy implication). *Let G be a d -regular graph on n vertices and suppose every adjacency eigenvalue on $\mathbf{1}^\perp$ has absolute value at most λ . Then for all $S, T \subseteq V(G)$,*

$$\left| e_G(S, T) - \frac{d}{n} |S| |T| \right| \leq \lambda \sqrt{|S| \left(1 - \frac{|S|}{n}\right) |T| \left(1 - \frac{|T|}{n}\right)} \leq \lambda \sqrt{|S| |T|}.$$

The ordered edge count is used, so an edge contained in $S \cap T$ contributes two ordered pairs when appropriate.

Proof. Let $u = n^{-1/2} \mathbf{1}$. Because G is d -regular, $Au = du$. Decompose

$$\mathbf{1}_S = \frac{|S|}{\sqrt{n}} u + x, \quad \mathbf{1}_T = \frac{|T|}{\sqrt{n}} u + y,$$

with $x, y \perp u$. The cross terms vanish because $Au = du$ and $x, y \perp u$, so

$$e_G(S, T) = \mathbf{1}_S^T A \mathbf{1}_T = \frac{d}{n} |S| |T| + x^T A y.$$

The restriction of A to $u^\perp$ has operator norm at most λ ; hence

$$|x^T A y| \leq \lambda \|x\|_2 \|y\|_2.$$

Direct computation gives

$$\|x\|_2^2 = |S| - \frac{|S|^2}{n} = |S| \left(1 - \frac{|S|}{n}\right),$$

and analogously for y . Substitution proves the first inequality, and the second follows by dropping the two factors bounded by 1. $\square$

Worked check. Taking $S = T$ an independent set gives $e_G(S, S) = 0$, so the weaker form yields

$$\frac{d}{n}|S|^2 \leq \lambda|S|,$$

and hence $|S| \leq \lambda n/d$ when $d > 0$. Thus spectral mixing immediately limits large independent sets.

Remark 3.2 (The converse is separate). A converse to expander mixing is subtler: discrepancy hypotheses can control the nontrivial spectrum, but quantitative converses incur additional losses under common formulations. The series keeps that theorem separate rather than silently treating the forward Cauchy–Schwarz argument as an equivalence.

Definition 3.3 (Centered adjacency operator and nonprincipal spectrum). For an n -vertex graph with adjacency matrix A and edge-density parameter p , the *centered adjacency operator* is $B = A - pJ$, where J is the all-ones matrix. For a d -regular graph, the *nonprincipal spectrum* is the spectrum of A restricted to $\mathbf{1}^\perp$; its spectral radius is

$$\lambda(G) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A|_{\mathbf{1}^\perp}\}.$$

The centered operator and nonprincipal spectrum coincide on $\mathbf{1}^\perp$ when $p = d/n$; no claim is made that $A - pJ$ annihilates $\mathbf{1}$ for a nonregular graph unless p is chosen accordingly.

Theorem 3.4 (Expander mixing lemma). *Let G be a d -regular graph on n vertices and let $\lambda = \lambda(G)$ be the nonprincipal spectral radius of Definition 3.3. For all $S, T \subseteq V(G)$,*

$$\left| e_G(S, T) - \frac{d}{n}|S||T| \right| \leq \lambda \sqrt{|S||T|}.$$

Equivalently, after dividing by n^2 , the ordered edge-density discrepancy is at most

$$\frac{\lambda}{n} \sqrt{\frac{|S|}{n} \frac{|T|}{n}}.$$

Quantitative converses from discrepancy to spectrum are separate theorems and may incur losses.

Proof. This is the second inequality of Theorem 3.1. Its proof decomposes the two indicator vectors into their $\mathbf{1}$ component and orthogonal component and applies the operator norm λ on $\mathbf{1}^\perp$. $\square$

Chapter 4

Fixed-Subgraph Counting from Cut Discrepancy

The Chung–Graham–Wilson theorem used a telescoping argument once. We isolate that mechanism because it becomes a prototype for graphon counting lemmas, regularity methods, and later transference arguments.

Theorem 4.1 (Fixed-subgraph counting in quasirandom graphs). *Fix $p \in [0, 1]$ and a finite simple graph H . If an n -vertex graph $G = G_n$ satisfies*

$$\Delta_p(G) = \|A - pJ\|_{\square} = o(n^2),$$

then

$$\text{hom}(H, G) = p^{e(H)} n^{v(H)} + o(n^{v(H)}).$$

The implicit o -term may depend on the fixed graph H but not on the vertex map being counted.

Proof. Write $r = v(H)$ and $m = e(H)$, and enumerate the edges as $e_1, \dots, e_m$. If $m = 0$, every map $V(H) \rightarrow V(G)$ is a homomorphism and the formula is the exact identity $n^r = n^r$, so assume $m \geq 1$.

For a map $\phi : V(H) \rightarrow V(G)$ and an edge $e = ab$, set

$$A_e(\phi) = A_{\phi(a), \phi(b)}.$$

Then

$$\text{hom}(H, G) = \sum_{\phi} \prod_{j=1}^m A_{e_j}(\phi).$$

The algebraic telescoping identity

$$\prod_{j=1}^m z_j - p^m = \sum_{k=1}^m \left(\prod_{j < k} z_j \right) (z_k - p) \left(\prod_{j > k} p \right)$$

holds for arbitrary scalars z_j . Apply it with $z_j = A_{e_j}(\phi)$ and sum over all maps ϕ . It is enough to show that each of the m resulting sums is $o(n^r)$.

Fix k and write $e_k = ab$. First fix the images of all $r - 2$ vertices other than a, b . Once those images are fixed, every factor corresponding to an edge not equal to ab either becomes a constant in $[0, 1]$, depends only on $x = \phi(a)$, or depends only on $y = \phi(b)$. Because H is simple, no second edge depends simultaneously on both x and y . Thus the remaining summand has the form

$$c f(x) g(y) (A_{xy} - p)$$

with $0 \leq c \leq 1$ and $0 \leq f, g \leq 1$. By Lemma 2.4, the absolute value of the sum over x, y is at most $\Delta_p(G)$.

There are at most n^{r-2} assignments of the other vertices, so the absolute value of the entire k th telescoping contribution is at most

$$n^{r-2} \Delta_p(G) = o(n^r).$$

Summing over the fixed number m of edges gives

$$\text{hom}(H, G) - p^m n^r = o(n^r),$$

which is the desired formula. □

Worked check. For $H = K_3$, the proof replaces the three edge factors one at a time. When, for example, the edge 12 is replaced by $A - p$, fixing the image of vertex 3 leaves two bounded functions of the images of vertices 1 and 2. The cut norm controls their correlation with the centered edge 12.

Chapter 5

Regularity: Weak and Strong

Quasirandomness describes graphs that already look uniform. Regularity is a structure theorem for an arbitrary dense graph: after a bounded partition, most pairs of classes behave uniformly at the scale relevant to counting fixed subgraphs. We begin with a particularly transparent matrix version.

Theorem 5.1 (Frieze–Kannan weak regularity, cut-matrix form). *Let A be the adjacency matrix of an n -vertex graph and let $0 < \varepsilon \leq 1$. There are sets $S_r, T_r \subseteq [n]$ and real numbers α_r , for $1 \leq r \leq R$ with $R \leq \lceil \varepsilon^{-2} \rceil$, such that*

$$A = \sum_{r=1}^R \alpha_r \mathbf{1}_{S_r} \mathbf{1}_{T_r}^\top + E \quad \text{and} \quad \|E\|_\square \leq \varepsilon n^2.$$

The statement is existential: finding a cut attaining or approximating the cut norm is a separate algorithmic problem.

Proof. Set $E_0 = A$. If $\|E_t\|_\square \leq \varepsilon n^2$, stop. Otherwise choose nonempty S, T with

$$\left| \sum_{i \in S, j \in T} (E_t)_{ij} \right| > \varepsilon n^2.$$

Put

$$\alpha = \frac{\sum_{S \times T} (E_t)_{ij}}{|S||T|}, \quad C = \alpha \mathbf{1}_S \mathbf{1}_T^\top, \quad E_{t+1} = E_t - C.$$

This is the orthogonal projection of E_t onto the one-dimensional span of $\mathbf{1}_S \mathbf{1}_T^\top$ for the Frobenius inner product. Hence

$$\|E_{t+1}\|_F^2 = \|E_t\|_F^2 - \frac{(\sum_{S \times T} (E_t)_{ij})^2}{|S||T|} < \|E_t\|_F^2 - \varepsilon^2 n^2,$$

because $|S||T| \leq n^2$.

Initially $\|A\|_F^2 \leq n^2$. The nonnegative quantity $\|E_t\|_F^2$ can therefore decrease by more than $\varepsilon^2 n^2$ fewer than ε^{-2} times. The procedure stops after at most $\lceil \varepsilon^{-2} \rceil$ steps, and telescoping the definitions of E_t gives the required decomposition. $\square$

Worked check. If A itself is the indicator of a complete bipartite rectangle $S \times T$, one cut matrix recovers it exactly. At the other extreme, the theorem makes no assertion that the sets S_r, T_r form a partition: weak regularity is a low-complexity cut decomposition, not yet Szemerédi regularity.

Definition 5.2 (Regular pair and equitable regular partition). For nonempty vertex sets X, Y , write

$$d(X, Y) = \frac{e(X, Y)}{|X||Y|}.$$

When X and Y are disjoint, the pair (X, Y) is ε -regular if every $X' \subseteq X$, $Y' \subseteq Y$ with $|X'| \geq \varepsilon|X|$ and $|Y'| \geq \varepsilon|Y|$ satisfies

$$|d(X', Y') - d(X, Y)| \leq \varepsilon.$$

An *equitable ε -regular partition* is a disjoint union

$$V = V_0 \dot{\cup} V_1 \dot{\cup} \cdots \dot{\cup} V_k$$

for which $|V_1| = \cdots = |V_k|$, $|V_0| \leq \varepsilon|V|$, and at most εk^2 unordered pairs $\{i, j\}$, $1 \leq i < j \leq k$, fail to be ε -regular. The set V_0 is the exceptional class.

Positive example. A complete or empty bipartite pair is ε -regular for every $\varepsilon > 0$.

Nonexample. Split $X = X_1 \dot{\cup} X_2$ and $Y = Y_1 \dot{\cup} Y_2$ into equal halves and put all edges between X_1, Y_1 and X_2, Y_2 , with no cross edges. The global density is $1/2$, while $d(X_1, Y_1) = 1$; the pair is not ε -regular for any $\varepsilon < 1/2$.

For a partition (not necessarily equitable) $\mathcal{P} = (V_0; V_1, \dots, V_k)$ of an n -vertex graph define its *energy*

$$q(\mathcal{P}) = \frac{1}{n^2} \sum_{i,j=1}^k |V_i||V_j|d(V_i, V_j)^2.$$

We omit the exceptional class from the energy. Since every density lies in $[0, 1]$, one always has $0 \leq q(\mathcal{P}) \leq 1$.

Lemma 5.3 (Equitable refinement energy increment). *Fix $0 < \varepsilon \leq 1/4$. There is an integer $k_*(\varepsilon)$ with the following property. Suppose $\mathcal{P} = (V_0; V_1, \dots, V_k)$ is equitable, $k \geq k_*(\varepsilon)$, $|V_0| \leq \varepsilon n$, and more than εk^2 cluster pairs are ε -irregular. If the common cluster size m is at least $2 \cdot 4^k$, then there is an equitable refinement $\mathcal{P}'$ such that*

$$q(\mathcal{P}') \geq q(\mathcal{P}) + \frac{\varepsilon^5}{2},$$

$$k < k' \leq 4k4^k, \quad |V'_0| \leq |V_0| + \frac{n}{2^k}.$$

Proof. For every irregular unordered pair $\{i, j\}$ choose witnesses $X_{ij} \subseteq V_i$ and $Y_{ij} \subseteq V_j$ with sizes at least εm and

$$|d(X_{ij}, Y_{ij}) - d(V_i, V_j)| > \varepsilon.$$

Within each V_i , refine simultaneously by all witness subsets involving i . At most $k - 1$ such subsets occur, so V_i is split into at most 2^k atoms. Call the resulting, not necessarily equitable, refinement $\mathcal{Q}$.

We first record its energy gain. For a fixed ordered pair (V_i, V_j) , let C_a and D_b be the atoms into which the two classes are split. The weighted variance identity gives

$$\sum_{a,b} \frac{|C_a||D_b|}{n^2} d(C_a, D_b)^2 - \frac{m^2}{n^2} d(V_i, V_j)^2 = \sum_{a,b} \frac{|C_a||D_b|}{n^2} (d(C_a, D_b) - d(V_i, V_j))^2.$$

Indeed, on expanding the square, the cross term vanishes because the weighted average of the refined densities is the original density. For an irregular pair, the atom refinement separates the witness sets, so the right-hand side is at least

$$\frac{|X_{ij}||Y_{ij}|}{n^2} \varepsilon^2 \geq \varepsilon^4 \frac{m^2}{n^2}.$$

More explicitly, inside $X_{ij} \times Y_{ij}$ the atom densities have weighted average $d(X_{ij}, Y_{ij})$, so Jensen's inequality shows that the sum of their squared deviations from $d(V_i, V_j)$ is at least the displayed witness contribution. The reverse ordered pair gives the same gain. Since there are more than εk^2 irregular unordered pairs,

$$q(\mathcal{Q}) - q(\mathcal{P}) > 2\varepsilon^5 \frac{k^2 m^2}{n^2}.$$

Because $km = n - |V_0| \geq (1 - \varepsilon)n \geq 3n/4$, the last quantity exceeds ε^5 .

It remains to restore equitability. Put

$$t = \left\lfloor \frac{m}{4^k} \right\rfloor.$$

The hypothesis on m gives $t \geq m/(2 \cdot 4^k)$. Split every atom of $\mathcal{Q}$ into sets of size exactly t , moving the remainder of each atom into the exceptional class. There are at most 2^k atoms inside each old V_i , so fewer than

$$2^k t \leq \frac{m}{2^k}$$

vertices are discarded from V_i . Thus fewer than $n/2^k$ vertices are added to V_0 .

Splitting atoms cannot decrease energy. Removing a set of R vertices from the nonexceptional classes can delete from the normalized energy only terms whose ordered pair touches one of those vertices; since all squared densities are at most 1, the loss is at most $2R/n$. Hence

$$q(\mathcal{P}') \geq q(\mathcal{Q}) - \frac{2}{2^k}.$$

Choose $k_*(\varepsilon)$ so that $2^{1-k_*} \leq \varepsilon^5/2$. Then the net gain is at least $\varepsilon^5/2$.

Finally, the new number of equal clusters is at most n/t . Since $|V_0| \leq n/4$ and $m = (n - |V_0|)/k$, we have $n/m \leq 2k$; using $t \geq m/(2 \cdot 4^k)$ yields

$$k' \leq \frac{n}{t} \leq 4k4^k.$$

For the lower bound, if R is the number of newly exceptional vertices, then $R < km/2^k$ and $t \leq m/4^k$, so

$$k' = \frac{km - R}{t} > k(1 - 2^{-k})4^k > k.$$

This completes the refinement estimate. □

Theorem 5.4 (Szemerédi regularity lemma). *For every $\varepsilon > 0$ and integer $m_0 \geq 1$, there are integers $M = M(\varepsilon, m_0)$ and $n_0 = n_0(\varepsilon, m_0)$ such that every graph on $n \geq n_0$ vertices admits an equitable ε -regular partition*

$$V = V_0 \dot{\cup} V_1 \dot{\cup} \cdots \dot{\cup} V_k$$

with

$$m_0 \leq k \leq M.$$

Proof. There is nothing to prove when $\varepsilon \geq 1/4$, because a result for $\min(\varepsilon, 1/4)$ is stronger. We therefore assume $0 < \varepsilon < 1/4$.

Let

$$L = \lceil 2\varepsilon^{-5} \rceil$$

and choose

$$k_0 \geq \max\{m_0, k_*(\varepsilon)\}$$

so large that

$$L2^{-k_0} \leq \frac{\varepsilon}{2}.$$

Define recursively $F(k) = 4k4^k$ and

$$M = F^{\circ L}(k_0).$$

Finally choose

$$n_0 \geq \max\left\{\frac{2k_0}{\varepsilon}, 4M4^M\right\}.$$

The first bound makes the initial exceptional class smaller than $\varepsilon n/2$. During the iteration $|V_0| \leq \varepsilon n < n/4$ and $k \leq M$, so every common cluster size satisfies $m = (n - |V_0|)/k \geq 3n/(4M) > 2 \cdot 4^M \geq 2 \cdot 4^k$, as required by the refinement lemma.

Start with any equitable partition into k_0 equal nonexceptional classes, putting fewer than k_0 leftover vertices into V_0 . The chosen lower bound on n_0 gives $|V_0| < k_0 \leq \varepsilon n/2$.

If the partition is ε -regular, stop. Otherwise apply Lemma 5.3. Each application raises the energy by at least $\varepsilon^5/2$, while energy is at most 1. Thus at most L applications are possible: $L + 1$ applications would force energy strictly greater than 1. The number of classes never exceeds M . Moreover, each refinement adds at most $n/2^k \leq n/2^{k_0}$ exceptional vertices, so throughout the procedure

$$|V_0| \leq \frac{\varepsilon n}{2} + L \frac{n}{2^{k_0}} \leq \varepsilon n.$$

Consequently the process must stop at an equitable ε -regular partition with $m_0 \leq k \leq M$. $\square$

Why the bound is enormous. The recurrence $k \mapsto 4k4^k$ is iterated on the order of ε^{-5} times. Nothing in this proof is intended to optimize the classical tower-type dependence; the point is that the number of parts is bounded solely in terms of ε and m_0 , not in terms of n .

Lemma 5.5 (Regular pairs have small rectangular cut discrepancy). *If (X, Y) is ε -regular of density d , then for all $S \subseteq X$ and $T \subseteq Y$,*

$$|e(S, T) - d|S||T|| \leq \varepsilon|X||Y|.$$

Proof. If $|S| \geq \varepsilon|X|$ and $|T| \geq \varepsilon|Y|$, regularity gives

$$|e(S, T) - d|S||T|| = |d(S, T) - d||S||T| \leq \varepsilon|X||Y|.$$

If, say, $|S| < \varepsilon|X|$, then each entry of the centered bipartite adjacency matrix lies in $[-1, 1]$, so

$$|e(S, T) - d|S||T|| \leq |S||T| < \varepsilon|X||Y|.$$

The case of small T is identical. $\square$

Theorem 5.6 (Counting lemma for regular graph systems). *Let H be a fixed simple graph on labelled vertices $[h]$, and let $V_1, \dots, V_h$ be pairwise disjoint vertex sets in a graph G . Suppose that for every $ij \in E(H)$ the pair (V_i, V_j) is ε -regular, of density d_{ij} . If N_H denotes the number of part-respecting labelled embeddings ϕ with $\phi(i) \in V_i$, then*

$$\left| N_H - \left(\prod_{ij \in E(H)} d_{ij} \right) \left(\prod_{i=1}^h |V_i| \right) \right| \leq e(H) \varepsilon \prod_{i=1}^h |V_i|.$$

In particular, if every required density is at least $d > 0$ and $e(H)\varepsilon < d^{e(H)}$, then at least

$$(d^{e(H)} - e(H)\varepsilon) \prod_i |V_i|$$

part-respecting embeddings exist.

Proof. For each required pair ij , let A_{ij} be its bipartite adjacency matrix and put

$$R_{ij} = A_{ij} - d_{ij}J.$$

Lemma 5.5 says that the rectangular cut norm of R_{ij} is at most $\varepsilon|V_i||V_j|$.

Order the $m = e(H)$ edges of H . In the product defining N_H , replace the adjacency factor for the edges one at a time by the constant factor d_{ij} . The difference between two consecutive products contains exactly one residual factor R_{ab} .

Fix the images of all vertices other than a, b . Every remaining factor incident with a depends only on the choice $x \in V_a$, and their product is a function $u(x) \in [0, 1]$; similarly the factors incident with b give $v(y) \in [0, 1]$. Factors involving neither vertex contribute a scalar in $[0, 1]$. Thus the sum over x, y is bounded in absolute value by

$$\left| \sum_{x \in V_a, y \in V_b} R_{ab}(x, y) u(x) v(y) \right|.$$

Pad R_{ab} by zeros outside $V_a \times V_b$. The layer-cake argument of Lemma 2.4 then applies verbatim to the resulting square matrix and bounds the last expression by $\varepsilon|V_a||V_b|$. Summing over the already fixed choices of the other $h - 2$ vertices shows that this telescoping step contributes at most

$$\varepsilon \prod_{i=1}^h |V_i|.$$

There are $m = e(H)$ steps. After the last replacement the product is constant and its total sum is exactly

$$\left(\prod_{ij \in E(H)} d_{ij} \right) \prod_i |V_i|.$$

The triangle inequality gives the stated error. The final assertion follows by replacing every d_{ij} by the lower bound d . $\square$

Worked check: triangles. For three disjoint classes whose three pair densities are all at least d , ε -regularity gives at least

$$(d^3 - 3\varepsilon)|V_1||V_2||V_3|$$

labelled triangles, provided the expression is positive. This is the exact form needed in the removal argument.

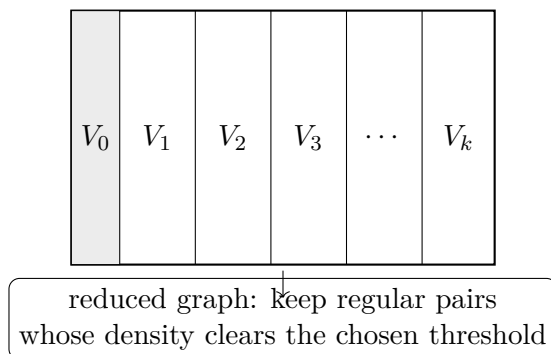


Figure 5.1: Regularity partition and reduced-graph interface. The exceptional class V_0 is small; the remaining clusters are equal-sized. Only most cluster pairs are regular, and a reduced graph requires a separately stated density threshold. Regularity alone does not assert that every reduced pair is dense.

Chapter 6

Triangle Removal

The regularity lemma becomes powerful when paired with the counting lemma: if a graph has to be changed substantially before all triangles disappear, then some regular triple of dense cluster pairs survives, forcing many triangles.

Definition 6.1 (Removal distance). Fix a finite graph H and $\varepsilon > 0$. An n -vertex graph G is ε -far from H -free in the non-induced deletion model if more than εn^2 edge deletions are required to destroy every injective copy of H . A *labelled embedding* is an injective vertex map preserving edges. In particular, the number of labelled embeddings of K_3 is six times the number of unordered triangles.

Positive example. A triangle-free graph is 0-far from triangle-free.

Nonexample. A single triangle in an otherwise empty n -vertex graph is not ε -far from triangle-free for any fixed $\varepsilon > 0$ once n is large: deleting one edge suffices.

Theorem 6.2 (Triangle removal lemma). *For every $\varepsilon > 0$ there exists $\delta > 0$ such that every n -vertex graph with at most δn^3 labelled triangle embeddings can be made triangle-free by deleting at most εn^2 edges.*

Proof. It is enough to treat $0 < \varepsilon < 1$. Put

$$d = \frac{\varepsilon}{8}$$

and choose a regularity parameter $\rho > 0$ so small that

$$\rho \leq \frac{\varepsilon}{8} \quad \text{and} \quad 3\rho \leq \frac{d^3}{2}.$$

Apply Theorem 5.4 with parameter ρ and with

$$m_0 \geq \frac{8}{\varepsilon}.$$

Let M and n_0 be the resulting bounds.

Suppose first that $n \geq n_0$, and take a regular partition

$$V = V_0 \dot{\cup} V_1 \dot{\cup} \cdots \dot{\cup} V_k, \quad m_0 \leq k \leq M,$$

with common nonexceptional size m . Form a subgraph G' by deleting four kinds of edges:

1. every edge incident with V_0 ;
2. every edge lying inside some V_i ;
3. every edge between an irregular pair (V_i, V_j) ;
4. every edge between a regular pair whose density is less than d .

The first operation deletes at most ρn^2 edges. The second deletes at most

$$k \binom{m}{2} \leq \frac{n^2}{2k} \leq \frac{\varepsilon n^2}{16}.$$

There are at most ρk^2 irregular unordered pairs, so the third deletes at most $\rho k^2 m^2 \leq \rho n^2$ edges. Finally the fourth deletes at most

$$d \binom{k}{2} m^2 \leq \frac{dn^2}{2} = \frac{\varepsilon n^2}{16}.$$

Thus the total number deleted is at most

$$2\rho n^2 + \frac{\varepsilon n^2}{8} \leq \frac{3\varepsilon n^2}{8} < \varepsilon n^2.$$

Assume G is ε -far from triangle-free. Then G' still contains a triangle. Its three vertices lie in three distinct nonexceptional classes V_i, V_j, V_ℓ , and each of the three corresponding pairs is ρ -regular of density at least d . The counting lemma therefore gives at least

$$(d^3 - 3\rho)m^3 \geq \frac{d^3}{2}m^3$$

part-respecting labelled triangles in G , for this fixed ordering of the three classes. Since $|V_0| \leq \rho n < n/2$ and $k \leq M$,

$$m = \frac{n - |V_0|}{k} \geq \frac{n}{2M}.$$

Consequently every ε -far graph on $n \geq n_0$ vertices has at least

$$\frac{d^3}{16M^3}n^3$$

labelled triangle embeddings.

Set

$$\delta = \min \left\{ \frac{d^3}{16M^3}, \frac{1}{2n_0^3} \right\}.$$

For $n \geq n_0$, the preceding lower bound proves the contrapositive. If $n < n_0$ and the graph has at most $\delta n^3 < 1$ labelled triangles, then it has no triangle at all, because the number of labelled embeddings is an integer. This covers every n and completes the proof. $\square$

What the proof actually uses. No numerical value of the enormous regularity bound $M(\rho, m_0)$ is needed. The proof uses only that M is finite and independent of n . This is why the removal lemma is qualitative despite being quantitatively explicit in principle.

6.1 The general graph removal lemma

The same regularity/counting mechanism removes any fixed finite pattern. A minor point that is easy to hide in a compressed proof is that two nonadjacent vertices of H may land in the same regularity class. We keep that possibility and count class-respecting homomorphisms first; only at the end do we subtract the noninjective maps.

Theorem 6.3 (Graph removal lemma). *Let H be a fixed finite simple graph with $h = v(H)$ vertices. For every $\varepsilon > 0$ there exists $\delta = \delta(H, \varepsilon) > 0$ such that every n -vertex graph G with at most*

$$\delta n^h$$

labelled embeddings of H can be made H -free by deleting at most εn^2 edges.

Proof. Write $q = e(H)$.

The edgeless case. Suppose first that $q = 0$. If $n < h$, then G is already H -free. If $n \geq h$, every injection $V(H) \hookrightarrow V(G)$ is an embedding, so the number of labelled embeddings is

$$(n)_h = n(n-1) \cdots (n-h+1).$$

Moreover

$$\frac{(n)_h}{n^h} = \prod_{j=0}^{h-1} \left(1 - \frac{j}{n}\right) \geq \prod_{j=0}^{h-1} \left(1 - \frac{j}{h}\right) = \frac{h!}{h^h}.$$

Thus choosing $\delta < h!/h^h$ makes the hypothesis impossible when $n \geq h$. This proves the theorem for edgeless H .

Henceforth $q \geq 1$. If $\varepsilon \geq 1/2$, deleting every edge of G uses at most $n^2/2 \leq \varepsilon n^2$ deletions and destroys every copy of H . We may therefore assume

$$0 < \varepsilon < \frac{1}{2}.$$

Set

$$d = \frac{\varepsilon}{4}, \quad B = \binom{h}{2}.$$

Choose $\rho > 0$ so small that

$$\rho \leq \min \left\{ \frac{\varepsilon}{16}, \frac{1}{2}, \frac{d^q}{4q} \right\},$$

and choose

$$m_0 \geq \frac{8}{\varepsilon}.$$

Apply Theorem 5.4 with regularity parameter ρ and lower bound m_0 . Let $M = M(\rho, m_0)$ and n_0 be the resulting bounds.

Suppose $n \geq n_0$ and take an equitable regular partition

$$V(G) = V_0 \dot{\cup} V_1 \dot{\cup} \cdots \dot{\cup} V_k, \quad m_0 \leq k \leq M,$$

whose nonexceptional classes all have size m . Form G' by deleting:

1. every edge incident with V_0 ;
2. every edge lying inside a nonexceptional class;

3. every edge between a ρ -irregular pair;
4. every edge between a ρ -regular pair of density less than d .

The four contributions are bounded respectively by

$$\rho n^2, \quad k \binom{m}{2} \leq \frac{n^2}{2k} \leq \frac{\varepsilon n^2}{16}, \quad \rho k^2 m^2 \leq \rho n^2, \quad d \binom{k}{2} m^2 \leq \frac{dn^2}{2} = \frac{\varepsilon n^2}{8}.$$

Hence fewer than

$$\left(2\rho + \frac{3\varepsilon}{16}\right) n^2 \leq \frac{5\varepsilon}{16} n^2 < \varepsilon n^2$$

edges are deleted.

Assume now, toward the contrapositive, that G is ε -far from H -free. Since fewer than εn^2 edges were deleted, G' still contains an injective copy ψ of H . For every vertex of H that is incident with an edge, let $c(a) \in [k]$ be the index of the class containing $\psi(a)$. Every edge $ab \in E(H)$ then satisfies $c(a) \neq c(b)$, and $(V_{c(a)}, V_{c(b)})$ is ρ -regular of density at least d . Isolated vertices of H may be assigned arbitrarily to any nonexceptional class; this defines $c(a)$ for all $a \in V(H)$.

We need a counting estimate in which distinct nonadjacent vertices of H are allowed to use the same class. Let N_c^{hom} be the number of maps

$$\phi : V(H) \longrightarrow V(G), \quad \phi(a) \in V_{c(a)},$$

that preserve every edge, without requiring ϕ to be injective. The telescoping proof of Theorem 5.6 applies verbatim to these independent vertex variables: disjointness of the vertex classes was used there only to make every counted map automatically injective. For completeness, order the q edges of H and replace their adjacency factors one at a time by the corresponding pair densities. In one replacement, after all variables except the endpoints a, b have been fixed, the remaining factors have the form

$$C u(x) v(y) (A_{c(a)c(b)}(x, y) - d_{ab}), \quad 0 \leq C, u, v \leq 1.$$

Lemma 5.5, followed by the layer-cake extension Lemma 2.4, bounds the sum over x, y by ρm^2 . There are m^{h-2} choices of the other variables. Summing the q telescoping errors gives

$$\left| N_c^{\text{hom}} - \left(\prod_{ab \in E(H)} d_{ab} \right) m^h \right| \leq q \rho m^h.$$

Since every $d_{ab} \geq d$,

$$N_c^{\text{hom}} \geq (d^q - q\rho) m^h \geq \frac{3d^q}{4} m^h.$$

Some of these maps may identify two nonadjacent vertices assigned to the same class. For a fixed unordered pair of vertices of H , there are at most m^{h-1} class-respecting maps identifying that pair. The union bound therefore shows that the number of noninjective class-respecting maps is at most

$$B m^{h-1}.$$

Consequently, whenever

$$m \geq \frac{4B}{d^q},$$

the graph G contains at least

$$\frac{d^q}{2} m^h$$

labelled embeddings of H .

Because $|V_0| \leq \rho n \leq n/2$ and $k \leq M$,

$$m = \frac{n - |V_0|}{k} \geq \frac{n}{2M}.$$

Thus, if

$$N = \max \left\{ n_0, \left\lceil \frac{8MB}{d^q} \right\rceil, h \right\},$$

then every ε -far graph on $n \geq N$ vertices has at least

$$c_H n^h, \quad c_H := \frac{d^q}{2^{h+1} M^h},$$

labelled embeddings of H .

Finally choose

$$\delta = \min \left\{ \frac{c_H}{2}, \frac{1}{2N^h} \right\}.$$

For $n \geq N$, the preceding lower bound proves the contrapositive. For $n < N$, the hypothesis gives fewer than one labelled embedding, because $\delta n^h < 1$; hence there is no copy of H at all. The theorem follows for every n . $\square$

Why homomorphisms appear inside an embedding proof. The reduced graph remembers clusters, not individual vertices. An injective copy of H in the cleaned graph can send two nonadjacent vertices to the same cluster. Counting all class-respecting maps makes the regular-pair calculation exact; the $O(m^{h-1})$ collision bound then recovers labelled embeddings. This is the labelled/injective conversion hidden in many short proofs of graph removal.

Chapter 7

Removal and One-Sided Property Testing

Graph removal has an algorithmic consequence that is qualitative but very powerful: a graph that is far from being H -free contains a positive density of labelled copies of H . Therefore a constant number of random local queries detects a violation with constant probability, independently of the ambient number of vertices.

Definition 7.1 (Dense-graph property tester). A *dense-graph property tester* receives the number n of vertices and oracle access to adjacency queries

$$(u, v) \mapsto \mathbf{1}_{\{uv \in E(G)\}}$$

for an unknown simple graph G on vertex set $[n]$. For a graph property $\mathcal{P}$ and $\varepsilon > 0$, a one-sided ε -tester must

- (i) accept every $G \in \mathcal{P}$ with probability 1;
- (ii) reject every graph that is ε -far from $\mathcal{P}$ with probability at least $2/3$;
- (iii) use at most $q(\varepsilon)$ adjacency queries, where $q(\varepsilon)$ is independent of n .

For H -freeness, distance is the non-induced edge-deletion distance from Definition 6.1.

Positive example. To test whether a graph is K_2 -free, sample random vertex pairs and query whether they are edges. If more than εn^2 edge deletions are needed, a random ordered pair is an edge with probability greater than 2ε .

Nonexample. An algorithm that examines the first 100 vertices only is not a tester for an arbitrary relabelling-invariant dense property: all violations may be concentrated outside those vertices. Random sampling, rather than a fixed location, is essential.

Theorem 7.2 (Removal implies one-sided testing). *Fix a finite simple graph H . For every $\varepsilon > 0$, H -freeness has a one-sided dense-graph tester whose query complexity depends only on H and ε , not on n .*

More precisely, if $h = v(H)$, $q = e(H)$, and $\delta = \delta(H, \varepsilon)$ is supplied by Theorem 6.3, then for $q \geq 1$ one may use

$$R = \left\lceil \frac{\log 3}{\delta} \right\rceil$$

independent trials and at most qR adjacency queries.

Proof. First suppose $q = 0$. If $n < h$, the graph is H -free, so accept. If $n \geq h$, every graph contains an injective copy of the edgeless graph H ; no edge deletion can change this, so reject. This is a zero-query one-sided tester.

Now assume $q \geq 1$. If $n < h$, accept, since no injective copy of H exists. For $n \geq h$, perform the following trial R times independently. Choose an ordered h -tuple

$$(x_1, \dots, x_h) \in [n]^h$$

uniformly with replacement. If the tuple has a repeated vertex, the trial does not reject. Otherwise query the q pairs $x_i x_j$ corresponding to edges $ij \in E(H)$. Reject immediately if every required edge is present. If no trial rejects, accept.

If G is H -free, no injective ordered tuple realizes H , so the algorithm accepts with probability 1.

Suppose instead that G is ε -far from H -free. The contrapositive of Theorem 6.3 says that G has more than

$$\delta n^h$$

labelled embeddings of H . Each labelled embedding is exactly one ordered h -tuple on which a trial rejects. Since there are n^h ordered tuples in total, one trial rejects with probability greater than δ . Therefore the probability that all R trials miss every copy is at most

$$(1 - \delta)^R \leq e^{-\delta R} \leq \frac{1}{3}.$$

Thus an ε -far graph is rejected with probability at least $2/3$. The number of queries is at most

$$qR = q \left\lceil \frac{\log 3}{\delta(H, \varepsilon)} \right\rceil,$$

which depends on H and ε but not on n . □

What is and is not quantitative here. The theorem proves constant-query testability for every fixed H , but the regularity proof can make $\delta(H, \varepsilon)$ extremely small. The much sharper question of when the query complexity is polynomial in $1/\varepsilon$ is separate; it is not silently inferred from graph regularity.

Worked check: one edge. Take $H = K_2$. Being ε -far from K_2 -free means that $e(G) > \varepsilon n^2$. A uniformly random ordered pair of vertices is a labelled K_2 embedding with probability

$$\frac{2e(G)}{n^2} > 2\varepsilon.$$

Thus $O(\varepsilon^{-1})$ trials already suffice in this special case, illustrating how a direct argument can be far sharper than the general regularity bound.

Theorem 7.3 (Induced removal and hereditary dense-property testability — source-owned). *For every finite simple graph H and every $\varepsilon > 0$ there is $\delta = \delta(H, \varepsilon) > 0$ such that every sufficiently large graph G which is ε -far, in normalized edge-edit distance, from having no induced copy of H contains at least $\delta |V(G)|^{|V(H)|}$ labelled induced copies of H . More generally, every hereditary dense graph property is testable with query complexity depending only on the distance parameter. The induced-removal theorem and the hereditary-property characterization are source-owned; no polynomial dependence of the query bound on $1/\varepsilon$ is asserted here.*

Chapter 8

Graphons, Cut Distance, and Subgraph Densities

Finite dense graphs become particularly transparent after they are embedded into a common analytic space. Graphons provide that space, while the cut norm measures precisely the rectangular discrepancy already encountered in the finite theory.

Definition 8.1 (Graphon, homomorphism density, and cut distance). A *graphon* is a symmetric measurable function

$$W : [0, 1]^2 \longrightarrow [0, 1],$$

where functions equal almost everywhere are identified. If H is a finite simple graph on labelled vertex set $[h]$, its *homomorphism density* in W is

$$t(H, W) = \int_{[0,1]^h} \prod_{ij \in E(H)} W(x_i, x_j) dx_1 \cdots dx_h.$$

For an integrable kernel $F : [0, 1]^2 \rightarrow \mathbb{R}$, define the cut norm

$$\|F\|_{\square} = \sup_{S, T \subseteq [0,1] \text{ measurable}} \left| \int_{S \times T} F(x, y) dx dy \right|.$$

If $\phi : [0, 1] \rightarrow [0, 1]$ is a measure-preserving bijection modulo null sets, put

$$W^{\phi}(x, y) = W(\phi(x), \phi(y)).$$

The *cut distance* is

$$\delta_{\square}(U, W) = \inf_{\phi} \|U - W^{\phi}\|_{\square}.$$

Graphons with cut distance zero are called *weakly isomorphic*; the metric graphon space is obtained after identifying those pairs.

The measurable layer-cake identity gives the continuous analogue of Lemma 2.4: whenever $0 \leq f, g \leq 1$ are measurable,

$$\left| \int_{[0,1]^2} F(x, y) f(x) g(y) dx dy \right| \leq \|F\|_{\square}.$$

Indeed $f(x) = \int_0^1 \mathbf{1}_{\{f(x) \geq s\}} ds$ and similarly for g ; Tonelli's theorem then expresses the weighted integral as an average of cut integrals.

Finite graphs as graphons. For a graph G on $[n]$, partition $[0, 1]$ into equal half-open intervals $I_1, \dots, I_n$ and let W_G be constant on each $I_i \times I_j$, equal to the adjacency entry A_{ij} . Then

$$t(H, W_G) = \frac{\text{hom}(H, G)}{n^{v(H)}}.$$

Thus graphon homomorphism density is literally the normalized finite homomorphism count.

Positive example. For the constant graphon $W \equiv p$,

$$t(H, W) = p^{e(H)}.$$

This is the exact density predicted by the random graph $G(n, p)$.

Nonexample. A measurable kernel that is not symmetric is not a graphon for undirected graphs, even if it takes values in $[0, 1]$; symmetry encodes the unordered nature of an edge.

Theorem 8.2 (Cut-norm counting lemma for graphons). *Let H be a finite simple graph with $m = e(H)$, and let U, W be graphons. Then*

$$|t(H, U) - t(H, W)| \leq m \|U - W\|_{\square}.$$

For $m = 0$, both homomorphism densities equal 1 and the right side is zero.

Proof. The case $m = 0$ is immediate, so assume $m \geq 1$ and enumerate the edges of H as

$$e_1, \dots, e_m.$$

For $0 \leq r \leq m$, let $P_r(x_1, \dots, x_h)$ be the product in which the factors belonging to $e_1, \dots, e_r$ are taken from U and the remaining edge factors are taken from W . Then

$$t(H, U) - t(H, W) = \sum_{r=1}^m \int_{[0,1]^h} (P_r - P_{r-1}).$$

Fix r and write $e_r = ab$. In $P_r - P_{r-1}$, the factor on the edge ab is

$$(U - W)(x_a, x_b),$$

while every other factor is a value of either U or W and hence lies in $[0, 1]$. Fix all variables except x_a, x_b . Because H is simple, no other edge depends simultaneously on both of these variables. The remaining product therefore factors as

$$c f(x_a) g(x_b) (U - W)(x_a, x_b),$$

where $0 \leq c \leq 1$ and $f, g : [0, 1] \rightarrow [0, 1]$ are measurable. By the layer-cake form of the cut norm recorded after Definition 8.1, the absolute value of the inner integral over (x_a, x_b) is at most

$$\|U - W\|_{\square}.$$

The remaining $h - 2$ variables range over a probability space of total measure one, so the entire r th telescoping term has the same bound. Summing the m terms proves

$$|t(H, U) - t(H, W)| \leq m \|U - W\|_{\square}.$$

□

Worked check: triangles. For $H = K_3$ the estimate is

$$|t(K_3, U) - t(K_3, W)| \leq 3\|U - W\|_{\square}.$$

The three terms correspond to replacing the three edge kernels one at a time. The proof is the continuum version of the telescoping argument already used for fixed-subgraph counting in a quasirandom finite graph.

Theorem 8.3 (Cut distance controls every fixed subgraph density). *For every finite simple graph H and graphons U, W ,*

$$|t(H, U) - t(H, W)| \leq e(H) \delta_{\square}(U, W).$$

Consequently, if

$$\delta_{\square}(W_n, W) \longrightarrow 0,$$

then for every fixed H ,

$$t(H, W_n) \longrightarrow t(H, W).$$

Proof. First note that homomorphism density is invariant under a measure-preserving relabeling. Indeed, if ϕ is measure preserving, then the product map $\phi^h : [0, 1]^h \rightarrow [0, 1]^h$ preserves Lebesgue measure. This is immediate on measurable rectangles and extends to all integrable functions by the usual simple-function approximation. Hence

$$t(H, W^{\phi}) = t(H, W).$$

Let $m = e(H)$ and fix $\eta > 0$. By the definition of the infimum in cut distance, there is a measure-preserving relabeling ϕ such that

$$\|U - W^{\phi}\|_{\square} \leq \delta_{\square}(U, W) + \eta.$$

Theorem 8.2 and relabeling invariance give

$$\begin{aligned} |t(H, U) - t(H, W)| &= |t(H, U) - t(H, W^{\phi})| \\ &\leq m\|U - W^{\phi}\|_{\square} \\ &\leq m(\delta_{\square}(U, W) + \eta). \end{aligned}$$

Letting $\eta \downarrow 0$ yields the asserted bound. Applying it to (W_n, W) proves the convergence statement. $\square$

Boundary of this proof unit. The forward implication above is elementary once the cut norm is fixed. The converse direction—that convergence of all finite homomorphism densities forces convergence in cut distance, together with compactness of graphon space modulo weak isomorphism—is substantially deeper. It is therefore stated as the source-owned interface QRCC-IV-THM-012 rather than being compressed into this elementary theorem.

Chapter 9

Quasirandom Tournaments

Definition 9.1 (Tournament quasirandomness tests). A *tournament* is an orientation of the complete graph: for every unordered pair of distinct vertices exactly one of $u \rightarrow v$ and $v \rightarrow u$ holds. A tournament sequence (T_n) is called *quasirandom in the Chung–Graham sense* when it satisfies any, hence all, of the equivalent conditions in Theorem 9.2. In particular, the balanced cut condition means that for all vertex sets A, B ,

$$e_{T_n}(A, B) - e_{T_n}(B, A) = o(n^2)$$

uniformly, where $e_T(A, B)$ counts ordered arcs from A to B and vertices in $A \cap B$ contribute only through pairs of distinct vertices.

Theorem 9.2 (Quasirandom tournament equivalences — source-owned). *For a tournament sequence with $n \rightarrow \infty$, the Chung–Graham theorem gives an equivalent family of quasirandom conditions, including uniform balanced directed cuts, the corresponding skew-adjacency spectral condition, and the correct asymptotic counts of every fixed tournament (equivalently a suitable finite set of cyclic/even configuration tests). Each normalization is the one induced by a uniformly random tournament. The implication chain among the full family of tournament properties is source-owned to Chung–Graham; the book does not substitute an undirected C_4 criterion for the directed configuration tests.*

Chapter 10

Uniform Hypergraphs and the Companion Interface

Definition 10.1 (Uniform hypergraph and relative quasirandomness). For $k \geq 2$, a k -uniform hypergraph on V is a set of k -element subsets of V . In the relative setting used by the companion additive-combinatorics volume, a k -uniform layer is tested relative to the $(k - 1)$ -uniform lower-dimensional complex supporting it; *relative quasirandomness* means that its balanced indicator has small discrepancy against the specified lower-dimensional test products. The precise complex, test family and parameter hierarchy are part of the companion theorem statement and must be imported together; ordinary graph cut discrepancy is not silently reused as a definition for $k \geq 3$.

Theorem 10.2 (Finite hypergraph regularity interface — companion-owned). *The selected finite hypergraph regularity statement is imported from the companion proof-obligation ledger at pointer IV.5.001. Its complex definition and complete parameter hierarchy are part of the import contract.*

Theorem 10.3 (Hypergraph removal interface — companion-owned). *The matching hypergraph removal theorem is imported from the same companion, pointer IV.5.003. Copy and edit-distance normalizations must agree with that theorem; no graph-level removal normalization is substituted.*

Theorem 10.4 (Hypergraph counting interface — companion-owned). *The counting lemma matched to the same regular-complex definition is imported from the companion, pointer IV.5.002. It may be used only together with the regularity interface of Theorem 10.2.*

Internalization and Reconstruction Guide: Volume IV

Purpose. Internalize the dense-graph equivalence principle: discrepancy, spectrum, small-subgraph counts, regularity, removal, testing, and graphon limits are different observational lenses on structured versus random-like graphs.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-004: Chernoff bounds.
2. QRCC-I-THM-005: Hoeffding inequality.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-IV-DEF-001 (Finite dense graphs and p-quasirandom sequences); QRCC-IV-DEF-002 (Homomorphism counts, labelled copies, and C4 density); QRCC-IV-DEF-003 (Adjacency spectrum, centered spectrum, and jumbledness); QRCC-IV-DEF-004 (Cut norm and dense graph discrepancy); QRCC-IV-DEF-011 (Centered adjacency operator and nonprincipal spectrum).

Chapter retrieval map

- Ch. 1: Random and Quasirandom Dense Graphs:** enter through QRCC-IV-DEF-001 (Finite dense graphs and p-quasirandom sequences); exit by being able to state QRCC-IV-THM-001 (Erdos–Renyi edge concentration) with its ownership boundary.
- Ch. 2: The Chung–Graham–Wilson Equivalences:** enter through QRCC-IV-DEF-002 (Homomorphism counts, labelled copies, and C4 density); exit by being able to state QRCC-IV-THM-002 (Chung–Graham–Wilson quasirandom equivalence theorem) with its ownership boundary.
- Ch. 3: Spectral Discrepancy and Expander Mixing:** enter through QRCC-IV-DEF-011 (Centered adjacency operator and nonprincipal spectrum); exit by being able to state QRCC-IV-THM-003 (Spectral discrepancy implication) with its ownership boundary.

- Ch. 4: Fixed-Subgraph Counting from Cut Discrepancy:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-IV-THM-006 (Fixed-subgraph counting in quasirandom graphs) with its ownership boundary.
- Ch. 5: Regularity: Weak and Strong:** enter through QRCC-IV-DEF-005 (Regular pair and regular partition); exit by being able to state QRCC-IV-THM-007 (Szemerédi regularity lemma) with its ownership boundary.
- Ch. 6: Triangle Removal:** enter through QRCC-IV-DEF-006 (Graph removal distance); exit by being able to state QRCC-IV-THM-009 (Triangle removal lemma) with its ownership boundary.
- Ch. 7: Removal and One-Sided Property Testing:** enter through QRCC-IV-DEF-007 (Dense-graph property tester); exit by being able to state QRCC-IV-THM-011 (Removal implies one-sided property testing) with its ownership boundary.
- Ch. 8: Graphons, Cut Distance, and Subgraph Densities:** enter through QRCC-IV-DEF-008 (Graphon and cut distance); exit by being able to state QRCC-IV-THM-012 (Compactness of graphon space modulo weak isomorphism) with its ownership boundary.
- Ch. 9: Quasirandom Tournaments:** enter through QRCC-IV-DEF-009 (Tournament quasirandomness); exit by being able to state QRCC-IV-THM-014 (Quasirandom tournament equivalences) with its ownership boundary.
- Ch. 10: Uniform Hypergraphs and the Companion Interface:** enter through QRCC-IV-DEF-010 (Uniform hypergraph and relative quasirandomness); exit by being able to state QRCC-IV-BB-015 (Hypergraph regularity lemma) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-IV-THM-001 — Erdős-Rényi edge concentration. Dependencies: QRCC-IV-DEF-001, QRCC-I-THM-004, QRCC-I-THM-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-IV-THM-003 — Spectral discrepancy implication. Dependencies: QRCC-IV-DEF-003. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-IV-THM-007 — Szemerédi regularity lemma. Dependencies: QRCC-IV-DEF-005, QRCC-IV-LEM-026. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-IV-THM-011 — Removal implies one-sided property testing. Dependencies: QRCC-IV-THM-010, QRCC-IV-DEF-007. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-IV-THM-014 — Quasirandom tournament equivalences. Dependencies: QRCC-IV-DEF-009, QRCC-IV-THM-002. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-IV-BB-015 — Hypergraph regularity lemma. Dependencies: QRCC-IV-DEF-010. *interface item: recover its precise statement, hypotheses, and role in the dependency chain.*

Common failure modes

- applying dense quasirandom equivalences to sparse graphs without a sparse model;
- mixing homomorphism densities, injective counts, and induced densities;
- reversing the quantifier order in regularity/removal statements;
- treating graphons pointwise rather than modulo measure-preserving equivalence;

Boundary and degenerate cases

- empty/complete graphs and vanishing density;
- finite graphs versus convergent dense graph sequences;
- induced versus non-induced removal;
- the graph case versus genuinely hypergraph-uniformity phenomena;

Retrieval and reconstruction drill

1. State **QRCC-IV-DEF-001** from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at **QRCC-IV-THM-001**. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For **QRCC-IV-THM-003**, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from **QRCC-IV-BB-015**, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume V replaces ordinary graph/Fourier uniformity by higher-order additive uniformity, the first systematic step toward nilsequence structure.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part V

Higher-Order Additive Quasirandomness

Chapter 1

Gowers Uniformity Begins with Fourier Analysis

Ownership convention for Volume V. The theorem proofs in this volume are not duplicated from the companion book *Additive Combinatorics: Foundations, Structure, Uniformity, and Inverse Theory*, Volumes I–V, Version 1.8 (24 August 2026). QRCC freezes the exact interface needed downstream: definitions, hypotheses, normalizations, theorem statements, and proof-location pointers. The three explicitly internal artifacts are developed here.

Definition 1.1 (Additive energy and balanced function). For finite subsets A, B of a finite abelian group G , the unnormalised additive energy is

$$E(A, B) = |\{(a_1, a_2, b_1, b_2) \in A^2 \times B^2 : a_1 + b_1 = a_2 + b_2\}|.$$

If $A \subseteq G$ has density $\alpha = |A|/|G|$, its balanced function is $f_A = 1_A - \alpha$. When a relative interval model is used later, the baseline is the relevant local density rather than a global constant. Normalised averages are written with $\mathbb{E}$; raw cardinality energy is never silently substituted for a normalised energy.

Definition 1.2 (Gowers norm and multiplicative derivative). For $d \geq 1$ and $f : G \rightarrow \mathbb{C}$,

$$\|f\|_{U^d(G)}^{2^d} = \mathbb{E}_{x, h_1, \dots, h_d \in G} \prod_{\omega \in \{0,1\}^d} \mathcal{C}^{|\omega|} f(x + \omega \cdot h), \quad \Delta_h f(x) = f(x + h) \overline{f(x)}.$$

Here $\mathcal{C}$ denotes complex conjugation. We use the companion convention $\|f\|_{U^1} = |\mathbb{E}f|$.

Theorem 1.3 (Fourier characterization of U^2). *For the normalized Fourier transform on a finite abelian group,*

$$\|f\|_{U^2}^4 = \sum_{\gamma \in \widehat{G}} |\widehat{f}(\gamma)|^4.$$

Consequently, if $|f| \leq 1$ and $\|f\|_{U^2} \geq \delta$, then some character γ satisfies $|\widehat{f}(\gamma)| \geq \delta^2$.

Remark 1.4 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, §4.4, Theorem 4.3 and Corollary 4.4 (physical pages 188–189 in the combined companion PDF). QRCC uses the same normalized finite-group Fourier convention.

Theorem 1.5 (Positivity and norm/seminorm property). *For $d \geq 1$ the cube average defining $\|f\|_{U^d}^{2^d}$ is nonnegative and real. For $d \geq 2$, $\|\cdot\|_{U^d}$ is a norm on functions on a finite abelian group; U^1 is the absolute mean and hence only a seminorm before quotienting constants of zero mean in the usual way.*

Remark 1.6 (Exact companion pointer). Additive Combinatorics v1.8, Volume IV, §2.1, Theorem 2.1 (combined PDF p. 349). The companion supplies the recursive proof and triangle inequality; QRCC does not duplicate it.

Interface check. The first theorem connects the ordinary Fourier layer of Volume I to higher-order uniformity. The second guarantees that later inverse statements use a genuine norm for $d \geq 2$ under the finite-group convention.

Chapter 2

Three-Term Progressions and Density Increment

Definition 2.1 (Three-term progression count). If 2 is invertible in the finite abelian group G , set

$$\Lambda_3(f_0, f_1, f_2) = \mathbb{E}_{x,d \in G} f_0(x) f_1(x+d) f_2(x+2d).$$

In interval models, a progression is nontrivial when $d \neq 0$ in the integer parametrization. In groups with torsion, distinctness of the three points is stated separately rather than inferred from $d \neq 0$.

Theorem 2.2 (Roth theorem in the selected companion model). *There are absolute constants C, N_0 such that, for $N \geq N_0$, every $A \subseteq [N]$ containing no nontrivial three-term arithmetic progression obeys*

$$\frac{|A|}{N} \leq \frac{C}{\sqrt{\log \log N}}.$$

In particular, every fixed-positive-density subset of $[N]$ contains a nontrivial three-term progression for all sufficiently large N .

Remark 2.3 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, Chapter 2, Theorem 2.1, beginning at combined PDF p. 178. This deliberately selected bound is pedagogical rather than best-current.

Theorem 2.4 (Fourier density-increment step). *In the local interval model of the companion, if $A \subseteq [N]$ has relative density α and is three-term-progression deficient at the scale required by the iteration, then the relative balanced function $f = 1_A - \alpha 1_{[N]}$ has a large nonzero Fourier coefficient. Linearizing that character on a subprogression yields a strictly larger relative density, with the length and density gains quantified in terms of α .*

Remark 2.5 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, Chapter 2: Lemma 2.2 gives the Fourier obstruction, followed by the character-linearization and density-increment steps in §§2.4–2.6. QRCC imports this complete local iteration interface rather than the obsolete v1.4 ledger pointer to a later “modern Roth” block.

Chapter 3

Cube Calculus

Definition 3.1 (Gowers cube multilinear form). For functions $(f_\omega)_{\omega \in \{0,1\}^d}$ on G , define

$$\Lambda_\square((f_\omega)) = \mathbb{E}_{x, h_1, \dots, h_d} \prod_{\omega \in \{0,1\}^d} \mathcal{C}^{|\omega|} f_\omega(x + \omega \cdot h).$$

The conjugation pattern is part of the definition and is retained at every Cauchy–Schwarz step.

Theorem 3.2 (Gowers–Cauchy–Schwarz). *For a finite abelian group,*

$$|\Lambda_\square((f_\omega))| \leq \prod_{\omega \in \{0,1\}^d} \|f_\omega\|_{U^d}.$$

Remark 3.3 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, §4.5 (combined PDF p. 189), with the same cube and conjugation convention.

Theorem 3.4 (Monotonicity). *For every $d \geq 1$ and every complex function on a finite abelian group,*

$$\|f\|_{U^d} \leq \|f\|_{U^{d+1}}.$$

Remark 3.5 (Exact companion pointer). Additive Combinatorics v1.8, Volume IV, §2.1, Proposition 2.3 (combined PDF p. 349).

Theorem 3.6 (Derivative identity). *For $d \geq 2$,*

$$\|f\|_{U^d}^{2^d} = \mathbb{E}_h \|\Delta_h f\|_{U^{d-1}}^{2^{d-1}}.$$

Remark 3.7 (Exact companion pointer). Additive Combinatorics v1.8, Volume IV, §2.1, Lemma 2.2 (combined PDF p. 349).

Chapter 4

Linear Systems, Complexity, and a Finite Laboratory

Definition 4.1 (Affine-linear systems and Cauchy–Schwarz complexity). Let F be a finite field, V a finite-dimensional F -vector space, and $L = (L_1, \dots, L_m)$ affine-linear forms $V^t \rightarrow V$ with coefficient vectors $a_i \in F^t$. The system has Cauchy–Schwarz complexity at most s at L_i if the remaining indices can be partitioned into $s + 1$ classes C_r such that a_i does not lie in the F -span of $\{a_j : j \in C_r\}$ for any r .

Theorem 4.2 (Generalized von Neumann principle). *Under the preceding vector-space hypotheses, if L has Cauchy–Schwarz complexity at most s at L_i and $|f_j| \leq 1$, then*

$$\left| \mathbb{E}_{x \in V^t} \prod_{j=1}^m f_j(L_j(x)) \right| \leq \|f_i\|_{U^{s+1}(V)}.$$

Remark 4.3 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, §5.4, Definition 5.3 and Theorem 5.4 (combined PDF pp. 193–195). This pointer supersedes the old generic progression-only pointer.

Theorem 4.4 (Four-term progressions are controlled by U^3). *If G is a finite abelian group on which 2 and 3 are invertible and $|f_j| \leq 1$, then*

$$\left| \mathbb{E}_{x,d} \prod_{j=0}^3 f_j(x + jd) \right| \leq \min_j \|f_j\|_{U^3(G)}.$$

Remark 4.5 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, Theorem 5.1 (combined PDF p. 192). The invertibility assumptions are retained explicitly.

Example 4.6 (Finite Gowers-norm laboratory). The executable `code/volume_05/gowers_norm_lab.py` performs exact rational cube counts for an indicator on $\mathbb{F}_5$, verifies the U^3 derivative identity, checks a four-term generalized-von-Neumann instance without floating-point comparison, and verifies by modular finite differences that linear phases have vanishing second derivative while quadratic phases have vanishing third derivative but need not have vanishing second derivative.

Chapter 5

The Szemerédi Density-Increment Architecture

Definition 5.1 (Density-increment datum). A density-increment datum records (i) a structured set, factor, or local model on which restriction is legal; (ii) a quantified increase from density α to at least $\alpha + \eta$; and (iii) explicit losses in size and complexity sufficient to iterate. A statement that merely says “pass to a structured subset” without these three pieces is not an iteration interface.

Theorem 5.2 (Gowers proof architecture for Szemerédi’s theorem). *For each $k \geq 3$ and $\alpha > 0$ there is $N_0(k, \alpha)$ such that every $A \subseteq [N]$ with $N \geq N_0$ and $|A| \geq \alpha N$ contains a nontrivial k -term arithmetic progression. The companion proof uses the Gowers-uniformity/density-increment mechanism with all localization and iteration losses made explicit.*

Remark 5.3 (Exact companion pointer). Additive Combinatorics v1.8 has a reader-level synthesis in Volume III, Chapter 6 (combined PDF pp. 197–201) and the detailed quantitative Gowers proof in Volume IV, Chapters 1–4 (beginning p. 347). QRCC points to the latter as proof owner and uses the former as the reading bridge.

Chapter 6

Quadratic Structure and the Failure of Linear Uniformity

Definition 6.1 (Quadratic phase and rank). Let V be a vector space over an odd prime field $\mathbb{F}_p$. A quadratic phase is $x \mapsto e_p(Q(x))$, where $Q : V \rightarrow \mathbb{F}_p$ has degree at most two. Its associated symmetric bilinear form is

$$B_Q(x, y) = Q(x + y) - Q(x) - Q(y) + Q(0),$$

and the rank of Q means rank B_Q . Cyclic local quadratic phases are a different model and are not identified with this global finite-vector-space definition.

Example 6.2 (Quadratic obstruction to U^2 control of four-term progressions). Take $G = \mathbb{F}_p$, $p > 3$, and $f(x) = e_p(ax^2)$ with $a \neq 0$. Completing the square gives $|\hat{f}(\xi)| = p^{-1/2}$ for every frequency, so ordinary Fourier coefficients vanish in size as $p \rightarrow \infty$. Yet the third additive derivative of ax^2 vanishes identically, hence $\|f\|_{U^3} = 1$. Thus linear uniformity can be tiny while quadratic structure is perfect, explaining why four-term patterns require the next Gowers level.

Theorem 6.3 (Quadratic phase correlation/rank interface). *For $Q : V \rightarrow \mathbb{F}_p$ quadratic with associated bilinear rank r , the normalized quadratic Gauss average has magnitude at most $p^{-r/2}$; consequently non-negligible quadratic bias forces bounded rank. Together with the finite-field U^3 inverse theorem, this is the quadratic correlation interface used by QRCC.*

Remark 6.4 (Exact companion pointer). Additive Combinatorics v1.8, Volume IV, §2.2, Theorem 2.5 and Corollary 2.6 (combined PDF pp. 349–350), together with Detailed Proof Appendix A for the inverse step.

Chapter 7

The U^3 Inverse Theorem

Definition 7.1 (Correlation inverse statement). An inverse theorem for U^d is a statement of the form

$$\|f\|_{U^d} \geq \delta, \quad \|f\|_\infty \leq 1 \quad \implies \quad |\mathbb{E} f \overline{\Phi}| \geq c(\delta),$$

where Φ belongs to a specified structured class of complexity bounded in terms of the displayed parameters. Ambient group, characteristic regime, classical versus non-classical polynomial language, and quantitative dependence are part of the theorem data.

Theorem 7.2 (Finite-vector-space U^3 inverse theorem). *Fix an odd prime p . For every $\delta > 0$ there exists $c_p(\delta) > 0$ such that, for every finite-dimensional $V = \mathbb{F}_p^n$ and every 1-bounded $f : V \rightarrow \mathbb{C}$ with $\|f\|_{U^3(V)} \geq \delta$, there is a quadratic polynomial $Q : V \rightarrow \mathbb{F}_p$ satisfying*

$$|\mathbb{E}_x f(x) e_p(-Q(x))| \geq c_p(\delta).$$

Remark 7.3 (Exact companion pointer). Additive Combinatorics v1.8, Detailed Proof Appendix A, Theorem A.1, beginning at combined PDF p. 541. The companion gives the complete additive-graph, symmetry, and integration argument.

Chapter 8

Higher-Order Inverse Theory

Definition 8.1 (Filtered nilmanifold and polynomial nilsequence). A filtered nilmanifold is G/Γ together with a rational filtration $G_\bullet$ compatible with commutators. A map $g : \mathbb{Z}^t \rightarrow G$ is polynomial relative to $G_\bullet$ when its iterated discrete derivatives land in the prescribed filtration levels. A polynomial nilsequence is $n \mapsto F(g(n)\Gamma)$, with degree, dimension, rationality/Mal’cev data and Lipschitz norm included in its complexity.

Theorem 8.2 (Green–Tao–Ziegler inverse theorem, qualitative interval form). *For each $s \geq 1$ and $\delta > 0$, every 1-bounded $f : [N] \rightarrow \mathbb{C}$ with $\|f\|_{U^{s+1}[N]} \geq \delta$ correlates by an amount bounded below in terms of (s, δ) with an s -step polynomial nilsequence of complexity bounded in terms of (s, δ) .*

Remark 8.3 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, §5.1–5.4, especially the qualitative interval inverse theorem used in the local-to-global construction, together with Detailed Proof Appendix D (GTZ reconstruction). The v1.8 preface records that the corrected GTZ proof infrastructure is internal to the companion.

Theorem 8.4 (Manners quantitative inverse interface). *In the quantitative inverse theorem formulation frozen by the companion, large U^{s+1} norm forces correlation with an s -step nilsequence while dimension, rationality, Lipschitz and correlation parameters are quantitatively propagated through the construction. QRCC uses exactly the parameter package stated by the companion and does not replace it by an unspecified “bounded complexity” claim when quantitative dependence is required.*

Remark 8.5 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, Chapters 7–11 (quantitative Manners proof sequence), with the comparison in §5.2. The companion retains Manners’ paper as the external provenance while supplying its selected proof dossier.

Theorem 8.6 (All-degree finite-field inverse theorem). *Fix a prime p , $s \geq 1$, and $\delta > 0$. There is $\varepsilon(p, s, \delta) > 0$ such that, for every finite-dimensional $\mathbb{F}_p$ -space V and every 1-bounded f with $\|f\|_{U^{s+1}(V)} \geq \delta$, there exists a possibly non-classical polynomial $P : V \rightarrow \mathbb{R}/\mathbb{Z}$ of degree at most s with*

$$|\mathbb{E}_x f(x) e(-P(x))| \geq \varepsilon(p, s, \delta).$$

Remark 8.7 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, Chapter 6, Theorem 6.1 (combined PDF p. 405), with the proof distributed through the subsequent finite-field chapters as recorded by Remark 6.2.

Theorem 8.8 (Gowers–Wolf true-complexity theorem: selected large-characteristic finite-field form). *For a fixed finite system of linear forms over $\mathbb{F}_p^n$ and fixed order k , in the sufficiently-large-characteristic regime of Gowers–Wolf, linear independence of the k th tensor powers of the forms is*

the criterion that degree- $(k - 1)$ uniformity suffices to control the associated multilinear average. The exact dependence on the system and the characteristic threshold is inherited from the primary theorem, not from the companion's expository discussion.

Remark 8.9 (Primary-source ownership). This row remains deliberately outside companion ownership. Source: W. T. Gowers and J. Wolf, *Linear forms and higher-degree uniformity for functions on $\mathbb{F}_p^n$* , Geom. Funct. Anal. 21 (2011), 36–69, DOI 10.1007/s00039-010-0106-3. The primary theorem proves the conjectured tensor-power criterion when p is sufficiently large; QRCC does not silently extend it to low characteristic.

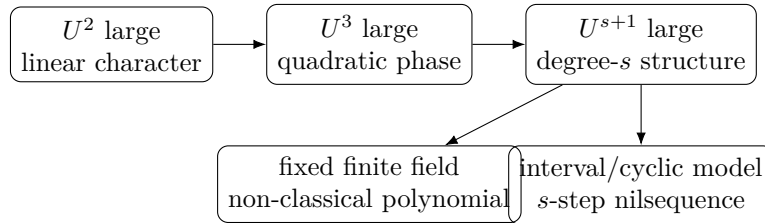


Figure 8.1: Higher-order structure ladder. The endpoint depends on the ambient group: nilsequences in interval/cyclic settings and polynomial phases in fixed finite fields.

Chapter 9

Nilmanifold Equidistribution, Factorization, and Counting

Definition 9.1 (Equidistribution and horizontal character). Fix a rational Mal’cev coordinate system on G/Γ . A finite orbit $(g(n)\Gamma)_{n \in \Omega}$ is δ -equidistributed if every Lipschitz test function F satisfies

$$\left| \mathbb{E}_{n \in \Omega} F(g(n)\Gamma) - \int_{G/\Gamma} F \, dm \right| \leq \delta \|F\|_{\text{Lip}}.$$

A horizontal character is a continuous homomorphism $\eta : G \rightarrow \mathbb{R}/\mathbb{Z}$ annihilating Γ ; its modulus/height is measured relative to the frozen Mal’cev basis.

Theorem 9.2 (Polynomial nilsequence representation interface). *The nilsequence package used by the higher-order inverse theorem consists of the companion’s nilpotent/Mal’cev geometry, filtered polynomial sequences, horizontal and vertical characters, quantitative equidistribution, factorization, and polynomial counting machinery. These are imported as a coherent package with one rationality convention; QRCC does not mix definitions from incompatible nilmanifold coordinate systems.*

Remark 9.3 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, Chapters 1–4 (combined PDF pp. 385–397), together with the detailed geometric appendices beginning after p. 540. This replaces the generic dependency `COMP:ADDITIVE_COMBINATORICS`.

Theorem 9.4 (Quantitative Leibman obstruction). *In the one- and multiparameter polynomial-orbit settings fixed by the companion, failure of the prescribed quantitative equidistribution yields a nontrivial horizontal character of controlled height whose composition with the polynomial orbit has controlled C^∞ norm. All bounds are measured in the same rational Mal’cev model used by the factorization theorem.*

Remark 9.5 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, Chapter 3 (quantitative equidistribution), with the complete quantitative Leibman proof in the detailed proof appendices.

Theorem 9.6 (Smooth–equidistributed–rational factorization). *For the companion’s polynomial sequence g , one has a quantitative factorization*

$$g = \varepsilon g' \gamma,$$

where ε is smooth, g' is quantitatively equidistributed in a controlled rational subnilmanifold, and γ is rational/periodic, with complexity losses explicitly bounded in the companion parameters.

Remark 9.7 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, Chapter 4, together with the factorization appendices in the detailed dossier.

Theorem 9.8 (Polynomial nilsequence counting). *For the boxes/convex-region regime and rationality data fixed by the companion, a totally equidistributed polynomial nilsequence has averages governed by Haar measure up to the stated Lipschitz, boundary, and equidistribution errors. QRCC uses only this frozen counting interface.*

Remark 9.9 (Exact companion pointer). Additive Combinatorics v1.8, Volume V, Chapter 4 (factorization and polynomial nilsequence counting), with the full counting theorem in the detailed proof appendices.

Chapter 10

Regularity, Dense Models, and Relative Counting

Definition 10.1 (Pseudorandom majorant and dense model). Let $\nu \geq 0$ on a finite probability space with $\mathbb{E}\nu = 1 + o(1)$, and fix a test/dual class $\mathcal{F}$. We call ν pseudorandom only relative to explicitly stated correlation/linear-forms conditions sufficient for the theorem at hand. A dense model for $0 \leq f \leq \nu$ is a function $0 \leq \tilde{f} \leq 1$ such that

$$|\mathbb{E}(f - \tilde{f})\phi| \leq \varepsilon \quad (\phi \in \mathcal{F}).$$

The class $\mathcal{F}$ is part of the data; “pseudorandom” is not an absolute adjective.

Theorem 10.2 (Arithmetic regularity/decomposition interface). *For the finite models selected by the companion, bounded functions admit a structured–uniform–small-error decomposition with the factor complexity, Gowers-uniformity threshold and L^2 error controlled by the displayed input parameters. The energy increment terminates because the relevant energy is bounded.*

Remark 10.3 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, Chapter 7, *Arithmetic Regularity: Structure, Uniformity, and Error* (combined PDF pp. 202–205).

Theorem 10.4 (Finite dense-model theorem). *Under the companion’s finite test-class closure and pseudorandomness hypotheses, every $0 \leq f \leq \nu$ has a dense model $0 \leq \tilde{f} \leq 1$ indistinguishable from f by the prescribed tests to the asserted error.*

Remark 10.5 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, §8.2, *A finite dense-model theorem* (combined PDF p. 206). QRCC does not silently replace the test-class hypotheses by an unrestricted statement.

Theorem 10.6 (Relative generalized von Neumann interface). *For the affine-linear systems and majorants satisfying the companion’s selected pseudorandomness hypotheses, the relevant multilinear average of functions dominated by ν is controlled by the appropriate relative Gowers-uniformity term plus the companion’s vanishing pseudorandomness error.*

Remark 10.7 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, §8.3, *Counting after modelling*, combined with the generalized von Neumann theorem of Volume III §5.4. The precise majorant condition is theorem-dependent as emphasized in §8.4.

Theorem 10.8 (Relative counting and relative Szemerédi interface). *With the same majorant/test hypotheses, dense-model replacement followed by the companion counting theorem transfers dense*

progression counts to the sparse function and yields the selected relative Szemerédi conclusion. This interface makes no prime-specific claim: prime majorants require their own number-theoretic verification elsewhere in the QRCC series.

Remark 10.9 (Exact companion pointer). Additive Combinatorics v1.8, Volume III, Chapter 8, especially §§8.2–8.4, together with the dense Szemerédi input from Chapter 6.

Exercises for the interface volume

1. Verify directly that $\|1_A\|_{U^2}^4$ is the normalized number of additive parallelograms in A .
2. For $p > 3$, compute the third additive derivative of x^2 over $\mathbb{F}_p$.
3. Show why a theorem about four-term progressions on $\mathbb{Z}/N\mathbb{Z}$ must state when 2 and 3 are invertible.
4. Give a two-form system of Cauchy–Schwarz complexity zero and verify Definition 5.3 of the companion in that case.
5. Explain why the local Roth balanced function is $1_A - \alpha 1_I$ rather than $1_A - \alpha$ on all of $\mathbb{Z}/p\mathbb{Z}$.
6. Trace the ownership path from a large U^3 norm to a quadratic phase, naming the exact companion appendix that owns the proof.
7. State the difference between a classical and a non-classical polynomial phase in the finite-field inverse theorem.
8. Explain why a horizontal character must be measured relative to a rational Mal’cev basis for a quantitative theorem.
9. In the factorization $g = \varepsilon g' \gamma$, identify which factor is smooth, equidistributed, and rational/periodic and why each role is distinct.
10. Give an example showing that a “pseudorandom majorant” cannot be defined independently of a test class.
11. Compare the endpoint structured objects in the interval GTZ theorem and the fixed-finite-field inverse theorem.
12. Explain why Gowers–Wolf true complexity remains primary-source-owned in QRCC even though the companion discusses true complexity.

Internalization and Reconstruction

Guide: Volume V

Purpose. Recognize why ordinary Fourier uniformity stops at linear structure and why higher Gowers norms, inverse theorems, nilsequences, regularity, and transference are the correct hierarchy for higher-order additive patterns.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-017: Convolution theorem.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-V-DEF-001 (Additive energy and balanced function); QRCC-V-DEF-003 (Gowers uniformity norm and multiplicative derivative); QRCC-V-DEF-002 (Three-term progression count); QRCC-V-DEF-011 (Gowers cube multilinear form); QRCC-V-DEF-004 (System of affine-linear forms and CS complexity).

Chapter retrieval map

- Ch. 1: Gowers Uniformity Begins with Fourier Analysis:** enter through QRCC-V-DEF-001 (Additive energy and balanced function); exit by being able to state QRCC-V-THM-001 (Fourier characterization of U_2) with its ownership boundary.
- Ch. 2: Three-Term Progressions and Density Increment:** enter through QRCC-V-DEF-002 (Three-term progression count); exit by being able to state QRCC-V-THM-002 (Roth theorem in selected finite models) with its ownership boundary.
- Ch. 3: Cube Calculus:** enter through QRCC-V-DEF-011 (Gowers cube multilinear form); exit by being able to state QRCC-V-THM-004 (Gowers–Cauchy–Schwarz inequality) with its ownership boundary.

- Ch. 4: Linear Systems, Complexity, and a Finite Laboratory:** enter through QRCC-V-DEF-004 (System of affine-linear forms and CS complexity); exit by being able to state QRCC-V-THM-003 (Generalized von Neumann inequality) with its ownership boundary.
- Ch. 5: The Szemerédi Density-Increment Architecture:** enter through QRCC-V-DEF-005 (Density increment datum); exit by being able to state QRCC-V-THM-006 (Gowers proof architecture for Szemerédi’s theorem) with its ownership boundary.
- Ch. 6: Quadratic Structure and the Failure of Linear Uniformity:** enter through QRCC-V-DEF-006 (Quadratic phase and rank); exit by being able to state QRCC-V-THM-008 (Quadratic phase correlation theorem) with its ownership boundary.
- Ch. 7: The U^3 Inverse Theorem:** enter through QRCC-V-DEF-007 (Correlation inverse statement); exit by being able to state QRCC-V-THM-007 (U^3 inverse theorem in finite-vector-space form) with its ownership boundary.
- Ch. 8: Higher-Order Inverse Theory:** enter through QRCC-V-DEF-008 (Filtered nilmanifold and polynomial nilsequence); exit by being able to state QRCC-V-BB-013 (Green–Tao–Ziegler inverse theorem) with its ownership boundary.
- Ch. 9: Nilmanifold Equidistribution, Factorization, and Counting:** enter through QRCC-V-DEF-009 (Equidistribution and horizontal character); exit by being able to state QRCC-V-BB-016 (Polynomial nilsequence representation interface) with its ownership boundary.
- Ch. 10: Regularity, Dense Models, and Relative Counting:** enter through QRCC-V-DEF-010 (Pseudorandom majorant and dense model); exit by being able to state QRCC-V-THM-009 (Arithmetic regularity and structured–uniform–error decomposition) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-V-THM-001 — Fourier characterization of U^2 . Dependencies: QRCC-V-DEF-003, QRCC-I-THM-017. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
2. QRCC-V-THM-004 — Gowers–Cauchy–Schwarz inequality. Dependencies: QRCC-V-DEF-003. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-V-THM-006 — Gowers proof architecture for Szemerédi’s theorem. Dependencies: QRCC-V-DEF-005, QRCC-V-THM-003, QRCC-V-THM-005. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-V-THM-007 — U^3 inverse theorem in finite-vector-space form. Dependencies: QRCC-V-DEF-007, QRCC-V-THM-008. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

5. QRCC-V-BB-016 — Polynomial nilsequence representation interface. Dependencies: QRCC-V-BB-013. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-V-THM-009 — Arithmetic regularity and structured-uniform-error decomposition. Dependencies: QRCC-V-DEF-010, QRCC-V-THM-005. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

Common failure modes

- expecting U2 control to govern patterns of true complexity above one;
- moving between finite-vector-space, cyclic, and integer settings without checking characteristic and local factors;
- using an inverse theorem as though it were a norm identity or a constructive algorithm with unspecified bounds;
- duplicating companion-owned proofs or weakening their hypotheses during paraphrase;

Boundary and degenerate cases

- characteristic two and other low-characteristic pathologies;
- complexity-one versus higher true-complexity systems;
- zero function, constant phase, and perfectly structured examples;
- quantitative nilsequence parameters and dense-model pseudorandomness thresholds;

Retrieval and reconstruction drill

1. State QRCC-V-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-V-THM-001. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-V-THM-004, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.

7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-V-THM-009, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume VI carries the structured-versus-uniform philosophy into noncommutative groups, where representation dimension replaces scalar Fourier coefficients.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part VI

Quasirandom Groups and Noncommutative Mixing

Chapter 1

Noncommutative Fourier Analysis on Finite Groups

Fourier analysis on an abelian group diagonalizes translations by scalar characters. For a nonabelian group the same idea survives, but the scalar frequencies are replaced by irreducible matrix representations. Keeping the normalization exact is essential: the mixing estimates later in the volume will read their quantitative constants directly from the representation blocks.

Definition 1.1 (Unitary representations and the noncommutative Fourier transform). Let G be a finite group. Throughout this volume

$$\langle f, h \rangle = \mathbb{E}_{x \in G} f(x) \overline{h(x)}, \quad \|f\|_2^2 = \mathbb{E}_x |f(x)|^2.$$

A finite-dimensional *unitary representation* is a homomorphism $\rho : G \rightarrow U(V_\rho)$. Its dimension is denoted d_ρ . Two representations ρ, σ are *equivalent* if there is a unitary isomorphism T satisfying $T\rho(g) = \sigma(g)T$ for every g .

For $f : G \rightarrow \mathbb{C}$, define its Fourier transform at ρ by

$$\widehat{f}(\rho) = \mathbb{E}_{x \in G} f(x) \rho(x)^*.$$

For a matrix M , M^* denotes conjugate transpose, $\|M\|_{\text{HS}}^2 = \text{tr}(MM^*)$, and $\|M\|_{\text{op}}$ is its Euclidean operator norm.

Positive example. The trivial representation $\mathbf{1}_G(g) = 1$ is unitary and $\widehat{f}(\mathbf{1}_G) = \mathbb{E}f$. Thus the ordinary mean is precisely the trivial Fourier block.

Nonexample. Choosing arbitrary unitary matrices U_g for the elements of G does not define a representation unless $U_{gh} = U_g U_h$ for every g, h .

Definition 1.2 (Irreducibility, characters, and the regular representation). A nonzero representation is *irreducible* if it has no nonzero proper invariant subspace. Its character is

$$\chi_\rho(g) = \text{tr } \rho(g).$$

Write $\text{Irr}(G)$ for a fixed set of representatives of the unitary equivalence classes of irreducible complex representations.

The *left regular representation* acts on $L^2(G)$ by

$$(L_g f)(x) = f(g^{-1}x).$$

The formula defining L_g is part of the definition; its decomposition into irreducibles is a theorem below, not an assumption.

Positive example. Every one-dimensional representation is irreducible. In particular, the trivial character is an irreducible representation.

Nonexample. If $|G| > 1$, the regular representation is reducible: the constant functions form a one-dimensional invariant subspace.

Definition 1.3 (Normalized convolution and the mean-zero subspace). For $f, g : G \rightarrow \mathbb{C}$, define

$$(f * g)(x) = \mathbb{E}_{y \in G} f(y)g(y^{-1}x).$$

Let

$$L_0^2(G) = \{f : G \rightarrow \mathbb{C} : \mathbb{E}f = 0\}.$$

Unless explicitly marked otherwise, the unadorned convolution $*$ and $L^q(G)$ norms in this volume use normalized counting measure. Chapter 9 will deliberately introduce a separately decorated counting-sum convolution $*_{\text{m}}$ and counting ℓ^q norms for probability masses. With the present normalized convention the convolution identity is not $\mathbf{1}_{\{e\}}$ but $|G|\mathbf{1}_{\{e\}}$.

Positive example. If $\delta_e = |G|\mathbf{1}_{\{e\}}$, then $\delta_e * f = f = f * \delta_e$. For a set A of density α , $\mathbf{1}_A - \alpha$ lies in $L_0^2(G)$.

Nonexample. The unnormalized sum $\sum_y f(y)g(y^{-1}x)$ is a different convolution and would change every constant in the Fourier and mixing formulas below.

Lemma 1.4 (Unitarization of finite-group representations). *Every finite-dimensional complex representation $\pi : G \rightarrow GL(V)$ of a finite group is equivalent to a unitary representation. More precisely, V admits a positive-definite Hermitian inner product invariant under every $\pi(g)$.*

Proof. Start with any positive-definite Hermitian inner product $(\cdot, \cdot)_0$ on V and average it over the group:

$$(v, w)_G = \mathbb{E}_{g \in G} (\pi(g)v, \pi(g)w)_0.$$

This form is Hermitian and positive definite: if $v \neq 0$, then every $\pi(g)v$ is nonzero, so each summand $(\pi(g)v, \pi(g)v)_0$ is positive. For $h \in G$, the change of variable $g' = gh$ gives

$$(\pi(h)v, \pi(h)w)_G = \mathbb{E}_g (\pi(gh)v, \pi(gh)w)_0 = \mathbb{E}_{g'} (\pi(g')v, \pi(g')w)_0 = (v, w)_G.$$

Choose an orthonormal basis for $(\cdot, \cdot)_G$. In this basis every $\pi(h)$ preserves the standard Hermitian inner product and is therefore unitary. The change of basis is an equivalence of representations. $\square$

Theorem 1.5 (Schur orthogonality and regular multiplicities). *Let $\rho, \sigma \in \text{Irr}(G)$. Then Schur's lemma holds: an intertwiner $T : V_\sigma \rightarrow V_\rho$ is zero if ρ and σ are inequivalent, while every endomorphism intertwining an irreducible ρ with itself is a scalar. In orthonormal bases,*

$$\mathbb{E}_{g \in G} \rho(g)_{ij} \overline{\sigma(g)_{kl}} = \begin{cases} d_\rho^{-1} \delta_{ik} \delta_{jl}, & \rho \simeq \sigma, \\ 0, & \rho \not\simeq \sigma. \end{cases}$$

Moreover, the left regular representation contains each $\rho \in \text{Irr}(G)$ with multiplicity d_ρ , and therefore

$$\sum_{\rho \in \text{Irr}(G)} d_\rho^2 = |G|.$$

Proof. Let $T : V_\sigma \rightarrow V_\rho$ intertwine the two representations. Its kernel is σ -invariant and its image is ρ -invariant. Irreducibility implies that a nonzero T is an isomorphism. Hence no nonzero intertwiner exists between inequivalent irreducibles. If $\rho = \sigma$ and T is an endomorphism, choose an eigenvalue λ of T over $\mathbb{C}$. Then $T - \lambda I$ is an intertwiner with nonzero kernel, so irreducibility forces $T = \lambda I$. This is Schur's lemma.

For an arbitrary linear map $S : V_\sigma \rightarrow V_\rho$, average its conjugates:

$$\mathcal{A}(S) = \mathbb{E}_{g \in G} \rho(g) S \sigma(g)^*.$$

A change of variable shows $\rho(h) \mathcal{A}(S) = \mathcal{A}(S) \sigma(h)$ for every h , so $\mathcal{A}(S)$ is an intertwiner. If $\rho \neq \sigma$, it is zero. If $\rho = \sigma$, Schur's lemma gives $\mathcal{A}(S) = cI$, and taking traces shows

$$c = \frac{\text{tr } S}{d_\rho}.$$

Choose S to be the matrix unit sending the ℓ th basis vector to the j th basis vector. Reading the (i, k) entry of the averaged matrix gives exactly the displayed matrix-coefficient orthogonality formula.

We next decompose the regular representation. Any finite-dimensional unitary representation is a direct sum of irreducibles: choose a nonzero invariant subspace W of minimal positive dimension. It is irreducible. Its orthogonal complement is invariant as well: if $y \in W^\perp$ and $w \in W$, then for every $g \in G$,

$$\langle \rho(g)y, w \rangle = \langle y, \rho(g)^* w \rangle = \langle y, \rho(g^{-1})w \rangle = 0,$$

because W is invariant. Thus $V = W \oplus W^\perp$, and induction on the dimension gives a direct sum of irreducibles. Matrix-coefficient orthogonality implies character orthogonality

$$\langle \chi_\rho, \chi_\sigma \rangle = \mathbb{E}_g \chi_\rho(g) \overline{\chi_\sigma(g)} = \delta_{\rho\sigma}.$$

If a unitary representation W decomposes as $\bigoplus_\sigma m_\sigma V_\sigma$, taking its character inner product with χ_ρ gives m_ρ .

For the left regular representation, the standard basis $\{\mathbf{1}_{\{x\}} : x \in G\}$ is permuted by left translation. Thus

$$\chi_{\text{reg}}(e) = |G|, \quad \chi_{\text{reg}}(g) = 0 \quad (g \neq e).$$

Consequently the multiplicity of ρ is

$$\langle \chi_{\text{reg}}, \chi_\rho \rangle = \frac{1}{|G|} |G| \overline{\chi_\rho(e)} = d_\rho.$$

Comparing dimensions of the regular representation now gives $|G| = \sum_\rho d_\rho^2$. □

Worked check. For an abelian group every irreducible has dimension 1, so the identity $\sum_\rho d_\rho^2 = |G|$ says that there are exactly $|G|$ characters. This recovers the familiar scalar Fourier basis.

Theorem 1.6 (Noncommutative Fourier inversion). *For every $f : G \rightarrow \mathbb{C}$ and $x \in G$,*

$$f(x) = \sum_{\rho \in \text{Irr}(G)} d_\rho \text{tr}(\hat{f}(\rho) \rho(x)).$$

Proof. Fix orthonormal bases in all irreducibles. By Theorem 1.5, the functions

$$\sqrt{d_\rho} \rho(\cdot)_{ij} \quad (\rho \in \text{Irr}(G), 1 \leq i, j \leq d_\rho)$$

are pairwise orthonormal in $L^2(G)$. Their number is $\sum_\rho d_\rho^2 = |G| = \dim L^2(G)$, so they form an orthonormal basis. The coefficient of $\sqrt{d_\rho} \rho(\cdot)_{ij}$ in the expansion of f is

$$\sqrt{d_\rho} \mathbb{E}_y f(y) \overline{\rho(y)_{ij}} = \sqrt{d_\rho} \widehat{f}(\rho)_{ji}.$$

Expanding in the basis and evaluating at x gives

$$f(x) = \sum_{\rho, i, j} d_\rho \widehat{f}(\rho)_{ji} \rho(x)_{ij} = \sum_\rho d_\rho \text{tr}(\widehat{f}(\rho) \rho(x)),$$

as claimed. $\square$

Theorem 1.7 (Noncommutative Plancherel). *For $f, h : G \rightarrow \mathbb{C}$,*

$$\langle f, h \rangle = \sum_{\rho \in \text{Irr}(G)} d_\rho \text{tr}(\widehat{f}(\rho) \widehat{h}(\rho)^*).$$

In particular,

$$\|f\|_2^2 = \sum_\rho d_\rho \|\widehat{f}(\rho)\|_{\text{HS}}^2.$$

Proof. Use the orthonormal matrix-coefficient basis from the inversion proof. Parseval's identity for that finite orthonormal basis gives

$$\langle f, h \rangle = \sum_{\rho, i, j} d_\rho \widehat{f}(\rho)_{ji} \overline{\widehat{h}(\rho)_{ji}}.$$

The inner sum is precisely $\text{tr}(\widehat{f}(\rho) \widehat{h}(\rho)^*)$. Taking $h = f$ gives the norm identity. $\square$

Theorem 1.8 (Convolution block diagonalization). *For the convolution convention of Definition 1.3,*

$$\widehat{f * g}(\rho) = \widehat{g}(\rho) \widehat{f}(\rho).$$

Under the Plancherel identification

$$L^2(G) \cong \bigoplus_{\rho \in \text{Irr}(G)} M_{d_\rho}(\mathbb{C}),$$

*the operator $C_f : g \mapsto f * g$ acts on the ρ block by right multiplication $X \mapsto X \widehat{f}(\rho)$. If the singular values of $\widehat{f}(\rho)$ are $s_1, \dots, s_{d_\rho}$, then the singular values of this block are $s_1, \dots, s_{d_\rho}$, each repeated d_ρ times.*

Proof. From $x = yz$ and unitarity,

$$\begin{aligned} \widehat{f * g}(\rho) &= \mathbb{E}_{x, y} f(y) g(y^{-1}x) \rho(x)^* \\ &= \mathbb{E}_{y, z} f(y) g(z) \rho(z)^* \rho(y)^* \\ &= \widehat{g}(\rho) \widehat{f}(\rho). \end{aligned}$$

Plancherel therefore sends C_f to right multiplication by $\widehat{f}(\rho)$ on each matrix block.

Write a singular-value decomposition $\widehat{f}(\rho) = U\Sigma V^*$. Left or right multiplication by a unitary is unitary for the Hilbert–Schmidt inner product, so right multiplication by $\widehat{f}(\rho)$ has the same singular values as right multiplication by $\Sigma = \text{diag}(s_1, \dots, s_d)$. On the matrix-unit basis,

$$E_{ij}\Sigma = s_j E_{ij}.$$

Thus each s_j occurs once for every row index i , hence with multiplicity $d = d_\rho$. The Plancherel inner product on the ρ block is d_ρ times the Hilbert–Schmidt inner product; multiplying an inner product by a positive scalar does not change singular values, so the same multiplicities hold for the actual Plancherel identification. $\square$

Worked check. When G is abelian, every block is 1×1 . The formula becomes the usual scalar identity $\widehat{f * g}(\chi) = \widehat{g}(\chi)\widehat{f}(\chi)$; scalar multiplication commutes, so the apparent order reversal disappears.

Kernel-side warning. Theorem 1.8 fixes the kernel on the left: the input g is sent to $f * g$, which becomes right matrix multiplication by $\widehat{f}(\rho)$. A right random walk instead fixes a kernel on the right, sending h to $h * f$, and therefore becomes left matrix multiplication. Definition 5.1 records both operators explicitly so that the two uses are never conflated.

Chapter 2

Representation Dimension as a Mixing Parameter

The representation dimensions now become quantitative randomness parameters. The trivial Fourier block records only the mean; if every other block is large, Plancherel forces any individual nontrivial block to carry only a small fraction of the total L^2 mass.

Definition 2.1 (D -quasirandom group). For an integer $D \geq 1$, a finite group G is D -quasirandom if every nontrivial irreducible complex representation of G has dimension at least D . The one-dimensional trivial representation is excluded from this minimum. For a family of groups, D may depend on the family parameter.

Positive example. Every nonabelian finite simple group is at least 2-quasirandom. Indeed, a nontrivial one-dimensional representation would have a proper normal kernel; simplicity would force it to be faithful, embedding the nonabelian group into the abelian group $\mathbb{C}^\times$, a contradiction. In particular, A_5 is 2-quasirandom.

Nonexample. A nontrivial finite abelian group has nontrivial one-dimensional characters, so it is not D -quasirandom for any $D > 1$.

Lemma 2.2 (Quasirandom L^2 convolution mixing). *Let G be D -quasirandom. If $f, g : G \rightarrow \mathbb{C}$ and at least one of f, g has mean zero, then*

$$\|f * g\|_2 \leq D^{-1/2} \|f\|_2 \|g\|_2.$$

Proof. Suppose first that $\mathbb{E}f = 0$. Then $\widehat{f}(\mathbf{1}_G) = 0$. By convolution block diagonalization and Plancherel,

$$\|f * g\|_2^2 = \sum_{\rho \neq \mathbf{1}_G} d_\rho \|\widehat{g}(\rho) \widehat{f}(\rho)\|_{\text{HS}}^2.$$

For each nontrivial ρ ,

$$\|\widehat{g}(\rho)\|_{\text{op}}^2 \leq \|\widehat{g}(\rho)\|_{\text{HS}}^2 \leq \frac{\|g\|_2^2}{d_\rho} \leq \frac{\|g\|_2^2}{D},$$

where the middle inequality uses the fact that the single ρ term in Plancherel is at most the full sum. Hence

$$\begin{aligned} \|f * g\|_2^2 &\leq \frac{\|g\|_2^2}{D} \sum_{\rho \neq \mathbf{1}_G} d_\rho \|\widehat{f}(\rho)\|_{\text{HS}}^2 \\ &\leq D^{-1} \|g\|_2^2 \|f\|_2^2. \end{aligned}$$

Taking square roots proves the claim.

If instead $\mathbb{E}g = 0$, use

$$\|\widehat{g}(\rho)\widehat{f}(\rho)\|_{\text{HS}} \leq \|\widehat{g}(\rho)\|_{\text{HS}}\|\widehat{f}(\rho)\|_{\text{op}}$$

and bound $\|\widehat{f}(\rho)\|_{\text{op}} \leq \|f\|_2/\sqrt{d_\rho}$. The same calculation gives the identical estimate. $\square$

Worked check. For $D = 1$ the lemma reduces to the normalized Young inequality $\|f * g\|_2 \leq \|f\|_2\|g\|_2$ in this finite setting. The gain $D^{-1/2}$ appears only after the trivial Fourier block has been removed.

Chapter 3

Product-Free Sets

A set is product-free when the most elementary multiplicative pattern is missing completely. In a highly quasirandom group this omission is already expensive: representation-theoretic mixing forces the set to be sparse.

Definition 3.1 (Product-free set). A subset $A \subseteq G$ is *product-free* if there is no ordered triple $(a, b, c) \in A^3$ satisfying $ab = c$. Repeated elements are allowed by the equation. We write

$$\alpha = \frac{|A|}{|G|}$$

for the density of A .

Positive example. In the group $C_2 = \{e, a\}$, the singleton $\{a\}$ is product-free because $a^2 = e \notin \{a\}$.

Nonexample. Every nonempty subset containing the identity fails to be product-free: if $a \in A$, then $ea = a$ gives a forbidden triple.

Theorem 3.2 (Gowers product-free-set bound). *If G is D -quasirandom and $A \subseteq G$ is product-free, then*

$$\frac{|A|}{|G|} \leq D^{-1/3}.$$

Proof. Let $\alpha = |A|/|G|$ and put $f = \mathbf{1}_A - \alpha$. Then $\mathbb{E}f = 0$ and

$$\|f\|_2^2 = \alpha(1 - \alpha) \leq \alpha.$$

Because A is product-free,

$$0 = \mathbb{E}_{x,y} \mathbf{1}_A(x) \mathbf{1}_A(y) \mathbf{1}_A(xy).$$

Expand each indicator as $\alpha + f$. Every term containing one or two copies of f vanishes: for example, x and y are independent uniform variables, and for fixed x the product xy is uniform as y varies. Hence

$$0 = \alpha^3 + \mathbb{E}_{x,y} f(x) f(y) f(xy).$$

The last average equals

$$\mathbb{E}_z (f * f)(z) f(z),$$

so Cauchy–Schwarz and Lemma 2.2 give

$$\alpha^3 \leq \|f * f\|_2 \|f\|_2 \leq D^{-1/2} \|f\|_2^3 \leq D^{-1/2} \alpha^{3/2}.$$

If $\alpha = 0$ there is nothing to prove. Otherwise divide by $\alpha^{3/2}$ and raise to the power $2/3$ to obtain $\alpha \leq D^{-1/3}$. $\square$

Worked consequence. If a family G_n is D_n -quasirandom with $D_n \rightarrow \infty$, every product-free subset has density tending to zero. This is the qualitative phenomenon behind Gowers’s negative answer to the Babai–Sós question for suitable families of finite groups.

Worked comparison: density $1/2$ in an abelian group versus vanishing density in a quasirandom family. Consider the additive cyclic group $C_{2m} = \mathbb{Z}/(2m)\mathbb{Z}$ and the set

$$A_{\text{odd}} := \{1, 3, 5, \dots, 2m - 1\}.$$

It has exactly m elements, hence density $1/2$. The sum of two odd residue classes is even, so there are no $a, b, c \in A_{\text{odd}}$ with $a + b = c$. In multiplicative notation this is a product-free set of density $1/2$. The group is abelian, so its quasirandomness parameter is only $D = 1$; Theorem 3.2 then gives merely $\alpha \leq 1$, which does not obstruct this example.

Now take $G_p = \text{SL}_2(\mathbb{F}_p)$ with $p \geq 5$. By Lemma 7.15, G_p is $(p - 1)/2$ -quasirandom. Therefore every product-free $A \subseteq G_p$ satisfies

$$\frac{|A|}{|G_p|} \leq \left(\frac{2}{p - 1} \right)^{1/3}. \quad (3.1)$$

For example, at $p = 101$ the right side is

$$50^{-1/3} \approx 0.2714,$$

whereas the cyclic example keeps density exactly 0.5 for every m . The contrast is not caused by group size alone. It is the absence of low-dimensional nontrivial representations—quantified by D —that forces multiplicative patterns to appear in dense subsets.

Chapter 4

Triple-Product Mixing

The product-free theorem is one extreme case of a quantitative counting principle. We now state the principle directly for arbitrary functions and then recover its set form.

Definition 4.1 (Triple-product correlation). For functions $f_1, f_2, f_3 : G \rightarrow \mathbb{C}$, define

$$\Lambda(f_1, f_2, f_3) = \mathbb{E}_{x,y \in G} f_1(x) f_2(y) f_3(xy).$$

For a function f , its centered part is

$$f^0 = f - \mathbb{E}f.$$

For subsets A, B, C , the quantity $\Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C)$ is the normalized number of ordered solutions $x \in A, y \in B, xy \in C$.

Positive example. $\Lambda(1, 1, 1) = 1$. If A, B, C behaved independently under multiplication, the corresponding indicator correlation would be close to the product of their densities.

Nonexample. If $H < G$ is a proper subgroup of density α , then $\Lambda(\mathbf{1}_H, \mathbf{1}_H, \mathbf{1}_H) = \alpha^2$, not α^3 : conditioning $x, y \in H$ forces $xy \in H$. Large proper subgroups are thus strong obstructions to product mixing.

Theorem 4.2 (Triple-product mixing in a quasirandom group). *Let G be D -quasirandom. If at least one of $f_1, f_2, f_3 : G \rightarrow \mathbb{C}$ has mean zero, then*

$$|\Lambda(f_1, f_2, f_3)| \leq D^{-1/2} \|f_1\|_2 \|f_2\|_2 \|f_3\|_2.$$

Consequently, if $A, B, C \subseteq G$ have densities α, β, γ , then

$$|\Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C) - \alpha\beta\gamma| \leq D^{-1/2} \sqrt{\alpha\beta\gamma}.$$

Proof. By the convolution convention,

$$\Lambda(f_1, f_2, f_3) = \mathbb{E}_z (f_1 * f_2)(z) f_3(z) = \langle f_1 * f_2, \overline{f_3} \rangle.$$

If f_1 or f_2 has mean zero, Cauchy–Schwarz followed by Lemma 2.2 gives the desired bound immediately.

Suppose instead that $\mathbb{E}f_3 = 0$. Write $f_i = a_i + f_i^0$ for $i = 1, 2$. Since convolution with a constant averages the other factor,

$$f_1 * f_2 = a_1 a_2 + f_1^0 * f_2^0.$$

The constant term has zero correlation with f_3 , so

$$\Lambda(f_1, f_2, f_3) = \langle f_1^0 * f_2^0, \overline{f_3} \rangle.$$

Apply the convolution lemma and use $\|f_i^0\|_2 \leq \|f_i\|_2$ to obtain the same estimate.

For the set form, put $f_A = \mathbf{1}_A - \alpha$ and similarly define f_B, f_C . Expanding the three indicators, all terms with fewer than three centered factors vanish by uniformity of x , y , and xy . Therefore

$$\Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C) - \alpha\beta\gamma = \Lambda(f_A, f_B, f_C).$$

Each centered indicator satisfies

$$\|f_A\|_2^2 = \alpha(1 - \alpha) \leq \alpha,$$

and similarly for B, C . The functional estimate gives the displayed set bound. $\square$

Worked check. Taking $A = B = C$ product-free makes the left correlation zero. The set form then yields $\alpha^3 \leq D^{-1/2}\alpha^{3/2}$ and hence reproduces $\alpha \leq D^{-1/3}$.

Chapter 5

Convolution Mixing and Set Counts

The Fourier estimate of Chapter 2 can be packaged in the form used by Babai–Nikolov–Pyber. Before doing so, we freeze a kernel-side convention that will also be used by the random-walk argument. This matters only in the noncommutative setting: fixing the kernel on the left and fixing it on the right are genuinely different operators.

Definition 5.1 (Convolution operators and singular-value contraction). Let G be a finite group and let $f : G \rightarrow \mathbb{C}$. Under the normalized convolution of Definition 1.3, define the *left-kernel* and *right-kernel* convolution operators by

$$\mathcal{L}_f h := f * h, \quad \mathcal{R}_f h := h * f.$$

The names refer to the side on which the fixed kernel f appears; they avoid the ambiguous phrases “left convolution” and “right convolution.” Put

$$f^\sharp(x) := \overline{f(x^{-1})}.$$

Both $\mathcal{L}_f$ and $\mathcal{R}_f$ preserve $L_0^2(G)$. Their mean-zero operator contractions are

$$\sigma_0^L(f) := \|\mathcal{L}_f|_{L_0^2(G)}\|_{\text{op}}, \quad \sigma_0^R(f) := \|\mathcal{R}_f|_{L_0^2(G)}\|_{\text{op}}.$$

For a linear operator A on a finite-dimensional Hilbert space with orthonormal basis (e_j) , its Hilbert–Schmidt norm is

$$\|A\|_{\text{HS}(\text{op})}^2 := \sum_j \|Ae_j\|_2^2.$$

This notation is intentionally distinct from the matrix Hilbert–Schmidt norm $\|M\|_{\text{HS}}$ of Definition 1.1.

Lemma 5.2 (Exact convolution-operator dictionary). *For every $f : G \rightarrow \mathbb{C}$:*

(i) *on the ρ Fourier block,*

$$\widehat{\mathcal{L}_f h}(\rho) = \widehat{h}(\rho) \widehat{f}(\rho), \quad \widehat{\mathcal{R}_f h}(\rho) = \widehat{f}(\rho) \widehat{h}(\rho); \quad (5.1)$$

(ii)

$$\mathcal{L}_f^* = \mathcal{L}_{f^\sharp}, \quad \mathcal{R}_f^* = \mathcal{R}_{f^\sharp}; \quad (5.2)$$

(iii) *if $|G| > 1$, then*

$$\sigma_0^L(f) = \sigma_0^R(f) = \max_{\substack{\rho \in \text{Irr}(G) \\ \rho \neq \mathbf{1}_G}} \|\widehat{f}(\rho)\|_{\text{op}}; \quad (5.3)$$

for the trivial group all three quantities are understood as 0;

(iv) the full and mean-zero Hilbert–Schmidt norms satisfy

$$\|\mathcal{L}_f\|_{\text{HS}(\text{op})}^2 = \|\mathcal{R}_f\|_{\text{HS}(\text{op})}^2 = \|f\|_2^2, \quad (5.4)$$

$$\|\mathcal{L}_f|_{L_0^2}\|_{\text{HS}(\text{op})}^2 = \|\mathcal{R}_f|_{L_0^2}\|_{\text{HS}(\text{op})}^2 = \|f - \mathbb{E}f\|_2^2. \quad (5.5)$$

In particular, if $f^\sharp = f$, both convolution operators are self-adjoint.

Proof. The first identity in (5.1) is Theorem 1.8. Applying that theorem with the two factors interchanged gives

$$\widehat{h * f}(\rho) = \widehat{f}(\rho)\widehat{h}(\rho),$$

which is the second identity.

A direct change of variables gives

$$\widehat{f^\sharp}(\rho) = \widehat{f}(\rho)^*. \quad (5.6)$$

Indeed, with $z = x^{-1}$ and using unitarity,

$$\widehat{f^\sharp}(\rho) = \mathbb{E}_x \overline{f(x^{-1})} \rho(x)^* = \mathbb{E}_z \overline{f(z)} \rho(z) = \widehat{f}(\rho)^*.$$

Under Plancherel, taking the Hilbert-space adjoint of right multiplication by $\widehat{f}(\rho)$ gives right multiplication by $\widehat{f}(\rho)^*$, and the analogous statement holds for left multiplication. Equation (5.6) therefore proves (5.2).

If the singular values of $\widehat{f}(\rho)$ are $s_1, \dots, s_{d_\rho}$, Theorem 1.8 shows that right multiplication by this matrix has these singular values, each repeated d_ρ times. The same statement for left multiplication follows by using matrix units after a singular-value decomposition on the opposite side. Consequently both block operator norms equal $\|\widehat{f}(\rho)\|_{\text{op}}$. The trivial block is exactly the constant functions because $\widehat{f}(\mathbf{1}_G) = \mathbb{E}f$. Removing that block gives (5.3).

Summing the squares of all block singular values yields

$$\|\mathcal{L}_f\|_{\text{HS}(\text{op})}^2 = \sum_{\rho} d_{\rho} \|\widehat{f}(\rho)\|_{\text{HS}}^2 = \|f\|_2^2$$

by Plancherel; the same computation holds for $\mathcal{R}_f$. Removing the trivial block subtracts $|\mathbb{E}f|^2$, so

$$\|f\|_2^2 - |\mathbb{E}f|^2 = \|f - \mathbb{E}f\|_2^2,$$

which proves (5.5). Finally, (5.2) gives self-adjointness when $f^\sharp = f$. $\square$

Operator form of quasirandom mixing. If G is D -quasirandom, then Plancherel gives for every nontrivial ρ

$$\|\widehat{f}(\rho)\|_{\text{op}} \leq \|\widehat{f}(\rho)\|_{\text{HS}} \leq D^{-1/2} \|f - \mathbb{E}f\|_2.$$

Hence

$$\sigma_0^{\text{L}}(f) = \sigma_0^{\text{R}}(f) \leq D^{-1/2} \|f - \mathbb{E}f\|_2. \quad (5.7)$$

This is the operator-norm version of the convolution estimate used below.

Object	Physical-side action	Fourier-side action
$\mathcal{L}_f$	$h \mapsto f * h$	$X \mapsto X \widehat{f}(\rho)$
$\mathcal{R}_f$	$h \mapsto h * f$	$X \mapsto \widehat{f}(\rho) X$
normalized *	$ G ^{-1} \sum_y$	exact identities (5.1)
probability $*_{\text{m}}$	$\sum_y$	converted by $F_\nu = G \nu$ in Chapter 9

We now state the set/function mixing estimate in the form later arguments call.

Theorem 5.3 (Babai–Nikolov–Pyber convolution inequality, normalized form). *Let G be a finite D -quasirandom group. If $f, g : G \rightarrow \mathbb{C}$ and $\mathbb{E}f = 0$, then*

$$\|f * g\|_2 \leq D^{-1/2} \|f\|_2 \|g\|_2.$$

The same conclusion holds if $\mathbb{E}g = 0$ instead. Consequently, for arbitrary f, g ,

$$\|f * g - (\mathbb{E}f)(\mathbb{E}g)\|_2 \leq D^{-1/2} \|f - \mathbb{E}f\|_2 \|g - \mathbb{E}g\|_2.$$

Proof. The first two assertions are exactly Lemma 2.2. For the centered form, write $f = a + f^0$ and $g = b + g^0$, where $a = \mathbb{E}f$, $b = \mathbb{E}g$ and both centered parts have mean zero. Since normalized convolution with a constant averages the other factor,

$$f * g = ab + f^0 * g^0.$$

Applying the first assertion to $f^0 * g^0$ gives the displayed inequality. $\square$

Normalization check. If X, Y are probability distributions on a group of order n and $U \equiv 1/n$, set $f = n(X - U)$ and $g = n(Y - U)$, now regarded as functions under normalized counting measure. Then the preceding inequality is the same representation-degree L^2 gain as the classical unnormalized-distribution form after translating the two norm conventions. Keeping this translation visible prevents factors of n from being lost when probabilistic and harmonic-analysis formulations are compared.

Corollary 5.4 (Set mixing from representation dimension). *Let $A, B, C \subseteq G$ and let*

$$N(A, B, C) = |\{(a, b, c) \in A \times B \times C : ab = c\}|.$$

Then

$$\left| N(A, B, C) - \frac{|A||B||C|}{|G|} \right| \leq D^{-1/2} \sqrt{|G||A||B||C|}.$$

In particular, if

$$|A||B||C| > \frac{|G|^3}{D},$$

then there is at least one solution $ab = c$ with $a \in A$, $b \in B$, $c \in C$.

Proof. Put $n = |G|$ and let α, β, γ be the three densities. With $f_A = \mathbf{1}_A - \alpha$ and similarly for B, C , the lower-order terms in

$$\Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C)$$

vanish, so

$$\Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C) - \alpha\beta\gamma = \langle f_A * f_B, \overline{f_C} \rangle.$$

Cauchy–Schwarz and Theorem 5.3 give

$$|\Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C) - \alpha\beta\gamma| \leq D^{-1/2} \|f_A\|_2 \|f_B\|_2 \|f_C\|_2 \leq D^{-1/2} \sqrt{\alpha\beta\gamma}.$$

Because $N(A, B, C) = n^2 \Lambda(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C)$, multiplying by n^2 gives

$$D^{-1/2} n^2 \sqrt{\frac{|A||B||C|}{n^3}} = D^{-1/2} \sqrt{n|A||B||C|},$$

which is the claimed error term.

If $|A||B||C| > n^3/D$, then $\alpha\beta\gamma > D^{-1}$. Hence $\alpha\beta\gamma > D^{-1/2} \sqrt{\alpha\beta\gamma}$, so the normalized solution count is strictly positive. Therefore $N(A, B, C) > 0$. $\square$

Worked check. For $A = B = C = S$, the sufficient condition becomes $|S| > |G|/D^{1/3}$. Thus every set above the product-free threshold has a solution $xy = z$ inside itself, exactly matching the preceding Gowers bound.

Exact finite-group laboratory. The executable source file `code/volume_06/noncommutative_lab.py` uses only Python's standard library and exact rational arithmetic. It implements the counting-sum probability convolution, right-walk matrices, involution, return probabilities, normalized-versus-counting L^2 conversion, and exact product powers on S_3 . Its self-tests include

$$P_\nu P_\eta = P_{\nu *_{\text{m}} \eta}, \quad P_\nu^* = P_{\nu^\vee}, \quad \text{tr}(P_\nu^m) = |G| \nu^{*_{\text{m}} m}(e),$$

and

$$\|F_\nu - 1\|_{L^2}^2 = |G| \|\nu - u\|_{\ell^2}^2.$$

For the uniform law on the three transpositions of S_3 , the program recovers

$$\|\nu\|_{\ell^2}^2 = \frac{1}{3}, \quad |\text{supp}(\nu^{*_{\text{m}} 2})| = 3, \quad \text{tr}(P_\nu^2) = 2,$$

so the non-flattening coset example of Chapter 9 is checked independently of the prose calculation. For the identity-containing set $\{e, \tau, \sigma\}$, with τ a transposition and σ a 3-cycle, the exact product-power sizes are

$$3, 6, 6, 6,$$

showing saturation of S_3 at the second product power.

Chapter 6

Approximate Subgroups and Multiplicative Energy

The representation-theoretic estimates of the preceding chapters become useful for expansion only after one can recognize when a probability measure, or a finite set carrying much multiplicative structure, is concentrating near an algebraic object. In an abelian group this role is played by additive energy, Ruzsa calculus, and approximate groups. None of those ideas is intrinsically commutative. What changes in a nonabelian group is that the order of every factor must be tracked and that small doubling alone no longer controls all longer products.

This chapter develops the exact finite-group tools needed later in the Bourgain–Gamburd flattening mechanism. There are two logically different routes. Large multiplicative energy gives large subsets with a small product set (Balog–Szemerédi–Gowers), whereas a deterministic small-tripling hypothesis directly produces an approximate group. We keep the routes separate so that later applications do not silently use a stronger hypothesis than they possess.

6.1 Approximate groups, energy, and Ruzsa distance

Definition 6.1 (Approximate group and control). Let G be a group and let $K \geq 1$. A finite nonempty set $H \subseteq G$ is a K -approximate group if

$$e \in H, \quad H^{-1} = H,$$

and there is a finite set $X \subseteq G$ with $|X| \leq K$ such that

$$H^2 \subseteq XH.$$

If $A, B \subseteq G$ are finite and nonempty and $L \geq 1$, we say that A is L -controlled by B if

$$|B| \leq L|A|$$

and A is contained in the union of at most L left translates of B and also in the union of at most L right translates of B . Equivalently, there are finite $X, Y \subseteq G$, $|X|, |Y| \leq L$, such that

$$A \subseteq XB \cap BY.$$

First examples. Every finite subgroup is a 1-approximate group. More generally, if g has order greater than $4N$ and

$$H_N = \{g^j : -N \leq j \leq N\},$$

then H_N is symmetric, contains e , and

$$H_N^2 \subseteq g^{-N} H_N \cup g^N H_N,$$

so H_N is a 2-approximate group. Thus an approximate group need not be a genuine subgroup: it is a finite set whose square can be covered by few translates of itself.

Remark 6.2 (One-sided covering is enough). Because H is symmetric, $H^2 \subseteq XH$ implies after inversion that $H^2 \subseteq HX^{-1}$. We therefore do not build a second covering condition into the definition. When a symmetric covering set is convenient, one may replace X by $X \cup X^{-1}$ at a factor-two cost.

Definition 6.3 (Multiplicative energy and noncommutative Ruzsa distance). Let $A, B \subseteq G$ be finite and nonempty. For $x \in G$ put

$$r_{A,B}(x) := |\{(a, b) \in A \times B : ab = x\}|.$$

The *multiplicative energy* of the ordered pair (A, B) is

$$E(A, B) := \sum_{x \in G} r_{A,B}(x)^2 = |\{(a, b, a', b') \in A \times B \times A \times B : ab = a'b'\}|.$$

The (left-invariant) *Ruzsa distance* is

$$d(A, B) := \log \frac{|AB^{-1}|}{|A|^{1/2}|B|^{1/2}}.$$

The word “distance” is conventional: although d is symmetric and obeys a triangle inequality, one can have $d(A, A) > 0$.

The energy normalization above is deliberately unnormalized. It is related to the normalized convolution convention of Definition 1.3, but keeping $E(A, B)$ as a literal count makes every power of $|A|$ and $|B|$ visible in the combinatorial argument.

Lemma 6.4 (Noncommutative Ruzsa triangle and covering). *Let $A, B, C \subseteq G$ be finite and nonempty. Then*

$$|AC^{-1}||B| \leq |AB^{-1}||BC^{-1}|. \quad (6.1)$$

Consequently

$$d(A, C) \leq d(A, B) + d(B, C). \quad (6.2)$$

Moreover, if $L \geq 1$, then:

(i) *if $|AB| \leq L|A|$, there is $X \subseteq B$ with $|X| \leq L$ and*

$$B \subseteq A^{-1}AX;$$

(ii) *if $|BA| \leq L|A|$, there is $X \subseteq B$ with $|X| \leq L$ and*

$$B \subseteq XAA^{-1}.$$

Proof. For every $z \in AC^{-1}$ choose once and for all one representation $z = a_z c_z^{-1}$ with $a_z \in A$ and $c_z \in C$. Define

$$\Phi : (AC^{-1}) \times B \longrightarrow (AB^{-1}) \times (BC^{-1}), \quad \Phi(z, b) = (a_z b^{-1}, b c_z^{-1}).$$

The product of the two output coordinates is z . Thus the output determines z ; once z is known, the chosen a_z is known and the first coordinate then determines b . Hence Φ is injective, proving (6.1). Dividing by $|A|^{1/2}|C|^{1/2}|B|$ and taking logarithms gives (6.2).

For the covering statement assume first that $|AB| \leq L|A|$. Choose $X \subseteq B$ maximal subject to the right translates Ax , $x \in X$, being pairwise disjoint. They all lie in AB , so

$$|X||A| \leq |AB| \leq L|A|,$$

and therefore $|X| \leq L$. By maximality, for every $b \in B$ the set Ab meets some Ax , $x \in X$. Thus $a_1 b = a_2 x$ for suitable $a_1, a_2 \in A$, which gives $b \in A^{-1}AX$. This proves (i). Applying the same argument to pairwise disjoint left translates xA proves (ii). $\square$

Two useful corollaries. Taking $B = A^{-1}$ in (6.2) gives

$$d(A, A) \leq 2d(A, A^{-1}). \quad (6.3)$$

Also, every product of two nonempty finite sets has cardinality at least the cardinality of either factor: fixing one factor gives an injective translate. We will use this elementary observation repeatedly when a denominator in the Ruzsa distance is discarded.

6.2 A quantitative graph lemma

The graph-theoretic heart of Balog–Szemerédi–Gowers is that a dense bipartite graph contains large vertex subsets joined by many short paths. We record a fully quantitative length-three version. Its constants are not optimized; their role is to make every subsequent polynomial loss explicit.

Lemma 6.5 (Many length-three paths in a dense bipartite graph). *Let $\Gamma \subseteq A \times B$ be a bipartite graph with*

$$|A| = n, \quad |B| = m, \quad |E(\Gamma)| \geq \frac{nm}{Q}, \quad Q \geq 1.$$

Then there are $A' \subseteq A$ and $B' \subseteq B$ such that

$$|A'| \geq \frac{n}{4\sqrt{2}Q}, \quad |B'| \geq \frac{m}{4Q}, \quad (6.4)$$

and every $(a, b) \in A' \times B'$ is joined by at least

$$\frac{nm}{2^{15}Q^6} \quad (6.5)$$

ordered paths

$$a - b_1 - a_1 - b$$

of length three in Γ .

Proof. Let

$$A_1 := \left\{ a \in A : \deg(a) \geq \frac{m}{2Q} \right\}.$$

The vertices outside A_1 account for at most $nm/(2Q)$ edges, so

$$e(A_1, B) \geq \frac{nm}{2Q}. \quad (6.6)$$

In particular $A_1 \neq \emptyset$.

For $a, a' \in A_1$ let

$$\text{codeg}(a, a') := |N(a) \cap N(a')|.$$

Put

$$\varepsilon := \frac{1}{64Q^2}, \quad \theta := \frac{\varepsilon m}{16Q^2} = \frac{m}{1024Q^4},$$

and call the ordered pair (a, a') *bad* if its codegree is less than θ .

Choose b_0 uniformly from B and set $N = N(b_0) \cap A_1$. From (6.6),

$$\mathbb{E}|N| \geq \frac{n}{2Q}, \quad \mathbb{E}|N|^2 \geq \frac{n^2}{4Q^2}. \quad (6.7)$$

Let Z be the number of bad ordered pairs lying in $N \times N$. A fixed bad pair belongs to $N \times N$ with probability $\text{codeg}(a, a')/m < \varepsilon/(16Q^2)$; hence

$$\mathbb{E}Z < \frac{\varepsilon|A_1|^2}{16Q^2} \leq \frac{\varepsilon n^2}{16Q^2}. \quad (6.8)$$

Combining (6.7) and (6.8),

$$\mathbb{E} \left(|N|^2 - \frac{2Z}{\varepsilon} \right) \geq \frac{n^2}{8Q^2}.$$

Therefore some choice of b_0 yields a set $A_2 := N$ satisfying

$$|A_2| \geq \frac{n}{2\sqrt{2}Q}, \quad Z \leq \frac{\varepsilon}{2}|A_2|^2. \quad (6.9)$$

For $a \in A_2$ let $z(a)$ be the number of $a' \in A_2$ for which (a, a') is bad, and define

$$A' := \{a \in A_2 : z(a) \leq \sqrt{\varepsilon}|A_2|\}.$$

Since $\sum_{a \in A_2} z(a) = Z$, at most $(\sqrt{\varepsilon}/2)|A_2|$ vertices are removed. In particular $|A'| \geq |A_2|/2$, which gives the first inequality in (6.4).

Every vertex of $A_2 \subseteq A_1$ has degree at least $m/(2Q)$. Set

$$B' := \left\{ b \in B : \deg_{A_2}(b) \geq \frac{|A_2|}{4Q} \right\}.$$

The edges incident to $B \setminus B'$ contribute less than $|A_2|m/(4Q)$, whereas $e(A_2, B) \geq |A_2|m/(2Q)$. Thus $|B'| \geq m/(4Q)$.

Fix $a \in A'$ and $b \in B'$. The vertex b has at least $|A_2|/(4Q)$ neighbors $a_1 \in A_2$. At most

$$\sqrt{\varepsilon}|A_2| = \frac{|A_2|}{8Q}$$

of them form a bad pair with a . Hence at least $|A_2|/(8Q)$ choices of a_1 are good. Each good pair (a, a_1) has at least $\theta = m/(1024Q^4)$ common neighbors b_1 . Therefore the number of length-three paths from a to b is at least

$$\frac{|A_2|}{8Q} \frac{m}{1024Q^4} \geq \frac{nm}{2^{14}\sqrt{2}Q^6} \geq \frac{nm}{2^{15}Q^6},$$

which is (6.5). □

6.3 Large energy produces structured subsets

Theorem 6.6 (Noncommutative Balog–Szemerédi–Gowers, explicit finite form). *Let A, B be finite nonempty subsets of an arbitrary group, put $n = |A|$ and $m = |B|$, and let $K \geq 1$. Suppose*

$$E(A, B) \geq \frac{n^{3/2}m^{3/2}}{K}. \quad (6.10)$$

Then there are subsets $A' \subseteq A$ and $B' \subseteq B$ such that

$$|A'| \geq \frac{n}{8\sqrt{2}K}, \quad |B'| \geq \frac{m}{8K}, \quad (6.11)$$

and

$$|A'B'| \leq 2^{24}K^9\sqrt{nm}. \quad (6.12)$$

Moreover,

$$|A'(A')^{-1}| \leq 2^{51}K^{19}n, \quad |(B')^{-1}B'| \leq 2^{52}K^{19}m. \quad (6.13)$$

Proof. Write $r(x) = r_{A,B}(x)$ and set

$$t := \frac{\sqrt{nm}}{2K}.$$

Call an edge $(a, b) \in A \times B$ *popular* when $r(ab) \geq t$, and let Γ be the resulting bipartite graph. Since

$$E(A, B) = \sum_{(a,b) \in A \times B} r(ab),$$

the nonpopular pairs contribute less than

$$nm t = \frac{n^{3/2}m^{3/2}}{2K}.$$

By (6.10), the popular pairs therefore contribute at least the same amount. For every x , however,

$$r(x) \leq \min(n, m) \leq \sqrt{nm}.$$

It follows that

$$|E(\Gamma)| \geq \frac{nm}{2K}. \quad (6.14)$$

Apply Lemma 6.5 with $Q = 2K$. We obtain subsets A', B' satisfying (6.11), and every $(a, b) \in A' \times B'$ is joined by at least

$$\frac{nm}{2^{21}K^6} \quad (6.15)$$

length-three paths $a - b_1 - a_1 - b$ in the popular graph.

Fix $c \in A'B'$ and choose one representation $c = ab$ with $a \in A'$, $b \in B'$. A path $a - b_1 - a_1 - b$ gives three popular products

$$x_1 = ab_1, \quad x_2 = a_1b_1, \quad x_3 = a_1b$$

with

$$c = x_1x_2^{-1}x_3. \quad (6.16)$$

For fixed endpoints a, b , the triple (x_1, x_2, x_3) determines the path, so distinct paths give distinct product triples. Each x_i has at least t representations as an element of AB . Consequently, if

$$F := \mathbf{1}_A *_u \mathbf{1}_B *_u \mathbf{1}_{B^{-1}} *_u \mathbf{1}_{A^{-1}} *_u \mathbf{1}_A *_u \mathbf{1}_B,$$

where $*_u$ denotes the unnormalized convolution

$$(f *_u g)(x) := \sum_{y \in G} f(y)g(y^{-1}x),$$

then (6.15)–(6.16) give

$$F(c) \geq \frac{nm}{2^{21}K^6} \left(\frac{\sqrt{nm}}{2K} \right)^3 = \frac{(nm)^{5/2}}{2^{24}K^9}. \quad (6.17)$$

On the other hand, the ℓ^1 mass of F is exactly n^3m^3 . Summing (6.17) over $c \in A'B'$ proves (6.12).

Put $P = A'B'$. For every $x \in A'(A')^{-1}$ choose one representation $x = aa'^{-1}$. Varying $b \in B'$ gives $|B'|$ distinct pairs

$$(ab, a'b) \in P \times P$$

with quotient x . Pairs belonging to different quotients are disjoint, so

$$|A'(A')^{-1}| |B'| \leq |P|^2.$$

Using (6.11) and (6.12),

$$|A'(A')^{-1}| \leq \frac{2^{48}K^{18}nm}{m/(8K)} = 2^{51}K^{19}n.$$

Similarly, varying $a \in A'$ in

$$b^{-1}b' = (ab)^{-1}(ab')$$

gives

$$|(B')^{-1}B'| |A'| \leq |P|^2 \leq 2^{48}K^{18}nm,$$

and therefore

$$|(B')^{-1}B'| \leq 2^{51}\sqrt{2}K^{19}m \leq 2^{52}K^{19}m.$$

□

Worked extremal check. If $A = B = H$ is a finite subgroup, then $r_{H,H}(x) = |H|$ for $x \in H$ and zero otherwise, so $E(H, H) = |H|^3$. Thus (6.10) holds with $K = 1$. The structured conclusion is not mysterious in this case: one may take $A' = B' = H$, and $A'B' = H$. The theorem is useful precisely because it recovers a polynomially weakened version of this behavior from energy alone, without assuming that A or B is already close to a subgroup.

Remark 6.7 (Comparison with the canonical theorem). Tao's finite specialization gives the same energy normalization and the same subset-size scales $|A|/(8\sqrt{2}K)$ and $|B|/(8K)$, with a sharper $K^{O(1)}$ product-set estimate (in the displayed proof, an $O(K^8)$ loss). The explicit exponents in (6.12)–(6.13) are intentionally conservative: they come from the self-contained length-three-path argument above and are sufficient for the later flattening interface.

6.4 Small tripling produces an approximate group

Large energy and small tripling are related but should not be conflated. The next theorem needs no energy hypothesis and no application of Theorem 6.6; Ruzsa's triangle and covering arguments are enough.

Theorem 6.8 (Small tripling implies explicit approximate-group control). *Let A be a finite nonempty subset of an arbitrary group and let $K \geq 1$. Assume*

$$|A^3| \leq K|A|. \quad (6.18)$$

Put

$$H_0 := A \cup A^{-1} \cup \{e\}, \quad H := H_0^3.$$

Then:

(i) for every $r \geq 3$ and every choice $\epsilon_1, \dots, \epsilon_r \in \{-1, +1\}$,

$$|A^{\epsilon_1} \dots A^{\epsilon_r}| \leq K^{7(r-2)}|A|; \quad (6.19)$$

(ii)

$$|H| \leq 27K^7|A|; \quad (6.20)$$

(iii) H is a 3^7K^{35} -approximate group;

(iv) A is $27K^7$ -controlled by H .

Proof. We first establish all signed words of length at most three. From (6.18), fixing one factor at a time gives

$$|A| \leq |A^2| \leq |A^3| \leq K|A|. \quad (6.21)$$

Hence

$$d(A, A^{-1}) = \log \frac{|A^2|}{|A|} \leq \log K \quad (6.22)$$

and

$$d(A^2, A^{-1}) = \log \frac{|A^3|}{|A^2|^{1/2}|A|^{1/2}} \leq \log K. \quad (6.23)$$

By the triangle inequality, $d(A^2, A) \leq 2 \log K$. Since $|A^2| \leq K|A|$, this gives

$$|A^2 A^{-1}| \leq K^2 |A^2|^{1/2} |A|^{1/2} \leq K^3 |A|. \quad (6.24)$$

Consequently $d(A, AA^{-1}) \leq 3 \log K$. Combining this with (6.22),

$$d(AA^{-1}, A^{-1}) \leq 4 \log K.$$

Also, by (6.3),

$$|AA^{-1}| \leq K^2|A|.$$

Applying the same argument to A^{-1} gives the companion bound

$$|A^{-1}A| \leq K^2|A|. \quad (6.25)$$

Therefore

$$|AA^{-1}A| \leq K^4 |AA^{-1}|^{1/2} |A|^{1/2} \leq K^5 |A|. \quad (6.26)$$

It follows that $d(A, A^{-1}A) \leq 5 \log K$, and another use of (6.22) gives

$$d(A^{-1}, A^{-1}A) \leq 6 \log K.$$

Using the second bound in (6.25),

$$|A^{-1}A^{-1}A| \leq K^6 |A|^{1/2} |A^{-1}A|^{1/2} \leq K^7 |A|. \quad (6.27)$$

Together with (6.18), (6.24), (6.26), and their inverse-set versions, this shows that every signed word of length at most three has size at most $K^7 |A|$.

Now suppose $r \geq 4$ and that (6.19) is known for all shorter lengths. Let

$$X = A^{\epsilon_1} \dots A^{\epsilon_{r-2}}, \quad Y = A^{-\epsilon_r} A^{-\epsilon_{r-1}}.$$

Then $XY^{-1} = A^{\epsilon_1} \dots A^{\epsilon_r}$. Applying the cardinality form (6.1) with the middle set equal to A gives

$$|XY^{-1}| |A| \leq |XA^{-1}| |AY^{-1}|. \quad (6.28)$$

The first factor on the right is a signed word of length $r-1$ and the second one a signed word of length three. Hence, by induction and the base case,

$$|XY^{-1}| |A| \leq K^{7(r-3)} |A| K^7 |A|,$$

which is exactly (6.19).

Every element of H_0^3 lies in one of at most 3^3 words obtained by choosing A , A^{-1} , or $\{e\}$ in each position. After deleting identity factors, every such word has length at most three, so the preceding bounds give

$$|H| = |H_0^3| \leq 3^3 K^7 |A| = 27 K^7 |A|.$$

The same argument with seven factors and (6.19) gives

$$|H_0^7| \leq 3^7 K^{35} |A|. \quad (6.29)$$

Since $|H_0| \geq |A|$ and $H^2 = H_0^6$,

$$|H_0 H^2| = |H_0^7| \leq 3^7 K^{35} |H_0|.$$

Apply Lemma 6.4(i) to the pair (H_0, H^2) . There is $Y \subseteq H^2$ with $|Y| \leq 3^7 K^{35}$ such that

$$H^2 \subseteq H_0^{-1} H_0 Y = H_0^2 Y \subseteq H Y.$$

The set H is symmetric, so taking inverses gives

$$H^2 \subseteq Y^{-1} H.$$

Thus H is a $3^7 K^{35}$ -approximate group.

Finally $A \subseteq H$ and (6.20) holds. Since $A \subseteq eH \cap He$, the set A lies in one left and one right translate of H . Hence it is $27 K^7$ -controlled by H . $\square$

Why tripling, not merely doubling? In a nonabelian group a set may have small $|A^2|$ while $|A^3|$ is much larger. For instance, if H is a finite subgroup and x lies outside its normalizer, the set $A = H \cup \{x\}$ can have $|A^2|$ comparable to $|A|$ while the mixed products appearing in A^3 probe the larger double coset HxH . The hypothesis (6.18) controls precisely the extra order-sensitive products needed in the proof.

Worked nonabelian check. Let $G = S_3$ and

$$A = \{e, (12), (23)\}.$$

Then $A = A^{-1}$, while

$$A^2 = \{e, (12), (23), (123), (132)\}$$

has five elements. The missing transposition appears at the next step because $(12)(23)(12) = (13)$, so $A^3 = S_3$ and

$$|A^3| = 6 = 2|A|.$$

Thus Theorem 6.8 applies with $K = 2$. Here $H_0 = A$ and $H = H_0^3 = S_3$, which is in fact a 1-approximate group. The theorem's polynomial constants are therefore intentionally worst-case guarantees, not predictions of the optimal control parameter in a particular small group.

Interface to product growth. Theorem 6.8 does not classify approximate subgroups of $\mathrm{SL}_2(\mathbb{F}_p)$. It performs the combinatorial reduction: small tripling forces a controlled approximate group. The next chapter's Helfgott frontier must then use the algebraic structure of $\mathrm{SL}_2(\mathbb{F}_p)$ to show that a generating set cannot remain in such a small-growth regime unless it becomes very large. Keeping this boundary explicit prevents the deep algebraic input from being hidden inside Balog–Szemerédi–Gowers terminology.

6.5 The global approximate-group structure theorem

For the next source-owned theorem, the *rank* of a finitely generated group means the least number of generators, and the *step* of a nilpotent group is the length of its lower central series minus one. The notation $O_K(1)$ means a finite quantity depending only on K ; no effective formula for that dependence is asserted.

Theorem 6.9 (Breuillard–Green–Tao structure theorem — source-owned). *Let A be a finite K -approximate group in a group G in the sense of Definition 6.1. Then there exist a subgroup $G_0 \leq G$ and a finite normal subgroup $H \trianglelefteq G_0$ such that*

- (i) *A is covered by $O_K(1)$ left translates of G_0 ;*
- (ii) *G_0/H is finitely generated and nilpotent, with rank and step $O_K(1)$;*
- (iii) *A^4 contains H and a generating set of G_0 .*

In particular G_0 is finite-by-nilpotent and virtually nilpotent.

Source and quantitative boundary. This is Theorem 1.6 of Breuillard–Green–Tao, *The structure of approximate groups*, Publ. Math. IHÉS 116 (2012), 115–221, DOI 10.1007/s10240-012-0043-9. The theorem is qualitative: its $O_K(1)$ constants depend only on K , but the general proof does not supply an effective bound for the covering constant. The stronger coset-nilprogression formulation in their Theorem 2.10 is therefore not paraphrased here as an effective polynomial-control theorem. This distinction is important when approximate-group structure is later compared with algorithmic growth bounds.

Chapter 7

Product Growth, Escape, and Generating Sets

The representation-theoretic mixing results from the preceding chapters become substantially more powerful when they are combined with a mechanism that forces a generating set to leave algebraic obstructions and enlarge under multiplication. This chapter fixes the product-set language that will be used for that later growth theory. We deliberately separate the definitions from the deeper subgroup-escape and product-growth theorems: the notation itself is elementary, while the quantitative escape estimates for $\mathrm{SL}_2(\mathbb{F}_p)$ require additional group structure.

Definition 7.1 (Product powers, generation, and product growth). Let G be a finite group and let $A \subseteq G$. Put

$$A^0 := \{e\}, \quad A^m := \{a_1 a_2 \cdots a_m : a_1, \dots, a_m \in A\} \quad (m \geq 1).$$

The subgroup generated by A is

$$\langle A \rangle := \bigcup_{m \geq 0} (A \cup A^{-1})^m.$$

We say that A *generates* G if $\langle A \rangle = G$.

For a real number $\delta > 0$, we say that A has *δ -product growth at scale 3* if

$$|A^3| \geq |A|^{1+\delta}.$$

A product theorem for a specified family of finite groups is a theorem which, under explicitly stated size and subgroup hypotheses, forces a dichotomy of the form

$$A^3 = G \quad \text{or} \quad |A^3| \geq |A|^{1+\delta},$$

with the allowed range of A , the value or uniformity of δ , and every subgroup exception made explicit.

Positive example. In the cyclic group $G = \mathbb{Z}/7\mathbb{Z}$ written additively, $A = \{0, 1\}$ generates G . Its three-fold sumset is

$$3A = \{0, 1, 2, 3\},$$

so $|3A| = 4 > |A|$. This illustrates genuine growth without asserting any uniform exponent for this small example.

Nonexample. If $H < G$ is a proper subgroup and $A = H$, then $A^m = H$ for every $m \geq 1$. Thus A has no product growth at all and cannot generate G . This is the basic obstruction that quantitative escape statements must rule out when one passes from arbitrary subsets to generating subsets.

Remark 7.2. The condition $A^3 = G$ is not the negation of product growth; both may hold simultaneously. A product theorem asserts that at least one of the two useful alternatives holds. In nonabelian applications one often first replaces A by $A \cup A^{-1} \cup \{e\}$ so that symmetry and the identity are available. Any such normalization will be stated explicitly when it is used.

7.1 Logarithmic girth and quantitative subgroup escape

The next theorem supplies the subgroup-escape input needed by the later flattening argument. There is an important hypothesis to make visible at the outset. A bare assertion that a set generates $\mathrm{SL}_2(\mathbb{F}_p)$ does not by itself provide the local tree structure used in the Bourgain–Gamburd escape mechanism. We therefore work with a fixed-degree symmetric generating set and assume logarithmic girth.

Fix $k \geq 2$. Write $F_k = \langle \tilde{s}_1, \dots, \tilde{s}_k \rangle$ for the free group with its standard symmetric alphabet, and let $B_{F_k}(R)$ denote its reduced-word ball of radius R . If

$$S = \{s_1, s_1^{-1}, \dots, s_k, s_k^{-1}\} \subseteq G \quad (|S| = 2k),$$

we use *girth* to mean the length of the shortest nonempty reduced word in the corresponding free alphabet that evaluates to the identity in G . Let

$$\mu = \frac{1}{2k} \sum_{s \in S} \delta_s, \quad (\nu *_{\mathrm{m}} \eta)(x) = \sum_{y \in G} \nu(y) \eta(y^{-1}x).$$

The decoration $*_{\mathrm{m}}$ marks ordinary counting-sum probability convolution, in contrast with the normalized function convolution $*$ of Definition 1.3; Chapter 9 records the full probability-mass convention. Write $\mu^{*_{\mathrm{m}} m}$ for the m -fold probability convolution power.

Lemma 7.3 (Pointwise decay on the free group). *Let $\tilde{\mu}$ be the uniform probability measure on the $2k$ standard generators and inverses of F_k , where $k \geq 2$. Put*

$$\rho_k := \frac{\sqrt{2k-1}}{k} < 1.$$

Then for every $m \geq 0$ and every $u, v \in F_k$,

$$\tilde{p}^{(m)}(u, v) := \Pr_u[X_m = v] \leq \rho_k^m.$$

Proof. The Cayley graph of F_k is the d -regular tree with $d = 2k$. Let A be its adjacency operator on ℓ^2 and root the tree at the identity. Orient every edge away from the root. For finitely supported f and $c = (d-1)^{-1/2}$, the inequality

$$2|ab| \leq c|a|^2 + c^{-1}|b|^2$$

gives

$$|\langle Af, f \rangle| \leq \sum_{x \rightarrow y} (c|f(x)|^2 + c^{-1}|f(y)|^2).$$

Every nonroot vertex occurs once as a child and $d - 1$ times as a parent, so its total coefficient is

$$c^{-1} + (d - 1)c = 2\sqrt{d - 1}.$$

The root has coefficient $dc = d/\sqrt{d - 1} \leq 2\sqrt{d - 1}$. Hence

$$\|A\|_{2 \rightarrow 2} \leq 2\sqrt{d - 1}.$$

The transition operator is $P = A/d$, and therefore

$$\|P\|_{2 \rightarrow 2} \leq \frac{2\sqrt{d - 1}}{d} = \frac{\sqrt{2k - 1}}{k} = \rho_k.$$

Finally,

$$\tilde{p}^{(m)}(u, v) = \langle \delta_v, P^m \delta_u \rangle \leq \|P\|_{2 \rightarrow 2}^m \|\delta_u\|_2 \|\delta_v\|_2 \leq \rho_k^m.$$

Density of finitely supported functions justifies the operator bound on all of ℓ^2 . $\square$

Why the decay is useful. The estimate is pointwise, not merely an L^2 statement. If a finite Cayley graph has no nontrivial reduced relation of length at most $2m$, then every endpoint reached in m steps has the same path-count as its unique lift in the free group. Thus the same ρ_k^m atom bound holds for the finite walk up to that radius.

Lemma 7.4 (A metabelian-law subset of a free ball is small). *Let F be a nonabelian free group and let $R \geq 1$. Suppose $\Sigma \subseteq B_F(R)$ satisfies*

$$[[g_1, g_2], [g_3, g_4]] = 1 \quad \text{for every } g_1, g_2, g_3, g_4 \in \Sigma,$$

where $[x, y] = xyx^{-1}y^{-1}$. Then

$$|\Sigma| \leq (8R + 1)(4R + 2).$$

Proof. We first record the elementary centralizer fact needed below. A nontrivial element w of a free group acts on its Cayley tree as a translation along a unique bi-infinite axis. If z commutes with w , then z preserves this axis. It cannot reverse the axis orientation, since conjugating a nontrivial translation by an orientation-reversing tree isometry changes that translation to its inverse. Because the free-group action on vertices is free, the resulting action of $C_F(w)$ on the oriented axis embeds in the integer translation group. Thus $C_F(w)$ is cyclic. If c is a primitive generator of this centralizer, its translation length $\ell_T(c)$ is a positive integer and $\ell_T(c^n) = |n|\ell_T(c)$. Hence $c^n \in B_F(L)$ forces $|n| \leq L$, so a cyclic subgroup meets a word ball of radius L in at most $2L + 1$ elements.

Set

$$T := \{[g, h] : g, h \in \Sigma\}.$$

Every element of T lies in $B_F(4R)$, and the assumed double-commutator law says that the elements of T commute pairwise. If $T \neq \{1\}$, choose a nontrivial $t_0 \in T$; then $T \subseteq C_F(t_0)$, which is cyclic. The same bound is trivial if $T = \{1\}$. Consequently

$$|T| \leq 8R + 1.$$

Fix $a \in T$ and count pairs $(g, h) \in \Sigma^2$ with $[g, h] = a$. If $g \neq 1$ and one solution h_0 exists, then any other solution h satisfies

$$h_0^{-1}h \in C_F(g).$$

Since $h, h_0 \in B_F(R)$, this centralizer element lies in $B_F(2R)$, giving at most $4R + 1$ possibilities for h . The exceptional row $g = 1$ contributes at most $|\Sigma|$ pairs, and only when $a = 1$. Hence every commutator fiber contains at most

$$(4R + 2)|\Sigma|$$

pairs. Summing the fibers over $a \in T$ gives

$$|\Sigma|^2 \leq |T|(4R + 2)|\Sigma|.$$

If Σ is nonempty, cancellation of $|\Sigma|$ and the bound on $|T|$ prove the claim; the empty case is immediate. $\square$

Lemma 7.5 (Central-lift-safe Dickson consequence). *Let $p \geq 5$ be prime and let $H < \mathrm{SL}_2(\mathbb{F}_p)$ be proper. Then at least one of the following holds:*

- (i) $|H| \leq 120$;
- (ii) H is metabelian, equivalently $H'' = [H', H'] = 1$.

Proof. We use Dickson's classical classification in its projective form. For a proper subgroup of $\mathrm{PSL}_2(\mathbb{F}_p)$, the nonexceptional cases are contained in a Borel subgroup or in the normalizer of a split or nonsplit torus; the exceptional projective images are A_4, S_4, A_5 and therefore have order at most 60.

Let $\pi : \mathrm{SL}_2(\mathbb{F}_p) \rightarrow \mathrm{PSL}_2(\mathbb{F}_p)$ be the quotient by $Z = \{\pm I\}$. First note that a proper H cannot satisfy $\pi(H) = \mathrm{PSL}_2(\mathbb{F}_p)$. Indeed, if it did, then either $Z \subseteq H$, forcing $H = \mathrm{SL}_2(\mathbb{F}_p)$, or $H \cap Z = \{I\}$, making H an index-two subgroup. The latter is impossible because $\mathrm{SL}_2(\mathbb{F}_p)$ is perfect for $p \geq 5$. For completeness, choose $a \in \mathbb{F}_p^\times$ with $a^2 \neq 1$. Conjugation by $\mathrm{diag}(a, a^{-1})$ shows that every upper unipotent matrix is a commutator, and the same holds for every lower unipotent matrix; these two unipotent families generate $\mathrm{SL}_2(\mathbb{F}_p)$.

Thus $\pi(H)$ is proper. If it is exceptional, then

$$|H| = |H \cap Z| |\pi(H)| \leq 2 \cdot 60 = 120.$$

If $\pi(H)$ lies in a Borel subgroup, then H lies in its full inverse image, which is the upper-triangular subgroup after conjugacy. Its commutator subgroup lies in the abelian unipotent radical, so it is metabelian. If $\pi(H)$ lies in a torus normalizer, then H lies in the inverse image of that normalizer; it has an abelian torus subgroup of index at most two, so again its commutator subgroup lies in an abelian group. Hence $H'' = 1$. $\square$

Remark 7.6 (Why the constant is 120, not 60). Projective formulations of Dickson's list have exceptional groups of order at most 60. In $\mathrm{SL}_2(\mathbb{F}_p)$ one must allow their central double covers. The safe threshold for the present argument is therefore 120. This small bookkeeping point matters because a binary exceptional subgroup need not be metabelian.

Theorem 7.7 (Quantitative escape from proper subgroups of $\mathrm{SL}_2(\mathbb{F}_p)$). *Fix $k \geq 2$ and $\tau > 0$. Put*

$$a_k := \log_{2k} \left(\frac{k^2}{2k-1} \right) > 0, \quad \beta := \frac{\tau a_k}{512}.$$

Then there exists $p_0 = p_0(k, \tau)$ such that the following holds for every prime $p \geq p_0$. Let

$$G = \mathrm{SL}_2(\mathbb{F}_p), \quad S = \{s_1, s_1^{-1}, \dots, s_k, s_k^{-1}\}$$

be a symmetric generating set of cardinality $2k$ satisfying

$$\text{girth}(G, S) \geq \tau \log_{2k} p,$$

and let μ be uniform on S . Define

$$r_p := \left\lfloor \frac{\tau}{64} \log_{2k} p \right\rfloor.$$

For every proper subgroup $H < G$, every $a \in G$, and every integer $\ell \geq r_p$,

$$\mu^{*m\ell}(aH) \leq p^{-\beta}.$$

Consequently, for every such H and ℓ ,

$$\mu^{*m2\ell}(H) \leq p^{-\beta}.$$

Proof. Write $r = r_p$ and $\rho = \rho_k$. By increasing p_0 we may assume $r \geq 1$ and

$$32r < \tau \log_{2k} p \leq \text{girth}(G, S). \quad (7.1)$$

The evaluation map from F_k to G is therefore injective on every word ball needed below: two words of length at most $2r$ representing the same group element would produce a nontrivial relation of length at most $4r$. Likewise, any word of length at most $32r$ which evaluates to the identity must already be trivial in F_k .

Because paths of length r in the finite Cayley graph have exactly the same endpoint identifications as paths in the free Cayley tree, Lemma 7.3 gives

$$\|\mu^{*mr}\|_{\ell^\infty} \leq \rho^r.$$

Since

$$\rho^2 = \frac{2k-1}{k^2} = (2k)^{-a_k}$$

and $r \geq (\tau/64) \log_{2k} p - 1$, we obtain

$$\|\mu^{*mr}\|_{\ell^\infty} \leq \rho^{-1} p^{-\tau a_k / 128} = \rho^{-1} p^{-4\beta}. \quad (7.2)$$

We claim first that, once p_0 is large enough,

$$\mu^{*mr}(aH) \leq p^{-\beta} \quad (7.3)$$

for every proper H and every coset aH . Suppose instead that a coset violates (7.3), and let

$$\Sigma := \text{supp}(\mu^{*mr}) \cap aH.$$

By (7.2),

$$|\Sigma| \geq \frac{\mu^{*mr}(aH)}{\|\mu^{*mr}\|_{\ell^\infty}} > \rho p^{3\beta}. \quad (7.4)$$

If $|H| \leq 120$, this is impossible for all sufficiently large p . We may therefore apply Lemma 7.5 and assume that $H'' = 1$.

Choose $x_0 \in \Sigma$ and set $T = x_0^{-1}\Sigma \subseteq H$. Each element of T has word length at most $2r$ in G . By (7.1) it has a unique lift to $B_{F_k}(2r)$, and distinct elements of T have distinct lifts. Call the lifted set $\tilde{T}$. For any four $\tilde{t}_1, \dots, \tilde{t}_4 \in \tilde{T}$, the word

$$[[\tilde{t}_1, \tilde{t}_2], [\tilde{t}_3, \tilde{t}_4]]$$

has length at most $32r$. Its image in G is the identity because $T \subseteq H$ and $H'' = 1$. By (7.1) the free word itself is therefore trivial. Lemma 7.4, applied with radius $2r$, yields

$$|\Sigma| = |\tilde{T}| \leq (16r + 1)(8r + 2). \quad (7.5)$$

The right side is $O_{k,\tau}((\log p)^2)$, whereas (7.4) grows like the positive power $p^{3\beta}$. This contradiction proves (7.3).

Now let $\ell \geq r$. For any left coset aH , associativity of convolution gives

$$\begin{aligned} \mu^{*\mathfrak{m}\ell}(aH) &= \sum_{y \in G} \mu^{*\mathfrak{m}(\ell-r)}(y) \mu^{*\mathfrak{m}r}(y^{-1}aH) \\ &\leq \sup_{b \in G} \mu^{*\mathfrak{m}r}(bH) \leq p^{-\beta}, \end{aligned}$$

which proves the asserted coset estimate at every later time. Finally,

$$\mu^{*\mathfrak{m}2\ell}(H) = \sum_{y \in G} \mu^{*\mathfrak{m}\ell}(y) \mu^{*\mathfrak{m}\ell}(y^{-1}H) \leq p^{-\beta} \sum_{y \in G} \mu^{*\mathfrak{m}\ell}(y) = p^{-\beta}.$$

This is the subgroup nonconcentration required by the later flattening interface. $\square$

Worked constant check. For $k = 2$, the free-walk factor is $\rho_2 = \sqrt{3}/2$ and $a_2 = \log_4(4/3) > 0$. The theorem deliberately retains only $\beta = \tau a_2/512$ of this decay. The large numerical loss pays for a transparent radius $r_p \leq (\tau/64) \log_4 p$ and is not intended to be sharp.

Hypothesis audit. The logarithmic-girth assumption is not decorative: it is what lets the finite walk inherit the free-tree atom bound and what turns the metabelian law in a proper subgroup into an actual free-group relation. The factor 64 was chosen conservatively so that the longest double-commutator word used above has length strictly below the available girth. No claim of optimality is made for β .

Generating without controlled local geometry is insufficient for this proof. The old registry wording mentioned only a generating set. That is too weak for the argument just given. For example, let $M < G$ be a large maximal proper subgroup and choose $g \notin M$. Then

$$S' = (M \setminus \{e\}) \cup \{g, g^{-1}\}$$

is symmetric and generates G , but the uniform one-step measure on S' is concentrated overwhelmingly on M as $|M| \rightarrow \infty$. The present theorem therefore records the fixed-degree logarithmic-girth hypothesis explicitly, rather than hiding it inside the phrase “early random walk.”

7.2 Helfgott growth at intermediate scales

The subgroup-escape theorem above is a random-walk statement. The second algebraic input needed by the Bourgain–Gamburd machine is different: a symmetric generating set which is still well below the size of the whole group cannot have persistently small tripling. We now prove the exact intermediate-scale form needed later.

There are two normalization points to keep visible. First, throughout this section the set A is assumed symmetric and to contain the identity. Hence our exact product power A^m is also the ball of words of length at most m , which is the convention used in Helfgott’s original proof. Second, the only deep scalar input is isolated explicitly below. All subsequent centralizer, escape, pivot, and trace-counting steps are proved in the manuscript.

Theorem 7.8 (Deep input: finite-field trace-form growth). *For every $\eta > 0$ there are constants $C_\eta \geq 1$ and $\kappa_\eta > 0$ with the following property. Let p be prime, put $E = \mathbb{F}_{p^2}$, let $\alpha, \beta \in E^\times$, and let $X \subset E^\times$ satisfy*

$$C_\eta < |X| < p^{1-\eta}.$$

Then

$$\left| \left\{ \alpha(xy + x^{-1}y^{-1}) + \beta(x^{-1}y + xy^{-1}) : x, y \in X^{20} \right\} \right| > |X|^{1+\kappa_\eta}, \quad (7.6)$$

where X^{20} denotes the twenty-fold multiplicative product set in $E^\times$.

Status of the input. This is the $q = p^2$ specialization of Proposition 3.3 in Helfgott’s *Growth and generation in $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$* , Annals of Mathematics 167 (2008), 601–623. Its proof is the finite-field sum–product layer of the argument. We keep it as a visibly labelled deep input rather than silently folding a sum–product theorem into the group proof. No later step in this section uses an additional unlabelled external result.

Mechanism and downstream use. In the primary proof, the trace form is forced to grow by combining finite-field sum–product with additive-combinatorial energy/Ruzsa reductions; the unusual restriction $|X| < p^{1-\eta}$ is essential because $\mathbb{F}_p^\times$, viewed inside $\mathbb{F}_{p^2}^\times$, is a genuine subfield obstruction near size p . The coefficients α, β are allowed to be arbitrary nonzero scalars. Later we use the theorem only with $\alpha = ad$ and $\beta = -bc$ from a generic pivot matrix; its sole downstream purpose in this chapter is to turn a large torus slice into a larger trace set.

Lemma 7.9 (Bounded-word escape from finitely many subspaces). *Let Γ act linearly on a d -dimensional vector space V over a field K . Let $A \subseteq \Gamma$ be finite, symmetric, contain the identity, and generate Γ . Suppose*

$$W = W_1 \cup \dots \cup W_s$$

is a union of s proper linear subspaces of V , and let $\mathcal{O} = \Gamma x$ be an orbit not contained in W . There are integers $m = m(d, s) \geq 1$ and constants $\vartheta = \vartheta(d, s) > 0$ such that, for every $x_0 \in \mathcal{O}$,

$$|\{g \in A^m : gx_0 \notin W\}| \geq \max\{1, \vartheta|A|\}. \quad (7.7)$$

Proof. We first prove a bounded escape list. For a finite union U of proper subspaces, order its complexity lexicographically by

$$(d_U, s_U),$$

where d_U is the largest dimension of a component and s_U is the number of components of that dimension. We claim, by induction on this pair, that there are words $q_1, \dots, q_L$ in A of length at most R , with L, R depending only on d and the number of components of U , such that for every $y \in \mathcal{O}$ at least one $q_i y$ lies outside U .

The zero-complexity case is immediate. For the induction step let U_+ be the union of the components of maximal dimension. If $\mathcal{O} \cap U_+ = \emptyset$, delete U_+ and use induction. Otherwise, because $\mathcal{O}$ is not contained in U_+ , along a word in the generators joining a point of $\mathcal{O} \cap U_+$ to a point outside U_+ there is a first exit. Thus there are $y_0 \in \mathcal{O} \cap U_+$ and $a \in A$ with $ay_0 \notin U_+$. The union

$$U' := U \cap a^{-1}U$$

has strictly smaller complexity: no maximal-dimensional component of U_+ can survive unchanged in both factors. Its number of components is bounded in terms of the old number alone. By

induction choose $q'_1, \dots, q'_{L'}$ escaping U' . For every y , some $q'_i y$ lies outside U' , hence either $q'_i y \notin U$ or $aq'_i y \notin U$. The list

$$q'_1, \dots, q'_{L'}, aq'_1, \dots, aq'_{L'}$$

therefore escapes U . This completes the bounded-list induction.

Apply the resulting list to $U = W$. Fix $x_0 \in \mathcal{O}$. For every $a \in A$, apply the list to ax_0 and choose an index $i(a)$ for which $q_{i(a)}ax_0 \notin W$. The elements $q_{i(a)}a$ lie in A^{R+1} . A fixed group element has at most L representations of the form $q_i a$ with $i \in \{1, \dots, L\}$, because $a = q_i^{-1}g$ is then determined by i . Hence at least $|A|/L$ distinct elements of A^{R+1} escape W . Taking $m = R + 1$ and $\vartheta = 1/L$, and retaining one escaping word when $|A|/L < 1$, proves (7.7). $\square$

Lemma 7.10 (Regular semisimple abundance and torus extraction). *Let $p \geq 5$, let $G = \mathrm{SL}_2(\mathbb{F}_p)$, and let $A \subseteq G$ be finite, symmetric, contain e , and generate G . If $|A| \geq 32$, then A^2 contains at least $|A|/8$ elements whose trace is not ± 2 .*

Moreover, writing $\mathrm{Tr}(A) = \{\mathrm{tr}(a) : a \in A\}$, there is a subset $V \subseteq A^4$ contained in the centralizer of one regular semisimple element, and hence simultaneously diagonalizable over $\mathbb{F}_{p^2}$, such that

$$|V| \geq \frac{(|\mathrm{Tr}(A)| - 2)|A|}{8|A^6|}. \quad (7.8)$$

Proof. Call an element *regular* when its trace is not ± 2 . If already at least $|A|/8$ elements of A are regular, the first assertion follows from $A \subseteq A^2$. Otherwise more than $7|A|/8$ elements of A have trace ± 2 . Since at most two of these are central, choose a noncentral one, call it g . It has a unique eigenline $L \subset \mathbb{F}_p^2$.

Let B be the elements $h \in A$ with trace ± 2 which preserve L . First suppose $|B| \leq |A|/2$. More than $3|A|/8$ trace- ± 2 elements h do not preserve L . In a basis with first vector on L ,

$$g = s \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad s \in \{\pm 1\}, \quad t \neq 0,$$

and such an h has $h_{21} \neq 0$. Therefore

$$\mathrm{tr}(g^{\pm 1}h) = s(\mathrm{tr}(h) \pm th_{21}).$$

The two displayed values cannot both belong to $\{\pm 2\}$: their average is $s\mathrm{tr}(h) \in \{\pm 2\}$, and in characteristic at least five the only way two elements of $\{\pm 2\}$ have that average is for them to be equal, which would force $th_{21} = 0$. Thus at least one of $gh, g^{-1}h$ is regular. Choosing one such product for every h , any output has at most two preimages, so A^2 contains more than $3|A|/16 > |A|/8$ regular elements.

Now suppose $|B| > |A|/2$. Since the stabilizer of L is a proper subgroup and A generates G , choose $x \in A$ which does not preserve L . In the same basis $x_{21} \neq 0$, while every $h \in B$ has the form

$$h = s \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \quad (s = \pm 1).$$

Then

$$\mathrm{tr}(hx) = s(\mathrm{tr}(x) + ux_{21}).$$

There are at most two values of u for which the parenthesis is ± 2 , and at most two choices of s for each u . Hence at most four elements of B produce a nonregular hx . Since $|A| \geq 32$, more than $|A|/8$ distinct products $hx \in A^2$ are regular. This proves the first assertion.

For the second assertion, let $R \subseteq A^2$ be the set of all regular elements. We have $|R| \geq |A|/8$, and because $A \subseteq A^2$,

$$|\mathrm{Tr}(R)| \geq |\mathrm{Tr}(A)| - 2. \quad (7.9)$$

We use the following elementary class-centralizer count. If S is any finite nonempty subset of a group and Λ_S is the set of conjugacy classes meeting S , then some $z \in S$ satisfies

$$|C_G(z) \cap S^{-1}S| \geq \frac{|\Lambda_S||S|}{|SSS^{-1}|}. \quad (7.10)$$

Indeed, choose one representative z from each class meeting S . For a fixed z , the map $a \mapsto aza^{-1}$, $a \in S$, has fibers of size at most $|C_G(z) \cap S^{-1}S|$. The conjugates arising from distinct chosen classes are disjoint and all lie in SSS^{-1} ; summing the resulting lower bounds and averaging proves (7.10).

Apply (7.10) to $S = R$. Different trace values lie in different conjugacy classes, so $|\Lambda_R| \geq |\mathrm{Tr}(R)|$. Since $R^{-1}R \subseteq A^4$ and $RRR^{-1} \subseteq A^6$, (7.9) gives a regular $z \in R$ for which

$$V := C_G(z) \cap R^{-1}R$$

satisfies (7.8). A regular element of $\mathrm{SL}_2(\mathbb{F}_p)$ has two distinct eigenlines over $\mathbb{F}_{p^2}$, and every matrix commuting with it preserves those eigenlines. Thus all elements of V are diagonal in one common eigenbasis over $\mathbb{F}_{p^2}$. $\square$

Lemma 7.11 (Generic pivot and cubic torus amplification). *There is an absolute integer $m_{\mathrm{piv}} \geq 1$ with the following property. Let $p \geq 5$, let $A \subseteq \mathrm{SL}_2(\mathbb{F}_p)$ be finite, symmetric, contain e , and generate the group. Suppose $V \subseteq \mathrm{SL}_2(\mathbb{F}_p)$ is simultaneously diagonalizable over $E = \mathbb{F}_{p^2}$ with distinct common eigenlines L_1, L_2 . Then there is $g \in A^{m_{\mathrm{piv}}}$ such that*

$$gL_i \neq L_j \quad (i, j \in \{1, 2\}). \quad (7.11)$$

For every such g ,

$$|VgVg^{-1}V| \geq \frac{1}{2} \left(\frac{|V|}{4} - 5 \right) |V|^2. \quad (7.12)$$

In particular, if $|V| \geq 80$, then

$$|VgVg^{-1}V| \geq \frac{1}{16} |V|^3. \quad (7.13)$$

Proof. Work in the four-dimensional E -vector space $M_2(E)$, with $\mathrm{SL}_2(\mathbb{F}_p)$ acting by left multiplication. For $i, j \in \{1, 2\}$ let

$$W_{ij} := \{T \in M_2(E) : TL_i \subseteq L_j\}.$$

These are proper linear subspaces. The orbit of I is the matrix group $G = \mathrm{SL}_2(\mathbb{F}_p)$. It is not contained in $W = \bigcup_{i,j} W_{ij}$. To see this, if the condition $hL_i = L_j$ has one solution in G , its solution set is a coset of the stabilizer of L_i . For an $\mathbb{F}_p$ -rational line, choosing it as the first coordinate line identifies the stabilizer with the upper-triangular determinant-one matrices, of order $p(p-1)$. If the line is nonrational, an $\mathbb{F}_p$ -matrix stabilizing it also stabilizes its Frobenius-conjugate line; in the resulting eigenbasis its eigenvalue $\lambda \in \mathbb{F}_{p^2}^\times$ satisfies $\lambda^{p+1} = 1$, so the stabilizer has at most $p+1$ elements. Thus every one of the four conditions contains at most $p(p-1)$ elements, whereas

$$|G| = p(p^2 - 1) > 4p(p-1)$$

for $p \geq 5$. Lemma 7.9, with $d = 4$ and $s = 4$, therefore supplies a word length m_{piv} independent of p, A, V and an element $g \in A^{m_{\mathrm{piv}}}$ satisfying (7.11).

Choose the common eigenbasis. Then

$$V = \left\{ D_r := \begin{pmatrix} r & 0 \\ 0 & r^{-1} \end{pmatrix} : r \in X \right\}$$

for some $X \subset E^\times$, and (7.11) says that if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $abcd \neq 0$. A direct multiplication gives

$$gD_rg^{-1} = \begin{pmatrix} rad - r^{-1}bc & (r^{-1} - r)ab \\ (r - r^{-1})cd & r^{-1}ad - rbc \end{pmatrix}. \quad (7.14)$$

The product of the off-diagonal entries is $-(r - r^{-1})^2 abcd$. The map $r \mapsto (r - r^{-1})^2$ has fibers of size at most four. Moreover the two diagonal entries in (7.14) can both avoid being nonzero only for at most four values of r , since their simultaneous vanishing forces $r^2 - r^{-2} = 0$. Hence there is a subset $U \subseteq gVg^{-1}$ for which all four entries relevant below are nonzero and

$$|\{h_{12}h_{21} : h \in U\}| \geq |V|/4 - 5. \quad (7.15)$$

Fix $h \in U$. For $D_s, D_t \in V$ consider

$$F_h(s, t) = D_s h D_t = \begin{pmatrix} sth_{11} & st^{-1}h_{12} \\ s^{-1}th_{21} & s^{-1}t^{-1}h_{22} \end{pmatrix}.$$

Because the four displayed entries of h are nonzero, a fixed output has at most two preimages (s, t) : the ratios of two possible pairs must satisfy $uv = 1$ and $u/v = 1$, leaving only the simultaneous sign ambiguity. In addition, $F_h(s, t)_{12}F_h(s, t)_{21} = h_{12}h_{21}$, so outputs coming from chosen h 's with distinct values of $h_{12}h_{21}$ are disjoint. Choosing one h for every value in (7.15) yields

$$|VgVg^{-1}V| \geq \frac{1}{2}(|V|/4 - 5)|V|^2,$$

which is (7.12). If $|V| \geq 80$, then $|V|/4 - 5 \geq 3|V|/16$, and (7.13) follows from the weaker constant $1/16$. $\square$

Lemma 7.12 (Uniform trace abundance in bounded word length). *There are absolute constants $m_{\text{tr}} \geq 1$ and $c_{\text{tr}} > 0$ such that, for every prime $p \geq 5$ and every finite symmetric generating set $A \subseteq \text{SL}_2(\mathbb{F}_p)$ containing e ,*

$$|\text{Tr}(A^{m_{\text{tr}}})| \geq c_{\text{tr}}|A|^{1/3}. \quad (7.16)$$

Proof. Sets A of bounded absolute size can be absorbed by decreasing c_{tr} , so assume $|A| \geq 32$. By Lemma 7.10, choose a regular element $h \in A^2$. Over $E = \mathbb{F}_{p^2}$ it has distinct eigenlines L_1, L_2 and, in their basis,

$$h = \begin{pmatrix} r & 0 \\ 0 & r^{-1} \end{pmatrix}, \quad r \neq r^{-1}. \quad (7.17)$$

In $M_2(E)$ let W_i be the linear subspace of matrices preserving L_i . The orbit of I under $G = \text{SL}_2(\mathbb{F}_p)$ is not contained in $W_1 \cup W_2$. Indeed, the stabilizer of an $\mathbb{F}_p$ -rational line has order $p(p-1)$, while a nonrational line and its Frobenius conjugate are stabilized together and give a norm-one torus of order at most $p+1$. Thus each stabilizer has order at most $p(p-1)$, whereas $|G| = p(p^2-1) > 2p(p-1)$. Lemma 7.9 therefore gives absolute m and $\vartheta > 0$ and a set

$$X := A^m \setminus (W_1 \cup W_2)$$

with $|X| \geq \vartheta|A|$. In the eigenbasis (7.17), every $x \in X$ has $x_{12}x_{21} \neq 0$.

Put

$$D := \{(x_{11}, x_{22}) : x \in X\}.$$

We need two elementary trace counts. First,

$$|\mathrm{Tr}(XX^{-1})| \geq \frac{|X|}{2|D|}. \quad (7.18)$$

Indeed, choose a fiber of the map $x \mapsto (x_{11}, x_{22})$ of size at least $|X|/|D|$, and fix one matrix x_0 in that fiber. If the common diagonal entries are a, d , then $bc = ad - 1 \neq 0$ for every matrix in the fiber. Parameterizing the upper-right entry by $t \neq 0$ forces the lower-left entry to be $(ad - 1)/t$. For fixed x_0 , the equation $\mathrm{tr}(x_0x^{-1}) = \lambda$ is therefore a quadratic equation in t , and has at most two solutions. This proves (7.18).

Second,

$$|\mathrm{Tr}(hX)| \geq \frac{|D|}{|\mathrm{Tr}(X)|}. \quad (7.19)$$

Partition D by the value $t = a + d$. Some trace fiber contains at least $|D|/|\mathrm{Tr}(X)|$ diagonal pairs. On such a fiber,

$$\mathrm{tr}(hx) = ra + r^{-1}d = (r - r^{-1})a + r^{-1}t,$$

which is injective in a because $r \neq r^{-1}$. Choosing one x for each diagonal pair proves (7.19).

Let $M = 2m + 2$. The three sets XX^{-1} , hX , and X all lie in A^M . Multiplying (7.18), (7.19), and the identity $|\mathrm{Tr}(X)| = |\mathrm{Tr}(X)|$ gives

$$|\mathrm{Tr}(A^M)|^3 \geq |\mathrm{Tr}(XX^{-1})| |\mathrm{Tr}(hX)| |\mathrm{Tr}(X)| \geq \frac{|X|}{2} \geq \frac{\vartheta|A|}{2}.$$

Thus (7.16) holds with $m_{\mathrm{tr}} = M$ and $c_{\mathrm{tr}} = (\vartheta/2)^{1/3}$. $\square$

Lemma 7.13 (Trace size forces product size). *There is an absolute integer $m_{\mathrm{amp}} \geq 1$ such that the following holds. Let $p \geq 5$ and let $B \subseteq \mathrm{SL}_2(\mathbb{F}_p)$ be finite, symmetric, contain e , generate the group, and satisfy $|B| \geq 32$. Put*

$$Q(B) := \frac{(|\mathrm{Tr}(B)| - 2)|B|}{8|B^6|}. \quad (7.20)$$

If $Q(B) \geq 80$, then

$$|B^{m_{\mathrm{amp}}}| \geq \frac{1}{16}Q(B)^3. \quad (7.21)$$

Proof. Lemma 7.10 gives a simultaneously diagonalizable $V \subseteq B^4$ with $|V| \geq Q(B)$. Apply Lemma 7.11 with $A = B$. Its pivot belongs to $B^{m_{\mathrm{piv}}}$, and hence

$$VgVg^{-1}V \subseteq B^{12+2m_{\mathrm{piv}}}.$$

Since $|V| \geq 80$, (7.13) gives

$$|B^{12+2m_{\mathrm{piv}}}| \geq \frac{1}{16}|V|^3 \geq \frac{1}{16}Q(B)^3.$$

Take $m_{\mathrm{amp}} = 12 + 2m_{\mathrm{piv}}$. $\square$

Theorem 7.14 (Helfgott product growth in $\mathrm{SL}_2(\mathbb{F}_p)$: symmetric intermediate-scale form). *Fix $0 < \delta < 3$. There are constants $c_\delta > 0$ and $\varepsilon_\delta > 0$ such that, for every prime $p \geq 5$ and every finite set*

$$A \subseteq G := \mathrm{SL}_2(\mathbb{F}_p)$$

satisfying

$$e \in A = A^{-1}, \quad \langle A \rangle = G, \quad |A| < p^{3-\delta}, \quad (7.22)$$

one has

$$|A^3| \geq c_\delta |A|^{1+\varepsilon_\delta}. \quad (7.23)$$

The primes 2 and 3 are excluded only to keep the regular-semisimple and small subgroup bookkeeping uniform; they are finite exceptional cases and can be absorbed by decreasing c_δ if an all-primes statement is desired.

Proof. Set

$$K := \frac{|A^3|}{|A|} \geq 1.$$

By Theorem 6.8, applied to the present symmetric set, every fixed product power satisfies the convenient coarse estimate

$$|A^r| \leq K^{7r} |A| \quad (r \geq 1). \quad (7.24)$$

For $r \geq 3$ this is weaker than (6.19), and the cases $r = 1, 2$ follow from $A \subseteq A^2 \subseteq A^3$.

Let $m_0 = m_{\mathrm{tr}}$ and $c_0 = c_{\mathrm{tr}}$ be the constants in Lemma 7.12. Put $B = A^{m_0}$. Then

$$|\mathrm{Tr}(B)| \geq c_0 |A|^{1/3}. \quad (7.25)$$

We may assume $|A|$ exceeds a sufficiently large absolute constant depending on δ ; otherwise (7.23) follows from $|A^3| \geq |A|$ after decreasing c_δ . Lemma 7.10, (7.24), and $|B| \geq |A|$ give a torus subset $V_0 \subseteq B^4 = A^{4m_0}$ with

$$|V_0| \geq c_1 |A|^{1/3} K^{-C_1}, \quad C_1 := 42m_0, \quad (7.26)$$

for an absolute $c_1 > 0$. Here we used $|B^6| = |A^{6m_0}| \leq K^{42m_0} |A|$ and absorbed the harmless subtraction of 2 in (7.25).

Apply the deep input Theorem 7.8 with $\eta = \delta/3$, and write $C = C_{\delta/3}$ and $\kappa = \kappa_{\delta/3} > 0$. If the right side of (7.26) is at most $2C + 2$, then

$$K^{C_1} \geq c_\delta |A|^{1/3},$$

and (7.23) already follows with a positive exponent depending only on δ . Hence assume the right side of (7.26) is larger than $2C + 2$. Choose a subset $V \subseteq V_0$ of cardinality

$$N := \lfloor c_1 |A|^{1/3} K^{-C_1} \rfloor > C. \quad (7.27)$$

After reducing c_1 if necessary, $N \leq |A|^{1/3}$. By (7.22),

$$N < p^{1-\delta/3}. \quad (7.28)$$

Choose the common eigenbasis of V over $E = \mathbb{F}_{p^2}$ and write

$$V = \left\{ \mathrm{diag}(x, x^{-1}) : x \in X \right\}, \quad |X| = N.$$

Lemma 7.11 supplies $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in A^{m_{\text{piv}}}$ whose four entries in this basis are nonzero. In particular $ad, -bc \in E^\times$. If $x, y \in X^{20}$, the corresponding diagonal matrices lie in $V^{20} \subseteq A^{80m_0}$, and direct multiplication gives

$$\text{tr} \left(\begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix} g \begin{pmatrix} y & 0 \\ 0 & y^{-1} \end{pmatrix} g^{-1} \right) = ad(xy + x^{-1}y^{-1}) - bc(x^{-1}y + xy^{-1}). \quad (7.29)$$

Thus, with

$$M := 160m_0 + 2m_{\text{piv}},$$

(7.6), (7.27)–(7.29) imply

$$|\text{Tr}(A^M)| > N^{1+\kappa}. \quad (7.30)$$

Notice that the left side is a trace set in $\mathbb{F}_p$, even though the diagonalization and the scalar calculation took place in E .

Put $B_1 = A^M$ and $C_2 := 42M$. From (7.20), (7.24), and (7.30), after increasing the bounded-size threshold for A if necessary,

$$Q(B_1) \geq \frac{1}{16} N^{1+\kappa} K^{-C_2}. \quad (7.31)$$

If $Q(B_1) < 80$, then (7.31) and the lower bound $N \geq (c_1/2)|A|^{1/3}K^{-C_1}$ coming from (7.27) give

$$K^{C_2+C_1(1+\kappa)} \geq c_\delta |A|^{(1+\kappa)/3},$$

which again implies (7.23) with a positive δ -dependent exponent.

It remains to consider $Q(B_1) \geq 80$. Lemma 7.13 and (7.31) yield

$$|A^{Mm_{\text{amp}}}| \geq c_2 N^{3(1+\kappa)} K^{-3C_2} \quad (7.32)$$

with an absolute $c_2 > 0$. On the other hand, (7.24) gives

$$|A^{Mm_{\text{amp}}}| \leq K^{7Mm_{\text{amp}}} |A|. \quad (7.33)$$

Using again $N \geq (c_1/2)|A|^{1/3}K^{-C_1}$ in (7.32), then comparing with (7.33), gives

$$K^{D_\delta} \geq c_\delta |A|^\kappa, \quad (7.34)$$

where

$$D_\delta := 7Mm_{\text{amp}} + 3C_2 + 3C_1(1+\kappa)$$

is finite and depends on δ only through κ . Hence

$$K \geq c'_\delta |A|^{\kappa/D_\delta}.$$

Taking $\varepsilon_\delta > 0$ to be the minimum of the positive exponents obtained in the three cases above, and decreasing c_δ to cover the bounded exceptional range, proves (7.23). $\square$

What has and has not been proved. The theorem is the symmetric intermediate-scale specialization of part (a) of Helfgott's 2008 Key Proposition. The primary paper states the more general unsymmetrized form and separately proves a bounded-word generation statement for sufficiently large sets. The present normalization is deliberately the one consumed by the forthcoming flattening argument: its structured sets are symmetric approximate groups, and only power growth below the ambient-group scale is needed there. We therefore do not disguise the separate large-set covering branch as an $A^3 = G$ assertion.

Worked small-prime check. For $p = 5$, let

$$u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \ell = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad A = \{I, u, u^{-1}, \ell, \ell^{-1}\}.$$

The upper and lower unipotent subgroups generate $\mathrm{SL}_2(\mathbb{F}_5)$, so A is a symmetric generating set. Writing a matrix as the four-tuple (a, b, c, d) and multiplying modulo 5, direct enumeration gives

$$|A| = 5, \quad |A^2| = 17, \quad |A^3| = 43, \quad |A^4| = 91, \quad |A^5| = 117, \quad |A^6| = 120 = |G|.$$

Thus this small example shows substantial early product growth and then bounded-word saturation. It is intentionally a normalization check: the theorem is asymptotic and its constants are not calibrated to make the $p = 5$ inequality numerically sharp. The displayed exact powers coincide with the corresponding bounded word balls because $I \in A = A^{-1}$.

Source-parity audit. Helfgott's Key Proposition (a) assumes $|A| < p^{3-\delta}$ and gives $|AAA| > c|A|^{1+\varepsilon}$, with c, ε depending only on δ . Proposition 3.3 is exactly the trace-form input isolated in Theorem 7.8; Proposition 4.1, Lemma 4.2, Corollaries 4.3–4.6, Lemma 4.7, Proposition 4.8, Lemma 4.9, and Proposition 4.10 supply the primary proof architecture reconstructed above. The use of Theorem 6.8 replaces Helfgott's short noncommutative Ruzsa power-reduction lemma by the stronger explicit small-tripling control already proved in Chapter 6; it changes no hypothesis or conclusion of the symmetric specialization.

7.2.1 The concrete quasirandomness parameter for $\mathrm{SL}_2(\mathbb{F}_p)$

The product theorem above controls structured sets, while the convolution estimate of Chapter 5 is parametrized by the least dimension of a nontrivial complex representation. For $\mathrm{SL}_2(\mathbb{F}_p)$ the needed lower bound has a short elementary proof from the action of the split torus on an upper-unipotent character.

Lemma 7.15 (Representation-degree lower bound for $\mathrm{SL}_2(\mathbb{F}_p)$). *Let $p \geq 5$ be prime and $G = \mathrm{SL}_2(\mathbb{F}_p)$. Every nontrivial irreducible complex representation of G has dimension at least*

$$\frac{p-1}{2}.$$

Consequently G is $(p-1)/2$ -quasirandom in the sense of Definition 2.1.

Proof. Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a nontrivial irreducible complex representation. Put

$$U := \left\{ u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{F}_p \right\} \cong (\mathbb{F}_p, +).$$

The group U is cyclic, generated by u_1 , and $\rho(u_1)^p = I$. Hence the minimal polynomial of $\rho(u_1)$ divides $X^p - 1$, which has no repeated root over $\mathbb{C}$; therefore $\rho(u_1)$ is diagonalizable and $\rho|_U$ is a direct sum of one-dimensional characters. Fix $\zeta = e^{2\pi i/p}$ and write

$$\psi_a(u_t) = \zeta^{at} \quad (a \in \mathbb{F}_p)$$

for all characters of U .

At least one nontrivial character ψ_a , $a \neq 0$, occurs in V . Indeed, if only ψ_0 occurred, then every element of U would act trivially. Hence the kernel of ρ , being normal, would contain every conjugate

of U . The upper and lower unipotent subgroups are conjugate in G and they generate $\mathrm{SL}_2(\mathbb{F}_p)$ by elementary row operations. Thus ρ would be trivial, a contradiction.

Choose a nonzero vector v in the ψ_a -weight space. For $r \in \mathbb{F}_p^\times$ put

$$d_r = \begin{pmatrix} r & 0 \\ 0 & r^{-1} \end{pmatrix}.$$

The conjugation identity

$$d_r u_t d_r^{-1} = u_{r^2 t}$$

implies

$$\rho(u_t)\rho(d_r)v = \rho(d_r)\rho(u_{r^{-2}t})v = \zeta^{ar^{-2}t}\rho(d_r)v.$$

Therefore $\rho(d_r)v$ lies in the $\psi_{ar^{-2}}$ -weight space. As r runs through $\mathbb{F}_p^\times$, the parameters ar^{-2} run through one coset of the subgroup of nonzero squares. There are exactly $(p-1)/2$ such parameters, and distinct characters of the finite abelian group U have linearly independent weight spaces. Hence

$$\dim V \geq \frac{p-1}{2}.$$

This is precisely the asserted quasirandomness bound. $\square$

Interface to flattening. Combining Lemma 7.15 with the normalized convolution inequality of Theorem 5.3 gives, for mean-zero $f : G \rightarrow \mathbb{C}$,

$$\|f * g\|_2 \leq \sqrt{\frac{2}{p-1}} \|f\|_2 \|g\|_2.$$

Thus the next chapter no longer needs an unspecified representation-degree parameter when the ambient group is $\mathrm{SL}_2(\mathbb{F}_p)$: the only remaining work is to combine this concrete spectral-mixing gain with subgroup escape and the intermediate-scale product theorem without losing control of the probability-measure normalizations.

Chapter 8

Product Theorems Beyond Rank One

The preceding chapter proved the rank-one product-growth theorem needed for $\mathrm{SL}_2(\mathbb{F}_p)$. Before returning to the Bourgain–Gamburd flattening mechanism, it is useful to record the family-level theorem that explains why the same architecture extends far beyond rank one. This chapter is deliberately short: the bounded-rank product theorem is a deep algebraic input, not a proof that should be hidden inside a few lines of exposition. The next chapter will use only the internally audited rank-one theorem (Theorem 7.14).

Definition 8.1 (Finite simple groups of Lie type and bounded rank). A *finite simple group of Lie type* is a finite simple group obtained from the rational points of a connected simple algebraic group over a finite field, including the standard twisted families, after passing to the appropriate simple central quotient. Its *Lie rank* is the rank of the underlying root system.

A family (G_i) has *bounded rank* if there is an integer r such that every G_i has Lie rank at most r . Any theorem stated uniformly for such a family must keep the dependence on r visible and must separately account for the finitely many small groups for which the usual simple-group parametrization degenerates.

Why the rank parameter matters. Uniform product-growth exponents cannot be expected to remain independent of rank. In high-rank matrix groups one can construct generating sets whose first few product powers grow only mildly. Thus “bounded rank” is a mathematical hypothesis, not merely an organizational convenience.

Theorem 8.2 (Bounded-rank product theorem — deep input). *For every integer $r \geq 1$ there are constants $c_r > 0$ and $\varepsilon_r > 0$ with the following property. Let L be a finite simple group of Lie type of Lie rank at most r , and let $A \subseteq L$ be a finite generating set. Then*

$$A^3 = L \quad \text{or} \quad |A^3| \geq c_r |A|^{1+\varepsilon_r}.$$

The constants depend only on the rank bound; the finitely many exceptional small groups may be absorbed by changing c_r .

Ownership boundary. Theorem 8.2 is retained as an owner-C deep input. Its proof uses the geometry and subgroup structure of algebraic groups in a way that is substantially deeper than the rank-one trace/pivot reconstruction of Chapter 7. The source theorem of Pyber–Szabó states precisely the growth-or-covering alternative above, with the growth exponent and implied constant depending only on the Lie rank. Recording the theorem here serves two purposes: it exposes the general family interface, and it prevents the reader from mistaking the special SL_2 proof for the only possible algebraic input to the flattening machine.

Rank one versus bounded rank. For $L = \mathrm{PSL}_2(\mathbb{F}_p)$ (and, after keeping track of the central cover, for $\mathrm{SL}_2(\mathbb{F}_p)$), Chapter 7 contains a much more explicit intermediate-scale theorem tailored to the probability-measure argument. In the next chapter we therefore use Theorem 7.14, not the black-box theorem above. The bounded-rank theorem becomes relevant when the same analytic mechanism is transported to other fixed-rank families.

Source-parity note. Pyber–Szabó’s growth theorem is stated for a generating set A of a finite simple group of Lie type of rank r and concludes that either A^3 is the whole group or $|A^3| \gg_r |A|^{1+\varepsilon_r}$. The displayed QRC statement does not add symmetry, a density hypothesis, or a uniform-in-rank exponent that is absent from the source.

Chapter 9

From Product Growth to L^2 Flattening and Spectral Mixing

We now assemble the Bourgain–Gamburd mechanism. Representation theory is not the remaining difficulty: Chapter 2 and Lemma 7.15 already supply the required quasirandomness. The task is to prove that an intermediate random-walk law cannot keep a large ℓ^2 norm. If convolution fails to reduce that norm, a dyadic decomposition produces two level sets with large multiplicative energy. Balog–Szemerédi–Gowers and approximate-group structure then create a structured set. Helfgott growth rules out a structured set that generates the whole group, while subgroup escape rules out one trapped in a proper subgroup.

Because two normalization conventions meet here, we begin by fixing them in one place. Every subsequent power of $|G|$ can then be checked mechanically.

Definition 9.1 (Probability-mass convolution, flattening, and escape). Let G be a finite group of order N . A *probability mass* is a function $\nu : G \rightarrow [0, 1]$ such that $\sum_x \nu(x) = 1$. For probability masses (or arbitrary functions) define the counting-sum convolution

$$(\nu *_{\mathfrak{m}} \eta)(x) := \sum_{y \in G} \nu(y) \eta(y^{-1}x),$$

and the involution

$$\nu^\vee(x) := \nu(x^{-1}).$$

We call ν *symmetric* when $\nu = \nu^\vee$. Counting norms are written

$$\|\nu\|_{\ell^q(G)} := \left(\sum_{x \in G} |\nu(x)|^q \right)^{1/q}, \quad \|\nu\|_{\ell^\infty(G)} := \max_x |\nu(x)|.$$

The notation $\|\cdot\|_{L^q(G)}$ remains reserved for normalized counting measure, as in Definition 1.3.

For a symmetric probability mass ν , its one-step ℓ^2 -*flattening ratio* is

$$\mathfrak{f}_2(\nu) := \frac{\|\nu *_{\mathfrak{m}} \nu\|_{\ell^2}}{\|\nu\|_{\ell^2}}.$$

For a proper subgroup $H < G$ define the coset-concentration parameter

$$\mathfrak{c}_H(\nu) := \sup_{a \in G} \nu(aH), \quad \nu(E) := \sum_{x \in E} \nu(x).$$

A family of walk laws *escapes proper subgroups at exponent* β on a specified time range if $\mathfrak{c}_H(\nu) \leq p^{-\beta}$ there for every proper H .

Lemma 9.2 (Normalization dictionary). *Let $N = |G|$, let $u \equiv N^{-1}$ be the uniform probability mass, and for a probability mass ν put*

$$F_\nu := N\nu.$$

Then:

$$(i) \quad \mathbb{E}_x F_\nu(x) = 1 \text{ and } F_u = 1;$$

(ii)

$$F_{\nu *_{\mathfrak{m}} \eta} = F_\nu * F_\eta,$$

where the convolution on the right is the normalized convolution of Definition 1.3;

(iii)

$$\|F_\nu\|_{L^2(G)} = N^{1/2} \|\nu\|_{\ell^2(G)}, \quad \|F_\nu - 1\|_{L^2(G)} = N^{1/2} \|\nu - u\|_{\ell^2(G)};$$

(iv)

$$\|\nu\|_{\ell^2}^2 = (\nu^\vee *_{\mathfrak{m}} \nu)(e),$$

and therefore, when ν is symmetric,

$$\|\nu\|_{\ell^2}^2 = (\nu *_{\mathfrak{m}} \nu)(e).$$

Proof. The first assertion is

$$\mathbb{E}_x F_\nu(x) = \frac{1}{N} \sum_x N\nu(x) = 1.$$

For the convolution identity,

$$(F_\nu * F_\eta)(x) = \frac{1}{N} \sum_y N\nu(y) N\eta(y^{-1}x) = N(\nu *_{\mathfrak{m}} \eta)(x).$$

This is $F_{\nu *_{\mathfrak{m}} \eta}(x)$. The norm conversion follows from

$$\|F_\nu\|_{L^2}^2 = \frac{1}{N} \sum_x N^2 |\nu(x)|^2 = N \|\nu\|_{\ell^2}^2,$$

and the centered version is identical after replacing ν by $\nu - u$. Finally,

$$(\nu^\vee *_{\mathfrak{m}} \nu)(e) = \sum_y \nu(y^{-1}) \nu(y^{-1}) = \sum_z \nu(z)^2.$$

If ν is symmetric, $\nu^\vee = \nu$. □

Lemma 9.3 (Right random-walk, trace, and singular-value dictionary). *Let G have order N , let ν, η be probability masses, and define the right random-walk operator*

$$(P_\nu h)(x) := \sum_{s \in G} \nu(s) h(xs). \tag{9.R1}$$

Then:

$$(i) \text{ with } F_\nu = N\nu \text{ as in Lemma 9.2,}$$

$$P_\nu = \mathcal{R}_{F_\nu^\vee}, \quad P_\nu P_\eta = P_{\nu *_{\mathfrak{m}} \eta}; \tag{9.R2}$$

(ii) $P_\nu^* = P_{\nu^\vee}$, the operator commutes with every left regular translation, and it preserves both the constants and $L_0^2(G)$;

(iii) on the ρ Fourier block,

$$\widehat{P_\nu h}(\rho) = M_\nu(\rho) \widehat{h}(\rho), \quad M_\nu(\rho) := \sum_{s \in G} \nu(s) \rho(s); \quad (9.R3)$$

thus

$$\|P_\nu|_{L_0^2}\|_{\text{op}} = \max_{\rho \neq \mathbf{1}_G} \|M_\nu(\rho)\|_{\text{op}}; \quad (9.R4)$$

(iv) for every integer $m \geq 1$,

$$\text{tr}(P_\nu^m) = N \nu^{*m}(e); \quad (9.R5)$$

(v) if ν is symmetric, then P_ν is self-adjoint, $M_\nu(\rho)$ is Hermitian, and

$$\text{tr}(P_\nu^{2m}) = N \|\nu^{*m}\|_{\ell^2}^2. \quad (9.R6)$$

Proof. By the definitions of normalized convolution and $F_{\nu^\vee}$,

$$(h * F_{\nu^\vee})(x) = \sum_y h(y) \nu^\vee(y^{-1}x) = \sum_s \nu(s) h(xs),$$

after the substitution $s = x^{-1}y$ followed by inversion; this proves the first identity in (9.R2). More transparently, setting $t = y^{-1}x$ gives

$$(h * F_{\nu^\vee})(x) = \sum_t \nu(t^{-1}) h(xt^{-1}) = \sum_s \nu(s) h(xs).$$

Successive right increments multiply in chronological order, so

$$P_\nu P_\eta h(x) = \sum_{s,t} \nu(s) \eta(t) h(xst) = P_{\nu * \eta} h(x),$$

proving the second identity.

Definition 5.1 and Lemma 5.2 give

$$P_\nu^* = \mathcal{R}_{F_{\nu^\vee}}^* = \mathcal{R}_{F_\nu} = P_{\nu^\vee}.$$

Right translations commute with left translations, proving the commutation claim. The constant function 1 is fixed by P_ν ; preservation of L_0^2 follows either from the adjoint identity or directly from $\mathbb{E} P_\nu h = \mathbb{E} h$.

For the Fourier formula, Lemma 5.2 gives left matrix multiplication by $\widehat{F_{\nu^\vee}}(\rho)$. A change of variables shows

$$\widehat{F_{\nu^\vee}}(\rho) = \sum_s \nu(s) \rho(s) = M_\nu(\rho),$$

which proves (9.R3), and (9.R4) follows from the mean-zero block formula (5.3).

By (9.R2), $P_\nu^m = P_{\nu^{*m}}$. In the point-mass basis, the diagonal coefficient at every x is the probability that the product of the m right increments is e , namely $\nu^{*m}(e)$. Summing the N equal diagonal coefficients proves (9.R5).

If $\nu = \nu^\vee$, the adjoint identity makes P_ν self-adjoint, and (9.R3) makes each $M_\nu(\rho)$ Hermitian. Finally symmetry is preserved by convolution, and Lemma 9.2 gives

$$\nu^{*2m}(e) = \|\nu^{*m}\|_{\ell^2}^2.$$

Combining this with (9.R5) proves (9.R6). $\square$

A small nonabelian obstruction. Let $G = S_3$ and let ν be uniform on its three transpositions. Then ν is symmetric and

$$\|\nu\|_{\ell^2} = \frac{1}{\sqrt{3}}.$$

The product of two equal transpositions is e , while the six ordered products of distinct transpositions split equally between the two 3-cycles. Hence $\nu *_{\text{m}} \nu$ is uniform on A_3 and again has ℓ^2 norm $1/\sqrt{3}$. There is no flattening. This is not a paradox: the original measure is supported entirely on the odd coset of the proper subgroup A_3 , so its coset-concentration parameter is 1. The example shows exactly why a flattening theorem must contain a subgroup/coset nonconcentration hypothesis.

9.1 The structural extraction used by the flattening proof

The explicit BSG theorem of Chapter 6 already proves that large multiplicative energy produces large subsets with controlled product sets. The next standard result packages the additional noncommutative Ruzsa-covering steps that place those subsets inside translates of one approximate group. We isolate this conversion as a deep combinatorial input rather than silently folding it into the word “BSG.”

Theorem 9.4 (Energy-to-approximate-group concentration — deep input). *There is an absolute constant $C_T \geq 1$ with the following property. Let A, B be finite nonempty subsets of an arbitrary group and let $K \geq 1$. If*

$$E(A, B) \geq K^{-1}|A|^{3/2}|B|^{3/2},$$

then there are a number

$$Q \leq C_T K^{C_T},$$

a Q -approximate group H , and elements x, y of the ambient group such that

$$Q^{-1}\sqrt{|A||B|} \leq |H| \leq Q\sqrt{|A||B|}, \tag{9.1}$$

and

$$|A \cap xH| \geq Q^{-1}|A|, \quad |B \cap Hy| \geq Q^{-1}|B|. \tag{9.2}$$

Source boundary. Theorem 9.4 is the finite-discrete specialization of the polynomial equivalence in Tao’s Theorem 5.4. The notation $K^{O(1)}$ in the source means precisely that one may choose one absolute constant C_T and enlarge it to dominate every displayed polynomial loss, yielding (9.1)–(9.2). Pass 16 proves the BSG subset extraction and the small-tripling-to-approximate-group theorem internally; this black box retains only the remaining structural packaging needed for the measure-level argument.

9.2 The L^2 flattening proposition in $\text{SL}_2(\mathbb{F}_p)$

Lemma 9.5 (Bourgain–Gamburd ℓ^2 flattening, rank-one form). *Fix γ with $0 < \gamma < 3/4$. There are $\varepsilon_f = \varepsilon_f(\gamma) > 0$ and $p_0(\gamma)$ such that the following holds for every prime $p \geq p_0$. Let*

$$G = \text{SL}_2(\mathbb{F}_p)$$

and let ν be a symmetric probability mass on G satisfying

$$\|\nu\|_{\ell^\infty} \leq p^{-\gamma}, \tag{9.3}$$

$$\|\nu\|_{\ell^2} > p^{-3/2+\gamma}, \quad (9.4)$$

and

$$(\nu *_{\mathfrak{m}} \nu)(H) \leq p^{-\gamma} \quad \text{for every proper subgroup } H < G. \quad (9.5)$$

Then

$$\|\nu *_{\mathfrak{m}} \nu\|_{\ell^2} \leq p^{-\varepsilon_f} \|\nu\|_{\ell^2}. \quad (9.6)$$

Proof. We prove the contrapositive quantitatively. Let C_T be the absolute constant in Theorem 9.4. Put

$$C_A := 6C_T + 1.$$

Apply Theorem 7.14 with $\delta = \gamma$ and denote its growth exponent by $\kappa_\gamma > 0$ and its multiplicative constant by $c_\gamma > 0$. Choose a positive number θ so small that

$$(C_A + 3)\theta < \frac{\gamma}{2}, \quad (C_A + 8)\theta < \gamma, \quad 2C_A\theta < \frac{\kappa_\gamma\gamma}{4}. \quad (9.7)$$

We shall show that, for all sufficiently large p , failure of (9.6) with $\varepsilon_f = \theta$ contradicts (9.5).

Assume therefore that

$$\|\nu *_{\mathfrak{m}} \nu\|_{\ell^2} > p^{-\theta} \|\nu\|_{\ell^2}. \quad (9.8)$$

Set

$$J := \lceil 12 \log_2 p \rceil$$

and define dyadic level sets

$$A_j := \{x \in G : 2^{-j} < \nu(x) \leq 2^{-j+1}\}, \quad 1 \leq j \leq J,$$

together with

$$\tilde{\nu} := \sum_{j=1}^J 2^{-j} \mathbf{1}_{A_j}.$$

Put $q(x) := 2^{-J}$ for every $x \in G$. The dyadic construction gives the pointwise inequalities

$$\tilde{\nu} \leq \nu \leq 2\tilde{\nu} + q.$$

Since $|G| < p^3$ and $2^{-J} \leq p^{-12}$,

$$\|q\|_{\ell^2} = |G|^{1/2} 2^{-J} < p^{-10}.$$

All functions here are nonnegative, so

$$\nu *_{\mathfrak{m}} \nu \leq 4\tilde{\nu} *_{\mathfrak{m}} \tilde{\nu} + 2\tilde{\nu} *_{\mathfrak{m}} q + 2q *_{\mathfrak{m}} \tilde{\nu} + q *_{\mathfrak{m}} q$$

pointwise. Because $\|\tilde{\nu}\|_{\ell^1} \leq 1$, both mixed terms are constant functions of size at most 2^{-J} , while $q *_{\mathfrak{m}} q$ is constant of size $|G|2^{-2J}$. Their ℓ^2 norms are $O(p^{-10})$. In view of (9.4), these errors are negligible compared with $p^{-\theta} \|\nu\|_{\ell^2}$. Hence, after increasing $p_0(\gamma, \theta)$, (9.8) implies

$$\|\tilde{\nu} *_{\mathfrak{m}} \tilde{\nu}\|_{\ell^2} \geq \frac{1}{8} p^{-\theta} \|\nu\|_{\ell^2} \geq p^{-2\theta} \|\tilde{\nu}\|_{\ell^2}. \quad (9.9)$$

The last inequality uses $\tilde{\nu} \leq \nu$ and $p^\theta \geq 8$. We shall also use the reverse norm comparison supplied by $\nu \leq 2\tilde{\nu} + q$: for sufficiently large p ,

$$\|\tilde{\nu}\|_{\ell^2} \geq \frac{1}{4} \|\nu\|_{\ell^2} \geq p^{-\theta} \|\nu\|_{\ell^2}. \quad (9.9a)$$

For large p we also have $J^2 \leq p^\theta$. Expanding the left side of (9.9) and using the triangle inequality, there are indices j_1, j_2 for which, after exchanging them if necessary, $j_1 \geq j_2$ and

$$J^2 2^{-j_1-j_2} \|\mathbf{1}_{A_{j_1}} *_{\mathbf{m}} \mathbf{1}_{A_{j_2}}\|_{\ell^2} \geq \|\tilde{\nu} *_{\mathbf{m}} \tilde{\nu}\|_{\ell^2}. \quad (9.10)$$

The exchange is legitimate because the level sets are symmetric and hence $E(A, B) = E(B, A)$ for these two sets. Write

$$A := A_{j_1}, \quad B := A_{j_2}, \quad n := |A|, \quad m := |B|, \quad a := 2^{-j_1}, \quad b := 2^{-j_2}.$$

Since $an \leq 1$ and $bm \leq 1$, $n \leq a^{-1}$ and $m \leq b^{-1}$. Furthermore

$$\|\tilde{\nu}\|_{\ell^2} \geq \max(a\sqrt{n}, b\sqrt{m})$$

and

$$\|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2} \leq \sqrt{nm} \min(\sqrt{n}, \sqrt{m}).$$

Equations (9.9)–(9.10) and $J^2 \leq p^\theta$ first give the common lower bound

$$ab \|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2} \geq p^{-3\theta} \|\tilde{\nu}\|_{\ell^2}.$$

Use the Young bounds

$$\|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2} \leq n\sqrt{m}, \quad \|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2} \leq m\sqrt{n}.$$

Comparing the first with $\|\tilde{\nu}\|_{\ell^2} \geq b\sqrt{m}$ yields $an \geq p^{-3\theta}$; comparing the second with $\|\tilde{\nu}\|_{\ell^2} \geq a\sqrt{n}$ yields $bm \geq p^{-3\theta}$. Thus

$$\min(an, bm) \geq p^{-3\theta}. \quad (9.11)$$

Moreover the two level sets contribute disjointly to the square norm, so

$$\|\tilde{\nu}\|_{\ell^2}^2 \geq a^2 n + b^2 m \geq 2ab\sqrt{nm}.$$

In particular $\|\tilde{\nu}\|_{\ell^2} \geq (ab)^{1/2} n^{1/4} m^{1/4}$. Another use of the common lower bound above, followed by $n \leq a^{-1}$ and $m \leq b^{-1}$, gives

$$\|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2} \geq p^{-3\theta} n^{3/4} m^{3/4}. \quad (9.12)$$

Therefore the multiplicative energy satisfies

$$E(A, B) = \|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2}^2 \geq p^{-6\theta} n^{3/2} m^{3/2}. \quad (9.13)$$

Apply Theorem 9.4 with $K = p^{6\theta}$. After enlarging p_0 once more, its polynomial constant may be absorbed into one power of p^θ , so the resulting approximate-group parameter obeys

$$Q \leq p^{C_A \theta}. \quad (9.14)$$

We obtain a Q -approximate group H and $x \in G$ such that

$$|H| \leq Q\sqrt{nm} \leq Q2^{j_1} \quad (9.15)$$

and

$$|A \cap xH| \geq Q^{-1}n. \quad (9.16)$$

By (9.11), (9.14), and (9.16),

$$\nu(xH) \geq a|A \cap xH| \geq Q^{-1}an \geq p^{-(C_A+3)\theta}. \quad (9.17)$$

Combining this with the atom bound (9.3) gives the lower size estimate

$$|H| \geq p^{\gamma - (C_A + 3)\theta}. \quad (9.18)$$

We also need an upper size estimate. From (9.10), Young's inequality, and $bm \leq 1$,

$$a\sqrt{n} \geq ab\|\mathbf{1}_A *_{\mathbf{m}} \mathbf{1}_B\|_{\ell^2} \geq p^{-3\theta}\|\tilde{\nu}\|_{\ell^2}.$$

By (9.9a) and (9.4), for sufficiently large p , $\|\tilde{\nu}\|_{\ell^2} \geq p^{-\theta}\|\nu\|_{\ell^2}$. Since $a\sqrt{n} \leq a^{1/2}$, we obtain

$$2^{j_1} = a^{-1} \leq p^{3-2\gamma+8\theta}. \quad (9.19)$$

Thus (9.14)–(9.15) yield

$$|H| \leq p^{3-2\gamma+(C_A+8)\theta} < p^{3-\gamma} \quad (9.20)$$

by (9.7).

Because H is a Q -approximate group, $H^2 \subseteq XH$ for some $|X| \leq Q$, and hence

$$|H^3| \leq Q^2|H| \leq p^{2C_A\theta}|H|. \quad (9.21)$$

Suppose that H is not contained in a proper subgroup. Then $\langle H \rangle = G$. The set H is symmetric and contains e , so Theorem 7.14, (9.20), and (9.18) imply

$$|H^3| \geq c_\gamma|H|^{1+\kappa_\gamma} \geq c_\gamma p^{\kappa_\gamma(\gamma-(C_A+3)\theta)}|H|. \quad (9.22)$$

By (9.7), the exponent in (9.22) is eventually larger than $2C_A\theta$, contradicting (9.21). Therefore H is contained in a proper subgroup $H_0 < G$.

Finally, (9.7) and (9.17) give

$$\nu(xH_0) \geq \nu(xH) > p^{-\gamma/2}. \quad (9.23)$$

For $a, b \in xH_0$ we have $a^{-1}b \in H_0$. Since ν is symmetric, those pairs contribute to the mass of $\nu *_{\mathbf{m}} \nu$ on H_0 :

$$(\nu *_{\mathbf{m}} \nu)(H_0) \geq \nu(xH_0)^2 > p^{-\gamma},$$

contradicting (9.5). This contradiction proves (9.6). $\square$

Remark 9.6 (Where the constants live). The proof does not claim an optimized numerical value of $\varepsilon_f(\gamma)$. It gives an effective choice once two source constants are fixed: the absolute polynomial exponent C_T in Theorem 9.4 and the growth exponent κ_γ in Theorem 7.14. Equation (9.7) is the complete dependency: no additional p -dependent loss is hidden in the flattening exponent.

9.3 Starting the flattening iteration from logarithmic girth

Lemma 9.7 (Escape initializes every dyadic flattening step). *Fix $k \geq 2$ and $\tau > 0$, and let the constants $\beta > 0$ and r_p be those of Theorem 7.7. Put*

$$\gamma := \min\{\beta, 1/4\}. \quad (9.24)$$

For every sufficiently large p , let $S \subseteq \mathrm{SL}_2(\mathbb{F}_p)$ be a symmetric $2k$ -element generating set with

$$\mathrm{girth}(\mathrm{Cay}(G, S)) \geq \tau \log_{2k} p,$$

and let μ be uniform on S . Then for every integer $\ell \geq r_p$, the symmetric law $\nu = \mu^{\ell}$ satisfies*

$$\|\nu\|_{\ell^\infty} \leq p^{-\gamma} \quad (9.25)$$

and

$$(\nu *_{\mathbf{m}} \nu)(H) \leq p^{-\gamma} \quad \text{for every proper } H < G. \quad (9.26)$$

Proof. Symmetry is preserved by convolution. Apply Theorem 7.7 first to the proper subgroup $\{e\}$ and its cosets: every atom of $\mu^{*m\ell}$ is at most $p^{-\beta}$. This proves (9.25). The same theorem gives directly

$$\mu^{*m2\ell}(H) \leq p^{-\beta}$$

for every proper H , which is (9.26). Since $\gamma \leq \beta$, weakening the right sides to $p^{-\gamma}$ is legitimate. $\square$

Lemma 9.8 (Finite flattening iteration reaches the return-probability scale). *Under the hypotheses of Lemma 9.7, let $\varepsilon_f = \varepsilon_f(\gamma)$ be supplied by Lemma 9.5, and put*

$$J_0 := \left\lceil \frac{3/2 - \gamma}{\varepsilon_f} \right\rceil + 1, \quad L_p := 2^{J_0} r_p. \quad (9.27)$$

Then, for all sufficiently large p ,

$$\|\mu^{*mL_p}\|_{\ell^2} \leq p^{-3/2+\gamma}. \quad (9.28)$$

Moreover $L_p = O_{k,\tau}(\log p)$.

Proof. Set

$$\nu_j := \mu^{*m(2^j r_p)}, \quad 0 \leq j \leq J_0.$$

Each time $\|\nu_j\|_{\ell^2} > p^{-3/2+\gamma}$, Lemma 9.7 supplies the two nonconcentration hypotheses of Lemma 9.5, so

$$\|\nu_{j+1}\|_{\ell^2} = \|\nu_j * \mu\|_{\ell^2} \leq p^{-\varepsilon_f} \|\nu_j\|_{\ell^2}. \quad (9.29)$$

If the target bound is reached earlier, there is nothing to prove: Young's inequality with a probability mass shows that subsequent convolution powers cannot increase the ℓ^2 norm. Otherwise (9.29) holds at every preceding step. Since $\|\nu_0\|_{\ell^2} \leq \|\nu_0\|_{\ell^1} = 1$,

$$\|\nu_{J_0}\|_{\ell^2} \leq p^{-J_0 \varepsilon_f} < p^{-3/2+\gamma}$$

by (9.27). This proves (9.28).

Finally Theorem 7.7 defines

$$r_p = \left\lfloor \frac{\tau}{64} \log_{2k} p \right\rfloor.$$

The integer J_0 depends only on k, τ through γ and $\varepsilon_f(\gamma)$, so $L_p = O_{k,\tau}(\log p)$. $\square$

9.4 Spectral gap and mixing

Corollary 9.9 (Growth-to-flattening and uniform spectral mixing). *Fix $k \geq 2$ and $\tau > 0$. There are constants $\rho_{k,\tau} < 1$, $C_{k,\tau} > 0$, and $p_0(k, \tau)$ such that the following holds. For every prime $p \geq p_0$, let*

$$G = \mathrm{SL}_2(\mathbb{F}_p)$$

and let $S \subseteq G$ be a symmetric $2k$ -element generating set with

$$\mathrm{girth}(\mathrm{Cay}(G, S)) \geq \tau \log_{2k} p.$$

Let P be the simple-random-walk operator

$$(Pf)(x) = \frac{1}{2k} \sum_{s \in S} f(xs).$$

Then every eigenvalue λ of P on the orthogonal complement of the constants satisfies

$$|\lambda| \leq \rho_{k,\tau}. \quad (9.30)$$

Consequently, if $q_n^{(x)}$ is the n -step probability mass started at x and u is uniform on G , then

$$\|q_n^{(x)} - u\|_{\text{TV}} \leq \frac{1}{2}|G|^{1/2}\rho_{k,\tau}^n, \quad (9.31)$$

and in particular

$$t_{\text{mix}}(\eta) \leq C_{k,\tau}(\log p + \log \eta^{-1}) \quad (0 < \eta < 1/2). \quad (9.32)$$

Proof. Let μ be uniform on S , take γ from (9.24), and let L_p be the time in Lemma 9.8. By Lemma 9.3, symmetry gives

$$\mu^{*m2L_p}(e) = \|\mu^{*mL_p}\|_{\ell^2}^2 \leq p^{-3+2\gamma}. \quad (9.33)$$

The same lemma identifies P with the right random-walk operator and gives the trace identity directly. Therefore

$$\text{tr}(P^{2L_p}) = |G|\mu^{*m2L_p}(e) < p^3p^{-3+2\gamma} = p^{2\gamma}. \quad (9.34)$$

The operator P is self-adjoint because $S = S^{-1}$ and it commutes with the left regular action. Since S generates G , the Cayley graph is connected, so the eigenspace for eigenvalue 1 consists exactly of the constants. Let

$$\rho := \max\{|\lambda| : \lambda \text{ is a nonconstant eigenvalue of } P\}.$$

Choose an eigenvalue with absolute value ρ . Its eigenspace is a nonzero left-invariant subrepresentation of the regular representation, and it contains no trivial constituent because every trivial vector is constant. By Lemma 7.15, its dimension is at least

$$D = \frac{p-1}{2}.$$

Since all terms in the even-power trace are nonnegative,

$$D\rho^{2L_p} \leq \text{tr}(P^{2L_p}) < p^{2\gamma}.$$

Hence

$$\rho^{2L_p} \leq \frac{2p^{2\gamma}}{p-1} \leq 4p^{-(1-2\gamma)}. \quad (9.35)$$

By construction $\gamma \leq 1/4$. For sufficiently large p the right side of (9.35) is at most $p^{-(1-2\gamma)/2}$.

From (9.27) and the definition of r_p there is a constant $C_0 = C_0(k, \tau)$ such that

$$L_p \leq C_0 \log p. \quad (9.36)$$

Taking logarithms in (9.35) and using (9.36) gives

$$\rho \leq \exp\left(-\frac{1-2\gamma}{4C_0}\right) =: \rho_{k,\tau} < 1,$$

which proves (9.30). Notice that this bounds the *absolute* nontrivial spectrum, so a possible negative eigenvalue is controlled as well.

For mixing, regard probability masses as column vectors with counting ℓ^2 norm. Because μ is symmetric, Lemma 9.3 gives $P = P^*$, so the same matrix that averages test functions also evolves probability-mass columns. The mean-zero vector $\delta_x - u$ has norm $\sqrt{1 - |G|^{-1}} < 1$. The spectral theorem and (9.30) give

$$\|q_n^{(x)} - u\|_{\ell^2} \leq \rho_{k,\tau}^n.$$

Cauchy–Schwarz converts this to total variation:

$$\|q_n^{(x)} - u\|_{\text{TV}} = \frac{1}{2} \|q_n^{(x)} - u\|_{\ell^1} \leq \frac{1}{2} |G|^{1/2} \|q_n^{(x)} - u\|_{\ell^2},$$

which is (9.31). Since $|G| = p(p^2 - 1) < p^3$, it is enough to take

$$n \geq \frac{\frac{3}{2} \log p + \log(1/(2\eta))}{-\log \rho_{k,\tau}},$$

and (9.32) follows after enlarging $C_{k,\tau}$. □

What has actually been proved. The corollary is a uniform expansion/mixing statement for the fixed-degree, logarithmic-girth family specified above. It is *not* a theorem for an arbitrary generating set of $\text{SL}_2(\mathbb{F}_p)$. The hypothesis is exactly the one needed to initialize subgroup escape in Theorem 7.7.

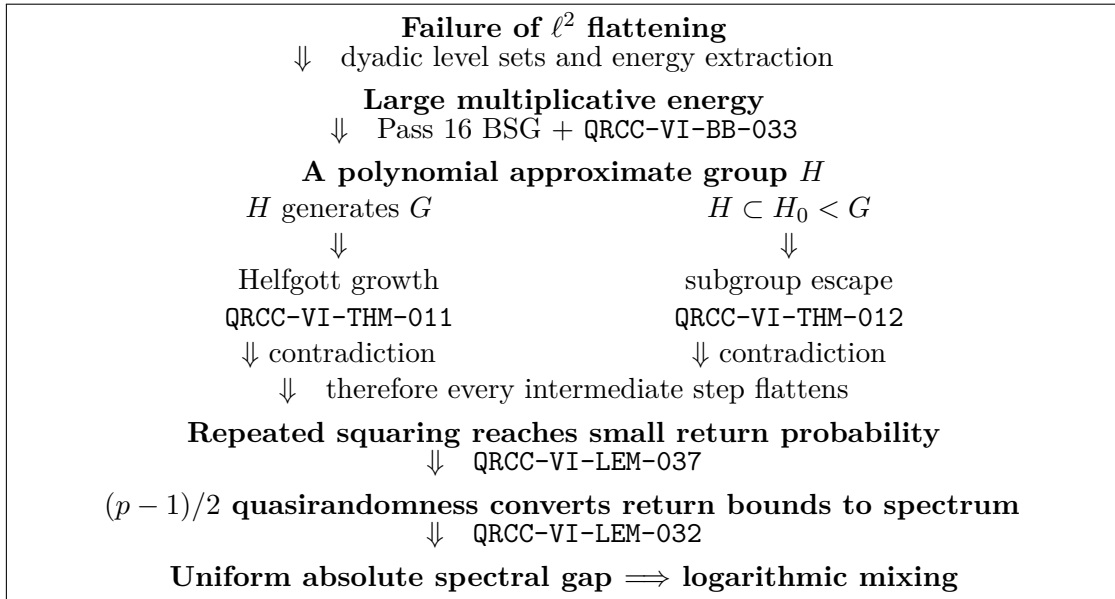


Figure 9.1: The rank-one Bourgain–Gamburd growth-to-mixing machine. Each arrow is owned by an explicit audited QRCC interface; the only owner-C structural arrow inside the flattening step is QRCC-VI-BB-033.

The complete mechanism. The logical chain is now explicit:

$$\begin{aligned}
 & \text{failure of } \ell^2 \text{ flattening} \implies \text{high energy} \implies \text{approximate group} \\
 \implies & \begin{cases} \text{generates } G & \Rightarrow \text{contradict Helfgott growth,} \\ \text{lies in a proper subgroup} & \Rightarrow \text{contradict escape,} \end{cases}
 \end{aligned}$$

so repeated squaring reaches the return-probability scale; the concrete representation-degree lower bound then turns that return bound into a spectral gap. This is the point at which quasirandomness, additive-combinatorial structure, algebraic product growth, and random-walk geometry genuinely meet.

Source-parity audit. Bourgain–Gamburd Proposition 2 uses the same three analytic hypotheses as Lemma 9.5: an atom bound, an intermediate ℓ^2 scale, and small two-step mass on every proper subgroup. Their proof performs the dyadic decomposition, obtains high multiplicative energy, invokes noncommutative BSG/approximate-group structure, and applies Helfgott growth to force the structured set into a proper subgroup. Their subsequent proof of the main expansion theorem combines the resulting return-probability estimate with the lower bound on nontrivial representation multiplicity. The QRCC proof reproduces those transitions with the Pass-15 escape theorem, the Pass-17 intermediate-scale Helfgott theorem, and the internally proved $(p-1)/2$ representation-degree bound.

Chapter 10

Random Generation in Symmetric Groups

Random generation is a compact test of how quasirandomness, subgroup structure, and probabilistic reasoning interact. This chapter closes the registered application with one exact family and one exact asymptotic theorem; it does not import the classification of finite simple groups as an unnamed black box.

Definition 10.1 (Random generation probability). Let G be a finite group and let $g_1, \dots, g_k$ be random elements with a specified joint law. Their *generation probability* is

$$\Pr(\langle g_1, \dots, g_k \rangle = G).$$

When the elements are independent and uniform this law is stated explicitly. For a family (G_n) , every asymptotic generation statement specifies the family, the number of random generators, and the parameter $n \rightarrow \infty$.

For the theorem below, S_n is the symmetric group on $\{1, \dots, n\}$ and $A_n \leq S_n$ is its alternating subgroup of even permutations. If σ, τ are independent uniform elements of S_n , set

$$p_n := \Pr(\langle \sigma, \tau \rangle \in \{A_n, S_n\}). \quad (10.1)$$

Theorem 10.2 (Eberhard–Virchow quantitative Dixon theorem — source-owned). *For every fixed $\varepsilon > 0$,*

$$p_n = 1 - \frac{1}{n} + O_\varepsilon(n^{-2+\varepsilon}) \quad (n \rightarrow \infty). \quad (10.2)$$

In particular, two independent uniform permutations generate either A_n or S_n with probability tending to one.

Historical and proof-ownership audit. Dixon proved the qualitative limit in 1969; the corrected DOI of his paper is 10.1007/BF01110210. Babai proved Dixon’s sharper conjectured asymptotic using consequences of the classification of finite simple groups. Eberhard and Virchow gave an elementary proof of (10.2) in *Combinatorica* 39 (2019), 273–288, DOI 10.1007/s00493-017-3629-5. Their argument uses character theory and character estimates and remains source-owned here.

Parity is part of the statement. The event in (10.1) is deliberately “ A_n or S_n ”, not merely “ S_n ”. If both generators are even, their subgroup lies in A_n ; if the subgroup contains A_n and at least one generator is odd, it is S_n . Thus omitting the parity alternative would change the limiting probability.

Finite sanity check. For $n = 2$, $A_2 = \{e\}$ and $S_2 = \{e, (12)\}$. All four ordered pairs of permutations generate either A_2 or S_2 , so $p_2 = 1$. This small case is not intended to approximate (10.2); it simply checks the event and the parity convention exactly.

10.1 Reader checkpoint and exercises

Chapter checkpoint. Random generation statements are meaningful only after the sampling law, group family, parity convention, and asymptotic parameter have been fixed. The deep content of Theorem 10.2 is the subgroup and character-theoretic control of the exceptional pairs, not the definition of generation probability itself.

Exercises.

1. Compute the generation probability for one uniform generator of C_m .
2. Verify the $n = 2$ calculation above directly from the four ordered pairs.
3. Prove the parity assertion: a subgroup of S_n containing A_n is either A_n or S_n .
4. Explain why “two random permutations generate S_n with probability tending to one” is false without a parity qualification.
5. Read (10.2) as an exceptional-pair estimate and identify the leading $1/n$ obstruction described in the source.
6. Compare the logical content of Dixon’s qualitative limit with the quantitative refinement (10.2); state exactly what additional rate information the latter supplies.

Internalization and Reconstruction

Guide: Volume VI

Purpose. Translate abelian Fourier intuition into representation-theoretic mixing, then connect quasirandomness, product growth, approximate groups, flattening, and random generation without silently commuting noncommuting objects.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

This is a foundation volume. Its prerequisite ladder is internal: begin with the first definitions and reconstruct each later family from those conventions rather than importing unstated external notation. Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-VI-DEF-001 (Unitary representation and noncommutative Fourier transform); QRCC-VI-DEF-002 (Irreducibility, character, and regular representation); QRCC-VI-DEF-006 (Normalized convolution and mean-zero L2 space); QRCC-VI-DEF-003 (D-quasirandom group); QRCC-VI-DEF-004 (Product-free set).

Chapter retrieval map

- Ch. 1: Noncommutative Fourier Analysis on Finite Groups:** enter through QRCC-VI-DEF-001 (Unitary representation and noncommutative Fourier transform); exit by being able to state QRCC-VI-THM-001 (Noncommutative Fourier inversion) with its ownership boundary.
- Ch. 2: Representation Dimension as a Mixing Parameter:** enter through QRCC-VI-DEF-003 (D-quasirandom group); exit by being able to state QRCC-VI-LEM-018 (Quasirandom L2 convolution mixing) with its ownership boundary.
- Ch. 3: Product-Free Sets:** enter through QRCC-VI-DEF-004 (Product-free set); exit by being able to state QRCC-VI-THM-004 (Gowers product-free-set bound) with its ownership boundary.
- Ch. 4: Triple-Product Mixing:** enter through QRCC-VI-DEF-005 (Triple-product correlation); exit by being able to state QRCC-VI-THM-005 (Triple-product mixing theorem) with its ownership boundary.

- Ch. 5: Convolution Mixing and Set Counts:** enter through QRCC-VI-DEF-012 (Convolution operator and singular-value contraction); exit by being able to state QRCC-VI-COR-007 (Mixing from representation dimension) with its ownership boundary.
- Ch. 6: Approximate Subgroups and Multiplicative Energy:** enter through QRCC-VI-DEF-007 (Approximate group and control); exit by being able to state QRCC-VI-BB-010 (Breuillard–Green–Tao approximate-group structure theorem) with its ownership boundary.
- Ch. 7: Product Growth, Escape, and Generating Sets:** enter through QRCC-VI-DEF-008 (Product growth and generating set); exit by being able to state QRCC-VI-THM-011 (Helfgott product growth in $SL_2(\mathbb{F}_p)$: symmetric intermediate-scale form) with its ownership boundary.
- Ch. 8: Product Theorems Beyond Rank One:** enter through QRCC-VI-DEF-009 (Finite simple group family and rank parameter); exit by being able to state QRCC-VI-THM-013 (Product theorem for bounded-rank finite simple groups) with its ownership boundary.
- Ch. 9: From Product Growth to $\text{\texttt{\textbackslash texorpdfstring{\$L^2\$}}$:** enter through QRCC-VI-DEF-010 (Probability-mass convolution, flattening, and escape); exit by being able to state QRCC-VI-COR-014 (Growth-to-flattening and uniform spectral mixing) with its ownership boundary.
- Ch. 10: Random Generation in Symmetric Groups:** enter through QRCC-VI-DEF-011 (Random generation probability); exit by being able to state QRCC-VI-THM-015 (Random-generation application) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-VI-THM-001 — Noncommutative Fourier inversion. Dependencies: QRCC-VI-DEF-001, QRCC-VI-DEF-002, QRCC-VI-THM-017. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-VI-THM-004 — Gowers product-free-set bound. Dependencies: QRCC-VI-DEF-003, QRCC-VI-DEF-004, QRCC-VI-LEM-018. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-VI-COR-007 — Mixing from representation dimension. Dependencies: QRCC-VI-THM-006, QRCC-VI-DEF-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-VI-THM-011 — Helfgott product growth in $SL_2(\mathbb{F}_p)$: symmetric intermediate-scale form. Dependencies: QRCC-VI-DEF-008, QRCC-VI-THM-009, QRCC-VI-BB-026, QRCC-VI-LEM-028. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-VI-COR-014 — Growth-to-flattening and uniform spectral mixing. Dependencies: QRCC-VI-DEF-010, QRCC-VI-LEM-032, QRCC-VI-LEM-037. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

6. QRCC-VI-THM-015 — Random-generation application. Dependencies: QRCC-VI-DEF-011, QRCC-VI-DEF-009. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

Common failure modes

- replacing matrix-valued Fourier coefficients by scalars in a nonabelian group;
- forgetting the trivial representation when measuring mixing;
- treating qualitative BGT $O_K(1)$ bounds as explicit effective constants;
- using product-growth theorems outside their rank, symmetry, or scale hypotheses;

Boundary and degenerate cases

- abelian groups where all irreducibles have dimension one;
- the trivial representation and constant functions;
- tiny sets where product-growth alternatives are vacuous;
- fixed-rank versus growing-rank simple groups;

Retrieval and reconstruction drill

1. State QRCC-VI-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-VI-THM-001. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-VI-THM-004, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-VI-THM-015, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume VII turns mixing statements into graph spectra, random-walk algorithms, explicit expanders, and coding constructions.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part VII

Expander Graphs, Random Walks, and Graph Products

Chapter 1

Expansion in Three Languages

Volume VI ended with spectral expansion for a particular family of Cayley walks. We now forget how the graph was constructed and study expansion as a graph-theoretic object in its own right. Three languages will be kept in parallel throughout this volume:

- (i) the *operator language*: normalized adjacency and its spectrum;
- (ii) the *probabilistic language*: contraction and mixing of the natural random walk;
- (iii) the *combinatorial language*: edge and vertex boundaries.

The point of the first four chapters is not merely that these viewpoints are related, but that every normalization and every exceptional case can be translated explicitly from one language to another.

Unless stated otherwise, Chapters 1–5 use a finite undirected loopless d -regular multigraph $G = (V, E)$ with $n := |V| \geq 2$ and $d \geq 1$. Parallel edges are allowed and are counted with multiplicity. Thus the adjacency matrix A is real symmetric and $A(u, v)$ is the multiplicity of the edge $\{u, v\}$.

Definition 1.1 (Normalized adjacency and spectral expansion). The *normalized adjacency operator*, equivalently the one-step Markov operator, is

$$(Pf)(x) := \frac{1}{d} \sum_{y \in V} A(x, y) f(y), \quad P = \frac{1}{d} A.$$

We equip functions $f : V \rightarrow \mathbb{C}$ with the normalized inner product

$$\langle f, g \rangle_V := \frac{1}{n} \sum_{x \in V} f(x) \overline{g(x)}, \quad \|f\|_{2,V}^2 = \langle f, f \rangle_V.$$

The constant function is denoted $\mathbf{1}$, and

$$L_0^2(V) := \{f : \langle f, \mathbf{1} \rangle_V = 0\}.$$

Since P is self-adjoint, its eigenvalues are real. We order them as

$$1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq -1.$$

The *one-sided spectral gap* and the *absolute nontrivial spectral radius* are

$$\gamma(G) := 1 - \lambda_2, \quad \lambda_{\text{abs}}(G) := \|P|_{L_0^2(V)}\|_{\text{op}} = \max\{\lambda_2, -\lambda_n\}.$$

The *absolute spectral gap* is

$$\gamma_{\text{abs}}(G) := 1 - \lambda_{\text{abs}}(G).$$

A family of fixed-degree graphs is a *one-sided spectral expander family* if $\inf \gamma(G) > 0$, and an *absolute spectral expander family* if $\inf \gamma_{\text{abs}}(G) > 0$.

The two gaps deliberately have different names. The one-sided gap is the quantity controlled by Cheeger inequalities. The absolute gap is the quantity that controls an ordinary non-lazy random walk. A connected bipartite graph has $\lambda_n = -1$, so its one-sided gap can be large while $\gamma_{\text{abs}} = 0$.

Connectivity and bipartiteness. Because P is stochastic, $P\mathbf{1} = \mathbf{1}$. The multiplicity of the eigenvalue 1 is the number of connected components: an eigenfunction with eigenvalue 1 must be constant along every edge. If G is connected, then -1 is an eigenvalue exactly when G is bipartite. Indeed, the sign function of a bipartition is sent to its negative; conversely equality in $\|Pf\|_2 \leq \|f\|_2$ at eigenvalue -1 forces neighboring values to have opposite phase, and connectedness propagates a two-coloring.

Definition 1.2 (Edge conductance and vertex expansion). For $S \subseteq V$, let

$$\partial_E S := E(S, V \setminus S)$$

be the multiset of edges with exactly one endpoint in S . For $0 < |S| \leq n/2$ define its *conductance*

$$\Phi(S) := \frac{|\partial_E S|}{d|S|}, \quad \Phi(G) := \min_{0 < |S| \leq n/2} \Phi(S).$$

Let

$$\partial_V S := \{y \in V \setminus S : y \text{ has a neighbor in } S\}$$

and define the *vertex-expansion ratio*

$$h_V(S) := \frac{|\partial_V S|}{|S|}, \quad h_V(G) := \min_{0 < |S| \leq n/2} h_V(S).$$

Since each vertex outside S is incident with at most d boundary edges,

$$|\partial_E S| \leq d|\partial_V S|, \quad \text{hence} \quad h_V(G) \geq \Phi(G). \quad (1.1)$$

The normalization $d|S|$ makes $0 \leq \Phi(G) \leq 1$ and is the one that matches the Markov operator P without an extra factor of d .

Corollary 1.3 (Positive expansion forces logarithmic diameter). *If G is connected and $\Phi(G) = \phi > 0$, then*

$$\text{diam}(G) \leq 2 \left\lceil \frac{\log n}{\log(1 + \phi)} \right\rceil. \quad (1.2)$$

In particular, for every fixed $\phi > 0$, a family with $\Phi(G) \geq \phi$ has diameter $O_\phi(\log n)$.

Proof. Fix $v \in V$ and let $B_r(v)$ be the ball of graph radius r . Whenever $|B_r(v)| \leq n/2$, Definition 1.2 and (1.1) give

$$|B_{r+1}(v)| \geq |B_r(v)| + |\partial_V B_r(v)| \geq (1 + \phi)|B_r(v)|.$$

Thus after

$$R := \left\lceil \frac{\log n}{\log(1 + \phi)} \right\rceil$$

steps every ball has more than $n/2$ vertices: if it crossed the $n/2$ threshold earlier there is nothing to prove, and otherwise repeated growth gives $|B_R(v)| \geq (1 + \phi)^R \geq n$. For any u, v , the two sets $B_R(u)$ and $B_R(v)$ therefore intersect. A vertex in their intersection lies on a walk of length at most $2R$ from u to v , proving (1.2). $\square$

Sanity checks. A cycle C_n has $\Phi(C_n) \asymp n^{-1}$ and diameter $\asymp n$, so the logarithmic conclusion correctly fails when the expansion constant decays. At the opposite extreme, K_n has conductance bounded below by roughly $1/2$ and diameter 1; (1.2) is intentionally a robust expansion bound, not an equality formula.

Chapter 2

Spectral Contraction, Mixing of Edges, and the Tree Barrier

The normalized adjacency matrix is simultaneously a self-adjoint operator and a transition matrix. The spectral theorem therefore converts its nontrivial eigenvalues into exact contraction rates. The same one-line decomposition also yields the expander-mixing inequality. We then turn to a subtler question: how small can the second eigenvalue of a large fixed-degree graph actually be? The Alon–Boppana theorem shows that the infinite regular tree sets the asymptotic barrier.

Theorem 2.1 (Spectral contraction of a regular random walk). *Let G be as in Definition 1.1. For every $f \in L_0^2(V)$ and every integer $t \geq 0$,*

$$\|P^t f\|_{2,V} \leq \lambda_{\text{abs}}(G)^t \|f\|_{2,V}. \quad (2.1)$$

For $t \geq 1$, equality holds precisely when the spectral support of f is contained in eigenspaces whose eigenvalues have absolute value $\lambda_{\text{abs}}(G)$.

If G is connected, then $\lambda_{\text{abs}}(G) < 1$ exactly when G is nonbipartite. In a connected bipartite graph the bipartition sign function s satisfies $Ps = -s$, so no non-lazy L^2 contraction is possible on all of $L_0^2(V)$.

Proof. Choose an orthonormal eigenbasis $v_1 = \mathbf{1}, v_2, \dots, v_n$ for the normalized inner product. Since $f \perp \mathbf{1}$,

$$f = \sum_{i=2}^n a_i v_i, \quad P^t f = \sum_{i=2}^n a_i \lambda_i^t v_i.$$

Consequently

$$\|P^t f\|_{2,V}^2 = \sum_{i=2}^n |a_i|^2 |\lambda_i|^{2t} \leq \lambda_{\text{abs}}^{2t} \sum_{i=2}^n |a_i|^2.$$

This proves (2.1), and the equality criterion follows from equality term by term. The final assertion is the connectivity/bipartiteness discussion after Definition 1.1. $\square$

Why the absolute value matters. For a connected bipartite expander the positive nontrivial spectrum may be well inside $(-1, 1)$, while -1 remains present. A theorem written only with “the second eigenvalue” can therefore be perfectly correct for Cheeger expansion and simultaneously false as a mixing theorem. We will keep λ_2 and λ_{abs} separate throughout the volume.

For subsets $S, T \subseteq V$, define the ordered edge count

$$e^{\rightarrow}(S, T) := \sum_{x \in S} \sum_{y \in T} A(x, y).$$

If $S = T$, an undirected edge internal to S contributes twice. This is the convention naturally represented by the quadratic form of A .

Corollary 2.2 (Expander mixing in normalized and edge-count form). *For all $S, T \subseteq V$,*

$$\left| e^{\rightarrow}(S, T) - \frac{d|S||T|}{n} \right| \leq d\lambda_{\text{abs}} \sqrt{|S||T| \left(1 - \frac{|S|}{n}\right) \left(1 - \frac{|T|}{n}\right)}. \quad (2.2)$$

In particular,

$$\left| e^{\rightarrow}(S, T) - \frac{d|S||T|}{n} \right| \leq d\lambda_{\text{abs}} \sqrt{|S||T|}. \quad (2.3)$$

Equivalently, if $s = |S|/n$ and $t = |T|/n$, then

$$|\langle \mathbf{1}_S, P\mathbf{1}_T \rangle_V - st| \leq \lambda_{\text{abs}} \sqrt{s(1-s)t(1-t)}. \quad (2.4)$$

Proof. Write

$$f_S := \mathbf{1}_S - s\mathbf{1}, \quad f_T := \mathbf{1}_T - t\mathbf{1}.$$

Both functions are mean zero, $P\mathbf{1} = \mathbf{1}$, and

$$\|f_S\|_{2,V}^2 = s(1-s), \quad \|f_T\|_{2,V}^2 = t(1-t).$$

The constant and mean-zero parts are orthogonal, so

$$\langle \mathbf{1}_S, P\mathbf{1}_T \rangle_V - st = \langle f_S, Pf_T \rangle_V.$$

Cauchy–Schwarz and Theorem 2.1 at $t = 1$ give (2.4). Since

$$\langle \mathbf{1}_S, P\mathbf{1}_T \rangle_V = \frac{e^{\rightarrow}(S, T)}{nd},$$

multiplying (2.4) by nd proves (2.2); dropping the two factors at most 1 gives (2.3). $\square$

A useful equality audit. The centered form (2.2) is sharper than the frequently quoted (2.3). For $S = V$ its right-hand side is zero, as it must be because $e^{\rightarrow}(V, T) = d|T|$ exactly. This is a quick way to detect a missing centering term in a purported mixing proof.

Theorem 2.3 (Alon–Boppana lower bound, with a diameter form). *Let $d \geq 3$ and let G be a connected d -regular graph of diameter $\Delta \geq 4$. Put*

$$k := \left\lfloor \frac{\Delta}{2} \right\rfloor - 1.$$

Then the second adjacency eigenvalue $\mu_2(A) = d\lambda_2(P)$ satisfies

$$\mu_2(A) \geq 2\sqrt{d-1} \cos\left(\frac{\pi}{k+1}\right). \quad (2.5)$$

Consequently, for every sequence of d -regular graphs G_j with $|V(G_j)| \rightarrow \infty$,

$$\liminf_{j \rightarrow \infty} \lambda_2(P_{G_j}) \geq \frac{2\sqrt{d-1}}{d}. \quad (2.6)$$

For $d = 2$, the same asymptotic conclusion holds because every connected 2-regular graph is a cycle.

Proof. If G is disconnected then $\lambda_2(P) = 1$ and (2.6) is immediate, so the only interesting case is connected.

Step 1: the comparison tree. Let $T_{d,k}$ be the rooted tree of depth k in which the root has d children and every non-leaf non-root vertex has $d-1$ children. Let μ be the Perron eigenvalue of its adjacency matrix and let $g_i > 0$ be the value of a positive radial Perron eigenvector on level i ; for the boundary calculation below, set $g_{k+1} := 0$.

The subspace of radial functions has an orthonormal level basis in which the adjacency matrix is tridiagonal. The root-to-level-one coefficient is $\sqrt{d}$, and all later off-diagonal coefficients are $\sqrt{d-1}$. Delete the root coordinate. The remaining $k \times k$ principal submatrix is the path matrix with off-diagonal entries $\sqrt{d-1}$, whose top eigenvalue is

$$2\sqrt{d-1} \cos\left(\frac{\pi}{k+1}\right).$$

Cauchy interlacing therefore gives

$$\mu \geq 2\sqrt{d-1} \cos\left(\frac{\pi}{k+1}\right). \quad (2.7)$$

We also need that $g_0 \geq g_1 \geq \dots \geq g_k > 0$. Since $T_{d,k}$ is not d -regular, $\mu < d$. The root equation $\mu g_0 = d g_1$ gives $g_1 < g_0$. If $g_{i-1} \geq g_i$, then

$$(d-1)g_{i+1} = \mu g_i - g_{i-1} \leq (\mu-1)g_i \leq (d-1)g_i,$$

so induction proves the monotonicity.

Step 2: two disjoint metric balls. Choose vertices s, t at distance Δ . Let

$$S_i := \{x : \text{dist}(s, x) = i\}, \quad T_i := \{x : \text{dist}(t, x) = i\}, \quad 0 \leq i \leq k.$$

Because $2k+1 < \Delta$, the two radius- k balls are disjoint and no edge joins them.

Define f by

$$f(x) = \begin{cases} c_1 g_i, & x \in S_i, \\ -c_2 g_i, & x \in T_i, \\ 0, & \text{otherwise,} \end{cases} \quad (2.8)$$

where $c_1, c_2 > 0$ are chosen so that $\sum_x f(x) = 0$. Such a choice is possible because both balls are nonempty.

We claim that

$$(Af)(x) \geq \mu f(x) \quad (x \in \cup_i S_i), \quad (Af)(x) \leq \mu f(x) \quad (x \in \cup_i T_i). \quad (2.9)$$

Consider $x \in S_i$ with $i > 0$. If p neighbors lie in S_{i-1} and q lie in S_i , then $p \geq 1$ and at most $d-p-q$ lie in S_{i+1} ; neighbors outside the ball contribute zero when $i = k$. Since the sequence g_i is nonincreasing,

$$(Af)(x) \geq c_1(pg_{i-1} + qg_i + (d-p-q)g_{i+1}) \geq c_1(g_{i-1} + (d-1)g_{i+1}) = \mu f(x).$$

The root case follows from $d g_1 = \mu g_0$. The T_i inequalities are the same calculation after multiplying by the negative coefficient $-c_2$.

Because f vanishes off the two balls, multiplying (2.9) by $f(x)$ reverses the second inequality and gives in both cases

$$f(x)(Af)(x) \geq \mu f(x)^2.$$

Summing over V yields

$$\frac{\langle f, Af \rangle}{\langle f, f \rangle} \geq \mu.$$

The function f is orthogonal to constants, so the Rayleigh–Ritz characterization of the second adjacency eigenvalue gives $\mu_2(A) \geq \mu$. Combining this with (2.7) proves (2.5).

Step 3: pass to large graphs. A d -regular graph of diameter Δ has

$$n \leq 1 + d \sum_{j=0}^{\Delta-1} (d-1)^j,$$

so for fixed $d \geq 3$, $n \rightarrow \infty$ forces $\Delta \rightarrow \infty$. Then $k \rightarrow \infty$ and the cosine in (2.5) tends to 1. Division by d proves (2.6). $\square$

Ramanujan scale. The constant $2\sqrt{d-1}/d$ is therefore not an artifact of a particular construction. It is the normalized spectral radius of the infinite d -regular tree and an asymptotic lower barrier for λ_2 . A d -regular graph whose nontrivial *adjacency* eigenvalues all have absolute value at most $2\sqrt{d-1}$ is called Ramanujan. Chapter 5 will separate the elementary barrier above from the much deeper constructions that meet it.

Chapter 3

From Spectral Contraction to Random-Walk Mixing

Theorem 2.1 is an L^2 statement about functions. A Markov chain is usually described instead by probability distributions and total variation. On a regular graph the stationary law is uniform, so the translation between the two languages is exact and elementary.

Let $u(x) = 1/n$ be the uniform probability mass. If μ is any probability mass on V , define its density relative to stationarity by

$$F_\mu(x) := n\mu(x).$$

Then $\mathbb{E}_V F_\mu = 1$ and

$$\|\mu - u\|_{\text{TV}} := \frac{1}{2} \sum_x |\mu(x) - u(x)| = \frac{1}{2} \mathbb{E}_V |F_\mu - 1|. \quad (3.1)$$

For the walk started from x , write $\mu_t^{(x)}$ for the law after t steps.

Corollary 3.1 (Spectral total-variation mixing bound). *For every vertex x and every integer $t \geq 0$,*

$$\|\mu_t^{(x)} - u\|_{\text{TV}} \leq \frac{1}{2} \sqrt{n-1} \lambda_{\text{abs}}(G)^t. \quad (3.2)$$

Hence, if G is connected and nonbipartite and $\gamma_{\text{abs}} = 1 - \lambda_{\text{abs}} > 0$, then

$$t_{\text{mix}}(\varepsilon) \leq \left\lceil \frac{\log(\sqrt{n-1}/(2\varepsilon))}{-\log \lambda_{\text{abs}}} \right\rceil \leq \left\lceil \frac{\log(\sqrt{n-1}/(2\varepsilon))}{\gamma_{\text{abs}}} \right\rceil, \quad (3.3)$$

where $t_{\text{mix}}(\varepsilon)$ is the least t for which the left-hand side of (3.2) is at most ε for every starting vertex.

For the lazy operator

$$P_L := \frac{1}{2}(I + P),$$

a connected graph is automatically aperiodic and

$$\lambda_{\text{abs}}(P_L|_{L_0^2}) = \frac{1 + \lambda_2(P)}{2} = 1 - \frac{\gamma(G)}{2}. \quad (3.4)$$

Thus the same bound applies with absolute gap $\gamma(G)/2$ even when G is bipartite.

Proof. Because P is symmetric, evolution of a probability column and action of P on a function use the same matrix. Therefore

$$F_{\mu_t^{(x)}} - 1 = P^t(F_{\delta_x} - 1).$$

The initial centered density has exact norm

$$\|F_{\delta_x} - 1\|_{2,V}^2 = \frac{1}{n}((n-1)^2 + (n-1)) = n-1.$$

By Cauchy–Schwarz in (3.1) and Theorem 2.1,

$$\|\mu_t^{(x)} - u\|_{\text{TV}} \leq \frac{1}{2} \|F_{\mu_t^{(x)}} - 1\|_{2,V} \leq \frac{1}{2} \sqrt{n-1} \lambda_{\text{abs}}^t,$$

which is (3.2). Solving for t gives the first inequality in (3.3), and $-\log r \geq 1-r$ for $0 < r < 1$ gives the second.

The eigenvalues of P_L are $(1 + \lambda_i)/2 \in [0, 1]$. Among the nonconstant eigenvalues the largest is therefore $(1 + \lambda_2)/2$, proving (3.4). $\square$

Pointwise versus distributional mixing. Equation (3.2) is uniform over the starting vertex and has the familiar $O(\log n)$ cost caused by a point mass initially having L^2 density $\sqrt{n-1}$ away from stationarity. If the initial distribution already has smaller L^2 discrepancy, the same proof replaces $\sqrt{n-1}$ by its exact initial centered-density norm.

Executable spectral-walk laboratory. The reader-facing program `code/volume_07/spectral_walk_lab.py` constructs cycles, complete graphs, and complete bipartite graphs; diagonalizes their normalized adjacency matrices numerically with a standard-library Jacobi method; simulates random walks; brute-forces conductance on small graphs; and compares observed total variation and edge discrepancy with (2.2), (3.2), and the Cheeger bounds of Chapter 4. Its built-in self-test is intended as a normalization regression test: changing a factor of d , using λ_2 instead of λ_{abs} for a bipartite walk, or counting internal edges only once causes a failure.

Theorem 3.2 (Scalar expander-walk Chernoff bound). *Let G be a finite undirected regular graph with normalized adjacency P and*

$$\lambda := \lambda_{\text{abs}}(G) < 1, \quad \gamma := 1 - \lambda.$$

Start the random walk $X_0, X_1, \dots$ from the uniform stationary law. For any function $f : V \rightarrow [0, 1]$, write

$$\mu := \mathbb{E}_V f, \quad S_t := \sum_{j=0}^{t-1} f(X_j).$$

Then for every integer $t \geq 1$ and every $0 < \varepsilon \leq 1$,

$$\Pr\left(\frac{S_t}{t} - \mu \geq \varepsilon\right) \leq \exp\left(-\frac{\gamma \varepsilon^2 t}{96}\right), \quad (3.5)$$

and hence

$$\Pr\left(\left|\frac{S_t}{t} - \mu\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{\gamma \varepsilon^2 t}{96}\right). \quad (3.6)$$

The constant 96 is deliberately conservative; the point of the theorem is the dimension-free exponent $\Theta(\gamma \varepsilon^2 t)$.

Proof. We give the moment-generating-operator proof because it pinpoints exactly where spectral expansion substitutes for independence. All Hilbert-space norms and inner products below are normalized by the uniform measure on V .

Fix $0 \leq r \leq 1$ and let D denote multiplication by

$$e^{rf/2}.$$

First we prove the operator estimate

$$\|DPD\|_{\text{op}} \leq 1 + r\mu + \frac{24r^2}{\gamma}. \quad (3.7)$$

Let x be a unit vector. Multiplying by a scalar phase if necessary, write

$$x = a\mathbf{1} + y, \quad a \geq 0, \quad y \perp \mathbf{1}, \quad a^2 + b^2 = 1, \quad b := \|y\|_2. \quad (3.8)$$

Put

$$h := e^{rf/2} - 1, \quad s := e^{r/2} - 1.$$

Convexity of $u \mapsto e^{ru/2}$ on $[0, 1]$ gives

$$0 \leq h \leq sf, \quad \mathbb{E}h \leq s\mu, \quad \|h\|_2 \leq s\sqrt{\mu}. \quad (3.9)$$

If Q is orthogonal projection onto $\mathbf{1}^\perp$, then

$$\begin{aligned} |\mathbb{E}(Dx)| &\leq a(1 + s\mu) + s\sqrt{\mu}b, \\ \|QDx\|_2 &= \|Q(x + hx)\|_2 \leq b + s. \end{aligned} \quad (3.10)$$

Since P fixes constants and has operator norm at most λ on $\mathbf{1}^\perp$,

$$|\langle Dx, P(Dx) \rangle| \leq |\mathbb{E}(Dx)|^2 + \lambda \|QDx\|_2^2. \quad (3.11)$$

We estimate the two terms while retaining the spectral penalty γb^2 . The two mixed terms are absorbed using the explicit Young bounds

$$4s\sqrt{\mu}b \leq \frac{\gamma}{4}b^2 + \frac{16s^2\mu}{\gamma}, \quad 2\lambda sb \leq \frac{\gamma}{4}b^2 + \frac{4s^2}{\gamma},$$

where we also used $0 \leq \lambda \leq 1$. Hence (3.10) gives

$$\begin{aligned} |\mathbb{E}(Dx)|^2 &\leq a^2 + 2s\mu + 2s^2\mu + \frac{\gamma}{4}b^2 + \frac{16s^2\mu}{\gamma}, \\ \lambda(b + s)^2 &\leq \lambda b^2 + \frac{\gamma}{4}b^2 + \lambda s^2 + \frac{4s^2}{\gamma}. \end{aligned} \quad (3.12)$$

Because $a^2 + \lambda b^2 = 1 - \gamma b^2$, adding (3.12) yields

$$|\langle x, DPDx \rangle| \leq 1 + 2s\mu + \frac{23s^2}{\gamma}. \quad (3.13)$$

For $0 \leq r \leq 1$,

$$2s \leq r + \frac{r^2}{2}, \quad s^2 \leq r^2.$$

Since $\gamma \leq 1$, (3.13) implies (3.7). Notice that (3.11) bounds the *absolute* Rayleigh quotient, so (3.7) is an operator-norm bound even when P has negative eigenvalues.

We now turn (3.7) into a tail bound. Stationarity and reversibility give the exact transfer-matrix identity

$$\mathbb{E} e^{rS_t} = \langle D\mathbf{1}, (DPD)^{t-1} D\mathbf{1} \rangle. \quad (3.14)$$

The endpoint factor obeys a one-variable Hoeffding estimate. Convexity gives

$$\|D\mathbf{1}\|_2^2 = \mathbb{E} e^{rf} \leq 1 - \mu + \mu e^r.$$

If

$$\psi(r) := \log(1 - \mu + \mu e^r),$$

then $\psi(0) = 0$, $\psi'(0) = \mu$, and $\psi''(r) \leq 1/4$, whence

$$\|D\mathbf{1}\|_2^2 \leq \exp\left(r\mu + \frac{r^2}{8}\right). \quad (3.15)$$

Combining (3.7), (3.14), (3.15), and $\log(1+u) \leq u$ gives

$$\mathbb{E} e^{rS_t} \leq \exp\left(r\mu t + \frac{24tr^2}{\gamma}\right). \quad (3.16)$$

Therefore Markov's inequality yields

$$\Pr(S_t \geq (\mu + \varepsilon)t) \leq \exp\left(-r\varepsilon t + \frac{24tr^2}{\gamma}\right). \quad (3.17)$$

Choose

$$r := \frac{\gamma\varepsilon}{48} \leq 1.$$

Then (3.17) becomes (3.5). Applying the same upper-tail estimate to $1 - f$, whose stationary mean is $1 - \mu$, gives the lower tail; the union bound proves (3.6). $\square$

Comparison with independence. For independent samples the exponent is $\Theta(\varepsilon^2 t)$. Along an expander walk, correlations cost one factor of the inverse relaxation scale: (3.6) has exponent $\Theta(\gamma\varepsilon^2 t)$. No factor depending on $|V|$ appears because the walk is started in stationarity. This scalar theorem is the interface used later for pseudorandomness; matrix-valued expander-walk concentration remains a separate frontier.

Chapter 4

Cheeger Inequalities and Exact Model Examples

Spectral gap and edge conductance are equivalent at the level of constant factors, but the two directions have different proofs. The easy direction uses one test function in the Rayleigh quotient. The hard direction turns a low-energy function into a cut by sweeping through its level sets. We give the full finite proof because it is the prototype of many local-to-global arguments later in the series.

For a real function $f : V \rightarrow \mathbb{R}$, the normalized Laplacian is

$$L := I - P,$$

and its Dirichlet form is

$$\langle f, Lf \rangle_V = \frac{1}{dn} \sum_{\{x,y\} \in E} (f(x) - f(y))^2, \quad (4.1)$$

where parallel edges are counted with multiplicity. On $L_0^2(V)$ the smallest eigenvalue of L is

$$\gamma(G) = 1 - \lambda_2(P). \quad (4.2)$$

Theorem 4.1 (Cheeger inequalities for finite regular graphs). *For every finite undirected d -regular graph,*

$$\boxed{\frac{\Phi(G)^2}{2} \leq \gamma(G) \leq 2\Phi(G).} \quad (4.3)$$

Equivalently, in adjacency-eigenvalue normalization,

$$\frac{d - \mu_2(A)}{2} \leq \frac{|\partial_E S|}{|S|} \quad \text{for every } 0 < |S| \leq n/2,$$

and

$$\min_{0 < |S| \leq n/2} \frac{|\partial_E S|}{|S|} \leq \sqrt{2d(d - \mu_2(A))}.$$

The theorem involves the one-sided gap γ , not the absolute gap.

Proof. Upper bound $\gamma \leq 2\Phi$. Fix S with $0 < |S| \leq n/2$, put $s = |S|/n$, and let

$$f := \mathbf{1}_S - s\mathbf{1}.$$

Then $f \perp \mathbf{1}$ and $f(x) - f(y)$ is nonzero exactly across the cut, with absolute value 1. Hence (4.1) gives

$$\langle f, Lf \rangle_V = \frac{|\partial_E S|}{dn}, \quad \|f\|_{2,V}^2 = s(1-s).$$

The variational characterization of γ yields

$$\gamma \leq \frac{|\partial_E S|/(dn)}{s(1-s)} = \frac{|\partial_E S|}{d|S|(1-s)} \leq 2\Phi(S),$$

because $s \leq 1/2$. Minimize over S .

A sweep inequality. We next prove the following statement. If $g \geq 0$, $g \not\equiv 0$, and $|\text{supp } g| \leq n/2$, then

$$\Phi(G)^2 \leq 2 \frac{\sum_{\{x,y\} \in E} (g(x) - g(y))^2}{d \sum_x g(x)^2}. \quad (4.4)$$

For $t \geq 0$, let

$$S_t := \{x : g(x)^2 \geq t\}.$$

Every S_t has size at most $n/2$. The layer-cake identity gives

$$\sum_{\{x,y\} \in E} |g(x)^2 - g(y)^2| = \int_0^\infty |\partial_E S_t| dt \geq \Phi(G) d \int_0^\infty |S_t| dt = \Phi(G) d \sum_x g(x)^2. \quad (4.5)$$

On the other hand, Cauchy–Schwarz and $|a^2 - b^2| = |a - b||a + b|$ give

$$\sum_E |g(x)^2 - g(y)^2| \leq \left(\sum_E (g(x) - g(y))^2 \right)^{1/2} \left(\sum_E (g(x) + g(y))^2 \right)^{1/2}.$$

Since $(a + b)^2 \leq 2(a^2 + b^2)$ and G is d -regular,

$$\sum_E (g(x) + g(y))^2 \leq 2d \sum_x g(x)^2.$$

Combining this with (4.5) and squaring proves (4.4).

Lower bound $\Phi^2/2 \leq \gamma$. Let f be a real eigenfunction satisfying $Lf = \gamma f$ and $f \perp \mathbf{1}$. Both the positive and negative sets are nonempty. Replace f by $-f$ if necessary so that

$$|\{x : f(x) > 0\}| \leq n/2,$$

and put $g = f_+ := \max(f, 0)$. The scalar truncation map $r \mapsto r_+$ is monotone and 1-Lipschitz, so for every edge

$$(g(x) - g(y))(f(x) - f(y)) \geq (g(x) - g(y))^2. \quad (4.6)$$

Using the bilinear version of (4.1),

$$\langle g, Lg \rangle_V \leq \langle g, Lf \rangle_V = \gamma \langle g, f \rangle_V = \gamma \|g\|_{2,V}^2.$$

Thus the Rayleigh quotient of g is at most γ . Applying the sweep inequality (4.4) gives $\Phi^2 \leq 2\gamma$, completing the proof. $\square$

Why laziness is not part of Cheeger. Cheeger controls $1 - \lambda_2$, so a connected bipartite graph may satisfy a strong Cheeger inequality while its ordinary walk is periodic. Passing to $P_L = (I + P)/2$ removes the negative endpoint and turns the one-sided gap γ into the absolute lazy-walk gap $\gamma/2$, exactly as in (3.4).

Worked comparison: cycle, complete graph, and complete bipartite graph. Three families isolate the three phenomena that are easiest to conflate.

(a) **Cycle C_n .** Here $d = 2$ and

$$\lambda_j(P) = \cos \frac{2\pi j}{n}, \quad 0 \leq j < n.$$

Hence

$$\gamma(C_n) = 1 - \cos \frac{2\pi}{n} \asymp n^{-2}.$$

If n is even then -1 is an eigenvalue and $\lambda_{\text{abs}} = 1$; the non-lazy walk is periodic. If n is odd then $\lambda_{\text{abs}} = \cos(\pi/n) = 1 - \Theta(n^{-2})$. A consecutive block of $\lfloor n/2 \rfloor$ vertices has two boundary edges, so $\Phi(C_n) = \Theta(n^{-1})$. Thus the lower Cheeger bound has the correct quadratic scale $\gamma \asymp \Phi^2$.

(b) **Complete graph K_n .** Now $d = n - 1$ and the nonconstant eigenvalue is $-1/(n - 1)$ with multiplicity $n - 1$. Therefore

$$\lambda_{\text{abs}}(K_n) = \frac{1}{n - 1}, \quad \gamma(K_n) = \frac{n}{n - 1}.$$

For $|S| = s \leq n/2$,

$$\Phi(S) = \frac{s(n - s)}{(n - 1)s} = \frac{n - s}{n - 1},$$

so the minimum occurs at $s = \lfloor n/2 \rfloor$. When n is even, $2\Phi(K_n) = n/(n - 1) = \gamma(K_n)$: the easy Cheeger inequality is exact.

(c) **Complete bipartite graph $K_{m,m}$.** The normalized spectrum is

$$1, \underbrace{0, \dots, 0}_{2m-2 \text{ times}}, -1.$$

Thus $\gamma = 1$ but $\lambda_{\text{abs}} = 1$. When m is even, choosing $m/2$ vertices in each side gives $\Phi = 1/2$, and the upper Cheeger bound $\gamma \leq 2\Phi$ is again exact. Yet a walk started on one side alternates sides forever. This is the canonical warning that one-sided expansion and non-lazy mixing are not identical notions.

Chapter 5

Random Expanders and the Near-Ramanujan Benchmark

Alon–Boppana says that the normalized nontrivial spectrum of a large d -regular graph cannot lie substantially inside $[-2\sqrt{d-1}/d, 2\sqrt{d-1}/d]$ on the positive side. A striking theorem of Friedman says that a uniformly random regular graph essentially attains this barrier on *both* sides. We record that theorem as a deep input and then deduce the elementary probabilistic-existence consequence needed by the rest of the series.

Theorem 5.1 (Friedman’s second-eigenvalue theorem — deep input). *Fix an integer $d \geq 3$ and $\varepsilon > 0$. Let $G_{n,d}$ be uniformly random among simple d -regular graphs on the labelled vertex set $[n]$, with nd even. If*

$$d = \mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$$

are the adjacency eigenvalues, then

$$\Pr\left(\max\{\mu_2, |\mu_n|\} > 2\sqrt{d-1} + \varepsilon\right) \longrightarrow 0 \quad (n \rightarrow \infty). \quad (5.1)$$

Equivalently, in normalized adjacency convention,

$$\lambda_{\text{abs}}(G_{n,d}) \leq \frac{2\sqrt{d-1} + \varepsilon}{d}$$

with probability tending to 1.

Ownership boundary. The selective-trace/tangle analysis behind (5.1) is not reproduced here. The theorem is kept as an owner-C deep input. Its role in this volume is to set the sharp random benchmark and to make the probabilistic existence of constant-degree expanders immediate. This separation prevents a short existence corollary from being mistaken for a proof of Friedman’s theorem.

Theorem 5.2 (Probabilistic existence of constant-degree absolute expanders). *For every fixed integer $d \geq 3$ there is a constant $\delta_d > 0$ and an infinite sequence of finite simple d -regular graphs G_j with $|V(G_j)| \rightarrow \infty$ such that*

$$\lambda_{\text{abs}}(G_j) \leq 1 - \delta_d. \quad (5.2)$$

Indeed, a uniformly random simple d -regular graph has (5.2) with probability tending to 1 for any

$$0 < \delta_d < 1 - \frac{2\sqrt{d-1}}{d}. \quad (5.3)$$

Proof. For $d \geq 3$,

$$2\sqrt{d-1} < d$$

because $(d-2)^2 > 0$. Choose δ_d satisfying (5.3), and then choose $\varepsilon > 0$ so small that

$$\frac{2\sqrt{d-1} + \varepsilon}{d} \leq 1 - \delta_d.$$

Theorem 5.1 says that a uniformly random simple d -regular graph satisfies this absolute spectral bound with probability approaching 1. Hence such graphs exist for every sufficiently large admissible n , and selecting one for an unbounded sequence of n proves the theorem. $\square$

What is optimal and what is not. For fixed d , Friedman gives

$$\lambda_{\text{abs}} \leq \frac{2\sqrt{d-1}}{d} + o(1)$$

for almost every graph, while Theorem 2.3 forces

$$\lambda_2 \geq \frac{2\sqrt{d-1}}{d} - o(1)$$

for every large graph. Thus the random model is asymptotically optimal at the Alon–Boppana scale. The next construction problem is different: produce such expansion *deterministically* with an efficient neighbor function. The registered Margulis–Gabber–Galil and Ramanujan constructions are left source-visible but unpromoted in this pass; Chapter 6 will instead begin with the combinatorial zig-zag route, whose operator proof depends only on the spectral dictionary already established.

5.1 Two classical explicit algebraic families

The probabilistic theorem above proves existence but does not by itself give a neighbor algorithm. We now put two classical algebraic constructions on the same convention ledger. Their roles are deliberately separated. For the Margulis–Gabber–Galil family the affine neighbor rule is elementary and the spectral estimate is isolated as a theorem-strength source input. For the Lubotzky–Phillips–Sarnak family both the arithmetic construction and the Ramanujan estimate are retained as one owner-C package, because the latter rests on arithmetic representation theory far beyond the present spectral chapter.

Theorem 5.3 (Gabber–Galil spectral estimate — deep input). *Let $m \geq 2$, put $V_m = (\mathbb{Z}/m\mathbb{Z})^2$, and define*

$$T_1 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (5.4)$$

For $v \in V_m$ let the eight labelled edges out of v be generated by the four affine bijections

$$A_1(v) = T_1 v, \quad A_2(v) = T_2 v, \quad A_3(v) = T_1 v + e_1, \quad A_4(v) = T_2 v + e_2 \quad (5.5)$$

and their inverses, with multiplicity retained. If A_m denotes the unnormalized adjacency operator of this 8-regular undirected multigraph, then every eigenvalue on the orthogonal complement of the constants satisfies

$$|\mu| \leq 5\sqrt{2}. \quad (5.6)$$

Equivalently,

$$\lambda_{\text{abs}}(M_m) \leq \frac{5\sqrt{2}}{8} < 1. \quad (5.7)$$

Ownership boundary. The affine rule (5.4)–(5.5) is elementary. The harmonic-analysis estimate (5.6), due to Gabber and Galil for this standard Margulis-type family, is kept as an owner-C input. In particular, no sentence below should be read as a new proof of the $5\sqrt{2}$ constant.

Algorithm 5.4 (Margulis–Gabber–Galil explicit expander family). *For every $m \geq 2$, let M_m be the labelled multigraph of Theorem 5.3. Then:*

1. M_m has exactly m^2 vertices and degree 8;
2. its rotation map is computable from $(x, y) \in V_m$ and a label in $[8]$ using $O(\log m)$ -bit modular arithmetic;
3. the family has the uniform absolute spectral bound (5.7), and hence, by Cheeger’s inequality,

$$\Phi(M_m) \geq \frac{1}{2} \left(1 - \frac{5\sqrt{2}}{8} \right) > 0. \quad (5.8)$$

Consequently $(M_m)_{m \geq 1}$ is a strongly explicit family of constant-degree expanders, in this volume’s port-multigraph convention.

Proof. Every T_i has determinant 1 modulo m , so each A_i is a bijection of V_m . Pair label i with the inverse label of A_i ; this gives an involutive rotation map and hence an undirected 8-regular port multigraph, including the correct multiplicities when two labelled moves coincide. The formulas use only additions, subtractions, doubling, and reduction modulo m , so a neighbor query uses $O(\log m)$ work bits.

Theorem 5.3 gives (5.7). In particular the graph is connected and nonbipartite for every nontrivial instance in the spectral sense used here. Applying the lower Cheeger inequality from Theorem 4.1,

$$\Phi(G) \geq \frac{1 - \lambda_2(G)}{2},$$

and using $\lambda_2 \leq \lambda_{\text{abs}}$ gives (5.8). The constants are independent of m . $\square$

A multiplicity check at $m = 3$. Take $v = (1, 2)$. The eight labelled destinations, in the order $A_1, A_1^{-1}, A_2, A_2^{-1}, A_3, A_3^{-1}, A_4, A_4^{-1}$, are

$$(2, 2), (0, 2), (1, 1), (1, 0), (0, 2), (2, 2), (1, 2), (1, 2). \quad (5.9)$$

Thus parallel edges and self-loops really occur. Treating (5.5) as a simple set of neighbors would silently change the Markov operator and invalidate the spectral normalization; the port-multigraph convention prevents that error.

Theorem 5.5 (Lubotzky–Phillips–Sarnak Ramanujan graphs — deep input). *Let $p \neq q$ be primes with*

$$p \equiv q \equiv 1 \pmod{4}, \quad q > 2\sqrt{p}. \quad (5.10)$$

Choose $\iota \in \mathbb{F}_q$ with $\iota^2 = -1$. There are exactly $p + 1$ integer solutions

$$p = a_0^2 + a_1^2 + a_2^2 + a_3^2, \quad a_0 > 0 \text{ odd}, \quad a_1, a_2, a_3 \text{ even}, \quad (5.11)$$

and to each attach the projective class of

$$\begin{pmatrix} a_0 + \iota a_1 & a_2 + \iota a_3 \\ -a_2 + \iota a_3 & a_0 - \iota a_1 \end{pmatrix} \in \text{PGL}_2(\mathbb{F}_q). \quad (5.12)$$

These $p + 1$ classes form a symmetric generating set and define an explicit $(p + 1)$ -regular Cayley graph $X^{p,q}$ with the following two branches.

1. If the Legendre symbol $\left(\frac{p}{q}\right) = +1$, then $X^{p,q}$ is a nonbipartite Cayley graph of $\text{PSL}_2(\mathbb{F}_q)$, hence has $q(q^2 - 1)/2$ vertices.
2. If $\left(\frac{p}{q}\right) = -1$, then $X^{p,q}$ is a bipartite Cayley graph of $\text{PGL}_2(\mathbb{F}_q)$, hence has $q(q^2 - 1)$ vertices; its two sides are the two cosets of $\text{PSL}_2(\mathbb{F}_q)$.

If μ is an adjacency eigenvalue other than the trivial eigenvalue $p + 1$ and, in the bipartite branch, the trivial eigenvalue $-(p + 1)$, then

$$|\mu| \leq 2\sqrt{p}. \quad (5.13)$$

Thus the nontrivial normalized spectrum satisfies

$$|\lambda| \leq \frac{2\sqrt{p}}{p + 1}. \quad (5.14)$$

Arithmetic ownership. The matrix rule (5.12) makes the Cayley generators explicit, but the Ramanujan estimate (5.13) is the deep part of the LPS theorem and is not reproved here. The two projective groups and the bipartite branch are stated separately because suppressing that dichotomy would incorrectly count the vertices and would also misclassify the eigenvalue $-(p + 1)$ as nontrivial. The former unresolved companion pointer has therefore been retired: this result is owned directly by the verified LPS source gate.

Worked arithmetic instance. Take $p = 5$ and $q = 13$. Since $5^2 \equiv -1 \pmod{13}$ we may take $\iota = 5$. The six solutions in (5.11) are

$$(1, \pm 2, 0, 0), \quad (1, 0, \pm 2, 0), \quad (1, 0, 0, \pm 2).$$

Moreover 5 is a quadratic nonresidue modulo 13, so this is the bipartite $\text{PGL}_2(\mathbb{F}_{13})$ branch: the graph is 6-regular on

$$13(13^2 - 1) = 2184$$

vertices, and every nontrivial adjacency eigenvalue has absolute value at most $2\sqrt{5}$. Its normalized Ramanujan bound is therefore $\lambda \leq 2\sqrt{5}/6 \approx 0.7454$.

Three explicitness levels. The chapter now distinguishes three logically different facts. Friedman shows that near-optimal expanders are abundant. Margulis–Gabber–Galil gives an elementary neighbor rule with a uniform but non-Ramanujan spectral constant. LPS gives a highly structured arithmetic neighbor rule together with the optimal Ramanujan bound, at the cost of a genuinely deep arithmetic spectral input. These distinctions become important in the logspace connectivity construction, where only one fixed constant-size expander is needed and exhaustive search is therefore sufficient.

Chapter 6

Rotation Maps and the Zig–Zag Product

The previous chapters treated a regular graph through its adjacency operator. For explicit constructions one needs a second representation: given a vertex and a local edge label, compute the opposite endpoint and the reverse label. This *rotation-map* model makes parallel edges and self-loops harmless, and it is the natural interface for graph products.

Throughout this chapter, a D -regular port graph on $[N]$ means an involution

$$\text{Rot}_G : [N] \times [D] \longrightarrow [N] \times [D].$$

If $\text{Rot}_G(v, i) = (w, j)$, port i at v is paired with port j at w . A fixed point represents a self-loop port. The normalized adjacency operator averages uniformly over the D ports. Thus “degree” in this chapter is the number of ports, which removes the usual convention ambiguity about how a loop contributes to degree.

Definition 6.1 (Rotation maps and combinatorial graph products). Let G be a D -regular port graph on $[N]$.

- (a) The *square* G^2 has the same vertex set and degree D^2 ; a label (i_1, i_2) means two successive G -steps. On normalized adjacency operators,

$$P_{G^2} = P_G^2. \tag{6.1}$$

- (b) If G_1, G_2 have parameters (N_1, D_1) and (N_2, D_2) , their *tensor product* $G_1 \otimes G_2$ has vertex set $[N_1] \times [N_2]$, degree $D_1 D_2$, and performs one step in each factor. Hence

$$P_{G_1 \otimes G_2} = P_{G_1} \otimes P_{G_2}. \tag{6.2}$$

- (c) Suppose G_1 is D_1 -regular on $[N_1]$ and G_2 is D_2 -regular on $[D_1]$. The *zig-zag product* $G_1 \text{ Z } G_2$ has vertex set $[N_1] \times [D_1]$ and degree D_2^2 . Its edge label is a pair $(i, j) \in [D_2]^2$. Starting from (v, k) , compute

$$\begin{aligned} (k', i') &= \text{Rot}_{G_2}(k, i), \\ (w, \ell') &= \text{Rot}_{G_1}(v, k'), \\ (\ell, j') &= \text{Rot}_{G_2}(\ell', j), \end{aligned} \tag{6.3}$$

and output

$$\text{Rot}_{G_1 \text{ Z } G_2}((v, k), (i, j)) = ((w, \ell), (j', i')). \tag{6.4}$$

Thus one product step is *zig inside the old cloud, jump between clouds, zag inside the new cloud*.

- (d) The *replacement product* $G_1 \text{ R } G_2$ uses the same cloud vertex set $[N_1] \times [D_1]$ and has degree $D_2 + 1$. For a label $i \in [D_2 + 1]$, set

$$\text{Rot}_{G_1 \text{ R } G_2}((v, k), i) = \begin{cases} ((v, m), j), & 1 \leq i \leq D_2, \quad (m, j) = \text{Rot}_{G_2}(k, i), \\ ((w, \ell), D_2 + 1), & i = D_2 + 1, \quad (w, \ell) = \text{Rot}_{G_1}(v, k). \end{cases} \quad (6.5)$$

Thus D_2 labels move inside the current cloud and one distinguished label crosses the G_1 edge indexed by the current port.

An (N, D, λ) -graph in this chapter means a D -regular port graph on N vertices whose normalized adjacency operator has absolute nontrivial spectral radius at most λ .

The definition in (6.3)–(6.4) is algorithmic, not merely existential: one query to Rot_{G_1} and two queries to Rot_{G_2} compute one zig-zag edge.

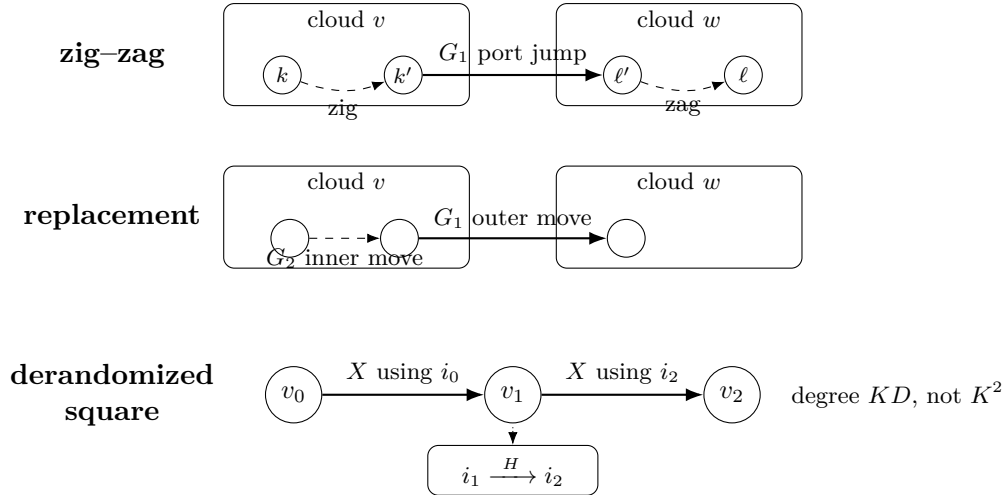


Figure 6.1: A rotation-map atlas for the three graph products used in this volume. Zig-zag spends two cheap inner moves around one expensive outer move; replacement chooses between an inner move and one outer port crossing; derandomized squaring takes two outer steps but recycles the first port label through an auxiliary expander before choosing the second. The drawing is schematic: ports, not Euclidean geometry, determine the graph.

Theorem 6.2 (Zig-zag spectral theorem). *Let G_1 be an (N_1, D_1, λ_1) -graph and G_2 a (D_1, D_2, λ_2) -graph, with $0 \leq \lambda_1, \lambda_2 < 1$. Then $G_1 \text{ Z } G_2$ is an $(N_1 D_1, D_2^2, \lambda)$ -graph satisfying*

$$\lambda \leq \lambda_1 + \lambda_2 + \lambda_2^2. \quad (6.6)$$

Moreover $\lambda < 1$. Quantitatively, if $0 < \lambda_2 < 1$, then

$$\lambda \leq 1 - \min \left\{ \frac{1 - \lambda_1}{9}, \frac{(1 - \lambda_1)^2 (1 - \lambda_2^2)}{9 \lambda_2^2} \right\} < 1, \quad (6.7)$$

while for $\lambda_2 = 0$ one has $\lambda \leq \lambda_1$. One product rotation query uses one query to G_1 and two to G_2 .

Proof. Work in the normalized Hilbert space on $[N_1] \times [D_1]$. Let $\mathcal{U}$ be the *parallel* subspace of functions that are constant inside each cloud $\{v\} \times [D_1]$, and let Q be orthogonal projection onto $\mathcal{U}$. Write

$$\alpha = \alpha^\parallel + \alpha^\perp, \quad \alpha^\parallel := Q\alpha, \quad \alpha^\perp := (I - Q)\alpha.$$

Let $B := I_{N_1} \otimes P_{G_2}$ be the operator for a zig or a zag. Let $\tilde{A}$ be the permutation operator induced by Rot_{G_1} on the $N_1 D_1$ ports. Because the rotation map is an involution, $\tilde{A}$ is self-adjoint and unitary. The normalized adjacency of the zig-zag product is exactly

$$M = B\tilde{A}B. \quad (6.8)$$

In particular M is self-adjoint.

The inner operator fixes the parallel subspace pointwise and contracts the perpendicular subspace:

$$B\alpha^\parallel = \alpha^\parallel, \quad \|B\alpha^\perp\|_2 \leq \lambda_2 \|\alpha^\perp\|_2. \quad (6.9)$$

There is one further structural identity. If a parallel vector is identified with its cloud-value function $a : [N_1] \rightarrow \mathbb{C}$, then cloud-averaging after the outer permutation gives exactly one G_1 step. Therefore, for $\alpha^\parallel \perp \mathbf{1}$,

$$|\langle \tilde{A}\alpha^\parallel, \alpha^\parallel \rangle| \leq \lambda_1 \|\alpha^\parallel\|_2^2. \quad (6.10)$$

Now take $\alpha \perp \mathbf{1}$ with $\|\alpha\|_2 = 1$, and set

$$p := \|\alpha^\parallel\|_2, \quad q := \|\alpha^\perp\|_2, \quad p^2 + q^2 = 1.$$

Using (6.8), moving the first B across the inner product, and applying (6.9)–(6.10) together with Cauchy–Schwarz gives

$$\begin{aligned} |\langle M\alpha, \alpha \rangle| &= |\langle \tilde{A}B\alpha, B\alpha \rangle| \\ &\leq \lambda_1 p^2 + 2\lambda_2 pq + \lambda_2^2 q^2. \end{aligned} \quad (6.11)$$

Since $2pq \leq 1$, (6.6) follows.

It remains to record why the product is genuinely expanding whenever both inputs are. If $\lambda_2 = 0$, then $B\alpha^\perp = 0$ and (6.11) immediately gives $|\langle M\alpha, \alpha \rangle| \leq \lambda_1$. Assume $0 < \lambda_2 < 1$ and put

$$\delta := \frac{1 - \lambda_1}{3\lambda_2}.$$

If $q \leq \delta$, then (6.11), $p \leq 1$, and $1 - \lambda_1 \leq 1$ give

$$|\langle M\alpha, \alpha \rangle| \leq \lambda_1 + \frac{2}{3}(1 - \lambda_1) + \frac{1}{9}(1 - \lambda_1)^2 \leq 1 - \frac{1 - \lambda_1}{9}. \quad (6.12)$$

If $q > \delta$, use only that $\tilde{A}$ is unitary:

$$\begin{aligned} |\langle M\alpha, \alpha \rangle| &\leq \|B\alpha\|_2^2 \\ &= p^2 + \|B\alpha^\perp\|_2^2 \\ &\leq 1 - (1 - \lambda_2^2)q^2 \\ &< 1 - \frac{(1 - \lambda_1)^2(1 - \lambda_2^2)}{9\lambda_2^2}. \end{aligned} \quad (6.13)$$

Taking the worse of (6.12) and (6.13) proves (6.7). Since M is self-adjoint, the Rayleigh characterization on $\mathbf{1}^\perp$ turns these bounds into the claimed absolute nontrivial eigenvalue bounds. The rotation query count is read directly from (6.3). $\square$

Theorem 6.3 (Replacement-product spectral theorem). *Under the hypotheses of Theorem 6.2, let $f(\lambda_1, \lambda_2) < 1$ be any valid absolute spectral bound for the zig-zag product $G_1 \mathbf{Z} G_2$. Then $G_1 \mathbf{R} G_2$ is an $(N_1 D_1, D_2 + 1, g)$ -graph with*

$$g \leq (p + (1 - p)f(\lambda_1, \lambda_2))^{1/3} < 1, \quad p := \frac{D_2^2}{(D_2 + 1)^3}. \quad (6.14)$$

A replacement-product rotation query uses exactly one query to either Rot_{G_1} or Rot_{G_2} .

Proof. Let $\tilde{A}$ be the outer port-permutation operator and let $\tilde{B} := I_{N_1} \otimes P_{G_2}$ be the inner-cloud walk. The replacement walk chooses the unique outer move with probability $1/(D_2 + 1)$ and one of the D_2 inner moves otherwise, so

$$R := P_{G_1 \mathbf{R} G_2} = \frac{\tilde{A} + D_2 \tilde{B}}{D_2 + 1}. \quad (6.15)$$

Expand the cube. The word $\tilde{B}\tilde{A}\tilde{B}$ occurs with coefficient D_2^2 and is exactly the zig-zag Markov operator from (6.8). Every other word in the expansion is a product of transition/permutation operators corresponding to a three-step regular port walk. The remaining multiset of words is closed under reversal, with the same coefficient on a word and its adjoint, because $\tilde{A}$ and $\tilde{B}$ are self-adjoint. Consequently their normalized weighted average is a self-adjoint stochastic operator C (and hence the Markov operator of an undirected regular weighted port graph) with $\|C\|_{\text{op}} \leq 1$. Hence

$$R^3 = p P_{G_1 \mathbf{Z} G_2} + (1 - p)C, \quad p = \frac{D_2^2}{(D_2 + 1)^3}. \quad (6.16)$$

Both summands fix constants, and on $\mathbf{1}^\perp$ their operator norms are at most $f(\lambda_1, \lambda_2)$ and 1, respectively. Thus

$$\|R^3|_{\mathbf{1}^\perp}\|_{\text{op}} \leq pf(\lambda_1, \lambda_2) + (1 - p). \quad (6.17)$$

The operator R is self-adjoint, so its absolute nontrivial spectral radius cubed equals the left side of (6.17). Taking cube roots proves (6.14). The query count follows immediately from Definition 6.1. $\square$

Why the decomposition is the whole idea. The parallel component is already uniform inside every cloud, so the outer G_1 step is what can contract it. The perpendicular component may carry no useful information from G_1 , but each of the two G_2 steps contracts its within-cloud oscillation. The zig-zag product works because every vector is an orthogonal sum of exactly these two failure modes.

Algorithm 6.4 (Recursive constant-degree expander construction). *Assume a seed (D^8, D, α) -graph H with*

$$\alpha \leq \frac{1}{5} \quad (6.18)$$

and an efficiently computable rotation map. Define

$$G_1 := H^2, \quad G_2 := H \otimes H, \quad (6.19)$$

and for $t > 2$, with

$$a := \left\lceil \frac{t-1}{2} \right\rceil, \quad b := \left\lfloor \frac{t-1}{2} \right\rfloor,$$

set

$$G_t := (G_a \otimes G_b)^2 \mathbf{Z} H. \quad (6.20)$$

Then, for every $t \geq 1$,

$$|V(G_t)| = D^{8t}, \quad \deg(G_t) = D^2, \quad \lambda_{\text{abs}}(G_t) \leq \frac{2}{5}. \quad (6.21)$$

Furthermore one rotation-map query to G_t can be evaluated using $O(t^2)$ oracle queries to Rot_H and time polynomial in t and $\log D$. For a fixed seed H , this is polynomial in $\log |V(G_t)|$.

Proof. The size and degree identities are inductive. For $G_1 = H^2$ the vertex count is D^8 and the degree is D^2 ; for $G_2 = H \otimes H$ they are D^{16} and D^2 . If (6.21) holds below t , then $a + b = t - 1$, so $G_a \otimes G_b$ has $D^{8(t-1)}$ vertices and degree D^4 . Squaring keeps the vertex set and changes the degree to D^8 , which is exactly the number of vertices of H . The zig-zag product therefore has D^{8t} vertices and degree D^2 .

For the spectrum, write $\lambda_t := \lambda_{\text{abs}}(G_t)$ and $\mu_t := \max_{1 \leq s \leq t} \lambda_s$. Equations (6.1)–(6.2) give

$$\lambda_1 \leq \alpha^2, \quad \lambda_2 \leq \alpha,$$

and, for $t > 2$, the outer graph in (6.20), before zig-zagging, has absolute nontrivial eigenvalue at most μ_{t-1}^2 . The basic bound (6.6) therefore gives

$$\lambda_t \leq \mu_{t-1}^2 + \alpha + \alpha^2. \quad (6.22)$$

Since $\alpha \leq 1/5$,

$$\alpha + \alpha^2 \leq \frac{6}{25}.$$

If $\mu_{t-1} \leq 2/5$, then

$$\lambda_t \leq \frac{4}{25} + \frac{6}{25} = \frac{2}{5}.$$

The base cases satisfy the same bound, so induction proves the spectral part of (6.21).

Finally let $Q(t)$ be the number of seed-rotation queries needed for one G_t rotation. A zig-zag step makes two calls to H . The squared tensor outer step makes two tensor-product calls, each of which makes one call to G_a and one to G_b . Thus, for an absolute constant C_0 ,

$$Q(t) \leq 2Q(a) + 2Q(b) + C_0. \quad (6.23)$$

Because $a + b = t - 1$ and $|a - b| \leq 1$,

$$a^2 + b^2 \leq \frac{(t-1)^2 + 1}{2}.$$

A direct induction in (6.23) yields $Q(t) = O(t^2)$; the arithmetic on labels has the same polynomial overhead. Since $\log |V(G_t)| = 8t \log D$, the fixed-seed family is strongly explicit. $\square$

Source boundary. The product definition, the parallel/perpendicular proof architecture, and the tensor-balanced recursion above are the Reingold–Vadhan–Wigderson interfaces. The manuscript proves the displayed spectral and recursion bounds locally. The replacement-product comparison is also proved locally from the cube identity (6.16); no graph-product spectral theorem in this chapter is left black-boxed.

6.1 The sharp zig-zag estimate needed at vanishing gap

The local proof of Theorem 6.2 is deliberately robust, but its explicit strict bound deteriorates quadratically as $1 - \lambda_1 \rightarrow 0$. Reingold's connectivity amplification needs a linear comparison of spectral gaps. We therefore record the sharper RVW estimate as a separate owner-C interface rather than silently strengthening the local Pass-20 theorem.

Theorem 6.5 (Sharp RVW zig-zag eigenvalue bound — source-gated input). *Under the hypotheses of Theorem 6.2,*

$$\lambda_{\text{abs}}(G_1 \text{ Z } G_2) \leq F(\lambda_1, \lambda_2), \quad (6.24)$$

where

$$F(x, y) := \frac{1}{2}(1 - y^2)x + \frac{1}{2}\sqrt{(1 - y^2)^2x^2 + 4y^2}. \quad (6.25)$$

Consequently

$$1 - \lambda_{\text{abs}}(G_1 \text{ Z } G_2) \geq \frac{1}{2}(1 - \lambda_2^2)(1 - \lambda_1). \quad (6.26)$$

In particular, if $\lambda_2 \leq 1/2$, then

$$1 - \lambda_{\text{abs}}(G_1 \text{ Z } G_2) \geq \frac{3}{8}(1 - \lambda_1). \quad (6.27)$$

Verification and the elementary corollary. Formula (6.25) is the improved RVW eigenvalue theorem and is retained as the single source-gated input in this section. The gap estimate itself is immediate and is included here to pin down the constant used later. Since $x \leq 1$,

$$\sqrt{(1 - y^2)^2x^2 + 4y^2} \leq \sqrt{(1 - y^2)^2 + 4y^2} = 1 + y^2.$$

Substituting this into (6.25) gives

$$F(x, y) \leq \frac{1}{2}(1 - y^2)x + \frac{1}{2}(1 + y^2) = 1 - \frac{1}{2}(1 - y^2)(1 - x),$$

which is (6.26); (6.27) follows from $1 - y^2 \geq 3/4$.

Chapter 7

Derandomized Squaring

Ordinary graph squaring improves a spectral radius from λ to λ^2 , but it also squares the degree. Derandomized squaring keeps the first random port of the outer walk and replaces the second independent port by a short step in an auxiliary expander on the port labels. The resulting degree is only the old degree times the auxiliary degree, while its spectral behavior remains close to true squaring.

We use the two-way-label formulation, which is exactly the rotation-map language of Chapter 6. This avoids the artificial “consistent one-way labelling” restriction and, for undirected inputs, preserves undirectedness automatically.

Theorem 7.1 (Derandomized squaring in rotation-map form). *Let X be an (N, K, λ) -graph with rotation map Rot_X , and let H be a (K, D, μ) -graph with rotation map Rot_H , where $0 \leq \lambda, \mu < 1$. Define $X \text{ DS } H$ on vertex set $[N]$ with edge labels $(i, j) \in [K] \times [D]$ by*

$$\begin{aligned} (v_1, i_1) &= \text{Rot}_X(v_0, i_0), \\ (i_2, j_1) &= \text{Rot}_H(i_1, j_0), \\ (v_2, i_3) &= \text{Rot}_X(v_1, i_2), \end{aligned} \tag{7.1}$$

and

$$\text{Rot}_{X \text{ DS } H}(v_0, (i_0, j_0)) = (v_2, (i_3, j_1)). \tag{7.2}$$

Then $X \text{ DS } H$ is an undirected KD -regular port graph and

$$\lambda_{\text{abs}}(X \text{ DS } H) \leq f(\lambda, \mu) := (1 - \mu)\lambda^2 + \mu = 1 - (1 - \lambda^2)(1 - \mu) \leq \lambda^2 + \mu. \tag{7.3}$$

In particular, if $\lambda = 1 - \gamma$, $0 < \gamma < 1/4$, and $\mu \leq 1/100$, then

$$1 - \lambda_{\text{abs}}(X \text{ DS } H) \geq \frac{3}{2} \gamma. \tag{7.4}$$

One rotation query to $X \text{ DS } H$ uses two queries to Rot_X and one query to Rot_H .

Proof. The maps in the three lines of (7.1) are permutations of the appropriate port spaces. More explicitly, the induced map on $[N] \times [K] \times [D]$ is

$$(\text{Rot}_X \times I_D) \circ (I_N \times \text{Rot}_H) \circ (\text{Rot}_X \times I_D). \tag{7.5}$$

Because the two input rotation maps are involutions, (7.5) is an involution: the same three involutions appear in reverse order, which here is the same order. Thus (7.2) defines an undirected port graph. The label set has size KD , proving the degree assertion.

We now prove the spectral estimate. Put normalized inner products on $\ell^2([N])$ and $\ell^2([N] \times [K])$. Define the isometric lift

$$(Lf)(v, i) := f(v) \quad (7.6)$$

and let its adjoint be cloud averaging,

$$(L^*F)(v) := \frac{1}{K} \sum_{i=1}^K F(v, i). \quad (7.7)$$

Then

$$L^*L = I, \quad LL^* = I_N \otimes J_K, \quad (7.8)$$

where J_K is orthogonal projection onto the constant functions on $[K]$. Let $\tilde{A}$ be the permutation operator induced by Rot_X on $[N] \times [K]$. Cloud averaging before and after one port permutation is exactly one X -step, so

$$L^*\tilde{A}L = P_X. \quad (7.9)$$

Let $B = P_H$. If $\mu > 0$, spectral decomposition on constants and their orthogonal complement gives

$$B = (1 - \mu)J_K + \mu C, \quad \|C\|_{\text{op}} \leq 1. \quad (7.10)$$

Indeed C acts as the identity on constants and as B/μ on $\mathbf{1}^\perp$. If $\mu = 0$, the same argument below is read with $B = J_K$ and without the C term. Write

$$\tilde{B} := I_N \otimes B, \quad \tilde{C} := I_N \otimes C.$$

The normalized adjacency operator of the derandomized square is

$$M = L^*\tilde{A}\tilde{B}\tilde{A}L. \quad (7.11)$$

Using (7.8)–(7.10),

$$\begin{aligned} M &= (1 - \mu)L^*\tilde{A}(LL^*)\tilde{A}L + \mu L^*\tilde{A}\tilde{C}\tilde{A}L \\ &= (1 - \mu)P_X^2 + \mu D, \end{aligned} \quad (7.12)$$

where

$$D := L^*\tilde{A}\tilde{C}\tilde{A}L, \quad \|D\|_{\text{op}} \leq 1. \quad (7.13)$$

The last inequality follows because L and L^* have norm 1, $\tilde{A}$ is unitary, and $\|\tilde{C}\| \leq 1$.

If $f \perp \mathbf{1}$, then $P_X^2 f \perp \mathbf{1}$ and

$$\|P_X^2 f\|_2 \leq \lambda^2 \|f\|_2.$$

Hence (7.12)–(7.13) imply

$$\|Mf\|_2 \leq ((1 - \mu)\lambda^2 + \mu)\|f\|_2, \quad (7.14)$$

which is (7.3). Since the combinatorial rotation map (7.2) is an involution, M is self-adjoint, so this operator-norm estimate is exactly an absolute nontrivial eigenvalue estimate.

Finally, if $\lambda = 1 - \gamma$ and $\mu \leq 1/100$, then

$$\begin{aligned} 1 - f(1 - \gamma, \mu) &= (1 - \mu)(1 - (1 - \gamma)^2) \\ &= (1 - \mu)\gamma(2 - \gamma) \\ &> \frac{99}{100} \cdot \frac{7}{4} \gamma > \frac{3}{2} \gamma, \end{aligned} \quad (7.15)$$

proving (7.4). The query count is visible in (7.1). $\square$

Randomness economy. A true square of a K -regular graph has K^2 ports and chooses two independent labels. The derandomized square has only KD ports: it chooses one genuinely random outer label and uses a D -regular expander on $[K]$ to recycle that randomness into the second label. When $D \ll K$ and μ is small, (7.3) shows that the spectral effect is still close to squaring.

Source reconciliation. The two-way-label rotation rule (7.1)–(7.2) is the form that remains well-defined and undirected under iteration. The combinatorial definition has exactly $K \cdot D$ edge labels, hence degree KD . The full-version Rozenman–Vadhan PDF displays KD^2 in its Theorems 4.4 and 6.3, while its Definitions 4.1 and 6.2, Proposition 4.2, the algorithmic degree recurrence, and the authors’ own overview all give degree KD . The manuscript follows the combinatorial definition and records the displayed theorem degree as a source typographical inconsistency rather than propagating it.

Boundary before deterministic logspace connectivity. Repeated derandomized squaring is one of the two classical routes to undirected s – t connectivity in deterministic logspace. Turning the spectral recurrence above into a full USTCON algorithm additionally requires a logspace-uniform supply of auxiliary expanders at the needed sizes and degrees, a degree-regularization reduction, and a recursive neighbor procedure whose workspace is proved to be $O(\log N)$. Those interfaces are substantial enough that QRCC-VII-THM-021 remains open rather than being compressed into a one-paragraph corollary.

7.1 Reingold’s connectivity amplification

We can now close the logspace-connectivity boundary announced at the end of the preceding section. The proof below keeps four ledgers separate: (i) a fixed auxiliary expander exists; (ii) every transformation phase improves an arbitrarily small positive spectral gap; (iii) connected components are never merged; and (iv) a rotation query to the recursively defined final graph still uses only logarithmic workspace.

Lemma 7.2 (A fixed auxiliary expander with Reingold’s size–degree relation). *There exists an even absolute constant d and an undirected d -regular Cayley multigraph H on exactly*

$$K := d^{16} \tag{7.16}$$

vertices such that

$$\lambda_{\text{abs}}(H) \leq \frac{1}{2}. \tag{7.17}$$

Because d and K are absolute constants, one may fix one such rotation table once and for all and hardwire it into a logspace machine.

Proof. Let $d = 2r$ be an even integer, eventually chosen sufficiently large, and let $K = d^{16}$. Choose $s_1, \dots, s_r$ independently and uniformly from the cyclic group $\mathbb{Z}/K\mathbb{Z}$, and make a labelled Cayley multigraph whose $2r = d$ ports at x lead to

$$x + s_1, x - s_1, \dots, x + s_r, x - s_r.$$

This graph is undirected even when some choices produce loops or parallel edges.

For a nontrivial character $\chi_a(x) = \exp(2\pi i ax/K)$, the corresponding normalized eigenvalue is

$$\Lambda_a = \frac{1}{r} \sum_{j=1}^r \cos\left(\frac{2\pi a s_j}{K}\right). \tag{7.18}$$

Each summand lies in $[-1, 1]$ and has mean zero. Hoeffding's inequality gives

$$\Pr\{|\Lambda_a| > 1/2\} \leq 2e^{-r/8} = 2e^{-d/16}. \quad (7.19)$$

There are $K - 1 < d^{16}$ nontrivial characters, so a union bound gives

$$\Pr\{\lambda_{\text{abs}}(H) > 1/2\} < 2d^{16}e^{-d/16}. \quad (7.20)$$

The right side is smaller than 1 for all sufficiently large even d . Hence at least one labelled generator list has (7.17). After choosing the least such list in any fixed lexicographic order, its finite rotation table is an absolute constant and may be built into the machine description. $\square$

Definition 7.3 (Reingold main transformation). Fix d, K, H as in Lemma 7.2. For a K -regular port graph G_0 on M vertices define

$$\ell := 2 \left\lceil \log_2(KM^2) \right\rceil \quad (7.21)$$

and recursively

$$G_i := (G_{i-1} \text{ Z } H)^8, \quad 1 \leq i \leq \ell. \quad (7.22)$$

The final graph is

$$\mathcal{T}(G_0, H) := G_\ell. \quad (7.23)$$

Since $\deg(G_{i-1} \text{ Z } H) = d^2$ and the eighth power has degree $d^{16} = K$, every G_i is again K -regular. Its vertex set is naturally

$$V(G_i) = V(G_0) \times [K]^i. \quad (7.24)$$

Lemma 7.4 (Gap amplification and component preservation). *Let G_0 be a K -regular port multigraph on M vertices. Assume that a connected component C of G_0 is nonbipartite and that every vertex of C carries at least one designated self-loop port. Then the corresponding component*

$$C_i := C \times [K]^i \subseteq V(G_i) \quad (7.25)$$

is a connected component of G_i for every i , and

$$\lambda_{\text{abs}}(G_\ell|_{C_\ell}) \leq \frac{1}{2}. \quad (7.26)$$

No two distinct connected components of G_0 are ever joined by the transformation.

Proof. We first quantify the initial gap using only results already proved in this volume. Write $m_C = |C|$. Connectivity implies that every nonempty $S \subset C$ with $|S| \leq m_C/2$ has at least one crossing edge, hence

$$\Phi(C) \geq \frac{2}{Km_C}. \quad (7.27)$$

The upper Cheeger inequality in Theorem 4.1 therefore gives

$$1 - \lambda_2(C) \geq \frac{\Phi(C)^2}{2} \geq \frac{2}{K^2 m_C^2}. \quad (7.28)$$

The designated self-loop port allows us to write the normalized adjacency as

$$P_C = \frac{1}{K}I + \frac{K-1}{K}Q \quad (7.29)$$

for a self-adjoint Markov operator Q . Since the spectrum of Q lies in $[-1, 1]$,

$$\lambda_{\min}(P_C) \geq -1 + \frac{2}{K}. \quad (7.30)$$

Combining (7.28)–(7.30) and $m_C \leq M$ gives the convenient uniform estimate

$$\lambda_0 := \lambda_{\text{abs}}(C) \leq 1 - \frac{2}{K^2 M^2}. \quad (7.31)$$

Now suppose λ_i is the absolute nontrivial spectral radius on the component descended from C . The sharp gap estimate (6.27), followed by the eighth power, yields

$$\lambda_{i+1} \leq \left(1 - \frac{3}{8}(1 - \lambda_i)\right)^8 < \left(1 - \frac{1}{3}(1 - \lambda_i)\right)^8. \quad (7.32)$$

For $0 \leq x < 1/2$ the last expression is at most $(5/6)^8 < 1/2$. For $1/2 \leq x \leq 1$,

$$\left(1 - \frac{1}{3}(1 - x)\right)^4 = \left(\frac{x+2}{3}\right)^4 \leq x. \quad (7.33)$$

Indeed

$$81x - (x+2)^4 = (1-x)(x^3 + 9x^2 + 33x - 16),$$

and the cubic is positive and increasing on $[1/2, 1]$. Squaring (7.33) therefore shows

$$\lambda_{i+1} \leq \max\{\lambda_i^2, 1/2\}. \quad (7.34)$$

Induction gives

$$\lambda_\ell \leq \max\{\lambda_0^{2^\ell}, 1/2\}. \quad (7.35)$$

By (7.21), $2^\ell \geq (KM^2)^2 = K^2 M^4$. Using $1 - u \leq e^{-u}$ in (7.31),

$$\lambda_0^{2^\ell} \leq \exp\left(-\frac{2 \cdot 2^\ell}{K^2 M^2}\right) \leq e^{-2M^2} < \frac{1}{2}, \quad (7.36)$$

so (7.26) follows.

It remains to justify the component ledger. A zig-zag step uses exactly one outer edge of G_i , hence cannot cross from one G_i -component to another; the two H steps merely change port coordinates. Taking an eighth power also cannot create an edge between distinct components. Conversely, on C_i the sharp spectral estimate is < 1 after the first phase, hence the descended graph is connected and nonbipartite; induction gives that $C \times [K]^i$ is exactly one component, not merely a union of components. This proves (7.25) and the no-merging assertion. $\square$

Lemma 7.5 (Logspace evaluation of the transformed rotation map). *For fixed d, K, H , suppose the rotation map of a K -regular graph G_0 on M vertices is computable in $O(\log M)$ space. Then a rotation query to $G_\ell = \mathcal{T}(G_0, H)$, with ℓ as in (7.21), can be evaluated in $O(\log M)$ workspace and polynomial time.*

Proof. A vertex of G_i has the form

$$(v, a_1, \dots, a_i), \quad v \in V(G_0), \quad a_j \in [K], \quad (7.37)$$

so its complete name uses $\log M + i \log K = O(\log M)$ bits. A port of G_i is an element of $[K] = [d]^{16}$, which we regard as a 16-tuple of H -labels.

To evaluate one G_i rotation, scan those 16 labels in order. Each pair of consecutive small labels performs one zig step, one recursive G_{i-1} rotation, and one zag step; eight such triples evaluate one edge of $(G_{i-1} \text{ Z } H)^8$. At the end reverse the sixteen return labels, as required by the rotation convention for graph powers.

The important space point is that recursion does not copy the whole vertex name at every level. Allocate once the shared array

$$v, a_1, \dots, a_\ell, \quad (7.38)$$

which already costs only $O(\log M)$ bits. At recursion depth i , mutate the relevant prefix in place. Each stack frame stores only a constant-range phase counter in [16] and a constant number of H -labels; the rotation maps are involutions, so no discarded copy of an intermediate configuration is needed. The recursion depth is at most $\ell = O(\log M)$, hence the stack also costs $O(\log M)$ bits. The base call is one G_0 rotation query and uses $O(\log M)$ space by hypothesis.

The recursion tree has constant branching and depth $O(\log M)$, so it has $M^{O(1)}$ nodes. Thus the same implementation is polynomial-time as well as logspace. $\square$

Theorem 7.6 (Reingold: undirected connectivity is in deterministic logspace). *There is a deterministic algorithm that, given the adjacency matrix of an undirected graph G on N vertices and vertices $s, t \in V(G)$, decides whether s and t lie in the same connected component using*

$$O(\log N)$$

work space. Equivalently,

$$\text{USTCON} \in \text{L}. \quad (7.39)$$

The algorithm described below also runs in polynomial time.

Proof. Step 1: regularize without changing connectivity. Fix the constants d, K, H from Lemma 7.2. Replace G by a K -regular port multigraph G_{reg} on $[N] \times [N]$. For $(v, w) \in [N]^2$ define the first three rotation rules by

$$\begin{aligned} \text{Rot}_{\text{reg}}((v, w), 1) &= ((v, w + 1 \bmod N), 2), \\ \text{Rot}_{\text{reg}}((v, w), 2) &= ((v, w - 1 \bmod N), 1), \\ \text{Rot}_{\text{reg}}((v, w), 3) &= ((w, v), 3) \quad \text{if } \{v, w\} \in E(G). \end{aligned}$$

When $\{v, w\} \notin E(G)$, label 3 is the self-loop $\text{Rot}_{\text{reg}}((v, w), 3) = ((v, w), 3)$, and every label $4, \dots, K$ is likewise a self-loop. Thus ports 1 and 2 are explicitly paired return labels, while every other displayed label is its own return port. The formulas define an involutive rotation map and are computable by scanning one adjacency-matrix entry and doing $O(\log N)$ -bit arithmetic.

For each connected component S of G , the set $S \times [N]$ is exactly one connected component of G_{reg} . Indeed each fiber $\{v\} \times [N]$ is a cycle, and the label-3 edges connect the v - and w -fibers exactly when $\{v, w\} \in E(G)$. Labels $4, \dots, K$ give a designated self-loop at every vertex. In particular every component is nonbipartite.

Step 2: amplify every component separately. Apply Definition 7.3 with $M = N^2$ to obtain

$$G_{\text{exp}} = \mathcal{T}(G_{\text{reg}}, H). \quad (7.40)$$

Its vertex count is

$$N^2 K^\ell = N^{O(1)}, \quad (7.41)$$

because K is constant and $\ell = O(\log N)$. By Lemma 7.4, each original component S gives exactly the component

$$S \times [N] \times [K]^\ell \quad (7.42)$$

of G_{exp} , and the absolute nontrivial spectral radius on that component is at most $1/2$. Thus if

$$s' = (s, 1, 1, \dots, 1), \quad t' = (t, 1, 1, \dots, 1), \quad (7.43)$$

then s' and t' are connected in G_{exp} exactly when s and t are connected in G .

Step 3: exploit logarithmic diameter deterministically. On a component C' of G_{exp} , Cheeger's lower inequality gives

$$\Phi(C') \geq \frac{1 - \lambda_2(C')}{2} \geq \frac{1}{4}. \quad (7.44)$$

Corollary 1.3 therefore gives the explicit bound

$$\text{diam}(C') \leq 2 \left\lceil \frac{\log |C'|}{\log(5/4)} \right\rceil = O(\log N). \quad (7.45)$$

Set

$$L := 2 \left\lceil \frac{\log |V(G_{\text{exp}})|}{\log(5/4)} \right\rceil. \quad (7.46)$$

Enumerate all K^L port strings of length L beginning at s' and accept iff one reaches t' . Because K is constant and $|V(G_{\text{exp}})|$ is polynomial in N , there are only $N^{O(1)}$ strings. A string itself uses $L \log K = O(\log N)$ bits.

By Lemma 7.5, each requested rotation of G_{exp} is evaluable in $O(\log N)$ workspace. Reuse that workspace for each successive edge of each candidate path; keep only the current vertex, the path counter, and the current label string. The total workspace remains $O(\log N)$. The enumeration accepts exactly in the connected case by (7.42)–(7.46), completing the proof. $\square$

A three-vertex regularization trace. Let the input be the path 1–2–3. In the fiber over vertex 2, the labels 1 and 2 cycle through $(2, 1), (2, 2), (2, 3)$. Label 3 sends $(2, 1)$ to $(1, 2)$ and $(2, 3)$ to $(3, 2)$, while $(2, 2)$ receives a label-3 self-loop because the input has no loop at 2. Thus the three fibers are joined exactly according to the original path, while labels $4, \dots, K$ make every lifted vertex lazy. Choosing $s = 1, t = 3$, the canonical lifts in (7.43) remain in one component throughout every zig-zag/powering phase; if the edge $\{2, 3\}$ is deleted, the two corresponding fiber blocks are never joined by any later phase. This finite trace is the component-preservation claim in Lemma 7.4 in its smallest nontrivial form.

Why this is a logspace proof rather than an existence proof. The spectral argument alone only says that a short path exists in the final component. The algorithmic content is the combination of three facts: H is a fixed finite table, the transformed rotation map is computable in logspace by Lemma 7.5, and all short label strings can be enumerated with an $O(\log N)$ counter. This is precisely the uniformity bridge that is lost if one writes “power until the graph expands” without tracking the neighbor procedure.

Source reconciliation. The phase rule (7.22), the exponent 8, the fixed relation $K = d^{16}$, componentwise action, and the shared-array recursion are the interfaces of Reingold's primary proof. The present manuscript replaces its separate inverse-polynomial initial-gap lemma by the already established Cheeger inequality plus the designated-loop bound (7.29)–(7.31), and proves

the existence of the one fixed auxiliary graph directly by the elementary probabilistic argument in Lemma 7.2. The only source-gated quantitative ingredient left in the amplification is the sharp RVW zig-zag estimate of Theorem 6.5.

Chapter 8

Expander Codes and Linear-Time Bit Flipping

Expansion can be turned into coding distance by a counting principle: a small set of nonzero variables touches too many local constraints for every touched constraint to contain a nonzero local codeword. With the parity-check local code, the same counting principle also drives a remarkably simple decoder. This chapter makes both mechanisms quantitative.

Definition 8.1 (Tanner/expander code on a biregular constraint graph). Let $B = (L \sqcup R, E)$ be a finite (c, d) -biregular bipartite graph: $|L| = n$, every variable vertex $v \in L$ has degree c , every constraint vertex $a \in R$ has degree d , and hence

$$|R| = \frac{cn}{d}. \quad (8.1)$$

For $S \subseteq L$, write $N(S) \subseteq R$ for its constraint neighborhood. We say that B has (α, η) *left expansion* if

$$0 < |S| \leq \alpha n \implies |N(S)| > \eta c |S|. \quad (8.2)$$

Let $C_0 \leq \mathbb{F}_2^d$ be a binary linear code. After fixing an ordering of the d variable neighbors of every $a \in R$, the *Tanner code* $C(B, C_0) \leq \mathbb{F}_2^L$ consists of all words $x = (x_v)_{v \in L}$ such that the local word

$$x|_{N(a)} \in C_0 \quad \text{for every } a \in R. \quad (8.3)$$

When B satisfies a specified small-set expansion hypothesis, we also call $C(B, C_0)$ an *expander code*.

Theorem 8.2 (Distance and rate of a Tanner expander code). *Let B be as in Definition 8.1, and let C_0 have length d , dimension k_0 , and minimum distance $d_0 \geq 2$. Suppose B has (α, η) left expansion with*

$$\eta d_0 \geq 1. \quad (8.4)$$

Then every nonzero word in $C(B, C_0)$ has Hamming weight strictly larger than αn . Consequently the global relative distance is $> \alpha$. Moreover

$$\dim C(B, C_0) \geq n - \frac{cn}{d}(d - k_0), \quad (8.5)$$

so its rate satisfies

$$R(C(B, C_0)) \geq 1 - c \left(1 - \frac{k_0}{d}\right). \quad (8.6)$$

(The rate lower bound is informative when the right side is positive.)

Proof. The rate estimate is rank counting. Each of the $|R| = cn/d$ local constraints is a copy of C_0 and can impose at most $d - k_0$ independent linear equations. Thus the total codimension is at most $(cn/d)(d - k_0)$, proving (8.5)–(8.6).

For distance, suppose for contradiction that $x \neq 0$ is a global codeword with support

$$S := \{v \in L : x_v = 1\}, \quad 0 < |S| \leq \alpha n.$$

Exactly $c|S|$ edges leave S . Every constraint $a \in N(S)$ sees a nonzero local word $x|_{N(a)} \in C_0$, so the local minimum-distance condition forces at least d_0 of its incident variables to lie in S . Counting the same S – $N(S)$ edges from the right gives

$$c|S| \geq d_0|N(S)|. \quad (8.7)$$

But (8.2) and (8.4) imply

$$d_0|N(S)| > d_0\eta c|S| \geq c|S|, \quad (8.8)$$

contradicting (8.7). Therefore no such nonzero codeword exists. $\square$

Parity-check specialization. Let

$$C_{\text{even}} := \left\{ z \in \mathbb{F}_2^d : \sum_{i=1}^d z_i = 0 \right\}. \quad (8.9)$$

This local code has dimension $d - 1$ and minimum distance 2. Hence any (α, η) left expander with $\eta \geq 1/2$ gives a global code of relative distance $> \alpha$ and rate at least $1 - c/d$. More expansion, namely $\eta > 3/4$, is enough for a linear-time adversarial-error decoder.

Algorithm 8.3 (Sipser–Spielman sequential bit-flip decoder). *Let B be a (c, d) -biregular constraint graph with*

$$0 < |S| \leq \alpha n \implies |N(S)| > \frac{3c}{4}|S|. \quad (8.10)$$

Use the parity-check local code C_{even} from (8.9). Given a received word $y \in \mathbb{F}_2^L$, mark a constraint unsatisfied when the parity of its neighboring variable bits is odd. While some variable is incident to strictly more unsatisfied than satisfied neighboring constraints, flip that variable bit.

If $y = x + e$ for a codeword $x \in C(B, C_{\text{even}})$ and

$$|\text{supp}(e)| \leq \frac{\alpha n}{2}, \quad (8.11)$$

then the algorithm terminates at x . For fixed c, d , it can be implemented in $O(n)$ time and $O(n)$ storage in the standard adjacency-list RAM model.

Proof. Measure all parities relative to the transmitted codeword x ; since x satisfies every constraint, a constraint is unsatisfied exactly when it sees an odd number of currently erroneous variables. Let S be the current error set and put $v := |S|$. Assume first that

$$0 < v \leq \alpha n. \quad (8.12)$$

Let u be the number of unsatisfied constraints in $N(S)$ and s the number of satisfied constraints in $N(S)$. Thus

$$u + s = |N(S)| > \frac{3c}{4}v. \quad (8.13)$$

Every unsatisfied constraint in $N(S)$ contains at least one error, whereas every satisfied constraint in $N(S)$ contains at least two errors: it is touched by S and its error parity is even. Counting the cv edges leaving S therefore yields

$$cv \geq u + 2s. \quad (8.14)$$

Combining (8.13) and (8.14),

$$cv \geq 2(u + s) - u > \frac{3cv}{2} - u,$$

so

$$u > \frac{cv}{2}. \quad (8.15)$$

Since every unsatisfied constraint counted by u touches at least one erroneous variable, (8.15) says that the average erroneous variable is incident to more than $c/2$ unsatisfied constraints. Hence at least one *erroneous* variable is eligible to be flipped.

The actual algorithm is not told which variables are erroneous and may occasionally flip a correct variable. We therefore need one invariant to keep (8.12) available. Let U denote the total number of unsatisfied constraints. Flipping a variable incident to $r > c/2$ unsatisfied constraints makes those r constraints satisfied and the other $c - r$ constraints unsatisfied, so

$$\Delta U = (c - r) - r = c - 2r \leq -1. \quad (8.16)$$

Thus U strictly decreases at every iteration. Initially, if $v_0 \leq \alpha n/2$, then

$$U_0 \leq cv_0 \leq \frac{c\alpha n}{2}. \quad (8.17)$$

If the current error-set size ever reached αn for the first time, (8.15) would give

$$U \geq u > \frac{c\alpha n}{2},$$

contradicting $U \leq U_0$ from (8.16)–(8.17). Therefore the current error set always has size $< \alpha n$.

If the algorithm terminated with a nonempty error set, (8.12)–(8.15) would produce an erroneous variable incident to more unsatisfied than satisfied constraints, contradicting the stopping rule. Hence the final error set is empty and the output is x .

Finally, U is a nonnegative integer and decreases by at least one per flip. There are at most $|R| = cn/d = O(n)$ constraints, so there are $O(n)$ flips. For fixed c, d , maintaining each constraint parity and the affected incident-variable counters costs constant work per neighboring pointer, giving total time $O(n)$ after $O(n)$ initialization. $\square$

Why the 3/4 threshold appears. The two inequalities driving the proof are

$$u + s > \rho cv, \quad cv \geq u + 2s.$$

They imply $u > (2\rho - 1)cv$. To force the strict majority $u > cv/2$ needed by bit flipping, one needs $2\rho - 1 > 1/2$, namely $\rho > 3/4$. Thus the familiar constant is not decorative: it is exactly the point at which unique-neighbor expansion beats the local majority threshold in this elementary decoder analysis.

Scope. The proof above is the simple binary parity-check expander code. General Tanner codes with stronger local codes admit more sophisticated local correction rules and better constants; those refinements are deliberately separated from the clean counting mechanism proved here.

Chapter 9

High-Dimensional Expansion: Links, Cohomology, and Agreement

Graphs have several equivalent-looking expansion languages because their first nontrivial topology and their random walk both live on vertices and edges. In dimension two and above this coincidence disappears. Local spectral expansion, coboundary expansion, cosystolic expansion, high-order walk mixing, agreement testing, and overlap are genuinely different notions. This chapter fixes them separately before recording any implication between them.

Throughout, X is a finite pure d -dimensional simplicial complex. Thus $X(i)$ denotes the set of i -dimensional faces, each of cardinality $i + 1$, and every face lies in at least one member of $X(d)$. We also put $X(-1) = \{\emptyset\}$. A probability distribution π_d on $X(d)$ with full support is fixed. The induced distribution π_i on $X(i)$ is obtained by sampling $\tau \sim \pi_d$ and then choosing an i -face of τ uniformly. Every norm and random walk below uses these induced measures unless another coefficient measure is displayed explicitly.

Definition 9.1 (High-dimensional expansion dictionary). Let $-1 \leq i < d$ and $\sigma \in X(i)$.

(a) The *link* of σ is

$$X_\sigma := \{\eta : \eta \cap \sigma = \emptyset, \eta \cup \sigma \in X\},$$

with its probability measure obtained by conditioning a random top face on containing σ and then deleting σ .

(b) For $f : X(i) \rightarrow \mathbb{R}$, the *upper operator* and *lower operator* are the conditional expectations

$$(U_i f)(\tau) := \mathbb{E}[f(\sigma) \mid \sigma \subset \tau], \quad \tau \in X(i+1), \quad (9.1)$$

and, for $g : X(i+1) \rightarrow \mathbb{R}$,

$$(D_{i+1} g)(\sigma) := \mathbb{E}[g(\tau) \mid \tau \supset \sigma]. \quad (9.2)$$

With the π_i inner products, $D_{i+1} = U_i^*$. The lazy *up-down* and *down-up* walks on i -faces are

$$P_i^{\uparrow\downarrow} := D_{i+1} U_i, \quad P_i^{\downarrow\uparrow} := U_{i-1} D_i \quad (9.3)$$

(the second expression being used for $i \geq 0$, with the evident augmented convention at $i = 0$).

(c) If X_σ has dimension at least one, let P_σ be normalized adjacency on the weighted one-skeleton of the link. Write

$$\rho_+(X_\sigma) := \sup_{h \perp \mathbf{1}} \frac{\langle P_\sigma h, h \rangle}{\|h\|_2^2}, \quad \rho_{\text{abs}}(X_\sigma) := \|P_\sigma|_{\mathbf{1}^\perp}\|_{\text{op}}. \quad (9.4)$$

The complex is a *one-sided γ local spectral expander* if every such link is connected and $\rho_+(X_\sigma) \leq \gamma$. It is a *two-sided γ local spectral expander*, or a γ -HD expander in the Dinur–Kaufman convention, if $\rho_{\text{abs}}(X_\sigma) \leq \gamma$ for every such link. Equivalently, if $\mathcal{L}_\sigma = I - P_\sigma$ is the normalized Laplacian, a one-sided adjacency bound $\rho_+ \leq \gamma$ is a Laplacian-gap bound $\lambda_1(\mathcal{L}_\sigma) \geq 1 - \gamma$.

- (d) Fix a coefficient field $\mathbb{F}$ and let $C^i(X; \mathbb{F})$ be the i -cochains. The simplicial coboundary is $\delta_i : C^i \rightarrow C^{i+1}$, with

$$Z^i := \ker \delta_i, \quad B^i := \text{im } \delta_{i-1}, \quad H^i(X; \mathbb{F}) := Z^i / B^i. \quad (9.5)$$

For $\mathbb{F}_2$ we use the weighted support norm

$$\|f\| := \pi_i\{\sigma : f(\sigma) = 1\}, \quad \text{dist}(f, S) := \min_{g \in S} \|f - g\|. \quad (9.6)$$

Here the augmented convention $C^{-1} = \mathbb{F}_2$ makes B^0 the two constant 0-cochains. The i th *coboundary-expansion constant* is

$$h_{\text{cob}}^i(X; \mathbb{F}_2) := \min_{f \notin B^i} \frac{\|\delta_i f\|}{\text{dist}(f, B^i)}. \quad (9.7)$$

Consequently positive coboundary expansion forces $H^i(X; \mathbb{F}_2) = 0$. The complex is an (ε, μ) *i-cosystolic expander* if

$$\|\delta_i f\| \geq \varepsilon \text{dist}(f, Z^i) \quad \text{for every } f \in C^i, \quad (9.8)$$

and every nontrivial cocycle $z \in Z^i \setminus B^i$ has $\|z\| \geq \mu$. Thus cosystolic expansion allows nonzero cohomology but requires nontrivial cohomology classes to have macroscopic support.

- (e) Fix $k \geq 0$ and a distribution $\mathcal{D}$ on pairs $(s_1, s_2) \in X(k)^2$. A *Boolean local assignment* is a family $f = (f_s)_{s \in X(k)}$ with $f_s : s \rightarrow \{0, 1\}$. Its agreement probability is

$$\text{agree}_{\mathcal{D}}(f) := \Pr_{(s_1, s_2) \sim \mathcal{D}} [f_{s_1}|_{s_1 \cap s_2} = f_{s_2}|_{s_1 \cap s_2}]. \quad (9.9)$$

An agreement theorem asserts, under stated hypotheses on X and $\mathcal{D}$, that agreement at least $1 - \varepsilon$ forces a global $F : X(0) \rightarrow \{0, 1\}$ whose restrictions equal f_s on all but $O(\varepsilon)$ of the k -faces.

- (f) For completeness, the measure convention for overlap is also fixed. The complex has *c-geometric overlap* if for every map $\phi : X(0) \rightarrow \mathbb{R}^d$ there is $z \in \mathbb{R}^d$ such that

$$\Pr_{\tau \sim \pi_d} [z \in \text{conv}(\phi(v) : v \in \tau)] \geq c. \quad (9.10)$$

It has *c-topological overlap* if the same conclusion holds for every continuous $\Phi : |X| \rightarrow \mathbb{R}^d$, with the event $z \in \Phi(|\tau|)$. Neither notion is identified with spectral, coboundary, or agreement expansion by definition.

Why the signs matter. A large normalized-Laplacian gap controls the largest nonconstant adjacency eigenvalue but says nothing about an eigenvalue near -1 . High-order random walk and agreement arguments often need an absolute link bound. For that reason (9.4) keeps one-sided and two-sided local spectral expansion distinct, just as Chapter 1 did for ordinary graphs.

Theorem 9.2 (Oppenheim–Garland local-to-global vanishing — deep input). *Let X be a pure d -dimensional simplicial complex equipped with the homogeneous top-face weight. Assume X and every positive-dimensional link are connected. Suppose every one-dimensional link of a codimension-two face has normalized-Laplacian spectral gap at least*

$$\alpha > \frac{d-1}{d}. \quad (9.11)$$

Then, for every $0 \leq i \leq d-1$,

$$\tilde{H}^i(X; \mathbb{R}) = 0. \quad (9.12)$$

More precisely, the weighted real cochain spaces admit the orthogonal Hodge-type decomposition asserted by Oppenheim’s local spectral theorem, and the nonzero spectrum of each corresponding upper Laplacian is bounded away from zero by a constant depending only on d, i , and α .

Ownership and normalization. Theorem 9.2 is retained as an owner-C theorem-strength input. The local-to-global spectral argument is Garland’s method and is not compressed into a few lines here. In Oppenheim’s convention the hypothesis is a lower bound on normalized-Laplacian gaps. Equation (9.4) translates it to an upper bound $1 - \alpha < 1/d$ on the corresponding one-sided normalized adjacency eigenvalues. No conclusion about $\mathbb{F}_2$ coboundary expansion follows merely from (9.12); changing coefficient field is a genuine mathematical change.

Theorem 9.3 (Dinur–Kaufman high-order walks and agreement — deep input). *Let X be a pure d -dimensional complex which is a two-sided γ local spectral expander in the sense of Definition 9.1. For $0 \leq k < d$, let $A_{k,k+1} = D_{k+1}U_k$ be the one-up walk on k -faces. Then the Dinur–Kaufman decreasing-differences theorem gives*

$$\lambda_2(A_{k,k+1}) \leq 1 - \frac{1}{k+1} + O(k\gamma), \quad (9.13)$$

and, more generally, if $k+t \leq d$, the walk which goes from a k -face up to a random $(k+t)$ -face and independently back down to a k -face has

$$\lambda_2(A_{k,k+t}) \leq \frac{k+1}{k+t+1} + O(tk\gamma). \quad (9.14)$$

For the agreement consequence, assume in addition that $k^2 < d$ and $\gamma < 1/d$. Let $\mathcal{D}_{k,2k}$ be the distribution obtained by choosing a random $2k$ -face r according to the induced face measure and, conditioned on r , choosing $s_1, s_2 \in X(k)$ independently among its k -faces. If a Boolean local assignment satisfies

$$\text{agree}_{\mathcal{D}_{k,2k}}(f) \geq 1 - \varepsilon, \quad (9.15)$$

then there is a global $F : X(0) \rightarrow \{0, 1\}$ such that

$$\Pr_{s \sim \pi_k} [f_s = F|_s] \geq 1 - C_{d,k}\varepsilon, \quad (9.16)$$

for a constant depending only on the fixed dimensions. One may take $F(x)$ to be the majority value of $f_s(x)$ over incident k -faces.

Scope of the agreement input. The estimate (9.16) is not a consequence of ordinary graph mixing alone. Its proof uses the near-complete-complex high-order walk estimates, a double sampler, and an agreement theorem inside a complete complex. We therefore keep the complete argument source-owned while exposing the exact local-view distribution and the dimensional hypotheses required by the reduction.

Corollary 9.4 (Face-incidence sampler from an up operator). *Let $0 \leq k < \ell \leq d$, and let*

$$U_{k \rightarrow \ell} := U_{\ell-1} \cdots U_k : L^2(X(k), \pi_k) \longrightarrow L^2(X(\ell), \pi_\ell).$$

Assume

$$\|U_{k \rightarrow \ell} h\|_2 \leq \sigma \|h\|_2 \quad (h \perp \mathbf{1}). \quad (9.17)$$

For any set $A \subseteq X(k)$ of measure $a := \pi_k(A)$, define the local density seen from an ℓ -face by

$$Z(\tau) := \Pr[\sigma \in A \mid \sigma \subset \tau].$$

Then

$$\mathbb{E}Z = a, \quad \text{Var}(Z) \leq \sigma^2 a(1-a), \quad (9.18)$$

and hence, for every $\eta > 0$,

$$\Pr_{\tau \sim \pi_\ell} (|Z(\tau) - a| \geq \eta) \leq \frac{\sigma^2 a(1-a)}{\eta^2}. \quad (9.19)$$

Proof. Put $h = \mathbf{1}_A - a\mathbf{1}$. By construction of the face measures, conditional expectation gives

$$Z - a = U_{k \rightarrow \ell} h, \quad \mathbb{E}Z = a.$$

Moreover $h \perp \mathbf{1}$ and $\|h\|_2^2 = a(1-a)$. Therefore (9.17) yields (9.18), and Chebyshev's inequality gives (9.19). $\square$

When Theorem 9.3 controls the down-up operator $U_{k \rightarrow \ell}^* U_{k \rightarrow \ell}$ by θ on constants-perpendicular, (9.17) holds with $\sigma \leq \sqrt{\theta}$. This is the exact bridge from high-order spectral mixing to the sampler language used in Volume VIII.

A complete two-dimensional calculation. Let $X = \partial\Delta^3$ be the boundary of a tetrahedron, with uniform measure on its four triangles. Then

$$|X(0)| = 4, \quad |X(1)| = 6, \quad |X(2)| = 4,$$

and the induced measures are uniform in every dimension. The link of a vertex is K_3 , whose normalized adjacency has spectrum $1, -\frac{1}{2}, -\frac{1}{2}$; the link of the empty face has one-skeleton K_4 , with spectrum $1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}$. Thus X is a two-sided $1/2$ local spectral expander.

Over $\mathbb{F}_2$, consider 1-cochains as subsets of the six edges. The space B^1 consists of the eight cuts of K_4 . A direct enumeration of the 2^6 edge sets shows that every $f \notin B^1$ satisfies

$$\frac{\|\delta_1 f\|}{\text{dist}(f, B^1)} = 3. \quad (9.20)$$

For example, a single edge has distance $1/6$ from B^1 and coboundary supported on the two incident triangles, of norm $1/2$. Hence

$$h_{\text{cob}}^1(\partial\Delta^3; \mathbb{F}_2) = 3. \quad (9.21)$$

The equality $H^1(\partial\Delta^3; \mathbb{F}_2) = 0$ is visible here as the fact that the only 1-cocycles are those eight cuts. This tiny example also illustrates why the numerical value of a coboundary constant depends on the face-measure normalization: the ratio 3 is correct for the probability norms in (9.6), not for unnormalized Hamming counts.

Dated frontier: sparse Abelian high-dimensional expanders. Dikstein–Liu–Wigderson prove the following exact construction theorem in CCC 2025. For every $\lambda, \varepsilon > 0$ and every integer $d \geq 2$, there is an infinite family of d -dimensional Cayley λ -local spectral expanders with vertex set $\mathbb{F}_2^n$ and degree, in the paper’s Cayley-complex convention, at most 2^{n^ε} . The result is source-owned and dated 2026-09-03; it is not used as a hidden prerequisite for the local-spectral, coboundary, or agreement theorems above. The stable publication is LIPIcs CCC 2025, Article 7, DOI 10.4230/LIPIcs.CCC.2025.7.

Boundary of the present treatment. Local spectral expansion is computationally convenient because it reduces a high-dimensional condition to graph spectra in links. Coboundary and cosystolic expansion carry genuinely topological information, while agreement expansion is tailored to local consistency testing. Theorems 9.2 and 9.3 deliberately do not collapse these notions into a single definition. Modern constructions and separation theorems are retained as research-frontier items rather than used as silent dependencies.

Chapter 10

Linear-Algebraic and Operator-Amplification Frontiers

This short closing chapter records two exact research interfaces that extend the expander viewpoint beyond ordinary adjacency matrices. Both are source-owned dated frontiers, not additional QRCC proofs.

10.1 Linear-algebraic notions of expansion

For a finite-dimensional vector space V and a bounded tuple of linear maps $A_1, \dots, A_d : V \rightarrow V$, *dimension expansion* asks that every suitably small subspace U gain a fixed multiplicative amount of dimension under $U + \sum_i A_i U$. *Quantum expansion* is the spectral analogue for the associated completely positive operator on matrices. The 2025 paper of Li–Qiao–Wigderson–Wigderson–Zhang introduces a corresponding *dimension edge expansion* notion and compares it with Hastings’s *quantum edge expansion*; exact constants and normalization conventions remain those of the source.

The verified theorem map is:

- (i) quantum expansion implies dimension expansion, but the converse is false;
- (ii) dimension edge expansion is equivalent to dimension expansion;
- (iii) quantum edge expansion is equivalent to quantum expansion;
- (iv) quantum edge expansion implies dimension edge expansion, and the implication is strict.

Thus the graph-theoretic coincidence of vertex, edge, and spectral expansion does not survive unchanged in linear algebra. Source: *Theory of Computing* 21(1) (2025), 1–28, DOI <https://doi.org/10.4086/toc.2025.v021a001>.

10.2 Operator amplification to almost-Ramanujan expansion

Jeronimo–Mittal–Roy–Wigderson give an efficient local transformation from an arbitrary bounded-degree expander to an expander with an almost-optimal tradeoff between target spectral expansion λ and degree d :

$$d \leq \frac{1}{\lambda^{2+o(1)}} \quad (10.1)$$

in the source’s asymptotic convention as $\lambda \rightarrow 0$. New neighbors are obtained within radius $O(\log(1/\lambda))$ in the original graph. The construction and the operator-amplification proof are

source-owned; QRCC uses this row only as a dated comparison point for spectral amplification. Source: *SIAM Journal on Computing*, special section FOCS 2022, DOI <https://doi.org/10.1137/22M1538739>.

10.3 Reader checkpoint

The two frontiers separate three ideas that are easy to conflate: changing the ambient object from vertex sets to subspaces, changing the expansion notion from dimension to operator spectrum, and algorithmically amplifying an ordinary graph's spectrum. None is used to replace the elementary expander machinery proved earlier in this volume.

Internalization and Reconstruction Guide: Volume VII

Purpose. Treat spectral expansion as a reusable engine connecting random walks, mixing, concentration, explicit products, connectivity algorithms, codes, and high-dimensional expansion.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-IV-THM-004: Expander mixing lemma.
2. QRCC-I-THM-016: Parseval/Plancherel theorem.
3. QRCC-I-THM-014: Coupling characterization of total variation.
4. QRCC-I-THM-004: Chernoff bounds.
5. QRCC-III-THM-007: Singleton bound.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-VII-DEF-001 (Normalized adjacency and spectral expansion); QRCC-VII-DEF-002 (Edge expansion and conductance); QRCC-VII-DEF-013 (Rotation map and combinatorial graph products); QRCC-VII-DEF-032 (Reingold main transformation); QRCC-VII-DEF-031 (Tanner code and left small-set expansion).

Chapter retrieval map

- Ch. 1: Expansion in Three Languages:** enter through QRCC-VII-DEF-001 (Normalized adjacency and spectral expansion); exit by being able to state QRCC-VII-COR-007 (Expansion implies logarithmic diameter) with its ownership boundary.
- Ch. 2: Spectral Contraction, Mixing of Edges, and the Tree Barrier:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VII-COR-004 (Expander mixing from spectral contraction) with its ownership boundary.

- Ch. 3: From Spectral Contraction to Random-Walk Mixing:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VII-COR-006 (Spectral mixing-time bound) with its ownership boundary.
- Ch. 4: Cheeger Inequalities and Exact Model Examples:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VII-THM-005 (Cheeger inequalities for finite regular graphs) with its ownership boundary.
- Ch. 5: Random Expanders and the Near-Ramanujan Benchmark:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VII-ALG-011 (Margulis–Gabber–Galil explicit expander family) with its ownership boundary.
- Ch. 6: Rotation Maps and the Zig–Zag Product:** enter through QRCC-VII-DEF-013 (Rotation map and combinatorial graph products); exit by being able to state QRCC-VII-ALG-017 (Recursive construction of constant-degree expanders) with its ownership boundary.
- Ch. 7: Derandomized Squaring:** enter through QRCC-VII-DEF-032 (Reingold main transformation); exit by being able to state QRCC-VII-THM-016 (Derandomized squaring) with its ownership boundary.
- Ch. 8: Expander Codes and Linear-Time Bit Flipping:** enter through QRCC-VII-DEF-031 (Tanner code and left small-set expansion); exit by being able to state QRCC-VII-ALG-020 (Sipser–Spielman bit-flip decoder) with its ownership boundary.
- Ch. 9: High-Dimensional Expansion: Links, Cohomology, and Agreement:** enter through QRCC-VII-DEF-022 (Local spectral and coboundary expansion); exit by being able to state QRCC-VII-BB-023 (Garland-type local-to-global cohomology criterion) with its ownership boundary.
- Ch. 10: Linear-Algebraic and Operator-Amplification Frontiers:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VII-F-026 (Linear-algebraic notions of expansion) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-VII-COR-007 — Expansion implies logarithmic diameter. Dependencies: QRCC-VII-DEF-002. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-VII-COR-006 — Spectral mixing-time bound. Dependencies: QRCC-VII-THM-003, QRCC-I-THM-014. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-VII-ALG-011 — Margulis–Gabber–Galil explicit expander family. Dependencies: QRCC-VII-BB-032, QRCC-VII-THM-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-VII-THM-016 — Derandomized squaring. Dependencies: QRCC-VII-DEF-013,

QRCC-VII-THM-003. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

5. QRCC-VII-BB-023 — Garland-type local-to-global cohomology criterion. Dependencies: QRCC-VII-DEF-022. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-VII-F-025 — Sparse Abelian high-dimensional expanders. Dependencies: QRCC-VII-DEF-022. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- mixing adjacency eigenvalues with normalized-walk eigenvalues;
- ignoring the -1 eigenvalue obstruction in bipartite graphs;
- identifying conductance, vertex expansion, and edge expansion without degree factors;
- assuming graph-product eigenvalue formulas survive an altered rotation-map convention;

Boundary and degenerate cases

- disconnected and bipartite regular graphs;
- irregular graphs requiring stationary-measure normalization;
- degree-one or tiny expanders where asymptotic slogans fail;
- ordinary spectral expansion versus high-dimensional local-to-global conditions;

Retrieval and reconstruction drill

1. State QRCC-VII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-VII-COR-007. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-VII-COR-006, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.

7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-VII-F-025, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume VIII uses expansion, limited independence, and distance amplification to manufacture randomness from weak sources.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part VIII

Weak Sources, Extractors, and Pseudorandomness

Chapter 1

Weak Sources and Limited Independence

A distribution can contain substantial entropy without looking uniformly random. Extractor theory asks when a short independent seed can turn that uncertainty into bits that are statistically indistinguishable from uniform. Before constructing extractors, we fix the metric and entropy conventions and build two exact limited-independence families that will recur throughout the volume.

Definition 1.1 (Min-entropy, statistical distance, and seeded extraction). For a random variable X on a finite set Ω , its *min-entropy* is

$$H_\infty(X) := -\log_2 \max_{x \in \Omega} \Pr[X = x]. \quad (1.1)$$

An (n, k) *source* is a random variable on $\{0, 1\}^n$ with $H_\infty(X) \geq k$. A source is *flat* if it is uniform on its support.

For distributions P, Q on the same finite set,

$$\Delta(P, Q) := \frac{1}{2} \sum_x |P(x) - Q(x)| = \max_A |P(A) - Q(A)| \quad (1.2)$$

is their *statistical distance*. We write $P \approx_\varepsilon Q$ if $\Delta(P, Q) \leq \varepsilon$.

A function

$$\text{Ext} : \{0, 1\}^n \times \{0, 1\}^s \rightarrow \{0, 1\}^m$$

is a (k, ε) *seeded extractor* if for every (n, k) source X , independent of a uniform seed $S \sim U_s$,

$$\text{Ext}(X, S) \approx_\varepsilon U_m. \quad (1.3)$$

It is *strong* if

$$(S, \text{Ext}(X, S)) \approx_\varepsilon (U_s, U_m). \quad (1.4)$$

All extractor errors in this volume are statistical distances, not squared ℓ^2 errors or distinguishing advantages with an extra factor of two.

Elementary consequences. Min-entropy at least k is exactly the point-mass bound $\Pr[X = x] \leq 2^{-k}$. It implies the collision bound

$$\text{cp}(X) := \sum_x \Pr[X = x]^2 \leq 2^{-k}, \quad (1.5)$$

because each summand is at most $2^{-k} \Pr[X = x]$. The converse need not hold with the same parameter; collision entropy and min-entropy are deliberately not identified.

Algorithm 1.2 (Pairwise-independent affine family over a finite field). *Let q be a prime power and identify the sample alphabet with $\mathbb{F}_q$. Choose $A, B \in \mathbb{F}_q$ independently and uniformly and set*

$$h_{A,B}(x) := Ax + B. \quad (1.6)$$

For every two distinct inputs $x \neq y$, the pair $(h_{A,B}(x), h_{A,B}(y))$ is uniform on $\mathbb{F}_q^2$. Hence the family is pairwise independent.

Proof. Fix $x \neq y$ and target values u, v . The equations

$$Ax + B = u, \quad Ay + B = v$$

have the unique solution

$$A = \frac{u - v}{x - y}, \quad B = u - Ax.$$

Thus exactly one of the q^2 seeds (A, B) produces (u, v) , so the pair is uniform. $\square$

Algorithm 1.3 (r -wise independence from low-degree polynomials). *Let $x_1, \dots, x_N$ be distinct elements of $\mathbb{F}_q$, so $N \leq q$. Choose independent uniform coefficients $a_0, \dots, a_{r-1} \in \mathbb{F}_q$ and put*

$$P(T) = a_0 + a_1T + \dots + a_{r-1}T^{r-1}, \quad Y_j = P(x_j). \quad (1.7)$$

Then $Y_1, \dots, Y_N$ are r -wise independent and each Y_j is uniform on $\mathbb{F}_q$.

Proof. Choose any $t \leq r$ distinct indices $j_1, \dots, j_t$. The map from the coefficient vector $(a_0, \dots, a_{r-1})$ to $(P(x_{j_1}), \dots, P(x_{j_t}))$ has rank t : its first t columns form a Vandermonde matrix with determinant

$$\prod_{1 \leq u < v \leq t} (x_{j_v} - x_{j_u}) \neq 0.$$

Every target vector in $\mathbb{F}_q^t$ therefore has exactly q^{r-t} preimages among the q^r coefficient vectors. Its probability is q^{-t} , which is precisely joint uniformity. $\square$

Seed accounting. For $q = 2^\ell$, Algorithm 1.2 uses 2ℓ random bits to generate any requested coordinate, while Algorithm 1.3 uses $r\ell$ bits for r -wise independence. These are exact independence statements, not asymptotic small-bias approximations.

Chapter 2

Small Bias and a Coding-Theoretic Dictionary

Limited independence controls low-dimensional marginals. Small-bias spaces instead control every nontrivial linear parity. The basic equivalence with a balanced linear code is exact and gives a useful translation between pseudorandomness and coding theory.

Theorem 2.1 (Small-bias spaces and linear-code balance). *Let $S = (x_1, \dots, x_N)$ be a multiset in $\mathbb{F}_2^n$, and let X be uniform on the indexed multiset. Form the $n \times N$ matrix*

$$G = [x_1 \ \cdots \ x_N]$$

and the binary linear code

$$C_S := \{a^T G : a \in \mathbb{F}_2^n\} \subseteq \mathbb{F}_2^N. \quad (2.1)$$

For every $a \in \mathbb{F}_2^n$,

$$\mathbb{E}(-1)^{a \cdot X} = 1 - 2 \frac{\text{wt}(a^T G)}{N}. \quad (2.2)$$

Consequently S is ε -biased, meaning

$$|\mathbb{E}(-1)^{a \cdot X}| \leq \varepsilon \quad (a \neq 0), \quad (2.3)$$

if and only if every nonzero codeword produced by a nonzero a has relative weight in

$$\left[\frac{1 - \varepsilon}{2}, \frac{1 + \varepsilon}{2} \right]. \quad (2.4)$$

If G has full row rank, this says exactly that every nonzero codeword of C_S is ε -balanced around relative weight $1/2$.

Proof. The j th coordinate of $a^T G$ is $a \cdot x_j$. Therefore

$$\mathbb{E}(-1)^{a \cdot X} = \frac{1}{N} \sum_{j=1}^N (-1)^{a \cdot x_j} = \frac{N - \text{wt}(a^T G) - \text{wt}(a^T G)}{N},$$

which is (2.2). Taking absolute values gives the equivalence between (2.3) and (2.4). Full row rank merely guarantees that $a \neq 0$ produces a nonzero codeword rather than the zero word. $\square$

Corollary 2.2 (Probabilistic existence of small-bias multisets). *For every $n \geq 1$ and $0 < \varepsilon \leq 1$, there is an ε -biased multiset $S \subseteq \mathbb{F}_2^n$ of size*

$$N \leq \left\lceil \frac{2(n+1) \ln 2}{\varepsilon^2} \right\rceil. \quad (2.5)$$

Proof. Choose $x_1, \dots, x_N$ independently and uniformly. For fixed $a \neq 0$, the signs $(-1)^{a \cdot x_j}$ are independent unbiased $\{\pm 1\}$ variables. Hoeffding's inequality gives

$$\Pr \left(\left| \frac{1}{N} \sum_j (-1)^{a \cdot x_j} \right| > \varepsilon \right) \leq 2e^{-N\varepsilon^2/2}.$$

A union bound over fewer than 2^n nonzero a makes the failure probability strictly below one for the choice (2.5). $\square$

What the theorem does and does not say. Uniformity on a linear subspace of $\mathbb{F}_2^n$ is generally *not* a small-bias distribution: every vector in the orthogonal subspace gives bias one. The coding dictionary instead regards the sampled points as columns of a generator matrix and asks all nonzero linear tests to produce balanced *codewords*. This orientation prevents a common but serious “code versus dual code” reversal.

Chapter 3

Universal Hashing and the Leftover Hash Lemma

Universal hashing is the cleanest first extractor because its proof exposes the entire analytic mechanism: min-entropy controls collision probability, universal hashing controls output collisions, and Cauchy–Schwarz converts an ℓ^2 estimate into statistical distance. The seed remains visible in the conclusion, so the result is automatically strong.

A family $\mathcal{H}$ of functions $h : \Omega \rightarrow \{0, 1\}^m$ is *2-universal* if, for every $x \neq x'$,

$$\Pr_{H \sim \mathcal{H}}[H(x) = H(x')] \leq 2^{-m}. \quad (3.1)$$

Theorem 3.1 (Strong leftover hash lemma). *Let X be a random variable on a finite set Ω with $H_\infty(X) \geq k$, and let H be uniform on a finite 2-universal family $\mathcal{H}$ of functions $\Omega \rightarrow \{0, 1\}^m$, independent of X . Then*

$$\Delta((H, H(X)), (H, U_m)) \leq \frac{1}{2} \sqrt{2^{m-k}}. \quad (3.2)$$

In particular, if

$$m \leq k - 2 \log_2(1/\varepsilon), \quad (3.3)$$

then the joint distribution in (3.2) is $\varepsilon/2$ -close to uniform and therefore also ε -close to uniform.

Proof. For fixed $h \in \mathcal{H}$, let $p_h(z) = \Pr[h(X) = z]$. Its collision probability is

$$\text{cp}(p_h) = \sum_z p_h(z)^2 = \Pr[h(X) = h(X') \mid h],$$

where X' is an independent copy of X . Averaging over H and separating the cases $X = X'$ and $X \neq X'$ gives

$$\begin{aligned} \mathbb{E}_H \text{cp}(p_H) &\leq \Pr[X = X'] + 2^{-m} \Pr[X \neq X'] \\ &\leq 2^{-k} + 2^{-m}. \end{aligned} \quad (3.4)$$

Here (1.5) was used in the last line.

Let $M = |\mathcal{H}|$ and write $P(h, z) = M^{-1}p_h(z)$ for the joint law of $(H, H(X))$, while $Q(h, z) = M^{-1}2^{-m}$ is uniform. Then

$$\begin{aligned} \sum_{h,z} (P(h, z) - Q(h, z))^2 &= \frac{1}{M^2} \sum_h (\text{cp}(p_h) - 2^{-m}) \\ &\leq \frac{2^{-k}}{M}. \end{aligned} \quad (3.5)$$

Cauchy–Schwarz on the $M2^m$ pairs (h, z) yields

$$\sum_{h,z} |P - Q| \leq \sqrt{M2^m} \sqrt{\frac{2^{-k}}{M}} = \sqrt{2^{m-k}}.$$

Dividing by two proves (3.2). Substituting (3.3) gives the final claim. $\square$

Proposition 3.2 (Exact affine-source check). *Let X be uniform on an affine subspace $x_0 + V \subseteq \mathbb{F}_2^n$ of dimension k . Choose a uniformly random linear map $A : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and an independent uniform $b \in \mathbb{F}_2^m$, and output $AX + b$. This affine family is 2-universal, so Theorem 3.1 applies. Moreover, conditioned on A , the distribution $AX + b$ is uniform on the affine subspace $Ax_0 + b + A(V)$, and its collision probability is exactly*

$$2^{-\text{rank}(A|_V)}. \quad (3.6)$$

Proof. For $x \neq x'$, $A(x - x')$ is uniform on $\mathbb{F}_2^m$, so $\Pr[Ax + b = Ax' + b] = 2^{-m}$. Thus the family is 2-universal. The image of a uniform vector of V under A is uniform on $A(V)$ with every image point having the same number of preimages. Translation by $Ax_0 + b$ proves the conditional description, and a uniform distribution on a space of size $2^{\text{rank}(A|_V)}$ has collision probability (3.6). $\square$

Interpretation. The random offset b is not needed for universality, but it makes every conditional output visibly affine-uniform. Rank deficiency is the only source of nonuniformity for a flat affine source, while the general leftover hash proof requires no algebraic structure at all.

Chapter 4

Why a Seed Is Necessary

The leftover hash lemma uses an independent seed. This is not merely a convenient proof device: for one completely arbitrary weak source, a deterministic map can be defeated by concentrating the source inside one of its large fibers.

Theorem 4.1 (Deterministic extraction from one arbitrary weak source is impossible). *Let*

$$f : \{0, 1\}^n \rightarrow \{0, 1\}^m, \quad 1 \leq m \leq n,$$

be deterministic, and let $0 \leq k < n$. Put

$$r := \left\lceil 2^{k+m-n} \right\rceil. \quad (4.1)$$

There is a flat (n, k) source X for which $f(X)$ is supported on at most r outputs. Consequently

$$\Delta(f(X), U_m) \geq 1 - \frac{r}{2^m}. \quad (4.2)$$

If $k \leq n - m$, then $r = 1$, so $f(X)$ may be chosen constant and

$$\Delta(f(X), U_m) = 1 - 2^{-m}. \quad (4.3)$$

In particular, no deterministic function extracts even one nearly uniform bit from every (n, k) source for any $k \leq n - 1$.

Proof. Order the 2^m fibers of f by decreasing size. The union of the largest r fibers has size at least the fraction $r/2^m$ of the whole domain, hence at least

$$\frac{r}{2^m} 2^n = r 2^{n-m} \geq 2^k$$

points by the definition of r . Choose any T of exactly 2^k points inside that union and let X be uniform on T . Then $H_\infty(X) = k$ and $f(X)$ is supported on at most r values.

If a distribution P is supported on $R \subseteq \{0, 1\}^m$ with $|R| \leq r$, the variational characterization (1.2) gives

$$\Delta(P, U_m) \geq P(R) - U_m(R) \geq 1 - \frac{r}{2^m},$$

which proves (4.2). When $k \leq n - m$, the exponent in (4.1) is at most zero, so $r = 1$ and (4.3) follows. Taking $m = 1$ gives the last assertion for every $k < n$. $\square$

Algorithm 4.2 (Weak-source and hashing laboratory). *The executable laboratory is the source file*

`code/volume_08/extractor_lab.py`

It performs exact-rational checks of:

- (i) statistical distance and collision probability;*
- (ii) the large-fiber obstruction in Theorem 4.1;*
- (iii) pairwise universality of all linear maps $\mathbb{F}_2^3 \rightarrow \mathbb{F}_2$;*
- (iv) the strong leftover-hash inequality on every distribution in a finite test family with the advertised min-entropy bound.*

It uses only the Python standard library and fails loudly if a factor of 1/2, a seed normalization, or a collision term is changed incorrectly.

The boundary created by the theorem. The obstruction is specific to a single arbitrary weak source and a seedless deterministic map. It does not rule out deterministic extraction from structured sources (affine, bit-fixing, samplable, or multiple independent sources), nor does it rule out seeded extraction. Those are separate models and will be treated with their assumptions visible.

Chapter 5

Extractor Graphs and the Probabilistic Method

A strong seeded extractor has a particularly transparent graph interpretation. The seed is not hidden: it labels the outgoing edge, so the right vertex is the pair consisting of the seed and the extracted output. This viewpoint makes the probabilistic existence theorem almost identical to a discrepancy estimate for a random left-regular bipartite graph.

Write

$$N := 2^n, \quad K := 2^k, \quad D := 2^d, \quad M := 2^m,$$

and identify the corresponding bit strings with $[N], [K], [D], [M]$ when convenient.

Theorem 5.1 (Strong extractor–graph equivalence). *Let $0 \leq k \leq n$ be integers and let*

$$\text{Ext} : [N] \times [D] \longrightarrow [M].$$

Form the D -left-regular labeled bipartite multigraph G_{Ext} with left side $[N]$, right side $[D] \times [M]$, and the seed- s edge

$$x \longmapsto (s, \text{Ext}(x, s)). \quad (5.1)$$

The following are equivalent.

- (i) *Ext is a strong (k, ε) seeded extractor.*
- (ii) *For every $A \subseteq [N]$ of cardinality K , if one chooses $x \in A$ and $s \in [D]$ independently and uniformly, the endpoint of the edge (5.1) is ε -close in statistical distance to the uniform law on $[D] \times [M]$.*
- (iii) *For every such A and every $T \subseteq [D] \times [M]$,*

$$\left| \frac{e(A, T)}{KD} - \frac{|T|}{DM} \right| \leq \varepsilon, \quad (5.2)$$

where $e(A, T)$ counts labeled edges from A into T with multiplicity.

Thus it is enough to test a strong extractor on flat sources of entropy exactly k .

Proof. For $X = U_A$ and an independent uniform seed S , the random endpoint in (5.1) is exactly

$$(S, \text{Ext}(X, S)).$$

The variational characterization (1.2) of statistical distance therefore makes (ii) and (iii) equivalent.

Condition (i) immediately implies (ii), because U_A is an (n, k) source. For the converse, consider the polytope

$$\mathcal{P}_K := \left\{ p \in \mathbb{R}^N : p_x \geq 0, \sum_x p_x = 1, p_x \leq \frac{1}{K} \right\}. \quad (5.3)$$

These are exactly the probability vectors of (n, k) sources. An extreme point of $\mathcal{P}_K$ cannot have two coordinates strictly between 0 and $1/K$: shifting a sufficiently small amount of mass from one to the other in both directions would express it as the midpoint of two distinct points of $\mathcal{P}_K$. Since the coordinates sum to one, every extreme point must therefore have exactly K coordinates equal to $1/K$ and all others zero. In other words, the extreme points are precisely the flat laws U_A with $|A| = K$.

Every point of the finite-dimensional polytope (5.3) is a convex combination of its extreme points. The map

$$P_X \mapsto P_{(S, \text{Ext}(X, S))}$$

is affine, and statistical distance to a fixed distribution is convex. Hence the ε bound for all flat extreme points implies the same bound for every (n, k) source. Thus (ii) implies (i). $\square$

Why the seed belongs on the right. If the right vertex were only $\text{Ext}(x, s)$, condition (5.2) would describe an ordinary seeded extractor. Retaining the seed coordinate is exactly what converts the graph statement into the *strong* extractor condition. This convention will matter again when extractor graphs are converted into averaging samplers.

Theorem 5.2 (Probabilistic existence of near-optimal strong extractors). *Let $0 \leq k \leq n$ be integers and $0 < \varepsilon \leq 1$. Suppose powers of two $D = 2^d$ and $M = 2^m$ satisfy*

$$D \geq \frac{4(1 + (n - k) \ln 2)}{\varepsilon^2}, \quad M \leq \frac{\varepsilon^2 K}{4 \ln 2}. \quad (5.4)$$

Then there exists a strong (k, ε) extractor

$$\text{Ext} : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^m. \quad (5.5)$$

Consequently, whenever the desired output length is nonnegative, one may choose parameters

$$d = \log_2(n - k + 1) + 2 \log_2(1/\varepsilon) + O(1), \quad m \geq k - 2 \log_2(1/\varepsilon) - 3. \quad (5.6)$$

The constants implicit in the seed-length $O(1)$ are absolute.

Proof. Choose every value $\text{Ext}(x, s)$ independently and uniformly in $[M]$. Fix a K -set $A \subseteq [N]$ and a set $T \subseteq [D] \times [M]$. For each $(x, s) \in A \times [D]$, let

$$Y_{x,s} := \mathbf{1}_{\{(s, \text{Ext}(x,s)) \in T\}}.$$

The KD variables are independent, take values in $[0, 1]$, and satisfy

$$\mathbb{E} \sum_{x,s} Y_{x,s} = K \sum_{s \in [D]} \frac{|T_s|}{M} = \frac{K|T|}{M}, \quad (5.7)$$

where $T_s := \{y : (s, y) \in T\}$. Hoeffding's inequality gives

$$\Pr \left[\left| e(A, T) - \frac{K|T|}{M} \right| > \varepsilon KD \right] \leq 2 \exp(-2\varepsilon^2 KD). \quad (5.8)$$

There are at most

$$\binom{N}{K} \leq \left(\frac{eN}{K}\right)^K = \exp(K(1 + (n-k) \ln 2)) \quad (5.9)$$

choices for A , and exactly 2^{DM} choices for T . By the union bound, the probability that (5.2) fails for at least one pair (A, T) is at most

$$\exp(\ln 2 + K(1 + (n-k) \ln 2) + DM \ln 2 - 2\varepsilon^2 KD). \quad (5.10)$$

The first inequality in (5.4) makes the second term in the exponent at most $\varepsilon^2 KD/4$, while the second inequality makes the third term at most $\varepsilon^2 KD/4$. Moreover the first inequality gives $\varepsilon^2 KD \geq 4K \geq 4$, so $\ln 2 < \varepsilon^2 KD/4$. Thus the exponent in (5.10) is strictly negative. Some choice of Ext therefore satisfies (5.2) simultaneously for every A, T , and Theorem 5.1 makes it a strong extractor.

For the parameter form, take D to be the least power of two satisfying the first inequality in (5.4). Since $1 + (n-k) \ln 2 \leq n-k+1$, this gives the seed-length estimate in (5.6). Take M to be the largest power of two not exceeding the second bound in (5.4). Whenever that bound is at least one,

$$m \geq k - 2 \log_2(1/\varepsilon) - \log_2(4 \ln 2) - 1 \geq k - 2 \log_2(1/\varepsilon) - 3,$$

which proves (5.6). $\square$

What existence does not provide. The random table above has exponential description size. The theorem fixes the information-theoretic target parameters, but it is not an explicit construction. The later condenser and Trevisan/GUV chapters are precisely about replacing this random table by efficiently computable structure.

5.1 Necessary parameters: what no seeded extractor can beat

The probabilistic construction above is useful only if its parameter scale is close to necessary. There are two complementary obstructions. The first is an exact support argument that is elementary in our strong convention. The second is the sharper information-theoretic lower bound of Radhakrishnan–Ta-Shma, which we retain as a theorem-strength source input and translate to the seed–output pair used throughout this volume.

Theorem 5.3 (Radhakrishnan–Ta-Shma extractor lower bounds — deep input). *There is an absolute constant $C_{\text{RT}} > 0$ with the following property. Let*

$$E : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^r$$

be a (k, ε) seeded extractor in statistical distance, with $0 < \varepsilon \leq 1/4$ and $0 \leq k < n$. Then

$$d \geq r - k + 2 \log_2(1/\varepsilon) - C_{\text{RT}}. \quad (5.11)$$

If, in addition, $r \geq k + 1$, then

$$d \geq \log_2(n - k) + 2 \log_2(1/\varepsilon) - C_{\text{RT}}. \quad (5.12)$$

The additive constant is universal; no explicit value for it is fixed in this source interface.

Ownership. The counting lower bound behind Theorem 5.3 is a deep source-owned argument and is not compressed here into an informal proof. The two consequences below, including the conversion to the strong seed–output convention, are proved locally.

Theorem 5.4 (Necessary output and seed lengths for strong extractors). *Let*

$$\text{Ext} : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^m$$

be a strong (k, ε) extractor, where $0 < \varepsilon < 1$ and $K = 2^k$. Then the exact support obstruction gives

$$M = 2^m \leq \frac{K}{1 - \varepsilon}, \quad m \leq k + \log_2 \frac{1}{1 - \varepsilon}. \quad (5.13)$$

If $0 < \varepsilon \leq 1/4$ and $k < n$, then for the universal constant in Theorem 5.3,

$$m \leq k - 2 \log_2(1/\varepsilon) + C_{\text{RT}}. \quad (5.14)$$

Moreover, whenever $d + m \geq k + 1$,

$$d \geq \log_2(n - k) + 2 \log_2(1/\varepsilon) - C_{\text{RT}}. \quad (5.15)$$

Thus the nonconstructive existence theorem Theorem 5.2 has the optimal seed and entropy-loss scales up to absolute additive constants.

Proof. Fix a flat k -source $X = U_A$, $|A| = K$, and let S be uniform on $[D]$. The support of

$$P := (S, \text{Ext}(X, S))$$

has cardinality at most KD . If U denotes the uniform law on $[D] \times [M]$, then, with $B = \text{supp } P$,

$$\varepsilon \geq \Delta(P, U) \geq P(B) - U(B) \geq 1 - \frac{KD}{DM} = 1 - \frac{K}{M}.$$

This proves (5.13).

For the sharper bounds, define

$$E'(x, s) := (s, \text{Ext}(x, s)) \in \{0, 1\}^{d+m}. \quad (5.16)$$

Strongness says exactly that E' is an ordinary (k, ε) seeded extractor with seed length d and output length $r = d + m$. Applying (5.11) to E' gives

$$d \geq (d + m) - k + 2 \log_2(1/\varepsilon) - C_{\text{RT}},$$

and cancelling d yields (5.14). If $d + m \geq k + 1$, applying (5.12) to E' gives (5.15). $\square$

Parameter checkpoint. For constant error, (5.14) says that a strong extractor can output at most $k + O(1)$ bits, while (5.15) forces $d \geq \log_2(n - k) - O(1)$ in the nontrivial-output regime. As the error decreases, both barriers acquire the same $2 \log_2(1/\varepsilon)$ scale seen in the probabilistic construction.

5.2 Weak designs and Trevisan reconstruction

The probabilistic extractor in Theorem 5.2 has an exponentially large truth table. Trevisan's construction replaces that table by two deterministic objects: a list-decodable binary code and a family of small, weakly overlapping views of one short seed. We keep those two roles separate. The code is retained as a theorem-strength source input, while the weak design and the entire reconstruction/counting argument are proved here.

Theorem 5.5 (Binary list-decodable code for Trevisan reconstruction — deep input). *For every integer $n \geq 1$ and every $0 < \delta < 1/2$, there is a polynomial-time injective encoding*

$$\text{EC}_{n,\delta} : \{0,1\}^n \longrightarrow \{0,1\}^{\bar{n}} \quad (5.17)$$

with $\bar{n} = \text{poly}(n, 1/\delta)$, which may be taken to be a power of two, such that every Hamming ball of radius $(1/2 - \delta)\bar{n}$ contains at most $1/\delta^2$ codewords from the image of $\text{EC}_{n,\delta}$.

Ownership. This is the coding-theoretic input used in Trevisan's original extractor. The polynomial construction and list-size theorem are source-owned. All seed-design and reconstruction steps below are local.

Algorithm 5.6 (Weak combinatorial designs by conditional expectation). *Let $\ell, m \geq 1$ and $\rho > 1$. Put*

$$d := \ell \left\lceil \frac{\ell}{\ln \rho} \right\rceil. \quad (5.18)$$

There is a deterministic algorithm, polynomial in m, d and the bit lengths of the exact rational quantities it manipulates, that constructs ℓ -subsets

$$S_1, \dots, S_m \subseteq [d] \quad (5.19)$$

satisfying, for every i ,

$$\sum_{j < i} 2^{|S_i \cap S_j|} \leq \rho(i-1) \leq \rho(m-1). \quad (5.20)$$

In particular, for $\rho = 2$ one may take $d \leq 2\ell^2$.

Proof. Assume $S_1, \dots, S_{i-1}$ have already been chosen and let S be a uniformly random ℓ -subset of $[d]$. For any fixed ℓ -set T , expand

$$2^{|S \cap T|} = \prod_{t \in T} (1 + \mathbf{1}_{\{t \in S\}}) = \sum_{R \subseteq T} \mathbf{1}_{\{R \subseteq S\}}.$$

If $|R| = r$, then

$$\Pr[R \subseteq S] = \frac{(\ell)_r}{(d)_r} \leq \left(\frac{\ell}{d}\right)^r,$$

because $d \geq \ell$. Hence

$$\mathbb{E} 2^{|S \cap T|} \leq \sum_{r=0}^{\ell} \binom{\ell}{r} \left(\frac{\ell}{d}\right)^r = \left(1 + \frac{\ell}{d}\right)^{\ell} \leq e^{\ell^2/d} \leq \rho. \quad (5.21)$$

Summing over the previously chosen sets gives

$$\mathbb{E} \sum_{j < i} 2^{|S \cap S_j|} \leq \rho(i-1),$$

so some S satisfies (5.20).

For constructivity, choose the elements of S_i one at a time by the method of conditional expectations. If a partial set P of size s has been fixed and a candidate next element $e \notin P$ is considered, then for each previous S_j the conditional expectation is an exact hypergeometric sum:

$$2^a \sum_h 2^h \frac{\binom{\ell-a}{h} \binom{d-s-1-(\ell-a)}{\ell-s-1-h}}{\binom{d-s-1}{\ell-s-1}}, \quad a := |(P \cup \{e\}) \cap S_j|. \quad (5.22)$$

The average of these conditional expectations over all legal choices of e is the current conditional expectation. Choosing a minimizing e therefore never increases it. After ℓ choices the conditional expectation is the actual potential, proving (5.20). All binomial coefficients have polynomial bit length in the displayed parameters, so exact rational comparison gives a deterministic polynomial-time construction in the stated model. Finally, $\ell/\ln 2 < 2\ell$ gives $d \leq 2\ell^2$ when $\rho = 2$. $\square$

Lemma 5.7 (A statistical distinguisher exposes a predictable next bit). *Let $Z = (Z_1, \dots, Z_m)$ be a random variable on $\{0, 1\}^m$. If*

$$\Delta(Z, U_m) > \varepsilon,$$

then for some $1 \leq i \leq m$ there is a deterministic predictor $A : \{0, 1\}^{i-1} \rightarrow \{0, 1\}$ such that

$$\Pr[A(Z_1, \dots, Z_{i-1}) = Z_i] > \frac{1}{2} + \frac{\varepsilon}{m}. \quad (5.23)$$

Proof. For $0 \leq i \leq m$, let H_i have the first i coordinates distributed as the first i coordinates of Z and the remaining $m - i$ coordinates independent and uniform. Thus $H_0 = U_m$ and $H_m = Z$. The triangle inequality gives

$$\varepsilon < \Delta(H_m, H_0) \leq \sum_{i=1}^m \Delta(H_i, H_{i-1}),$$

so one summand exceeds ε/m . For that i , conditioning on a prefix $z \in \{0, 1\}^{i-1}$ shows

$$\Delta(H_i, H_{i-1}) = \mathbb{E}_z \left| \Pr[Z_i = 1 \mid Z_{<i} = z] - \frac{1}{2} \right|. \quad (5.24)$$

Choose $A(z)$ to be the more likely conditional bit. Its conditional success probability is $1/2$ plus the absolute value in (5.24); averaging proves (5.23). $\square$

Lemma 5.8 (Weak-design reconstruction family). *Let $C : \{0, 1\}^\ell \rightarrow \{0, 1\}$ be any Boolean truth table, let $S_1, \dots, S_m \subseteq [d]$ have cardinality ℓ , and define*

$$G_C(y)_j := C(y|_{S_j}), \quad y \in \{0, 1\}^d. \quad (5.25)$$

Fix i and a predictor $A : \{0, 1\}^{i-1} \rightarrow \{0, 1\}$. There is a family $\mathcal{F}_i$ of Boolean functions on $\{0, 1\}^\ell$ with

$$|\mathcal{F}_i| \leq 2^{\sum_{j < i} 2^{|S_i \cap S_j|}} \quad (5.26)$$

such that if

$$\Pr_y[A(G_C(y)_1, \dots, G_C(y)_{i-1}) = G_C(y)_i] > \frac{1}{2} + \eta, \quad (5.27)$$

then some $F \in \mathcal{F}_i$ agrees with C on more than a $1/2 + \eta$ fraction of its inputs.

Proof. Write $u = y|_{S_i}$. After the coordinates outside S_i are fixed, each earlier bit $C(y|_{S_j})$ becomes some Boolean function

$$P_j : \{0, 1\}^{S_i \cap S_j} \rightarrow \{0, 1\}$$

of the overlap coordinates of u . There are at most $2^{|S_i \cap S_j|}$ possibilities for P_j . Therefore all functions of the form

$$F(u) = A(P_1(u|_{S_i \cap S_1}), \dots, P_{i-1}(u|_{S_i \cap S_{i-1}})) \quad (5.28)$$

form a family of size at most the product of those numbers, which is exactly (5.26).

Average the success indicator in (5.27) first over the coordinates outside S_i and then over u . Some fixed outside assignment has conditional success greater than $1/2 + \eta$. The corresponding tuple $(P_1, \dots, P_{i-1})$ produces an F in (5.28) with that same agreement against C on uniform u , proving the lemma. $\square$

Theorem 5.9 (Trevisan extraction in the transparent $\rho = 2$ regime). *Let $n, k, m \geq 1$ and $0 < \varepsilon < 1$. Suppose*

$$k \geq 2m + 3 \log_2 \frac{m}{\varepsilon}. \quad (5.29)$$

Set $\delta := \varepsilon/(2m)$ and choose the code from Theorem 5.5. Write $\bar{n} = 2^\ell$ and choose a weak $(\ell, 2)$ design $S_1, \dots, S_m \subseteq [d]$ from Algorithm 5.6, so that $d \leq 2\ell^2$. For $x \in \{0, 1\}^n$, identify its codeword with a truth table

$$C_x : \{0, 1\}^\ell \rightarrow \{0, 1\},$$

and define

$$\text{Tre}(x, y)_i := C_x(y|_{S_i}), \quad 1 \leq i \leq m. \quad (5.30)$$

Then Tre is a (k, ε) seeded extractor. Moreover

$$d \leq 2\ell^2 = O\left(\log^2 \frac{nm}{\varepsilon}\right), \quad (5.31)$$

where the hidden constant depends only on the fixed polynomial in the source-owned code length.

Proof. Let X be any (n, k) source and $Y \sim U_d$ independent. Suppose, toward a contradiction, that

$$\Delta(\text{Tre}(X, Y), U_m) > \varepsilon.$$

Lemma 5.7 supplies an index i and a predictor A whose average success is greater than $1/2 + \varepsilon/m$. For each fixed x , write s_x for its success probability over Y , and define

$$B := \left\{ x : s_x > \frac{1}{2} + \frac{\varepsilon}{2m} \right\}. \quad (5.32)$$

If $q := \Pr[X \in B]$, then

$$\mathbb{E} s_X \leq q + (1 - q) \left(\frac{1}{2} + \frac{\varepsilon}{2m} \right).$$

Since the left side exceeds $1/2 + \varepsilon/m$, this forces

$$q > \frac{\varepsilon}{m - \varepsilon} > \frac{\varepsilon}{m}. \quad (5.33)$$

Now fix $x \in B$. Lemma 5.8, together with the weak $(\ell, 2)$ bound, gives a family of at most

$$2^{2(m-1)} \quad (5.34)$$

predictors, one of which agrees with the truth table C_x on more than a $1/2 + \delta$ fraction of inputs. By the list-decoding property in Theorem 5.5, each such predictor can be close to at most $1/\delta^2 = (2m/\varepsilon)^2$ codewords. Injectivity of the code therefore gives

$$|B| \leq 2^{2(m-1)} \left(\frac{2m}{\varepsilon} \right)^2. \quad (5.35)$$

The hypothesis (5.29) is exactly strong enough to imply

$$|B| 2^{-k} \leq \frac{\varepsilon}{m}. \quad (5.36)$$

Indeed, taking base-two logarithms of (5.35) and comparing with $k - \log_2(m/\varepsilon)$ reduces (5.36) to (5.29). But min-entropy at least k gives $\Pr[X \in B] \leq |B|2^{-k}$, contradicting (5.33). Hence $\text{Tre}(X, Y) \approx_\varepsilon U_m$.

Finally, the source-owned code has $\bar{n} = \text{poly}(n, 1/\delta) = \text{poly}(n, m/\varepsilon)$, so $\ell = O(\log(nm/\varepsilon))$. Algorithm 5.6 with $\rho = 2$ yields (5.31). $\square$

What was deliberately not hidden. The selected theorem extracts roughly one half of the available entropy. The near-all-entropy refinements of Raz–Reingold–Vadhan use weaker designs with ρ closer to one and sharper bookkeeping. Those refinements are source-verified in the bibliography, but (5.29)–(5.36) give a complete local reconstruction proof with every family-size and list-size factor visible.

5.3 Dated efficiency frontier: nearly-linear-time seeded extraction

Doron–Ribeiro construct strong seeded extractors for n -bit sources of min-entropy k and target error ε with seed length $d = O(\log(n/\varepsilon))$ and output length $m = (1 - \eta)k$ for every fixed $\eta > 0$, evaluable in $O(n \log^c n)$ time for a constant c in their model, with a one-time preprocessing qualification in the smallest-entropy/error regimes. They also give a Trevisan instantiation that is truly linear-time in the RAM model in its stated output regime. This is a dated implementation frontier, not a strengthening silently used in Theorem 5.9. Source: IEEE Transactions on Information Theory 71(11) (2025), 9028–9054, DOI 10.1109/TIT.2025.3605160.

Chapter 6

Condensers, Lossless Expansion, and Entropy Compression

An extractor asks for an almost uniform output. A condenser asks for less: it preserves, or nearly preserves, min-entropy while moving the source into a shorter ambient space. This weaker target is often much easier to realize explicitly, and a lossless condenser is exactly the probabilistic analogue of a bipartite graph whose small left sets have almost no edge collisions.

Throughout this chapter

$$N = 2^n, \quad K = 2^k, \quad D = 2^d, \quad M = 2^m,$$

and the seed is independent and uniform on $[D]$.

Definition 6.1 (Condensers, losslessness, deficiency, and dispersers). Let

$$\text{Con} : [N] \times [D] \longrightarrow [M].$$

- (a) For $k' \geq 0$, Con is a $k \rightarrow_\varepsilon k'$ *condenser* if for every (n, k) source X , $\text{Con}(X, S)$ is ε -close to some distribution Y on $[M]$ with $H_\infty(Y) \geq k'$.
- (b) The *entropy loss* of this ordinary condenser is

$$\ell := (k + d) - k', \tag{6.1}$$

and its *output deficiency* is

$$\Delta_{\text{out}} := m - k'. \tag{6.2}$$

It is *lossless* when $k' = k + d$, so $\ell = 0$.

- (c) A *strong* $k \rightarrow_\varepsilon k'$ *condenser* is one for which $(S, \text{Con}(X, S))$ is ε -close to a joint distribution on $[D] \times [M]$ of min-entropy at least $d + k'$. In this selected strong convention k' measures the retained output entropy after the public seed is exposed, and the corresponding conditional-output loss is $k - k'$.
- (d) The neighbor graph of Con has left side $[N]$, right side $[M]$, and one labeled edge $x \xrightarrow{s} \text{Con}(x, s)$ for every $s \in [D]$. We say it is an $(= K, A)$ *vertex expander* if

$$|\Gamma(B)| \geq A|B| \quad \text{for every } B \subseteq [N], \quad |B| = K. \tag{6.3}$$

(e) A function $\text{Disp} : [N] \times [D] \rightarrow [M]$ is a (k, ε) *dispenser* if for every $B \subseteq [N]$ with $|B| \geq K$,

$$|\Gamma(B)| \geq (1 - \varepsilon)M. \quad (6.4)$$

A dispenser controls only support. A condenser controls the output probability distribution, and a lossless condenser controls it nearly down to the collision scale.

The sign in (6.1) is worth fixing explicitly: the input source and seed carry at most $k + d$ bits of min-entropy together, so *loss* is $(k + d) - k'$, not the reverse.

Proposition 6.2 (Lossless condenser–near-lossless expansion equivalence). *Assume $M \geq KD$. Then $\text{Con} : [N] \times [D] \rightarrow [M]$ is a $k \rightarrow_\varepsilon (k + d)$ lossless condenser if and only if its neighbor graph is an*

$$(= K, (1 - \varepsilon)D) \quad (6.5)$$

vertex expander; equivalently,

$$|\Gamma(A)| \geq (1 - \varepsilon)KD \quad (|A| = K). \quad (6.6)$$

For a flat source U_A , the exact distance from its random-neighbor law to the convex set of $(k + d)$ -sources is

$$\frac{KD - |\Gamma(A)|}{KD}. \quad (6.7)$$

Proof. Fix $A \subseteq [N]$, $|A| = K$, and let P_A be the law of $\text{Con}(U_A, S)$. Put $r := KD$ and write c_y for the number of labeled edges from A to $y \in [M]$. Thus $P_A(y) = c_y/r$ and $\sum_y c_y = r$.

Suppose first that P_A is ε -close to a distribution Q with $H_\infty(Q) \geq \log_2 r$. Since $Q(y) \leq 1/r$,

$$1 = P_A(\Gamma(A)) \leq Q(\Gamma(A)) + \varepsilon \leq \frac{|\Gamma(A)|}{r} + \varepsilon,$$

which gives (6.6).

Conversely, assume (6.6). The excess mass of P_A above the cap $1/r$ is

$$\sum_y \left(P_A(y) - \frac{1}{r} \right)_+ = \frac{1}{r} \sum_{c_y \geq 1} (c_y - 1) = \frac{r - |\Gamma(A)|}{r}. \quad (6.8)$$

For every probability law Q satisfying $Q(y) \leq 1/r$,

$$\Delta(P_A, Q) = \sum_y (P_A(y) - Q(y))_+ \geq \sum_{c_y \geq 2} \left(\frac{c_y}{r} - \frac{1}{r} \right) = \frac{r - |\Gamma(A)|}{r}. \quad (6.9)$$

Because $M \geq r$, after capping each occupied coordinate at $1/r$ we may distribute the removed mass among coordinates below the cap, obtaining a probability law Q_A with $Q_A(y) \leq 1/r$ for every y . The total mass moved is exactly the quantity in (6.8), so equality holds in (6.9). Hence

$$\Delta(P_A, Q_A) = \frac{r - |\Gamma(A)|}{r} \leq \varepsilon.$$

This proves (6.7) and the condenser property for flat k -sources.

Finally, the set of (n, k) source laws is the convex hull of the flat K -source laws by the polytope argument in Theorem 5.1. The set of $(k + d)$ -source laws on $[M]$ is convex, and distance to a convex set in ℓ^1 is convex. Hence the same ε bound extends from flat sources to every (n, k) source. $\square$

Theorem 6.3 (Probabilistic existence of near-optimal lossless condensers). *Let $0 < \varepsilon \leq 1/2$ and $0 \leq k \leq n$. Suppose $D = 2^d$ satisfies*

$$D \geq \frac{2(1 + (n - k) \ln 2)}{\varepsilon \ln 2}. \quad (6.10)$$

Let $M = 2^m$ be any power of two satisfying

$$M \geq \frac{2eKD}{\varepsilon}. \quad (6.11)$$

Then there exists a $k \rightarrow_\varepsilon (k + d)$ lossless condenser

$$\text{Con} : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^m. \quad (6.12)$$

In particular one may choose

$$d = \log_2(n - k + 1) + \log_2(1/\varepsilon) + O(1), \quad (6.13)$$

while the least admissible power of two M satisfies

$$m < k + d + \log_2(1/\varepsilon) + \log_2(4e). \quad (6.14)$$

Thus nonconstructive lossless condensing needs only logarithmic seed and $O(\log(1/\varepsilon))$ ambient output deficiency.

Proof. Choose every table entry $\text{Con}(x, s)$ independently and uniformly from $[M]$. Fix a K -set A and expose the $r := KD$ labeled endpoints from A in any fixed order. Call position j a *repeat* if its endpoint has appeared earlier. The number of repeats is exactly $r - |\Gamma(A)|$.

Put $t := \lceil \varepsilon r \rceil$. For any fixed set J of t positions, conditional on every earlier exposure, the probability that a designated $j \in J$ is a repeat is at most r/M . Hence

$$\Pr[J \text{ consists entirely of repeats}] \leq \left(\frac{r}{M}\right)^t.$$

A union bound over the choices of J , followed by $\binom{r}{t} \leq (er/t)^t$ and $t \geq \varepsilon r$, gives

$$\Pr[|\Gamma(A)| < (1 - \varepsilon)r] \leq \binom{r}{t} \left(\frac{r}{M}\right)^t \leq \left(\frac{er}{\varepsilon M}\right)^t \leq 2^{-t} \leq 2^{-\varepsilon KD}, \quad (6.15)$$

where (6.11) is stronger than needed in the penultimate step.

There are at most

$$\binom{N}{K} \leq \exp(K(1 + (n - k) \ln 2))$$

choices for A . Therefore the probability that some K -set violates (6.6) is at most

$$\exp(K(1 + (n - k) \ln 2) - \varepsilon KD \ln 2) < 1 \quad (6.16)$$

by (6.10). Some table consequently satisfies (6.6) for every A . Since (6.11) implies $M \geq KD$, Proposition 6.2 turns it into a lossless condenser.

Taking D to be the least power of two satisfying (6.10) gives (6.13). Taking M to be the least power of two satisfying (6.11) gives $M < 4eKD/\varepsilon$, which is exactly (6.14). $\square$

Theorem 6.4 (Guruswami–Umans–Vadhan lossless condenser — deep input). *Fix a constant $\alpha \in (0, 1)$. For every positive integer n , every $1 \leq k \leq n$, and every $\varepsilon > 0$, there is an explicit*

$$k \rightarrow_\varepsilon (k + d)$$

lossless condenser

$$C : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^m \quad (6.17)$$

with

$$d = \left(1 + \frac{1}{\alpha}\right) (\log_2 n + \log_2 k + \log_2(1/\varepsilon)) + O_\alpha(1), \quad (6.18)$$

and

$$m \leq 2d + (1 + \alpha)k. \quad (6.19)$$

The construction is polynomial-time explicit in the standard finite-field representation used by Guruswami–Umans–Vadhan.

Ownership and comparison with the probabilistic target. Theorem 6.4 is retained as owner-C. Its proof uses the Parvaresh–Vardy/list-decoding construction and is not reproduced here. The locally proved Theorem 6.3 supplies the clean information-theoretic benchmark; the GUV theorem supplies explicitness with parameters polynomially close to that benchmark and, crucially, preserves all $k + d$ bits of source-plus-seed min-entropy up to statistical error.

A collision picture. For a flat source A , a lossless condenser does not demand that all KD labeled edges land at different right vertices. It permits at most an ε fraction of collision loss in the precise sense

$$KD - |\Gamma(A)| \leq \varepsilon KD.$$

Proposition 6.2 says that this combinatorial statement and the min-entropy statement are two descriptions of the same phenomenon.

Chapter 7

Averaging Samplers and the Extractor Dictionary

A seeded extractor and an averaging sampler can be the same table viewed in opposite directions. An extractor fixes a weak random source over table rows and asks whether a random column produces a nearly uniform output. A sampler fixes a test function on the output universe and asks for how many rows the average over columns is atypical. The equivalence is elementary once the failure-set and min-entropy conventions are written exactly.

Definition 7.1 (Averaging samplers, Boolean samplers, and seed labels). Let Ω be a finite set and let

$$\text{Samp} : [R] \longrightarrow \Omega^D, \quad \text{Samp}(x) = (z_1(x), \dots, z_D(x)).$$

For $f : \Omega \rightarrow [0, 1]$ put

$$\mu(f) := \mathbb{E}_{z \sim U_\Omega} f(z), \quad \text{Av}_f(x) := \frac{1}{D} \sum_{s=1}^D f(z_s(x)). \quad (7.1)$$

The map is a (δ, ε) *averaging sampler* if

$$\Pr_{x \sim U_R} [\text{Av}_f(x) > \mu(f) + \varepsilon] \leq \delta \quad (7.2)$$

for every $f : \Omega \rightarrow [0, 1]$. It is a *Boolean averaging sampler* if (7.2) is required only for indicator functions $f = \mathbf{1}_T$. Applying (7.2) to $1 - f$ gives the corresponding lower-tail estimate, so the one-sided convention loses no information when the full averaging condition is available.

If $\text{Ext} : [N] \times [D] \rightarrow [M]$, its ordinary row sampler is

$$\text{Samp}_{\text{out}}(x)_s := \text{Ext}(x, s). \quad (7.3)$$

Its *seed-labelled row sampler* has universe $[D] \times [M]$ and

$$\text{Samp}_{\text{lab}}(x)_s := (s, \text{Ext}(x, s)). \quad (7.4)$$

The labelled version is the sampler counterpart of a strong extractor.

Theorem 7.2 (Extractor-to-sampler conversion). *Let $N = 2^n$, $K = 2^k$, and $\text{Ext} : [N] \times [D] \rightarrow [M]$.*

(i) If Ext is an ordinary (k, ε) extractor, then Samp_{out} is a

$$(K/N, \varepsilon) \quad (7.5)$$

averaging sampler on $[M]$.

(ii) If Ext is a strong (k, ε) extractor, then Samp_{lab} is a $(K/N, \varepsilon)$ averaging sampler on $[D] \times [M]$.

In both cases the sample complexity is exactly D .

Proof. We prove (i); the proof of (ii) is identical with the labelled universe. Fix $f : [M] \rightarrow [0, 1]$ and let

$$B_f := \left\{ x \in [N] : \frac{1}{D} \sum_{s \in [D]} f(\text{Ext}(x, s)) > \mu(f) + \varepsilon \right\}. \quad (7.6)$$

If $|B_f| \geq K$, choose $A \subseteq B_f$ with $|A| = K$ and let $X = U_A$. Then

$$\mathbb{E}f(\text{Ext}(X, S)) > \mu(f) + \varepsilon. \quad (7.7)$$

But statistical distance at most ε from U_M implies

$$|\mathbb{E}f(\text{Ext}(X, S)) - \mathbb{E}f(U_M)| \leq \varepsilon \quad (7.8)$$

for every $[0, 1]$ -valued f , contradicting (7.7). Hence $|B_f| < K$, and a uniform row lies in B_f with probability $< K/N$, proving (7.5).

For (ii), replace $f(y)$ by an arbitrary $F(s, y) : [D] \times [M] \rightarrow [0, 1]$ and use the strong extractor comparison between $(S, \text{Ext}(X, S))$ and $U_D \times U_M$. $\square$

Theorem 7.3 (Sampler-to-extractor conversion). *Let $\text{Samp} : [N] \rightarrow [M]^D$ be a (δ, ε) Boolean averaging sampler, and define*

$$\text{Ext}(x, s) := \text{Samp}(x)_s. \quad (7.9)$$

If

$$k' \geq \log_2 \frac{\delta N}{\varepsilon}, \quad (7.10)$$

then Ext is an ordinary $(k', 2\varepsilon)$ extractor. In particular, when $\delta = K/N = 2^k/N$, it is a

$$(k + \log_2(1/\varepsilon), 2\varepsilon) \quad (7.11)$$

extractor.

The same assertion holds in the strong convention: if a seed-labelled sampler of the form (7.4) is a (δ, ε) Boolean averaging sampler on $[D] \times [M]$, then the associated Ext is a strong $(k', 2\varepsilon)$ extractor under (7.10).

Proof. Let X be any source on $[N]$ with $H_\infty(X) \geq k'$. Fix $T \subseteq [M]$ and put $\mu(T) = |T|/M$. By the Boolean sampler property, the bad-row set

$$B_T := \left\{ x : \Pr_{s \sim U_D} [\text{Ext}(x, s) \in T] > \mu(T) + \varepsilon \right\} \quad (7.12)$$

has cardinality at most δN . Since every atom of X has mass at most $2^{-k'}$, (7.10) gives

$$\Pr[X \in B_T] \leq \delta N 2^{-k'} \leq \varepsilon. \quad (7.13)$$

Therefore

$$\begin{aligned} \Pr[\text{Ext}(X, S) \in T] &\leq \Pr[X \in B_T] + \Pr[\text{Ext}(X, S) \in T \mid X \notin B_T] \\ &\leq \varepsilon + \mu(T) + \varepsilon = \mu(T) + 2\varepsilon. \end{aligned} \quad (7.14)$$

For two probability laws P, Q on a finite set, $\Delta(P, Q) = \max_T (P(T) - Q(T))$. Taking the maximum over T in (7.14) proves $\Delta(\text{Ext}(X, S), U_M) \leq 2\varepsilon$.

For the strong statement, use test sets $T \subseteq [D] \times [M]$ and the seed-labelled rows (7.4). The same calculation gives statistical distance at most 2ε between $(S, \text{Ext}(X, S))$ and $U_D \times U_M$. $\square$

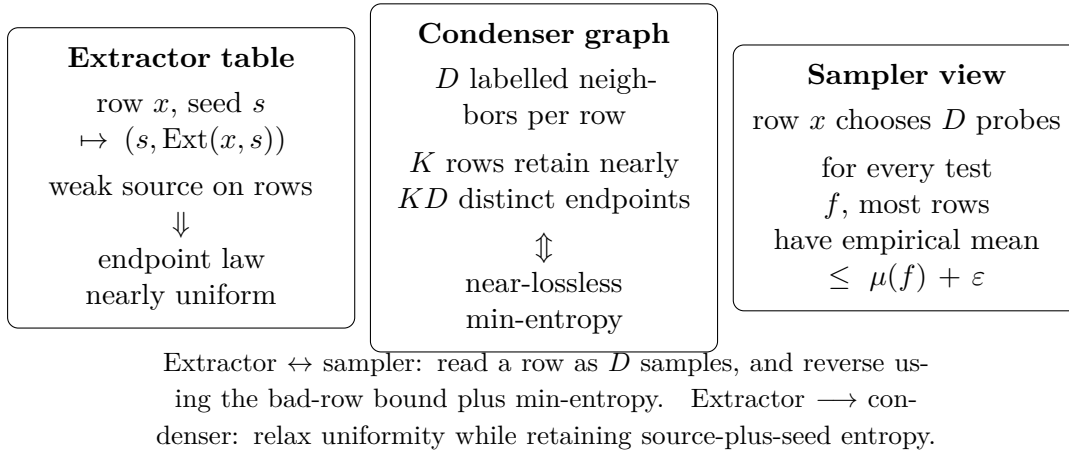


Figure 7.1: Three views of one left-regular table. Extractors control endpoint statistics, lossless condensers control collision loss and retained min-entropy, and samplers control empirical averages for most rows. Seed labels on the extractor side encode the strong convention.

The one-line dictionary. Theorem 7.2 says that extractor discrepancy forbids too many bad rows. Theorem 7.3 reverses the argument: a Boolean bad-row bound plus a min-entropy cap prevents a weak source from placing too much probability on the exceptional rows. The extra $\log_2(1/\varepsilon)$ bits in (7.11) are exactly the entropy needed to make the bad-set mass at most ε .

Chapter 8

Explicit Small Bias and Two Independent Sources

The seeded-extractor theory developed so far assumes one short independent uniform seed. Two complementary constructions show what happens when the randomness model changes. First, the Naor–Naor theorem makes the small-bias spaces of Chapter 2 fully explicit. Second, if two independent weak sources are available, the mod-two inner product already extracts one nearly uniform bit once their collision entropies cross the ambient-dimension threshold. We then use the probabilistic method to show that arbitrary functions can do much better nonconstructively at logarithmic-plus entropy.

Theorem 8.1 (Naor–Naor explicit small-bias generator — deep input). *There is a universal constant $C > 0$ and an explicit deterministic map*

$$\text{SB}_{n,\varepsilon} : \{0,1\}^r \rightarrow \{0,1\}^n, \quad r \leq C(\log_2 n + \log_2(1/\varepsilon) + 1), \quad (8.1)$$

for every $n \geq 1$ and $0 < \varepsilon < 1$, such that, for uniform $R \sim U_r$ and every nonzero $a \in \mathbb{F}_2^n$,

$$\left| \mathbb{E}(-1)^{a \cdot \text{SB}_{n,\varepsilon}(R)} \right| \leq \varepsilon. \quad (8.2)$$

The coordinates of the output are computable in time polynomial in $\log n$ and $\log(1/\varepsilon)$ in the standard explicit-construction model.

Relation to Chapter 2. Theorem 2.1 proved the exact equivalence between small bias and balanced nonzero words in the associated binary linear-code picture, while Corollary 2.2 gave a probabilistic sample space of size $O(n/\varepsilon^2)$. Theorem 8.1 is the genuinely explicit step: its seed has only $O((\log n + \log(1/\varepsilon)))$ bits. We retain its finite-field construction as source-owned rather than replacing it by an informal sketch.

Definition 8.2 (Two-source extractor and collision entropy). The *collision entropy* of a finite random variable X is

$$H_2(X) := -\log_2 \sum_x \Pr[X = x]^2. \quad (8.3)$$

For integers k_1, k_2 , a deterministic function

$$\text{Ext}_2 : \{0,1\}^{k_1} \times \{0,1\}^{k_2} \rightarrow \{0,1\}^m$$

is a $((k_1, k_2), \varepsilon)$ *two-source extractor* if, for every pair of *independent* sources X, Y with $H_\infty(X) \geq k_1$ and $H_\infty(Y) \geq k_2$,

$$\text{Ext}_2(X, Y) \approx_\varepsilon U_m. \quad (8.4)$$

Independence is part of the model and will never be inferred merely from the marginal entropy bounds.

The elementary inequality (1.5) gives $H_2(X) \geq H_\infty(X)$; collision entropy is therefore the right intermediate quantity for the Fourier proof below.

Theorem 8.3 (Inner product as a two-source extractor). *Let X, Y be independent random variables on $\mathbb{F}_2^n$. Then*

$$\left| \mathbb{E}(-1)^{\langle X, Y \rangle} \right| \leq 2^{\frac{n - H_2(X) - H_2(Y)}{2}}. \quad (8.5)$$

Consequently, for the bit $B := \langle X, Y \rangle \pmod{2}$,

$$\Delta(B, U_1) \leq \frac{1}{2} 2^{\frac{n - H_\infty(X) - H_\infty(Y)}{2}}. \quad (8.6)$$

In particular, if $0 < \varepsilon \leq 1/2$ and

$$H_\infty(X) + H_\infty(Y) \geq n + 2 \log_2(1/\varepsilon) - 2, \quad (8.7)$$

then inner product is an ε -extractor for the two sources.

Proof. Write $p(x) = \Pr[X = x]$ and $q(y) = \Pr[Y = y]$ and define the unnormalized Walsh transform

$$\tilde{q}(x) := \sum_y q(y) (-1)^{\langle x, y \rangle}.$$

Independence gives

$$\mathbb{E}(-1)^{\langle X, Y \rangle} = \sum_x p(x) \tilde{q}(x).$$

Cauchy–Schwarz and character orthogonality yield

$$\left| \sum_x p(x) \tilde{q}(x) \right| \leq \left(\sum_x p(x)^2 \right)^{1/2} \left(\sum_x |\tilde{q}(x)|^2 \right)^{1/2} = 2^{n/2} \left(\sum_x p(x)^2 \right)^{1/2} \left(\sum_y q(y)^2 \right)^{1/2},$$

which is (8.5) by Definition 8.2. Since $H_2 \geq H_\infty$, the same bound holds with the min-entropies on the right. For any random bit B ,

$$\Delta(B, U_1) = \frac{1}{2} |\Pr[B = 0] - \Pr[B = 1]| = \frac{1}{2} |\mathbb{E}(-1)^B|,$$

which proves (8.6). Solving the inequality in (8.6) for an error at most ε gives (8.7). $\square$

Subspace stress test. If X is uniform on a subspace $U \leq \mathbb{F}_2^n$ and Y is uniform on $V \leq \mathbb{F}_2^n$, then character orthogonality gives the exact identity

$$\mathbb{E}(-1)^{\langle X, Y \rangle} = \frac{|U \cap V^\perp|}{|U|}. \quad (8.8)$$

Thus $U = V^\perp$ gives bias one even though $\dim U + \dim V = n$. The excess-entropy margin in (8.7) is therefore not a cosmetic artifact of the proof.

Theorem 8.4 (Probabilistic existence of two-source extractors). *Let $0 \leq k_1, k_2 \leq n$, set $N := 2^n$, $K_i := 2^{k_i}$, and let $0 < \varepsilon \leq 1$. Suppose a power of two $M = 2^m$ satisfies*

$$K_2 \geq \frac{4(1 + (n - k_1) \ln 2)}{\varepsilon^2}, \quad K_1 \geq \frac{4(1 + (n - k_2) \ln 2)}{\varepsilon^2}, \quad (8.9)$$

and

$$M \leq \frac{\varepsilon^2 K_1 K_2}{4 \ln 2}. \quad (8.10)$$

Then there exists an $((k_1, k_2), \varepsilon)$ two-source extractor

$$F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^m. \quad (8.11)$$

In the symmetric case $k_1 = k_2 = k$, it is enough that

$$k \geq \log_2(n + 1) + 2 \log_2(1/\varepsilon) + 3, \quad (8.12)$$

and then one may choose, whenever the right side is nonnegative,

$$m \geq 2k - 2 \log_2(1/\varepsilon) - 3. \quad (8.13)$$

Proof. Choose the table values $F(x, y)$ independently and uniformly in $[M]$. Fix sets $A, B \subseteq [N]$ with $|A| = K_1$, $|B| = K_2$, and fix $T \subseteq [M]$. The random variables $\mathbf{1}_{\{F(x, y) \in T\}}$, $(x, y) \in A \times B$, are independent, with mean $|T|/M$. Hoeffding therefore gives

$$\Pr \left[\left| \frac{e_F(A, B; T)}{K_1 K_2} - \frac{|T|}{M} \right| > \varepsilon \right] \leq 2e^{-2\varepsilon^2 K_1 K_2}. \quad (8.14)$$

The number of triples (A, B, T) is at most

$$\binom{N}{K_1} \binom{N}{K_2} 2^M \leq \exp(K_1(1 + (n - k_1) \ln 2) + K_2(1 + (n - k_2) \ln 2) + M \ln 2). \quad (8.15)$$

By (8.9), the first two terms in the exponent are each at most $\varepsilon^2 K_1 K_2 / 4$, and by (8.10) so is the third. Moreover (8.9) implies $\varepsilon^2 K_1 K_2 / 4 > \ln 2$. Combining (8.14)–(8.15), the union bound is strictly below one. Hence some table satisfies, simultaneously for every A, B, T ,

$$\left| \Pr[F(U_A, U_B) \in T] - \frac{|T|}{M} \right| \leq \varepsilon. \quad (8.16)$$

By the variational characterization (1.2), $F(U_A, U_B)$ is ε -close to uniform.

It remains to pass from flat to arbitrary sources. The polytope argument in Theorem 5.1 writes every min-entropy- k_i source as a convex combination of flat K_i -sources. Because X and Y are independent, their product law is the corresponding convex combination of product flat laws $U_A \otimes U_B$. The map to $F(X, Y)$ is affine and statistical distance to uniform is convex, so (8.16) proves (8.11).

For the symmetric parameter form, (8.12) gives $K \geq 8(n + 1)/\varepsilon^2$, which implies both inequalities in (8.9). Choose M to be the largest power of two not exceeding the right side of (8.10). Since $\log_2(4 \ln 2) < 3$, the one-bit loss from rounding down to a power of two is absorbed by the displayed constant 3, yielding (8.13). $\square$

Constructive versus nonconstructive. The inner-product theorem is explicit but needs the sum of source entropies to exceed n . The random table in Theorem 8.4 works already when each entropy is only logarithmic in n and $1/\varepsilon$, but has exponential description size. The next three source-owned theorems record, separately and without parameter blending, three landmarks on the difficult constructive road between these endpoints.

Theorem 8.5 (Raz asymmetric strong two-source extractor — deep input). *Let $0 < \delta < 1/2$, let $n_1, n_2, m \in \mathbb{N}$, and let the min-entropy thresholds satisfy $0 \leq b_1 < n_1$ and $0 \leq b_2 \leq n_2$. Suppose*

$$n_1 \geq 6 \log_2 n_1 + 2 \log_2 n_2, \quad (8.17)$$

$$b_1 \geq \left(\frac{1}{2} + \delta\right) n_1 + 3 \log_2 n_1 + \log_2 n_2, \quad b_2 \geq 5 \log_2(n_1 - b_1), \quad (8.18)$$

and

$$m \leq \delta \min \left\{ \frac{n_1}{8}, \frac{b_2}{40} \right\} - 1. \quad (8.19)$$

Then there is an explicit map, in Raz's uniform finite-field circuit model,

$$\text{Raz} : \{0, 1\}^{n_1} \times \{0, 1\}^{n_2} \longrightarrow \{0, 1\}^m, \quad (8.20)$$

such that for every pair of independent (n_1, b_1) - and (n_2, b_2) -sources X_1, X_2 ,

$$\Delta((X_i, \text{Raz}(X_1, X_2)), (X_i, U_m)) \leq 2^{-3m/2-1}, \quad i = 1, 2. \quad (8.21)$$

In particular it is strong in both inputs. Moreover, for every fixed $0 < \delta < 1/2$, the same source gives an explicit equal-length family with

$$b_1 = \left(\frac{1}{2} + \delta\right) n, \quad b_2 = O_\delta(\log n), \quad m = \Omega_\delta(n), \quad \Delta(\text{Raz}(X_1, X_2), U_m) = n^{-\Omega_\delta(1)}. \quad (8.22)$$

The constants hidden in (8.22) are source-owned and are not silently identified with the constants in (8.17)–(8.21).

Normalization audit for Raz. Raz defines “ γ -close” using the full ℓ^1 distance, whereas Definition 1.1 uses half the ℓ^1 distance. Thus the source's exact error $\gamma = 2^{-3m/2}$ becomes the QRCC bound $2^{-3m/2-1}$ in (8.21). The asymmetric hypothesis is equally important: one source is just above entropy rate $1/2$, while the other can have only logarithmic entropy. Corollary (8.22) is obtained in the source by using the short strong output as a seed for a seeded extractor; it is not obtained by pretending that (8.19) itself permits linear m when $b_2 = O(\log n)$.

Theorem 8.6 (Bourgain below-half two-source extractor — deep input). *There are universal constants $\gamma, c_1, c_2 > 0$ such that, for all sufficiently large n , there is a polynomial-time computable map*

$$\text{Bou}_n : \{0, 1\}^n \times \{0, 1\}^n \longrightarrow \{0, 1\}^{m_n}, \quad m_n \geq c_1 n, \quad (8.23)$$

with the following property. If X, Y are independent and

$$H_\infty(X), H_\infty(Y) \geq \left(\frac{1}{2} - \gamma\right) n, \quad (8.24)$$

then

$$\Delta(\text{Bou}_n(X, Y), U_{m_n}) \leq 2^{-c_2 n}. \quad (8.25)$$

The constants are absolute but not numerically fixed by the theorem-strength source statement used here.

What Bourgain actually breaks. The point of Theorem 8.6 is the strict crossing of the one-half entropy-rate barrier for *both* independent sources, while retaining linear output and exponentially small error. The primary proof passes through finite-field sum-product and exponential-sum estimates; the standard bitstring formulation above is recorded in Rao’s canonical exposition of Bourgain’s construction. Later literature often summarizes the threshold numerically as about $0.499n$, but the verified theorem used here quantifies it as $(1/2 - \gamma)n$ for an unspecified universal $\gamma > 0$. We therefore do not hard-code 0.499 as an exact Bourgain constant.

Theorem 8.7 (Chattopadhyay–Zuckerman polylog-entropy extractor — deep input). *There is a sufficiently large universal constant $C_1 > 0$ such that for every $n \in \mathbb{N}$, every $0 < \varepsilon < 1$, and every integer $k \leq n$ with*

$$k \geq C_1 (\log_2(n/\varepsilon))^{56}, \quad (8.26)$$

there is an explicit function

$$2\text{Ext}_{n,k,\varepsilon} : \{0,1\}^n \times \{0,1\}^n \longrightarrow \{0,1\}, \quad (8.27)$$

computable in time $\text{poly}(n, 1/\varepsilon)$, such that for independent (n, k) -sources X, Y ,

$$\Delta(2\text{Ext}_{n,k,\varepsilon}(X, Y), U_1) \leq \varepsilon. \quad (8.28)$$

The source also notes that a one-bit two-source extractor is strong with similar parameters, by Barak’s argument.

Parameter reading. Theorem 8.7 is a one-bit theorem; no many-bit output is being imported here. For example, taking $\varepsilon = n^{-a}$ with fixed $a > 0$ turns (8.26) into

$$k \geq C_1 (a + 1)^{56} (\log_2 n)^{56}, \quad (8.29)$$

and the evaluation time remains polynomial in n . This is the first explicit theorem in the present spine that reaches polylogarithmic min-entropy for both sources. Its proof machinery—in particular the non-malleable-extractor reduction and the construction of almost-balanced resilient functions—remains source-owned and is not replaced by a slogan such as “correlation breaking.”

Three regimes, not one synthetic theorem. Raz is asymmetric and can tolerate a logarithmic second source when the first source has rate above one half. Bourgain is symmetric and moves both sources a fixed amount below one half while keeping linear output and exponential error. Chattopadhyay–Zuckerman is symmetric at polylogarithmic entropy but, in the verified theorem above, outputs one bit and pays polynomial running time in $1/\varepsilon$. These hypotheses and conclusions are deliberately kept separate because no implication among the three parameter packages is being asserted merely from their historical ordering.

Chapter 9

Affine, Leakage-Resilient, and Non-Malleable Extraction

The seeded and two-source interfaces of the preceding chapters do not cover all structured weak sources that arise in derandomization and cryptography. This chapter isolates three different robustness mechanisms and keeps their hypotheses separate. First, affine sources have algebraic support and admit seedless deterministic extraction. Second, leakage-resilient extraction asks for independent-source extraction that remains uniform after a restricted multiparty transcript is revealed, and this yields extraction from a basic adversarial-source model with only three good blocks. Third, non-malleable seeded extraction protects one extractor evaluation against a related seed evaluation. None of these models is silently identified with another.

9.1 Binary affine sources and deterministic extraction

Definition 9.1 (Binary affine source and affine extractor). Identify $\{0, 1\}^n$ with the vector space $\mathbb{F}_2^n$. If $V \leq \mathbb{F}_2^n$ is a linear subspace of dimension r and $a \in \mathbb{F}_2^n$, the uniform law on the coset $a + V$ is an r -dimensional binary affine source. Its min-entropy is exactly r .

For integers $0 \leq k \leq n$, a deterministic function

$$\text{AExt} : \mathbb{F}_2^n \longrightarrow \{0, 1\}^m$$

is a (k, ε) binary affine extractor if

$$\Delta(\text{AExt}(X), U_m) \leq \varepsilon$$

for every affine source X supported on a coset of dimension at least k . There is no seed and no independence hypothesis beyond the internal uniformity of the affine source.

Theorem 9.2 (Bourgain binary affine extractor — deep input). *For every fixed $0 < \delta < 1$ there are constants $c_\delta > 0$ and $n_0(\delta)$ and an explicit family*

$$\text{AExt}_n : \mathbb{F}_2^n \longrightarrow \mathbb{F}_2, \quad n \geq n_0(\delta),$$

such that, whenever X is uniform on an affine subspace of $\mathbb{F}_2^n$ of dimension strictly greater than δn ,

$$\Delta(\text{AExt}_n(X), U_1) \leq 2^{-c_\delta n}.$$

Thus every fixed positive affine entropy rate admits an explicit deterministic one-bit extractor with exponentially small error. The constants and the algebraic construction remain source-owned.

Normalization and source audit. The convenient later formulation of Bourgain’s construction measures multi-output error pointwise in ℓ_∞ . Its discussion explicitly records the original construction as a one-output-bit family with error $2^{-\Omega(n)}$. For a distribution Z on one bit,

$$\Delta(Z, U_1) = \left| \Pr[Z = 1] - \frac{1}{2} \right|,$$

so the pointwise one-bit error and the statistical-distance convention of this book agree exactly. We therefore import only this one-bit theorem. We do *not* reinterpret the later linear-output ℓ_∞ statement as a linear-output total-variation theorem.

Why the theorem is nontrivial. No fixed nonzero linear functional can serve as a universal affine extractor: if $\ell : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is linear, then the uniform law on any large affine subspace contained in a level set of ℓ makes $\ell(X)$ constant. Bourgain instead uses an algebraic construction whose analysis is based on finite-field sum-product input and multilinear exponential-sum bounds. Those estimates are the deep source-owned part of Theorem 9.2; the present manuscript freezes only the binary field, affine-dimension threshold, output length, error scale, and explicitness interface.

9.2 Adversarial sources and number-on-forehead leakage

Definition 9.3 (Basic adversarial source). Let $N, K, n, k \in \mathbb{N}$ with $K \leq N$. A random vector

$$X = (X_1, \dots, X_N) \in (\{0, 1\}^n)^N$$

is an (N, K, n, k) *basic adversarial source* if the blocks $X_1, \dots, X_N$ are mutually independent and there is a set $G \subseteq [N]$, $|G| \geq K$, such that $H_\infty(X_i) \geq k$ for every $i \in G$. The blocks outside G carry no entropy promise and may even be deterministic, but they remain independent of every other block.

The independence clause is part of the model, not a convenience. A more general adversarial-source notion allows bad blocks to depend on good blocks; that correlated-bad-source regime is deliberately excluded from the theorem below.

Definition 9.4 (NOF protocol and leakage-resilient extractor). Fix $N, n, \mu \in \mathbb{N}$. A deterministic map

$$\Pi : (\{0, 1\}^n)^N \longrightarrow \{0, 1\}^\mu$$

is a *number-on-forehead (NOF) protocol of communication μ* if, for each transcript position $i \in [\mu]$, there are functions

$$S_i : \{0, 1\}^{i-1} \rightarrow [N], \quad g_i : \{0, 1\}^{i-1} \times (\{0, 1\}^n)^{N-1} \rightarrow \{0, 1\}$$

such that, writing $\Pi_{<i}(x) = (\Pi_1(x), \dots, \Pi_{i-1}(x))$ and letting x_{-j} denote the tuple with block j omitted,

$$\Pi_i(x) = g_i(\Pi_{<i}(x), x_{-S_i(\Pi_{<i}(x))}).$$

Thus the speaker at step i may depend on the previous transcript, and the speaker may inspect every input block except the one on its own forehead. The protocol is *non-adaptive* when each S_i is constant and each g_i ignores the previous transcript.

A function

$$\text{LRE} : (\{0, 1\}^n)^N \rightarrow \{0, 1\}^m$$

is a *leakage-resilient extractor* for *min-entropy* k , *error* ε , and μ -bit *NOF leakage* if, for every collection of mutually independent n -bit sources $X_1, \dots, X_N$ with $H_\infty(X_i) \geq k$ for all i , and every such Π ,

$$\Delta((\text{LRE}(X_1, \dots, X_N), \Pi(X_1, \dots, X_N)), (U_m, \Pi(X_1, \dots, X_N))) \leq \varepsilon,$$

where U_m is independent of the displayed transcript.

Theorem 9.5 (Chattopadhyay–Goodman leakage-resilient and three-good-source extraction — verified frontier). *There exist universal constants $C, \gamma > 0$ with the following two properties whenever $n \geq k \geq (\log_2 n)^C$.*

1. *There is an explicit three-source map*

$$\text{LRE}_n : (\{0, 1\}^n)^3 \rightarrow \{0, 1\}^m, \quad m = \lceil k^\gamma \rceil,$$

that is leakage-resilient against NOF protocols of communication $\mu = \lceil k^\gamma \rceil$, with statistical-distance error

$$\varepsilon = 2^{-\lceil k^\gamma \rceil}.$$

2. *For every $N \geq 3$, there is an explicit map*

$$\text{AdvExt}_{N,n} : (\{0, 1\}^n)^N \rightarrow \{0, 1\}^m, \quad m = \lceil k^\gamma \rceil,$$

that extracts from every $(N, 3, n, k)$ basic adversarial source with the same error $2^{-\lceil k^\gamma \rceil}$.

The constants are universal and source-owned.

Source-owned reduction architecture. The first part is the paper’s three-source LRE theorem. The second part is obtained there by applying that LRE to every triple of input blocks and XORing the outputs. After three good blocks are fixed as the distinguished triple and all remaining independent blocks are conditioned to constants, the XOR of the other triple terms decomposes into functions each missing one of the three good inputs; the NOF-leakage guarantee and data processing then give the adversarial-source extractor. This architecture is recorded to make the model transition visible, but the low-entropy LRE construction itself remains a source-owned modern input.

Model boundary. The weak non-malleable extractor used internally in the 2025 construction is a multi-independent-source object and is not Definition 9.6, which concerns a single weak source with a tampered public seed. Accordingly, Theorem 9.5 depends on the adversarial-source and NOF/LRE definitions above, not on the seeded non-malleability theorem below. This prevents a notation-level similarity from becoming a false logical dependency.

9.3 Non-malleable extraction under seed tampering

A strong seeded extractor guarantees that its output remains uniform when the public seed is revealed. That is not enough when an active adversary can replace the seed by a related value and can thereby expose a second extractor output computed from the same weak source. Non-malleable extraction asks that the original output remain uniform even after this related output is revealed. The fixed-point-free condition below is essential: if the tampering map were allowed to leave the seed unchanged, the adversary would be handed the very output that is supposed to remain hidden.

Definition 9.6 (Non-malleable extractor and t -tampering extension). Let $n, k, d, m \in \mathbb{N}$ and $0 < \varepsilon < 1$. A map

$$\mathcal{A} : \{0, 1\}^d \rightarrow \{0, 1\}^d$$

is *fixed-point-free* if $\mathcal{A}(s) \neq s$ for every $s \in \{0, 1\}^d$.

A function

$$\text{nmExt} : \{0, 1\}^n \times \{0, 1\}^d \rightarrow \{0, 1\}^m \quad (9.1)$$

is a (k, ε) *non-malleable extractor* if, for every (n, k) -source X , every independent seed $S \sim U_d$, and every fixed-point-free $\mathcal{A}$,

$$\Delta((\text{nmExt}(X, S), \text{nmExt}(X, \mathcal{A}(S)), S), (U_m, \text{nmExt}(X, \mathcal{A}(S)), S)) \leq \varepsilon, \quad (9.2)$$

where U_m in the second distribution is independent of all other displayed variables.

More generally, for $t \geq 1$, let $\mathcal{A} = (\mathcal{A}_1, \dots, \mathcal{A}_t)$ with each $\mathcal{A}_i : \{0, 1\}^d \rightarrow \{0, 1\}^d$ fixed-point-free. The function is a (k, ε) *t -non-malleable extractor* if

$$\begin{aligned} \Delta((\text{nmExt}(X, S), \text{nmExt}(X, \mathcal{A}_1(S)), \dots, \text{nmExt}(X, \mathcal{A}_t(S)), S), \\ (U_m, \text{nmExt}(X, \mathcal{A}_1(S)), \dots, \text{nmExt}(X, \mathcal{A}_t(S)), S)) \leq \varepsilon. \end{aligned} \quad (9.3)$$

No independence is assumed among the tampered seeds; they are all arbitrary functions of the same public seed S .

Strong extraction is a marginal consequence. Projecting the two distributions in (9.2) onto the first and third coordinates and using monotonicity of statistical distance under deterministic maps gives

$$\Delta((\text{nmExt}(X, S), S), (U_m, S)) \leq \varepsilon. \quad (9.4)$$

Thus non-malleability is genuinely stronger than the strong seeded-extractor condition; “strong” is not an additional adjective that needs to be inserted into Definition 9.6.

What is not part of the definition. Modern extractor constructions often use *correlation breakers with advice*, advice generators, or related primitives. Those are construction interfaces: an advice string is supplied so that correlated executions can be forced into distinguishable branches. Advice is therefore not silently built into the baseline non-malleable-extractor definition (9.2). Keeping these notions separate prevents a common but substantive mismatch between an extractor’s security game and the machinery used to construct it.

Theorem 9.7 (Cohen–Raz–Segev short-seed non-malleable extractor — deep input). *Fix a constant $0 < \delta < 1/2$. For every pair of integers n, d satisfying*

$$2.01 \log_2 n \leq d \leq n, \quad (9.5)$$

there is an explicit function

$$\text{nmExt}_{n,d} : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^m, \quad m = \Omega_\delta(d), \quad (9.6)$$

which, in the statistical-distance convention of this book, is a (k, ε) non-malleable extractor with

$$k = \left(\frac{1}{2} + \delta\right)n, \quad \varepsilon = 2^{-m-1}. \quad (9.7)$$

The hidden constant in $m = \Omega_\delta(d)$ is source-owned. The construction is unconditional.

Normalization audit and parameter regimes. The source calls two laws ε -close when their *full* ℓ^1 distance is at most ε . Its Theorem 1.6 gives error 2^{-m} in that convention, hence (9.7) has QRCC statistical distance at most 2^{-m-1} . Taking $d = \lceil 2.01 \log_2 n \rceil$ yields a logarithmic seed, output $\Omega_\delta(\log n)$, and polynomially small error. Taking $d = n$ yields output $\Omega_\delta(n)$ and exponentially small error. The constant 2.01 is not conceptually special: the source notes that it can be replaced by any fixed constant strictly larger than 2.

Generalized source theorem. The same paper proves a t -tampering version. In its full- ℓ^1 error normalization, if n, d, m, t and $0 < \delta < 1/2$ obey

$$d \geq \frac{23}{\delta}tm + 2\log_2 n, \quad n \geq \frac{160}{\delta}tm, \quad \delta \geq \frac{10\log_2(nd)}{n}, \quad (9.8)$$

then there is an explicit $((1/2 + \delta)n, 2^{-m})$ t -non-malleable extractor. We record this stronger statement only as a source-owned extension; the one-tampering theorem above is the selected interface needed by the next privacy-amplification chapter.

Cryptographic role. The non-malleable condition addresses the exact obstacle absent from passive privacy amplification: after seeing S , an active adversary may replace it by $\mathcal{A}(S)$ and thereby cause a correlated extractor evaluation on the same weak secret. Theorem 9.7 says that, above source entropy rate $1/2 + \delta$, this related output can be exposed without destroying near-uniformity of the original output. Turning that primitive into a full active privacy-amplification protocol additionally requires authentication, reconciliation of transcript-dependent entropy, and an explicit security experiment. Those protocol steps are not smuggled into the present theorem; they form the next proof unit.

9.4 Dated frontier: almost-logarithmic bit-fixing entropy

An *oblivious bit-fixing source* on $\{0, 1\}^n$ has exactly k coordinates that are independent uniform bits, while the remaining $n - k$ coordinates are fixed constants chosen in advance. Doron–Fridman construct explicit extractors supporting $k = \tilde{O}(\log n)$ and outputting almost all of the source entropy with low error. The asymptotic tilde and error/output tradeoffs are kept at source strength rather than expanded into unsupported constants here. Source: APPROX/RANDOM 2025, LIPIcs 353, Article 33, DOI 10.4230/LIPIcs.APPROX/RANDOM.2025.33. This dated result is not a prerequisite for the non-malleable extractor theorem in the present chapter.

Chapter 10

Low-Complexity Extraction and Privacy Amplification

This chapter closes two different interfaces that are easy to blur. The first asks how much extraction can be carried out by shallow circuits or by functions that inspect only a small part of the weak source. The second turns extraction into a cryptographic key-agreement primitive by placing the public transcript inside the security experiment. Circuit depth, locality, and explicitness are therefore fixed first; privacy amplification begins only after those computational conventions have been separated from the information-theoretic extractor definition.

10.1 Constant-depth extraction and two locality conventions

Definition 10.1 (AC^0 extraction and locality conventions). An AC^0 circuit family has polynomial size, constant depth, and unbounded-fan-in AND and OR gates together with fan-in-one negations. When a source theorem says that an extractor family is *explicit*, we record separately what the source proves about constructing the circuit descriptions; we use the stronger adjective *uniform* only when the source itself does.

Let

$$E : \{0, 1\}^n \times \{0, 1\}^d \longrightarrow \{0, 1\}^m$$

and fix a seed s . Following the locality convention used by Goldreich–Viola–Wigderson, E is *whole-residual t -local* if the vector-valued residual map $x \mapsto E(x, s)$ depends, as a whole, on at most t coordinates of x . Following the small-locality convention used by Cheng–Li, E has *per-output locality ℓ* if, for every output coordinate j , the Boolean function $x \mapsto E(x, s)_j$ depends on at most ℓ source coordinates; the dependency set may vary with j and s . The first condition with parameter t implies the second with parameter t , but the converse need not hold.

The distinction is quantitative, not terminological. For example, the map $x \mapsto (x_1, \dots, x_m)$ has per-output locality one, while its whole output depends on m source coordinates. Consequently a small per-output locality bound must never be rewritten as a small whole-residual locality bound.

Theorem 10.2 (AC^0 and locality extraction interface — verified frontier). *The following two source-verified interfaces hold.*

1. **Constant entropy rate, whole-residual locality.** *For every fixed constants $\delta, \beta > 0$ and every error function $\varepsilon(n) = 1/\text{poly}(n)$, there is an explicit strong $(\delta n, \varepsilon(n))$ extractor*

$$E_n : \{0, 1\}^n \times \{0, 1\}^{\beta m(n)} \longrightarrow \{0, 1\}^{m(n)}, \quad m(n) = \Theta(\log n),$$

that is whole-residual $O(\log n)$ -local. It is computable by a uniform AC^0 family; the source's residual-function/selection construction gives depth three and polynomial size.

2. **Entropy** $n/\text{polylog } n$, **per-output locality**. For every fixed $c \in \mathbb{N}$, every entropy function $k(n) = \Omega(n/\log^c n)$, and every $\varepsilon(n) = 1/\text{poly}(n)$, there is an explicit strong $(k(n), \varepsilon(n))$ extractor

$$\text{Ext}_n : \{0, 1\}^n \times \{0, 1\}^{d(n)} \longrightarrow \{0, 1\}^{m(n)}$$

with

$$d(n) = O(\log n), \quad m(n) = k(n)^{\Omega(1)},$$

computable by polynomial-size AC^0 circuits of depth $c + 10$ and having per-output locality $O(\log^{c+5} n)$. The source proves an explicit construction; QRCC does not strengthen that statement to a particular circuit uniformity convention not stated there.

All hidden constants are allowed to depend on the fixed asymptotic constants in the corresponding clause, but not on n .

Structured-source companion and the remaining dated frontier. Cheng–Li also prove an explicit deterministic AC^0 extractor for oblivious bit-fixing sources: for fixed $a, c \in \mathbb{N}$ and $\gamma \in (0, 1]$, min-entropy $k = \Omega(n/\log^a n)$ permits error $2^{-\log^c n}$, output length $(1 - \gamma)k$, and depth $O(a + c + 1)$. Here an oblivious bit-fixing source has a fixed set of independent uniform coordinates and all remaining coordinates fixed to constants. This 2018 theorem is not the dated low-entropy frontier ‘QRCC-VIII-F-027’, whose substantially smaller entropy regime remains a separate source gate.

What the interface does not say. The second clause applies to arbitrary (n, k) sources, but only in the stated high-entropy regime and with a public seed. The deterministic structured-source companion has no seed but needs the bit-fixing hypothesis. Thus neither clause licenses the slogan that deterministic AC^0 circuits extract from arbitrary weak sources. Nor does membership in AC^0 by itself supply either of the locality bounds in Definition 10.1.

10.2 Privacy amplification with an active public channel

Extraction becomes cryptography only after the public transcript is made part of the security statement. Against a passive eavesdropper a strong extractor already suffices: one party sends a fresh seed and both parties extract from their shared weak secret. An active adversary can instead replace that seed, so agreement needs a second ingredient. We first fix the security experiment, prove the finite-transcript entropy loss used in the privacy argument, and construct a one-time information-theoretic authenticator. We then give a locally auditable two-round compiler from non-malleable extraction, strong extraction, and that authentication interface.

Definition 10.3 (Privacy amplification with an active public channel). Alice and Bob initially share the same sample of an (n, k) source $W \in \{0, 1\}^n$. They use independent private coins and communicate on a public channel completely controlled by an information-theoretic adversary Eve. Their candidate outputs are

$$R_A, R_B \in \{0, 1\}^m \cup \{\perp\}.$$

A party may enter a *KeyDerived* state when it has produced a candidate key whose privacy is already guaranteed, or a *KeyConfirmed* state when it has accepted a candidate key as agreeing with the other party. Let V_E be Eve's complete public view, including her private coins; equivalently, one may condition on those coins and analyze deterministic adversarial strategies.

An $(n, k, m; \varepsilon_{\text{priv}}, \varepsilon_{\text{auth}})$ *privacy-amplification protocol* satisfies, for every (n, k) source W :

1. *Correctness.* If Eve forwards every message unchanged, one party reaches KeyDerived, the other reaches KeyConfirmed, and $R_A = R_B$.
2. *Privacy.* For $P \in \{A, B\}$, whenever $\Pr[\text{KeyDerived}_P] > 0$,

$$\Delta((R_P, V_E \mid \text{KeyDerived}_P), (U_m, V_E \mid \text{KeyDerived}_P)) \leq \varepsilon_{\text{priv}}. \quad (10.1)$$

3. *Authenticity.*

$$\Pr[(\text{KeyConfirmed}_A \vee \text{KeyConfirmed}_B) \wedge R_A \neq R_B] \leq \varepsilon_{\text{auth}}. \quad (10.2)$$

The *entropy loss* is $k - m$. Communication complexity counts honest public bits, and round complexity counts alternating honest-message rounds. When both errors are at most ε , we abbreviate the two-error notation to an (n, k, m, ε) protocol.

The asymmetric state convention is useful rather than cosmetic. A party can know that its own key is private before it knows that the other party obtained the same key. Authenticity is therefore an agreement-on-confirmation property, not a claim that every active execution must terminate without an abort.

Definition 10.4 (Information-theoretic one-time message authentication). For message length r , key length ℓ , and tag length τ , let

$$\{\text{MAC}_K : \{0, 1\}^r \rightarrow \{0, 1\}^\tau\}_{K \in \{0, 1\}^\ell}$$

be a deterministic family. It is ε_{MAC} -*secure for one-time authentication* if, for every fixed message $M \in \{0, 1\}^r$ and every (possibly randomized) adversary that, after receiving the single tag $\text{MAC}_K(M)$ for uniform K , outputs (M', σ') , one has

$$\Pr[M' \neq M \wedge \sigma' = \text{MAC}_K(M')] \leq \varepsilon_{\text{MAC}}. \quad (10.3)$$

The adversary is computationally unbounded. Side information independent of K may be fixed before the experiment and then averaged over afterwards.

This is a purely information-theoretic primitive. No computational MAC, one-way function, or pseudorandom-function assumption is being introduced in Volume VIII.

Lemma 10.5 (Finite-transcript min-entropy leakage). *Let X, Z be finite random variables, suppose $H_\infty(X) \geq k$, and assume $|\text{supp } Z| \leq 2^r$. For every $0 < \eta \leq 1$,*

$$\Pr_{z \leftarrow Z}[H_\infty(X \mid Z = z) \geq k - r - \log_2(1/\eta)] \geq 1 - \eta. \quad (10.4)$$

Only values z of positive probability are used in the conditional min-entropy.

Proof. For every z with $\Pr[Z = z] > 0$ and every x ,

$$\Pr[X = x \mid Z = z] = \frac{\Pr[X = x, Z = z]}{\Pr[Z = z]} \leq \frac{2^{-k}}{\Pr[Z = z]}. \quad (10.5)$$

Call z bad if the conditional min-entropy in (10.4) is smaller than the displayed threshold. Then some x has conditional probability greater than $2^{-k+r}\eta$, and (10.5) forces

$$\Pr[Z = z] < \eta 2^{-r}.$$

There are at most 2^r possible values of z , so the total probability of all bad values is strictly below η . This proves (10.4). $\square$

Why the logarithmic failure term is present. A transcript with r bits can cost r bits of min-entropy for most conditioned transcripts, but not literally for every transcript: a very rare transcript can identify the source almost completely. The extra $\log_2(1/\eta)$ buys a $1 - \eta$ high-probability statement and will remain a separate privacy-error term below.

Theorem 10.6 (Polynomial one-time MAC over a binary extension field). *Let $r \geq 1$ and $0 < \varepsilon_{\text{MAC}} < 1$. Choose an integer*

$$\tau \geq \left\lceil \log_2 \left(\frac{r}{\varepsilon_{\text{MAC}}} \right) \right\rceil \quad (10.6)$$

and put $q = 2^\tau$. There is an explicit ε_{MAC} -secure one-time MAC for r -bit messages with tag length τ and key length 2τ .

Proof. By the finite-field theorem ‘QRCC-III-THM-001’, fix a representation of $\mathbb{F}_q$. Put $c = \lceil r/\tau \rceil$. Pad an r -bit message M by zeros to $c\tau$ bits, split it into field elements $M_0, \dots, M_{c-1} \in \mathbb{F}_q$, and form

$$P_M(T) = \sum_{i=0}^{c-1} M_i T^i. \quad (10.7)$$

For a uniform key $(A, B) \in \mathbb{F}_q^2$, define

$$\text{MAC}_{A,B}(M) = AP_M(A) + B \in \mathbb{F}_q. \quad (10.8)$$

The fixed binary encoding of $\mathbb{F}_q$ identifies the key and tag with 2τ and τ bits, respectively.

Fix the authenticated message M . For every $a \in \mathbb{F}_q$, the uniform choice of B makes (10.8) uniform on $\mathbb{F}_q$. Hence, even after the observed tag is fixed to a value t , the conditional distribution of A remains uniform and

$$B = t - AP_M(A). \quad (10.9)$$

Fix also the adversary’s randomness and its response (M', t') to t , where $M' \neq M$. A successful forgery is equivalent, using (10.9), to

$$A(P_{M'}(A) - P_M(A)) = t' - t. \quad (10.10)$$

Because the padded coefficient vectors of distinct r -bit messages are distinct, $P_{M'} - P_M$ is a nonzero polynomial of degree at most $c - 1$. Thus the polynomial obtained by moving the right side of (10.10) to the left is nonzero and has degree at most c . It has at most c roots in $\mathbb{F}_q$. Conditional on every observed tag and every adversarial coin fixing, the forgery probability is therefore at most

$$\frac{c}{q} \leq \frac{r}{2^\tau} \leq \varepsilon_{\text{MAC}}, \quad (10.11)$$

where $c \leq r$ because $r, \tau \geq 1$. Averaging proves (10.3). Field arithmetic and polynomial evaluation are effective, so the family is explicit. $\square$

A passive baseline. Let $\text{Ext} : \{0, 1\}^n \times \{0, 1\}^d \rightarrow \{0, 1\}^m$ be a strong $(k, \varepsilon_{\text{Ext}})$ extractor. Against a passive eavesdropper Alice samples $S \sim U_d$, sends S , and both parties output $\text{Ext}(W, S)$. Exact agreement is immediate and

$$\Delta((\text{Ext}(W, S), S), (U_m, S)) \leq \varepsilon_{\text{Ext}}. \quad (10.12)$$

Thus the communication is d and the entropy loss is $k - m$. In particular, Theorem 3.1 gives the concrete leftover-hash guarantee

$$\varepsilon_{\text{Ext}} \leq \frac{1}{2} \sqrt{2^{m-k}}, \quad (10.13)$$

so $m \leq k - 2\log_2(1/\varepsilon)$ gives error at most $\varepsilon/2$. Nothing in this passive argument prevents Eve from replacing S by a related seed.

Theorem 10.7 (Two-round active privacy amplification from non-malleable extraction). *Let $0 < \eta \leq 1$, and suppose the following three information-theoretic objects are available:*

1. a $(k, \varepsilon_{\text{nm}})$ non-malleable extractor

$$\text{nmExt} : \{0, 1\}^n \times \{0, 1\}^{d_1} \rightarrow \{0, 1\}^\ell, \quad d_1 \geq 1,$$

in the fixed-point-free sense of Definition 9.6;

2. a strong $(k_\star, \varepsilon_{\text{Ext}})$ extractor

$$\text{Ext} : \{0, 1\}^n \times \{0, 1\}^{d_2} \rightarrow \{0, 1\}^m, \quad k_\star := k - d_1 - \ell - \log_2(1/\eta) \geq 0; \quad (10.14)$$

3. an ε_{MAC} -secure one-time MAC for d_2 -bit messages with ℓ -bit keys and τ -bit tags, in the sense of Definition 10.4.

Then the following two-round protocol is an

$$(n, k, m; \eta + \varepsilon_{\text{Ext}}, 2\varepsilon_{\text{nm}} + \varepsilon_{\text{MAC}}) \quad (10.15)$$

privacy-amplification protocol. Its honest communication is $d_1 + d_2 + \tau$ bits and its entropy loss is $k - m$.

Protocol. *Alice samples $Y \sim U_{d_1}$, sends Y , and computes $K = \text{nmExt}(W, Y)$. If Bob receives Y' , he computes $K' = \text{nmExt}(W, Y')$, samples an independent $S' \sim U_{d_2}$, sends*

$$(S', \sigma') = (S', \text{MAC}_{K'}(S')), \quad (10.16)$$

enters KeyDerived, and outputs $R_B = \text{Ext}(W, S')$. If Alice receives (S, σ) and $\sigma = \text{MAC}_K(S)$, she enters KeyConfirmed and outputs $R_A = \text{Ext}(W, S)$; otherwise she outputs $\perp$.

Proof. It is enough to analyze deterministic Eve: a randomized strategy is a convex combination obtained by fixing Eve's private coins, which are included in her view.

Correctness. For a passive Eve, $Y' = Y$ and $(S, \sigma) = (S', \sigma')$. Hence $K' = K$, the verification equation holds, Bob reaches KeyDerived, Alice reaches KeyConfirmed, and $R_A = \text{Ext}(W, S') = R_B$.

Privacy. For fixed Eve, Y' is a deterministic function of Y , while her second message to Alice is a deterministic function of what she has seen. Thus it suffices to retain

$$V_E = (Y, S', \sigma') \quad (10.17)$$

in the public view; all further adversarially generated values are deterministic functions of this tuple. Give Eve the additional variable K' and set $Z = (Y, K')$. Its support has size at most $2^{d_1+\ell}$. Lemma 10.5 shows that, except on a set of Z -values of probability at most η ,

$$H_\infty(W \mid Z = z) \geq k_\star. \quad (10.18)$$

Bob's S' is uniform and independent of (W, Z) . On every good value z , the strong-extractor property therefore gives

$$(\text{Ext}(W, S'), S' \mid Z = z) \approx_{\varepsilon_{\text{Ext}}} (U_m, S' \mid Z = z). \quad (10.19)$$

Averaging over good values and charging the entire bad mass gives error at most $\eta + \varepsilon_{\text{Ext}}$. Finally, $\sigma' = \text{MAC}_{K'}(S')$ is a deterministic function of (Z, S') , so data processing allows it to be appended

without increasing statistical distance. Bob is the only party that enters KeyDerived in this protocol, proving (10.1) with the first error in (10.15).

Authenticity. Let F be Eve's deterministic first-message tampering map, so $Y' = F(Y)$. If Alice confirms while $S = S'$, then automatically $R_A = R_B$. It remains to bound acceptance of a forged message $S \neq S'$. Partition according to

$$B_0 = \{F(Y) = Y\}, \quad B_1 = \{F(Y) \neq Y\}. \quad (10.20)$$

On B_0 , $K' = K$. By (9.4), non-malleable extraction implies the strong marginal bound

$$(K, Y) \approx_{\varepsilon_{\text{nm}}} (U_\ell, Y). \quad (10.21)$$

Append the independent seed S' , compute the authentic tag on S' , and run Eve's deterministic second-message strategy. Statistical distance remains at most ε_{nm} . In the corresponding ideal experiment the MAC key is uniform; after the one legitimate tag on S' , Definition 10.4 bounds a valid tag on any different S by ε_{MAC} . Consequently

$$\Pr[\text{forge} \wedge B_0] \leq \varepsilon_{\text{nm}} + \Pr[B_0]\varepsilon_{\text{MAC}}. \quad (10.22)$$

For B_1 , define a globally fixed-point-free map F^* by

$$F^*(y) = \begin{cases} F(y), & F(y) \neq y, \\ y \oplus e_1, & F(y) = y, \end{cases} \quad (10.23)$$

where $e_1 \neq 0$ exists because $d_1 \geq 1$. Put $K^* = \text{nmExt}(W, F^*(Y))$. Definition 9.6 gives

$$(K, K^*, Y) \approx_{\varepsilon_{\text{nm}}} (U_\ell, K^*, Y). \quad (10.24)$$

On B_1 we have $K^* = K'$. Append the independent S' , its tag under K' , and Eve's deterministic second-message strategy. In the ideal experiment the target key U_ℓ is uniform independently of the complete side information. Conditional on that side information, one may even grant Eve an unused legitimate tag under U_ℓ on S' ; an adversary that ignores this extra tag is covered by the one-time MAC experiment. Hence

$$\Pr[\text{forge} \wedge B_1] \leq \varepsilon_{\text{nm}} + \Pr[B_1]\varepsilon_{\text{MAC}}. \quad (10.25)$$

Adding (10.22) and (10.25), and using $\Pr[B_0] + \Pr[B_1] = 1$, gives

$$\Pr[\text{KeyConfirmed}_A \wedge R_A \neq R_B] \leq 2\varepsilon_{\text{nm}} + \varepsilon_{\text{MAC}}. \quad (10.26)$$

Only Alice can enter KeyConfirmed, so this is exactly (10.2). The communication and entropy-loss formulas are immediate from the protocol. $\square$

Source comparison and the fixed-point issue. The Cohen–Raz–Segev presentation of the Dodis–Wichs compiler states the sharper authenticity term $\varepsilon_{\text{nm}} + \varepsilon_{\text{MAC}}$ and communication $d_1 + d_2 + \tau$. Its proof separates the events $Y' = Y$ and $Y' \neq Y$. Definition 9.6, however, deliberately freezes the baseline non-malleability game to *globally* fixed-point-free tampering maps. Conditioning an arbitrary F on $F(Y) \neq Y$ does not leave Y uniform, so this book does not silently apply the definition after that conditioning. The global replacement F^* in (10.23) is valid without changing the frozen definition, at the cost of charging the non-malleability error once on each branch. Formula (10.15) is therefore the locally proved conservative compiler; the sharper source bound is retained as a source statement, not asserted as a consequence of a stronger hidden definition.

Instantiation with the completed Volume-VIII spine. Theorem 9.7 supplies an unconditional non-malleable extractor above entropy rate $1/2 + \delta$ with source-owned output length $\ell_0 = \Omega_\delta(d_1)$. Truncating its output to any $\ell \leq \ell_0$ preserves non-malleability by data processing. Whenever $2\tau \leq \ell_0$, Theorem 10.6 therefore supplies the required $\ell = 2\tau$ information-theoretic authentication key after truncation. The strong-extractor slot can already be filled, without any additional cryptographic assumption, by the 2-universal leftover-hash interface of Theorem 3.1. More communication-efficient strong extractors improve the protocol parameters but are not needed for the logical compiler proved here.

What is now separated. The active protocol exposes four costs rather than compressing them into one opaque “security parameter”: transcript conditioning costs η in privacy; the final strong extractor contributes ε_{Ext} ; non-malleable extraction contributes at most $2\varepsilon_{\text{nm}}$ in the locally frozen fixed-point-free interface; and information-theoretic authentication contributes ε_{MAC} . Entropy loss is exactly $k - m$ and communication is exactly $d_1 + d_2 + \tau$ before any asymptotic instantiation.

Internalization and Reconstruction Guide: Volume VIII

Purpose. Separate source entropy from algorithmic randomness and learn the extractor design pattern: identify a weak-source model, quantify distance, spend a seed or additional source, and track robustness under leakage or tampering.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-III-THM-001: Classification of finite fields.
2. QRCC-I-THM-004: Chernoff bounds.
3. QRCC-I-THM-015: Fourier inversion on finite abelian groups.
4. QRCC-III-THM-012: Johnson bound.
5. QRCC-I-THM-001: Union bound.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-VIII-DEF-001 (Min-entropy and statistical extraction); QRCC-VIII-DEF-002 (Condensers, dispersers, and lossless condensers); QRCC-VIII-DEF-003 (Averaging samplers); QRCC-VIII-DEF-036 (Two-source extractor and collision entropy); QRCC-VIII-DEF-022 (Non-malleable extractor).

Chapter retrieval map

- Ch. 1: Weak Sources and Limited Independence:** enter through QRCC-VIII-DEF-001 (Min-entropy and statistical extraction); exit by being able to state QRCC-VIII-ALG-005 (Pairwise-independent affine hash family) with its ownership boundary.
- Ch. 2: Small Bias and a Coding-Theoretic Dictionary:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VIII-THM-007 (Small-bias spaces and linear-code duality) with its ownership boundary.

- Ch. 3: Universal Hashing and the Leftover Hash Lemma:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VIII-THM-008 (Strong leftover hash lemma) with its ownership boundary.
- Ch. 4: Why a Seed Is Necessary:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VIII-THM-004 (Impossibility of deterministic extraction from one arbitrary weak source) with its ownership boundary.
- Ch. 5: Extractor Graphs and the Probabilistic Method:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-VIII-ALG-013 (Weak combinatorial designs for reconstruction) with its ownership boundary.
- Ch. 6: Condensers, Lossless Expansion, and Entropy Compression:** enter through QRCC-VIII-DEF-002 (Condensers, dispersers, and lossless condensers); exit by being able to state QRCC-VIII-PROP-034 (Lossless condenser–near-lossless expansion equivalence) with its ownership boundary.
- Ch. 7: Averaging Samplers and the Extractor Dictionary:** enter through QRCC-VIII-DEF-003 (Averaging samplers); exit by being able to state QRCC-VIII-THM-012 (Extractor-to-sampler conversion) with its ownership boundary.
- Ch. 8: Explicit Small Bias and Two Independent Sources:** enter through QRCC-VIII-DEF-036 (Two-source extractor and collision entropy); exit by being able to state QRCC-VIII-BB-037 (Naor–Naor explicit small-bias generator) with its ownership boundary.
- Ch. 9: Affine, Leakage-Resilient, and Non-Malleable Extraction:** enter through QRCC-VIII-DEF-022 (Non-malleable extractor); exit by being able to state QRCC-VIII-BB-021 (Bourgain binary affine-source extractor) with its ownership boundary.
- Ch. 10: Low-Complexity Extraction and Privacy Amplification:** enter through QRCC-VIII-DEF-041 (Privacy amplification with an active public channel); exit by being able to state QRCC-VIII-LEM-043 (Finite-transcript min-entropy leakage) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-VIII-ALG-005 — Pairwise-independent affine hash family. Dependencies: QRCC-III-THM-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-VIII-THM-008 — Strong leftover hash lemma. Dependencies: QRCC-VIII-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-VIII-ALG-013 — Weak combinatorial designs for reconstruction. Dependencies: no earlier formal dependency. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

4. QRCC-VIII-THM-012 — Extractor-to-sampler conversion. Dependencies: QRCC-VIII-DEF-003, QRCC-VIII-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-VIII-BB-021 — Bourgain binary affine-source extractor. Dependencies: QRCC-VIII-DEF-045. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-VIII-F-026 — Nearly-linear-time seeded extractors with short seeds. Dependencies: QRCC-VIII-THM-014. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- using Shannon entropy where min-entropy is required;
- forgetting whether an extractor is strong in the seed or in a source;
- omitting side information from statistical-distance statements;
- confusing probabilistic existence with an explicit efficiently computable construction;

Boundary and degenerate cases

- entropy $k=0$ and $k=n$ endpoints;
- seed length zero and deterministic-extraction impossibility;
- independent versus correlated multi-source inputs;
- passive leakage versus active tampering/non-malleability;

Retrieval and reconstruction drill

1. State QRCC-VIII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-VIII-ALG-005. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-VIII-THM-008, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.

7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-VIII-F-026, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume IX asks how pseudorandom objects eliminate random bits from algorithms themselves.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part IX

Randomized Complexity and Derandomization

Chapter 1

Randomized Classes and the Finite Minimax Principle

Randomness enters complexity theory through two logically different objects. A *uniform randomized algorithm* is one fixed machine that receives fresh coin flips on every input; a *nonuniform* simulation may instead freeze a different auxiliary object at every input length. Later derandomization arguments move between these viewpoints, so the machine, promise, error, and random-string conventions are fixed before any theorem is used.

1.1 Language, promise, and randomness conventions

Definition 1.1 (Randomized complexity classes). A polynomial-time randomized decision machine is a deterministic polynomial-time predicate

$$M(x, r) \in \{0, 1\}, \quad r \in \{0, 1\}^{r(n)}, \quad n = |x|,$$

where $r(n)$ is polynomially bounded and r is uniform. By padding unused coin flips we may, and henceforth do, require an exact length $r(n) \geq 1$ for every input length. The machine is *uniform*: the same finite machine and the same polynomial bounds apply at all lengths.

For a language $L \subseteq \{0, 1\}^*$:

1. $L \in BPP$ if some such M satisfies

$$x \in L \implies \Pr_r[M(x, r) = 1] \geq \frac{2}{3}, \quad x \notin L \implies \Pr_r[M(x, r) = 1] \leq \frac{1}{3}.$$

This is *two-sided bounded error*.

2. $L \in RP$ if

$$x \in L \implies \Pr_r[M(x, r) = 1] \geq \frac{1}{2}, \quad x \notin L \implies \Pr_r[M(x, r) = 1] = 0.$$

Equivalently, false acceptance is impossible and only yes-instances may suffer error. Put $coRP := \{L : \bar{L} \in RP\}$.

3. $L \in ZPP$ if there is a randomized machine that halts almost surely, always returns the correct membership bit, and has expected running time bounded by a polynomial in $|x|$. Such a Las Vegas machine may read fresh fair coins on line; no a priori finite random-string length is imposed on an individual run.

A *promise problem* is a pair $\Pi = (\Pi_{\text{yes}}, \Pi_{\text{no}})$ of disjoint sets. It belongs to *Promise-BPP* if one uniform polynomial-time randomized machine obeys the same 2/3 versus 1/3 inequalities on the two promised sets. Its behavior outside $\Pi_{\text{yes}} \cup \Pi_{\text{no}}$ is unrestricted. Thus a language class is not silently identified with its promise analogue.

More generally, an error bound $\varepsilon(n)$ means that each promised input of length n is answered incorrectly with probability at most $\varepsilon(n)$. One-sided error means that one side has error exactly zero; two-sided error permits error on both sides.

A *nonuniform* algorithm may receive an advice string a_n depending only on the input length. Polynomial-length deterministic advice is the *P/poly* model recorded in QRCC-II-DEF-008 and QRCC-II-THM-013. Advice depending on the individual input, or on the random coins drawn during that individual run, is not length advice.

Positive example. If a deterministic polynomial-time decider ignores one padded random bit, it is simultaneously a BPP, RP, and coRP machine under the preceding convention. The padding changes neither its language nor its running time by more than a constant.

Nonexample. A promise-BPP machine that is correct only on two disjoint promised sets does not by itself define a BPP language: its behavior off the promise may violate both bounded-error inequalities.

1.2 A finite cost matrix and Yao's principle

For the theorem below, no asymptotic machine model is needed. The point is to freeze the direction of the two optimizations once and for all. An input distribution is chosen to make every deterministic algorithm expensive; a randomized algorithm is chosen to make every individual input inexpensive.

Theorem 1.2 (Finite Yao minimax principle). *Let $\mathcal{X}$ and $\mathcal{A}$ be nonempty finite sets and let*

$$c : \mathcal{A} \times \mathcal{X} \longrightarrow [0, C]$$

be a cost function. Write $\Delta(S)$ for the probability simplex on a finite set S . Then

$$\min_{\rho \in \Delta(\mathcal{A})} \max_{x \in \mathcal{X}} \mathbb{E}_{a \sim \rho} c(a, x) = \max_{\mu \in \Delta(\mathcal{X})} \min_{a \in \mathcal{A}} \mathbb{E}_{x \sim \mu} c(a, x). \quad (1.1)$$

Consequently:

1. *if some distribution μ makes every deterministic a have expected cost at least L , then every randomized algorithm ρ has some input x on which its expected cost is at least L ;*
2. *if for every input distribution μ there is a deterministic a with expected cost at most U , then one randomized algorithm ρ has expected cost at most U on every input.*

The finiteness of both pure strategy sets is part of the statement.

Proof. Index $\mathcal{A} = \{a_1, \dots, a_s\}$ and $\mathcal{X} = \{x_1, \dots, x_t\}$ and put $C_{ij} = c(a_i, x_j)$. The left side of (1.1) is the optimal value of the finite linear program

$$\begin{aligned} & \text{minimize} && z \\ & \text{subject to} && \sum_{i=1}^s \rho_i C_{ij} \leq z \quad (1 \leq j \leq t), \\ & && \sum_{i=1}^s \rho_i = 1, \quad \rho_i \geq 0. \end{aligned} \quad (1.2)$$

Indeed, for a fixed distribution ρ , the least admissible z is exactly the largest expected cost over pure inputs.

The finite linear-programming dual can be written

$$\begin{aligned} & \text{maximize} && v \\ & \text{subject to} && \sum_{j=1}^t \mu_j C_{ij} \geq v \quad (1 \leq i \leq s), \\ & && \sum_{j=1}^t \mu_j = 1, \quad \mu_j \geq 0. \end{aligned} \tag{1.3}$$

For a fixed μ , the largest admissible v is $\min_i \sum_j \mu_j C_{ij}$, which is the right side of (1.1). Both programs are feasible and bounded because $0 \leq C_{ij} \leq C$. Finite-dimensional strong linear-programming duality therefore gives equality of (1.2) and (1.3), proving (1.1).

For the first consequence, the right side of (1.1) is at least L , hence so is the left side; therefore every ρ has $\max_x \mathbb{E}_{a \sim \rho} c(a, x) \geq L$. For the second, the right side is at most U , so an optimizing ρ on the left has every input cost at most U . $\square$

Worked 2×2 check. Let $c(a_1, x_1) = c(a_2, x_2) = 0$ and $c(a_1, x_2) = c(a_2, x_1) = 1$. Mixing the two algorithms equally makes each input cost $1/2$, so the left side of (1.1) is at most $1/2$. The uniform input distribution makes each deterministic algorithm cost $1/2$, so the right side is at least $1/2$. Both values are therefore exactly $1/2$.

What the theorem does not say. Equation (1.1) is a finite zero-sum cost identity. When it is instantiated with decision trees, protocols, or bounded resource classes, the admissible pure strategies and the payoff must encode the desired correctness convention. The slogan “hard distribution implies randomized lower bound” is valid only after those sets and costs have been fixed.

Chapter 2

Error Reduction by Independent Repetition

The constants $2/3$, $1/3$, and $1/2$ in the randomized-class definitions are conventional. What matters is a constant gap from the decision threshold and independent fresh randomness. This chapter records the exact amplification cost needed by the nonuniform and polynomial-hierarchy arguments that follow.

Theorem 2.1 (Error reduction by independent repetition). *Let M be a randomized polynomial-time machine using exactly $r(n)$ random bits on length- n inputs.*

1. **BPP.** *If M has two-sided error at most $1/3$, then for every $0 < \delta < 1$ and every odd integer*

$$t \geq 18 \ln(1/\delta), \quad (2.1)$$

the majority of t independent runs has error at most δ . It uses exactly $tr(n)$ random bits and at most t times the running time of M .

2. **RP.** *If M has no false accepts and accepts every yes-instance with probability at least $1/2$, then the OR of t independent runs has no false accepts and yes-instance error at most 2^{-t} .*
3. **coRP.** *In the equivalent convention in which yes-instances are accepted surely and no-instances are accepted with probability at most $1/2$, the AND of t independent runs keeps yes-instances error-free and has no-instance error at most 2^{-t} .*

In the one-sided clauses it therefore suffices to take $t \geq \lceil \log_2(1/\delta) \rceil$.

Proof. For the BPP clause, fix an input and let X_i be the indicator that the i th independent run is wrong. Then the X_i are independent, $0 \leq X_i \leq 1$, and

$$\mathbb{E} \sum_{i=1}^t X_i \leq \frac{t}{3}.$$

If the majority is wrong, at least $t/2$ runs are wrong, hence

$$\sum_{i=1}^t (X_i - \mathbb{E} X_i) \geq \frac{t}{6}.$$

The Hoeffding inequality from Theorem 2.5 gives

$$\Pr[\text{majority wrong}] \leq \exp\left(-\frac{2(t/6)^2}{t}\right) = e^{-t/18} \leq \delta, \quad (2.2)$$

where the final inequality is (2.1). Disjoint $r(n)$ -bit seed blocks make the runs independent and give the stated randomness and time costs.

For RP, a no-instance is rejected on every run, while a yes-instance is rejected on all t runs with probability at most $(1/2)^t$. Taking OR gives the claim. The coRP statement is the complementary calculation: yes-instances are accepted on every run, while a no-instance survives the AND with probability at most $(1/2)^t$. $\square$

Randomness accounting. Amplification here does not recycle random bits: the seed is the concatenation of t independent seeds. Later space-bounded pseudorandom generators are interesting precisely because they try to simulate many random choices from a much shorter seed.

Worked numerical check. At $t = 181$, (2.2) is at most $e^{-181/18} < 4.4 \times 10^{-5}$. An RP machine repeated 20 times has one-sided error at most $2^{-20} < 10^{-6}$.

Chapter 3

Fixing Randomness and the Polynomial Hierarchy

Amplification makes a single input very unlikely to be answered incorrectly. Two different counting arguments then exploit that small error. Adleman's argument takes a union bound over all inputs of one length and freezes one globally good random string, producing nonuniformity. The Sipser–Gács–Lautemann argument instead covers the random-string cube by a polynomial number of translates and converts the size gap into two alternating quantifiers.

3.1 Adleman's fixing argument

Theorem 3.1 (Adleman's theorem: $BPP \subseteq P/poly$). *Every language in BPP has polynomial-size circuits. Equivalently,*

$$BPP \subseteq P/poly. \quad (3.1)$$

More explicitly, if the original BPP decider uses $r(n)$ random bits and polynomial time, then after $O(n)$ independent repetitions there is, for every input length n , one polynomial-length seed a_n on which the amplified decider is correct simultaneously for all 2^n inputs of that length.

Proof. Let $L \in BPP$ and let M be a decider in the convention of Definition 1.1. Apply Theorem 2.1 with

$$\delta_n = 2^{-n-2}.$$

Choosing the least admissible odd repetition count gives $t(n) = O(n+1)$, so the amplified machine M' still runs in polynomial time and uses a polynomial number $m(n) = t(n)r(n)$ of random bits.

For a fixed length- n input x , let

$$B_x := \{a \in \{0,1\}^{m(n)} : M'(x,a) \neq \mathbf{1}_L(x)\}.$$

A uniform seed belongs to B_x with probability at most 2^{-n-2} . Hence, by the union bound, Theorem 1.2,

$$\Pr_a[\exists x \in \{0,1\}^n : a \in B_x] \leq 2^n 2^{-n-2} = \frac{1}{4} < 1. \quad (3.2)$$

Therefore some seed a_n lies in none of the B_x . Fix one such seed for each length. The deterministic advice machine

$$D(x, a_n) := M'(x, a_n)$$

runs in polynomial time, has polynomial advice length $m(n)$, and is correct on every length- n input. The circuit/advice characterization QRCC-II-THM-013 now gives $L \in P/poly$. $\square$

Historical normalization. Adleman's 1978 Theorem I is stated for his one-sided random-polynomial-time class R and proves polynomial-size circuits by a nonconstructive fixing argument. The two-sided BPP statement (3.1) is proved here locally: error reduction first supplies the exponentially small per-input error needed for the same finite-length union-bound principle. The original theorem is not being quoted as though it literally used the modern BPP convention.

Proposition 3.2 (Finite-length randomness-fixing checkpoint). *Suppose an amplified randomized decider on four-bit inputs has error at most $2^{-6} = 1/64$ on each individual input. Then at least three quarters of its random strings are correct simultaneously on all 16 four-bit inputs. In particular, a globally correct length-four advice string exists.*

Proof. For each of the 16 inputs, at most a $1/64$ fraction of seeds are bad. The union of all bad-seed sets therefore has density at most

$$16 \cdot \frac{1}{64} = \frac{1}{4}.$$

Thus at least a $3/4$ fraction of seeds avoid every bad set simultaneously. $\square$

What fixing does not provide. The proof of Theorem 3.1 establishes existence of a_n but gives no uniform polynomial-time procedure for finding it. That is exactly why the conclusion is P/poly rather than P .

3.2 Translation covering and two alternations

Theorem 3.3 (Sipser–Gács–Lautemann containment).

$$BPP \subseteq \Sigma_2^P \cap \Pi_2^P. \quad (3.3)$$

More precisely, for every $L \in BPP$ there are polynomially bounded functions $m(n), k(n)$ and a polynomial-time predicate R such that

$$x \in L \iff \exists t_1, \dots, t_{k(n)} \in \{0, 1\}^{m(n)} \forall z \in \{0, 1\}^{m(n)} R(x, t_1, \dots, t_{k(n)}, z) = 1. \quad (3.4)$$

Proof. Fix $L \in BPP$ and a decider M using exactly $r_0(n) \geq 1$ random bits on length- n inputs. Put

$$q(n) := n + r_0(n) + 10. \quad (3.5)$$

Use the explicit majority construction from Theorem 2.1 with the least odd integer

$$t(n) \geq 18q(n) \ln 2.$$

The resulting machine D has error

$$\varepsilon(n) \leq 2^{-q(n)} \quad (3.6)$$

and uses

$$m(n) = t(n)r_0(n) \quad (3.7)$$

random bits. Since $q(n) \geq 12$, $t(n) < 18q(n) \ln 2 + 2 < 15q(n)$, so

$$m(n) < 15q(n)r_0(n). \quad (3.8)$$

For the remainder of the proof fix $x \in \{0, 1\}^n$ and set

$$S_x := \{z \in \{0, 1\}^{m(n)} : D(x, z) = 1\}, \quad N := 2^{m(n)}.$$

Thus

$$x \in L \implies |S_x| \geq (1 - \varepsilon)N, \quad x \notin L \implies |S_x| \leq \varepsilon N. \quad (3.9)$$

Write $S \oplus u := \{s \oplus u : s \in S\}$ and choose

$$k(n) := \left\lfloor \frac{m(n)}{q(n)} \right\rfloor + 1. \quad (3.10)$$

Then $qk > m$ and, by (3.8),

$$k < 15r_0(n) + 1. \quad (3.11)$$

We need two elementary counting facts.

Dense sets admit a polynomial translation cover. Assume $|S_x| \geq (1 - \varepsilon)N$ and choose $T_1, \dots, T_k$ independently and uniformly from $\{0, 1\}^m$. For a fixed z , translation invariance gives

$$\Pr[z \notin \bigcup_{i=1}^k (S_x \oplus T_i)] = \left(1 - \frac{|S_x|}{N}\right)^k \leq \varepsilon^k.$$

A union bound over all N points gives

$$\Pr[\text{some } z \text{ is uncovered}] \leq N\varepsilon^k \leq 2^{m-qk} < 1. \quad (3.12)$$

Hence some fixed shifts $t_1, \dots, t_k$ cover the entire cube.

Sparse sets cannot admit such a cover. If $|S_x| \leq \varepsilon N$, then for every fixed shifts $t_1, \dots, t_k$,

$$\left| \bigcup_{i=1}^k (S_x \oplus t_i) \right| \leq k\varepsilon N. \quad (3.13)$$

By $r_0 \geq 1$, $r_0 \leq 2^{r_0}$; hence (3.5), (3.6), and (3.11) imply

$$k\varepsilon < (15r_0 + 1)2^{-(n+r_0+10)} \leq 16 \cdot 2^{r_0} \cdot 2^{-(n+r_0+10)} = 2^{-n-6} < 1. \quad (3.14)$$

Thus the union in (3.13) is a strict subset of the cube.

Combining the two facts,

$$x \in L \iff \exists t_1, \dots, t_k \forall z \bigvee_{i=1}^k D(x, z \oplus t_i) = 1. \quad (3.15)$$

The disjunction in (3.15) is part of one deterministic polynomial-time predicate: $k(n)$ and $m(n)$ are polynomially bounded, so no third quantified block is introduced by the index i . Equation (3.15) is therefore the Σ_2^P form (3.4) by QRCC-II-THM-011. Hence $L \in \Sigma_2^P$.

Finally BPP is closed under complement by flipping the output bit in Definition 1.1. Applying the just-proved Σ_2^P containment to $\bar{L}$ yields $L \in \Pi_2^P$. This proves (3.3). $\square$

Quantifier checkpoint. The expression

$$\exists(t_1, \dots, t_k) \forall z \bigvee_{i=1}^k D(x, z \oplus t_i)$$

has only two quantified blocks. The polynomial-size OR over i is evaluated inside the deterministic matrix predicate. Replacing that OR by an unnecessary existentially quantified index after z would obscure the Σ_2^P normalization.

Source boundary. The theorem is traditionally called the Sipser–Gács–Lautemann containment. The reader uses Lautemann’s translation/counting architecture but fixes its own exact amplification exponent so that both the covering and noncovering inequalities are strict for every length under the padded $r_0(n) \geq 1$ convention.

Chapter 4

Space-Bounded Tests and Pseudorandom Generators

Independent repetition spends a fresh random string on every trial. Space-bounded computation admits a different economy: a short seed can be expanded into a long coin sequence that no small-space computation can distinguish from truly random coins. The model boundary matters. A finite layered branching program is a convenient nonuniform picture, while the classical theorems of Nisan and of Nisan–Zuckerman are stated for uniform space-bounded machines with online access to random coins. We keep those two interfaces explicit.

Definition 4.1 (Layered branching programs and space-bounded tests). A *binary layered branching program of length R* is a tuple

$$B = (V_0, V_1, \dots, V_R, v_0, F, \delta_1, \dots, \delta_R), \quad (4.1)$$

where $v_0 \in V_0$, $F \subseteq V_R$, and

$$\delta_i : V_{i-1} \times \{0, 1\} \longrightarrow V_i \quad (1 \leq i \leq R).$$

Its width is $W = \max_i |V_i|$. For $y = y_1 \cdots y_R \in \{0, 1\}^R$ the program follows $v_i = \delta_i(v_{i-1}, y_i)$ and accepts iff $v_R \in F$. We write $B(y) \in \{0, 1\}$ for this acceptance indicator. A *space- S branching-program test* has width at most 2^S .

For a map $G : \{0, 1\}^s \rightarrow \{0, 1\}^R$, we say that G ε -fools B if

$$|\Pr[B(U_R) = 1] - \Pr[B(G(U_s)) = 1]| \leq \varepsilon. \quad (4.2)$$

For a class of tests, (4.2) is required for every member of the class.

The uniform machine version is obtained from a randomized machine whose random tape is read online, one fresh bit at a time. On a fixed external input, the machine configurations immediately before successive coin reads form layers; if at most S bits of mutable work configuration are present, there are at most $2^{O(S)}$ configurations per layer. Conversely, a family of layered programs whose transition map $(i, v, b) \mapsto \delta_i(v, b)$ is uniformly computable can be evaluated by storing the current vertex and the layer counter. Thus a width- 2^S program costs $S + O(\log R)$ bits under this uniform-access interpretation.

Uniformity checkpoint. An arbitrary nonuniform table of R transition functions need not itself be accessible to an S -space Turing machine. Consequently, “width 2^S ” and “uniform space S ” are not silently identified in what follows. The classical source theorems use the machine formulation; layered programs are the finite-state image of that model after an external input is fixed.

A width-four finite test. Let $R = 6$, let every layer have states $\{0, 1, 2, 3\}$, start in state 0, and set

$$\delta_i(a, b) = a + b \pmod{4}. \quad (4.3)$$

Accept only state 0 at the last layer. Thus $B(y) = 1$ exactly when the Hamming weight $|y|$ is divisible by 4. Under six uniform coins,

$$\Pr[B(U_6) = 1] = \frac{\binom{6}{0} + \binom{6}{4}}{2^6} = \frac{16}{64} = \frac{1}{4}. \quad (4.4)$$

This small test will be reused by the executable checkpoint below.

Theorem 4.2 (Nisan’s generator for space-bounded computation; source-verified). *There is a universal constant $C > 0$ such that, for $R \geq 2$ and $S \geq 1$, there is an explicitly given map*

$$G_{R,S} : \{0, 1\}^s \longrightarrow \{0, 1\}^R, \quad s \leq CS \log(R + 2), \quad (4.5)$$

with the following property. For every randomized space- S algorithm in the online-random-tape model and every fixed external input, replacing its R uniform coin tosses by $G_{R,S}(U_s)$ changes its acceptance probability by at most

$$\varepsilon = 2^{-S}. \quad (4.6)$$

The generator is computable in time polynomial in R and S and in $O(S \log R)$ work space. If the seed is instead placed on a separate read-only tape with two-way access, the generator can be supplied online and the simulation carried out in $O(S)$ working space.

In the nontrivial regime in which R dominates S , the primary theorem gives the sharper displayed seed bound $O(S \log(R/S))$. Formula (4.5) is the safe all-regime normalization: when $R < 2S$, using the original R random bits already costs $O(S)$ bits, while for $R \geq 2S$ the source bound implies $O(S \log R)$.

Source-owned proof boundary. Theorem 4.2 is imported from Nisan’s 1992 theorem rather than marked as a local proof. The construction first builds pseudorandom block generators for finite-state machines using universal hashing, then concatenates the blocks and converts them to the online space model. Its analysis is a matrix/hybrid argument across recursively recycled blocks. These ingredients explain the seed accounting and the 2^{-S} error, but this chapter does not pretend that the preceding paragraph is a replacement proof of the source construction.

Simulation consequence. If a randomized space- S computation uses R random bits, exhaustive choice of a uniform seed for (4.5) reduces the number of random bits to $O(S \log R)$. This is a randomness reduction, not by itself a deterministic simulation in space S : enumerating all seeds gives a deterministic simulation only when the acceptance-gap/error convention permits it and the seed can be streamed in the required space model.

Theorem 4.3 (Nisan–Zuckerman linear randomness in space; source-verified deep input). *For every constant $\nu > 0$ and every polynomial p , there is an explicitly given pseudorandom generator*

$$G : \{0, 1\}^{O(S)} \longrightarrow \{0, 1\}^{p(S)} \quad (4.7)$$

for space S with distinguishing parameter

$$2^{-S^{1-\nu}}, \quad (4.8)$$

computable in time $\text{poly}(S)$ and space $O(S)$.

Consequently, fix any constant $\gamma > 0$. If a randomized algorithm on length- n inputs runs in space $S = S(n) \geq \log n$, time $T = T(n) \geq n$, and uses only $\text{poly}(S)$ random bits, then it can be simulated with $O(S)$ random bits in space S and time $T + \text{poly}(S)$; the distribution of its output is within

$$\exp(-S^{1-\gamma}) \quad (4.9)$$

in statistical distance of the original output distribution.

Hypothesis checkpoint. The phrase “randomness is linear in space” has a crucial input hypothesis: the original computation uses only $\text{poly}(S)$ random bits. Theorem 4.3 does not say that an arbitrary R -coin space- S computation can always be compressed to $O(S)$ coins with the same parameters. The first clause is the source’s generator theorem; the second is its simulation theorem, including $S \geq \log n$ and $T \geq n$.

Relation to extractors. The Nisan–Zuckerman construction uses an extractor as its main recycling tool, but the theorem is kept as a source-owned deep input here. In particular, it is not made logically dependent on the much later GUV condenser or on the extractor-to-sampler theorem from Volume VIII. This preserves both historical and mathematical dependency direction.

Executable branching-program checkpoint. The script `code/volume_09/branching_prg_1ab.py` evaluates finite layered programs exactly by exhaustive enumeration. For the width-four test (4.3), it verifies the uniform value $1/4$ and compares it with the 3-seed-bit linear distribution

$$G(s)_i = \langle v_i, s \rangle \pmod{2}, \quad (v_1, \dots, v_6) = (100, 010, 001, 110, 101, 011). \quad (4.10)$$

The six output bits in (4.10) are pairwise independent, but the program accepts with probability $1/2$, so its exact fooling error on this test is $1/4$. This is intentionally a counterexample to the idea that low-wise independence alone implements Nisan’s theorem. The finite linear map is a pedagogical test distribution, not an implementation of either source-owned generator.

Chapter 5

Computational Pseudorandomness and the Next-Bit Test

The space-bounded generators of the preceding chapter fool a specified information-theoretic test class. Cryptographic pseudorandomness changes the quantifiers: the output is compared with a uniform string against every feasible distinguisher, and security is asymptotic in a security parameter. The model boundary is fixed before the hybrid argument. In this chapter the primary convention is *nonuniform polynomial-size Boolean circuits*; the uniform probabilistic-polynomial-time convention is recorded separately and is not silently identified with it.

5.1 Computational indistinguishability and generators

Definition 5.1 (Computational pseudorandomness and next-bit prediction). Let U_ℓ denote the uniform distribution on $\{0, 1\}^\ell$. A distribution ensemble $X = \{X_n\}_{n \geq 1}$ has polynomially bounded output length $m(n)$ if X_n is supported on $\{0, 1\}^{m(n)}$ and $m(n) \leq n^{O(1)}$.

For a Boolean circuit D on $m(n)$ input bits, define its distinguishing advantage by

$$\text{Adv}_{X,n}^{\text{dist}}(D) := \left| \Pr[D(X_n) = 1] - \Pr[D(U_{m(n)}) = 1] \right|. \quad (5.1)$$

At a fixed length n , X_n is (S, ε) -pseudorandom if (5.1) is at most ε for every circuit of size at most S . Asymptotically, X is *computationally pseudorandom* if for every polynomial-size circuit family $D = \{D_n\}$ and every polynomial q ,

$$\text{Adv}_{X,n}^{\text{dist}}(D_n) < \frac{1}{q(n)} \quad (5.2)$$

for all sufficiently large n . Thus the default convention in this chapter is nonuniform: D_n may contain advice depending on n , but not on the particular sampled string.

A *uniform pseudorandom generator* is one deterministic polynomial-time algorithm defining maps

$$G_n : \{0, 1\}^{s(n)} \longrightarrow \{0, 1\}^{m(n)}, \quad (5.3)$$

where $s(n), m(n) \leq n^{O(1)}$, $m(n) > s(n)$ for all sufficiently large n , and the ensemble $G_n(U_{s(n)})$ is computationally pseudorandom in the preceding circuit sense. The difference $m(n) - s(n)$ is the *stretch*. Efficient computability of G and indistinguishability of its output are separate requirements.

For a distribution X_n on $m = m(n)$ bits, a predictor for position $i \in \{1, \dots, m\}$ is a Boolean circuit $P : \{0, 1\}^{i-1} \rightarrow \{0, 1\}$. Its next-bit advantage is

$$\text{Adv}_{X,n,i}^{\text{pred}}(P) := \Pr[P(X_{n,1} \cdots X_{n,i-1}) = X_{n,i}] - \frac{1}{2}. \quad (5.4)$$

The ensemble passes the *nonuniform next-bit test* if for every polynomial S , every polynomial q , all sufficiently large n , every $i \leq m(n)$, and every predictor of size at most $S(n)$, the quantity in (5.4) is smaller than $1/q(n)$.

The *uniform* versions replace circuit families by one probabilistic polynomial-time algorithm receiving $(1^n, y)$, and for prediction also the position i . Unless explicitly stated otherwise, Theorem 5.2 uses the nonuniform circuit convention above.

Model checkpoint. Equation (5.1) is an acceptance-probability gap, not statistical distance of the underlying distributions. A computationally unbounded test can always detect the support deficit of a stretching deterministic generator by enumerating all seeds. Pseudorandomness asserts only that the chosen feasible test class cannot exploit that deficit with nonnegligible advantage.

Uniform versus nonuniform. Nonuniform circuit security implies security against uniform polynomial-time distinguishers. The converse is not built into the definition. The hybrid proof below works in either model provided that the model is closed under hardwiring polynomially many constants and a constant number of calls to the distinguisher; the reader theorem is stated quantitatively for circuits so that no hidden uniformity conversion is needed.

5.2 Yao's universal next-bit test

Theorem 5.2 (Yao next-bit characterization; quantitative circuit form). *Let X be any distribution on $\{0, 1\}^m$, with $m \geq 1$.*

1. *If a circuit $P : \{0, 1\}^{i-1} \rightarrow \{0, 1\}$ predicts X_i from $X_1 \cdots X_{i-1}$ with success probability $1/2 + \alpha$, then the test*

$$D_P(y) := \mathbf{1}\{P(y_1 \cdots y_{i-1}) = y_i\} \quad (5.5)$$

distinguishes X from U_m with advantage exactly α and has size $|P| + O(1)$.

2. *Conversely, if a Boolean circuit D of size S satisfies*

$$|\Pr[D(X) = 1] - \Pr[D(U_m) = 1]| = \delta, \quad (5.6)$$

then for some $i \in \{1, \dots, m\}$ there is a deterministic circuit predictor $P_i : \{0, 1\}^{i-1} \rightarrow \{0, 1\}$ of size at most $2S + O(1)$ whose success probability is at least

$$\frac{1}{2} + \frac{\delta}{m}. \quad (5.7)$$

Consequently, for every polynomially bounded length function $m(n)$, a distribution ensemble $X_n \in \{0, 1\}^{m(n)}$ is computationally pseudorandom against polynomial-size circuit families if and only if it passes the nonuniform next-bit test. In particular, the same equivalence applies to the output ensemble of a generator from Definition 5.1.

Proof. For the first implication, under a uniform input $Y \sim U_m$ the bit Y_i is independent of its prefix. Hence

$$\Pr[D_P(U_m) = 1] = \frac{1}{2}, \quad \Pr[D_P(X) = 1] = \frac{1}{2} + \alpha,$$

which gives distinguishing advantage α .

For the converse, choose $\sigma \in \{-1, +1\}$ so that

$$\sigma(\mathbb{E}D(X) - \mathbb{E}D(U_m)) = \delta. \quad (5.8)$$

For $0 \leq j \leq m$, define the hybrid

$$H_j = (X_1, \dots, X_j, U_{j+1}, \dots, U_m), \quad (5.9)$$

where the uniform suffix is independent of X . Thus $H_0 = U_m$ and $H_m = X$. Telescoping gives

$$\delta = \sum_{j=1}^m \sigma(\mathbb{E}D(H_j) - \mathbb{E}D(H_{j-1})). \quad (5.10)$$

Therefore some i satisfies

$$\sigma(\mathbb{E}D(H_i) - \mathbb{E}D(H_{i-1})) \geq \frac{\delta}{m}. \quad (5.11)$$

Put $Z = (X_1, \dots, X_{i-1})$, $B = X_i$, and let $R \sim U_{m-i}$ be independent. Write

$$d_b(Z, R) := D(Z, b, R), \quad b \in \{0, 1\}.$$

For the moment allow a predictor to use the fresh suffix R and one fair tie bit. When $\sigma = +1$, it predicts 1 if $(d_0, d_1) = (0, 1)$, predicts 0 if $(d_0, d_1) = (1, 0)$, and guesses uniformly on a tie. When $\sigma = -1$, reverse these two non-tie answers. A direct two-case calculation yields

$$\Pr[P(Z; R) = B] - \frac{1}{2} = \sigma(\mathbb{E}D(H_i) - \mathbb{E}D(H_{i-1})). \quad (5.12)$$

By (5.11), the average success of this randomized predictor is at least $1/2 + \delta/m$. Averaging over its finite internal random tape, there is one fixed choice of the suffix $R = r^*$ and of the tie bit whose deterministic predictor has at least the same success probability. Hardwire these constants. The resulting circuit uses two copies of D plus $O(1)$ comparison/output gates, so its size is at most $2S + O(1)$. This proves (5.7).

For the asymptotic statement, a nonnegligible next-bit advantage immediately gives a nonnegligible distinguishing advantage by the first implication. Conversely, if a polynomial-size distinguisher has nonnegligible gap $\delta(n)$ on infinitely many lengths, then (5.7) gives predictors of polynomial size and advantage $\delta(n)/m(n)$ on those lengths. Since $m(n)$ is polynomially bounded, that advantage is still nonnegligible. This is exactly the equivalence asserted. $\square$

Finite hybrid checkpoint. Let A, B be independent uniform bits and put

$$X = (A, B, A \oplus B). \quad (5.13)$$

The parity test $D(y) = \mathbf{1}\{y_1 \oplus y_2 \oplus y_3 = 0\}$ accepts X with probability 1 and U_3 with probability $1/2$, so $\delta = 1/2$. The first two hybrids in (5.9) are still uniform, while the last replacement changes the acceptance probability by $1/2$. Accordingly the predictor $P(y_1, y_2) = y_1 \oplus y_2$ predicts the third bit perfectly: its next-bit advantage is $1/2$, stronger than the guaranteed $\delta/m = 1/6$. The script `code/volume_09/computational_prg_lab.py` verifies these probabilities exactly by exhaustive enumeration.

Source and normalization note. Yao’s 1982 FOCS paper develops computational indistinguishability of source ensembles, polynomial statistical tests, and pseudorandom-number generators. Blum and Micali’s 1984 journal paper then records explicitly, as Yao’s theorem, that passing the next-bit test is equivalent to passing every polynomial-size statistical test. Their formulation is nonuniform and circuit based, with inverse-polynomial prediction and distinguishing gaps. Theorem 5.2 is the modern finite quantitative form used by this book and is proved locally; it is not presented as a verbatim transcription of the notation in either historical paper.

What the theorem does not provide. The next-bit characterization is a test equivalence. It does not construct a pseudorandom generator from scratch, does not prove that one-way functions exist, and does not convert uniform security into nonuniform security. Those are separate interfaces in the later chapters.

5.3 One-way functions: interface only

The next hardness layer uses an inversion primitive, but it is important not to smuggle a full cryptographic equivalence into the definition. We therefore freeze only the classical one-way-function interface needed later in the series.

Definition 5.3 (Length-regular one-way-function interface). A *classical length-regular function family* is a sequence

$$f_n : \{0, 1\}^n \longrightarrow \{0, 1\}^{m(n)} \quad (5.14)$$

defined by one deterministic polynomial-time algorithm, where the length function $m(n)$ is polynomially related to n : for some constant $c \geq 1$ and all sufficiently large n ,

$$n^{1/c} \leq m(n) \leq n^c. \quad (5.15)$$

For an inverter $A_n : \{0, 1\}^{m(n)} \rightarrow \{0, 1\}^n$, its inversion success at length n is

$$\text{Succ}_{f,n}^{\text{inv}}(A_n) := \Pr_{x \leftarrow U_n} [f_n(A_n(f_n(x))) = f_n(x)]. \quad (5.16)$$

Thus success means finding *some* preimage of the sampled image, not necessarily recovering the original x .

The family $f = \{f_n\}$ is a *one-way function family* in the default nonuniform convention of this volume if it is efficiently computable and, for every polynomial-size Boolean-circuit family $A = \{A_n\}$,

$$\text{Succ}_{f,n}^{\text{inv}}(A_n) \quad (5.17)$$

is negligible in n . The uniform classical variant instead quantifies over probabilistic polynomial-time algorithms receiving 1^n and $f_n(x)$. No quantum adversary is included in either convention here.

Boundary checkpoint. The definition is the strong one-wayness convention: inversion success itself must be negligible. Weak one-way functions, regular functions, one-way permutations, distributional one-wayness, and black-box reductions are deferred to Volume XVI. In particular, Definition 5.3 neither asserts that one-way functions exist nor proves that they imply pseudorandom generators. The latter implication requires the later hard-core-bit and iteration machinery and is kept out of the present interface.

Chapter 6

Mild Average Hardness and Hard-Core Sets

Worst-case circuit lower bounds say that every small circuit fails somewhere. Average-case hardness says that every small circuit fails on a noticeable fraction of the uniform input space. Impagliazzo's hard-core lemma turns this mild average hardness into a dense region on which substantially smaller circuits cannot predict the target much better than a fair coin. This chapter freezes the density, advantage, and circuit-loss conventions before the later XOR and Nisan–Wigderson amplification steps.

6.1 Average hardness, hard-core sets, and dense measures

Definition 6.1 (Average circuit hardness and hard-core advantage). Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and let C be a Boolean circuit on n input bits. Write

$$\text{succ}_{U_n}(C, f) := \Pr_{x \leftarrow U_n} [C(x) = f(x)], \quad \text{err}_{U_n}(C, f) := 1 - \text{succ}_{U_n}(C, f). \quad (6.1)$$

For $0 < \delta < 1/2$, we say that f is δ -hard for circuits of size S if every circuit C of size at most S satisfies

$$\text{err}_{U_n}(C, f) \geq \delta. \quad (6.2)$$

If $H \subseteq \{0, 1\}^n$ is nonempty, its density is $\rho(H) = |H|/2^n$, and

$$\text{Adv}_H(C, f) := \Pr_{x \leftarrow U_H} [C(x) = f(x)] - \frac{1}{2}. \quad (6.3)$$

For $0 < \eta < 1/2$, the set H is an (η, S') hard-core set for f if every circuit of size at most S' has

$$\text{Adv}_H(C, f) \leq \eta. \quad (6.4)$$

A distribution D on $\{0, 1\}^n$ is δ -dense if

$$\max_x D(x) \leq \frac{1}{\delta 2^n}. \quad (6.5)$$

Thus U_H is δ -dense whenever $|H| \geq \delta 2^n$. Equivalently, a *measure* is a function $M : \{0, 1\}^n \rightarrow [0, 1]$ with mass

$$\mu(M) := \mathbb{E}_{x \leftarrow U_n} M(x), \quad (6.6)$$

and, when $\mu(M) > 0$, it induces

$$D_M(x) := \frac{M(x)}{2^n \mu(M)}. \quad (6.7)$$

If $\mu(M) \geq \delta$, then D_M is δ -dense. Flat hard-core sets and fuzzy hard-core measures are kept distinct: every flat set gives a measure, but a general measure need not be uniform on a subset.

Normalization checkpoint. Hardness in (6.2) is an error probability under the uniform input distribution, while (6.3) is prediction success minus $1/2$ after conditioning on the hard core. Hence “ $1/2 - \eta$ hard on H ” and “prediction advantage at most η on H ” are the same statement. We do not use absolute correlation in place of (6.3); because the circuit class is closed under output complementation, a predictor with negative correlation can be complemented.

6.2 Impagliazzo’s hard-core lemma

Theorem 6.2 (Impagliazzo hard-core lemma; original density form). *There is a universal constant $c > 0$ with the following property. Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let $0 < \delta < 1/2$ and $0 < \eta < 1/2$, and let $n \leq S < 2^n/n$. Suppose every Boolean circuit C of size at most S satisfies*

$$\Pr_{x \leftarrow U_n} [C(x) = f(x)] \leq 1 - \delta. \quad (6.8)$$

Then there is a set $H \subseteq \{0, 1\}^n$ with

$$|H| \geq \delta 2^n \quad (6.9)$$

such that every circuit C of size at most

$$S' := \lfloor c\eta^2\delta^2S \rfloor \quad (6.10)$$

satisfies

$$\Pr_{x \leftarrow U_H} [C(x) = f(x)] \leq \frac{1}{2} + \eta. \quad (6.11)$$

Source normalization. Impagliazzo’s preliminary 1994 statement uses a parameter ε and says that a δ -hard Boolean function has a set of size at least $\delta 2^n$ on which it is $\frac{1}{2}(1 - \varepsilon)$ -hard for circuits of size $d\varepsilon^2\delta^2S$, for an absolute constant d . The published FOCS 1995 paper is the primary source. Setting $\varepsilon = 2\eta$ turns the source success threshold $(1 + \varepsilon)/2$ into (6.11); the factor four is absorbed into the universal constant c . The source definition of δ -hardness uses the non-strict success bound $\Pr[C(x) = f(x)] \leq 1 - \delta$, matching (6.8). We deliberately retain this original $\eta^2\delta^2S$ loss rather than silently replacing it with later improved hard-core lemmas.

Source-owned proof architecture. The primary proof first works with the fuzzy measure in Definition 6.1. Assuming that every measure of mass at least δ admits a predictor with noticeable advantage, it repeatedly chooses such predictors while down-weighting inputs on which their accumulated signed margin is already comfortably positive. If the measure mass ever falls below δ , the majority of the predictors found so far already computes f on more than a $1 - \delta$ fraction of all inputs. If the process continues for the full boosting horizon, the accumulated margin becomes positive on every input, again producing a circuit contradicting (6.8). This gives a hard-core measure with the $\eta^2\delta^2S$ resource loss. The source then converts the measure to a flat set by independent sampling and a counting bound over circuits.

This chapter imports that construction as a source-owned theorem. The finite minimax principle from Theorem 1.2 gives a useful modern game-theoretic interpretation of the same phenomenon, but it is not presented as a replacement proof for Impagliazzo's boosting-and-sampling argument.

Toy hard-core checkpoint. Take the restricted hypothesis class consisting only of the two constant functions on $\{0, 1\}^3$, and let $f(x_1, x_2, x_3) = x_1 \oplus x_2 \oplus x_3$. Each constant predictor succeeds on exactly four of the eight inputs, so the whole cube itself is a density-one hard core with prediction advantage 0. The example is intentionally only a finite model of the definition: the theorem concerns the much richer class of all Boolean circuits up to the stated size bound.

What the theorem does not say. The set H need not be efficiently recognizable, samplable by a small circuit, or independent of f . Nor does the theorem preserve the original circuit size S : the resource loss in (6.10) is part of the claim. Later strengthened hard-core lemmas improve this loss or the representation of the hard core, but those refinements are separate source gates. Chapter 7 will use only the frozen interface (6.9)–(6.11) when turning mild hardness into stronger average-case hardness.

Chapter 7

Hardness Amplification: XOR and the Exponential-Time Bridge

Chapter 6 isolates a dense region on which a mildly hard Boolean function is close to unpredictable. The first task here is to turn that local unpredictability into hardness of an XOR on independent inputs. The second task is to record the stronger Impagliazzo–Wigderson bridge that keeps the total input length linear while converting a worst-case exponential circuit lower bound in E into the exponentially small prediction advantage needed by hardness-versus-randomness generators.

7.1 Independent XOR amplification

For $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and $k \geq 1$, define

$$f^{\oplus k}(x_1, \dots, x_k) := f(x_1) \oplus \dots \oplus f(x_k). \quad (7.1)$$

The input distribution in this section is always the product distribution U_n^k ; no pseudorandomly correlated replacement for the k inputs is being used in Theorem 7.1.

Theorem 7.1 (Yao-style XOR amplification; local hard-core corollary). *Let $c > 0$ be the universal constant from Theorem 6.2. Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let $0 < \delta < 1/2$ and $0 < \varepsilon < 1/2$, let $k \geq 1$, and let $n \leq S < 2^n/n$. Suppose every Boolean circuit C of size at most S satisfies*

$$\Pr_{x \leftarrow U_n} [C(x) = f(x)] \leq 1 - \delta. \quad (7.2)$$

Assume moreover that

$$\varepsilon > 2(1 - \delta)^k. \quad (7.3)$$

Set

$$S_{\text{hc}} := \left\lfloor \frac{c\varepsilon^2\delta^2 S}{16} \right\rfloor. \quad (7.4)$$

If $S_{\text{hc}} \geq 1$, then every circuit D of size at most $S_{\text{hc}} - 1$ satisfies

$$\Pr_{(x_1, \dots, x_k) \leftarrow U_n^k} [D(x_1, \dots, x_k) = f^{\oplus k}(x_1, \dots, x_k)] \leq \frac{1}{2} + \varepsilon. \quad (7.5)$$

If $S_{\text{hc}} = 0$, the size conclusion is understood as vacuous.

Proof. Apply Theorem 6.2 with hard-core advantage $\eta = \varepsilon/4$. There is a set $H \subseteq \{0, 1\}^n$ of density

$$\rho := |H|/2^n \geq \delta \quad (7.6)$$

such that every circuit of size at most S_{hc} predicts f on U_H with success at most $1/2 + \varepsilon/4$.

If $\rho = 1$, put $a := 0$ and ignore the complementary distribution below. If $\rho < 1$, let G be the uniform distribution on $\{0, 1\}^n \setminus H$. Then

$$U_n = \rho U_H + (1 - \rho)G. \quad (7.7)$$

Taking the k -fold product writes U_n^k as a convex combination indexed by subsets $J \subseteq [k]$: coordinates in J are sampled independently from U_H , and coordinates outside J from G . The all- G component has weight

$$a = (1 - \rho)^k \leq (1 - \delta)^k < \varepsilon/2. \quad (7.8)$$

Suppose for contradiction that a circuit D of size at most $S_{\text{hc}} - 1$ has success greater than $1/2 + \varepsilon$ in (7.5). Let M be the maximum success of D over the mixture components with $J \neq \emptyset$. Since the all- G component has success at most one,

$$\Pr[D = f^{\oplus k}] \leq a + (1 - a)M. \quad (7.9)$$

If $M \leq 1/2 + \varepsilon/4$, then using $a < \varepsilon/2$ gives

$$a + (1 - a)M \leq \frac{1}{2} + \frac{a}{2} + \frac{\varepsilon}{4} \leq \frac{1}{2} + \frac{\varepsilon}{2}, \quad (7.10)$$

contradicting the assumed success. Hence some nonempty component J has success greater than $1/2 + \varepsilon/4$.

Choose any $j \in J$. Average over all coordinates other than j under that fixed mixture component. There is a fixed tuple $(x_i)_{i \neq j}$ for which, when $x_j \leftarrow U_H$,

$$\Pr[D(x_1, \dots, x_k) = f^{\oplus k}(x_1, \dots, x_k)] > \frac{1}{2} + \frac{\varepsilon}{4}. \quad (7.11)$$

Hardwire those $k - 1$ strings and also the nonuniform advice bit

$$b := \bigoplus_{i \neq j} f(x_i). \quad (7.12)$$

The circuit

$$C(x) := D(x_1, \dots, x_{j-1}, x, x_{j+1}, \dots, x_k) \oplus b \quad (7.13)$$

has size at most $|D| + 1 \leq S_{\text{hc}}$ and predicts $f(x)$ on U_H with the success in (7.11), contradicting the hard-core guarantee. Therefore (7.5) holds. $\square$

Historical quantitative comparison. Yao's 1982 trapdoor-functions paper is the historical source of the XOR lemma. A standard exact circuit formulation, recorded for example as Theorem 19.2 in Arora–Barak, says that if every size- S circuit has success strictly below $1 - \delta$, then for every $\varepsilon > 2(1 - \delta)^k$ the XOR is hard against circuits of size $\varepsilon^2 S / (400n)$ at success threshold $1/2 + \varepsilon$. Theorem 7.1 is intentionally a different, conservative local corollary: it is proved entirely from the already frozen Theorem 6.2, so its resource loss retains the factor δ^2 inherited from Impagliazzo's original hard-core theorem. The historical sharper circuit loss is not silently substituted into the local proof.

Parameter checkpoint. Condition (7.3) is the independence gain. For example, if $\delta = 1/3$ and $k = 8$, then $2(1 - \delta)^k = 2(2/3)^8 < 0.079$; choosing $\varepsilon = 0.1$ makes the theorem nonvacuous once the starting circuit size is large enough for $S_{\text{hc}} \geq 1$. The theorem does not claim that the k inputs can be sampled from a shorter seed. That derandomized direct-product issue is exactly the extra content needed for the linear-input-length Impagliazzo–Wigderson bridge below.

7.2 Worst-case to strong average-case hardness inside E

Write $E := \text{DTIME}(2^{O(n)})$. For a Boolean function family $g = (g_m)_{m \geq 1}$ define its size- T prediction advantage at length m by

$$\text{Adv}_T(g_m) := \max_{|C| \leq T} \left(\Pr_{x \leftarrow U_m} [C(x) = g_m(x)] - \frac{1}{2} \right). \quad (7.14)$$

This is a nonuniform circuit quantity; the fact that $g \in E$ is a separate uniform computability statement.

Theorem 7.2 (Impagliazzo–Wigderson worst-case-to-average-case bridge). *Suppose there is a Boolean language/function $f \in E$ and a constant $\alpha > 0$ such that, for all sufficiently large n , every Boolean circuit computing f_n exactly has size at least $2^{\alpha n}$. Then there are a Boolean function $g \in E$ and a constant $\beta > 0$ such that, for all sufficiently large m ,*

$$\text{Adv}_{2^{\beta m}}(g_m) \leq 2^{-\beta m}. \quad (7.15)$$

Equivalently, every circuit of size at most $2^{\beta m}$ predicts g_m on a uniform input with success at most $1/2 + 2^{-\beta m}$.

Source normalization and ownership. This theorem is the average-hardness interface extracted from the 1997 Impagliazzo–Wigderson route, not a claim that the classical independent-input XOR lemma alone preserves linear input length. Their proceedings paper fixes $E = \text{DTIME}(2^{O(n)})$ and proves the following chain.

First, their Theorem 4 converts worst-case circuit complexity $2^{\Omega(n)}$ for a Boolean $f \in E$ into a Boolean $h \in E$ whose length- cn slice has success at most $1 - n^{-2}$ for circuits of size $2^{\Omega(n)}$, with only a constant-factor input-length increase. Their Theorem 5 raises this to constant average hardness: from the $1 - n^{-2}$ regime one obtains a Boolean $g \in E$ whose $2n$ -bit slice has success at most $2/3$ against exponentially large circuits. The main derandomized direct-product theorem (their Theorem 9) then constructs, for every fixed $0 < \gamma < 1$, constants $\gamma' > 0$ and $c > 0$ and a polynomial-time map

$$G : \{0, 1\}^{cn} \longrightarrow (\{0, 1\}^n)^n \quad (7.16)$$

that turns constant hardness against size $2^{\gamma n}$ circuits into exponentially small success for the associated direct-product task against size $2^{\gamma' n}$ circuits. Their Goldreich–Levin reduction then converts that vector-valued direct-product hardness into a Boolean predicate with exponentially small prediction advantage. Constant-factor changes of input length preserve an exponential circuit exponent, and taking the minimum of the resulting positive exponents yields a single β on the transformed length sequence. A standard padding step fills the bounded gaps between those lengths: define an intermediate-length slice by applying the preceding hard slice to a fixed-length prefix and ignoring the surplus bits. Any predictor on the padded slice can have the surplus bits hardwired by averaging, producing a predictor of no larger size and no smaller success on the hard slice. Because

the retained prefix length is a constant fraction of the padded length, the exponential size and advantage exponents only change by a constant factor. This gives (7.15) for all sufficiently large m .

The coding/self-reducibility start, hard-core step, direct-product generator, and Booleanization are therefore kept as source-owned components of Theorem 7.2. Theorem 7.1 records the classical independent-input mechanism locally, but it is not presented as a proof of the linear-blowup theorem (7.15).

Why the input-length accounting matters. If one applies the classical XOR lemma to inverse-polynomial hardness using $k = \text{poly}(n)$ fully independent copies, the new input length can be polynomially larger than n . Exponential hardness $2^{\Omega(n)}$ in the old length then becomes only $2^{n^{\Omega(1)}}$ when expressed in the new length, which is insufficient for the strongest polynomial-time NW simulation. Impagliazzo–Wigderson’s derandomized direct product keeps the total encoded input length $O(n)$, preserving a true $2^{\Omega(m)}$ hardness exponent.

Boundary checkpoint. Theorem 7.2 does not itself prove $P = BPP$. It produces the strongly average-hard Boolean function that the next Nisan–Wigderson chapters consume. The combinatorial design, reconstruction lemma, generator theorem, and final deterministic simulation remain separate proof interfaces.

Chapter 8

Nisan–Wigderson Designs and Reconstruction

The Nisan–Wigderson generator evaluates one hard Boolean function on many nearly disjoint views of a short seed. This chapter separates the two pieces that are often compressed into one slogan: an explicit set-system construction and a reconstruction theorem showing that a distinguisher for the generator would predict the hard function.

8.1 Polynomial-graph designs

Call a family $\mathcal{S} = \{S_1, \dots, S_M\}$ of subsets of a universe $[d]$ an (ℓ, a) -design if every block has size ℓ and $|S_i \cap S_j| \leq a$ whenever $i \neq j$.

Theorem 8.1 (Finite-field polynomial design). *Let q be a prime power and let $1 \leq r \leq q$. There is an explicit $(q, r-1)$ -design on a universe of size q^2 with exactly q^r blocks. More concretely, identify the universe with $\mathbb{F}_q \times \mathbb{F}_q$. For every polynomial*

$$p(t) = a_0 + a_1 t + \dots + a_{r-1} t^{r-1} \in \mathbb{F}_q[t] \quad (8.1)$$

of degree $< r$, set

$$S_p := \{(x, p(x)) : x \in \mathbb{F}_q\}. \quad (8.2)$$

Then $|S_p| = q$, there are q^r distinct blocks, and for $p \neq p'$,

$$|S_p \cap S_{p'}| \leq r-1. \quad (8.3)$$

Hence any desired first $M \leq q^r$ blocks give an explicit $(q, r-1)$ -design with M blocks and universe size q^2 .

Proof. Existence and arithmetic of $\mathbb{F}_q$ are supplied by the finite-field classification theorem proved in Volume III. For a fixed polynomial p , the first coordinate in (8.2) runs through all q field elements, so $|S_p| = q$. There are q independent choices for each of the r coefficients in (8.1), so there are q^r such polynomials and hence q^r blocks.

If $p \neq p'$, then a point lies in both graphs exactly when $p(x) = p'(x)$. The nonzero polynomial $p - p'$ has degree at most $r-1$, so it has at most $r-1$ roots in the field. This proves (8.3). Enumerating coefficient vectors in a fixed lexicographic order makes the construction explicit and lets us retain any prescribed first $M \leq q^r$ blocks. $\square$

Primary-source comparison. Nisan–Wigderson use the same graph-of-a-polynomial construction. In their notation, for output length M and block size q , choosing degree at most $\lceil \log_q M \rceil - 1$ supplies at least M blocks, pairwise intersections bounded by that degree, on a q^2 -point universe. Their Lemma 2.5 specializes this to a $(\log M, q)$ -type design with universe $O(q^2)$ and also records the low-space constructibility of the incidence matrix.

8.2 Reconstruction from a distinguisher

Given an (ℓ, a) -design $\mathcal{S} = \{S_1, \dots, S_M\}$ on $[d]$ and $f : \{0, 1\}^\ell \rightarrow \{0, 1\}$, define the Nisan–Wigderson map

$$\text{NW}_{f, \mathcal{S}}(y) := (f(y|_{S_1}), \dots, f(y|_{S_M})) \in \{0, 1\}^M. \quad (8.4)$$

The restriction $y|_{S_i}$ is ordered by the increasing order of the elements of S_i .

Theorem 8.2 (Nisan–Wigderson reconstruction lemma). *There is an absolute constant $c_0 > 0$ with the following property. Let $\mathcal{S}$ be an (ℓ, a) -design with M blocks on $[d]$, let $f : \{0, 1\}^\ell \rightarrow \{0, 1\}$, and let D be a Boolean circuit of size s such that*

$$\left| \Pr_{y \leftarrow U_d} [D(\text{NW}_{f, \mathcal{S}}(y)) = 1] - \Pr_{u \leftarrow U_M} [D(u) = 1] \right| \geq \varepsilon. \quad (8.5)$$

Then there is a Boolean circuit C on ℓ input bits of size at most

$$2s + c_0 M a 2^a + c_0 \quad (8.6)$$

(with the term $a 2^a$ interpreted as 1 when $a = 0$) such that

$$\Pr_{x \leftarrow U_\ell} [C(x) = f(x)] \geq \frac{1}{2} + \frac{\varepsilon}{M}. \quad (8.7)$$

All hardwiring in C is nonuniform advice; no efficient algorithm for finding that advice is asserted.

Proof. For $0 \leq i \leq M$, define a hybrid distribution H_i as follows. Sample $y \leftarrow U_d$. Put the first i coordinates equal to

$$f(y|_{S_1}), \dots, f(y|_{S_i}),$$

and fill coordinates $i + 1, \dots, M$ with independent uniform bits. Thus $H_0 = U_M$ and $H_M = \text{NW}_{f, \mathcal{S}}(U_d)$.

Choose a sign $\sigma \in \{-1, +1\}$ so that the signed endpoint difference is at least ε . Telescoping gives an index $i \in [M]$ for which

$$\sigma(\mathbb{E}D(H_i) - \mathbb{E}D(H_{i-1})) \geq \frac{\varepsilon}{M}. \quad (8.8)$$

Write $x = y|_{S_i}$. Average (8.8) over the seed coordinates outside S_i and over the independent suffix bits used after coordinate i . There is a fixed assignment z to $[d] \setminus S_i$ and a fixed suffix $r \in \{0, 1\}^{M-i}$ for which the signed gap remains at least ε/M .

For every $j < i$, once z is fixed the bit $f(y|_{S_j})$ is a Boolean function $b_j(x)$ of only the coordinates of x indexed by $S_i \cap S_j$. There are at most a such coordinates. An arbitrary Boolean function of at most a inputs has a DNF of size $O(a 2^a)$; therefore all $i - 1$ prefix bits $b_1(x), \dots, b_{i-1}(x)$ can be implemented with total size at most $c_0 M a 2^a$ after increasing the absolute constant c_0 .

For $b \in \{0, 1\}$ put

$$d_b(x) := D(b_1(x), \dots, b_{i-1}(x), b, r). \quad (8.9)$$

Evaluate both $d_0(x)$ and $d_1(x)$. If they differ, output the bit b that maximizes $\sigma d_b(x)$; if they agree, output an independent fair bit. For each fixed x , the predictor's success excess over $1/2$ is exactly

$$\sigma \left(d_{f(x)}(x) - \frac{d_0(x) + d_1(x)}{2} \right). \quad (8.10)$$

Averaging (8.10) over $x \leftarrow U_\ell$ gives precisely the signed hybrid gap in (8.8), hence advantage at least ε/M .

The only randomness in this predictor is the tie-breaking bit. Averaging over its two possible fixed values shows that one hardwired value preserves at least the same average success. The deterministic predictor uses two copies of D , the prefix circuitry, and constant additional logic, yielding (8.6). $\square$

Reconstruction checkpoint. The cost 2^a is not cosmetic. Once the seed outside S_i is hardwired, an earlier output bit depends on the current ℓ -bit input only through the intersection $S_i \cap S_j$. The reconstruction may therefore hardwire the truth table of that restricted function using 2^a values. If intersections were large, the reconstructed predictor could be larger than the hardness assumption allows.

8.3 The generator theorem

Theorem 8.3 (Nisan–Wigderson generator theorem; quantitative circuit form). *Let $\mathcal{S}$ be an (ℓ, a) -design with M blocks on $[d]$ and let $f : \{0, 1\}^\ell \rightarrow \{0, 1\}$. Suppose every circuit C of size at most T satisfies*

$$\Pr_{x \leftarrow U_\ell} [C(x) = f(x)] \leq \frac{1}{2} + \gamma. \quad (8.11)$$

Let $s \geq 1$ satisfy

$$2s + c_0 M a 2^a + c_0 \leq T, \quad (8.12)$$

where c_0 is the absolute constant from Theorem 8.2. Then every Boolean circuit D of size at most s satisfies

$$|\Pr[D(\text{NW}_{f,\mathcal{S}}(U_d)) = 1] - \Pr[D(U_M) = 1]| \leq M\gamma. \quad (8.13)$$

In particular, if $M\gamma \leq \varepsilon$, the map $\text{NW}_{f,\mathcal{S}} : \{0, 1\}^d \rightarrow \{0, 1\}^M$ is an (s, ε) computational pseudorandom generator in the fixed-length nonuniform sense of Definition 5.1.

Proof. If some size- s circuit had distinguishing advantage greater than $M\gamma$, Theorem 8.2 with $\varepsilon > M\gamma$ would yield a predictor for f of size at most T and success greater than $1/2 + \gamma$, contradicting (8.11). This proves (8.13), and the final statement is exactly the fixed-length pseudorandomness convention from Definition 5.1. $\square$

Primary-source comparison. Nisan–Wigderson's original Lemma 2.4 chooses output length M , intersection bound $\log M$, a size- M distinguisher, and a hard function whose hardness parameter exceeds M^2 . Their hybrid predictor gains advantage greater than $1/M^2$, then hardwires the seed coordinates outside the selected block and implements every previous output bit by a DNF/CNF on at most $\log M$ variables. Theorems 8.2 and 8.3 expose the same mechanism with the distinguisher size, hybrid gap, intersection bound, and truth-table cost left as independent parameters.

Current boundary. Theorems 7.2 and 8.3 are now the two mathematical halves needed for the final Impagliazzo–Wigderson simulation theorem: one supplies an exponentially hard-on-average function in E , and the other turns such hardness plus a design into pseudorandom bits. What remains is the global parameter synthesis that chooses the hard-function input length and design parameters so the seed can be enumerated in polynomial time while the reconstructed circuit remains below the hardness threshold.

Chapter 9

The Impagliazzo–Wigderson Simulation

The preceding chapter isolates the Nisan–Wigderson reconstruction theorem from the combinatorial design that feeds it. The remaining issue is quantitative: when the hard-function input length is only $\Theta(\log n)$, the design itself must live on an $O(\log n)$ -point universe so that every generator seed can be enumerated in polynomial time. We first prove that logarithmic-universe design regime directly, and then close the final hardness-to-randomness synthesis.

9.1 A small-universe design in the logarithmic regime

The graph-of-a-polynomial construction of Theorem 8.1 is optimal for many parameter ranges but is not the design we use below: with block size $\ell = \Theta(\log n)$ its q^2 universe can have size $\Theta(\log^2 n)$. The following elementary greedy argument is the logarithmic-universe regime already highlighted by Nisan–Wigderson.

Proposition 9.1 (Greedy small-universe design). *Let $M \geq 2$, let*

$$a := \lceil \log_2 M \rceil, \tag{9.1}$$

and let $\ell \geq a + 1$. If

$$d := \left\lceil \frac{2e\ell^2}{a+1} \right\rceil, \tag{9.2}$$

then one can deterministically construct M subsets $S_1, \dots, S_M \subseteq [d]$, each of cardinality ℓ , such that

$$|S_i \cap S_j| \leq a \quad (i \neq j). \tag{9.3}$$

Moreover, when $d = O(\log M)$ the construction can be carried out in time polynomial in M by lexicographically scanning all ℓ -subsets of $[d]$.

Proof. Fix an already chosen ℓ -set $S \subseteq [d]$, and choose an ℓ -set T uniformly from all $\binom{d}{\ell}$ possibilities. If $|S \cap T| \geq a + 1$, then T contains some fixed $(a + 1)$ -subset of S . For any such subset $R \subseteq S$,

$$\Pr[R \subseteq T] = \frac{(\ell)_{a+1}}{(d)_{a+1}} \leq \left(\frac{\ell}{d}\right)^{a+1}. \tag{9.4}$$

A union bound over the $\binom{\ell}{a+1}$ choices of R gives

$$\Pr[|S \cap T| \geq a+1] \leq \binom{\ell}{a+1} \left(\frac{\ell}{d}\right)^{a+1} \leq \left(\frac{e\ell^2}{d(a+1)}\right)^{a+1} \leq 2^{-(a+1)}. \quad (9.5)$$

Suppose $S_1, \dots, S_{j-1}$ have already been chosen. The fraction of ℓ -sets violating (9.3) with at least one previous block is at most

$$(j-1)2^{-(a+1)} < M2^{-(a+1)} \leq \frac{1}{2}, \quad (9.6)$$

because $2^a \geq M$. Thus a legal next set exists. Repeatedly choose the first legal set in lexicographic order. When $d = O(\log M)$ there are at most $2^d = M^{O(1)}$ candidate subsets, and every intersection test is polynomial in d and M , proving the algorithmic assertion. $\square$

Source comparison. Section 2.3 of the original Nisan–Wigderson hardness-versus-randomness paper uses precisely the nearly-disjoint-set regime needed here. For block size $m = C \log n$, their construction discussion records a polynomial-time greedy design with logarithmic-size universe and logarithmic pairwise intersections. Proposition 9.1 gives a self-contained counting proof with explicit dependence $d = O(\ell^2/a)$; it does not replace the finite-field design of Chapter 8, which remains useful in other parameter ranges.

9.2 The final hardness-to-randomness synthesis

Theorem 9.2 (Impagliazzo–Wigderson derandomization theorem). *Suppose a Boolean function family $f = (f_m)_{m \geq 1}$ is the characteristic family of a language in E , and suppose there is a constant $\alpha > 0$ such that, for all sufficiently large m , every Boolean circuit computing f_m exactly has size at least $2^{\alpha m}$. Then*

$$BPP = P. \quad (9.7)$$

More quantitatively, for every fixed integer $\kappa \geq 1$ there are constants $D = D(\kappa) > 0$ and n_0 and a deterministic polynomial-time computable family

$$G_n : \{0, 1\}^{d(n)} \longrightarrow \{0, 1\}^{M(n)}, \quad M(n) = n^\kappa, \quad d(n) \leq D \log_2 n, \quad (9.8)$$

defined for $n \geq n_0$, such that every Boolean circuit C of size at most n^κ on $M(n)$ input bits satisfies

$$\left| \Pr[C(G_n(U_{d(n)})) = 1] - \Pr[C(U_{M(n)}) = 1] \right| \leq \frac{1}{12}. \quad (9.9)$$

The constants and the generator family may depend on the fixed hard function f and on κ , but not on the individual circuit C .

Proof. Apply Theorem 7.2. There are a Boolean function $g \in E$ and a constant $\beta > 0$ such that, for every sufficiently large input length ℓ , every circuit of size at most $2^{\beta \ell}$ predicts g_ℓ on a uniform input with success at most

$$\frac{1}{2} + 2^{-\beta \ell}. \quad (9.10)$$

Fix $\kappa \geq 1$. Let c_0 be the reconstruction constant from Theorem 8.2. Choose a constant $C > 0$ so large that

$$C > \kappa + 2, \quad \beta C > 2\kappa + 4. \quad (9.11)$$

For a sufficiently large simulation length n , put

$$M := n^\kappa, \quad a := \lceil \log_2 M \rceil, \quad \ell := \lceil C \log_2 n \rceil. \quad (9.12)$$

Then $\ell \geq a + 1$. Apply Proposition 9.1. It supplies an (ℓ, a) -design $\mathcal{S}_n = \{S_1, \dots, S_M\}$ on a universe $[d]$ with

$$d \leq \frac{2e\ell^2}{a+1} + 1 = O(\log n). \quad (9.13)$$

Because $d = O(\log n)$ and $M = n^\kappa$, the lexicographic greedy construction runs in polynomial time in n .

Set

$$\gamma := 2^{-\beta\ell}, \quad T := 2^{\beta\ell}. \quad (9.14)$$

For all sufficiently large n , (9.10) is precisely the hardness hypothesis of Theorem 8.3. Also $2^a < 2M$, so

$$2n^\kappa + c_0 M a 2^a + c_0 \leq 2M + 2c_0 M^2 a + c_0 \leq n^{2\kappa+2} \leq 2^{\beta\ell} = T \quad (9.15)$$

for all sufficiently large n , where the penultimate inequality merely absorbs the logarithmic factor a and the fixed constants. Moreover,

$$M\gamma \leq n^{\kappa-\beta C} \leq n^{-4} < \frac{1}{12}. \quad (9.16)$$

Therefore Theorem 8.3, applied to g_ℓ and $\mathcal{S}_n$, gives

$$G_n(y) := \text{NW}_{g_\ell, \mathcal{S}_n}(y) \quad (9.17)$$

and proves (9.9) for every size- n^κ circuit.

It remains to verify uniform evaluation. Because $g \in E$, there is a constant c_g and a deterministic algorithm evaluating g_ℓ in time at most $2^{c_g \ell}$ for all sufficiently large ℓ . Under (9.12), this is $n^{O(1)}$. Constructing $\mathcal{S}_n$ is polynomial-time, and evaluating all $M = n^\kappa$ output coordinates in (9.17) is still polynomial-time. Thus the family G_n is uniformly polynomial-time computable. Equation (9.13) gives $d(n) = O(\log n)$, proving the quantitative generator statement.

Now let $L \in BPP$. Fix a randomized polynomial-time machine A deciding L with the usual $2/3$ versus $1/3$ acceptance gap. Choose one fixed integer κ large enough that on every length- n input, for all large n , (i) A uses at most n^κ random bits and (ii) after hardwiring the input x , the acceptance predicate as a function of its random tape is computed by a Boolean circuit of size at most n^κ . Condition (ii) follows from the uniform-computation-to-circuit simulation theorem recorded in Volume II; unused random-input coordinates can be appended so that the circuit is viewed as a test on exactly $M = n^\kappa$ bits.

For $x \in \{0, 1\}^n$, let C_x be this acceptance circuit. By (9.9),

$$x \in L \implies \Pr_z[C_x(G_n(z)) = 1] \geq \frac{2}{3} - \frac{1}{12} = \frac{7}{12}, \quad (9.18)$$

whereas

$$x \notin L \implies \Pr_z[C_x(G_n(z)) = 1] \leq \frac{1}{3} + \frac{1}{12} = \frac{5}{12}. \quad (9.19)$$

Enumerate all $2^{d(n)} = n^{O(1)}$ seeds z , evaluate $G_n(z)$, run A on x with the corresponding padded random tape, and accept exactly when more than half of the seeds accept. Equations (9.18)–(9.19) prove correctness. The number of seeds, the generator evaluation time, and the running time of A are all polynomial in n , so this is a deterministic polynomial-time algorithm for L . Hence $BPP \subseteq P$; the reverse inclusion is immediate by using no randomness. This proves (9.7). $\square$

Parameter ledger. The proof uses four scales and no hidden fifth parameter. The randomized computation contributes the polynomial exponent κ ; the IW bridge contributes the average-hardness exponent β ; we choose a hard-function input length $\ell = C \log n$ with $\beta C > 2\kappa + 4$; and the greedy design then has intersection $a = \Theta(\log n)$ and universe $d = O(\log n)$. The reconstruction cost is only $n^{O(\kappa)}$, while the available hard-circuit budget is $n^{\beta C}$. This is the precise point at which exponential hardness on $\Theta(\log n)$ -bit inputs buys enough room to fool every polynomial-size test while leaving only polynomially many generator seeds.

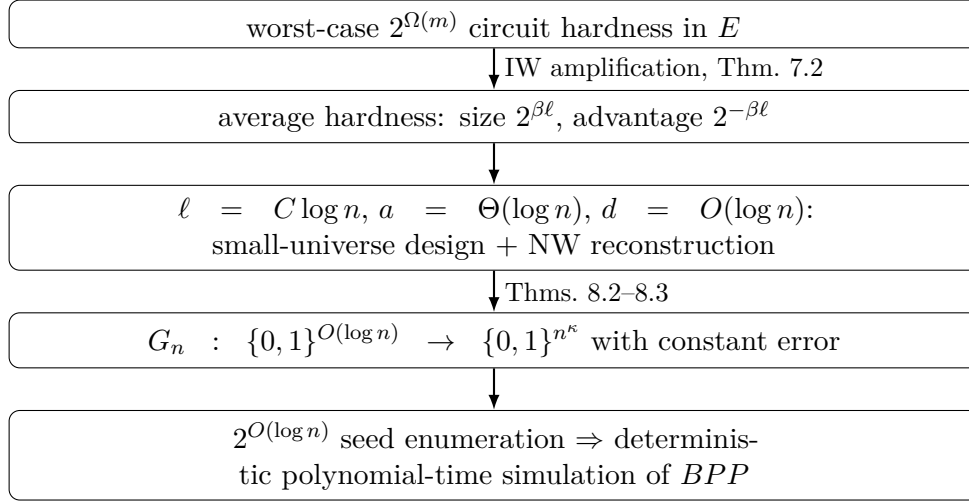


Figure 9.1: Hardness-to-randomness dependency pipeline. The quantitative bottleneck is the simultaneous choice $\ell = \Theta(\log n)$ and $d = O(\log n)$. The finite-field q^2 design alone would not give the final seed bound in this regime.

Primary-source comparison and ownership. Impagliazzo–Wigderson’s Theorem 2 states the same final consequence: if a Boolean function in E has circuit complexity $2^{\Omega(n)}$, then $P = BPP$. Their Theorem 3 records the Nisan–Wigderson hardness-versus-randomness interface that exponential average hardness in E already suffices for this simulation. The proof above is QRCC-owned parameter synthesis: it combines the source-verified IW average-hardness bridge from Theorem 7.2, the locally proved quantitative NW reconstruction from Chapter 8, and the explicit greedy small-universe design proved at the start of this chapter. No claim is made that the q^2 polynomial-graph design alone has the needed logarithmic seed.

Chapter 10

Hitting Sets, One-Sided Derandomization, and Pseudodeterminism

A pseudorandom generator controls acceptance probabilities for every test in a class. A hitting set asks for less: it only requires that every sufficiently dense accepting set contain at least one generated point. That weaker notion is exactly what one-sided randomized algorithms need. This chapter fixes the quantifiers and then records a selected verifiable-search consequence of the hardness-to-randomness generator from Chapter 9.

10.1 Hitting sets and black-box one-sided derandomization

Definition 10.1 (Hitting set and hitting-set generator). Let $\mathcal{C}_m$ be a family of Boolean tests on m bits and let $0 < \rho \leq 1$. A set $H \subseteq \{0, 1\}^m$ is a ρ -hitting set for $\mathcal{C}_m$ if every $D \in \mathcal{C}_m$ satisfying

$$\Pr_{u \leftarrow U_m} [D(u) = 1] \geq \rho \quad (10.1)$$

has some $h \in H$ with $D(h) = 1$.

A map $\text{HGen} : \{0, 1\}^d \rightarrow \{0, 1\}^m$ is a ρ -hitting-set generator for $\mathcal{C}_m$ when its image is a ρ -hitting set. The generator is *explicitly enumerable* when all 2^d outputs can be generated in time polynomial in the ambient input length relevant to the application.

A black-box one-sided derandomization replaces the random tape of a one-sided randomized algorithm by every point in such a hitting set and accepts if at least one replacement accepts. The definition does not require the uniform distribution on the hitting set to approximate acceptance probabilities.

Theorem 10.2 (PRGs imply hitting sets; hitting sets derandomize one-sided error). *Let $G : \{0, 1\}^d \rightarrow \{0, 1\}^m$ satisfy*

$$|\Pr[D(G(U_d)) = 1] - \Pr[D(U_m) = 1]| \leq \varepsilon \quad (10.2)$$

for every $D \in \mathcal{C}_m$.

1. *If $\rho > \varepsilon$, then $\text{Im}(G)$ is a ρ -hitting set for $\mathcal{C}_m$.*

2. Let A be a one-sided randomized decision algorithm whose length- n random tape has length $m(n)$, such that on yes-instances its acceptance probability is at least ρ and on no-instances it is zero. If an explicitly enumerable ρ -hitting set H_n hits every fixed-input acceptance test of A , then enumerating H_n gives a deterministic decision algorithm with running time $|H_n|$ times the polynomial cost of one run of A , plus the cost of enumerating H_n .

In particular, an $O(\log n)$ -seed PRG with error strictly below the one-sided acceptance density yields polynomial-time black-box derandomization.

Proof. For the first assertion, suppose $D \in \mathcal{C}_m$ has uniform acceptance probability at least ρ but rejects every point in $\text{Im}(G)$. Then $\Pr[D(G(U_d)) = 1] = 0$, so (10.2) would have error at least ρ , contrary to $\rho > \varepsilon$.

For the second assertion, on a no-instance every random tape rejects, so every point of H_n rejects. On a yes-instance the accepting random tapes have uniform density at least ρ , hence the hitting property supplies at least one $h \in H_n$ on which A accepts. Exhaustive enumeration is therefore correct. If $d = O(\log n)$, then $|H_n| \leq 2^d = n^{O(1)}$, giving polynomial running time when the generator/enumerator and A are polynomial-time. $\square$

Why the converse is weaker. Hitting only records whether a dense accepting set is met at least once. It does not control the frequency with which the generator lands in that set, nor does it say anything about tests whose accepting sets are below the density threshold. Thus an arbitrary hitting-set generator is not automatically a computational PRG in the sense of Definition 5.1; two-sided fooling is strictly more information than the black-box property used in Theorem 10.2.

10.2 Pseudodeterministic search

Definition 10.3 (Pseudodeterministic algorithm). Let $R(x, y)$ be a search relation. A randomized search algorithm A is *pseudodeterministic with error ε* for R if, for every input x in the domain of R , there exists a distinguished valid output y_x such that

$$R(x, y_x) = 1 \quad \text{and} \quad \Pr[A(x) = y_x] \geq 1 - \varepsilon. \quad (10.3)$$

The distinguished output may depend on x but not on the random coins. A deterministic search algorithm is therefore a zero-error special case. Validity alone is not pseudodeterminism: an algorithm that frequently outputs many different valid witnesses need not satisfy (10.3).

Theorem 10.4 (Verifiable-search compiler under exponential hardness in E). *Assume the circuit-hardness hypothesis of Theorem 9.2. Let $R(x, y)$ be a polynomially balanced relation decidable in deterministic polynomial time. Suppose a randomized polynomial-time algorithm A has the following property: for every $x \in \text{Dom}(R)$,*

$$\Pr_r[R(x, A(x; r)) = 1] \geq \frac{2}{3}, \quad (10.4)$$

and the number of random bits and output length are polynomially bounded in $|x|$. Then there is a deterministic polynomial-time algorithm B such that $R(x, B(x)) = 1$ for every $x \in \text{Dom}(R)$. In particular, B is a zero-error pseudodeterministic algorithm for R .

One canonical choice is: enumerate the hardness-to-randomness generator seeds, retain every valid output produced by A , and return the lexicographically least retained output.

Proof. Choose κ large enough that on length- n inputs the random tape length, the running time of A , the output length, and the Boolean-circuit size of the fixed-input validity test

$$D_x(r) := R(x, A(x; r)) \quad (10.5)$$

are all at most n^κ for all sufficiently large n . Pad the random tape with ignored coordinates to length $M = n^\kappa$.

The quantitative part of Theorem 9.2 supplies a uniformly polynomial-time generator $G_n : \{0, 1\}^{O(\log n)} \rightarrow \{0, 1\}^M$ that fools every size- n^κ test with error at most $1/12$. For $x \in \text{Dom}(R)$, (10.4) therefore implies

$$\Pr_z[D_x(G_n(z)) = 1] \geq \frac{2}{3} - \frac{1}{12} = \frac{7}{12} > 0. \quad (10.6)$$

Hence at least one generator seed produces a valid output. There are only $n^{O(1)}$ seeds. Enumerate them all, evaluate $A(x; G_n(z))$, verify the result with R , and return the lexicographically least valid output found. The generator, A , and the verifier are polynomial-time, so the entire search is polynomial-time. The returned witness is a deterministic function of x , hence it is canonical and satisfies Definition 10.3 with error zero. The finitely many input lengths below the generator threshold can be absorbed into the machine description. $\square$

Scope checkpoint. The theorem is intentionally restricted to polynomially balanced search relations with polynomial-time verification and a randomized algorithm whose success is visible through that verifier. It does not claim that the language equality $BPP = P$ alone derandomizes arbitrary search problems, nor that every randomized search procedure can be made pseudodeterministic without an explicit hitting/fooling mechanism for its success predicate.

10.3 Open conjectures and dated source maps

The unconditional equality $BPP = P$ remains open: the equality proved in Chapter 9 is conditional on exponential circuit hardness in E . The corresponding promise-class question must also be kept separate from the language equality. The two source maps below record what is known in two nearby directions without turning either direction into an unconditional solution of these conjectures.

Extremely rare error: the Goldreich–Wigderson source map

For a circuit class $\mathcal{C}$ and a bound $B : \mathbb{N} \rightarrow \mathbb{N}$, Goldreich and Wigderson define the $(\mathcal{C}, B)$ -search problem as follows: given an n -input circuit $C \in \mathcal{C}$ promised to evaluate to 1 on all but at most $B(n)$ inputs, find one $x \in \{0, 1\}^n$ with $C(x) = 1$. The parameter $B(n)$ counts *bad random strings*; it is not an error probability unless divided by 2^n .

The primary 2013/2014 source establishes three boundaries that are useful here. First, if a probabilistic log-space algorithm has at most $\exp(\text{poly log } n)$ erroneous random pads on each length- n input, then the language is in L ; the proof retains only $\log_2 B(n) + 2 = \text{polylog}(n)$ random bits and invokes the Nisan–Zuckerman space generator. Second, for every constants $c < 1$ and $d \in \mathbb{N}$ and every polynomial p , the

$$(AC_{d,p}^0, 2^{n^c})\text{-search problem} \quad (10.7)$$

is solvable in deterministic polynomial time. In the stronger black-box form, a deterministic polynomial-time algorithm outputs a polynomial-size set $S_n \subseteq \{0, 1\}^n$ that hits every promised circuit, and every such circuit accepts at least two-thirds of S_n . Thus the result handles every fixed-depth polynomial-size AC^0 family when the number of bad strings is 2^{n^c} for any fixed $c < 1$.

Third, the analogous statement for $AC^0[2]$ is *not* proved unconditionally. The source instead proves a hardness-of-the-quantified- problem implication: if, for every fixed depth and polynomial size bound, one can solve an $(AC^0[2], 2^{n^c})$ -search problem for some constant $c > 0$, then one obtains deterministic polynomial-time search for an $AC^0[2]$ circuit accepted by a majority of its inputs. This is precisely why “extremely small error” is a separate quantified derandomization program, not a theorem saying that arbitrary BPP algorithms can currently be unconditionally derandomized.

Status boundary. This row is a source-verified dated frontier, not a new downstream proof interface. The unconditional AC^0 theorem and the log-space consequence are recorded at their source strength; the $AC^0[2]$ statement is recorded as a conditional reduction, and no claim about unrestricted $P/poly$ is inferred.

Uniform, non-black-box, and instance-wise hardness versus randomness

A representative modern revision of the hardness-versus-randomness paradigm is the Chen–Tell theorem. Their hardness object is a *multi-output* function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$, and hardness is uniform and almost-all-inputs: a probabilistic algorithm of the stated running time must fail to compute $f(x)$ with the required constant success probability on every sufficiently long input x , rather than merely fail on one input per length.

The selected source theorem is the following exact qualitative parameter regime. There is a sufficiently large universal constant $c > 1$ such that if there exists

$$f : \{0, 1\}^n \longrightarrow \{0, 1\}^n \quad (10.8)$$

computed by deterministic log-space-uniform circuits of polynomial size and depth n^2 , but not computable by uniform probabilistic time n^c on almost all inputs, then

$$\text{Promise-BPP} = \text{Promise-P} . \quad (10.9)$$

The depth hypothesis can be replaced by a fixed depth n^k at the cost of a larger constant in the hardness exponent. The source also proves a converse necessity statement at the level relevant to the conceptual map: if $\text{Promise-BPP} = \text{Promise-P}$, then for every fixed c there is a deterministic polynomial-time multi-output function that is hard for probabilistic time n^c on almost all inputs.

The resulting derandomizer is intentionally *non-black-box*. It uses a targeted generator whose output set depends on the input being derandomized and uses the circuit computing the hard function inside the reconstruction. This is different from the Chapter 9 NW/IW black-box seed-enumeration route. Moreover, “instance-wise” refers to the reconstruction guarantee: failure of the targeted generator on a particular input supplies a reconstruction for the hard object associated with that same input. The source develops additional superfast average-case tradeoffs under extra hypotheses; those results are not silently folded into (10.8)–(10.9).

Status boundary. This row is a dated theorem map anchored to the Chen–Tell 2021/2024 result. It does not assert that all uniform, white-box, targeted, or instance-wise hardness-versus-randomness theorems are equivalent, and no downstream theorem in the current volume depends on this map.

Internalization and Reconstruction Guide: Volume IX

Purpose. Understand derandomization as a chain from randomized algorithms to distributional viewpoints, pseudorandom generators, hardness amplification, reconstruction, and hardness-versus-randomness tradeoffs.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-II-THM-024: Uniform computation has polynomial-size circuits.
2. QRCC-I-THM-005: Hoeffding inequality.
3. QRCC-II-THM-013: Circuit characterization of $P/poly$.
4. QRCC-II-THM-011: Polynomial-hierarchy quantifier characterization.
5. QRCC-III-THM-009: Gilbert–Varshamov bound.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-IX-DEF-001 (Randomized complexity classes); QRCC-IX-DEF-008 (Branching programs and space-bounded tests); QRCC-IX-DEF-006 (Computational pseudorandom generator); QRCC-IX-DEF-029 (One-way-function interface); QRCC-IX-DEF-010 (Average circuit hardness and hard-core advantage).

Chapter retrieval map

- Ch. 1: Randomized Classes and the Finite Minimax Principle:** enter through QRCC-IX-DEF-001 (Randomized complexity classes); exit by being able to state QRCC-IX-THM-005 (Yao minimax principle) with its ownership boundary.
- Ch. 2: Error Reduction by Independent Repetition:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-IX-THM-002 (Error reduction by independent repetition) with its ownership boundary.

- Ch. 3: Fixing Randomness and the Polynomial Hierarchy:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-IX-THM-003 (Adleman theorem BPP subset P/poly) with its ownership boundary.
- Ch. 4: Space-Bounded Tests and Pseudorandom Generators:** enter through QRCC-IX-DEF-008 (Branching programs and space-bounded tests); exit by being able to state QRCC-IX-BB-010 (Nisan–Zuckerman randomness-efficient space simulation) with its ownership boundary.
- Ch. 5: Computational Pseudorandomness and the Next-Bit Test:** enter through QRCC-IX-DEF-006 (Computational pseudorandom generator); exit by being able to state QRCC-IX-THM-007 (Yao next-bit characterization) with its ownership boundary.
- Ch. 6: Mild Average Hardness and Hard-Core Sets:** enter through QRCC-IX-DEF-010 (Average circuit hardness and hard-core advantage); exit by being able to state QRCC-IX-THM-011 (Impagliazzo hard-core lemma) with its ownership boundary.
- Ch. 7: Hardness Amplification: XOR and the Exponential-Time Bridge:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-IX-THM-012 (XOR and direct-product hardness amplification) with its ownership boundary.
- Ch. 8: Nisan–Wigderson Designs and Reconstruction:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-IX-ALG-013 (Combinatorial designs for Nisan–Wigderson) with its ownership boundary.
- Ch. 9: The Impagliazzo–Wigderson Simulation:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-IX-THM-017 (Impagliazzo–Wigderson derandomization theorem) with its ownership boundary.
- Ch. 10: Hitting Sets, One-Sided Derandomization, and Pseudodeterminism:** enter through QRCC-IX-DEF-018 (Hitting-set generator and black-box derandomization); exit by being able to state QRCC-IX-THM-019 (PRG-to-hitting-set and hitting-set-to-one-sided derandomization) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-IX-THM-005 — Yao minimax principle. Dependencies: no earlier formal dependency. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-IX-THM-003 — Adleman theorem BPP subset P/poly. Dependencies: QRCC-IX-THM-002, QRCC-II-THM-013. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-IX-THM-007 — Yao next-bit characterization. Dependencies: QRCC-IX-DEF-006. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

4. QRCC-IX-THM-012 — XOR and direct-product hardness amplification. Dependencies: QRCC-IX-THM-011. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-IX-THM-017 — Impagliazzo–Wigderson derandomization theorem. Dependencies: QRCC-IX-THM-015, QRCC-IX-THM-016, QRCC-II-THM-024. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-IX-CONJ-024 — BPP equals P. Dependencies: QRCC-IX-DEF-001. *open frontier: separate the conjecture from the strongest proved substitute and never route a later theorem through the conjecture.*

Common failure modes

- switching between average-over-input and worst-case-over-input error without notice;
- using a one-sided hitting set as though it derandomized a two-sided algorithm;
- hiding nonuniform advice inside a supposedly uniform generator;
- forgetting the quantitative hardness and stretch needed by a hardness-versus-randomness theorem;

Boundary and degenerate cases

- zero-error algorithms versus bounded error;
- seed length as large as the original randomness budget;
- uniform versus advice-taking simulations;
- decision versus search/pseudodeterministic outputs;

Retrieval and reconstruction drill

1. State QRCC-IX-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-IX-THM-005. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-IX-THM-003, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.

6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-IX-CONJ-024, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume X studies the circuit models against which derandomization hardness is measured and records the known lower-bound barriers.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part X

Boolean Circuit Complexity and Barriers

Chapter 1

Circuit Models, Formulas, and Basis Robustness

Circuit lower bounds are meaningful only after the computational model is frozen. A change of a fixed bounded-fan-in basis should cost only constants, whereas replacing an unbounded-fan-in gate by fan-in-two gates can change the depth asymptotically. This chapter records the conventions used throughout Volume X and proves exactly the bounded-fan-in robustness that later arguments may invoke.

1.1 Circuit and formula conventions

Definition 1.1 (Boolean circuits, formulas, and standard families). A *Boolean basis* $\mathcal{B}$ is a finite set of Boolean functions. For $b \in \mathcal{B}$, write $a(b)$ for its arity and put

$$r_{\mathcal{B}} := \max(2, \max_{b \in \mathcal{B}} a(b)).$$

The basis is *functionally complete* if every Boolean function of every finite arity has a formula over $\mathcal{B}$ (with constants allowed at leaves).

A single-output $\mathcal{B}$ -circuit on variables $x_1, \dots, x_n$ is a finite directed acyclic graph. Its source vertices are labeled by input variables or constants 0, 1; every non-source vertex is labeled by a gate $b \in \mathcal{B}$ and has $a(b)$ ordered incoming wires; one vertex is designated as the output. Repeated wires into different input ports are allowed. The *size* $|C|$ is the number of non-source gates. The *depth* of an input or constant is 0, the depth of a gate is one plus the largest depth of an input to that gate, and the depth of the circuit is the depth of its output. Gate fan-in is its arity; fan-out is the number of outgoing wire occurrences. A multi-output circuit has several designated outputs and uses the same size/depth convention.

A *formula* is a rooted tree whose internal vertices are gates and whose leaves are labeled by variables or constants. Thus a computed gate value is never shared by two parents. Equivalently, a single-output circuit becomes a formula when every non-output gate has fan-out at most one after repeated input occurrences are duplicated. Formula size is the number of internal gates and formula depth is the longest root-to-leaf gate count.

A circuit family $\{C_n\}_{n \geq 1}$ *computes* a language L if C_n has n input bits and $C_n(x) = 1$ exactly for $x \in L \cap \{0, 1\}^n$. It is *polynomial-time uniform* if one deterministic polynomial-time transducer outputs a standard encoding of C_n from 1^n , and *log-space uniform* if the transducer can instead be implemented in deterministic logarithmic work space. A family with no such construction

requirement is *nonuniform*. These are construction conventions; a size/depth upper bound alone does not imply uniformity.

The standard classes used below have polynomial size unless stated otherwise:

1. AC^0 consists of constant-depth circuits over unbounded-fan-in $\wedge, \vee$ and fan-in-one $\neg$ gates.
2. $AC^0[m]$, for fixed $m \geq 2$, additionally permits an unbounded-fan-in MOD_m gate. In this volume the bare gate $\text{MOD}_m(y_1, \dots, y_r)$ outputs 1 exactly when $\sum_i y_i \equiv 0 \pmod{m}$. Residue- a variants are obtained by appending constants, and the complementary convention is obtained by one NOT gate. Thus these convention changes do not alter the class. Finally, $ACC^0 := \bigcup_{m \geq 2} AC^0[m]$.
3. TC^0 permits unbounded-fan-in threshold gates, whose output is 1 when a fixed integer-weighted sum of the inputs meets a fixed threshold.
4. NC^1 uses bounded-fan-in Boolean gates and depth $O(\log n)$.

Uniform versions of these classes are named explicitly when uniformity is needed; otherwise the circuit-class notation is nonuniform.

A *monotone circuit* uses only monotone gates (in particular, the usual basis is $\wedge, \vee$ with no input negations) and therefore computes a coordinatewise monotone Boolean function. A *deterministic branching program* is a directed graph with a start node, terminal 0/1 nodes, and nonterminal nodes labeled by input variables, each having outgoing edges labeled 0 and 1; on input x the corresponding edge is followed. A layered branching program has edges only between consecutive layers; its length is the number of transitions and its width is the maximum number of nodes in a layer. Branching programs are recorded here to fix terminology, but the first two chapters use only circuits and formulas.

Convention checks. A formula may contain many leaves labeled by the same variable; those are separate occurrences, not shared computed gates. Conversely, one circuit gate may feed arbitrarily many later gates unless a fan-out restriction is stated. The word “bounded fan-in” always means a constant independent of n ; it does not mean that an n -input AND is a single bounded-fan-in gate.

1.2 Fixed bounded-fan-in bases are robust

Theorem 1.2 (Circuit-basis robustness and normal forms). *Let $\mathcal{B}$ and $\mathcal{B}'$ be fixed finite functionally complete bounded-fan-in Boolean bases. There are constants $c_s, c_d \geq 1$, depending only on the two bases, with the following properties.*

1. *Every $\mathcal{B}$ -circuit of size s and depth D has an equivalent $\mathcal{B}'$ -circuit of size at most $c_s s$ and depth at most $c_d D$. The same gate-by-gate replacement maps formulas to formulas with the same bounds.*
2. *In particular, every such circuit may be converted with constant-factor size and depth overhead to the De Morgan basis $\{\wedge_2, \vee_2, \neg\}$.*
3. *After that conversion, a size- s , depth- D fan-in-two circuit can be made strictly layered—every wire goes from layer i to layer $i+1$ —with depth D and size $O(sD)$ by inserting identity buffers. Consequently an arbitrary fixed-basis circuit has a strict layered fan-in-two realization of depth $O(D)$ and size $O(sD)$.*

No corresponding constant-depth conclusion is asserted when an unbounded-fan-in gate is replaced by fan-in-two gates: balancing an n -input AND or OR can require depth $\Theta(\log n)$.

Proof. Because $\mathcal{B}'$ is functionally complete, each fixed gate function $b \in \mathcal{B}$ has some $\mathcal{B}'$ -formula F_b computing it. The set $\mathcal{B}$ is finite, so

$$c_s := \max_{b \in \mathcal{B}} |F_b|, \quad c_d := \max_{b \in \mathcal{B}} \text{depth}(F_b) \quad (1.1)$$

are finite constants after replacing a zero-size projection, if present, by a harmless identity realization.

Take a topological order of the gates of a $\mathcal{B}$ -circuit C and replace every gate labeled b by a fresh copy of F_b , wiring the roots and leaves to the same predecessors and successors. Each old gate creates at most c_s new gates, so the new size is at most $c_s s$. Any input-output path of C meets at most D old gates; inside each replacement gadget it meets at most c_d gates. The new depth is therefore at most $c_d D$. If C is a formula, use a fresh gadget at each tree vertex. No gate value is shared by two parents, so the result remains a formula. This proves (1). Taking $\mathcal{B}' = \{\wedge_2, \vee_2, \neg\}$ proves (2).

For (3), let C already have fan-in at most two. Give each gate its usual depth level. If a wire goes from a vertex at level i to a gate at level $j > i + 1$, subdivide that wire by $j - i - 1$ identity buffers, placing one at each intermediate level. An identity buffer can be realized by $u \mapsto u \vee u$ (or $u \wedge u$), so it costs one fan-in-two De Morgan gate. The new depth is still D and every wire joins adjacent layers. There are at most $2s$ incoming gate-wires, and each contributes fewer than D buffers, giving at most $s + 2s(D - 1) = O(sD)$ gates.

Finally, the last warning is a model distinction rather than an omitted optimization. A binary tree computing an AND of n independent inputs has $\lceil \log_2 n \rceil$ depth, whereas an unbounded-fan-in AND is one AC^0 gate. Thus bounded-basis robustness cannot be used to erase the bounded/unbounded fan-in distinction. $\square$

Worked normalization check. The NAND basis is functionally complete: $\neg x = \text{NAND}(x, x)$ and $x \wedge y = \neg \text{NAND}(x, y)$, while OR follows from De Morgan. Conversely a NAND is one AND followed by one NOT. Hence NAND and De Morgan circuits simulate one another with constant overhead, exactly as the theorem predicts.

Scope checkpoint. The theorem concerns fixed finite bounded-fan-in bases. It does not identify AC^0 , ACC^0 , TC^0 , and NC^1 , and it does not turn a nonuniform family into a uniform one. Later lower bounds must retain the model, uniformity, and fan-in convention stated in their own hypotheses.

Chapter 2

Counting Circuits and the Shannon Barrier

There are doubly exponentially many Boolean functions on n bits but far fewer small circuits. The resulting lower bound is nonconstructive: it says that almost every truth table is hard, without identifying an explicit hard function. This gap between abundance and explicitness is the first barrier that later circuit-lower-bound methods must overcome.

2.1 A topological circuit count

Theorem 2.1 (Shannon counting lower bound in the QRCC circuit model). *Fix a finite bounded-fan-in Boolean basis $\mathcal{B}$. There is a constant $c_{\mathcal{B}} > 0$ such that, for all sufficiently large n , the fraction of Boolean functions $f : \{0, 1\}^n \rightarrow \{0, 1\}$ computable by $\mathcal{B}$ -circuits of size at most*

$$s_n = \left\lfloor c_{\mathcal{B}} \frac{2^n}{n} \right\rfloor \quad (2.1)$$

is at most

$$2^{-2^{n-1}}. \quad (2.2)$$

In particular, almost every n -input Boolean function requires $\Omega_{\mathcal{B}}(2^n/n)$ gates.

Proof. Let $q = |\mathcal{B}|$ and put $r = \max(2, \max_{b \in \mathcal{B}} a(b))$. We deliberately overcount circuits. For a circuit with exactly t noninput gates, number the gates in a topological order $1, \dots, t$. At gate j , choose a gate type in at most q ways and choose each of at most r ordered input ports from the n variables, the two constants, or earlier gates. Replacing “earlier” by the larger common pool of $n + t + 2$ possible sources only increases the count. Finally choose the output from the same pool. Hence the number N_t of Boolean functions that can be represented by size- t circuits is at most the number of such labeled descriptions:

$$N_t \leq (n + t + 2)(q(n + t + 2)^r)^t. \quad (2.3)$$

Different descriptions may compute the same function, so (2.3) remains a valid upper bound.

Summing over $0 \leq t \leq s$ and using monotonicity of the right side gives

$$N_{\leq s} \leq (s + 1)(n + s + 2)(q(n + s + 2)^r)^s. \quad (2.4)$$

Choose, for example,

$$c_{\mathcal{B}} = \frac{1}{8r}. \quad (2.5)$$

For all sufficiently large n , the value $s = s_n$ satisfies $n + s + 2 \leq 2^n$ and $\log_2 q \leq rn$. Taking base-two logarithms in (2.4), we obtain

$$\begin{aligned} \log_2 N_{\leq s} &\leq \log_2(s+1) + \log_2(n+s+2) + s(\log_2 q + r \log_2(n+s+2)) \\ &\leq o(2^n) + 2rn s \\ &\leq o(2^n) + \frac{1}{4} 2^n \leq \frac{1}{2} 2^n \end{aligned} \tag{2.6}$$

for all sufficiently large n . Thus $N_{\leq s} \leq 2^{2^{n-1}}$.

There are exactly 2^{2^n} Boolean functions on n bits, one for each truth table of length 2^n . Dividing the preceding bound by this total gives

$$\frac{N_{\leq s}}{2^{2^n}} \leq 2^{2^{n-1}-2^n} = 2^{-2^{n-1}},$$

which proves (2.2). Since the exceptional fraction tends to zero, a uniformly random truth table requires more than $c_B 2^n/n$ gates with probability tending to one. $\square$

Why this is a counting theorem, not an explicit lower bound. The proof does not name a hard function. It shows that the set of small circuit descriptions is too small to cover the truth-table space. Deciding which individual truth tables lie outside that set can itself be difficult, and later chapters therefore require methods that exploit structure of a specific explicit function.

Historical ownership. Shannon's 1949 switching-circuit work is the historical source for the counting/synthesis viewpoint. The precise finite-basis DAG convention, constant c_B , and estimate (2.2) above are the QRCC local proof; they are not presented as a verbatim theorem statement from Shannon's circuit model.

2.2 A constructive near-matching upper bound

Theorem 2.2 (Lupanov upper bound: local constant-factor reconstruction). *There is an absolute constant $C > 0$ such that every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ has a De Morgan circuit of size at most*

$$C \frac{2^n}{n} \tag{2.7}$$

for all sufficiently large n .

The classical Lupanov synthesis theorem is sharper: in the same bounded-fan-in Boolean-circuit setting the worst-case size is

$$(1 + o(1)) \frac{2^n}{n}. \tag{2.8}$$

The proof below is QRCC-owned and deliberately proves only the constant-factor form (2.7); the source audit records (2.8) as the sharper classical comparison.

Proof. All logarithms in this proof are base two. The construction has two reusable pieces.

A bank of all small Boolean functions. Let $B(k)$ be the number of gates in one multi-output De Morgan circuit whose outputs include *every* Boolean function of variables $y_1, \dots, y_k$. From a bank for $k-1$ variables, precompute $\neg y_k$ once and, for every ordered pair (g_0, g_1) of $(k-1)$ -variable Boolean functions already present in the bank, add the three-gate multiplexer

$$(\neg y_k \wedge g_0) \vee (y_k \wedge g_1). \tag{2.9}$$

Every k -variable Boolean function is obtained from exactly one such ordered pair of cofactors. Since there are $2^{2^{k-1}}$ functions on $k-1$ variables, there are 2^{2^k} ordered pairs. Consequently

$$B(k) \leq B(k-1) + 1 + 3 \cdot 2^{2^k}, \quad (2.10)$$

and hence

$$B(k) = O(2^{2^k}). \quad (2.11)$$

The important point is that the entire bank is computed once and its outputs may have arbitrary fan-out.

A shared Shannon multiplexer tree. Choose, for sufficiently large n ,

$$k = \lfloor \log_2(n - \log_2 n) \rfloor, \quad t = n - k. \quad (2.12)$$

Regard $x_{t+1}, \dots, x_n$ as the final k variables. For every assignment $a \in \{0, 1\}^t$ to the first t variables, the residual function

$$f_a(x_{t+1}, \dots, x_n) := f(a, x_{t+1}, \dots, x_n)$$

is one of the outputs of the k -variable bank. Starting from these 2^t outputs, reconstruct f by a binary tree of multiplexers, one Shannon variable per level. Sharing the t input negations, this costs at most $3(2^t - 1) + t$ additional gates.

It remains to bound the two terms. From the definition of k ,

$$2^k \leq n - \log_2 n,$$

so

$$2^{2^k} \leq 2^{n - \log_2 n} = \frac{2^n}{n}. \quad (2.13)$$

Also $2^k > (n - \log_2 n)/2$, whence

$$2^t = \frac{2^n}{2^k} < \frac{2^{n+1}}{n - \log_2 n} = O\left(\frac{2^n}{n}\right). \quad (2.14)$$

Combining (2.11), (2.13), and (2.14) proves (2.7). $\square$

Where the $1/n$ saving comes from. A truth-table construction pays separately for essentially every one of the 2^n rows. Here the final $k \simeq \log n$ variables are handled by a single shared bank of all their possible cofactors. Choosing $2^k \simeq n$ makes both the bank cost 2^{2^k} and the multiplexer-tree cost 2^{n-k} of order $2^n/n$. Lupanov's finer local-coding decomposition removes the remaining constant-factor slack and yields the sharp coefficient $1 + o(1)$.

Calibration with Shannon counting. Theorems 2.1 and 2.2 together show that the maximum bounded-fan-in circuit complexity of an n -input Boolean function is $\Theta(2^n/n)$. Only the lower theorem is nonconstructive: Theorem 2.2 gives an explicit synthesis procedure from the truth table.

Chapter 3

Explicit Formula Lower Bounds

Circuit sharing destroys many elementary counting arguments, but formulas are trees. Two classical methods exploit that tree structure in complementary ways. Nechiporuk counts how many distinct residual functions one block of variables can induce. Khrapchenko counts adjacent 0- and 1-inputs that a literal leaf must separate. Both methods yield explicit superlinear lower bounds, unlike the nonconstructive Shannon count.

3.1 Nechiporuk's subfunction method

For this section let $\mathcal{B}_2$ denote the full binary basis consisting of all sixteen Boolean functions of two arguments. The *leafsize* $\ell(F)$ of a $\mathcal{B}_2$ -formula is its number of variable/constant leaves. If V is a set of variables, write $\ell_V(F)$ for the number of leaves labeled by variables from V .

Given a Boolean function f on variable set X and $V \subseteq X$, a *V-subfunction* is obtained by fixing every variable in $X \setminus V$ to constants. Let $r_V(f)$ denote the number of distinct *nonconstant* V -subfunctions of f .

Theorem 3.1 (Nechiporuk subfunction lower bound). *Let f be a Boolean function on variable set X , and let $V_1, \dots, V_m \subseteq X$ be pairwise disjoint. Then every $\mathcal{B}_2$ -formula F computing f satisfies*

$$\ell(F) \geq \frac{1}{4} \sum_{i=1}^m \log_2(\max\{1, r_{V_i}(f)\}). \quad (3.1)$$

Consequently a De Morgan formula of QRCC gate-size s satisfies

$$s + 1 \geq \frac{1}{4} \sum_{i=1}^m \log_2(\max\{1, r_{V_i}(f)\}). \quad (3.2)$$

In particular, there is an explicit family ED_N on $M = 2N \log_2 N$ input bits, for powers of two N , whose De Morgan formula size is $\Omega(M^2 / \log M)$.

Proof. We first prove the block estimate. Fix $V \subseteq X$ and let $l = \ell_V(F)$. If $l = 0$, no nonconstant V -subfunction exists. Assume $l \geq 1$. Inside the formula tree, retain the l leaves labeled by variables from V and every path from those leaves to the root. Compress every maximal path whose internal vertices have only one retained child into a single edge. The resulting skeleton is a rooted binary tree with l leaves, at most $l - 1$ branching vertices, and at most $2l - 1$ compressed path segments.

Now fix all variables outside V . Every discarded subtree becomes a constant. Along one compressed path segment, each original binary gate has one continuing input and one fixed input, so the whole segment acts on its continuing value as one of the four unary Boolean maps

$$0, \quad 1, \quad z, \quad \neg z. \quad (3.3)$$

The branch-gate labels and the V -leaf labels are fixed by F itself. Hence all variation among V -subfunctions is described by choosing at most one of four unary maps on each of at most $2l - 1$ segments. Therefore

$$r_V(f) \leq 4^{2l-1} < 16^l, \quad (3.4)$$

and thus

$$\ell_V(F) = l \geq \frac{1}{4} \log_2 r_V(f) \quad (3.5)$$

whenever $r_V(f) > 0$. Summing (3.5) over pairwise disjoint V_i is valid because every formula leaf belongs to at most one block:

$$\ell(F) \geq \sum_i \ell_{V_i}(F),$$

which proves (3.1).

For (3.2), start from any De Morgan formula in the convention of Definition 1.1. Push negations to the variables by De Morgan's laws and remove double negations. Regard a negated variable as a literal leaf. The remaining internal gates are binary AND/OR gates, hence a full-binary formula with at most $s + 1$ leaves. Applying (3.1) gives (3.2).

It remains to give an explicit family. Let $N = 2^k$. The input to ED_N consists of N blocks

$$X_1, \dots, X_N \in \{0, 1\}^{2k},$$

so each block names one element of a universe of size N^2 and the total input length is $M = 2kN = 2N \log_2 N$. Define $\text{ED}_N = 1$ exactly when the N named elements are pairwise distinct.

Take V_i to be the $2k$ bits of block X_i . Fix the other $N - 1$ blocks to pairwise distinct universe elements forming a set S . The induced subfunction on X_i is

$$g_S(z) = 1 \iff z \notin S. \quad (3.6)$$

Distinct $(N - 1)$ -subsets $S \subseteq [N^2]$ give distinct nonconstant subfunctions. Thus

$$r_{V_i}(\text{ED}_N) \geq \binom{N^2}{N-1} \geq \left(\frac{N^2}{N-1} \right)^{N-1} \geq N^{N-1}. \quad (3.7)$$

Using all N blocks in (3.1),

$$\ell(\text{ED}_N) \geq \frac{1}{4} N(N-1) \log_2 N. \quad (3.8)$$

Since $M = 2N \log_2 N$ and $\log N = \Theta(\log M)$, (3.8) is $\Omega(M^2 / \log M)$. Equation (3.2) transfers the same asymptotic lower bound to QRCC De Morgan gate-size. $\square$

Why formula sharing matters. The proof charges each variable occurrence to only one disjoint block. In a general circuit one computed subfunction can fan out to many later gates, so the same leaf-occurrence accounting is unavailable. Nechiporuk's method is therefore a formula lower-bound method, not a general-circuit lower bound in disguise.

Source comparison. Nechiporuk’s 1966 result introduced the explicit near-quadratic phenomenon. The common modern form counts all distinct block subfunctions and obtains the same $\frac{1}{4} \sum_i \log_2 s_i$ scale over the full binary basis. The local proof above counts nonconstant subfunctions so that the zero-occurrence case is literal and no hidden additive constant is needed; the element-distinctness application has only nonconstant subfunctions of the type used in (3.6), so the exponent is unchanged.

3.2 Khrapchenko’s adjacency method

A *literal-leaf De Morgan formula* is a binary AND/OR formula whose leaves are x_i , $\neg x_i$, or constants. Let $L_{\text{lit}}(f)$ be its minimum number of leaves. Every QRCC De Morgan formula of gate-size s normalizes to such a formula with at most $s + 1$ leaves.

For disjoint sets $A \subseteq f^{-1}(1)$ and $B \subseteq f^{-1}(0)$ define

$$E(A, B) := \{(a, b) \in A \times B : d_H(a, b) = 1\}. \quad (3.9)$$

Theorem 3.2 (Khrapchenko adjacency bound). *For every Boolean function f and every nonempty $A \subseteq f^{-1}(1)$, $B \subseteq f^{-1}(0)$,*

$$L_{\text{lit}}(f) \geq \frac{|E(A, B)|^2}{|A||B|}. \quad (3.10)$$

Hence any QRCC De Morgan formula of gate-size s obeys

$$s \geq \frac{|E(A, B)|^2}{|A||B|} - 1. \quad (3.11)$$

For parity $\text{PARITY}_n = x_1 \oplus \cdots \oplus x_n$ this gives

$$L_{\text{lit}}(\text{PARITY}_n) \geq n^2, \quad s_{\text{QRCC}}(\text{PARITY}_n) \geq n^2 - 1. \quad (3.12)$$

When n is a power of two, the literal-leaf lower bound n^2 is attained by a balanced recursive construction.

Proof. Let F be a literal-leaf De Morgan formula for f . We first associate with its leaves a partition of $A \times B$ into rectangles. Start with the whole pair (A, B) at the root. At an OR gate $G \vee H$, every $b \in B$ makes both children zero, while every $a \in A$ makes at least one child one. Assign each a deterministically to one child that is one and route the pair (a, b) to that child. This partitions the current rectangle according only to its A -coordinate. At an AND gate $G \wedge H$, every $a \in A$ makes both children one, while every $b \in B$ makes at least one child zero; choose a zero child deterministically from b , partitioning according only to the B -coordinate. Recursing reaches the leaves and partitions $A \times B$ into at most $L := \ell(F)$ disjoint rectangles

$$R_j = S_j \times T_j. \quad (3.13)$$

At a leaf labeled by a literal on coordinate i , every $(a, b) \in S_j \times T_j$ differs in that literal’s value, hence in coordinate i .

Let $c_j = |E(A, B) \cap R_j|$, $s_j = |S_j|$, and $t_j = |T_j|$. Inside a leaf rectangle all pairs differ in one fixed coordinate i . For each $a \in S_j$ there is at most one $b \in T_j$ at Hamming distance one from a (and conversely), so

$$c_j \leq \min(s_j, t_j) \leq \sqrt{s_j t_j}. \quad (3.14)$$

Because the rectangles partition $A \times B$,

$$\sum_j s_j t_j = |A| |B|, \quad \sum_j c_j = |E(A, B)|. \quad (3.15)$$

Cauchy–Schwarz and (3.14) give

$$|E(A, B)|^2 = \left(\sum_j c_j \right)^2 \leq L \sum_j c_j^2 \leq L \sum_j s_j t_j = L |A| |B|, \quad (3.16)$$

which proves (3.10). The normalization discussed before the theorem yields (3.11).

For parity take $A = \text{PARITY}_n^{-1}(1)$ and $B = \text{PARITY}_n^{-1}(0)$. Both sets have size 2^{n-1} , and flipping any one of the n coordinates changes parity. Therefore

$$|E(A, B)| = n 2^{n-1},$$

and (3.10) gives $L_{\text{lit}} \geq n^2$.

For the matching construction when $n = 2^r$, compute parity P_m and its complement Q_m recursively on two disjoint halves of the variables. With $m = 2u$, write P_u^L, Q_u^L for the left-half formulas and P_u^R, Q_u^R for the right-half formulas. Then

$$P_m = (P_u^L \wedge Q_u^R) \vee (Q_u^L \wedge P_u^R), \quad Q_m = (P_u^L \wedge P_u^R) \vee (Q_u^L \wedge Q_u^R). \quad (3.17)$$

Every displayed occurrence is a fresh formula copy; there is no sharing. Starting from one literal leaf at $m = 1$, the parity leaf count therefore satisfies $L(2u) = 4L(u)$, and hence $L(n) = n^2$ for powers of two. $\square$

Method boundary. The adjacency ratio in (3.10) is at most quadratic in n for Hamming-distance one edges. Thus Khrapchenko’s method beautifully certifies the exact quadratic parity leafsize in the power-of-two case, but by itself it cannot produce a superquadratic formula lower bound. Later formula methods must bring in additional structure.

3.3 Information complexity and the composition frontier

The Gavinsky–Meir–Weinstein–Wigderson source map

The modern information-complexity route to formula lower bounds is most safely recorded through the theorem it actually proves, rather than through a generic “information complexity gives stronger formula lower bounds” slogan. Let R_g denote the Karchmer–Wigderson relation of a nonconstant Boolean function $g : \{0, 1\}^m \rightarrow \{0, 1\}$, let U_n denote the n -bit universal relation, and let $R_{g \diamond U_n}$ be the composition relation defined in the source. If $L(g)$ is the De Morgan formula leafsize of g , then the Gavinsky–Meir–Weinstein–Wigderson main theorem gives

$$C(R_{g \diamond U_n}) \geq \log_2 L(g) + n - O\left(\left(1 + \frac{m}{n}\right) \log m\right). \quad (3.18)$$

The same lower bound applies to the logarithm of the number of leaves of any protocol for $R_{g \diamond U_n}$. In particular, this establishes the desired additive composition behaviour, up to the displayed loss, when the inner problem is the universal relation.

The conceptual contribution relevant to the present volume is the use of *information complexity*: the logarithm of formula size is reinterpreted through the information transmitted by protocols for Karchmer–Wigderson relations, and the chain rule permits information to be charged across the independent inner instances. This is a genuine step toward the full Karchmer–Raz–Wigderson composition conjecture, but it does not prove that conjecture for arbitrary $g \diamond f$.

Source boundary. Equation (3.18) is source-owned and is not locally reproved here. The 2017 journal paper proves composition with the universal relation; its introduction also recalls the earlier Andreev–Håstad parity-composition bound $L(g \diamond \text{PARITY}_n) = L(g)n^2 / \text{polylog}(m)$. That recalled parity theorem is not promoted here as a new consequence of the information-complexity argument. Thus this frontier records the exact composition theorem and method boundary, not an invented stand-alone formula lower-bound exponent.

Chapter 4

Random Restrictions and the Switching Lemma

Random restrictions are the depth-reduction engine behind the classical AC^0 lower bounds. The point is not merely that many variables are fixed: a small-width DNF or CNF typically becomes a function of small *decision-tree depth*, and a shallow decision tree can be written in both normal forms. This chapter fixes the probability space and depth conventions before any switching argument is used downstream.

4.1 Restriction, width, and decision-tree conventions

A *restriction* on variables $x_1, \dots, x_n$ is a vector $\rho \in \{0, 1, *\}^n$. If $\rho_i \in \{0, 1\}$, then x_i is fixed to that value; if $\rho_i = *$, then x_i remains a live variable. For a Boolean function or formula F , write $F \upharpoonright \rho$ for the restricted function. If ρ and π have disjoint sets of live coordinates on which π supplies values, then $\rho\pi$ denotes their compatible composition.

For $0 \leq p \leq 1$, the QRCC *p-random restriction* R_p is the product distribution in which, independently for every coordinate,

$$\Pr[\rho_i = *] = p, \quad \Pr[\rho_i = 0] = \Pr[\rho_i = 1] = \frac{1-p}{2}. \quad (4.1)$$

Thus p is the survival probability, not the fixing probability.

A *width- w DNF* is an OR of conjunctions of at most w literals; a *width- w CNF* is an AND of disjunctions of at most w literals. Repeated literals, contradictory DNF terms, and tautological CNF clauses may be deleted without increasing width, so in the switching proof every surviving term is taken to contain each variable at most once.

A deterministic *decision tree* is a rooted binary tree whose internal nodes query live input variables and whose two outgoing edges are labeled 0 and 1; leaves are labeled by output bits. Its depth is the largest number of queries on a root-to-leaf path. For a Boolean function g , let $D(g)$ denote the minimum depth of a deterministic decision tree computing g . In particular, $D(g) = 0$ exactly for constant g . A depth- d decision tree gives both a width- d DNF, by OR-ing the root-to-1 paths, and a width- d CNF, by AND-ing clauses that exclude the root-to-0 paths. This last observation is what turns a decision-tree simplification statement into a DNF/CNF *switch*.

4.2 A complete two-variable restriction table

Consider the width-2 DNF

$$F(x_1, x_2) = (x_1 \wedge x_2) \vee (\neg x_1 \wedge \neg x_2), \quad (4.2)$$

which is the equality function on two bits. Put $q = (1 - p)/2$. All nine restrictions, their residual functions, minimum decision-tree depths, and R_p weights are as follows.

ρ_1	ρ_2	$F \upharpoonright \rho$	$D(F \upharpoonright \rho)$	weight
0	0	1	0	q^2
0	1	0	0	q^2
0	*	$\neg x_2$	1	pq
1	0	0	0	q^2
1	1	1	0	q^2
1	*	x_2	1	pq
*	0	$\neg x_1$	1	pq
*	1	x_1	1	pq
*	*	F	2	p^2

Fix the displayed term order in (4.2), and within the first term query x_1 before x_2 . Under the all-star restriction the canonical tree first selects $x_1 \wedge x_2$ and queries both variables. The branch 11 already accepts; on the other three leaves the second term is either satisfied or falsified by the two queried bits. Hence the canonical tree has depth two, matching the minimum depth of equality. The table gives the exact tail probabilities

$$\Pr[D(F \upharpoonright R_p) \geq 1] = 4pq + p^2 = 2p - p^2, \quad \Pr[D(F \upharpoonright R_p) \geq 2] = p^2. \quad (4.3)$$

For example, at $p = 1/20$ the depth-two failure probability is exactly $1/400$. The source-sharp switching estimate $(5pw)^2 = 1/4$ and the locally proved estimate below $(8pw/(1-p))^2 = 256/361$ are deliberately much looser on this tiny instance; the content of the lemma is the uniform exponential tail for arbitrarily many terms and variables.

4.3 The canonical decision tree

Let

$$F = C_1 \vee \cdots \vee C_m \quad (4.4)$$

be a DNF with a fixed ordering of its terms and a fixed ordering of the variables inside each term. Given a restriction ρ , define the *canonical decision tree* $T(F, \rho)$ recursively.

1. If every C_j is falsified by ρ , output 0.
2. Otherwise let C_j be the first term not falsified by ρ . If it is already satisfied, output 1.
3. Otherwise query, in their fixed order, all variables of C_j that are still live under ρ . For each assignment π returned on those queries, continue recursively with $T(F, \rho\pi)$ unless C_j has become satisfied, in which case that branch outputs 1.

The tree computes $F \upharpoonright \rho$, so

$$D(F \upharpoonright \rho) \leq \text{depth } T(F, \rho). \quad (4.5)$$

The canonical tree is an analysis device; $D(\cdot)$ always denotes the minimum decision-tree depth, not necessarily the depth of this particular tree.

4.4 A weighted encoding proof

Theorem 4.1 (Switching lemma in the QRCC product-restriction convention). *Let F be a width- w DNF or width- w CNF, let $t \geq 1$, and let $0 \leq p < 1$. Under the product restriction R_p of (4.1),*

$$\Pr[D(F \upharpoonright R_p) \geq t] \leq \min \left\{ 1, \left(\frac{8pw}{1-p} \right)^t \right\}. \quad (4.6)$$

Consequently, whenever $p \leq 1/9$,

$$\Pr[D(F \upharpoonright R_p) \geq t] \leq (9pw)^t. \quad (4.7)$$

Equivalently, outside an event of probability at most the right-hand side of (4.6), a width- w DNF becomes both a width- $(t-1)$ DNF and a width- $(t-1)$ CNF after restriction; the dual statement holds for a width- w CNF.

Proof. The case $p = 0$ is immediate because every variable is fixed, so assume $0 < p < 1$. It suffices to prove the DNF case. Fix the term and variable orderings used in the canonical tree. Let $\mathcal{S}$ be the set of restrictions ρ for which $T(F, \rho)$ has depth at least t . By (4.5), the event in (4.6) is contained in $\mathcal{S}$.

For a restriction ρ , write

$$\mu_p(\rho) = p^{|\text{Stars}(\rho)|} \left(\frac{1-p}{2} \right)^{n-|\text{Stars}(\rho)|} \quad (4.8)$$

for its R_p probability. For each $\rho \in \mathcal{S}$, choose the first root-to-node path of length t in $T(F, \rho)$ under the fixed lexicographic ordering of branch labels. Along its first t queries, suppose the canonical procedure encounters terms

$$C_{j_1}, \dots, C_{j_r}$$

and, from those terms, consecutive nonempty blocks of queried variables

$$B_1, \dots, B_r, \quad |B_1| + \dots + |B_r| = t. \quad (4.9)$$

The last block may be only an initial segment of the live variables of its term. Let $\pi_i \in \{0, 1\}^{B_i}$ be the branch assignment on B_i . Because C_{j_i} is a conjunction and has not been falsified before its queries begin, there is a unique assignment $\sigma_i \in \{0, 1\}^{B_i}$ that makes every queried literal of C_{j_i} true. Put

$$\hat{\rho} := \rho \sigma_1 \dots \sigma_r. \quad (4.10)$$

Exactly the t starred coordinates in the disjoint blocks B_i are filled, so

$$|\text{Stars}(\hat{\rho})| = |\text{Stars}(\rho)| - t \quad (4.11)$$

and therefore

$$\mu_p(\hat{\rho}) = \left(\frac{1-p}{2p} \right)^t \mu_p(\rho). \quad (4.12)$$

We now record enough auxiliary data to recover ρ from $\hat{\rho}$. For each queried variable in B_i , record its position among the at most w variables of C_{j_i} and one delimiter bit marking whether it is the last queried variable of the block B_i . Across all t queries this location/delimiter string has at most $(2w)^t$ possibilities. Also record all t branch bits in $\pi_1 \dots \pi_r$, contributing 2^t possibilities. Thus the total auxiliary alphabet has size at most

$$(4w)^t. \quad (4.13)$$

The map

$$\rho \mapsto (\hat{\rho}, \text{location/delimiter code}, \pi_1 \cdots \pi_r) \quad (4.14)$$

is injective on $\mathcal{S}$. Indeed, start from $\hat{\rho}$. The first term not falsified by $\hat{\rho}$ is exactly C_{j_1} : all earlier terms were already falsified by fixed bits of ρ , and (4.10) changes only coordinates that were stars of ρ ; meanwhile C_{j_1} is not falsified because σ_1 satisfies its queried literals. The first block of the location/delimiter code therefore recovers B_1 , and the first branch bits recover π_1 . Replacing σ_1 by π_1 in $\hat{\rho}$ yields

$$\rho \pi_1 \sigma_2 \cdots \sigma_r.$$

The first term not falsified by this restriction is exactly C_{j_2} for the same reason. Iterating recovers every B_i and π_i . Once all t affected coordinates are known, replacing their σ_i values by stars recovers the original ρ . This proves injectivity.

Fix one auxiliary code a , and let $\mathcal{S}_a$ be the restrictions in $\mathcal{S}$ carrying that code. By injectivity, the corresponding $\hat{\rho}$ are distinct. Summing (4.12) over $\rho \in \mathcal{S}_a$ and using that the total R_p weight of all restrictions is one gives

$$\sum_{\rho \in \mathcal{S}_a} \mu_p(\rho) \leq \left(\frac{2p}{1-p} \right)^t. \quad (4.15)$$

There are at most $(4w)^t$ possible codes, hence

$$\Pr[R_p \in \mathcal{S}] = \sum_{\rho \in \mathcal{S}} \mu_p(\rho) \leq (4w)^t \left(\frac{2p}{1-p} \right)^t = \left(\frac{8pw}{1-p} \right)^t. \quad (4.16)$$

Together with (4.5) this proves (4.6). If $p \leq 1/9$, then $8/(1-p) \leq 9$, proving (4.7).

For a CNF F , the negation $\neg F$ is a width- w DNF, restriction commutes with negation, and $D(g) = D(\neg g)$. The same bound follows. Finally, when $D(F \upharpoonright \rho) \leq t-1$, the path-to-leaf representations of a decision tree give both normal forms of width at most $t-1$, proving the switching formulation. $\square$

Source-sharp comparison. The standard modern Håstad switching lemma is commonly normalized in the same product space (4.1) as

$$\Pr[D(F \upharpoonright R_p) \geq t] \leq (5pw)^t. \quad (4.17)$$

Håstad's original restriction distribution is exactly (4.1), while the canonical decision-tree formulation and weighted encoding proofs were streamlined in later expositions. QRCC records (4.17) as the sharper source-owned constant and does *not* claim that the local encoding above proves 5 in place of $8/(1-p)$. All subsequent local arguments may invoke the completely proved bound (4.6); if a later quantitative statement needs the constant 5, that use must be marked as source-owned explicitly.

Width, depth, and degree are different parameters. Width refers to the number of literals in a DNF term or CNF clause. Decision-tree depth counts adaptive input queries. Polynomial degree is yet another complexity measure. A depth- d decision tree implies DNF and CNF width at most d , but a width- w DNF need not itself have decision-tree depth w . The switching lemma is precisely the probabilistic statement that this missing implication usually becomes true, with controlled loss, after a random restriction.

Dependency checkpoint. The proof uses only the circuit/formula conventions recorded under QRCC-X-DEF-001, elementary product probabilities, and a finite injective encoding. No Chernoff bound, Fourier theorem, polynomial approximation theorem, or later AC^0 lower bound is used. This keeps the switching lemma available as the first depth-reduction input to the next chapter.

Chapter 5

Parity versus Constant-Depth Circuits

The switching lemma becomes a circuit lower bound only after one explains how to apply it simultaneously to all gates in a layer and how to preserve enough unfixed variables to keep parity hard. This chapter carries out that passage at a deliberately qualitative scale. The goal is the classical class separation $\text{PARITY} \notin AC^0$ for polynomial-size, constant-depth circuits. The sharper size exponent is postponed to QRCC-X-THM-008; no later quantitative theorem is used here.

5.1 Normalization and the depth-reduction picture

Write

$$\text{PARITY}_n(x_1, \dots, x_n) := x_1 \oplus \dots \oplus x_n. \quad (5.1)$$

Under any restriction ρ , parity on the remaining live variables is either parity or its complement:

$$\text{PARITY}_n \upharpoonright \rho = b \oplus \bigoplus_{i:\rho_i=*} x_i \quad (5.2)$$

for a fixed bit b determined by the coordinates already assigned. Thus a restriction can reduce the number of variables, but it never turns parity on two or more live variables into an AND or an OR of literals.

Before using random restrictions we normalize the circuit. Starting from the AC^0 convention of Definition 1.1, compute alongside each gate its complement. De Morgan's laws then move every negation to an input literal with at most a factor-two increase in the number of AND/OR gates. Next flatten an AND feeding an AND, and similarly an OR feeding an OR. To make the remaining AND/OR skeleton strictly layered, one must be slightly careful because circuit size counts gates rather than wires. For each signed input literal and each existing gate output, create at most one delayed copy at each later level; unrestricted fan-out lets that single delayed copy serve all wires that need the signal at that level. Each delayed copy is a fan-in-one AND or OR chosen to match the prescribed layer type. Thus, from a circuit of original gate size S on n inputs and depth at most d , the normalized AND/OR skeleton has at most

$$2S + (d - 1)(2S + 2n) \leq 2d(S + n) \quad (5.0)$$

non-input AND/OR gates. This gives a layered circuit in which

1. layers are numbered $1, \dots, e$ from the inputs to the output, with $e \leq d$;
2. every wire in the AND/OR skeleton goes from layer i to layer $i + 1$;

3. each layer contains only AND gates or only OR gates; and
4. the gate types alternate from one layer to the next.

Input negations are henceforth absorbed into the word *literal*; they are not counted as an additional AND/OR layer in the normalized representation. The shared-delay construction, rather than one buffer per wire, is what keeps the normalization controlled under the gate-count size convention.

The useful direction of the switching lemma is easiest to see at the first layer. An unbounded OR of literals is a width-one DNF, even if its fan-in is huge. After a suitable random restriction it becomes a shallow decision tree, and hence a CNF with small clause width. The outer AND of that CNF has the same type as the original second layer and can be merged into it. The circuit has not yet lost a layer, but its new first-layer gates have small fan-in. After that preprocessing, the first two layers form small-width CNFs or DNFs; a second switching step reverses their order, allowing one genuine layer to collapse. Repeating this operation reaches depth two.

Why the first layer is not counted during switching. A depth- m decision tree may expand to as many as 2^m leaves when written as a CNF or DNF. The number of newly created first-layer gates can therefore grow. This does not damage the induction: at every later switching step we union-bound only over the old gates in the next layer, and their number never increases. What must remain bounded is the *fan-in* of the newly created first-layer gates, not their count.

5.2 The qualitative parity lower bound

Theorem 5.1 (Fixed-depth polynomial-size parity lower bound). *Fix integers $d \geq 2$ and $k \geq 1$. There is $n_0 = n_0(d, k)$ such that, for every $n \geq n_0$, no depth-at-most- d AC^0 circuit of size at most n^k computes PARITY_n . Consequently*

$$\text{PARITY} \notin AC^0. \quad (5.3)$$

More explicitly, one may take any n_0 large enough that, with

$$m_n := \left\lceil \log_2(64d^2n^{k+3}) \right\rceil, \quad \mu_n := \frac{n}{18(18m_n)^{d-2}}, \quad (5.4)$$

one has $\mu_n \geq 4m_n$ for every $n \geq n_0$.

Proof. Suppose, toward a contradiction, that a depth-at-most- d circuit C of size at most n^k computes parity. Apply the normalization above, and let $e \leq d$ be the depth of the resulting alternating layered circuit. It has at most

$$M := 4dn^k \quad (5.5)$$

non-input gates. If $e \leq 1$, the circuit is a single AND or OR of literals (up to a literal output), which cannot compute parity once $n \geq 2$. Thus assume $2 \leq e \leq d$. Put $m = m_n$. By (5.4),

$$2^{-m} \leq \frac{1}{64d^2n^{k+3}}. \quad (5.6)$$

We expose the restriction in independent stages, each sampled independently on the coordinates that remain live after the preceding stages.

Stage 0: bound the first-layer fan-in. Assume first that layer 1 contains OR gates; the AND case is the dual argument. Every such OR is a width-one DNF. Apply an independent $R_{1/18}$ restriction. The local switching bound (4.7), with $w = 1$ and $t = m$, gives for each layer-1 gate

$$\Pr[D(g \upharpoonright R_{1/18}) \geq m] \leq (9/18)^m = 2^{-m}. \quad (5.7)$$

There are at most M such gates. Except on an event of probability at most $M2^{-m}$, every restricted first-layer gate therefore has decision-tree depth at most $m - 1$. Write each one as a CNF of clause width at most $m - 1$, hence at most m . Its outer AND merges with the AND gates of layer 2. We obtain an equivalent depth- e circuit whose first layer consists of OR gates of fan-in at most m . If the original first layer contained AND gates, use the CNF form of the switching lemma and obtain first-layer AND gates of fan-in at most m instead.

One genuine depth-reduction stage. Now the first-layer fan-in is at most m . Consider any gate g in layer 2. Together with all first-layer gates feeding it, g computes a width-at-most- m CNF or DNF on the input literals. Apply an independent restriction

$$R_{1/(18m)}. \quad (5.8)$$

Since $1/(18m) \leq 1/9$, (4.7) gives

$$\Pr[D(g \upharpoonright R_{1/(18m)}) \geq m] \leq \left(9 \cdot \frac{1}{18m} \cdot m\right)^m = 2^{-m}. \quad (5.9)$$

There are at most M gates in this layer. Outside an event of probability at most $M2^{-m}$, every one of them becomes a decision tree of depth at most $m - 1$, hence can be written in the opposite normal form with bottom width at most m . The outer gate of that normal form has the same type as layer 3 and merges with it. Thus one layer disappears, while the new first-layer fan-in is still at most m . The number of gates above the first layer has not increased.

Repeat this genuine depth-reduction step $e - 2$ times. Including Stage 0, there are $e - 1 \leq d - 1$ switching stages in total. At every stage the conditional failure probability is at most $M2^{-m}$, regardless of the previously exposed restrictions. A union bound therefore gives

$$\Pr[\text{some switching stage fails}] \leq (d - 1)M2^{-m} < \frac{1}{16n^3} < \frac{1}{16}, \quad (5.10)$$

where (5.5)–(5.6) were used.

How many variables remain live? The composition of independent QRCC product restrictions is again a QRCC product restriction, with survival probability equal to the product of the stage survival probabilities. Since the actual normalized depth is e , the complete staged restriction has survival probability

$$P_e = \frac{1}{18} \left(\frac{1}{18m} \right)^{e-2} = \frac{1}{18(18m)^{e-2}} \geq \frac{1}{18(18m)^{d-2}}. \quad (5.11)$$

If L is the number of live variables at the end, then $L \sim \text{Bin}(n, P_e)$. Writing $\mu_e = nP_e$, (5.11) gives

$$\mathbb{E}L = \mu_e \geq \mu_n. \quad (5.12)$$

The lower-tail Chernoff bound of Theorem 2.4, with $\delta = 1/2$, gives

$$\Pr[L < \mu_n/2] \leq \Pr[L < \mu_e/2] \leq e^{-\mu_e/8} \leq e^{-\mu_n/8} \leq e^{-m/2} < \frac{1}{16}, \quad (5.13)$$

because $\mu_n \geq 4m$ and $m \geq 6$. Combining (5.10) and (5.13), there is at least one complete restriction for which every switching stage succeeds and

$$L \geq \mu_n/2 \geq 2m. \quad (5.14)$$

For that restriction, C has become a depth-two circuit whose first-layer gates have fan-in at most m , while by (5.2) the computed function is parity or its complement on L live variables.

Depth two cannot finish the job. A depth-two alternating circuit is a DNF or a CNF. Suppose a satisfiable DNF term omits some live variable x_i . Choose an assignment satisfying the term and then flip only x_i . The term remains true, so the DNF outputs 1 on both assignments, while parity flips. Therefore every satisfiable term of a DNF computing parity or its complement must contain all L live variables. The CNF statement follows by the same argument or by negation. Thus every nonconstant first-layer gate in a depth-two representation of parity has fan-in at least L . But our restricted circuit has first-layer fan-in at most m , while $L \geq 2m > m$, a contradiction.

Finally, for fixed d and k one has $m_n = O_{d,k}(\log n)$, whereas $\mu_n = n/(\log n)^{d-2}$ up to a constant depending only on d, k . Therefore $\mu_n \geq 4m_n$ for all sufficiently large n , so an admissible $n_0(d, k)$ exists. Since every polynomial-size constant-depth AC^0 family has some fixed d and polynomial exponent k , (5.3) follows. $\square$

Worked parameter check. For a fixed depth $d = 3$ and size bound n^2 , (5.4) has $m_n = \Theta(\log n)$. The staged restriction first keeps each variable with probability $1/18$ and then with probability $1/(18m_n)$, so the expected number of final live variables is

$$\mu_n = \frac{n}{324m_n}. \quad (5.15)$$

This eventually dominates $4m_n$ because $n/(\log n)^2 \rightarrow \infty$. At the same time every union-bound failure term is at most 2^{-m_n} , which was chosen much smaller than the reciprocal polynomial number of gates. The two requirements—many live variables and simultaneous switching—are therefore compatible rather than competing accidents.

Historical and quantitative boundary. Furst–Saxe–Sipser and Ajtai established the qualitative separation of parity from constant-depth polynomial-size AND/OR/NOT circuits. The proof above is a QRCC-local derivation using the later switching-lemma interface from Chapter 4 and the already proved Chernoff bound. It is not claimed to be a reconstruction of either original historical proof. Håstad’s later work sharpened the random-restriction method to near-optimal quantitative lower bounds. Those exact depth-versus-size exponents and their chosen normalization belong to QRCC-X-THM-008, not to the statement of the present theorem.

5.3 Restriction laboratory checkpoint

The executable file `code/volume_10/restriction_lab.py` implements a finite exact check of the Chapter 4 interface. It enumerates small truth-table-distinct width-two DNFs on three variables, enumerates all product restrictions for $p = 1/20$, computes the minimum residual decision-tree depth by exhaustive dynamic programming, and checks every resulting tail against both the locally proved bound (4.6) and the source-owned comparison (4.17). It also reproduces the equality-DNF tails $2p - p^2$ and p^2 from the previous chapter. The laboratory intentionally contains no Fourier or LMN component; that former scope has been removed so the code depends only on the switching lemma already proved.

Dependency checkpoint. The local proof of Theorem 5.1 uses only the circuit-model convention QRCC-X-DEF-001, the switching lemma QRCC-X-THM-006, the multiplicative Chernoff bound QRCC-I-THM-004, elementary De Morgan/layering normalization, and the trivial depth-two parity obstruction proved inside the argument. In particular, it does not use QRCC-X-THM-008, LMN Fourier concentration, or Razborov–Smolensky polynomial approximation.

Chapter 6

Near-Optimal Parity Lower Bounds

Chapter 5 used a deliberately coarse schedule: it only had to beat a polynomial number of gates while leaving a polylogarithmically controlled set of live variables. The present chapter keeps the same local switching-lemma interface but reorganizes the restriction argument so that the correct fixed-depth exponential scale becomes visible. The key invariant is the one used in Håstad's induction: control the bottom fan-in and, separately, the number of gates whose subcircuits still have depth at least two.

6.1 Source normalization and the quantitative target

The primary source uses unbounded-fan-in AND/OR circuits with negations moved to input literals. Depth is the longest input-to-output gate path, equivalently the number of gate levels after adjacent equal connectives are collapsed, and size is the total number of AND/OR gates. In that source model Håstad's parity theorem states that, for an absolute constant n_0 , no depth- d parity circuit of size

$$2^{h_d n^{1/(d-1)}}, \quad h_d := \left(\frac{1}{10}\right)^{d/(d-1)}, \quad (6.1)$$

exists when $n > n_0^d$. The accompanying strengthened induction bounds both the bottom fan-in and the number of subcircuits of depth at least two by the same scale $\frac{1}{10}n^{1/(d-1)}$.

Equation (6.1) is recorded as a *source-owned sharp normalization*. The QRCC-local switching theorem proved in Chapter 4 has the convenient corollary

$$\Pr[D(F \upharpoonright R_p) \geq t] \leq (9pw)^t \quad (p \leq 1/9) \quad (6.2)$$

for a width- w DNF or CNF. We do not claim that (6.2) reproduces the sharper constant in Håstad's source proof. Instead we prove the same optimal exponent $1/(d-1)$ with an explicit conservative QRCC-local coefficient

$$c_d := 800^{-d/(d-1)}. \quad (6.3)$$

This distinction makes the source comparison exact without assigning the source's constant to a proof that does not establish it.

6.2 The strengthened induction invariant

Put

$$a := \frac{1}{100}, \quad s_r(N) := \left\lfloor aN^{1/(r-1)} \right\rfloor \quad (r \geq 2). \quad (6.4)$$

A gate is said to be *at distance at least two from the inputs* if it is on layer 2 or higher in a strict alternating layering. Thus the subcircuit rooted at that gate has depth at least two.

Lemma 6.1 (QRCC strengthened parity induction). *For each fixed integer $r \geq 2$ there is N_r such that, whenever $N \geq N_r$, no strictly layered alternating depth- r AND/OR circuit computing parity or its complement on N variables can simultaneously satisfy*

1. *bottom fan-in at most $s_r(N)$; and*
2. *at most $2^{s_r(N)}$ gates at distance at least two from the inputs.*

Input negations, when present, are regarded as signed literals and do not form an additional layer.

Proof. We induct on r .

Base case $r = 2$. A depth-two alternating circuit is a DNF or a CNF. As proved directly in Chapter 5, every satisfiable DNF term computing parity or its complement must mention all N variables; dually every relevant CNF clause must mention all N variables. Since $s_2(N) = \lfloor N/100 \rfloor < N$, the bottom-fan-in condition is already impossible. The second invariant is not needed in the base case.

Induction step. Fix $r \geq 3$ and suppose the claim holds for depth $r-1$. Let C be a counterexample at depth r on N variables. For compactness write

$$B := N^{1/(r-1)}, \quad X := N^{(r-2)/(r-1)}, \quad p := \frac{2}{B}, \quad s := s_r(N) = \lfloor aB \rfloor. \quad (6.5)$$

Take N large enough that $p \leq 1/9$, $s \geq 1$, and all thresholds needed for the induction hypothesis below hold.

Consider a gate in layer 2. Together with the bottom gates feeding it, the gate computes a DNF or CNF of width at most s . Apply an independent product restriction R_p . By (6.2), with target decision-tree depth s ,

$$\Pr[D(g \upharpoonright R_p) \geq s] \leq (9ps)^s \leq (18a)^s = (0.18)^s. \quad (6.6)$$

There are at most 2^s layer-2 gates, because every such gate is at distance at least two from the inputs. Hence

$$\Pr[\text{some layer-2 switching fails}] \leq 2^s (0.18)^s = (0.36)^s. \quad (6.7)$$

The number L of live variables has distribution $\text{Bin}(N, p)$ and mean

$$\mu = pN = 2X. \quad (6.8)$$

The Chernoff bound from Theorem 2.4 gives

$$\Pr[L < X] \leq \Pr[L < \mu/2] \leq e^{-\mu/8} = e^{-X/4}. \quad (6.9)$$

For all sufficiently large N , the sum of (6.7) and (6.9) is less than one. Therefore some restriction makes every layer-2 switch succeed and leaves at least X live variables.

Let

$$M := \lceil X \rceil. \quad (6.10)$$

If more than M variables remain live, fix arbitrary extras. Further restriction cannot increase decision-tree depth, so every switched layer-2 subcircuit still has decision-tree depth at most $s-1$.

Write each such tree in the opposite normal form with bottom width at most s ; its outer gate has the same connective as layer 3 and merges into that layer. One circuit layer therefore disappears.

The resulting depth- $(r - 1)$ circuit computes parity or its complement on exactly M variables. Its new bottom fan-in is at most s , and

$$M^{1/(r-2)} \geq X^{1/(r-2)} = B, \quad s \leq s_{r-1}(M). \quad (6.11)$$

Moreover every gate at distance at least two in the new circuit descends from an old gate at distance at least three. Consequently their number is at most $2^s \leq 2^{s_{r-1}(M)}$. Once N is large enough that $M \geq N_{r-1}$, this contradicts the induction hypothesis. The induction closes. $\square$

6.3 Parameter ledger for the induction

The entire quantitative induction can be read from the following table. The “objects” column is exactly the family over which a union bound is taken; new bottom gates created by a decision-tree expansion are never added to that count.

Stage	Objects union-bounded	Current width	Survival	Target tree depth	Quantitative effect
Induction $r \rightarrow r - 1$	old layer-2 gates, at most 2^s	$s = \lfloor B/100 \rfloor$	$p = 2/B$	s	switching failure at most $(0.36)^s$; expected live variables $2X$
Initial pre- processing	old bottom gates, at most M_C	1	$q = 1/100$	$t = s_e(m)$	switching failure at most $M_C(0.09)^t$; expected live variables $n/100$

Here $B = N^{1/(r-1)}$, $X = N^{(r-2)/(r-1)}$ in the induction row. In the preprocessing row, e is the actual normalized depth, $m = \lfloor n/200 \rfloor$, and M_C is the normalized gate count. The proof below verifies that $M_C \leq 2^t$ at the claimed size scale.

6.4 The near-optimal fixed-depth lower bound

Theorem 6.2 (Near-optimal fixed-depth parity lower bound). *Fix an integer $d \geq 2$ and define*

$$c_d := 800^{-d/(d-1)}. \quad (6.12)$$

There is $n_0 = n_0(d)$ such that for every $n \geq n_0$, every depth-at-most- d AC^0 circuit, in the gate-count convention of Definition 1.1, that computes PARITY_n has size

$$|C| > 2^{c_d n^{1/(d-1)}}. \quad (6.13)$$

Thus the exponent $n^{1/(d-1)}$ agrees with the near-optimal exponent in Håstad’s parity lower bound, while the explicit coefficient c_d in (6.12) is the conservative constant proved by the QRCC-local switching interface.

Proof. Suppose instead that C has depth at most d and size

$$S \leq 2^{c_d n^{1/(d-1)}}. \quad (6.14)$$

Normalize exactly as in Chapter 5: move negations to signed input literals, flatten adjacent equal connectives, and use shared delayed copies to obtain a strict alternating circuit of actual depth $e \leq d$. Its number of non-input AND/OR gates satisfies

$$M_C \leq 2d(S + n). \quad (6.15)$$

If $e \leq 1$, parity is impossible for $n \geq 2$, so assume $2 \leq e \leq d$.

Apply a product restriction with survival

$$q := \frac{1}{100}, \quad m := \left\lfloor \frac{n}{200} \right\rfloor, \quad t := s_e(m) = \left\lfloor \frac{1}{100} m^{1/(e-1)} \right\rfloor. \quad (6.16)$$

Each bottom gate is a width-one DNF or CNF. By (6.2),

$$\Pr[D(g \upharpoonright R_q) \geq t] \leq (9q)^t = (0.09)^t. \quad (6.17)$$

We first verify that the normalized gate count is small enough for a simultaneous switch. For n large enough, $m \geq n/400$ and

$$t \geq \frac{1}{200} 400^{-1/(d-1)} n^{1/(d-1)}. \quad (6.18)$$

Indeed $e \leq d$ makes $m^{1/(e-1)} \geq m^{1/(d-1)}$, and the factor $1/200$ absorbs the floor in (6.16). On the other hand, for sufficiently large n , $\log_2 n \leq c_d n^{1/(d-1)}$ and $\log_2(4d) \leq c_d n^{1/(d-1)}$, so (6.14)–(6.15) give

$$\log_2 M_C \leq 3c_d n^{1/(d-1)}. \quad (6.19)$$

The choice (6.12) has enough slack because

$$3 \cdot 800^{-d/(d-1)} < \frac{1}{200} 400^{-1/(d-1)}. \quad (6.20)$$

Thus $M_C \leq 2^t$ for all sufficiently large n . A union bound in (6.17) therefore yields

$$\Pr[\text{some bottom gate fails to switch}] \leq M_C (0.09)^t \leq (0.18)^t. \quad (6.21)$$

The live-variable count L has mean $n/100$. Since $m \leq n/200$, Chernoff gives

$$\Pr[L < m] \leq e^{-n/800}. \quad (6.22)$$

For large n , the sum of (6.21) and (6.22) is less than one, so some restriction simultaneously switches every bottom gate and leaves at least m live variables. Fix arbitrary extras if necessary, leaving exactly m . As in Stage 0 of Chapter 5, write each residual bottom decision tree in the opposite normal form and merge its outer connective into layer 2. The result is still depth e , computes parity or its complement on m variables, has bottom fan-in at most $t = s_e(m)$, and has at most $M_C \leq 2^t$ gates at distance at least two from the inputs.

For n sufficiently large, also $m \geq N_e$. The resulting circuit is therefore forbidden by Lemma 6.1. This contradiction proves (6.13). $\square$

Why the exponent is the right one. The induction restriction in (6.5) keeps a fraction $\Theta(N^{-1/(r-1)})$ of the variables. It therefore maps N variables to $\Theta(N^{(r-2)/(r-1)})$ live variables, precisely the change needed to turn the scale $N^{1/(r-1)}$ into the next-depth scale $M^{1/(r-2)}$. This identity, displayed in (6.11), is the quantitative core that the coarse logarithmic schedule of Chapter 5 intentionally did not try to expose.

Worked depth-three check. Take $r = 3$ in the strengthened induction. Then

$$B = \sqrt{N}, \quad X = \sqrt{N}, \quad p = \frac{2}{\sqrt{N}}, \quad s = \left\lfloor \frac{\sqrt{N}}{100} \right\rfloor. \quad (6.23)$$

For each old second-layer gate, the switching failure is at most $(0.18)^s$; union-bounding over at most 2^s such gates gives $(0.36)^s$. Meanwhile the expected number of live variables is $2\sqrt{N}$, so Chernoff leaves at least $\sqrt{N}$ variables except with probability at most $e^{-\sqrt{N}/4}$. The residual depth-two circuit has bottom fan-in at most $\sqrt{N}/100$, whereas a depth-two parity representation needs bottom fan-in at least the number of live variables. This is the depth-three instance of the general induction with every scale visible.

Source-sharp comparison. The QRCC theorem (6.13) proves the correct fixed-depth exponent using only the locally established switching constant. Håstad's primary theorem is sharper in its own source normalization: it gives the coefficient $h_d = (1/10)^{d/(d-1)}$ and a threshold $n > n_0^d$ for an absolute n_0 . The primary source also records the matching-scale upper bound $n2^{n^{1/(d-1)}}$. Hence the power $1/(d-1)$ is optimal, while constant optimization is deliberately not part of the present local reconstruction. No average-case correlation statement is being asserted here; that is a separate theorem type and must not be merged into this result ID.

Dependency and ownership checkpoint. The local proof of Theorem 6.2 uses only the circuit-model conventions of QRCC-X-DEF-001, the QRCC-local switching lemma QRCC-X-THM-006, the Chernoff bound QRCC-I-THM-004, and the elementary depth-two parity obstruction reproved inside the strengthened induction. It does not use LMN Fourier concentration, Razborov–Smolensky polynomial approximation, a correlation theorem, or the source-owned sharp switching constant. Håstad's theorem is used to freeze the comparison statement and its model normalization, not as a hidden proof step in (6.13).

Chapter 7

Fourier Concentration for Constant-Depth Circuits

The parity lower bounds in Chapters 5–6 are worst-case statements: they show that one particular Boolean function cannot be represented by a small constant-depth circuit. The Linial–Mansour–Nisan theorem is structurally different. It says that *every* small constant-depth circuit has most of its Fourier energy on low-degree characters. The bridge is random restriction: switching makes the restricted circuit a shallow decision tree, a shallow decision tree has low Fourier degree, and an averaging identity transfers that restricted low-degree statement back to the original Fourier tail.

7.1 Fourier normalization and the source statement

We use the sign cube already fixed in Volume I. For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$f(x) = \sum_{A \subseteq [n]} \hat{f}(A) \chi_A(x), \quad \chi_A(x) := \prod_{i \in A} x_i, \quad (7.1)$$

where the Fourier coefficients are normalized with respect to the uniform probability measure. Parseval therefore reads

$$\sum_{A \subseteq [n]} \hat{f}(A)^2 = \mathbb{E}[f^2]. \quad (7.2)$$

For an integer $T \geq 0$ define the high-degree Fourier mass

$$W^{>T}(f) := \sum_{|A| > T} \hat{f}(A)^2. \quad (7.3)$$

If $C : \{0, 1\}^n \rightarrow \{0, 1\}$ is a Boolean circuit, its sign encoding is

$$f_C(x) := (-1)^{C((1-x_1)/2, \dots, (1-x_n)/2)}. \quad (7.4)$$

Changing between bit and sign coordinates merely relabels the two values of each variable; decision-tree depth and the product-restriction survival parameter are unchanged.

The primary bibliographic route is Linial–Mansour–Nisan, *Constant Depth Circuits, Fourier Transform, and Learnability*, Journal of the ACM 40(3), 607–620 (1993), with a preliminary FOCS 1989 version. Its main structural conclusion is that the Fourier spectrum of an AC^0 function is

concentrated on low degrees. For a stable modern normalization, one may state the quantitative conclusion as follows: a depth- d , size- s circuit has Fourier tail at most ε above degree

$$O_d\left(\log(s/\varepsilon)^{d-1} \log(1/\varepsilon)\right) \quad (0 < \varepsilon \leq 1/2). \quad (7.5)$$

The source gate records (7.5) as the canonical quantitative statement. We do not freeze a disputed numerical prefactor from secondary transcriptions of the original main lemma. Instead, the theorem below gives an explicit QRCC cutoff derived entirely from the switching estimate already proved in Chapter 4. This is enough to recover (7.5), with transparent constants and with the QRCC gate-count convention kept visible.

7.2 From restrictions to the original Fourier tail

A restriction on the sign cube is written as $\rho \in \{-1, 1, *\}^n$. Under R_p , each coordinate is live with probability p and is otherwise fixed uniformly to -1 or 1 , independently across coordinates. This is exactly the sign-coordinate image of the product restriction used in Chapter 4.

Lemma 7.1 (Restriction-to-spectrum bridge). *Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, let $0 < p \leq 1$, and let $k \geq 1$ be an integer. If*

$$\eta := \Pr_{\rho \sim R_p} [D(f \upharpoonright \rho) \geq k], \quad (7.6)$$

then

$$W^{>3k/p}(f) \leq 3\eta. \quad (7.7)$$

Here a nonintegral cutoff means that the sum is over sets A with $|A| > 3k/p$.

Proof. First freeze the live-coordinate set $J \subseteq [n]$ and average only over the uniform fixing $y \in \{-1, 1\}^{[n] \setminus J}$. If $T \subseteq J$, then the Fourier coefficient of the restricted function is

$$f \upharpoonright (J, y)(T) = \sum_{B \subseteq [n] \setminus J} \widehat{f}(T \cup B) \chi_B(y). \quad (7.8)$$

Orthogonality of the characters in the fixed coordinates gives

$$\mathbb{E}_y [f \upharpoonright (J, y)(T)^2] = \sum_{B \subseteq [n] \setminus J} \widehat{f}(T \cup B)^2. \quad (7.9)$$

Summing (7.9) over $|T| \geq k$ and then choosing each coordinate of J independently with probability p yields the exact identity

$$\mathbb{E}_{\rho \sim R_p} \left[\sum_{|T| \geq k} \widehat{f \upharpoonright \rho}(T)^2 \right] = \sum_{A \subseteq [n]} \Pr[|A \cap J| \geq k] \widehat{f}(A)^2. \quad (7.10)$$

Now fix A with $|A| > 3k/p$. For $X := |A \cap J| \sim \text{Bin}(|A|, p)$ we have $\mu := \mathbb{E}X > 3k$. The lower-tail Chernoff bound from Theorem 2.4, with $k = (1 - \delta)\mu$ and hence $\delta > 2/3$, gives

$$\Pr[X < k] \leq \exp(-\delta^2 \mu / 2) < e^{-2k/3} < \frac{2}{3}. \quad (7.11)$$

Thus $\Pr[X \geq k] > 1/3$. Restricting the right side of (7.10) to $|A| > 3k/p$ therefore shows

$$\frac{1}{3} W^{>3k/p}(f) \leq \mathbb{E}_\rho \left[\sum_{|T| \geq k} \widehat{f \upharpoonright \rho}(T)^2 \right]. \quad (7.12)$$

It remains to bound the expectation. A Boolean function computed by a decision tree of depth strictly less than k has multilinear degree strictly less than k : recursively, a query of x_i writes the function as $\frac{1+x_i}{2}f_+ + \frac{1-x_i}{2}f_-$ and increases degree by at most one. Hence the summand in (7.12) is zero when $D(f \upharpoonright \rho) < k$. In the complementary event it is at most one by Parseval. Its expectation is therefore at most η , proving (7.7). $\square$

The proof has separated three quantities that must not be conflated: $D(f \upharpoonright \rho)$ is the *restricted decision-tree depth*; the restricted Fourier degree is bounded by that depth; and $W^{>T}(f)$ is the *original* high-degree Fourier mass. Identity (7.10), not an informal identification of these notions, is the transfer mechanism.

7.3 A bounded-bottom collapse lemma

For the next lemma, a circuit is *analysis-layered* if its nonliteral gates occupy alternating AND/OR layers, every gate-to-gate wire joins consecutive gate layers, and a signed input literal may feed any layer directly. Direct literals are harmless: inside a DNF or CNF they are simply width-one terms or clauses.

Lemma 7.2 (Simultaneous switching to a decision tree). *Let G be an analysis-layered alternating circuit of actual depth $e \geq 2$. Suppose it has at most $M \geq 2$ nonliteral gates and bottom fan-in at most $w \geq 1$. For $0 < \eta \leq 1$ put*

$$\ell := \left\lceil \log_2 \frac{2M}{\eta} \right\rceil, \quad k := \left\lceil \log_2 \frac{2}{\eta} \right\rceil, \quad (7.13)$$

and

$$p := \frac{1}{18w} \left(\frac{1}{18\ell} \right)^{e-2}. \quad (7.14)$$

Then

$$\Pr_{\rho \sim R_p} [D(G \upharpoonright \rho) \geq k] \leq \eta. \quad (7.15)$$

Proof. If $e = 2$, the whole circuit is a width- w DNF or CNF. Formula (7.14) gives $p = 1/(18w)$, so the local switching estimate directly yields

$$\Pr[D(G \upharpoonright R_p) \geq k] \leq (9pw)^k = 2^{-k} \leq \frac{\eta}{2} < \eta. \quad (7.16)$$

Assume henceforth that $e \geq 3$.

A product restriction with survival p can be exposed sequentially: first use survival $q_0 := 1/(18w)$ and then use survival $q_1 := 1/(18\ell)$ exactly $e - 2$ times. The final live probability is the product $q_0 q_1^{e-2} = p$, and conditioned on a coordinate becoming fixed its sign remains uniform. Thus the composite restriction has distribution R_p .

Under the first exposure, every layer-2 gate together with its bottom inputs is a width- w DNF or CNF. The local switching estimate gives

$$\Pr[D(H \upharpoonright R_{q_0}) \geq \ell] \leq (9q_0 w)^\ell = 2^{-\ell}. \quad (7.17)$$

Whenever switching succeeds, represent the resulting decision tree in the opposite normal form of width at most ℓ . Its outer connective then matches the next circuit layer and can be flattened into that layer, removing one nonliteral layer.

For the next $e - 3$ exposures, repeat this operation on the next old gate layer using survival q_1 and width ℓ . The same calculation gives failure at most $2^{-\ell}$ for each gate that is switched. Crucially, the union bound is *not* over M fresh gates at every stage. The target roots at successive stages are distinct old nonliteral gates of G ; new terms or clauses created from a decision tree lie on the new bottom layer and are not new target roots. Consequently at most M target gates are ever union-bounded, and the initial plus intermediate layer-collapse stages fail with probability at most

$$M2^{-\ell} \leq \frac{\eta}{2}. \quad (7.18)$$

After these $e - 2$ layer collapses the restricted circuit has depth at most two and bottom width at most ℓ . Use the final, still-unused q_1 exposure to switch this last DNF or CNF to decision-tree depth $< k$. Again by the local switching estimate,

$$\Pr[D \geq k] \leq (9q_1\ell)^k = 2^{-k} \leq \frac{\eta}{2}. \quad (7.19)$$

The union of (7.18) and (7.19) has probability at most η , proving the claim. $\square$

7.4 Trimming wide bottom gates

The collapse lemma assumes bounded bottom fan-in, whereas AC^0 does not. The standard trimming step costs only a small approximation error.

Start with a QRCC depth-at-most- d circuit C of size S . Push negations to signed input literals by computing each AND/OR gate value together with its complement; this uses at most $2S$ AND/OR gates. Flatten adjacent equal connectives. If a gate output is needed several gate levels later, create one shared unary identity copy at each intervening gate level and let all later uses share it. At most $d - 1$ such copies are required per gate, so the analysis-layered form has at most

$$M := 2dS \quad (7.20)$$

nonliteral gates. Unlike the strict physical layering used in Chapter 5, input literals are allowed to feed any gate layer directly; therefore no $+n$ input-buffer term is required in (7.20).

Before measuring bottom fan-in, delete duplicate literals, simplify constants, and simplify a bottom AND containing both x_i and $\neg x_i$ to 0 (dually, a bottom OR containing both to 1). Every remaining bottom AND of fan-in r is 1 on exactly a 2^{-r} fraction of uniform inputs, and every remaining bottom OR is 0 on exactly a 2^{-r} fraction. Replace each bottom AND of fan-in $> w$ by 0 and each bottom OR of fan-in $> w$ by 1. If G is the resulting circuit, then the union bound gives

$$\Pr[C \neq G] \leq M2^{-w}. \quad (7.21)$$

This is an approximation step, not an equivalent-circuit normalization.

7.5 The Linial–Mansour–Nisan concentration theorem

Theorem 7.3 (Linial–Mansour–Nisan Fourier concentration; explicit QRCC form). *Fix an integer $d \geq 2$. Let C be a depth-at-most- d AC^0 circuit on n inputs of QRCC gate-count size $S \geq 1$, and let $f := f_C : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be its sign encoding. For $0 < \varepsilon \leq 1$, put*

$$M := 2dS, \quad w := \left\lceil \log_2 \frac{16M}{\varepsilon} \right\rceil, \quad (7.22)$$

$$\ell := \left\lceil \log_2 \frac{24M}{\varepsilon} \right\rceil, \quad k := \left\lceil \log_2 \frac{24}{\varepsilon} \right\rceil, \quad (7.23)$$

and

$$T := \left\lceil 54wk(18\ell)^{d-2} \right\rceil. \quad (7.24)$$

Then

$$\boxed{W^{>T}(f) = \sum_{|A|>T} \widehat{f}(A)^2 \leq \varepsilon.} \quad (7.25)$$

In particular, for fixed d and $0 < \varepsilon \leq 1/2$,

$$T = O_d\left(\log(S/\varepsilon)^{d-1} \log(1/\varepsilon)\right), \quad (7.26)$$

with harmless additive constants inside the logarithms. Thus polynomial-size fixed-depth circuits have ε -Fourier concentration at polylogarithmic degree.

Proof. Use the analysis-layered normalization above, with at most $M = 2dS$ gates, and trim bottom fan-in at the value w from (7.22). Let g be the sign encoding of the trimmed circuit G . Equation (7.21) gives

$$\alpha := \Pr[f \neq g] \leq M2^{-w} \leq \frac{\varepsilon}{16}, \quad \|f - g\|_2^2 = 4\alpha \leq \frac{\varepsilon}{4}. \quad (7.27)$$

Let $e \leq d$ be the actual nonliteral depth of G . If $e \leq 1$, then g is constant, a literal, or a single AND/OR of at most w literals, hence has Fourier degree at most $w \leq T$. In that case $W^{>T}(f) \leq \|f - g\|_2^2 \leq \varepsilon/4$, so assume $e \geq 2$.

Apply Lemma 7.2 with $\eta := \varepsilon/12$. Its parameters are exactly the ℓ, k in (7.23), and its survival probability is

$$p = \frac{1}{18w} \left(\frac{1}{18\ell} \right)^{e-2}. \quad (7.28)$$

Therefore

$$\Pr_{\rho \sim R_p} [D(g \upharpoonright \rho) \geq k] \leq \frac{\varepsilon}{12}. \quad (7.29)$$

The restriction-to-spectrum bridge gives

$$W^{>3k/p}(g) \leq \frac{\varepsilon}{4}. \quad (7.30)$$

Since $e \leq d$,

$$\frac{3k}{p} = 54wk(18\ell)^{e-2} \leq 54wk(18\ell)^{d-2} \leq T, \quad (7.31)$$

so $W^{>T}(g) \leq \varepsilon/4$.

Let $P_{>T}$ denote orthogonal Fourier projection onto characters of degree strictly greater than T . Contractivity of an orthogonal projection and (7.27) give

$$\begin{aligned} W^{>T}(f) &= \|P_{>T}f\|_2^2 \\ &\leq 2\|P_{>T}g\|_2^2 + 2\|P_{>T}(f - g)\|_2^2 \\ &\leq 2 \cdot \frac{\varepsilon}{4} + 2 \cdot \frac{\varepsilon}{4} = \varepsilon. \end{aligned} \quad (7.32)$$

This proves the explicit claim. Finally, w and ℓ are $O(\log(S/\varepsilon))$, while $k = O(1 + \log(1/\varepsilon))$. For $\varepsilon \leq 1/2$ the additive 1 is absorbed by $\log(1/\varepsilon)$, yielding (7.26). $\square$

7.6 A two-variable Fourier/restriction check

Consider

$$h(x_1, x_2) = \frac{1 + x_1 + x_2 - x_1x_2}{2}. \quad (7.33)$$

This is the sign encoding (7.4) of the two-input AND: it equals -1 only at $(-1, -1)$. Its four Fourier coefficients have magnitudes $1/2$, so

$$W^{>1}(h) = \widehat{h}(\{1, 2\})^2 = \frac{1}{4}, \quad \deg(h) = 2. \quad (7.34)$$

Under R_p , the restricted function is nonconstant exactly when neither coordinate is fixed to $+1$ and at least one coordinate is live. Hence

$$\Pr[D(h \upharpoonright \rho) \geq 1] = p. \quad (7.35)$$

The expected nonconstant Fourier mass is different: when exactly one coordinate is live and the other is fixed to -1 , the restricted function is a literal and contributes mass 1 ; when both are live, the nonconstant mass is $3/4$. Therefore

$$\mathbb{E}_\rho \left[\sum_{|T| \geq 1} \widehat{h \upharpoonright \rho(T)}^2 \right] = p(1 - p) + \frac{3}{4}p^2 = p - \frac{1}{4}p^2. \quad (7.36)$$

Identity (7.10) gives the same answer directly: the two singleton coefficients contribute $p/2$, while the degree-two coefficient contributes $(2p - p^2)/4$. Thus even in this smallest example, restricted decision-tree depth, restricted Fourier mass, and the original degree-two tail are three distinct objects linked by a precise averaging identity.

7.7 Ownership and dependency checkpoint

The locally proved theorem uses exactly four mathematical interfaces already closed in the series:

1. the Boolean Fourier characters and normalization from Volume I, Definition 7.1;
2. Parseval from Volume I, Theorem 7.2;
3. the Chernoff lower-tail estimate from Volume I, Theorem 2.4; and
4. the local switching lemma from Volume X, Theorem 4.1.

No Razborov–Smolensky polynomial approximation theorem, no correlation lower bound, and no later barrier result is used. The LMN primary/canonical source owns the historical theorem and its sharper source conventions; equations (7.22)–(7.25), including the explicit constants $16, 24, 54, 18$, are QRCC-local consequences of the certified $(9pw)^t$ switching interface. Håstad’s later sharpening of Fourier concentration is likewise not merged into ‘QRCC-X-THM-009’.

Chapter 8

Polynomial Approximation over Finite Fields

Random restrictions simplify AC^0 circuits by making their Boolean structure shallow. The Razborov–Smolensky method uses a different kind of simplification: over a finite field of the same characteristic as the modular gates, every gate can be replaced exactly or probabilistically by a low-degree polynomial. Composing those gate polynomials gives a low-degree polynomial for the whole circuit with small pointwise error.

The distinction between three polynomial notions is important throughout this chapter. An *exact polynomial representation* agrees with a Boolean function on every Boolean input. An ε -*error probabilistic polynomial* is a distribution on polynomials for which every fixed Boolean input is computed correctly with probability at least $1 - \varepsilon$ over the sampled polynomial. A *deterministic ε -approximant with respect to a distribution μ* is one fixed polynomial whose disagreement probability under $x \sim \mu$ is at most ε . The second notion is strictly the interface proved below; the third follows from it by averaging.

8.1 Source normalization and the modular gate convention

The primary historical sources are A. A. Razborov, *Lower bounds on the size of bounded depth circuits over a complete basis with logical addition*, Mathematical Notes 41(4), 333–338 (1987), and R. Smolensky, *Algebraic methods in the theory of lower bounds for Boolean circuit complexity*, STOC 1987, 77–82. Razborov’s argument gives the characteristic-two polynomial-approximation method, and Smolensky’s paper sets the method over arbitrary positive characteristic and combines it with a separate algebraic obstruction for modular functions of incompatible characteristic.

There is a harmless convention difference worth freezing explicitly. Smolensky’s bare MOD_p gate is written in the complementary form that is zero when the number of ones is divisible by p . By Definition 1.1, QRCC instead writes $\text{MOD}_p = 1$ on the zero residue class. The two gates differ by one NOT, so the circuit class and asymptotic lower bounds are unchanged. Under the QRCC convention, if $y_i \in \{0, 1\}$ then

$$\text{MOD}_p(y_1, \dots, y_m) = 1 - \left(\sum_{i=1}^m y_i \right)^{p-1} \quad \text{in } \mathbb{F}_p. \quad (8.1)$$

Indeed, a nonzero element of $\mathbb{F}_p$ has $(p - 1)$ st power one, while zero has $(p - 1)$ st power zero. This is the exact algebraic part of the method. The only approximation needed for the theorem below is at AND and OR gates.

A stable modern statement of Razborov's characteristic-two approximation lemma is that a depth- d $AC^0[2]$ circuit with s internal gates has an ε -error probabilistic polynomial of degree at most $O(\log(s/\varepsilon)^d)$; under the usual base-two normalization the classical gate-by-gate construction gives the explicit scale $\lceil \log_2(s/\varepsilon) \rceil^d$. We prove the corresponding prime- p statement directly, with all constants visible, rather than importing that theorem as a black box.

8.2 One unbounded gate from random linear tests

Fix a prime p . For a vector $a = (a_1, \dots, a_m) \in \mathbb{F}_p^m$, write

$$L_a(y) := \sum_{i=1}^m a_i y_i. \quad (8.2)$$

If $y \in \mathbb{F}_p^m$ is nonzero and a is uniform in $\mathbb{F}_p^m$, then $L_a(y)$ is uniform in $\mathbb{F}_p$: choose an index i_0 with $y_{i_0} \neq 0$, freeze every coefficient except a_{i_0} , and observe that $a_{i_0} \mapsto L_a(y)$ is a bijection of $\mathbb{F}_p$. Therefore

$$\Pr_a[L_a(y) = 0] = \frac{1}{p} \quad (y \neq 0). \quad (8.3)$$

Take independent uniform vectors $a^{(1)}, \dots, a^{(t)} \in \mathbb{F}_p^m$ and define

$$Q_{\vee, t}(y) := 1 - \prod_{j=1}^t (1 - L_{a^{(j)}}(y)^{p-1}). \quad (8.4)$$

For $y = 0$, this polynomial is always zero. For $y \neq 0$, Fermat's identity makes every factor in the product either zero or one, so $Q_{\vee, t}(y) = 0$ exactly when all t random linear forms vanish. Hence

$$\Pr[Q_{\vee, t}(y) \neq \text{OR}(y)] \leq p^{-t} \quad \text{for every } y \in \{0, 1\}^m, \quad (8.5)$$

and the total degree is at most $(p-1)t$, independently of the fan-in m . The analogous AND polynomial is

$$Q_{\wedge, t}(y) := 1 - Q_{\vee, t}(1 - y_1, \dots, 1 - y_m), \quad (8.6)$$

with the same degree and error. Constants and literals are exact degree-zero or degree-one polynomials, NOT is $1 - y$, and the QRCC zero-residue MOD_p gate is exactly (8.1).

Why the random test is fan-in independent. The construction does not sample input coordinates and hope to see a one. That would require a degree/error bound depending on the Hamming weight. Instead it hashes the entire nonzero vector to one field element. Equation (8.3) says that every nonzero vector, sparse or dense, is missed with exactly probability $1/p$ in one trial.

8.3 Circuit composition and pointwise error

For the theorem, the size S is the number of noninput gates, exactly as in Definition 1.1; the depth is the number of gate levels along a longest input-output path. No alternation or fan-in normalization is needed. NOT gates may occur anywhere. Every unbounded gate has finite fan-in because the circuit itself is finite.

For a sampled gate polynomial, composition means literal substitution in $\mathbb{F}_p[x_1, \dots, x_n]$. After all substitutions, powers x_i^r with $r \geq 1$ may be reduced to x_i modulo $x_i^2 - x_i$. This *multilinearization* preserves the polynomial's values on $\{0, 1\}^n$ and never increases total degree.

Theorem 8.1 (Razborov–Smolensky polynomial approximation; explicit QRCC form). *Let p be a prime, let $d \geq 1$, and let C be a depth-at-most- d $AC^0[p]$ circuit on n inputs with $S \geq 1$ noninput gates. For $0 < \varepsilon \leq 1/2$, set*

$$t := \left\lceil \log_p \frac{S}{\varepsilon} \right\rceil, \quad R := (p-1)t. \quad (8.7)$$

Then there is a distribution $\mathcal{P}_C$ on multilinear polynomials $P \in \mathbb{F}_p[x_1, \dots, x_n]$ of degree at most

$$\deg P \leq R^d = \left((p-1) \left\lceil \log_p \frac{S}{\varepsilon} \right\rceil \right)^d \quad (8.8)$$

such that for every fixed Boolean input $x \in \{0, 1\}^n$,

$$\Pr_{P \sim \mathcal{P}_C} [P(x) \neq C(x)] \leq \varepsilon. \quad (8.9)$$

Consequently, for every probability distribution μ on $\{0, 1\}^n$, there exists one deterministic multilinear polynomial P_μ of the same degree satisfying

$$\Pr_{x \sim \mu} [P_\mu(x) \neq C(x)] \leq \varepsilon. \quad (8.10)$$

For fixed p, d this degree is $O_{p,d}(\log(S/\varepsilon)^d)$; in particular, polynomial-size fixed-depth $AC^0[p]$ circuits admit polylogarithmic-degree constant-error approximants over $\mathbb{F}_p$.

Proof. Put $\delta := \varepsilon/S$. At every AND or OR gate g , independently sample a local degree-at-most- R polynomial from (8.4) or (8.6), using the common value t from (8.7). Since $p^{-t} \leq \delta$, for every fixed Boolean vector of exact child values the local gate polynomial disagrees with the gate on that vector with probability at most δ . At a NOT gate use $1 - y$ exactly, and at a MOD_p gate use (8.1) exactly. Constants and input variables are left unchanged.

Process the circuit in topological order. For each gate g , substitute the already constructed child polynomials into the sampled local polynomial for g ; call the result P_g . If g has circuit depth j , then

$$\deg P_g \leq R^j. \quad (8.11)$$

This is immediate for inputs at depth zero. NOT does not increase degree. A modular gate multiplies the maximum child degree by at most $p-1 \leq R$, and an AND or OR local polynomial has total degree at most R , so after substitution its degree is at most R times the largest child degree. Induction on j proves (8.11), and the output polynomial therefore has degree at most R^d . Multilinearize it at the end; Boolean-cube values are unchanged and the degree can only fall.

It remains to justify the error in the presence of sharing. Fix an input $x \in \{0, 1\}^n$. For every AND/OR gate g , let $B_g(x)$ be the event that its *local* sampled gate polynomial is wrong when evaluated on the exact Boolean values of g 's children under x . These exact child values depend only on C and x , not on the random polynomials, so

$$\Pr[B_g(x)] \leq \delta. \quad (8.12)$$

If none of the events $B_g(x)$ occurs, topological induction shows that every composed polynomial $P_g(x)$ equals the exact gate value $g(x)$: the claim is true at the inputs, exact NOT and MOD_p preserve it, and at an AND/OR gate the local polynomial is correct on the exact child vector. There are at most S AND/OR gates, hence the union bound from Theorem 1.2 gives

$$\Pr[P_C(x) \neq C(x)] \leq \sum_g \Pr[B_g(x)] \leq S\delta = \varepsilon. \quad (8.13)$$

This proves the pointwise probabilistic-polynomial statement.

Finally fix any distribution μ on the Boolean cube. Averaging (8.9) over $x \sim \mu$ and interchanging the two finite expectations gives

$$\mathbb{E}_{P \sim \mathcal{P}_C} \Pr_{x \sim \mu} [P(x) \neq C(x)] \leq \varepsilon. \quad (8.14)$$

At least one polynomial in the support of $\mathcal{P}_C$ therefore has μ -error at most ε , proving (8.10). $\square$

Worked check over $\mathbb{F}_3$. For a two-input OR and one random test, choose $a = (1, 2) \in \mathbb{F}_3^2$. Then

$$Q_{\vee,1}(y_1, y_2) = (y_1 + 2y_2)^2. \quad (8.15)$$

On $(0, 0), (1, 0), (0, 1)$ it takes the correct OR values 0, 1, 1, while on $(1, 1)$ it is 0 because $1 + 2 = 0$ in $\mathbb{F}_3$. For any fixed nonzero Boolean input, exactly one third of the random coefficient vectors give a zero linear form, in agreement with (8.5). Two independent tests reduce that pointwise error to $1/9$ while doubling the degree from 2 to 4. The three-input QRCC modular gate is instead exact:

$$\text{MOD}_3(y_1, y_2, y_3) = 1 - (y_1 + y_2 + y_3)^2. \quad (8.16)$$

The example exhibits the two roles cleanly: characteristic-3 modular gates are algebraically easy, whereas unbounded AND/OR gates are only probabilistically easy at low degree.

8.4 The incompatible-modulus obstruction

We now close the second half of the Razborov–Smolensky argument. Lowercase $\text{res}_q(x)$ will denote the full residue $\sum_i x_i \pmod{q} \in \mathbb{Z}_q$, whereas the uppercase QRCC gate MOD_q remains the Boolean zero-residue indicator from Definition 1.1. Distinguishing these two functions is essential: the algebraic obstruction naturally sees all q residue classes, while the circuit is assumed to compute only one Boolean residue test.

Fix distinct primes p, q . Fermat’s congruence $p^{q-1} \equiv 1 \pmod{q}$ shows that q divides $p^{q-1} - 1$. By the finite-field existence theorem and cyclicity of the multiplicative group, the field

$$K := \mathbb{F}_{p^{q-1}} \quad (8.17)$$

contains an element ω of multiplicative order exactly q . Since $|K| = p^{q-1} \geq q$, fix any injection $\psi : \mathbb{Z}_q \hookrightarrow K$. Lagrange interpolation gives a polynomial $M \in K[z]$ of degree at most $q - 1$ such that

$$M(\psi(r)) = \omega^r \quad (r \in \mathbb{Z}_q). \quad (8.18)$$

On the two-point set $\{1, \omega\}$ the linear polynomial

$$J(y) := \frac{y - 1}{\omega - 1} \quad (8.19)$$

satisfies $J(1) = 0$ and $J(\omega) = 1$.

Lemma 8.2 (Low-degree residue obstruction). *Let $m \geq 1$ and let $Q \in K[x_1, \dots, x_m]$ have total degree D . If*

$$D \leq \frac{\sqrt{m}}{8(q-1)}, \quad (8.20)$$

then, under the uniform distribution on $\{0, 1\}^m$,

$$\Pr[Q(x) \neq \psi(\text{res}_q(x))] \geq \frac{75}{512}. \quad (8.21)$$

Proof. Let

$$G := \{x \in \{0, 1\}^m : Q(x) = \psi(\text{res}_q(x))\} \quad (8.22)$$

be the agreement set. Make the bijective coordinate change $y_i = \omega^{x_i} \in \{1, \omega\}$ and put

$$R(y_1, \dots, y_m) := M(Q(J(y_1), \dots, J(y_m))). \quad (8.23)$$

Then

$$D' := \deg R \leq (q-1)D \leq \frac{\sqrt{m}}{8}. \quad (8.24)$$

If $x \in G$ and $y_i = \omega^{x_i}$, equations (8.18)–(8.23) give

$$R(y) = \omega^{\text{res}_q(x)} = \omega^{\sum_i x_i} = \prod_{i=1}^m y_i. \quad (8.25)$$

Let $H \subseteq \{1, \omega\}^m$ be the image of G ; hence $|H| = |G|$ and (8.25) holds on H .

Every function $f : H \rightarrow K$ is the restriction of a multilinear polynomial on the two-point cube. Indeed, for $a \in H$ the product of the coordinatewise linear indicators of $y_i = a_i$ is a delta function at a , and the delta functions span all functions on H . Expand such a polynomial as a linear combination of monomials $y_I := \prod_{i \in I} y_i$.

Put

$$t_* := \frac{m + D'}{2}. \quad (8.26)$$

If $|I| > t_*$, then on H equation (8.25) and $y_i^q = 1$ give

$$y_I = \left(\prod_{i=1}^m y_i \right) \prod_{i \notin I} y_i^{q-1} = R(y) \prod_{i \notin I} y_i^{q-1}. \quad (8.27)$$

Each monomial of R involves at most D' distinct variables. Moreover, any one-variable function on $\{1, \omega\}$, in particular $y \mapsto y^{q-1}$, agrees there with a linear polynomial. Therefore the right side of (8.27), restricted to H , is a linear combination of multilinear monomials involving at most

$$D' + (m - |I|) < t_* \quad (8.28)$$

variables. Thus every function $H \rightarrow K$ is spanned by the multilinear monomials of degree at most $\lfloor t_* \rfloor$. If

$$N_* := \sum_{j=0}^{\lfloor t_* \rfloor} \binom{m}{j}, \quad (8.29)$$

then there are at most $|K|^{N_*}$ functions in this span, while there are exactly $|K|^{|H|}$ functions $H \rightarrow K$. Consequently

$$|G| = |H| \leq N_*. \quad (8.30)$$

It remains to turn (8.30) into a constant gap. Let $B \sim \text{Bin}(m, 1/2)$ and write $W := 2B - m = \sum_{i=1}^m \xi_i$ for independent uniform signs $\xi_i \in \{-1, 1\}$. Direct expansion gives

$$\mathbb{E}W^2 = m, \quad \mathbb{E}W^4 = 3m^2 - 2m. \quad (8.31)$$

For the nonnegative random variable $Y = W^2$, Cauchy–Schwarz gives the Paley–Zygmund estimate

$$\Pr[Y \geq \theta \mathbb{E}Y] \geq (1 - \theta)^2 \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y^2}. \quad (8.32)$$

For completeness, split $\mathbb{E}Y$ over the event $A = \{Y \geq \theta \mathbb{E}Y\}$ and its complement:

$$(1 - \theta)\mathbb{E}Y \leq \mathbb{E}[Y\mathbf{1}_A] \leq \sqrt{\mathbb{E}Y^2 \Pr(A)},$$

which is exactly (8.32). With $\theta = 1/16$, (8.31) yields

$$\Pr\left[|W| \geq \frac{\sqrt{m}}{4}\right] \geq \frac{75}{256}. \quad (8.33)$$

The distribution of W is symmetric, so

$$\Pr\left[B \geq \frac{m}{2} + \frac{\sqrt{m}}{8}\right] \geq \frac{75}{512}. \quad (8.34)$$

By (8.24), $t_* \leq m/2 + \sqrt{m}/16$. Therefore (8.29)–(8.34) imply

$$\frac{|G|}{2^m} \leq \Pr[B \leq t_*] \leq 1 - \frac{75}{512},$$

which proves (8.21). $\square$

Why the characteristic mismatch is used exactly once. The only place where $p \neq q$ is essential is the existence of an element of order q in a finite extension of characteristic p . If $p = q$, then $p^r - 1 \not\equiv 0 \pmod{p}$ for every r , so no finite field of characteristic p has an element of multiplicative order p . This is the algebraic fault line that makes MOD_p easy in characteristic p while leaving MOD_q hard when $q \neq p$.

8.5 The distinct-prime modular lower bound

Corollary 8.3 (Razborov–Smolensky distinct-prime modular lower bound). *Fix distinct primes p, q and an integer depth $d \geq 1$. Let $n > q$, put $m := n - q$, and suppose a depth-at-most- d $AC^0[p]$ circuit C of QRCC gate size $S \geq 1$ computes the QRCC zero-residue function MOD_q on n bits. Set*

$$\delta := \frac{75}{1024q}, \quad A_m := \frac{m^{1/(2d)}}{(p-1)(8(q-1))^{1/d}}. \quad (8.35)$$

Then

$$\boxed{S > \frac{75}{1024qp} p^{A_m}}. \quad (8.36)$$

In particular, for fixed p, q, d ,

$$S = \exp\left(\Omega_{p,q,d}(n^{1/(2d)})\right), \quad (8.37)$$

and therefore

$$\text{MOD}_q \notin AC^0[p]. \quad (8.38)$$

Proof. For each residue $r \in \mathbb{Z}_q$, restrict the final q input positions of C to constants. For $r = 0$ set all of them to zero; for $1 \leq r \leq q - 1$, set exactly $q - r$ of them to one and the rest to zero. The resulting circuit C_r has at most S noninput gates, depth at most d , and on $x \in \{0, 1\}^m$ satisfies

$$C_r(x) = 1 \iff \text{res}_q(x) = r. \quad (8.39)$$

Apply Theorem 8.1 to each C_r , with the uniform input distribution and error δ . We obtain deterministic multilinear polynomials $P_r \in \mathbb{F}_p[x_1, \dots, x_m]$ of degree at most

$$D := \left((p-1) \left\lceil \log_p \frac{S}{\delta} \right\rceil \right)^d \quad (8.40)$$

such that P_r disagrees with C_r on at most a δ fraction of the Boolean cube.

Embed $\mathbb{F}_p$ in the extension field K from (8.17), retain the fixed injection $\psi : \mathbb{Z}_q \hookrightarrow K$, and define

$$Q(x) := \sum_{r=0}^{q-1} \psi(r) P_r(x) \in K[x_1, \dots, x_m]. \quad (8.41)$$

Whenever all q polynomials P_r are correct at x , exactly one of them is one, namely $P_{\text{res}_q(x)}$, and all the others are zero. Hence $Q(x) = \psi(\text{res}_q(x))$. The union bound gives

$$\Pr_x[Q(x) \neq \psi(\text{res}_q(x))] \leq q\delta = \frac{75}{1024}. \quad (8.42)$$

The low-degree residue obstruction says that error below $75/512$ is impossible when $D \leq \sqrt{m}/(8(q-1))$. Thus

$$D > \frac{\sqrt{m}}{8(q-1)}. \quad (8.43)$$

Taking d th roots in (8.40)–(8.43) gives

$$\left\lceil \log_p \frac{S}{\delta} \right\rceil > A_m. \quad (8.44)$$

Since $\lceil u \rceil \leq u + 1$, equation (8.44) implies $\log_p(S/\delta) > A_m - 1$, and therefore

$$S > \delta p^{A_m-1} = \frac{75}{1024qp} p^{A_m},$$

which is (8.36). For fixed p, q, d , $m = n - q = \Theta(n)$, so (8.37) follows; polynomial-size constant-depth families are eventually smaller than this bound, proving (8.38). $\square$

Worked characteristic-mismatch check: $p = 2, q = 3$. Here $K = \mathbb{F}_{2^2} = \mathbb{F}_4$, whose multiplicative group has order three. Thus any nonidentity element ω has order three and $1 + \omega + \omega^2 = 0$. The exact MOD₂ gate polynomial is the parity polynomial in characteristic two, while the residue map modulo three is converted by $x_i \mapsto \omega^{x_i}$ into $\omega^{\sum_i x_i} = \prod_i \omega^{x_i}$. The obstruction lemma says that a polynomial over $\mathbb{F}_4$ agreeing too often with this product cannot have degree $o(\sqrt{m})$; the corollary then converts that degree obstruction into the depth- d size exponent $m^{1/(2d)}$.

8.6 Source boundary and dependency checkpoint

Smolensky's 1987 primary theorem states the same qualitative distinct-prime separation, and in his depth- k normalization obtains the historical $\exp(\Omega(n^{1/(2k)}))$ gate scale for incompatible modular functions. The proof above reconstructs a conservative explicit QRCC constant from the already certified approximation theorem and an elementary dimension count; it does not attribute the coefficient in (8.36) to the source.

Theorem 8.1 depends only on the Volume-X circuit model, the union bound ‘QRCC-I-THM-001’, and finite-field cyclicity ‘QRCC-III-THM-002’. Corollary 8.3 additionally uses the existence of finite extensions ‘QRCC-III-THM-003’, Lagrange interpolation ‘QRCC-III-ALG-023’, and the Cauchy–Schwarz specialization of ‘QRCC-I-LEM-035’. It does not use switching lemmas, LMN Fourier concentration, Håstad’s parity theorem, monotone-circuit lower bounds, or threshold/ACC machinery. The next Volume-X proof frontier is therefore ‘QRCC-X-THM-010’, the Razborov monotone approximation method.

Chapter 9

Monotone Approximation and the Clique Lower Bound

The finite-field method of the preceding chapter exploits algebraic cancellation. Monotone circuits admit no negations and therefore suggest a very different lower-bound strategy: replace the exact lattice of monotone accepting sets by a deliberately coarser lattice, simulate the circuit in that lattice, and charge every discrepancy to one gate. Razborov introduced this *method of approximations* in his 1985 monotone lower bounds. Alon and Boppana gave a self-contained lattice formulation and sharpened the clique application to an exponential lower bound. This chapter reconstructs that route locally.

Two levels must be kept separate. The first theorem is an abstract circuit-to-lattice statement and has no graph-theoretic hypothesis. The second theorem chooses a specific closure lattice on small vertex sets and proves that its join and meet defects cannot collectively hide the clique function unless the circuit is large. In particular, the vague companion-combinatorics placeholder formerly attached to this block is unnecessary: every combinatorial lemma used below is proved here.

9.1 Accepting families and legitimate approximation lattices

Put $B_N := \{0, 1\}^N$, ordered coordinatewise. If $f : B_N \rightarrow \{0, 1\}$ is monotone, write

$$A(f) := \{x \in B_N : f(x) = 1\}. \quad (9.1)$$

Thus $A(f)$ is upward closed. Let $\mathcal{A}_N$ be the set of all upward-closed subsets of B_N , ordered by inclusion.

A *legitimate lattice* $\mathcal{K} \subseteq \mathcal{A}_N$ is a subposet which is itself a lattice under the inherited inclusion order and contains $\emptyset$, B_N , and every coordinate accepting family $A(x_i) = \{x : x_i = 1\}$. We denote its internal join and meet by $M \sqcup N$ and $M \sqcap N$. Because these are respectively the least upper bound and greatest lower bound *inside* $\mathcal{K}$,

$$M \cup N \subseteq M \sqcup N, \quad M \cap N \subseteq M \sqcap N. \quad (9.2)$$

The corresponding *join defect* and *meet defect* are

$$\Delta_{\vee}(M, N) := (M \sqcup N) \setminus (M \cup N), \quad \Delta_{\wedge}(M, N) := (M \cap N) \setminus (M \sqcap N). \quad (9.3)$$

They are exactly the false positives introduced by the approximate OR and the false negatives introduced by the approximate AND.

For a monotone f , define its *lattice distance* $\rho(f, \mathcal{K})$ to be the least integer $t \geq 0$ for which there exist $M, M_1, N_1, \dots, M_t, N_t \in \mathcal{K}$ such that

$$M \subseteq A(f) \cup \bigcup_{i=1}^t \Delta_{\vee}(M_i, N_i) \quad (9.4)$$

and

$$A(f) \subseteq M \cup \bigcup_{i=1}^t \Delta_{\wedge}(M_i, N_i). \quad (9.5)$$

Finally $L^+(f)$ denotes the minimum number of noninput gates in a *fan-in-two* monotone $\{\wedge, \vee\}$ -circuit computing f , in the size convention of Definition 1.1. This is the classical Razborov–Alon–Boppana model used throughout the present chapter; no lower bound for unbounded-fan-in monotone gates is asserted here.

Theorem 9.1 (Razborov monotone approximation method; legitimate-lattice form). *For every monotone Boolean function $f : B_N \rightarrow \{0, 1\}$ and every legitimate lattice $\mathcal{K} \subseteq \mathcal{A}_N$,*

$$\boxed{L^+(f) \geq \rho(f, \mathcal{K}).} \quad (9.6)$$

More precisely, if a monotone circuit for f has t gates, then simulating that circuit inside $\mathcal{K}$ produces an element $M \in \mathcal{K}$ for which (9.4)–(9.5) hold using at most one join or meet defect for each circuit gate.

Proof. Take a monotone straight-line program for f with t fan-in-two AND/OR gates. Run the same straight-line program in $\mathcal{K}$: replace every input x_i by $A(x_i)$, constants by $\emptyset$ and B_N , every OR by $\sqcup$, and every AND by $\sqcap$. For a circuit gate g , let $M_g \in \mathcal{K}$ be the element obtained in this simulation and let f_g be the exact Boolean function computed at g .

We prove by induction along a topological order that, after the first j gates, there are at most j recorded defects such that

$$M_g \subseteq A(f_g) \cup \bigcup \Delta_{\vee}, \quad A(f_g) \subseteq M_g \cup \bigcup \Delta_{\wedge} \quad (9.7)$$

for every already constructed gate g . Inputs and constants have no defect.

Suppose first that the new gate is $g = u \vee v$. Then

$$M_g = M_u \sqcup M_v = (M_u \cup M_v) \cup \Delta_{\vee}(M_u, M_v). \quad (9.8)$$

Substituting the first induction inclusion for M_u and M_v gives the first half of (9.7), with the new join defect added. For the other direction,

$$A(f_g) = A(f_u) \cup A(f_v) \subseteq M_u \cup M_v \cup \bigcup \Delta_{\wedge} \subseteq M_g \cup \bigcup \Delta_{\wedge}, \quad (9.9)$$

so OR creates no new meet defect.

If instead $g = u \wedge v$, then $M_g = M_u \sqcap M_v \subseteq M_u \cap M_v$. A point of M_g which lies in none of the previously recorded join defects lies in both $A(f_u)$ and $A(f_v)$, hence in $A(f_g)$. Thus the first inclusion in (9.7) persists. Conversely,

$$\begin{aligned} A(f_g) &= A(f_u) \cap A(f_v) \\ &\subseteq (M_u \cap M_v) \cup \bigcup \Delta_{\wedge} \\ &= M_g \cup \Delta_{\wedge}(M_u, M_v) \cup \bigcup \Delta_{\wedge}, \end{aligned} \quad (9.10)$$

so AND creates at most its one new meet defect.

At the output gate, (9.7) is exactly (9.4)–(9.5) with at most t defect pairs. Therefore $\rho(f, \mathcal{K}) \leq t$. Minimizing over monotone circuits gives (9.6). $\square$

Worked defect example. The theorem is useful precisely because the internal lattice operations need not be exact set union and intersection. The clique lattice constructed below makes this visible already on three vertices. Take $l = r = 2$ and the two edge cores $X = \{1, 2\}$ and $Y = \{2, 3\}$. The families generated by X and Y represent the predicates “edge 12 is present” and “edge 23 is present.” Their internal meet has no common two-vertex generator, whereas the exact intersection contains every graph carrying both edges; those graphs form a meet defect. For the join, the two cores have intersection $\{2\}$; the closure rule below therefore reaches a singleton and then the empty core, so the internal join is the family of all graphs. Graphs carrying neither edge are then join defects. The circuit simulation is an approximation, but Theorem 9.1 guarantees that every such discrepancy is charged to a gate rather than appearing mysteriously at the output.

9.2 The small-core clique lattice

Fix a vertex set $V = [m]$ and integers $l, r \geq 2$. Put

$$\mathcal{V}_l := \{W \subseteq V : |W| \leq l\}. \quad (9.11)$$

For $W, W_1, \dots, W_r \in \mathcal{V}_l$, say that $W_1, \dots, W_r$ *r-imply* W , written $W_1, \dots, W_r \vdash W$, when

$$W_i \cap W_j \subseteq W \quad (1 \leq i < j \leq r). \quad (9.12)$$

For $\mathcal{C} \subseteq \mathcal{V}_l$, write $\mathcal{C} \vdash W$ if some r members of $\mathcal{C}$ *r-imply* W . Repetitions are allowed. A family $\mathcal{C}$ is *closed* if every W implied by $\mathcal{C}$ already belongs to $\mathcal{C}$; in addition, as a harmless normalization, a closed family containing a singleton also contains $\emptyset$. The *closure* $\mathcal{C}^*$ is the intersection of all closed families containing $\mathcal{C}$.

Repeated witnesses show that closed families are upward closed inside $\mathcal{V}_l$: if $X \in \mathcal{C}$, $X \subseteq W$, and $|W| \leq l$, then r copies of X imply W . The extra singleton rule simply records that a clique on zero or one vertices imposes no edge requirement.

For $W \subseteq V$, let K_W denote the complete graph on W . For $\mathcal{C} \subseteq \mathcal{V}_l$, define

$$[\mathcal{C}] := \{G \text{ on } V : \text{some } W \in \mathcal{C} \text{ has } K_W \subseteq G\}. \quad (9.13)$$

Thus $[\mathcal{C}]$ is a monotone family of edge vectors. Define

$$\mathcal{K}(m, r, l) := \{[\mathcal{C}] : \mathcal{C} \subseteq \mathcal{V}_l \text{ is closed}\}. \quad (9.14)$$

Lemma 9.2 (Clique-lattice operations). *$\mathcal{K}(m, r, l)$ is a legitimate lattice. If $\mathcal{A}, \mathcal{B}$ are closed, then its internal join and meet are*

$$[\mathcal{A}] \sqcup [\mathcal{B}] = [(\mathcal{A} \cup \mathcal{B})^*], \quad [\mathcal{A}] \sqcap [\mathcal{B}] = [\mathcal{A} \cap \mathcal{B}]. \quad (9.15)$$

Proof. The empty family represents no graph, while any closed family containing $\emptyset$ represents all graphs. The coordinate predicate for an edge $e = \{u, v\}$ is represented by the closure of $\{\{u, v\}\}$; every set added to this closure still contains u and v , so the represented graphs are exactly those containing e . Hence the required constants and coordinates belong to the family.

If $[\mathcal{D}]$ is an upper bound for $[\mathcal{A}]$ and $[\mathcal{B}]$, then for every $W \in \mathcal{A} \cup \mathcal{B}$ the graph K_W belongs to $[\mathcal{D}]$. Consequently some $D \in \mathcal{D}$ satisfies $K_D \subseteq K_W$. If $|D| \geq 2$ this forces $D \subseteq W$, and upward closure gives $W \in \mathcal{D}$. If $|D| \leq 1$, closedness forces $\emptyset \in \mathcal{D}$, hence every member of $\mathcal{V}_l$ lies in $\mathcal{D}$, so again $W \in \mathcal{D}$. Thus $\mathcal{A} \cup \mathcal{B} \subseteq \mathcal{D}$, hence $(\mathcal{A} \cup \mathcal{B})^* \subseteq \mathcal{D}$. This proves the join formula. The same K_W test shows that any common lower bound $[\mathcal{D}]$ of $[\mathcal{A}]$ and $[\mathcal{B}]$ has $\mathcal{D} \subseteq \mathcal{A} \cap \mathcal{B}$, proving the meet formula. $\square$

The implication relation is a relaxed sunflower condition. A classical sunflower has pairwise intersections equal to one common core; here we only need all pairwise intersections to lie *inside* a prospective core. No external sunflower theorem will be used.

Lemma 9.3 (Few small minimal cores). *Let $r \geq 2$ and $k \geq 0$. Suppose $\mathcal{F}$ is a family of sets of size at most k with the following property: there do not exist $W, W_1, \dots, W_r \in \mathcal{F}$ and a proper subset $U \subsetneq W$ such that $W_i \cap W_j \subseteq U$ for all $i < j$. Then*

$$|\mathcal{F}| \leq (r-1)^k. \quad (9.16)$$

Consequently, if $\mathcal{A} \subseteq \mathcal{V}_l$ is closed, then for every $k \leq l$ it has at most $(r-1)^k$ inclusion-minimal members of size at most k .

Proof. We prove (9.16) by induction on r . For $r = 2$, two distinct members W_1, W_2 would violate the property by taking $U = W_1 \cap W_2$ and choosing as W whichever of W_1, W_2 properly contains U . Hence $|\mathcal{F}| \leq 1 = (r-1)^k$.

Assume $r \geq 3$ and choose $D \in \mathcal{F}$. For each $C \subseteq D$, let

$$\mathcal{F}_C := \{W \setminus C : W \in \mathcal{F}, W \cap D = C\}. \quad (9.17)$$

If $\mathcal{F}_C$ violated the stated property with $r-1$ witnesses and a proper core $U' \subsetneq W'$, adjoining C to every set and using D as the r th witness would violate the property of $\mathcal{F}$. Hence $\mathcal{F}_C$ satisfies the induction hypothesis with parameter $k - |C|$. Therefore

$$\begin{aligned} |\mathcal{F}| &= \sum_{C \subseteq D} |\mathcal{F}_C| \leq \sum_{C \subseteq D} (r-2)^{k-|C|} \\ &\leq \sum_{i=0}^k \binom{k}{i} (r-2)^{k-i} = (r-1)^k. \end{aligned} \quad (9.18)$$

For the consequence, let $\mathcal{F}$ be the inclusion-minimal members of $\mathcal{A}$ having size at most k . A forbidden configuration would give $\mathcal{A} \vdash U$, so closedness would put U in $\mathcal{A}$, contrary to the minimality of W . Apply (9.16). $\square$

9.3 Closure growth and random multipartite graphs

For a nonclosed family $\mathcal{C}_i$, choose an inclusion-minimal $W_{i+1} \in \mathcal{V}_l \setminus \mathcal{C}_i$ that is forced either by the r -implication rule or by the singleton-to-empty normalization, and put

$$\mathcal{C}_{i+1} := \mathcal{C}_i \cup \{W \in \mathcal{V}_l : W_{i+1} \subseteq W\}. \quad (9.19)$$

Starting from $\mathcal{C}_0 = \mathcal{C}$, this *closure algorithm* terminates at $\mathcal{C}^*$.

Lemma 9.4 (Closure algorithm length). *For every $\mathcal{C} \subseteq \mathcal{V}_l$, the closure algorithm terminates at $\mathcal{C}^*$ after at most*

$$|\mathcal{V}_l| \leq 2m^l \quad (9.20)$$

iterations.

Proof. Each iteration adds the newly selected set W_{i+1} , so no set can be selected twice and the number of iterations is at most $|\mathcal{V}_l|$. Moreover, for $m \geq 2$ and $l \geq 1$,

$$|\mathcal{V}_l| = \sum_{k=0}^l \binom{m}{k} \leq \sum_{k=0}^l m^k \leq 2m^l. \quad (9.21)$$

It remains only to identify the terminal family. Inductively $\mathcal{C}_i \subseteq \mathcal{C}^*$: an implied set belongs to every closed superset of $\mathcal{C}_i$, and once such a set is added, all of its supersets of size at most l are themselves implied by repeated copies. The singleton normalization is also part of the definition of closedness. When the algorithm stops, the current family is closed and contains the original $\mathcal{C}$, hence it contains $\mathcal{C}^*$. The reverse inclusion was just proved, so the terminal family is exactly $\mathcal{C}^*$. $\square$

Let $g \geq l$. A *random g -coloring* O assigns each vertex of V independently and uniformly a color in $[g]$. A set W is *properly colored* if all its vertices receive distinct colors. Write $G(O)$ for the complete g -partite graph whose edges join pairs of vertices of different colors.

Lemma 9.5 (One implication under a random coloring). *If $\mathcal{A} \subseteq \mathcal{V}_l$ and $\mathcal{A} \vdash W$, then*

$$\Pr[W \text{ is properly colored and no } A \in \mathcal{A} \text{ is properly colored}] \leq \left(1 - \frac{(g)_l}{g^l}\right)^r, \quad (9.23)$$

where $(g)_l = g(g-1)\cdots(g-l+1)$.

Proof. Choose witnesses $W_1, \dots, W_r \in \mathcal{A}$ implying W . Condition on the *entire color assignment* on W , restricted to assignments in which W is properly colored. The events that the W_i fail to be properly colored are then mutually independent, because the sets $W_i \setminus W$ are pairwise disjoint by (9.12). Their conditional probabilities depend only on the cardinalities below, not on the particular distinct colors used on W ; hence the same product bound remains valid after forgetting the particular proper coloring of W . Put $p_i = |W_i \cap W|$ and $q_i = |W_i \setminus W|$. Conditioned on a proper coloring of W , the q_i outside vertices must receive distinct colors avoiding the p_i already used colors, so

$$\Pr[W_i \text{ properly colored} \mid W \text{ properly colored}] = \frac{(g-p_i)_{q_i}}{g^{q_i}} \geq \frac{(g)_l}{g^l}, \quad (9.24)$$

because $p_i + q_i \leq l$. Multiplying the r conditional failure probabilities proves (9.23). $\square$

Lemma 9.6 (Closure error on a multipartite negative input). *For every $\mathcal{C} \subseteq \mathcal{V}_l$,*

$$\Pr[G(O) \in [\mathcal{C}^*] \setminus [\mathcal{C}]] \leq 2m^l \left(1 - \frac{(g)_l}{g^l}\right)^r. \quad (9.25)$$

Proof. Run the closure algorithm and let $W_1, \dots, W_p$ be its selected sets. The event on the left says that some W_i is the first newly added set that is properly colored while no set in the preceding family is properly colored. These first-hit events are disjoint. Before W_i is added by the r -implication rule, the current family implies it, so the preceding lemma bounds its first-hit probability by the parenthesized term in (9.25). A singleton-to-empty normalization step creates no additional first-hit event: if the singleton was already in the initial family, then $[\mathcal{C}]$ was already the family of all graphs; if it was added during the algorithm, then that singleton is properly colored under every coloring, so the first-hit event occurred when the singleton itself was added. Hence every nonzero first-hit probability is charged to an r -implication step. Finally $p \leq 2m^l$ by (9.20). $\square$

9.4 The exponential monotone lower bound for CLIQUE

Let $\text{CLIQUE}(m, s)$ be the monotone Boolean function on the $\binom{m}{2}$ edge variables of a graph on $[m]$ which is one exactly when the graph contains a clique on s vertices.

Theorem 9.7 (Alon–Boppana/Razborov exponential monotone CLIQUE lower bound). *Assume $m \geq 256$ and*

$$4 \leq s \leq \frac{1}{8} \left(\frac{m}{\log_2 m} \right)^{2/3}. \quad (9.26)$$

Set

$$l := \lfloor \sqrt{s} \rfloor, \quad r := \lceil 8\sqrt{s} \log_2 m \rceil, \quad h := \left\lceil \frac{l+1}{2} \right\rceil. \quad (9.27)$$

Then every monotone circuit computing $\text{CLIQUE}(m, s)$ has size at least

$$L^+(\text{CLIQUE}(m, s)) \geq \frac{1}{8} \left(\frac{m}{s(r-1)} \right)^h > \frac{1}{8} \left(\frac{m}{8s^{3/2} \log_2 m} \right)^h. \quad (9.28)$$

In particular, for

$$s_m := \left\lfloor \frac{1}{8} \left(\frac{m}{\log_2 m} \right)^{2/3} \right\rfloor \quad (9.29)$$

and all sufficiently large m ,

$$L^+(\text{CLIQUE}(m, s_m)) = \exp\left(\Omega\left((m/\log m)^{1/3}\right)\right). \quad (9.30)$$

Equivalently, if $N = \binom{m}{2}$ is the number of input bits, the specialization is $\exp(\Omega(N^{1/6}/(\log N)^{1/3}))$.

Proof. Use the legitimate lattice $\mathcal{K} = \mathcal{K}(m, r, l)$ from (9.14), and write $f = \text{CLIQUE}(m, s)$. Put $t = \rho(f, \mathcal{K})$. By the definition of lattice distance there are $M, M_i, N_i \in \mathcal{K}$ for $1 \leq i \leq t$ satisfying (9.4)–(9.5). We prove

$$t \geq \frac{1}{8} \left(\frac{m}{s(r-1)} \right)^h. \quad (9.31)$$

The approximation theorem will then imply the same lower bound for $L^+(f)$.

The parameter restriction gives

$$a := \frac{s(r-1)}{m} < \frac{8s^{3/2} \log_2 m}{m} \leq \frac{1}{2\sqrt{2}} < \frac{1}{2}. \quad (9.32)$$

Write $M = [\mathcal{A}]$ with $\mathcal{A}$ closed. There are two cases.

Case 1: M is not the family of all graphs. Every s -clique graph is a positive input for f , so (9.5) says that every one of the $\binom{m}{s}$ possible s -cliques lies in M or in one of the meet defects $\Delta_\wedge(M_i, N_i)$.

First, M contains fewer than half of those clique graphs. Since M is not all graphs, $\mathcal{A}$ contains no set of size at most one. Every s -clique represented by M contains an inclusion-minimal member of $\mathcal{A}$. For each $2 \leq k \leq l$, there are at most $(r-1)^k$ minimal members of size k , and each lies in exactly $\binom{m-k}{s-k}$ of the s -subsets. Hence

$$\begin{aligned} \frac{|M \cap \{s\text{-clique graphs}\}|}{\binom{m}{s}} &\leq \sum_{k=2}^l (r-1)^k \frac{\binom{m-k}{s-k}}{\binom{m}{s}} \\ &\leq \sum_{k=2}^l \left(\frac{s(r-1)}{m} \right)^k \leq \sum_{k=2}^{\infty} 2^{-k} = \frac{1}{2}. \end{aligned} \quad (9.33)$$

Second, fix i and write $M_i = [\mathcal{A}_i]$ and $N_i = [\mathcal{B}_i]$ with $\mathcal{A}_i, \mathcal{B}_i$ closed. By (9.15),

$$\Delta_\wedge(M_i, N_i) = ([\mathcal{A}_i] \cap [\mathcal{B}_i]) \setminus [\mathcal{A}_i \cap \mathcal{B}_i]. \quad (9.34)$$

If the clique graph on an s -set Z lies in this defect, then Z contains minimal $X \in \mathcal{A}_i$ and $Y \in \mathcal{B}_i$ but contains no member of $\mathcal{A}_i \cap \mathcal{B}_i$. If $|X \cup Y| \leq l$, upward closure of both families would put $X \cup Y$ in their intersection, a contradiction. Thus $|X \cup Y| > l$, so at least one of X, Y has size at least $h = \lceil (l+1)/2 \rceil$. Using the minimal-core bound for both families,

$$\begin{aligned} \frac{|\Delta_\wedge(M_i, N_i) \cap \{s\text{-clique graphs}\}|}{\binom{m}{s}} &\leq 2 \sum_{k=h}^l (r-1)^k \frac{\binom{m-k}{s-k}}{\binom{m}{s}} \\ &\leq 2 \sum_{k=h}^l a^k \leq 4a^h. \end{aligned} \quad (9.35)$$

Since M covers at most one half of the clique graphs, (9.5) and (9.35) give

$$1 \leq \frac{1}{2} + 4ta^h, \quad (9.36)$$

which is exactly (9.31).

Case 2: M is the family of all graphs. Every complete $(s-1)$ -partite graph is a negative input for f . Therefore (9.4) says that every such graph belongs to at least one join defect. Write $M_i = [\mathcal{A}_i]$, $N_i = [\mathcal{B}_i]$, and $\mathcal{C}_i = \mathcal{A}_i \cup \mathcal{B}_i$. From (9.15),

$$\Delta_\vee(M_i, N_i) = [\mathcal{C}_i^*] \setminus [\mathcal{C}_i]. \quad (9.37)$$

Take a random $(s-1)$ -coloring O and its complete multipartite graph $G(O)$. We need two elementary parameter bounds. With $g = s-1$ and $l = \lfloor \sqrt{s} \rfloor$, the probability that l independent colors are all distinct satisfies

$$\frac{(g)_l}{g^l} \geq \frac{1}{3}. \quad (9.38)$$

Indeed, the union bound on a color collision gives

$$1 - \frac{(g)_l}{g^l} \leq \binom{l}{2} \frac{1}{g} \leq \frac{s}{2(s-1)} \leq \frac{2}{3}. \quad (9.39)$$

Because $r \geq 8l \log_2 m$ and $2/3 < 2^{-1/2}$,

$$2m^l \left(\frac{2}{3}\right)^r < 2m^l 2^{-r/2} \leq 2m^{-3l} \leq m^{-2l}, \quad (9.40)$$

where the last inequality uses $m^l \geq 2$. The closure-error lemma therefore gives, for every i ,

$$\Pr[G(O) \in \Delta_\vee(M_i, N_i)] \leq m^{-2l}. \quad (9.41)$$

If (9.31) failed, then certainly $t < m^l$: the base $m/[s(r-1)]$ is at most m and $h \leq l$. A union bound would therefore give

$$\Pr \left[G(O) \in \bigcup_{i=1}^t \Delta_\vee(M_i, N_i) \right] \leq tm^{-2l} < m^{-l} < 1. \quad (9.42)$$

Thus some complete $(s - 1)$ -partite graph would lie in none of the join defects, contradicting the conclusion from (9.4). This proves (9.31) in the second case as well.

By Theorem 9.1, $L^+(f) \geq t$, proving the first inequality in (9.28). Since $r - 1 < 8\sqrt{s} \log_2 m$, the second inequality follows.

Finally take $s = s_m$ from (9.29). Then the base in the second expression of (9.28) is at least 2, while $h = \Omega(\sqrt{s}) = \Omega((m/\log m)^{1/3})$. Hence $L^+(\text{CLIQUE}(m, s_m)) \geq 2^{\Omega((m/\log m)^{1/3})}$, which is (9.30). Since $N = \binom{m}{2} = \Theta(m^2)$ and $\log N = \Theta(\log m)$, the input-bit formulation follows. $\square$

A finite parameter check. Take $m = 2^{24}$ and choose the theorem's extremal scale $s = \lfloor \frac{1}{8}(m/\log_2 m)^{2/3} \rfloor = 984$. Then $l = 31$, $r = 6023$, and $h = 16$; the base $m/[s(r - 1)]$ in (9.28) is about 2.83. The negative test objects are complete $(s - 1)$ -partite graphs. The proof does not sample arbitrary negative graphs: the coloring construction guarantees they contain no s -clique, while (9.25) says that one lattice join defect captures only a tiny fraction of them. On the positive side, the test objects are the $\binom{m}{s}$ clique graphs themselves, and (9.33)–(9.35) count exactly how many can be hidden by M and by one meet defect. The two distributions are therefore serving different halves of the distance definition, not being mixed into a single average-error statement.

9.5 Source and hypothesis checkpoint

Razborov's 1985 note *Lower bounds on the monotone complexity of some Boolean functions* introduced the approximation method and obtained the first superpolynomial monotone lower bounds for CLIQUE. Alon and Boppana, *The monotone circuit complexity of Boolean functions*, *Combinatorica* 7(1) (1987), 1–22, restate the legitimate-lattice inequality as their Theorem 2.1 and give the self-contained closure/coloring argument leading to their Lemma 3.8 and Theorem 3.9. Their source theorem covers $3 \leq s \leq \frac{1}{4}(m/\log_2 m)^{2/3}$ and yields the same $\exp(\Omega((m/\log m)^{1/3}))$ specialization. QRCC uses the slightly smaller local range $s \leq \frac{1}{8}(m/\log_2 m)^{2/3}$ and the larger coloring parameter $r = \lceil 8\sqrt{s} \log_2 m \rceil$. This deliberately trades source-sharp constants for the elementary closure bound $|\mathcal{V}_l| \leq 2m^l$; the exponent and the exponential specialization are unchanged. The additional $s \geq 4$ and $m \geq 256$ restrictions merely remove small exceptional cases.

The proof is entirely monotone. It does not lower-bound unrestricted Boolean circuits for CLIQUE, does not imply $P \neq NP$, and does not use the finite-field approximation machinery of Chapter 8. The graph variables are edge indicators; m is the number of vertices and $N = \binom{m}{2}$ the number of circuit inputs. The words “exponential lower bound” in this chapter mean exponential in a positive power of the natural parameter, specifically (9.30), not a $2^{\Omega(N)}$ lower bound in the number of edge variables.

Chapter 10

Depth-Two Threshold Circuits and Parity

The preceding chapters separated constant-depth AND/OR circuits, modular circuits in incompatible characteristics, and monotone circuits. Threshold gates form a different frontier. A single threshold gate already computes majority, and arbitrary real weights make the model substantially stronger than the bounded-depth Boolean bases treated earlier. Consequently this chapter fixes the model before stating any lower bound and keeps three notions separate: a selected explicit-function theorem that will be proved locally, a dated research frontier, and the stronger ACC^0 algorithm-to-lower-bound results treated separately in Chapter 11.

10.1 The depth-two LTF model

A *linear threshold function* (LTF) on Boolean inputs is a function

$$h(x_1, \dots, x_n) = \mathbf{1} \left[\sum_{i=1}^n a_i x_i \geq \theta \right], \quad a_i, \theta \in \mathbb{R}. \quad (10.1)$$

No bound on the magnitude or bit complexity of the real weights is imposed in the lower-bound model. A depth-two LTF $\circ$ LTF circuit has g bottom LTF gates fed by the input variables and one output LTF gate. Following the source convention, the output gate may also receive input variables directly. The *gate count* is $g+1$. The *wire count* W counts every input-to-bottom, bottom-to-output, and optional input-to-output wire occurrence. Repeated wires into one threshold gate may be combined algebraically; hence the number of distinct variables on which a bottom gate depends is at most its incoming-wire count.

This arbitrary-real model is deliberately more permissive than the TC^0 convention in Definition 1.1, where threshold gates are recorded with integer weights. On a fixed finite Boolean cube every LTF has an integer-weight representation after a sufficiently fine perturbation and rescaling, but the required integers need not have polynomially many bits. Thus no efficient-weight claim is hidden in the lower bound below. The matching upper construction, by contrast, will use integer weights of magnitude at most one at the output and unit weights at the bottom gates.

We use the uniform approximation convention

$$\text{err}_{\text{unif}}(C, f) := \Pr_{x \sim \{0,1\}^n} [C(x) \neq f(x)]. \quad (10.2)$$

The selected theorem concerns error at most $1/100$, not exact computation and not correlation under an unspecified distribution.

10.2 Dated restricted-circuit frontier

The following ledger is a source checkpoint dated **2026-08-29**. It is not a single theorem and the rows are not interchangeable.

Model	Verified position used by QRCC
Depth-two arbitrary-weight THR $\circ$ THR	Kane–Williams (STOC 2016) prove superlinear explicit average-case lower bounds and the tight PARITY benchmark reconstructed below. Chen–Tal–Wang (ECCC TR26-039 / STOC 2026) prove that for every $\varepsilon > 0$ there is a function in E^{NP} requiring $n^{2.5-\varepsilon}$ -size THR $\circ$ THR circuits.
Explicit depth- d TC^0 , $d > 2$	Hatami–Hoza–Tal–Tell prove, for $d = o(\log \log n)$, an explicit function computed by a uniform depth- $(d+1)$ AC^0 circuit with $O(n)$ wires but not by depth- d TC^0 circuits with $n^{1+\gamma}$ wires, where $\gamma = 2^{-\Theta(d)}$; the lower bound is average-case as well.
ACC^0	Williams’ nonuniform ACC^0 separation is an algorithm-to-lower-bound result for $NEXP$ and belongs to the Chapter-11 campaign. It is not used here as a direct explicit-function threshold lower bound.

The 2026 depth-two theorem does not subsume the explicit-depth hierarchy row: its hard function is placed in E^{NP} and the model is specifically depth two. Conversely, the depth- d hierarchy result does not supply an $n^{2.5-\varepsilon}$ depth-two gate bound. Restricted subclasses such as THR $\circ$ MAJ or bounded-weight threshold circuits can have much stronger lower bounds and are not silently identified with arbitrary-weight THR $\circ$ THR.

10.3 A local Littlewood–Offord estimate

The lower bound will leave one random variable alive in each of two large blocks. The key question is whether a threshold gate remains genuinely nontrivial. We reconstruct the required anti-concentration estimate so that the proof does not depend on the currently open hypercontractive entry QRCC-I-THM-020.

Lemma 10.1 (Interval Littlewood–Offord bound). *Let $X_1, \dots, X_N$ be independent uniform bits and $L(X) = \sum_i a_i X_i$. Suppose at least $k \geq 1$ coefficients satisfy $|a_i| \geq \Delta > 0$. For every closed interval I of length at most Δ ,*

$$\Pr[L(X) \in I] \leq \frac{4}{\sqrt{k}}. \quad (10.3)$$

Proof. Condition on every coordinate except any chosen k whose coefficient magnitudes are at least Δ . Replacing X_i by $1 - X_i$ when $a_i < 0$ only shifts the sum by a constant and makes the selected coefficients positive. Hence the remaining outcomes are indexed by subsets of $[k]$, and moving from a subset A to a strict superset B increases the selected sum by at least Δ .

For a half-open interval J of length Δ , the subsets whose sums lie in J therefore form an antichain. The same is true for a singleton. For completeness, choose a uniform random permutation of $[k]$. A fixed set A of size r occurs as the first r elements with probability $1/\binom{k}{r}$, and an antichain can occur as an initial segment at most once along any permutation. Hence

$$\sum_{A \in \mathcal{F}} \binom{k}{|A|}^{-1} \leq 1$$

for every antichain $\mathcal{F}$, which gives $|\mathcal{F}| \leq \binom{k}{\lfloor k/2 \rfloor}$.

We also record the needed central-binomial estimate rather than hiding it. If $k = 2r$, put $p_r = 4^{-r} \binom{2r}{r}$. Since $p_{r+1}/p_r = (2r+1)/(2r+2)$ and

$$\left(\frac{2r+1}{2r+2} \right)^2 \leq \frac{r+1}{r+2},$$

induction from $p_1 = 1/2$ gives $p_r \leq 1/\sqrt{r+1}$. For odd $k = 2r+1$ the central mass is $((2r+1)/(2r+2))p_r < p_r$. Thus in both cases

$$2^{-k} \binom{k}{\lfloor k/2 \rfloor} \leq \frac{2}{\sqrt{k}}. \quad (10.4)$$

A closed interval of length Δ is the union of one half-open interval of that length and its right endpoint. Applying the antichain estimate to the two pieces gives (10.3). $\square$

For an LTF $h(x) = \mathbf{1}[L(x) \geq \theta]$, let S be the support of its nonzero coefficients and $u = |S|$. For $i \in S$, define the rank

$$k_i := |\{j \in S : |a_j| \geq |a_i|\}|. \quad (10.5)$$

After ordering the support by decreasing coefficient magnitude, k_i is at least the position of i in that order, and therefore

$$\sum_{i \in S} \frac{1}{\sqrt{k_i}} \leq \sum_{r=1}^u \frac{1}{\sqrt{r}} \leq 2\sqrt{u}. \quad (10.6)$$

Let $\mathcal{P}$ be a partition of $[n]$ into m equal-sized parts. A restriction $\rho \sim R_{\mathcal{P}}$ chooses one coordinate uniformly in each part to remain live and assigns independent uniform bits to all other coordinates. Write B for the random live set.

Lemma 10.2 (Forcing one LTF to a constant). *For every n -input LTF with support size u ,*

$$\Pr_{\rho \sim R_{\mathcal{P}}} [h \upharpoonright \rho \text{ is nonconstant}] \leq \frac{12m\sqrt{u}}{n} \leq \frac{12m}{\sqrt{n}}. \quad (10.7)$$

Proof. First fix the live set B and randomize only the coordinates outside B . The contribution of the fixed coordinates will be denoted $L'(x)$. The gate can remain nonconstant only if $L'(x)$ lies in the threshold window swept out by the live variables. That window has length $\sum_{i \in B \cap S} |a_i|$ and can be partitioned into intervals I_i of length $|a_i|$, one for each $i \in B \cap S$.

Let k'_i be the number of support coordinates outside B whose coefficient magnitude is at least $|a_i|$. There are two cases. If some $i \in B \cap S$ has at least $k'_i/2$ support coordinates of rank at most k_i already in B , then

$$\sum_{j \in B \cap S} \frac{1}{\sqrt{k'_j}} \geq \frac{1}{2}, \quad (10.8)$$

so the trivial probability bound 1 is at most six times this sum. Otherwise $k'_i \geq k_i/2$ for every $i \in B \cap S$. Lemma 10.1 and a union bound over the intervals give

$$\Pr[h \upharpoonright \rho \text{ nonconstant} \mid B] \leq \sum_{i \in B \cap S} \frac{4}{\sqrt{k'_i}} \leq 6 \sum_{i \in B \cap S} \frac{1}{\sqrt{k_i}}. \quad (10.9)$$

Thus (10.9) holds in both cases.

Each coordinate is live with probability m/n . Averaging (10.9) over B and using (10.6),

$$\Pr[h \upharpoonright \rho \text{ nonconstant}] \leq 6 \frac{m}{n} \sum_{i \in S} \frac{1}{\sqrt{k_i}} \leq \frac{12m\sqrt{u}}{n}, \quad (10.10)$$

which proves the claim. $\square$

Call a restricted bottom gate *trivial* if it is a constant or a Boolean function of a single live variable. Such a function is one of $0, 1, x_i, 1 - x_i$ and hence is affine in that bit before it is substituted into the top LTF.

Lemma 10.3 (Forcing a sparse LTF to one live input). *If an LTF depends on at most u inputs, then*

$$\Pr_{\rho \sim R_{\mathcal{P}}} [h \upharpoonright \rho \text{ is not trivial}] \leq \frac{12u m^{3/2}}{n^{3/2}}. \quad (10.11)$$

In particular, for $m = 2$ this probability is at most $34u/n^{3/2}$.

Proof. Nontriviality requires both that the restricted gate be nonconstant and that at least two variables from its support S survive. The fixed-live-set argument (10.9) therefore gives

$$\Pr[\text{nontrivial}] \leq 6 \sum_{i \in S} \frac{1}{\sqrt{k_i}} \Pr[i \in B, |B \cap S| \geq 2]. \quad (10.12)$$

For a fixed $i \in S$,

$$\Pr[i \in B, |B \cap S| \geq 2] \leq \min \left\{ \frac{m}{n}, u \frac{m^2}{n^2} \right\} \leq \frac{\sqrt{u} m^{3/2}}{n^{3/2}}. \quad (10.13)$$

Indeed two distinct coordinates in the same partition block can never both survive, while coordinates in different blocks survive together with probability $(m/n)^2$; the first inequality follows by a union bound, and the second from $\min\{a, b\} \leq \sqrt{ab}$. Substituting (10.6) into (10.12) gives (10.11). Since $12 \cdot 2^{3/2} < 34$, the stated two-block bound follows. $\square$

10.4 The two-bit obstruction

The terminal obstruction is elementary but essential: parity on two bits is not linearly separable.

Lemma 10.4 (Two-bit parity versus one LTF). *Every LTF on two Boolean inputs disagrees with $x_1 \oplus x_2$ on at least one of the four inputs. The same is true for its complement $1 \oplus x_1 \oplus x_2$. Hence its uniform agreement with either function is at most $3/4$.*

Proof. Suppose $\mathbf{1}[w_1x_1 + w_2x_2 \geq \theta]$ agreed with XOR everywhere. The positive points $(1, 0)$ and $(0, 1)$ imply $w_1, w_2 \geq \theta$, while the negative point $(0, 0)$ gives $\theta > 0$ and the negative point $(1, 1)$ gives $w_1 + w_2 < \theta$. But $w_1 + w_2 \geq 2\theta > \theta$, a contradiction.

For XNOR, positivity at $(0, 0)$ gives $\theta \leq 0$, negativity at $(1, 0)$ and $(0, 1)$ gives $w_1, w_2 < \theta$, and positivity at $(1, 1)$ requires $w_1 + w_2 \geq \theta$. Instead $w_1 + w_2 < 2\theta \leq \theta$, again a contradiction. $\square$

Theorem 10.5 (Tight average-case PARITY bounds for depth-two LTF circuits). *Let $n \geq 5$. Suppose an arbitrary-real-weight LTF $\circ$ LTF circuit C agrees with PARITY_n on at least 99% of the*

uniform inputs. If g is its number of bottom gates and W its total wire count in the model fixed above, then

$$\boxed{g > \frac{\sqrt{n-1}}{100}, \quad W > \frac{(n-1)^{3/2}}{150}.} \quad (10.14)$$

Conversely, for every $n \geq 100$ there is an integer-weight depth-two LTF circuit agreeing with parity on at least 99% of the inputs and using at most

$$10\sqrt{n} + 3 \quad \text{gates} \quad \text{and} \quad (n+1)(10\sqrt{n} + 2) \quad \text{wires}. \quad (10.15)$$

Thus the 99%-average-case gate and wire complexities are respectively $\Theta(\sqrt{n})$ and $\Theta(n^{3/2})$; the lower bounds hold for arbitrary real weights, while the displayed upper bound uses small integer weights.

Proof. We first prove the lower bound for even n . Partition $[n]$ into two equal blocks and draw $\rho \sim R_{\mathcal{P}}$, leaving exactly two variables live. Let A be the event that every bottom gate becomes trivial. If a bottom gate has support size u_j , Lemma 10.2 with $m = 2$ gives

$$\Pr[\text{gate } j \text{ nontrivial}] \leq \Pr[\text{gate } j \text{ nonconstant}] \leq \frac{24}{\sqrt{n}}. \quad (10.16)$$

Hence, by the union bound,

$$\Pr[A^c] \leq \frac{24g}{\sqrt{n}}. \quad (10.17)$$

The sparse version instead gives

$$\Pr[A^c] \leq \sum_j \frac{34u_j}{n^{3/2}} \leq \frac{34W}{n^{3/2}}, \quad (10.18)$$

because the sum of the bottom support sizes is at most the total wire count. Consequently either assumption

$$g \leq \frac{\sqrt{n}}{100} \quad \text{or} \quad W \leq \frac{n^{3/2}}{150} \quad (10.19)$$

would imply $\Pr[A] \geq 0.76$.

On event A , every bottom output is 0, 1, a live bit, or the complement of a live bit. Substituting these affine unary functions into the output LTF, and retaining any raw live inputs that feed the output directly, leaves a single LTF of the two surviving variables.

For a fixed restriction ρ , let $e(\rho)$ be the fraction of the four completions of the two live bits on which the restricted circuit disagrees with the restricted target. By (5.2), that target is XOR or XNOR. The joint experiment “draw ρ , then fill its two live bits uniformly” produces a uniform n -bit input, so

$$\mathbb{E}_{\rho}[e(\rho)] \leq \frac{1}{100}. \quad (10.20)$$

Lemma 10.4 gives $e(\rho) \geq 1/4$ whenever A occurs. Markov’s inequality in this deterministic form therefore yields

$$\Pr[A] \leq 4\mathbb{E}[e(\rho)] \leq 0.04, \quad (10.21)$$

contradicting the lower bound 0.76 obtained from either inequality in (10.19). Thus for even n , $g > \sqrt{n}/100$ and $W > n^{3/2}/150$.

If n is odd, average the error over the two slices $x_n = 0$ and $x_n = 1$. One slice has error at most $1/100$. The restricted target on that slice is PARITY on $n-1$ bits or its complement, and the

preceding proof treats the two functions symmetrically. Fixing one input can only decrease gate supports and wire count. Applying the even case to $n - 1$ proves (10.14).

For the upper bound let $X = \sum_{i=1}^n x_i$ and put

$$A_n = \left\lceil \frac{n}{2} - 5\sqrt{n} \right\rceil, \quad B_n = \left\lfloor \frac{n}{2} + 5\sqrt{n} \right\rfloor. \quad (10.22)$$

For every integer $k \in [A_n, B_n + 1]$, use a bottom threshold gate

$$L_k = \mathbf{1}[X \geq k]. \quad (10.23)$$

There are at most $10\sqrt{n} + 2$ such gates. Since $L_k - L_{k+1} = \mathbf{1}[X = k]$, a top LTF with coefficients in $\{-1, 0, 1\}$ computes

$$H(x) = \mathbf{1} \left[\sum_{\substack{A_n \leq k \leq B_n \\ k \text{ odd}}} (L_k - L_{k+1}) \geq 1 \right]. \quad (10.24)$$

Thus H agrees exactly with PARITY whenever $A_n \leq X \leq B_n$. Under the uniform input distribution, $\mathbb{E}X = n/2$ and $\text{Var}(X) = n/4$. Chebyshev's inequality gives

$$\Pr[|X - n/2| \geq 5\sqrt{n}] \leq \frac{1}{100}. \quad (10.25)$$

Hence H has error at most $1/100$. The bottom gates use unit input weights, the top weights are $-1, 0, 1$, the total gate count is at most $10\sqrt{n} + 3$, and the total number of input-to-bottom plus bottom-to-output wires is at most $(n + 1)(10\sqrt{n} + 2)$. This proves (10.15). $\square$

Worked restriction calculation. Take an even n and a bottom gate depending on u distinct inputs. Under the two-block restriction, Lemma 10.3 gives

$$\Pr[\text{that gate remains genuinely two-variable}] \leq \frac{34u}{n^{3/2}}. \quad (10.26)$$

For example, a dense gate with $u = n$ survives nontrivially with probability at most $34/\sqrt{n}$, while a gate with only $u = \sqrt{n}$ relevant inputs has the stronger bound $34/n$. This is why the proof needs both restriction lemmas: the first charges the number of bottom gates, and the second charges their aggregate supports, hence wires.

10.5 Source and hypothesis checkpoint

Kane and Williams, *Super-Linear Gate and Super-Quadratic Wire Lower Bounds for Depth-Two and Depth-Three Threshold Circuits*, STOC 2016, 633–643, DOI 10.1145/2897518.2897636, define LTFs using arbitrary real weights and allow the top LTF of an LTF $\circ$ LTF circuit to read original inputs directly. Their Theorem 1.4 states that MAJ $\circ$ MAJ circuits agreeing with PARITY on 99% of inputs have $\Theta(\sqrt{n})$ gate complexity and $\Theta(n^{3/2})$ wire complexity, and that the lower bounds hold even for LTF $\circ$ LTF. Their Lemma 1.1 and Lemma 3.1 supply the random-restriction estimates in asymptotic notation.

QRCC reconstructs those ingredients through Lemmas 10.1–10.3 and therefore owns the conservative explicit constants $1/100$ and $1/150$ in (10.14). They are not claimed as source constants. The matching upper construction is the same central-Hamming-window architecture as the source proof, with the concrete choice $5\sqrt{n}$ supplied locally by Chebyshev.

The hypercontractive theorem QRCC-I-THM-020 is not used. The formal imports are only the circuit model, the already audited union bound, Markov inequality, and Chebyshev inequality; Sperner/Littlewood–Offord anti-concentration is reconstructed in this chapter. No claim about general polynomial-size TC^0 or ACC^0 follows from Theorem 10.5. The dated frontier above records stronger and newer results in different models rather than weakening those model distinctions.

Chapter 11

SAT Algorithms and Nonuniform ACC Lower Bounds

The lower bounds in Chapters 5–10 attack a restricted circuit model by exhibiting a function that the model cannot represent efficiently. Williams’ algorithmic method reverses the direction of travel. One first designs a SAT algorithm that saves even a modest amount over exhaustive search on a circuit class, then turns that saving into a contradiction with a uniform time hierarchy under the assumption that the same circuit class computes too many languages nonuniformly.

This reversal creates several opportunities for accidental overstatement. A SAT algorithm is useful only on the exact syntactic class in which the verification circuit remains; a saving measured on $n + c \log n$ variables is not the same as a saving measured on n variables; and the $NEXP$ lower bound uses an easy-witness theorem that is logically separate from the E^{NP} -based meta-theorem. We therefore freeze all three interfaces before proving either registered theorem.

11.1 Complexity and circuit conventions

For this chapter,

$$E^{NP} := \bigcup_{a \geq 1} \text{DTIME}(2^{an})^{NP}, \quad NEXP := \bigcup_{a \geq 1} \text{NTIME}(2^{n^a}). \quad (11.1)$$

We also write $\text{NTIME}[2^n]$ for languages accepted in $O(2^n)$ nondeterministic time on the fixed robust machine model used by the source theorem. The square-bracket notation is only a reminder that this is the particular exponential scale used in the hierarchy contradiction; it is not a new complexity-class operation.

A circuit class $\mathcal{C}$ will be called *Williams-robust in this chapter* when the following quantitative closure rules hold.

1. Every polynomial-size constant-depth AC^0 family is a $\mathcal{C}$ family.
2. $\mathcal{C}$ is closed under composition of circuit families. In the uses below, putting r copies of circuits in parallel costs the sum of their sizes, and feeding their outputs through a fixed AC^0 checker changes depth by only $O(1)$.
3. Fixing input bits or output bits of a $\mathcal{C}$ circuit leaves a $\mathcal{C}$ circuit with no larger size or depth.

The ordinary gate-set classes $AC^0[m]$ satisfy these rules. The purpose of stating them is not to invent a new complexity class but to expose exactly where the generic lower-bound argument uses closure.

Let $S, d : \mathbb{N} \rightarrow \mathbb{N}$ be monotone nondecreasing, with $S(n) \geq n$ and $d(n) \geq 1$. The problem $\mathcal{C}$ -CIRCUIT-SAT receives an explicitly described $\mathcal{C}$ circuit and asks whether some Boolean assignment makes its output 1. A running-time hypothesis below is always measured in terms of the displayed parameter n , even when the instance has $n + c \log n$ variables. This convention is essential: the $O(\log n)$ address overhead must already be paid for by the SAT saving.

A closure nonexample. The class consisting of a *single* LTF gate is not Williams-robust: the composition of two LTF families is generally a depth-two threshold circuit, not one LTF. Thus a fast SAT algorithm for single threshold gates cannot be inserted into the theorem below merely because the verifier happens to use threshold gates. The verifier must remain in the same class after all copies and compositions are made.

11.2 Two source-owned interfaces

Two classical inputs are used with their exact hypotheses visible. They are source-owned inputs in this chapter; their independent QRCC proof campaigns remain separate.

Strong nondeterministic time hierarchy. For time-constructible t_1, t_2 satisfying

$$t_1(n+1) = o(t_2(n)), \quad (11.2)$$

there is a unary language in $\text{NTIME}(t_2)$ that is not in $\text{NTIME}(t_1)$. This is the form attributed to Žak in Williams' Theorem 2.1 and is the source-frozen statement required from the registered entry QRCC-II-THM-003. In particular, for every fixed $c > 0$, after harmless integer rounding,

$$t_2(n) = 2^n, \quad t_1(n) = \frac{2^n}{n^c} \quad (11.3)$$

satisfy (11.2), because $t_1(n+1)/t_2(n) = 2/(n+1)^c \rightarrow 0$.

Locally accessible succinct Cook–Levin reduction. There is a constant c_0 such that every $L \in \text{NTIME}[2^n]$ has a polynomial-time reduction which, on $x \in \{0, 1\}^n$, produces a Boolean circuit C_x with at most $n + c_0 \log n$ address inputs and $n^{O(1)}$ gates. Its truth table encodes a 3-CNF F_x of size $2^n n^{O(1)}$, and

$$x \in L \iff F_x \text{ is satisfiable.} \quad (11.4)$$

Moreover the gate/wire description of C_x , and each requested clause bit of F_x , are computable from $(x, \text{address})$ in polynomial time. This is the strong local Cook–Levin interface recorded as Fact 1/Theorem 3.3 in Williams' ACC paper, building on Turlakis and Fortnow–Lipton–van Melkebeek–Viglas. The locality, rather than mere NEXP -completeness of SUCCINCT-3SAT , is what lets the verifier below stay small.

11.3 Verifying a nonuniform compression

The succinct reduction circuit C_x is an unrestricted Boolean circuit. A $\mathcal{C}$ -SAT algorithm cannot be called on a verifier containing C_x unless we first replace C_x by an equivalent $\mathcal{C}$ circuit. The replacement exists nonuniformly under the circuit upper-bound assumption, but need not be constructible uniformly. Nondeterminism plus the SAT algorithm certifies a guessed replacement.

Lemma 11.1 (Verified compression). *Let $\mathcal{C}$ be Williams-robust. Suppose every language in P has nonuniform $\mathcal{C}$ circuits of depth at most $d(n)$ and size at most $S(n)$, where*

$$n \leq S(n) \leq 2^{n/4}. \quad (11.5)$$

There are constants $c_1, b_1, e_1 > 0$ with the following property. If $\mathcal{C}$ -CIRCUIT-SAT on circuits having at most $n + c_1 \log n$ inputs, depth at most $2d(3n) + b_1$, and size $O(nS(2n) + S(3n))$ is decidable in $O(2^n/n^{c_1})$ time, then for every $L \in \text{NTIME}[2^n]$ there is a nondeterministic procedure which, on input $x \in \{0, 1\}^n$, either rejects or outputs a $\mathcal{C}$ circuit $\tilde{C}_x$ equivalent to the succinct reduction circuit C_x of (11.4), with size at most $S(2n)$ and depth at most $d(2n)$. Every nonrejecting branch outputs a correct circuit, at least one nonrejecting branch exists, and the procedure runs in

$$O(2^n/n^{c_1}) + S(3n)n^{e_1} \quad (11.6)$$

time.

Proof. Fix x and index the $n^{O(1)}$ gates of C_x topologically. The *gate-information function*

$$(x, j) \mapsto (\text{type of gate } j, \text{ indices of its at most two predecessors}) \quad (11.7)$$

is polynomial-time computable. Each of its $O(\log n)$ output bits therefore has a $\mathcal{C}$ circuit of size at most $S(2n)$ and depth at most $d(2n)$ for all sufficiently large n . Putting these copies in parallel gives a $\mathcal{C}$ circuit $D(x, j)$ of size $O(S(2n) \log n)$ and depth at most $d(2n)$. The nondeterministic procedure guesses D , explicitly evaluates it on every polynomially many gate indices j , and compares the resulting gate list with the uniformly constructed C_x . An incorrect D is rejected in $S(2n)n^{O(1)}$ time.

The *gate-value function*

$$(x, i, j) \mapsto \text{value of gate } j \text{ when } C_x \text{ is evaluated on address } i \quad (11.8)$$

is also in P . Hence there exists a $\mathcal{C}$ circuit $E(x, i, j)$ of size at most $S(3n)$ and depth at most $d(3n)$ computing it. Guess such an E . To certify the guess, build a circuit $\text{BAD}(i, j)$ as follows. Use the already verified $D(x, j)$ to obtain the gate type and predecessor indices j_1, j_2 , evaluate $E(x, i, j), E(x, i, j_1), E(x, i, j_2)$, and let a fixed AC^0 checker output 1 exactly when those values violate the Boolean semantics of gate j . For an input gate the checker instead compares $E(x, i, j)$ with the corresponding bit of i .

Because $\mathcal{C}$ contains AC^0 and is closed under composition, BAD is a $\mathcal{C}$ circuit. It has $n + O(\log n)$ inputs,

$$\text{size}(\text{BAD}) = O(nS(2n) + S(3n)), \quad \text{depth}(\text{BAD}) \leq 2d(3n) + O(1). \quad (11.9)$$

Choose c_1, b_1 large enough to absorb the fixed address and checker constants. Run the assumed SAT algorithm on BAD and reject if it is satisfiable.

If the SAT call declares BAD unsatisfiable, then for every address i every claimed gate value obeys the correct local gate equation. Since D gives the actual topological wiring of C_x , induction over the gate order forces all values claimed by E to equal the true evaluation of C_x .

One final certification keeps the output compression small enough for the meta-theorem. For this fixed x , the Boolean function $i \mapsto C_x(i)$ is in P (the machine may have x hardwired into its finite control). Hence the assumed nonuniform P upper bound supplies a $\mathcal{C}$ circuit $\tilde{C}_x$ on the $n + O(\log n)$ address bits, of size at most $S(2n)$ and depth at most $d(2n)$ for all sufficiently large n . Guess such a circuit and, with j_* the output-gate index of C_x , form

$$\text{DIFF}(i) := \tilde{C}_x(i) \oplus E(x, i, j_*). \quad (11.10)$$

The XOR is a fixed AC^0 checker. Thus DIFF is a $\mathcal{C}$ circuit of size $O(S(2n) + S(3n))$ and depth at most $2d(3n) + O(1)$. A final call to the assumed SAT algorithm accepts this guess only when DIFF is unsatisfiable. Every surviving $\tilde{C}_x$ is therefore equivalent to C_x and has the advertised $S(2n)$ size bound.

Correct circuits D, E , and $\tilde{C}_x$ exist by the assumed nonuniform P upper bound, so at least one branch survives. The constant number of SAT calls still costs $O(2^n/n^{c_1})$. All explicit guessing, circuit construction, and direct evaluation costs $S(3n)n^{O(1)}$. Since $S(3n) \leq 2^{3n/4}$, this overhead is asymptotically below the displayed SAT budget after enlarging the fixed constants. This proves (11.6). $\square$

The proof illustrates why mere existence of a nonuniform replacement is not enough. The SAT algorithm is used as a *universal consistency checker* for the guessed gate-value circuit. This is the precise point at which class closure under composition is load-bearing.

Theorem 11.2 (A SAT algorithm implies a nonuniform circuit lower bound). *Let $\mathcal{C}$ be Williams-robust, and let S, d be monotone nondecreasing with*

$$n \leq S(n) \leq 2^{n/4}, \quad d(n) \geq 1. \quad (11.11)$$

There are constants $c, b, K > 0$ such that the following implication holds. If $\mathcal{C}$ -CIRCUIT-SAT instances having at most $n + c \log n$ variables, depth at most $2d(3n) + b$, and size at most

$$K(nS(2n) + S(3n)) \quad (11.12)$$

can be decided deterministically in

$$O(2^n/n^c) \quad (11.13)$$

time for all sufficiently large n , then E^{NP} does not have nonuniform $\mathcal{C}$ circuits simultaneously bounded by depth $d(n)$ and size $S(n)$.

Proof. Assume, for contradiction, that every language in E^{NP} has such $\mathcal{C}$ circuits. In particular $P \subseteq E^{NP}$ supplies the hypothesis of Lemma 11.1. Increase c, b, K once so that all constants from the local Cook–Levin reduction and that lemma are covered.

Choose, using the source-owned nondeterministic hierarchy, a language

$$L \in \text{NTIME}[2^n] \setminus \text{NTIME}(2^n/n^c). \quad (11.14)$$

On input x , run the verified-compression procedure to obtain on every surviving branch a $\mathcal{C}$ circuit $\tilde{C}_x$ equivalent to the succinct reduction C_x .

We next justify the succinct witness circuit used by the branch. Consider the Boolean function $A(x, a)$ that, when F_x is satisfiable, returns bit a of the lexicographically first satisfying assignment, and returns 0 otherwise. There are at most $2^{n+O(\log n)}$ assignment positions. A deterministic machine with an NP oracle can recover the lexicographically first assignment by prefix search: for each successive bit it asks whether a satisfying completion of the current prefix exists. Writing the exponentially long queries and performing exponentially many such steps still costs $2^{O(n)}$. Therefore

$$A \in E^{NP}. \quad (11.15)$$

By the contradictory circuit assumption, for each input length there is a $\mathcal{C}$ circuit W_x of size at most $S(3n)$ and depth at most $d(3n)$ encoding these witness bits. The nondeterministic algorithm simply guesses a circuit of those bounds; correctness will be checked globally rather than bit by bit.

Build a $\mathcal{C}$ circuit $V_x(i)$ that outputs 1 exactly when clause i of the decompressed formula is *violated* by the assignment encoded by the guessed W_x . Recovering the $O(n)$ bits of the clause uses $O(n)$ copies of $\tilde{C}_x$; the three referenced assignment bits use three copies of W_x ; a constant-depth Boolean checker handles signs and the final three-way OR. Hence

$$\text{size}(V_x) = O(nS(2n) + S(3n)), \quad \text{depth}(V_x) \leq 2d(3n) + O(1), \quad (11.16)$$

and V_x has at most $n + O(\log n)$ address inputs.

Run the hypothesized $\mathcal{C}$ -SAT algorithm on V_x and accept exactly when it is unsatisfiable. If $x \in L$, choose the correct compression branch and the genuine witness circuit guaranteed by (11.15); then no clause is violated and an accepting branch exists. If $x \notin L$, the decompressed 3-CNF is unsatisfiable, so every guessed assignment violates some clause and every branch rejects. Thus this is a correct nondeterministic algorithm for L .

The SAT calls cost $O(2^n/n^c)$. Circuit guessing and explicit bookkeeping cost at most

$$S(3n)n^{O(1)} \leq 2^{3n/4}n^{O(1)},$$

which is smaller for all large n . We have placed L in $\text{NTIME}(2^n/n^c)$, contradicting (11.14). The assumed circuit upper bound for E^{NP} is therefore impossible. $\square$

Worked quantitative check. If $S(n) = n^k$ and $d(n) = d_0$ are fixed polynomial-size/constant-depth bounds, then the verifier in (11.12) has polynomial size and constant depth. Suppose more strongly that a SAT algorithm on N variables runs in $2^{N-N^\alpha} \text{poly}(N)$ time for some $\alpha > 0$. With $N = n + c \log n$,

$$2^{N-N^\alpha} \text{poly}(N) = 2^n n^{O(1)} 2^{-N^\alpha} \leq \frac{2^n}{n^c} \quad (11.17)$$

for every fixed required c and all sufficiently large n . Thus a stretched-exponential SAT saving easily absorbs the logarithmic address increase in the meta-theorem.

11.4 ACC to symmetric-AND circuits

We now specialize to the QRCC convention $ACC^0 = \bigcup_{m \geq 2} AC^0[m]$ from Definition 1.1. For a fixed pair (d, m) an $ACC[d, m]$ circuit here means a depth-at-most- d circuit over unbounded AND, OR, NOT, and the zero-residue MOD_m gate. Switching to the complementary modular convention costs one NOT and therefore has no bearing on the lower bound.

A SYM^+ circuit has bottom AND gates on input variables and a top symmetric gate: its output is an arbitrary Boolean function of the *number* of bottom ANDs that evaluate to 1. The top symmetric function is given explicitly enough that its value on each count $0, \dots, s$ can be computed in $\text{poly}(s)$ time.

The following is a deep, source-owned representation theorem rather than a consequence of the Razborov–Smolensky approximation theorem of Chapter 8. For fixed d, m there is an integer $q = q(d, m) \geq 1$ and an algorithm which transforms every size- s $ACC[d, m]$ circuit into an equivalent SYM^+ circuit of size

$$s^{O((\log s)^q)}, \quad (11.18)$$

with bottom AND fan-in $O((\log s)^q)$, in time polynomial in the output size. This is Williams' Lemma 4.1, whose source lineage is Yao, Beigel–Tarui, and Allender–Gore. We use (11.18) exactly as a source-owned translation interface.

The second ingredient is elementary enough to reconstruct.

Lemma 11.3 (All-input evaluation of SYM^+). *Let G be a SYM^+ circuit on n variables with s bottom AND gates. If the top symmetric function can be tabulated in $\text{poly}(s)$ time, then the complete truth table of G can be computed in*

$$(2^n + \text{poly}(s)) \text{poly}(n) \quad (11.19)$$

time.

Proof. Normalize each bottom AND by deleting repeated variables and discarding an identically zero gate if necessary. Let $G_j \subseteq [n]$ be the support of the j th remaining AND and define

$$f(S) := |\{j : G_j = S\}|, \quad S \subseteq [n]. \quad (11.20)$$

The table of f has 2^n entries and can be initialized in $O(2^n + s \text{poly}(n))$ time.

For an assignment whose set of 1-coordinates is T , precisely the ANDs with $G_j \subseteq T$ are satisfied. Hence their number is the subset zeta transform

$$g(T) = \sum_{S \subseteq T} f(S). \quad (11.21)$$

Compute all values by Yates' dynamic program. Starting with $g_0 = f$, for $i = 1, \dots, n$ set

$$g_i(T) = \begin{cases} g_{i-1}(T) + g_{i-1}(T \setminus \{i\}), & i \in T, \\ g_{i-1}(T), & i \notin T. \end{cases} \quad (11.22)$$

An induction on i gives $g_n = g$, and the table takes $O(n2^n)$ additions of $O(\log s)$ -bit integers.

Finally tabulate the top symmetric function once on counts $0, 1, \dots, s$ in $\text{poly}(s)$ time. Each assignment T then requires only a lookup at count $g(T)$. This proves (11.19). Notice that the factor $\text{poly}(s)$ is additive preprocessing, not a factor paid for each of the 2^n assignments. $\square$

Proposition 11.4 (A locally derived ACC-SAT saving). *For every fixed $d \geq 2$ and $m \geq 2$ there are constants $0 < \varepsilon < \delta < 1$ such that satisfiability of depth- d $\text{ACC}[d, m]$ circuits on n inputs and of size at most 2^{n^ε} is decidable deterministically in*

$$2^{n - \Omega(n^\delta)} \quad (11.23)$$

time.

Proof. Apply the source-owned translation (11.18) at depth $d + 1$ and let $q \geq 1$ be a corresponding exponent. Put

$$\delta := \frac{1}{4(q+1)}, \quad \varepsilon := \frac{\delta}{2}, \quad \ell := \lfloor n^\delta \rfloor. \quad (11.24)$$

Given C of size $s \leq 2^{n^\varepsilon}$, form a new circuit C' on $n - \ell$ variables by taking the OR of all 2^ℓ restrictions obtained by assigning the first ℓ inputs in every possible way. Then C is satisfiable exactly when C' is, while C' has depth at most $d + 1$ and size

$$r \leq s2^\ell \leq 2^{2n^\delta} \quad (11.25)$$

for all sufficiently large n .

Translate C' to SYM^+ . For a constant $A = A(d, m)$ its size r' satisfies

$$\log_2 r' \leq A(\log_2 r)^{q+1} \leq A2^{q+1} n^{1/4}. \quad (11.26)$$

Thus $r' = 2^{O(n^{1/4})} = 2^{o(n)}$ and the translation itself takes $2^{O(n^{1/4})}$ time. Lemma 11.3 evaluates the translated circuit on all $2^{n-\ell}$ remaining assignments in

$$2^{n-\ell} \text{poly}(n) + 2^{O(n^{1/4})} = 2^{n-\Omega(n^\delta)}. \quad (11.27)$$

We accept exactly when one value is 1. Since $\varepsilon = \delta/2$, the claimed strict inequality $\varepsilon < \delta$ is explicit. $\square$

The constants in (11.24) are deliberately conservative QRCC choices. The source theorem optimizes the parameter selection differently; only the existence of a stretched-exponential saving is needed below.

11.5 From universal witnesses to the NEXP separation

Two further source-owned statements isolate the nonuniform witness step. First, SUCCINCT-BOUNDED-HALTING is *NEXP*-complete under polynomial-size AC^0 reductions and itself lies in $\text{NTIME}[2^n]$ under the standard binary time-bound encoding. Second, the Impagliazzo–Kabanets–Wigderson easy-witness theorem, in the form used by Williams, says:

$$NEXP \subseteq P/\text{poly} \implies \text{every } NEXP \text{ language has polynomial-size universal witness circuits.} \quad (11.28)$$

A universal witness circuit is required to work for the chosen correct exponential-time verifier: for each yes-instance it encodes an accepting witness bit by bit. Statement (11.28) is Williams' Theorem 5.3 and is not silently identified with Theorem 11.2.

The following transfer lemma is elementary and will convert those unrestricted witness circuits into one fixed *ACC* model.

Lemma 11.5 (Circuit-evaluation transfer). *Let $\mathcal{C}$ be any nonuniform circuit class. Suppose the Boolean CIRCUIT-EVAL language has $\mathcal{C}$ circuits of size at most $R(N)$. Then every Boolean circuit family $\{B_n\}$ of size $T(n)$ has an equivalent $\mathcal{C}$ family of size*

$$R(n + O(T(n) \log T(n))). \quad (11.29)$$

If the evaluator family has fixed depth and fixed modulus in $AC^0[m]$, the converted family has the same fixed depth and modulus.

Proof. Encode a size- $T(n)$ circuit B_n by $O(T(n) \log T(n))$ bits describing its gates and wires. On input $(\langle B_n \rangle, x)$, CIRCUIT-EVAL returns $B_n(x)$. Take the $\mathcal{C}$ circuit for that evaluator input length and hardwire the description $\langle B_n \rangle$. Only the n input bits x remain live. Hardwiring cannot increase size or depth and does not change a modular gate's modulus, proving (11.29). $\square$

Theorem 11.6 (Williams: *NEXP* is not contained in nonuniform ACC^0). *There is a language in $\text{NTIME}[2^n]$ that has no polynomial-size nonuniform ACC^0 circuit family. Consequently*

$$\boxed{NEXP \not\subseteq ACC^0}. \quad (11.30)$$

Here an ACC^0 family has one fixed constant depth and one fixed modulus $m \geq 2$ (up to the harmless convention changes from Chapter 1), while its size may be any polynomial in the input length.

Proof. Assume the contrary:

$$\text{NTIME}[2^n] \subseteq \text{ACC}^0 \quad (11.31)$$

with polynomial-size nonuniform families. Since the *NEXP*-complete SUCCINCT-BOUNDED-HALTING problem lies in $\text{NTIME}[2^n]$, it has a polynomial-size $\text{ACC}[d_0, m_0]$ family for some fixed d_0, m_0 . Composing this family with the source-owned AC^0 reductions from arbitrary *NEXP* languages shows

$$\text{NEXP} \subseteq \text{ACC}^0 \subseteq P/\text{poly}. \quad (11.32)$$

The universal-witness theorem (11.28) therefore supplies polynomial-size unrestricted witness circuits for the succinct 3-SAT instances appearing in the local Cook–Levin reduction.

We next put all helper circuits and witnesses into one fixed modular class. The language CIRCUIT-EVAL lies in $P \subseteq \text{NTIME}[2^n]$, so (11.31) gives it polynomial-size $\text{ACC}[d_1, m_1]$ circuits for fixed d_1, m_1 . By Lemma 11.5, every polynomial-size unrestricted circuit family—in particular the universal witness family from (11.28) and the polynomial-time gate-information and gate-value families used in Lemma 11.1—has an equivalent polynomial-size $\text{ACC}[d_1, m_1]$ family. Thus all pieces of the verifier can be chosen with one fixed modulus and constant depth.

Proposition 11.4 now gives a deterministic SAT algorithm for every fixed depth obtained by composing those pieces. On $N = n + c \log n$ variables and polynomial-size circuits its running time is

$$2^{N - \Omega(N^\delta)} = 2^n n^{O(1)} 2^{-\Omega(n^\delta)} \leq \frac{2^n}{n^c} \quad (11.33)$$

for every desired fixed c and all sufficiently large n .

Choose from the strong nondeterministic hierarchy a language satisfying

$$L \in \text{NTIME}[2^n] \setminus \text{NTIME}(2^n/n^c).$$

Run the verified-compression construction of Lemma 11.1, now with $\mathcal{C} = \text{AC}^0[m_1]$ and polynomial S . Guess the transferred $\text{ACC}[d_1, m_1]$ universal witness circuit for the resulting succinct 3-CNF, and build the same bad-clause circuit V_x used in the proof of Theorem 11.2. Equation (11.33) decides whether V_x is satisfiable inside $O(2^n/n^c)$ time. A correct witness yields an accepting branch exactly on $x \in L$, whereas an unsatisfiable decompressed formula causes every guessed witness to leave some bad clause. Hence

$$L \in \text{NTIME}(2^n/n^c), \quad (11.34)$$

contradicting the choice of L . Therefore (11.31) is false. Since $\text{NTIME}[2^n] \subseteq \text{NEXP}$, the stated *NEXP* consequence follows. $\square$

Why Chapter 8 is not a dependency. The *ACC*-to-*SYM*⁺ translation (11.18) is a different algebraic representation theorem from the Razborov–Smolensky probabilistic-polynomial approximation proved in Chapter 8. The proof above never invokes *QRCC-X-THM-012*; keeping that edge would falsely suggest that the known *ACC* lower bound follows from the distinct-prime modular separation. The Pass-44 dependency ledger therefore removes that spurious edge.

Worked savings comparison. Suppose the final verification circuit has $N = n + 7 \log_2 n$ inputs and the fixed-depth *ACC* algorithm saves $N^{1/20}$ in the exponent. Then

$$\frac{2^{N - N^{1/20}}}{2^n/n^{100}} = n^{107} 2^{-N^{1/20}} \longrightarrow 0. \quad (11.35)$$

Thus even a very small stretched-exponential saving eventually dominates *every* polynomial factor generated by succinct addressing, construction, and machine-model simulation. This asymptotic absorption is the quantitative hinge between the SAT algorithm and the hierarchy theorem.

11.6 Source and hypothesis checkpoint

Ryan Williams, *Improving Exhaustive Search Implies Superpolynomial Lower Bounds*, SIAM Journal on Computing 42(3), 1218–1244 (2013), DOI 10.1137/10080703X, supplies the original algorithm-to-lower-bound framework and explicitly records the strong nondeterministic time hierarchy form used here. Williams, *Non-Uniform ACC Circuit Lower Bounds*, JACM 61(1), Article 2 (2014), DOI 10.1145/2559903, strengthens the generic $\mathcal{C}$ -SAT theorem, proves the ACC SAT algorithm, and proves the nonuniform ACC lower bound for $\text{NTIME}[2^n]$.

The exact source split in this chapter is:

1. the strong nondeterministic hierarchy (Žak, with the earlier Seiferas–Fischer–Meyer lineage) is source-owned;
2. the locally accessible succinct Cook–Levin reduction of Williams’ Fact 1/Theorem 3.3 is source-owned;
3. Williams’ Lemma 4.1 supplies the source-owned representation theorem $\text{ACC} \rightarrow \text{SYM}^+$. Its lineage runs through Yao, Beigel–Tarui, and Allender–Gore;
4. the universal-witness theorem, Williams’ Theorem 5.3 with Impagliazzo–Kabanets–Wigderson lineage, is source-owned;
5. Lemma 11.1, the succinct-witness argument for E^{NP} , Lemma 11.3, the parameter derivation in Proposition 11.4, and Lemma 11.5 are reconstructed locally.

This separation matters. Theorem 11.2 is a generic algorithm-to-lower-bound theorem whose conclusion concerns E^{NP} under a circuit upper-bound assumption. Theorem 11.6 additionally uses universal witnesses to reach the $\text{NEXP}/\text{NTIME}[2^n]$ separation. Neither theorem is a direct explicit-function lower bound like the PARITY, modular, or CLIQUE theorems of the preceding chapters, and neither uses the Razborov–Smolensky theorem as a hidden substitute for the $\text{ACC} \rightarrow \text{SYM}^+$ translation.

Chapter 12

Three Barriers to General Circuit Lower Bounds

The preceding chapters prove substantial circuit lower bounds, but only after choosing a restricted computational model. This final chapter of the present Volume-X foundations asks a different question: why have the same styles of argument not resolved the central unrestricted problems? Three classical answers isolate three different kinds of proof behavior.

The distinctions matter. *Relativization* asks whether an argument survives after both sides receive the same arbitrary oracle. *Natural proofs* concern efficiently recognizable, large properties of Boolean truth tables and are conditional on strong pseudorandomness. *Algebrization* asks whether an argument survives a richer oracle experiment in which one side may query low-degree polynomial extensions. The three tests are related, but they are not interchangeable.

12.1 Relativization: the first barrier

We use the oracle-machine convention of QRCC-II-DEF-010. Thus P^A and NP^A are the deterministic and nondeterministic polynomial-time classes with query access to the same oracle A . An argument *relativizes* when its reasoning continues to hold uniformly for every oracle A .

Theorem 12.1 (Baker–Gill–Solovay relativization barrier). *There exist recursive oracles A and B for which*

$$P^A = NP^A, \quad P^B \neq NP^B. \quad (12.1)$$

Consequently, no proof whose P -versus- NP reasoning relativizes uniformly can settle either $P = NP$ or $P \neq NP$.

Proof. The oracle constructions themselves were proved earlier as QRCC-II-THM-016; we reuse that audited theorem rather than duplicate its finite-extension diagonalization here. Its equality oracle may be taken to be a PSPACE-complete language such as TQBF, giving

$$P^A = PSPACE = NP^A, \quad (12.2)$$

while its separation oracle B diagonalizes against every clocked deterministic polynomial-time oracle machine using the unary language

$$L_B = \{1^n : \exists y \in \{0, 1\}^n \text{ with } y \in B\} \in NP^B. \quad (12.3)$$

For each deterministic oracle machine, a fresh length is chosen at which the machine sees only negative answers; after it halts, an unqueried witness is inserted exactly when needed to make that machine's answer wrong. The finite extension is never changed later, so the diagonalization is permanent.

Now suppose a proof of $P = NP$ relativized. Applying the same proof relative to B would yield $P^B = NP^B$, contradicting (12.1). Likewise, a relativizing proof of $P \neq NP$ applied relative to A would yield $P^A \neq NP^A$, again contradicting (12.1). Hence either resolution must use a step that fails to relativize. $\square$

What the theorem does not say. The theorem is a barrier to a *proof behavior*, not evidence that $P = NP$ or $P \neq NP$ is more likely. Nor does it say that diagonalization is useless: time- and space-hierarchy theorems are powerful precisely because their target statements are compatible with relativization. The obstruction is specific to a question whose answer changes between relativized worlds.

A finite stage of the separation construction. Suppose a clocked machine M_i makes at most $q_i(n)$ oracle queries on 1^n . Choose a new length n with $q_i(n) < 2^n$ and answer every new length- n query by 0. If M_i accepts, leave every length- n string out of the oracle. If it rejects, at least one $y \in \{0, 1\}^n$ was not queried; insert such a y . In both cases the already observed computation is unchanged and becomes wrong about (12.3). The full proof merely schedules these finite injuries so that every machine is treated once and every oracle bit is eventually fixed.

12.2 Natural properties of truth tables

Let

$$\mathcal{F}_n := \{f : \{0, 1\}^n \rightarrow \{0, 1\}\}. \quad (12.4)$$

A property is a sequence $\Gamma = (\Gamma_n)_{n \geq 1}$ with $\Gamma_n \subseteq \mathcal{F}_n$. The input used to test membership in Γ_n is the full 2^n -bit truth table, so “polynomial-time constructive” means polynomial in 2^n , not polynomial in n .

Definition 12.2 (Natural property against a circuit class). Let $\mathcal{C}$ be a Boolean circuit model. A property $\Gamma = (\Gamma_n)$ is *P-natural and useful against $\mathcal{C}$* when the following three conditions hold.

1. **Constructivity.** There is a deterministic algorithm and a constant $a > 0$ that, given the full truth table of $f \in \mathcal{F}_n$, decides whether $f \in \Gamma_n$ in time at most 2^{an} for all sufficiently large n .
2. **Largeness.** There are constants $b > 0$ and n_0 such that

$$\frac{|\Gamma_n|}{|\mathcal{F}_n|} \geq 2^{-bn} \quad (12.5)$$

for every $n \geq n_0$.

3. **Usefulness.** For every constant $k > 0$, there is n_k such that for all $n \geq n_k$, no function in Γ_n is computable by an n^k -size circuit from $\mathcal{C}$.

The constants a, b belong to the property; usefulness quantifies over every fixed polynomial circuit-size exponent.

The preceding definition deliberately separates three ideas that are often compressed into the adjective “natural.” A property may be easy to test without being large, large without being useful, or useful without having any known truth-table algorithm.

Definition 12.3 (Natural-proof-strong pseudorandom function ensemble). Let $\mathcal{C}$ be a circuit model. A keyed ensemble

$$\mathbf{F} = \{F_{n,s} : \{0,1\}^n \rightarrow \{0,1\} : s \in \{0,1\}^{r(n)}\} \quad (12.6)$$

is *natural-proof strong in $\mathcal{C}$* if:

1. there is a constant k such that every $F_{n,s}$ has an n^k -size $\mathcal{C}$ circuit; and
2. for every constants $a, b > 0$, all sufficiently large n satisfy

$$\left| \Pr_s[D_n^{F_{n,s}} = 1] - \Pr_R[D_n^R = 1] \right| < 2^{-bn} \quad (12.7)$$

for every oracle circuit D_n of size at most 2^{an} , where R is a uniformly random member of $\mathcal{F}_n$.

This is the strong exponential-security form needed by the truth-table constructivity convention in Definition 12.2. It is an assumption, not a theorem asserting that such an ensemble exists.

Theorem 12.4 (Razborov–Rudich natural-proofs barrier). *If a natural-proof-strong pseudorandom function ensemble exists in $\mathcal{C}$, then no property can simultaneously be P -constructive, large, and useful against polynomial-size $\mathcal{C}$ circuits in the sense of Definition 12.2.*

Proof. Assume that $\Gamma = (\Gamma_n)$ is such a property. Let a be a constructivity exponent and b a largeness exponent, so that for all large n a truth-table algorithm A_n decides Γ_n in time at most 2^{an} and

$$\Pr_{R \leftarrow \mathcal{F}_n} [R \in \Gamma_n] \geq 2^{-bn}. \quad (12.8)$$

Let k be the polynomial circuit-size exponent of the strong PRF ensemble. By usefulness, for all sufficiently large n ,

$$F_{n,s} \notin \Gamma_n \quad \text{for every key } s. \quad (12.9)$$

Turn the truth-table tester into an oracle distinguisher D_n . On oracle input $G : \{0,1\}^n \rightarrow \{0,1\}$, query G at all 2^n Boolean inputs, write down its truth table, run A_n , and accept exactly when the table lies in Γ_n . For a constant $a' > \max\{a, 1\}$ and all large n , this procedure has oracle-circuit size at most $2^{a'n}$. Equations (12.8)–(12.9) give

$$\Pr_R[D_n^R = 1] \geq 2^{-bn}, \quad \Pr_s[D_n^{F_{n,s}} = 1] = 0. \quad (12.10)$$

Thus D_n distinguishes the strong PRF ensemble from a random function with advantage at least 2^{-bn} , contradicting (12.7) with parameters a' and b . No such property exists. $\square$

Two diagnostic nonexamples. The property “the truth table contains between $0.49 \cdot 2^n$ and $0.51 \cdot 2^n$ ones” is constructive and overwhelmingly large, but it is not useful against polynomial-size circuits: many simple functions are balanced. At the other extreme, “ f has circuit complexity greater than n^{100} ” is useful by definition and is large by counting for large n , but no polynomial-in- 2^n recognition algorithm is known. The natural-proofs barrier applies only when all three conditions are present together.

Conditionality. Unlike relativization, Theorem 12.4 has a cryptographic hypothesis. Its conclusion is therefore not “natural proofs are impossible” unconditionally. It says that a sufficiently efficient and sufficiently large useful property would itself be a statistical distinguisher against the strong pseudorandomness assumption.

12.3 Algebraic relativization

Relativization gives both machines the same black box. Arithmetization can break that symmetry: an algorithm may exploit evaluations of a low-degree polynomial extension away from Boolean points. Aaronson and Wigderson’s algebrization framework makes that extra power part of the oracle experiment.

Definition 12.5 (Extension oracle and algebrization). Write an oracle as a sequence $A = (A_m)_{m \geq 1}$ with $A_m : \{0, 1\}^m \rightarrow \{0, 1\}$. A *bounded-degree finite-field extension* $\tilde{A}$ consists, for every m and every finite field $\mathbb{F}$, of a polynomial

$$\tilde{A}_{m, \mathbb{F}} : \mathbb{F}^m \rightarrow \mathbb{F} \quad (12.11)$$

that agrees with A_m on $\{0, 1\}^m$ and whose multidegree is bounded by one absolute constant independent of m and $\mathbb{F}$.

For standard oracle classes $\mathcal{C}, \mathcal{D}$:

- an inclusion $\mathcal{C} \subseteq \mathcal{D}$ *algebrizes* if $\mathcal{C}^A \subseteq \mathcal{D}^{\tilde{A}}$ for every A and every bounded-degree extension $\tilde{A}$ of A ;
- a separation $\mathcal{C} \not\subseteq \mathcal{D}$ *algebrizes* if $\mathcal{C}^{\tilde{A}} \not\subseteq \mathcal{D}^A$ for every such pair $(A, \tilde{A})$.

For classes with more than polynomial time, a bracket [poly] is used when queries are restricted to polynomially many input coordinates, as in the Aaronson–Wigderson convention.

The asymmetry in Definition 12.5 is intentional. For an inclusion, the simulating class receives the stronger algebraic oracle; for a separation, the class being lower-bounded receives it. This choice captures arithmetization-based simulations while still providing meaningful oracle barriers.

12.3.1 Two elementary algebraic-oracle lemmas

For $z \in \{0, 1\}^n$, define the Boolean interpolation polynomial

$$\delta_z(x_1, \dots, x_n) := \prod_{j: z_j=1} x_j \prod_{j: z_j=0} (1 - x_j). \quad (12.12)$$

On Boolean inputs, $\delta_z(w)$ is 1 when $w = z$ and 0 otherwise.

Lemma 12.6 (Multiquadratic hiding at finitely many queries). *Let $\mathbb{F}$ be a field and let $y_1, \dots, y_t \in \mathbb{F}^n$ with $t < 2^n$. Then for at least $2^n - t$ Boolean points $w \in \{0, 1\}^n$ there is a multiquadratic polynomial $p : \mathbb{F}^n \rightarrow \mathbb{F}$ satisfying*

$$p(y_i) = 0 \quad (i \in [t]), \quad p(w) = 1, \quad p(z) = 0 \quad (z \in \{0, 1\}^n, z \neq w). \quad (12.13)$$

If finitely many fields are considered simultaneously and the total number of queried points over all fields is $t < 2^n$, one Boolean point w can be chosen so that (12.13) is realizable over every field at once.

Proof. Write an arbitrary multilinear polynomial in the Boolean interpolation basis,

$$m(x) = \sum_{z \in \{0, 1\}^n} c_z \delta_z(x). \quad (12.14)$$

The requirements $m(y_i) = 0$ are t homogeneous linear equations in the 2^n coefficients c_z . Row-reduce the system, set every free variable to 1, and solve for the at most t pivot variables. At least $2^n - t$ of the coefficients therefore equal 1. For any Boolean point w with $c_w = 1$, set

$$p(x) := m(x)\delta_w(x). \quad (12.15)$$

Each variable has degree at most two. Since $m(y_i) = 0$, all queried values vanish; on Boolean points, δ_w kills every point other than w and $p(w) = m(w) = c_w = 1$. This proves the one-field claim.

For a field containing $t_{\mathbb{F}}$ queried points, at most $t_{\mathbb{F}}$ Boolean points fail the one-field conclusion. The union of these bad sets over the finitely many queried fields has size at most $\sum_{\mathbb{F}} t_{\mathbb{F}} = t < 2^n$, so a common good w exists. $\square$

Lemma 12.7 (A PSPACE oracle has a PSPACE multilinear extension). *If every Boolean slice A_m is computable in space polynomial in m , then its unique multilinear extension over every effectively represented finite field is computable in polynomial space.*

Proof. For $y \in \mathbb{F}^m$, the unique multilinear extension is

$$\tilde{A}_{m,\mathbb{F}}(y) = \sum_{z \in \{0,1\}^m} A_m(z)\delta_z(y). \quad (12.16)$$

Enumerate z with an m -bit counter. For each z , evaluate $A_m(z)$ using polynomial workspace, compute $\delta_z(y)$ by streaming through the coordinates, and add the term to one running field accumulator. The exponential number of summands affects time but not workspace. Field elements and field operations have polynomial-size encodings under the oracle model. Hence the entire evaluation uses polynomial space. $\square$

12.3.2 Collapse and separation with algebraic oracles

Lemma 12.8 (An algebraic-oracle collapse pair). *There exist an oracle A and a multilinear extension $\tilde{A}$ such that*

$$NP^{\tilde{A}} \subseteq P^A. \quad (12.17)$$

Proof. Take A to be a PSPACE-complete language, sliced by input length, and take $\tilde{A}$ to be its unique multilinear extension. By Lemma 12.7, queries to $\tilde{A}$ can be answered in polynomial space. Therefore

$$NP^{\tilde{A}} \subseteq NPSPACE = PSPACE. \quad (12.18)$$

Since A is PSPACE-complete, $P^A = PSPACE$, yielding (12.17). $\square$

Lemma 12.9 (An algebraic-oracle separation pair). *There exist an oracle B and a multiquadratic extension $\tilde{B}$ such that*

$$NP^B \not\subseteq P^{\tilde{B}}. \quad (12.19)$$

Indeed the separating language can be taken to be

$$L_B := \{1^n : \exists w \in \{0,1\}^n \text{ with } B_n(w) = 1\}. \quad (12.20)$$

Proof. The language L_B is always in NP^B : guess w and make one Boolean oracle query. We build $(B, \tilde{B})$ in stages so that no deterministic polynomial-time $\tilde{B}$ -oracle machine decides it.

Enumerate all clocked deterministic polynomial-time algebraic-oracle machines $M_1, M_2, \dots$. At the start of stage i , only finitely many input lengths have been fixed. Choose a fresh target length

n_i larger than every length touched earlier and so large that the clock of M_i on 1^{n_i} is strictly less than 2^{n_i} .

Simulate $M_i(1^{n_i})$. Queries to already fixed lengths receive their permanent answers. Every query to an as-yet-unfixed algebraic slice is answered by 0. Since the machine is clocked, it makes finitely many queries. Let b_i be the maximum of n_i and every input length mentioned by a query in this simulation. After the target slice is handled below, permanently set every still-unfixed non-target length $m \leq b_i$ to the zero Boolean function with the zero polynomial extension.

At the target length there are two cases. If M_i accepted, again set the Boolean slice B_{n_i} and every algebraic extension at that length to zero. Then $1^{n_i} \notin L_B$, so the machine is wrong.

If M_i rejected, collect its target-length queries by field. Their total number is less than 2^{n_i} . Lemma 12.6 gives a Boolean point w_i and, for every queried field, a multiquadratic extension that is zero on every point the machine queried but equals 1 at w_i and 0 on all other Boolean points. Set $B_{n_i}(w_i) = 1$ and $B_{n_i}(z) = 0$ for $z \neq w_i$. For unqueried fields use the multilinear polynomial δ_{w_i} . Thus the completed extension agrees with every answer seen by M_i , so its rejecting computation is unchanged, while now $1^{n_i} \in L_B$.

At either branch, M_i fails. Future stages choose target lengths larger than b_i , hence never alter any oracle answer used in this diagonalization. Every deterministic polynomial-time algebraic-oracle machine is eventually treated, proving (12.19). The construction is effective: each stage performs finite simulations and finite-field Gaussian elimination. Since all lengths at most b_i are permanently fixed at stage i and $b_i \infty$, membership in the Boolean oracle can be decided by simulating stages until the relevant length is covered. Thus B may be chosen recursive as well. $\square$

Theorem 12.10 (Aaronson–Wigderson algebrization barrier for P versus NP). *There are bounded-degree algebraic-oracle pairs witnessing both directions needed to block an algebrizing resolution of P versus NP :*

$$\exists(A, \tilde{A}) \quad NP^{\tilde{A}} \subseteq P^A, \quad (12.21)$$

$$\exists(B, \tilde{B}) \quad NP^B \not\subseteq P^{\tilde{B}}. \quad (12.22)$$

Consequently, proving either $P \neq NP$ or $P = NP$ requires a step that does not algebrize in the sense of Definition 12.5.

Proof. Equation (12.21) is Lemma 12.8; equation (12.22) is Lemma 12.9.

The statement $P \neq NP$ is the separation $NP \not\subseteq P$. If a proof of that separation algebrized, then by definition we would have $NP^{\tilde{A}} \not\subseteq P^A$ for every oracle-extension pair, contradicting (12.21). The statement $P = NP$ can be expressed as the inclusion $NP \subseteq P$, since $P \subseteq NP$ is unconditional. If a proof of that inclusion algebrized, then $NP^B \subseteq P^{\tilde{B}}$ would hold for every pair, contradicting (12.22). Therefore either resolution needs a non-algebrizing ingredient. $\square$

Where the new difficulty enters. In the ordinary BGS separation, answering an unqueried Boolean point later is enough to fool the simulated machine. An algebraic-oracle machine can query non-Boolean field points that contain global information about many Boolean values. Lemma 12.6 is exactly the extra step needed: fewer than 2^n algebraic queries still leave a Boolean spike that can be hidden while preserving all queried polynomial values.

12.4 Comparing the three barriers

Proposition 12.11 (Barrier comparison). *The three barriers above have the following precise relationship.*

1. *Relativization and algebrization are oracle-lifting tests; naturalness is a truth-table-property test for circuit lower-bound arguments.*
2. *For the standard oracle classes in this chapter, a statement that relativizes also satisfies the corresponding algebrized statement-level lifting, because A can be recovered by querying $\tilde{A}$ on Boolean points. The converse fails: arithmetization-based results such as the $PSPACE \subseteq IP$ direction are nonrelativizing in the ordinary same-oracle sense yet algebrize.*
3. *Naturalness has no definitional implication to or from either oracle notion. A natural-property obstruction is conditional on strong pseudorandomness and concerns three separate conditions: largeness, constructivity, and usefulness. BGS and Aaronson–Wigderson instead construct oracle worlds.*
4. *Therefore overcoming one barrier does not, by itself, certify that the other two have been overcome.*

Proof. Item 1 follows immediately by comparing Definitions 12.2 and 12.5.

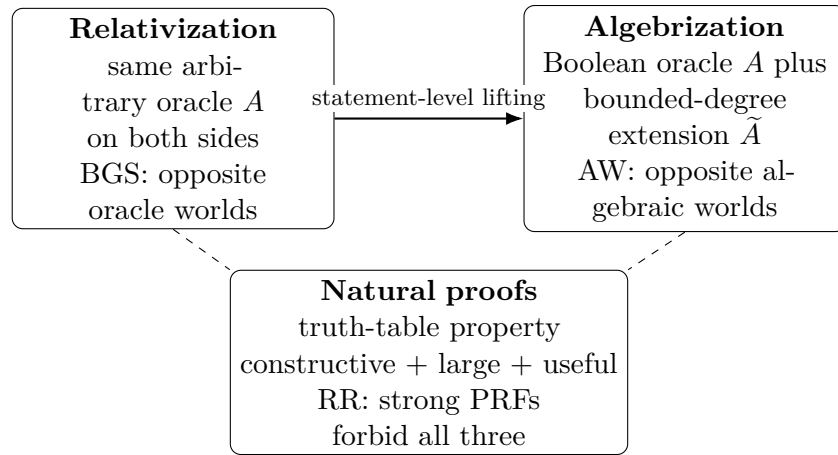
For item 2, suppose first that an inclusion $\mathcal{C} \subseteq \mathcal{D}$ relativizes, so $\mathcal{C}^A \subseteq \mathcal{D}^A$ for every A . Any $\mathcal{D}^A$ machine can simulate its Boolean-oracle queries using $\tilde{A}$ at Boolean points, hence $\mathcal{D}^A \subseteq \mathcal{D}^{\tilde{A}}$ and therefore

$$\mathcal{C}^A \subseteq \mathcal{D}^{\tilde{A}}. \quad (12.23)$$

For a relativizing separation $\mathcal{C}^A \not\subseteq \mathcal{D}^A$, the same hard language lies in $\mathcal{C}^{\tilde{A}}$ because $\tilde{A}$ also reveals A on Boolean points; thus $\mathcal{C}^{\tilde{A}} \not\subseteq \mathcal{D}^A$. So ordinary relativization implies the corresponding algebrized statement. Aaronson and Wigderson’s algebrizing simulation of arithmetization-based interactive-proof results supplies the converse counterexample at the level of proof techniques: ordinary relativization is not required for algebrization.

Item 3 is a type distinction. Theorem 12.4 starts from a set Γ_n of 2^n -bit truth tables and derives an oracle distinguisher against a PRF. Theorems 12.1 and 12.10 instead quantify over external computational oracles and say nothing about the density of a truth-table property. None of the three definitions supplies the missing data needed to infer another one. Item 4 follows. $\square$

Barrier	Object varied	Obstruction	What it does <i>not</i> say
Relativization	arbitrary oracle A	$P^A = NP^A$ for one oracle and $P^B \neq NP^B$ for another	nonrelativizing methods cannot work
Natural proofs	truth-table property Γ_n	large constructive usefulness distinguishes strong PRFs from random functions	every circuit lower-bound proof is natural, or that the barrier is unconditional
Algebrization	A together with low-degree $\tilde{A}$	both an algebraic collapse pair and an algebraic separation pair	every algebraic argument fails, or that naturalness has been addressed



Dashed lines mean “no definitional implication.” Overcoming one test does not automatically overcome the others.

Figure 12.1: The three barrier tests act on different mathematical objects. The solid arrow records the elementary statement-level implication discussed in Proposition 12.11; it is not an equivalence of proof techniques.

12.5 What remains open after the barriers

The barrier theorems do not close the central circuit questions. In particular, the registered conjectures

$$NP \not\subseteq P/poly \quad \text{and} \quad NEXP \not\subseteq P/poly \quad (12.24)$$

remain open. Chapters 5–11 should therefore be read as a map of successful lower-bound mechanisms in restricted models, while Chapter 12 records the known reasons those mechanisms cannot simply be extrapolated without new ideas.

A useful way to remember the chapter is by the question each barrier asks:

- **Relativization:** would the proof still work if every machine received the same arbitrary black box?
- **Natural proofs:** does the proof secretly give an efficient test that accepts many truth tables while rejecting every easy one?
- **Algebrization:** would the proof still work if low-degree extensions of the oracle were available to the simulating or lower-bounded side in the Aaronson–Wigderson sense?

A future unrestricted lower-bound proof may evade these tests in different ways; passing one of them is not a substitute for checking the others.

Internalization and Reconstruction Guide: Volume X

Purpose. Learn lower bounds as model-sensitive statements: counting, formulas, restrictions, Fourier/polynomial approximations, monotonicity, SAT algorithms, and barrier theorems each attack a different notion of circuit simplicity.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-004: Chernoff bounds.
2. QRCC-III-THM-002: Cyclicity of finite-field multiplicative groups.
3. QRCC-I-THM-001: Union bound.
4. QRCC-II-THM-003: Nondeterministic time hierarchy theorem.
5. QRCC-I-DEF-007: Boolean characters, degree, and influence.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-X-DEF-001 (Boolean circuit and formula models); QRCC-X-DEF-018 (Natural property against a circuit class); QRCC-X-DEF-020 (Algebraic relativization/algebrization); QRCC-X-DEF-030 (Strong pseudorandom-function assumption used by natural proofs).

Chapter retrieval map

- Ch. 1: Circuit Models, Formulas, and Basis Robustness:** enter through QRCC-X-DEF-001 (Boolean circuit and formula models); exit by being able to state QRCC-X-THM-001 (Circuit-basis robustness and normal forms) with its ownership boundary.
- Ch. 2: Counting Circuits and the Shannon Barrier:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-002 (Shannon counting lower bound) with its ownership boundary.

- Ch. 3: Explicit Formula Lower Bounds:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-004 (Nechiporuk formula lower-bound method) with its ownership boundary.
- Ch. 4: Random Restrictions and the Switching Lemma:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-006 (Switching lemma) with its ownership boundary.
- Ch. 5: Parity versus Constant-Depth Circuits:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-007 (Parity lower bound for AC0) with its ownership boundary.
- Ch. 6: Near-Optimal Parity Lower Bounds:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-008 (Håstad near-optimal AC0 lower bounds) with its ownership boundary.
- Ch. 7: Fourier Concentration for Constant-Depth Circuits:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-009 (Linial–Mansour–Nisan Fourier concentration) with its ownership boundary.
- Ch. 8: Polynomial Approximation over Finite Fields:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-COR-013 (MOD_q not in AC0[p] for distinct primes) with its ownership boundary.
- Ch. 9: Monotone Approximation and the Clique Lower Bound:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-010 (Razborov monotone approximation method) with its ownership boundary.
- Ch. 10: Depth-Two Threshold Circuits and Parity:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-014 (Tight average-case PARITY bounds for depth-two linear-threshold circuits) with its ownership boundary.
- Ch. 11: SAT Algorithms and Nonuniform ACC Lower Bounds:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-X-THM-015 (SAT algorithms imply circuit lower bounds) with its ownership boundary.
- Ch. 12: Three Barriers to General Circuit Lower Bounds:** enter through QRCC-X-DEF-018 (Natural property against a circuit class); exit by being able to state QRCC-X-PROP-022 (Comparison of relativization, naturalization, and algebrization) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-X-THM-001 — Circuit-basis robustness and normal forms. Dependencies: QRCC-X-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-X-THM-004 — Nechiporuk formula lower-bound method. Dependencies: QRCC-X-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

3. QRCC-X-THM-007 — Parity lower bound for AC0. Dependencies: QRCC-X-THM-006, QRCC-I-THM-004. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-X-THM-009 — Linial–Mansour–Nisan Fourier concentration. Dependencies: QRCC-I-DEF-007, QRCC-I-THM-018, QRCC-I-THM-004, QRCC-X-THM-006. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-X-THM-010 — Razborov monotone approximation method. Dependencies: QRCC-X-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-X-F-026 — Information-complexity composition frontier. Dependencies: QRCC-X-DEF-001, QRCC-XII-DEF-011. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- comparing lower bounds across different bases, fan-in conventions, or uniformity notions;
- using a formula lower bound as a circuit lower bound;
- treating an average-case lower bound as worst-case without an amplification step;
- reading relativization/natural-proofs/algebrization barriers as impossibility theorems for all proof techniques;

Boundary and degenerate cases

- depth one/two edge cases;
- bounded versus unbounded fan-in;
- monotone versus unrestricted circuits;
- modular gates in matching versus distinct characteristic;

Retrieval and reconstruction drill

1. State QRCC-X-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-X-THM-001. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-X-THM-004, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.

5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-X-F-026, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XI moves from Boolean circuits to arithmetic circuits, where PIT becomes the canonical derandomization/lower-bound interface.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XI

Arithmetic Complexity, Polynomial Identity Testing, and Algebraic Structure

Chapter 1

Arithmetic Circuits: Degree, Homogeneity, and Division

Boolean circuits compute functions on a finite cube. Arithmetic circuits instead compute polynomials, or rational functions when division is allowed. The change looks small syntactically but introduces several notions of degree, several reasonable notions of “the same computation,” and a crucial distinction between an identity test that can inspect a circuit and one that only sees evaluations. These conventions will be used throughout Volume XI, so we freeze them before stating structural theorems.

The two main results of this chapter are deliberately elementary but powerful. First, a circuit can be split into its low-degree homogeneous pieces with only a quadratic overhead in the cutoff degree. Second, if a division circuit computing a degree- d polynomial is regular at one expansion point, then all of its division gates can be removed with polynomial overhead. The second proof is a finite Taylor-series computation: no analytic convergence is involved.

How to read this chapter. There are four layers, and they should not be mentally collapsed. The first section fixes the syntax and semantics of circuits, formulas, ABPs, formal degree, and black-box versus white-box PIT. The second gives the homogeneous component construction that will be reused whenever a semantic degree bound must be turned into syntactic control. The third removes divisions once a regular center is supplied. The final section is qualitatively different: it uses characteristic-zero factor lifting to show that low circuit complexity is closed under taking polynomial factors.

A useful dependency chain is

circuit conventions $\longrightarrow$ homogeneous extraction $\longrightarrow$ division elimination $\longrightarrow$ factor closure.

Only the first two arrows are purely gate-local. The last theorem needs algebraic factorization input and a characteristic hypothesis. Keeping that boundary visible will prevent several common mistakes later in the volume.

Learning goals. After this chapter, a reader should be able to distinguish semantic from formal degree, explain exactly what black-box access hides, carry out truncated homogeneous arithmetic gate by gate, state why a regular center makes formal series inversion legal, and identify the point at which characteristic zero enters factor closure.

1.1 The arithmetic models

Fix a field F and variables $x = (x_1, \dots, x_n)$.

Definition 1.1 (Arithmetic circuits, formulas, ABPs, degree, and PIT). The following conventions are in force in Volume XI.

1. A *division-free arithmetic circuit over F* is a finite directed acyclic graph with a designated output gate. Source gates are variables x_i or field constants $c \in F$. Every internal gate has fan-in two and is labelled $+$, $-$, or $\times$. Its semantics is the polynomial in $F[x]$ obtained in topological order. Subtraction is retained as notation; it can be replaced by addition and multiplication by -1 .
2. A *circuit with division* may additionally contain gates u/v . Such a gate denotes the corresponding element of the rational-function field $F(x)$ and is legal only when the rational function computed by v is nonzero. For $a \in F^n$, the circuit is *defined at a* if every division encountered in its numerical evaluation at a has nonzero denominator. We call such an a a *regular center* for the circuit.
3. The *size* of a circuit is its total number of gates, including source gates. Its *depth* is the maximum number of internal arithmetic gates on a source-to-output path. Later results that use another standard convention will say so explicitly. These choices differ from operation-count conventions only by at most the number of sources and do not affect polynomial-size classes.
4. A *formula* is an arithmetic circuit whose underlying directed graph is a rooted tree directed toward the output. Thus a computed subexpression is not shared: if it is used twice, it occurs twice in the formula.
5. For a division-free circuit, the *formal degree* $\text{fdeg}(v)$ of a gate is defined recursively by

$$\begin{aligned} \text{fdeg}(c) &= 0, & \text{fdeg}(x_i) &= 1, \\ \text{fdeg}(u \pm v) &= \max\{\text{fdeg}(u), \text{fdeg}(v)\}, & \text{fdeg}(uv) &= \text{fdeg}(u) + \text{fdeg}(v). \end{aligned} \quad (1.1)$$

The *semantic degree* is the total degree of the polynomial actually computed. Always

$$\deg p_v \leq \text{fdeg}(v), \quad (1.2)$$

but cancellation can make the inequality strict.

6. A division-free circuit is *syntactically homogeneous* if every gate v is assigned a nonnegative integer $\delta(v)$ so that variables have label 1, nonzero constants have label 0, a zero gate may be assigned any label, addition/subtraction gates have two children of the same label and keep that label, and multiplication gates receive the sum of their children's labels. Then every gate computes either 0 or a homogeneous polynomial of degree $\delta(v)$.
7. An *algebraic branching program* (ABP) is a layered directed acyclic graph

$$V_0 \sqcup V_1 \sqcup \dots \sqcup V_L,$$

with one source $s \in V_0$, one sink $t \in V_L$, edges only from V_i to V_{i+1} , and every edge labelled by an affine linear form over F . It computes

$$\sum_{\pi: s \rightsquigarrow t} \prod_{e \in \pi} \ell_e(x). \quad (1.3)$$

Its length is L , width is $\max_i |V_i|$, and its graph size is the number of vertices plus nonzero labelled edges. No circuit-ABP equivalence is being built into this definition.

8. *Polynomial identity testing* (PIT) asks whether the output polynomial of a division-free arithmetic circuit is identically zero. In *white-box PIT* the algorithm receives the circuit graph and constants. In *black-box PIT* it receives only evaluation access to the polynomial, together with whatever variable/degree/evaluation-field bounds are stated by the theorem. Black-box PIT is therefore a stronger access restriction, not a synonym for randomized PIT.

Formal degree can overestimate the polynomial degree. The circuit

$$C(x, y) = (x + y)^2 - (x^2 + 2xy + y^2) \quad (1.4)$$

computes the zero polynomial, whose semantic degree is conventionally $-\infty$ (or may simply be called zero), while the output has formal degree 2. More generally, formal degree records what the syntax could have produced before cancellation. This is why later balancing and depth-reduction arguments often control formal degree rather than only the degree of the final output.

Division is a rational-function operation. The expression $(1 - x^3)/(1 - x)$ is a legal rational function over every field, because $1 - x$ is not the zero polynomial. The resulting rational function equals the polynomial $1 + x + x^2$, but the displayed division circuit is not defined at $x = 1$. This distinction between rational-function equality and pointwise circuit evaluation is exactly what the next theorem must respect.

1.2 Homogeneous-component extraction

For $p \in F[x]$, write $H_j[p]$ for its homogeneous component of total degree j , so that

$$p = \sum_{j \geq 0} H_j[p]. \quad (1.5)$$

Only finitely many summands are nonzero.

Theorem 1.2 (Homogeneous-component extraction). *Let C be a division-free arithmetic circuit over an arbitrary field F , of size s , and let $r \geq 0$. There is a division-free circuit $C_{\leq r}$ of size*

$$O(s(r + 1)^2) \quad (1.6)$$

that simultaneously outputs

$$H_0[p_C], H_1[p_C], \dots, H_r[p_C]. \quad (1.7)$$

Moreover each output $H_j[p_C]$ is produced by a syntactically homogeneous subcircuit of formal degree j . The construction is uniform and uses no field size or characteristic hypothesis.

Proof roadmap. Attach to every gate the vector of homogeneous components through degree d . Addition is coordinatewise, while multiplication becomes a truncated convolution. The gate invariant says that coordinate k is exactly the homogeneous degree- k part of the polynomial at that gate. Counting the convolution pairs gives the quadratic $(d + 1)^2$ overhead; no interpolation or field-size hypothesis is needed.

Hypothesis check. The input circuit is division-free because homogeneous components of a quotient are not obtained by the same finite convolution rule. The cutoff d is a syntactic output request, not a promise that the original circuit has formal degree d .

Proof. For every gate v of C , create tagged gates

$$(v, 0), (v, 1), \dots, (v, r), \quad (1.8)$$

whose intended values are

$$p_{v,j} := H_j[p_v]. \quad (1.9)$$

We construct them in a topological order.

If v is a constant c , set $(v, 0) = c$ and $(v, j) = 0$ for $j > 0$. If $v = x_i$, set $(v, 1) = x_i$ and all other tagged components to zero. For an addition or subtraction gate $v = u \pm w$, define

$$p_{v,j} = p_{u,j} \pm p_{w,j} \quad (0 \leq j \leq r). \quad (1.10)$$

For a product gate $v = uw$, use the homogeneous Cauchy product

$$p_{v,j} = \sum_{i=0}^j p_{u,i} p_{w,j-i} \quad (0 \leq j \leq r). \quad (1.11)$$

Equations (1.10)–(1.11) are identities in the polynomial ring, so induction on the circuit proves (1.9) at every gate.

The same induction gives syntactic homogeneity: every summand in (1.11) has label $i + (j - i) = j$, so the sum can be formed using only degree- j gates. A zero component is represented by a zero gate tagged with the required degree.

It remains to count gates. An addition/subtraction gate creates $r + 1$ new arithmetic gates. At a multiplication gate, component j uses $j + 1$ multiplications and at most j additions. Summing over $j = 0, \dots, r$ gives

$$\sum_{j=0}^r (2j + 1) = (r + 1)^2. \quad (1.12)$$

Thus each original gate causes at most $O((r + 1)^2)$ gates, proving (1.6). $\square$

Why interpolation is only a comparison route. Suppose additionally that $\deg p_C \leq D$ and F contains $D + 1$ distinct scalars $\lambda_0, \dots, \lambda_D$. Then

$$p_C(\lambda_i x) = \sum_{j=0}^D \lambda_i^j H_j[p_C](x). \quad (1.13)$$

The Vandermonde matrix $(\lambda_i^j)_{i,j}$ is invertible, so each homogeneous component is an F -linear combination of $D + 1$ scaled copies of C . This route is useful when preserving the internal depth of the original circuit matters, but it needs enough field points and a degree bound on the entire polynomial. The gate-by-gate proof above needs neither condition and can extract only the first r components even when higher-degree terms are present.

A three-degree worked gate. If a multiplication gate computes

$$(a_0 + a_1 + a_2)(b_0 + b_1 + b_2),$$

where a_i, b_i are homogeneous of degree i , its first three components are

$$a_0b_0, \quad a_0b_1 + a_1b_0, \quad a_0b_2 + a_1b_1 + a_2b_0. \quad (1.14)$$

This finite convolution is the basic operation both in homogenization and in the division-elimination proof below.

1.3 Division elimination from a regular center

The classical theorem is often stated over an infinite field, where one first chooses a point avoiding all denominators. For the local theorem it is cleaner to expose the true hypothesis: we assume such a point has already been supplied. This removes every field-size issue from the proof.

Theorem 1.3 (Strassen-style division elimination at a regular center). *Let C be an arithmetic circuit over a field F using $+, -, \times, /$, of size s . Suppose that*

1. *the rational function at the output equals a polynomial $f \in F[x_1, \dots, x_n]$ of total degree at most d ; and*
2. *$a \in F^n$ is a regular center for C .*

Then f has a division-free arithmetic circuit over F of size

$$O(s(d+1)^2) \quad (1.15)$$

and formal degree at most d . Given C , a , and d , the transformed circuit is obtained by a deterministic gate-by-gate construction using only field arithmetic and the inverses of the nonzero denominator values encountered at a .

Proof roadmap. Translate the supplied regular center to the origin. At each gate keep only homogeneous terms through degree d . A denominator has nonzero constant term, so its truncated reciprocal is determined recursively from the equation $WQ = 1$ degree by degree. The same gate invariant as in homogeneous extraction then proves that the output truncation equals the full polynomial, because the promised semantic degree is at most d .

Hypothesis check. The regular center is the only reason reciprocal series can be started: its constant denominator coefficient must be invertible. The output must actually be a polynomial of degree at most d ; otherwise truncation computes only a finite jet, not the full rational function.

Proof. Translate the variables by writing

$$x = a + y. \quad (1.16)$$

For each gate v of C , let $R_v(y) \in F(y)$ be the rational function computed at that gate after the translation. Because the original circuit is defined at a , every R_v is regular at $y = 0$. We shall construct, for every gate v , polynomials

$$R_{v,0}, R_{v,1}, \dots, R_{v,d} \quad (1.17)$$

that are the first $d + 1$ homogeneous terms of the formal Taylor expansion of R_v at the origin. No convergence statement is being used: all identities are identities in the ring of truncated formal power series

$$F[[y_1, \dots, y_n]]/(\text{terms of total degree } > d). \quad (1.18)$$

For constants and translated variables the components are immediate:

$$c = (c) + 0 + \dots, \quad a_i + y_i = (a_i) + (y_i) + 0 + \dots. \quad (1.19)$$

Addition and multiplication use exactly the recurrences (1.10)–(1.11), truncated after degree d .

Consider now a division gate $v = u/w$. Write

$$W(y) = \sum_{j=0}^d W_j(y) + O(> d) \quad (1.20)$$

for the already constructed expansion of its denominator. Regularity gives

$$W_0 = W(0) = R_w(0) \in F^\times. \quad (1.21)$$

Define homogeneous polynomials $Q_0, \dots, Q_d$ recursively by

$$Q_0 = W_0^{-1}, \quad (1.22)$$

$$Q_k = -W_0^{-1} \sum_{i=1}^k W_i Q_{k-i} \quad (1 \leq k \leq d). \quad (1.23)$$

For every $k \leq d$, the degree- k component of $W \sum_{j=0}^d Q_j$ is therefore 1 when $k = 0$ and 0 otherwise. Hence

$$\left(\sum_{i=0}^d W_i \right) \left(\sum_{j=0}^d Q_j \right) \equiv 1 \pmod{> d}, \quad (1.24)$$

so $\sum_{j=0}^d Q_j$ is the inverse of W through total degree d . If

$$U(y) = \sum_{i=0}^d U_i(y) + O(> d),$$

set

$$R_{v,k} = \sum_{i=0}^k U_i Q_{k-i}. \quad (1.25)$$

This is exactly the degree- k component of U/W . Thus a topological induction constructs the correct truncated expansion at every gate.

At the output gate we obtain

$$f(a + y) = \sum_{k=0}^d R_{\text{out},k}(y), \quad (1.26)$$

with no omitted term because f has degree at most d . Finally substitute $y_i = x_i - a_i$. Each $R_{\text{out},k}$ had formal degree k in y , and the substitution $y_i = x_i - a_i$ has formal degree 1, so the resulting division-free circuit has formal degree at most d and computes $f(x)$.

The size estimate follows from the same finite convolutions used in Theorem 1.2. Addition/subtraction costs $O(d)$ per original gate; multiplication costs $O((d+1)^2)$; the inverse recurrence (1.23) and the final quotient convolution (1.25) together cost $O((d+1)^2)$ per division gate. Thus the total size is $O(s(d+1)^2)$. $\square$

Geometric-series form of the same proof. For a single quotient h/g with $g(0) = 1$, put $z = 1 - g$, which has zero constant term. In truncated power series,

$$\frac{h}{g} = \frac{h}{1-z} \equiv h(1+z+z^2+\cdots+z^d) \pmod{>d}, \quad (1.27)$$

because every monomial of z^{d+1} has degree at least $d+1$. This is the usual Strassen geometric-series picture. Recurrence (1.23) is its gate-local version and avoids first converting the whole circuit to a single quotient.

Worked example: cancellation behind a division. Take

$$f(x) = \frac{1-x^3}{1-x}. \quad (1.28)$$

The circuit is defined at $a = 0$, and as a rational function its output equals the degree-2 polynomial $1+x+x^2$. Since $1/(1-x) \equiv 1+x+x^2 \pmod{x^3}$,

$$(1-x^3)(1+x+x^2) \equiv 1+x+x^2 \pmod{x^3}. \quad (1.29)$$

Because the promised output degree is 2, truncation loses nothing. Notice that the original numerator has degree 3: division elimination is controlled by the degree of the *polynomial output*, not by the unreduced degrees of numerator and denominator expressions.

Where infinite fields enter the classical statement. If F is infinite and every division gate is legal as a rational function, then the finitely many polynomial numerators and denominators whose nonvanishing is needed define a nonempty Zariski-open set. A point $a \in F^n$ can therefore be chosen outside their zero sets, and Theorem 1.3 applies. Over a finite field the same conclusion may require passing to an extension field and simulating extension-field arithmetic. The local theorem above does not need that machinery because its regular center is an explicit hypothesis.

1.4 Factor closure in characteristic zero

The next result is deeper than homogenization or division elimination. Its algorithmic origin is Kaltofen's straight-line-program factorization theorem, but for circuit complexity it is useful to separate two questions: *does a small circuit for f imply a small circuit for a factor?*, and *can such a factor circuit be found efficiently?* We prove the first question nonuniformly in characteristic zero. The proof follows the modern Newton–resultant reconstruction of the classical closure theorem and uses the homogeneous-component machinery already proved in this chapter.

Theorem 1.4 (Kaltofen factor closure, characteristic-zero form). *Let F be a field of characteristic zero. Let*

$$f \in F[x_1, \dots, x_n] \setminus \{0\}$$

have total degree at most D and a division-free arithmetic circuit of size s . If $g \in F[x_1, \dots, x_n]$ divides f , then g has a division-free arithmetic circuit of size

$$\text{poly}(s, n, D). \quad (1.30)$$

The polynomial hidden in poly is universal: it does not depend on the particular field F , on the coefficients of f , or on the chosen factor g . The assertion is nonuniform and existential; it does not claim that the factor circuit is deterministically recoverable from the input circuit.

Proof roadmap. First isolate a coprime factor block and lift its coefficients degree by degree; the same nonsingular linear system is reused while the residual changes. Then reduce an irreducible factor occurring with multiplicity e to its perfect power and extract the underlying factor. Finally multiply the selected irreducible-factor circuits to obtain an arbitrary divisor.

Hypothesis check. Characteristic zero enters at perfect-power extraction through the scalar multiplicity e . Coprime lifting itself is more general. The theorem is a nonuniform circuit-closure statement; it should not be read as a claim that the local proof supplies a deterministic polynomial-time factorization algorithm.

We prove the theorem through two closure lemmas. They are stated separately because they isolate exactly where coprimality and multiplicity enter.

Lemma 1.5 (Coprime factor lifting). *Let F have characteristic zero. Suppose $f = gh$ with $g, h \in F[x_1, \dots, x_n]$ nonconstant,*

$$\gcd(g, h) = 1, \quad \deg f \leq D,$$

and f has a circuit of size s . Then both g and h have circuits of size $\text{poly}(s, n, D)$.

Proof. A nonsingular linear change of coordinates and a scaling by field constants do not change circuit size by more than a polynomial factor. Since F is infinite, choose a linear change so that, with $x = (x_1, \dots, x_{n-1})$ and distinguished variable $y = x_n$, the three polynomials are monic in y :

$$\begin{aligned} f(x, y) &= y^d + \sum_{r=0}^{d-1} f_r(x) y^r, \\ g(x, y) &= y^{d_1} + \sum_{i=0}^{d_1-1} g_i(x) y^i, \\ h(x, y) &= y^{d_2} + \sum_{j=0}^{d_2-1} h_j(x) y^j, \end{aligned} \tag{1.31}$$

where $d = d_1 + d_2 \leq D$. Indeed, the leading coefficient in a generic direction is the highest homogeneous part evaluated at that direction, so one avoids only finitely many proper algebraic zero sets. The coordinate change is an automorphism, hence it preserves coprimality.

We first record a coefficient-extraction fact. Tag each gate of the circuit for f by its y -degree $0, \dots, d$, exactly as in the proof of Theorem 1.2. Addition is componentwise and multiplication uses the one-variable convolution

$$[y^r](uv) = \sum_{i=0}^r [y^i]u [y^{r-i}]v. \tag{1.32}$$

Thus all coefficient polynomials $f_0, \dots, f_{d-1}$ have a simultaneous circuit of size $\text{poly}(s, D)$.

Introduce unknown tuples

$$u = (u_0, \dots, u_{d_1-1}), \quad w = (w_0, \dots, w_{d_2-1})$$

and, for $0 \leq r < d$, define

$$Q_r(u, w) = \sum_{i+j=r} u_i w_j - f_r, \tag{1.33}$$

where we use the conventions $u_{d_1} = w_{d_2} = 1$ and coefficients outside the allowed ranges are zero. Then

$$Q_r(\mathbf{g}, \mathbf{h}) = 0 \quad (0 \leq r < d), \tag{1.34}$$

for $\mathbf{g} = (g_0, \dots, g_{d_1-1})$ and $\mathbf{h} = (h_0, \dots, h_{d_2-1})$.

Let

$$J(u, w) = \left(\frac{\partial Q_r}{\partial (u, w)_c} \right)_{r,c} \quad (1.35)$$

be the $d \times d$ Jacobian matrix. At $(\mathbf{g}, \mathbf{h})$ this is, up to row/column ordering, the Sylvester matrix of the two monic polynomials g and h viewed as polynomials in y . Hence

$$\det J(\mathbf{g}, \mathbf{h}) = \pm \text{Res}_y(g, h) \neq 0, \quad (1.36)$$

because g and h are coprime. The determinant is therefore a nonzero polynomial in x . Translate the x -coordinates once more so that

$$A := J(\mathbf{g}(0), \mathbf{h}(0)) \in F^{d \times d} \quad (1.37)$$

is nonsingular.

Write $\mathbf{m} = (x_1, \dots, x_{n-1})$. We now reconstruct all g_i, h_j by increasing total x -degree. Suppose for some $k \geq 1$ we have circuit approximations

$$\tilde{\mathbf{g}}^{(k)} \equiv \mathbf{g} \pmod{\mathbf{m}^k}, \quad \tilde{\mathbf{h}}^{(k)} \equiv \mathbf{h} \pmod{\mathbf{m}^k}. \quad (1.38)$$

Let p_i, q_j be the homogeneous degree- k corrections. Since each Q_r is bilinear in (u, w) , Taylor expansion modulo $\mathbf{m}^{k+1}$ gives

$$A \begin{pmatrix} p \\ q \end{pmatrix} = - \begin{pmatrix} H_k[Q_0(\tilde{\mathbf{g}}^{(k)}, \tilde{\mathbf{h}}^{(k)})] \\ \vdots \\ H_k[Q_{d-1}(\tilde{\mathbf{g}}^{(k)}, \tilde{\mathbf{h}}^{(k)})] \end{pmatrix}. \quad (1.39)$$

Terms quadratic in the corrections have degree at least $2k \geq k+1$ and vanish in this congruence. Since A^{-1} consists only of field constants, equation (1.39) uniquely expresses every p_i, q_j as an F -linear combination of the degree- k homogeneous parts on the right.

Theorem 1.2 computes those homogeneous parts with polynomial overhead. Consequently one Newton step from precision k to precision $k+1$ increases the circuit size by $\text{poly}(s, n, D)$, with a polynomial independent of k . The base case $k=1$ consists only of the field constants $g_i(0), h_j(0)$. After $D+1$ steps we have

$$\tilde{g}_i \equiv g_i \pmod{\mathbf{m}^{D+1}}, \quad \tilde{h}_j \equiv h_j \pmod{\mathbf{m}^{D+1}}. \quad (1.40)$$

Every coefficient polynomial in (1.31) has degree at most D , so the congruences are equalities. Reassembling the powers of y proves the claimed $\text{poly}(s, n, D)$ circuits for g and h . $\square$

Lemma 1.6 (Perfect-power extraction). *Let F have characteristic zero. If $f = g^e$, with $e \geq 1$, $\deg f \leq D$, and f has circuit size s , then g has circuit size $\text{poly}(s, n, D)$.*

Proof. The constant case is trivial. Otherwise introduce a new variable z and put

$$\Phi(x, z) = z^e - f(x) = (z - g(x))(z^{e-1} + z^{e-2}g + \dots + g^{e-1}). \quad (1.41)$$

A repeated-squaring circuit computes z^e , so Φ has size $s + O(\log D)$. The two displayed factors are coprime in $F[x][z]$: substituting $z = g(x)$ in the second factor gives

$$e g(x)^{e-1} \neq 0, \quad (1.42)$$

where characteristic zero is essential. Lemma 1.5 therefore gives a $\text{poly}(s, n, D)$ circuit for $z - g(x)$; setting $z = 0$ and multiplying by -1 gives such a circuit for g . $\square$

Proof of Theorem 1.4. Factor f in the UFD $F[x]$ as

$$f = c \prod_{i=1}^r p_i^{e_i}, \quad (1.43)$$

with distinct monic irreducibles p_i . For each i the primary block $p_i^{e_i}$ is coprime to $f/p_i^{e_i}$. If the cofactor is nonconstant, Lemma 1.5 gives a circuit of size $\text{poly}(s, n, D)$ for $p_i^{e_i}$. If the cofactor is a field constant, then $p_i^{e_i}$ is already a scalar multiple of f and has size $O(s)$. In either case Lemma 1.6 gives a circuit of the same polynomial order for p_i itself.

Any divisor has the form

$$g = c' \prod_{i=1}^r p_i^{a_i}, \quad 0 \leq a_i \leq e_i. \quad (1.44)$$

There are at most D nonconstant irreducible factors counted with multiplicity, and every exponent is at most D . Repeated squaring and a binary product tree combine the corresponding factor circuits with only polynomial additional size. This proves (1.30). Reversing the preliminary linear changes of variables also costs only $O(n)$ affine gates per input occurrence and hence preserves a polynomial bound. $\square$

A worked lifting step. Take

$$f(x, y) = (y - x)(y + 1) = y^2 + (1 - x)y - x. \quad (1.45)$$

Here $g = y - x$, $h = y + 1$, and the coefficient equations are

$$Q_0(u_0, w_0) = u_0 w_0 + x, \quad Q_1(u_0, w_0) = u_0 + w_0 - (1 - x). \quad (1.46)$$

At $x = 0$ the solution is $(u_0, w_0) = (0, 1)$ and

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad (1.47)$$

The degree-one residual at the constant approximation is $(x, x)^T$, so the correction solves

$$A \begin{pmatrix} p \\ q \end{pmatrix} = - \begin{pmatrix} x \\ x \end{pmatrix}, \quad (p, q) = (-x, 0). \quad (1.48)$$

Thus one Newton step already recovers $u_0 = -x$ and $w_0 = 1$, hence the exact factors. In larger degree the same linear system is reused at every homogeneous precision level; only the residual vector changes.

Why small characteristic is not silently included. Kaltofen's 1989 algorithmic theorem treats straight-line-program inputs over the usual coefficient fields and, in positive characteristic p , explicitly separates the case in which p divides an irreducible factor's multiplicity. In that case the white-box output can naturally be a circuit for a p^k -th power of the factor rather than for the factor itself. The local proof above also shows the obstruction transparently: equation (1.42) fails when $p \mid e$. Modern characteristic-zero closure proofs therefore do not justify an unrestricted small-characteristic version by mere formal substitution. PASS47 records the broader algorithmic source statement, but the QRCC theorem proved here is exactly the characteristic-zero circuit-closure theorem.

Source ownership and algorithmic strength. The proof above is a nonuniform circuit-existence reconstruction. Kaltofen’s primary result is stronger in a different direction: for coefficient fields covered by the algorithm it gives randomized polynomial-time factorization of a straight-line program and returns straight-line representations of the factors (or the appropriate positive-characteristic powers), with explicit control of failure probability and coefficient representation. Conversely, the local proof does not claim a deterministic factorization algorithm. These two claims must not be conflated when VP is introduced in the next chapter.

1.5 Dated frontier: constant-depth polynomial and structured-matrix algebra

The published Andrews–Wigderson theorem should not be paraphrased as a blanket constant-depth algorithm for arbitrary matrix rank, determinant, characteristic polynomial, or inverse. Its verified interface is instead the following. Over the source’s field model, polynomial-size constant-depth $\text{AC}_{\mathbb{F}}^0$ arithmetic formulae compute polynomial GCD (also for any number of input polynomials), discriminant, resultant, Bézout coefficients, squarefree decomposition, and inversion of structured matrices such as Sylvester and Bézout matrices. The techniques also extend to multivariate polynomial inputs represented by constant-depth arithmetic circuits for problems including GCD and squarefree decomposition.

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1.6 Reader checkpoint and exercises

Chapter checkpoint. The reusable idea of the chapter is *normalization without changing the polynomial*. Homogeneous extraction turns semantic degree information into a controlled circuit; a regular center turns rational gate operations into finite formal-series arithmetic; and, in characteristic zero, factor lifting shows that a small circuit cannot hide an intrinsically hard polynomial factor. The first two tools will be used immediately in the definition and manipulation of VP/VNP families.

Exercises.

1. Give a division-free circuit whose output has semantic degree 1 but whose formal degree is at least 8. Identify the cancellation responsible for the gap, and run the degree-1 homogeneous-component construction on it.
2. For $C(x, y) = (1+x+y)^3$, compute by hand the vectors stored at the addition and multiplication gates when Theorem 1.2 is run with cutoff $d = 2$. Verify that the discarded degree-3 component never contaminates the lower coordinates.
3. Consider the division circuit $(x^2 - 1)/(x - 1)$. Determine all regular centers over a field with more than two elements, perform the first two homogeneous reciprocal steps around one regular center, and recover the polynomial $x + 1$.
4. Explain why “the final rational function is a polynomial” does not imply that every point is a regular center for the circuit that computes it. Give a one-variable example other than the preceding exercise.

5. Work through the coefficient system for the factorization $y^2 - (1+x)^2 = (y - (1+x))(y + (1+x))$ around $x = 0$. Identify the linear system that would be reused in a degree-by-degree lifting proof.
6. Over characteristic $p > 0$, compare the formal derivatives of $u^p - v^p$ with its factorization $(u - v)^p$. Use the example to explain, in your own words, why the multiplicity step in the local factor-closure proof cannot simply copy the characteristic-zero argument.

Chapter 2

Valiant's Classes: VP, VNP, and Projections

Arithmetic circuits become a complexity theory only after we organize individual polynomials into families. This chapter fixes that family-level language before we discuss determinant universality or permanent completeness. The conventions are nonuniform and work over an arbitrary fixed field: no algorithm for printing the circuit family is silently required. Uniform variants will be named explicitly when they are needed.

The central distinction parallels, but is not identical to, the Boolean P – NP distinction. A family in VP has polynomially bounded degree and polynomial-size arithmetic circuits. A family in VNP is obtained by summing a VP polynomial over a polynomial number of Boolean witness bits. The witness sum has exponentially many terms, so it is a definition of a class rather than a polynomial-size circuit construction.

How to read this chapter. The chapter moves from *objects* to *reductions*. First freeze what a polynomial family is and what “polynomial size” means in the nonuniform setting. Next learn the witness-sum definition of VNP literally: it is a sum over Boolean strings, not an instruction to enumerate those strings by a small circuit. Finally, keep p-projections separate from more permissive algebraic substitutions. Chapter 3 will rely on precisely that rigidity when the permanent is shown complete.

The minimum dependency picture is

$$\text{VP}_e \subseteq \text{VP} \subseteq \text{VNP}, \quad \text{p-projection} \implies \text{downward closure},$$

with Theorem 1.2 supplying formal-degree cleanup for a VP witness when needed. Nothing here assumes that the circuit family can be printed efficiently.

Learning goals. A reader should be able to test whether a proposed family satisfies the p-family degree/variable bounds, write and manipulate a VNP witness sum, decide whether a substitution is genuinely a projection, and explain the difference between a nonuniform existence statement and a uniform construction.

2.1 Families and reductions

Fix once and for all a field F .

Definition 2.1 (p-families, VP, VP_e , VNP, weak skewness, and p-projections). The following conventions are in force throughout the remainder of Volume XI.

1. A *p-family* over F is a sequence

$$(f_n)_{n \geq 1}, \quad f_n \in F[x_1, \dots, x_{v(n)}], \quad (2.1)$$

for which both the variable bound $v(n)$ and the total degree $\deg f_n$ are bounded by polynomials in n . Dummy variables are allowed, so two p-families may always be padded to a common polynomial variable bound.

2. The class VP_F consists of p-families (f_n) for which there is a polynomial $s(n)$ such that f_n has a division-free arithmetic circuit over F of size at most $s(n)$. This is a *nonuniform* definition: the circuit for length n need only exist. We write VP when F is fixed by context.
3. The class $VP_{e,F}$ (Valiant's polynomial-size expressions) is the subclass of p-families computable by arithmetic formulas of polynomial size. Thus

$$VP_{e,F} \subseteq VP_F \quad (2.2)$$

by definition. The subscript e records expression/formula complexity; it is not an exponent.

4. A p-family (f_n) belongs to VNP_F if there exist a polynomially bounded nonnegative integer $m(n)$ and a p-family

$$g_n(x, y) \in VP_F, \quad y = (y_1, \dots, y_{m(n)}), \quad (2.3)$$

such that

$$f_n(x) = \sum_{b \in \{0,1\}^{m(n)}} g_n(x, b). \quad (2.4)$$

The family (g_n) is a *VP witness family* for (f_n) . We allow $m(n) = 0$; then the Boolean cube has one empty string and the sum contains one term.

5. A division-free arithmetic circuit is *weakly skew* if at every multiplication gate $u \times v$, at least one of the two input subcircuits is vertex-disjoint from the remainder of the circuit except for the edge entering that multiplication gate. Equivalently, one factor at every multiplication can be viewed as a private subcomputation. Weak skewness is a syntactic restriction; no equivalence with ABPs or determinants is part of this definition.
6. If $f \in F[x_1, \dots, x_r]$ and $G \in F[z_1, \dots, z_N]$, then f is a *projection* of G if there is a substitution

$$z_j \mapsto a_j \in F \cup \{x_1, \dots, x_r\} \quad (2.5)$$

for every j such that the resulting polynomial is f . Repetition of target variables is allowed; arbitrary linear forms or monomials are not.

For p-families (f_n) and (G_N) , we say (f_n) is a *p-projection* of (G_N) , written

$$(f_n) \leq_p (G_N), \quad (2.6)$$

if there is a polynomially bounded index map $N = t(n)$ such that f_n is a projection of $G_{t(n)}$ for every sufficiently large n . Finite initial exceptions may be absorbed by changing finitely many circuits.

Three distinctions worth freezing. First, the degree bound in a p-family is semantic: it is a bound on the polynomial itself. A particular circuit may have much larger formal degree because of cancellation. The homogenization theorem from Chapter 1 lets us replace such a circuit by one with polynomially bounded formal degree whenever that bookkeeping is needed.

Second, VNP is not defined by “guessing field elements.” The sum in (2.4) is over the Boolean cube $\{0, 1\}^{m(n)}$, even when F is infinite. Consequently the definition makes sense over every field, including fields of characteristic two.

Third, p-projections are deliberately restrictive. For example, the substitution $z_1 \mapsto x_1 + x_2$ is not a projection. This rigidity is what makes a p-projection completeness theorem for the permanent mathematically strong.

A small witness-sum example. Let

$$g_n(x, y) = \prod_{i=1}^n ((1 - y_i) + y_i x_i). \quad (2.7)$$

For a Boolean vector b , the factor indexed by i equals 1 when $b_i = 0$ and x_i when $b_i = 1$. Hence

$$\sum_{b \in \{0,1\}^n} g_n(x, b) = \sum_{S \subseteq [n]} \prod_{i \in S} x_i = \prod_{i=1}^n (1 + x_i). \quad (2.8)$$

This example happens to lie already in VP; its purpose is to make the Boolean-witness semantics literal before we use witness sums to describe much harder families.

2.2 Basic closure calculus

For closure under a polynomial number of operations, we need one harmless nonuniform convention. A doubly indexed collection

$$\{f_{n,i} : 1 \leq i \leq k(n)\} \quad (2.9)$$

is a *polynomial bundle of VP families* if $k(n)$ is polynomially bounded and one polynomial simultaneously bounds the number of variables, degrees, and circuit sizes of all $f_{n,i}$. A polynomial bundle of VNP families is defined similarly, with one polynomial simultaneously bounding the witness lengths and the VP-witness circuit data. This is a size-bound convention, not a uniformity requirement.

Theorem 2.2 (Basic closure properties of VP and VNP). *Over every fixed field F the following statements hold.*

1.

$$\text{VP}_{e,F} \subseteq \text{VP}_F \subseteq \text{VNP}_F. \quad (2.10)$$

2. VP_F and VNP_F are closed downward under p -projections.

3. If $\{f_{n,i} : 1 \leq i \leq k(n)\}$ is a polynomial bundle in VP_F , then

$$S_n = \sum_{i=1}^{k(n)} f_{n,i}, \quad P_n = \prod_{i=1}^{k(n)} f_{n,i} \quad (2.11)$$

form p -families in VP_F . The analogous statement holds for a polynomial bundle in VNP_F .

4. Every VNP_F family admits a witness representation (2.4) in which all witness polynomials are computed by polynomial-size division-free circuits of polynomially bounded formal degree. Moreover, witness strings may be padded to any larger polynomially bounded length without changing the summed polynomial.

All claims are nonuniform and require no characteristic hypothesis.

Proof roadmap. Handle each closure operation at the witness level. VP inclusion in VNP uses zero witness bits. Projections are pushed directly into the witnessing circuits. Polynomially many products use concatenated witness blocks, while polynomially many sums require an additional Boolean selector block. The last step invokes homogeneous-component extraction to replace a low semantic-degree witness circuit by one of low formal degree.

Hypothesis check. The bundle condition is what supplies one polynomial bound for all summands or factors at a fixed input length. Nonuniformity permits the selector codewords to be hardwired; characteristic independence comes from using only 0, 1, and Boolean identities, never division by the size of the cube.

Proof. The inclusion $\text{VP}_{e,F} \subseteq \text{VP}_F$ is immediate because a formula is a circuit. If $(f_n) \in \text{VP}_F$, take zero witness bits in (2.4) and $g_n = f_n$. This proves $\text{VP}_F \subseteq \text{VNP}_F$ without introducing a spurious factor such as $2^{m(n)}$.

For p-projections, substitute each source variable by the prescribed target variable or field constant directly in the circuit. Size cannot increase and total degree remains polynomially bounded. If

$$f_n(x) = \sum_{b \in \{0,1\}^{m(n)}} g_n(x, b) \quad (2.12)$$

is a VNP witness representation, applying the projection only to the x variables gives the projected family with the projected g_n as its VP witness. Thus both classes are closed under p-projections.

Now take a VP polynomial bundle. Place the $k(n)$ circuits side by side. A binary addition tree computes S_n , and a binary product tree computes P_n , using $O(k(n))$ additional gates. The degree of S_n is at most the largest input degree, while

$$\deg P_n \leq \sum_{i=1}^{k(n)} \deg f_{n,i}, \quad (2.13)$$

which is still polynomial because both $k(n)$ and the common degree bound are polynomial. Hence both families lie in VP.

For VNP, write

$$f_{n,i}(x) = \sum_{b \in \{0,1\}^{m_i(n)}} g_{n,i}(x, b), \quad (2.14)$$

with all bounds uniform across the bundle. Products are easiest: concatenate all witness blocks and use

$$G_n(x, b^{(1)}, \dots, b^{(k)}) = \prod_{i=1}^{k(n)} g_{n,i}(x, b^{(i)}). \quad (2.15)$$

The VP product closure already proved shows that (G_n) is a VP family, and summing over the concatenated Boolean cube gives exactly $\prod_i f_{n,i}$. The total witness length is a polynomial.

For sums, first pad every witness block to a common length

$$M(n) = \max_i m_i(n). \quad (2.16)$$

If $g_{n,i}$ originally uses only the first m_i bits, replace it by

$$\tilde{g}_{n,i}(x, y) = g_{n,i}(x, y_1, \dots, y_{m_i}) \prod_{j=m_i+1}^{M(n)} (1 - y_j). \quad (2.17)$$

On the Boolean cube, the added product forces every unused bit to zero, so

$$\sum_{y \in \{0,1\}^{M(n)}} \tilde{g}_{n,i}(x, y) = f_{n,i}(x). \quad (2.18)$$

This padding identity is characteristic-free.

Let $r(n) = \lceil \log_2 k(n) \rceil$ and assign each $i \leq k(n)$ a distinct binary code $c(i) \in \{0,1\}^{r(n)}$. Define the Boolean selector polynomial

$$\chi_i(z) = \prod_{j:c(i)_j=1} z_j \prod_{j:c(i)_j=0} (1 - z_j). \quad (2.19)$$

For Boolean z , exactly one $\chi_i(z)$ is one when $z = c(i)$ and all are zero; codewords not assigned to an index contribute zero. Therefore the VP family

$$H_n(x, z, y) = \sum_{i=1}^{k(n)} \chi_i(z) \tilde{g}_{n,i}(x, y) \quad (2.20)$$

satisfies

$$\sum_{z \in \{0,1\}^{r(n)}} \sum_{y \in \{0,1\}^{M(n)}} H_n(x, z, y) = \sum_{i=1}^{k(n)} f_{n,i}(x). \quad (2.21)$$

Its size and degree are polynomially bounded, so polynomial bundles in VNP are closed under sums as well.

It remains to clean the witness representation. Suppose $g_n \in \text{VP}$ has semantic degree at most $D(n) = n^{O(1)}$ and circuit size $s(n) = n^{O(1)}$, but its given circuit has large formal degree. Apply Theorem 1.2 with cutoff $D(n)$. Summing the extracted homogeneous components $H_0[g_n], \dots, H_{D(n)}[g_n]$ gives another circuit for g_n of size

$$O(s(n)(D(n) + 1)^2) \quad (2.22)$$

and formal degree at most $D(n)$. Thus the witness circuit may always be taken to have polynomial formal degree.

Finally, if a representation uses $m(n)$ witness bits and $M(n) \geq m(n)$ is any larger polynomially bounded length, the same zero-forcing factor as in (2.17) pads it to exactly $M(n)$ bits while leaving its Boolean sum unchanged. This proves all four assertions. $\square$

Why the selector proof is not a uniformity assumption. At each n the list of $k(n)$ selector codewords and the corresponding circuits may be hardwired nonuniformly. The proof only asserts existence of a polynomial-size circuit at each length. If a later theorem requires a uniform VP or VNP family, it must separately specify how those circuits and codewords are generated.

Why arbitrary-field closure works. The padding and selector polynomials use only the constants 0 and 1 and the identities $z \in \{0, 1\}$. They never divide by 2, count the Boolean cube as a nonzero field element, or appeal to interpolation. Consequently the proof is valid unchanged in characteristic two.

2.3 Historical normalization and the next completeness questions

Valiant's 1979 paper predates the now-standard circuit-language notation VP and VNP. It formulates the central ideas using polynomially bounded families, simple substitutions/projections, and “p-definability,” where a polynomial is obtained by summing a polynomial-size expression over Boolean substitutions. Definition 2.1 is the standard modern circuit formulation of those ideas, not a claim that the symbols VP and VNP appear verbatim in the STOC paper.

The primary source also makes the restrictive nature of projection explicit: source indeterminates may be replaced only by field constants or target indeterminates. It then identifies the permanent as a central p-definable family. Chapter 3 will upgrade these historical interfaces to the two structural results needed by the rest of Volume XI: polynomial-size determinant circuits and VNP-completeness of the permanent under p-projections. Neither completeness statement is being smuggled into the definitions or the elementary closure theorem proved here.

2.4 Reader checkpoint and exercises

Chapter checkpoint. VP and VNP are classes of *families*, not isolated polynomials. VNP uses a Boolean witness sum whose witness polynomial is itself in VP. A p-projection allows only constants and target variables, so a completeness theorem under p-projections is stronger than one under arbitrary substitutions. All closure statements in this chapter are nonuniform and characteristic-free.

Exercises.

1. Decide which substitutions are projections: $z_1 \mapsto x_2$, $z_2 \mapsto 3$, $z_3 \mapsto x_1 + x_2$, $z_4 \mapsto x_1x_2$, and $z_5 \mapsto x_1$. For each rejected case, state exactly which clause of Definition 2.1 fails.
2. Write an explicit VP witness $g(x, y)$ whose Boolean sum equals $1 + x + x^2 + x^3$. Use at most two witness bits and verify the sum term by term.
3. Reprove the padding identity used in the closure theorem when the ground field has characteristic two. Point out why no factor equal to 2 is ever divided out.
4. Let $f_{n,1}, \dots, f_{n,n^2}$ be a polynomial bundle in VNP. Write the selector construction for their sum with a witness block of length $\lceil 2 \log_2 n \rceil$ and explain what happens to unused selector codewords.
5. Give an example of a nonuniform circuit family for which the definition of VP does not itself provide an efficient algorithm that prints the n th circuit. What additional condition would a uniform variant need?
6. Explain why the exponential number of terms in a VNP witness sum does not contradict the statement that the witness polynomial belongs to VP.

Chapter 3

Determinant Circuits and Weakly-Skew Universality

The determinant is the first genuinely structural polynomial in arithmetic complexity. It has exponentially many Leibniz terms, yet it admits compact circuits without division over every field. More importantly, determinant is not merely an efficiently computable example: it is the universal polynomial for a robust restricted circuit model. Algebraic branching programs, weakly skew circuits, and polynomial-size affine determinantal representations all encode the same family-level notion of computation.

This chapter proves those facts in an order that keeps the interfaces separate. First we give a self-contained Samuelson–Berkowitz recurrence for the characteristic polynomial. Then we prove the circuit–ABP equivalence and the cycle-cover construction that turns a layered ABP into a determinant. The final section then treats the permanent separately: VNP-completeness requires a Boolean-witness summation gadget and, in Valiant’s original projection theorem, a characteristic-not-two hypothesis. Keeping that argument separate prevents determinant universality from being conflated with permanent hardness.

How to read this chapter. There are two universality stories. The determinant story is structural: weakly skew circuits, algebraic branching programs, and affine determinants are three polynomially equivalent presentations of essentially the same computation. The permanent story is a completeness story: Boolean witness sums can be compiled into cycle-cover weights, giving VNP-hardness under the very restrictive p-projections of Chapter 2.

Do not merge the two stories. Efficient computability of the determinant does not make every VP circuit a projection of a same-size determinant, and permanent hardness has the explicit characteristic-not-two boundary proved later in the chapter. The useful mental map is

$$\text{weakly skew circuit} \leftrightarrow \text{ABP} \longrightarrow \text{det}, \quad \text{VNP} \longrightarrow \text{Boolean witness sum} \longrightarrow \text{per}.$$

Learning goals. By the end, a reader should be able to trace a small ABP through the determinant construction, identify exactly where weak skewness is used, explain why cycle covers encode formula parse choices, and locate the characteristic-two failure in the permanent-hardness reduction rather than attributing it to VNP membership.

3.1 A division-free determinant recurrence

Let A be an $n \times n$ matrix whose entries are independent variables over a field F . Write

$$\chi_A(z) = \det(zI_n - A) = z^n + c_1(A)z^{n-1} + \cdots + c_n(A). \quad (3.1)$$

The coefficient $c_j(A)$ is homogeneous of total degree j in the entries of A .

We begin with the block identity that drives the recursion. Partition

$$A = \begin{pmatrix} a & r \\ c & B \end{pmatrix}, \quad (3.2)$$

where B is $m \times m$ and $m = n - 1$. If

$$\chi_B(z) = z^m + b_1z^{m-1} + \cdots + b_m, \quad (3.3)$$

put $b_0 = 1$. The adjugate identity

$$(zI_m - B) \operatorname{adj}(zI_m - B) = \chi_B(z)I_m \quad (3.4)$$

shows, by comparing coefficients of powers of z , that

$$\operatorname{adj}(zI_m - B) = \sum_{k=0}^{m-1} z^{m-1-k} Q_k, \quad Q_k = B^k + b_1B^{k-1} + \cdots + b_kI_m. \quad (3.5)$$

The block determinant identity therefore gives

$$\chi_A(z) = (z - a)\chi_B(z) - r \operatorname{adj}(zI_m - B)c. \quad (3.6)$$

Define scalars

$$q_0 = 1, \quad q_1 = -a, \quad q_{k+2} = -rB^k c \quad (0 \leq k \leq m-1). \quad (3.7)$$

If

$$\chi_A(z) = z^{m+1} + d_1z^m + \cdots + d_{m+1}, \quad (3.8)$$

then (3.5)–(3.7) imply the triangular convolution

$$\boxed{d_j = \sum_{i=0}^j q_{j-i} b_i} \quad (0 \leq j \leq m+1), \quad (3.9)$$

where $b_i = 0$ for $i > m$. Formula (3.9) is the local form of the Samuelson–Berkowitz recurrence that we will use.

Theorem 3.1 (Division-free determinant circuits). *Over every field F , the generic determinant polynomial*

$$\operatorname{Det}_n(X) = \det(X) \quad (3.10)$$

has a division-free arithmetic circuit of size $O(n^4)$ and formal degree at most n . More strongly, one circuit of size $O(n^4)$ simultaneously computes all coefficients of $\det(zI_n - X)$, with $c_j(X)$ produced by a syntactically homogeneous subcircuit of degree j .

In addition, Det_n has a polynomial-size weakly skew circuit; the explicit construction below gives the conservative bound $O(n^6)$. No field-size, characteristic, or division hypothesis is used.

Proof roadmap. The determinant circuit is built from a division-free recurrence rather than Gaussian elimination with pivots. The recurrence computes characteristic polynomial coefficients of successively enlarged principal blocks and then extracts the determinant. The two things to monitor are that every operation is polynomially bounded and that no division by a variable-dependent pivot is introduced.

Hypothesis check. Because the construction is division-free, it works over every field. The claim is an upper bound for the determinant family; universality for weakly skew circuits is a separate theorem below.

Proof. We recurse on the matrix dimension using (3.9). Suppose the coefficient vector $(b_0, \dots, b_m)$ of the trailing principal block B has already been computed. To obtain the new scalars in (3.7), compute

$$v_0 = c, \quad v_{k+1} = Bv_k \quad (0 \leq k \leq m-2), \quad (3.11)$$

and then

$$q_{k+2} = -rv_k. \quad (3.12)$$

Each matrix–vector multiplication uses $O(m^2)$ arithmetic gates, and there are $O(m)$ such multiplications. Thus all q_j cost $O(m^3)$ gates. The convolution (3.9) uses $O(m^2)$ additional multiplication/addition gates. Summing these costs over dimensions $1, 2, \dots, n$ gives

$$\sum_{m < n} O(m^3) = O(n^4). \quad (3.13)$$

The construction contains no division.

The degree bookkeeping is exact. By induction b_i is homogeneous of degree i . Every coordinate of $B^k c$ has degree $k+1$, hence q_{k+2} has degree $k+2$, while q_0 and q_1 have degrees 0 and 1. Every summand $q_{j-i}b_i$ in (3.9) therefore has degree j . Choosing balanced binary addition trees inside each fixed degree gives a syntactically homogeneous circuit for every coefficient d_j , and no gate has formal degree larger than n .

Finally,

$$c_n(X) = \det(-X) = (-1)^n \det(X), \quad (3.14)$$

so the last characteristic-polynomial coefficient yields the determinant after one scalar multiplication.

For the weakly skew refinement, observe that each scalar q_k in (3.7) is computed by a skew circuit: every multiplication in the repeated matrix–vector products multiplies a previously computed polynomial by a fresh copy of a matrix entry. In the convolution for d_j , keep the recursively constructed coefficient circuit for the b_i ’s as the reusable side. For every product $q_{j-i}b_i$, make a fresh private copy of the skew circuit computing q_{j-i} . Thus every multiplication has one private input subcircuit. At dimension $m+1$ there are $O(m^2)$ such products, and a fresh q -circuit costs at most $O(m^3)$ gates. Hence this deliberately nonoptimized simulation adds $O(m^5)$ gates at that stage and

$$\sum_{m < n} O(m^5) = O(n^6) \quad (3.15)$$

overall. The result is weakly skew and still division-free. $\square$

A 2×2 check. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the trailing block is $B = (d)$, so $(b_0, b_1) = (1, -d)$ and

$$(q_0, q_1, q_2) = (1, -a, -bc).$$

Equation (3.9) gives

$$d_1 = -d - a, \quad d_2 = ad - bc,$$

hence

$$\chi_A(z) = z^2 - (a + d)z + (ad - bc), \quad (3.16)$$

as expected.

Source checkpoint. Berkowitz's 1984 paper proves substantially sharper parallel bounds: over an arbitrary commutative ring with identity, determinant, characteristic polynomial, and adjoint admit uniform polynomial-size circuits of depth $O(\log^2 n)$. Theorem 3.1 intentionally claims only the simpler $O(n^4)$ size bound that follows from the recurrence above. The local proof therefore does not inherit Berkowitz's sharper processor count or depth bound without proving them.

3.2 From weakly skew circuits to branching programs

The weakly skew condition is exactly what permits multiplication to be modeled by concatenating path computations. To make the induction precise, it is convenient to allow a weakly skew circuit to have several designated output vertices. When an output multiplication gate is removed, its private factor subcircuit is separated from the remainder; gates in that private piece will no longer need vertices in the branching program used for the remainder.

Lemma 3.2 (Weakly skew circuit to ABP). *Let C be a weakly skew circuit with s gates over F . There is an acyclic ABP with at most $2s + 1$ edges whose edge labels are variables or field constants and whose source-sink path sum is the polynomial computed by C . After subdividing edges, it can be made layered with $O(s^2)$ graph size.*

Proof. We prove a slightly stronger induction. For a weakly skew circuit with possibly several current outputs, construct one ABP with a source s_0 and, for every output that remains reusable by the rest of the circuit, a vertex v_g whose s_0 - v_g path sum is the polynomial at g .

Take one output gate g . If g is an input, add one edge from the source to v_g with the corresponding variable or constant label.

If $g = u + v$ or $u - v$, remove g . Its children become current outputs of the smaller circuit. By induction there are vertices v_u, v_v . Add a new vertex v_g and edges

$$v_u \longrightarrow v_g \quad \text{of weight } 1, \quad v_v \longrightarrow v_g \quad \text{of weight } \pm 1. \quad (3.17)$$

The new path sum is $p_u \pm p_v$.

If $g = uv$, choose the private child guaranteed by weak skewness, say u . Removing g separates the private subcircuit C_u from the remaining circuit C_2 containing v . By induction obtain ABPs Γ_u and Γ_2 . Let s_u be the source of Γ_u and let v_v be the vertex in Γ_2 computing p_v . Identify s_u with v_v . Every path that now reaches the old u -output vertex consists uniquely of a path contributing p_v followed by a path contributing p_u , so its total path sum is $p_v p_u$. Use that vertex as v_g . The private

piece has no connection back to any other reusable gate, so this concatenation cannot corrupt the path sums already represented in Γ_2 .

Each addition/subtraction output contributes at most two new edges, while a multiplication step merely identifies one source with one existing vertex. Thus the total edge count is at most $2s + 1$.

The resulting ABP is acyclic but need not be layered. Let $d(w)$ be the length of a longest source- w path. Whenever an edge participates in a shorter path, subdivide it by enough new vertices, put the original label on one segment and label the other new segments by 1. Since an acyclic graph with $O(s)$ edges has longest path $O(s)$, at most $O(s)$ subdivisions per original edge suffice, giving $O(s^2)$ graph size. All source-sink path products are unchanged. $\square$

3.3 From branching programs back to circuits and determinants

For the reverse circuit simulation, the affine labels in our ABP definition are useful but must be charged explicitly. If

$$\ell(x) = a_0 + a_1x_1 + \cdots + a_rx_r, \quad (3.18)$$

a fresh formula for ℓ uses $O(r + 1)$ gates. We make a private copy of this formula every time an edge label is multiplied into a path-sum polynomial.

Lemma 3.3 (Layered ABP to weakly skew circuit). *Let Γ be a layered ABP of graph size M over variables $x_1, \dots, x_r$, with affine-linear edge labels. Then Γ has an equivalent weakly skew circuit of size $O(M(r + 1))$. If every edge label is already a variable or a field constant, the size is $O(M)$.*

Proof. For a vertex v , let P_v denote the sum of the weights of all source- v paths. Set $P_s = 1$. Layer by layer,

$$P_v = \sum_{u \rightarrow v} P_u \ell_{uv}(x). \quad (3.19)$$

For every incoming edge make a fresh private formula computing ℓ_{uv} . The multiplication $P_u \ell_{uv}$ is therefore weakly skew with the affine-label formula as its private factor. Sum the incoming contributions with a binary addition tree. Previously computed P_u gates may be shared freely. The stated size bounds follow by summing over all edges. $\square$

The determinant representation is even more direct. Because a layered ABP has all source-sink paths of one length, all relevant cycle covers have one common sign.

Lemma 3.4 (Layered ABP as one affine determinant). *Let Γ be a layered ABP with N vertices and source-sink path length L , computing f . There is an $N \times N$ matrix $M_\Gamma(x)$ whose entries are affine linear forms such that*

$$\det M_\Gamma(x) = f(x). \quad (3.20)$$

If the ABP has no parallel edges and its labels are variables or field constants, then f is a projection of Det_N . In general, subdividing parallel edges gives a variable/constant-labelled determinant of dimension $N + O(M)$, where M is the ABP graph size.

Proof. Start with the weighted adjacency matrix of Γ . Add an edge

$$t \longrightarrow s \quad (3.21)$$

of weight $(-1)^L$ and put a self-loop of weight 1 at every vertex other than s and t . Call the weighted adjacency matrix of the resulting graph M_Γ .

Any nonzero cycle cover must use the edge $t \rightarrow s$, because s has no other incoming edge. The same cycle must then follow a directed source–sink path of Γ ; every vertex outside that path must use its self-loop because the original ABP is acyclic. Conversely every source–sink path gives exactly one such cycle cover. If a path has L edges, its nontrivial cycle has length $L + 1$, so the permutation sign contributed by that cycle is $(-1)^L$. The back edge supplies another factor $(-1)^L$. Hence every cycle cover contributes exactly the weight of its corresponding ABP path, and summing them proves (3.20).

When every original edge label is a variable or a constant and there are no parallel edges, the matrix entries already lie in $F \cup \{x_1, \dots, x_r\}$. If parallel edges are present, subdivide each but one of them through a fresh vertex, putting the old label on the first segment and 1 on the second. The path sum is unchanged, the graph remains acyclic before the back edge is added, and only $O(M)$ vertices are introduced. The resulting determinant is therefore a projection of the generic determinant of polynomially related dimension. $\square$

Theorem 3.5 (ABPs, weakly skew circuits, and determinant universality). *Fix a field F . For a p -family (f_n) , the following conditions are equivalent up to polynomial overhead.*

1. (f_n) is computed by polynomial-size weakly skew arithmetic circuits.
2. (f_n) is computed by polynomial-size layered ABPs with affine-linear edge labels.
3. There are polynomially bounded dimensions $N(n)$ and affine-linear matrices $A_n(x)$ of size $N(n) \times N(n)$ such that

$$f_n(x) = \det A_n(x). \quad (3.22)$$

Moreover, after polynomially expanding every affine ABP edge into a variable/constant-labelled gadget, every such family is a p -projection of the generic determinant family. Conversely, p -projections of determinant have polynomial-size weakly skew circuits.

Proof roadmap. Prove the equivalence by explicit translations. A weakly skew multiplication has a private input subcomputation, which lets an ABP splice that factor without duplicating the whole shared DAG. Conversely, dynamic programming over a layered ABP computes path sums by a weakly skew circuit. Finally, add one feedback edge from sink to source and use the determinant expansion to isolate the path contributions.

Hypothesis check. Weak skewness is essential only in the circuit-to-ABP direction. The ABP-to-determinant construction uses affine matrix entries and yields a determinant of polynomially larger dimension; it is not claiming that arbitrary circuits are weakly skew.

Proof. Lemma 3.2 gives $(1) \Rightarrow (2)$ and Lemma 3.3 gives $(2) \Rightarrow (1)$. Since a p -family has only polynomially many variables, the factor $r + 1$ in the affine-label simulation remains polynomial.

Lemma 3.4 gives $(2) \Rightarrow (3)$ directly. For the p -projection refinement, replace every affine edge

$$u \longrightarrow v \quad \text{labelled} \quad a_0 + \sum_{j=1}^r a_j x_j \quad (3.23)$$

by parallel two-edge paths through fresh intermediate vertices. One path has labels $a_0, 1$, and for every nonzero a_j one path has labels a_j, x_j . Doing this for every edge doubles the layer length and increases graph size by at most a factor $O(r + 1)$. All new edge labels are variables or field constants, and the source–sink path sum is unchanged. Applying Lemma 3.4 now gives a determinant

whose generic matrix variables are replaced only by target variables or field constants: precisely a p-projection.

For (3) $\Rightarrow$ (1), run the weakly skew recurrence from the proof of Theorem 3.1 directly on the affine matrix $A_n(x)$. Every place where that construction uses a matrix entry inside a private q -factory receives a fresh private formula for the corresponding affine form. The reusable characteristic-coefficient side is left shared. Since there are only polynomially many variables and matrix positions, the overhead stays polynomial and weak skewness is preserved. The p-projection case is simpler still, because the substituted entries are only variables or field constants. $\square$

A three-layer example. Suppose an ABP has source s , intermediate vertices u, v , and sink t , with edge labels

$$s \rightarrow u : a, \quad s \rightarrow v : b, \quad u \rightarrow t : c, \quad v \rightarrow t : d.$$

It computes

$$f = ac + bd. \tag{3.24}$$

Order the vertices (s, u, v, t) . Since every source–sink path has length two, the back edge $t \rightarrow s$ has weight 1. Adding loops at u, v gives

$$M = \begin{pmatrix} 0 & a & b & 0 \\ 0 & 1 & 0 & c \\ 0 & 0 & 1 & d \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{3.25}$$

The only nonzero cycle covers are

$$s \rightarrow u \rightarrow t \rightarrow s \quad \text{with the loop at } v,$$

and

$$s \rightarrow v \rightarrow t \rightarrow s \quad \text{with the loop at } u,$$

so

$$\det M = ac + bd. \tag{3.26}$$

This example is the full determinant-universality proof in miniature.

Historical normalization. Toda’s 1992 arithmetic-circuit work identifies determinant with polynomial-size skew and related weakly-skew models. Malod–Portier later gave a particularly clean circuit-hierarchy treatment and the weakly-skew/ABP equivalence. Our ABP convention is slightly more permissive at the edge level because it allows affine linear forms; equation (3.23) shows explicitly that this changes only a polynomial factor and does not alter the p-projection class.

3.4 Permanent completeness under p-projections

The permanent family is

$$\text{Perm}_n(X) = \sum_{\sigma \in S_n} \prod_{i=1}^n x_{i, \sigma(i)}. \tag{3.27}$$

There are two logically different parts to its completeness theorem. Membership in VNP is characteristic-free. Hardness is where the Boolean witness sum must be compiled into cycle covers, and the final couple-removal gadget uses 2^{-1} .

Lemma 3.6 (Permanent membership in VNP). *Over every field F , the permanent family belongs to VNP_F .*

Proof. Use Boolean variables b_{ij} , one for each position of an $n \times n$ matrix. For row i and column j , put

$$R_i(b) = \sum_{k=1}^n b_{ik} \prod_{\ell \neq k} (1 - b_{i\ell}), \quad C_j(b) = \sum_{k=1}^n b_{kj} \prod_{\ell \neq k} (1 - b_{\ell j}). \quad (3.28)$$

On the Boolean cube, R_i is one exactly when row i contains one selected entry, and C_j has the analogous column meaning. Hence

$$I(b) = \prod_{i=1}^n R_i(b) \prod_{j=1}^n C_j(b) \quad (3.29)$$

is the indicator of permutation matrices. Give every selected entry its matrix weight by

$$W(X, b) = \prod_{i,j=1}^n ((1 - b_{ij}) + b_{ij}x_{ij}). \quad (3.30)$$

Then

$$\text{Perm}_n(X) = \sum_{b \in \{0,1\}^{n^2}} I(b)W(X, b). \quad (3.31)$$

The witness polynomial has polynomial size and polynomial degree. Equations (3.28)–(3.31) use only 0, 1, +, −, ×, so the proof is valid in every characteristic. $\square$

The hardness direction starts from the modern Definition 2.1, whose VNP witness is a VP circuit rather than a formula. The next lemma is the missing normalization bridge.

Lemma 3.7 (Circuit witnesses admit a formula-sum normal form). *Let C be a division-free fan-in-two arithmetic circuit of size s and formal degree at most D , computing $p_C(x)$. There are $M = \text{poly}(s, D)$ Boolean variables z and an arithmetic formula $\Phi_C(x, z)$ of size $\text{poly}(s, D)$ such that*

$$p_C(x) = \sum_{z \in \{0,1\}^M} \Phi_C(x, z). \quad (3.32)$$

The construction works over every field.

Proof. We first replace C by a polynomial-size *multiplicatively disjoint* circuit, meaning that the two input subcircuits of every multiplication gate are vertex-disjoint. Apply Theorem 1.2 through degree D and, for this proof only, copy the subcircuit for each homogeneous output so that the $D+1$ components do not share gates. This costs at most $O(s(D+1)^3)$ gates. Degree-zero subcircuits may be evaluated to field constants.

Consider one positive-degree syntactically homogeneous component of degree d . For every gate v of positive syntactic degree $\delta(v)$ and every contiguous interval $I \subseteq [d]$ of length $\delta(v)$, create a copy v^I . At an addition or subtraction gate, both children inherit the same interval. At a multiplication $v = uw$ with both child degrees positive, split I into consecutive intervals

$$I = I_u \dot{\cup} I_w, \quad |I_u| = \delta(u), \quad |I_w| = \delta(w), \quad (3.33)$$

and connect v^I to u^{I_u} and w^{I_w} . If one child has degree zero, use a fresh constant leaf for that child. Descendants of the two children of a multiplication gate carry subintervals of the disjoint sets I_u

and I_w ; therefore they cannot share a copied vertex. At most $d \leq D$ interval copies are made per gate. Summing the separately copied homogeneous outputs produces a multiplicatively disjoint circuit C^* of a conservative size

$$S = O(s(D+1)^4) \quad (3.34)$$

computing p_C .

We now sum the parse trees of C^* by a formula over auxiliary Boolean bits. Trim gates that do not reach the output o . Give each gate v an activity bit a_v , and each parent-to-child wire e a selection bit z_e . For Boolean variables $t_1, \dots, t_k$, define

$$Z(t) = \prod_{i=1}^k (1 - t_i), \quad O(t) = \sum_{i=1}^k t_i \prod_{j \neq i} (1 - t_j). \quad (3.35)$$

Thus Z indicates “exactly zero” and O indicates “exactly one.” For every nonoutput gate u , let $E^+(u)$ be the selected-wire bits on wires from parents of u down to u , and impose

$$P_u = (1 - a_u)Z(E^+(u)) + a_uO(E^+(u)). \quad (3.36)$$

This says that an inactive gate has no selected parent and an active nonoutput gate has exactly one selected parent.

If v is an addition or subtraction gate with child-wire bits r_v, s_v , put

$$A_v = (1 - a_v)(1 - r_v)(1 - s_v) + a_v(r_v(1 - s_v) + (1 - r_v)s_v), \quad (3.37)$$

while at a multiplication gate put

$$M_v = (1 - a_v)(1 - r_v)(1 - s_v) + a_v r_v s_v. \quad (3.38)$$

Together with a_o , the product of all factors (3.36)–(3.38) is a Boolean indicator $J_C(a, z)$. It is one precisely when the selected wires form a legal parse tree of C^* . Multiplicative disjointness is exactly what prevents two branches created by a multiplication from reconverging at one circuit gate; if they did, that gate would lie below both multiplication children. Consequently each legal parse tree has one unique Boolean encoding.

For an input gate v labelled by a variable or field constant λ_v , use

$$L_v = (1 - a_v) + a_v \lambda_v. \quad (3.39)$$

At a subtraction gate let s_v be the bit selecting its right child and use the sign factor $1 - 2s_v$. Define

$$\Phi_C(x, a, z) = J_C(a, z) \prod_{v \text{ input}} L_v \prod_{v \text{ subtraction}} (1 - 2s_v). \quad (3.40)$$

On an invalid Boolean assignment this formula is zero. On a valid one it is exactly the monomial, including signs, contributed by the corresponding parse tree. Induction on circuit gates gives the familiar expansion of a circuit polynomial as the sum of its parse-tree weights, proving (3.32).

There are $O(S)$ activity and wire bits. Formula (3.35) has quadratic size in the number of arguments, and the sum of all circuit fan-outs is $O(S)$, so (3.36)–(3.40) have polynomial size. In characteristic two the factor $1 - 2s_v$ becomes 1, exactly as subtraction becomes addition; no division or field-size hypothesis was used. $\square$

Corollary 3.8 (Modern VNP has a formula witness sum). *If $(f_n) \in \text{VNP}_F$, then for polynomially many Boolean variables w there is a polynomial-size formula family $\Phi_n(x, w)$ satisfying*

$$f_n(x) = \sum_{w \in \{0,1\}^{q(n)}} \Phi_n(x, w). \quad (3.41)$$

Proof. By Theorem 2.2, choose the VP witness circuits with polynomial formal degree. Apply Lemma 3.7 to each witness circuit and concatenate its new parse-tree bits with the original VNP witness bits. Interchanging the two finite Boolean sums gives (3.41). $\square$

We next compile a formula into a cycle-cover sum without using determinant signs.

Lemma 3.9 (A formula as an ordinary permanent). *Let $\Phi(x, w)$ be an arithmetic formula of polynomial size. There is a polynomial-size square matrix $A(x, w)$ whose entries lie in $F \cup \{x_i, w_j\}$ and such that*

$$\text{per } A(x, w) = \Phi(x, w). \quad (3.42)$$

Proof. A formula is weakly skew. The construction in Theorem 3.5 therefore gives a polynomial-size layered ABP; expand affine labels into constant/variable-labelled two-edge paths exactly as in the p-projection part of that theorem. Let s, t be its source and sink. Add a back edge $t \rightarrow s$ of weight 1 and self-loops of weight 1 at every other vertex. Because the ABP is acyclic, every nonzero cycle cover consists of one source–sink path, the back edge, and loops on all unused vertices. The permanent has no permutation sign, so the weight of that cover is exactly the ABP path weight. Summing covers gives (3.42). Parallel labelled edges may be subdivided through fresh vertices before taking the adjacency matrix, so each matrix entry is literally one variable or one field constant. $\square$

To sum a witness variable that occurs in several matrix positions, we temporarily use a constrained version of the permanent.

Definition 3.10 (Coupled permanent). Let A be a square matrix, viewed as a complete directed weighted graph. A *couple* is an unordered pair $\{e, e'\}$ of directed edges with distinct tails and distinct heads. A set Q of couples is required to use no directed edge in more than one couple. For such a set Q , define

$$\text{per}(A; Q) = \sum_{\sigma \in S_N: e \in \sigma \iff e' \in \sigma \text{ for every } \{e, e'\} \in Q} \prod_{i=1}^N A_{i, \sigma(i)}. \quad (3.43)$$

Thus an admissible cycle cover must use both edges of every couple or neither. For $Q = \emptyset$ this is the ordinary permanent.

Lemma 3.11 (Boolean selector for repeated witness occurrences). *Suppose y occurs in exactly the matrix positions $e_1, \dots, e_m$ of A , and none of those edges belongs to an existing couple in Q . There is a matrix $C = \gamma(A, y)$, obtained by adding $2m + 2$ vertices, and a set P of m new couples such that*

$$\text{per}(C; Q \cup P) = \text{per}(A; Q)|_{y=0} + y \text{per}(A; Q)|_{y=1}. \quad (3.44)$$

Every occurrence of y in the old matrix is replaced by 1, while C has exactly one new occurrence of y .

Proof. Adjoin vertices

$$u_0, u_1, \dots, u_{m+1}, v_1, \dots, v_m. \quad (3.45)$$

Give every new vertex a self-loop of weight 1. Add $u_0 \rightarrow u_1$ of weight y and $u_{m+1} \rightarrow u_0$ of weight 1. For each $1 \leq k \leq m$, add the direct edge

$$d_k : u_k \longrightarrow u_{k+1} \quad (3.46)$$

of weight 1 and the two-edge detour

$$u_k \longrightarrow v_k \longrightarrow u_{k+1} \quad (3.47)$$

with both weights 1. There are no other edges between the old graph and this selector component. Replace the weight y on every old occurrence edge e_k by 1, and set

$$P = \{\{d_k, e_k\} : 1 \leq k \leq m\}. \quad (3.48)$$

(The case $m = 0$ uses only u_0, u_1 and the two opposite edges just described.)

The selector component has exactly two types of cycle cover. First, every new vertex may take its self-loop. Second, there is one large cycle

$$u_0 \rightarrow u_1 \rightsquigarrow u_2 \rightsquigarrow \dots \rightsquigarrow u_{m+1} \rightarrow u_0, \quad (3.49)$$

where at stage k one independently chooses the direct edge d_k or the detour through v_k ; an unused v_k takes its self-loop. The all-loop cover has weight 1, while every large-cycle cover has weight y . Acyclicity of all edges in (3.46)–(3.47) except the single closing edge proves that there are no other covers.

Now fix an old cycle cover τ of A satisfying Q , and let $M(\tau) \subseteq [m]$ be the indices for which τ uses e_k . Coupling forces the selector to use d_k exactly for $k \in M(\tau)$. If $M(\tau) \neq \emptyset$, there is exactly one compatible selector cover: the large cycle choosing direct edges precisely on $M(\tau)$, and its extra weight is y . If $M(\tau) = \emptyset$, there are two compatible selector covers: the all-loop cover of weight 1 and the all-detour large cycle of weight y . Because all old y -weights were replaced by 1, summing first over covers with $M(\tau) = \emptyset$ and then over all old covers gives exactly the right side of (3.44). $\square$

The one-occurrence selector in miniature. When $m = 1$, the selector has an all-loop cover, a large cycle using d_1 , and a large cycle using the detour through v_1 . If the old permanent is written as $q_0 + yq_1$, coupling d_1 to the unique y -edge leaves contributions $q_0 + yq_0 + yq_1 = q(0) + yq(1)$. Setting the sole new y -edge to 1 therefore performs the Boolean sum $q(0) + q(1)$.

The last local gadget removes a couple entirely. This is the only point at which characteristic different from two enters.

Lemma 3.12 (Three-vertex couple remover). *Assume $\text{char } F \neq 2$. Let Q be a set of couples in a matrix G and let $\{e, e'\} \notin Q$ be one additional couple, with $e = (u, v)$ and $e' = (u', v')$, where $u \neq u'$ and $v \neq v'$. One can add three vertices and obtain a matrix H such that*

$$\text{per}(H; Q) = \text{per}(G; Q \cup \{\{e, e'\}\}). \quad (3.50)$$

All new entries are field constants.

Proof. The coupler matrix on new vertices a, b, c is

$$K = \begin{pmatrix} -1 & 1 & \frac{1}{2} \\ 1 & 1 & -\frac{1}{2} \\ 1 & 1 & -\frac{1}{2} \end{pmatrix}, \quad \text{rows and columns ordered } a, b, c. \quad (3.51)$$

Direct expansion gives the transfer signature

$$\begin{aligned} \text{per } K &= 1, & \text{per } K(b, c \mid a, c) &= 1, \\ \text{per } K(b \mid a) &= 0, & \text{per } K(b \mid c) &= 0, \\ \text{per } K(c \mid a) &= 0, & \text{per } K(c \mid c) &= 0, \end{aligned} \quad (3.52)$$

where $K(U \mid V)$ means deletion of rows U and columns V .

Construct H from G by setting the two old edge weights $H(u, v) = H(u', v') = 0$, adjoining K , and adding four unit edges

$$u \rightarrow a, \quad u' \rightarrow c, \quad b \rightarrow v, \quad c \rightarrow v'. \quad (3.53)$$

All other cross-entries between G and K are zero.

A cycle cover uses equally many edges from the old vertices into K as from K back to the old vertices. Hence the four external edges in (3.53) can occur only in six balanced patterns. If none occur, the internal completion factor is $\text{per } K = 1$ and the corresponding old cover uses neither e nor e' . If all four occur, the remaining internal entry is $K(b, c \mid a, c)$, whose permanent is 1; this reproduces an old cover using both e and e' . The four remaining balanced patterns have internal factors

$$\text{per } K(b \mid a), \quad \text{per } K(b \mid c), \quad \text{per } K(c \mid a), \quad \text{per } K(c \mid c), \quad (3.54)$$

all zero by (3.52). Thus exactly the old covers using both coupled edges or neither survive, with unchanged weight, which is (3.50). The constants $\pm \frac{1}{2}$ exist precisely because 2 is invertible in F . $\square$

Theorem 3.13 (Valiant completeness of the permanent). *Let F be a fixed field.*

1. *The permanent family belongs to VNP_F in every characteristic.*
2. *If $\text{char } F \neq 2$, then the permanent family is VNP_F -complete under p -projections: for every $(f_n) \in \text{VNP}_F$ there is a polynomially bounded index map $N = t(n)$ such that f_n is a projection of Perm_N .*

Proof roadmap. Membership is the easy half: a permutation is a polynomial-size Boolean witness and its selected matrix-entry product is in VP. Hardness is longer. Compile a VP witness circuit into a formula-sum normal form, encode formula evaluation by cycle covers, eliminate repeated witness occurrences with Boolean selectors, and finally remove the resulting equality couples by a constant-size gadget. The last gadget is exactly where 2^{-1} is used.

Hypothesis check. Permanent membership in VNP is characteristic-free. The p -projection hardness proved here assumes $\text{char } F \neq 2$. In characteristic two the permanent equals the determinant, so silently dropping the hypothesis would turn the theorem into an unproved VP-versus-VNP collapse statement.

Proof. Part 1 is Lemma 3.6. For hardness, fix $(f_n) \in \text{VNP}_F$. Theorem 2.2 gives a witness family whose circuits have polynomial size and polynomial formal degree. Corollary 3.8 therefore gives

$$f_n(x) = \sum_{w \in \{0,1\}^{q(n)}} \Phi_n(x, w) \quad (3.55)$$

with $q(n)$ and the formula size polynomially bounded. By Lemma 3.9,

$$\Phi_n(x, w) = \text{per } A_n(x, w), \quad (3.56)$$

where A_n has polynomial dimension and every entry is a target variable, a witness variable, or a field constant.

Process $w_1, \dots, w_{q(n)}$ in order with Lemma 3.11. After processing w_i , substitute the single new marker occurrence of w_i by the constant 1. Equation (3.44) then replaces the current polynomial by its Boolean sum over w_i while recording equality constraints as couples. The selector does not introduce new occurrences of any unprocessed witness variable. Since the initial matrix has polynomially many positions, the total number of witness occurrences, added vertices, and couples remains polynomial. At the end we have a polynomial-size matrix $B_n(x)$ and a polynomial-size set Q_n of couples such that

$$f_n(x) = \text{per}(B_n(x); Q_n), \quad (3.57)$$

and all matrix entries lie in $F \cup \{x_1, \dots, x_{v(n)}\}$.

Assume now $\text{char } F \neq 2$. Remove the couples of Q_n one at a time with Lemma 3.12. Each step adds only three vertices and field constants and preserves the constrained permanent. We finish with a polynomial-size matrix $D_n(x)$ satisfying

$$f_n(x) = \text{per } D_n(x), \quad D_n(x)_{ij} \in F \cup \{x_1, \dots, x_{v(n)}\}. \quad (3.58)$$

Replacing the variables of the generic permanent matrix by these entries is exactly a projection in the sense of Definition 2.1. The matrix dimension is polynomially bounded in n , so (3.58) is a p-projection reduction. $\square$

Characteristic-two boundary. The membership proof and the circuit-to-formula normalization are characteristic-free. The hardness construction above stops exactly at Lemma 3.12, because its signature uses 2^{-1} . In characteristic two the permutation sign disappears, so $\text{Perm}_n = \text{Det}_n$; by Theorem 3.1 this family is already in VP. Thus a characteristic-two VNP-completeness statement under p-projections would imply $\text{VP}_F = \text{VNP}_F$. No such collapse is asserted here.

Historical normalization. Valiant's original completeness theorem is formulated for polynomial-size expressions defining Boolean-cube sums and proves permanent hardness over fields of characteristic different from two. The proof above makes the modern interface explicit rather than identifying that historical formula model with Definition 2.1 by fiat: Lemma 3.7 compiles a VP circuit witness into a formula plus extra Boolean parse-tree bits; Lemmas 3.11 and 3.12 then give a fully local witness-summing/cycle-cover reconstruction.

3.5 Reader checkpoint and exercises

Chapter checkpoint. The determinant is simultaneously easy and universal for a restricted circuit model: it has polynomial-size division-free circuits, and affine determinants capture ABPs and weakly skew circuits with polynomial overhead. The permanent plays a different role: over characteristic different from two it is complete for VNP under p-projections. The cycle-cover gadgets are therefore not a proof that determinant and permanent have comparable complexity.

Exercises.

1. Build the layered ABP for $(x + y)z + x$ and compute the source-to-sink path sum directly. Then follow the ABP-to-determinant construction and identify the cycle covers corresponding to the two summands.
2. In the weakly-skew-circuit-to-ABP proof, mark the exact edge where a private multiplication input is spliced into the path computation. Explain what may go wrong if both multiplication inputs are shared elsewhere.
3. For a generic 3×3 matrix, verify directly that the determinant and permanent differ only by permutation signs. Then specialize to characteristic two and explain why the two polynomials coincide.
4. Show that the permanent family is in VNP by writing the Boolean witness conditions for a permutation matrix: one selected entry in every row and every column. Your witness polynomial should vanish on Boolean strings that do not encode permutations.
5. Give a substitution that would prove hardness under affine projections but is *not* a p-projection. Explain why the permanent theorem must avoid using such a substitution.
6. Draw a dependency diagram with separate arrows for “efficient determinant circuit,” “weakly skew universality,” and “permanent VNP-completeness.” Annotate every arrow with its characteristic assumption.

Chapter 4

The Chasm at Depth Four

A general arithmetic circuit is a directed acyclic graph with potentially many alternating additions and multiplications. Lower bounds for that model are hard partly because a proof must understand an arbitrary pattern of reuse and depth. The depth-four theorem says that, after paying a controlled subexponential size cost, the depth can be collapsed to just four alternating layers. The theorem is therefore not merely a circuit simplification: it turns lower bounds for a rigid normal form into a possible route toward lower bounds for unrestricted circuits.

This chapter proves the reduction locally. Two interfaces are kept explicit. First, Theorem 1.2 extracts homogeneous components over an arbitrary field, so no interpolation or characteristic assumption is hidden. Second, the balancing step is reconstructed with gate quotients and degree frontiers. Once every multiplication decomposes a degree- k polynomial into a sum of products of factors of degree at most $k/2$, the actual collapse to depth four is a short potential argument.

How to read this chapter. Depth reduction is an *upper-bound normal-form theorem*. It does not prove a lower bound by itself. Its value is that any sufficiently strong lower bound against the resulting depth-four class would propagate back to general circuits. The proof has three conceptual stages: homogenize, balance by cutting at a degree frontier, and then expand only the low-degree pieces.

The free parameter t is the bookkeeper. Small t makes bottom multiplication simple but creates more factors at the upper product layer; large t does the opposite. The tradeoff

$$S^{O(t+d/t)}$$

is therefore more informative than the special choice $t \asymp \sqrt{d}$. Chapter 5 will choose a lower-bound measure tailored to the bounded-bottom normal form produced here.

Learning goals. A reader should be able to distinguish ordinary depth from the flattened $\Sigma\Pi\Sigma\Pi$ convention, explain the roles of bottom and top-product fan-in, derive the $t + d/t$ tradeoff from the balanced decomposition, and say why the local proof is characteristic-free.

4.1 The depth-four normal form

For this chapter only, it is convenient to use the standard flattened notation for a layered arithmetic circuit. A

$$\Sigma\Pi\Sigma\Pi$$

circuit has a top sum layer, a product layer, another sum layer, and a bottom product layer, followed by variables and field constants. The fan-ins in this notation need not be two. We count gates and nonzero wires; replacing any bounded-fan-in product used during the proof by a constant-size binary tree is therefore harmless. The phrase *depth four* always refers to the four alternating internal layers in the flattened normal form, not to such a binary implementation.

If every bottom product gate has fan-in at most t , write $\Sigma\Pi\Sigma\Pi^{[t]}$. The *top fan-in* is the number of children of the top Σ gate. The *top-product fan-in* is the largest fan-in of a product gate in the second layer. A depth-four circuit is *syntactically homogeneous* if every product term below the top sum has the same total degree and every intermediate sum combines terms of one degree.

The parameter

$$S := \max\{2, s, n, d\} \quad (4.1)$$

will absorb the harmless polynomial overhead of homogenization and balancing. All constants hidden in $O(\cdot)$ below are absolute and independent of the field.

Theorem 4.1 (Depth reduction to depth four). *Let F be an arbitrary field, let $f \in F[x_1, \dots, x_n]$ have total degree at most $d \geq 1$, and suppose that a division-free arithmetic circuit of size s computes f . Put $S = \max\{2, s, n, d\}$. For every integer t with $1 \leq t \leq d$, there is a $\Sigma\Pi\Sigma\Pi^{[t]}$ circuit for f such that*

1. *its top fan-in is $S^{O(d/t)}$;*
2. *its top-product fan-in is at most*

$$5 + 4 \left\lceil \frac{8d}{t} \right\rceil = O(d/t); \quad (4.2)$$

and

3. *its total depth-four size is*

$$S^{O(t+d/t)}. \quad (4.3)$$

If f is homogeneous of degree d , the resulting depth-four circuit can be chosen syntactically homogeneous. For general f , each top summand belongs to one homogeneous component, so the same bounds hold without asserting global homogeneity.

In particular, with $t = \lceil \sqrt{d} \rceil$,

$$\text{size}_{\Sigma\Pi\Sigma\Pi}(f) \leq S^{O(\sqrt{d})}, \quad (4.4)$$

and both product layers have fan-in $O(\sqrt{d})$.

Proof roadmap. First use Chapter 1 to work with homogeneous pieces. On a homogeneous circuit, introduce gate quotients and repeatedly cut a computation at the first frontier where degree drops by a constant factor. This produces a balanced circuit in which long source-to-output chains cannot persist. For a chosen threshold t , expand only factors of degree at most t into monomials and flatten the remaining sums/products. Counting frontier choices gives the $S^{O(t+d/t)}$ size bound.

Parameter check. Do not set $t = \sqrt{d}$ too early. The theorem exports the full tradeoff because later lower bounds may apply only to a particular bottom-degree range. The special choice $t = \lceil \sqrt{d} \rceil$ is merely the balanced corollary.

The rest of the chapter proves the theorem. We start with the one piece of structure that makes the collapse possible.

4.2 Gate quotients and degree frontiers

Let C be a syntactically homogeneous circuit. Write $[u]$ for the polynomial computed at a gate u and $\delta(u)$ for its syntactic degree. At every multiplication gate $u = u_L u_R$, interchange the two children if necessary so that

$$\delta(u_L) \leq \delta(u_R). \quad (4.5)$$

This choice will be called *right-heavy*.

For two gates u, v , define the *gate quotient* $[u : v]$ recursively. If $u = v$, set $[u : v] = 1$. If u is a source different from v , set $[u : v] = 0$. For addition and subtraction gates put

$$[u_1 \pm u_2 : v] = [u_1 : v] \pm [u_2 : v], \quad (4.6)$$

and for a right-heavy product gate put

$$[u_L u_R : v] = [u_L] [u_R : v]. \quad (4.7)$$

Thus the quotient follows the heavier child of a product. Whenever the recursive expression is not the syntactic zero, it is homogeneous of degree

$$\delta([u : v]) = \delta(u) - \delta(v). \quad (4.8)$$

The notation is purely algebraic: no division is being performed.

For a real threshold $m > 0$, define the multiplication frontier

$$\mathcal{F}_m = \{w = w_L w_R : \delta(w) > m, \delta(w_L) \leq m, \delta(w_R) \leq m\}. \quad (4.9)$$

Only gates reachable from the gate under discussion can contribute, because all other gate quotients vanish.

Lemma 4.2 (Frontier identities). *Let C be right-heavy and syntactically homogeneous.*

1. *If $\delta(u) > m \geq 1$, then*

$$[u] = \sum_{w \in \mathcal{F}_m} [u : w] [w]. \quad (4.10)$$

2. *If $\delta(u) > m > \delta(v)$ and $m \geq 1$, then*

$$[u : v] = \sum_{w \in \mathcal{F}_m} [u : w] [w : v]. \quad (4.11)$$

Zero summands may of course be deleted.

Proof. Both identities follow by the same structural induction. We spell it out because the choice of the heavy child is what prevents a hidden depth argument.

For (4.10), an addition/subtraction gate passes the claim to its two children via (4.6). Suppose $u = u_L u_R$. If $u \in \mathcal{F}_m$, the term $w = u$ contributes $[u : u][u] = [u]$, and no deeper frontier gate of threshold m can occur below a child of degree at most m . If $u \notin \mathcal{F}_m$, then, because $\delta(u) > m$, at least one child has degree greater than m ; right-heaviness forces $\delta(u_R) > m$. The induction hypothesis at u_R gives

$$[u] = [u_L][u_R] = \sum_{w \in \mathcal{F}_m} [u_L][u_R : w][w] = \sum_{w \in \mathcal{F}_m} [u : w][w]. \quad (4.12)$$

A nonzero source has syntactic degree at most 1, so under $\delta(u) > m \geq 1$ there is no nonzero source base case; a tagged zero satisfies the identity trivially.

For (4.11), $u = v$ is excluded by $m > \delta(v)$. Addition/subtraction again follows from (4.6). At a multiplication gate, if $u \in \mathcal{F}_m$, the frontier term $w = u$ is exactly $[u : v]$. Otherwise, because $\delta(u) > m$, right-heaviness forces $\delta(u_R) > m$, so induction and (4.7) give

$$[u : v] = [u_L][u_R : v] = \sum_{w \in \mathcal{F}_m} [u_L][u_R : w][w : v] = \sum_{w \in \mathcal{F}_m} [u : w][w : v]. \quad (4.13)$$

This closes the induction. $\square$

4.3 A degree-halving balanced circuit

The frontier identities can now be turned into a circuit whose product factors all lose a constant fraction of the parent degree.

Lemma 4.3 (Balanced gate-quotient decomposition). *Let C be a syntactically homogeneous fan-in-two circuit of size s_0 over F . There is a circuit of size $s_0^{O(1)}$ containing the output polynomial of C and the needed gate quotients, with the following flattened description. Every positive-degree gate-label g of degree $k > 1$ used in the construction has an identity*

$$g = \sum_{i=1}^{\sigma} c_i \prod_{j=1}^{r_i} g_{ij}, \quad r_i \leq 5, \quad (4.14)$$

where $\sigma \leq s_0^{O(1)}$, $c_i \in F$, every nonzero product term is homogeneous of degree k , and every positive-degree factor satisfies

$$\deg g_{ij} \leq \frac{k}{2}, \quad \sum_{j=1}^{r_i} \deg g_{ij} = k. \quad (4.15)$$

Degree-zero factors are absorbed into c_i . Degree-0 and degree-1 labels are base gates. The construction uses no division and is valid in every characteristic.

Proof. There are only s_0 gate values $[u]$ and at most s_0^2 ordered gate quotients $[u : v]$, so it is enough to give a polynomial-size expansion for each label. We process degree levels increasingly; at a fixed level k , gate-value expansions are constructed before quotient expansions of degree difference k . The strict frontier drop below guarantees that quotient factors created in a gate-value expansion have smaller degree, while a quotient expansion can refer to a same-degree gate value whose expansion has already been constructed.

First take a gate value $[u]$ of degree $k > 1$ and set $m = k/2$. Since $k \geq 2$, this threshold satisfies $m \geq 1$. By Lemma 4.2,

$$[u] = \sum_{w \in \mathcal{F}_{k/2}} [u : w][w_L][w_R]. \quad (4.16)$$

For every nonzero term,

$$\deg[u : w] = k - \delta(w) < k/2, \quad \delta(w_L), \delta(w_R) \leq k/2. \quad (4.17)$$

Thus a gate value needs at most three degree-halved factors per product term.

Now take a quotient $[u : v]$ of degree difference

$$k := \delta(u) - \delta(v) > 1 \quad (4.18)$$

and put

$$m := \frac{\delta(u) + \delta(v)}{2}. \quad (4.19)$$

Here $m \geq k/2 \geq 1$. The quotient frontier identity and the product recurrence at each frontier gate give

$$[u : v] = \sum_{w \in \mathcal{F}_m} [u : w] [w_L] [w_R : v]. \quad (4.20)$$

The first and third factors have degree at most $k/2$. Indeed,

$$\deg[u : w] = \delta(u) - \delta(w) < k/2, \quad (4.21)$$

while $\delta(w_R) \leq m$ gives

$$\deg[w_R : v] = \delta(w_R) - \delta(v) \leq k/2 \quad (4.22)$$

whenever that quotient is nonzero.

The middle factor $[w_L]$ requires one extra check. In a nonzero quotient term, the recursion for $[w_R : v]$ reaches v , hence $\delta(w_R) \geq \delta(v)$. Therefore

$$\delta(w_L) = \delta(w) - \delta(w_R) \leq \delta(u) - \delta(v) = k. \quad (4.23)$$

If $\delta(w_L) \leq k/2$, keep it. Otherwise $\delta(w_L) \geq 2$, so the gate-value frontier identity (4.10) applies once to $[w_L]$ at the admissible threshold $\delta(w_L)/2 \geq 1$:

$$[w_L] = \sum_{p \in \mathcal{F}_{\delta(w_L)/2}} [w_L : p] [p_L] [p_R]. \quad (4.24)$$

Every new factor has degree at most $\delta(w_L)/2 \leq k/2$. Substituting (4.24) into (4.20) creates at most five factors in a product term.

There are at most s_0 choices for each frontier gate. Hence (4.16) has at most s_0 terms and the once-expanded form of (4.20) at most s_0^2 terms. Across $O(s_0^2)$ labels, this is polynomial size. Subtraction signs and scalar factors can be absorbed into the coefficients c_i . The entire construction uses only addition, subtraction, and multiplication, proving the field-independent claim. $\square$

What the balancing lemma actually buys. The important conclusion is not merely logarithmic depth. Equation (4.14) provides a reusable local expansion in which the degree of every positive-degree factor is at most half the degree being expanded. The depth-four reduction can therefore ignore the original circuit depth completely and work only with the multiset of factor degrees in one summand.

4.4 Collapsing the balanced circuit

We first prove the homogeneous case of Theorem 4.1. Suppose that f is homogeneous of degree d . If $d = 1$, write the homogeneous linear form directly as one middle sum of its at most n variable monomials (with trivial product gates); this is already a $\Sigma\Pi\Sigma\Pi^{[1]}$ circuit satisfying every stated

bound. Hence assume $d > 1$. Apply Theorem 1.2 with cutoff d and retain the degree- d output. It is syntactically homogeneous and has size

$$s_0 = O\left(s(d+1)^2\right) = S^{O(1)}. \quad (4.25)$$

Apply Lemma 4.3. Let

$$\sigma := s_0^{O(1)} = S^{O(1)} \quad (4.26)$$

bound the number of summands in every balanced expansion (4.14).

Expand the output once using (4.14). Thereafter, in every current product summand, choose a factor of largest degree; if its degree is greater than t , replace it in place by its balanced expansion (4.14). Repeat until every factor has degree at most t . Each replacement branches into at most σ summands.

The only issue is to prove that a branch cannot continue for too many replacements. Call a positive-degree factor *bad* if its degree is greater than

$$\frac{t}{8}. \quad (4.27)$$

Consider the expansion of a largest factor of degree $k > t$. The replacement has at most five positive-degree factors, each of degree at most $k/2$, and their degrees sum to k . The largest replacement factor therefore has degree at least $k/5$. More importantly, after removing one largest replacement factor, the remaining factors have total degree at least $k/2$; among at most four of them, one has degree at least $k/8$. Consequently at least two replacement factors have degree $> t/8$. The old factor was itself bad, so every replacement increases the number of bad factors in that branch by at least one.

Every summand remains homogeneous of total degree d . Hence it contains fewer than

$$\frac{8d}{t} \quad (4.28)$$

bad factors. A branch therefore performs at most

$$L := \left\lceil \frac{8d}{t} \right\rceil \quad (4.29)$$

replacements after the initial output expansion. The number of final product summands is at most

$$\sigma^{L+1} = S^{O(d/t)}. \quad (4.30)$$

This is the claimed top fan-in.

A product term begins with at most five factors. Replacing one factor by at most five can increase the number of factors by at most four. Thus the final second-layer product fan-in is at most

$$5 + 4L = 5 + 4 \left\lceil \frac{8d}{t} \right\rceil = O(d/t). \quad (4.31)$$

This makes the top-product bound explicit.

It remains to flatten each terminal factor. A homogeneous polynomial of degree $r \leq t$ in n variables has at most

$$\binom{n+r-1}{r} \leq (n+t)^t \leq (2S)^t = S^{O(t)} \quad (4.32)$$

monomials. Write each terminal factor as the sum of those monomials. This is an existential normal-form expansion; no efficient coefficient-extraction algorithm is being claimed. Every monomial becomes a bottom product gate of fan-in exactly its degree, hence at most t . Combining (4.30)–(4.32), and counting all gates and nonzero wires, gives

$$S^{O(d/t)} S^{O(t)} = S^{O(t+d/t)}. \quad (4.33)$$

All rewrites preserve homogeneity, so the homogeneous case is proved.

The nonhomogeneous case. Now let only $\deg f \leq d$ be known. Theorem 1.2 produces

$$f = \sum_{k=0}^d f_k, \quad f_k := H_k[f], \quad (4.34)$$

with all f_k represented by syntactically homogeneous circuits of total size $S^{O(1)}$. For $k \geq 1$, apply the homogeneous reduction to f_k with

$$t_k := \min\{t, k\}. \quad (4.35)$$

If $k \leq t$, then $k/t_k = 1$ and its depth-four size is $S^{O(k+1)} \leq S^{O(t)}$. If $k > t$, its size is $S^{O(t+k/t)} \leq S^{O(t+d/t)}$. The same dichotomy gives top fan-in $S^{O(d/t)}$ and top-product fan-in $O(d/t)$ after all top sums are merged. The degree-zero component is a field constant. Since there are at most $d+1 \leq S+1$ components, the extra factor is absorbed by the same asymptotic bounds. This proves all three assertions of Theorem 4.1.

Finally choose $t = \lceil \sqrt{d} \rceil$. Then

$$t + \frac{d}{t} = O(\sqrt{d}), \quad \frac{d}{t} = O(\sqrt{d}), \quad (4.36)$$

which gives (4.4) and the two $O(\sqrt{d})$ product-fan-in bounds.

4.5 A worked reduction and the literature boundary

Worked circuit. Consider the homogeneous degree-four circuit

$$f = (x_1 + x_2)(x_3 + x_4)(x_5 + x_6)(x_7 + x_8). \quad (4.37)$$

A binary implementation may first form

$$A = (x_1 + x_2)(x_3 + x_4), \quad B = (x_5 + x_6)(x_7 + x_8), \quad f = AB. \quad (4.38)$$

Take $t = 1$. The output factors A, B have degree 2, hence are expanded once; their children are the four degree-one linear forms. No factor now has degree larger than t . In flattened notation the result is

$$f = (x_1 + x_2)(x_3 + x_4)(x_5 + x_6)(x_7 + x_8), \quad (4.39)$$

viewed as a top sum of one product of four sums of bottom monomials. Thus it is a $\Sigma\Pi\Sigma\Pi^{[1]}$ circuit with top fan-in 1, top-product fan-in 4, and bottom product fan-in 1. The point of the general theorem is that exactly this expansion can be performed even when the degree-halved factors are shared gates inside a large DAG rather than visibly nested formulas.

Why the argument is characteristic-free. No interpolation, division, root extraction, or averaging appears anywhere in the proof. Homogeneous extraction is gate-local; the frontier identities are polynomial identities with coefficients already in F ; and the terminal monomial expansion only reads coefficients of a polynomial. Therefore the normal form is available over finite fields and in small characteristic exactly as stated.

Source and strength boundary. Agrawal and Vinay introduced the depth-four “chasm”; Koiran sharpened the size analysis for polynomial-size circuits; and Tavenas obtained a still sharper bound. A convenient standard tradeoff formulation is precisely the one used above: for $0 < t \leq d$, homogeneous depth-four size $s^{O(t+d/t)}$ with top fan-in $s^{O(d/t)}$, after the usual convention that s also dominates the number of variables and polynomial preprocessing factors. Our parameter S makes that convention explicit and our proof also spells out the nonhomogeneous componentwise interface.

Tavenas’ sharper analysis gives, in its parameter regime, depth-four size

$$\exp\left(O\left(\sqrt{d \log(ds) \log n}\right)\right). \quad (4.40)$$

Equation (4.40) is recorded as a literature improvement, not as a stronger local claim proved by the present potential argument. The conservative $S^{O(t+d/t)}$ theorem is the exact dependency exported to the lower-bound chapters that follow.

4.6 Reader checkpoint and exercises

Chapter checkpoint. Depth four is a bridge rather than a destination. Homogenization controls formal degree, frontier decompositions balance shared DAGs without unfolding them into formulas, and low-degree expansion produces a $\Sigma\Pi\Sigma\Pi^{[t]}$ representation of size $S^{O(t+d/t)}$. The parameter t records exactly how an eventual lower bound against bounded-bottom circuits would feed back into a lower bound for general circuits.

Exercises.

1. Minimize $t + d/t$ over positive real t and then explain why the integer choice $t = \lceil \sqrt{d} \rceil$ changes the bound only by a constant factor in the exponent.
2. For the worked polynomial (4.37), repeat the reduction with $t = 2$. Record bottom product fan-in, top-product fan-in, and top fan-in, and compare with the $t = 1$ presentation.
3. Construct a small homogeneous circuit DAG with a shared gate that is used by two later products. Explain why naively expanding the DAG as a formula may cause duplication, and indicate how the frontier method avoids that step.
4. Verify one of the frontier identities in the chapter on a concrete three-level circuit by expanding both sides as polynomials.
5. Suppose a lower-bound theorem ruled out size $\exp(o(d/t))$ for $\Sigma\Pi\Sigma\Pi^{[t]}$ circuits. Using the local depth reduction, write the inequality that would have to be compared with $S^{O(t+d/t)}$ before claiming a general-circuit consequence.
6. Compare the local $S^{O(t+d/t)}$ bound with the sharper literature bound (4.40) in the regime $s = n^{10}$ and $d = n$. Which logarithmic factors remain in the sharper exponent?

Chapter 5

Partial Derivatives as a Lower-Bound Measure

Depth reduction turns a general circuit into a rigid normal form, but a normal form is useful only if we have a quantity that cannot grow too quickly inside that form and is nevertheless large for an explicit polynomial. The classical partial-derivative method provides exactly such a quantity: collect the polynomials obtained by differentiating the target and take the dimension of their linear span.

There is one characteristic issue that is easy to hide if one writes only ordinary derivatives. Repeated ordinary differentiation introduces factorials, which can vanish in positive characteristic. We therefore use *Hasse derivatives*. They agree with divided ordinary derivatives in characteristic zero, and for multilinear polynomials they agree with the usual iterated first-order partial derivatives in every characteristic. This choice makes the calculus in this chapter valid over an arbitrary field.

How to read this chapter. This chapter is a model case for the general lower-bound method

choose a complexity measure $\longrightarrow$ upper-bound it on easy circuits
 $\longrightarrow$ lower-bound it on an explicit polynomial.

For ordinary partial derivatives the measure is $D_k(f)$ or $D(f)$. For the bounded-bottom depth-four model, derivatives alone lose too much information, so monomial shifts are added and the measure becomes a shifted-partial span. The point of every dimension estimate should be read through this upper-bound/lower-bound comparison.

There are also two proof-ownership levels. The elementary Hasse-calculus and the depth-three determinant lower bound are proved locally. The later bounded-bottom depth-four theorem is a source-owned deep input whose exact hypotheses are reconstructed and audited. In particular, “deep input” does not mean “heuristic”; it means the proof is owned by the cited literature rather than rebranded as a new elementary QRCC theorem.

Learning goals. A reader should be able to compute small Hasse-derivative spaces in any characteristic, use subadditivity/submultiplicativity to upper-bound a circuit, count determinant/permanent derivative directions, explain why shifts enlarge the measure, and match every parameter in the bounded-bottom theorem to the depth-four normal form of Chapter 4.

5.1 Hasse derivative spaces

Fix a field F and variables $x = (x_1, \dots, x_N)$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{N}^N$, write $|\alpha| = \sum_i \alpha_i$ and $z^\alpha = \prod_i z_i^{\alpha_i}$. The *Hasse derivative* $\partial^{[\alpha]} f \in F[x]$ is defined by the coefficient identity

$$f(x+z) = \sum_{\alpha \in \mathbb{N}^N} \partial^{[\alpha]} f(x) z^\alpha. \quad (5.1)$$

Thus $\partial^{[0]} f = f$. If F has characteristic zero, then

$$\partial^{[\alpha]} f = \frac{1}{\alpha!} \partial^\alpha f, \quad \alpha! := \prod_i \alpha_i!, \quad (5.2)$$

but equation (5.1), not (5.2), is the definition used here.

For $k \geq 0$, define the order- k derivative space and its dimension by

$$\mathcal{D}_k(f) := \text{span}_F \{ \partial^{[\alpha]} f : |\alpha| = k \}, \quad D_k(f) := \dim_F \mathcal{D}_k(f). \quad (5.3)$$

The all-orders derivative space is

$$\mathcal{D}(f) := \text{span}_F \{ \partial^{[\alpha]} f : \alpha \in \mathbb{N}^N \}, \quad D(f) := \dim_F \mathcal{D}(f). \quad (5.4)$$

If f is homogeneous of degree d , then $\mathcal{D}_k(f)$ is homogeneous of degree $d - k$ (or zero), so spaces of different orders lie in different homogeneous degrees. Consequently

$$D(f) = \sum_{k=0}^d D_k(f) \quad \text{for homogeneous } f. \quad (5.5)$$

The product notation below is linear-algebraic. For subspaces $U, V \subseteq F[x]$, write

$$UV := \text{span}_F \{ uv : u \in U, v \in V \}. \quad (5.6)$$

In particular, $\dim(UV) \leq (\dim U)(\dim V)$.

Lemma 5.1 (Hasse derivative calculus). *For all $f, g \in F[x]$ and every $k \geq 0$,*

$$\mathcal{D}_k(f+g) \subseteq \mathcal{D}_k(f) + \mathcal{D}_k(g), \quad (5.7)$$

and

$$\mathcal{D}_k(fg) \subseteq \sum_{i=0}^k \mathcal{D}_i(f) \mathcal{D}_{k-i}(g). \quad (5.8)$$

Hence

$$D_k(f+g) \leq D_k(f) + D_k(g), \quad (5.9)$$

$$D_k(fg) \leq \sum_{i=0}^k D_i(f) D_{k-i}(g), \quad (5.10)$$

and, for the all-orders spaces,

$$D(f+g) \leq D(f) + D(g), \quad D(fg) \leq D(f)D(g). \quad (5.11)$$

All statements hold over every field.

Proof. Linearity of the coefficient extraction in (5.1) gives (5.7). For the product, multiply the two finite expansions

$$f(x+z) = \sum_{\beta} \partial^{[\beta]} f(x) z^{\beta}, \quad g(x+z) = \sum_{\gamma} \partial^{[\gamma]} g(x) z^{\gamma}. \quad (5.12)$$

The coefficient of z^{α} is

$$\partial^{[\alpha]}(fg) = \sum_{\beta+\gamma=\alpha} \partial^{[\beta]} f \partial^{[\gamma]} g. \quad (5.13)$$

When $|\alpha| = k$, every summand has $|\beta| = i$ and $|\gamma| = k-i$ for some i , proving (5.8). Taking dimensions and using $\dim(U+V) \leq \dim U + \dim V$ and $\dim(UV) \leq (\dim U)(\dim V)$ gives (5.9)–(5.10).

Finally, every Hasse derivative of a product belongs to the span of products of one Hasse derivative of f and one of g , so $\mathcal{D}(fg) \subseteq \mathcal{D}(f)\mathcal{D}(g)$. This gives (5.11). No factorial or division has been used. $\square$

5.2 The depth-three upper bound and determinant lower bound

A homogeneous depth-three circuit of degree d will mean a flattened circuit of the form

$$C = \sum_{q=1}^T \prod_{j=1}^d \ell_{qj}(x), \quad (5.14)$$

where every ℓ_{qj} is a homogeneous linear form over F . Zero product summands may be deleted, and every remaining scalar coefficient may be absorbed into one linear form of its product. The integer T is the top fan-in. This is a syntactically homogeneous $\Sigma\Pi\Sigma$ model; the theorem below does *not* claim the same bound for arbitrary inhomogeneous depth-three circuits.

Theorem 5.2 (Partial-derivative dimension lower bound). *Let F be an arbitrary field. Assume $d \geq 1$ in the first part and $m \geq 1$ in the determinant part.*

1. *If a homogeneous degree- d polynomial f is computed by a homogeneous $\Sigma\Pi\Sigma$ circuit (5.14) of top fan-in T , then for every $0 \leq k \leq d$,*

$$D_k(f) \leq T \binom{d}{k}, \quad (5.15)$$

and therefore

$$D(f) \leq T 2^d. \quad (5.16)$$

Equivalently, every such circuit satisfies the lower bounds

$$T \geq \frac{D_k(f)}{\binom{d}{k}} \quad (0 \leq k \leq d), \quad T \geq \frac{D(f)}{2^d}. \quad (5.17)$$

2. *Let $X = (x_{ij})_{1 \leq i, j \leq m}$ be a generic $m \times m$ matrix. For $0 \leq k \leq m$,*

$$D_k(\det X) = \binom{m}{k}^2, \quad (5.18)$$

and hence

$$D(\det X) = \sum_{k=0}^m \binom{m}{k}^2 = \binom{2m}{m}. \quad (5.19)$$

Consequently every homogeneous degree- m $\Sigma\Pi\Sigma$ circuit for $\det_m$ has

$$T \geq \binom{m}{k} \quad \text{for every } 0 \leq k \leq m. \quad (5.20)$$

In particular,

$$T \geq \binom{m}{\lfloor m/2 \rfloor} \geq \frac{2^m}{m+1}. \quad (5.21)$$

Thus homogeneous depth-three circuits for the determinant require exponential top fan-in, and therefore exponential gate/wire size, over every field.

Proof roadmap. For one product of d linear forms, an order- k Hasse derivative is spanned by products obtained by deleting k factors, so at most $\binom{d}{k}$ directions appear. Sum this bound over the top fan-in. For the determinant, differentiate with respect to entries in distinct rows and columns: the surviving minors are linearly independent and can be counted exactly. Comparing the upper and lower dimensions forces large top fan-in.

Hypothesis check. Homogeneity is part of the circuit model being lower-bounded, not a cosmetic normalization. Hasse derivatives remove any characteristic restriction from the calculus. The determinant calculation uses its multilinear monomial structure, so repeated differentiation in one matrix entry contributes zero.

Proof. Fix one product gate

$$P = \prod_{j=1}^d \ell_j(x). \quad (5.22)$$

Because ℓ_j is homogeneous linear,

$$\ell_j(x+z) = \ell_j(x) + \ell_j(z). \quad (5.23)$$

To extract a term of total z -degree k from $P(x+z) = \prod_j (\ell_j(x) + \ell_j(z))$, choose exactly k of the d factors to contribute their z -term. Therefore every order- k Hasse derivative of P is a linear combination of the at most $\binom{d}{k}$ polynomials

$$\prod_{j \notin S} \ell_j(x), \quad S \subseteq [d], \quad |S| = k. \quad (5.24)$$

Hence

$$D_k(P) \leq \binom{d}{k}. \quad (5.25)$$

Applying the sum inequality (5.9) to the T product gates in (5.14) gives (5.15). Summing over k and using (5.5) yields

$$D(f) \leq T \sum_{k=0}^d \binom{d}{k} = T2^d, \quad (5.26)$$

which proves the first part.

Now expand the determinant as

$$\det X = \sum_{\pi \in S_m} \operatorname{sgn}(\pi) \prod_{i=1}^m x_{i,\pi(i)}. \quad (5.27)$$

It is multilinear in the m^2 variables. An order- k Hasse derivative is zero unless the differentiated variables occupy k distinct rows and k distinct columns. If they form a partial matching

$$Q = \{(i, \sigma(i)) : i \in I\}, \quad |I| = |\sigma(I)| = k, \quad (5.28)$$

and write $\partial^{[Q]}$ for the Hasse multi-index derivative having entry 1 on the positions of Q and 0 elsewhere. Then coefficient extraction from (5.27) gives

$$\partial^{[Q]} \det X = \varepsilon_Q \det X_{[m] \setminus I, [m] \setminus \sigma(I)}, \quad \varepsilon_Q \in \{1, -1\}. \quad (5.29)$$

In characteristic two the two displayed signs coincide, which causes no problem. Conversely, for every choice of a remaining row set R and column set C of size $m - k$, choose any bijection between their complements and (5.29) produces the minor $\det X_{R,C}$ up to a nonzero sign. Thus $\mathcal{D}_k(\det X)$ is exactly the span of all $(m - k) \times (m - k)$ minors.

Those minors are linearly independent over every field. Indeed, every monomial in $\det X_{R,C}$ uses variables from exactly the row set R and exactly the column set C . If $(R, C) \neq (R', C')$, the two minors therefore have disjoint monomial supports. There are

$$\binom{m}{m-k}^2 = \binom{m}{k}^2 \quad (5.30)$$

such minors, proving (5.18). Since different derivative orders have different homogeneous degrees, (5.5) gives the sum in (5.31). For completeness, Vandermonde's identity here is just coefficient extraction from $(1+z)^m(1+z)^m = (1+z)^{2m}$: the coefficient of z^m is $\sum_{k=0}^m \binom{m}{k} \binom{m}{m-k} = \sum_k \binom{m}{k}^2$ on the left and $\binom{2m}{m}$ on the right. Hence

$$D(\det X) = \sum_{k=0}^m \binom{m}{k}^2 = \binom{2m}{m}, \quad (5.31)$$

which is (5.19).

Apply (5.17) with $d = m$ and (5.18):

$$T \geq \frac{\binom{m}{k}^2}{\binom{m}{k}} = \binom{m}{k}. \quad (5.32)$$

Finally the largest binomial coefficient is at least the average of the $m+1$ coefficients in $(1+1)^m$, so

$$\binom{m}{\lfloor m/2 \rfloor} \geq \frac{1}{m+1} \sum_{k=0}^m \binom{m}{k} = \frac{2^m}{m+1}. \quad (5.33)$$

This proves (5.20)–(5.21). □

5.3 A two-by-two calculation

For

$$f = \det \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} = x_{11}x_{22} - x_{12}x_{21}, \quad (5.34)$$

the derivative dimensions are visible without any asymptotics:

$$D_0(f) = 1, \quad \mathcal{D}_1(f) = \text{span}\{x_{22}, x_{21}, x_{12}, x_{11}\}, \quad D_1(f) = 4, \quad (5.35)$$

and

$$\mathcal{D}_2(f) = \text{span}\{1\}, \quad D_2(f) = 1. \quad (5.36)$$

Thus $D(f) = 6 = \binom{4}{2}$. A product of two linear forms has first-derivative dimension at most $\binom{2}{1} = 2$, so any homogeneous depth-three circuit for f needs at least $4/2 = 2$ product gates. The standard expression in (5.34) uses exactly two, so the order-one derivative bound is tight in this example.

5.4 What this measure proves—and what it does not

Homogeneity is a real hypothesis. The proof of (5.15) uses that every product gate consists of exactly d homogeneous linear factors. In an arbitrary depth-three circuit, affine constants and cancellations between different degrees can make the degree- d part of one large product expand into many homogeneous products. The theorem therefore exports a lower bound for the homogeneous $\Sigma\Pi\Sigma$ model, not for unrestricted depth three.

Why Hasse derivatives matter. Ordinary repeated derivatives are sufficient in characteristic zero, but the identity between an order- r ordinary derivative and coefficient extraction contains the factor $r!$. Hasse derivatives build coefficient extraction into the definition, so the product rule (5.13) and the determinant-minor calculation remain valid when the characteristic is small. For the determinant, which is multilinear, the relevant Hasse derivatives are in fact the familiar iterated first-order partial derivatives.

Source boundary. Nisan and Wigderson made the span of partial derivatives a systematic arithmetic circuit lower-bound measure. Their motivating depth-three argument records the same three structural facts used here: linearity under scaling, subadditivity under sums, multiplicativity of the all-orders derivative dimension under products, and consequently the upper bound $s2^d$ for a depth-three circuit with top fan-in s and product fan-in at most d . That motivating section is stated over characteristic zero and applies the measure to elementary symmetric polynomials.

The present theorem makes two deliberate local normalizations. First, it uses Hasse derivatives so the calculus is characteristic-free. Second, it keeps the order- k spaces separate, yielding the sharper interface (5.15), and works the determinant-minor rank calculation explicitly. These are local deductions, not claims that the Nisan–Wigderson motivating theorem was stated in exactly this form. Shifted partial derivatives, which are needed for depth-four lower bounds, form a strictly stronger measure and are taken up in the next source-gated block below.

5.5 Shifted partials and the bounded-bottom depth-four barrier

The order- k derivative space can be too small because different derivatives may collide. The shifted-partial method enlarges the space before measuring it: after differentiating, one is allowed to multiply by low-degree monomials. This retains the linear-algebraic flavor of the preceding theorem, but it gives many more opportunities for the hard polynomial to expose distinct leading monomials.

For $k, \ell \geq 0$, define the *shifted order- k Hasse derivative space* by

$$\mathcal{S}_{k, \leq \ell}(f) := \text{span}_F \left\{ x^\beta \partial^{[\alpha]} f : |\alpha| = k, |\beta| \leq \ell \right\}, \quad (5.37)$$

and write

$$\Gamma_{k, \ell}(f) := \dim_F \mathcal{S}_{k, \leq \ell}(f). \quad (5.38)$$

We deliberately allow shifts of degree *at most* ℓ . This is the normalization used in the depth-four argument below. By linearity,

$$\Gamma_{k,\ell}(f + g) \leq \Gamma_{k,\ell}(f) + \Gamma_{k,\ell}(g). \quad (5.39)$$

The first ingredient is the circuit-side upper bound. Its useful feature is that the internal polynomial F may be arbitrarily complicated; only the number and degrees of its polynomial inputs matter.

Lemma 5.3 (Low-degree-generator upper bound). *Let $Q = (Q_1, \dots, Q_m)$ with $Q_i \in F[x_1, \dots, x_N]$ and $\deg Q_i \leq t$. If*

$$T(x) = F(Q_1(x), \dots, Q_m(x)) \quad (5.40)$$

for an arbitrary $F \in F[u_1, \dots, u_m]$, then for every $k, \ell \geq 0$,

$$\Gamma_{k,\ell}(T) \leq \binom{m+k}{k} \binom{N + (t-1)k + \ell}{N}. \quad (5.41)$$

Consequently, if

$$C = \sum_{i=1}^s F_i(Q_{i1}, \dots, Q_{im}) \quad (5.42)$$

with every $\deg Q_{ij} \leq t$, then

$$\Gamma_{k,\ell}(C) \leq s \binom{m+k}{k} \binom{N + (t-1)k + \ell}{N}. \quad (5.43)$$

All of these inequalities hold over every field.

Proof. Set

$$R_i(x, z) := Q_i(x + z) - Q_i(x). \quad (5.44)$$

Every monomial of R_i has positive z -degree. More precisely, the coefficient of z^α in R_i is $\partial^{[\alpha]} Q_i(x)$ and has x -degree at most $t - |\alpha|$. Expand F in Hasse derivatives in its own variables:

$$F(Q(x) + R(x)) = \sum_{\gamma \in \mathbb{N}^m} (\partial_u^{[\gamma]} F)(Q(x)) R(x)^\gamma. \quad (5.45)$$

To contribute to a coefficient of total z -degree k , a term R^γ can have only $|\gamma| \leq k$, since every factor from every R_i has positive z -degree. If a particular coefficient of R^γ has total z -degree k , its x -degree is at most

$$t|\gamma| - k \leq (t-1)k. \quad (5.46)$$

It follows that every order- k Hasse derivative of T lies in the span of polynomials

$$(\partial_u^{[\gamma]} F)(Q(x)) x^\delta, \quad |\gamma| \leq k, \quad |\delta| \leq (t-1)k. \quad (5.47)$$

After multiplying by all shifts of degree at most ℓ , it is enough to use $|\delta| \leq (t-1)k + \ell$. There are

$$\#\{\gamma \in \mathbb{N}^m : |\gamma| \leq k\} = \binom{m+k}{k} \quad (5.48)$$

possible Hasse derivatives of F , and

$$\#\{x^\delta : |\delta| \leq (t-1)k + \ell\} = \binom{N + (t-1)k + \ell}{N} \quad (5.49)$$

possible monomial shifts. Their products span $\mathcal{S}_{k,\leq \ell}(T)$, proving (5.41). Equation (5.43) follows from (5.39). No ordinary chain rule, factorial inverse, or characteristic assumption has been used. $\square$

The hard-polynomial side is a leading-monomial count. Let

$$\text{Perm}_n(X) = \sum_{\pi \in S_n} \prod_{i=1}^n x_{i, \pi(i)} \quad (5.50)$$

in the $N = n^2$ variables x_{ij} . Order the variables row by row,

$$x_{11} \succ x_{12} \succ \cdots \succ x_{1n} \succ x_{21} \succ \cdots \succ x_{nn}, \quad (5.51)$$

and use the induced lexicographic monomial order. A sequence

$$(x_{i_1 j_1}, \dots, x_{i_r j_r}) \quad (5.52)$$

is called *r-increasing* if $i_1 < \cdots < i_r$ and $j_1 < \cdots < j_r$.

A nonzero order- k Hasse derivative of the permanent is an $(n-k) \times (n-k)$ permanental minor. Under (5.51), the leading monomial of such a minor is the product of the entries on its increasing diagonal. Therefore distinct monomials of degree at most $n-k+\ell$ that contain an $(n-k)$ -increasing sequence give distinct leading monomials among the shifted order- k derivatives. Choosing one shifted derivative for each such leading monomial gives a linearly independent family.

Lemma 5.4 (Two-diagonal permanent count). *For every $0 \leq k \leq n$ and every $\ell \geq 0$,*

$$\Gamma_{k, \ell}(\text{Perm}_n) \geq \binom{n+k}{2k} \binom{n^2 + \ell - 2k}{\ell}. \quad (5.53)$$

The same lower bound holds for $\det_n$ over every field.

Proof. Put $r = n - k$ and consider only the $2n - 1$ variables on the main diagonal and the diagonal immediately above it:

$$\mathcal{A}_n := \{x_{ii} : 1 \leq i \leq n\} \cup \{x_{i, i+1} : 1 \leq i < n\}. \quad (5.54)$$

Read them in the path order

$$x_{11}, x_{12}, x_{22}, x_{23}, \dots, x_{n-1, n}, x_{nn}. \quad (5.55)$$

An r -increasing sequence inside $\mathcal{A}_n$ is exactly a choice of r nonadjacent vertices of this path. Hence the number of such sequences is

$$\binom{(2n-1) - r + 1}{r} = \binom{2n-r}{r}. \quad (5.56)$$

Fix one such sequence Q . Every chosen vertex of the path except possibly the last vertex has a distinct immediate successor in (5.55), so Q has at least $r - 1$ distinct *companions* in $\mathcal{A}_n \setminus Q$. Multiply the squarefree monomial corresponding to Q by an arbitrary monomial using only

- variables outside $\mathcal{A}_n$,
- the r variables of Q themselves, and
- any fixed $r - 1$ companions of Q ,

with additional degree at most $d - r$. The support on $\mathcal{A}_n$ then has Q as its lexicographically leading r -increasing sequence among the increasing sequences contained entirely in $\mathcal{A}_n$: if such a sequence uses a companion, replacing that companion by its predecessor in Q produces a lexicographically larger increasing sequence. Variables outside $\mathcal{A}_n$ do not change this restricted support. Thus the canonical leading sequence inside $\mathcal{A}_n$ recovers Q , and different choices of Q produce disjoint collections of monomials.

There are at least

$$n^2 - (2n - 1) + r + (r - 1) = n^2 - 2n + 2r \quad (5.57)$$

variables available for the multiplier. The number of multiplier monomials of degree at most $d - r$ is therefore at least

$$\binom{n^2 - 2n + 2r + d - r}{d - r} = \binom{n^2 - 2n + d + r}{d - r}. \quad (5.58)$$

Multiplying by the number (5.56) of choices for Q gives that the number of monomials of degree at most d containing an r -increasing sequence is at least

$$\binom{2n - r}{r} \binom{n^2 - 2n + d + r}{d - r}. \quad (5.59)$$

Now set $d = r + \ell = n - k + \ell$ and $r = n - k$. Equations (5.56)–(5.59) become

$$\binom{n + k}{n - k} \binom{n^2 + \ell - 2k}{\ell} = \binom{n + k}{2k} \binom{n^2 + \ell - 2k}{\ell}, \quad (5.60)$$

which proves (5.53).

For the determinant, the same differentiated partial matchings produce complementary determinant minors up to a nonzero sign. Their leading diagonal monomials are the same as for the permanent minors, so the identical leading monomial count applies. In characteristic two the signs merge, but they do not vanish. $\square$

We can now state the source-frozen depth-four lower bound. The formulation first records the stronger low-degree-generator statement and then extracts the homogeneous depth-four circuit corollary. The asymptotic condition $t = o(n)$ means that $n/t \rightarrow \infty$; it includes the depth-reduction point $t = \lceil \sqrt{n} \rceil$.

Theorem 5.5 (Shifted-partial lower bound at bounded bottom degree). *Let F be an arbitrary field, let $t = t(n) \geq 1$ be integer-valued with $t = o(n)$, and fix a constant $c \geq 1$.*

Suppose that, for all sufficiently large n ,

$$\text{Perm}_n(X) = \sum_{i=1}^s F_i(Q_{i1}, \dots, Q_{im}) \quad (5.61)$$

where $m \leq cn/t$, every F_i is an arbitrary polynomial in m variables, and every $Q_{ij} \in F[X]$ has degree at most t . Then

$$s \geq \exp(\Omega_c(n/t)). \quad (5.62)$$

The same conclusion holds with Perm_n replaced by $\det_n$.

Consequently, if a homogeneous depth-four circuit

$$C = \sum_{i=1}^s \prod_j Q_{ij} \quad (5.63)$$

computes Perm_n or $\det_n$, every Q_{ij} is homogeneous of degree at most t , and every top product has total degree n , then

$$s \geq \exp(\Omega(n/t)). \quad (5.64)$$

In particular, homogeneous $\Sigma\Pi\Sigma\Pi^{[\lceil\sqrt{n}\rceil]}$ circuits for the permanent or determinant have top fan-in, and hence total size, at least

$$\exp(\Omega(\sqrt{n})). \quad (5.65)$$

There is no additional sparsity, monomial-support, or per-variable individual-degree hypothesis on the polynomials Q_{ij} beyond their total degree bound $\deg Q_{ij} \leq t$.

Proof roadmap and ownership boundary. Read this theorem as the strengthened version of the same measure argument. The shifted-partial dimension of a polynomial generated from few low-degree inputs has a controlled upper bound; the permanent/determinant has many independent shifted derivatives; an explicit parameter choice makes the second quantity exponentially larger unless the top fan-in is $\exp(\Omega(n/t))$. The chapter reconstructs the parameter comparison so the interface can be checked locally, while the theorem remains source-owned at the registered deep-input level.

Hypothesis check. The bottom objects Q_{ij} are bounded in *total degree*; no bounded support or multilinearity assumption is being added. The range $\log n \leq t \leq n/(4 \log n)$ and the bound on the number of generators are load-bearing in the quantitative comparison.

Proof. We prove the permanent statement; Lemma 5.4 gives the same proof for the determinant. Put

$$r := \frac{n}{t}. \quad (5.66)$$

Since $t = o(n)$, we have $r \rightarrow \infty$. Fix

$$\varepsilon := \frac{1}{32e^2(2c+1)}, \quad k := \lfloor \varepsilon r \rfloor, \quad \ell := n^2 t. \quad (5.67)$$

For all sufficiently large n ,

$$\frac{\varepsilon r}{2} \leq k \leq \varepsilon r. \quad (5.68)$$

Lemma 5.3 applied to (5.61), with $N = n^2$, gives

$$\Gamma_{k,\ell}(\text{Perm}_n) \leq s \binom{m+k}{k} \binom{n^2 + \ell + (t-1)k}{n^2}. \quad (5.69)$$

Lemma 5.4 gives

$$\Gamma_{k,\ell}(\text{Perm}_n) \geq \binom{n+k}{2k} \binom{n^2 + \ell - 2k}{\ell}. \quad (5.70)$$

Therefore

$$s \geq \frac{\binom{n+k}{2k}}{\binom{m+k}{k}} \cdot \frac{\binom{n^2 + \ell - 2k}{\ell}}{\binom{n^2 + \ell + (t-1)k}{n^2}}. \quad (5.71)$$

We bound the two ratios elementarily. The standard inequalities

$$\binom{A}{B} \geq \left(\frac{A}{B}\right)^B, \quad \binom{A}{B} \leq \left(\frac{eA}{B}\right)^B \quad (5.72)$$

give, using $k \leq \varepsilon n/t$, $m \leq cn/t$, and (5.68),

$$\binom{n+k}{2k} \geq \left(\frac{t}{2\varepsilon}\right)^{2k}, \quad (5.73)$$

and

$$\binom{m+k}{k} \leq \left[e \left(\frac{2c}{\varepsilon} + 1 \right) \right]^k. \quad (5.74)$$

For the shift ratio, set $N = n^2$ and $L = Nt = \ell$. Factorials give the exact identity

$$\frac{\binom{N+L-2k}{L}}{\binom{N+L+(t-1)k}{N}} = \frac{N!}{(N-2k)!} \frac{(L+(t-1)k)!}{L!} \frac{(N+L-2k)!}{(N+L+(t-1)k)!}. \quad (5.75)$$

For sufficiently large n , $2k/N \leq 1/2$. Using $\log(1-u) \geq -2u$ for $0 \leq u \leq 1/2$,

$$\frac{N!}{(N-2k)!} \geq N^{2k} \exp\left(-\frac{4k^2}{N}\right). \quad (5.76)$$

Also

$$\frac{(L+(t-1)k)!}{L!} \geq L^{(t-1)k}. \quad (5.77)$$

There are $(t+1)k$ factors in the denominator represented by the last ratio in (5.75), and each is at most $N+L+(t-1)k$. Since $N+L = N(t+1)$ and $\log(1+u) \leq u$,

$$\frac{(N+L+(t-1)k)!}{(N+L-2k)!} \leq [N(t+1)]^{(t+1)k} \exp\left(\frac{(t-1)k^2}{N}\right). \quad (5.78)$$

Because $k \leq \varepsilon n/t$, the two exponential error terms in (5.76) and (5.78) have total exponent at most $5\varepsilon^2$. Hence

$$\frac{\binom{N+L-2k}{L}}{\binom{N+L+(t-1)k}{N}} \geq e^{-5\varepsilon^2} \left(\frac{t^{t-1}}{(t+1)^{t+1}} \right)^k. \quad (5.79)$$

Finally

$$\frac{t^{t-1}}{(t+1)^{t+1}} = \frac{1}{t^2(1+1/t)^{t+1}} \geq \frac{1}{2e t^2}, \quad (5.80)$$

because $(1+1/t)^t \leq e$ and $1+1/t \leq 2$.

Substituting (5.73), (5.74), and (5.79)–(5.80) into (5.71) yields

$$s \geq e^{-5\varepsilon^2} \left[\frac{1}{8e^2\varepsilon(2c+\varepsilon)} \right]^k. \quad (5.81)$$

Our choice of ε gives

$$8e^2\varepsilon(2c+\varepsilon) \leq \frac{1}{4}, \quad (5.82)$$

so

$$s \geq e^{-5\varepsilon^2} 4^k = \exp(\Omega_c(r)) = \exp(\Omega_c(n/t)), \quad (5.83)$$

using (5.68). This proves the stronger representation theorem.

It remains to pass from (5.63) to (5.61). Delete constant factors by absorbing them into the coefficient of the top summand. Inside one degree- n product, greedily group the remaining

homogeneous factors so that every new block, except possibly the final block, has degree strictly larger than $t/2$ and at most t . This is possible because each original factor has degree at most t : a factor of degree $> t/2$ forms its own block, while factors of degree at most $t/2$ are accumulated until the block first exceeds $t/2$. Thus the number of grouped factors is at most

$$\frac{2n}{t} + 1. \quad (5.84)$$

Since $n/t \rightarrow \infty$, this is at most $3n/t$ for all sufficiently large n . Each top summand is therefore a product $F_i(u) = \prod_h u_h$ of at most $3n/t$ polynomials of degree at most t ; pad with unused inputs if necessary and apply the first part with $c = 3$. Equations (5.64)–(5.65) follow. $\square$

5.6 A finite shifted-partial calculation

The role of the shift can already be seen in a three-variable polynomial. Let

$$f(x, y, z) = x(y + z) = xy + xz. \quad (5.85)$$

Its order-one derivative space is

$$\mathcal{D}_1(f) = \text{span}\{y + z, x, x\}, \quad D_1(f) = 2, \quad (5.86)$$

so the y - and z -derivatives collide. With shifts of degree at most one, we also obtain

$$x^2, xy, xz, y^2 + yz, yz + z^2 \quad (5.87)$$

in degree two, and these five polynomials are linearly independent over every field. Since degree-one and degree-two pieces are independent of one another,

$$\Gamma_{1,1}(f) = 2 + 5 = 7. \quad (5.88)$$

The example is intentionally tiny: shifting does not make the duplicate $\partial_y f = \partial_z f$ different; rather, it enlarges the ambient span so that products of the surviving derivatives with monomials expose many more independent directions.

5.7 Source boundary for the depth-four theorem

Gupta, Kamath, Kayal, and Saptharishi prove the bounded-bottom depth-four lower bound for the permanent and determinant over arbitrary fields. Their $\Sigma\Pi\Sigma\Pi^{[\text{hom}]}(t)$ model writes each top summand as a product of homogeneous polynomials of degree at most t , and their main consequence is top fan-in $\exp(\Omega(n/t))$; at $t = \sqrt{n}$ this is $\exp(\Omega(\sqrt{n}))$. They also isolate the stronger representation in which each summand is an arbitrary polynomial $F_i(Q_{i1}, \dots, Q_{im})$ of $m = O(n/t)$ low-degree generators. That stronger form is the source-facing statement frozen in Theorem 5.5.

The local reconstruction differs from the primary exposition in one harmless but useful normalization. The source writes ordinary partial derivatives and chain-rule expansions. Here all derivatives are Hasse derivatives. The hard permanent/determinant derivatives are multilinear, so the two notions coincide on the distinct-variable derivatives that matter; on the circuit side, coefficient extraction in (5.45) gives the needed composition upper bound without any characteristic restriction. Thus the reader proof makes the source's advertised arbitrary-field boundary transparent rather than relying on implicit factorial cancellations.

No later bounded-support, projected-shifted-partial, or random-restriction variant is imported into this theorem. In particular, (5.64) assumes only a bound on the *total degree* of each bottom polynomial Q_{ij} ; it does not assume bounded monomial support, bounded individual degree of the ambient variables, multilinearity of the circuit, or sparsity of the Q_{ij} . Conversely, the lower bound is only $\exp(\Omega(n/t))$. At the depth-reduction point it gives $\exp(\Omega(\sqrt{n}))$, while Theorem 4.1 produces a depth-four upper bound with an additional logarithmic base-size factor. This is why the result *approaches* the chasm without by itself proving a general arithmetic circuit lower bound.

5.8 Reader checkpoint and exercises

Chapter checkpoint. A lower-bound measure succeeds only when the same quantity is simultaneously small on the circuit model and large on the target polynomial. Hasse derivatives make the elementary dimension method characteristic-free; shifted partials enlarge the measure enough to attack bounded-bottom depth-four circuits. The final quantitative theorem has an explicit model and parameter window, and it reaches an $\exp(\Omega(\sqrt{n}))$ lower bound at the natural depth-reduction scale without closing the general arithmetic-circuit problem.

Exercises.

1. Compute every nonzero Hasse derivative of $f(x) = x^p$ over characteristic p . Compare with ordinary formal differentiation and explain why (5.1) is the safer definition.
2. For $f = (x_1 + x_2)(x_3 + x_4)$, compute $D_0(f)$, $D_1(f)$, and $D_2(f)$. Check the product upper bound from Lemma 5.1 numerically.
3. For the 3×3 determinant, list the order-one Hasse derivatives and identify them with signed 2×2 minors. Use distinct monomials to justify linear independence.
4. Apply Theorem 5.2 with $k = \lfloor d/2 \rfloor$ to a hypothetical homogeneous depth-three circuit for a target with known $D_k(f)$. Write the resulting top-fan-in lower bound before making any asymptotic approximation.
5. Recompute the shifted-partial span of $f = x(y + z)$ from (5.85)–(5.88) with shift degree zero. Which new independent directions appear only after degree-one shifts are allowed?
6. Put $t = \lceil \sqrt{n} \rceil$ into the bounded-bottom theorem and derive the $\exp(\Omega(\sqrt{n}))$ top-fan-in consequence. Then state precisely why this does not by itself contradict the general depth-four upper bound from Chapter 4.

Transition to PIT. Chapters 1–5 asked how circuits can be normalized and how restricted models can be lower-bounded. The next block changes the question: instead of proving that a circuit is large, we ask whether a presented circuit computes the zero polynomial. That shift leads first to randomized identity testing, then to structured deterministic PIT, hardness-versus-randomness, noncommutative methods, and finally invariant-theoretic scaling algorithms.

Chapter 6

Polynomial Identity Testing by Random Evaluation

Polynomial identity testing asks a yes–no question with an asymmetry that is ideal for randomization. If a polynomial is identically zero, every evaluation vanishes. If it is nonzero, a single random evaluation should usually expose a nonzero value. The entire argument rests on one elementary fact: a polynomial of degree D cannot vanish on too large a fraction of a Cartesian grid unless it is the zero polynomial.

There are two field issues that should not be blurred. First, the probability bound itself is valid over *every* field, provided the random coordinates are drawn from a finite subset of that field. Second, a small finite base field may contain too few evaluation points to make the bound useful. A circuit over $\mathbb{F}_q$ can then be evaluated over an extension field, but a black box that accepts queries only from $\mathbb{F}_q^n$ cannot: the nonzero polynomial $x^q - x$ vanishes at every base-field point. This chapter keeps those access models separate.

How to read this chapter. The chapter has two PIT stories with different resources. Sections 1–3 prove randomized identity testing from Schwartz–Zippel and then make the finite-field implementation literal. Sections 4–7 switch to deterministic *black-box* PIT for read-once oblivious ABPs (ROABPs), where derandomization is possible only because the polynomial has a strong coefficient-space structure. Random sampling and basis isolation solve different problems; neither subsumes the other for unrestricted circuits.

Keep three access models separate while reading:

white-box circuit access, base-field black-box access, extension-field evaluation access.

A small finite field is harmless for an explicitly given circuit that can be evaluated in a constructed extension, but it is a genuine obstruction for a black box restricted to base-field query points.

Learning goals. A reader should be able to prove Schwartz–Zippel by induction, identify its one-sided error direction, explain why $x^q - x$ defeats base-field black-box sampling, compute coefficient dimension across an ROABP cut, and describe how basis-isolating weights convert hidden-order structure into a deterministic hitting-set search.

6.1 The univariate root bound

We begin with the only algebraic fact needed in the probabilistic proof.

Lemma 6.1 (Univariate root bound). *Let F be a field and let $0 \neq h \in F[X]$ have degree $r \geq 0$. Then h has at most r distinct roots in F .*

Proof. Induct on r . A nonzero constant has no root. If $r \geq 1$ and a is a root, polynomial division gives

$$h(X) = (X - a)q(X), \quad \deg q = r - 1. \quad (6.1)$$

Every root of h different from a is a root of q , so the induction hypothesis gives at most $1 + (r - 1) = r$ roots. $\square$

The field hypothesis is essential here: over a ring with zero divisors a nonzero polynomial can have more roots than its degree. No infinitude or characteristic assumption is needed.

6.2 Schwartz–Zippel

Theorem 6.2 (Schwartz–Zippel lemma). *Let F be a field, let $S \subseteq F$ be a nonempty finite set, and let*

$$0 \neq f \in F[x_1, \dots, x_n]$$

have total degree $D \geq 0$. Choose $A_1, \dots, A_n$ independently and uniformly from S . Then

$$\Pr[f(A_1, \dots, A_n) = 0] \leq \frac{D}{|S|}. \quad (6.2)$$

Consequently the probability is at most $\min\{1, D/|S|\}$. The statement includes $n = 0$: in that case a nonzero polynomial is a nonzero constant and the probability is 0.

Proof roadmap. Induct on the number of variables. View f as a univariate polynomial in the last variable and let g be its leading coefficient. A bad sample occurs either because the first $n - 1$ coordinates kill g , or because they leave a nonzero univariate polynomial whose final coordinate lands on one of its roots. The degree splits as $(D - r) + r$, which is exactly why total degree closes the induction.

Hypothesis check. The coordinates must be independent for the conditioning step. The field may be finite and may have arbitrary characteristic. The useful numerical regime is $|S| > D$, but the inequality itself remains correct when $D/|S| \geq 1$.

Proof. We induct on n . The case $n = 0$ was just noted. Assume $n \geq 1$ and write f as a polynomial in the final variable:

$$f(x_1, \dots, x_n) = \sum_{j=0}^r g_j(x_1, \dots, x_{n-1})x_n^j, \quad g_r \neq 0. \quad (6.3)$$

Here $0 \leq r \leq D$, and every monomial of $g_r x_n^r$ has total degree at most D , so

$$\deg g_r \leq D - r. \quad (6.4)$$

Put $A' = (A_1, \dots, A_{n-1})$ and let

$$E = \{g_r(A') = 0\}. \quad (6.5)$$

By the induction hypothesis,

$$\Pr(E) \leq \frac{\deg g_r}{|S|} \leq \frac{D - r}{|S|}. \quad (6.6)$$

Condition on a value $A' = a'$ for which E does not occur. Then

$$X \mapsto f(a', X)$$

is a nonzero univariate polynomial of degree exactly r . By Lemma 6.1, at most r elements of S are roots, hence

$$\Pr[f(a', A_n) = 0 \mid A' = a'] \leq \frac{r}{|S|}. \quad (6.7)$$

Therefore

$$\begin{aligned} \Pr[f(A) = 0] &\leq \Pr(E) + \Pr[f(A) = 0 \text{ and } E^c] \\ &\leq \frac{D-r}{|S|} + \frac{r}{|S|} = \frac{D}{|S|}. \end{aligned} \quad (6.8)$$

This closes the induction. $\square$

Why total degree is the correct induction invariant. The leading coefficient g_r may still involve all of the first $n-1$ variables. What makes the recurrence close is precisely (6.4): using degree r in x_n leaves at most $D-r$ total degree for its coefficient. Bounding each variable degree separately would also give a valid theorem, but would lose the clean $D/|S|$ parameter.

The bound can be exact. Take

$$S = \{0, 1, 2, 3, 4\} \subset \mathbb{Q}, \quad h(X) = (X-1)(X-3). \quad (6.9)$$

Then $D = 2$ and exactly two of the five sample points are roots, so

$$\Pr[h(A) = 0] = \frac{2}{5} = \frac{D}{|S|}. \quad (6.10)$$

More generally, whenever $T \subseteq S$ has $|T| = D \leq |S|$, the polynomial $\prod_{a \in T} (X - a)$ attains equality.

When the displayed bound exceeds one. If $D \geq |S|$, (6.2) remains true but may be numerically useless. This is not a proof defect. Over $\mathbb{F}_q$ the nonzero degree- q polynomial

$$X^q - X \quad (6.11)$$

vanishes at every element of $\mathbb{F}_q$, so base-field sampling fails with probability one and the bound $q/q = 1$ is sharp.

6.2.1 A useful nonuniform-coordinate variant

For the machine implementation below it is convenient to sample finite-field elements from binary strings. Exact uniformity is unnecessary.

Lemma 6.3 (Schwartz–Zippel with bounded point masses). *Let $f \neq 0$ have total degree D over a field K . Suppose $A_1, \dots, A_n$ are independent K -valued random variables and*

$$\sup_{a \in K} \Pr[A_i = a] \leq \rho \quad (1 \leq i \leq n). \quad (6.12)$$

Then

$$\Pr[f(A_1, \dots, A_n) = 0] \leq D\rho. \quad (6.13)$$

Proof. Repeat the proof of Theorem 6.2. The induction bound on the leading coefficient is $(D-r)\rho$. Conditional on a good choice of the first $n-1$ coordinates, the final univariate polynomial has at most r roots, and those roots have total probability at most $r\rho$. Adding the two terms gives $D\rho$. $\square$

6.3 Randomized identity testing

Corollary 6.4 (Randomized PIT and the coRP consequence). *Let F be a field and let $f \in F[x_1, \dots, x_n]$ have total degree at most D .*

1. *Let K/F be a field extension and let $S \subseteq K$ be finite and nonempty. If evaluation of f is available on K^n , then one independent uniform sample $A \in S^n$ gives the black-box test*

$$\text{ZERO} \iff f(A) = 0. \quad (6.14)$$

If $f = 0$, the test always returns ZERO. If $f \neq 0$, its false-zero probability is at most $D/|S|$. With r independent repetitions, returning ZERO only when every evaluation vanishes, the false-zero probability is at most

$$\left(\frac{D}{|S|}\right)^r. \quad (6.15)$$

2. *Fix a finite field $F = \mathbb{F}_q$. Under the standard binary encoding of division-free arithmetic circuits over this fixed field, the language*

$$\text{PIT}_{\mathbb{F}_q} = \{C : C \text{ computes the zero polynomial}\} \quad (6.16)$$

belongs to coRP. The algorithm uses an explicitly verified extension field when the base field is too small. Its running time is polynomial in the circuit encoding length; no unit-cost assumption for exponentially large integers is being used.

Proof roadmap. For extension-field evaluation, a nonzero polynomial stays nonzero after the coefficient field is embedded in the extension, so Schwartz–Zippel applies verbatim. For the literal fixed-finite-field *coRP* statement, first bound the formal degree from the circuit, construct and verify an extension of size at least a constant multiple of that degree, sample encoded extension elements with bounded point mass, and repeat enough times to obtain one-sided constant error.

Access-model check. The *coRP* machine has the circuit, so it can reinterpret its gates over the extension field. A black box accepting only points of $\mathbb{F}_q^n$ cannot be silently upgraded to extension-field access; $x^q - x$ is the canonical warning.

Proof. For part 1, embed the coefficients of f from F into K . An injective field embedding cannot turn a nonzero coefficient into zero, so a nonzero polynomial over F remains nonzero over K . Theorem 6.2 gives the one-sample error bound. Independent repetition multiplies the probability of the only possible error.

We now prove the literal *coRP* statement in part 2. Let C have size s . Compute its formal degree Δ gate by gate using the recurrence in Definition 1.1. These degrees are stored in binary. With binary multiplication gates,

$$\Delta \leq 2^s, \quad (6.17)$$

so the degree bookkeeping itself uses only $O(s)$ bits. If $\Delta = 0$, evaluate C once over $\mathbb{F}_q$ and decide deterministically. Hence assume $\Delta \geq 1$.

Choose the least $m \geq 1$ such that

$$N := q^m \geq 8\Delta. \quad (6.18)$$

Then $m = O(s)$ because q is fixed. We shall construct a verified irreducible polynomial $h \in \mathbb{F}_q[T]$ of degree m using only polynomially many random bits and deterministic finite-field arithmetic.

Let $I_q(m)$ denote the number of monic irreducible degree- m polynomials over $\mathbb{F}_q$. The standard factorization

$$T^{q^m} - T = \prod_{d|m} \prod_{\substack{g \text{ monic irreducible} \\ \deg g = d}} g(T) \quad (6.19)$$

follows from the classification/subfield theory of finite fields: the roots are exactly $\mathbb{F}_{q^m}$, the derivative is -1 , and an irreducible polynomial of degree d has its roots in $\mathbb{F}_{q^m}$ exactly when $d \mid m$. Comparing degrees gives

$$q^m = \sum_{d|m} d I_q(d). \quad (6.20)$$

For $m \geq 2$, every proper divisor of m is at most $m/2$, while $d I_q(d) \leq q^d$ by (6.20) applied at degree d . Hence

$$m I_q(m) \geq q^m - \sum_{j=1}^{\lfloor m/2 \rfloor} q^j \geq \frac{q^m}{2}. \quad (6.21)$$

The last inequality is immediate for $m = 2, 3$; for $m \geq 4$ the geometric sum is at most $2q^{m/2} \leq q^m/2$. For $m = 1$, $I_q(1) = q$. Thus in all cases

$$\frac{I_q(m)}{q^m} \geq \frac{1}{2m}. \quad (6.22)$$

A uniform binary string can sample the $N = q^m$ monic degree- m candidates with small, controlled bias. Put $b = \lceil \log_2 N \rceil$, interpret a uniform b -bit integer modulo N , and then expand the resulting index in base q to obtain the m lower coefficients of a monic polynomial. Since $N \leq 2^b < 2N$, every candidate receives between $1/2^b$ and $2/2^b$ probability. Therefore the probability of drawing an irreducible candidate is at least

$$\frac{I_q(m)}{2N} \geq \frac{1}{4m}. \quad (6.23)$$

Irreducibility can be checked deterministically in polynomial time. A monic h of degree m is irreducible exactly when

$$T^{q^m} \equiv T \pmod{h} \quad (6.24)$$

and

$$\gcd(h, T^{q^i} - T) = 1 \quad (1 \leq i \leq \lfloor m/2 \rfloor). \quad (6.25)$$

Indeed, an irreducible degree- m polynomial satisfies these conditions. In the other direction, (6.24) makes h a squarefree divisor of $T^{q^m} - T$. If it were reducible, one irreducible factor would have degree at most $m/2$, and that factor would violate (6.25) at its degree. The residues $T^{q^i} \bmod h$ are obtained successively by fixed- q powering, and Euclidean gcd computation is polynomial in m . Thus the test uses polynomial time for fixed q .

Try $8m$ independent candidates. By (6.23), the probability that all fail is at most

$$\left(1 - \frac{1}{4m}\right)^{8m} \leq e^{-2} < \frac{1}{4}. \quad (6.26)$$

If all candidates fail, the PIT machine returns ZERO. This convention preserves one-sidedness.

Otherwise fix a verified irreducible h and represent

$$K = \mathbb{F}_q[T]/(h), \quad |K| = N. \quad (6.27)$$

Sample each coordinate of $A \in K^n$ by the same b -bit modulo- N map. Every field element has probability at most $2/N$, so Lemma 6.3 and (6.18) give, for every nonzero output polynomial of C ,

$$\Pr[C(A) = 0 \mid h \text{ was found}] \leq \frac{2\Delta}{N} \leq \frac{1}{4}. \quad (6.28)$$

Evaluate the circuit gate by gate in K and return ZERO exactly when the value is zero.

If C computes the zero polynomial, every successful extension-field evaluation is zero, and construction failure also returns ZERO; hence a yes-instance is accepted with probability 1. If C computes a nonzero polynomial, a union bound over extension-construction failure and false-zero evaluation gives

$$\Pr[\text{false acceptance}] < \frac{1}{4} + \frac{1}{4} = \frac{1}{2}. \quad (6.29)$$

This is exactly the *coRP* convention of Definition 1.1.

Finally, all costs are polynomial in the input length. We have $m = O(s)$; there are $O(m)$ candidate trials; each irreducibility test uses polynomially many operations on degree- m polynomials over the fixed field; sampling uses polynomially many bits; and one circuit evaluation uses $O(s)$ operations in K , each representable by polynomially many base-field operations. This proves part 2. $\square$

Why the small-field extension is not a base-field black-box trick. Suppose the oracle answers queries only at points of $\mathbb{F}_q^n$. Then the one-variable black boxes for 0 and $X^q - X$ return exactly the same value on every legal query, even though one polynomial is zero and the other is not. Thus no amount of randomness can solve unrestricted PIT in that oracle model. The extension-field step in the second part of Corollary 6.4 is available because a circuit description can be reinterpreted over an extension field. A genuinely black-box theorem must state extension-field query access as part of the oracle interface.

Arithmetic operations versus bit operations. Equation (6.14) is an algebraic statement: once an evaluation point is supplied, a size- s circuit takes $O(s)$ operations in the sampling field. The *coRP* statement is stronger and therefore fixes a concrete effective setting, namely a fixed finite base field. Its proof accounts for binary random bits, extension construction, polynomial arithmetic, and the $O(s)$ -bit formal-degree bound. No claim is made here that an arbitrary field presented as part of the input has a uniform polynomial-time encoding or extension-field constructor.

6.4 ROABPs, coefficient dimension, and the unknown-order problem

Random evaluation solves unrestricted PIT only probabilistically. A hitting set for a restricted model does more: it replaces the random grid by a deterministic set that intersects the nonzero locus of *every* polynomial in the model. The first structured model we treat is the read-once oblivious algebraic branching program (ROABP).

A width- w ROABP in variable order $\pi(1), \dots, \pi(n)$ can be written in matrix form as

$$f(x) = u^\top D_1(x_{\pi(1)}) D_2(x_{\pi(2)}) \cdots D_n(x_{\pi(n)}) v, \quad (6.30)$$

where $u, v \in F^w$ and each $D_i(X) \in F[X]^{w \times w}$. We say the individual-degree bound is δ when every entry of every D_i has degree at most δ . The order is *unknown* in the black-box model when the hitting set is required to depend only on (n, w, δ, F) , not on π or on the matrices in (6.30).

For a partition of the variables as $y \sqcup z$, expand

$$f(y, z) = \sum_{\alpha} y^{\alpha} f_{\alpha}(z) \quad (6.31)$$

and define the coefficient dimension

$$\text{coeffDim}_{y|z}(f) := \dim_F \text{span}\{f_{\alpha}(z)\}_{\alpha}. \quad (6.32)$$

If y is a prefix of the hidden ROABP order, then

$$\text{coeffDim}_{y|z}(f) \leq w. \quad (6.33)$$

Indeed, after multiplying the prefix matrices, expand the resulting row vector $u^{\top} \prod_{i \leq r} D_i$ into its y -monomials. Every coefficient polynomial $f_{\alpha}(z)$ is a scalar linear combination of the w coordinates of the suffix vector $\prod_{i > r} D_i v$. Thus all coefficient polynomials lie in a common w -dimensional space.

When the sampling field contains more than the y -individual-degree bound, multivariate interpolation identifies (6.32) with the span dimension of the specializations $f(a, z)$ as a ranges over a sufficiently large grid. This is the usual evaluation-dimension intuition for ROABPs. The black-box construction below, however, must work without knowing which subsets are prefixes of the hidden order. Basis isolation supplies exactly that order-oblivious mechanism.

6.5 Basis-isolating weights

Let $\mathcal{A}$ be a finite-dimensional F -algebra and write

$$D(x) = \sum_m D_m m \in \mathcal{A}[x] \quad (6.34)$$

with coefficient vectors $D_m \in \mathcal{A}$. A weight assignment $\omega : \{x_1, \dots, x_n\} \rightarrow \mathbb{N}$ extends additively to monomials.

Definition 6.5 (Basis-isolating weight assignment). The assignment ω is *basis isolating* for D if there is a set S of monomials such that

1. $\{D_m : m \in S\}$ is a basis for the coefficient space of D ;
2. the weights $\omega(m)$, $m \in S$, are pairwise distinct; and
3. for every $m \notin S$,

$$D_m \in \text{span}\{D_{m'} : m' \in S, \omega(m') < \omega(m)\}. \quad (6.35)$$

The point of the definition is that an arbitrary nonzero scalar projection of D cannot cancel its least surviving isolated coefficient.

Lemma 6.6 (Basis isolation leaves a nonzero univariate image). *Let $L : \mathcal{A} \rightarrow F$ be linear and suppose $f = L(D) \neq 0$. If ω is basis isolating for D , then*

$$g_{\omega}(t) := f(t^{\omega(x_1)}, \dots, t^{\omega(x_n)}) \quad (6.36)$$

is a nonzero polynomial in $F[t]$.

Proof. Let S be the isolated basis. Some $m \in S$ has $L(D_m) \neq 0$, since otherwise (6.35) would force $L(D_m) = 0$ for every coefficient of D , contrary to $f \neq 0$. Choose m_* of minimum ω -weight among the basis monomials with $L(D_{m_*}) \neq 0$.

The coefficient of $t^{\omega(m_*)}$ in (6.36) is the sum of $L(D_m)$ over all m with $\omega(m) = \omega(m_*)$. No other basis monomial contributes at that weight. If $m \notin S$ has that weight, (6.35) expresses D_m using basis coefficients of strictly smaller weight, all of whose L -images vanish by the choice of m_* . Hence every such m contributes zero and

$$[t^{\omega(m_*)}]g_\omega = L(D_{m_*}) \neq 0. \quad (6.37)$$

□

We next need a small family of weight functions that can separate any polynomially bounded collection of monomial pairs. The following version keeps the construction completely independent of the hidden ROABP order.

Lemma 6.7 (Universal modular Kronecker family). *Fix $n \geq 1$, $\delta \geq 0$, and an integer $P \geq 1$. Put*

$$M = \max\{2, \delta + 1, nP + 1\}. \quad (6.38)$$

There is a deterministically enumerable family $\mathcal{U}$ of fewer than $2M$ weight assignments

$$\eta : \{x_1, \dots, x_n\} \longrightarrow \{0, 1, \dots, 2M - 1\} \quad (6.39)$$

with the following property: for every collection of at most P pairs (m, m') of distinct monomials of individual degree at most δ , some $\eta \in \mathcal{U}$ satisfies $\eta(m) \neq \eta(m')$ for every prescribed pair.

Proof. By Bertrand's postulate choose a prime p with $M < p < 2M$. For each $a \in \mathbb{F}_p$, define

$$\eta_a(x_i) = \text{the least residue of } a^{i-1} \pmod{p}. \quad (6.40)$$

These p assignments form $\mathcal{U}$.

For a fixed pair $m = x^e \neq m' = x^{e'}$, consider

$$h_{m,m'}(Z) = \sum_{i=1}^n (e_i - e'_i) Z^{i-1} \in \mathbb{F}_p[Z]. \quad (6.41)$$

Because $p > \delta$, at least one nonzero integer coefficient $e_i - e'_i \in [-\delta, \delta]$ stays nonzero modulo p . Thus (6.41) is a nonzero polynomial of degree at most $n - 1$ and has at most $n - 1$ roots. If $\eta_a(m) = \eta_a(m')$ as ordinary integers, then their difference is also zero modulo p , so $h_{m,m'}(a) = 0$. Consequently each prescribed pair rules out at most $n - 1$ choices of a . Fewer than $Pn < p$ choices are ruled out in total, leaving one assignment that separates every pair.

The family is deterministic: one may find a prime in $(M, 2M)$ by enumerating that interval and testing divisibility, then enumerate all residues a . Since our complexity bounds are polynomial in the numerical degree parameter δ , this elementary construction is polynomial in the parameters used here. Bertrand's postulate is used only to keep the modulus $O(M)$. □

6.6 Order-oblivious weight construction

The matrix product in an ROABP is a polynomial over the algebra $\mathcal{A} = M_w(F)$, whose vector-space dimension is

$$k = w^2. \quad (6.42)$$

The next lemma is the heart of the deterministic test.

Lemma 6.8 (Order-oblivious basis-isolating family). *Let $1 \leq d \leq n$ and let $x = X_1 \sqcup \cdots \sqcup X_d$ be an unknown partition into nonempty blocks of at most n variables. Let*

$$D(x) = P_1(X_1)P_2(X_2) \cdots P_d(X_d) \in \mathcal{A}[x], \quad (6.43)$$

where $\dim_F \mathcal{A} \leq k$, every P_i has at most s monomials, and all monomials have individual degree at most δ . Define

$$P_* := 2n(s^2 + k^4), \quad M_* := \max\{2, \delta + 1, nP_* + 1\}, \quad Q := 2M_*, \quad (6.44)$$

$$L := \lceil \log_2 \max\{1, d\} \rceil, \quad B := n(\delta + 1)Q + 1, \quad (6.45)$$

and

$$W := Q \frac{B^{L+1} - 1}{B - 1}. \quad (6.46)$$

There is a family $\mathcal{W}$ depending only on (n, k, s, δ) , not on the partition, factor order, or coefficients, such that

$$|\mathcal{W}| \leq Q^{L+1}, \quad \max_{\omega \in \mathcal{W}, i} \omega(x_i) \leq W, \quad (6.47)$$

and at least one $\omega \in \mathcal{W}$ is basis isolating for D .

Proof. Pad (6.43) by identity factors until the number of factors is 2^L ; this is less than $2d \leq 2n$ and changes neither the polynomial nor the variable set. Apply Lemma 6.7 with the pair budget P_* . It gives one universal candidate family $\mathcal{U}$, of size $< Q$ and variable weights $< Q$, that can be reused at every round.

We construct a lexicographic tuple of weights $(\eta_0, \eta_1, \dots, \eta_L)$, each chosen from $\mathcal{U}$. At round 0, the total number of pairs of support monomials that must be distinguished inside the factors is at most

$$2n \binom{s}{2} < 2ns^2 \leq P_*. \quad (6.48)$$

Choose η_0 that separates all those pairs. In each factor, order the support monomials by η_0 and greedily keep a maximal linearly independent set of coefficients. At most k monomials survive in each factor. Every coefficient discarded from a factor is a linear combination of coefficients of smaller η_0 -weight. By bilinearity of multiplication in $\mathcal{A}$, a global product coefficient containing such a discarded coefficient is a linear combination of global product coefficients of strictly smaller first weight.

Now pair adjacent surviving factors. Before the greedy step in any new block there are at most k^2 candidate product monomials. Across all blocks, the number of pairs that must be distinguished is at most

$$2n \binom{k^2}{2} < 2nk^4 \leq P_*. \quad (6.49)$$

Choose $\eta_1 \in \mathcal{U}$ that separates those pairs, order monomials by the lexicographic pair (η_0, η_1) , and greedily keep a coefficient basis in each block. Again at most k monomials per block survive,

and every discarded coefficient is spanned by coefficients of strictly smaller lexicographic weight. Multiplying by fixed coefficients in blocks to the left or right preserves that linear dependence.

Repeat the same operation. At round r there are half as many blocks as at round $r - 1$; pairing two retained bases creates at most k^2 candidate monomials per block, so (6.49) continues to cover all required pairs. Future weights are lower-priority coordinates and therefore cannot disturb any greedy choice already made. After L pairing rounds only one block remains. Its final greedy basis has at most k coefficients, all with distinct lexicographic weight, and every coefficient discarded at any earlier stage is in the span of coefficients of strictly smaller lexicographic weight. Repeated substitution of these dependence relations leaves only the final basis. Thus the selected tuple is basis isolating in lexicographic form.

It remains to encode the tuple by one integer weight. For any monomial of individual degree at most δ , its weight under a single η_r is less than $n\delta Q < B$. Define

$$\omega(x_i) = \sum_{r=0}^L \eta_r(x_i) B^{L-r}. \quad (6.50)$$

Base B therefore preserves lexicographic comparison of monomial-weight tuples. Hence ω is basis isolating. Enumerating all $(L + 1)$ -tuples from $\mathcal{U}$ gives at most Q^{L+1} candidates, and (6.50) gives the bound (6.46)–(6.47). The family is universal: the hidden partition and order are used only to prove that one of the enumerated tuples works, not to enumerate the family itself. $\square$

Theorem 6.9 (Deterministic black-box PIT for unknown-order ROABPs). *Let F be a field, let $n, w \geq 1$ and $\delta \geq 0$, and let $\mathcal{C}$ be the class of n -variate polynomials computed by width- w ROABPs in an arbitrary and unknown variable order, with individual degree at most δ . Set*

$$s := \delta + 1, \quad k := w^2, \quad (6.51)$$

and let Q, L, B, W be the quantities in (6.44)–(6.46) with $d = n$. Put

$$R := n\delta W. \quad (6.52)$$

If K/F is any field extension with $|K| > R$, choose any $T \subseteq K$ with $|T| = R + 1$. There is a deterministically enumerable hitting set $H \subseteq K^n$ for $\mathcal{C}$ satisfying

$$|H| \leq Q^{L+1}(R + 1) = (nw(\delta + 1))^{O(\log n)}. \quad (6.53)$$

The set depends only on (n, w, δ) and the chosen evaluation field, not on the ROABP variable order. In particular, when $|F| > R$ this is a deterministic base-field black-box PIT algorithm of quasipolynomial running time $(nw(\delta + 1))^{O(\log n)}$.

Proof roadmap. An ROABP of width w has coefficient spaces of dimension at most w across every prefix cut. Choose weights so that a basis of each relevant coefficient space acquires distinct least weights; then a nonzero basis coefficient survives in the univariate specialization. A universal modular Kronecker family supplies candidate separators without knowing the hidden variable order, and repeated pairing reduces the global search to a quasipolynomial family of evaluations.

Hypothesis check. The theorem exploits read-once oblivious structure, bounded width, and bounded individual degree; it is not a hitting set for arbitrary ABPs. The order is unknown to the algorithm but exists in the promise. The final univariate test also requires enough evaluation-field elements, so the small-field boundary is stated explicitly rather than hidden in asymptotic notation.

Proof. Write an unknown-order ROABP as in (6.30) and set

$$D(x) = D_1(x_{\pi(1)}) \cdots D_n(x_{\pi(n)}) \in M_w(F)[x]. \quad (6.54)$$

Each factor is supported on at most $s = \delta + 1$ powers of its one variable, so Lemma 6.8 gives a universal family $\mathcal{W}$ with at least one basis-isolating assignment for D . Let $\Lambda(A) = u^\top A v$. Then $f = \Lambda(D)$. If $f \neq 0$, Lemma 6.6 says that for the good assignment ω ,

$$g_\omega(t) = f(t^{\omega(x_1)}, \dots, t^{\omega(x_n)}) \quad (6.55)$$

is nonzero. Every monomial of f has exponent at most δ in each variable, so by (6.47)

$$\deg g_\omega \leq n\delta W = R. \quad (6.56)$$

Embed the coefficients from F into K ; nonzeroness is preserved. By the univariate root bound, some $a \in T$ satisfies $g_\omega(a) \neq 0$.

Therefore the explicit set

$$H = \left\{ (a^{\omega(x_1)}, \dots, a^{\omega(x_n)}) : \omega \in \mathcal{W}, a \in T \right\} \quad (6.57)$$

hits every nonzero polynomial in $\mathcal{C}$. Its size is bounded by $Q^{L+1}(R+1)$. From (6.44)–(6.46), Q and B are polynomial in n, w, δ , while $L = O(\log n)$; hence both W and the hitting-set size are $(nw(\delta+1))^{O(\log n)}$. Enumeration is deterministic by the explicit separator-family construction and repeated squaring for the powers in (6.57). No step uses the hidden order π . $\square$

Small fields and black-box access. The theorem deliberately states an evaluation-field hypothesis instead of hiding it. If F itself has more than R elements, take $K = F$. Over a small finite field one may instead evaluate an explicit ROABP/circuit over a large enough extension field, but a base-field-only black box cannot in general be queried there. The obstruction $X^q - X$ from (6.11) still applies. Thus Theorem 6.9 is a literal base-field black-box theorem only under its displayed field-size condition, and otherwise an extension-query hitting-set theorem.

6.7 A width-two checkpoint

The coefficient-dimension picture can already be seen in two 2×2 matrices. Over any field let

$$u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad D_x(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad D_y(y) = \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix}. \quad (6.58)$$

Then

$$f(x, y) = u^\top D_x(x) D_y(y) v = 1 - x + xy. \quad (6.59)$$

As a polynomial in the prefix variable x ,

$$f(x, y) = 1 + x(y - 1), \quad (6.60)$$

so the two coefficient polynomials 1 and $y - 1$ are linearly independent and

$$\text{coeffDim}_{x|y}(f) = 2. \quad (6.61)$$

Equivalently, the specializations $f(0, y) = 1$ and $f(1, y) = y$ span a two-dimensional space. A zero polynomial would have coefficient dimension zero at every cut. This is the width witness behind white-box/evaluation-space PIT.

For the black-box viewpoint, the simple weights $\omega(x) = 1$ and $\omega(y) = 2$ give

$$f(t, t^2) = 1 - t + t^3 \neq 0. \quad (6.62)$$

The general theorem does not know that this particular weight works. Its basis-isolation family deterministically enumerates a quasipolynomial list of candidate weights guaranteed to contain some successful one, even when the order x, y is hidden.

6.8 Source and attribution boundary

The randomized polynomial-identity method predates the modern complexity-class language. DeMillo and Lipton gave an early probabilistic algebraic-program identity-testing argument in 1978. Schwartz developed fast probabilistic algorithms for polynomial identities in the 1979 EUROSAM proceedings and the 1980 *Journal of the ACM* version. Zippel’s adjacent EUROSAM paper developed probabilistic algorithms for sparse polynomials. The multivariate degree-over-grid estimate is now conventionally called the Schwartz–Zippel lemma (often also Schwartz–Zippel–DeMillo–Lipton).

Theorem 6.2 is not presented as a verbatim quotation from one historical paper. PASS53 owns the exact total-degree formulation, the $n = 0$ convention, the bounded-point-mass extension, and the explicit induction. Likewise, Corollary 6.4 is a local complexity-theoretic synthesis: its fixed-finite-field extension construction and exact one-sided error budget are proved here rather than attributed to the historical sources. This ownership boundary is important because the phrase “PIT is in *coRP*” otherwise suppresses both the coefficient/field representation and the small-field access model.

For deterministic ROABP PIT, the primary theorem frozen here is Agrawal, Gurjar, Korwar, and Saxena, *Hitting-Sets for ROABP and Sum of Set-Multilinear Circuits*, SIAM Journal on Computing 44(3) (2015), 669–697, DOI 10.1137/140975103; the ECCC/arXiv version is the 2014 report. Their Theorem 1 gives a $\text{poly}(n, w, \delta)^{\log n}$ -time black-box hitting set for width- w ROABPs of individual degree δ even when the variable order is unknown. Section 3 proves the result through coefficient spaces over the matrix algebra, basis-isolating weight assignments, a Kronecker-style separator family, and repeated pairing of factors.

Theorem 6.9 follows that proof architecture but owns its normalization. In particular, (6.44)–(6.57) expose one conservative universal parameter package, explain why the candidate family is independent of the hidden order, and state explicitly how many evaluation-field elements are needed after the monomial-to-univariate substitution. The primary paper’s asymptotic theorem does not isolate this field-size issue in its theorem sentence; its proof says to try enough values of the final univariate parameter. QRCC therefore does not silently claim base-field black-box PIT when the base field is too small. The earlier Forbes–Shpilka theorem gives a known-order quasipolynomial hitting set by a different recursive span-preserving generator; it is historical context rather than the source theorem used for PASS54.

6.9 Reader checkpoint and exercises

Chapter checkpoint. Randomized PIT needs only a degree bound and enough evaluation points; structured deterministic PIT replaces randomness by algebraic information about the coefficient spaces. Small fields create an access-model issue rather than a failure of Schwartz–Zippel. Unknown-order ROABPs remain tractable because a universal family of basis-isolating weights can search over the hidden order with quasipolynomial overhead.

Exercises.

1. For $h(X) = \prod_{a \in T} (X - a)$ with $T \subseteq S$, verify directly that Schwartz–Zippel is tight when $|T| = D \leq |S|$.
2. Prove the bounded-point-mass version for two variables without citing the induction in the text. Identify exactly where independence of the two random variables is used.
3. Over $\mathbb{F}_q$, show that $X^q - X$ is nonzero as a formal polynomial but vanishes on every base-field input. Explain why this does not prevent an explicit circuit from being tested over $\mathbb{F}_{q^m}$.
4. If a circuit has formal degree $\Delta \leq 2^s$, find a sufficient extension degree m so that $q^m \geq 8\Delta$. Bound m in terms of s when q is fixed.
5. For the width-two polynomial $1 - x + xy$ in (6.59), compute the coefficient spaces across both possible variable orders. Which dimensions certify that width two is sufficient?
6. Explain why a hitting set for known-order ROABPs is not automatically a hitting set for unknown-order ROABPs. What part of the basis-isolation construction removes the need to know the order?

Chapter 7

Derandomization Forces Lower Bounds

Polynomial identity testing is not merely an algorithmic problem. Kabanets and Impagliazzo discovered that a sufficiently strong deterministic solution would force a major circuit lower bound. This is a useful warning about the status of PIT: a general polynomial-time derandomization is not expected to be an isolated algorithmic trick, because it would resolve a central lower-bound problem at the same time.

The theorem has two logically different escape routes. Either Boolean circuits already fail to capture $NEXP$, or the permanent fails to have polynomial-size arithmetic circuits. We keep both alternatives visible and also keep the coefficient model visible: the local proof below uses *division-free arithmetic circuits over $\mathbb{Z}$ whose integer constants are charged by their bit lengths*. This is enough for the registered QRCC theorem. The primary Kabanets–Impagliazzo paper proves a stronger rational-circuit variant with division after a separate constant-division elimination argument.

How to read this chapter. This is a *consequence theorem*: deterministic PIT is the hypothesis, not the conclusion. The proof assumes both advertised lower-bound alternatives fail and then uses PIT to certify guessed permanent circuits. That converts an exact permanent oracle into nondeterministic certificates, after which three source-owned complexity-class interfaces force the contradiction.

Complexity-class recall. Here $P/poly$ denotes nonuniform polynomial-size Boolean circuits; $NEXP$ is nondeterministic time $2^{n^{O(1)}}$; NP is nondeterministic polynomial time; MA is Merlin–Arthur; PH is the polynomial hierarchy; and $\#P$ is the counting class whose canonical complete functions count accepting NP computation paths. These classes were defined earlier in the series; they are recalled here because every arrow in the final containment chain carries a different source ownership.

Learning goals. A reader should be able to explain how the downward self-reduction makes a candidate permanent circuit checkable by PIT, why exact permanent values have polynomial bit length on Boolean matrices, reconstruct the class-containment contradiction, and state precisely which stronger rational-circuit variant is left to the primary Kabanets–Impagliazzo source.

7.1 Three source-owned classical interfaces

The synthesis uses three complexity-theoretic inputs that are classical but not reproved here.

1. **Easy witnesses.** If $NEXP \subseteq P/poly$, then $NEXP = MA$. This is the Impagliazzo–Kabanets–Wigderson easy-witness consequence used explicitly by the primary proof.
2. **Toda plus Valiant.** The integer permanent is $\#P$ -complete and $PH \subseteq P^{\#P}$, hence

$$PH \subseteq P^{\text{Perm}}. \quad (7.1)$$

Moreover $MA \subseteq PH$ in the source’s complexity-class normalization.

3. **Nondeterministic time hierarchy.** $NEXP \not\subseteq NP$. Equivalently, there is a nondeterministic exponential-time language outside every polynomial nondeterministic time bound.

The first and second items are used only for the Boolean-complexity implication. The arithmetic portion of the proof is elementary once deterministic PIT is available.

7.2 Certifying a permanent circuit with PIT

For an $r \times r$ variable matrix $X = (x_{ij})$, write $X^{(j)}$ for the $(r-1) \times (r-1)$ matrix obtained by deleting row 1 and column j . The permanent is characterized recursively by

$$\text{Perm}_1(x) = x, \quad \text{Perm}_r(X) = \sum_{j=1}^r x_{1j} \text{Perm}_{r-1}(X^{(j)}). \quad (7.2)$$

This recurrence is a self-correction interface: if candidate circuits satisfy it as formal polynomial identities at every size, induction forces them to be the permanent.

Lemma 7.1 (Downward self-reduction gives a PIT certificate). *Assume deterministic polynomial-time identity testing for division-free arithmetic circuits over $\mathbb{Z}$, with circuit size including the bit lengths of integer constants. Given a polynomial-size circuit C_r in the r^2 variables x_{ij} , one can verify in deterministic polynomial time, using polynomially many PIT calls, whether $C_r \equiv \text{Perm}_r$ provided candidate circuits $C_1, \dots, C_{r-1}$ are supplied as part of the certificate and all have polynomial size.*

Proof. Test

$$C_1(x) - x \equiv 0 \quad (7.3)$$

and, for every $2 \leq i \leq r$, test

$$C_i(X) - \sum_{j=1}^i x_{1j} C_{i-1}(X^{(j)}) \equiv 0. \quad (7.4)$$

Variable deletion/renaming and the sum in (7.4) enlarge the circuit only polynomially. If every identity holds, induction on i and (7.2) gives $C_i \equiv \text{Perm}_i$ for every $i \leq r$. Conversely the genuine permanent circuits satisfy every test. $\square$

The certificate is useful only if a verified circuit can be evaluated without creating exponentially long intermediate integers. Modular arithmetic removes that problem.

Lemma 7.2 (Permanent graph enters NP). *Assume deterministic polynomial-time PIT as above and assume that Perm_r has polynomial-size division-free arithmetic circuits over $\mathbb{Z}$ for every r . Then the graph language*

$$G_{\text{Perm}} := \{(M, v) : M \in \{0, 1\}^{r \times r}, v = \text{Perm}(M)\} \quad (7.5)$$

belongs to NP .

Proof. On input (M, v) , guess polynomial-size circuits $C_1, \dots, C_r$ and verify them by Lemma 7.1. At least one correct guess exists by hypothesis, and an accepted guess necessarily computes the permanent as a formal integer polynomial.

Let

$$b := \lceil r \log_2 r \rceil + 1, \quad N := 2^b. \quad (7.6)$$

For a 0–1 matrix, $0 \leq \text{Perm}(M) \leq r! < r^r < N$. Evaluate the accepted circuit $C_r(M)$ gate by gate modulo N . Every residue has $O(r \log r)$ bits, so the evaluation is polynomial time even if the corresponding unreduced integer intermediates would be enormous. Because $C_r(M) = \text{Perm}(M)$ as an integer and the latter lies in $[0, N)$, the residue recovers the exact answer. Accept precisely when it equals the supplied binary integer v . $\square$

Why the modulus may be composite. PIT has already certified an identity over $\mathbb{Z}$. The modulus in the last step is used only for bounded-bit evaluation, not for identity testing. Thus 2^b is perfectly legitimate and no primality algorithm is hidden in the proof. The stronger Kabanets–Impagliazzo treatment of circuits with divisions needs a prime avoiding a nonzero denominator constant; that extra issue is absent in the division-free QRCC theorem.

Lemma 7.3 (An exact permanent oracle with NP certificates collapses its polynomial-time oracle class). *If $G_{\text{Perm}} \in NP$, then*

$$P^{\text{Perm}} \subseteq NP. \quad (7.7)$$

Proof. Fix a deterministic polynomial-time oracle machine A^{Perm} . A nondeterministic simulation guesses, in order, the exact binary answer to each oracle query together with an NP witness that the guessed value is the permanent of the queried 0–1 matrix. Only polynomially many queries of polynomial dimension occur, and each answer has polynomial bit length because it is at most $r!$. A false answer has no valid graph witness, so every accepting branch follows the unique genuine oracle computation. A branch using all correct answers exists. Hence the oracle computation is simulated in NP . $\square$

7.3 The PIT-to-lower-bounds theorem

Theorem 7.4 (Kabanets–Impagliazzo PIT-to-lower-bounds theorem). *Suppose arithmetic-circuit identity testing over $\mathbb{Z}$ for explicitly encoded division-free circuits is decidable deterministically in polynomial time. Then at least one of the following holds:*

$$\boxed{NEXP \not\subseteq P/\text{poly}},$$

or

$$\boxed{\{\text{Perm}_n\} \text{ has no polynomial-size division-free arithmetic circuits over } \mathbb{Z}.}$$

(7.8)

The first alternative is a Boolean circuit lower bound; the second is an arithmetic circuit lower bound. No implication between the two alternatives is asserted.

Proof roadmap. Argue by contradiction. Assume both lower-bound alternatives fail. Small permanent circuits plus deterministic PIT make the permanent recurrence certifiable, placing exact permanent evaluation on Boolean matrices inside an NP-verifiable interface. Combine this with the easy-witness theorem and Toda/Valiant to obtain

$$NEXP \subseteq MA \subseteq PH \subseteq P^{\text{Perm}} \subseteq NP,$$

contradicting the nondeterministic time hierarchy.

Hypothesis check. The local theorem uses explicitly encoded division-free integer circuits, with constant bit lengths charged to size. The coefficient model matters both for PIT running time and for the permanent recurrence check. The stronger source variant allowing rational circuits with division is not inferred for free.

Proof. Assume for contradiction that both alternatives fail. Thus $NEXP \subseteq P/\text{poly}$ and the permanent has polynomial-size arithmetic circuits of the registered type.

By the easy-witness theorem,

$$NEXP = MA \subseteq PH. \quad (7.9)$$

Valiant's $\#P$ -completeness of permanent together with Toda's theorem gives $PH \subseteq P^{\text{Perm}}$. Lemmas 7.2 and 7.3 therefore imply

$$NEXP \subseteq PH \subseteq P^{\text{Perm}} \subseteq NP. \quad (7.10)$$

But $NP \subseteq NEXP$ trivially, so (7.10) would give $NEXP = NP$, contradicting the nondeterministic time hierarchy. Hence at least one alternative in (7.8) must hold. $\square$

A useful contrapositive reading. If one simultaneously believes that $NEXP \subseteq P/\text{poly}$ and that permanent has polynomial-size arithmetic circuits, then general deterministic PIT cannot run in polynomial time. The theorem does not identify which of the two lower bounds is true; it proves that derandomization cannot avoid proving at least one of them.

7.4 A tiny recurrence check

For

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the recursion (7.2) reads

$$\text{Perm}_2(X) = ad + bc. \quad (7.11)$$

Suppose a guessed circuit claims instead to compute $ad - bc$. The PIT certificate tests

$$(ad - bc) - (ad + bc) = -2bc. \quad (7.12)$$

Over $\mathbb{Z}$ this is not the zero polynomial, so the candidate is rejected. The example also illustrates why the coefficient domain matters: reducing the same identity modulo characteristic two would erase the sign distinction. Theorem 7.4 is specifically an integer-circuit theorem.

7.5 Source and ownership boundary

The primary source is Valentine Kabanets and Russell Impagliazzo, *Derandomizing Polynomial Identity Tests Means Proving Circuit Lower Bounds*, Computational Complexity 13 (2004), 1–46, DOI 10.1007/s00037-004-0182-6; ECCC TR02-055 is the full prepublication version. The source proves the dichotomy in (7.8), and in fact treats polynomial-size rational arithmetic circuits with divisions as well. Its Section 3 proves the permanent self-reduction-to-ACIT interface and uses modular evaluation to control intermediate integer bit length; Section 4 derives the lower bound from derandomization.

PASS55 locally reconstructs the division-free core needed by the QRCC circuit model. The easy-witness implication $NEXP \subseteq P/poly \Rightarrow NEXP = MA$, Valiant’s $\#P$ -completeness theorem, Toda’s $PH \subseteq P^{\#P}$ theorem, and the nondeterministic hierarchy remain source-owned complexity inputs. In particular, the phrase “PIT implies lower bounds” should not be read as an elementary consequence of Schwartz–Zippel: the permanent verification lemma and the complexity-class chain are essential.

7.6 Reader checkpoint and exercises

Chapter checkpoint. The theorem is a bridge from derandomization to lower bounds, not a direct construction of a hard polynomial. PIT makes a downward-self-reducible permanent circuit verifiable; classical complexity inclusions then convert the simultaneous failure of both lower-bound alternatives into the impossible collapse $NEXP \subseteq NP$. Every arrow in that argument has its own model and source boundary.

Exercises.

1. Write the full first-row permanent recurrence for a 3×3 matrix and list the PIT identities needed to certify candidate circuits C_1, C_2, C_3 .
2. For a Boolean $r \times r$ matrix, prove that its permanent lies between 0 and $r!$ and therefore has $O(r \log r)$ output bits.
3. Starting from the negation of both alternatives in (7.8), reconstruct the containment chain ending in $NEXP \subseteq NP$. Annotate each arrow with the lemma or source-owned theorem that supplies it.
4. Explain why the 2×2 determinant candidate in (7.12) fails over $\mathbb{Z}$ but passes that single recurrence check after reduction modulo two. Why is this not a contradiction to the theorem?
5. State the contrapositive of Theorem 7.4 in words. Which pair of simultaneous upper-bound beliefs would force polynomial-time PIT to fail?
6. The primary source treats rational circuits with division. List the extra issues that must be controlled before the local division-free integer proof can be promoted to that stronger coefficient model.

Chapter 8

Noncommutative Rank Measures and White-Box PIT

Removing commutativity makes some algebraic problems easier to lower-bound and identity-test. Words have a unique left-to-right order, so a cut after k letters creates a canonical coefficient matrix. Nisan used the ranks of these matrices to characterize homogeneous noncommutative ABP size and obtained exponential lower bounds for determinant and permanent. Raz and Shpilka later turned the same linear-algebraic rigidity into deterministic polynomial-time white-box PIT.

Throughout this chapter F is an arbitrary field and $F\langle x_1, \dots, x_N \rangle$ is the free associative algebra. Two words are equal only when their complete ordered variable sequences agree.

How to read this chapter. Noncommutativity supplies a canonical notion of position. Cutting every word after k letters produces a coefficient matrix $M_k(f)$, and its rank measures how many independent prefix states an ABP needs at that cut. The first half of the chapter turns this observation into an exact ABP-size formula and an exponential formula lower bound. The second half uses essentially the same finite-dimensional span invariant algorithmically: instead of expanding a polynomial into exponentially many words, white-box PIT maintains bases of the reachable coefficient spaces layer by layer.

The key contrast with Chapter 6 is access. Raz–Shpilka PIT reads the ABP edge labels, so it is deterministic polynomial-time *white-box* PIT. It does not provide a black-box hitting set for general noncommutative ABPs.

Learning goals. A reader should be able to construct $M_k(f)$ for a small noncommutative polynomial, prove that an ABP layer factors this matrix, compute the subset-block ranks of noncommutative determinant/permanent, and run the layerwise coordinate basis algorithm on a small ABP without expanding all words.

8.1 Nisan’s coefficient matrices

Let f be homogeneous of degree d . For $0 \leq k \leq d$, define $M_k(f)$ as the $N^k \times N^{d-k}$ matrix whose rows are degree- k words u , whose columns are degree- $(d-k)$ words v , and whose entry is

$$M_k(f)[u, v] = [uv]f, \tag{8.1}$$

the coefficient of the concatenated word uv in f .

Lemma 8.1 (Cut rank lower-bounds an ABP layer). *Let a homogeneous degree- d noncommutative ABP have a layer V_k after exactly k variable-labelled steps and compute f . Then*

$$\text{rank } M_k(f) \leq |V_k|. \quad (8.2)$$

Consequently every such ABP has at least $\sum_{k=0}^d \text{rank } M_k(f)$ vertices when size is counted as the sum of layer widths.

Proof. For $v \in V_k$, let p_v be the degree- k polynomial computed from the source to v , and let q_v be the degree- $(d-k)$ polynomial computed from v to the sink. Every source-sink path crosses exactly one $v \in V_k$, so

$$f = \sum_{v \in V_k} p_v q_v. \quad (8.3)$$

Let a_v be the coefficient column of p_v indexed by degree- k words and $b_v^\top$ the coefficient row of q_v indexed by degree- $(d-k)$ words. Concatenation is unique in the free algebra, hence

$$M_k(f) = \sum_{v \in V_k} a_v b_v^\top. \quad (8.4)$$

The right side has rank at most $|V_k|$. □

For completeness we also record the matching upper bound, which is the precise form of Nisan's characterization used below.

Lemma 8.2 (Minimal homogeneous ABP size). *For homogeneous f of degree d , there is a homogeneous noncommutative ABP whose layer- k width is exactly $r_k := \text{rank } M_k(f)$. Hence the minimum layered-vertex size equals*

$$B(f) = \sum_{k=0}^d r_k. \quad (8.5)$$

Proof. For each k , consider the vector space L_k of left residuals obtained by fixing a degree- k prefix:

$$f_u := \sum_{|v|=d-k} [uv] f v. \quad (8.6)$$

The coefficient rows of these residuals are exactly the rows of $M_k(f)$, so $\dim L_k = r_k$. Choose a basis $g_{k,1}, \dots, g_{k,r_k}$ of L_k .

Decompose every residual according to its first remaining letter. The left-letter coefficient maps induced by this decomposition give linear maps

$$T_{k,j} : L_k \longrightarrow L_{k+1} \quad (1 \leq j \leq N). \quad (8.7)$$

For $g \in L_k$ write

$$g = \sum_{j=1}^N x_j \Delta_j(g), \quad \Delta_j(g) \in L_{k+1}. \quad (8.7a)$$

Then $T_{k,j} = \Delta_j$. Write the matrices of these maps in the chosen bases. Starting from the coordinate vector of $f \in L_0$, reading a prefix $x_{i_1} \cdots x_{i_k}$ applies $T_{k-1,i_k} \cdots T_{0,i_1}$ and produces the coordinates of the corresponding residual. The unique basis vector of $L_d = F \cdot 1$ supplies the final scalar functional. Turning the matrix entries into edge weights therefore gives a layered ABP whose path coefficient for each word is exactly its coefficient in f . Hence it computes f and has widths r_k . Together with Lemma 8.1, this gives (8.5). □

8.2 Permanent and determinant have one rank unit per column-subset block

Define the row-ordered noncommutative permanent and determinant by

$$\text{ncPerm}_n = \sum_{\sigma \in S_n} x_{1,\sigma(1)} x_{2,\sigma(2)} \cdots x_{n,\sigma(n)}, \quad (8.8)$$

$$\text{ncDet}_n = \sum_{\sigma \in S_n} \text{sgn}(\sigma) x_{1,\sigma(1)} x_{2,\sigma(2)} \cdots x_{n,\sigma(n)}. \quad (8.9)$$

These are formal noncommutative polynomials even when the underlying matrix entries are named as in the commutative determinant.

Lemma 8.3 (Exact cut ranks). *For every field F , every $0 \leq k \leq n$, and either polynomial $f \in \{\text{ncPerm}_n, \text{ncDet}_n\}$,*

$$\text{rank } M_k(f) = \binom{n}{k}. \quad (8.10)$$

In characteristic two the two polynomials coincide, and the same formula remains valid.

Proof. A nonzero row of $M_k(f)$ is a word using exactly the variables from rows $1, \dots, k$ and some k -element column set $S \subseteq [n]$. A nonzero column must use rows $k+1, \dots, n$ and the complementary column set $[n] \setminus S$. Thus, after deleting zero rows/columns and grouping by S , $M_k(f)$ is block diagonal with one block for each $S \in \binom{[n]}{k}$.

For the permanent, every entry inside a surviving block is 1: any prefix bijection $[k] \rightarrow S$ concatenated with any suffix bijection $\{k+1, \dots, n\} \rightarrow [n] \setminus S$ is one permutation. Hence each block is an all-ones matrix of rank one.

For the determinant, fix S . If α is the prefix bijection and β the suffix bijection, the sign of the assembled permutation factors as

$$\text{sgn}(\sigma) = \varepsilon_S a_\alpha b_\beta, \quad (8.11)$$

where $\varepsilon_S \in \{\pm 1\}$ accounts for the crossings between S and its complement, while a_α and b_β are the signs relative to the increasing orders inside the two sets. The block is therefore the nonzero outer product $\varepsilon_S(a_\alpha)_\alpha(b_\beta)_\beta^\top$ and again has rank one. Over characteristic two the signs are all 1.

There are exactly $\binom{n}{k}$ blocks, proving (8.10). $\square$

Lemma 8.4 (Formula-to-ABP conversion). *A noncommutative formula with s internal gates can be converted to a source–sink noncommutative ABP with at most $2s + 2$ vertices computing the same polynomial. If the output is homogeneous of degree d , homogeneous-component splitting yields a homogeneous layered ABP with only a polynomial factor in s and d ; for the formula lower bound below it is enough to use the standard series–parallel construction directly on a syntactically homogeneous formula.*

Proof. Associate to a leaf a single labelled edge. For an addition gate, put the two child ABPs in parallel between a common source and sink. For an ordered multiplication gate $g = h_1 h_2$, concatenate the sink of the ABP for h_1 with the source of the ABP for h_2 . Because a formula is a tree, no subgraph is shared and the number of vertices is linear in the number of formula gates. The order of concatenation preserves noncommutative multiplication. $\square$

Theorem 8.5 (Nisan’s noncommutative formula lower bound). *Over every field, any syntactically homogeneous noncommutative formula for ncPerm_n or ncDet_n has size $2^{\Omega(n)}$. More precisely, every homogeneous noncommutative ABP for either polynomial has layered-vertex size exactly*

$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad (8.12)$$

and hence every formula has size at least $(2^n - 2)/2$ under the conversion normalization of Lemma 8.4.

Proof roadmap. Nisan’s exact ABP-size characterization says that the minimum homogeneous ABP size is the sum of the cut ranks $\text{rank } M_k(f)$. For the row-ordered noncommutative permanent and determinant, the k th matrix breaks into one rank-one block for each k -subset of columns, giving rank $\binom{n}{k}$. Summing over k yields 2^n . A homogeneous formula converts to an ABP with only linear overhead, so the same rank count forces exponential formula size.

Hypothesis check. The formula is syntactically homogeneous in the registered theorem. The rank calculation works over every field, including characteristic two; in that characteristic determinant and permanent signs coincide, but the word-order rank argument still gives the same exponential noncommutative lower bound.

Proof. Lemma 8.3 and Nisan’s characterization (8.5) give the exact ABP size 2^n . A formula of size s would yield an ABP with at most $2s + 2$ vertices, so $2s + 2 \geq 2^n$. $\square$

Why this is genuinely noncommutative. The coefficient matrix $M_k(f)$ uses the unique cut between the k th and $(k + 1)$ st letters. In the commutative ring a monomial has many reorderings and there is no analogous canonical word cut. Thus Theorem 8.5 is not a corollary of Chapter 5’s commutative partial-derivative method; it is a separate lower-bound mechanism.

8.3 White-box PIT for noncommutative ABPs

A layered homogeneous noncommutative ABP has vertices $V_0 = \{s\}, V_1, \dots, V_d = \{t\}$. Every edge from V_{i-1} to V_i carries a homogeneous linear form

$$\ell_e = \sum_{j=1}^N a_{e,j} x_j. \quad (8.13)$$

For $v \in V_i$, let p_v be the polynomial computed by all source– v paths. The ABP is zero exactly when $p_t = 0$. Expanding the polynomials is impossible in general, but their span can be represented recursively by coordinates.

Lemma 8.6 (Injective last-letter decomposition). *If $q_1, \dots, q_N$ are homogeneous noncommutative polynomials of the same degree, then*

$$\sum_{j=1}^N q_j x_j = 0 \iff q_1 = \dots = q_N = 0. \quad (8.14)$$

Proof. Words ending in different variables are disjoint basis elements of the free algebra. The coefficient of every word ux_j on the left is exactly the coefficient of u in q_j . $\square$

Algorithm 8.7 (Layerwise span PIT). *Maintain for each layer i a basis $B_i = (b_{i,1}, \dots, b_{i,r_i})$ of $\text{span}\{p_v : v \in V_i\}$ together with coordinates of every p_v in that basis. Start with $B_0 = (1)$. Suppose coordinates are known at layer $i - 1$. For each $v \in V_i$, write*

$$p_v = \sum_{u \in V_{i-1}} p_u \ell_{uv} = \sum_{j=1}^N q_{v,j} x_j. \quad (8.15)$$

Each $q_{v,j}$ is a known linear combination of B_{i-1} . Encode p_v by the concatenated coordinate vector

$$c_v := ([q_{v,1}]_{B_{i-1}}, \dots, [q_{v,N}]_{B_{i-1}}) \in F^{Nr_{i-1}}. \quad (8.16)$$

Row-reduce the vectors $\{c_v : v \in V_i\}$, choose pivot vectors as B_i , and store the resulting coordinates. At the final layer output ZERO iff the coordinate vector of p_t is zero.

Lemma 8.8 (Correctness and size of the span representation). *Algorithm 8.7 maintains exactly the polynomial span at every layer, with*

$$r_i \leq |V_i|. \quad (8.17)$$

It uses polynomially many field operations in the ABP description size, N , and d .

Proof. By induction, the coordinates of every $q_{v,j}$ in (8.15) are exact. Lemma 8.6 says that the linear map

$$(q_1, \dots, q_N) \mapsto \sum_j q_j x_j \quad (8.18)$$

is injective. Therefore a linear relation among the encoded vectors c_v is exactly the same linear relation among the polynomials p_v . Gaussian elimination consequently chooses a genuine basis and computes exact coordinates. The vector length is $Nr_{i-1} \leq N|V_{i-1}|$ and there are $|V_i|$ vectors, so standard elimination at each of d layers costs only a polynomial number of field operations. At the sink, its coordinate is zero exactly when the output polynomial is zero. $\square$

Theorem 8.9 (Raz–Shpilka deterministic white-box PIT). *There is a deterministic polynomial-time white-box identity test for layered homogeneous noncommutative algebraic branching programs over an arbitrary field. Given the ABP graph and all edge coefficients, the algorithm decides whether its formal polynomial in $F\langle x_1, \dots, x_N \rangle$ is zero by layerwise linear algebra. A general noncommutative ABP/formula of polynomial formal degree is reduced to this form by the standard homogeneous-component layering with polynomial overhead.*

Proof roadmap. Process the ABP layer by layer. Represent each vertex polynomial in coordinates relative to a basis of the current layer’s span. Multiplying by an edge label creates a tuple of previous-layer coordinate vectors indexed by the last letter; injectivity of last-letter decomposition turns polynomial equality into ordinary linear dependence among those tuples. Gaussian elimination keeps a basis of polynomial dimension, and the sink coordinate is zero exactly when the output polynomial is zero.

Access-model check. Every edge coefficient is read explicitly, so the algorithm is white-box. No claim is made that evaluation access alone reveals the coordinate bases used in the proof. Homogeneous components handle nonhomogeneous inputs with polynomial overhead in formal degree.

Proof. For the homogeneous layered input, use Algorithm 8.7 and Lemma 8.8. For a general polynomial-formal-degree ABP, split each vertex state into its homogeneous components and route a degree- r edge contribution to the appropriate degree layer; affine labels are split into constant and linear parts. The number of states grows by at most a factor polynomial in the formal degree. Test every output homogeneous component. The polynomial is zero iff all components are zero. $\square$

White-box really matters. The algorithm uses every edge coefficient to construct (8.16). It is therefore not a black-box hitting set. Chapter 6’s ROABP theorem solves a different problem: it receives only evaluation access but exploits the much stronger read-once-oblivious structure and pays quasipolynomial time.

8.4 A three-layer worked trace

Take variables x, y and the width-two ABP whose first layer computes

$$p_a = x + y, \quad p_b = x - y. \quad (8.19)$$

With basis $B_1 = (p_a, p_b)$, let the sink polynomial be

$$p_t = p_a x + p_b y. \quad (8.20)$$

The last-letter blocks are $q_x = p_a$ and $q_y = p_b$, so

$$c_t = ((1, 0), (0, 1)) \neq 0. \quad (8.21)$$

The algorithm declares the output nonzero without expanding $x^2 + yx + xy - y^2$. If the second term in (8.20) were replaced by $-p_a x$, then the two x contributions would cancel at the coordinate-vector level and the zero output would be detected by Gaussian elimination.

8.5 Source and ownership boundary

Noam Nisan’s *Lower Bounds for Non-Commutative Computation*, STOC 1991, 410–418, DOI 10.1145/103418.103462, introduces the matrices $M_k(f)$ and proves that minimum homogeneous noncommutative ABP size is the sum of their ranks. The paper derives tight exponential bounds for determinant/permanent and the resulting formula lower bounds. PASS55 reconstructs the row-ordered permanent/determinant block-rank calculation in Lemma 8.3, including the characteristic-two boundary.

Ran Raz and Amir Shpilka, *Deterministic Polynomial Identity Testing in Non Commutative Models*, Computational Complexity 14 (2005), 1–19, DOI 10.1007/s00037-005-0188-8, give deterministic polynomial-time white-box PIT for noncommutative formulas and, in particular, noncommutative ABPs. PASS55 uses the ABP specialization and spells out the layerwise coordinate-basis invariant. The source’s broader formula and pure set-multilinear-circuit results are historical context, not silently promoted as additional QRCC theorems here.

8.6 Reader checkpoint and exercises

Chapter checkpoint. The coefficient matrix $M_k(f)$ is both a lower-bound witness and an algorithmic state space. Its rank forces width at an ABP cut; for noncommutative permanent/determinant

those ranks sum to 2^n . In the PIT direction, explicit edge labels let the algorithm maintain only a basis of the same reachable span, turning equality in the free algebra into ordinary Gaussian elimination.

Exercises.

1. For $f = xy + yx + xz$ in $F\langle x, y, z \rangle$, write $M_1(f)$ explicitly and compute its rank over an arbitrary field.
2. Given a width- w homogeneous ABP, derive the factorization of $M_k(f)$ through the vector of polynomials computed at layer k . Conclude $\text{rank } M_k(f) \leq w$.
3. For $n = 3$, identify the column-subset blocks contributing to $M_1(\text{ncPerm}_3)$ and $M_2(\text{ncPerm}_3)$. Verify the ranks 3 and 3 without expanding all six words into a single list.
4. Convert the noncommutative formula $(x + y)(z + x) + yz$ into a series-parallel ABP. Count vertices and check that multiplication order is preserved.
5. Run one layer of Algorithm 8.7 on a width-two ABP of your choice. Write the coordinate tuples that are sent to Gaussian elimination.
6. Explain, by reference to the data used by the algorithm, why the theorem is white-box. What information would be unavailable if you were given only an evaluation oracle?

Chapter 9

Operator Scaling, Noncommutative Rank, and the Null Cone

The previous chapter treated noncommutative polynomials by finite-dimensional coefficient spaces. A more surprising bridge appears for a symbolic matrix

$$L_A(x) = x_1 A_1 + \cdots + x_m A_m.$$

Its invertibility over the free skew field, a geometric invariant-theory null cone, a linear-algebraic obstruction called a shrunk subspace, and positivity of an analytic determinant potential all describe the same full-rank phenomenon. Operator scaling turns that equivalence into an algorithm.

The algebraic equivalences and the analytic algorithm live over slightly different coefficient models. We therefore fix them explicitly: structural orbit-closure statements are made over $\mathbb{C}$; the Turing-model algorithm receives rational (after clearing denominators, integral) matrices. No claim of a polynomial-time analytic algorithm over an arbitrary field is intended.

How to read this chapter. This chapter is a dictionary followed by an algorithm. The dictionary says that five apparently different full-rank conditions—free-skew-field invertibility, absence of shrunk subspaces, null-cone nonmembership, rank non-decrease, and positive capacity—coincide at the relevant boundary. Operator scaling then alternately fixes the two marginals while capacity acts as a potential. The scalar Sinkhorn–Knopp theorem at the end is the commutative diagonal model that makes the same normalization mechanism visible without noncommuting matrices.

A useful dependency chain is

$$\begin{aligned} &\text{shrunk subspaces} / \text{null cone} / \text{ncrk} \iff \text{cap} > 0 \\ &\longrightarrow \text{scaling progress} \longrightarrow \text{deterministic full-rank decision.} \end{aligned}$$

The equivalence arrow uses deep invariant-theory input; the progress and normalization calculations around it are locally exposed. Do not read the algorithm as computing every possible sub-full noncommutative rank value: its registered promise is the full-versus-non-full decision problem.

Learning goals. A reader should be able to translate a matrix tuple among the linear-pencil, completely-positive-map, capacity, and null-cone languages; perform one operator normalization step; explain why a shrunk subspace drives capacity to zero; separate mathematical convergence from rational bit-complexity control; and state the support/total-support/full-indecomposability distinctions in scalar matrix scaling.

9.1 The matrix-space dictionary

Definition 9.1 (Matrix spaces, noncommutative rank, shrunk subspaces, and capacity). Let $A = (A_1, \dots, A_m)$ with $A_i \in M_n(\mathbb{C})$.

1. The *linear pencil* is

$$L_A(x) = \sum_{i=1}^m x_i A_i \in M_n(\mathbb{C}\langle x_1, \dots, x_m \rangle). \quad (9.1)$$

Its *noncommutative rank*, $\text{ncrk}(L_A)$, is its rank over the free skew field $\mathbb{C}\langle\langle x \rangle\rangle$. It has *full noncommutative rank* when this rank is n .

2. A pair of subspaces $U, W \leq \mathbb{C}^n$ is a *c-shrunk subspace certificate* when

$$A_i U \subseteq W \quad (1 \leq i \leq m), \quad \dim U - \dim W = c > 0. \quad (9.2)$$

3. The left–right action of $G = SL_n(\mathbb{C}) \times SL_n(\mathbb{C})$ is

$$(B, C) \cdot A = (BA_1 C^{-1}, \dots, BA_m C^{-1}). \quad (9.3)$$

The tuple is in the *null cone* when 0 lies in the Zariski/equivalently Euclidean closure of its G -orbit.

4. The associated completely positive operator and its Hilbert–Schmidt adjoint are

$$T_A(X) = \sum_i A_i X A_i^*, \quad T_A^*(Y) = \sum_i A_i^* Y A_i. \quad (9.4)$$

It is *rank non-decreasing* when $\text{rank } T_A(X) \geq \text{rank } X$ for every positive semidefinite X .

5. Its *capacity* is

$$\text{cap}(T_A) := \inf\{\det T_A(X) : X \succ 0, \det X = 1\}. \quad (9.5)$$

The tuple/operator is *doubly stochastic* when $T_A(I) = T_A^*(I) = I$. Its discrepancy is

$$\text{ds}(T_A) := \text{tr}(T_A(I) - I)^2 + \text{tr}(T_A^*(I) - I)^2. \quad (9.6)$$

For analytic scaling we also allow $A_i^{B,C} = BA_i C$ with arbitrary invertible B, C ; using C instead of C^{-1} does not change the family of left–right orbits when the determinant-one restriction is imposed.

Why positivity appears. The free-skew-field rank is algebraic, while (9.5) is an optimization over positive definite matrices. The connection is not notation: it is the content of the Gurvits/operator-scaling theorem below.

9.2 Structural equivalence

Lemma 9.2 (Shrunk subspaces are exactly rank-decrease witnesses). *The tuple A has a shrunk subspace iff T_A is rank decreasing on some positive semidefinite matrix.*

Proof. If $A_i U \subseteq W$ with $\dim W < \dim U$, take the orthogonal projection P_U . Then

$$\text{range } T_A(P_U) = \text{span}_i A_i U \subseteq W, \quad (9.7)$$

so $\text{rank } T_A(P_U) < \text{rank } P_U$.

Conversely, let $X \succeq 0$ and put $U = \text{range } X$. Since X is positive definite on U ,

$$\text{range } T_A(X) = \text{span}_i A_i U =: W. \quad (9.8)$$

If T_A decreases rank, then $\dim W < \dim U$, giving a shrunk subspace. $\square$

Lemma 9.3 (A shrunk subspace forces zero capacity). *If A has a shrunk subspace, then $\text{cap}(T_A) = 0$.*

Proof. Let $p = \dim U$, $q = \dim W < p$, and let P be the orthogonal projection onto U . For $t > 1$ set

$$X_t = t^{n-p}P + t^{-p}(I - P). \quad (9.9)$$

The exponents cancel, so $\det X_t = 1$. Write

$$T_A(X_t) = t^{n-p}R + t^{-p}S, \quad (9.10)$$

where $R = T_A(P)$ has rank at most q and $S = T_A(I - P) \succeq 0$. In a basis whose first q vectors span $\text{range } R$, multilinearity of the determinant (or the singular-value min-max bound) gives

$$\det T_A(X_t) = O\left(t^{q(n-p)}t^{-p(n-q)}\right) = O(t^{n(q-p)}) \longrightarrow 0. \quad (9.11)$$

Taking the infimum in (9.5) gives zero capacity. $\square$

Two converse implications are deep and are frozen as primary-source inputs. Cohn/Fortin–Reutenauer theory identifies full free-skew-field rank with absence of a shrunk subspace. Invariant theory of the left–right action identifies the same shrunk-subspace obstruction with null-cone membership. Gurvits’ capacity theorem identifies rank non-decrease with positive capacity.

Theorem 9.4 (Full noncommutative rank, null cone, and positive capacity). *For $A \in M_n(\mathbb{C})^m$, the following are equivalent:*

- (i) $\text{ncrk}(L_A) = n$;
- (ii) A has no shrunk subspace;
- (iii) A is not in the null cone of the $SL_n \times SL_n$ left–right action;
- (iv) T_A is rank non-decreasing;
- (v) $\text{cap}(T_A) > 0$.

Proof roadmap and ownership boundary. Start from the elementary linear obstruction: a c -shrunk subspace witnesses rank decrease and can be used to build determinant-one test matrices along which capacity tends to zero. The converse equivalences among full noncommutative rank, null-cone nonmembership, rank non-decrease, and positive capacity use the source-owned Cohn–Fortin–Reutenauer/Gurvits/invariant-theory lineage. The theorem packages these interfaces at the exact full-rank boundary needed by the algorithm.

Hypothesis check. This is a full-versus-non-full statement. It does not identify the exact value of a sub-full noncommutative rank. The structural formulation is over $\mathbb{C}$, where the orbit-closure and positivity languages used here are fixed.

Proof. The equivalence (i) $\Leftrightarrow$ (ii) is the Cohn–Fortin–Reutenauer noncommutative-rank characterization, and (ii) $\Leftrightarrow$ (iii) is the left–right null-cone theorem of the matrix-semi-invariant literature. These are source-owned algebraic inputs. Lemma 9.2 proves (ii) $\Leftrightarrow$ (iv) locally. Gurvits’ capacity theorem gives (iv) $\Leftrightarrow$ (v); the easy direction (ii fails) $\Rightarrow$ (v fails) is also the explicit degeneration in Lemma 9.3. Chaining the equivalences proves the result. $\square$

Field boundary. The free-skew-field/shrunk-subspace equivalence has versions over much more general fields. The theorem above intentionally uses $\mathbb{C}$ because null-cone orbit closure, conjugate transpose, positive semidefinite matrices, and capacity then share one literal setting. The later algorithm accepts rational input and uses this complex analytic characterization as its correctness bridge.

9.3 Alternating normalization

If $T_A(I)$ is nonsingular, define the *left normalization*

$$B = T_A(I)^{-1/2}, \quad A'_i = BA_i, \quad (9.12)$$

so that $T_{A'}(I) = I$. If $T_{A'}^*(I)$ is nonsingular, define the *right normalization*

$$C = T_{A'}^*(I)^{-1/2}, \quad A'_i = A_iC, \quad (9.13)$$

so that $T_{A'}^*(I) = I$.

Algorithm 9.5 (Operator scaling). *Given a tuple $A = (A_1, \dots, A_m)$:*

1. *If either $T_A(I)$ or $T_A^*(I)$ is singular, output RANK-DECREASING.*
2. *Right-normalize once. Thereafter alternate left and right normalization.*
3. *After each normalization compute $\text{ds}(T)$. If it is at most the chosen correctness threshold $\eta \leq 1/(n+1)$, output RANK-NONDECREASING. Otherwise continue for the prescribed iteration budget.*

Away from the null cone, every exact normalization step is well defined.

Algorithm reading guide. One iteration does not attempt to solve the entire invariant-theory problem. It checks a marginal, normalizes the offending side by the inverse square root of that positive-definite marginal, and repeats. The structural theorem says when positive capacity exists; the capacity theorem says why repeated local normalizations cannot keep making substantial progress forever. The two pieces are logically separate and should be cited separately.

Well-definedness. If A is outside the null cone, Theorem 9.4 gives positive capacity. If $T_A(I)$ were singular, choose $X = I$ in (9.5) to get capacity zero. If $T_A^*(I)$ were singular, its kernel contains a nonzero vector v with $A_i v = 0$ for all i , so $U = \langle v \rangle$, $W = 0$ is a shrunk subspace, again contradicting Theorem 9.4. Normalization by invertible matrices preserves the absence of shrunk subspaces, so the argument repeats at every iterate. $\square$

9.4 Capacity as a potential

Lemma 9.6 (Scaling law). *For invertible B, C and $A'_i = BA_iC$,*

$$\text{cap}(T_{A'}) = |\det B|^2 |\det C|^2 \text{cap}(T_A). \quad (9.14)$$

Proof. We have $T_{A'}(X) = BT_A(CXC^*)B^*$. If $\det X = 1$, then $Y = CXC^*$ has determinant $|\det C|^2$. Writing $Y = |\det C|^{2/n} Z$ gives $\det Z = 1$ and

$$\det T_{A'}(X) = |\det B|^2 |\det C|^2 \det T_A(Z). \quad (9.15)$$

Taking infima, and applying the inverse change of variables for the reverse inequality, proves equality. $\square$

Lemma 9.7 (One-sided normalization never decreases capacity). *Suppose $T(I) = I$ and $S = T^*(I) \succ 0$. Right normalization changes capacity by*

$$\text{cap}(T_R) = \frac{\text{cap}(T)}{\det S} \geq \text{cap}(T). \quad (9.16)$$

The analogous statement holds with left and right interchanged.

Proof. The scaling law gives the equality. Moreover

$$\text{tr } S = \text{tr } T^*(I) = \text{tr } T(I) = n. \quad (9.17)$$

AM–GM on the positive eigenvalues of S yields $\det S \leq 1$. $\square$

The next elementary estimate gives a conservative polynomial convergence bound. The primary GGOW analysis obtains sharper constants; sharpness is not needed here.

Lemma 9.8 (Quantitative progress from marginal discrepancy). *If $S \succ 0$, $\text{tr } S = n$, and $\Delta = \text{tr}(S - I)^2$, then*

$$-\log \det S \geq \frac{\Delta}{2n^2}. \quad (9.18)$$

Hence a normalization step correcting the marginal S multiplies capacity by at least $\exp(\Delta/(2n^2))$.

Proof. Every eigenvalue $0 < \lambda \leq n$. The function $g(x) = x - 1 - \log x$ has $g(1) = g'(1) = 0$ and $g''(x) = x^{-2} \geq n^{-2}$ on $(0, n]$. Strong convexity gives

$$\lambda - 1 - \log \lambda \geq \frac{(\lambda - 1)^2}{2n^2}. \quad (9.19)$$

Summing and using $\sum(\lambda - 1) = 0$ proves (9.18). Combine with (9.16). $\square$

Lemma 9.9 (Small discrepancy certifies rank non-decrease — Gurvits criterion). *If $T(I) = I$ and*

$$\text{tr}(T^*(I) - I)^2 \leq \frac{1}{n+1}, \quad (9.20)$$

then T is rank non-decreasing. The same holds with T and T^ exchanged.*

Proof. This is the quantitative Gurvits rank-nondecrease criterion used in the primary operator-scaling analysis. Its proof converts a hypothetical rank-decrease/shrunk-subspace witness into orthogonal projections and applies a matrix Cauchy–Schwarz estimate to force squared marginal discrepancy strictly above the displayed threshold. PASS55 treats that projection estimate as a source-owned analytic lemma rather than replacing it by an unverified local constant. The algorithm reserves a constant-factor gap below this threshold when finite-precision truncation is used. $\square$

Ownership of the discrepancy criterion. Lemma 9.9 is source-owned at its quantitative constant. The capacity progress estimate (9.18), by contrast, is a deliberately weaker QRCC-local substitute for GGOW’s sharper robust AM–GM bound.

Theorem 9.10 (Capacity monotonicity and polynomial convergence of operator scaling). *Let $A_1, \dots, A_m \in M_n(\mathbb{Z})$ have entries of absolute value at most $M \geq 1$, and suppose T_A is rank non-decreasing. After the initial right normalization, the published capacity lower bound gives*

$$\text{cap}(T) \geq \exp(-4n \log(Mmn)). \quad (9.23)$$

Along exact alternating operator scaling, capacity is nondecreasing. If every iterate had discrepancy larger than $1/(n+1)$, Lemma 9.8 would increase capacity by a factor at least

$$\exp\left(\frac{1}{2n^2(n+1)}\right) \quad (9.24)$$

per step. Since every one-sided normalized iterate has capacity at most 1, within

$$O(n^4 \log(Mmn)) \quad (9.25)$$

steps some iterate has discrepancy at most $1/(n+1)$ and is certified rank non-decreasing by Lemma 9.9.

Moreover the GGOW truncation analysis supplies a polynomial number of bits of precision per normalization that preserves a constant-factor gap between the accept/reject discrepancy thresholds. Therefore the same convergence is implementable with polynomial Turing bit complexity on rational/integer input.

Proof roadmap. Capacity transforms predictably under left/right scaling and never decreases under a one-sided normalization. If a marginal is noticeably far from the identity, a strong-convexity estimate gives a definite multiplicative capacity increase. A source-owned lower bound prevents positive capacity from being arbitrarily tiny on the integral input normalization. Therefore too many large-discrepancy iterations are impossible, and the process reaches a small- discrepancy rank-nondecreasing certificate in polynomially many iterations.

Complexity check. Analytic convergence alone is not a Turing-time algorithm: matrix entries may grow and irrational operations must be approximated. The registered theorem uses the source-owned truncation and bit-complexity bridge for rational/integer input; the local potential estimate is intentionally conservative.

Proof. Monotonicity is Lemma 9.7. For a one-sided normalized operator, choosing $X = I$ in (9.5) and applying AM–GM to the other marginal shows capacity at most 1. If discrepancy exceeds $1/(n+1)$, exactly one marginal is already the identity and the other has squared Frobenius distance above that threshold, so (9.18) gives (9.24). Starting from (9.23), more than $O(n^4 \log(Mmn))$ such multiplicative gains would force capacity above 1. Thus a small-discrepancy iterate appears and Lemma 9.9 certifies rank non-decrease.

The lower bound (9.23) and the polynomial-precision truncation stability are the two deep quantitative source inputs from the GGOW analysis. Their Section 4 proves that truncating every normalization to polynomially many bits changes the discrepancy by less than the reserved constant-factor gap throughout the polynomial iteration budget. Combining that source-owned bit lemma with the local potential argument gives the Turing-model statement. $\square$

9.5 Deterministic full-rank/null-cone decision

Algorithm 9.11 (Deterministic polynomial-time full noncommutative-rank/null- cone decision). *There is a deterministic polynomial-time Turing algorithm which, on rational input $A_1, \dots, A_m \in M_n(\mathbb{Q})$, decides the equivalent questions*

$$\text{nckr}(L_A) = n \iff A \notin \text{NullCone} \iff \text{cap}(T_A) > 0. \quad (9.26)$$

The algorithm clears denominators, runs the truncated operator-scaling algorithm for the polynomial iteration budget, and returns FULL when a sufficiently small discrepancy is reached and SINGULAR otherwise. The published GGOW bit-complexity theorem guarantees polynomial running time in n, m and the input bit length.

Algorithm reading guide. The decision procedure is a wrapper around scaling with certified stopping conditions. Small discrepancy gives a full-rank/rank-nondecreasing certificate; source-owned lower bounds plus the iteration budget certify the opposite case when such a point cannot be reached. The output is therefore a decision of full versus non-full noncommutative rank/null-cone membership, not an exact numerical noncommutative-rank computation below n .

Proof. Clearing denominators multiplies all A_i by one nonzero scalar and therefore does not change full noncommutative rank, shrunk subspaces, or null-cone membership. If a marginal becomes singular, Algorithm 9.5 has an immediate rank-decrease certificate. If the input is full, Theorem 9.4 gives positive capacity and Theorem 9.10 guarantees that truncated scaling reaches the small-discrepancy acceptance threshold in polynomially many polynomial-bit steps. Conversely, a small-discrepancy acceptance is rank non-decreasing by Lemma 9.9, hence full by Theorem 9.4. If the polynomial budget expires without such an iterate, the positive-capacity case has been excluded, so the tuple is in the null cone and has non-full noncommutative rank. $\square$

Decision versus exact rank. The registered algorithm decides *fullness*, which is the symbolic noncommutative singularity/null-cone problem. Computing the exact rank below n requires an additional reduction/blow-up argument and is not silently included in this row.

9.6 A complete two-by-two scaling example

Proposition 9.12 (Operator scaling for two two-by-two matrices). *Let*

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}. \quad (9.27)$$

Then $L_A = \text{diag}(x_1, 2x_2)$ has full noncommutative rank, $\text{cap}(T_A) = 4$, and one left normalization sends the tuple to

$$A'_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A'_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad (9.28)$$

which is doubly stochastic and has capacity 1.

Proof. For $X = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \succ 0$,

$$T_A(X) = \begin{pmatrix} a & 0 \\ 0 & 4c \end{pmatrix}, \quad \det T_A(X) = 4ac \geq 4 \det X. \quad (9.29)$$

Under $\det X = 1$, equality is attained at $X = I$, so capacity is 4. Also

$$T_A(I) = T_A^*(I) = \text{diag}(1, 4). \quad (9.30)$$

Left normalization by $B = \text{diag}(1, 1/2)$ gives (9.28). The new operator maps I to I and so does its adjoint. Its capacity is 1 by the scaling law, or directly from $\det T_{A'}(X) = ac \geq \det X = 1$. $\square$

A contrasting shrunk example is the single tuple $A_1 = \text{diag}(1, 0)$. Taking $U = \mathbb{C}^2$ and $W = \langle e_1 \rangle$ gives a 1-shrunk subspace; $T(I)$ is singular and the algorithm rejects before attempting a matrix square root.

9.7 The correspondence map

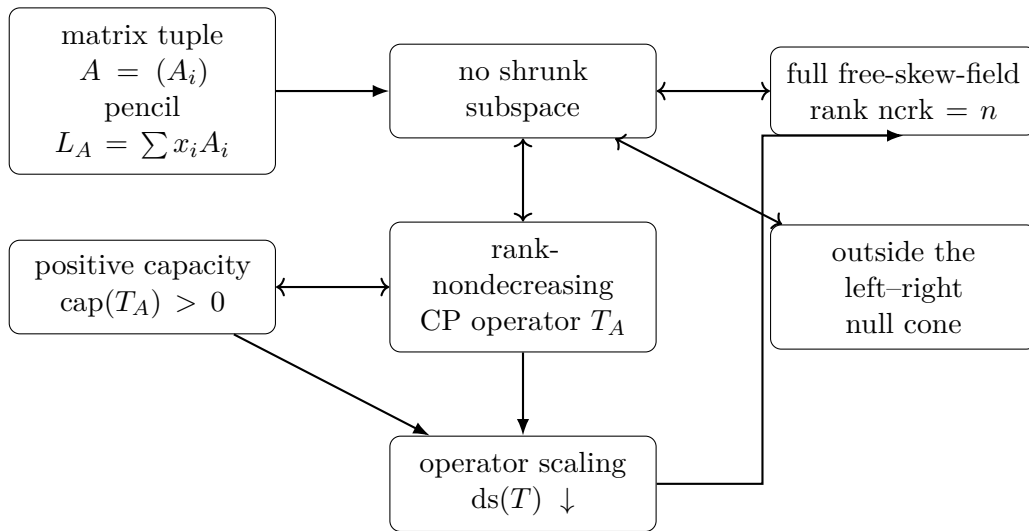


Figure 9.1: Invariant-theory/algorithm correspondence. Double arrows are exact structural equivalences in Theorem 9.4; downward arrows feed the analytic potential into the scaling algorithm.

Figure 9.1 is meant as a dependency map, not a proof by picture. In particular, the algebraic equivalence between the null cone and a shrunk subspace is a deep invariant-theory theorem; the algorithmic arrow from capacity to scaling uses the quantitative lower bound and truncation theorem.

9.8 Scalar matrix scaling: the Sinkhorn–Knopp theorem

Operator scaling has a classical commutative ancestor. To keep the zero-pattern hypotheses visible, we state the scalar terminology before the theorem rather than hiding it inside the convergence proof.

Definition 9.13 (Support terminology for matrix scaling). Let $A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{n \times n}$.

1. A *positive diagonal* of A is a permutation $\sigma \in S_n$ such that $a_{i,\sigma(i)} > 0$ for every i . The matrix has *support* if it has at least one positive diagonal.
2. The matrix has *total support* if $A \neq 0$ and every positive entry a_{ij} belongs to at least one positive diagonal.
3. A nonnegative matrix is *doubly stochastic* if every row sum and every column sum equals 1.
4. The matrix A is *diagonally scalable* if there are diagonal matrices D_r, D_c with strictly positive diagonal entries such that $D_r A D_c$ is doubly stochastic.
5. For $n \geq 2$, the matrix is *fully indecomposable* if it contains no $s \times (n - s)$ all-zero submatrix for $1 \leq s < n$; a nonzero 1×1 matrix is fully indecomposable by convention.
6. The *support graph* $G(A)$ is the bipartite graph with row vertices $r_1, \dots, r_n$, column vertices $c_1, \dots, c_n$, and edge $r_i c_j$ exactly when $a_{ij} > 0$.

7. If all row sums are positive, the *row normalization* is

$$\mathcal{R}(A) = \text{diag}(r_1^{-1}, \dots, r_n^{-1})A, \quad r_i = \sum_j a_{ij}. \quad (9.31)$$

If all column sums are positive, the *column normalization* is

$$\mathcal{C}(A) = A \text{diag}(c_1^{-1}, \dots, c_n^{-1}), \quad c_j = \sum_i a_{ij}. \quad (9.32)$$

The difference between support and total support is essential. Support means that at least one perfect matching survives in the zero pattern; total support means that every positive edge belongs to some perfect matching.

Lemma 9.14 (Total support supplies a relative-interior stochastic point). *If A has total support and $S = \{(i, j) : a_{ij} > 0\}$, then there exists a doubly stochastic matrix H such that*

$$h_{ij} > 0 \iff (i, j) \in S. \quad (9.33)$$

Proof. For each $(i, j) \in S$, choose a positive diagonal of A containing that entry, and let $P^{(i,j)}$ be its permutation matrix. The average

$$H = \frac{1}{|S|} \sum_{(i,j) \in S} P^{(i,j)} \quad (9.34)$$

is doubly stochastic because every summand is. No summand uses a zero entry of A , while the summand indexed by (i, j) uses that entry. Hence (9.33) holds. $\square$

Lemma 9.15 (Entropy minimization produces the diagonal scaling). *Suppose A has total support and let*

$$\mathcal{P}_S = \left\{ X = (x_{ij}) \geq 0 : x_{ij} = 0 \text{ off } S, \sum_j x_{ij} = 1, \sum_i x_{ij} = 1 \right\}. \quad (9.35)$$

Then the functional

$$\Phi_A(X) = \sum_{(i,j) \in S} x_{ij} \log \frac{x_{ij}}{a_{ij}}, \quad 0 \log 0 := 0, \quad (9.36)$$

has a minimizer $B = (b_{ij}) \in \mathcal{P}_S$ which is positive on every edge of S , and there are positive vectors u, v such that

$$b_{ij} = u_i a_{ij} v_j \quad (1 \leq i, j \leq n). \quad (9.37)$$

Moreover a doubly stochastic matrix of the form $D_r A D_c$ is unique.

Proof. The polytope $\mathcal{P}_S$ is nonempty by Lemma 9.14, compact, and Φ_A is continuous on it, so a minimizer B exists.

We first show that $b_{ij} > 0$ on all of S . Let H be the matrix from Lemma 9.14. If some allowed coordinate of B vanished, then $B_t = (1-t)B + tH$ would remain in $\mathcal{P}_S$. Every coordinate that is positive at B changes its contribution to Φ_A by $O(t)$, whereas a vanishing coordinate with $h_{ij} > 0$ contributes

$$th_{ij} \log \frac{th_{ij}}{a_{ij}} = th_{ij} \log t + O(t). \quad (9.38)$$

Thus the one-sided directional derivative at $t = 0$ is $-\infty$, contradicting minimality. Hence B lies in the relative interior of $\mathcal{P}_S$.

Let $Z = (z_{ij})$ be supported on S with every row sum and column sum zero. For sufficiently small $|t|$, $B + tZ \in \mathcal{P}_S$, so first-order optimality gives

$$\sum_{(i,j) \in S} z_{ij} \left(\log \frac{b_{ij}}{a_{ij}} + 1 \right) = 0. \quad (9.39)$$

The orthogonal complement of the space of such Z is the row space of the row/column incidence map. Consequently there are real numbers α_i, β_j such that, on S ,

$$\log \frac{b_{ij}}{a_{ij}} + 1 = \alpha_i + \beta_j. \quad (9.40)$$

Setting $u_i = e^{\alpha_i}$ and $v_j = e^{\beta_j - 1}$ gives (9.37).

It remains to prove uniqueness of the doubly stochastic scaling, not merely of the chosen minimizer. Let $C = (c_{ij}) = D_r A D_c$ be any doubly stochastic positive diagonal scaling. On S we have $\log(c_{ij}/a_{ij}) = \rho_i + \gamma_j$. Therefore for any $X \in \mathcal{P}_S$,

$$\begin{aligned} \Phi_A(X) - \Phi_A(C) &= \sum_S x_{ij} \log \frac{x_{ij}}{c_{ij}} + \sum_S (x_{ij} - c_{ij})(\rho_i + \gamma_j) \\ &= \sum_S x_{ij} \log \frac{x_{ij}}{c_{ij}} \geq 0. \end{aligned} \quad (9.41)$$

The last inequality is Gibbs' inequality applied after dividing both matrices by their common total mass n , and equality holds only for $X = C$. Thus every doubly stochastic diagonal scaling is the unique minimizer of Φ_A and must equal B . $\square$

Lemma 9.16 (Full indecomposability and support-graph connectedness). *For a matrix with total support, full indecomposability is equivalent to connectedness of $G(A)$.*

Proof. Suppose first that $G(A)$ is disconnected and let one connected component have row set R and column set C . Because every edge in the component belongs to a positive diagonal, any positive diagonal matches the vertices of this component internally; hence $|R| = |C| =: s$. There are no edges from R to C^c , so $A[R, C^c]$ is an $s \times (n - s)$ zero submatrix. Thus A is not fully indecomposable.

Conversely, suppose $A[R, C] = 0$ with $|R| = s$ and $|C| = n - s$. Since total support implies support, choose a positive diagonal. The rows in R cannot use columns in C , so they are matched bijectively to the s columns of C^c . If there were a positive entry $a_{ij} > 0$ with $i \notin R$ and $j \in C^c$, total support would place that entry on a positive diagonal. That diagonal would leave only $s - 1$ columns of C^c for the s rows of R , an impossibility because those rows have no edges into C . Hence there are no edges from R^c to C^c either. The vertex sets $R \cup C^c$ and $R^c \cup C$ therefore form distinct connected components of $G(A)$. $\square$

Theorem 9.17 (Sinkhorn–Knopp matrix scaling in the total-support regime). *Let $A \in \mathbb{R}_{\geq 0}^{n \times n}$ have total support. Then:*

- (i) *there exist diagonal matrices D_r, D_c with strictly positive diagonal entries such that $B = D_r A D_c$ is doubly stochastic;*
- (ii) *the doubly stochastic matrix B is unique;*
- (iii) *the pair (D_r, D_c) is unique up to $D_r \mapsto \lambda D_r$, $D_c \mapsto \lambda^{-1} D_c$ with $\lambda > 0$ if and only if A is fully indecomposable;*

(iv) beginning with A and alternately normalizing all rows and all columns, with either choice of the first normalization, the matrices converge entrywise to B .

The classical theorem is asymptotic: no polynomial iteration bound or Turing bit-complexity claim is included here.

Proof roadmap. Total support supplies a doubly stochastic point in the relative interior of the matrix's support face. Minimize a strictly convex entropy/KL functional on that face; the first-order equations produce positive row and column scaling factors. The same relative-entropy potential decreases under alternating row and column normalization and forces convergence to the unique scaled doubly stochastic matrix. Support-graph connectedness then identifies when the two diagonal factors themselves are unique up to a common scalar gauge.

Hypothesis check. Total support is the local hypothesis for positive diagonal scalability. The classical source proves the stronger support-only convergence statement for the alternating iteration; that extension remains source-owned here. Full indecomposability is needed only for uniqueness of the factors, not for uniqueness of the doubly stochastic matrix.

Proof. Parts (i) and (ii) are Lemma 9.15.

For (iii), suppose $B = D_r A D_c = D'_r A D'_c$. Write $p_i = (D'_r)_{ii}/(D_r)_{ii}$ and $q_j = (D'_c)_{jj}/(D_c)_{jj}$. For every $(i, j) \in S$,

$$p_i q_j = 1. \quad (9.42)$$

Along a connected component of $G(A)$ this forces every row ratio p_i to be one common constant and every column ratio q_j to be its reciprocal. By Lemma 9.16, full indecomposability means there is only one component, giving precisely the displayed global scalar freedom. If there are at least two components, one component can be rescaled by an independent positive scalar and its reciprocal without changing B , so global scalar uniqueness fails.

For (iv), first row-normalize A and write the alternating sequence as

$$X^{(0)} = \mathcal{R}(A), \quad Y^{(k)} = \mathcal{C}(X^{(k)}), \quad X^{(k+1)} = \mathcal{R}(Y^{(k)}). \quad (9.43)$$

Total support guarantees that no row or column is zero, and diagonal scaling preserves the zero pattern, so every step is defined. Every $X^{(k)}$ is row stochastic, every $Y^{(k)}$ is column stochastic, and both have total mass n .

Use the already constructed B as a reference and define

$$\mathcal{D}(B\|X) = \sum_{(i,j) \in S} b_{ij} \log \frac{b_{ij}}{x_{ij}}. \quad (9.44)$$

Because B and every half-iterate have total mass n , Gibbs' inequality gives $\mathcal{D}(B\|X) \geq 0$. If X is row stochastic with column sums $c_1, \dots, c_n$, then

$$\mathcal{D}(B\|\mathcal{C}(X)) - \mathcal{D}(B\|X) = \sum_{j=1}^n \log c_j \leq 0, \quad (9.45)$$

where the equality uses the column sums of B and the inequality uses $\sum_j c_j = n$ and AM–GM. Similarly, if Y is column stochastic with row sums $r_1, \dots, r_n$,

$$\mathcal{D}(B\|\mathcal{R}(Y)) - \mathcal{D}(B\|Y) = \sum_{i=1}^n \log r_i \leq 0. \quad (9.46)$$

Thus the divergence is nonincreasing along every half-step and bounded below. The nonnegative decrements in (9.45)–(9.46) tend to zero. Since a positive vector with coordinate sum n satisfies $-\sum_j \log c_j \geq 0$ with equality only at $(1, \dots, 1)$, compactness gives

$$c_j^{(k)} \rightarrow 1 \quad \text{and} \quad r_i^{(k)} \rightarrow 1 \quad \text{for every } i, j. \quad (9.47)$$

Hence consecutive half-iterates differ entrywise by $o(1)$.

It remains to identify the only possible limit. The row-stochastic matrices $X^{(k)}$ lie in the compact cube $[0, 1]^{n \times n}$. Let a subsequence converge to C . Equation (9.47) makes C doubly stochastic. Moreover no entry in S can tend to zero: if $x_{ij}^{(k)} \rightarrow 0$ while $b_{ij} > 0$, the corresponding summand in $\mathcal{D}(B \| X^{(k)})$ tends to $+\infty$, whereas all other summands are bounded below by the finite constants $b_{uv} \log b_{uv}$; this contradicts the uniform upper bound supplied by monotonicity. Thus C has exactly the same positive support as A .

Each $X^{(k)}$ has the form $\text{diag}(u^{(k)})A\text{diag}(v^{(k)})$ with positive factors. On every support edge, $u_i^{(k)}v_j^{(k)} = x_{ij}^{(k)}/a_{ij}$ converges along the chosen subsequence to a positive limit. In each connected component of $G(A)$, fix one row factor to 1 by the harmless gauge transformation $u \mapsto \lambda u$, $v \mapsto \lambda^{-1}v$. Propagating along a spanning tree of the component shows successively that all adjusted row and column factors converge to positive limits. Hence $C = \tilde{D}_r A \tilde{D}_c$ for positive diagonal matrices $\tilde{D}_r, \tilde{D}_c$. Part (ii) forces $C = B$.

Every convergent subsequence therefore has the same limit B , so $X^{(k)} \rightarrow B$. Equation (9.47) then also gives $Y^{(k)} \rightarrow B$. This is the claimed convergence when rows are normalized first. The column-first version follows by the same argument with rows and columns interchanged (equivalently, by applying the proof to A^T). $\square$

The stronger support-only convergence statement. Sinkhorn and Knopp prove more than the registered theorem. For an arbitrary nonnegative square matrix, alternating row/column normalization converges to a doubly stochastic limit if and only if the matrix has support. When support is total, the limit is the positive diagonal scaling from Theorem 9.17; when support is not total, the limit cannot be written as $D_r A D_c$ with positive diagonal D_r, D_c . This stronger zero-pattern statement is recorded at primary-source strength, while the locally owned proof above deliberately freezes the total-support regime needed for the operator-scaling comparison.

A zero-pattern obstruction. Consider

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \quad (9.48)$$

It has support because the main diagonal is positive, but it does not have total support: the entry a_{12} belongs to no positive diagonal. Hence no positive diagonal scaling can be doubly stochastic. Nevertheless the source-strength Sinkhorn–Knopp theorem says that alternating normalization converges to I_2 . In fact, after a column normalization followed by a row normalization, the row-stochastic iterates have the form

$$X_k = \begin{pmatrix} 1 - \frac{1}{2k+1} & \frac{1}{2k+1} \\ 0 & 1 \end{pmatrix} \longrightarrow I_2 \quad (k \geq 1). \quad (9.49)$$

Thus a positive entry can disappear only in the limit; positive diagonal scalings themselves never change the zero pattern.

A worked total-support scaling. For

$$A = \begin{pmatrix} 1 & 4 \\ 9 & 16 \end{pmatrix}, \quad (9.50)$$

all entries are positive, so total support and full indecomposability are automatic. The matrices

$$D_r = \text{diag}\left(\frac{2}{5}, \frac{1}{15}\right), \quad D_c = \text{diag}\left(1, \frac{3}{8}\right) \quad (9.51)$$

give

$$D_r A D_c = \begin{pmatrix} 2/5 & 3/5 \\ 3/5 & 2/5 \end{pmatrix}, \quad (9.52)$$

which is doubly stochastic. The theorem says that every alternating row/column normalization sequence converges to this same matrix, regardless of the scalar gauge chosen for the two diagonal factors.

Exact relation to operator scaling. The scalar algorithm is a coordinate-Kraus restriction of operator scaling, not an identification of the two input models. Given $A = (a_{ij}) \geq 0$, define

$$K_{ij} = \sqrt{a_{ij}} e_i e_j^*. \quad (9.53)$$

For the completely positive map $T(X) = \sum_{i,j} K_{ij} X K_{ij}^*$,

$$T(I) = \text{diag}\left(\sum_j a_{1j}, \dots, \sum_j a_{nj}\right), \quad T^*(I) = \text{diag}\left(\sum_i a_{i1}, \dots, \sum_i a_{in}\right). \quad (9.54)$$

Thus left/right operator normalization by positive diagonal matrices sends the coefficient matrix to $a_{ij} \mapsto b_i^2 a_{ij} c_j^2$ and, after taking square roots of the scalar normalization factors, reproduces Sinkhorn row/column scaling. General operator scaling allows arbitrary matrix Kraus operators and noncommuting marginals, so the matrix-scaling theorem is an instructive diagonal model, not a replacement for Theorems 9.4 and 9.10.

9.9 Executable laboratory

Proposition 9.18 (PIT and operator-scaling laboratory). *The executable file*

`code/volume_11/
pit_operator_scaling_lab.py`

contains deterministic self-tests for three interfaces used in Chapters 6–9:

1. *a small-grid polynomial-identity checkpoint that distinguishes a true zero expression from a nonzero polynomial;*
2. *layerwise coordinate-basis PIT for a small noncommutative ABP;*
3. *the exact diagonal 2×2 operator-scaling example (9.27)–(9.30), checking the marginals and capacity transformation.*

Running the file with `-self-test` exits successfully only if every checkpoint passes.

9.10 Source and ownership boundary

The primary algorithmic source for this chapter is Ankit Garg, Leonid Gurvits, Rafael Oliveira, and Avi Wigderson, *Operator Scaling: Theory and Applications*, Foundations of Computational Mathematics 20 (2020), 223–290; arXiv:1511.03730. The paper gives a deterministic polynomial-time algorithm over rational input for noncommutative symbolic-matrix singularity, develops capacity and Algorithm G, proves an explicit capacity lower bound for integral rank-nondecreasing operators, and supplies the truncation/bit-complexity analysis.

Theorem 9.4 also uses the older structural lineage surveyed and assembled there: Cohn/Fortin–Reutenauer for full noncommutative rank versus shrunk subspaces; Gurvits for rank non-decrease and capacity; and the matrix-semi-invariant left–right-action theorem for the null cone versus shrunk subspaces. PASS55 locally proves the easy shrunk-subspace/rank-decrease direction, the explicit capacity degeneration, the scaling law, monotonicity, and a conservative progress inequality. It does *not* relabel the deep invariant-theory equivalence, the integral capacity lower bound (9.23), or the polynomial-precision truncation theorem as new elementary QRCC proofs.

The scalar matrix-scaling theorem in this chapter is source-audited against Richard Sinkhorn and Paul Knopp, *Concerning Nonnegative Matrices and Doubly Stochastic Matrices*, Pacific Journal of Mathematics 21(2) (1967), 343–348, DOI 10.2140/pjm.1967.21.343. Their theorem separates total support (existence of a positive diagonal scaling) from support (convergence of the alternating normalization itself), proves uniqueness of the scaled doubly stochastic matrix, and identifies full indecomposability as exactly the extra hypothesis for uniqueness of the diagonal factors up to one scalar. The QRCC proof reconstructs the total-support scaling/convergence part through a relative-entropy potential and records the support-only extension at explicit source-owned strength.

The local convergence exponent in (9.25) is intentionally weaker than the sharper GGOW iteration analysis. Its purpose is to make polynomiality visible from one elementary strong-convexity inequality while preserving the published source as the authority for the bit-complexity bridge.

9.11 Reader checkpoint and exercises

Chapter checkpoint. Operator scaling works because a structural invariant-theory obstruction and an analytic potential meet at the same full-rank boundary. Shrunk subspaces expose failure, positive capacity survives scaling, marginal discrepancy forces progress, and rational bit-complexity control turns convergence into a deterministic decision algorithm. Sinkhorn scaling is the diagonal commutative shadow of this mechanism, with a finer support/total-support classification specific to nonnegative matrices.

Exercises.

1. For a single matrix $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, find a shrunk-subspace certificate and compute $T_A(I)$ and $T_A^*(I)$. Which full-rank conditions in Theorem 9.4 fail?
2. Prove the determinant scaling law for capacity directly from $T_{A^{B,C}}(X)$ by the change of variables used in the chapter.
3. Starting from the exact two-by-two tuple in Proposition 9.12, perform one left normalization and one right normalization and verify the new marginals.
4. Explain why “small discrepancy” is a useful stopping condition but does not by itself prove polynomial Turing running time without entry-size/truncation control.

5. Give a matrix with support but not total support, and a matrix with total support but not full indecomposability. For each, state which conclusion of the Sinkhorn–Knopp theorem survives.
6. For the positive matrix in (9.50), verify the diagonal factors in (9.51) and the doubly stochastic result (9.52). Multiply one factor by $\lambda > 0$ and the other by λ^{-1} to see the scalar gauge.
7. Derive (9.54) from the coordinate Kraus operators $K_{ij} = \sqrt{a_{ij}}e_i e_j^*$. Identify the precise point at which the general operator-scaling model becomes more expressive than scalar matrix scaling.
8. Run the Volume XI laboratory in self-test mode:

```
code/volume_11/pit_operator_scaling_lab.py -self-test.
```

Then add a small Sinkhorn row/column normalization check locally and compare its stopping criterion with the operator-scaling discrepancy test. This coding exercise is illustrative; it does not replace either convergence proof.

Chapter 10

Tensor Methods, Geometric Complexity Theory, and Asymptotic Spectra

The preceding chapter turned a concrete invariant-theory null-cone problem into an algorithm by operator scaling. This final chapter widens the lens in two orthogonal directions. First, orbit closures and their representation-theoretic coordinate rings provide the language in which geometric complexity theory is formulated. Second, direct sums and tensor products turn complexity parameters into semiring rank problems, and Strassen’s asymptotic spectrum converts their amortized versions into a dual family of monotone homomorphisms.

The ownership boundary is especially important here. The invariant-theory vocabulary below is defined locally. The robust torus-orbit theorem and the asymptotic-spectrum duality theorem are stated at verified source strength but are not relabelled as new QRCC proofs. The broader GCT program remains a dated research frontier, but its reader-facing source map is now explicitly verified rather than left as a generic source gate.

How to read this chapter. The final chapter is deliberately more source-facing than Chapters 1–9. Its first thread supplies invariant-theory vocabulary and one verified robust torus-orbit frontier. Its second thread builds the abstract semiring/preorder language of asymptotic spectra, checks tensor restriction as the motivating example, and then imports Strassen duality at exact source strength. The last sections are maps of what the framework organizes and of what geometric complexity theory still leaves open.

There are therefore three status labels to keep mentally distinct:

local definition/proof, verified source-owned theorem, dated/open frontier gate.

A source-owned theorem is rigorous and load-bearing, but its proof remains with the cited source. A frontier gate is neither a theorem nor permission to use a folklore strengthening downstream.

Learning goals. A reader should be able to define orbit closure, null cone, invariant ring, and highest-weight occurrence in the local normalization; explain the precise number-theoretic status of the robust torus-orbit theorem; verify the Strassen axioms in the integer and tensor examples; interpret spectral points as dual monotone semiring homomorphisms; and explain why duality does not itself classify the spectrum or solve a GCT separation.

10.1 Orbit closures, invariants, and highest-weight data

We first recall the algebraic-geometric conventions used in the registered definition. If $S \subseteq \mathbb{C}^N$, let

$$I(S) := \{f \in \mathbb{C}[x_1, \dots, x_N] : f(s) = 0 \text{ for every } s \in S\}.$$

Its *Zariski closure* is the common zero set of $I(S)$; a subset is *Zariski closed* when it equals its own Zariski closure. For a Zariski-closed subgroup $G \leq GL_N(\mathbb{C})$, a function on G is called *regular* when it is the restriction of a polynomial in the matrix entries and $(\det g)^{-1}$. If V is finite dimensional, its polynomial coordinate ring is $\mathbb{C}[V] = \text{Sym}(V^*)$, the symmetric algebra of the dual vector space, equivalently the algebra of polynomial functions on V . Let $\mathbb{P}(V)$ denote the set of one-dimensional linear subspaces of V . For $X \subseteq \mathbb{P}(V)$, $I(X)$ denotes the homogeneous ideal of homogeneous polynomials that vanish on every line represented by a point of X .

For the highest-weight notation later in the definition we use only $GL_r(\mathbb{C})$ and make the convention explicit. Let $D_r \leq GL_r(\mathbb{C})$ be the diagonal subgroup and let U_r be the subgroup of upper-triangular matrices with every diagonal entry equal to 1. For an integer tuple $\lambda = (\lambda_1, \dots, \lambda_r)$ with $\lambda_1 \geq \dots \geq \lambda_r$, write $t^\lambda = \prod_i t_i^{\lambda_i}$ for $t = \text{diag}(t_1, \dots, t_r) \in D_r$. Suppose $GL_r(\mathbb{C})$ acts linearly on W and, after a basis of W is chosen, the matrix entries of the action are regular functions of the group element. Define the λ -highest-weight space by

$$W_\lambda^{\text{hw}} := \{w \in W : uw = w \text{ for every } u \in U_r, tw = t^\lambda w \text{ for every } t \in D_r\}.$$

We say that λ *occurs* in W when this space is nonzero, and its *highest-weight multiplicity* is $\dim W_\lambda^{\text{hw}}$. This definition is sufficient for the occurrence/multiplicity vocabulary used below; no highest-weight classification theorem is being imported into the definition.

Definition 10.1 (Orbit closure, invariant ring, highest-weight data, and moment polytope). All objects in this definition are over $\mathbb{C}$.

1. A *complex linear algebraic group* is a Zariski-closed subgroup $G \leq GL_N(\mathbb{C})$. A finite-dimensional *rational representation* of G is a homomorphism

$$\rho : G \longrightarrow GL(V) \tag{10.1}$$

whose matrix entries are regular functions on G after bases are chosen. We call G *reductive* when, in every finite-dimensional rational representation, every G -invariant linear subspace has a G -invariant linear complement. A *reductive group action* in this chapter is a rational representation of such a group.

2. For $v \in V$, its *orbit* is $Gv = \{\rho(g)v : g \in G\}$ and its *orbit closure* $\overline{Gv}$ is its Zariski closure. The *null cone* is

$$\mathcal{N}(V) = \{v \in V : 0 \in \overline{Gv}\}. \tag{10.2}$$

The *invariant ring* is

$$\mathbb{C}[V]^G = \{f \in \mathbb{C}[V] : f(\rho(g)v) = f(v) \text{ for every } g \in G, v \in V\}. \tag{10.3}$$

3. For the highest-weight and moment-set language, specialize to $G = GL_r(\mathbb{C})$ and a nonzero vector $v \in V$. Let $[v] \in \mathbb{P}(V)$ denote the one-dimensional subspace spanned by v , and put

$$X_v := \overline{G \cdot [v]} \subseteq \mathbb{P}(V), \tag{10.4}$$

where the bar denotes projective Zariski closure, equivalently the common zero set of the homogeneous ideal $I(G \cdot [v])$. Its *homogeneous coordinate ring* is

$$R(X_v) = \bigoplus_{d \geq 0} R_d(X_v) := \text{Sym}(V^*)/I(X_v). \quad (10.5)$$

The action on polynomial functions is $(g \cdot f)(x) = f(g^{-1}x)$. Because X_v is G -stable, this action preserves $I(X_v)$ and therefore makes $R(X_v)$ a graded G -module. The *highest-weight data* of X_v is

$$\mathcal{H}(v) := \{(d, \lambda) : d \geq 1, \lambda_1 \geq \dots \geq \lambda_r, R_d(X_v)_\lambda^{\text{hw}} \neq 0\}. \quad (10.6)$$

The corresponding highest-weight multiplicity is $\dim R_d(X_v)_\lambda^{\text{hw}}$.

4. In the algebraic highest-weight normalization used here, define

$$\Delta(v) := \overline{\left\{ \frac{\lambda}{d} : (d, \lambda) \in \mathcal{H}(v) \right\}} \subseteq \mathbb{R}^r, \quad (10.7)$$

where the closure is Euclidean. We use the conventional name *moment polytope* for $\Delta(v)$. The fact that this closed set has the expected convex/polyhedral properties is a theorem of the surrounding theory, not part of this definition and not used below. Likewise, symplectic moment-map conventions can differ by standard identifications or scalar normalizations and are not silently mixed with (10.7).

Definition boundary. Definition 10.1 supplies vocabulary only. It does not, by definition alone, assert finite generation of invariant rings, a Hilbert–Mumford criterion, a moment-map characterization of $\Delta(v)$, convexity/polyhedrality of (10.7), or any complexity bound for orbit-closure membership. Whenever such a theorem is needed, it must have a separate theorem statement and source/proof owner. This distinction prevents geometric language from becoming an unregistered theorem import.

10.2 A verified robust torus-orbit frontier

The first modern frontier can be stated without invoking general reductive groups. Let $M = (m_{\ell j}) \in \mathbb{Z}^{d \times n}$ and write $m_j \in \mathbb{Z}^d$ for its j th column. The algebraic torus $T = (\mathbb{C}^\times)^d$ acts diagonally on $(\mathbb{C}^\times)^n$ by

$$(t \cdot v)_j = t^{m_j} v_j, \quad t^{m_j} := \prod_{\ell=1}^d t_\ell^{m_{\ell j}}. \quad (10.8)$$

Write $O_v = T \cdot v$. With Log the componentwise principal complex logarithm, define

$$\delta_{\log}(v, w) := \min_{\alpha \in 2\pi i \mathbb{Z}^n} \|\text{Log } v - \text{Log } w - \alpha\|_2, \quad \delta_{\log}(O_v, O_w) := \inf_{x \in O_v, y \in O_w} \delta_{\log}(x, y). \quad (10.9)$$

For the rational input model below, entries of M have binary bit length at most B and coordinates of $v, w \in (\mathbb{Q}(i)^\times)^n$ have bit length at most b . Define $\text{sep}_T(d, n, B, b)$ to be the minimum positive value of $\delta_{\log}(O_v, O_w)$ over such inputs with distinct orbits. The *algebraic-torus separation hypothesis* is the assertion

$$\text{sep}_T(d, n, B, b) \geq \exp(-\text{poly}(d, n, B, b)). \quad (10.10)$$

We also freeze the number-theoretic statement that appears in the source. If $v_1, \dots, v_n \in \mathbb{Q}(i)^\times$ and $e_1, \dots, e_n \in \mathbb{Z}$, put

$$\Lambda(v, e) = \sum_{j=1}^n e_j \operatorname{Log} v_j. \quad (10.11)$$

The *linear-forms-in-logarithms hypothesis* used here says that whenever $\Lambda(v, e) \neq 0$,

$$|\Lambda(v, e)| \geq \exp(-\operatorname{poly}(n, b, B)), \quad (10.12)$$

where b bounds the bit lengths of the v_j and B those of the e_j .

Theorem 10.2 (Bürgisser–Doğan–Makam–Walter–Wigderson robust torus-orbit frontier — verified source theorem). *For the algebraic torus model (10.8)–(10.12), the following source-owned statements hold.*

1. *There is a universal constant $c > 0$ for which approximating $\delta_{\log}(O_v, O_w)$ within factor*

$$\gamma = n^{c/\log \log n} \quad (10.13)$$

is NP-hard.

2. *If the separation hypothesis (10.10) holds, then the orbit distance can be approximated in polynomial time within a factor $\gamma = \exp(\operatorname{poly}(d, n, B))$.*
3. *The algebraic-torus separation hypothesis (10.10) is equivalent to the linear-forms-in-logarithms hypothesis (10.12). The corresponding compact-torus separation hypothesis is equivalent to the restriction of (10.12) to $|v_1| = \dots = |v_n| = 1$.*

How to read this source-owned theorem. The theorem has three layers: hardness of robust torus-orbit approximation, an algorithm whose running time depends on an explicit separation parameter, and a number-theoretic equivalence controlling that separation. The third layer uses the paper’s exact Number-Theoretic Hypothesis 1.11; the ordinary *abc* conjecture is context, not a drop-in replacement. Downstream use must therefore cite the exact numbered alternative needed, not the chapter title’s informal “*abc*” shorthand.

What the “*abc*” label does and does not mean. The 2024 source relates (10.12) tightly to the classical *abc* landscape: known Baker–Wüstholz–Matveev bounds give weaker lower bounds, while suitable stronger linear-forms estimates (together with their p -adic analogues) imply strong *abc*-type consequences. The source itself calls (10.12) a number-theoretic hypothesis/version connected to *abc*. Therefore QRCC does *not* replace Theorem 10.2(3) by the stronger and incorrect slogan “ordinary *abc* if and only if robust torus-orbit distance is in polynomial time.”

Proof architecture, source-owned. The hardness direction reduces closest vector problems to orbit-distance approximation through an efficient lattice-lifting theorem. The algorithmic direction reduces logarithmic orbit distance to lattice and invariant computations, with the separation parameter controlling the precision needed to distinguish distinct orbits. The equivalence in part (3) identifies this separation parameter with quantitative lower bounds for nonzero linear forms in principal complex logarithms. These reductions are recorded to expose the mechanism, but their proofs remain with the cited CCC 2024 paper.

10.3 Strassen preorders and asymptotic vocabulary

We now fix every hypothesis needed by the duality theorem. A *commutative semigroup* is a set with an associative, commutative binary operation; a *commutative monoid* is a commutative semigroup equipped with an identity element. To match the modern source exactly, a *positive commutative semiring* here means a commutative additive semigroup $(R, +)$ together with a commutative multiplicative monoid $(R, \cdot, 1)$ satisfying both distributive laws; a zero element is not required. Positive integers are embedded by $n = \underbrace{1 + \cdots + 1}_{n \text{ times}}$.

Definition 10.3 (Strassen preorder, asymptotic rank, and spectrum). Let R be a positive commutative semiring.

1. A relation $\leq_P$ is a *preorder* if it is reflexive and transitive. It is a *semiring preorder* if $a \leq_P b$ and $c \leq_P d$ imply $a + c \leq_P b + d$ and $ac \leq_P bd$.
2. A semiring preorder P is a *Strassen preorder* in the modern strong-Archimedean convention when
 - (a) for positive integers m, n , one has $m \leq_P n$ exactly when $m \leq n$ in $\mathbb{N}$;
 - (b) $a \leq_P a + b$ for every $a, b \in R$;
 - (c) for every $a \in R$ there is $n \in \mathbb{N}_{\geq 1}$ such that $1 \leq_P a \leq_P n$.
3. The P -rank and P -subrank are

$$R_P(a) := \min\{n \geq 1 : a \leq_P n\}, \quad Q_P(a) := \max\{n \geq 1 : n \leq_P a\}. \quad (10.17)$$

Their asymptotic versions are

$$\tilde{R}_P(a) := \inf_{k \geq 1} R_P(a^k)^{1/k}, \quad \tilde{Q}_P(a) := \sup_{k \geq 1} Q_P(a^k)^{1/k}. \quad (10.18)$$

4. The *asymptotic preorder* $\leq_{\tilde{P}}$ is defined by $a \leq_{\tilde{P}} b$ when there is a function $f : \mathbb{N}_{\geq 1} \rightarrow \mathbb{N}$ with $f(k)/k \rightarrow 0$ and

$$a^k \leq_P b^k 2^{f(k)} \quad \text{for every } k \geq 1. \quad (10.19)$$

Here $2^{f(k)}$ denotes the corresponding embedded positive integer in R ; it is the $2^{o(k)}$ overhead in the source definition.

5. A *spectral point* is a map $\phi : R \rightarrow \mathbb{R}_{\geq 1}$ satisfying

$$\phi(a + b) = \phi(a) + \phi(b), \quad \phi(ab) = \phi(a)\phi(b), \quad \phi(1) = 1, \quad (10.20)$$

and monotonicity $a \leq_P b \Rightarrow \phi(a) \leq \phi(b)$. The *asymptotic spectrum* is the set X_P of all spectral points.

6. An element $a \in R$ is *gapped* if either $2 \leq_P a^k$ for some $k \geq 1$, or there is a spectral point $\phi \in X_P$ with $\phi(a) = 1$.

The strong Archimedean axiom in Definition 10.3 is intentionally the simplified modern convention. Strassen's original "good preorder" formalism uses a weaker relative Archimedean condition. The theorem below is stated only in the modern convention just defined, so no equivalence between the two formalisms is being assumed without proof.

10.4 Tensor restriction as a Strassen preorder

Asymptotic spectra are abstract, but the tensor example shows why the abstraction is useful. Fix a field F . A *finite three-tensor* is an element $T \in U \otimes V \otimes W$ for finite-dimensional F -vector spaces U, V, W ; a *simple tensor* has the form $u \otimes v \otimes w$. The *tensor rank* $R(T)$ is the least r for which T is a sum of r simple tensors. Two tensors are identified when invertible linear maps on the three factors carry one to the other. Contracting one tensor factor gives a linear map, for example $U^* \rightarrow V \otimes W$; the rank of such a map is called a *matrix-flattening rank*.

For tensors S and T , write

$$S \leq_{\text{res}} T \quad (10.14)$$

if there are linear maps on the three tensor factors sending T to S ; this is called *tensor restriction*. Direct sum gives addition and tensor (Kronecker) product gives multiplication. The multiplicative unit is the one-dimensional diagonal tensor $\langle 1 \rangle = 1 \otimes 1 \otimes 1$, and

$$\langle r \rangle := \sum_{i=1}^r e_i \otimes e_i \otimes e_i \quad (10.15)$$

represents the positive integer r .

Lemma 10.4 (The tensor-restriction semiring satisfies the Strassen axioms). *On isomorphism classes of nonzero finite three-tensors, direct sum, tensor product, and $\leq_{\text{res}}$ form a commutative positive semiring with a Strassen preorder in the sense of Definition 10.3. Its rank relative to the embedded integers is ordinary tensor rank, and its subrank is*

$$Q(T) = \max\{r : \langle r \rangle \leq_{\text{res}} T\}. \quad (10.16)$$

Proof. Restriction maps compose and are preserved by direct sums and tensor products, so $\leq_{\text{res}}$ is a compatible semiring preorder. Projection from a direct sum gives $T \leq_{\text{res}} T \oplus S$. If $T \neq 0$, choose one nonzero coordinate of T , project each factor onto that coordinate, and rescale one map; this gives $\langle 1 \rangle \leq_{\text{res}} T$. Every finite tensor has a finite rank decomposition, which is exactly a restriction $T \leq_{\text{res}} \langle r \rangle$ for some r .

It remains only to check that the embedded integers carry their ordinary order. If $r \leq s$, coordinate projection gives $\langle r \rangle \leq_{\text{res}} \langle s \rangle$. Conversely, restriction cannot increase the rank of any matrix flattening, while every flattening of $\langle r \rangle$ has rank r . Hence $\langle r \rangle \leq_{\text{res}} \langle s \rangle$ implies $r \leq s$. Finally, the least r with $T \leq_{\text{res}} \langle r \rangle$ is precisely the least number of simple tensors in a rank decomposition of T , which is ordinary tensor rank. The subrank formula is its definition in the opposite direction. $\square$

10.5 Strassen duality

Theorem 10.5 (Deep input: Strassen asymptotic-spectrum duality). *Let R be a positive commutative semiring equipped with a Strassen preorder P as in Definition 10.3. Then for every $a, b \in R$:*

$$\tilde{R}_P(a) = \max_{\phi \in X_P} \phi(a), \quad (10.21)$$

$$a \leq_{\tilde{P}} b \iff \phi(a) \leq \phi(b) \quad \text{for every } \phi \in X_P. \quad (10.22)$$

If a is gapped, then in addition

$$\tilde{Q}_P(a) = \min_{\phi \in X_P} \phi(a). \quad (10.23)$$

The maximum and, in the gapped case, the minimum are attained.

Hypothesis check before using duality. The semiring must be positive and commutative, the preorder must satisfy the stated Strassen axioms, and spectral points must be monotone semiring homomorphisms normalized at 1. Formula (10.22) uses the exact $2^{o(k)}$ asymptotic overhead from the definition above. The minimum formula for asymptotic subrank additionally requires the element to be gapped; omitting that condition would strengthen the source theorem.

Source-owned proof architecture. The modern proof first closes the preorder under subexponential overhead, extends suitable closed preorders to maximal/total ones, and identifies the resulting fractional rank functional with a monotone semiring homomorphism. This yields the spectral characterization of the asymptotic preorder and of asymptotic rank. Compactness of the spectrum turns the extremal suprema and infima into a maximum and minimum. For subrank, the source isolates the additional gappedness condition before identifying the fractional quantity with $\tilde{Q}_P$. None of those separation/extension arguments is reproved here; Theorem 10.5 is an explicitly source-owned deep input.

Worked normalization check. Take $R = \mathbb{N}_{\geq 1}$ with ordinary addition, multiplication, and order. It satisfies the Strassen axioms. Any spectral point must obey $\phi(n) = n\phi(1) = n$, so X_P contains only the identity map. Hence $R_P(n) = Q_P(n) = n$ and

$$\tilde{R}_P(n) = \tilde{Q}_P(n) = n, \quad (10.24)$$

while (10.21)–(10.23) reduce to the same equality. This elementary example checks the direction of the preorder and the normalization of spectral points before applying the theorem to the tensor-restriction semiring above.

Companion boundary. The theorem no longer depends on an unresolved abstract-algebra companion pointer. The semiring, preorder, rank, subrank, asymptotic preorder, spectrum, and gap hypotheses used in its statement are all defined immediately above, and statement ownership is the verified Strassen/Wigderson–Zuiddam source lineage. A future companion-volume treatment may prove a more general ordered-semiring theorem, but it is not load-bearing for Volume XI.

10.6 Dated asymptotic-spectrum frontier

The following is a source checkpoint dated **2026-08-31**, based on the published Wigderson–Zuiddam synthesis. It is intentionally an application map, not a claim that all relevant spectra have been classified.

Setting	Role of the asymptotic-spectrum framework
Tensors and matrix multiplication	Direct sum and tensor product produce the basic semiring; spectral points dualize asymptotic rank, organize matrix-multiplication bounds, and support structural/barrier questions about tensor methods.
Graphs and communication	Suitable semirings/preorders encode asymptotic graph parameters and zero-error/communication-type quantities, showing that Strassen duality is not tensor-specific.
Probability and quantum information	Monotone multiplicative/additive functionals become spectral points for resource-like preorders, connecting asymptotic conversion rates to dual functionals.
General structure	The modern synthesis develops connectivity, convexity, density, and extension questions for spectra; determining the full spectrum in major examples remains a substantive mathematical problem rather than an automatic consequence of Theorem 10.5.

In particular, the duality theorem tells us how an already-known spectrum controls asymptotic rank; it does *not* by itself identify all spectral points. That distinction is the main frontier discipline needed when the framework is applied to matrix multiplication, tensor networks, graph parameters, or quantum resource theories.

10.7 Geometric complexity theory: verified frontier map

The algebraic-complexity results of the earlier chapters motivate a geometric reformulation but do not solve it. Let $\det_n$ be the degree- n determinant polynomial, and after fixing an embedding of the permanent variables and a homogenizing variable z , write a padded permanent as

$$z^{n-m} \text{per}_m. \quad (10.25)$$

The group $GL_{n^2}(\mathbb{C})$ acts on degree- n forms by invertible linear changes of variables. The Mulmuley–Sohoni GCT program studies lower bounds through orbit and orbit-closure geometry and seeks explicit representation-theoretic obstructions to the relevant containments.

The dated source map now records three different facts that must not be collapsed into one claim.

- (i) Mulmuley–Sohoni’s GCT framework reduces selected algebraic-complexity lower-bound questions to geometric-invariant-theory obstruction problems; this is a programmatic reduction, not a lower bound by itself.
- (ii) Bürgisser–Ikenmeyer–Panova prove that the proposed *occurrence-obstruction* route cannot separate the determinant orbit closure from the padded-permanent orbit closure in their setting. Their theorem explicitly does not rule out the more general *multiplicity-obstruction* approach.
- (iii) Derksen–Makam prove exponential lower bounds on the maximal degree of a system of generators (and of invariants defining the null cone) for two invariant-ring actions relevant to GCT: $SL(V)$ acting on $S^3(V)^{\oplus 3}$, and $SL(V) \times SL(W) \times SL(Z)$ acting on $(V \otimes W \otimes Z)^{\oplus 5}$.

The stable sources are Mulmuley–Sohoni, SIAM J. Comput. 31(2) (2001), <https://doi.org/10.1137/S009753970038715X>; Bürgisser–Ikenmeyer–Panova, J. Amer. Math.

Soc. 32 (2019), 163–193, <https://doi.org/10.1090/jams/908>; and Derksen–Makam, J. Combin. Algebra 6 (2022), 353–405, <https://doi.org/10.4171/JCA/66>.

Unresolved target. None of these results proves permanent versus determinant, $VP \neq VNP$, or a comparable permanent lower bound. PASS108 closes the *source gate* by replacing folklore summaries with this exact theorem/barrier map; it does not close the underlying GCT research program.

10.8 Source and ownership boundary

Definition 10.1 is local QRCC vocabulary in the finite-dimensional complex setting. It fixes an algebraic highest-weight normalization of the moment polytope and deliberately does not import Hilbert–Mumford, Kempf–Ness, finite-generation, or moment-polytope membership theorems without separate ownership.

Theorem 10.2 is source-owned by Peter Bürgisser, Mahmut Levent Doğan, Visu Makam, Michael Walter, and Avi Wigderson, *Complexity of Robust Orbit Problems for Torus Actions and the abc-Conjecture*, 39th Computational Complexity Conference (CCC 2024), LIPIcs 300, Article 14, DOI 10.4230/LIPIcs.CCC.2024.14. QRCC uses the paper’s algebraic logarithmic-distance model, its NP-hard approximation regime, its separation-controlled algorithm, and its exact equivalence between the algebraic separation hypothesis and Number-Theoretic Hypothesis 1.11. The ordinary *abc* conjecture is not substituted for that exact hypothesis.

Theorem 10.5 follows the modern canonical formulation in Avi Wigderson and Jeroen Zuiddam, *Asymptotic Spectra: Theory, Applications and Extensions*, Bulletin of the American Mathematical Society 63 (2026), 199–331, DOI 10.1090/bull/1880, especially their definitions of Strassen preorder/rank/subrank and their duality summary. The original lineage is Volker Strassen, *The asymptotic spectrum of tensors*, Journal für die reine und angewandte Mathematik 384 (1988), 102–152. QRCC adopts the stronger simplified Archimedean convention of the modern source and preserves the gappedness hypothesis on the subrank formula.

The dated frontier QRCC-XI-F-026 is an expository map of the same 2026 Wigderson–Zuiddam synthesis; it does not turn survey examples into new QRCC theorems. The GCT item QRCC-XI-F-029 now has a verified dated source map. The permanent-versus-determinant target itself remains open, and every stronger theorem-level claim is deferred to an exact primary source.

10.9 Reader checkpoint and exercises

Chapter checkpoint. The final chapter supplies a controlled interface to two broad research programs. Invariant theory packages orbit-closure questions into algebraic and representation-theoretic data; asymptotic spectra dualize an already-specified Strassen preorder through monotone semiring homomorphisms. The verified source theorems are strong, but neither the robust torus-orbit result nor Strassen duality resolves permanent versus determinant, classifies all spectral points, or resolves the GCT lower-bound program.

Exercises.

1. Let $\mathbb{C}^*$ act on $\mathbb{C}^2$ by $t \cdot (x, y) = (tx, t^{-1}y)$. Describe the orbit of $(1, 1)$ and determine whether 0 lies in its affine orbit closure. Identify a simple invariant polynomial.

2. For a diagonal torus action with three integer weights on a one-dimensional weight basis, compute the convex hull of the normalized occurring weights. Use this toy model to explain the “moment polytope” vocabulary without invoking a moment-map theorem.
3. Restate Theorem 10.2 without using the phrase “the *abc* conjecture.” Name the exact number-theoretic hypothesis that controls the equivalence and explain why this wording is safer.
4. Verify all Strassen axioms for $R = \mathbb{N}_{\geq 1}$ with ordinary order. Show directly that the only spectral point is the identity map.
5. For a rank-one tensor $T = u \otimes v \otimes w$, compute its tensor rank, subrank, asymptotic rank, and asymptotic subrank in the restriction semiring.
6. Explain why $\tilde{R}_P(a) = \max_{\phi \in X_P} \phi(a)$ is useful only after enough spectral points are known. Give the logical difference between a duality theorem and a classification of X_P .
7. Identify the extra hypothesis required for the asymptotic-subrank minimum formula. Construct a checklist that would prevent you from applying (10.23) to an element whose gappedness has not been established.
8. Using the GCT source-gate section, write a two-column table separating “defined locally” from “still theorem/source dependent” for orbit closure, highest-weight occurrence, occurrence obstructions, multiplicity obstructions, and permanent-versus-determinant noncontainment.

End-of-volume perspective. Volume XI began with the syntax of arithmetic circuits and ended with geometric and asymptotic languages for their hardest open problems. Along the way, normal forms enabled lower-bound measures, PIT connected algebraic structure to derandomization, noncommutativity supplied exact rank methods, and invariant theory turned full symbolic-matrix rank into an algorithmic scaling problem. The reader should now be able to distinguish what is locally proved, what is imported at verified source strength, and what remains an explicit frontier. That distinction is the prerequisite for using Volume XI safely in Volume XII and later cryptographic/complexity interfaces.

Internalization and Reconstruction Guide: Volume XI

Purpose. Move from arithmetic circuit normal forms to determinant/permanent universality, depth reduction, lower-bound measures, polynomial identity testing, scaling, invariant theory, and geometric-complexity frontiers.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-010: Gibbs inequality and nonnegativity of relative entropy.
2. QRCC-IX-DEF-001: Randomized complexity classes.
3. QRCC-III-THM-022: Subfields and Frobenius theorem.
4. QRCC-IX-DEF-018: Hitting-set generator and black-box derandomization.
5. QRCC-X-THM-015: SAT algorithms imply circuit lower bounds.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XI-DEF-001 (Arithmetic circuit models); QRCC-XI-DEF-006 (VP, VNP, and p-projections); QRCC-XI-DEF-018 (Matrix spaces, noncommutative rank, shrunk subspaces, and capacity); QRCC-XI-DEF-024 (Orbit closure, invariants, and moment polytope).

Chapter retrieval map

- Ch. 1: Arithmetic Circuits: Degree, Homogeneity, and Division:** enter through QRCC-XI-DEF-001 (Arithmetic circuit models); exit by being able to state QRCC-XI-THM-004 (Elimination of division gates) with its ownership boundary.
- Ch. 2: Valiant's Classes: VP, VNP, and Projections:** enter through QRCC-XI-DEF-006 (VP, VNP, and p-projections); exit by being able to state QRCC-XI-THM-006 (Basic closure properties of VP and VNP) with its ownership boundary.

- Ch. 3: Determinant Circuits and Weakly-Skew Universality:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XI-THM-007 (Determinant has polynomial-size arithmetic circuits) with its ownership boundary.
- Ch. 4: The Chasm at Depth Four:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XI-THM-010 (Depth reduction to depth four) with its ownership boundary.
- Ch. 5: Partial Derivatives as a Lower-Bound Measure:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XI-BB-013 (Shifted-partial-derivative lower bounds) with its ownership boundary.
- Ch. 6: Polynomial Identity Testing by Random Evaluation:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XI-COR-003 (Polynomial identity testing is in coRP) with its ownership boundary.
- Ch. 7: Derandomization Forces Lower Bounds:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XI-THM-015 (Kabanets–Impagliazzo PIT-to-lower-bounds theorem) with its ownership boundary.
- Ch. 8: Noncommutative Rank Measures and White-Box PIT:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XI-THM-012 (Nisan noncommutative formula lower bound) with its ownership boundary.
- Ch. 9: Operator Scaling, Noncommutative Rank, and the Null Cone:** enter through QRCC-XI-DEF-018 (Matrix spaces, noncommutative rank, shrunk subspaces, and capacity); exit by being able to state QRCC-XI-ALG-020 (Operator-scaling algorithm) with its ownership boundary.
- Ch. 10: Tensor Methods, Geometric Complexity Theory, and Asymptotic Spectra:** enter through QRCC-XI-DEF-024 (Orbit closure, invariants, and moment polytope); exit by being able to state QRCC-XI-BB-025 (Strassen asymptotic-spectrum duality) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XI-THM-004 — Elimination of division gates. Dependencies: QRCC-XI-DEF-001, QRCC-XI-THM-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XI-THM-007 — Determinant has polynomial-size arithmetic circuits. Dependencies: QRCC-XI-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-XI-BB-013 — Shifted-partial-derivative lower bounds. Dependencies: QRCC-XI-THM-010, QRCC-XI-THM-011. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XI-THM-015 — Kabanets–Impagliazzo PIT-to-lower-bounds theorem. Dependencies: QRCC-XI-COR-003, QRCC-XI-THM-008, QRCC-XI-THM-015. *local proof: reconstruct the proof*

from the hypotheses and named dependencies before rereading the certified argument.

5. QRCC-XI-ALG-020 — Operator-scaling algorithm. Dependencies: QRCC-XI-DEF-018. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-XI-F-027 — Constant-depth arithmetic circuits for linear-algebra problems. Dependencies: QRCC-XI-THM-007, QRCC-XI-DEF-001. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- confusing formal degree with semantic polynomial degree;
- using division gates at points where the denominator may vanish;
- mixing white-box and black-box PIT models;
- stating algebraic-geometric or GCT frontiers without their field/characteristic and closure conventions;

Boundary and degenerate cases

- constant/zero polynomials and zero divisors;
- small-characteristic failures of characteristic-zero arguments;
- division-free versus regular-center division elimination;
- orbit closure versus orbit membership and exact versus approximate scaling;

Retrieval and reconstruction drill

1. State QRCC-XI-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XI-THM-004. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XI-THM-007, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.

7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XI-F-027, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XII externalizes computation into communication, queries, streams, and data structures, providing lower-bound currencies used throughout complexity.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XII

Communication, Query, Streaming, and Data Complexity

Chapter 1

Deterministic Communication: Rectangles, Fooling Sets, and Rank

Communication complexity asks how much information two parties must exchange when an input is split between them. The local computation performed by either party is free; only communicated bits are charged. This deliberately simple model isolates an information bottleneck that will later reappear in randomized protocols, discrepancy, information complexity, lifting theorems, streaming, and data-structure lower bounds.

This first chapter fixes the model carefully and proves three elementary lower bounds. Every transcript of a deterministic protocol cuts out a combinatorial rectangle. A fooling set forces many rectangles. A communication matrix of large algebraic rank also forces many rectangles because each monochromatic leaf contributes a rank-one indicator matrix. These are different witnesses for the same geometric obstruction: a shallow protocol cannot carve the input matrix into enough simple pieces.

How to read this chapter. There are four layers. Definition 1.1 freezes the communication and error conventions. Theorem 1.2 is the structural bridge from protocols to rectangles. Theorems 1.4 and 1.5 then turn two different rectangle obstructions into lower bounds. The final section is a dated research frontier: the log-rank conjecture is not used as a theorem anywhere downstream.

A useful dependency chain is

$$\text{protocol tree} \longrightarrow \text{rectangle partition} \longrightarrow \begin{cases} \text{fooling-set obstruction,} \\ \text{rank obstruction.} \end{cases}$$

The first arrow is purely combinatorial. The fooling-set branch is also purely combinatorial; the rank branch additionally uses only elementary matrix rank.

Learning goals. After this chapter, a reader should be able to define deterministic, randomized, one-way, simultaneous, nondeterministic, and distributional communication complexity with the error convention explicit; prove that every deterministic transcript cylinder is a rectangle; derive deterministic and nondeterministic fooling-set bounds; prove the log-rank lower bound over an arbitrary field; and separate that proved lower bound from the still-open log-rank conjecture.

1.1 Models and conventions

Fix finite nonempty sets X and Y and a Boolean function

$$f : X \times Y \longrightarrow \{0, 1\}.$$

Alice receives $x \in X$ and Bob receives $y \in Y$. Local computation is free. All logarithms in this chapter are base 2 unless a different base is displayed.

Definition 1.1 (Communication protocols and complexity measures). The following conventions are in force in Volume XII.

1. A *deterministic two-party protocol* is a finite rooted binary tree. Every internal node is owned by Alice or Bob. At an Alice node, the outgoing bit is a deterministic function of x and of the transcript leading to that node; at a Bob node it is a deterministic function of y and the preceding transcript. Every leaf carries an output bit. The protocol *computes* f *exactly* when the leaf reached on every (x, y) is labelled $f(x, y)$. Its *cost* is the maximum number of communicated bits on a root-to-leaf path. The deterministic communication complexity $D(f)$ is the minimum cost of an exact deterministic protocol for f .
2. For a node or transcript prefix τ , its *transcript cylinder* is

$$C_\tau := \{(x, y) \in X \times Y : \text{the execution on } (x, y) \text{ reaches } \tau\}.$$

A *combinatorial rectangle* is a subset $A \times B$ with $A \subseteq X$ and $B \subseteq Y$. It is *b-monochromatic* when $f(x, y) = b$ throughout $A \times B$. A *monochromatic rectangle partition* is a disjoint family of monochromatic rectangles whose union is $X \times Y$. Write $\chi(f)$ for the least number of rectangles in such a partition.

3. The *communication matrix* of f is the 0/1 matrix

$$M_f = (f(x, y))_{x \in X, y \in Y}.$$

For a field K , write $r_K(f) := \text{rank}_K(M_f)$ after viewing the entries 0, 1 in K . The real rank is abbreviated $r(f) := r_{\mathbb{R}}(f)$.

4. A *public-coin randomized protocol* has access to a common random string R visible to both parties; after fixing $R = r$, its communication is a deterministic protocol. A *private-coin randomized protocol* instead gives Alice and Bob independent private random strings R_A, R_B . Its communication cost is the maximum number of transmitted bits over every input and every coin outcome. Under the *worst-case error convention*, error at most ε means

$$\Pr[\Pi(x, y) \neq f(x, y)] \leq \varepsilon \quad \text{for every } (x, y) \in X \times Y.$$

The minimum public-coin and private-coin costs are denoted $R_\varepsilon^{\text{pub}}(f)$ and $R_\varepsilon^{\text{priv}}(f)$, respectively. Unless stated otherwise, “bounded error” means $\varepsilon = 1/3$.

5. In an *Alice-to-Bob one-way protocol*, Alice sends one message $m(x)$ and Bob outputs a deterministic or randomized function of the message, his input y , and the allowed coins; only the message length is charged. In a *simultaneous-message protocol*, Alice and Bob send one message each to a referee, who outputs the answer; the sum of the two message lengths is charged. The direction, error convention, and coin model must be stated whenever one of these restricted models is used.

6. For $b \in \{0, 1\}$, let $C_b(f)$ be the minimum number of b -monochromatic rectangles whose union contains every input in $f^{-1}(b)$. We use the standard rectangle-cover normalization

$$N^b(f) := \lceil \log_2 \max\{1, C_b(f)\} \rceil$$

for b -nondeterministic communication complexity. Equivalently, a certificate names one rectangle in a fixed b -monochromatic cover and the local membership tests certify that the input lies in it. The $b = 1$ measure is often called nondeterministic complexity and the $b = 0$ measure co-nondeterministic complexity. This geometric convention is the one used in the fooling-set theorem below.

7. If μ is a probability distribution on $X \times Y$, the μ -distributional deterministic complexity with error ε is

$$D_\varepsilon^\mu(f) := \min\{\text{cost}(P) : P \text{ deterministic and } \Pr_{(x,y) \sim \mu} [P(x,y) \neq f(x,y)] \leq \varepsilon\}.$$

This is an average-error requirement under a fixed input distribution, unlike the worst-case input requirement in the randomized measures above.

Why the output convention matters. A leaf of our deterministic protocol has a fixed output label. Thus a party cannot silently use its private input after the last communicated bit to choose between two outputs. This convention makes every exact leaf rectangle monochromatic and keeps the rectangle theorem literally true. Other standard output conventions differ by at most one final bit; later quantitative statements will retain their intended asymptotic strength, but the present proofs use the fixed-leaf convention exactly.

Randomness is not yet being used. The public/private-coin and distributional notions are defined now so later chapters do not use them before definition. None is needed in the three proofs of this chapter. In particular, no form of Yao's minimax principle is being smuggled into a deterministic rectangle argument.

1.2 Protocol trees are rectangle partitions

Theorem 1.2 (Protocol rectangles and monochromatic partitions). *Let P be a deterministic protocol for $f : X \times Y \rightarrow \{0, 1\}$. Every transcript cylinder of P is a combinatorial rectangle. Consequently, if P computes f exactly, then its nonempty leaf cylinders form a monochromatic rectangle partition of $X \times Y$. If the cost of P is at most c , the partition has at most 2^c rectangles. Hence*

$$D(f) \geq \lceil \log_2 \chi(f) \rceil. \tag{1.1}$$

Proof roadmap. Induct on the transcript length. The root cylinder is $X \times Y$. At an Alice node, the next message bit filters only the set of possible x values; at a Bob node, it filters only the y values. Therefore the rectangle invariant survives one step. Leaves are disjoint and exhaustive because the protocol is deterministic. Exact correctness then makes each leaf monochromatic, and a binary tree of depth at most c has at most 2^c leaves.

Proof. At the root the transcript cylinder is $X \times Y$, a rectangle. Suppose a node with transcript prefix τ has cylinder $A \times B$.

If Alice owns the node, her next bit is a function $a_\tau(x)$ of x and the fixed prefix τ . For $e \in \{0, 1\}$ set

$$A_e := \{x \in A : a_\tau(x) = e\}.$$

The child reached by message e therefore has cylinder $A_e \times B$. If Bob owns the node, the same argument gives a child cylinder $A \times B_e$ for a suitable $B_e \subseteq B$. Induction on the transcript length proves that every transcript cylinder is a rectangle.

Every input follows exactly one root-to-leaf path, so the nonempty leaf cylinders are pairwise disjoint and cover $X \times Y$. If P computes f exactly, all inputs reaching a leaf have the leaf's fixed output bit; hence each leaf cylinder is monochromatic. A binary tree of depth at most c has at most 2^c leaves, so every exact cost- c protocol yields a monochromatic rectangle partition with at most 2^c members. Thus $\chi(f) \leq 2^c$, and minimizing c gives (1.1). $\square$

A tiny worked protocol. Let $X = Y = \{0, 1\}$ and $f(x, y) = x \wedge y$. Alice first sends x . If $x = 0$, the protocol may stop at a 0-leaf. If $x = 1$, Bob sends y and the protocol stops at the corresponding leaf. The three nonempty leaf rectangles are

$$\{0\} \times \{0, 1\}, \quad \{1\} \times \{0\}, \quad \{1\} \times \{1\}.$$

The example is intentionally about geometry, not optimality: every partial transcript restricts only rows or only columns, and every exact leaf is monochromatic.

What the theorem does not say. An arbitrary monochromatic rectangle partition need not be realizable by a protocol of cost $\lceil \log_2 \chi(f) \rceil$. Protocol partitions have a recursive row/column-cut structure. The theorem needs only the easy direction “protocol $\Rightarrow$ partition,” which is enough for lower bounds.

1.3 Fooling sets

Definition 1.3 (Fooling set). Fix $b \in \{0, 1\}$. A set $S \subseteq f^{-1}(b)$ is a *b-fooling set* if, for every two distinct $(x, y), (x', y') \in S$, at least one crossed input changes the value:

$$f(x, y') = 1 - b \quad \text{or} \quad f(x', y) = 1 - b. \quad (1.2)$$

Theorem 1.4 (Fooling-set lower bound). *If S is a b -fooling set for f , then every b -monochromatic rectangle contains at most one element of S . Consequently*

$$C_b(f) \geq |S|, \quad N^b(f) \geq \lceil \log_2 |S| \rceil, \quad D(f) \geq \lceil \log_2 |S| \rceil. \quad (1.3)$$

Here the inequalities are nontrivial when $S \neq \emptyset$.

Proof roadmap. The crossed-input condition is exactly what prevents two fooling-set points from occupying one monochromatic rectangle: rectangles contain both crossed corners whenever they contain two opposite corners. The nondeterministic bound follows from a cover count; the deterministic bound follows because the b -labelled leaves of an exact protocol form such a cover and the entire protocol has at most 2^c leaves.

Proof. Suppose a b -monochromatic rectangle $A \times B$ contained two distinct fooling-set elements (x, y) and (x', y') . Then $x, x' \in A$ and $y, y' \in B$, so both crossed pairs (x, y') and (x', y) also lie in $A \times B$. Since the rectangle is b -monochromatic, both crossed values equal b , contradicting (1.2). Thus each b -monochromatic rectangle contains at most one element of S .

Any b -monochromatic cover of $f^{-1}(b)$ must therefore use at least $|S|$ rectangles, proving $C_b(f) \geq |S|$ and the stated bound on $N^b(f)$. If a deterministic exact protocol has cost c , Theorem 1.2 provides at most 2^c monochromatic leaves in total. Its b -leaves cover $f^{-1}(b)$, so $2^c \geq |S|$. This gives the deterministic bound in (1.3). $\square$

Example: equality. For $n \geq 1$, define

$$\text{EQ}_n(x, y) = \mathbf{1}[x = y], \quad x, y \in \{0, 1\}^n.$$

The diagonal

$$S = \{(x, x) : x \in \{0, 1\}^n\}$$

is a 1-fooling set of size 2^n : for $x \neq x'$, both crossed pairs are unequal. Hence

$$D(\text{EQ}_n) \geq n, \quad N^1(\text{EQ}_n) \geq n. \quad (1.4)$$

No randomized conclusion follows from this argument; later chapters will show that public randomness changes equality dramatically.

Example: disjointness. Identify strings with subsets of $[n]$ and put

$$\text{DISJ}_n(A, B) = \mathbf{1}[A \cap B = \emptyset].$$

Then

$$S = \{(A, [n] \setminus A) : A \subseteq [n]\} \quad (1.5)$$

is a 1-fooling set. For $A \neq B$, either $A \setminus B$ or $B \setminus A$ is nonempty. In the first case $A \cap ([n] \setminus B) \neq \emptyset$; in the second case $B \cap ([n] \setminus A) \neq \emptyset$. At least one crossed pair is therefore a 0-input. Again $|S| = 2^n$, so $D(\text{DISJ}_n) \geq n$ and $N^1(\text{DISJ}_n) \geq n$.

1.4 The rank lower bound

Theorem 1.5 (Log-rank lower bound). *Let K be any field and let $r_K(f) = \text{rank}_K(M_f)$. If $r_K(f) \geq 1$, then*

$$D(f) \geq \lceil \log_2 r_K(f) \rceil. \quad (1.6)$$

Equivalently, without excluding the identically-zero function,

$$D(f) \geq \lceil \log_2 \max\{1, r_K(f)\} \rceil. \quad (1.7)$$

For a 0/1 communication matrix, the rational and real ranks agree, whereas finite-field ranks may differ from them and from one another.

Proof roadmap. Use only the 1-labelled leaves. The indicator matrix of a rectangle $A \times B$ is the outer product $\mathbf{1}_A \mathbf{1}_B^\top$ and hence has rank at most one over every field. Because the exact protocol's 1-leaves disjointly partition the 1-entries of M_f , the whole communication matrix is the sum of those rank-one indicators. Rank subadditivity then turns the number of 1-leaves into a lower bound on communication.

Proof. Let P be an exact deterministic protocol of cost c . By Theorem 1.2, its leaf cylinders are monochromatic rectangles. Let $\mathcal{L}_1$ be the set of leaves labelled 1. For $\ell \in \mathcal{L}_1$, write its rectangle as $A_\ell \times B_\ell$. Over K let

$$R_\ell := \mathbf{1}_{A_\ell} \mathbf{1}_{B_\ell}^\top.$$

This matrix has rank at most one. The 1-leaf rectangles are pairwise disjoint and their union is exactly $f^{-1}(1)$, so entrywise

$$M_f = \sum_{\ell \in \mathcal{L}_1} R_\ell. \quad (1.8)$$

No characteristic issue occurs in (1.8): every 1-entry occurs in exactly one summand. Subadditivity of matrix rank gives

$$r_K(f) \leq \sum_{\ell \in \mathcal{L}_1} \text{rank}_K(R_\ell) \leq |\mathcal{L}_1| \leq 2^c. \quad (1.9)$$

If $r_K(f) \geq 1$, taking base-2 logarithms yields $c \geq \log_2 r_K(f)$, and integrality of c proves (1.6). The zero-matrix case gives (1.7).

Finally, M_f has rational entries. The rank of a rational matrix is the largest order of a nonzero minor; a rational minor is nonzero over $\mathbb{Q}$ if and only if it is nonzero over $\mathbb{R}$. Thus $r_{\mathbb{Q}}(f) = r_{\mathbb{R}}(f)$. Reducing determinants modulo a prime can annihilate a nonzero integer numerator, so no analogous equality with arbitrary finite fields is asserted. $\square$

Worked check: equality. The matrix of EQ_n is the identity matrix of order 2^n . Therefore it has rank 2^n over every field, and Theorem 1.5 again gives $D(\text{EQ}_n) \geq n$.

Worked check: disjointness. For DISJ_n , reindex a column B by its complement $T = [n] \setminus B$. The resulting matrix is

$$Z_{A,T} = \mathbf{1}[A \subseteq T]. \quad (1.10)$$

Order all subsets of $[n]$ by any linear extension of inclusion. Then $Z_{A,T}$ can be nonzero only when the row of A occurs no later than the column of T , and every diagonal entry is 1. Thus Z is triangular with determinant 1. Column complementation is only a permutation, so M_{DISJ_n} has rank 2^n over every field. The rank method therefore also yields the n -bit deterministic lower bound for disjointness.

Rank versus rectangle counting. Theorem 1.5 is not a separate protocol geometry theorem. It is an algebraic way to certify that many rank-one rectangle indicators are needed. A fooling set and high rank can behave very differently on a particular function, so later chapters will treat them as distinct lower-bound techniques.

1.5 The log-rank conjecture: dated frontier, not a theorem

Log-rank conjecture — OPEN PROBLEM, checkpoint 2026-08-31. With the 0/1 matrix convention of Definition 1.1, the log-rank conjecture asks whether there are universal constants $C, c > 0$ such that for every finite Boolean communication problem

$$D(f) \leq C(1 + \log_2 r(f))^c. \quad (1.11)$$

The additive 1 is only a normalization for constant-rank edge cases; for $r(f) \geq 2$ this is the usual assertion that deterministic communication is polylogarithmic in real rank.

Theorem 1.5 supplies the universal lower inequality $D(f) \geq \log_2 r(f)$. As of the checkpoint date, the best refereed general upper bound recorded by the project is

$$D(f) = O\left(\sqrt{r(f)}\right), \quad (1.12)$$

due to Sudakov and Tomon. On the separation side, a 2026 preprint of Zhao Song constructs explicit Boolean matrices with

$$D(f) = \Omega\left(\frac{(\log r(f))^2}{\log \log r(f)}\right). \quad (1.13)$$

Equation (1.13) is recorded as a *preprint frontier result*, not as part of the locally proved QRCC theorem layer. A 2026 CCC paper of Hambardzumyan, Lovett, and Shirley gives new equivalent formulations, including one through signed rectangle rank; equivalence reformulations do not resolve (1.11).

Convention boundary. The literature also studies sign rank, approximate rank, approximate trace norm, and randomized or quantum variants of log-rank questions. They are different matrix parameters. No theorem in this chapter substitutes one of those ranks for the exact 0/1 real rank $r(f)$ unless a later statement explicitly proves the required comparison.

1.6 Source and ownership boundary

The historical communication-model source gate is Andrew Chi-Chih Yao, *Some Complexity Questions Related to Distributive Computing (Preliminary Report)*, STOC 1979, 209–213, DOI 10.1145/800135.804414. The local rectangle proof above is self-contained; the citation fixes historical/source ownership rather than replacing the proof.

For the fooling-set method, the project uses Alfred V. Aho, Jeffrey D. Ullman, and Mihalis Yannakakis, *On Notions of Information Transfer in VLSI Circuits*, STOC 1983, 133–139, DOI 10.1145/800061.808742, as the primary historical source gate. Theorem 1.4 is again proved locally at the exact rectangle-cover normalization fixed here.

For the rank lower bound, the project uses Kurt Mehlhorn and Erik Meineche Schmidt, *Las Vegas Is Better than Determinism in VLSI and Distributed Computing (Extended Abstract)*, STOC 1982, 330–337, DOI 10.1145/800070.802208. The arbitrary-field proof in Theorem 1.5 is local and explicitly distinguishes real, rational, and finite-field rank.

The log-rank conjecture source gate begins with László Lovász and Michael E. Saks, *Lattices, Möbius Functions and Communication Complexity*, FOCS 1988, 81–90. Its stable DOI is 10.1109/SFCS.1988.21924. The dated 2026 frontier additionally uses Benny Sudakov and István Tomon, *Matrix Discrepancy and the Log-Rank Conjecture*, Mathematical Programming 212 (2025),

567–579, DOI 10.1007/s10107-024-02117-9; Lianna Hambardzumyan, Shachar Lovett, and Morgan Shirley, *The Log-Rank Conjecture: New Equivalent Formulations*, CCC 2026, LIPIcs 383, Article 7, DOI 10.4230/LIPIcs.CCC.2026.7; and Zhao Song, *Alphabet-Preserving Lifting for the Log-Rank Conjecture*, arXiv:2608.01812 (2026), which is explicitly treated as a preprint at this checkpoint.

1.7 Reader checkpoint and exercises

What should now be stable. A deterministic transcript is not merely a string of bits: fixing it separates the possible Alice inputs from the possible Bob inputs, hence gives a rectangle. That invariant is the engine behind all three lower bounds in this chapter. Fooling sets obstruct monochromatic rectangles combinatorially. Rank obstructs their indicator matrices algebraically. The log-rank conjecture asks whether the rank obstruction is, up to a polynomial in its logarithm, already the whole story for deterministic communication.

1. **Transcript invariant.** Write out the induction in Theorem 1.2 for a protocol whose first three speakers are Alice, Bob, Alice. Identify explicitly the row set and column set after each bit.
2. **Partition lower bound.** Let P be a deterministic cost- c protocol. Prove directly, without invoking the theorem as a black box, that the number of nonempty leaves is at most 2^c even when different root-to-leaf paths have different lengths.
3. **Fooling-set geometry.** Give a 1-fooling set for EQ_2 and list all crossed pairs for two distinct members. Then explain why a set of 1-inputs need not be a fooling set.
4. **Disjointness.** Verify carefully that (1.5) is a 1-fooling set. Which part of the argument fails if one replaces $[n] \setminus A$ by A ?
5. **Rank-one rectangles.** Over an arbitrary field K , prove that the indicator matrix of every nonempty rectangle has rank exactly one. What is the rank of the empty rectangle indicator?
6. **Equality by rank.** Use the communication matrix directly to recover the lower bound $D(\text{EQ}_n) \geq n$. Compare the information used by this proof with the diagonal fooling-set proof.
7. **Disjointness determinant.** For $n = 2$, write the matrix in (1.10) with subsets ordered as $\emptyset, \{1\}, \{2\}, \{1, 2\}$. Compute its determinant and identify the triangular pattern used in the general proof.
8. **Field dependence.** Find a small 0/1 matrix whose rank over $\mathbb{R}$ differs from its rank over $\mathbb{F}_2$. Explain why Theorem 1.5 remains valid over both fields even though the numerical lower bounds may differ.
9. **Model discipline.** Explain why a public-coin protocol with worst-case error $1/3$ is not the same object as a deterministic protocol having error $1/3$ under one distribution μ . Identify which quantifier changes.
10. **Frontier reading.** Make a four-row table containing: the proved rank lower bound, the 2025 refereed general upper bound, the 2026 explicit preprint separation, and the log-rank conjectured upper bound. Label each row “proved locally,” “verified source theorem,” “preprint frontier,” or “open conjecture.”

Transition to Chapter 2. Deterministic rectangle geometry makes equality expensive. Randomness changes that example dramatically. The next chapter will introduce the public-coin and private-coin comparison, prove Newman's theorem in a finite-domain normalization, and use equality as the first clean demonstration that randomized communication can beat deterministic communication by an exponential amount in the input length.

Chapter 2

Public Coins, Private Coins, and Newman's Theorem

Randomized communication protocols can obtain striking savings by hashing a large input down to a short fingerprint. At first sight, however, public randomness looks much stronger than private randomness: a public string gives Alice and Bob an arbitrarily long synchronized random object for free, whereas private coins do not create any shared random seed. Newman's theorem shows that on a finite input domain this apparent advantage is small. A public-coin protocol can be simulated with private coins after sending only the index of one seed in a short, hardwired list.

How to read this chapter. Theorem 2.1 is the structural result. Its proof is a finite sampling argument: sample several public tapes, use Hoeffding's inequality to show that one list is simultaneously good on every input, hardwire that list, and communicate only a uniformly sampled index. The equality example then makes the mechanism concrete. It also exposes the one-bit effect of the fixed-output leaf convention from Definition 1.1.

Learning goals. After this chapter, a reader should be able to distinguish public from private coins at the level of a protocol definition; reproduce Newman's sampling and union-bound proof with all parameters explicit; explain why the overhead is logarithmic in the input length rather than in the size of the input space; and analyze deterministic and randomized fingerprinting protocols for equality.

2.1 From a common random tape to a short seed list

Recall that $R_\varepsilon^{\text{pub}}(f)$ and $R_\varepsilon^{\text{priv}}(f)$ use worst-case input and worst-case communication, with probability only over the allowed coins. For a public-coin protocol Π , fixing the public random tape $R = r$ leaves a deterministic protocol Π_r . The key point is that a finite input domain requires only a small finite multiset of such deterministic protocols to approximate the original error probability simultaneously on every input.

Theorem 2.1 (Newman's public-versus-private randomness theorem). *Let X, Y be finite nonempty sets, let $f : X \times Y \rightarrow \{0, 1\}$, let $0 \leq \varepsilon < 1$, and let $0 < \delta \leq 1 - \varepsilon$. Put $N := |X||Y|$, and let t be the least power of 2 satisfying*

$$2t\delta^2 \geq \ln(2N). \quad (2.1)$$

Then

$$R_{\varepsilon+\delta}^{\text{priv}}(f) \leq R_{\varepsilon}^{\text{pub}}(f) + \log_2 t. \quad (2.2)$$

In particular,

$$R_{\varepsilon+\delta}^{\text{priv}}(f) \leq R_{\varepsilon}^{\text{pub}}(f) + O\left(\log(1 + \log_2 N) + \log \frac{1}{\delta}\right). \quad (2.3)$$

If $X, Y \subseteq \{0, 1\}^n$, then

$$R_{\varepsilon+\delta}^{\text{priv}}(f) \leq R_{\varepsilon}^{\text{pub}}(f) + O\left(\log(n+1) + \log \frac{1}{\delta}\right). \quad (2.4)$$

All hidden constants are absolute.

Proof roadmap. Take an optimal public-coin protocol of cost c . Independently sample t public tapes. For a fixed input, the empirical fraction of tapes on which the protocol errs exceeds $\varepsilon + \delta$ with probability at most $e^{-2t\delta^2}$ by Hoeffding's inequality. A union bound over all N inputs shows that with positive probability the same sampled list is good for every input. Fix one such list once and for all. Alice privately chooses a uniform index in the list and sends that index to Bob; both parties then run the corresponding deterministic protocol. Choosing t to be a power of two makes the uniform index exact with $\log_2 t$ private bits and $\log_2 t$ communicated bits, without any rounding convention.

Proof. Let

$$c := R_{\varepsilon}^{\text{pub}}(f),$$

and choose a public-coin protocol Π of communication at most c and pointwise error at most ε . Independently sample t public tapes $R_1, \dots, R_t$ according to the public-coin distribution of Π . For a fixed input $z = (x, y) \in X \times Y$, define

$$Z_i(z) := \mathbf{1}[\Pi_{R_i}(x, y) \neq f(x, y)].$$

The random variables $Z_1(z), \dots, Z_t(z)$ are independent, lie in $[0, 1]$, and satisfy $\mathbb{E}Z_i(z) \leq \varepsilon$. Therefore Theorem 2.5 gives

$$\Pr\left[\frac{1}{t} \sum_{i=1}^t Z_i(z) > \varepsilon + \delta\right] \leq e^{-2t\delta^2}. \quad (2.5)$$

Taking a union bound over the N possible inputs,

$$\Pr\left[\exists z \in X \times Y : \frac{1}{t} \sum_{i=1}^t Z_i(z) > \varepsilon + \delta\right] \leq Ne^{-2t\delta^2} \leq \frac{1}{2} \quad (2.6)$$

by (2.1). Hence there exists a fixed list of public tapes $r_1, \dots, r_t$ such that, simultaneously for every input z ,

$$\frac{1}{t} |\{i : \Pi_{r_i}(z) \neq f(z)\}| \leq \varepsilon + \delta. \quad (2.7)$$

Hardwire this list into a new private-coin protocol. Alice uses her private coins to choose I uniformly from $[t]$ and sends the binary representation of I to Bob. Because t is a power of two, this costs exactly $\log_2 t$ bits. Both parties then execute the deterministic protocol Π_{r_I} on their inputs. The hardwired list is part of the protocol description and therefore costs no communication. By (2.7), for every input the new protocol errs with probability at most $\varepsilon + \delta$, and its worst-case communication is at most $c + \log_2 t$. This proves (2.2).

For the asymptotic form, let

$$A := \frac{\ln(2N)}{2\delta^2}.$$

The least power of two at least A satisfies $t < 2 \max\{1, A\}$. Thus

$$\log_2 t < 1 + \log_2 \max\left\{1, \frac{\ln(2N)}{2\delta^2}\right\} = O\left(\log(1 + \log_2 N) + \log \frac{1}{\delta}\right),$$

which is (2.3). If $X, Y \subseteq \{0, 1\}^n$, then $N \leq 2^{2n}$, so $\log(1 + \log_2 N) = O(\log(n + 1))$, giving (2.4). $\square$

Why this is an existence theorem rather than a shared-randomness trick. The good list $r_1, \dots, r_t$ is selected in the proof and then fixed as part of the protocol description. It is not freshly shared randomness. The only fresh random choice is Alice's private index I , and communicating I is exactly what synchronizes Bob with her choice. This distinction is the conceptual core of the theorem.

Hypothesis boundary. Finiteness of $X \times Y$ is what makes the union bound in (2.6) possible. The quantitative overhead depends on $\log \log |X \times Y|$, not on the communication cost of the original protocol. The theorem also permits arbitrary interaction inside Π ; the seed-reduction argument does not require one-way communication. Finally, the proof uses the additive Hoeffding inequality, not Yao's minimax principle.

Source boundary. Newman's 1991 paper is the historical source for the public-versus-private randomness equivalence. The exact finite-domain normalization (2.1)–(2.4), including the power-of-two seed-list choice and the fixed-output leaf convention, is proved independently here. No claim is made that these displayed constants are the paper's original notation or optimal constants.

2.2 Equality as a fingerprinting laboratory

For $n \geq 1$, recall

$$\text{EQ}_n(x, y) = \mathbf{1}[x = y], \quad x, y \in \{0, 1\}^n.$$

The same function behaves very differently under exact deterministic and public-coin communication. We use a two-bit linear fingerprint so that the error calculation is completely explicit.

Proposition 2.2 (Equality under deterministic, public, and private coins). *Under the fixed-output leaf convention of Definition 1.1, for every $n \geq 1$:*

1.

$$D(\text{EQ}_n) = n + 1. \tag{2.8}$$

2. *There is a public-coin protocol of cost 3 and worst-case error at most $1/4$. In particular,*

$$R_{1/3}^{\text{pub}}(\text{EQ}_n) \leq 3. \tag{2.9}$$

3. *Let t_n be the least power of two satisfying*

$$t_n \geq 72(2n + 1) \ln 2. \tag{2.10}$$

Then Newman's theorem gives

$$R_{1/3}^{\text{priv}}(\text{EQ}_n) \leq 3 + \log_2 t_n = O(\log(n+1)). \quad (2.11)$$

The example does not assert a matching private-coin lower bound; that requires a separate lower-bound argument and is not imported here.

Proof. For the deterministic lower bound, the diagonal $\{(x, x) : x \in \{0, 1\}^n\}$ is a 1-fooling set. More precisely, every 1-monochromatic leaf rectangle contains at most one diagonal input, so an exact protocol needs at least 2^n distinct 1-leaves. Since $n \geq 1$ there also exists a 0-input, hence at least one 0-leaf. Thus the protocol has at least $2^n + 1$ nonempty leaves. A binary protocol of cost c has at most 2^c leaves, so

$$c \geq \lceil \log_2(2^n + 1) \rceil = n + 1.$$

For the matching upper bound, Alice sends her n -bit string x and Bob sends one final bit telling whether $x = y$. The last bit is needed only because our leaves carry fixed output labels. This proves (2.8).

For the public-coin protocol, regard x, y as column vectors over $\mathbb{F}_2$. The public coins choose independently and uniformly

$$R \in \mathbb{F}_2^{2 \times n}, \quad b \in \mathbb{F}_2^2,$$

and define the two-bit hash

$$h_{R,b}(z) := Rz + b \in \mathbb{F}_2^2. \quad (2.12)$$

This is a pairwise-independent hash family. Indeed, fix $x \neq y$ and target values $u, v \in \mathbb{F}_2^2$. With $d = x - y \neq 0$, the conditions $h_{R,b}(x) = u$ and $h_{R,b}(y) = v$ are equivalent to

$$Rd = u - v, \quad b = u - Rx. \quad (2.13)$$

For either row of R , prescribing its inner product with nonzero d leaves exactly 2^{n-1} choices. Hence exactly 2^{2n-2} matrices R satisfy the first equation, and then b is forced. Out of 2^{2n+2} seeds (R, b) , each ordered pair (u, v) therefore occurs with probability $1/16$. Thus $(h_{R,b}(x), h_{R,b}(y))$ is uniform on $\mathbb{F}_2^2 \times \mathbb{F}_2^2$, the same two-point interpolation phenomenon used in Algorithm 1.2.

Alice sends the two bits $h_{R,b}(x)$. Bob compares them with $h_{R,b}(y)$ and sends one final decision bit, which becomes the fixed label of the reached leaf. If $x = y$, the fingerprints always agree. If $x \neq y$, pairwise independence gives

$$\Pr[h_{R,b}(x) = h_{R,b}(y)] = \frac{1}{4}. \quad (2.14)$$

The communication is therefore 3 bits and the worst-case error is at most $1/4$, proving (2.9).

Finally apply Theorem 2.1 with $\varepsilon = 1/4$, $\delta = 1/12$, and $N = 2^{2n}$. Condition (2.1) becomes

$$\frac{t}{72} \geq (2n+1) \ln 2,$$

which is exactly (2.10). The simulated private-coin protocol has error at most $1/3$ and communication at most $3 + \log_2 t_n$, proving (2.11). $\square$

What the example teaches. The public seed in (2.12) may be much longer than the message, but public random bits are free in the model. Newman's theorem does not ask Alice to transmit that long seed. It replaces the original seed distribution by a hardwired list of only polynomially many seeds and transmits an $O(\log n)$ -bit list index. This is why the private-coin overhead is logarithmic in n .

Convention check. In a convention where Bob may output locally after Alice's last message, the exact deterministic upper bound above would be n , and the public fingerprint would use only the two hash bits. The present volume deliberately retains the fixed-output leaf convention from Chapter 1, so both protocols carry one final output bit. The asymptotic separation is unchanged.

2.3 Chapter checkpoint

The chapter establishes the following chain without invoking any later randomized lower-bound technology:

$$\text{public coins} \xrightarrow[\text{sample} + \text{union bound}]{\text{Newman}} \text{private coins with } O(\log n) \text{ overhead,}$$

and equality supplies the first concrete fingerprinting application. The proof is deliberately one-sided: it is an upper-bound simulation theorem. To prove that randomized protocols themselves must communicate many bits for other functions, the next chapters introduce distributional reasoning, discrepancy, and corruption/rectangle methods.

Exercises

1. **Seed-list arithmetic.** Show directly that if t is the least power of two satisfying (2.1), then

$$\log_2 t \leq 1 + \max \left\{ 0, \log_2 \frac{\ln(2N)}{2\delta^2} \right\}.$$

Deduce (2.3) with an explicit absolute constant.

2. **Why one fixed list matters.** For each input z , Hoeffding shows that most sampled lists are good for z . Explain why choosing a different good list for each z would not define a valid communication protocol, and identify the exact role of the union bound.
3. **One-bit fingerprints.** Replace the two-row matrix R in (2.12) by one random row and one random offset. Compute the collision probability for $x \neq y$. Repeat the protocol independently k times and determine its communication and one-sided error under the fixed-leaf convention.
4. **Zero communication cannot solve equality.** Prove that for $n \geq 1$ no public-coin zero-communication protocol can compute EQ_n with worst-case error below $1/2$. Conclude that its constant public-coin upper bound is nontrivial even though this chapter does not optimize the constant.
5. **Domain-sensitive Newman bound.** Suppose $|X| \leq 2^a$ and $|Y| \leq 2^b$, where a and b need not be equal. Rewrite (2.3) in terms of $a + b$ and δ . Compare this with the symmetric n -bit form (2.4).
6. **Error budget.** Starting from the $1/4$ -error public protocol for equality, choose a different $\delta \in (0, 3/4]$ and write the exact seed-list condition needed for a private protocol of error at most $1/4 + \delta$. Explain the tradeoff between communication overhead and allowed error increase.

Chapter 3

Discrepancy: Rectangle Bias and the Inner-Product Lower Bound

A deterministic protocol of small cost partitions the communication matrix into few rectangles. For randomized lower bounds, monochromaticity is usually too rigid: after the coins are fixed, a protocol may make errors on a small fraction of the inputs under a chosen distribution. The right question is therefore not whether a rectangle is perfectly monochromatic, but whether any rectangle can carry substantial signed bias.

This chapter turns that idea into the discrepancy method. The basic theorem is short once the conventions are frozen: the correlation of a deterministic protocol with the target function is a sum of at most 2^c signed rectangle correlations. If every rectangle has small correlation, a low-cost protocol cannot have noticeable advantage. The inner-product function then supplies a canonical example because its sign matrix is Hadamard: row orthogonality gives a sharp enough spectral estimate on every rectangle to force linear randomized communication. The final section turns from signed cancellation to the one-sided corruption viewpoint: a low-error protocol must contain a large leaf rectangle that is nearly monochromatic on at least one chosen output side.

How to read this chapter. First freeze the sign and distribution normalization. Then Theorem 3.1 proves the discrepancy lower bound directly; no minimax theorem is needed because a worst-case randomized protocol can simply have its random tapes fixed after averaging its error under one chosen input distribution. Theorem 3.2 computes the Hadamard orthogonality of inner product and feeds it into the first theorem. The one-sided corruption method is then developed separately in Theorem 3.3, so signed cancellation and one-sided density remain visibly different lower-bound mechanisms. The final theorem reconstructs Razborov’s hard rectangle lemma for set disjointness and feeds it into that corruption interface, closing the chapter’s main randomized lower-bound example.

Learning goals. After this chapter, a reader should be able to define distributional rectangle discrepancy with the sign convention explicit; convert randomized worst-case error into one deterministic protocol with small μ -average error by fixing coins; derive the exact factor $1 - 2\varepsilon$ in the discrepancy bound; prove Hadamard row orthogonality for inner product over $\mathbb{F}_2$; use the spectral norm to obtain an $\Omega(n)$ randomized communication lower bound; prove a one-sided corruption bound by extracting a large nearly monochromatic leaf rectangle from a low-error protocol; and reconstruct the random-partition/entropy lemma that yields $R_{1/3}(\text{DISJ}_n) = \Omega(n)$.

3.1 Signed rectangles and discrepancy

Fix finite nonempty X, Y , a Boolean function $f : X \times Y \rightarrow \{0, 1\}$, and a probability distribution μ on $X \times Y$. The *sign matrix* of f is

$$F(x, y) := (-1)^{f(x, y)} \in \{-1, +1\}.$$

For a rectangle $R = A \times B$, its μ -signed mass is

$$\mu_F(R) := \sum_{(x, y) \in R} \mu(x, y) F(x, y),$$

and the μ -discrepancy of f is

$$\text{disc}_\mu(f) := \max_{A \subseteq X, B \subseteq Y} \left| \sum_{(x, y) \in A \times B} \mu(x, y) (-1)^{f(x, y)} \right|. \quad (3.1)$$

The empty rectangle contributes zero and causes no convention issue. Because μ has at least one positive-mass atom and singleton sets are rectangles, $\text{disc}_\mu(f) > 0$.

The normalization in (3.1) includes the probability weight μ itself. Some sources instead use unnormalized rectangle sums or encode Boolean values by $\{0, 1\}$; constants then look different. All statements below use (3.1).

Theorem 3.1 (Discrepancy lower bound for randomized communication). *Let X, Y be finite nonempty sets, let $f : X \times Y \rightarrow \{0, 1\}$, let μ be a probability distribution on $X \times Y$, and let $0 \leq \varepsilon < 1/2$. Then*

$$D_\varepsilon^\mu(f) \geq \log_2 \left(\frac{1 - 2\varepsilon}{\text{disc}_\mu(f)} \right). \quad (3.2)$$

Moreover the same lower bound holds for both worst-case randomized models:

$$R_\varepsilon^{\text{pub}}(f), \quad R_\varepsilon^{\text{priv}}(f) \geq \log_2 \left(\frac{1 - 2\varepsilon}{\text{disc}_\mu(f)} \right). \quad (3.3)$$

Consequently one may maximize the right-hand side over all input distributions μ .

Proof roadmap. Take a deterministic cost- c protocol whose μ -error is at most ε . Its leaf cylinders are at most 2^c disjoint rectangles by Theorem 1.2, even though they need not now be monochromatic for f . The protocol's own output is constant on each leaf. Encoding target and output as signs turns agreement minus disagreement into the correlation $1 - 2\eta$, where η is the error. Each leaf contributes at most one discrepancy unit in absolute value, hence $1 - 2\varepsilon \leq 2^c \text{disc}_\mu(f)$. For a randomized protocol, first average its pointwise error over μ and then fix a random tape that does no worse than that average.

Proof. Let P be a deterministic protocol of cost at most c , and let

$$\eta := \Pr_{(x, y) \sim \mu} [P(x, y) \neq f(x, y)] \leq \varepsilon.$$

Write

$$G(x, y) := (-1)^{P(x, y)}.$$

Since $F(x, y)G(x, y)$ equals $+1$ on correct inputs and -1 on incorrect inputs,

$$\mathbb{E}_{(x,y) \sim \mu}[F(x, y)G(x, y)] = 1 - 2\eta \geq 1 - 2\varepsilon. \quad (3.4)$$

By Theorem 1.2, the nonempty leaf cylinders of P form a disjoint rectangle partition $R_1, \dots, R_m$ with $m \leq 2^c$. The fixed label of leaf R_j makes G constant on that rectangle; write this sign as $\sigma_j \in \{-1, +1\}$. Therefore

$$\begin{aligned} 1 - 2\varepsilon &\leq \sum_{j=1}^m \sigma_j \sum_{(x,y) \in R_j} \mu(x, y) F(x, y) \\ &\leq \sum_{j=1}^m \left| \sum_{(x,y) \in R_j} \mu(x, y) F(x, y) \right| \\ &\leq m \operatorname{disc}_\mu(f) \leq 2^c \operatorname{disc}_\mu(f). \end{aligned}$$

Because $\operatorname{disc}_\mu(f) > 0$, taking base-2 logarithms gives $c \geq \log_2((1 - 2\varepsilon)/\operatorname{disc}_\mu(f))$. Minimizing over all deterministic P with μ -error at most ε proves (3.2).

Now let Π be either a public-coin or private-coin protocol with worst-case error at most ε and communication at most c . Let ρ denote the complete outcome of all random tapes used by Π ; after fixing ρ , the protocol Π_ρ is deterministic and still has communication at most c . If $e_\rho(x, y)$ is its error indicator, then

$$\begin{aligned} \mathbb{E}_\rho \Pr_{(x,y) \sim \mu}[e_\rho(x, y) = 1] &= \mathbb{E}_{(x,y) \sim \mu} \Pr_\rho[\Pi_\rho(x, y) \neq f(x, y)] \\ &\leq \varepsilon. \end{aligned}$$

Hence some fixed tape outcome ρ_* yields a deterministic cost- c protocol with μ -average error at most ε . Applying (3.2) to Π_{ρ_*} proves (3.3). Since (3.3) holds for every μ , its right-hand side may be maximized over μ . $\square$

Why minimax is not load-bearing here. The last paragraph uses only the elementary fact that some value of a random variable is no larger than its expectation. It proves a lower bound on a worst-case randomized protocol from any single hard distribution μ . Yao's minimax theorem becomes essential when one wants an equality between optimized randomized and distributional quantities; that stronger optimization identity is already proved elsewhere in the series as 'QRCC-IX-THM-005', but it is not a hidden dependency of Theorem 3.1.

A 2×2 discrepancy check. For one-bit inner product, $f(x, y) = xy$ on $\{0, 1\}^2$, and under the uniform distribution the sign matrix is

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The first row is a rectangle of signed mass $2/4 = 1/2$, so discrepancy is at least $1/2$. A direct check of the nine choices of nonempty row/column subsets shows that no rectangle has larger normalized signed mass. Thus $\operatorname{disc}_{\text{unif}}(f) = 1/2$. The example is intentionally small: it checks the probability normalization in (3.1), not the asymptotic strength of the method.

Boundary with corruption. Discrepancy is a two-sided signed-correlation measure. A corruption or rectangle bound instead asks whether a sufficiently large rectangle can be too close to one output value under a chosen distribution. That one-sided density notion has different hypotheses and bookkeeping and is owned by the separately registered ‘QRCC-XII-THM-006’; it is not being claimed as a corollary merely by changing terminology.

3.2 The Hadamard witness: inner product modulo two

For $n \geq 1$, define

$$\text{IP}_n(x, y) := \sum_{i=1}^n x_i y_i \pmod{2}, \quad x, y \in \mathbb{F}_2^n. \quad (3.5)$$

Its sign matrix is

$$H_n = (H_n(x, y))_{x, y \in \mathbb{F}_2^n}, \quad H_n(x, y) := (-1)^{x \cdot y}. \quad (3.6)$$

For a real matrix A , its Euclidean operator norm is

$$\|A\|_{2 \rightarrow 2} := \max_{v \neq 0} \frac{\|Av\|_2}{\|v\|_2}. \quad (3.7)$$

A square $\{\pm 1\}$ matrix is called a *Hadamard matrix* when its rows are pairwise orthogonal. Equation (3.6) is the Sylvester/Hadamard matrix associated with the bilinear form $x \cdot y$ over $\mathbb{F}_2$.

Theorem 3.2 (Inner-product discrepancy and randomized lower bound). *Let $n \geq 1$, and let μ_n be the uniform distribution on $\mathbb{F}_2^n \times \mathbb{F}_2^n$. The sign matrix H_n satisfies*

$$H_n H_n^\top = 2^n I, \quad \|H_n\|_{2 \rightarrow 2} = 2^{n/2}, \quad (3.8)$$

and

$$\text{disc}_{\mu_n}(\text{IP}_n) \leq 2^{-n/2}. \quad (3.9)$$

Hence for every $0 \leq \varepsilon < 1/2$,

$$R_\varepsilon^{\text{pub}}(\text{IP}_n), \quad R_\varepsilon^{\text{priv}}(\text{IP}_n) \geq \frac{n}{2} + \log_2(1 - 2\varepsilon). \quad (3.10)$$

In particular, at error $1/3$ both randomized communication complexities are $\Omega(n)$; together with the exact deterministic protocol that sends the input and one final output bit, they are $\Theta(n)$ under the fixed-output leaf convention.

Proof roadmap. Two distinct rows of H_n are orthogonal because a nonzero linear functional on $\mathbb{F}_2^n$ takes the values 0 and 1 equally often. This gives (3.8). For a rectangle $A \times B$, its unnormalized signed sum is $\mathbf{1}_A^\top H_n \mathbf{1}_B$. Cauchy–Schwarz and the operator norm bound this by $2^{n/2} \sqrt{|A||B|} \leq 2^{3n/2}$. Dividing by the 2^{2n} input pairs yields (3.9), and Theorem 3.1 finishes the communication lower bound.

Proof. For $x, x' \in \mathbb{F}_2^n$,

$$(H_n H_n^\top)(x, x') = \sum_{y \in \mathbb{F}_2^n} (-1)^{(x+x') \cdot y}. \quad (3.11)$$

If $x = x'$, every summand is 1 and the sum is 2^n . If $x \neq x'$, put $d = x + x' \neq 0$ and choose a coordinate j with $d_j = 1$. Pair each y with $y + e_j$. The two terms in every pair have opposite signs,

so the sum in (3.11) is 0. Therefore $H_n H_n^\top = 2^n I$. All singular values of H_n are consequently $2^{n/2}$, proving the operator-norm identity in (3.8).

Let $A, B \subseteq \mathbb{F}_2^n$ and let $\mathbf{1}_A, \mathbf{1}_B$ be their indicator column vectors. The unnormalized sign sum over the rectangle is

$$S(A, B) := \sum_{x \in A, y \in B} (-1)^{x \cdot y} = \mathbf{1}_A^\top H_n \mathbf{1}_B.$$

By Cauchy–Schwarz and (3.8),

$$\begin{aligned} |S(A, B)| &\leq \|\mathbf{1}_A\|_2 \|H_n \mathbf{1}_B\|_2 \\ &\leq \sqrt{|A|} 2^{n/2} \sqrt{|B|} \\ &\leq 2^{3n/2}, \end{aligned}$$

where the last step uses $|A|, |B| \leq 2^n$. Under μ_n each input pair has mass 2^{-2n} , so

$$\left| \sum_{x \in A, y \in B} \mu_n(x, y) (-1)^{\text{IP}_n(x, y)} \right| \leq 2^{-n/2}.$$

Taking the maximum over rectangles proves (3.9).

Apply Theorem 3.1 with $\mu = \mu_n$:

$$R_\varepsilon^{\text{pub}}(\text{IP}_n), R_\varepsilon^{\text{priv}}(\text{IP}_n) \geq \log_2((1 - 2\varepsilon)2^{n/2}) = \frac{n}{2} + \log_2(1 - 2\varepsilon),$$

which is (3.10). For $\varepsilon = 1/3$, the right-hand side is $n/2 - \log_2 3$, hence linear in n . Conversely Alice can send all n bits of x , after which Bob computes $x \cdot y$ and sends the one final output bit required by our fixed-output convention. Thus $D(\text{IP}_n) \leq n + 1$, and the bounded-error public- and private-coin costs are both $\Theta(n)$. $\square$

Source boundary. Chor and Goldreich’s 1988 paper explicitly develops the Hadamard-submatrix imbalance method, proves the relevant Hadamard rectangle-sum estimate, identifies inner product modulo two with the Sylvester/Hadamard construction, and derives linear probabilistic communication lower bounds. PASS61 uses that paper for primary provenance and theorem-architecture comparison. Equations (3.1)–(3.10), including the exact modern discrepancy normalization and the worst-case public/private statement, are proved locally rather than attributed verbatim to the historical notation.

3.3 One-sided corruption and large rectangles

Discrepancy controls signed cancellation on every rectangle. Corruption asks a different question: after choosing one output value b , how large can a rectangle be while containing only a small conditional mass of inputs with the opposite value? The distinction matters for functions such as disjointness, where the hard rectangle lemma is naturally one-sided.

Fix $b \in \{0, 1\}$ and $0 \leq \theta \leq 1$. A rectangle $R \subseteq X \times Y$ is called (b, θ) -monochromatic under μ if

$$\mu(R \cap f^{-1}(1 - b)) \leq \theta \mu(R). \quad (3.12)$$

Thus at most a θ fraction of the μ -mass of R has the wrong value for the proposed label b . Define the largest such rectangle mass by

$$\text{mono}_{\mu, \theta}^b(f) := \max\{\mu(R) : R \text{ is a rectangle satisfying (3.12)}\}, \quad (3.13)$$

and write

$$\alpha_b := \mu(f^{-1}(b)). \quad (3.14)$$

The case $\theta = 0$ means exactly b -monochromatic. The parameter in (3.13) is deliberately side-specific; taking a maximum or minimum over b without saying which side is protected can erase the hypothesis needed by an application.

Theorem 3.3 (One-sided corruption/rectangle lower bound). *Let X, Y be finite nonempty sets, let $f : X \times Y \rightarrow \{0, 1\}$, let μ be a probability distribution on $X \times Y$, and let $0 \leq \varepsilon < 1/2$. Fix $b \in \{0, 1\}$ and $0 < \theta \leq 1$, and put*

$$\kappa_b := \alpha_b - \varepsilon - \frac{\varepsilon}{\theta}. \quad (3.15)$$

If $\kappa_b > 0$, then

$$D_\varepsilon^\mu(f) \geq \log_2 \left(\frac{\kappa_b}{\text{mono}_{\mu, \theta}^b(f)} \right). \quad (3.16)$$

The same lower bound holds for both worst-case randomized models $R_\varepsilon^{\text{pub}}(f)$ and $R_\varepsilon^{\text{priv}}(f)$. Consequently, if for both $b \in \{0, 1\}$ one has $\alpha_b \geq \alpha$ and $\text{mono}_{\mu, \theta}^b(f) \leq m$, with $\alpha - \varepsilon - \varepsilon/\theta > 0$, then all three complexities are at least

$$\log_2 \left(\frac{\alpha - \varepsilon - \varepsilon/\theta}{m} \right). \quad (3.17)$$

For the exact-error endpoint, if $\varepsilon = 0$, $\alpha_b > 0$, and $\theta = 0$, then

$$D_0^\mu(f), \quad R_0^{\text{pub}}(f), \quad R_0^{\text{priv}}(f) \geq \log_2 \left(\frac{\alpha_b}{\text{mono}_{\mu, 0}^b(f)} \right). \quad (3.18)$$

Proof roadmap. Consider the leaves labeled b in a deterministic cost- c protocol. Their union has mass at least $\alpha_b - \varepsilon$: at most ε mass of true b -inputs can be lost to leaves labeled $1 - b$. Call a b -leaf *bad* if more than a θ fraction of its mass consists of $(1 - b)$ -inputs. Since all false-positive mass over every b -leaf is at most ε , the bad leaves have total mass at most ε/θ . Hence the good b -leaves carry at least κ_b mass. There are at most 2^c leaves, so one good leaf has mass at least $\kappa_b/2^c$ and is a (b, θ) -monochromatic rectangle. Rearranging gives (3.16). As in the discrepancy theorem, worst-case randomized protocols need only have their coins fixed after averaging under this one distribution μ .

Proof. Let P be a deterministic protocol of cost at most c and μ -error at most ε . By Theorem 1.2, its nonempty leaf cylinders are disjoint rectangles and there are at most 2^c of them. Let $\mathcal{L}_b$ denote the set of leaves whose fixed output label is b , and put $U_b := \bigcup_{L \in \mathcal{L}_b} L$.

The μ -mass of true b -inputs sent to leaves labeled $1 - b$ is at most the total error ε . Therefore

$$\mu(U_b) \geq \alpha_b - \varepsilon. \quad (3.19)$$

Split $\mathcal{L}_b$ into good and bad leaves, where L is good when

$$\mu(L \cap f^{-1}(1 - b)) \leq \theta \mu(L)$$

and bad otherwise. If M_{bad} is the total μ -mass of the bad b -leaves, then disjointness of the leaves gives

$$\varepsilon \geq \sum_{L \in \mathcal{L}_b} \mu(L \cap f^{-1}(1 - b)) > \theta M_{\text{bad}},$$

so $M_{\text{bad}} \leq \varepsilon/\theta$. Combining this with (3.19), the good b -leaves have total mass at least

$$\alpha_b - \varepsilon - \frac{\varepsilon}{\theta} = \kappa_b. \quad (3.20)$$

When $\kappa_b > 0$, at least one good leaf L_* therefore satisfies

$$\mu(L_*) \geq \frac{\kappa_b}{2^c}.$$

The leaf L_* is a rectangle and is (b, θ) -monochromatic, hence by (3.13)

$$\text{mono}_{\mu, \theta}^b(f) \geq \frac{\kappa_b}{2^c}.$$

The left-hand side is positive, and rearranging proves (3.16). The balanced consequence (3.17) follows by applying this bound to either side and inserting the common estimates $\alpha_b \geq \alpha$ and $\text{mono}_{\mu, \theta}^b(f) \leq m$.

For a public- or private-coin protocol of worst-case error at most ε , fix the complete random tape exactly as in the proof of Theorem 3.1. Some fixing has μ -average error at most ε and no larger communication cost, so the deterministic argument applies verbatim.

Finally suppose $\varepsilon = 0$ and $\theta = 0$. Every b -input must end at a b -leaf and no $(1 - b)$ -input may end there. Thus the exact b -monochromatic leaf rectangles have total mass α_b ; one has mass at least $\alpha_b/2^c$. This proves (3.18), including the randomized statement after zero-error coin fixing. $\square$

Why the one-sided normalization matters. The numerator in (3.16) contains two distinct losses. The term ε allows true b -inputs to be sent to the wrong output side; the term ε/θ discards b -leaves whose conditional corruption is too high. A hard-distribution application must therefore control both the prior mass α_b and the maximum mass of a nearly b -monochromatic rectangle. No signed cancellation is involved.

Exact AND sanity check. Let $f(x, y) = x \wedge y$ on $\{0, 1\}^2$ under the uniform distribution. For $b = 1$, $\alpha_1 = 1/4$ and the largest exactly 1-monochromatic rectangle is the singleton $\{1\} \times \{1\}$ of mass $1/4$, so (3.18) gives the expected trivial lower bound 0. For $b = 0$, $\alpha_0 = 3/4$, while the largest exactly 0-monochromatic rectangle has mass $1/2$ (a full row or column avoiding $(1, 1)$). Hence (3.18) gives $D_0^\mu(f) \geq \log_2(3/2)$ and therefore, by integrality, cost at least one. The example is deliberately not tight; it checks the side-specific mass convention and shows that the two output sides can carry very different corruption information.

Source and minimax boundary. Lee and Shraibman's canonical lower-bound survey develops the corruption method in its randomized-communication chapter: an almost-monochromatic rectangle is measured by its μ -mass and its conditional opposite-label mass, and the corruption theorem derives a communication lower bound by separating protocol leaves into sufficiently clean and overly corrupted rectangles. The survey records the historical lineage through Yao and later reformulations. PASS62 uses this as a theorem-strength normalization/provenance gate, but proves the b -symmetric form (3.16) independently. In particular, the randomized consequence above again uses direct coin fixing for one fixed μ ; the full Yao minimax equality is not load-bearing.

3.4 Razborov's hard distribution for set disjointness

Recall that

$$\text{DISJ}_n(A, B) = 1 \iff A \cap B = \emptyset, \quad A, B \subseteq [n].$$

The deterministic fooling-set proof from Chapter 1 already gives a linear exact lower bound, but it says nothing about bounded-error protocols. The missing ingredient is a distribution under which every sufficiently large rectangle containing many disjoint pairs must also contain a definite proportion of pairs with a unique intersection. Razborov's proof obtains exactly this statement by conditioning on a random partition and then using entropy to show that the conditioning cannot be badly unbalanced too often.

For a finite set U and integer k , write $\binom{U}{k}$ for the family of all k -element subsets of U .

Theorem 3.4 (Razborov disjointness lower bound). *There are absolute constants $c > 0$ and n_0 such that for every $n \geq n_0$,*

$$R_{1/3}^{\text{pub}}(\text{DISJ}_n), \quad R_{1/3}^{\text{priv}}(\text{DISJ}_n) \geq cn. \quad (3.21)$$

More precisely, let $N = 4m - 1$ with $m \geq 260$, and let μ_0 and μ_1 be the uniform distributions on

$$\Omega_0 := \{(A, B) : A, B \in \binom{[N]}{m}, |A \cap B| = 0\}, \quad \Omega_1 := \{(A, B) : A, B \in \binom{[N]}{m}, |A \cap B| = 1\}, \quad (3.22)$$

respectively. Put $\delta_N := 2^{-N/100}$. Then every rectangle $Q \subseteq \binom{[N]}{m} \times \binom{[N]}{m}$ satisfies

$$\mu_1(Q) \geq \frac{1}{45}\mu_0(Q) - \frac{2}{3}\delta_N. \quad (3.23)$$

If

$$\rho := \frac{3}{4}\mu_0 + \frac{1}{4}\mu_1, \quad \theta := \frac{1}{272}, \quad (3.24)$$

then, with output side $b = 1$ for disjoint pairs,

$$\text{mono}_{\rho, \theta}^1(\text{DISJ}_N) \leq 45\delta_N. \quad (3.25)$$

Consequently, for $\varepsilon_0 := 10^{-3}$,

$$R_{\varepsilon_0}^{\text{pub}}(\text{DISJ}_N), \quad R_{\varepsilon_0}^{\text{priv}}(\text{DISJ}_N) \geq \frac{N}{100} - \log_2\left(\frac{45000}{477}\right). \quad (3.26)$$

Proof roadmap. Generate a disjoint pair and a uniquely intersecting pair from the same random partition $[N] = Z_A \sqcup Z_B \sqcup \{i\}$ with $|Z_A| = |Z_B| = 2m - 1$. Conditional on the partition, Alice and Bob choose independent uniform m -subsets of $Z_A \cup \{i\}$ and $Z_B \cup \{i\}$. For a fixed rectangle, compare the probability that a side of the rectangle is hit when the distinguished point i is forced out versus forced in. A partition is called bad when forcing i in loses more than a factor three, apart from an exponentially small slack. If too many choices of i were bad, a dense family of m -subsets would have at least one fifth of its coordinate marginals below $1/4$; binary entropy then gives an upper bound on the family that contradicts its assumed density. A weighted averaging step shows that bad partitions carry at most four fifths of the disjoint rectangle mass. On the remaining partitions, the two factor-three comparisons multiply to give (3.23). Finally (3.23) bounds large nearly-1-monochromatic rectangles under ρ , Theorem 3.3 gives a small-constant-error linear lower bound, and 125 independent repetitions amplify a hypothetical $1/3$ -error protocol to that error level at only constant-factor cost.

Proof. We first prove the rectangle comparison (3.23). Choose uniformly a partition

$$t = (Z_A, Z_B, \{i\}) \quad (3.27)$$

of $[N]$ into parts of sizes $2m-1, 2m-1, 1$. Conditional on t , let A be a uniform member of $\binom{Z_A \cup \{i\}}{m}$ and let B be an independent uniform member of $\binom{Z_B \cup \{i\}}{m}$. Since each of the two underlying sets has size $2m$,

$$\Pr[i \in A \mid t] = \Pr[i \in B \mid t] = \frac{1}{2}. \quad (3.28)$$

Let A_0, A_1 denote the conditional versions of A given $i \notin A$ and $i \in A$, respectively; define B_0, B_1 analogously. By permutation symmetry, (A_0, B_0) is uniform on Ω_0 and (A_1, B_1) is uniform on Ω_1 .

Fix a rectangle $Q = \mathcal{A} \times \mathcal{B}$. For a partition t , define

$$\begin{aligned} p_A(t) &:= \Pr[A \in \mathcal{A} \mid t], & p_{A,v}(t) &:= \Pr[A_v \in \mathcal{A} \mid t], \\ p_B(t) &:= \Pr[B \in \mathcal{B} \mid t], & p_{B,v}(t) &:= \Pr[B_v \in \mathcal{B} \mid t] \quad (v \in \{0, 1\}). \end{aligned} \quad (3.29)$$

Equation (3.28) gives

$$p_A = \frac{1}{2}(p_{A,0} + p_{A,1}), \quad p_B = \frac{1}{2}(p_{B,0} + p_{B,1}). \quad (3.30)$$

Also, p_A and $p_{B,0}$ depend only on Z_B , while p_B and $p_{A,0}$ depend only on Z_A . Indeed, A ranges over the m -subsets of $[N] \setminus Z_B$, whereas B_0 ranges over the m -subsets of Z_B ; the other pair is symmetric. Averaging the remaining split shows

$$\mathbb{E}[p_{A,0} \mid Z_B] = p_A, \quad \mathbb{E}[p_{B,0} \mid Z_A] = p_B. \quad (3.31)$$

For example, after Z_B is fixed, i is uniform in the $2m$ -set $U = [N] \setminus Z_B$. Every $S \in \binom{U}{m}$ occurs as A_0 for exactly the m choices $i \in U \setminus S$, so the averaged law of A_0 is uniform on $\binom{U}{m}$, which is the law of A under the same conditioning.

Call t *A-bad* when

$$p_{A,1}(t) < \frac{1}{3}p_{A,0}(t) - \delta_N, \quad (3.32)$$

and define *B-bad* symmetrically. We claim that, for every fixed Z_B ,

$$\Pr[t \text{ is } A\text{-bad} \mid Z_B] < \frac{1}{5}, \quad (3.33)$$

with the symmetric statement after fixing Z_A .

Fix Z_B and put $U = [N] \setminus Z_B$, so $|U| = 2m$. The value p_A is now constant. If $p_A < \delta_N$, then (3.30) gives $p_{A,0} \leq 2p_A < 2\delta_N$, and the right-hand side of (3.32) is negative; thus no partition is *A-bad*. Assume therefore $p_A \geq \delta_N$ and let

$$\mathcal{S} := \mathcal{A} \cap \binom{U}{m}.$$

Choose W uniformly from $\mathcal{S}$, and for $j \in U$ put $q_j := \Pr[j \in W]$. Then

$$p_A = \frac{|\mathcal{S}|}{\binom{2m}{m}}, \quad p_{A,1} = 2p_A q_i, \quad p_{A,0} = 2p_A(1 - q_i). \quad (3.34)$$

Hence *A-badness* implies $q_i < \frac{1}{3}(1 - q_i)$ and therefore $q_i < 1/4$.

Suppose contrary to (3.33) that at least one fifth of the $2m$ choices of i are A -bad. Regard W as its $2m$ -bit incidence vector. Let

$$h_2(q) := -q \log_2 q - (1-q) \log_2(1-q)$$

with the usual continuous values at $q = 0, 1$. Since W is uniform on $\mathcal{S}$,

$$\begin{aligned} H(W) &= \log_2 |\mathcal{S}| \\ &\geq \log_2 \binom{2m}{m} - \frac{N}{100} \\ &\geq 2m - \log_2(2m+1) - \frac{4m-1}{100}. \end{aligned} \tag{3.35}$$

The last inequality uses $\binom{2m}{m} \geq 2^{2m}/(2m+1)$. Entropy subadditivity and monotonicity of h_2 on $[0, 1/2]$ give, because at least $2m/5$ coordinate marginals are below $1/4$,

$$H(W) \leq \sum_{j \in U} h_2(q_j) \leq \frac{8m}{5} + \frac{2m}{5} h_2(1/4) < 1.925m. \tag{3.36}$$

For $m \geq 260$, the right-hand side of (3.35) is strictly larger than $1.925m$, a contradiction. This proves (3.33) and its symmetric counterpart.

Let $\chi_A(t), \chi_B(t)$ be the indicators of A -badness and B -badness, and let $\chi(t)$ indicate their union. From (3.30), $p_{A,0} \leq 2p_A$ and $p_{B,0} \leq 2p_B$. Conditioning on Z_B , using that $p_{B,0}$ and p_A are fixed, then applying (3.31) and (3.33), gives

$$\begin{aligned} \mathbb{E}[p_{A,0} p_{B,0} \chi_A \mid Z_B] &\leq 2p_{B,0} p_A \Pr[\chi_A = 1 \mid Z_B] \\ &< \frac{2}{5} p_{B,0} p_A = \frac{2}{5} \mathbb{E}[p_{A,0} p_{B,0} \mid Z_B]. \end{aligned} \tag{3.37}$$

The symmetric estimate for χ_B and $\chi \leq \chi_A + \chi_B$ imply

$$\mathbb{E}[p_{A,0} p_{B,0} \chi] \leq \frac{4}{5} \mathbb{E}[p_{A,0} p_{B,0}]. \tag{3.38}$$

On a nonbad partition, (3.32) and its B -analogue hold in the reverse direction. For $x, y \geq 0$ one has $(x - \delta_N)_+(y - \delta_N)_+ \geq xy - \delta_N(x + y)$. Applying this with $x = p_{A,0}/3$ and $y = p_{B,0}/3$ yields

$$p_{A,1} p_{B,1} \geq \frac{1}{9} p_{A,0} p_{B,0} - \frac{\delta_N}{3} (p_{A,0} + p_{B,0}) \geq \frac{1}{9} p_{A,0} p_{B,0} - \frac{2}{3} \delta_N. \tag{3.39}$$

Therefore, using (3.38),

$$\begin{aligned} \mu_1(Q) &= \mathbb{E}[p_{A,1} p_{B,1}] \\ &\geq \frac{1}{9} \mathbb{E}[p_{A,0} p_{B,0} (1 - \chi)] - \frac{2}{3} \delta_N \\ &\geq \frac{1}{45} \mathbb{E}[p_{A,0} p_{B,0}] - \frac{2}{3} \delta_N \\ &= \frac{1}{45} \mu_0(Q) - \frac{2}{3} \delta_N, \end{aligned}$$

which is (3.23).

We now convert the rectangle comparison into the form needed by Theorem 3.3. Under ρ from (3.24), disjoint pairs have output 1 and prior mass $\alpha_1 = 3/4$, while unique-intersection pairs have output 0. Equation (3.23) gives every rectangle Q the inequality

$$\rho(Q \cap \text{DISJ}_N^{-1}(0)) \geq \frac{1}{135} \rho(Q \cap \text{DISJ}_N^{-1}(1)) - \frac{1}{6} \delta_N. \quad (3.40)$$

If Q is $(1, \theta)$ -monochromatic for $\theta = 1/272$, then its 0-mass is at most $\theta \rho(Q)$ and its 1-mass is at least $(1 - \theta) \rho(Q)$. Substituting these two inequalities into (3.40) gives

$$\left(\frac{1}{135} (1 - \theta) - \theta \right) \rho(Q) \leq \frac{1}{6} \delta_N.$$

The coefficient in parentheses is $1/270$, so $\rho(Q) \leq 45 \delta_N$ and (3.25) follows.

Take $\varepsilon_0 = 1/1000$. For side $b = 1$ and $\theta = 1/272$, the numerator in Theorem 3.3 is

$$\kappa_1 = \frac{3}{4} - \frac{1}{1000} - \frac{272}{1000} = \frac{477}{1000} > 0. \quad (3.41)$$

Combining (3.25) with that theorem gives (3.26) for both public and private coins.

It remains to return to the standard error $1/3$. Suppose a public- or private-coin protocol has cost c and worst-case error at most $1/3$. Run it independently 125 times with fresh allowed coins and output the majority bit. For each fixed input, the 125 correctness indicators are independent, have mean at least $2/3$, and Hoeffding's inequality gives majority-error at most

$$\exp\left(-2 \cdot 125 \cdot (1/6)^2\right) = e^{-125/18} < 10^{-3}. \quad (3.42)$$

The amplified protocol has cost at most $125c$. Hence (3.26) implies

$$R_{1/3}^{\text{pub}}(\text{DISJ}_N), \quad R_{1/3}^{\text{priv}}(\text{DISJ}_N) \geq \frac{1}{125} \left(\frac{N}{100} - \log_2 \frac{45000}{477} \right) = \Omega(N). \quad (3.43)$$

Finally, for arbitrary sufficiently large n , choose $m = \lfloor (n+1)/4 \rfloor$ and $N = 4m - 1$. Then $N \leq n$ and $N \geq n - 3$. Restricting DISJ_n to inputs supported on the first N coordinates gives DISJ_N , so communication complexity cannot decrease under this restriction. Equation (3.43) therefore proves (3.21). $\square$

What the entropy step is really doing. The family $\mathcal{S}$ in the proof can be exponentially large only if most coordinates have reasonably balanced inclusion probabilities. A bad conditioning forces the distinguished coordinate's inclusion probability below $1/4$. If this happened on a constant fraction of coordinates, subadditivity would cap the entropy of a uniform member of $\mathcal{S}$ strictly below the entropy forced by $|\mathcal{S}| \geq 2^{-N/100} \binom{2m}{m}$. The argument is therefore not a generic “large rectangle” counting trick: it is a statement that dense slices of the middle layer cannot be highly biased in too many coordinate directions at once.

Source and normalization boundary. Razborov's 1992 paper proves linear distributional complexity for disjointness under the mixture that puts weight $3/4$ on uniformly random disjoint m -sets and weight $1/4$ on uniformly random pairs with a unique intersection. Its main lemma compares every rectangle's unique-intersection probability with its disjoint probability up to an exponentially small additive term, using the same random-partition and entropy architecture reconstructed above. PASS63 keeps that architecture and the factor-three badness threshold, but owns the explicit finite normalization $m \geq 260$, $\delta_N = 2^{-N/100}$, the displayed $1/45$ constant, the reduction through Theorem 3.3, and the 125-fold amplification calculation. These local constants are therefore proof-owned here rather than represented as a verbatim theorem statement from the source.

3.5 Chapter checkpoint

The chapter now contains two complementary generic rectangle lower-bound mechanisms and one complete hard-distribution application. For discrepancy,

$$\begin{aligned} \text{small randomized cost} &\implies \text{one low-error deterministic fixing under } \mu \\ &\implies \text{few signed rectangles} \implies \text{large discrepancy,} \end{aligned}$$

and Hadamard orthogonality supplies $\text{disc}_{\text{unif}}(\text{IP}_n) \leq 2^{-n/2}$. For corruption, the same low-cost protocol geometry instead forces a large leaf rectangle with small *conditional opposite-label mass*; there is no signed cancellation. Both generic randomized consequences use direct coin fixing, not a hidden minimax step. Razborov's random-partition/entropy lemma now supplies an exponentially small one-sided rectangle bound for unique set disjointness; the corruption theorem and constant repetition then give $R_{1/3}^{\text{pub}}(\text{DISJ}_n), R_{1/3}^{\text{priv}}(\text{DISJ}_n) = \Omega(n)$. Chapter 3 is therefore proof-closed at its registered theorem layer.

Exercises

1. **Sign-convention audit.** Let p_R be the μ -mass of 0-inputs in a rectangle R and q_R the μ -mass of 1-inputs. Show that its signed mass is $p_R - q_R$. Rewrite (3.1) entirely in terms of these two masses and explain how the formula changes if the sign convention is $(-1)^{1-f}$.
2. **The factor $1 - 2\varepsilon$.** For a deterministic protocol with μ -error exactly η , prove directly that $\mathbb{E}_\mu[FG] = 1 - 2\eta$. Give a two-input example showing why replacing $1 - 2\varepsilon$ by $1 - \varepsilon$ would be false.
3. **Fixing private coins.** Write the private random tapes as a pair (R_A, R_B) . Starting only from the pointwise error bound, justify every exchange of finite sums needed to show that some fixed pair (r_A, r_B) has μ -average error at most ε . Identify where finiteness is used.
4. **Exact one-bit discrepancy.** Complete the 2×2 calculation following Theorem 3.1 and prove that the uniform discrepancy of IP_1 is exactly $1/2$. Compare this with the general spectral upper bound $2^{-1/2}$.
5. **Hadamard submatrices.** Let H be an $m \times m$ Hadamard matrix, so $HH^\top = mI$. Prove that every $r \times s$ submatrix has signed sum at most $\sqrt{mrs}$ in absolute value. Identify this as the abstract form of the estimate used for inner product.
6. **Constant-error dependence.** Fix any $\varepsilon_0 < 1/2$. Use (3.10) to give explicit constants $c_{\varepsilon_0} > 0$ and n_0 such that $R_{\varepsilon_0}^{\text{pub}}(\text{IP}_n) \geq c_{\varepsilon_0}n$ for all $n \geq n_0$.
7. **Biased input distributions.** Construct a distribution μ concentrated on a single input pair and compute $\text{disc}_\mu(f)$. Explain why Theorem 3.1 then gives no useful lower bound, illustrating why the choice of hard distribution matters.
8. **Corruption mass bookkeeping.** In the proof of Theorem 3.3, let E_b be the error mass on leaves labeled b and F_b the mass of true b -inputs sent to leaves labeled $1 - b$. Prove $E_b + F_b \leq \varepsilon$, and use this sharper identity to rederive (3.19). Explain why the displayed theorem keeps the simpler two-loss form rather than optimizing constants across the two error directions.
9. **Choice of corruption threshold.** Suppose a hard-distribution lemma proves $\text{mono}_{\mu, \theta}^b(f) \leq 2^{-\gamma n}$ for some constants $\gamma > 0$ and $\theta > 0$, while $\alpha_b \geq a > 0$. Determine an explicit constant error range $0 \leq \varepsilon < \varepsilon_0(a, \theta)$ for which Theorem 3.3 yields an $\Omega(n)$ lower bound.

10. **Discrepancy versus corruption.** For a (b, θ) -monochromatic rectangle R , express its signed mass in terms of $\mu(R)$ and the opposite-label mass. Show that a large nearly monochromatic rectangle forces a large signed rectangle mass when $\theta < 1/2$, but explain why a small discrepancy upper bound and a one-sided corruption lemma package their hypotheses differently and should not be identified as the same method.
11. **Razborov entropy threshold.** In the proof of Theorem 3.4, verify $h_2(1/4) < 0.812$ and check directly that for every $m \geq 260$,

$$2m - \log_2(2m + 1) - \frac{4m - 1}{100} > 1.925m.$$

Explain why these two inequalities are exactly what is needed to prove that fewer than one fifth of the distinguished coordinates can be bad.

12. **From rectangle comparison to corruption.** Starting only from (3.23), rederive (3.40) for the 3/4–1/4 mixture. Then solve symbolically for all θ such that a $(1, \theta)$ -monochromatic rectangle must have exponentially small mass. Check that $\theta = 1/272$ makes the coefficient exactly $1/270$.
13. **Amplification constants.** Suppose a protocol has pointwise error at most $1/3$. Determine the least odd integer r for which the Hoeffding estimate $\exp(-r/18) \leq 10^{-3}$ holds. Compare it with the convenient choice $r = 125$ used in the theorem and explain why optimizing this constant has no effect on the $\Omega(n)$ conclusion.

Chapter 4

Information Cost, Transcript Geometry, and Amortized Communication

Communication counts the bits that cross the channel. Information complexity asks a more refined question: how much of the parties' uncertainty about the inputs is actually removed by those bits? A transcript can be long and highly redundant, or short and highly revealing. The distinction becomes decisive in direct-sum arguments, where one wants to understand whether many independent copies can share communication overhead.

This chapter develops the elementary layer completely. We define internal and external information cost in the finite two-party model, prove the transcript factorization and posterior identities that make those quantities computable, decompose information bit by bit along an interactive protocol, and derive

$$0 \leq \text{IC}_{\mu}^{\text{int}}(\Pi) \leq \text{IC}_{\mu}^{\text{ext}}(\Pi) \leq \text{CC}(\Pi).$$

The reverse simulation direction is not elementary. It is isolated at the end of the chapter as the source-owned Braverman–Rao amortization theorem rather than being hidden inside the local proof.

How to read this chapter. Definition 4.1 freezes the transcript, randomness, logarithm-base, and zero-probability conventions first. Theorem 4.2 then has three layers. First, a fixed transcript has a multiplicatively factored likelihood, which is the randomized analogue of rectangle geometry. Second, mutual-information chain rules decompose the cost into one communicated bit at a time. Third, the protocol Markov property at an Alice or Bob node compares internal information, external information, and the entropy of that bit. Only after those local facts are closed do we state the deep compression/amortization interface ‘QRCC-XII-BB-013’.

Learning goals. After this chapter, a reader should be able to distinguish the public transcript from private random tapes; normalize information measured with natural logarithms into bits; compute internal and external information cost in small examples; prove transcript factorization and its posterior form; derive expected-KL and bitwise chain-rule identities; explain why internal cost can be strictly smaller than external cost; prove that the two agree on product input distributions; and state exactly what extra theorem is needed to turn low information into low amortized communication.

4.1 The public transcript and two information costs

Recall from Definition 1.1 that the input sets are finite. The information-theoretic primitives H, D, I use the natural logarithm convention of Definition 4.1. Since communication is counted in bits, every information cost below is divided by $\log 2$.

Definition 4.1 (Internal and external information cost). Let μ be a probability distribution on the finite set $X \times Y$, and let $(X, Y) \sim \mu$. A *finite-coin protocol* Π is a finite communication tree with a finite public random variable R , independent of (X, Y) , together with independent private random tapes for Alice and Bob. Equivalently, after fixing $R = r$ and a transcript prefix, the next transmitted bit is sampled from a finite kernel that depends only on the speaking party's input and that prefix. Private random tapes implement these local kernels and are not part of the public transcript.

Let M be the terminal communicated bit string and set

$$T := (R, M),$$

the *public transcript random variable*. Thus public randomness is included in T , while private randomness is not. The *internal information cost* and *external information cost* of Π under μ , measured in bits, are

$$\text{IC}_\mu^{\text{int}}(\Pi) := \frac{I(X; T \mid Y) + I(Y; T \mid X)}{\log 2}, \quad (4.1)$$

$$\text{IC}_\mu^{\text{ext}}(\Pi) := \frac{I(X, Y; T)}{\log 2}. \quad (4.2)$$

Because R is independent of (X, Y) , the same quantities may be written

$$\begin{aligned} I(X; T \mid Y) &= I(X; M \mid Y, R), & I(Y; T \mid X) &= I(Y; M \mid X, R), \\ I(X, Y; T) &= I(X, Y; M \mid R). \end{aligned}$$

For a Boolean function $f : X \times Y \rightarrow \{0, 1\}$ and $0 \leq \varepsilon < 1$, define the *distributional internal information complexity*

$$\text{IC}_\mu^{\text{int}}(f, \varepsilon) := \inf_{\Pi} \text{IC}_\mu^{\text{int}}(\Pi), \quad (4.3)$$

where the infimum ranges over finite-coin protocols whose fixed leaf output satisfies

$$\Pr_{(X, Y) \sim \mu, \text{coins}} [\Pi(X, Y) \neq f(X, Y)] \leq \varepsilon.$$

Define $\text{IC}_\mu^{\text{ext}}(f, \varepsilon)$ by replacing the internal cost in (4.3) with the external cost. Unless a superscript is needed, the abbreviation $\text{IC}_\mu(f, \varepsilon)$ means the internal quantity.

All conditional entropies and mutual informations ignore zero-probability conditioning fibers exactly as in Definition 4.1. No hidden “condition on a null transcript” convention is used.

Why public coins are included in the transcript. An external observer sees the public seed and every message, so $T = (R, M)$ is the natural public record. Since R is input-independent, including it costs no information by itself; it merely records which randomized protocol tree was selected. Writing the same cost conditionally on R is often more convenient in proofs.

Three one-bit sanity checks. Let $X, Y \in \{0, 1\}$.

1. If the protocol sends nothing, both information costs are zero.
2. If X and Y are independent fair bits and Alice sends $M = X$, then

$$\text{IC}_\mu^{\text{int}}(\Pi) = \text{IC}_\mu^{\text{ext}}(\Pi) = 1.$$

The one communicated bit is completely new both to Bob and to an outside observer.

3. If $X = Y$ is a fair bit and Alice again sends $M = X$, then

$$\text{IC}_\mu^{\text{int}}(\Pi) = 0, \quad \text{IC}_\mu^{\text{ext}}(\Pi) = 1.$$

Bob already knows Alice's input and Alice already knows Bob's, but an outside observer learns the common bit. This is the smallest example showing that internal and external information cost are genuinely different notions for correlated inputs.

4.2 Transcript factorization and posterior geometry

For a public seed r and a transcript string m , write

$$q_{r,m}(x, y) := \Pr[M = m \mid X = x, Y = y, R = r].$$

The local nature of communication forces this likelihood to factor into an Alice part and a Bob part. This is the probabilistic replacement for the deterministic statement that transcript cylinders are rectangles.

Theorem 4.2 (Basic information-complexity identities and inequalities). *Let Π be a finite-coin two-party protocol under an input distribution μ on $X \times Y$, and suppose its worst-case communication is at most c bits. Let $T = (R, M)$ be its public transcript.*

1. **Likelihood factorization and posterior form.** *For every r, m there exist nonnegative functions $a_{r,m} : X \rightarrow [0, 1]$ and $b_{r,m} : Y \rightarrow [0, 1]$ such that*

$$q_{r,m}(x, y) = a_{r,m}(x)b_{r,m}(y). \tag{4.4}$$

Whenever $\Pr[T = (r, m)] > 0$, the posterior input law is therefore

$$\Pr[X = x, Y = y \mid T = (r, m)] = \frac{\mu(x, y)a_{r,m}(x)b_{r,m}(y)}{\sum_{x', y'} \mu(x', y')a_{r,m}(x')b_{r,m}(y')}. \tag{4.5}$$

The same factorization holds for every positive-probability transcript prefix. Moreover,

$$(\log 2) \text{IC}_\mu^{\text{ext}}(\Pi) = \mathbb{E}_T \left[D(P_{XY|T} \| P_{XY}) \right], \tag{4.6}$$

$$(\log 2) \text{IC}_\mu^{\text{int}}(\Pi) = \mathbb{E}_{Y,T} \left[D(P_{X|Y,T} \| P_{X|Y}) \right] + \mathbb{E}_{X,T} \left[D(P_{Y|X,T} \| P_{Y|X}) \right]. \tag{4.7}$$

2. **Bitwise chain decomposition.** *For each fixed public seed, zero-pad every terminal transcript to length c and call the resulting bits $B_1, \dots, B_c$. Because terminal strings in a protocol*

tree are prefix-free, this padding is injective and hence loses no transcript information. Put $H_i = (R, B_1, \dots, B_{i-1})$. Then

$$(\log 2) \text{IC}_\mu^{\text{int}}(\Pi) = \sum_{i=1}^c [I(X; B_i \mid Y, H_i) + I(Y; B_i \mid X, H_i)], \quad (4.8)$$

$$(\log 2) \text{IC}_\mu^{\text{ext}}(\Pi) = \sum_{i=1}^c I(X, Y; B_i \mid H_i). \quad (4.9)$$

After a leaf has been reached, the padded bits are deterministic and contribute zero.

3. **Information is bounded by communication.** One always has

$$0 \leq \text{IC}_\mu^{\text{int}}(\Pi) \leq \text{IC}_\mu^{\text{ext}}(\Pi) \leq c. \quad (4.10)$$

If $\mu = \mu_X \times \mu_Y$ is a product distribution, then the middle inequality is an equality:

$$\text{IC}_\mu^{\text{int}}(\Pi) = \text{IC}_\mu^{\text{ext}}(\Pi). \quad (4.11)$$

4. **Problem-level consequences.** For every $0 \leq \varepsilon < 1$,

$$\text{IC}_\mu^{\text{int}}(f, \varepsilon) \leq \text{IC}_\mu^{\text{ext}}(f, \varepsilon) \leq D_\varepsilon^\mu(f). \quad (4.12)$$

In particular, information complexity is a lower-bound measure for distributional communication. The theorem does not assert that an arbitrary one-copy protocol can be compressed to communication equal to its information cost.

Proof roadmap. For (4.4), multiply the local probabilities of the bits sent by Alice and collect them into one factor depending on x ; collect Bob's factors separately. Bayes' rule then gives (4.5), while expanding mutual information as an average relative entropy gives (4.6)–(4.7). For (4.8)–(4.9), use the mutual-information chain rule along the padded message bits. At an Alice node, the next bit is conditionally independent of Y once X and the history are known; at a Bob node the symmetric statement holds. Comparing conditional entropies one bit at a time proves internal $\leq$ external. External information is at most transcript entropy, and a depth- c binary tree has at most 2^c leaves. Finally, when the prior is a product distribution, the factorized posterior remains a product after every transcript prefix, turning the one-bit inequality into equality.

Proof. Step 1: likelihood factorization. Fix r and a terminal transcript $m = (m_1, \dots, m_\ell)$. At each Alice-owned prefix $m_{<i}$, let $p_i^A(m_i \mid x, r, m_{<i})$ be the local probability of the transmitted bit; define p_i^B analogously at Bob-owned prefixes. Independence of the private tapes and the behavioral protocol rule give

$$\Pr[M = m \mid x, y, r] = \prod_{i: A \text{ speaks}} p_i^A(m_i \mid x, r, m_{<i}) \prod_{i: B \text{ speaks}} p_i^B(m_i \mid y, r, m_{<i}).$$

The two products are respectively $a_{r,m}(x)$ and $b_{r,m}(y)$, proving (4.4). The identical argument with a prefix in place of m proves the prefix version. Bayes' rule and the input-independence of R now give (4.5).

For completeness, expand one of the information terms directly. If A, B, C are finite random variables, then Definition 4.1 gives

$$I(A; B \mid C) = \sum_{b,c} \Pr[B = b, C = c] D(P_{A|B=b, C=c} \| P_{A|C=c}), \quad (4.13)$$

where only positive-probability fibers contribute. For the external identity, apply (4.13) with $A = (X, Y)$ and $B = T$, with no conditioning variable. For the two internal terms, apply it with $(A, B, C) = (X, T, Y)$ and then with $(A, B, C) = (Y, T, X)$. This proves (4.6) and (4.7). Gibbs' inequality, Theorem 4.5, then shows from these formulas that all displayed information costs are nonnegative.

Step 2: chain decomposition. For fixed r , the terminal bit strings are leaves of a binary tree and hence form a prefix-free family. Appending zeros to length c is therefore injective: if one leaf string were the prefix of the padded form of a different leaf, it would also be the prefix of that other leaf, impossible in a protocol tree. Thus, conditioned on R , M and $B^c = (B_1, \dots, B_c)$ determine one another.

Since R is independent of (X, Y) , the mutual-information chain rule from Theorem 4.6 yields

$$\begin{aligned} I(X; T \mid Y) &= I(X; B^c \mid Y, R) = \sum_{i=1}^c I(X; B_i \mid Y, H_i), \\ I(Y; T \mid X) &= \sum_{i=1}^c I(Y; B_i \mid X, H_i), \\ I(X, Y; T) &= I(X, Y; B^c \mid R) = \sum_{i=1}^c I(X, Y; B_i \mid H_i). \end{aligned}$$

Dividing by $\log 2$ proves (4.8)–(4.9).

Step 3: compare the two information costs one bit at a time. Fix i and a positive-probability history $H_i = h$. Suppose first that Alice speaks at that history. By the protocol rule,

$$B_i \perp\!\!\!\perp Y \mid (X, H_i = h),$$

so

$$I(Y; B_i \mid X, H_i = h) = 0$$

and

$$\begin{aligned} I(X; B_i \mid Y, H_i = h) &= H(B_i \mid Y, H_i = h) - H(B_i \mid X, H_i = h), \\ I(X, Y; B_i \mid H_i = h) &= H(B_i \mid H_i = h) - H(B_i \mid X, H_i = h). \end{aligned}$$

Conditioning reduces entropy by Theorem 4.4, so the second line is at least the first. If Bob speaks, the same argument with X and Y interchanged applies. If the protocol has already stopped, $B_i = 0$ deterministically and both contributions vanish. Averaging over histories and summing i proves

$$\text{IC}_\mu^{\text{int}}(\Pi) \leq \text{IC}_\mu^{\text{ext}}(\Pi).$$

For the upper bound by communication,

$$(\log 2) \text{IC}_\mu^{\text{ext}}(\Pi) = I(X, Y; M \mid R) \leq H(M \mid R). \quad (4.14)$$

For each r , the support of $M \mid R = r$ consists of at most 2^c leaves. If a finite random variable W has support size k , then Gibbs' inequality applied to its law P_W and the uniform law U_k gives

$$0 \leq D(P_W \parallel U_k) = \log k - H(W),$$

hence $H(W) \leq \log k$. Therefore $H(M \mid R = r) \leq c \log 2$ for every r , and averaging proves the last inequality in (4.10).

Step 4: product priors. Assume $\mu(x, y) = \mu_X(x)\mu_Y(y)$. The prefix version of (4.5) shows that, after every positive-probability history h , the conditional input law still factors:

$$P_{XY|H_i=h} = P_{X|H_i=h}P_{Y|H_i=h}. \quad (4.15)$$

At an Alice node, B_i is generated from X and h only. Combining (4.15) with this local channel rule makes B_i independent of Y given h , so

$$H(B_i | Y, h) = H(B_i | h).$$

Thus the internal and external contributions in Step 3 are equal. The Bob case is symmetric, and summing proves (4.11).

Step 5: optimize over protocols. The protocol-level inequality $\text{IC}_\mu^{\text{int}}(\Pi) \leq \text{IC}_\mu^{\text{ext}}(\Pi)$ survives taking the infimum over the same feasible protocols. Every deterministic protocol feasible for $D_\varepsilon^\mu(f)$ is also a finite-coin protocol, and by (4.10) its external information cost is at most its communication cost. Taking the minimum over those deterministic protocols proves (4.12). $\square$

Why the factorization is a “soft rectangle.” For a deterministic protocol, $q_{r,m}(x, y)$ is the indicator of a rectangle. In a randomized protocol it takes values in $[0, 1]$, but (4.4) still separates the Alice and Bob coordinates multiplicatively. Equation (4.5) says that observing a transcript reweights the prior by one row factor and one column factor. Many information-complexity arguments are probabilistic versions of rectangle arguments expressed through this posterior update.

Expected information versus worst-case communication. The left side of (4.10) is an average under μ and the protocol coins; c is a worst-case maximum over every input and every coin outcome. The inequality therefore has substantial slack in general. If one instead studies expected communication, additional coding issues appear; the source-owned compression theorem below is precisely about exploiting such slack over many independent copies.

A correlated-input edge case. The example $X = Y$ and $M = X$ shows that equality in the middle of (4.10) cannot be claimed without a product hypothesis. The protocol structure alone gives internal $\leq$ external; independence of the prior is what preserves independence under factorized posterior updates and forces equality.

4.3 Where compression actually enters

Theorem 4.2 proves only the easy direction: communication can reveal no more information than the number of communicated bits. Turning a protocol with information cost I into another protocol communicating about I bits is a much deeper interactive sampling problem. We therefore freeze the repeated-copy model before stating the source-owned result.

For $n \geq 1$, let f^n denote the coordinatewise task on independent inputs $((X_1, Y_1), \dots, (X_n, Y_n)) \sim \mu^n$. For $0 < \rho < 1$, define $D_\rho^{\mu,n}(f)$ to be the minimum worst-case communication cost of a deterministic protocol that outputs n bits and satisfies, for every coordinate j ,

$$\Pr_{\mu^n}[\text{output}_j \neq f(X_j, Y_j)] \leq \rho. \quad (4.16)$$

This is a coordinatewise distributional error requirement; it does not demand that all n coordinates be simultaneously correct with probability $1 - \rho$.

Theorem 4.3 (Braverman–Rao: information equals amortized communication). *Let $f : X \times Y \rightarrow \{0, 1\}$ be a finite two-party function, let μ be a distribution on $X \times Y$, and let $0 < \rho < 1$. In the conventions of Definition 4.1,*

$$\text{IC}_\mu^{\text{int}}(f, \rho) = \lim_{n \rightarrow \infty} \frac{D_\rho^{\mu, n}(f)}{n}. \quad (4.17)$$

This theorem is a source-owned deep input in QRCC: its complete interactive compression proof is not claimed locally in this chapter.

Source-owned compression architecture. The reverse inequality in (4.17) is powered by interactive transcript simulation. In the standard bounded-round formulation, if an r -round protocol has internal information cost I bits, then for every $\eta > 0$ one can simulate its transcript by a new interactive protocol with a high-probability good event and conditional expected communication

$$I + O(\sqrt{rI + 1}) + 2r \log_2(1/\eta), \quad (4.18)$$

while, on the good event, both parties output the same transcript with the target transcript law. Repeating many independent copies and controlling the simulation failures makes the additive overhead negligible per copy, yielding (4.17). Equation (4.18) is recorded only as proof architecture for the source-owned theorem; it is not a new locally certified QRCC theorem.

Why zero error is excluded here. The compression simulation introduces a small error/failure probability. The positive error hypothesis $0 < \rho < 1$ is therefore part of the interface, not cosmetic notation. The later prior-free zero-error theory has additional subtleties and is not imported into this distributional statement.

4.4 Dated frontier: distributed estimation

The information-cost viewpoint also motivates a newer communication problem in which the parties do not hold points x and y but probability laws from which such points may be sampled. Because the model and its error parameter differ from ordinary function computation, we freeze the definitions before recording the source-owned frontier.

Let $f : X \times Y \rightarrow [-1, 1]$ be a bounded function on finite sets. Alice receives a probability distribution p on X and Bob receives a probability distribution q on Y . The ε -distributed-estimation task is to output a real number Z satisfying

$$\Pr[|Z - \mathbb{E}_{x \sim p, y \sim q} f(x, y)| \leq \varepsilon] \geq 1 - \delta, \quad (4.19)$$

where the probability is over the public-coin protocol and $\delta < 1/2$ is the stated failure probability. Write $\bar{R}_{\varepsilon, \delta}(f)$ for the minimum worst-case communication of such a public-coin protocol, and $\bar{R}_{\varepsilon, \delta}^{\text{ow}}(f)$ for the one-way version. When $\delta = 1/3$ we suppress it. Write $R^{\text{ow}}(f)$ for the larger of the usual constant-error one-way communication complexities for computing $f(x, y)$ in the two possible speaking directions. In this paragraph only, $\tilde{O}$ hides logarithmic factors in $1/\varepsilon$ and no input-size factor.

Remark 4.4 (Verified 2026 frontier: communication complexity of distributed estimation). The 2025 preprint of Gopalan–Meka–Raghavendra–Singhal–Wigderson, accepted to FOCS 2026, proves among other results the following source-owned statements.

1. For every bounded f , a two-round debiasing protocol satisfies

$$\overline{R}_\varepsilon(f) \leq \tilde{O}\left(\frac{R^{\text{ow}}(f)}{\varepsilon}\right). \quad (4.20)$$

The naive independent-sampling protocol instead gives $\tilde{O}(R(f)/\varepsilon^2)$ communication.

2. For $\text{IP}_n(x, y) = (-1)^{\langle x, y \rangle}$, throughout the source's regime $\varepsilon > 2^{-\Theta(\sqrt{n})}$,

$$\overline{R}_\varepsilon(\text{IP}_n) = \Omega(n/\varepsilon), \quad \overline{R}_\varepsilon^{\text{ow}}(\text{IP}_n) = \Omega(n/\varepsilon^2). \quad (4.21)$$

Thus the linear dependence on $1/\varepsilon$ in the general interactive debiasing theorem is unimprovable in general, while one-way communication can retain the quadratic dependence.

3. For equality on n -bit strings, the paper gives

$$\overline{R}_\varepsilon(\text{EQ}_n) \leq \tilde{O}(\varepsilon^{-2/3}), \quad \overline{R}_\varepsilon(\text{EQ}_n) \geq \Omega\left(\min\{\varepsilon^{-2/3}, 2^n\}\right). \quad (4.22)$$

It also proves for every nonconstant Boolean f the spectral/rank lower bound

$$\overline{R}_\varepsilon(f) = \tilde{\Omega}\left(\min\left\{\varepsilon^{-2/3}, \frac{\text{rank}(f)}{\log(\text{rank}(f))}\right\}\right), \quad (4.23)$$

with the paper's communication-matrix convention.

These are recorded as a dated frontier, not reproved in QRCC and not used as hidden dependencies of the locally certified Chapter 4 theorems.

Publication-status boundary. At the 2026-09-01 checkpoint, the primary manuscript is the November 2025 preprint (arXiv 2511.21015). The authors' publication page records the work as accepted to FOCS 2026. PASS70 therefore labels the item “frontier verified” rather than treating an accepted conference paper as an already archival journal theorem. The displayed model, interaction restrictions, additive error, and ε -dependence are part of the source gate.

What Chapter 4 has and has not proved. The local layer now supplies every elementary identity needed to use information cost as a lower-bound potential: posterior factorization, KL interpretations, chain rules, internal-versus-external comparison, product equality, and information $\leq$ communication. The nontrivial compression direction and exact amortized equality are explicitly source-owned by Theorem 4.3. Later direct-sum and direct-product arguments may cite that theorem, but must not silently upgrade it to a one-copy identity.

Chapter checkpoint and exercises

The conceptual chain is now

$$\begin{aligned} \text{interactive transcript} &\implies \text{factorized likelihood} \implies \text{posterior/KL identities,} \\ &\quad \text{IC}^{\text{int}} \leq \text{IC}^{\text{ext}} \leq \text{CC}. \end{aligned}$$

with product priors collapsing the two information notions and Braverman–Rao closing the reverse direction only after amortization.

1. **Transcript factorization.** For a three-message protocol in which Alice, Bob, then Alice speak, write $q_m(x, y)$ explicitly as the product of the three local message probabilities and identify the two factors in (4.4).
2. **Posterior computation.** Let X, Y be independent fair bits. Alice sends X through a binary symmetric channel with crossover probability p , using private coins. Compute the posterior law of X after each possible transcript and verify (4.6) directly.
3. **Internal versus external.** Recompute the example $X = Y$ and $M = X$ from the definitions using natural logarithms first, then divide by $\log 2$. Identify the exact term that vanishes internally but not externally.
4. **Product equality.** Starting from (4.4), prove directly that a product prior remains product after conditioning on any positive-probability transcript prefix. Then recover (4.11) without quoting Step 4 verbatim.
5. **Entropy support bound.** Prove $H(W) \leq \log k$ for a random variable with support size at most k by applying Gibbs' inequality to the uniform law. Determine the equality condition.
6. **Variable transcript lengths.** Explain carefully why zero-padding a prefix-free leaf set is injective. Give a set of binary strings for which zero-padding would fail to be injective if prefix-freeness were dropped.
7. **Distributional lower bounds.** Let P be a deterministic protocol with μ -error at most ε . Show directly from Theorem 4.2 that both information complexities of f are at most $\text{cost}(P)$, and then take the appropriate infimum/minimum to recover (4.12).
8. **Amortization convention.** Contrast coordinatewise error in (4.16) with the requirement that all n outputs be correct simultaneously. Give a simple family of independent error events showing why the two requirements are quantitatively very different.
9. **Compression boundary.** Make a two-column dependency table separating every statement proved locally in Theorem 4.2 from the statements owned by Theorem 4.3. Explain why replacing the latter by the sentence “information equals communication” would be mathematically false at the one-copy level.

Transition to Chapter 5. Information cost is now a legitimate additive potential. The next chapter can use that potential to study direct-sum, direct-product, and parallel-repetition phenomena while keeping three quantities separate: one-copy communication, one-copy information, and amortized communication over repeated independent inputs.

Chapter 5

Direct Sums, Strong Direct Products, and the Cell-Probe Interface

Chapter 4 separated two facts that are often blurred together. Information cost is an additive potential, while turning information back into communication is a compression problem. This chapter uses that separation twice. First we prove, completely locally, that internal information complexity has an exact direct-sum law under independent repetition with a coordinatewise error requirement. We then place the much stronger statement that *all* coordinates are simultaneously correct with exponentially small success below linear communication where it belongs: as a source-owned theorem for the specific function Disjointness, not as a generic consequence of information additivity.

The second half of the chapter changes models. A static data structure hides a large stored object behind random-access probes. If Alice holds the query and Bob holds the stored object, every probe naturally becomes one short address message from Alice and one word-sized content message from Bob. Making this accounting exact is the standard bridge from cell-probe lower bounds to asymmetric communication lower bounds.

How to read this chapter. Theorem 5.1 is an information theorem: it has independent input pairs and coordinatewise error, and its proof is a chain-rule embedding argument. Theorem 5.2 is deliberately different: it is a worst-case public-coin strong direct-product theorem for Disjointness and is source-owned. Theorem 5.3 then leaves information complexity entirely and gives an exact protocol simulation of a static cell-probe scheme. Keeping these three interfaces separate prevents both a false generic direct-product claim and a hidden word-address cost in data-structure reductions.

Learning goals. After this chapter, a reader should be able to distinguish coordinatewise error from simultaneous success; derive the exact internal-information direct-sum identity; state the source-owned strong direct-product theorem without genericizing it; translate a static cell probe into address/content messages; and account for the final answer bit imposed by QRCC's fixed-leaf convention.

5.1 Coordinatewise repetition and exact information direct sum

Let $f : X \times Y \rightarrow \{0, 1\}$ be a finite two-party function and let μ be a distribution on $X \times Y$. For $k \geq 1$, write

$$f^{\otimes k}(x_1, \dots, x_k; y_1, \dots, y_k) := (f(x_1, y_1), \dots, f(x_k, y_k)).$$

Inputs to the repeated task are sampled as $((X_1, Y_1), \dots, (X_k, Y_k)) \sim \mu^k$. For $0 \leq \varepsilon < 1$, define

$$\text{IC}_{\mu^k}^{\text{int, coord}}(f^{\otimes k}, \varepsilon)$$

to be the infimum of the internal information cost, in the bit normalization of Definition 4.1, over finite-coin protocols that output k bits $Z_1, \dots, Z_k$ and satisfy

$$\Pr[Z_j \neq f(X_j, Y_j)] \leq \varepsilon \quad \text{for every } j \in [k]. \quad (5.1)$$

This is a *coordinatewise* error condition. It does not require the whole output vector to be correct with probability $1 - \varepsilon$.

Theorem 5.1 (Exact direct sum for internal information complexity). *For every finite Boolean function f , every distribution μ on its input pairs, every integer $k \geq 1$, and every $0 \leq \varepsilon < 1$,*

$$\text{IC}_{\mu^k}^{\text{int, coord}}(f^{\otimes k}, \varepsilon) = k \text{IC}_{\mu}^{\text{int}}(f, \varepsilon). \quad (5.2)$$

The equality is for internal information complexity and the coordinatewise repeated-copy model (5.1); it is not a claim that one-copy randomized communication complexity is additive.

Proof roadmap. The upper bound runs almost-optimal one-copy protocols independently. For the lower bound, start from an arbitrary k -copy protocol Π and extract from it, for each coordinate j , a one-copy protocol Π_j . The auxiliary coordinates are filled in so that the variables publicly revealed before running Π are independent of the real j th input. A forward chain rule for Alice's information and a reverse chain rule for Bob's information then show that the sum of the extracted one-copy information costs is at most the information cost of Π .

Proof. Write $I_{\mu}(f, \varepsilon) = \text{IC}_{\mu}^{\text{int}}(f, \varepsilon)$.

Upper bound. Fix $\delta > 0$ and choose a one-copy finite-coin protocol P of μ -error at most ε with

$$\text{IC}_{\mu}^{\text{int}}(P) \leq I_{\mu}(f, \varepsilon) + \delta.$$

Run k independent copies $P_1, \dots, P_k$, each on (X_i, Y_i) with independent coins, and concatenate their public transcripts $T_1, \dots, T_k$. The random triples (X_i, Y_i, T_i) are independent across i . Hence the conditional mutual-information chain rule from Theorem 4.6 gives

$$\begin{aligned} I(X^k; T^k \mid Y^k) &= \sum_{i=1}^k I(X_i; T_i \mid Y_i), \\ I(Y^k; T^k \mid X^k) &= \sum_{i=1}^k I(Y_i; T_i \mid X_i). \end{aligned}$$

After division by $\log 2$, the repeated protocol has internal information cost at most $k(I_{\mu}(f, \varepsilon) + \delta)$ and satisfies (5.1). Letting $\delta \downarrow 0$ proves the “ $\leq$ ” direction of (5.2).

Lower bound: extracting one coordinate. Let Π be any finite-coin protocol for the k -copy task satisfying (5.1), and let T denote its public transcript. Fix $j \in [k]$. We construct a one-copy protocol Π_j on a genuine input pair $(X_j, Y_j) \sim \mu$.

Before Π is run, use public randomness to sample

$$X_1, \dots, X_{j-1} \sim \mu_X, \quad Y_{j+1}, \dots, Y_k \sim \mu_Y$$

independently. For each $i < j$, Bob privately samples Y_i from the conditional law $\mu(\cdot \mid X_i)$; for each $i > j$, Alice privately samples X_i from $\mu(\cdot \mid Y_i)$. On zero-marginal fibers choose any fixed conditional law, since those fibers are never reached. Thus the assembled pairs (X_i, Y_i) are independent with common law μ . Alice knows every X_i , Bob knows every Y_i , and they can run Π exactly. Protocol Π_j outputs the j th output bit of Π .

Set

$$Z_j = (X_1, \dots, X_{j-1}, Y_{j+1}, \dots, Y_k).$$

This variable is public and independent of (X_j, Y_j) . Therefore the internal information cost of Π_j , before division by $\log 2$, is

$$J_j = I(X_j; T \mid Y_j, X_{<j}, Y_{>j}) + I(Y_j; T \mid X_j, X_{<j}, Y_{>j}). \quad (5.3)$$

Its error is exactly the j th coordinate error of Π , hence is at most ε .

We compare (5.3) with the full information revealed by Π . Since independent input pairs imply

$$X_j \perp Y_{<j} \mid (Y_j, X_{<j}, Y_{>j}),$$

the chain-rule identity

$$I(A; T \mid B, C) = I(A; T \mid B) + I(A; C \mid B, T) \quad \text{when } I(A; C \mid B) = 0 \quad (5.4)$$

shows that adding $Y_{<j}$ to the conditioning can only increase the first term in (5.3). Thus

$$I(X_j; T \mid Y_j, X_{<j}, Y_{>j}) \leq I(X_j; T \mid Y^k, X_{<j}). \quad (5.5)$$

Similarly,

$$Y_j \perp X_{>j} \mid (X_j, X_{<j}, Y_{>j})$$

gives

$$I(Y_j; T \mid X_j, X_{<j}, Y_{>j}) \leq I(Y_j; T \mid X^k, Y_{>j}). \quad (5.6)$$

Summing (5.5) in forward order and (5.6) in reverse order, and applying the conditional mutual-information chain rule, yields

$$\sum_{j=1}^k J_j \leq I(X^k; T \mid Y^k) + I(Y^k; T \mid X^k). \quad (5.7)$$

Each extracted protocol has μ -error at most ε , so $J_j/(\log 2) \geq I_\mu(f, \varepsilon)$. Dividing (5.7) by $\log 2$ gives

$$kI_\mu(f, \varepsilon) \leq \text{IC}_{\mu^k}^{\text{int}}(\Pi).$$

Taking the infimum over Π proves the “ $\geq$ ” direction and hence (5.2). $\square$

Why the auxiliary sampling is asymmetric. A tempting but incorrect shortcut is to make every auxiliary pair fully public. That produces a perfectly valid one-copy protocol, but the resulting conditioning contains both sides of every other input pair and no longer matches the two chain-rule directions needed for (5.7). Publicly exposing $X_{<j}$ and $Y_{>j}$, while privately completing the opposite halves, is exactly what makes the forward Alice chain and reverse Bob chain line up.

Direct sum is not strong direct product. Equation (5.2) says that information needed per coordinate cannot be shared away when the coordinates are independent and each coordinate must retain a fixed error guarantee. A strong direct-product theorem asks much more: if communication is below a constant fraction of the natural k -copy budget, must the probability of getting *all* answers correct fall exponentially in k ? Such a statement is not valid in arbitrary models/functions merely because (5.2) holds. For Disjointness, however, a theorem of Klauck gives exactly this stronger conclusion.

For a public-coin protocol producing k output bits, its *worst-case simultaneous success probability* is

$$\inf_{(x^k, y^k)} \Pr_{\text{public coins}} [\text{all } k \text{ output bits are correct on } (x^k, y^k)]. \quad (5.8)$$

Recall that $\text{DISJ}_n(x, y) = 1$ iff the two represented subsets of $[n]$ are disjoint; $\text{NDISJ}_n = 1 - \text{DISJ}_n$.

Theorem 5.2 (Klauck: strong direct product for Disjointness). *There are absolute constants $\alpha, \beta > 0$ such that for every $n, k \geq 1$, every public-coin randomized protocol that computes k copies of NDISJ_n using at most βkn communicated bits has worst-case simultaneous success probability at most*

$$2^{-\alpha k}. \quad (5.9)$$

The same conclusion holds for DISJ_n by complementing each output. This theorem is a source-owned deep input in QRCC; its intersection-sampling/linear-programming proof is not reconstructed locally here.

Ownership and relation to Chapter 3. Theorem 3.4 locally reconstructed the one-copy Razborov $\Omega(n)$ lower bound. Klauck's theorem strengthens the repeated-copy conclusion and uses a generalized intersection-sampling lemma together with a linear-programming dual. The source theorem itself explicitly warns that direct-sum and strong direct-product phenomena are not automatic for arbitrary distributional communication problems. QRCC therefore records (5.9) as a problem-specific deep input rather than deriving it from Theorem 5.1.

5.2 Static cell probes as asymmetric communication

We now freeze the data-structure model before stating the simulation. A finite *static Boolean data-structure problem* consists of a query set Q , a data set D , and a map

$$F : Q \times D \rightarrow \{0, 1\}.$$

A deterministic (S, w, t) *cell-probe scheme* consists of a preprocessing map

$$\phi : D \rightarrow (\{0, 1\}^w)^S$$

and a query algorithm which, on query $q \in Q$, may perform arbitrary computation for free and adaptively read at most t cells of the stored array $\phi(d)$. Cell addresses belong to $[S] = \{1, \dots, S\}$ and hence can be represented using $\lceil \log_2 S \rceil$ bits. The contents of a probed cell are exactly w bits. The query algorithm must output $F(q, d)$ after its probes.

A randomized (S, w, t, ε) scheme below means that preprocessing ϕ is still deterministic while the query algorithm may use input-independent random coins, makes at most t probes for every coin outcome, and has worst-case error at most ε . The random coins may be exposed as public coins in the communication simulation.

Theorem 5.3 (Cell-probe-to-asymmetric-communication simulation). *Let (Q, D, F) have a deterministic (S, w, t) static cell-probe scheme. In the associated two-party communication problem Alice receives $q \in Q$, Bob receives $d \in D$, and they must compute $F(q, d)$ under the fixed-leaf convention of Definition 1.1. There is a deterministic protocol with at most $2t + 1$ messages, Alice speaking first, in which*

$$\text{Alice sends at most } t\lceil \log_2 S \rceil + 1 \text{ bits in total,} \quad \text{Bob sends at most } tw \text{ bits in total.} \quad (5.10)$$

Each probe contributes one address message of $\lceil \log_2 S \rceil$ bits from Alice and one w -bit cell-content message from Bob; after the final probe, Alice sends the one-bit answer so that the reached communication leaf has a fixed output label.

If one uses the alternative standard convention in which Alice may output locally after the final Bob message, the final answer bit is unnecessary: the simulation has at most $2t$ messages, Alice sends $t\lceil \log_2 S \rceil$ bits, and Bob sends tw bits, exactly as in the MNSW formulation.

If the cell-probe scheme is randomized in the sense fixed above with worst-case error at most ε , the same construction gives a public-coin protocol with the corresponding message/communication bounds and worst-case error at most ε .

Proof roadmap. Bob can preprocess his data without communication, because local computation and memory construction are free in the communication model. Alice then emulates the query algorithm. Whenever the algorithm requests a cell, Alice sends its address; Bob reads that cell of $\phi(d)$ and returns its contents. The returned word determines the next adaptive address exactly as in the original query algorithm.

Proof. Bob computes the stored array $M = \phi(d) = (M_1, \dots, M_S)$ locally. Alice starts the query algorithm on q . Suppose its first requested address is $a_1 \in [S]$. Alice sends the binary representation of a_1 using $\lceil \log_2 S \rceil$ bits. Bob replies with the w -bit word M_{a_1} . Alice now has exactly the information that the cell-probe algorithm receives after its first probe, so she can compute the second requested address a_2 , send it, receive M_{a_2} , and continue.

After at most t probes, Alice's local simulation reaches the same Boolean output $F(q, d)$. Under QRCC's fixed-leaf convention she now sends that answer bit to Bob; both parties then reach the leaf labelled by that bit. Thus there are at most t address messages, t content messages, and one final answer message, giving (5.10) and at most $2t + 1$ messages. Under the output-at-Alice convention the last message is omitted. If the query algorithm stops early, the probe-simulation part stops early as well.

For a randomized query algorithm, expose its input-independent random tape as public randomness. Conditional on a fixed tape, the preceding deterministic simulation is exact, and averaging over the public tape gives precisely the same output distribution and therefore the same worst-case error bound ε . $\square$

Why only the randomized variant is claimed here. The old registry placeholder also mentioned a “nondeterministic variant” without fixing a certificate model or whether certificate length is charged. Those conventions are not interchangeable. Since no later registered Volume XII dependency currently needs such a variant, PASS65 removes that unsupported clause rather than silently choosing one. A certificate/cell-probe simulation should be registered only when the later lower bound fixes its exact nondeterministic model.

Worked example: one-probe set membership. Let the universe have U elements and store its characteristic vector in $S = \lceil U/w \rceil$ words of w bits. A membership query computes the unique word address containing the requested bit and makes one probe. Theorem 5.3 therefore gives one address message, one returned word, and under QRCC's fixed-leaf convention one final answer bit. Thus Alice sends $\lceil \log_2 S \rceil + 1$ bits and Bob returns w bits. With one bit per cell ($S = U, w = 1$), the total is one $\lceil \log_2 U \rceil$ -bit address, one content bit, and one final answer bit. The output-at-Alice convention omits only the last bit.

Worked example: binary-search predecessor toy model. Suppose m sorted w -bit keys are stored one per cell. Ignoring comparison computation, a standard binary search probes at most $t = \lceil \log_2 m \rceil$ cells. The communication simulation sends at most

$$\lceil \log_2 m \rceil^2 \text{ address bits from Alice, and } w \lceil \log_2 m \rceil \text{ content bits from Bob.}$$

The example is intentionally elementary: sophisticated predecessor structures change the probe bound and space, but every adaptive probe is still converted into the same address/content exchange.

Chapter checkpoint and exercises

The dependency picture is now

$$\begin{aligned} \text{independent copies} + \text{information chain rule} &\implies \text{exact information direct sum,} \\ \text{Disjointness} + \text{Klauck deep input} &\implies \text{strong direct product,} \\ \text{static cell probes} &\implies \text{asymmetric communication protocol.} \end{aligned}$$

The first and third arrows are locally proved; the middle arrow is explicitly source-owned.

1. **Conditioning monotonicity used correctly.** Prove identity (5.4) from the mutual-information chain rule. Explain why arbitrary conditioning need not increase conditional mutual information, and identify the conditional-independence hypothesis that makes the direction in (5.5) valid.
2. **Two-copy extraction.** Write the construction of Π_1 and Π_2 explicitly when $k = 2$. For each extracted protocol list which auxiliary variable is public, which is privately sampled by Alice, and which is privately sampled by Bob.
3. **Coordinatewise versus simultaneous error.** Suppose k independent one-copy protocols each fail with probability $1/10$. Compute the probability that all k outputs are correct. Explain why a theorem controlling (5.1) alone does not imply a strong direct-product statement such as (5.9).
4. **Complement invariance.** Show that a public-coin protocol for $\text{NDISJ}_n^{\otimes k}$ with a given simultaneous success probability and communication cost immediately yields one for $\text{DISJ}_n^{\otimes k}$ with the same parameters.
5. **Probe accounting.** For $S = 2^{20}$, $w = 64$, and $t = 7$, compute separately Alice's maximum number of bits, Bob's maximum number of bits, total messages, and total communication in Theorem 5.3, both with QRCC fixed-leaf outputs and with local output by Alice.
6. **Adaptive addresses.** Explain why Bob does not need to know the query in the cell-probe simulation. Give a three-probe decision tree in which the second and third addresses depend

on previous cell contents, and trace the corresponding six probe-simulation messages and the final fixed-leaf answer message.

7. **Randomized probes.** Prove that exposing the query algorithm's private, input-independent random tape as public randomness does not change its error probability. Identify what would have to be redefined if the preprocessing itself were randomized and its random seed were hidden in the stored representation.
8. **Lower-bound transfer.** Suppose every $2t$ -message communication protocol for a static problem must satisfy $A \geq a$ Alice-to-Bob bits or $B \geq b$ Bob-to-Alice bits. Use (5.10) to derive the corresponding necessary inequalities on t, S, w for any deterministic cell-probe scheme, keeping the final one-bit output correction explicit.

Transition to Chapter 6. Chapter 5 has supplied two composition principles and one model-reduction principle. The next registered workfront is a lifting theorem: a gadget will convert decision-tree hardness into communication hardness. Before that theorem can be source-owned safely, the gadget, composed search problem, and communication/query conventions in 'QRCC-XII-DEF-016' must be defined exactly.

Chapter 6

Query-to-Communication Lifting with Low-Discrepancy Gadgets

The first five chapters developed communication lower bounds directly inside the communication model: rectangles, discrepancy, corruption, information cost, direct products, and data-structure simulations. Lifting reverses the direction of attack. One first proves a lower bound in the much simpler decision-tree model and then composes the outer problem with a carefully chosen two-party gadget. A lifting theorem says that a cheap protocol for the composed problem would simulate back to a cheap decision tree, so query lower bounds become communication lower bounds almost automatically.

This chapter fixes that interface at source precision. Definition 6.1 defines total search problems, deterministic and randomized query complexity, search protocols, gadgets, block composition, and the exact uniform-discrepancy normalization used by the selected theorem. Theorem 6.2 is then retained as a source-owned deep input: the Chattopadhyay–Filmus–Koroth–Meir–Pitassi lifting theorem for logarithmic-size low-discrepancy gadgets. The inner-product gadget gives a concrete instance because Chapter 3 already proved its exponentially small discrepancy.

How to read this chapter. There are three layers. First freeze all models before the lifting statement. Second, separate the easy “query algorithm $\Rightarrow$ protocol” direction, which we prove locally, from the difficult “protocol $\Rightarrow$ query algorithm” simulation, which remains source-owned. Third, specialize the theorem to inner product and check one outer problem—parity—where both deterministic and bounded-error randomized query complexity can be established in a few lines. This keeps the lifting theorem usable without disguising its min-entropy simulation machinery as elementary folklore.

Learning goals. After this chapter, a reader should be able to define a total search problem and its query complexity; form the block composition $S \circ g^n$; translate the discrepancy normalization of Chapter 3 to the gadget convention; state exactly which gadget length, error shift, and randomness model the selected lifting theorem uses; prove the easy simulation upper bound; and apply the theorem to the inner-product gadget without silently dropping the $2^{-\eta b/8}$ randomized-error loss.

6.1 Search problems, query trees, gadgets, and composition

Definition 6.1 (Lifting gadget and composed search problem). Fix an integer $n \geq 1$ and a finite nonempty output set $\mathcal{O}$.

1. A *total search problem* on n input bits is a relation

$$S \subseteq \{0, 1\}^n \times \mathcal{O}$$

such that

$$S(z) := \{o \in \mathcal{O} : (z, o) \in S\} \neq \emptyset \quad \text{for every } z \in \{0, 1\}^n. \quad (6.1)$$

A Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is identified with the search problem $S_f(z) = \{f(z)\}$.

2. A *deterministic decision tree* for S is a finite rooted binary tree whose internal nodes are labelled by coordinates $i \in [n]$, whose two outgoing edges are labelled 0, 1, and whose leaves are labelled by outputs in $\mathcal{O}$. On input z , a node labelled i follows the edge z_i . The tree *solves* S when its leaf output belongs to $S(z)$ for every z . Its query cost is its maximum root-to-leaf number of queries, and

$$D^{\text{dt}}(S)$$

is the minimum such cost.

A *randomized decision tree* is a probability distribution over deterministic decision trees. Its cost is the maximum depth of a deterministic tree in its support. For $\varepsilon \geq 0$, it solves S with worst-case error at most ε when

$$\Pr[T(z) \in S(z)] \geq 1 - \varepsilon \quad \text{for every } z \in \{0, 1\}^n, \quad (6.2)$$

and $R_\varepsilon^{\text{dt}}(S)$ is the minimum such cost.

3. Let $\mathcal{X}, \mathcal{Y}$ be finite nonempty sets and let $F \subseteq (\mathcal{X} \times \mathcal{Y}) \times \mathcal{O}$ be a total two-party search problem. A deterministic communication protocol for F is the fixed-leaf protocol model of Definition 1.1, except that leaves carry labels in $\mathcal{O}$ rather than only bits. It solves F when the reached leaf output lies in $F(x, y)$ on every input. Its minimum worst-case communication is denoted $D^{\text{cc}}(F)$. A public-coin protocol is a distribution over such deterministic protocols, with the public random tape fixed before communication. The minimum cost for worst-case error at most ε is denoted $R_\varepsilon^{\text{cc}}(F)$.

4. A *Boolean lifting gadget* is a function

$$g : \Lambda \times \Lambda \longrightarrow \{0, 1\}$$

on a finite nonempty block alphabet Λ . Its n -fold coordinate map is

$$g^n(x, y) := (g(x_1, y_1), \dots, g(x_n, y_n)) \in \{0, 1\}^n, \quad (6.3)$$

for $x, y \in \Lambda^n$. The *block composition* of S with g is the two-party search problem

$$(S \circ g^n)(x, y) := S(g^n(x, y)). \quad (6.4)$$

Thus the protocol must output any valid solution of the outer search problem on the hidden gadget-output string.

5. When $\Lambda = \{0, 1\}^b$, let U, V be independent uniform random variables on Λ . The *uniform discrepancy* of g is

$$\text{disc}(g) := \max_{A, B \subseteq \Lambda} |\Pr[g(U, V) = 0, (U, V) \in A \times B] - \Pr[g(U, V) = 1, (U, V) \in A \times B]|. \quad (6.5)$$

Equivalently, (6.5) is exactly the Chapter 3 signed discrepancy $\text{disc}_{\mu_b}(g)$ for the uniform distribution μ_b on $\Lambda \times \Lambda$.

6. The concrete gadget used below is the *inner-product gadget*

$$\text{IP}_b(u, v) := \sum_{j=1}^b u_j v_j \pmod{2}, \quad u, v \in \{0, 1\}^b. \quad (6.6)$$

By Theorem 3.2,

$$\text{disc}(\text{IP}_b) \leq 2^{-b/2}. \quad (6.7)$$

Why total relations are convenient. The lifting theorem is naturally stated for search problems, not only Boolean functions. A partial function may be encoded as a total relation by allowing every output outside its promise. Equation (6.1) therefore avoids a separate “undefined input” branch in both query and communication protocols. When a later application really needs a promise-sensitive convention, that promise must be stated explicitly rather than inferred from the word “search.”

Three superscripts that should not be dropped. The superscripts dt and cc distinguish query cost from communication cost, while R in this chapter always means worst-case-error public-coin randomization. Chapter 1 also defined private-coin communication, but the selected randomized lifting theorem is public-coin. No private-coin lifting statement is being claimed here.

6.2 The selected low-discrepancy lifting theorem

Theorem 6.2 (Low-discrepancy query-to-communication lifting — deep input). *For every $0 < \eta \leq 1/2$ there is a constant*

$$c_\eta = O\left(\eta^{-2} \log \frac{1}{\eta}\right) \quad (6.8)$$

with the following property. Let $n \geq 2$, let $\Lambda = \{0, 1\}^b$ with

$$b \geq c_\eta \log_2 n, \quad (6.9)$$

and let $g : \Lambda \times \Lambda \rightarrow \{0, 1\}$ satisfy

$$\text{disc}(g) \leq 2^{-\eta b}. \quad (6.10)$$

Then every total search problem $S \subseteq \{0, 1\}^n \times \mathcal{O}$ satisfies

$$D^{\text{cc}}(S \circ g^n) = \Omega(D^{\text{dt}}(S) b). \quad (6.11)$$

Moreover, for every $\varepsilon > 0$, if

$$\varepsilon' := \varepsilon + 2^{-\eta b/8}, \quad (6.12)$$

then

$$R_\varepsilon^{\text{cc}}(S \circ g^n) = \Omega\left((R_{\varepsilon'}^{\text{dt}}(S) - O(1))b\right). \quad (6.13)$$

The constants hidden by $O(1)$ and $\Omega(\cdot)$ are absolute once η is fixed. The lower bounds (6.11)–(6.13) are source-owned; QRCC does not reproduce their full min-entropy simulation proof.

Exact source boundary. Chattopadhyay, Filmus, Korothe, Meir, and Pitassi prove the deterministic and randomized simulation theorems for low-discrepancy gadgets and derive (6.11)–(6.13) for search problems. Their deterministic simulation converts a protocol of communication C into a decision tree making $O(C/b)$ queries. Their randomized simulation produces a transcript distribution within $2^{-\eta b/8}$ of the appropriate gadget fiber and therefore incurs exactly the additive error in (6.12). The paper proves more—including round-preserving versions—but QRCC does not import those refinements into the registered theorem.

Proof architecture, without a hidden proof claim. The difficult direction maintains a structured rectangle of communication inputs while simulating the protocol. Low discrepancy supplies a uniform-marginals principle for high-min-entropy block distributions. When too much information accumulates on some coordinates, the simulator queries those outer bits and conditions on the corresponding gadget outputs. A deficiency accounting argument charges each newly queried coordinate against $\Omega(b)$ units of communication. In the randomized proof, messages and density-restoring parts are sampled rather than chosen arbitrarily, and a coupling argument controls the transcript-distribution error. These are the source proof’s core interfaces; their technical min-entropy and conditioning lemmas remain source-owned.

6.3 The easy direction is local

The lifting theorem supplies the hard lower direction. The converse simulation needs no deep machinery and is useful enough to record explicitly.

Proposition 6.3 (Decision-tree simulation by a gadget protocol). *Let $S \subseteq \{0, 1\}^n \times \mathcal{O}$ be a total search problem and let $g : \{0, 1\}^b \times \{0, 1\}^b \rightarrow \{0, 1\}$ be any gadget. Under QRCC’s fixed-leaf output convention,*

$$D^{\text{cc}}(S \circ g^n) \leq (b + 1)D^{\text{dt}}(S), \quad (6.14)$$

and for every $\varepsilon \geq 0$,

$$R_\varepsilon^{\text{cc}}(S \circ g^n) \leq (b + 1)R_\varepsilon^{\text{dt}}(S). \quad (6.15)$$

Proof. Fix a deterministic decision tree T for S . Alice and Bob simulate it. Whenever T queries coordinate i , Alice sends her entire block $x_i \in \{0, 1\}^b$ using b bits. Bob computes $g(x_i, y_i)$ and sends that one bit back. Both parties now know the answer to the query and hence the next node of T . When the simulated tree reaches a leaf, its label $o \in \mathcal{O}$ is a fixed function of the public transcript of query answers, so the communication protocol can use the same fixed leaf label. Each query costs $b + 1$ communicated bits, proving (6.14).

For a randomized decision tree, use its random tape as public communication randomness and then perform exactly the same deterministic simulation. For every input, the communication protocol has the same output distribution as the chosen randomized query tree. Hence the worst-case error is unchanged and (6.15) follows. $\square$

Why the “+1” belongs here. Sending only Alice’s b -bit block lets Bob evaluate g , but Alice would not know which branch of an adaptive outer decision tree to follow. Bob therefore sends the gadget output bit back. This is different from the Chapter 5 cell-probe convention issue: here the bit is needed to synchronize the next query, not merely to communicate a final answer.

Combining (6.11) and (6.14) gives

$$D^{\text{cc}}(S \circ g^n) = \Theta(D^{\text{dt}}(S)b) \quad (6.16)$$

throughout the low-discrepancy regime (6.8)–(6.10). In the randomized setting the upper bound (6.15) uses outer error ε , whereas the deep lower bound (6.13) uses the slightly larger ε' . One must not erase that mismatch unless a separate argument shows that the outer randomized query complexity is stable under the error change.

6.4 Inner product as a logarithmic lifting gadget

Set $g = \text{IP}_b$. Equation (6.7) allows Theorem 6.2 to be applied with $\eta = 1/2$. Consequently there is an absolute constant $C > 0$ such that, whenever

$$b \geq C \log_2 n, \quad (6.17)$$

every total search problem S obeys

$$D^{\text{cc}}(S \circ \text{IP}_b^n) = \Theta(D^{\text{dt}}(S)b), \quad (6.18)$$

and

$$R_\varepsilon^{\text{cc}}(S \circ \text{IP}_b^n) = \Omega\left((R_{\varepsilon+2^{-b/16}}^{\text{dt}}(S) - O(1))b\right). \quad (6.19)$$

The locally proved upper bound remains

$$R_\varepsilon^{\text{cc}}(S \circ \text{IP}_b^n) \leq (b+1)R_\varepsilon^{\text{dt}}(S). \quad (6.20)$$

Worked example: lifting parity. Let the outer Boolean search problem be

$$S_\oplus(z) = \left\{ \bigoplus_{i=1}^n z_i \right\}.$$

A deterministic decision tree must query all n coordinates: if a leaf is reached with some coordinate unqueried, flipping only that coordinate preserves the transcript but changes the required parity. Hence

$$D^{\text{dt}}(S_\oplus) = n. \quad (6.21)$$

The same maximum-depth lower bound holds for every randomized decision tree with worst-case error $\delta < 1/2$. Indeed, under the uniform distribution on z , every deterministic tree of depth less than n leaves at least one uniform bit unqueried at each leaf and therefore has average error exactly $1/2$. A distribution over such trees still has uniform-input average error $1/2$, so some input has error at least $1/2$. Thus

$$R_\delta^{\text{dt}}(S_\oplus) = n \quad (0 \leq \delta < 1/2). \quad (6.22)$$

Take $\varepsilon = 1/3$. If $b \geq 43$, then $1/3 + 2^{-b/16} < 1/2$. Together with (6.17), equations (6.19), (6.20), and (6.22) give

$$R_{1/3}^{\text{cc}}(S_\oplus \circ \text{IP}_b^n) = \Theta(nb). \quad (6.23)$$

Thus a one-line query lower bound lifts to a linear-in-total-block-size randomized communication lower bound once the gadget hypotheses have been certified.

6.5 Hypothesis and ownership audit

Low discrepancy is not optional. The selected theorem does not say that every nonconstant gadget lifts query complexity. Its hypothesis is the quantitative uniform discrepancy bound (6.10). Chapter 3 supplies that certificate for inner product, but another gadget needs its own proof or source gate.

Logarithmic gadget length is part of the theorem. The statement requires $b \geq c_\eta \log n$. Nothing here establishes the same lifting for constant-size gadgets. Replacing (6.9) by $b = O(1)$ would be a genuine strengthening, not a notational simplification.

Randomized means public coin. Equation (6.13) is a public-coin communication lower bound matched to randomized decision trees. Chapter 2 relates public and private coins for finite functions, but QRCC does not use Newman’s theorem to advertise a private-coin lifting theorem with the same exact parameters.

The error shift belongs to the source theorem. The $2^{-\eta b/8}$ term in (6.12) comes from the source’s transcript-distribution approximation. It is small when $b = \Omega(\log n)$ but is not zero. Applications such as parity may absorb it only after verifying that the relevant outer query lower bound persists at the enlarged error.

Search outputs remain relational. For a search problem, the output need not be unique. The lifting theorem guarantees that the simulated query tree finds some element of $S(z)$. Boolean decision problems are only the singleton-output special case.

6.6 Chapter checkpoint

The lifting interface is now closed at the registered level. All terminology in the deep theorem is defined before use; the source-owned lower direction has its discrepancy, gadget-size, public-coin, search-problem, and randomized-error parameters frozen; the easy upper simulation is proved locally under QRCC’s fixed-leaf convention; and the inner-product specialization is justified by the already proved Chapter 3 discrepancy bound. Later reductions may therefore cite Theorem 6.2 without reintroducing an undefined composition convention or silently strengthening the source.

Exercises.

1. For a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, verify directly that the search composition in (6.4) is the usual block-composed function $f(g(x_1, y_1), \dots, g(x_n, y_n))$.
2. Prove (6.15) carefully when the randomized decision tree is a finite distribution over deterministic trees. Identify exactly where public randomness is used.
3. Suppose g has deterministic communication complexity at most q in a protocol that reveals the gadget output to both parties. Show that the proof of Proposition 6.3 improves to $D^{\text{cc}}(S \circ g^n) \leq qD^{\text{dt}}(S)$.
4. Use Theorem 3.2 to rederive (6.7) from the Hadamard operator norm, explicitly matching the probability normalization in (6.5).
5. Let $S_{\text{OR}}(z) = \{z_1 \vee \dots \vee z_n\}$. Prove $D^{\text{dt}}(S_{\text{OR}}) = n$ and deduce from (6.18) that deterministic communication for $S_{\text{OR}} \circ \text{IP}_b^n$ is $\Theta(nb)$ whenever $b \geq C \log n$.
6. Explain why (6.19) by itself does not imply $R_{1/3}^{\text{cc}}(S \circ \text{IP}_b^n) = \Omega(R_{1/3}^{\text{dt}}(S)b)$ for an arbitrary search problem S . State one additional error-robustness hypothesis on S that would make such a conclusion valid.
7. The definition of a randomized decision tree charges the maximum depth over the support, not expected queries. Give an example showing why replacing this by expected query cost changes what must be proved in the parity argument (6.22).

Chapter 7

Polynomial, Approximate-Degree, and Sensitivity Methods for Query Complexity

Chapter 6 transferred decision-tree hardness into communication hardness. This chapter returns to the query model itself and develops three of its most useful structural languages: low-degree polynomials, dual witnesses for approximation, and local sensitivity on the Boolean cube. The order matters. We first freeze every query and Boolean-function measure used later. We then prove the classical polynomial method and its quantum cross-reference, derive the exact finite-dimensional dual characterization of approximate degree, and finally reconstruct Huang’s cube-spectral argument. The only source-owned bridge in the final sensitivity collapse is Tal’s inequality $\text{bs}(f) \leq \deg(f)^2$, which is registered separately rather than hidden inside Huang’s proof.

How to read this chapter. There are four layers. Definition 7.1 fixes the models and normalizations. Theorem 7.2 turns query algorithms into polynomials. Theorem 7.4 turns failure of low-degree approximation into an ℓ_1 -normalized dual witness. The last part proves the sharp “more than half a cube” spectral lemma and uses it to show $\deg(f) \leq s(f)^2$; Theorem 7.8 supplies the separately source-owned block-sensitivity/degree bridge, after which all the standard classical query measures collapse to one polynomial-equivalence class.

Learning goals. After this chapter, a reader should be able to distinguish exact degree from approximate degree, derive a degree bound from a randomized or quantum query algorithm, write and verify a dual polynomial, explain why parity has approximate degree n for every error below $1/2$, prove Huang’s signed-adjacency lemma, and trace exactly where the quartic sensitivity bound uses Tal’s source-owned theorem.

7.1 Decision trees, quantum queries, certificates, and Boolean measures

Definition 7.1 (Decision-tree and query measures). Let $n \geq 1$ and let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a total Boolean function. Write $[n] = \{1, \dots, n\}$.

1. The *deterministic query complexity* of f is

$$D(f) := D^{\text{dt}}(S_f), \quad S_f(x) := \{f(x)\}, \quad (7.1)$$

using the decision-tree model of Definition 6.1. For $0 \leq \varepsilon < 1$, the *randomized query complexity* is

$$R_\varepsilon(f) := R_\varepsilon^{\text{dt}}(S_f). \quad (7.2)$$

Thus the random tape is independent of the input, every tape value specifies a deterministic tree of the charged maximum depth, and the error is worst case over inputs.

2. A *T-query quantum algorithm* for f has a finite complex state vector whose coordinates are indexed by basis labels (i, b, z) with $i \in [n]$ an address, $b \in \{0, 1\}$ an answer bit, and z an arbitrary finite workspace label. Between queries it applies input-independent *unitary* matrices U , meaning $U^*U = I$, where U^* is conjugate transpose. A query applies the oracle

$$O_x|i, b, z\rangle = |i, b \oplus x_i, z\rangle, \quad (7.3)$$

where $\oplus$ denotes addition modulo two. At the end, a fixed subset of basis labels is declared accepting; the acceptance probability is the sum of squared absolute values of the corresponding amplitudes. The least T for which this probability differs from $f(x)$ by at most ε for every x is the *bounded-error quantum query complexity* $Q_\varepsilon(f)$. This chapter uses Q_ε only as a polynomial- method cross-reference; no quantum algorithm is needed elsewhere in Volume XII.

3. A set $S \subseteq [n]$ is a *certificate for f at x* if every $y \in \{0, 1\}^n$ satisfying $y_i = x_i$ for all $i \in S$ also satisfies $f(y) = f(x)$. Define

$$C(f, x) := \min\{|S| : S \text{ certifies } f \text{ at } x\}, \quad C(f) := \max_x C(f, x). \quad (7.4)$$

For $z \in \{0, 1\}$, let $C_z(f)$ be the maximum of $C(f, x)$ over inputs with $f(x) = z$, with the maximum of the empty set defined as 0.

4. For $B \subseteq [n]$, let x^B be the string obtained by flipping exactly the coordinates in B . The *sensitivity at x* and *sensitivity of f* are

$$s(f, x) := |\{i \in [n] : f(x) \neq f(x^{\{i\}})\}|, \quad s(f) := \max_x s(f, x). \quad (7.5)$$

The *block sensitivity at x* is the largest k for which there are pairwise disjoint nonempty blocks $B_1, \dots, B_k \subseteq [n]$ satisfying $f(x^{B_j}) \neq f(x)$ for every j ; set

$$\text{bs}(f) := \max_x \text{bs}(f, x). \quad (7.6)$$

5. Every real-valued function on $\{0, 1\}^n$ has a unique multilinear real polynomial representation. The *exact degree* $\deg(f)$ is the total degree of that representing polynomial. For $0 \leq \varepsilon < 1$, the *ε -approximate degree* is

$$\widetilde{\deg}_\varepsilon(f) := \min\{\deg p : p \in \mathbb{R}[x_1, \dots, x_n], |p(x) - f(x)| \leq \varepsilon \forall x \in \{0, 1\}^n\}. \quad (7.7)$$

The polynomial may always be multilinearized using $x_i^2 = x_i$ on the Boolean cube, without increasing its degree. We abbreviate $\widetilde{\deg}(f) := \widetilde{\deg}_{1/3}(f)$.

Why the multilinear representation is unique. For each $a \in \{0, 1\}^n$, the multilinear indicator

$$L_a(x) := \prod_{i:a_i=1} x_i \prod_{i:a_i=0} (1 - x_i) \quad (7.8)$$

equals 1 at a and 0 at every other Boolean input. Hence $\sum_a f(a)L_a(x)$ represents f . If a multilinear polynomial vanishes at every Boolean point, induction on n after writing it as $p_0(x_{<n}) + x_n p_1(x_{<n})$ shows that both p_0 and p_1 vanish identically. Thus the representation is unique.

Two error thresholds that should not be conflated. For $\{0, 1\}$ -valued functions, error $1/2$ is the trivial approximation threshold: the constant polynomial $1/2$ approximates every Boolean function to error $1/2$. Accordingly, the nontrivial bounded-error query regime below will always state $\varepsilon < 1/2$ when such a restriction is needed.

7.2 The polynomial method for query lower bounds

Theorem 7.2 (Polynomial method for randomized and quantum queries). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and let $T \geq 0$.*

1. *If a deterministic decision tree of depth at most T outputs a bit $A(x)$, then A is represented on the Boolean cube by a real multilinear polynomial of degree at most T .*
2. *If a randomized decision tree has maximum depth at most T , its acceptance probability*

$$p(x) := \Pr[A(x) = 1] \quad (7.9)$$

is a real multilinear polynomial of degree at most T .

3. *If a quantum query algorithm in the model of Definition 7.1 makes at most T queries, its acceptance probability is a real multilinear polynomial of degree at most $2T$.*

Consequently, for every $0 \leq \varepsilon < 1$,

$$D(f) \geq \deg(f), \quad R_\varepsilon(f) \geq \widetilde{\deg}_\varepsilon(f), \quad Q_\varepsilon(f) \geq \frac{1}{2} \widetilde{\deg}_\varepsilon(f). \quad (7.10)$$

Proof roadmap. For a deterministic tree, each accepting leaf contributes the product of the literals that describe its root-to-leaf path. Randomization only averages those polynomials. In the quantum model, every oracle call can increase the degree of each complex amplitude by at most one; squaring amplitudes to obtain probabilities doubles the degree.

Proof. First fix a deterministic decision tree of depth at most T . Repeated queries to a coordinate along one root-to-leaf path may be deleted, so assume no path repeats a coordinate. If an accepting leaf ℓ is reached precisely when queried coordinates $i_1, \dots, i_t$ take prescribed values $a_1, \dots, a_t$, its path indicator is

$$P_\ell(x) = \prod_{j:a_j=1} x_{i_j} \prod_{j:a_j=0} (1 - x_{i_j}), \quad (7.11)$$

a multilinear polynomial of degree $t \leq T$. Exactly one leaf indicator equals 1 on each input, so summing (7.11) over accepting leaves gives the output bit $A(x)$ and has degree at most T . This proves the first claim.

For a randomized tree, condition on its input-independent random tape r . The first part gives a degree- T polynomial $P_r(x)$ for the acceptance indicator under tape r . The vector space of degree-at-most- T multilinear polynomials is finite dimensional, so coefficientwise expectation gives

$$p(x) = \mathbb{E}_r P_r(x), \quad (7.12)$$

still of degree at most T . This proves the second claim.

For the quantum claim, write every amplitude immediately before the first query as a degree-0 complex polynomial in x . Suppose all amplitudes before a query have degree at most t . On a basis coordinate (i, b, z) , the oracle replaces the amplitude by

$$(1 - x_i)a_{i,b,z}(x) + x_i a_{i,b \oplus 1,z}(x), \quad (7.13)$$

which has degree at most $t+1$. An input-independent unitary forms fixed complex linear combinations of amplitudes and does not increase degree. By induction, after T queries every final amplitude has degree at most T . If $\mathcal{A}$ is the accepting set of basis labels, then

$$p(x) = \sum_{\alpha \in \mathcal{A}} |a_\alpha(x)|^2 = \sum_{\alpha \in \mathcal{A}} a_\alpha(x) \overline{a_\alpha(x)} \quad (7.14)$$

is a real polynomial of degree at most $2T$. Replacing powers x_i^k with x_i for $k \geq 1$ yields a multilinear polynomial agreeing with (7.14) on the Boolean cube and cannot increase degree.

If a deterministic tree computes f exactly, its polynomial is the unique multilinear representation of f , giving $T \geq \deg(f)$. If a randomized or quantum algorithm has worst-case error at most ε , its acceptance polynomial differs from $f(x)$ by at most ε pointwise. Equations (7.7), (7.12), and (7.14) therefore give the last two inequalities in (7.10). $\square$

Source boundary. Beals, Buhrman, Cleve, Mosca, and de Wolf establish the quantum acceptance-polynomial lemma with degree $2T$ in the black-box model. The classical degree- T statement above is the simpler decision-tree specialization and is proved independently here rather than attributed to the quantum paper. No converse from low-degree polynomials to efficient randomized or quantum algorithms is claimed.

Worked example: exact parity and conjunction. The unique multilinear polynomial for $\text{AND}_n(x) = x_1 \cdots x_n$ has degree n , so (7.10) gives $D(\text{AND}_n) \geq n$, matching the obvious n -query algorithm. For parity,

$$\text{PARITY}_n(x) = \frac{1 - \prod_{i=1}^n (1 - 2x_i)}{2}, \quad (7.15)$$

which also has exact degree n , hence deterministic query complexity n . The next section strengthens the parity statement to every randomized error below $1/2$.

7.3 Approximate-degree duality

Let $\mathcal{V}_d$ denote the real vector space of all functions $q : \{0, 1\}^n \rightarrow \mathbb{R}$ represented by multilinear polynomials of degree at most d . For functions u, v on the cube write

$$\langle u, v \rangle := \sum_{x \in \{0,1\}^n} u(x)v(x), \quad \|u\|_1 := \sum_x |u(x)|, \quad \|u\|_\infty := \max_x |u(x)|. \quad (7.16)$$

A function ψ has *pure high degree greater than d* when $\langle \psi, q \rangle = 0$ for every $q \in \mathcal{V}_d$.

Lemma 7.3 (Finite-dimensional strict separation). *Let C be a nonempty closed convex subset of a finite-dimensional real inner-product space and let $z \notin C$. Then there is a nonzero vector ψ such that*

$$\langle \psi, z \rangle > \sup_{c \in C} \langle \psi, c \rangle. \quad (7.17)$$

Proof. Choose $c_0 \in C$. A point of C farther from z than c_0 cannot minimize the distance to z , so compactness of the intersection of C with the corresponding closed Euclidean ball gives a closest point $y \in C$. Put $\psi = z - y \neq 0$. For any $c \in C$ and $0 \leq t \leq 1$, convexity gives $y + t(c - y) \in C$. Minimality of y implies that the derivative at $t = 0$ of $\|z - y - t(c - y)\|_2^2$ is nonnegative, hence $\langle \psi, c - y \rangle \leq 0$. Therefore $\langle \psi, c \rangle \leq \langle \psi, y \rangle$ for every $c \in C$, while $\langle \psi, z \rangle = \langle \psi, y \rangle + \|\psi\|_2^2$ is strictly larger. $\square$

Theorem 7.4 (Dual characterization of approximate degree). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let $0 \leq \varepsilon < 1$, and let $0 \leq d \leq n$. Then*

$$\widetilde{\deg}_\varepsilon(f) > d \quad (7.18)$$

if and only if there exists a function $\psi : \{0, 1\}^n \rightarrow \mathbb{R}$ such that

$$\|\psi\|_1 = 1, \quad \langle \psi, f \rangle > \varepsilon, \quad \langle \psi, q \rangle = 0 \quad \text{for every } q \in \mathcal{V}_d. \quad (7.19)$$

In particular, for every $n \geq 1$ and every $0 \leq \varepsilon < 1/2$,

$$\widetilde{\deg}_\varepsilon(\text{PARITY}_n) = n, \quad R_\varepsilon(\text{PARITY}_n) = n. \quad (7.20)$$

At the boundary $\varepsilon = 1/2$, the constant polynomial $1/2$ shows that $\widetilde{\deg}_{1/2}(\text{PARITY}_n) = 0$.

Proof roadmap. The primal feasibility question asks whether f belongs to the Minkowski sum of the low-degree subspace $\mathcal{V}_d$ and the ℓ_∞ ball of radius ε . If not, strict separation produces a normal vector ψ . Because the set contains the entire unbounded subspace $\mathcal{V}_d$, the separator must annihilate that subspace; the support function of the ℓ_∞ ball is exactly $\varepsilon\|\psi\|_1$.

Proof. Let

$$B_\varepsilon := \{u : \|u\|_\infty \leq \varepsilon\}, \quad C := \mathcal{V}_d + B_\varepsilon. \quad (7.21)$$

The set C is convex and closed. Indeed, if $v_k + b_k \rightarrow h$ with $v_k \in \mathcal{V}_d$ and $b_k \in B_\varepsilon$, compactness of B_ε gives a subsequence $b_k \rightarrow b \in B_\varepsilon$, whence $v_k \rightarrow h - b \in \mathcal{V}_d$ because finite-dimensional subspaces are closed. By definition, $f \in C$ exactly when a degree-at-most- d polynomial approximates f within ε .

Assume first that (7.18) holds, so $f \notin C$. Apply Lemma 7.3 to obtain a nonzero ψ with

$$\langle \psi, f \rangle > \sup_{v \in \mathcal{V}_d, b \in B_\varepsilon} \langle \psi, v + b \rangle. \quad (7.22)$$

If $\langle \psi, v_0 \rangle \neq 0$ for some $v_0 \in \mathcal{V}_d$, scaling v_0 by arbitrary positive or negative real numbers would make the supremum in (7.22) infinite. Hence ψ annihilates $\mathcal{V}_d$. The remaining supremum is

$$\sup_{\|b\|_\infty \leq \varepsilon} \langle \psi, b \rangle = \varepsilon \sum_x |\psi(x)| = \varepsilon \|\psi\|_1, \quad (7.23)$$

attained by taking $b(x) = \varepsilon \text{sgn}(\psi(x))$. Divide ψ by its nonzero ℓ_1 norm to obtain (7.19).

Conversely, suppose ψ satisfies (7.19) and that a $q \in \mathcal{V}_d$ obeys $\|f - q\|_\infty \leq \varepsilon$. Then

$$\langle \psi, f \rangle = \langle \psi, f - q \rangle \leq \sum_x |\psi(x)| |f(x) - q(x)| \leq \varepsilon \|\psi\|_1 = \varepsilon, \quad (7.24)$$

contradicting (7.19). This proves the equivalence.

For parity, define

$$\chi(x) := (-1)^{x_1 + \dots + x_n}, \quad \psi(x) := -2^{-n} \chi(x). \quad (7.25)$$

Then $\|\psi\|_1 = 1$. If a monomial $m_S(x) = \prod_{i \in S} x_i$ has $|S| \leq n-1$, choose $j \notin S$ and pair each x with $x^{\{j\}}$; the monomial is unchanged while χ changes sign. Hence $\langle \psi, m_S \rangle = 0$, and by linearity ψ annihilates every polynomial of degree at most $n-1$. On precisely the odd-parity inputs $\chi(x) = -1$, so

$$\langle \psi, \text{PARITY}_n \rangle = 2^{-n} \cdot 2^{n-1} = \frac{1}{2}. \quad (7.26)$$

For every $\varepsilon < 1/2$, (7.19) with $d = n-1$ gives approximate degree at least n ; the exact polynomial (7.15) gives the matching upper bound. The randomized-query identity in (7.20) follows from Theorem 7.2 and the trivial n -query upper bound. Finally the constant $1/2$ proves the stated boundary case. $\square$

What duality does and does not say. A dual witness certifies that every low-degree polynomial fails somewhere; it is not a probability distribution and need not be nonnegative. Its ℓ_1 normalization is what turns pointwise approximation error into the right-hand side of (7.24). The strict correlation in (7.19) corresponds exactly to the strict assertion $\deg_\varepsilon(f) > d$.

7.4 The cube-spectral lemma

For $m \geq 1$, the m -dimensional hypercube graph Q_m has vertex set $\{0, 1\}^m$ and an edge between two strings exactly when they differ in one coordinate. For a finite graph H , $\Delta(H)$ denotes its maximum vertex degree. If A is a real symmetric matrix, write its eigenvalues in nonincreasing order $\lambda_1(A) \geq \dots \geq \lambda_N(A)$. A *principal submatrix* is obtained by restricting a matrix to the same index set of rows and columns.

Lemma 7.5 (One-sided principal-submatrix interlacing). *Let A be a real symmetric $N \times N$ matrix and let B be the principal submatrix on a coordinate subspace of dimension m . Then*

$$\lambda_1(B) \geq \lambda_{N-m+1}(A). \quad (7.27)$$

Proof. By the spectral theorem, A has an orthonormal eigenbasis $v_1, \dots, v_N$. Let E be the span of $v_1, \dots, v_{N-m+1}$ and let W be the m -dimensional coordinate subspace on which B acts. Since $\dim E + \dim W = N + 1$, their intersection contains a nonzero vector w . Expanding w in the eigenbasis gives the Rayleigh-quotient bound

$$\frac{w^T A w}{w^T w} \geq \lambda_{N-m+1}(A). \quad (7.28)$$

Because $w \in W$, the same quotient is $w^T B w / w^T w$, and the largest eigenvalue of a real symmetric matrix is the maximum of its Rayleigh quotient. This proves (7.27). $\square$

Lemma 7.6 (Huang's signed cube matrix). *Define $A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and recursively*

$$A_m = \begin{pmatrix} A_{m-1} & I \\ I & -A_{m-1} \end{pmatrix}. \quad (7.29)$$

Then A_m is a symmetric $2^m \times 2^m$ matrix supported exactly on the edges of Q_m , every nonzero entry is ± 1 , and its eigenvalues are $\sqrt{m}$ and $-\sqrt{m}$, each with multiplicity 2^{m-1} .

Proof. The recursive block structure is exactly the decomposition of Q_m into two copies of Q_{m-1} joined by the coordinate- m perfect matching, so the support assertion follows inductively. Direct multiplication gives

$$A_m^2 = \begin{pmatrix} A_{m-1}^2 + I & 0 \\ 0 & A_{m-1}^2 + I \end{pmatrix}. \quad (7.30)$$

Since $A_1^2 = I$, induction gives $A_m^2 = mI$. Thus every eigenvalue is $\pm\sqrt{m}$. The trace is 0 (already from the zero diagonal), so the positive and negative eigenvalues have equal multiplicity, namely 2^{m-1} each. $\square$

Lemma 7.7 (More than half a cube has a high-degree vertex). *If an induced subgraph H of Q_m has more than 2^{m-1} vertices, then*

$$\Delta(H) \geq \sqrt{m}. \quad (7.31)$$

Proof. It is enough to choose any induced subgraph on exactly $2^{m-1} + 1$ of the vertices of H , because enlarging the vertex set cannot decrease the maximum degree below that of the chosen induced subgraph. Let A_H be the corresponding principal submatrix of A_m from Lemma 7.6. With $N = 2^m$ and $m_H = 2^{m-1} + 1$, Lemma 7.5 gives

$$\lambda_1(A_H) \geq \lambda_{2^{m-1}}(A_m) = \sqrt{m}. \quad (7.32)$$

Let v be an eigenvector for $\lambda_1(A_H)$ and choose a coordinate u with $|v_u|$ maximal. Since A_H is supported only on edges of H and its nonzero entries have absolute value 1,

$$\lambda_1(A_H)|v_u| = |(A_H v)_u| \leq \sum_{w \sim_H u} |v_w| \leq \deg_H(u)|v_u| \leq \Delta(H)|v_u|. \quad (7.33)$$

Because $v_u \neq 0$, (7.31) follows. $\square$

7.5 Sensitivity, degree, and the polynomial collapse

Theorem 7.8 (Tal’s block-sensitivity/degree inequality — deep input). *For every total Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,*

$$\text{bs}(f) \leq \deg(f)^2. \quad (7.34)$$

This inequality is source-owned. Its function-composition proof and the earlier Nisan–Szegedy approximation machinery on which it builds are not silently reproduced inside the sensitivity theorem below.

Why this bridge is registered separately. Huang’s 2019 paper proves the new spectral ingredient $s(f) \geq \sqrt{\deg(f)}$ and then combines it with the previously known block-sensitivity/degree inequality. Treating (7.34) as though it followed from the signed cube matrix would misstate the proof architecture. QRCC therefore certifies the spectral half locally and keeps Tal’s factor-one improvement as a visible source-owned input.

Theorem 7.9 (Sensitivity theorem and classical query-measure collapse). *For every total Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,*

$$s(f) \geq \sqrt{\deg(f)}, \quad \text{bs}(f) \leq s(f)^4. \quad (7.35)$$

Moreover,

$$s(f) \leq \text{bs}(f) \leq C(f) \leq s(f) \text{bs}(f), \quad (7.36)$$

and every nonconstant f satisfies

$$D(f) \leq C_0(f)C_1(f) \leq C(f)^2 \leq s(f)^{10}. \quad (7.37)$$

For every fixed $0 \leq \varepsilon < 1/2$,

$$(1 - 2\varepsilon)s(f) \leq R_\varepsilon(f) \leq D(f). \quad (7.38)$$

Together with $\deg(f) \leq D(f)$, these inequalities show that $s, \text{bs}, C, D, R_\varepsilon$, and exact polynomial degree are all polynomially related on total Boolean functions (with the constants in the randomized comparison allowed to depend on the fixed error $\varepsilon < 1/2$).

Proof roadmap. The new step is $\deg(f) \leq s(f)^2$. Restrict f to the variables of a highest-degree monomial, multiply its $\{\pm 1\}$ encoding by parity, and use the resulting nonzero parity correlation to obtain a set containing more than half the cube. Every edge inside that set is a sensitive edge of the restricted function. Lemma 7.7 then forces sensitivity at least the square root of the degree. Tal's separate deep input (7.34) gives the quartic block-sensitivity bound. The remaining query-measure relations are elementary and proved below.

Proof. If f is constant, then $s(f) = \deg(f) = \text{bs}(f) = 0$ and (7.35) is immediate, so assume f is nonconstant and let $d = \deg(f) \geq 1$. Choose a degree- d monomial in the unique multilinear representation of f , with variable set $S \subseteq [n]$, $|S| = d$. Fix every variable outside S arbitrarily. No monomial of degree greater than d exists, so the coefficient of the full monomial on S cannot receive a contribution from another term under this restriction. The resulting Boolean function $g : \{0, 1\}^d \rightarrow \{0, 1\}$ still has degree d , while $s(g) \leq s(f)$ because every sensitive edge of the restriction is a sensitive edge of f .

Set $h(x) = 1 - 2g(x) \in \{-1, 1\}$ and $\chi(x) = (-1)^{x_1 + \dots + x_d}$. If $a_{[d]}$ is the coefficient of $x_1 \cdots x_d$ in the multilinear polynomial for h , the Boolean coefficient-extraction identity (obtained by expanding the polynomial and summing termwise) gives

$$a_{[d]} = \sum_{x \in \{0, 1\}^d} (-1)^{d - |x|} h(x), \quad (7.39)$$

where $|x|$ is Hamming weight. Since $\deg(h) = d$, $a_{[d]} \neq 0$, and therefore

$$\mathbb{E}_x[h(x)\chi(x)] \neq 0. \quad (7.40)$$

Choose $\sigma \in \{-1, 1\}$ so that the average of $\sigma h(x)\chi(x)$ is positive, and let

$$H := \{x \in \{0, 1\}^d : \sigma h(x)\chi(x) = 1\}. \quad (7.41)$$

The quantity in braces is always ± 1 , so positive average means $|H| > 2^{d-1}$. If adjacent vertices x, y both lie in H , parity changes sign: $\chi(y) = -\chi(x)$. Equation (7.41) then forces $h(y) = -h(x)$, equivalently $g(y) \neq g(x)$. Thus every edge of the induced graph on H is sensitive for g , and

$$\Delta(H) \leq s(g) \leq s(f). \quad (7.42)$$

Lemma 7.7 gives $\Delta(H) \geq \sqrt{d}$, proving $s(f) \geq \sqrt{\deg(f)}$. The source-owned Theorem 7.8 now yields

$$\text{bs}(f) \leq \deg(f)^2 \leq s(f)^4, \quad (7.43)$$

which proves (7.35).

We next prove (7.36). Every sensitive coordinate is a sensitive singleton block, so $s(f) \leq \text{bs}(f)$. Let $B_1, \dots, B_k$ be a maximal collection of pairwise disjoint inclusion-minimal sensitive blocks at a fixed input x . Then $k \leq \text{bs}(f, x) \leq \text{bs}(f)$. Each minimal sensitive block has size at most $s(f)$: if B is minimal, then for every $i \in B$ the input $x^{B \setminus \{i\}}$ has the same function value as x , whereas x^B has the opposite value, so all coordinates in B are sensitive at x^B . Hence $|B| \leq s(f)$.

The union $U = \bigcup_j B_j$ is a certificate at x . Otherwise there is a y agreeing with x on U but with $f(y) \neq f(x)$. The difference set between x and y lies outside U and contains an inclusion-minimal sensitive block at x disjoint from all B_j , contradicting maximality. Therefore

$$C(f, x) \leq |U| \leq s(f) \text{bs}(f), \quad (7.44)$$

and maximizing over x proves (7.36), including the obvious $\text{bs}(f) \leq C(f)$.

For (7.37), we prove the standard certificate decision-tree bound by tracking how failed 0-certificates reduce the remaining 1-certificate complexity. At a current restriction, if no 0-input is compatible with the transcript, every completion has value 1 and the tree outputs 1. Otherwise choose a compatible 0-input y , take an original 0-certificate S for y of size at most $C_0(f)$, and query all its as-yet-unqueried coordinates. If every answer agrees with y on S , output 0.

Suppose instead that the round fails. View the function before this round as a function on its still-unqueried coordinates. For any 1-input z compatible with the new transcript, let T be a 1-certificate for z in that pre-round restricted function. The assignments specified by the 0-certificate S and by T , together with the already fixed transcript, cannot be compatible: otherwise one completion would be forced to have both output values. Since y was itself compatible with the old transcript, the conflict occurs at a previously unfixed coordinate of $S \cap T$. That coordinate has now been queried and fixed, so deleting it from T leaves a valid 1-certificate inside the new restricted subcube. Consequently the maximum 1-certificate complexity of the restricted function drops by at least one after every failed round. It starts at at most $C_1(f)$, so there can be at most $C_1(f)$ rounds, each querying at most $C_0(f)$ new coordinates. This constructs a decision tree with

$$D(f) \leq C_0(f)C_1(f) \leq C(f)^2. \quad (7.45)$$

Combining (7.44) and (7.43) gives $C(f) \leq s(f)^5$ and hence $D(f) \leq s(f)^{10}$.

Finally fix $0 \leq \varepsilon < 1/2$ and choose an input x with $s(f, x) = s(f)$. For a sensitive coordinate i , couple the randomized tree on x and $x^{\{i\}}$ using the same random tape. Unless the tree queries coordinate i , the two runs have identical transcripts and outputs. Since the correct output bits differ and each run errs with probability at most ε , the acceptance probabilities on the two inputs differ by at least $1 - 2\varepsilon$. Hence the probability, on input x , that the algorithm ever queries i is at least $1 - 2\varepsilon$. Summing these query indicators over the $s(f)$ sensitive coordinates shows that the expected number of queries on x is at least $(1 - 2\varepsilon)s(f)$. A randomized tree of maximum depth T makes at most T queries on every tape, so $T \geq (1 - 2\varepsilon)s(f)$. The upper bound $R_\varepsilon(f) \leq D(f)$ is immediate by using a deterministic tree. This proves (7.38) and the claimed polynomial equivalence. $\square$

Source parity and ownership audit. Huang's Annals paper proves the sharp induced-subgraph statement $\Delta(H) \geq \sqrt{m}$ for $2^{m-1} + 1$ cube vertices and records the Boolean-function consequence $s(f) \geq \sqrt{\deg(f)}$. PASS67 reconstructs both the signed-matrix argument and the required Boolean-function implication directly in (7.39)–(7.42). Huang then invokes the previously known relation between block sensitivity and degree; QRCC uses Tal's factor-one version (7.34) and labels it as a deep input. The elementary certificate and randomized-query consequences (7.36)–(7.38) are local additions.

Worked example: OR. For OR_n , the all-zero input has sensitivity and block sensitivity n , so $s = \text{bs} = n$. A 0-certificate must expose all n zeroes, while a 1-certificate needs one coordinate, hence $C_0 = n$, $C_1 = 1$, and $C = n$. Also $D = n$ and the exact polynomial $1 - \prod_i (1 - x_i)$ has degree n . Thus the elementary inequalities above can be tight at their left-hand ends even though the general powers in (7.35)–(7.37) are deliberately nonoptimal.

Chapter checkpoint and exercises

The Chapter 7 dependency picture is now

$$\begin{aligned} \text{query transcript} &\implies \text{low-degree acceptance polynomial}, \\ \text{failure of low-degree approximation} &\iff \ell_1 \text{ dual witness}, \\ \text{Huang signed cube matrix} &\implies \deg(f) \leq s(f)^2, \\ \text{Tal deep input } \text{bs}(f) \leq \deg(f)^2 &\implies \text{bs}(f) \leq s(f)^4. \end{aligned} \tag{7.46}$$

The first, second, and spectral implications are proved locally. The one source-owned arrow bridge is visibly registered as Theorem 7.8.

1. **Leaf polynomials.** For the three-bit majority decision tree that queries x_1, x_2 , and queries x_3 only when the first two bits differ, write every accepting-leaf polynomial (7.11), sum them, and multilinearize the result.
2. **Randomized acceptance.** A randomized algorithm chooses a uniformly random coordinate and outputs that bit. Compute its acceptance polynomial and identify its degree. For which symmetric Boolean functions can this be a useful approximation?
3. **Quantum degree bookkeeping.** Starting from degree-0 amplitudes, verify (7.13) for one oracle call and show explicitly that an input-independent unitary cannot increase degree. Explain why taking squared absolute values, rather than amplitudes themselves, is responsible for the factor 2.
4. **Weak duality first.** Assume ψ has $\|\psi\|_1 = 1$ and annihilates $\mathcal{V}_d$. Prove directly that $\langle \psi, f \rangle > \varepsilon$ rules out every degree- d approximant before reading the separation argument that proves the converse.
5. **Parity threshold.** Check every step of (7.25)–(7.26) for $n = 3$. Then explain why the same witness cannot prove anything at error $\varepsilon = 1/2$ and exhibit the degree-0 approximant at that boundary.
6. **Signed cube matrices.** Write A_2 and A_3 from (7.29), square them directly, and verify that replacing every -1 entry by $+1$ produces the ordinary cube adjacency matrix.
7. **Interlacing dimension count.** In Lemma 7.5, prove $\dim(E \cap W) \geq 1$ from the dimension formula. Explain why the index $N - m + 1$ becomes exactly 2^{m-1} when the principal submatrix has $2^{m-1} + 1$ rows.
8. **From cube expansion to sensitivity.** For a degree- d Boolean function, reproduce the restriction-to- d -variables argument and derive (7.39) by Möbius inversion. Verify that an edge inside (7.41) must be a sensitive edge.
9. **Minimal sensitive blocks.** Prove that every inclusion-minimal sensitive block has size at most $s(f)$, and then prove from maximality that the union of a maximal disjoint family of such blocks is a certificate. Deduce (7.44) without citing an external theorem.

10. **Certificate search.** Trace the proof of $D(f) \leq C_0(f)C_1(f)$ on OR_n and on parity. Which example makes the product bound tight, and which makes it very loose?
11. **Randomized sensitivity lower bound.** Formalize the same-random-tape coupling used in (7.38). Show that if coordinate i is never queried, the two executions on x and $x^{\{i\}}$ are identical, including their leaf output.
12. **Ownership boundary.** Explain why Lemma 7.7 alone yields $\deg(f) \leq s(f)^2$ but not the quartic block-sensitivity bound. Identify the precise extra statement supplied by Theorem 7.8.

Transition to Chapter 8. Chapter 7 has closed the query-complexity algebraic and local-sensitivity toolkit. The next registered block is property testing. Before using removal phenomena or an adaptivity separation, Chapter 8 must first freeze the oracle, distance, and success conventions for the BLR and dense-graph testing models; only then can the linearity test and graph-removal reduction be stated without ambiguity.

Chapter 8

Property Testing, Removal, and Adaptivity

Chapter 7 studied query algorithms that must compute a Boolean function on every input. Property testing changes the objective: distinguish objects having a property from objects that are far from that property while inspecting only a tiny part of the input. This chapter fixes the distance and oracle conventions before using them, proves the BLR linearity test by Fourier analysis, transports the already certified graph-removal machinery into the dense-graph testing model, and then records two source-owned adaptivity separations at deliberately different historical levels.

How to read this chapter. Definition 8.1 fixes generic oracle testing, normalized Hamming distance, one- versus two-sided error, adaptivity, and the dense adjacency-matrix model. Theorem 8.2 is fully local: a single three-query BLR trial has rejection probability at least the distance to linearity whenever that distance is below the trivial $1/2$ threshold, and independent repetition gives the usual $O(1/\varepsilon)$ tester. Theorem 8.3 is a model-and-reduction bridge from the Volume IV graph removal theorem. Theorem 8.4 and the dated frontier statement 8.5 are source-owned: their proofs contain substantial graph-testing constructions that are not reconstructed here.

Learning goals. After this chapter, a reader should be able to state a property tester without leaving its distance or success convention implicit; distinguish a three-query BLR *trial* from an amplified ε -tester; derive BLR soundness from Fourier inversion and Parseval; explain why graph removal yields one-sided constant-query testing for fixed subgraph freeness; and compare a classical natural $3/2$ adaptivity gap with the later near-quadratic dense-graph frontier.

8.1 Testing models, proximity, and dense graphs

Definition 8.1 (Property-testing and dense-graph models). 1. Let D be a finite nonempty set and Σ a finite nonempty alphabet. An *oracle object* is a function $f : D \rightarrow \Sigma$, and a nonempty *property* is a set $\mathcal{P} \subseteq \Sigma^D$. The normalized Hamming *distance* is

$$\text{dist}(f, g) := \frac{|\{z \in D : f(z) \neq g(z)\}|}{|D|}, \quad \text{dist}(f, \mathcal{P}) := \min_{g \in \mathcal{P}} \text{dist}(f, g). \quad (8.1)$$

For $0 < \varepsilon < 1$, the object f is ε -far from $\mathcal{P}$ when $\text{dist}(f, \mathcal{P}) > \varepsilon$.

2. An ε -tester for $\mathcal{P}$ is a randomized oracle algorithm that, for every $f \in \Sigma^D$, outputs either ACCEPT or REJECT and satisfies

$$\Pr[T^f \text{ accepts}] \geq \frac{2}{3} \quad (f \in \mathcal{P}), \quad \Pr[T^f \text{ rejects}] \geq \frac{2}{3} \quad (\text{dist}(f, \mathcal{P}) > \varepsilon). \quad (8.2)$$

The probability is over the tester's internal random tape, which is independent of the oracle. A tester has *one-sided error* when every $f \in \mathcal{P}$ is accepted with probability 1; otherwise the guarantee (8.2) is called *two-sided error*. Its *query complexity* is the maximum number of oracle values requested over all inputs and random tapes.

3. A tester is *nonadaptive* when, after its explicit parameters and random tape are fixed, its entire list of query locations is fixed before any oracle answer is seen. It is *adaptive* when later queries may depend on earlier answers. Thus nonadaptive testers are a special case of adaptive testers. When a family $\mathcal{P} = (\mathcal{P}_n)_{n \geq 1}$ has an input-size parameter n , its testing query complexity is the least worst-case query bound as a function of n and ε ; when the bound is independent of n we suppress n .
4. In the *dense simple-graph adjacency model*, an n -vertex graph $G = ([n], E)$ is represented by its symmetric adjacency predicate

$$A_G : [n] \times [n] \rightarrow \{0, 1\}, \quad A_G(u, u) = 0. \quad (8.3)$$

One adjacency query reveals $A_G(u, v)$. For two such graphs define the *adjacency-matrix distance*

$$\Delta_{\text{mat}}(G, H) := \frac{1}{n^2} |\{(u, v) \in [n]^2 : A_G(u, v) \neq A_H(u, v)\}|. \quad (8.4)$$

For a graph property $\Pi = (\Pi_n)$, meaning each Π_n is closed under vertex relabeling, G is ε -far from Π in this model when $\Delta_{\text{mat}}(G, H) > \varepsilon$ for every $H \in \Pi_n$. Because an undirected edge occupies two off-diagonal matrix entries, (8.4) differs by a factor of two from an unordered-edge edit normalization. The fixed- H removal theorem below therefore continues to use the explicit edge-deletion convention of Volume IV instead of silently identifying the two normalizations.

5. The *bi-clique collection property*, abbreviated BCC, consists of the finite simple graphs whose vertex set can be partitioned into sets $V_1, \dots, V_m$ such that each induced graph $G[V_i]$ is a complete bipartite graph (with either side allowed to be empty) and there are no edges between distinct V_i, V_j . Equivalently, every connected component is complete bipartite, including isolated vertices as degenerate components.

Two normalizations worth checking immediately. For a Boolean string of length N , changing one coordinate changes normalized Hamming distance by $1/N$. For an undirected n -vertex graph, toggling one ordinary edge changes exactly two entries of (8.3), hence changes Δ_{mat} by $2/n^2$. This factor does not change asymptotic exponents, but it matters when transferring an explicit εn^2 deletion threshold from a removal theorem.

Amplification convention. The success threshold $2/3$ in (8.2) is conventional. If independent repetitions of a one-sided trial each reject a far object with probability at least $p > 0$, then $R = \lceil (\log 3)/p \rceil$ repetitions miss every witness with probability at most $(1 - p)^R \leq e^{-pR} \leq 1/3$. Because all random choices can be sampled before any answer is read, independent repetition preserves nonadaptivity when each trial is nonadaptive.

8.2 The BLR linearity test

Identify $\{0, 1\}^n$ with the vector space $\mathbb{F}_2^n$. For $a, x \in \mathbb{F}_2^n$ write $a \cdot x = \sum_i a_i x_i \pmod{2}$ and

$$\ell_a(x) := a \cdot x. \quad (8.5)$$

The property of linear Boolean functions is $\text{Lin}_n := \{\ell_a : a \in \mathbb{F}_2^n\}$. These are linear maps fixing the origin; the affine translate $1 + \ell_a$ is not in Lin_n unless the constant term is zero.

Theorem 8.2 (BLR linearity test: local Fourier form). *Let $n \geq 1$ and $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$. A single BLR trial samples independent uniform $x, y \in \mathbb{F}_2^n$, queries $f(x), f(y), f(x + y)$, and rejects exactly when*

$$f(x) + f(y) \neq f(x + y) \pmod{2}. \quad (8.6)$$

Let ρ be its rejection probability.

1. If $f \in \text{Lin}_n$, then $\rho = 0$.
2. If $\rho < 1/2$, there exists $a \in \mathbb{F}_2^n$ such that

$$\text{dist}(f, \ell_a) \leq \rho. \quad (8.7)$$

Consequently, for every $0 < \varepsilon < 1/2$, if f is ε -far from Lin_n , then $\rho \geq \varepsilon$.

3. Therefore

$$R = \left\lceil \frac{\log 3}{\varepsilon} \right\rceil \quad (8.8)$$

independent BLR trials form a one-sided nonadaptive ε -tester using at most $3R = O(1/\varepsilon)$ oracle queries.

Proof roadmap. Pass to the sign function $g = (-1)^f$. The BLR acceptance bias is the cubic correlation $\mathbb{E}_{x,y} g(x)g(y)g(x + y)$. Fourier inversion turns this into a sum of cubes of Fourier coefficients. Parseval says their squared masses sum to one, so a large cubic sum forces one large positive coefficient. That coefficient is exactly the correlation with a linear character and hence decodes a nearby linear function.

Proof. If $f = \ell_a$, linearity gives $f(x) + f(y) = f(x + y)$ for every pair, proving completeness.

For soundness set

$$g(x) := (-1)^{f(x)}, \quad \chi_a(x) := (-1)^{a \cdot x}, \quad \widehat{g}(a) := \mathbb{E}_x g(x) \chi_a(x). \quad (8.9)$$

The characters χ_a are the characters of the finite abelian group $\mathbb{F}_2^n$. By Fourier inversion and character orthogonality from Theorem 6.3,

$$\begin{aligned} 1 - 2\rho &= \mathbb{E}_{x,y} g(x)g(y)g(x + y) \\ &= \sum_{a \in \mathbb{F}_2^n} \widehat{g}(a)^3. \end{aligned} \quad (8.10)$$

Indeed, after expanding all three copies of g , averaging over x and y kills every triple of frequencies except the one in which all three are equal. Parseval, Theorem 6.4, gives

$$\sum_a \widehat{g}(a)^2 = \mathbb{E}_x g(x)^2 = 1. \quad (8.11)$$

Let $M = \max_a \widehat{g}(a)$. Since $\widehat{g}(a)^3 \leq M\widehat{g}(a)^2$ term by term, (8.10)–(8.11) imply

$$1 - 2\rho \leq M. \quad (8.12)$$

When $\rho < 1/2$, choose a attaining M . Agreement of f with ℓ_a is equivalent to $g(x)\chi_a(x) = 1$, and disagreement is equivalent to -1 . Hence

$$\widehat{g}(a) = 1 - 2 \operatorname{dist}(f, \ell_a), \quad (8.13)$$

so (8.12) yields (8.7).

Now fix $0 < \varepsilon < 1/2$. If an ε -far f had $\rho < \varepsilon$, then $\rho < 1/2$ and (8.7) would exhibit a linear function within distance $< \varepsilon$, a contradiction. Thus one trial rejects with probability at least ε . Independent repetition and the amplification calculation following Definition 8.1 give (8.8). Every query triple is determined by the random tape before the oracle is read, so the amplified tester is nonadaptive. $\square$

Source boundary. Blum, Luby, and Rubinfeld introduced the self-testing/correcting framework and the three-query linearity consistency check. PASS68 uses that test as the historical source gate but proves the displayed Fourier normalization independently. In particular, the single trial in (8.6) is not mislabeled as a $2/3$ -sound ε -tester for arbitrary small ε ; the $2/3$ guarantee belongs to the repeated tester in (8.8).

Worked checks. For parity $f(x) = x_1 + \cdots + x_n \pmod{2}$, every trial accepts. At the opposite extreme, the constant-one function fails (8.6) on every pair, so $\rho = 1$; its nearest linear function is at distance $1/2$. This illustrates why the useful decoder statement is naturally separated at the threshold $\rho < 1/2$.

8.3 From local removal witnesses to dense-graph testing

Theorem 8.3 (Fixed-subgraph testing from removal). *Fix a finite simple graph H with $h = v(H)$ vertices and $q = e(H) \geq 1$. Use the Volume IV convention that an n -vertex graph G is ε -far from H -free when more than εn^2 edge deletions are required to destroy all non-induced copies of H . Let $\delta = \delta(H, \varepsilon) > 0$ be supplied by the graph removal theorem, Theorem 6.3. Then H -freeness has a one-sided nonadaptive tester using at most*

$$q \left\lceil \frac{\log 3}{\delta(H, \varepsilon)} \right\rceil \quad (8.14)$$

adjacency queries. The bound is independent of n ; no polynomial dependence of δ on ε is asserted.

Proof. Set $R = \lceil (\log 3)/\delta \rceil$. Before reading any oracle answer, independently sample R ordered h -tuples from $[n]^h$. For each tuple with distinct entries, query the q pairs corresponding to edges of H and reject if all required edges are present. Repeated vertices simply make that trial nonrejecting. All query locations are fixed by the initial randomness, so the tester is nonadaptive.

An H -free graph is accepted with probability one. If G is ε -far, Theorem 6.3 gives more than δn^h labelled embeddings of H . Hence one uniformly sampled ordered tuple is a rejecting witness with probability greater than δ . The probability that all R trials miss is at most

$$(1 - \delta)^R \leq e^{-\delta R} \leq \frac{1}{3}, \quad (8.15)$$

which proves the claim. $\square$

Relation to the earlier theorem. Theorem 7.2 already proved this removal-to-testing implication in Volume IV. The point of Theorem 8.3 is not to duplicate graph regularity: it transports the result into the generic testing language of Definition 8.1 and records explicitly that the tester is nonadaptive. The case $q = 0$ remains the elementary edgeless-graph endpoint handled in Volume IV.

Scope boundary: not all hereditary testing comes for free. Ordinary non-induced graph removal proves (8.14) for a fixed forbidden subgraph. It does not by itself imply the full induced/hereditary graph-property testing theorems, which require stronger induced or infinite removal machinery. The earlier registry wording that bundled “hereditary/subgraph-freeness” has therefore been narrowed at PASS68.

Worked check: one forbidden edge. For $H = K_2$, direct counting is much sharper than an abstract removal parameter. If more than εn^2 edge deletions are needed, then $e(G) > \varepsilon n^2$, so a random ordered pair exposes an edge with probability $2e(G)/n^2 > 2\varepsilon$. Thus $O(1/\varepsilon)$ random edge queries suffice.

8.4 Adaptivity: a natural separation and a later frontier

For functions of the proximity parameter, $\tilde{O}(\cdot)$ suppresses factors polynomial in $\log(1/\varepsilon)$. The next theorem is retained at source ownership: its adaptive upper bound and nonadaptive lower bound use detailed structural arguments for bi-clique collections that are not consequences of removal alone.

Theorem 8.4 (Goldreich–Ron bi-clique adaptivity separation — deep input). *In the dense adjacency-matrix testing model of Definition 8.1, the property BCC satisfies:*

1. *there is an adaptive one-sided tester with query complexity $\tilde{O}(\varepsilon^{-1})$;*
2. *every nonadaptive tester has query complexity $\Omega(\varepsilon^{-3/2})$, and this lower bound still holds when the input graph is promised to be bipartite.*

The asymptotic query complexities are in the standard dense-property-testing sense, uniform over graph size with ε as the proximity parameter.

Source ownership. Goldreich and Ron define bi-clique collection as a collection of isolated complete bipartite components and state the two bounds above as their Theorem 1.2. Their paper also makes the adjacency-matrix distance and adaptive/nonadaptive conventions explicit. PASS68 source-verifies that theorem but does not reproduce its long adaptive tester or its indistinguishability lower bound. The theorem is therefore marked a verified deep input rather than a locally proved result.

Why this belongs after removal rather than inside it. Theorem 8.3 obtains nonadaptive testers by finding local forbidden witnesses. Theorem 8.4 shows that a natural graph property can benefit polynomially from using answers to decide what to query next. Thus adaptivity is an algorithmic resource, not merely a presentation choice for the same random-sampling argument.

Theorem 8.5 (Goldreich–Wigderson near-quadratic dense-graph gap — dated frontier). *There exists a graph property Π of n -vertex graphs for which, in the dense adjacency-matrix model,*

1. an adaptive tester uses

$$O(\varepsilon^{-1}\sqrt{n} \log n) \quad (8.16)$$

queries;

2. every nonadaptive tester requires $\Omega(n)$ queries.

This is the exact Theorem 1.2 frontier statement selected from the Goldreich–Wigderson FOCS 2021 work; it is source-owned and is not used as a premise for the locally proved results of this chapter.

Frontier context. The same work develops a broader hierarchy: up to polylogarithmic factors, gaps between adaptive complexity $q(n)$ and nonadaptive complexity $q(n)^c$ can be realized for $c \in [1, 2]$ in the stated range of q . PASS68 records the clean near-quadratic theorem (8.16) as the stable frontier anchor rather than expanding the reader statement into all parameter variants. This keeps the 2011 natural BCC theorem and the 2021 near-quadratic existential theorem visibly distinct.

Chapter checkpoint and exercises

The Chapter 8 dependency picture is

$$\begin{aligned} \text{Fourier inversion} + \text{Parseval} &\implies \text{BLR distance decoder and tester,} \\ \text{graph removal} &\implies \text{one-sided nonadaptive fixed-}H \text{ tester,} \\ \text{Goldreich–Ron source theorem} &\implies \tilde{O}(\varepsilon^{-1}) \text{ versus } \Omega(\varepsilon^{-3/2}), \\ \text{Goldreich–Wigderson frontier} &\implies O(\varepsilon^{-1}\sqrt{n} \log n) \text{ versus } \Omega(n). \end{aligned} \quad (8.17)$$

The first two arrows are locally proved; the last two are explicitly source-owned.

1. **Distance conventions.** For two simple graphs differing in exactly m undirected edges, compute their adjacency-matrix distance (8.4). Compare it with m/n^2 and explain the factor two.
2. **BLR identity.** Starting from Fourier inversion on $\mathbb{F}_2^n$, derive (8.10) term by term. State exactly where independence and uniformity of x, y are used.
3. **Decoder inequality.** Show that $\sum_a \hat{g}(a)^3 \leq (\max_a \hat{g}(a)) \sum_a \hat{g}(a)^2$ even when some Fourier coefficients are negative. Deduce (8.7).
4. **Three queries versus an ε -tester.** For $\varepsilon = 10^{-3}$, estimate the number of BLR trials prescribed by (8.8). Explain why calling one trial a “three-query ε -tester with soundness $2/3$ ” would be false at this proximity scale.
5. **Nonadaptivity under repetition.** Write the full list of random pairs used by R BLR trials before any query is answered. Repeat the same exercise for the R graph-copy trials in Theorem 8.3.
6. **Removal endpoint.** Reprove the $H = K_2$ worked check directly. Why can it give a far better dependence on ε than substituting a generic regularity-based $\delta(H, \varepsilon)$?
7. **Induced versus non-induced.** Give an example showing that checking only the required edges of a sampled H detects non-induced copies, not induced copies. Identify what additional queries would certify an induced copy and why ordinary graph removal still does not automatically yield the full hereditary theorem.

8. **Adaptivity bookkeeping.** Restate the two clauses of Theorem 8.4 using the definitions of adaptive and nonadaptive testing. Explain why the bipartite-promise clause strengthens the lower bound rather than the upper bound.
9. **Compare the two separations.** Ignoring logarithmic factors and fixing constant ε , compare the BCC separation with (8.16). Which theorem concerns a natural fixed property with complexity expressed primarily through ε , and which concerns an existential family whose gap grows with n ?
10. **Ownership audit.** List the exact ingredients of this chapter that are proved locally and those that remain source-owned. Explain why citing Theorem 8.4 does not license a reader to treat the Goldreich–Ron lower-bound proof as reconstructed inside QRCC.

Transition to Chapter 9. Property testing extracts a global yes/no conclusion from sparse local access. The next registered block changes the resource again: a streaming algorithm sees a long update sequence through a tiny memory state. Chapter 9 will therefore begin by fixing the turnstile/ insertion-only stream convention and frequency vector before stating the AMS sketch or using communication lower bounds as streaming lower bounds.

Chapter 9

Streaming, Frequency Moments, and Linear Sketches

The previous chapters moved from communication to query complexity and property testing. Streaming algorithms impose a different resource restriction: the input arrives sequentially and only a small memory state may survive between updates. This chapter fixes the update, pass, space, and approximation conventions before any streaming theorem is stated. It then proves the classical AMS second-moment sketch, makes the communication simulation exact under QRCC’s fixed-leaf convention, records one source-owned historical lower bound for large frequency moments, and closes with a self-contained Johnson–Lindenstrauss argument and its deliberately limited linear-sketch interpretation.

How to read this chapter. Definition 9.1 is load bearing: insertion-only, strict-turnstile, and general-turnstile streams are not interchangeable, and a p -pass algorithm here always rereads the same stream in the same order. Algorithm 9.2 is fully local and works even in the general-turnstile model. Theorem 9.3 is the reduction theorem used when streaming lower bounds are imported from communication complexity. Theorem 9.4 is intentionally source-owned at the classical AMS exponent. Theorem 9.5 is again local, but its finite-set oblivious guarantee is not silently upgraded to adaptive-query robustness.

Learning goals. A reader should be able to write down a frequency vector from an update stream; distinguish stream length from total update magnitude; derive the AMS expectation and variance from four-wise independence; explain every memory-state message in a multipass communication simulation; state exactly what the classical AMS large-moment lower bound does and does not prove; and derive the Johnson–Lindenstrauss dimension from a one-vector Gaussian tail plus a union bound.

9.1 Streaming and sketching conventions

Definition 9.1 (Streaming, frequency-vector, pass, and linear-sketch models). Fix a universe $[n] = \{1, \dots, n\}$.

1. An *integer update stream* is a finite sequence

$$\sigma = ((a_1, \Delta_1), \dots, (a_m, \Delta_m)), \quad a_t \in [n], \quad \Delta_t \in \mathbb{Z} \setminus \{0\}. \quad (9.1)$$

Its prefix frequency vector is

$$f_i^{(t)} := \sum_{u \leq t: a_u = i} \Delta_u, \quad (9.2)$$

and its final frequency vector is $f = f^{(m)} \in \mathbb{Z}^n$. The *stream length* is m , while the *total update magnitude* is

$$L(\sigma) := \sum_{t=1}^m |\Delta_t|. \quad (9.3)$$

These two parameters need not agree.

2. The stream is *insertion only* when every $\Delta_t = 1$; it is *strict turnstile* when every prefix satisfies $f_i^{(t)} \geq 0$ for all i ; and it is *general turnstile* when no sign restriction is imposed on prefix or final frequencies.
3. For $q > 0$, the q th frequency moment is

$$F_q(f) := \sum_{i=1}^n |f_i|^q. \quad (9.4)$$

Also set $F_0(f) := |\{i : f_i \neq 0\}|$ and $F_\infty(f) := \max_i |f_i|$. In particular, $F_2(f) = \|f\|_2^2$.

4. A randomized p -pass streaming algorithm reads the same update sequence (9.1), in the same order, exactly p times. Between updates and between passes it retains only its current memory state. It uses s bits of space if the set of possible memory states has cardinality at most 2^s ; the input-independent random tape is not charged to this state bound, matching the public-randomness convention used in communication reductions. Unless explicitly stated otherwise, success probability means at least $2/3$ on every valid stream.
5. For $0 < \varepsilon < 1$, a number Z is a $(1 \pm \varepsilon)$ approximation to a nonnegative target T when

$$|Z - T| \leq \varepsilon T. \quad (9.5)$$

For a randomized algorithm the inequality is required with the stated success probability. When $T = 0$, (9.5) forces $Z = 0$ on the success event.

6. Let $\mathbb{F}$ be a field and let $A \in \mathbb{F}^{k \times n}$. The map $f \mapsto Af$ is a *linear sketch*. A randomized linear sketch is *oblivious* when the random matrix A is sampled independently of the vector or finite family of vectors whose norms or distances will later be estimated. An oblivious guarantee for a fixed family does not, by itself, imply correctness for vectors chosen adaptively after A or its outputs have been exposed.

Sanity checks. In an insertion-only stream, $L = m$ and f_i is the ordinary multiplicity of item i . In a general-turnstile stream the final vector may have negative coordinates even though every moment in (9.4) is nonnegative. The AMS estimator below uses only linear updates to a signed counter, so its algebra does not need nonnegative intermediate frequencies.

9.2 The AMS second-moment sketch

Algorithm 9.2 (AMS second-moment sketch with explicit amplification). *Let σ be a general-turnstile stream over $[n]$, let f be its final frequency vector, let $0 < \varepsilon < 1$, and let $0 < \delta < 1$. Choose*

$$r := \left\lceil \frac{8}{\varepsilon^2} \right\rceil, \quad q := \text{the least odd integer at least } 8 \log(1/\delta). \quad (9.6)$$

For every pair $(u, v) \in [q] \times [r]$, independently choose a four-wise independent family of unbiased signs

$$h_{u,v}(1), \dots, h_{u,v}(n) \in \{-1, 1\} \quad (9.7)$$

using an $O(\log n)$ -bit seed from Algorithm 1.3. Maintain

$$Z_{u,v} := \sum_{i=1}^n h_{u,v}(i) f_i \quad (9.8)$$

by the update rule $Z_{u,v} \leftarrow Z_{u,v} + \Delta_t h_{u,v}(a_t)$. At the end form

$$Y_u := \frac{1}{r} \sum_{v=1}^r Z_{u,v}^2 \quad (9.9)$$

and output the median Y of $Y_1, \dots, Y_q$. Then

$$\Pr[|Y - F_2(f)| > \varepsilon F_2(f)] \leq \delta. \quad (9.10)$$

Moreover the sketch is one pass and uses

$$O\left(\varepsilon^{-2} \log(1/\delta) (\log n + \log(L(\sigma) + 1))\right) \quad (9.11)$$

bits under exact integer counter storage.

Sign-family construction. Choose $2^d \geq n$, identify the universe with distinct points of $\mathbb{F}_{2^d}$, sample a uniform random polynomial of degree at most three, and apply any fixed nonzero $\mathbb{F}_2$ -linear coordinate map $\mathbb{F}_{2^d} \rightarrow \mathbb{F}_2$ to each evaluation. The four-wise independent uniform field evaluations supplied by Algorithm 1.3 become four-wise independent unbiased bits, hence signs. The seed has $4d = O(\log n)$ bits.

Proof roadmap. For one sign vector, $X = Z^2$ is unbiased for F_2 . Four-wise independence kills every mixed fourth-moment term except paired indices and gives $\text{Var}(X) \leq 2F_2^2$. Averaging r copies makes one group good with probability at least $3/4$ by Chebyshev. The median of q independent groups fails only if at least half the groups fail, which is exponentially unlikely by Hoeffding.

Proof. Fix one copy and abbreviate $h(i)$ by h_i . Since the signs are unbiased and pairwise independent,

$$\begin{aligned} \mathbb{E}Z^2 &= \mathbb{E}\left(\sum_i h_i f_i\right)^2 \\ &= \sum_i f_i^2 + 2 \sum_{i < j} f_i f_j \mathbb{E}[h_i h_j] = F_2(f). \end{aligned} \quad (9.12)$$

Writing $Z^2 = F_2 + 2 \sum_{i < j} f_i f_j h_i h_j$, four-wise independence shows that distinct unordered pairs have zero covariance. Therefore

$$\text{Var}(Z^2) = 4 \sum_{i < j} f_i^2 f_j^2 \leq 2 \left(\sum_i f_i^2 \right)^2 = 2F_2(f)^2. \quad (9.13)$$

For a group average Y_u of r independent copies,

$$\text{Var}(Y_u) \leq \frac{2F_2(f)^2}{r} \leq \frac{\varepsilon^2 F_2(f)^2}{4}. \quad (9.14)$$

If $F_2(f) = 0$, then $f = 0$ and every counter is exactly zero. Otherwise Chebyshev, Theorem 1.4, gives

$$\Pr[|Y_u - F_2(f)| > \varepsilon F_2(f)] \leq \frac{1}{4}. \quad (9.15)$$

Let B_u indicate failure of group u . The B_u are independent, lie in $[0, 1]$, and have expectation at most $1/4$. If the median fails then at least $q/2$ groups fail. Hoeffding, Theorem 2.5, yields

$$\Pr\left[\sum_{u=1}^q B_u \geq q/2\right] \leq e^{-q/8} \leq \delta, \quad (9.16)$$

which proves (9.10).

For space, each seeded sign family uses $O(\log n)$ bits. Every counter in (9.8) satisfies $|Z_{u,v}| \leq L(\sigma)$ at every time, so exact signed storage costs $O(\log(L(\sigma) + 1))$ bits. There are $rq = O(\varepsilon^{-2} \log(1/\delta))$ counters and seeds, proving (9.11). $\square$

Source boundary. Alon, Matias, and Szegedy prove the logarithmic-space F_2 phenomenon using exactly the four-wise-independent-sign estimator, the same unbiasedness/variance calculation, Chebyshev, and median amplification. PASS69 keeps that theorem as the historical source gate but derives the turnstile-compatible update rule, the explicit (ε, δ) constants in (9.6), and the exact $\log(L + 1)$ counter accounting locally.

Worked example. For $f = (3, -1, 2)$, $F_2 = 14$. With signs $(1, -1, 1)$ the counter is $Z = 6$ and the one-copy estimate is 36; with $(1, 1, -1)$ it is $Z = 0$. Individual copies can therefore be very noisy. The theorem is not a claim that every sign choice approximates F_2 : unbiasedness and the bounded fourth moment are what make averaging and medians work.

9.3 Communication from a bounded streaming state

Theorem 9.3 (Exact two-block communication simulation). *Let $g : X \times Y \rightarrow \{0, 1\}$ be a finite Boolean communication problem. Suppose Alice can map $x \in X$ to a stream block $\sigma_A(x)$ and Bob can map $y \in Y$ to a block $\sigma_B(y)$ so that a streaming problem on the concatenation*

$$\sigma(x, y) := \sigma_A(x) \circ \sigma_B(y) \quad (9.17)$$

has correct Boolean answer $g(x, y)$. If a deterministic p -pass streaming algorithm solves every such concatenated stream using s bits of memory, then there is a deterministic QRCC fixed-leaf protocol for g of communication cost at most

$$(2p - 1)s + 1. \quad (9.18)$$

If the streaming algorithm is randomized with worst-case error at most η , the same simulation, with the streaming random tape used as public randomness, gives a public-coin protocol of error at most η and cost at most (9.18).

Under the alternative convention in which Bob may output locally after the final simulation step, the final $+1$ disappears and the cost is $(2p - 1)s$.

Proof. Fix the streaming random tape; the deterministic case is the same argument with no tape. At the start of pass one the streaming state is the fixed initial state. Alice processes her block $\sigma_A(x)$ and sends the resulting s -bit state to Bob. Bob continues on $\sigma_B(y)$. If another pass remains, Bob

sends the resulting state back to Alice; the next pass then starts on $\sigma_A(x)$ with exactly the state that the streaming algorithm would have retained.

Thus each of the p passes contributes one Alice-to-Bob state message, and each of the first $p - 1$ passes contributes one Bob-to-Alice state message. The state communication is therefore

$$ps + (p - 1)s = (2p - 1)s. \quad (9.19)$$

After the last Bob block, Bob knows the streaming output. In QRCC's fixed output-labelled leaf model, the transcript must itself determine that output, so Bob sends one final answer bit. This gives (9.18). If local output by Bob is permitted, that bit is unnecessary.

For a randomized streaming algorithm, expose its input-independent random tape as public coins. Conditioned on the tape, the state transfer is exact, so the communication protocol has exactly the same error probability as the streaming algorithm on every input pair. $\square$

Reduction template. If a communication lower bound gives $R_\eta(g) \geq C$, then any p -pass s -space streaming algorithm fitting (9.17) must satisfy

$$(2p - 1)s + 1 \geq C. \quad (9.20)$$

The theorem does not manufacture the encoding (9.17); every streaming lower bound must still prove that its chosen stream blocks convert the communication promise into the required numerical gap.

One-pass convention check. For $p = 1$, Alice sends exactly one memory image to Bob. This is the classical one-way streaming reduction. The standard local-output communication convention counts s bits; QRCC counts $s + 1$ because the final leaf output must be part of the communication transcript. As in the cell-probe simulation of Chapter 5, the distinction is only one bit but is recorded rather than hidden.

9.4 A source-owned large-moment lower bound

Theorem 9.4 (Selected AMS lower bound for large frequency moments — source owned). *Fix an integer $k > 5$ and a constant $0 < \gamma < 1/2$. There is a constant $c_{k,\gamma} > 0$ such that every randomized one-pass insertion-only streaming algorithm which, on every stream of length at most n over universe $[n]$, outputs Z_k satisfying*

$$\Pr[|Z_k - F_k| > 0.1F_k] < \gamma \quad (9.21)$$

uses at least

$$c_{k,\gamma} n^{1-5/k} \quad (9.22)$$

bits of memory for all sufficiently large n .

Ownership boundary. This is Theorem 3.2 of Alon–Matias–Szegedy at its selected classical strength. Its proof uses a multiparty disjoint-versus-unique-intersection communication game and a nontrivial distributional lower bound. That multiparty argument is not reconstructed here and is therefore not disguised as a consequence of the elementary two-party simulation in Theorem 9.3. The statement is also not advertised as the modern optimal $p > 2$ frequency-moment bound: later work substantially sharpened the exponent and polylogarithmic factors.

Why the hypotheses matter. The theorem is insertion-only, one-pass, and fixed- $k > 5$. It gives a constant-relative-error lower bound with failure bounded by any fixed $\gamma < 1/2$. None of those qualifiers may be dropped when the theorem is imported later. In particular, the theorem does not itself state a turnstile lower bound, a multipass lower bound, or the optimal $n^{1-2/k}$ exponent.

9.5 Johnson–Lindenstrauss and the linear-sketch interface

Theorem 9.5 (Gaussian Johnson–Lindenstrauss lemma and fixed-family sketches). *Let $0 < \varepsilon < 1$, $0 < \delta < 1$, and let $P = \{x_1, \dots, x_N\} \subset \mathbb{R}^d$ with $N \geq 2$. Let $G \in \mathbb{R}^{k \times d}$ have independent standard normal entries and set $A = k^{-1/2}G$. If*

$$k \geq \frac{8}{\varepsilon^2} \log \frac{N(N-1)}{\delta}, \quad (9.23)$$

then with probability at least $1 - \delta$, simultaneously for every $i < j$,

$$(1 - \varepsilon)\|x_i - x_j\|_2^2 \leq \|A(x_i - x_j)\|_2^2 \leq (1 + \varepsilon)\|x_i - x_j\|_2^2. \quad (9.24)$$

More generally, for any fixed finite nonempty set $Q \subset \mathbb{R}^d$, if

$$k \geq \frac{8}{\varepsilon^2} \log \frac{2|Q|}{\delta}, \quad (9.25)$$

then the same random linear sketch satisfies, with probability at least $1 - \delta$,

$$|\|Az\|_2^2 - \|z\|_2^2| \leq \varepsilon\|z\|_2^2 \quad (z \in Q). \quad (9.26)$$

The probability is over an oblivious choice of A independent of P or Q . No adaptive-query robustness or finite-bit implementation claim is included.

Proof roadmap. For one fixed nonzero vector z , rotational invariance gives $k\|Az\|_2^2/\|z\|_2^2 \sim \chi_k^2$. We derive the needed chi-square tail directly from the Gaussian moment generating function, then union-bound over the fixed family of difference vectors.

Proof. Fix $z \neq 0$. Each row inner product $\langle G_j, z \rangle / \|z\|_2$ is an independent standard normal variable g_j . Hence

$$W := \frac{k\|Az\|_2^2}{\|z\|_2^2} = \sum_{j=1}^k g_j^2. \quad (9.27)$$

For $t < 1/2$, the one-dimensional Gaussian integral gives $\mathbb{E}e^{tg_j^2} = (1 - 2t)^{-1/2}$, and therefore

$$\mathbb{E}e^{tW} = (1 - 2t)^{-k/2}. \quad (9.28)$$

For the upper tail, Markov's inequality with $t = \varepsilon/(2(1 + \varepsilon))$ gives

$$\Pr[W \geq k(1 + \varepsilon)] \leq \exp\left[-\frac{k}{2}(\varepsilon - \log(1 + \varepsilon))\right] \leq e^{-k\varepsilon^2/8}, \quad (9.29)$$

because $\varepsilon - \log(1 + \varepsilon) \geq \varepsilon^2/4$ for $0 < \varepsilon < 1$. Applying the same argument to e^{-tW} and optimizing at $t = \varepsilon/(2(1 - \varepsilon))$ gives

$$\Pr[W \leq k(1 - \varepsilon)] \leq \exp\left[-\frac{k}{2}(-\varepsilon - \log(1 - \varepsilon))\right] \leq e^{-k\varepsilon^2/4}. \quad (9.30)$$

Thus every fixed z satisfies

$$\Pr \left[\left| \|Az\|_2^2 - \|z\|_2^2 \right| > \varepsilon \|z\|_2^2 \right] \leq 2e^{-k\varepsilon^2/8}. \quad (9.31)$$

Zero vectors satisfy the desired inequality deterministically.

For (9.26), apply the union bound, Theorem 1.2, to the at most $|Q|$ bad events. Condition (9.25) makes their total probability at most δ . For (9.24), take the family of the $\binom{N}{2}$ pairwise difference vectors. Since $2\binom{N}{2} = N(N-1)$, (9.23) makes the union-bound failure probability at most δ . $\square$

Linear-sketch interpretation and its boundary. For a fixed family Q , storing Az is a linear compression of each vector z , and (9.26) lets one recover every squared norm in that family to relative error ε . Pairwise distances are obtained by linearity because $A(x_i - x_j) = Ax_i - Ax_j$. This geometric theorem does not charge the bit precision of Gaussian entries or sketch coordinates, so it is not by itself a finite-space streaming theorem. Likewise, once a user chooses a new query vector as a function of the realized matrix A or prior answers, the fixed-family union bound above no longer applies automatically.

Source boundary. Johnson and Lindenstrauss proved the foundational $O(\varepsilon^{-2} \log N)$ Euclidean dimension reduction theorem. PASS69 uses that theorem for provenance but proves the displayed Gaussian normalization and constants independently. The random-matrix distribution and constants in (9.23)–(9.31) are therefore QRCC-owned proof choices, not presented as a verbatim quotation from the 1984 article.

9.6 Executable sketching and communication laboratory

Proposition 9.6 (Sketching and communication laboratory). *The executable file*

`code/volume_12/
sketching_communication_lab.py`

contains deterministic self-tests for four reader interfaces:

1. *a small equality fingerprint checkpoint;*
2. *exhaustive verification of AMS unbiasedness and the variance bound on a four-coordinate frequency vector;*
3. *exact counting of state messages in the two-block p -pass communication simulation;*
4. *a fixed-seed numerical Johnson–Lindenstrauss experiment, explicitly labelled empirical rather than a proof.*

Running the file with `-self-test` exits successfully only if all checkpoints pass.

Chapter checkpoint and exercises

The Chapter 9 ownership map is

$$\begin{aligned} &\text{four-wise signs} + \text{Chebyshev} + \text{Hoeffding} \implies \text{AMS } F_2 \text{ sketch,} \\ &\quad \text{bounded stream state} \implies \text{communication protocol,} \\ &\text{AMS multiparty lower bound} \implies \Omega(n^{1-5/k}) \text{ for fixed } k > 5, \\ &\text{Gaussian concentration} + \text{union bound} \implies \text{JL fixed-family embedding.} \end{aligned} \quad (9.32)$$

The first, second, and fourth arrows are proved locally. The third is the single source-owned deep arrow in the chapter.

1. **Turnstile accounting.** Give a length- m general-turnstile stream with $L(\sigma) \gg m$. Explain why the exact AMS counter space depends on $\log L$ rather than only $\log m$.
2. **Four-wise moment expansion.** Expand $\mathbb{E}(\sum_i h_i f_i)^4$ and classify the index patterns that survive. Recover (9.13) directly.
3. **Why pairwise independence is not enough.** Identify the fourth-moment terms whose cancellation requires independence of four distinct signs.
4. **Median amplification.** Replace the group failure probability $1/4$ by an arbitrary $\alpha < 1/2$ and derive a Hoeffding bound for the number of groups needed to reach failure δ .
5. **One-pass state transfer.** Set $p = 1$ in Theorem 9.3. Draw the exact message transcript under both the local-output and fixed-leaf conventions.
6. **Two passes.** For $p = 2$, list the three s -bit state messages in order and identify which party holds the state at each boundary. Where does the final $+1$ bit enter?
7. **Reduction gap design.** Suppose a stream statistic differs by only an additive 1 between disjoint and uniquely intersecting set inputs but has baseline value $\Theta(n)$. Explain why a constant-factor approximation need not distinguish the cases and why Theorem 9.3 alone is therefore insufficient.
8. **AMS lower-bound scope.** List every hypothesis in Theorem 9.4. Which changes would require a new source theorem rather than a cosmetic restatement?
9. **Gaussian moment generating function.** Evaluate the integral proving $\mathbb{E}e^{tg^2} = (1 - 2t)^{-1/2}$ for $t < 1/2$.
10. **Tail inequalities.** Prove $\varepsilon - \log(1 + \varepsilon) \geq \varepsilon^2/4$ and $-\varepsilon - \log(1 - \varepsilon) \geq \varepsilon^2/2$ for $0 < \varepsilon < 1$ by differentiation.
11. **JL pair counting.** Starting from (9.31), derive exactly why the factor inside the logarithm in (9.23) is $N(N - 1)/\delta$ rather than merely N/δ .
12. **Adaptive-query warning.** After seeing a rank- k matrix A with $k < d$, choose a nonzero vector z depending on A for which the norm estimate fails maximally. Explain why this does not contradict Theorem 9.5.

Transition to Chapter 10. Chapter 9 has supplied the stream-state simulation and the linear-sketch viewpoint needed for the final data-structure interface. Chapter 10 will first freeze the predecessor/cell-probe model used by its source-owned lower bound, then draw the reduction map among communication, query, streaming, and data-structure complexity without reversing any arrow or suppressing its quantitative loss.

Chapter 10

Data Structures, Predecessor Search, and the Reduction Map

The preceding chapters have moved among several models of limited access to information: decision trees inspect input coordinates, communication protocols exchange bits, streaming algorithms retain bounded state, and static data structures probe a bounded number of memory cells. These models are related, but the relationships are not equivalences. A correct reduction must say which object is transformed into which, which player owns which information, and what quantitative loss is paid.

This final chapter closes Volume XII in two steps. First, it freezes a concrete static-predecessor cell-probe model and records one sharp source-owned lower bound of Pătraşcu–Thorup at exactly the near-linear-space/randomized-query strength needed here. Second, it assembles the reduction map proved across the volume, with every arrow labelled by its direction and loss. The map is a guide to lower-bound transport, not a license to reverse reductions.

How to read this chapter. Definition 10.1 comes first because phrases such as “word size”, “near-linear space”, and “randomized cell-probe query” are not interchangeable conventions. Theorem 10.2 is then source-owned: its cell-probe-elimination proof is not reconstructed locally. Figure 10.1 finally places that theorem beside the local simulation theorems of Chapters 5, 6, and 9. The important negative lesson is visible in the figure as well: the predecessor lower bound lies beyond what the elementary communication simulation alone can prove.

Learning goals. After this chapter, a reader should be able to define static predecessor search and its sentinel output; convert between universe size and key length; distinguish words, cells, and total bit space; state the near-linear randomized predecessor lower bound with its parameter regime intact; explain why the theorem is not merely a communication lower bound; read every arrow in the reduction map in the correct direction; and derive a streaming or cell-probe lower bound from a communication lower bound only after checking the quantitative loss.

10.1 Static predecessor and the cell-probe convention

Definition 10.1 (Static predecessor and near-linear randomized cell probes). Let $U \geq 2$ be an integer and write

$$[U]_0 := \{0, 1, \dots, U - 1\}, \quad \ell := \lceil \log_2 U \rceil.$$

A *static predecessor instance* is a set $Y \subseteq [U]_0$ of $n \geq 1$ distinct keys. For a query $x \in [U]_0$, define

$$\text{pred}_Y(x) := \max(\{y \in Y : y \leq x\} \cup \{\perp\}), \quad (10.1)$$

where the sentinel $\perp$ is ordered below every element of $[U]_0$. Thus the answer is $\perp$ exactly when $x < \min Y$.

A static *w-bit cell-probe representation* stores the preprocessed set Y in an array of M cells, each containing exactly w bits. Its total stored bit space is

$$S_{\text{bit}} := Mw. \quad (10.2)$$

A query algorithm may perform arbitrary local computation for free and may adaptively read cells; its *query time* is the maximum number t of cell probes. Unless stated otherwise in this chapter, $w = \ell$ so one key fits in one word.

A *randomized t-probe query algorithm with worst-case error at most η* has an input-independent random tape, makes at most t probes for every tape and every pair (Y, x) , and returns the correct value $\text{pred}_Y(x)$ with probability at least $1 - \eta$ for every (Y, x) . The representation of Y is fixed before the query; the lower bound below concerns randomness in the query algorithm.

The universe is *polynomial* when $U = n^{\Theta(1)}$, equivalently $\ell = \Theta(\log n)$. Space is *near-linear* in the Pătraşcu–Thorup normalization when

$$S_{\text{bit}} = n \log^{O(1)} n = \tilde{O}(n) \quad \text{bits}. \quad (10.3)$$

Here $\tilde{O}(n)$ hides a fixed power of $\log n$ only. Because storing the n keys already requires $\Theta(n\ell)$ bits in the polynomial-universe regime, (10.3) is near-linear relative to input length as well as near-linear in the number of keys up to polylogarithmic factors.

Relation to the Chapter 5 simulation. The generic static cell-probe model of Theorem 5.3 has S cells and w bits per cell. In the present notation $S = M$ and $S_{\text{bit}} = Mw$. The predecessor theorem below uses total bit space because that is the normalization in its source. Converting between the two notations is therefore bookkeeping, not a new reduction.

10.2 A sharp source-owned predecessor lower bound

Theorem 10.2 (Pătraşcu–Thorup: randomization does not improve near-linear predecessor). *Fix constants $0 < \eta < 1/2$ and constants implicit in the relations*

$$\ell = w = \Theta(\log n), \quad S_{\text{bit}} = n \log^{O(1)} n. \quad (10.4)$$

For static predecessor search in the model of Definition 10.1, every fixed-probe-budget randomized query algorithm with worst-case error at most η requires

$$t = \Omega(\log \ell) = \Omega(\log \log n). \quad (10.5)$$

The constants in the $\Omega(\cdot)$ notation may depend on the fixed polynomial-universe, near-linear-space, and error constants. The bound matches the classical van Emde Boas $O(\log \ell)$ search time in this regime up to constant factors.

This is a source-owned deep input: QRCC does not claim a local proof of the cell-probe elimination argument.

Source boundary and proof architecture. The selected source is M. Pătraşcu and M. Thorup, *Randomization Does Not Help Searching Predecessors*, SODA 2007, pp. 555–564. Their randomized theorem extends the tight STOC 2006 deterministic time–space trade-off to fixed-running-time query algorithms with two-sided error bounded away from $1/2$. In the polynomial-universe regime $w = \ell = \Theta(\log n)$ and near-linear space $\tilde{O}(n)$ bits, the source explicitly identifies the van Emde Boas $\Theta(\log \ell) = \Theta(\log \log n)$ behavior as optimal.

The technical mechanism is a randomized cell-probe elimination lemma. Roughly, a hypothetical fast query scheme is converted recursively into a smaller predecessor instance while controlling both the fraction of rejected queries and the accumulated error. Iterating the elimination too many times would leave a zero-probe structure for a nontrivial predecessor instance, giving the contradiction. Those elimination and error-control lemmas are the hard content retained at the source boundary.

Why the communication simulation is not the proof. Theorem 5.3 converts a t -probe structure into an asymmetric protocol, but that route cannot distinguish polynomial from near-linear space sharply enough: changing the number of cells by a polynomial factor changes an address by only a constant factor. The original Pătraşcu–Thorup work emphasizes exactly this communication barrier. Consequently Theorem 10.2 depends on the cell-probe model definition above, not on the corruption theorem or the lifting theorem. Those communication tools remain useful for many other data-structure lower bounds, but they do not own this one.

Worked parameter check. Suppose $U = n^4$, so $\ell = \lceil 4 \log_2 n \rceil$, and suppose the representation uses $S_{\text{bit}} \leq n \log^3 n$ bits. For any fixed error target such as $\eta = 1/3$, Theorem 10.2 gives

$$t = \Omega(\log \ell) = \Omega(\log \log n).$$

This is an asymptotic statement: plugging a particular finite value of n into the right-hand side does not reveal the hidden constant. Its content is that no family of such randomized structures can have $o(\log \log n)$ worst-case probe complexity throughout this regime.

10.3 The reduction map

How to read the arrows. The top arrow is the lifting interface of Theorem 6.2: after fixing a gadget and composed search problem, query hardness becomes communication hardness with theorem-specific block-size and error losses. The lower-left arrow is Theorem 9.3: a p -pass s -bit streaming algorithm on a two-block stream yields a protocol of cost at most $(2p - 1)s + 1$ under the fixed-leaf convention. The lower-right arrow is Theorem 5.3: a t -probe data structure yields an asymmetric protocol with address and word costs kept separate. The information-complexity arrows combine Theorem 4.2 with the source-owned Braverman–Rao amortization theorem, Theorem 4.3.

A lower-bound transport example. Suppose a two-block streaming encoding of a communication problem is such that Alice’s input forms the first block and Bob’s input the second, and suppose every protocol for the resulting problem requires at least L bits. If a p -pass streaming algorithm used s bits, then Theorem 9.3 would give a protocol of cost at most $(2p - 1)s + 1$; hence

$$s \geq \frac{L - 1}{2p - 1}. \quad (10.6)$$

The reduction proves a space lower bound only after the stream encoding itself has been checked to preserve the promise and approximation gap. This is why Chapter 9 treated the state-transfer simulation separately from frequency-moment lower bounds.

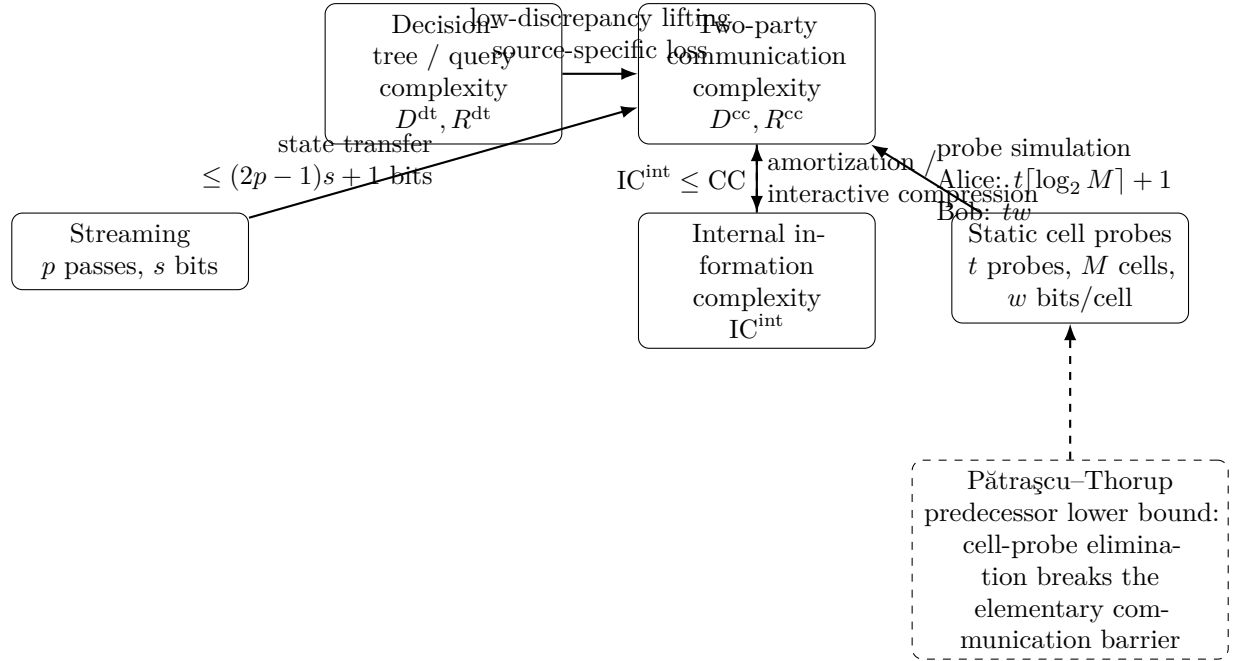


Figure 10.1: Model-aware reduction map for Volume XII. Solid arrows are local simulations or inequalities proved in the volume; dashed arrows mark source-owned deep interfaces. To transport a lower bound, read a simulation arrow contrapositively and preserve its quantitative label. No arrow may be reversed merely because the two models are adjacent in the diagram.

A data-structure transport example. If a Boolean static data-structure problem has an asymmetric communication lower bound saying that no protocol with Alice sending at most A bits and Bob sending at most B bits can solve it, then a t -probe (M, w) structure would contradict that bound whenever

$$t \lceil \log_2 M \rceil + 1 \leq A, \quad tw \leq B. \quad (10.7)$$

This can yield strong lower bounds when the communication theorem is sensitive to the two message budgets separately. It does not reproduce Theorem 10.2 in the near-linear predecessor regime; the dashed predecessor box in Figure 10.1 records that boundary explicitly.

Volume XII checkpoint and exercises

The final dependency picture is not a single chain but a collection of model-preserving bridges:

$$\begin{aligned}
 &\text{rectangles/discrepancy/corruption} \implies \text{communication lower bounds,} \\
 &\text{information cost + compression} \implies \text{direct-sum/amortized statements,} \\
 &\quad \text{query hardness + lifting} \implies \text{communication hardness,} \\
 &\quad \text{stream state or cell probes} \implies \text{communication protocols,} \\
 &\quad \text{cell-probe elimination} \implies \text{near-linear predecessor lower bounds.}
 \end{aligned} \quad (10.8)$$

The first four lines share communication complexity as a hub; the last deliberately does not reduce to that hub.

1. **Sentinel boundary.** For $Y = \{4, 9, 17\} \subseteq [32]_0$, compute $\text{pred}_Y(x)$ for $x \in \{0, 4, 8, 9, 31\}$ and identify the only query returning $\perp$.
2. **Bits versus cells.** If $w = 64$ and the data structure uses $M = n \log^2 n$ cells, write its total bit space and determine whether it is near-linear in the sense of Definition 10.1.
3. **Polynomial universe.** Show that $U = n^c$ for a fixed $c > 0$ gives $\ell = \Theta(\log n)$. Conversely, if $\ell \leq C \log_2 n$ for a fixed C , show that $U \leq 2n^C$.
4. **Predecessor source boundary.** List every hypothesis in Theorem 10.2. Explain why replacing near-linear space by arbitrary polynomial space would change the theorem's asymptotic conclusion.
5. **Why randomization is substantive.** Explain why a deterministic lower bound does not automatically imply the randomized lower bound in Theorem 10.2. Which part of the source theorem controls this issue?
6. **Streaming contrapositive.** Derive (10.6) carefully from Theorem 9.3. What changes under the convention that Bob may output locally?
7. **Cell-probe asymmetry.** For $M = 2^{20}$ and $w = 64$, compute the Alice/Bob communication budgets supplied by a t -probe scheme. Why can an asymmetric communication lower bound be stronger than a lower bound on total communication alone?
8. **Do not reverse the arrow.** Give a logical example showing why “algorithm in model A implies protocol in model B” does not imply “protocol in model B implies algorithm in model A.” Apply the lesson to one arrow of Figure 10.1.
9. **Lifting audit.** Using Theorem 6.2, list the gadget-size, discrepancy, and randomized-error conditions that must be preserved before converting a decision-tree lower bound into a communication lower bound.
10. **Information versus communication.** Explain why $\text{IC}^{\text{int}} \leq \text{CC}$ is a one-copy inequality, whereas equality in the reverse direction appears only after amortization in Theorem 4.3.
11. **Model-choice exercise.** For each of the following tasks, name the most direct lower-bound interface developed in Volume XII: (a) a two-party Boolean function with a Hadamard sign matrix; (b) an oracle property close to a linearity code; (c) a streaming frequency statistic; (d) a static predecessor data structure. Justify each choice in one sentence.
12. **Synthesis.** Starting from a hard decision-tree search problem, describe a hypothetical route to a communication lower bound by lifting and then to a streaming lower bound by a stream encoding. Mark every point where a promise, error parameter, block size, or message/space factor must be checked.

Student-facing completion of Volume XII

A reader who has completed this volume should now be able to move between four kinds of limited access without conflating them. Chapters 1–3 build deterministic and randomized communication lower bounds from rectangles, rank, discrepancy, corruption, and the Razborov disjointness argument. Chapters 4–6 add information complexity, direct-sum/direct-product interfaces, cell-probe simulation, and lifting. Chapters 7–8 move to query complexity and property testing. Chapter 9 adds streaming

and linear sketches, and this chapter closes the circle with static data structures and the model-aware reduction map.

The recurring discipline is as important as any one theorem: before transferring a bound, identify the input owner, output convention, error notion, resource unit, and direction of simulation. The same numerical symbol can mean bits of communication, probes, oracle queries, retained stream state, or information revealed; Volume XII treats those as related but distinct resources.

Volume-completion checklist. The formal theorem layer is closed except for items deliberately labelled open problems or verified frontiers. Every technical term used by the Chapter 10 formal statements has been defined before use. Each chapter has a reader roadmap and exercises, and the final reduction map supplies the cross-chapter synthesis. Substantial Volume XIII work should begin only from this certified Volume XII state.

Internalization and Reconstruction Guide: Volume XII

Purpose. Use communication as a common currency for distributed computation, query complexity, information, property testing, streaming, sketching, and cell-probe lower bounds; every reduction must preserve its model conventions.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-005: Hoeffding inequality.
2. QRCC-I-THM-026: Mutual-information chain rule.
3. QRCC-III-ALG-004: Euclidean polynomial arithmetic.
4. QRCC-I-THM-009: Conditioning reduces entropy.
5. QRCC-I-THM-010: Gibbs inequality and nonnegativity of relative entropy.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XII-DEF-001 (Communication protocols and complexity measures); QRCC-XII-DEF-011 (Internal and external information cost); QRCC-XII-DEF-016 (Lifting gadget and composed search problem); QRCC-XII-DEF-017 (Decision-tree and query measures); QRCC-XII-DEF-021 (Property-testing and dense-graph models).

Chapter retrieval map

Ch. 1: Deterministic Communication: Rectangles, Fooling Sets, and Rank: enter through QRCC-XII-DEF-001 (Communication protocols and complexity measures); exit by being able to state QRCC-XII-THM-002 (Protocol rectangles and monochromatic partitions) with its ownership boundary.

- Ch. 2: Public Coins, Private Coins, and Newman’s Theorem:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XII-THM-007 (Newman public-versus-private randomness theorem) with its ownership boundary.
- Ch. 3: Discrepancy: Rectangle Bias and the Inner-Product Lower Bound:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XII-THM-005 (Discrepancy lower bound for randomized communication) with its ownership boundary.
- Ch. 4: Information Cost, Transcript Geometry, and Amortized Communication:** enter through QRCC-XII-DEF-011 (Internal and external information cost); exit by being able to state QRCC-XII-BB-013 (Amortized communication equals information complexity) with its ownership boundary.
- Ch. 5: Direct Sums, Strong Direct Products, and the Cell-Probe Interface:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XII-THM-014 (Exact direct sum for internal information complexity) with its ownership boundary.
- Ch. 6: Query-to-Communication Lifting with Low-Discrepancy Gadgets:** enter through QRCC-XII-DEF-016 (Lifting gadget and composed search problem); exit by being able to state QRCC-XII-BB-015 (Low-discrepancy query-to-communication lifting theorem) with its ownership boundary.
- Ch. 7: Polynomial, Approximate-Degree, and Sensitivity Methods for Query Complexity:** enter through QRCC-XII-DEF-017 (Decision-tree and query measures); exit by being able to state QRCC-XII-THM-018 (Polynomial method for randomized query lower bounds) with its ownership boundary.
- Ch. 8: Property Testing, Removal, and Adaptivity:** enter through QRCC-XII-DEF-021 (Property-testing and dense-graph models); exit by being able to state QRCC-XII-THM-021 (BLR linearity test) with its ownership boundary.
- Ch. 9: Streaming, Frequency Moments, and Linear Sketches:** enter through QRCC-XII-DEF-024 (Streaming, frequency-vector, pass, and linear-sketch models); exit by being able to state QRCC-XII-ALG-024 (AMS second-moment sketch) with its ownership boundary.
- Ch. 10: Data Structures, Predecessor Search, and the Reduction Map:** enter through QRCC-XII-DEF-028 (Static predecessor and near-linear randomized cell-probe model); exit by being able to state QRCC-XII-BB-028 (Cell-probe predecessor/data-structure lower bound) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XII-THM-002 — Protocol rectangles and monochromatic partitions. Dependencies: QRCC-XII-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

2. QRCC-XII-THM-005 — Discrepancy lower bound for randomized communication. Dependencies: QRCC-XII-DEF-001, QRCC-XII-THM-002. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-XII-THM-014 — Exact direct sum for internal information complexity. Dependencies: QRCC-XII-THM-012, QRCC-I-THM-026. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-XII-THM-018 — Polynomial method for randomized query lower bounds. Dependencies: QRCC-XII-DEF-017. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-XII-ALG-024 — AMS second-moment sketch. Dependencies: QRCC-XII-DEF-024, QRCC-VIII-ALG-006, QRCC-I-THM-003, QRCC-I-THM-005. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-XII-CONJ-016 — Log-rank conjecture. Dependencies: QRCC-XII-THM-004. *open frontier: separate the conjecture from the strongest proved substitute and never route a later theorem through the conjecture.*

Common failure modes

- dropping the error convention when converting randomized protocols to deterministic rectangles;
- mixing public-coin, private-coin, one-way, and two-way bounds;
- using a query-to-communication simulation without accounting for address and answer lengths;
- ignoring word size, adaptivity, preprocessing, or update/query conventions in data-structure reductions;

Boundary and degenerate cases

- constant functions and one-bit domains;
- zero communication/query cases;
- partial functions and promises;
- cell size/address space extremes in cell-probe simulations;

Retrieval and reconstruction drill

1. State QRCC-XII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XII-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XII-THM-005, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?

4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XII-CONJ-016, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XIII studies verification rather than direct computation, culminating in PCP and algebraic/semialgebraic proof systems.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XIII

Interactive Proofs, PCP, and Proof Complexity

Chapter 1

Interactive Proofs: Private Coins, Public Coins, and Proximity

An interactive proof replaces a static certificate by a conversation. The prover may be computationally unbounded, while the verifier is randomized polynomial time. The point is not that the prover is trusted: soundness quantifies over every prover strategy. This distinction is what makes interactive proofs useful later for arithmetization, sum-check, $\text{IP} = \text{PSPACE}$, multiple provers, and PCP theory.

This chapter fixes the protocol conventions before using them. It then proves the classical Goldwasser–Sipser transformation from private verifier coins to public coins. The proof is worth including because its hashing argument is a prototype of a recurring complexity-theoretic move: replace an inaccessible probability by an interactive certificate that a finite set is large. The chapter closes with a source-owned interactive-proof-of-proximity frontier; no IPP theorem is used as an internal lemma yet.

How to read this chapter. Definition 1.1 distinguishes messages, exchange rounds, verifier coins, transcripts, and the quantifiers in completeness and soundness. Theorem 1.4 then has three layers: error amplification without adding interaction stages, an approximate lower-bound hashing lemma, and a public-coin simulation. The final frontier paragraph defines what “proximity” means before recording a theorem with sublinear query complexity.

Learning goals. After this chapter, a reader should be able to write the acceptance probability of an interactive protocol with all prover/verifier quantifiers visible; distinguish private-coin and public-coin protocols; explain why public coins do not weaken polynomial-round interactive proofs; prove the finite-set hashing lemma used in that conversion; and state an interactive proof of proximity with its distance, query, communication, round, and soundness parameters explicit.

1.1 Interactive proof conventions

Definition 1.1 (Interactive proof systems, coins, transcripts, and round conventions). Fix a language $L \subseteq \{0, 1\}^*$. The following conventions are used throughout Volume XIII.

1. An *interactive protocol* $\Pi = (P, V)$ consists of an interactive prover P and an interactive verifier V . The verifier is a probabilistic polynomial-time algorithm. The prover has no computational bound unless the text explicitly replaces a proof system by an argument system. Both parties

receive the common input x ; each next message may depend on that party's local state and on the complete preceding transcript.

2. A *move* is one transmitted message. An *exchange round* is one verifier message followed by one prover response. Whenever a cited source uses a different convention, the theorem statement names the convention rather than silently changing a factor of two. Polynomially many dummy bits or empty messages may be appended to make message lengths and the number of moves fixed on each input length.
3. A *private-coin verifier* has a random tape R_V hidden from the prover except for information revealed through verifier messages. In a *public-coin* or *Arthur–Merlin* protocol, each randomized verifier message is the fresh random string drawn at that stage, possibly followed by deterministic computation from the input and transcript. Thus all verifier randomness used to choose a challenge is visible to the prover when the challenge is sent. Public-coin protocols are a special case of private-coin protocols.
4. For fixed strategies P, V , input x , and verifier random tape r , write $\text{tr}_{P,V}(x; r)$ for the resulting *transcript*, namely the ordered list of messages together with the final verifier decision. With R_V uniform, the induced random variable $\text{Tr}_{P,V}(x) := \text{tr}_{P,V}(x; R_V)$ is the *transcript distribution*. If the prover itself is randomized, its coins are included in the probability space; because an unbounded prover can fix its best random tape, randomization never improves the supremum in information-theoretic soundness.
5. The protocol is a (c, s) -*interactive proof* for L if

$$\begin{aligned} x \in L &\implies \exists P_{\text{hon}} \quad \Pr[V^{P_{\text{hon}}}(x) \text{ accepts}] \geq c(|x|), \\ x \notin L &\implies \forall P^* \quad \Pr[V^{P^*}(x) \text{ accepts}] \leq s(|x|). \end{aligned}$$

The probability is over the verifier's coins and any explicitly allowed prover coins. Unless another pair is displayed, *bounded error* means $c \geq 2/3$ and $s \leq 1/3$. *Perfect completeness* means $c = 1$. Soundness quantified over every, even unbounded, prover is information-theoretic (and is sometimes informally called statistical soundness); *computational soundness* quantifies only over a stated efficient prover class and defines an argument rather than an interactive proof.

6. The phrase *honest-verifier* modifies properties of the verifier's view or transcript: it means the property is required only when the verifier follows its prescribed algorithm. A *malicious-verifier* version quantifies over a specified class of deviating verifiers. This modifier does not weaken the soundness quantifier above. It becomes important later for zero-knowledge-type properties, and is defined here so the term is not used before explanation.
7. Write $\text{IP}[q]$ for languages having bounded-error private-coin interactive proofs with at most $q(n)$ moves in the stated move convention, polynomial total communication, and a polynomial-time verifier. Write $\text{AM}[q]$ for the corresponding public-coin class. The unsubscripted class IP permits polynomially many moves.

A two-message example. For graph nonisomorphism, the verifier may privately choose one of two input graphs, randomly relabel it, and ask the prover which graph was chosen. On a nonisomorphic pair an unbounded prover can answer perfectly; on an isomorphic pair the hidden bit is information-theoretically concealed. This example is only motivational here. The transformation below explains why hidden verifier coins are not necessary for the language-recognition power of polynomial-round interactive proofs.

1.2 Two finite-probability lemmas

The first lemma is an error-reduction fact. Its only nontrivial point is that a cheating prover in parallel copies may correlate its answers across copies. The proof therefore does not assume the acceptance indicators are independent.

Lemma 1.2 (Parallel amplification without additional moves). *Let Π be a $(2/3, 1/3)$ -interactive proof using q moves. For every integer $t \geq 1$, run t independent verifier copies in parallel, batching all messages belonging to the same move, and accept by majority. The resulting protocol still uses q moves, multiplies communication and verifier randomness by t , and has completeness and soundness errors at most $\exp(-\Omega(t))$. Consequently, for every polynomial $p(n)$, choosing $t = O(p(n))$ makes both errors at most $2^{-p(n)}$ with polynomial overhead and no increase in move count.*

Proof. For completeness, use the product honest-prover strategy. The verifier coins in the t copies are independent, so the t acceptance indicators are independent Bernoulli variables of mean at least $2/3$; the Chernoff bound of Volume I gives an exponentially small lower-tail probability.

For soundness we first record a tensorization fact. If $S \subseteq [t]$, then for every joint cheating strategy in the parallel protocol,

$$\Pr[\text{every copy in } S \text{ accepts}] \leq (1/3)^{|S|}. \quad (1.1)$$

Indeed, view a finite interactive protocol as a game tree whose chance nodes are verifier coin draws and whose maximization nodes are prover messages. For the event that all copies in S accept, backward induction on the remaining moves shows that the value of the product game is the product of the one-copy values: expectation at independent chance nodes tensorizes, and at a prover node maximizing over a tuple of answers cannot exceed the product of the separate maxima because the continuation payoff is already a product. Copies outside S can be fixed as auxiliary side information and do not change the one-copy soundness bound. This proves (1.1).

Let X_i be the acceptance indicator of copy i , and put $X = \sum_i X_i$. For $\lambda > 0$, expand the product

$$e^{\lambda X} = \prod_{i=1}^t (1 + (e^\lambda - 1)X_i).$$

Using (1.1) term by term gives

$$\mathbb{E} e^{\lambda X} \leq \sum_{S \subseteq [t]} (e^\lambda - 1)^{|S|} 3^{-|S|} = \left(1 + \frac{e^\lambda - 1}{3}\right)^t.$$

This is exactly the moment-generating-function upper bound for a binomial $\text{Bin}(t, 1/3)$ random variable. The same Chernoff calculation therefore bounds $\Pr[X \geq t/2]$ by $\exp(-\Omega(t))$. Taking a sufficiently large constant multiple of $p(n)$ completes the amplification claim. $\square$

The second lemma is the approximate set-size test used by Goldwasser and Sipser. All vector spaces in the lemma are over $\mathbb{F}_2$.

Lemma 1.3 (Approximate lower-bound hashing). *Let $C \subseteq \{0, 1\}^k$ be nonempty and let integers $b, k, \ell > 0$ satisfy $\ell > \max\{b, 8\}$. Independently choose ℓ uniformly random linear maps $h_i : \mathbb{F}_2^k \rightarrow \mathbb{F}_2^b$, set $H(C) = \bigcup_{i=1}^\ell h_i(C)$, and independently choose a multiset $Z = (z_1, \dots, z_{\ell^2})$ of uniform strings from $\mathbb{F}_2^b$. Then:*

1. if $b = 2 + \lceil \log_2 |C| \rceil$, then

$$\Pr[|H(C)| \geq |C|/\ell] \geq 1 - 2^{-\ell}, \quad \Pr[H(C) \cap Z \neq \emptyset] \geq 1 - 2^{-\ell/8};$$

2. always $|H(C)| \leq \ell|C|$; and if $|C| \leq 2^b/d$ for $d > 0$, then

$$\Pr[H(C) \cap Z \neq \emptyset] \leq \ell^3/d.$$

Proof. For distinct $x, y \in \mathbb{F}_2^k$, a uniformly random linear map satisfies $\Pr[h_i(x) = h_i(y)] = 2^{-b}$, because $h_i(x - y) = 0$ imposes b independent linear conditions on the random matrix describing h_i . Under the first hypothesis, $2^b \geq 4|C|$. Fix $x \in C$. By a union bound, the probability that $h_i(x)$ collides with some $h_i(y)$, $y \neq x$, is at most $|C|2^{-b} \leq 1/4$. Independence of the h_i 's makes the probability that this happens for every i at most $4^{-\ell}$. A second union bound over $x \in C$ is at most $|C|4^{-\ell} \leq 2^{-\ell}$, using $|C| < 2^{b-1} < 2^\ell$.

Outside that failure event, every $x \in C$ is isolated by at least one h_i . Hence $\sum_i |h_i(C)| \geq |C|$, so some $h_i(C)$, and therefore $H(C)$, has size at least $|C|/\ell$. Also $b = 2 + \lceil \log_2 |C| \rceil$ implies $2^b < 8|C|$, so on this event a uniform $z \in \mathbb{F}_2^b$ lies in $H(C)$ with probability at least $1/(8\ell)$. Therefore

$$\Pr[H(C) \cap Z = \emptyset] \leq 2^{-\ell} + \left(1 - \frac{1}{8\ell}\right)^{\ell^2} \leq 2^{-\ell} + e^{-\ell/8} \leq 2^{-\ell/8},$$

where the last elementary inequality holds for $\ell > 8$.

The deterministic bound $|H(C)| \leq \ell|C|$ is immediate. If $|C| \leq 2^b/d$, then $|H(C)|/2^b \leq \ell/d$. Each of the ℓ^2 independent points in Z therefore hits $H(C)$ with probability at most ℓ/d , and a union bound gives ℓ^3/d . $\square$

1.3 Private coins versus public coins

Theorem 1.4 (Goldwasser–Sipser public-coin transformation). *Goldwasser and Sipser state, in their original message-count convention, that for every polynomial bound Q ,*

$$\text{IP}[Q] \subseteq \text{AM}[Q + 2]. \quad (1.2)$$

Since every public-coin protocol is already a private-coin protocol, (1.2) implies equality of the unrestricted polynomial-round language classes, $\text{IP} = \text{AM}$. The additive constant in a notation such as $\text{IP}[Q]$ is convention-sensitive; Definition 1.1 therefore keeps our single-message and exchange-round counts explicit rather than silently identifying them with the 1986 notation.

More concretely, a bounded-error private-coin protocol with polynomially many moves, polynomial-length messages, and polynomially many verifier random bits can be transformed into a public-coin protocol with two additional moves in that normalization, polynomial communication and randomness overhead, completeness at least $2/3$, and soundness at most $1/3$. The theorem does not claim that the computational complexity of an honest prover is preserved.

Proof. Fix an input w of length n . The zero-exchange case is trivial, so assume $g \geq 1$. Pad the protocol, without changing its language, so that it has exactly g verifier–prover exchange pairs, every message has length m , and the verifier uses exactly ℓ private random bits. By Lemma 1.2, while keeping the same interaction stages we may assume the completeness and soundness errors are at most

$$e \leq \ell^{-12g^2}, \quad (1.3)$$

after increasing m and ℓ polynomially. Pad further so that $\ell > \max\{g, m, 80\}$.

Write the verifier's i -th message as $x_i \in \{0, 1\}^m$, the prover response as $y_i \in \{0, 1\}^m$, and the transcript prefix after i pairs as $s_i = x_1 \# y_1 \# \cdots \# x_i \# y_i$. The public-coin verifier Arthur and prover Merlin use the following protocol.

Arthur's protocol. Merlin first sends an integer b_1 . For each $1 \leq i \leq g$, after Arthur has received b_i and s_{i-1} , Arthur samples the hash experiment of Lemma 1.3 with domain $\{0, 1\}^m$, output length b_i , and parameter ℓ , and sends all matrices defining H_i and all points in Z_i . Merlin responds with x_i, y_i and, unless $i = g$, the next integer b_{i+1} . Arthur rejects unless $x_i \in H_i^{-1}(Z_i)$.

After the g -th response, Merlin sends an integer b_{g+1} . Arthur performs one final hash experiment on the private-randomness space $\{0, 1\}^\ell$, now with output length b_{g+1} . Merlin responds with a random tape $r \in \{0, 1\}^\ell$, and Arthur rejects unless $r \in H_{g+1}^{-1}(Z_{g+1})$. Finally Arthur deterministically checks that

$$V(w, r, s_{i-1}) = x_i \quad (1 \leq i \leq g), \quad V(w, r, s_g) = \text{accept}, \quad (1.4)$$

and that

$$\sum_{i=1}^{g+1} b_i \geq \ell - g \log_2 \ell. \quad (1.5)$$

Every randomized Arthur message consists solely of freshly sampled public randomness, so this is an Arthur–Merlin protocol. Hash matrices and target points have polynomial total length because g, m, ℓ are polynomially bounded and every $b_i \leq \max\{m, \ell\} + 2$ may be range-checked.

Completeness. Assume $w \in L$, and fix an honest private-coin prover P whose amplified verifier accepts with probability at least $1 - e$, hence at least $2/3$. Let

$$a_0 := \{r \in \{0, 1\}^\ell : (V * P)(w, r) \text{ accepts}\},$$

so $|a_0| \geq (2/3)2^\ell$. Suppose s_{i-1} has been fixed. For each possible verifier message $x \in \{0, 1\}^m$, let

$$a_x := \{r : (V * P)(w, r) \text{ accepts via } s_{i-1} \# x\}.$$

The nonempty a_x 's partition the current set a_{i-1} . Group these sets into at most ℓ dyadic classes according to the interval $(2^{d-1}, 2^d]$ containing $|a_x|$. Let $\mathcal{Y}_i$ be a class whose union is largest, put $C_i := \{x : a_x \in \mathcal{Y}_i\}$, and send

$$b_i = 2 + \lceil \log_2 |C_i| \rceil. \quad (1.6)$$

By Lemma 1.3, except with probability at most $2^{-\ell/8}$, the hash challenge contains some $x_i \in C_i$; Merlin returns such an x_i , the honest response $y_i = P(s_{i-1} \# x_i)$, and we set $a_i = a_{x_i}$.

Because $|\bigcup \mathcal{Y}_i| \geq |a_{i-1}|/\ell$, all members of $\mathcal{Y}_i$ differ in size by at most a factor two, and $2|C_i| \leq 2^{b_i-1}$,

$$|a_i| \geq \frac{|a_{i-1}|}{\ell 2^{b_i}}. \quad (1.7)$$

At the end Merlin sends $b_{g+1} = 2 + \lceil \log_2 |a_g| \rceil$, and the final application of the hashing lemma finds an $r \in a_g \cap H_{g+1}^{-1}(Z_{g+1})$ except with probability at most $2^{-\ell/8}$. Iterating (1.7) and using $\log_2 |a_0| > \ell - 1$ gives

$$\log_2 |a_g| > (\ell - 1) - g \log_2 \ell - \sum_{i=1}^g b_i.$$

Since $b_{g+1} > 1 + \log_2 |a_g|$, condition (1.5) follows. A union bound over the $g + 1$ hash experiments gives failure probability at most $(g + 1)2^{-\ell/8} < 1/3$. Thus completeness is at least $2/3$.

Soundness. Assume $w \notin L$, and fix an arbitrary Merlin strategy. For a transcript prefix s , define

$$\alpha(s) := \max_{P^*} \Pr_r[(V * P^*)(w, r) \text{ accepts via } s], \quad (1.8)$$

where the maximum is over all continuations of the private-coin prover after s . In particular, $\alpha(s_0) \leq e$ for the empty prefix s_0 . For each possible next verifier message x , choose y_x attaining the best continuation. The verifier random tapes partition by that next message, so

$$\alpha(s_i) = \sum_x \alpha(s_i \# x \# y_x). \quad (1.9)$$

Consequently, for every $c > 0$, the heavy-message set

$$X_c := \{x : \alpha(s_i \# x \# y_x) \geq \alpha(s_i)/c\}$$

has cardinality at most c .

During public round $i + 1$, let the exceptional event E_{i+1} be that Merlin's chosen extension satisfies

$$\alpha(s_{i+1}) \geq \frac{\alpha(s_i)}{2^{b_{i+1}}/d}. \quad (1.10)$$

Taking $c = 2^{b_{i+1}}/d$, the set of messages capable of causing (1.10) has size at most $2^{b_{i+1}}/d$. Part 2 of Lemma 1.3 therefore gives $\Pr[E_{i+1}] \leq \ell^3/d$, regardless of Merlin's strategy after seeing the public hash challenge.

For the final hash round, let E_{g+1} be the event that Arthur receives an accepting private random tape even though

$$2^\ell \alpha(s_g) \leq 2^{b_{g+1}}/d. \quad (1.11)$$

The set of accepting random tapes has at most $2^{b_{g+1}}/d$ elements under (1.11), so the same hashing lemma again yields $\Pr[E_{g+1}] \leq \ell^3/d$. Choose

$$d := 3(g+1)\ell^3. \quad (1.12)$$

A union bound makes the probability of any exceptional event at most $1/3$.

Suppose no exceptional event occurs and Arthur nevertheless accepts. From the negations of (1.10), followed by acceptance in the final round and the negation of (1.11),

$$2^\ell \alpha(s_0) > \prod_{i=1}^{g+1} \frac{2^{b_i}}{d}.$$

Taking logarithms gives

$$\ell + \log_2 \alpha(s_0) > \sum_{i=1}^{g+1} b_i - (g+1) \log_2 d. \quad (1.13)$$

Because $\ell > \max\{g, 80\}$, equation (1.12) implies $(g+1) \log_2 d \leq 10g \log_2 \ell$. Combining this with $\alpha(s_0) \leq e \leq \ell^{-12g^2}$ yields from (1.13)

$$\sum_{i=1}^{g+1} b_i < \ell - 2g \log_2 \ell < \ell - g \log_2 \ell,$$

contradicting Arthur's acceptance condition (1.5). Therefore Arthur can accept only when an exceptional event occurs, and soundness is at most $1/3$.

The public simulation has only a fixed number of boundary messages beyond the original exchange pattern. Under the original Goldwasser–Sipser message accounting this is precisely the $Q + 2$ bound in (1.2); under the single-message/exchange-round convention fixed in this book it is an $O(1)$ additive overhead. Polynomial-time hash generation and the deterministic checks in (1.4) establish the verifier-time bound. This proves the claimed transformation. $\square$

Why the theorem is about recognition power, not efficient provers. The construction asks Merlin to find preimages witnessing that certain implicitly defined sets are large. An unbounded prover can do this. Nothing in the proof gives a polynomial-time method for an honest prover to find those witnesses, so no efficient-prover preservation claim is made.

1.4 Interactive proofs of proximity: a source-verified frontier

For a string $x \in \Sigma^n$ and a set $S \subseteq \Sigma^n$, define the fractional Hamming distance

$$\text{dist}(x, S) := \min_{y \in S} \frac{1}{n} |\{j \in [n] : x_j \neq y_j\}|.$$

An *interactive proof of ε -proximity* for a language $L \subseteq \bigcup_n \Sigma^n$ gives the verifier random-access oracle access to the input x . It must accept every $x \in L$ with the stated completeness probability and reject, against every cheating prover, every x satisfying $\text{dist}(x, L \cap \Sigma^n) > \varepsilon$ with the stated soundness probability. Inputs within distance ε of the language need not satisfy either decision promise. *Query complexity* counts oracle positions of x read by the verifier and is separate from communication and verifier running time.

Remark 1.5 (Source-verified IPP frontier). Rothblum–Vadhan–Wigderson (STOC 2013, DOI 10.1145/2488608.2488709) prove the following parameterized theorem. Let L be computed by log-space-uniform fan-in-two circuits of depth $D(n)$ and size $S(n)$, and let $\varepsilon = \varepsilon(n)$, with the parameters computable in $O(\log n)$ space. There is an interactive proof of ε -proximity with perfect completeness, soundness error $1/2$, query complexity

$$(1/\varepsilon)^{1+o(1)},$$

communication

$$\varepsilon n (1/\varepsilon)^{o(1)} \text{poly}(D),$$

round complexity $O(\log n + D \log S)$, honest-prover time $\text{poly}(S)$, and verifier time

$$O\left(\left((1/\varepsilon) + \varepsilon n\right)^{1+o(1)} \text{poly}(D) \log S\right).$$

In particular, for $L \in \text{NC}$ and $\varepsilon \geq n^{-1/2}$, the query complexity, communication, and verifier running time are all $n^{1/2+o(1)}$, with polylogarithmically many rounds. This item is frontier-owned: the theorem is source-verified here but its long construction is not silently imported as a QRCC-local proof.

A 2026 source refresh by Goldreich revisits the Rothblum–Vadhan–Wigderson framework and its constant-round limitations. The refresh changes no ownership decision here: Chapter 1 records the exact theorem interface, while any downstream use must cite QRCC-XIII-F-029 as a source-owned frontier rather than a locally reconstructed lemma.

Dependency checkpoint. The only prior-series quantitative input used in the local Goldwasser–Sipser proof is standard Chernoff concentration from Volume I; the hashing collision calculation is proved inside this chapter. The IPP result is source-owned and has no downstream proof obligation discharged by this chapter. The next local workfront is arithmetization of Boolean formulas and quantified Boolean formulas, followed by sum-check.

Chapter 2

Arithmetization: Turning Boolean Quantifiers into Low-Degree Algebra

Interactive proofs become powerful when a discrete assertion is replaced by a polynomial identity that can be tested at random field points. For quantified Boolean formulas, however, there is a second difficulty beyond the initial Boolean-to-polynomial translation: existential and universal quantifiers multiply polynomial expressions, so naively eliminating many quantifiers makes degrees explode. The degree-reduction operation in this chapter is the mechanism that prevents that explosion while preserving every value on the Boolean cube.

How this chapter fits the proof of $IP = PSPACE$. Volume II theorem QRCC-II-THM-022 already establishes that true quantified Boolean formulas are $PSPACE$ -complete. Theorem 2.6 below replaces a TQBF instance by a polynomial operator chain of only quadratic length. Each operator can later be checked by asking for one bounded-degree univariate polynomial and then evaluating it at a fresh random field point. Chapter 3 isolates the corresponding random-polynomial soundness principle in the sum-check protocol; Chapter 4 assembles the pieces into the complete interactive proof.

Learning goals. After this chapter, a reader should be able to arithmetize each Boolean connective; prove semantic preservation on $\{0, 1\}^n$; distinguish equality as formal polynomials from equality only on the Boolean cube; define the multilinearization operator; prove that it preserves Boolean-cube values without increasing total degree; and write an explicit low-degree operator chain for a fully quantified Boolean formula.

2.1 Boolean formulas as polynomial evaluation programs

Let F be a field. We identify the Boolean values false and true with $0, 1 \in F$. If $a, b \in \{0, 1\}$, then

$$\neg a = 1 - a, \quad a \wedge b = ab, \quad a \vee b = a + b - ab. \quad (2.1)$$

The right-hand sides make sense for arbitrary field elements even though their Boolean meaning is only promised at inputs in $\{0, 1\}$.

Definition 2.1 (Arithmetization of a Boolean formula). Let $\varphi(x_1, \dots, x_n)$ be a Boolean formula built from variables, constants, $\neg, \wedge, \vee$. Its *arithmetization over F* is the arithmetic formula

$A_\varphi \in F[X_1, \dots, X_n]$ defined recursively by

$$\begin{aligned} A_0 &= 0, & A_1 &= 1, & A_{x_i} &= X_i, \\ A_{\neg\psi} &= 1 - A_\psi, & A_{\psi \wedge \theta} &= A_\psi A_\theta, & A_{\psi \vee \theta} &= A_\psi + A_\theta - A_\psi A_\theta. \end{aligned} \quad (2.2)$$

The arithmetic formula itself, rather than a fully expanded list of monomials, is the default representation. Thus $A_\varphi(a)$ can be evaluated by one bottom-up pass through φ .

Lemma 2.2 (Boolean semantic preservation). *For every $a \in \{0, 1\}^n$,*

$$A_\varphi(a) = 1 \iff \varphi(a) = \text{true}, \quad (2.3)$$

and otherwise $A_\varphi(a) = 0$. If the formula has at most s variable occurrences and connective gates, then every individual degree and the total degree of the expanded polynomial A_φ are at most s .

Proof. The semantic assertion follows by structural induction using the three identities in (2.1). For the degree bound, constants have degree 0 and a variable has degree 1. Negation does not increase degree. At an AND or OR gate the only potentially higher-degree term is the product of the two child polynomials, whose degree is at most the sum of the two child degrees. Charging one unit to each leaf occurrence and connective gives the coarse but sufficient bound s . $\square$

Formal equality versus Boolean-cube equality. The polynomial produced by (2.2) need not be multilinear. For example, $x \wedge \neg x$ arithmetizes to $X - X^2$, which is not the zero polynomial but vanishes at $X = 0, 1$. Interactive proofs evaluate polynomials away from Boolean points, so this distinction must remain visible rather than being silently quotiented out.

2.2 Multilinear degree reduction

Definition 2.3 (Single-variable multilinearization). For $p \in F[X_1, \dots, X_n]$ and $1 \leq i \leq n$, define

$$(L_i p)(X_1, \dots, X_n) := (1 - X_i)p|_{X_i=0} + X_i p|_{X_i=1}. \quad (2.4)$$

Thus $L_i p$ is affine in X_i . For a set $S = \{i_1, \dots, i_t\}$, a *multilinearization sequence on S* means applying the indicated L_i 's one at a time; we never assume that this operation is free inside an interactive proof.

Lemma 2.4 (Degree-reduction invariants). *For every polynomial p :*

1. $L_i p$ agrees with p at every point whose i -th coordinate lies in $\{0, 1\}$, and hence on the entire Boolean cube;
2. $\deg_{X_i}(L_i p) \leq 1$;
3. for $j \neq i$, $\deg_{X_j}(L_i p) \leq \deg_{X_j} p$;
4. $\deg(L_i p) \leq \deg p$.

Repeatedly applying $L_1, \dots, L_n$ therefore produces a multilinear polynomial agreeing with p on $\{0, 1\}^n$.

Proof. At $X_i = 0$, equation (2.4) equals $p|_{X_i=0}$, and at $X_i = 1$ it equals $p|_{X_i=1}$, proving the first assertion. The second and third are immediate from (2.4).

For total degree, write

$$p = p_0 + X_i p_1 + X_i^2 p_2 + \dots + X_i^r p_r,$$

where the p_j 's do not contain X_i . Then

$$L_i p = p_0 + X_i(p_1 + p_2 + \cdots + p_r). \quad (2.5)$$

Every monomial of $X_i p_j$ on the right has degree at most the degree of the corresponding monomial of $X_i^j p_j$ in p . Hence total degree does not increase. $\square$

2.3 Quantifier operators

Suppose p is a polynomial in variables that include X_i . Define

$$\begin{aligned} (\mathcal{A}_i p) &:= p|_{X_i=0} p|_{X_i=1}, \\ (\mathcal{E}_i p) &:= p|_{X_i=0} + p|_{X_i=1} - p|_{X_i=0} p|_{X_i=1}. \end{aligned} \quad (2.6)$$

The letter $\mathcal{A}$ stands for universal AND and $\mathcal{E}$ for existential OR. Both operators eliminate X_i .

Lemma 2.5 (Quantifier semantics on Boolean prefixes). *Assume $p(a, b) \in \{0, 1\}$ whenever all its arguments are Boolean, where b denotes the coordinate X_i . Then for every Boolean assignment a to the remaining variables,*

$$\begin{aligned} (\mathcal{A}_i p)(a) &= 1 \iff p(a, 0) = p(a, 1) = 1, \\ (\mathcal{E}_i p)(a) &= 1 \iff p(a, 0) = 1 \text{ or } p(a, 1) = 1. \end{aligned} \quad (2.7)$$

If every remaining variable has individual degree at most d in p , then before any multilinearization its individual degree in $\mathcal{A}_i p$ or $\mathcal{E}_i p$ is at most $2d$.

Proof. On inputs in $\{0, 1\}$, the two formulas in (2.6) are exactly the AND and OR identities in (2.1). The degree bound follows because substitution cannot increase degree and the only new multiplication multiplies two substituted copies of p . $\square$

Theorem 2.6 (Low-degree arithmetization of TQBF). *Let*

$$\Phi = Q_1 x_1 Q_2 x_2 \cdots Q_n x_n \varphi(x_1, \dots, x_n) \quad (2.8)$$

be a fully quantified Boolean formula, where each $Q_i \in \{\forall, \exists\}$, and let the quantifier-free formula φ have encoding size $s \geq n$. Over every field F , there is an explicitly specified sequence of

$$M = \sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (2.9)$$

polynomial operators that starts from A_φ and ends at the constant truth value of Φ . The sequence is obtained, for $i = n, n-1, \dots, 1$, by applying the quantifier operator $\mathcal{A}_i$ or $\mathcal{E}_i$ corresponding to Q_i , followed by

$$L_1, L_2, \dots, L_{i-1}. \quad (2.10)$$

It has the following properties.

1. *After the block for quantifier Q_i , the current polynomial is independent of $X_i, \dots, X_n$, is multilinear in $X_1, \dots, X_{i-1}$, and on Boolean prefixes equals the truth value of the quantified suffix $Q_i x_i \cdots Q_n x_n \varphi$.*

2. Every univariate restriction encountered when checking one operator has degree at most $D := 2s$. The first quantifier step in fact needs degree at most s ; after its linearization block all later quantifier inputs are multilinear and all later temporary degrees are at most 2.
3. The initial polynomial $A_\varphi(r)$ can be evaluated at any $r \in F^n$ using $O(s)$ field operations from the formula representation; no expanded monomial table is required.

Consequently the complete algebraic description needed by the interactive protocol has polynomial length, polynomial degree bounds, and polynomial-time verifier evaluation even though the prover may manipulate much larger symbolic expressions.

Proof architecture. The proof is an induction from the innermost quantifier outward. A quantifier operator computes AND or OR of the two Boolean branches. That operation may square degrees in the variables that remain. The immediately following L_j 's restore individual degree one while preserving all Boolean-prefix values. The first block is the only place where the original formula degree s can appear.

Proof. Set $p_{n+1} := A_\varphi$. Process the variables from n down to 1. Suppose the current polynomial before processing x_i agrees, on every Boolean assignment to $x_1, \dots, x_i$, with the truth value of

$$Q_{i+1}x_{i+1} \cdots Q_n x_n \varphi$$

under that prefix. This is true at $i = n$ by Lemma 2.2. Apply $\mathcal{A}_i$ if $Q_i = \forall$ and $\mathcal{E}_i$ if $Q_i = \exists$. Lemma 2.5 shows that, on every Boolean assignment to the remaining prefix $x_1, \dots, x_{i-1}$, the new polynomial is exactly the truth value after applying Q_i . Now apply $L_1, \dots, L_{i-1}$. By Lemma 2.4, none of the Boolean-prefix values changes and the result is multilinear in all remaining variables. This proves the semantic and multilinearity invariant by induction. After $i = 1$, no variables remain, so the constant produced is precisely the truth value of Φ .

There is one quantifier operator and $i - 1$ linearizations in the block for x_i , hence the number of operators is $\sum_{i=1}^n i = M$.

For the degree bound, Lemma 2.2 gives individual degree at most s in A_φ . The innermost quantifier therefore examines a univariate of degree at most s , and its output has remaining individual degrees at most $2s$. Every one of the following linearization checks consequently sees degree at most $2s$. After that first block the current polynomial is multilinear. Every subsequent quantifier can raise remaining individual degrees only from 1 to 2, after which its linearization block returns them to 1. Thus the uniform bound $D = 2s$ is valid for every operator check.

Finally, (2.2) is an arithmetic formula with size linear in the Boolean formula representation. Evaluating it bottom-up at $r \in F^n$ uses $O(s)$ additions, subtractions, and multiplications. No coefficient expansion is performed. $\square$

2.3.1 A complete arithmetization example

Consider

$$\Phi = \forall x \exists y [(x \vee y) \wedge (\neg x \vee y)]. \quad (2.11)$$

The matrix is logically equivalent to y , but we deliberately do not simplify it before arithmetization. Equation (2.2) gives

$$p(X, Y) = (X + Y - XY)(1 - X + XY). \quad (2.12)$$

For the existential quantifier,

$$p(X, 0) = X - X^2, \quad p(X, 1) = 1,$$

so

$$(\mathcal{E}_Y p)(X) = (X - X^2) + 1 - (X - X^2) \cdot 1 = 1. \quad (2.13)$$

Its multilinearization in X is still 1. The universal step therefore gives

$$(\mathcal{A}_X 1) = 1 \cdot 1 = 1, \quad (2.14)$$

which certifies that Φ is true. The intermediate term $X - X^2$ is the useful warning: off the Boolean cube, an arithmetized false branch need not be the zero polynomial. Degree reduction preserves the Boolean semantics without pretending otherwise.

Hypothesis audit. Nothing in Theorem 2.6 requires a finite field. Finiteness becomes necessary only when the verifier later samples a random point. The theorem also does not claim that the intermediate polynomials have compact expanded coefficient lists; only the verifier's initial evaluation and all degree bounds are polynomially controlled. This distinction is essential because the prover is computationally unbounded in an interactive proof.

Chapter 3

The Sum-Check Protocol

The sum-check protocol turns one exponentially large sum over a Boolean cube into a sequence of univariate consistency checks. The verifier never evaluates the entire sum. Instead, each round fixes one variable at a fresh random field point. A dishonest prover can survive a false claim only when two distinct low-degree univariate polynomials happen to agree at that random point.

Learning goals. After this chapter, a reader should be able to write every prover message in sum-check, identify the true partial-sum polynomial hidden behind that message, prove perfect completeness and the exact $\sum_i d_i/|F|$ soundness bound, audit communication and verifier work, and execute a complete small-field transcript including an optimal cheating strategy for that transcript.

3.1 Statement and protocol

Let F be a finite field. The existence and basic arithmetic of finite fields are fixed in Volume III, in particular QRCC-III-THM-001; the root-counting principle used below is a one-variable case of Schwartz–Zippel, Theorem 6.2.

Theorem 3.1 (Sum-check over the Boolean cube). *Let*

$$g \in F[X_1, \dots, X_m], \quad d_i := \deg_{X_i} g, \quad (3.1)$$

and suppose the verifier can evaluate $g(r_1, \dots, r_m)$ at a chosen field point in time T_g . Given a claimed value $H \in F$ for

$$H_{\text{true}} := \sum_{b \in \{0,1\}^m} g(b), \quad (3.2)$$

there is an m -round public-coin interactive protocol with:

1. *perfect completeness;*
2. *soundness error at most*

$$\frac{d_1 + \dots + d_m}{|F|}, \quad (3.3)$$

more precisely, this quantity may be replaced by its minimum with 1;

3. *prover-to-verifier communication of $\sum_{i=1}^m (d_i + 1)$ field elements when each univariate message is sent by its coefficient list, plus m verifier field challenges;*
4. *verifier work $O(\sum_i d_i + m) + T_g$ field operations apart from field-element sampling.*

If every individual degree is at most d , the soundness error is at most $md/|F|$.

Protocol. For $1 \leq i \leq m$, after the verifier has chosen $r_1, \dots, r_{i-1} \in F$, the honest prover's i -th polynomial is

$$q_i(T) := \sum_{b_{i+1}, \dots, b_m \in \{0,1\}} g(r_1, \dots, r_{i-1}, T, b_{i+1}, \dots, b_m). \quad (3.4)$$

It has degree at most d_i . In round 1, the prover sends a claimed polynomial $p_1(T)$ of degree at most d_1 ; the verifier checks

$$p_1(0) + p_1(1) = H. \quad (3.5)$$

If the check passes, the verifier samples uniform $r_1 \in F$. In round $i > 1$, the prover sends $p_i(T)$ of degree at most d_i , and the verifier checks

$$p_i(0) + p_i(1) = p_{i-1}(r_{i-1}), \quad (3.6)$$

then samples uniform $r_i \in F$. After round m , the verifier accepts only if

$$p_m(r_m) = g(r_1, \dots, r_m). \quad (3.7)$$

A message with degree exceeding its declared bound is rejected immediately.

Proof architecture. The honest partial sums satisfy the same endpoint identities as the prover messages. For soundness, maintain one invariant: immediately before a fresh random challenge, the prover's current scalar claim is false. If the prover sends a polynomial satisfying the endpoint check, that polynomial cannot equal the true partial-sum polynomial. Their nonzero difference has degree at most d_i , so a random field point makes the false claim accidentally become true with probability at most $d_i/|F|$. Summing these failure probabilities over the rounds proves (3.3).

Proof. For completeness, assume $H = H_{\text{true}}$ and let the prover send the true polynomials q_i from (3.4). Summing first over $X_i = 0, 1$ gives

$$q_i(0) + q_i(1) = q_{i-1}(r_{i-1}) \quad (3.8)$$

for $i > 1$, while the same identity at $i = 1$ gives (3.5). At the final round there are no Boolean variables left to sum, so

$$q_m(r_m) = g(r_1, \dots, r_m).$$

Every check passes for every verifier random string.

For soundness, assume $H \neq H_{\text{true}}$. Before round 1, the scalar claim is therefore false. More generally, suppose that immediately before round i , the scalar on the right side of (3.5) or (3.6) differs from the corresponding true scalar. Let q_i be the true polynomial (3.4). If a prover polynomial p_i passes the endpoint check and nevertheless equalled q_i as a formal polynomial, then its endpoint sum would equal the true preceding scalar, contradicting the invariant. Hence

$$p_i - q_i \neq 0, \quad \deg(p_i - q_i) \leq d_i. \quad (3.9)$$

A nonzero univariate polynomial of degree at most d_i has at most d_i roots in a field. Therefore

$$\Pr_{r_i \leftarrow F} [p_i(r_i) = q_i(r_i)] \leq \frac{d_i}{|F|}. \quad (3.10)$$

Unless this exceptional event occurs, the scalar claim entering the next round remains false. If no exceptional event occurs in any round, the final claimed value $p_m(r_m)$ differs from the true value $g(r_1, \dots, r_m)$, so (3.7) rejects. The union bound over rounds gives (3.3).

A degree- d_i univariate coefficient list has $d_i + 1$ field elements. Endpoint evaluation of the lists and evaluation at the challenges costs $O(\sum_i d_i + m)$ field operations, followed by one evaluation of g , establishing the communication and verifier-work bounds. $\square$

3.1.1 The reusable one-polynomial consistency step

The following observation is extracted because Chapter 4 uses exactly the same soundness mechanism for quantifier and multilinearization operators, even when the endpoint relation is a product rather than a sum.

Lemma 3.2 (Random-point reduction for an endpoint relation). *Let $q \in F[T]$ have degree at most d , let $\Lambda : F^2 \rightarrow F$ be any fixed function that the verifier can evaluate, and let*

$$c_{\text{true}} = \Lambda(q(0), q(1)).$$

If $c \neq c_{\text{true}}$ and a prover sends $p \in F[T]$ of degree at most d satisfying

$$c = \Lambda(p(0), p(1)),$$

then $p \neq q$ and, for uniform $r \in F$,

$$\Pr[p(r) = q(r)] \leq d/|F|. \quad (3.11)$$

Proof. If $p = q$, the displayed endpoint relation would imply $c = c_{\text{true}}$, a contradiction. Thus $p - q$ is a nonzero polynomial of degree at most d , and the univariate root bound gives (3.11). $\square$

3.2 A complete transcript over $\mathbb{F}_7$

Example 3.3 (Truthful and optimally cheating sum-check transcript). Work over $F = \mathbb{F}_7$ with

$$g(X_1, X_2) = X_1X_2 + 2X_1 + 3X_2 + 1. \quad (3.12)$$

The four Boolean evaluations are

$$g(0, 0) = 1, \quad g(0, 1) = 4, \quad g(1, 0) = 3, \quad g(1, 1) = 0,$$

so the true sum is

$$H_{\text{true}} = 1 + 4 + 3 + 0 = 1 \pmod{7}. \quad (3.13)$$

The honest first polynomial is

$$q_1(T) = g(T, 0) + g(T, 1) = 5T + 5. \quad (3.14)$$

Indeed $q_1(0) + q_1(1) = 5 + 3 = 1$. Suppose the verifier chooses $r_1 = 3$. Then $q_1(3) = 6$, and the honest second polynomial is

$$q_2(T) = g(3, T) = 6T. \quad (3.15)$$

Its endpoint sum is $0 + 6 = 6$, as required. If the verifier now chooses $r_2 = 4$, then

$$q_2(4) = 3 = g(3, 4),$$

so the transcript accepts.

Now suppose a cheating prover tries to establish the false initial claim $H' = 2$. An optimal first-round strategy for this instance is

$$p_1(T) = q_1(T) + T = 6T + 5. \quad (3.16)$$

It passes the false endpoint check because

$$p_1(0) + p_1(1) = 5 + 4 = 2.$$

The discrepancy $p_1 - q_1 = T$ vanishes exactly when $r_1 = 0$, an event of probability $1/7$. If that happens, the cheating prover has landed on the true scalar and can continue honestly, accepting with probability 1.

If $r_1 \neq 0$, the scalar discrepancy is r_1 . In round two the prover may send

$$p_2(T) = q_2(T) + r_1 T, \tag{3.17}$$

where now $q_2(T) = g(r_1, T)$. Equation (3.17) has exactly the required endpoint-sum discrepancy r_1 , and its difference from the true polynomial vanishes exactly at $r_2 = 0$. Hence this strategy accepts with exact probability

$$\frac{1}{7} + \frac{6}{7} \cdot \frac{1}{7} = \frac{13}{49}. \tag{3.18}$$

This is optimal for the false claim $H' = 2$: in round one two distinct degree-one polynomials can agree at at most one of the seven challenges; conditioned on missing that point, the final two distinct degree-one polynomials can again agree at at most one challenge. Thus no cheating strategy can exceed (3.18). The theorem's general union bound gives $2/7$, which is valid but not tight in this concrete transcript.

Hypothesis audit. The theorem requires a field, not merely a finite commutative ring, because a nonzero polynomial over a ring with zero divisors may have more roots than its degree. The Boolean cube can be replaced by a general product set with the corresponding endpoint sums, but that stronger variant is not needed here. Soundness is information-theoretic: the prover may be computationally unbounded and may choose each message adaptively after seeing all previous verifier challenges.

Chapter 4

From Arithmetization to $\text{IP} = \text{PSPACE}$

The central theorem of interactive-proof complexity says that polynomial-time randomized verification with interaction captures exactly polynomial space. The difficult containment is $\text{PSPACE} \subseteq \text{IP}$. Because TQBF is already known to be PSPACE -complete, it is enough to verify a quantified Boolean formula. Chapter 2 has turned such a formula into a polynomial-length chain of low-degree operators. The verifier now checks that chain backwards, replacing each claimed polynomial value by one value of the preceding polynomial at a fresh random point.

The reverse containment $\text{IP} \subseteq \text{PSPACE}$ is conceptually different and much easier: a polynomial-space machine can traverse the exponentially large protocol tree depth-first and compute the optimal prover acceptance probability exactly.

Learning goals. After this chapter, a reader should be able to write the Shamir-style verifier step for a universal quantifier, existential quantifier, and multilinearization operator; prove the false-claim invariant; choose a polynomial-size field giving constant soundness; account for verifier time, communication, and randomness; and prove the reverse containment by an exact polynomial-space protocol-tree evaluation.

4.1 Checking one arithmetization operator backwards

Fix a TQBF instance Φ with n quantified variables and quantifier-free formula size s . Let

$$f_0 = A_\varphi, \quad f_1, \dots, f_M, \quad M = \frac{n(n+1)}{2}, \quad (4.1)$$

be the forward operator chain from Theorem 2.6; thus f_M is the constant truth value of Φ . Every univariate restriction needed below has degree at most

$$D = 2s. \quad (4.2)$$

The verifier runs the chain backwards, beginning with the claim $f_M = 1$.

Suppose the current claim is that $f_t(a) = c$, where a records whatever field coordinates are currently relevant, and $f_t = T(f_{t-1})$ for one operator T . Let

$$q(Z) = f_{t-1}(a \text{ with the operator variable replaced by } Z). \quad (4.3)$$

The prover sends a coefficient list for a polynomial $p(Z)$ of degree at most D . The verifier uses the appropriate endpoint relation:

$$T = \mathcal{A}_i : \quad c = p(0)p(1), \quad (4.4)$$

$$T = \mathcal{E}_i : \quad c = p(0) + p(1) - p(0)p(1), \quad (4.5)$$

$$T = L_i : \quad c = (1 - a_i)p(0) + a_i p(1). \quad (4.6)$$

If the displayed check fails, the verifier rejects. Otherwise it samples a fresh uniform $r \in F$, replaces the relevant coordinate by r , and continues with the new scalar claim $p(r)$ about f_{t-1} .

Equations (4.4)–(4.6) are different endpoint relations, but their soundness is identical. If the incoming scalar claim is false and the prover passes the endpoint check, then the sent polynomial cannot be the true q . Lemma 3.2, extracted from the sum-check analysis, shows that a fresh random point accidentally turns the false claim into a true one with probability at most $D/|F|$.

Theorem 4.1 (Shamir: IP = PSPACE). *Under the interactive-proof conventions of Definition 1.1,*

$$\text{IP} = \text{PSPACE}. \quad (4.7)$$

More explicitly:

1. *every true quantified Boolean formula of length N has an information-theoretically sound interactive proof with a polynomial-time verifier, $O(N^2)$ interaction stages, polynomial total communication, polynomial randomness, perfect completeness, and soundness at most $1/3$;*
2. *every language having a bounded-error polynomial-round interactive proof is decidable in deterministic polynomial space.*

Consequently, because Volume II theorem QRCC-II-THM-022 proves TQBF PSPACE-complete under polynomial-time many-one reductions, every language in PSPACE has such an interactive proof.

Proof architecture. For $\text{PSPACE} \subseteq \text{IP}$, choose a prime field large enough that a union bound over all M random polynomial reductions is below $1/3$. A false TQBF starts the backwards check with a false scalar claim $1 \neq f_M = 0$. Unless one random challenge hits a root of a nonzero discrepancy polynomial, falsity propagates all the way back to f_0 , whose value the verifier computes directly. For the reverse containment, fix all message lengths and verifier randomness by padding, and evaluate the finite protocol game tree using maxima at prover nodes and averages at verifier-randomness nodes.

Proof. 1. $\text{PSPACE} \subseteq \text{IP}$. Volume II theorem QRCC-II-THM-022 reduces every PSPACE language to TQBF, so it suffices to give an interactive proof for TQBF. Let Φ have n quantified variables and matrix size s , both at most its encoding length N . If $n = 0$, the verifier simply evaluates the Boolean formula, so assume $n \geq 1$. Form the operator chain (4.1). There are

$$M = \frac{n(n+1)}{2} \leq N^2$$

operator checks, and every relevant univariate degree is at most $D = 2s \leq 2N$.

Choose

$$k := \lceil \log_2(6MD + 2) \rceil, \quad F := \mathbb{F}_{2^k}. \quad (4.8)$$

Then $|F| = 2^k \geq 6MD + 2$ and $k = O(\log N)$. This is a uniform polynomial-time field choice, not merely an existence assertion. Volume III theorem QRCC-III-THM-003 gives a field of size 2^k , and

QRCC-III-THM-002 implies the existence of an irreducible degree- k polynomial over $\mathbb{F}_2$ by taking the minimal polynomial of a multiplicative generator. The verifier can enumerate the $2^k = \text{poly}(N)$ monic degree- k candidates and use the deterministic irreducibility test QRCC-III-ALG-024 to find one. It then represents F as $\mathbb{F}_2[T]/(h)$; Euclidean polynomial arithmetic QRCC-III-ALG-004 makes every field operation polynomial time. Thus a field element uses $k = O(\log N)$ bits.

The verifier begins with the assertion $f_M = 1$ and runs the backwards checks (4.4)–(4.6) until reaching $f_0 = A_\varphi$. At the end, for the accumulated field point $a \in F^n$, the verifier evaluates $A_\varphi(a)$ directly by a bottom-up pass through the quantifier-free Boolean formula and accepts exactly when the final scalar claim equals this value.

If Φ is true, then $f_M = 1$. The honest prover sends at every stage the true restriction polynomial q from (4.3). The endpoint relation is then an identity by the definition of the corresponding operator, and the new scalar claim after the random challenge is true. Induction over the reverse chain shows that every check passes. Completeness is therefore perfect.

Now suppose Φ is false. Then the true constant is $f_M = 0$, so the verifier's initial claim 1 is false. Condition on the history up to any stage at which the current scalar claim is still false. If a prover message passes the appropriate endpoint check, Lemma 3.2 shows that the new claim becomes true with probability at most

$$D/|F|.$$

This holds adaptively for every cheating prover strategy: the polynomial may depend on the entire previous transcript, but the next field challenge is fresh and uniform. If no such accidental repair occurs in any of the M stages, the final claim about $A_\varphi(a)$ remains false and the verifier's direct evaluation rejects. Hence

$$\Pr[\text{accept false } \Phi] \leq \frac{MD}{|F|} \leq \frac{1}{6} < \frac{1}{3}. \quad (4.9)$$

The slack in (4.8) is intentional; no amplification is needed.

Each prover message contains at most $D + 1 = O(N)$ field elements, and there are $M = O(N^2)$ messages. The verifier sends one field challenge per stage. Thus total communication and randomness are polynomial. Checking each coefficient list, evaluating its endpoints and random point, and evaluating A_φ at the end are all polynomial-time operations. This proves TQBF $\in$ IP, and polynomial-time reductions transfer the protocol to every language in PSPACE.

2. IP $\subseteq$ PSPACE. Let $L \in$ IP. The public-coin transformation of Theorem 1.4, together with its amplification step, gives a bounded-error polynomial-round public-coin protocol for L with polynomial communication and verifier randomness. It is important to make this conversion before the tree calculation: maximizing separately below hidden private-coin branches would incorrectly allow the prover to condition on coins that its transcript does not reveal.

By the padding convention in Definition 1.1, on inputs of length N we may assume fixed polynomial bounds on the number and length of messages and on the total number $R(N)$ of public verifier random bits, with a common schedule on every branch. Because the verifier challenges are public, the transcript identifies every verifier-randomness branch that the prover is allowed to know.

For a fixed input x , consider the finite game tree of this public-coin protocol. At a prover node, its value is the maximum of the values of all legal prover-message children. At a verifier random node, its value is the average of the children corresponding to the uniformly random challenge. A leaf has value 0 or 1 according to the verifier's deterministic final decision. Backward induction therefore makes the root value exactly

$$\sup_{P^*} \Pr[V^{P^*}(x) \text{ accepts}]. \quad (4.10)$$

A deterministic machine computes this value by depth-first recursion. It enumerates possible prover messages one at a time and keeps the largest child value; it enumerates verifier challenges one at a time and accumulates their values. Because the total number of verifier random bits is $R(N)$, every subtree value can be represented exactly with denominator dividing $2^{R(N)}$, so its numerator uses only $O(R(N))$ bits. A recursion frame stores the current transcript prefix, a message or randomness counter, one running maximum or sum, and polynomially many control bits. The transcript length and recursion depth are polynomial in N ; multiplying two polynomial bounds still gives polynomial space. Time can be exponential or worse, which is irrelevant for **PSPACE**.

If $x \in L$, completeness makes the root value at least $2/3$; if $x \notin L$, soundness makes it at most $1/3$. Comparing the exactly computed rational with $1/2$ therefore decides L in deterministic polynomial space. Hence $\text{IP} \subseteq \text{PSPACE}$. Combining the two containments proves (4.7). $\square$

4.2 What sum-check contributes, and what it does not

Theorem 3.1 and the proof above share the same algebraic soundness engine: a false scalar claim forces a false bounded-degree univariate polynomial, and a fresh random field point catches that discrepancy except with probability bounded by degree divided by field size. Ordinary sum-check uses the linear endpoint relation $c = p(0) + p(1)$. TQBF additionally needs the multiplicative universal relation (4.4) and the Boolean OR relation (4.5). Calling the entire TQBF protocol merely “an application of sum-check” would therefore erase a real mathematical distinction. The correct dependency is that the sum-check chapter isolates and proves the random low-degree consistency mechanism before the Shamir operator chain uses its general endpoint-relation form.

Parameter checkpoint. With $n, s \leq N$, we have $M \leq N^2$, $D \leq 2N$, and may choose the extension field $|F| = 2^k = O(N^3)$ from (4.8). A field element then requires $O(\log N)$ bits. The crude communication bound from coefficient lists is $O(MD \log |F|) = O(N^3 \log N)$ bits, already polynomial. The theorem does not optimize communication; its purpose is the class equality.

Hypothesis audit. The containment $\text{PSPACE} \subseteq \text{IP}$ uses the prior **PSPACE**-completeness of TQBF, finite-field root bounds, and the explicitly reconstructed arithmetization chain. It does not use a computational assumption. The reverse containment uses information-theoretic soundness; for an argument system, maximizing over all prover messages would not model the restricted cheating class. Finally, the proof relies on polynomially bounded interaction and verifier randomness. Allowing an exponential transcript would invalidate the polynomial-space traversal bound.

Chapter 5

Multiprover Interactive Proofs and the Low-Degree Interface

A second prover changes the verification problem in a way that a faster single prover does not. The verifier can ask two separated agents questions whose answers must be compatible with one pre-agreed global object, while neither prover can adapt its answer to the question seen by the other. This chapter makes that separation condition precise, proves the elementary exponential-time upper bound for classical multiprover protocols, and then freezes two deep source-owned inputs: the Babai–Fortnow–Lund characterization $\text{MIP} = \text{NEXP}$, and a quantitative Arora–Sudan low-degree test. The latter is presented as a proof-oracle consistency theorem; it is not retroactively identified with the historically different multilinearity test used inside BFL.

Learning goals. After this chapter, a reader should be able to distinguish a classical multiprover strategy from a single prover with two names; write the local-view constraint mathematically; explain why shared classical randomness does not increase the optimal classical value; simulate a polynomial-resource multiprover protocol in nondeterministic exponential time; state exactly what is imported from BFL; and run the line-vs-point low-degree test on a concrete polynomial.

5.1 Classical separated provers

Definition 5.1 (Classical multiprover interactive proof). Fix an input $x \in \{0, 1\}^n$. A *classical k -prover interactive protocol* consists of one probabilistic polynomial-time verifier V and unbounded provers $P_1, \dots, P_k$ connected to the verifier by separate classical channels. Before the protocol starts the provers may agree on an arbitrary joint strategy. After the protocol starts they do not communicate with one another.

Formally, at every prover move, the message sent by P_i is a function only of x , the pre-agreed strategy data, and the *local transcript* visible on prover i 's own channel. It may not depend on messages or verifier questions sent only to another prover. The verifier may use private randomness and may choose later questions adaptively from its own full transcript. All verifier computation, verifier randomness, round count, and total communication are bounded by polynomials in n , unless a different bound is stated explicitly.

For functions $c, s : \mathbb{N} \rightarrow [0, 1]$ with $c(n) > s(n)$, such a protocol has completeness c and soundness

s for a language L when

$$\begin{aligned} x \in L &\implies \exists(P_1, \dots, P_k) \quad \Pr[V^{P_1, \dots, P_k}(x) \text{ accepts}] \geq c(n), \\ x \notin L &\implies \forall(P_1^*, \dots, P_k^*) \quad \Pr[V^{P_1^*, \dots, P_k^*}(x) \text{ accepts}] \leq s(n). \end{aligned}$$

The probability is over the verifier's coins and any explicitly allowed shared classical randomness. Shared classical randomness does not increase the supremal cheating probability: conditioning on its value gives a convex combination of deterministic joint strategies, so one deterministic strategy in the support is at least as good as the mixture.

Write MIP for the class obtained with a polynomially bounded number of classical provers, polynomial interaction resources, and fixed constants $c = 2/3$, $s = 1/3$. A *one-round k -prover game* is the special case in which the verifier samples a question tuple $(q_1, \dots, q_k)$, sends q_i to P_i , receives one answer a_i from each, and applies a deterministic acceptance predicate to the questions, answers, input, and verifier coins. For two provers we also say *two-prover one-round game*.

This definition is purely classical: no shared entangled state, quantum message, commuting-operator strategy, or nonclassical correlation is allowed. Those correlation models belong to the later Volume-XV entangled-prover interface and must not be inferred from the notation MIP here.

Why the local transcript matters. If both provers were permitted to see the union of their transcripts, they could simply behave as one prover and the model would collapse back to an ordinary interactive proof. The mathematical resource is therefore not “two answers” but the information constraint separating the answer functions.

5.2 The easy exponential-time containment

The deep theorem below is an equality, but one direction needs no low-degree testing.

Proposition 5.2 (Classical multiprover protocols are in NEXP). *Under Definition 5.1,*

$$\text{MIP} \subseteq \text{NEXP}.$$

The same conclusion holds if the number of provers is any polynomial in the input length.

Proof. Fix a protocol for inputs of length n . Pad every message and every interaction schedule to fixed polynomial bounds. A deterministic strategy for prover P_i is then a table assigning a next message to each possible local transcript at which P_i moves. A local transcript has polynomial length, so there are at most $2^{p(n)}$ table entries for some polynomial p , and each entry has polynomial length. Thus all prover strategy tables together have size $2^{\text{poly}(n)}$.

A nondeterministic exponential-time machine guesses those tables. It then enumerates every verifier random tape, simulates the protocol using table lookup for the provers, and counts the number of accepting tapes. There are at most $2^{r(n)}$ tapes for a polynomial r , and each simulation takes polynomial time plus exponential-size-table lookup time that can still be implemented within $2^{\text{poly}(n)}$. Hence the exact acceptance numerator is computable in deterministic exponential time after the strategy guess.

If $x \in L$, some joint strategy has acceptance probability at least $2/3$, so some guess is accepted by the nondeterministic simulator. If $x \notin L$, every joint strategy has acceptance probability at most $1/3$, so no guessed tables pass the $2/3$ threshold. The constant gap is more than enough for exact integer comparison. Therefore $L \in \text{NEXP}$. $\square$

5.3 The BFL characterization

Theorem 5.3 (Babai–Fortnow–Lund: $\text{MIP} = \text{NEXP}$). *For the classical multiprover model of Definition 5.1,*

$$\text{MIP} = \text{NEXP}. \quad (5.1)$$

Moreover, the hard containment $\text{NEXP} \subseteq \text{MIP}$ already holds with two noncommunicating classical provers and a randomized polynomial-time verifier using polynomially many interaction steps and polynomial total communication. The constants $2/3$ and $1/3$ in our class convention are a normalization of a constant completeness–soundness gap; standard repetition changes the constants without changing the language class.

Source boundary. Equation (5.1) and the two-prover classical model are source-owned by László Babai, Lance Fortnow, and Carsten Lund, *Non-Deterministic Exponential Time has Two-Prover Interactive Protocols*, *Computational Complexity* 1 (1991), 3–40, DOI 10.1007/BF01200056. Proposition 5.2 proves the easy containment locally. The converse is retained as a deep input: QRCC does not compress the BFL multilinearity analysis into a fictitious one-page proof.

Proof architecture of the deep direction. The BFL construction starts from an exponentially long nondeterministic computation and replaces its exponentially large witness/tableau by values of a low-complexity algebraic extension over a finite field. Algebraic identities turn global correctness into polynomial identities that can be sampled at random points, continuing the arithmetization philosophy of Chapters 2–4. The new obstacle is that an untrusted oracle may answer different queries as though they came from incompatible global polynomials. BFL therefore develops a multilinearity verification procedure and self-correction mechanism. The second prover supplies a separated consistency check: because it does not see the first prover’s query, it cannot tailor its answer to every local inconsistency. Once the verifier has reliable access to the encoded multilinear object, polynomial extrapolation and interactive algebraic checks verify the encoded exponential computation with constant soundness error. This architecture is the conceptual bridge from $IP = \text{PSPACE}$ to PCPs.

A toy consistency game. Suppose two provers claim to share a Boolean table $T : \{0, 1\}^m \rightarrow \{0, 1\}$. The verifier chooses independent-looking queries but, with some probability, secretly sends the same point z to both provers and checks equality of the returned bits. A single prover seeing both questions could always coordinate two incompatible local stories after learning the test. Two separated provers must precommit through their answer functions. This toy check is far weaker than BFL’s multilinearity test, but it isolates the information-theoretic resource exploited by the actual construction.

5.4 A selected finite-field low-degree test

The next theorem is a later low-degree-testing interface chosen because its oracle model and quantitative decoding conclusion are explicit enough to be used safely downstream. It should not be confused with the historically earlier BFL multilinearity test.

Let F be a finite field of size q . An affine line in F^m is a set

$$\ell = \{u + tv : t \in F\}, \quad v \neq 0.$$

Fix one canonical parametrization for every affine line. A *degree- d line oracle* B assigns to each line ℓ a univariate polynomial $B_\ell(t) \in F[t]$ of degree at most d . Given $f : F^m \rightarrow F$, the *line-vs-point low-degree test* samples a uniformly random affine line ℓ , samples a uniformly random point $x = u + av$ on that line, reads the line polynomial B_ℓ and the value $f(x)$, and accepts exactly when

$$B_\ell(a) = f(x). \quad (5.2)$$

Write $\delta_d(f, B)$ for its acceptance probability and

$$\text{agr}(f, P) := q^{-m} |\{x \in F^m : f(x) = P(x)\}|. \quad (5.3)$$

Here “degree d ” means total degree at most d .

Theorem 5.4 (Arora–Sudan line-vs-point low-degree test, selected form). *There exist positive integers c_0, c_1, c_2, c_3 , independent of F, m, d, f , such that the following statements hold for every function $f : F^m \rightarrow F$ and every integer $d > 0$.*

1. *If f has agreement at least $\rho > 0$ with some degree- d polynomial, then there is a degree- d line oracle B with $\delta_d(f, B) \geq \rho$.*

2. *If*

$$q > c_0 \left(\frac{dm}{\rho} \right)^{c_1} \quad (5.4)$$

and some degree- d line oracle B satisfies $\delta_d(f, B) \geq \rho$, then there is a degree- d polynomial $P : F^m \rightarrow F$ with

$$\text{agr}(f, P) \geq \frac{\rho^{c_3}}{c_2}. \quad (5.5)$$

3. *In the stronger near-success form, let $\varepsilon > 0$. If a degree- d line oracle satisfies $\delta_d(f, B) \geq \rho$ and*

$$q > \frac{64 \cdot 4^{c_3}}{\varepsilon^{c_3+3} \rho^{c_3-1}} + c_0 \left(\frac{4dm}{\varepsilon \rho} \right)^{c_1}, \quad (5.6)$$

then some degree- d polynomial P satisfies

$$\text{agr}(f, P) \geq \rho - \varepsilon. \quad (5.7)$$

The tester uses one point query to f and one line-oracle query returning the $d + 1$ coefficients of a univariate polynomial; a direct canonical implementation uses $O(m \log q)$ random bits.

Source boundary. Items 1–2 are Theorem 1 and item 3 is the second strong form (Theorem 4) in Sanjeev Arora and Madhu Sudan, *Improved Low-Degree Testing and Its Applications*, ECCC TR97-003 (1997), later published in *Combinatorica* 23(3), 365–426 (2003), DOI 10.1007/s00493-003-0025-0. The constants are intentionally left as the absolute positive integers supplied by the primary theorem; QRCC does not invent numerical values for them.

What the theorem says. The oracle B is allowed to lie differently on different lines. High acceptance nevertheless forces those local univariate claims to organize around one genuine global low-degree polynomial. Statement (5.5) is already a decoding theorem from a nontrivial test success rate. Under the larger-field hypothesis (5.6), essentially all of the acceptance mass is explained by one global polynomial, losing only ε in agreement.

Proof architecture. The analysis first bounds how many degree- d polynomials can have substantial agreement with the same function. It then shows that, except on a controlled exceptional set, successful line claims must be explained by this short candidate list. The stronger form converts line-wise success into point-wise agreement by averaging and concentration, while the difficult core reconstructs a global polynomial from sufficiently coherent local restrictions. The primary proof uses noisy polynomial reconstruction and an algebraic irreducibility argument; those steps remain inside this black-box boundary.

Worked completeness check over $\mathbb{F}_5$. Take

$$f(x, y) = x^2 + xy + y^2$$

and the line $\ell(t) = (1 + t, 2 + 3t)$. Restriction to the line gives

$$f(1 + t, 2 + 3t) = 2 + 4t + 3t^2 \pmod{5}. \quad (5.8)$$

Thus the honest line oracle stores $B_\ell(t) = 2 + 4t + 3t^2$. If the sampled point corresponds to $a = 2$, then $\ell(2) = (3, 3)$ and both sides of (5.2) equal $2 \in \mathbb{F}_5$. The same identity holds at every point of every line because a total-degree-two polynomial restricts to a univariate degree-at-most-two polynomial. This is the completeness mechanism behind item 1.

5.5 What Chapter 6 may now assume

The chapter has established three distinct interfaces that should not be conflated:

1. Definition 5.1 fixes the information pattern of classical separated provers.
2. Theorem 5.3 supplies the deep NEXP-power of two classical provers, with the easy reverse containment proved locally.
3. Theorem 5.4 supplies a later quantitative low-degree proof-oracle consistency theorem suitable for the PCP transition.

Chapter 6 can therefore define PCP verifiers without using undefined oracle terminology and can state its gap-amplification chain with the low-degree ingredient visible rather than hidden behind an unexplained appeal to “arithmetization.”

Chapter 6

Probabilistically Checkable Proofs, Gap Amplification, and Label Cover

Interactive proofs let a verifier interrogate a prover over time. A probabilistically checkable proof moves the burden into a static oracle: the verifier tosses only a few random coins and reads only a few proof locations. The point is not that a constant number of symbols can certify an arbitrary theorem by magic. Rather, the proof is encoded so redundantly that a false global claim forces a constant density of locally detectable inconsistencies. This chapter fixes the verifier model, makes the PCP–constraint-system dictionary exact, proves the graph-powering amplification step, and then closes the full Dinur-style iteration through preprocessing and alphabet-reducing composition. The endpoint is the PCP theorem itself, followed by a concrete projection-game interface and a first constant-gap MAX-3SAT consequence.

Powering alone is not the PCP theorem. Degree regularization, alphabet reduction, and composition must prevent the encoded instance from growing too fast while the gap is repeatedly amplified. The chapter therefore keeps a parameter ledger after every transformation and proves the reusable one-step recurrence explicitly. This makes the polynomial-size invariant visible rather than hiding it inside the slogan “gap amplification proves PCP.”

Learning goals. After this chapter, a reader should be able to define $\text{PCP}_{c,s}[r, q; \Gamma]$ with all probability and oracle conventions explicit; remove constant-query adaptivity by expanding the finite answer tree; convert a nonadaptive PCP verifier into a gap-CSP and back; explain why clause sampling from an arbitrary satisfiable-versus-unsatisfiable formula has only an inverse-size gap; construct the powered constraint graph and prove its $\sqrt{t}$ -scale gap amplification from spectral mixing and a second-moment argument; explain why a constant-relative-distance code and an assignment tester restore a fixed alphabet; iterate the resulting one-step amplifier without superpolynomial growth; derive $\text{NP} = \text{PCP}[O(\log n), O(1)]$; and convert the constant-gap CSP first to Label Cover and then to a constant-gap MAX-3SAT instance.

6.1 The PCP verifier model

Definition 6.1 (Probabilistically checkable proof and gap-CSP). Let $L \subseteq \{0, 1\}^*$, let Γ be a finite alphabet, and let $r, q : \mathbb{N} \rightarrow \mathbb{N}$. A Γ -valued PCP verifier for L is a probabilistic polynomial-time oracle machine V which, on an input $x \in \{0, 1\}^n$, uses at most $r(n)$ random bits and makes at most

$q(n)$ oracle queries to a finite proof string

$$\pi \in \Gamma^{\ell(n)}.$$

A query names an index in $[\ell(n)]$ and returns the single symbol stored there. Queries may be adaptive unless *nonadaptive* is stated explicitly; repeated reads count as repeated queries. The verifier may perform arbitrary polynomial-time computation between queries.

For functions $c, s : \mathbb{N} \rightarrow [0, 1]$ with $c(n) > s(n)$, write

$$L \in \text{PCP}_{c,s}[r, q; \Gamma]$$

when

$$\begin{aligned} x \in L &\implies \exists \pi \quad \Pr[V^\pi(x) \text{ accepts}] \geq c(n), \\ x \notin L &\implies \forall \pi \quad \Pr[V^\pi(x) \text{ accepts}] \leq s(n), \end{aligned}$$

where the probability is only over the verifier's random coins. The notation $\text{PCP}[r, q]$ means the binary-alphabet convention $\Gamma = \{0, 1\}$, perfect completeness $c = 1$, and some fixed soundness at most $1/2$; changing one fixed constant soundness threshold below one to another is a constant-factor amplification issue, not a change of class.

For a fixed random string R , the verifier's *local answer string* is the ordered tuple of oracle symbols read on that random branch. A nonadaptive verifier has *free-bit complexity at most f* if, for every input and random string, at most 2^f local answer strings make the verifier accept. Its *robustness* on a proof π is the expected relative Hamming distance, over R , between the observed local answer string and the nearest local answer string that would make the branch accept; when no accepting local string exists, the distance is defined to be one.

A *finite q -ary constraint system* consists of a finite variable set U , an alphabet Σ , and a finite multiset E of constraints. Each constraint $e \in E$ names at most q variables and specifies a relation $C_e \subseteq \Sigma^{|e|}$. For an assignment $\sigma : U \rightarrow \Sigma$, let

$$\text{UNSAT}_\sigma(G) := \frac{1}{|E|} |\{e \in E : \sigma|_e \notin C_e\}|, \quad \text{UNSAT}(G) := \min_\sigma \text{UNSAT}_\sigma(G). \quad (6.1)$$

The *gap-CSP promise problem with gap $\varepsilon > 0$* asks one to distinguish $\text{UNSAT}(G) = 0$ from $\text{UNSAT}(G) \geq \varepsilon$. For a binary constraint graph, every constraint has arity two and the multiset E is viewed as an undirected multigraph; parallel edges and self-loops are allowed, and each edge carries its own binary relation.

Accessible proof length. The explicit length $\ell(n)$ is useful bookkeeping, but a small-randomness verifier cannot meaningfully access an arbitrary number of proof cells. If $|\Gamma| = s$, an adaptive verifier with at most r random bits and q queries has at most

$$2^r (1 + s + \cdots + s^{q-1}) \leq q 2^r s^{q-1} \quad (6.2)$$

query occurrences across all random strings and answer-tree nodes. Cells never appearing in this finite tree may be deleted. In particular, $r = O(\log n)$, $q = O(1)$, and constant alphabet imply polynomial relevant proof length.

Proposition 6.2 (Constant-query adaptivity can be removed). *Let V be a Γ -valued PCP verifier making at most q adaptive queries. There is a nonadaptive verifier with the same randomness, completeness, and soundness that makes at most*

$$1 + s + \cdots + s^{q-1} \leq qs^{q-1}, \quad s = |\Gamma|, \quad (6.3)$$

distinct oracle queries on each random string.

Proof. Fix the input and a verifier random string. Before seeing any answers, expand the deterministic query procedure into its depth-at-most- q answer tree. Every internal node is labelled by the proof location queried there, and each outgoing edge is labelled by one of the s possible answers. Query nonadaptively the union of all locations labelling internal nodes. The returned symbols determine a unique root-to-leaf trajectory, after which the new verifier applies exactly the original leaf decision. Thus every proof and random string has exactly the same acceptance bit as before. A depth- q s -ary tree has at most the number of internal nodes in (6.3). $\square$

Proposition 6.3 (The PCP-gap-CSP dictionary). *For a nonadaptive Γ -valued PCP verifier using exactly r random bits and at most q queries, there is a q -ary constraint system on the proof locations such that, for every proof π ,*

$$\Pr[V^\pi(x) \text{ rejects}] = \text{UNSAT}_\pi(G_x). \quad (6.4)$$

Conversely, a polynomial-size q -ary constraint system can be checked by sampling one uniformly random constraint, querying its variables, and testing its relation.

Proof. For each random string R , create one constraint on the proof locations queried on branch R . Its accepting relation is exactly the set of local answer strings for which the deterministic post-query computation accepts. Taking all 2^r random strings with multiplicity makes the fraction of violated constraints equal the rejection probability, proving (6.4). In the reverse direction, a uniformly sampled constraint is rejected exactly when the assignment violates it, so the rejection probability is its unsatisfied fraction. $\square$

Why a gap must be amplified. If a 3CNF formula has m clauses, the naive verifier that samples one clause and queries its three assignment bits has perfect completeness. But a false formula may have an assignment violating only one clause, so its rejection probability can be merely $1/m$. The PCP theorem requires a constant gap while retaining only logarithmic randomness and constant query complexity. The rest of this chapter isolates the transformation that increases the gap.

6.2 Constraint graphs and the selected power operation

For the powering argument it is convenient to use a transition convention. Write

$$G = (V, E^\rightarrow, \Sigma, \{C_e\}_{e \in E^\rightarrow}).$$

This is a finite symmetric d -regular constraint multigraph with $d \geq 2$: every vertex has exactly d outgoing constraint arcs with multiplicity, every arc (u, v) has a reverse arc (v, u) carrying the reversed relation, and self-loop arcs are allowed. We measure UNSAT using the uniform distribution on these oriented transition constraints. This is only a bookkeeping convention; an ordinary undirected constraint edge is represented by its two orientations. Fix one distinguished self-loop transition at every vertex. Let

$$P := A/d$$

be the normalized random-walk matrix, acting on $L^2(V, \pi)$ for the uniform stationary measure π . Assume the graph is connected and

$$\|Pf\|_2 \leq \rho \|f\|_2 \quad \text{whenever } \mathbb{E}_\pi f = 0, \quad 0 \leq \rho < 1. \quad (6.5)$$

This is exactly the spectral-contraction interface proved earlier as ‘QRCC-VII-THM-003’.

For an even integer t , put $m = t/2$ and choose a fixed constant $a > 0$, to be specified in the proof. Set

$$R_t := m + \lceil a\sqrt{m} \rceil, \quad \Gamma_R(u) := \{v \in V : \text{dist}_G(u, v) \leq R_t\}. \quad (6.6)$$

A *local opinion at u* is an assignment $\omega_u : \Gamma_R(u) \rightarrow \Sigma$. The powered alphabet at u is the set of all such opinions; using a fixed padding/order convention gives the uniform bound

$$|\text{powered alphabet}| \leq |\Sigma|^{1+d+\dots+d^{R_t}} \leq |\Sigma|^{2d^{R_t}}. \quad (6.7)$$

The *selected t -step power $G^{(t)}$* has the same vertex set. Every length- t walk

$$W = (v_0, v_1, \dots, v_t)$$

creates one oriented powered constraint, with multiplicity, from v_0 to v_t ; the reversed walk supplies the reverse constraint. Given endpoint opinions ω_{v_0} and ω_{v_t} , this powered constraint accepts precisely when there exists an assignment τ on $\Gamma_R(v_0) \cup \Gamma_R(v_t)$ which agrees with both opinions wherever they are defined and satisfies every base constraint $C_{\{v_i, v_{i+1}\}}$ along the walk. Since every path vertex is within distance at most m of one endpoint, all tested path constraints are within this union. The padded radius in (6.6) is slightly more generous than the minimal neighborhood used in the canonical powering construction; it makes the middle-window consistency coordinates literally available at both endpoints and does not change the constant- t asymptotics used later.

The powered transition multigraph is d^t -regular. Hence the number of oriented transition constraints is

$$|E^\rightarrow(G^{(t)})| = |V|d^t = d^{t-1}|E^\rightarrow(G)|. \quad (6.8)$$

6.3 A walk-length stability lemma

The self-loop assumption has a concrete role: a walk whose length changes by $O(\sqrt{t})$ still samples comparable endpoint distributions. We isolate that fact before using it inside the amplification proof.

Lemma 6.4 (Binomial-overlap stability). *Fix $p \in (0, 1)$, an integer $s \geq 2$, and a constant $a > 0$. There exist constants $\eta = \eta(p, s, a) > 0$ and $m_0 = m_0(p, s, a)$ with the following property. If $m \geq m_0$, $|\ell - m| \leq a\sqrt{m}$, and $(h_k)_{k \geq 0} \subseteq [0, 1]$ satisfies*

$$\mathbb{E}[h_K] \geq \frac{1}{s}, \quad K \sim \text{Bin}(m, p), \quad (6.9)$$

then

$$\mathbb{E}[h_{K'}] \geq \eta, \quad K' \sim \text{Bin}(\ell, p). \quad (6.10)$$

Proof. Choose a constant $B > 0$, depending only on p, s , so that Chebyshev's inequality gives

$$\Pr(|K - pm| > B\sqrt{m}) \leq \frac{1}{2s} \quad (6.11)$$

for every sufficiently large m . Since $0 \leq h_k \leq 1$, (6.9) and (6.11) imply

$$\sum_{|k - pm| \leq B\sqrt{m}} \Pr[K = k] h_k \geq \frac{1}{2s}. \quad (6.12)$$

For $|\ell - m| \leq a\sqrt{m}$ and $|k - pm| \leq B\sqrt{m}$, both k and $\ell - k$ stay a linear distance from the endpoints $0, \ell$. Write $\delta = \ell - m$. When $\delta \geq 0$, the binomial mass formula gives the exact product

$$\frac{\Pr[K' = k]}{\Pr[K = k]} = \prod_{j=1}^{\delta} \frac{(1-p)(m+j)}{m-k+j} = \prod_{j=1}^{\delta} \left(1 + \frac{(k-pm) - pj}{(1-p)m - (k-pm) + j}\right). \quad (6.13)$$

Every displayed correction has absolute value at most $C/\sqrt{m}$, for a constant $C = C(p, a, B)$, and there are at most $a\sqrt{m}$ factors. Increasing m_0 makes every correction at most $1/2$ in absolute value. The elementary bound $\log(1+z) \geq z - z^2$ for $|z| \leq 1/2$ then shows that the logarithm of (6.13) is bounded below by a constant depending only on p, a, B . When $\delta < 0$, apply the same product calculation to the reciprocal ratio, noting that the central band for m is still an $O(\sqrt{\ell})$ band for $\ell = m + O(\sqrt{m})$. Consequently there is a $\tau = \tau(p, a, B) > 0$ such that

$$\Pr[K' = k] \geq \tau \Pr[K = k] \quad (6.14)$$

throughout the band in (6.12), once m is sufficiently large. Multiplying (6.12) by τ gives (6.10) with $\eta = \tau/(2s)$. $\square$

Remark 6.5. The product estimate in (6.13)–(6.14) is the elementary local-overlap form of the fact that two binomial laws with means separated by only $O(\sqrt{m})$ have constant overlap. No central limit theorem is needed, and all constants are uniform in the sequence (h_k) .

6.4 Gap amplification by powering

Theorem 6.6 (Expander-based graph-powering gap amplification). *Fix a degree $d \geq 2$, a finite alphabet Σ , and $\rho < 1$. There exist constants $a > 0$, $\beta > 0$, and an even t_0 , depending only on $d, |\Sigma|, \rho$, such that the following holds. Let G be a finite connected symmetric d -regular transition constraint multigraph with one distinguished self-loop transition at every vertex and satisfying the spectral contraction (6.5). For every even $t \geq t_0$, construct the padded-radius power $G^{(t)}$ by (6.6)–(6.8). Then:*

1. *if $\text{UNSAT}(G) = 0$, then $\text{UNSAT}(G^{(t)}) = 0$;*
2. *for every G ,*

$$\text{UNSAT}(G^{(t)}) \geq \beta\sqrt{t} \min\left\{\text{UNSAT}(G), \frac{1}{t}\right\}; \quad (6.15)$$

3. *the powered out-degree is d^t , the number of oriented transition constraints grows by the factor d^{t-1} , and the powered alphabet obeys (6.7).*

Thus for any fixed sufficiently large even t , powering is a linear-size constant-alphabet transformation and multiplies every sufficiently small unsatisfied fraction by $\Omega(\sqrt{t})$.

Proof. Completeness. If $\sigma : V \rightarrow \Sigma$ satisfies every base constraint, assign to each powered vertex u the local opinion $\omega_u = \sigma|_{\Gamma_R(u)}$. For every powered walk, the union assignment $\tau = \sigma$ witnesses acceptance. This proves item 1. The degree, edge-count, and alphabet claims were established in (6.7)–(6.8), so it remains to prove (6.15).

Popular opinions. Fix an arbitrary powered assignment $\tilde{\sigma}$. Put $s = |\Sigma|$ and $m = t/2$. For a base vertex v , run an m -step random walk starting at v , with endpoint U , and read the symbol

$\tilde{\sigma}(U)(v)$. This coordinate exists because $v \in \Gamma_R(U)$. Define $\sigma(v)$ to be a modal symbol of this random variable. Hence

$$\Pr[\tilde{\sigma}(U)(v) = \sigma(v)] \geq \frac{1}{s}. \quad (6.16)$$

Remove the one distinguished self-loop at every vertex. The remaining transition multigraph is $(d-1)$ -regular, so for a stochastic matrix Q ,

$$P = \frac{1}{d}I + \frac{d-1}{d}Q. \quad (6.17)$$

An ℓ -step P -walk therefore has the same endpoint distribution as a Q -walk of random length $K_\ell \sim \text{Bin}(\ell, p)$, where $p = (d-1)/d$. For fixed v and the chosen symbol $\sigma(v)$, let h_k be the probability that a k -step Q -endpoint U has $\tilde{\sigma}(U)(v) = \sigma(v)$. Equation (6.16) is exactly (6.9). By Lemma 6.4, after choosing a fixed $a > 0$ and increasing t_0 if necessary, there is an $\eta > 0$ such that for every $|\ell - m| \leq a\sqrt{m}$,

$$\Pr[\tilde{\sigma}(U_\ell)(v) = \sigma(v)] \geq \eta, \quad (6.18)$$

where U_ℓ is an ℓ -step random-walk endpoint started at v . The same η works for every vertex and every powered assignment.

Unsatisfied base edges and auxiliary thinning. Let $\mathcal{U} \subseteq E^\rightarrow$ be the multiset of oriented base transitions whose constraint is violated by σ , and set

$$u := \text{UNSAT}_\sigma(G).$$

If $u = 0$, (6.15) is trivial. Otherwise define

$$\theta := \min\left\{1, \frac{1}{tu}\right\}, \quad f := \theta u = \min\left\{u, \frac{1}{t}\right\}. \quad (6.19)$$

The Bernoulli variables introduced next are used only in the analysis and are not part of the verifier or the powered instance.

Sample a stationary length- t walk $W = (v_0, \dots, v_t)$. Let

$$I := \{i : |i - m| \leq \lfloor a\sqrt{m}/2 \rfloor\} \quad (6.20)$$

and put $L = |I| = \Theta(\sqrt{t})$. Independently for each $i \in I$, sample $M_i \sim \text{Bernoulli}(\theta)$. Write $B_i = 1$ when the base edge $\{v_i, v_{i+1}\}$ lies in $\mathcal{U}$.

Call position i a *hit* if $M_i = B_i = 1$ and both endpoint opinions agree with the popular assignment on that violated edge:

$$\tilde{\sigma}(v_0)(v_i) = \sigma(v_i), \quad \tilde{\sigma}(v_t)(v_{i+1}) = \sigma(v_{i+1}). \quad (6.21)$$

Every hit forces the powered constraint to reject: any union assignment agreeing with the two endpoint opinions gives the violated base edge exactly the values $(\sigma(v_i), \sigma(v_{i+1}))$. The coordinates in (6.21) are available because the middle window in (6.20) lies inside the padded radius (6.6).

Conditional on the oriented edge (v_i, v_{i+1}) , reversibility makes the backward segment from v_i to v_0 an i -step random walk, the forward segment from v_{i+1} to v_t a $(t-i-1)$ -step random walk, and the two segments independent. Both lengths differ from m by $O(\sqrt{m})$. Therefore (6.18) gives, uniformly over the conditioned edge,

$$\Pr[\text{both equalities in (6.21)} \mid B_i = 1] \geq \eta^2. \quad (6.22)$$

If N_i is the hit indicator and $N = \sum_{i \in I} N_i$, stationarity and $\Pr[B_i = 1] = u$ imply

$$\mathbb{E}N_i \geq \theta u \eta^2 = f \eta^2, \quad \mathbb{E}N \geq L f \eta^2. \quad (6.23)$$

Second moment from spectral mixing. Let

$$g(v) := \Pr[\{v, X_1\} \in \mathcal{U} \mid X_0 = v],$$

so $0 \leq g \leq 1$ and $\mathbb{E}_\pi g = u$. For $i < j$, stationarity, reversibility, and the symmetry of the violated-edge indicator give

$$\Pr[B_i = B_j = 1] = \langle g, P^{j-i-1} g \rangle_\pi. \quad (6.24)$$

Write $g = u + g_0$ with $\mathbb{E}_\pi g_0 = 0$. The audited spectral contraction ‘QRCC-VII-THM-003’ and $\|g_0\|_2^2 \leq \mathbb{E}g^2 \leq u$ yield

$$\Pr[B_i = B_j = 1] \leq u^2 + u \rho^{j-i-1}. \quad (6.25)$$

Set $Z := \sum_{i \in I} M_i B_i$. Since every hit is marked and violated, $0 \leq N \leq Z$. Using independence of the marks, (6.25), and $\theta^2 u \leq \theta u = f$,

$$\begin{aligned} \mathbb{E}Z^2 &\leq Lf + 2 \sum_{i < j} (f^2 + f \rho^{j-i-1}) \\ &\leq Lf + L^2 f^2 + \frac{2Lf}{1-\rho}. \end{aligned} \quad (6.26)$$

Because $f \leq 1/t$ and $L = \Theta(\sqrt{t})$, we have $Lf \leq 1$ after enlarging t_0 , so $L^2 f^2 \leq Lf$. Thus for a constant $C = C(\rho, a)$,

$$\mathbb{E}N^2 \leq \mathbb{E}Z^2 \leq CLf. \quad (6.27)$$

A hit occurs with amplified probability. Cauchy–Schwarz in the form $\Pr[N > 0] \geq (\mathbb{E}N)^2 / \mathbb{E}N^2$, together with (6.23) and (6.27), gives

$$\Pr[N > 0] \geq \frac{\eta^4}{C} Lf \geq \beta \sqrt{t} \min\left\{u, \frac{1}{t}\right\} \quad (6.28)$$

for some $\beta > 0$ depending only on d, s, ρ . For each fixed powered walk W , the auxiliary marks can create a hit only if that powered constraint already rejects. Hence $\Pr_{W,M}[N > 0] \leq \Pr_W[\text{powered constraint rejects}]$. Thus (6.28) lower-bounds the rejection fraction of the arbitrary powered assignment $\tilde{\sigma}$. Finally, $u = \text{UNSAT}_\sigma(G) \geq \text{UNSAT}(G)$, and the map $z \mapsto \min\{z, 1/t\}$ is monotone. Minimizing over $\tilde{\sigma}$ proves (6.15). $\square$

Source boundary and relation to Dinur’s powering lemma. In Dinur’s primary paper, the powering lemma for an expander constraint graph gives a bound of the form

$$\text{UNSAT}(G^t) \geq \beta_2 \sqrt{t} \min\{\text{UNSAT}(G), 1/t\}, \quad (6.29)$$

with constants controlled by the regular expander parameters, while tracking the enlarged local alphabet. Theorem 6.6 reconstructs the amplification mechanism locally in a deliberately padded-radius variant: the extra $O(\sqrt{t})$ opinion radius makes every middle-window coordinate used in (6.21) explicit. The proof uses the same structural ingredients — popular local opinions, middle hits, random-walk mixing, and a second-moment estimate — but all constants in the QRCC statement are existential and depend only on the frozen $(d, |\Sigma|, \rho)$. No stronger quantitative constant is attributed to the source.

6.5 Parameter ledger for one powering step

For fixed even $t \geq t_0$, Theorem 6.6 has the following parameter effect:

quantity	after powering	(6.30)
vertices	$ V $	
oriented constraints	$ V d^t = d^{t-1} E^\rightarrow $	
degree	d^t	
alphabet	$\leq \Sigma ^{2d^{R_t}}$	
small gap δ	$\geq \beta\sqrt{t}\delta \quad (\delta \leq 1/t).$	

For a fixed t , the size blowup is constant, but the new alphabet depends exponentially on the fixed powering radius. Repeating powering without resetting the alphabet would therefore make the “constant” itself grow from one round to the next. The missing operation is a composition that re-encodes each large-alphabet local constraint over one fixed small alphabet while losing only a fixed factor in the gap.

6.6 Assignment testers: the one deep composition input

The composition interface is stronger than an ordinary PCP. It must reject in proportion to how far the *already supplied input bits* are from any satisfying input, because those input bits are the shared coordinates that keep the local gadgets for different outer constraints consistent.

Definition 6.7 (Assignment tester and proximity rejection). Let $\Phi : \{0, 1\}^X \rightarrow \{0, 1\}$ be a Boolean circuit on a finite set X of designated input bits. For $a : X \rightarrow \{0, 1\}$, define

$$\text{rdist}(a, \text{SAT}(\Phi)) := \min_{z \in \text{SAT}(\Phi)} \frac{|\{x \in X : a(x) \neq z(x)\}|}{|X|}, \quad (6.31)$$

with the extended-distance convention that this value is one when $\text{SAT}(\Phi) = \emptyset$. Empty outer relations are handled separately in Lemma 6.10, so the composition proof does not rely on attributing this convention to the primary assignment-tester source.

An *assignment tester* with alphabet Σ_{AT} , two distinguished Boolean symbols $\bar{0}, \bar{1} \in \Sigma_{\text{AT}}$, and rejection coefficient $\varepsilon_{\text{AT}} > 0$ maps Φ to a finite binary constraint system T_Φ whose vertex set contains X . Every vertex of T_Φ , including those in X , has the same alphabet Σ_{AT} . When an input assignment $a : X \rightarrow \{0, 1\}$ is inserted into the tester, we interpret 0 as $\bar{0}$ and 1 as $\bar{1}$. Writing $Y := V(T_\Phi) \setminus X$, the tester satisfies:

1. if $a \in \text{SAT}(\Phi)$, some extension $b : Y \rightarrow \Sigma_{\text{AT}}$ satisfies every constraint of T_Φ ;
2. if $a \notin \text{SAT}(\Phi)$, then for every $b : Y \rightarrow \Sigma_{\text{AT}}$,

$$\text{UNSAT}_{a \cup b}(T_\Phi) \geq \varepsilon_{\text{AT}} \text{rdist}(a, \text{SAT}(\Phi)). \quad (6.32)$$

Here $a \cup b$ uses the distinguished-symbol embedding on the coordinates in X . This uniform-alphabet convention is the one needed for composition: the Boolean blocks shared by several local tester gadgets are literally the same vertices of the final constraint graph.

Theorem 6.8 (Dinur constant-alphabet assignment tester — deep input). *There are an absolute constant $\varepsilon_{\text{AT}} > 0$, the fixed alphabet $\Sigma_{\text{AT}} = \{0, 1\}^3$, distinguished Boolean symbols $\bar{0} = 000$ and $\bar{1} = 111$, and an explicit assignment tester in the sense of Definition 6.7. The explicit construction may furthermore be chosen so that any tester constraint involving a designated input vertex rejects whenever that vertex carries a symbol other than $\bar{0}$ or $\bar{1}$.*

Source boundary. This is the assignment-tester theorem in Irit Dinur, *The PCP theorem by gap amplification*, *Journal of the ACM* 54(3), Article 12 (2007), especially Definition 2.2, Theorem 5.1, and the explicit construction in Section 7. The primary theorem deliberately does not need a useful running-time bound for the tester in the composition application. QRCC uses it only on circuits whose input size is an absolute constant after the powering parameter has been frozen. Hence the finitely many required tester outputs can be hardwired into the uniform reduction. No claim is made that Theorem 6.8 is proved by the Arora–Sudan theorem of Chapter 5; the two are compatible low-degree/PCP interfaces but have distinct source ownership.

6.7 A local preprocessing lemma

The powering theorem needs a regular expander with a self-loop at every vertex. We now build that normal form locally, using the already-audited strongly explicit Margulis–Gabber–Galil family ‘QRCC-VII-ALG-011’.

Lemma 6.9 (Degree control, expanderization, and self-loop preprocessing). *There exist absolute constants*

$$d \geq 3, \quad 0 \leq \rho < 1, \quad \beta_{\text{pre}} > 0, \quad C_{\text{pre}} > 0 \quad (6.33)$$

such that every finite binary constraint multigraph G over any finite alphabet Σ can be transformed in polynomial time into a symmetric d -regular transition constraint multigraph $\text{prep}(G)$ with the same alphabet, one distinguished null self-loop transition at every vertex, absolute random-walk contraction at most ρ , and

$$\beta_{\text{pre}} \text{UNSAT}(G) \leq \text{UNSAT}(\text{prep}(G)) \leq \text{UNSAT}(G), \quad \text{size}(\text{prep}(G)) \leq C_{\text{pre}} \text{size}(G). \quad (6.34)$$

All constants are independent of $|\Sigma|$ and of the input size.

Proof. We may symmetrize every binary constraint by including its reversed transition with the reversed relation; this does not change its violated fraction. Delete isolated variables. If no constraint remains, output one fixed satisfiable regular instance and the lemma is immediate. Let $m = |E^{\rightarrow}| > 0$, and let $k_v \geq 1$ be the number of outgoing transitions from v .

Cloud degree reduction. For every integer $h \geq 1$, choose

$$q(h) := \max\{2, \lceil \sqrt{h} \rceil\}, \quad N(h) := q(h)^2.$$

Then $h \leq N(h) \leq 4h$. By ‘QRCC-VII-ALG-011’, the graph $M_{q(h)}$ has $N(h)$ vertices, degree $D_0 = 8$, a strongly explicit rotation map, and

$$\lambda_0 := \frac{5\sqrt{2}}{8} < 1 \quad (6.35)$$

as an absolute nontrivial spectral bound.

Replace v by a cloud of $N_v := N(k_v)$ copies carrying a copy of $M_{q(k_v)}$. Assign a distinct cloud copy (v, e) to every outgoing base transition e from v . Every expander transition inside a cloud receives the equality relation. If $e = (v, w)$ and $\bar{e} = (w, v)$ are reverse base transitions, join (v, e) and $(w, \bar{e})$ by the original relation and its reverse. Every unused cloud copy receives one null external self-loop transition. Finally add one more null self-loop transition at every cloud copy; this will be the distinguished self-loop used by powering. The resulting graph H is

$$d_1 := D_0 + 2 = 10 \quad (6.36)$$

regular. Since

$$N_0 := |V(H)| = \sum_v N_v \leq 4 \sum_v k_v = 4m, \quad (6.37)$$

its size is linear in the input constraint count.

Fix any assignment τ to the cloud graph and let $\sigma(v)$ be a modal symbol in the cloud of v . Put δ_v for the fraction of copies of v that disagree with this modal symbol. If the symbol frequencies in that cloud are p_a , then $1 - \sum_a p_a^2 \geq 1 - \max_a p_a = \delta_v$. Spectral contraction of the cloud expander therefore gives

$$\begin{aligned} \Pr_{(x,y) \text{ cloud transition}} [\tau(x) \neq \tau(y)] &\geq (1 - \lambda_0) \left(1 - \sum_a p_a^2 \right) \\ &\geq (1 - \lambda_0) \delta_v. \end{aligned} \quad (6.38)$$

Let B be the number of violated cloud-equality transitions and $Q := \sum_v N_v \delta_v$. Equation (6.38) yields

$$Q \leq \frac{B}{D_0(1 - \lambda_0)}. \quad (6.39)$$

Let F be the base transitions violated by the decoded assignment σ . For every $e = (v, w) \in F$, either the corresponding external transition is violated by τ , or the active copy (v, e) disagrees with $\sigma(v)$, or the active copy $(w, \bar{e})$ disagrees with $\sigma(w)$. Active copies are distinct over outgoing incidences, so if U is the total number of violated transitions of H ,

$$|F| \leq U + 2Q \leq U \left(1 + \frac{2}{D_0(1 - \lambda_0)} \right) =: c_0 U. \quad (6.40)$$

Since $|F|/m \geq \text{UNSAT}(G)$, (6.37)–(6.40) give a constant-factor gap lower bound for H . Conversely, copying an optimal base assignment to every cloud violates only those external transitions corresponding to violated base constraints, so the gap never increases above $\text{UNSAT}(G)$.

Global expanderization. Choose $q = \max\{2, \lceil \sqrt{N_0} \rceil\}$ and $N = q^2 \leq 4N_0$. Pad H to N vertices by giving each new dummy vertex d_1 null self-loop transitions. Superimpose a copy of M_q on all N vertices and put the null relation on its eight transitions. The final transition degree is

$$d = d_1 + D_0 = 18. \quad (6.41)$$

All newly added constraints are always satisfiable, so they only dilute the already established gap by another constant factor. Combining (6.37)–(6.41), one may take for example

$$\beta_{\text{pre}} := \frac{1}{288c_0} > 0. \quad (6.42)$$

The normalized walk operator has the convex decomposition

$$P_{\text{prep}} = \frac{d_1}{d} P_{\text{pad}} + \frac{D_0}{d} P_{M_q}.$$

On the orthogonal complement of the constants, P_{pad} has norm at most one and P_{M_q} has norm at most λ_0 . Hence

$$\|P_{\text{prep}} f\|_2 \leq \rho \|f\|_2, \quad \rho := \frac{d_1 + D_0 \lambda_0}{d} < 1. \quad (6.43)$$

The Margulis rotation map is strongly explicit, so all cloud and global edges can be generated in polynomial time. This proves every claim in (6.34). $\square$

6.8 Alphabet reduction by composition

The error-correcting code in the next lemma is where constant relative distance matters. A one-hot code would have relative distance tending to zero with the outer alphabet and would make the gap loss depend on the alphabet. Instead, the binary specialization of the Gilbert–Varshamov theorem ‘QRCC-III-THM-009’ gives absolute constants $R_0, \rho_0 > 0$ and, for every sufficiently large block length, a binary code of rate at least R_0 and relative distance at least ρ_0 . For any fixed finite outer alphabet we choose a long enough member and hardwire an injection

$$e : \Sigma \hookrightarrow \{0, 1\}^\ell, \quad \ell = O(\log |\Sigma|), \quad \text{rdist}(e(a), e(a')) \geq \rho_0 \quad (a \neq a'). \quad (6.44)$$

Lemma 6.10 (Composition restores a fixed alphabet). *Let $\mathcal{P}$ be the assignment tester of Theorem 6.8. There is an absolute constant*

$$\beta_{\text{comp}} := \frac{\varepsilon_{\text{AT}} \rho_0}{4} > 0 \quad (6.45)$$

such that for every fixed finite alphabet Σ and every binary constraint graph G over Σ , the composed graph $G \circ \mathcal{P}$ is a graph over the fixed tester alphabet $\Sigma_{\text{AT}} = \{0, 1\}^3$, can be constructed in time linear in $\text{size}(G)$, and has

$$\text{size}(G \circ \mathcal{P}) \leq C_{\text{comp}}(|\Sigma|) \text{size}(G), \quad (6.46)$$

and satisfies

$$\beta_{\text{comp}} \text{UNSAT}(G) \leq \text{UNSAT}(G \circ \mathcal{P}) \leq \text{UNSAT}(G). \quad (6.47)$$

Here $C_{\text{comp}}(|\Sigma|)$ may depend on the fixed outer alphabet but not on the size of G . In particular, the composition output alphabet is exactly the eight-symbol tester alphabet.

Proof. Use the code (6.44). For every outer vertex v , create one shared Boolean block $[v]$ of ℓ bits. For every outer constraint c_e on (u, v) , form a Boolean circuit $\tilde{c}_e$ on $[u] \cup [v]$ which accepts exactly when the two blocks are legal codewords $e(a), e(b)$ and (a, b) satisfies c_e . Run the assignment tester $\mathcal{P}$ on every $\tilde{c}_e$, identifying the input bits $[v]$ across all tester gadgets incident on the same outer vertex and keeping all auxiliary coordinates otherwise fresh. Because $\ell = O(\log |\Sigma|)$, every local circuit has size depending only on the fixed quantity $|\Sigma|$. If an outer relation is empty, replace its local circuit by a fixed all-rejecting gadget over Σ_{AT} ; this case already contributes one full unit of local UNSAT and does not need assignment-tester soundness. For every nonempty relation, run the tester as above. There are only finitely many bounded-size local circuits. Replicate the constraint multiset of each resulting gadget to one common positive cardinality (for example the least common multiple of their positive cardinalities); replication preserves each gadget’s violated fraction exactly. Each tester output already uses the single alphabet Σ_{AT} . After identifying shared input vertices, the union of all gadgets therefore remains a constraint graph over exactly Σ_{AT} , and for fixed Σ all local gadgets are hardwired constants. This proves the linear-time construction and (6.46).

For the upper inequality in (6.47), take an optimal outer assignment σ and put the legal codeword $e(\sigma(v))$ on each block $[v]$. Whenever c_e is satisfied, assignment-tester completeness extends the two blocks to a completely satisfying inner gadget. Define the extension arbitrarily for an outer constraint that σ violates. Since every outer edge contributes the same number of inner constraints, the composed violated fraction is at most the original one.

For the lower inequality, take an arbitrary composed assignment. By the input-symbol enforcement in Theorem 6.8, if a shared coordinate in some block $[v]$ carries a symbol outside $\{\bar{0}, \bar{1}\}$, every incident tester constraint already rejects. Changing that coordinate to either distinguished Boolean symbol therefore cannot turn any formerly satisfied constraint into a violation. Repeating this operation, we may assume without increasing the violated fraction that every shared block is

genuinely Boolean. Identify $\bar{0}, \bar{1}$ with $0, 1$ and, for each v , decode $\sigma(v)$ as a codeword closest to the Boolean string written on $[v]$. Suppose an outer edge $e = (u, v)$ is violated by this decoded pair. If the outer relation is empty, its all-rejecting gadget has violated fraction one, which is at least β_{comp} . Otherwise we claim that the concatenated input string on $[u] \cup [v]$ is at relative distance at least $\rho_0/4$ from $\text{SAT}(\tilde{c}_e)$. Indeed, every satisfying input is $e(a)e(b)$ for a satisfying outer pair (a, b) . Such a pair cannot equal $(\sigma(u), \sigma(v))$. If, say, $a \neq \sigma(u)$, nearest-codeword decoding and the triangle inequality give

$$\rho_0 \ell \leq d_H(e(a), e(\sigma(u))) \leq d_H(e(a), [u]) + d_H([u], e(\sigma(u))) \leq 2d_H(e(a), [u]). \quad (6.48)$$

Thus at least $\rho_0 \ell/2$ bits differ in one of the two ℓ -bit blocks, which is relative distance at least $\rho_0/4$ on their concatenation. Assignment-tester soundness therefore forces at least $\varepsilon_{\text{AT}} \rho_0/4$ of that edge gadget's constraints to reject. Averaging over all outer edges violated by the decoded assignment proves the lower inequality in (6.47). $\square$

6.9 The one-step amplifier and the PCP theorem

We can now freeze all constants before the input arrives. By Theorem 6.8 and Lemma 6.10, composition resets the alphabet exactly to

$$\Sigma_0 := \{0, 1\}^3, \quad |\Sigma_0| = 8. \quad (6.49)$$

Apply Lemma 6.9 to an alphabet-eight graph. This fixes the degree d , the spectral constant ρ , and β_{pre} . With these parameters, Theorem 6.6 supplies constants $\beta_{\text{pow}} > 0$ and t_0 . Choose one sufficiently large even integer $t \geq t_0$ such that

$$\alpha := \beta_{\text{comp}} \beta_{\text{pow}} \beta_{\text{pre}} \sqrt{t} \geq 2. \quad (6.50)$$

The code in the composition step still has the same absolute relative-distance constant ρ_0 ; only its fixed block length changes with the now-frozen powered alphabet. Thus β_{comp} does not deteriorate when t is enlarged.

Lemma 6.11 (One complete Dinur-style amplification round). *For the constants frozen above there are absolute constants $C > 0$ and $\varepsilon_* > 0$ and a polynomial-time transformation T on binary constraint graphs over Σ_0 such that*

$$\text{size}(T(G)) \leq C \text{size}(G), \quad \text{UNSAT}(T(G)) \geq \min\{2 \text{UNSAT}(G), \varepsilon_*\}, \quad (6.51)$$

and satisfiable instances remain satisfiable. Every output graph again has alphabet Σ_0 .

Proof. Define

$$T(G) := (\text{prep}(G))^{(t)} \circ \mathcal{P}. \quad (6.52)$$

Write $\delta = \text{UNSAT}(G)$. Lemma 6.9, Theorem 6.6, and Lemma 6.10 give

$$\text{UNSAT}(T(G)) \geq \beta_{\text{comp}} \beta_{\text{pow}} \sqrt{t} \min\{\beta_{\text{pre}} \delta, 1/t\}. \quad (6.53)$$

If $\beta_{\text{pre}} \delta \leq 1/t$, (6.50) makes this at least 2δ . Otherwise it is at least

$$\varepsilon_* := \frac{\beta_{\text{comp}} \beta_{\text{pow}}}{\sqrt{t}} > 0. \quad (6.54)$$

This proves the gap recurrence. Preprocessing, powering, and composition each preserve perfect satisfiability. Their size factors are constant because d, t, Σ_0 , the powered alphabet, and the inner tester are all frozen independently of the input. Their product is the constant C . $\square$

A polynomial-gap starting graph. Let $\Phi = C_1 \wedge \cdots \wedge C_m$ be a 3CNF formula in the normal form produced by ‘QRCC-II-THM-006’, after deleting tautologies and repeated literals and treating an empty clause as a fixed no-instance. For every clause fix an ordering of its at most three distinct variables. Interpret the first coordinates of a symbol in $\Sigma_0 = \{0, 1\}^3$ as their truth values and ignore any unused coordinates; let $S_i \subseteq \Sigma_0$ be the symbols that satisfy the clause. Create one vertex c_i per clause and one anchor vertex a , all over Σ_0 . An anchor constraint (c_i, a) accepts exactly when the label of c_i lies in S_i , independently of the anchor label. For every pair of clauses sharing a variable, add a consistency constraint that accepts exactly when both local labels are satisfying and assign the same truth value to every shared variable.

A global satisfying truth assignment gives a satisfying graph assignment. Conversely, if every graph constraint is satisfied, the pairwise consistency constraints make the local clause values define one global truth assignment, and every anchor constraint says its clause is true. Thus the graph is satisfiable exactly when Φ is. It has at most

$$m + \binom{m}{2} \leq m^2 \quad (m \geq 2) \quad (6.55)$$

undirected constraints. Therefore an unsatisfiable instance has

$$\text{UNSAT}(G_0) \geq m^{-2}. \quad (6.56)$$

The construction takes polynomial time and already uses the reset alphabet Σ_0 .

Theorem 6.12 (PCP theorem via gap amplification).

$$\boxed{\text{NP} = \text{PCP}[O(\log n), O(1)]}. \quad (6.57)$$

More concretely, there are constants q and $\varepsilon > 0$ and a polynomial-time reduction from 3SAT instances of encoding length n to binary constant-alphabet constraint systems of size $n^{O(1)}$ such that satisfiable formulas map to satisfiable systems, whereas unsatisfiable formulas map to systems with UNSAT at least ε . Sampling one constraint and reading its two alphabet symbols yields a PCP verifier with logarithmic randomness and constant query complexity; encoding the constant alphabet in bits gives the binary PCP convention of Definition 6.1.

Proof. The inclusion $\text{NP} \subseteq \text{PCP}[O(\log n), O(1)]$ is the substantive direction. By ‘QRCC-II-THM-006’, it suffices to start from 3SAT. Build G_0 as above. If the input is a no-instance then $\delta_0 := \text{UNSAT}(G_0) \geq m^{-2}$. Define $G_{i+1} = T(G_i)$, with T from Lemma 6.11. Inductively,

$$\delta_i \geq \min\{2^i \delta_0, \varepsilon_*\}, \quad \text{size}(G_i) \leq C^i \text{size}(G_0). \quad (6.58)$$

Choose

$$k := \left\lceil \log_2 \frac{\varepsilon_*}{\delta_0} \right\rceil_+ = O(\log m). \quad (6.59)$$

Then $\text{UNSAT}(G_k) \geq \varepsilon_*$, while

$$\text{size}(G_k) \leq C^{O(\log m)} m^{O(1)} = m^{O(1)}. \quad (6.60)$$

Every transformation is polynomial time and preserves satisfiability, so this is the required constant-gap reduction.

Now use the PCP–CSP dictionary from Proposition 6.3. A verifier samples one of the polynomially many final constraints uniformly, reads the two Σ_0 symbols, and checks the binary relation. It has

perfect completeness, soundness at most $1 - \varepsilon_*$, randomness $O(\log n)$, and two constant-alphabet queries. Since $|\Sigma_0| = 8$, three proof bits encode one symbol, so at most six binary proof bits are read. A fixed number of independent repetitions, accepting only if every copy accepts, lowers soundness to at most $1/2$ while preserving $O(\log n)$ randomness and $O(1)$ queries.

For the reverse inclusion, let $L \in \text{PCP}[O(\log n), O(1)]$. By the accessible-proof-length bound (6.2), only polynomially many proof cells are ever queried. An NP machine guesses these cells and deterministically enumerates the polynomially many verifier random strings, computing the exact acceptance fraction in polynomial time; it accepts exactly when every random string accepts. Perfect completeness supplies such a guess on yes-instances, while soundness below one forbids one on no-instances. Thus $\text{PCP}[O(\log n), O(1)] \subseteq \text{NP}$, proving (6.57). $\square$

stage	size	alphabet / degree	gap	role
3SAT local CSP	$n^{O(1)}$	8, arbitrary degree	$\delta_0 \geq n^{-O(1)}$	start
preprocess	$\times C_{\text{pre}}$	8, fixed $d, \rho < 1$	$\geq \beta_{\text{pre}} \delta$	regularize / expand
power	$\times C_{\text{pow}}$	fixed large alphabet; degree d^t	$\geq \beta_{\text{pow}} \sqrt{t} \cdot \min\{\beta_{\text{pre}} \delta, 1/t\}$	amplify
compose	$\times C_{\text{comp}}$	reset to 8	$\times \beta_{\text{comp}}$	alphabet reduction
iterate $O(\log n)$	$n^{O(1)}$	always reset to 8	$\geq \varepsilon_*$	constant gap
sample constraint	polynomial proof	two symbols / six bits	reject $\geq \varepsilon_*$	PCP verifier

Figure 6.1: The PASS78 PCP pipeline. The table records the invariant that makes gap amplification a polynomial-size proof system rather than only a local improvement in soundness.

Why the iteration is the theorem. Powering alone is not the PCP theorem: it raises the gap and simultaneously enlarges the local alphabet. Composition alone is not the PCP theorem: it restores a small alphabet but does not create a constant gap. Equation (6.51) is the invariant that makes the alternation reusable, and (6.60) is the reason $O(\log n)$ repetitions remain polynomial size.

6.10 Projection games and Label Cover

The constant-gap CSP produced above can be put into the projection form used throughout hardness of approximation. We fix the exact game convention before stating the reduction.

Definition 6.13 (Projection game / Label Cover). A *projection game* is a bipartite multigraph $H = (U, W, E)$, finite left and right alphabets A, B , and, for every edge $e = (u, w)$, a projection map

$$\pi_e : A \rightarrow B.$$

A labeling (α, β) , with $\alpha : U \rightarrow A$ and $\beta : W \rightarrow B$, satisfies e when $\pi_e(\alpha(u)) = \beta(w)$. Its value is the maximum fraction of satisfied edges and

$$\text{UNSAT}(H) := 1 - \text{val}(H). \quad (6.61)$$

Parallel edges are allowed and counted with multiplicity. We use *Label Cover* synonymously with this bipartite projection-game convention.

Theorem 6.14 (Label Cover completeness from the PCP theorem). *There are absolute constants $|A|, |B|$ and $\gamma > 0$ and a polynomial-time reduction from 3SAT to Label Cover such that*

$$\begin{aligned} \Phi \text{ satisfiable} &\implies \text{UNSAT}(H_\Phi) = 0, \\ \Phi \text{ unsatisfiable} &\implies \text{UNSAT}(H_\Phi) \geq \gamma. \end{aligned} \quad (6.62)$$

The output has polynomial size and every constraint is a genuine projection.

Proof. Use the constant-gap binary constraint system $G = (V, E, \Sigma_0, C)$ produced in the proof of Theorem 6.12; let $\delta := \text{UNSAT}(G)$, so $\delta = 0$ in the yes-case and $\delta \geq \varepsilon_*$ in the no-case. Put $B := \Sigma_0$ and $A := [|\Sigma_0|^2]$, both absolute constant alphabets.

For every base variable $v \in V$, create a right vertex w_v . For every constraint $e = (u, v)$ whose relation $C_e \subseteq \Sigma_0^2$ is nonempty, create one left vertex z_e . Choose a surjection $\text{dec}_e : A \rightarrow C_e$ by listing all satisfying pairs and repeating the first pair as needed. Add two projection edges from z_e to w_u, w_v , mapping a left label to the first and second coordinates of dec_e , respectively.

If $C_e = \emptyset$, it is violated by every base assignment. Replace it by a disjoint fixed projection gadget with one left vertex, one right vertex, and two parallel edges whose projection maps are the two distinct constants $b_0, b_1 \in B$. Every labeling violates at least one of those two edges. Thus every base constraint contributes exactly two projection edges.

If G is satisfiable, no relation used by a satisfying assignment is empty; label every right vertex by the satisfying base symbol and each left vertex by an index decoding to the corresponding satisfying pair. All projection edges are satisfied.

Conversely, fix any Label-Cover labeling and read its right labels as an assignment $\sigma : V \rightarrow \Sigma_0$. For every nonempty base constraint violated by σ , the two projection edges incident on its left vertex cannot both be satisfied: if they were, the decoded left pair would equal $(\sigma(u), \sigma(v))$ and would belong to C_e . Every empty base constraint gadget also violates at least one of its two edges. Hence if $m = |E|$, at least δm of the $2m$ projection edges are violated, so

$$\text{UNSAT}(H_\Phi) \geq \frac{\delta}{2}. \quad (6.63)$$

Take $\gamma = \varepsilon_*/2$. The construction has linear size in G , hence polynomial size in the original formula. $\square$

6.11 A first approximation-hardness consequence

Corollary 6.15 (Constant-gap hardness for MAX-3SAT). *There exists an absolute constant $\eta > 0$ for which it is NP-hard to distinguish between*

1. *a satisfiable 3CNF formula, and*
2. *a 3CNF formula for which every assignment satisfies at most a $(1 - \eta)$ -fraction of the clauses.*

Consequently MAX-3SAT has no polynomial-time approximation scheme unless $P = NP$.

Proof. Start from the constant-gap Label-Cover instance of Theorem 6.14. Its two alphabets are absolute constants, so encode one left or right label by at most $b = O(1)$ Boolean variables. For every projection edge $e = (u, w)$, write the Boolean predicate P_e which accepts exactly when both endpoint bit strings are legal label encodings and $\pi_e(\alpha(u)) = \beta(w)$. Since b is constant, the truth table of P_e can be represented by a CNF of at most 2^{2b} clauses, each of width at most $2b$: for every rejecting local bit string, include the clause that excludes exactly that string.

Apply the clause-chain conversion from ‘QRCC-II-THM-006’ to every constant-width clause, using fresh extension variables separately for each edge gadget. There is therefore an absolute constant K such that every projection edge becomes at most K 3CNF clauses. If Label Cover is satisfiable, choose the corresponding endpoint encodings and then the local extension variables; all clauses are satisfied.

For an arbitrary Boolean assignment in the no-case, decode every valid endpoint bit string as its label and decode an invalid string arbitrarily. Whenever a projection edge is violated by the decoded labels, its predicate P_e is false: invalid encodings are rejected by definition, and valid encodings are accepted only for a satisfied projection. Therefore that edge's CNF is false, and the equisatisfiable 3CNF gadget has at least one unsatisfied clause for every setting of its fresh extension variables. Since at least a γ -fraction of Label-Cover edges are violated, at least $\gamma|E|$ clauses among at most $K|E|$ output clauses are false. Thus one may take

$$\eta := \frac{\gamma}{K} > 0. \tag{6.64}$$

The reduction is polynomial time and has perfect completeness. A PTAS run with accuracy parameter $\eta/2$ would distinguish the two cases, contradicting NP-hardness unless $P = NP$. $\square$

Where the proof frontier moves next. The PCP and first Label-Cover/approximation interfaces are now closed. Chapter 7 can therefore state Raz parallel repetition on the exact projection-game model of Definition 6.13; later in the volume, proof complexity can use constant-gap PCP/Label-Cover reductions without treating them as unexplained folklore.

Chapter 7

Parallel Repetition of Two-Prover Games

Chapter 6 produced a constant-gap projection game. A natural next question is whether one can make the soundness much smaller by asking several independent copies of the game at once and accepting only when every coordinate is won. Independent play would multiply the values immediately, but optimal provers in the repeated game are not required to play independently: each answer coordinate may depend on the prover's entire vector of questions. This cross-coordinate dependence is the reason parallel repetition is a theorem rather than a one-line calculation.

This chapter first fixes the finite classical game model and the repetition operation. It then source-owns Raz's deep exponential-decay theorem at a conservative asymptotic strength that is faithful to the original theorem and does not import the later sharper Holenstein exponent. All structural bookkeeping needed by the QRCC Label-Cover interface is proved locally, and the chapter ends by amplifying the constant soundness from Chapter 6 to any fixed target soundness.

Learning goals. After this chapter, a reader should be able to define the value of a finite two-prover one-round game, form its conjunctive parallel repetition, explain why correlated cross-coordinate strategies make the upper bound nontrivial, prove the product-strategy lower bound, verify that projection games remain projection games under repetition, state Raz's theorem with its answer-length dependence explicit, and use it to turn the PASS78 constant-gap Label-Cover reduction into a fixed-target soundness reduction without losing polynomial size.

7.1 Finite games and their repeated product

Definition 7.1 (Two-prover game, value, and parallel repetition). A *finite classical two-prover one-round game* is a tuple

$$G = (X, Y, A, B, \mu, V), \quad (7.1)$$

where X, Y are finite question sets, A, B are finite answer alphabets, μ is a probability distribution on $X \times Y$, and

$$V : X \times Y \times A \times B \longrightarrow \{0, 1\} \quad (7.2)$$

is the verifier predicate. The verifier samples $(x, y) \sim \mu$, sends only x to the first prover and only y to the second, receives answers $a \in A$ and $b \in B$, and accepts when $V(x, y, a, b) = 1$. The provers do not communicate after receiving their questions and have no entanglement or other nonclassical resource. Shared classical randomness is allowed, but it does not change the value because the

acceptance probability of a randomized joint strategy is a convex combination of deterministic strategy values.

The *value* is therefore

$$\text{val}(G) := \max_{f: X \rightarrow A, g: Y \rightarrow B} \Pr_{(x,y) \sim \mu} [V(x, y, f(x), g(y)) = 1]. \quad (7.3)$$

For an integer $k \geq 1$, the k -fold *conjunctive parallel repetition* $G^{\otimes k}$ samples k independent question pairs

$$(x_1, y_1), \dots, (x_k, y_k) \stackrel{\text{iid}}{\sim} \mu, \quad (7.4)$$

sends $x^k = (x_1, \dots, x_k)$ to the first prover and $y^k = (y_1, \dots, y_k)$ to the second, receives answer vectors $a^k \in A^k$ and $b^k \in B^k$, and accepts exactly when

$$\prod_{i=1}^k V(x_i, y_i, a_i, b_i) = 1. \quad (7.5)$$

Thus a repeated-game strategy is an arbitrary pair of functions $F : X^k \rightarrow A^k$ and $H : Y^k \rightarrow B^k$; its i -th answer is not required to depend only on the i -th question.

For answer-size bookkeeping put

$$s(G) := \max \{1, \lceil \log_2 |A| \rceil + \lceil \log_2 |B| \rceil\}. \quad (7.6)$$

This is the total answer length in bits up to an additive constant. A *projection game* is the special case of Definition 6.13 viewed as a game: the verifier samples an edge, asks for its two endpoint labels, and accepts exactly when the edge projection is satisfied.

The definition is compatible with the classical separated-prover convention of Definition 5.1. The new ingredient is not a new prover model, but the coordinate-product question distribution and the all-coordinates acceptance predicate.

Lemma 7.2 (Elementary repetition facts). *Let G be a finite classical two-prover game and let $k \geq 1$.*

1. *Product strategies give*

$$\text{val}(G^{\otimes k}) \geq \text{val}(G)^k. \quad (7.7)$$

In particular, perfect completeness is preserved.

2. *If G is a projection game with left/right alphabets A, B , then $G^{\otimes k}$ is again a projection game with alphabets A^k, B^k . For an original edge e with projection $\pi_e : A \rightarrow B$, the projection attached to an edge tuple $e^k = (e_1, \dots, e_k)$ is*

$$\pi_{e^k}(a_1, \dots, a_k) := (\pi_{e_1}(a_1), \dots, \pi_{e_k}(a_k)). \quad (7.8)$$

If the original verifier is uniform on a multiset of M edges, the repeated verifier is uniform on the M^k ordered edge tuples, counted with multiplicity.

Proof. Choose deterministic one-copy strategies f, g attaining $\text{val}(G)$; the maximum exists because the strategy sets are finite. In the repeated game use them coordinatewise:

$$F(x_1, \dots, x_k) = (f(x_1), \dots, f(x_k)), \quad H(y_1, \dots, y_k) = (g(y_1), \dots, g(y_k)).$$

The k question pairs are independent, so the events that these product strategies win their individual coordinates are independent and each has probability $\text{val}(G)$. The probability of winning all coordinates is therefore $\text{val}(G)^k$, proving (7.7).

For a projection game, an ordered edge tuple independently chooses one base projection in every coordinate. Once the left answer vector is fixed, equation (7.8) uniquely specifies the only right answer vector satisfying all k coordinate constraints. Hence the repeated predicate is again a projection predicate. Uniform independent sampling from M base edges is exactly uniform sampling from the M^k ordered tuples. This also proves the stated size bookkeeping. $\square$

Why the opposite inequality is hard. Equation (7.7) is a lower bound on the repeated value. It is not legitimate to reverse it by saying that the provers must play coordinatewise: the functions F, H see all of their respective questions. Raz’s theorem controls precisely the advantage potentially created by those cross-coordinate dependencies.

7.2 Raz’s parallel-repetition theorem

Theorem 7.3 (Raz parallel repetition, selected asymptotic form). *There are universal constants $c_R \geq 2$ and $C_R > 0$ such that the following holds. Let G be any finite classical two-prover one-round game in the sense of Definition 7.1. If*

$$\text{val}(G) \leq 1 - \varepsilon, \quad 0 < \varepsilon \leq \frac{1}{2}, \quad (7.9)$$

then for every integer $k \geq 1$,

$$\text{val}(G^{\otimes k}) \leq (1 - \varepsilon^{c_R})^{C_R k / s(G)}. \quad (7.10)$$

Equivalently, for every fixed finite answer alphabet and every fixed $\varepsilon > 0$, the value decays exponentially in k .

Source boundary and quantitative caution. This is a deep source-owned theorem. The canonical source is Ran Raz, *A Parallel Repetition Theorem*, *SIAM Journal on Computing* 27(3) (1998), 763–803, DOI 10.1137/S0097539795280895. The primary paper states exponential decay for arbitrary classical two-prover one-round games and makes the exponent depend only on the original error and the total number of possible answers, with logarithmic dependence on that answer count. Raz’s later formulation records the standard safe asymptotic envelope $(1 - \varepsilon^c)^{\Omega(k/s)}$ for a universal constant c . QRCC freezes that envelope as (7.10). It does *not* identify c_R with the later Holenstein exponent 3, and it does not claim strong repetition $(1 - \varepsilon)^{\Omega(k/s)}$, which is false for general games.

Proof architecture, not a substitute for the proof. Suppose a repeated strategy wins all coordinates with unexpectedly large probability. Condition on a suitably chosen set of already-won coordinates. Entropy/information inequalities show that for many remaining coordinates this conditioning cannot substantially distort both the local question distribution and the relevant answer information. One then extracts from the repeated strategy a one-copy strategy whose success is too large, contradicting (7.9). The quantitative argument must control how much conditioning can be paid for across coordinates; this is where the answer-length dependence enters. These ideas explain the theorem’s interface, but QRCC does not count this paragraph as a proof of Raz’s theorem.

7.2.1 A four-question example

Consider the classical game with $X = Y = A = B = \{0, 1\}$, uniform independent questions, and acceptance condition

$$a \oplus b = x \wedge y. \quad (7.11)$$

For deterministic answers $a = f(x)$, $b = g(y)$, the four required equations are

$$f(0) \oplus g(0) = 0, \quad f(0) \oplus g(1) = 0, \quad f(1) \oplus g(0) = 0, \quad f(1) \oplus g(1) = 1. \quad (7.12)$$

XORing all four left sides gives zero because every value of f and g occurs twice, while XORing the right sides gives one. Thus no deterministic strategy wins all four question pairs and $\text{val}(G) \leq 3/4$. The constant strategy $f = g = 0$ loses only on $(1, 1)$, so

$$\text{val}(G) = \frac{3}{4}. \quad (7.13)$$

Lemma 7.2 gives the elementary lower bound $\text{val}(G^{\otimes k}) \geq (3/4)^k$; Theorem 7.3 supplies a possibly weaker but universal exponential upper bound. The gap between those two statements is a useful reminder that parallel repetition controls optimal correlated repeated strategies, not only coordinatewise play.

7.3 Fixed-target Label-Cover soundness

Corollary 7.4 (Constant-target soundness amplification for Label Cover). *Fix any constant $\eta \in (0, 1)$. There is a polynomial-time reduction from 3SAT to a constant-alphabet projection game H_η such that*

$$\begin{aligned} \Phi \text{ satisfiable} &\implies \text{val}(H_\eta) = 1, \\ \Phi \text{ unsatisfiable} &\implies \text{val}(H_\eta) \leq \eta. \end{aligned} \quad (7.14)$$

For fixed η , both answer alphabets and the repetition count are absolute constants, and the output size is polynomial in $|\Phi|$.

Proof. Theorem 6.14 gives a polynomial-size constant-alphabet projection game H with perfect completeness and, on unsatisfiable formulas,

$$\text{val}(H) \leq 1 - \gamma \quad (7.15)$$

for one absolute $\gamma > 0$. Decrease γ , if necessary, so that $\gamma \leq 1/2$. Because the two answer alphabets in that theorem are fixed, so is $s_0 := s(H)$. Apply Theorem 7.3 with $\varepsilon = \gamma$. There is an absolute

$$\delta_0 := 1 - (1 - \gamma^{\text{CR}})^{C_R/s_0} > 0 \quad (7.16)$$

such that

$$\text{val}(H^{\otimes k}) \leq (1 - \delta_0)^k. \quad (7.17)$$

Choose

$$k_\eta := \left\lceil \frac{\log(1/\eta)}{-\log(1 - \delta_0)} \right\rceil. \quad (7.18)$$

This depends only on the fixed target η , not on Φ . Set $H_\eta := H^{\otimes k_\eta}$. Lemma 7.2 preserves projection form and perfect completeness, while (7.17)–(7.18) give the no-case bound.

If the base instance has N vertices and M edges, the explicit repeated instance has at most N^{k_η} question tuples on each side and M^{k_η} edge tuples. Since k_η is constant for fixed η , this is polynomial size and can be generated in polynomial time. Its alphabets A^{k_η}, B^{k_η} are likewise fixed constants. $\square$

What the corollary does and does not say. For every fixed target η , the reduction is polynomial-time and has constant alphabets. This is enough for many constant-factor hardness arguments. The statement does not assert a single polynomial-time construction when η is allowed to shrink arbitrarily with the input length; then the repetition count and the explicit product-instance size must be tracked as functions of η .

7.4 Transition to proof complexity

Parallel repetition completes the classical PCP/game layer needed in this volume. Chapter 8 changes questions: instead of asking how a verifier probabilistically checks a long proof, it asks how long particular *formal refutations* must be. The proof-system syntax, size, width, and tree-like versus DAG-like conventions must therefore be defined from scratch before any resolution lower bound is stated.

Chapter 8

Resolution Proofs, Width, and Exponential Lower Bounds

The previous chapters used probabilistic verification to make false statements locally visible. We now turn the point of view around: a propositional refutation is a deterministic certificate that a CNF has no satisfying assignment, and the question is how large such a certificate must be. Resolution is the basic test case. It has one local inference rule, it underlies the classical DPLL search paradigm, and its lower-bound theory already exhibits a central proof-complexity phenomenon: a short proof can be converted into a narrow proof, so proving that every proof must contain a wide clause automatically implies a size lower bound.

This chapter fixes the proof-system and resolution conventions first. It then proves resolution soundness and completeness and makes the tree-like-resolution/DPLL correspondence explicit. The Ben–Sasson–Wigderson size–width theorem is reconstructed in full from two restriction lemmas. We then combine this tradeoff with a locally proved sparse-expander bottleneck argument to recover an exponential lower bound for the ordinary pigeonhole principle. The final section source-freezes a classical average-case counterpart: Chvátal–Szemerédi’s exponential resolution lower bound for random fixed-density CNFs.

Learning goals. After this chapter, a reader should be able to distinguish a Cook–Reckhow proof system from a particular refutation calculus; write and restrict a resolution derivation; explain size, width, tree-likeness, and DPLL search trees without changing conventions mid-proof; prove resolution soundness and completeness; convert DPLL trees and tree-like resolution refutations in both directions; prove that a size- S resolution refutation over n variables yields width $w(F) + O(\sqrt{n \log S})$; turn a linear width lower bound into an exponential size lower bound; and explain why sparse boundary expansion makes the pigeonhole principle hard for resolution.

8.1 Proof systems and the resolution calculus

Definition 8.1 (Cook–Reckhow systems, resolution, width, and DPLL). A *propositional proof system* in the sense of Cook and Reckhow is a polynomial-time computable surjection

$$P : \{0, 1\}^* \longrightarrow \text{TAUT}, \tag{8.1}$$

where TAUT is a fixed reasonable binary encoding of propositional tautologies. A string π with $P(\pi) = \tau$ is a *P-proof* of τ , and $|\pi|$ is its size. A proof system P *polynomially simulates* a proof

system Q if there is a polynomial-time map T and a polynomial p such that

$$Q(\pi) = \tau \implies P(T(\pi)) = \tau, \quad |T(\pi)| \leq p(|\pi| + |\tau|). \quad (8.2)$$

Two systems are *polynomially equivalent* if each polynomially simulates the other.

For the concrete refutation calculus used below, a *literal* is a variable x or its negation $\neg x$. A *clause* is a finite disjunction of literals, identified with its set of literals; tautological clauses containing both x and $\neg x$ may be deleted. The empty clause is written $\perp$. A *CNF* F is a finite set of clauses interpreted conjunctively. Its *initial width* is

$$w(F) := \max_{C \in F} |C|. \quad (8.3)$$

A partial assignment ρ maps some variables to $\{0, 1\}$. The restriction $C \upharpoonright \rho$ deletes C if a literal of C is made true; otherwise it deletes all literals made false. The restricted CNF $F \upharpoonright \rho$ consists of the surviving restricted clauses.

The *resolution rule* is

$$\frac{C \vee x \quad D \vee \neg x}{C \vee D}, \quad (8.4)$$

where duplicate literals are removed and a tautological resolvent may be discarded. We also allow *weakening* $C \mapsto C \vee D$. A *resolution derivation from F* is a finite sequence of clauses in which every line is an initial clause of F , a weakening of an earlier line, or the resolvent of two earlier lines. A *resolution refutation* ends in $\perp$.

The *length* or *size* $S(\pi)$ of a derivation is its number of clause lines, including repeated initial clauses. For unsatisfiable F , let

$$S(F) := \min_{\pi \text{ refutes } F} S(\pi). \quad (8.5)$$

The *width* of π is $w(\pi) := \max_{C \in \pi} |C|$, and

$$W(F) := \min_{\pi \text{ refutes } F} w(\pi). \quad (8.6)$$

A refutation is *tree-like* if each occurrence of a derived clause is used as a premise at most once; equivalently, after duplicating initial leaves as needed, the inference DAG is a rooted binary tree whose root is $\perp$. Write $S_T(F)$ for minimum tree-like size.

A *DPLL contradiction search tree* for F is a rooted binary decision tree. Each internal node carries a variable not previously queried on that root-to-node path; its two outgoing edges set that variable to 0 and 1. Each leaf is labelled by an initial clause of F falsified by the root-to-leaf assignment. Its size is its number of nodes. Thus every root-to-leaf path is a certificate that the corresponding partial assignment already falsifies an input clause.

A proof system is *automatizable* if there is an algorithm which, on every tautology (or in a fixed refutation presentation, every contradiction) having a shortest proof of size s , outputs a proof in time polynomial in s and the input length. This term is included for later frontier statements; no automatizability assumption is used in the lower bounds below.

Why both levels of definition are needed. Equation (8.1) is the abstract complexity-theoretic notion: it lets us compare proof formalisms by simulation. Equations (8.3)–(8.6) are the syntax-level conventions needed to measure one specific calculus. Conflating them would make later phrases such as “resolution width” or “polynomial simulation” ambiguous.

Lemma 8.2 (Restrictions preserve resolution). *If π is a resolution derivation from F and ρ is a partial assignment, then after deleting satisfied lines and suppressing repetitions, the surviving restrictions of the lines of π form a resolution derivation from $F \upharpoonright \rho$. Neither size nor width increases.*

Proof. Initial clauses restrict to initial clauses or disappear. Weakening remains weakening after deleting satisfied literals. For a resolution step on x , if x is unset then the restricted conclusion is still a resolvent (possibly followed by weakening); if x is set, one parent is satisfied or the pivot disappears and the restricted conclusion is a weakening of the surviving parent. Deleting satisfied and duplicate lines gives the claimed derivation. Restriction only removes literals and lines. $\square$

8.2 Soundness, completeness, and DPLL

Theorem 8.3 (Resolution soundness, completeness, and tree search). *Let F be a finite CNF.*

1. *Every clause derived by resolution and weakening is a semantic consequence of F . Hence a resolution refutation exists only if F is unsatisfiable.*
2. *If F is unsatisfiable, then F has a resolution refutation.*
3. *DPLL contradiction search trees and tree-like resolution refutations polynomially simulate one another. With the conventions of Definition 8.1, each conversion has only constant-factor overhead in the number of tree nodes/lines. Thus DPLL and tree-like resolution are polynomially equivalent refutation systems.*

Proof. For soundness, suppose an assignment satisfies both premises $C \vee x$ and $D \vee \neg x$. If $x = 0$, then it must satisfy C ; if $x = 1$, it must satisfy D . In either case it satisfies $C \vee D$. Weakening is also sound. Induction over the derivation shows that every line follows from F , whereas no assignment satisfies $\perp$.

For completeness, induct on the number of variables. If there are none, unsatisfiability means that $\perp \in F$. Otherwise choose a variable x . Both restrictions $F \upharpoonright (x = 0)$ and $F \upharpoonright (x = 1)$ are unsatisfiable, for a satisfying assignment of either restriction would extend to one of F . By induction they have refutations. Lift the first refutation back to F by adjoining the literal x , which is false under $x = 0$, to every surviving line; this derives the unit clause x . Similarly, lifting the $x = 1$ refutation derives $\neg x$. Resolving these two units yields $\perp$.

For the search-tree correspondence, first start from a DPLL tree. At a leaf labelled by an input clause C , use C as the leaf of a tree-like derivation. Suppose an internal node queries x , with recursively constructed child derivations ending in clauses C_0, C_1 falsified by the partial assignments at the two children. Weaken them to $C_0 \vee x$ on the $x = 0$ branch and $C_1 \vee \neg x$ on the $x = 1$ branch, then resolve on x . The resulting clause is falsified by the assignment at the parent. At the root, the empty assignment falsifies only $\perp$, so the construction ends in a tree-like refutation. There are at most two weakening lines and one resolution line per internal search node.

Conversely, take a tree-like refutation and inspect its final inference. After removing vacuous weakenings, the last step resolves unit clauses x and $\neg x$. Query x at the root. Restrict the subtree deriving x by $x = 0$; Lemma 8.2 turns it into a tree-like refutation of $F \upharpoonright (x = 0)$. Restrict the subtree deriving $\neg x$ by $x = 1$ and obtain a refutation of $F \upharpoonright (x = 1)$. Recurse. The variable just fixed vanishes under restriction, so no root-to-leaf path queries it again. Every recursive step consumes one resolution node and restriction never increases the number of proof nodes. Leaves occur when a restricted input clause becomes empty, exactly the DPLL leaf condition. Hence the resulting DPLL tree has size linear in the original proof tree. $\square$

8.2.1 A complete four-clause example

Let

$$F_{\square} := (x \vee y) \wedge (x \vee \neg y) \wedge (\neg x \vee y) \wedge (\neg x \vee \neg y). \quad (8.7)$$

Resolving the first two clauses on y gives x ; resolving the last two gives $\neg x$; one more resolution step gives $\perp$. The refutation has width two. The matching DPLL tree queries x at the root and then y on each branch. Every one of the four complete assignments reaches a leaf labelled by the unique clause of (8.7) that it falsifies. This small example is the exact local picture repeated recursively in the proof of Theorem 8.3.

8.3 Short proofs are narrow

The next theorem is the bridge from semantic/combinatorial width lower bounds to syntactic size lower bounds. We prove it with explicit constants sufficient for every downstream application.

Lemma 8.4 (One-variable lifting and merging). *Let x be a variable and let $a \in \{0, 1\}$. Define the literal falsified by $x = a$ as*

$$\lambda_{x,a} := \begin{cases} x, & a = 0, \\ \neg x, & a = 1. \end{cases} \quad (8.8)$$

If $F \upharpoonright (x = a)$ derives a clause A in width at most q , then F derives $A \vee \lambda_{x,a}$ in width at most $q + 1$.

Consequently, if $F \upharpoonright (x = a)$ has a refutation of width at most $K - 1$ and $F \upharpoonright (x = 1 - a)$ has one of width at most K , then

$$W(F) \leq \max\{K, w(F)\}. \quad (8.9)$$

Proof. Take the restricted derivation and adjoin $\lambda_{x,a}$ to every line. If a restricted initial clause came from an original clause containing the opposite pivot literal, the lifted line is that original clause; otherwise it is a weakening of an original clause. Resolution steps remain resolution steps after adjoining the same literal to both parents and conclusion. This proves the first claim.

For the second, lift the first restricted refutation to derive the unit $\lambda_{x,a}$. Resolve that unit against every original clause containing the complementary literal. The resulting clauses are precisely the nontrivial clauses of $F \upharpoonright (x = 1 - a)$, with width at most $w(F)$. Append the assumed width- K refutation of that restriction. The maximum width is (8.9). $\square$

Theorem 8.5 (Ben-Sasson–Wigderson width–size tradeoff). *Let F be an unsatisfiable CNF on $n \geq 1$ variables. Then*

$$W(F) \leq w(F) + \lceil \log_2 S_T(F) \rceil \quad (8.10)$$

for tree-like resolution. For general resolution, if $S := S(F)$, put

$$d := \left\lceil \sqrt{2n \ln(S + 1)} \right\rceil. \quad (8.11)$$

Then

$$W(F) \leq w(F) + 2d. \quad (8.12)$$

In particular, with $\Delta_W := \max\{W(F) - w(F) - 2, 0\}$,

$$S(F) \geq \exp\left(\frac{\Delta_W^2}{8n}\right) - 1. \quad (8.13)$$

Thus a family of bounded-width CNFs on $O(n)$ variables with refutation width $\Omega(n)$ requires resolution size $\exp(\Omega(n))$.

Proof. For (8.10), induct on $b+n$ under the hypothesis $S_T(F) \leq 2^b$. The case $b = 0$ means that $\perp$ is already an input clause. Otherwise the final tree-like resolution step resolves two subtrees on some pivot x . One subtree has size at most half the total. Restrict the two subtrees by the assignments falsifying their final pivot units and use induction: one branch gains at most $b-1$ beyond initial width and the other at most b . Lemma 8.4 merges them without exceeding $w(F) + b$.

For the general case, let π be a minimum-size refutation of size S , and call a clause of π *fat* if its width is greater than d . If $d \geq n$, then every non-tautological clause has width at most $n \leq w(F) + 2d$, so assume $d < n$. Define

$$a := \left(1 - \frac{d}{2n}\right)^{-1} > 1. \quad (8.14)$$

We prove the following stronger claim by induction on $b+n$: if a resolution refutation of a restriction of F has fewer than a^b fat clauses, where fatness is still measured against the fixed threshold d , then that restriction has a refutation of width at most

$$d + w(F) + b. \quad (8.15)$$

For $b = 0$ there are no fat clauses, so the given refutation already has width at most $\max\{d, w(F)\}$.

For the induction step, let M be the number of fat clauses. Each contains more than d literals, so the total number of literal occurrences among fat clauses exceeds dM . There are at most $2n$ possible literals. Hence some literal ℓ occurs in more than

$$\frac{d}{2n}M \quad (8.16)$$

fat clauses. Set ℓ to true. Lemma 8.2 deletes every clause counted in (8.16), leaving fewer than

$$\left(1 - \frac{d}{2n}\right)M < a^{b-1} \quad (8.17)$$

fat clauses. The induction hypothesis gives width at most $d + w(F) + b - 1$ on this branch. Under the opposite assignment, the variable disappears and the number of fat clauses is still below a^b , so induction on the number of remaining variables gives width at most $d + w(F) + b$. Lemma 8.4 combines the two branches and preserves the latter bound. This proves (8.15).

Finally,

$$\ln a = -\ln\left(1 - \frac{d}{2n}\right) \geq \frac{d}{2n}, \quad (8.18)$$

so

$$a^d \geq \exp\left(\frac{d^2}{2n}\right) \geq S + 1 > S. \quad (8.19)$$

The number of fat clauses is at most $S < a^d$. Apply (8.15) with $b = d$ to obtain (8.12). Since $d \leq \sqrt{2n \ln(S+1)} + 1$, equation (8.12) implies

$$\Delta_W \leq 2\sqrt{2n \ln(S+1)},$$

which rearranges to (8.13). □

Width search. There are at most

$$\sum_{j=0}^q 2^j \binom{n}{j} \leq (1 + 2n)^q \quad (8.20)$$

non-tautological clauses of width at most q . Saturating this finite set under resolution gives a deterministic $n^{O(q)}$ -time search for a width- q refutation. Theorem 8.5 therefore explains why width is simultaneously a lower-bound measure and an algorithmic proof-search parameter.

8.4 The pigeonhole principle

For $n \geq 1$, the standard pigeonhole CNF PHP_n^{n+1} has variables $x_{i,j}$ for pigeons $i \in [n+1]$ and holes $j \in [n]$, with clauses

$$P_i := \bigvee_{j=1}^n x_{i,j}, \quad H_{i,i',j} := \neg x_{i,j} \vee \neg x_{i',j} \quad (i < i'). \quad (8.21)$$

The pigeon clauses demand that every pigeon enters a hole; the collision clauses forbid two pigeons from entering the same hole. The formula is unsatisfiable because an injection from $[n+1]$ into $[n]$ does not exist. Its large initial width n prevents a direct use of (8.13), so we first restrict it to a sparse constant-width subformula.

Lemma 8.6 (A sparse boundary-expanding pigeon graph exists). *For all sufficiently large n , there is a bipartite graph $G = (P, H; E)$ with $|P| = n+1$, $|H| = n$, every pigeon of degree exactly five, and the following property. If*

$$r := \left\lfloor \frac{n}{4096} \right\rfloor, \quad (8.22)$$

then every nonempty $S \subseteq P$ with $|S| \leq r$ has at least $|S|$ boundary holes, where

$$\partial S := \{h \in H : |N(h) \cap S| = 1\}. \quad (8.23)$$

Proof. Independently for each pigeon choose a uniformly random five-element subset of H . Fix $S \subseteq P$ with $|S| = s \leq r$. If $|\partial S| < s$, then, writing $N(S)$ for the neighborhood, the $5s$ incident edges satisfy

$$5s \geq |\partial S| + 2(|N(S)| - |\partial S|),$$

so $|N(S)| < 3s$. Hence all neighbors of S lie in some $T \subseteq H$ of size $3s$. For fixed S, T ,

$$\Pr[N(S) \subseteq T] = \left(\frac{\binom{3s}{5}}{\binom{n}{5}} \right)^s \leq \left(\frac{3s}{n} \right)^{5s}. \quad (8.24)$$

A union bound and $\binom{a}{b} \leq (ea/b)^b$ give

$$\begin{aligned} \Pr[\exists S, |S| = s, |\partial S| < s] &\leq \binom{n+1}{s} \binom{n}{3s} \left(\frac{3s}{n} \right)^{5s} \\ &\leq \left(18e^4 \frac{s}{n} \right)^s \leq \left(\frac{18e^4}{4096} \right)^s. \end{aligned} \quad (8.25)$$

Because $18e^4/4096 < 1/4$, summing (8.25) over $1 \leq s \leq r$ gives total failure probability less than $\sum_{s \geq 1} 4^{-s} = 1/3$. Thus a graph with the required property exists. $\square$

For later use, a *matching of a pigeon set* $S \subseteq P$ *into the holes* is an injective map $m : S \rightarrow H$ such that $(p, m(p)) \in E$ for every $p \in S$.

Lemma 8.7 (Finite Hall matching criterion). *Let $B = (L, R; E)$ be a finite bipartite graph. There is a matching of all vertices of L into R if and only if*

$$|N(X)| \geq |X| \quad \text{for every } X \subseteq L. \quad (8.26)$$

Proof. Necessity is immediate: a matching sends distinct vertices of X to distinct members of $N(X)$. For sufficiency, induct on $|L|$. The empty-left-side case is trivial, so assume $L \neq \emptyset$. If every nonempty proper $X \subsetneq L$ satisfies a strict inequality $|N(X)| > |X|$, choose any edge (u, v) , delete u, v , and apply induction: for $Y \subseteq L \setminus \{u\}$, deleting v removes at most one neighbor, so $|N'(Y)| \geq |N(Y)| - 1 \geq |Y|$. Reinsert (u, v) .

Otherwise some nonempty proper $X \subsetneq L$ has $|N(X)| = |X|$. Induction matches X into $N(X)$. In the residual bipartite graph on $L \setminus X$ and $R \setminus N(X)$, every $Y \subseteq L \setminus X$ has

$$|N_{\text{res}}(Y)| \geq |N(X \cup Y)| - |N(X)| \geq |X| + |Y| - |X| = |Y|.$$

Induction matches the residual left vertices as well, and the two matchings are disjoint. $\square$

For such a graph, let $G\text{-PHP}$ be the subformula of (8.21) using only variables $x_{p,h}$ for edges $(p, h) \in E$, with each pigeon clause restricted to its five neighbors and with collision clauses only for pairs of present edges meeting the same hole. This CNF has at most $5(n+1)$ variables and initial width five.

Lemma 8.8 (Boundary expansion forces width). *For the graph in Lemma 8.6,*

$$W(G\text{-PHP}) \geq \frac{r}{12}. \quad (8.27)$$

Proof. For a pigeon p , let A_p be the conjunction of its pigeon clause and all collision clauses in which a variable of pigeon p occurs. For a clause C , define

$$\mu(C) := \min\{|S| : S \subseteq P, \bigwedge_{p \in S} A_p \models C\}. \quad (8.28)$$

Every input clause has $\mu \leq 1$. If C is a resolvent of B, D , then $\mu(C) \leq \mu(B) + \mu(D)$; weakening cannot increase μ .

Every set of at most r pigeons can be matched to distinct holes. Indeed, every subset S' of such a set has $|N(S')| \geq |\partial S'| \geq |S'|$, so Hall's condition holds. Setting the matching-edge variable of each selected pigeon to one and all other edge variables to zero satisfies their A_p 's. Therefore

$$\mu(\perp) > r. \quad (8.29)$$

In any refutation choose the first clause C with $\mu(C) \geq r/2$. It is not an input clause. If it is obtained by resolution from earlier clauses, subadditivity gives $\mu(C) < r$; weakening also preserves the upper bound. Thus

$$\frac{r}{2} \leq s := \mu(C) < r. \quad (8.30)$$

Choose a minimum witness S of size s for (8.28).

Fix a boundary hole $h \in \partial S$ and let $p \in S$ be its unique neighbor in S . By minimality, $A_{S \setminus \{p\}} \not\models C$, so there is an assignment α satisfying all these conjuncts and falsifying C . Modify α

by setting $x_{p,h} = 1$, setting every other edge variable of pigeon p to zero, and setting every other edge variable incident with hole h to zero. The resulting assignment satisfies A_p and preserves all A_q for $q \in S \setminus \{p\}$: the boundary property says that no such q is adjacent to h , and changing other variables from one to zero can only help collision clauses. Hence the new assignment satisfies A_S and must satisfy C . Consequently at least one variable changed in this modification must occur in C .

A variable $x_{q,u}$ can be charged in this way to at most six boundary holes: at most five for which q is the unique boundary pigeon (because q has degree five), plus possibly the single hole u through the hole-clearing part of the modification. Therefore

$$|C| \geq \frac{|\partial S|}{6} \geq \frac{s}{6} \geq \frac{r}{12},$$

proving (8.26). □

Theorem 8.9 (Exponential resolution lower bound for the pigeonhole principle). *There is an absolute constant $c_{\text{PHP}} > 0$ and an integer n_0 such that for every $n \geq n_0$, every general resolution refutation of the standard pigeonhole formula PHP_n^{n+1} has size at least*

$$\exp(c_{\text{PHP}}n). \quad (8.31)$$

Proof. Choose G from Lemma 8.6. The sparse formula $G\text{-PHP}$ has

$$N_G \leq 5(n+1), \quad w(G\text{-PHP}) = 5, \quad (8.32)$$

and by Lemma 8.8 its refutation width is at least $r/12 = \Omega(n)$. The size lower bound (8.13) therefore gives

$$S(G\text{-PHP}) \geq \exp\left(\frac{(r/12 - 7)^2}{40(n+1)}\right) - 1 = \exp(\Omega(n)). \quad (8.33)$$

Now restrict the standard formula PHP_n^{n+1} by setting $x_{p,h} = 0$ for every nonedge $(p,h) \notin E(G)$. Its pigeon clauses become exactly the five-neighbor pigeon clauses of $G\text{-PHP}$; every collision clause containing a removed variable is satisfied and disappears, while the remaining collision clauses are exactly those of the sparse formula. Hence

$$\text{PHP}_n^{n+1} \upharpoonright \rho = G\text{-PHP}. \quad (8.34)$$

By Lemma 8.2, any resolution refutation of the standard formula restricts to one of the sparse formula without increasing size. Thus

$$S(\text{PHP}_n^{n+1}) \geq S(G\text{-PHP}) = \exp(\Omega(n)),$$

and an absolute positive constant may be fixed for all sufficiently large n . □

Historical location. Haken proved the first exponential general-resolution lower bound using the pigeonhole principle. The proof above follows the later width-based simplification: instead of carrying random restrictions through the entire lower-bound argument, it proves a linear width bottleneck for a sparse restriction and invokes Theorem 8.5 once.

8.5 Random CNFs: a source-owned classical theorem

Theorem 8.10 (Chvátal–Szemerédi random-CNF lower bound). *Fix positive integers k, c such that*

$$k \geq 3, \quad c 2^{-k} \geq 0.7. \quad (8.35)$$

For each n , sample cn clauses independently and uniformly with replacement from the set of all width- k clauses on n variables using k distinct variables and arbitrary signs, and let $F_{n,k,c}$ be their conjunction. Then there exists a constant $\varepsilon_{k,c} > 0$ such that, with probability tending to one as $n \rightarrow \infty$, $F_{n,k,c}$ is unsatisfiable and every resolution refutation of it has more than

$$(1 + \varepsilon_{k,c})^n \quad (8.36)$$

clauses. Equivalently, for every fixed admissible pair (k, c) , random formulas in this model require $\exp(\Omega(n))$ resolution size with high probability.

Source boundary. Theorem 8.10 is a deep source-owned input: Vašek Chvátal and Endre Szemerédi, *Many Hard Examples for Resolution*, *Journal of the ACM* 35(4) (1988), 759–768, DOI 10.1145/48014.48016. QRCC freezes the theorem at the explicit $c 2^{-k} \geq 0.7$ hypothesis stated in the primary paper rather than silently replacing it by a modern satisfiability-threshold formulation.

The proof architecture is closely related to the width picture developed above but is not reproduced here: a random clause hypergraph is shown to be locally sparse/expanding; any short refutation would have to avoid the resulting bottleneck after suitable restrictions; the combinatorics force an exponentially large refutation. Later width-based work makes this architecture more modular, but source verification is not counted as a local proof of the Chvátal–Szemerédi theorem.

Concrete parameter check. For $k = 3$ and $c = 6$, one has $c 2^{-k} = 6/8 = 0.75 \geq 0.7$. Hence a random 3-CNF with $6n$ independently sampled clauses lies in the theorem’s regime: with high probability it is unsatisfiable and requires exponentially many resolution lines. This example is only a hypothesis check; it does not identify the hidden constant $\varepsilon_{3,6}$.

8.6 What the chapter has established

The logical chain is now explicit:

$$\text{DPLL search} \longleftrightarrow \text{tree-like resolution} \subseteq \text{general resolution}, \quad (8.37)$$

while

$$\text{large mandatory width} \xRightarrow{\text{Theorem 8.5}} \text{large resolution size}. \quad (8.38)$$

The pigeonhole theorem demonstrates this mechanism deterministically and completely inside QRCC; Theorem 8.10 records a classical probabilistic family where exponential resolution complexity is typical. Chapter 9 moves from clauses to algebraic proof lines, where width is replaced by polynomial degree and the corresponding dual objects become algebraic certificates.

Chapter 9

Polynomial Calculus, Nullstellensatz, and Interpolation

Resolution reasons with clauses. Algebraic proof systems instead translate Boolean constraints into polynomial equations and measure the complexity of deriving the impossible equation $1 = 0$. This change of language is not cosmetic: clause width becomes polynomial degree, linear algebra becomes a proof-search tool, and lower bounds can be witnessed by dual linear functionals. At the same time, static algebraic certificates and dynamic polynomial-calculus derivations must be kept separate. A low-degree dynamic proof can exploit intermediate cancellations that are not visible in a single bounded-degree Nullstellensatz identity.

This chapter first fixes those conventions over an arbitrary field. It then gives an exact finite-dimensional dual description of degree-bounded polynomial calculus and works out a genuine degree lower bound on a small fractional-knapsack instance. Next, a finite Boolean form of the Nullstellensatz is proved directly by point indicators, with no algebraic-closure hypothesis and no external commutative-algebra theorem. The chapter then returns to resolution and proves feasible interpolation by constructing the separating circuit gate by gate from a refutation. The final section source-freezes the modern interface between restricted Ideal Proof System lower bounds and algebraic circuit lower bounds.

Learning goals. After this chapter, a reader should be able to define polynomial calculus and Nullstellensatz certificates without confusing their degree measures; compute the least degree-bounded polynomial-calculus closure by linear algebra; explain the dual separating-functional criterion; verify a concrete degree lower bound; multilinearize a polynomial modulo the Boolean ideal with no degree increase; construct a Boolean Nullstellensatz certificate for every unsatisfiable finite Boolean system; and extract a Boolean interpolant circuit of size linear in a resolution refutation.

9.1 Polynomial calculus and static algebraic certificates

Definition 9.1 (Polynomial calculus, multilinearization, and Nullstellensatz certificates). Let $\mathbb{F}$ be a field and let

$$\mathcal{Q} = \{f_1, \dots, f_m\} \subseteq \mathbb{F}[x_1, \dots, x_n] \tag{9.1}$$

be a finite set of polynomial equations, interpreted as $f_i = 0$. When the intended assignments are Boolean we adjoin the *Boolean axioms*

$$b_j := x_j^2 - x_j = 0, \quad 1 \leq j \leq n. \quad (9.2)$$

A *polynomial-calculus derivation* (PC derivation) from $\mathcal{Q}$ is a sequence of polynomials. A line may be an input polynomial f_i , a Boolean axiom b_j , or may follow from earlier lines by one of the rules

$$\frac{p}{\alpha p + \beta q} \quad (\alpha, \beta \in \mathbb{F}), \quad \frac{p}{x_j p} \quad (1 \leq j \leq n). \quad (9.3)$$

A *PC refutation* derives the constant polynomial 1. The *degree* of a PC derivation is the largest total degree of a polynomial appearing as a line. Its *monomial size* is the sum, over all lines after collecting like terms, of the numbers of monomials with nonzero coefficients. Unless a different measure is explicitly named, “PC degree” below means total degree in this dynamic derivation sense.

For $p \in \mathbb{F}[x_1, \dots, x_n]$, its *multilinearization* $\text{ML}(p)$ is the unique multilinear polynomial congruent to p modulo the Boolean ideal

$$\mathcal{B} := \langle x_1^2 - x_1, \dots, x_n^2 - x_n \rangle. \quad (9.4)$$

Equivalently, every positive power x_j^e in a monomial is replaced by x_j . Thus p and $\text{ML}(p)$ agree on $\{0, 1\}^n$.

A *Boolean Nullstellensatz certificate* for the infeasibility of (9.1)–(9.2) is a polynomial identity

$$1 = \sum_{i=1}^m h_i f_i + \sum_{j=1}^n q_j (x_j^2 - x_j), \quad (9.5)$$

where $h_i, q_j \in \mathbb{F}[x_1, \dots, x_n]$. Its *certificate degree* is

$$\max \left\{ \max_i \deg(h_i f_i), \max_j \deg(q_j (x_j^2 - x_j)) \right\}. \quad (9.6)$$

This is a static identity, whereas (9.3) is a dynamic derivation. Every degree- D Nullstellensatz certificate can be expanded monomial by monomial into a degree- D PC refutation, but the converse implication with the same degree is not part of the definition and is false in general.

Definition discipline. The adjective *Boolean* in (9.5) is essential. The classical Hilbert Nullstellensatz discusses common zeros over an algebraic closure. Here the Boolean axioms force the relevant quotient to be a finite function algebra on $\{0, 1\}^n$, so the certificate theorem below works over an arbitrary field and is proved directly.

9.2 Degree-bounded polynomial calculus as finite-dimensional linear algebra

For $d \geq 0$, let $\mathcal{P}_{\leq d}$ denote the finite-dimensional $\mathbb{F}$ -vector space of polynomials of total degree at most d . Given $\mathcal{Q}$ together with the Boolean axioms, define the *degree- d PC closure* inductively by

$$V_d^{(0)} := \text{span}_{\mathbb{F}} \{ q : q \in \mathcal{Q} \cup \{b_1, \dots, b_n\}, \deg q \leq d \}, \quad (9.7)$$

and

$$V_d^{(t+1)} := \text{span}_{\mathbb{F}} \left(V_d^{(t)} \cup \{ x_j p : p \in V_d^{(t)}, \deg p \leq d-1, 1 \leq j \leq n \} \right). \quad (9.8)$$

Because $\mathcal{P}_{\leq d}$ is finite-dimensional, this increasing chain stabilizes. Write

$$V_d(\mathcal{Q}) := \bigcup_{t \geq 0} V_d^{(t)}. \quad (9.9)$$

A linear functional $L : \mathcal{P}_{\leq d} \rightarrow \mathbb{F}$ separates the constant from the degree- d PC closure if

$$L(1) = 1, \quad L(p) = 0 \quad \text{for every } p \in V_d(\mathcal{Q}). \quad (9.10)$$

Theorem 9.2 (Polynomial-calculus duality and a degree lower-bound criterion). *Let $\mathbb{F}$ be a field, let $\mathcal{Q} \subseteq \mathbb{F}[x_1, \dots, x_n]$ be finite, and let $d \geq 0$.*

1. *A polynomial $p \in \mathcal{P}_{\leq d}$ belongs to $V_d(\mathcal{Q})$ if and only if it has a PC derivation from $\mathcal{Q}$ in which every line has degree at most d .*
2. *There is no degree- d PC refutation if and only if there exists a separating functional L satisfying (9.10).*
3. *Over every field of characteristic zero, the three-variable fractional-knapsack system*

$$x_1 + x_2 + x_3 - \frac{1}{2} = 0, \quad x_i^2 - x_i = 0 \quad (i = 1, 2, 3) \quad (9.11)$$

has no PC refutation of degree at most two. In particular its PC refutation degree is at least three.

Proof. For the first assertion, every operation used in (9.7)–(9.8) is a legal PC operation and never exceeds degree d , so every element of every $V_d^{(t)}$ is derivable in degree at most d . Conversely, induction over a degree- d PC derivation shows that each input line lies in $V_d^{(0)}$, linear combinations remain in the current span, and a multiplication line $x_j p$ with degree at most d has $\deg p \leq d - 1$ and therefore enters the next closure stage. Hence the stabilized space is exactly the set of degree- d derivable polynomials.

The first assertion says that a degree- d refutation exists exactly when $1 \in V_d(\mathcal{Q})$. If a separating functional exists, this membership is impossible. In the other direction, suppose $1 \notin V_d(\mathcal{Q})$. Choose a basis of $V_d(\mathcal{Q})$, extend it by 1, and then extend further to a basis of $\mathcal{P}_{\leq d}$. Define L to be zero on the first basis block, one on the basis vector 1, and zero on every remaining basis vector. Linear extension gives (9.10). This is ordinary finite-dimensional vector-space separation; no topological or positivity assumption is involved.

For the concrete lower bound, abbreviate

$$f := x_1 + x_2 + x_3 - \frac{1}{2}, \quad b_i := x_i^2 - x_i. \quad (9.12)$$

Inside $\mathcal{P}_{\leq 2}$, put

$$U := \text{span}\{f, b_1, b_2, b_3, x_1 f, x_2 f, x_3 f\}. \quad (9.13)$$

We first check that U is already the stabilized degree-two PC closure. Suppose a linear combination

$$af + \sum_i c_i b_i + \sum_i d_i x_i f \quad (9.14)$$

has degree at most one. Comparing the coefficients of x_i^2 gives $c_i + d_i = 0$, while comparing $x_i x_j$ for $i \neq j$ gives $d_i + d_j = 0$. With three indices and characteristic zero these equations force

$d_1 = d_2 = d_3 = 0$, hence also $c_1 = c_2 = c_3 = 0$. Thus the only degree-at-most-one elements of U are scalar multiples of f . Multiplying such an element by any variable produces a scalar multiple of one of the already listed $x_i f$. Therefore (9.13) contains the axioms and is closed under every multiplication allowed at degree two, so $V_2(\mathcal{Q}) = U$.

Now define $L : \mathcal{P}_{\leq 2} \rightarrow \mathbb{F}$ on the monomial basis by

$$L(1) = 1, \quad L(x_i) = L(x_i^2) = \frac{1}{6}, \quad L(x_i x_j) = -\frac{1}{24} \quad (i \neq j). \quad (9.15)$$

Then

$$L(f) = 3 \cdot \frac{1}{6} - \frac{1}{2} = 0, \quad L(b_i) = 0, \quad (9.16)$$

and for every i ,

$$L(x_i f) = \frac{1}{6} + 2 \left(-\frac{1}{24} \right) - \frac{1}{2} \cdot \frac{1}{6} = 0. \quad (9.17)$$

Hence L annihilates all of $U = V_2(\mathcal{Q})$ while $L(1) = 1$. The second assertion excludes every degree-two refutation. $\square$

What the dual object is—and is not. The functional in (9.10) annihilates the *dynamically generated* closure (9.7)–(9.9). It is not enough in general to annihilate only bounded-degree monomial multiples of the initial axioms; that would describe a static Nullstellensatz truncation and can miss cancellations created by a PC derivation. This distinction is why the closure operator appears explicitly in the theorem.

Source and historical boundary. Clegg, Edmonds, and Impagliazzo introduced the Gröbner and polynomial-calculus proof-search viewpoint in *STOC 1996*, DOI 10.1145/237814.237860. Impagliazzo, Pudlák, and Sgall later developed degree lower bounds by explicitly describing low-degree consequence spaces in *Computational Complexity* 8(2) (1999), 127–144, DOI 10.1007/S000370050024. The finite-dimensional closure theorem and the worked system (9.11) are proved locally above; QRCC does not source-own them.

9.3 Finite Boolean Nullstellensatz without an algebraic-closure hypothesis

The next lemma is the only algebraic reduction needed for the certificate theorem.

Lemma 9.3 (Degree-controlled multilinear reduction). *For every field $\mathbb{F}$ and every polynomial $p \in \mathbb{F}[x_1, \dots, x_n]$, there are polynomials $q_1, \dots, q_n$ such that*

$$p - \text{ML}(p) = \sum_{j=1}^n q_j(x_j^2 - x_j) \quad (9.18)$$

and every nonzero product $q_j(x_j^2 - x_j)$ has degree at most $\deg p$. Moreover, if a multilinear polynomial vanishes at every point of $\{0, 1\}^n$, then it is the zero polynomial. Consequently, if p vanishes on the Boolean cube, then p belongs to the Boolean ideal $\mathcal{B}$ with a representation satisfying the same degree bound.

Proof. For one variable and $e \geq 2$,

$$x^e - x = (x^2 - x)(1 + x + \cdots + x^{e-2}). \quad (9.19)$$

Applying (9.19) monomial by monomial and variable by variable replaces every positive exponent by one. Each subtraction has degree no larger than the monomial being reduced, proving (9.18) and the degree statement.

For injectivity on the Boolean cube, induct on n . Write a multilinear polynomial as $p = q + x_n r$, with q, r multilinear in $x_1, \dots, x_{n-1}$. Vanishing at $x_n = 0$ forces $q = 0$ by induction, and vanishing at $x_n = 1$ then forces $r = 0$. The final claim follows by applying this to $\text{ML}(p)$. $\square$

Theorem 9.4 (Finite Boolean Nullstellensatz certificate theorem). *Let $\mathbb{F}$ be an arbitrary field and let $f_1, \dots, f_m \in \mathbb{F}[x_1, \dots, x_n]$, with $D := \max_i \deg f_i$. The system*

$$f_1 = \cdots = f_m = 0, \quad x_j^2 - x_j = 0 \quad (1 \leq j \leq n) \quad (9.20)$$

has no solution in $\{0, 1\}^n$ if and only if it has a Boolean Nullstellensatz certificate (9.5). When it is infeasible, there is such a certificate of degree at most $n + D$. Consequently, every infeasible finite Boolean polynomial system has a PC refutation of degree at most $n + D$.

Proof. A certificate immediately implies infeasibility: at a common Boolean zero every term on the right of (9.5) vanishes, contradicting $1 = 0$.

For the converse, suppose (9.20) has no Boolean solution. For every $a = (a_1, \dots, a_n) \in \{0, 1\}^n$, choose one index $i(a)$ for which $f_{i(a)}(a) \neq 0$. Define the Boolean point indicator

$$\delta_a(x) := \prod_{j:a_j=1} x_j \prod_{j:a_j=0} (1 - x_j). \quad (9.21)$$

It is multilinear of degree at most n , equals one at a , and vanishes at every other Boolean point. For each input polynomial put

$$h_i(x) := \sum_{a:i(a)=i} \frac{\delta_a(x)}{f_i(a)}. \quad (9.22)$$

The denominators are nonzero field elements, so they are invertible. Let

$$g(x) := \sum_i h_i(x) f_i(x). \quad (9.23)$$

At every Boolean point a , exactly the δ_a term selected in (9.22) survives, and therefore $g(a) = 1$. Hence $g - 1$ vanishes on the Boolean cube. By Lemma 9.3,

$$g - 1 = \sum_j q_j (x_j^2 - x_j) \quad (9.24)$$

with every Boolean-ideal product of degree at most $\deg(g - 1) \leq n + D$. Rearranging (9.24) gives

$$1 = \sum_i h_i f_i - \sum_j q_j (x_j^2 - x_j), \quad (9.25)$$

a certificate of degree at most $n + D$.

Finally expand each multiplier in (9.25) into monomials. Starting from an axiom, repeated applications of the multiplication rule in (9.3) derive each monomial multiple without exceeding the degree of that product; linear combination then derives 1. Thus the same degree bound gives a PC refutation. $\square$

A two-point example. Over any field of characteristic different from two, the Boolean system

$$x - \frac{1}{2} = 0, \quad x^2 - x = 0 \quad (9.26)$$

is infeasible. Taking $h = 4x - 2$ gives

$$(4x - 2) \left(x - \frac{1}{2} \right) - 4(x^2 - x) = 1. \quad (9.27)$$

Thus the certificate degree is two. This tiny identity is the one-variable shadow of the point indicator construction above.

Source boundary and companion retirement. Beame, Impagliazzo, Krajíček, Pitassi, and Pudlák studied degree lower bounds in the Nullstellensatz proof system in *Proceedings of the London Mathematical Society* 73 (1996), 1–26, DOI 10.1112/plms/s3-73.1.1. The theorem above is the finite Boolean certificate existence statement needed by QRCC, proved directly from (9.21)–(9.24). Therefore the former `COMP:COMMUTATIVE_ALGEBRA` placeholder is not a live dependency of Theorem 9.4.

9.4 Feasible interpolation for resolution

Let $A(x, y)$ and $B(x, z)$ be CNFs. The tuple x consists of the variables common to the two formulas; y occurs only in A , z occurs only in B , and the three tuples are pairwise disjoint. For a clause C , write C_{xy} for the subclause containing only literals on $x \cup y$, and C_{xz} for the subclause containing only literals on $x \cup z$. Define two sets of common-variable assignments,

$$U_A := \{a : \exists y \ A(a, y) = 1\}, \quad U_B := \{a : \exists z \ B(a, z) = 1\}. \quad (9.28)$$

If $A \wedge B$ is unsatisfiable, then $U_A \cap U_B = \emptyset$. A Boolean circuit $I(x)$ *interpolates* this pair when

$$a \in U_A \implies I(a) = 1, \quad a \in U_B \implies I(a) = 0. \quad (9.29)$$

Theorem 9.5 (Feasible interpolation for resolution). *Let $A(x, y)$ and $B(x, z)$ be CNFs with variable partition as above, and suppose $A \wedge B$ has a resolution refutation π of S lines under Definition 8.1. Then one can construct from π , in time polynomial in its explicit size, a Boolean circuit $I(x)$ satisfying (9.29) and having $O(S)$ noninput gates over the basis $\{\wedge, \vee, \neg\}$.*

Consequently, if every Boolean circuit separating U_A from U_B has size at least L , then every resolution refutation of $A \wedge B$ has size $\Omega(L)$.

Proof. We associate a circuit $I_C(x)$ with every clause occurrence C in the refutation and maintain the two semantic invariants

$$A \models C_{xy} \vee I_C, \quad B \models C_{xz} \vee \neg I_C. \quad (9.30)$$

For an initial clause from A , set $I_C = 0$. The first implication in (9.30) is then the input clause itself, while the second is tautological because $\neg I_C = 1$. For an initial clause from B , set $I_C = 1$, and the roles reverse. A weakening keeps the same circuit because enlarging a clause can only make either implication easier to satisfy.

It remains to propagate the invariant through a resolution step. Suppose C is the resolvent of parent clauses C_1, C_2 , with partial interpolants I_1, I_2 .

If the pivot is a private y -variable, define

$$I_C := I_1 \vee I_2. \quad (9.31)$$

On the A side, semantic resolution of the two parent implications gives the first implication in (9.30). On the B side the pivot is absent from both projections: if the projected resolvent is false, both parent projections are false, so the parent invariants force both $\neg I_1$ and $\neg I_2$, which is $\neg I_C$.

If the pivot is a private z -variable, use instead

$$I_C := I_1 \wedge I_2. \quad (9.32)$$

The same argument with A and B interchanged proves (9.30).

Finally suppose the common pivot is x_j , occurring positively in C_1 and negatively in C_2 . Define the multiplexer

$$I_C := (\neg x_j \wedge I_1) \vee (x_j \wedge I_2). \quad (9.33)$$

If the projected resolvent is false and $x_j = 0$, the first A -side parent implication forces $I_1 = 1$, hence $I_C = 1$; if $x_j = 1$, the second forces $I_2 = 1$. This proves the first invariant. The identical case split on the B invariants forces $\neg I_1$ when $x_j = 0$ and $\neg I_2$ when $x_j = 1$, proving the second.

At the final empty clause, both projections are empty, so (9.30) reads

$$A \models I, \quad B \models \neg I. \quad (9.34)$$

Therefore any common assignment extendible to a satisfying assignment of A receives output one, and any one extendible to a satisfying assignment of B receives output zero, exactly (9.29).

Treat the circuits attached to proof lines as a shared DAG. Initial clauses contribute constants; a private pivot contributes one $\vee$ - or $\wedge$ -gate; and a common pivot contributes only a constant number of gates for (9.33). Thus the circuit has $O(S)$ gates and can be generated by a single pass over the proof. The lower-bound transfer is the contrapositive of this construction. $\square$

Worked interpolation trace. Consider

$$A(x, y) = (x \vee y) \wedge (x \vee \neg y), \quad B(x, z) = (\neg x \vee z) \wedge (\neg x \vee \neg z). \quad (9.35)$$

Resolving the two A -clauses on y derives x ; resolving the two B -clauses on z derives $\neg x$; resolving these units gives $\perp$. Initial A -clauses receive partial interpolant 0, so the private- y rule gives $0 \vee 0 = 0$ at the clause x . Initial B -clauses receive 1, so the private- z rule gives $1 \wedge 1 = 1$ at $\neg x$. The final common-pivot rule is

$$(\neg x \wedge 0) \vee (x \wedge 1) = x. \quad (9.36)$$

Indeed, $A(a, y)$ is satisfiable exactly when $a = 1$, while $B(a, z)$ is satisfiable exactly when $a = 0$.

Source boundary. Krajíček's *Interpolation theorems, lower bounds for proof systems, and independence results for bounded arithmetic*, *Journal of Symbolic Logic* 62(2) (1997), 457–486, DOI 10.2307/2275541, establishes feasible interpolation for resolution in the broader proof-complexity framework. The resolution construction (9.30)–(9.34), including its gate count, is proved locally here. No monotone-interpolation conclusion is asserted: such a strengthening requires additional polarity hypotheses that are not present in Theorem 9.5.

9.5 Frontier: algebraic circuit lower bounds as proof lower bounds

Theorem 9.6 (Dated algebraic-circuit/proof-complexity frontier). *Forbes, Shpilka, Tzameret, and Wigderson develop restricted subsystems of the Ideal Proof System (IPS) and two mechanisms that convert suitably strengthened algebraic circuit lower bounds into lower bounds for those restricted proof systems. The relevant strengthened notions include functional lower bounds—hardness of every polynomial agreeing with a target function on the Boolean cube—and lower bounds that hold for every nonzero multiple of an explicit polynomial. Their results yield unconditional lower bounds and separations for several restricted algebraic circuit classes, including sparse polynomials, depth-three powering formulas, read-once oblivious algebraic branching programs, and multilinear formulas.*

This registered frontier does not assert that an ordinary lower bound for an arbitrary algebraic circuit class automatically yields a lower bound for unrestricted IPS, Polynomial Calculus, Frege, or Extended Frege. The strengthened functional/multiple hypotheses and the chosen restricted IPS subsystem are part of the theorem interface.

Source ownership and date. The source is Michael A. Forbes, Amir Shpilka, Iddo Tzameret, and Avi Wigderson, *Proof Complexity Lower Bounds from Algebraic Circuit Complexity, Theory of Computing* 17(10) (2021), 1–88, DOI 10.4086/toc.2021.v017a010, a journal version of the CCC 2016 work. QRCC records this as a dated frontier/result map, not as a locally reconstructed proof of its restricted-IPS lower bounds.

9.6 What the chapter has established

The algebraic part of the volume now has three distinct proof-complexity interfaces:

$$\text{dynamic PC} \xrightleftharpoons[\text{finite-dimensional dual}]{\text{degree-}d \text{ closure}} \text{linear algebra}, \quad (9.37)$$

$$\text{Boolean infeasibility} \implies \text{static Nullstellensatz certificate} \implies \text{PC refutation}, \quad (9.38)$$

and

$$\text{short resolution refutation of } A(x, y) \wedge B(x, z) \implies \text{small separator on the common variables } x. \quad (9.39)$$

The first line supplies a degree-lower-bound method, the second supplies a complete finite Boolean certificate theorem, and the third turns circuit lower bounds into resolution lower bounds whenever a suitable interpolation pair is available. Chapter 10 will move from these algebraic and interpolation tools to semialgebraic proof systems, where positivity rather than ideal membership becomes the central semantic resource.

Chapter 10

Sum-of-Squares Proofs, Pseudoexpectations, and Semialgebraic Lower Bounds

Polynomial calculus derives equations. Sum-of-squares (SoS) reasoning adds an order structure: a square is nonnegative, and a polynomial inequality may be multiplied by a square without changing its direction. This extra positivity makes semidefinite programming appear naturally. It also changes the dual witness. A linear functional that merely annihilates low-degree equations is not enough; a genuine pseudoexpectation must additionally be nonnegative on every low-degree square.

The distinction matters mathematically. Chapter 9’s degree-bounded PC functional for a fractional subset-sum instance need not be positive. In the present chapter, positivity is encoded by a moment matrix, weak duality is proved in one line, and the converse is derived by finite-dimensional separation under an explicit closed-cone regularity hypothesis. The chapter then freezes a Grigoriev knapsack instance whose static Positivstellensatz/SoS degree is linear, records the Meka–Potechin–Wigderson average-case planted-clique lower bound at its exact round convention, and ends with a dated branch-and-cut frontier. A small exact laboratory checks the chapter’s moment matrix and the earlier resolution/PC interfaces computationally.

Learning goals. After this chapter, a reader should be able to distinguish equality and inequality constraints in an SoS certificate; translate positivity of a pseudoexpectation into a positive-semidefinite moment or localizing matrix; prove the easy pseudoexpectation-to-lower-bound direction without invoking SDP duality; state precisely the regularity needed for the converse separation theorem; verify the symmetric moments forced by a knapsack equation; explain what is deep in Grigoriev’s positivity argument; and read an average-case SoS lower bound without confusing hierarchy rounds, polynomial degree, objective value, and integrality gap.

10.1 The degree-bounded quadratic module and its dual

Definition 10.1 (Sum-of-squares proofs, moment matrices, and pseudoexpectations). Work over the ordered field $\mathbb{R}$. Fix variables $x = (x_1, \dots, x_n)$, equality constraints

$$g_1(x) = \dots = g_a(x) = 0, \tag{10.1}$$

inequality constraints

$$h_1(x) \geq 0, \dots, h_b(x) \geq 0, \tag{10.2}$$

and, when Boolean assignments are intended, the Boolean equations

$$x_i^2 - x_i = 0 \quad (1 \leq i \leq n). \quad (10.3)$$

The Boolean equations are equality constraints; they are not inequalities.

For an even degree bound $2r$, let $\mathcal{I}_{\leq 2r}$ be the vector space of all sums

$$\sum_{j=1}^a u_j g_j + \sum_{i=1}^n v_i (x_i^2 - x_i) \quad (10.4)$$

in which every displayed product has total degree at most $2r$. A polynomial σ is a *sum of squares of degree at most $2r$* if

$$\sigma = \sum_{t=1}^N q_t^2, \quad \deg q_t \leq r. \quad (10.5)$$

The *degree- $2r$ truncated quadratic module* is

$$\mathcal{M}_{2r} := \left\{ \sigma_0 + \sum_{j=1}^b \sigma_j h_j + \iota : \begin{array}{l} \sigma_0, \sigma_j \text{ are sums of squares,} \\ \deg \sigma_0 \leq 2r, \deg(\sigma_j h_j) \leq 2r, \\ \iota \in \mathcal{I}_{\leq 2r} \end{array} \right\}. \quad (10.6)$$

A *degree- $2r$ SoS proof* of $p \geq 0$ is a representation $p \in \mathcal{M}_{2r}$. A *degree- $2r$ SoS refutation* is a representation

$$-1 \in \mathcal{M}_{2r}. \quad (10.7)$$

Thus the degree counts every complete summand q_t^2 , $q_t^2 h_j$, and constraint multiple, not merely the multiplier before it is multiplied by its constraint.

A *degree- $2r$ pseudoexpectation* is a linear functional

$$\tilde{\mathbb{E}} : \mathbb{R}[x]_{\leq 2r} \longrightarrow \mathbb{R} \quad (10.8)$$

satisfying

$$\tilde{\mathbb{E}}[1] = 1, \quad \tilde{\mathbb{E}}[q^2] \geq 0 \quad (\deg q \leq r), \quad (10.9)$$

all degree-allowed equality identities

$$\tilde{\mathbb{E}}[u g_j] = 0, \quad \tilde{\mathbb{E}}[v(x_i^2 - x_i)] = 0, \quad (10.10)$$

and all degree-allowed inequality localizations

$$\tilde{\mathbb{E}}[q^2 h_j] \geq 0 \quad \text{whenever } \deg(q^2 h_j) \leq 2r. \quad (10.11)$$

It is a truncated linear functional, not a probability distribution; its “moments” may fail to extend to any honest distribution on $\mathbb{R}^n$.

Let $\mathcal{A}_r$ be the list of monomials of degree at most r . The *degree- $2r$ moment matrix* of $\tilde{\mathbb{E}}$ is

$$M_r(\alpha, \beta) := \tilde{\mathbb{E}}[x^{\alpha+\beta}], \quad \alpha, \beta \in \mathcal{A}_r. \quad (10.12)$$

For an inequality h , the *localizing matrix* is defined analogously on those monomials whose products with h remain within degree $2r$:

$$M_{r,h}(\alpha, \beta) := \tilde{\mathbb{E}}[x^{\alpha+\beta} h(x)]. \quad (10.13)$$

A real symmetric matrix A is *positive semidefinite*, written $A \succeq 0$, if $z^\top A z \geq 0$ for every real vector z .

Finally, the degree- $2r$ system has *closed-cone regularity* when $\mathcal{M}_{2r}$, viewed as a convex cone in the finite-dimensional coefficient space $\mathbb{R}[x]_{\leq 2r}$, is closed. This condition is stated explicitly because linear images of semidefinite cones need not be closed in complete generality. It is the only regularity hypothesis used in the converse direction of the duality theorem below.

Equality versus inequality constraints. For an equality $g = 0$, both ug and $-ug$ belong to the equality subspace, so a dual functional must annihilate it. For an inequality $h \geq 0$, only square multiples q^2h are automatically nonnegative, so the dual condition is positivity rather than equality. This asymmetry is the algebraic reason moment and localizing matrices play different roles.

Theorem 10.2 (SoS duality via pseudoexpectations). *Fix a degree $2r$ system as in Definition 10.1.*

1. *A normalized linear functional satisfies the square-positivity condition in (10.9) if and only if its moment matrix M_r is positive semidefinite. Likewise, (10.11) for a fixed inequality h_j is equivalent to positive semidefiniteness of the corresponding localizing matrix.*
2. *If a degree- $2r$ pseudoexpectation exists, then there is no degree- $2r$ SoS refutation. This implication requires no regularity hypothesis.*
3. *Conversely, suppose $\mathcal{M}_{2r}$ is closed. If there is no degree- $2r$ SoS refutation, then a degree- $2r$ pseudoexpectation exists.*

Hence, under closed-cone regularity, degree- $2r$ SoS nonrefutability is equivalent to feasibility of the degree- $2r$ pseudoexpectation constraints.

Proof. Write a polynomial q of degree at most r as $q = \sum_{\alpha \in \mathcal{A}_r} c_\alpha x^\alpha$. By linearity,

$$\tilde{\mathbb{E}}[q^2] = \sum_{\alpha, \beta} c_\alpha c_\beta \tilde{\mathbb{E}}[x^{\alpha+\beta}] = c^\top M_r c. \quad (10.14)$$

Thus positivity on every such square is exactly $M_r \succeq 0$. Replacing q^2 by $q^2 h_j$ gives the identical calculation for the localizing matrix.

For weak duality, assume a refutation

$$-1 = \sigma_0 + \sum_j \sigma_j h_j + \iota \quad (10.15)$$

exists and apply a pseudoexpectation. Equality terms in ι vanish by (10.10), the sum-of-squares terms have nonnegative pseudoexpectation by (10.9), and every localized inequality term is nonnegative by (10.11). Therefore

$$-1 = \tilde{\mathbb{E}}[-1] \geq 0, \quad (10.16)$$

a contradiction.

For the converse, the absence of a refutation says $-1 \notin \mathcal{M}_{2r}$. By hypothesis $\mathcal{M}_{2r}$ is a closed convex cone in a finite-dimensional real vector space. The finite-dimensional strong separation theorem therefore gives a nonzero linear functional L with

$$L(-1) < 0, \quad L(p) \geq 0 \quad \text{for all } p \in \mathcal{M}_{2r}. \quad (10.17)$$

In particular $L(1) > 0$; normalize by setting $\tilde{\mathbb{E}} := L/L(1)$. Every allowed square and localized square lies in $\mathcal{M}_{2r}$, so (10.9) and (10.11) follow. Moreover $\mathcal{I}_{\leq 2r}$ is a vector subspace contained in $\mathcal{M}_{2r}$. Hence for every $\iota \in \mathcal{I}_{\leq 2r}$, both ι and $-\iota$ lie in the cone; (10.17) forces $L(\iota) = 0$, which is exactly (10.10). Finally $\tilde{\mathbb{E}}[1] = 1$. Thus the normalized separator is a pseudoexpectation. $\square$

Worked degree-two pseudoexpectation. Consider the infeasible Boolean system

$$x_1 + x_2 = \frac{1}{2}, \quad x_1^2 = x_1, \quad x_2^2 = x_2. \quad (10.18)$$

Define a degree-two functional by

$$\tilde{\mathbb{E}}[1] = 1, \quad \tilde{\mathbb{E}}[x_1] = \tilde{\mathbb{E}}[x_2] = \tilde{\mathbb{E}}[x_1^2] = \tilde{\mathbb{E}}[x_2^2] = \frac{1}{4}, \quad \tilde{\mathbb{E}}[x_1 x_2] = -\frac{1}{8}. \quad (10.19)$$

The equation and its degree-one multiples vanish:

$$\tilde{\mathbb{E}}[x_1 + x_2 - 1/2] = 0, \quad \tilde{\mathbb{E}}[x_i(x_1 + x_2 - 1/2)] = 0. \quad (10.20)$$

On the monomial basis $(1, x_1, x_2)$, the moment matrix is

$$M_1 = \begin{pmatrix} 1 & 1/4 & 1/4 \\ 1/4 & 1/4 & -1/8 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}. \quad (10.21)$$

Its eigenvalues are $9/8, 3/8, 0$, so $M_1 \succeq 0$. Therefore (10.19) is a genuine degree-two pseudoexpectation and no degree-two SoS refutation exists. Notice the negative “moment” $\tilde{\mathbb{E}}[x_1 x_2]$: pseudoexpectations are not honest probability measures, even though every degree-one square has nonnegative pseudoexpectation.

Source and companion boundary. The moment-matrix/pseudoexpectation viewpoint is standard in the SoS/Lasserre literature; for a proof-complexity formulation, Meka, Potechin, and Wigderson explicitly use a positive-semidefinite linear mapping as a dual certificate for degree-bounded Positivstellensatz refutations. The exact closed-cone separation theorem above is proved locally. No unnamed convex-optimization theorem is needed beyond the elementary finite-dimensional strong separation theorem, so the former `COMP:CONVEX_OPTIMIZATION` placeholder is retired at this point rather than carried into later chapters.

10.2 A symmetric knapsack pseudoexpectation and Grigoriev’s degree barrier

For real ρ and an integer $s \geq 0$, write the falling factorial

$$(\rho)_s := \rho(\rho - 1) \cdots (\rho - s + 1), \quad (\rho)_0 := 1. \quad (10.22)$$

For a multilinear monomial $x_S := \prod_{i \in S} x_i$, the symmetric knapsack moments are

$$\tilde{\mathbb{E}}_\rho[x_S] := \frac{(\rho)_{|S|}}{(n)_{|S|}} = \frac{\binom{\rho}{|S|}}{\binom{n}{|S|}}. \quad (10.23)$$

The generalized binomial coefficient is the polynomial $\binom{\rho}{s} = (\rho)_s / s!$; no integrality of ρ is assumed.

Lemma 10.3 (The knapsack equation forces the symmetric moments). *Let a permutation-invariant truncated linear functional on multilinear monomials satisfy $\tilde{\mathbb{E}}[1] = 1$ and all degree-allowed multiples of $\sum_{i=1}^n x_i - \rho = 0$. Then its monomial moments are necessarily (10.23). Conversely, the values (10.23) satisfy every such degree-allowed knapsack identity and every Boolean identity after multilinear reduction.*

Proof. Permutation invariance makes $\widetilde{\mathbb{E}}[x_S] = a_s$ depend only on $s = |S|$, with $a_0 = 1$. Multiplying the knapsack equation by x_S , multilinearizing, and applying the functional gives

$$0 = sa_s + (n - s)a_{s+1} - \rho a_s. \quad (10.24)$$

Thus

$$a_{s+1} = \frac{\rho - s}{n - s} a_s, \quad (10.25)$$

and induction yields (10.23). Reading the recurrence in reverse verifies all allowed knapsack multiples for the explicitly defined functional. Boolean multiples vanish because $x_i^2 \equiv x_i$ under multilinearization. $\square$

Theorem 10.4 (Grigoriev-style linear SoS degree lower bound for symmetric knapsack). *For $n \geq 3$, put*

$$\rho_n := \frac{n}{2} + \frac{1}{4}. \quad (10.26)$$

The Boolean knapsack system

$$x_i^2 - x_i = 0 \quad (1 \leq i \leq n), \quad \sum_{i=1}^n x_i - \rho_n = 0 \quad (10.27)$$

is infeasible. Every static Positivstellensatz/SoS refutation in the equality-only convention of Definition 10.1 has degree at least $n + 1$.

The symmetric functional (10.23), with $\rho = \rho_n$, is the canonical dual candidate: its normalization, Boolean consistency, and knapsack consistency are local consequences of Lemma 10.3. The nontrivial positivity of the corresponding quadratic form through the linear-degree range is the deep Grigoriev input and is source-owned here.

Proof of the parameter specialization and local part. The number ρ_n is never an integer, whereas every Boolean sum $\sum_i x_i$ is integral, so (10.27) is infeasible.

Let

$$k := \left\lfloor \frac{n-1}{2} \right\rfloor. \quad (10.28)$$

Then $k < \rho_n < n - k$ and $k \geq (n - 2)/2$. Grigoriev's primary knapsack theorem states that in this regime the static Positivstellensatz degree is at least $n + 1$. In the equality-only case his static identity is exactly the present SoS convention: an ideal combination of the equations plus a sum of squares equals -1 . Therefore his theorem applies verbatim after the notation change.

The moment formula itself is not imported: Lemma 10.3 proves that the knapsack equation forces (10.23) and verifies all equality constraints. What remains deep is positivity on all low-degree squares. Grigoriev reduces precisely this step to nonnegativity of a highly symmetric quadratic form and proves the required spectral/combinatorial statement. Once that positivity is granted, Theorem 10.2(2) gives the lower bound immediately as well. QRCC therefore exposes the dual witness while leaving the difficult quadratic-form analysis with its primary source. $\square$

Why this theorem is not a polynomial-calculus statement. Chapter 9 measured dynamic PC degree over an arbitrary field. The present theorem is over the ordered real field and exploits positivity of squares. Its static SoS/Positivstellensatz degree cannot be identified with either PC degree or Boolean Nullstellensatz degree. The shared knapsack syntax is not a license to merge these proof systems.

Primary-source boundary. Dima Grigoriev, *Complexity of Positivstellensatz Proofs for the Knapsack*, *Computational Complexity* 10 (2001/2002), 139–154, DOI 10.1007/s00037-001-8192-0, proves the staircase lower bound. In particular, for an integer k with $k < \rho < n - k$, the static Positivstellensatz degree is at least $n + 1$ once $k \geq (n - 2)/2$. The source also makes explicit that the key step is nonnegativity of the relevant quadratic form. PASS82 source-owns that step rather than compressing it into the phrase “by symmetry.”

10.3 Average-case SoS lower bounds: planted clique

The preceding knapsack theorem is a deterministic proof-complexity lower bound. Average-case SoS lower bounds ask a different question: for a random instance, can a bounded number of hierarchy rounds certify the true optimum? The next theorem is intentionally source-owned because its proof requires delicate random-matrix estimates.

For a graph $G = ([n], E)$ and a real parameter K , the standard clique equations are

$$x_i^2 - x_i = 0, \quad x_i x_j = 0 \quad (\{i, j\} \notin E), \quad \sum_{i=1}^n x_i - K = 0. \quad (10.29)$$

The r -round convention below corresponds to moments through degree $2r$.

Theorem 10.5 (Meka–Potechin–Wigderson planted-clique/maximum-clique SoS lower bound; source-owned). *There is a universal constant $C > 1$ such that, in the Meka–Potechin–Wigderson round convention, with high probability over $G \sim G(n, 1/2)$, the clique system (10.29) has no degree- $2r$ Positivstellensatz/SoS refutation whenever*

$$K \leq \frac{n^{1/(2r)}}{C^r (\log n)^{1/r}}. \quad (10.30)$$

Equivalently, the natural r -round SoS relaxation for maximum clique has value at least the right-hand side of (10.30) with high probability. Since $\omega(G) = (2 + o(1)) \log_2 n$ with high probability, the source records the weaker but cleaner integrality-gap lower bound

$$\frac{n^{1/(2r)}}{C^r (\log n)^2}. \quad (10.31)$$

For the planted distribution $G(n, 1/2, t)$, monotonicity under adding clique edges gives the corresponding gap divided by t .

What is and is not proved here. The theorem’s proof is source-owned. Its architecture is nevertheless important: construct an explicit graph-dependent pseudoexpectation, form its moment matrix, decompose the expectation using Johnson-scheme symmetry, and control the random error by trace/norm estimates until the matrix is positive semidefinite with high probability. This is qualitatively different from the deterministic finite-dimensional separation argument in Theorem 10.2.

The theorem is also not the later nearly-tight planted-clique SoS lower bound. PASS82 freezes the 2015 Meka–Potechin–Wigderson theorem at its own r -round parameter regime and does not silently replace it by the stronger pseudo-calibration bounds developed later.

Primary-source boundary. Raghu Meka, Aaron Potechin, and Avi Wigderson, *Sum-of-Squares Lower Bounds for Planted Clique*, STOC 2015, 87–96, DOI 10.1145/2746539.2746600. Their Theorem 1.5 gives the no- $PS(r)$ -refutation range (10.30); Theorem 1.1 gives (10.31), and the appendix converts the dual certificate into the standard maximum-clique SDP value. The graph distribution, round convention, objective, probability qualifier, and logarithmic losses are all retained here.

10.4 Executable finite laboratory

Proposition 10.6 (Proof-system and pseudoexpectation laboratory). *The executable file `code/volume_13/proof_pseudoexpectation_lab.py` is a deterministic standard-library laboratory with exact-rational self-tests for three interfaces:*

1. *resolution closure on the four-clause contradiction $(x \vee y) \wedge (x \vee \neg y) \wedge (\neg x \vee y) \wedge (\neg x \vee \neg y)$;*
2. *degree-two polynomial-calculus closure for the three-variable fractional-knapsack example of Chapter 9, checking that the constant polynomial is not in the derived span;*
3. *the degree-two pseudoexpectation (10.19), checking all defining linear constraints and every principal minor of the exact moment matrix.*

Running the file with `-self-test` terminates with a fixed PASS token exactly when all three checkpoints succeed.

Proof. Each checkpoint exhausts a finite exact object. Resolution closure enumerates legal resolvents until stabilization and checks that the empty clause appears. The PC checkpoint uses rational Gaussian elimination on the degree-two monomial coefficient vectors and verifies that adding every legal variable multiple does not increase the stabilized rank to include the constant. For the SoS checkpoint, rational arithmetic verifies (10.20); for a real symmetric matrix, positive semidefiniteness is equivalent to nonnegativity of every principal minor, and the laboratory checks all principal minors of (10.21) exactly. No floating-point optimization result is used. $\square$

10.5 A dated branch-and-cut frontier

Proposition 10.7 (Branch-and-cut proof systems: verified 2026 frontier). *As of September 2026, the following distinctions are source-verified and must remain separate.*

1. *Stabbing Planes (SP) formalizes branching on integral linear inequalities and can simulate Cutting Planes; it is a proof-complexity abstraction of branch-and-cut rather than a theorem about every implementation of a mixed-integer solver.*
2. *For bounded-coefficient Stabbing Planes SP^* , Fleming–Göös–Impagliazzo–Pitassi–Robere–Tan–Wigderson give a translation into Cutting Planes. Combining that translation with known Cutting-Planes lower bounds yields exponential lower bounds for SP^* .*
3. *The same 2026 journal paper gives short Cutting-Planes refutations for unsatisfiable linear systems over prime finite fields and proves depth lower bounds for semantic Cutting Planes on Tseitin formulas. These facts rule out the obsolete slogan that Tseitin is simply exponentially hard for unrestricted Cutting Planes.*
4. *None of these statements is an exponential size lower bound for unrestricted Stabbing Planes with arbitrary coefficients, nor a blanket lower bound for unrestricted branch-and-cut algorithms.*

Coefficient bounds, semantic versus syntactic inference, size versus depth, and the allowed branching rule are part of the theorem statement.

Sources and date. The current canonical route is Noah Fleming, Mika Göös, Russell Impagliazzo, Toniann Pitassi, Robert Robere, Li-Yang Tan, and Avi Wigderson, *On the Power and Limitations of Branch and Cut*, *Theory of Computing* 22(2) (2026), 1–38, DOI 10.4086/toc.2026.v022a002, together with the original Stabbing-Planes model of Beame et al. (ITCS 2018). The journal version was published June 3, 2026. This item is a dated frontier map, not a locally reconstructed lower-bound proof.

10.6 What Chapter 10 has established

There are now three distinct dual objects in the proof-complexity part of Volume XIII. Resolution uses combinatorial search and width; polynomial calculus uses linear functionals annihilating a dynamically generated vector space; SoS uses positive linear functionals whose moment and localizing matrices are positive semidefinite. The extra positivity is exactly what allows SoS to reason about inequalities, and it is also exactly what makes lower bounds substantially harder.

The next chapter may therefore import the pseudoexpectation language without re-defining it. When a later theorem says that a low-degree SoS relaxation “believes” a false combinatorial object exists, that phrase means there is a normalized truncated functional satisfying (10.9)–(10.11), not that there is an actual probability distribution supported on false solutions.

Internalization and Reconstruction Guide: Volume XIII

Purpose. Follow the proof-verification ladder from interactive proofs and arithmetization through sum-check, PCP, parallel repetition, resolution, polynomial calculus, and sum-of-squares/pseudoexpectations.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-II-THM-022: TQBF is PSPACE-complete.
2. QRCC-II-THM-006: 3SAT NP-completeness.
3. QRCC-I-THM-004: Chernoff bounds.
4. QRCC-III-THM-001: Classification of finite fields.
5. QRCC-XI-THM-002: Schwartz–Zippel lemma.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XIII-DEF-001 (Interactive proof system); QRCC-XIII-DEF-006 (Multiprover interactive proof); QRCC-XIII-DEF-009 (Probabilistically checkable proof); QRCC-XIII-DEF-034 (Assignment tester and proximity rejection); QRCC-XIII-DEF-039 (Projection game / Label Cover).

Chapter retrieval map

- Ch. 1: Interactive Proofs: Private Coins, Public Coins, and Proximity:** enter through QRCC-XIII-DEF-001 (Interactive proof system); exit by being able to state QRCC-XIII-THM-002 (Goldwasser–Sipser public-coin transformation) with its ownership boundary.
- Ch. 2: Arithmetization: Turning Boolean Quantifiers into Low-Degree Algebra:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state

QRCC-XIII-THM-004 (Arithmetization of Boolean formulas and TQBF) with its ownership boundary.

- Ch. 3: The Sum-Check Protocol:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIII-THM-003 (Sum-check protocol) with its ownership boundary.
- Ch. 4: From Arithmetization to IP :** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIII-THM-005 (IP equals PSPACE) with its ownership boundary.
- Ch. 5: Multiprover Interactive Proofs and the Low-Degree Interface:** enter through QRCC-XIII-DEF-006 (Multiprover interactive proof); exit by being able to state QRCC-XIII-BB-007 (MIP equals NEXP) with its ownership boundary.
- Ch. 6: Probabilistically Checkable Proofs, Gap Amplification, and Label Cover:** enter through QRCC-XIII-DEF-009 (Probabilistically checkable proof); exit by being able to state QRCC-XIII-COR-014 (Constant-gap hardness for MAX-3SAT) with its ownership boundary.
- Ch. 7: Parallel Repetition of Two-Prover Games:** enter through QRCC-XIII-DEF-040 (Finite two-prover game and parallel repetition); exit by being able to state QRCC-XIII-BB-012 (Raz parallel repetition theorem) with its ownership boundary.
- Ch. 8: Resolution Proofs, Width, and Exponential Lower Bounds:** enter through QRCC-XIII-DEF-015 (Cook-Reckhow proof system and simulation); exit by being able to state QRCC-XIII-BB-019 (Random-CNF resolution lower bounds) with its ownership boundary.
- Ch. 9: Polynomial Calculus, Nullstellensatz, and Interpolation:** enter through QRCC-XIII-DEF-020 (Polynomial calculus and Nullstellensatz systems); exit by being able to state QRCC-XIII-THM-021 (Polynomial-calculus duality and degree lower-bound criterion) with its ownership boundary.
- Ch. 10: Sum-of-Squares Proofs, Pseudoexpectations, and Semialgebraic Lower Bounds:** enter through QRCC-XIII-DEF-024 (Sum-of-squares proof and pseudoexpectation); exit by being able to state QRCC-XIII-BB-027 (SoS lower bounds for planted clique) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XIII-THM-002 — Goldwasser-Sipser public-coin transformation. Dependencies: QRCC-I-THM-004, QRCC-XIII-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XIII-THM-003 — Sum-check protocol. Dependencies: QRCC-III-THM-001, QRCC-XI-THM-002. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-XIII-BB-007 — MIP equals NEXP. Dependencies: QRCC-XIII-DEF-006, QRCC-XIII-THM-003. *source-owned interface: reconstruct the exact statement, model,*

and dependency role; do not replace the cited deep proof by a slogan.

4. QRCC-XIII-BB-012 — Raz parallel repetition theorem. Dependencies: QRCC-XIII-DEF-040. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XIII-THM-021 — Polynomial-calculus duality and degree lower-bound criterion. Dependencies: QRCC-XIII-DEF-020. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-XIII-F-029 — Interactive proofs of proximity. Dependencies: QRCC-XIII-DEF-001. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- forgetting field-size and degree constraints in arithmetization or sum-check;
- confusing soundness error with proof-of-knowledge extraction;
- comparing PCP parameters without tracking gap, randomness, queries, and proof length together;
- treating pseudoexpectations as honest probability distributions;

Boundary and degenerate cases

- degree at or above field size;
- zero-round/one-round protocol edge cases;
- perfect versus constant soundness;
- degree truncation where pseudoexpectation positivity no longer applies;

Retrieval and reconstruction drill

1. State QRCC-XIII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XIII-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XIII-THM-003, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.

6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XIII-F-029, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XIV reinterprets hardness under distributions, fine-grained parameters, and meta-computational encodings.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XIV

Average-Case Complexity, Fine-Grained Complexity, and Meta-Complexity

Chapter 1

Distributional Problems and Average Polynomial Time

Worst-case polynomial time asks for one polynomial bound that holds on every input of a given size. Average-case complexity needs a different convention: exceptional inputs may take much longer, but the probability of seeing inputs with increasingly large running time must decrease at a polynomial rate. This chapter fixes the distributional model and proves the equivalence between the tail and fractional-moment formulations of that requirement.

How to read this chapter. Read the chapter in three layers. First fix the object being measured: a language together with one input distribution for each length. Second separate the two distribution-access conventions PComp and PSamp from the running-time notion AvgP. Third use Theorem 1.2 as a conversion tool between polynomial tails and positive fractional moments. The positive example and nonexample are not decoration: they explain why robust average polynomial time is deliberately weaker than ordinary expected polynomial time.

Learning goals. After this chapter, a reader should be able to define a distributional decision problem; distinguish point-probability computability from efficient sampling; state the QRCC polynomial-tail convention for AvgP; prove the tail/fractional-moment equivalence with the exponent loss visible; and give a runtime distribution that is in robust AvgP although its ordinary expectation is not polynomially bounded.

Definition 1.1 (Distributional problems, ensemble classes, AvgP, and average-case reductions). An *ensemble* is a sequence $D = \{D_n\}_{n \geq 1}$ of probability distributions on $\{0, 1\}^*$ for which there is a polynomial b such that $D_n(x) > 0$ implies $|x| \leq b(n)$. Its support is

$$\text{Supp}(D_n) := \{x : D_n(x) > 0\}.$$

A *distributional decision problem* is a pair (L, D) consisting of a language $L \subseteq \{0, 1\}^*$ and an ensemble D . Algorithms receive both a sample x and the ensemble parameter n ; the latter is not silently identified with $|x|$.

The ensemble is *polynomial-time samplable* if a randomized algorithm, on input 1^n , runs in time polynomial in n on every coin sequence and outputs exactly the law D_n . Fix once and for all an effectively enumerable length-lexicographic order $\preceq$ on $\{0, 1\}^*$. Put

$$F_{D_n}(x) := \sum_{y \preceq x} D_n(y).$$

The ensemble is *polynomial-time computable* if $F_{D_n}(x)$ is exactly computable in time polynomial in n for every x with $|x| \leq b(n)$, using the canonical rational encoding returned by the distribution algorithm. We denote the two ensemble classes by **PSamp** and **PComp**, respectively. These names are kept distinct: no converse from samplability to cumulative-probability computability is built into the notation.

Let A be a deterministic algorithm that is correct on every $x \in \text{Supp}(D_n)$, and let $T_A(x; n) \geq 1$ be its running time. We say that A has *average polynomial running time* with respect to D when there are a constant $\varepsilon > 0$ and a polynomial p such that, for every n and every real $t \geq 1$,

$$\Pr_{x \sim D_n} [T_A(x; n) \geq t] \leq \frac{p(n)}{t^\varepsilon}. \quad (1.1)$$

The class **AvgP** consists of the distributional decision problems admitting such an errorless deterministic algorithm. Bounded-error and failure-symbol variants are different classes and are not silently identified with **AvgP**.

For distributional problems (L, D) and (L', D') , a *domination-preserving average-case reduction*, written

$$(L, D) \leq_{\text{AvgP}} (L', D'),$$

is a deterministic map $f(x; n)$ together with polynomials m, q such that f is computable in time polynomial in n , and for every $x \in \text{Supp}(D_n)$,

$$x \in L \iff f(x; n) \in L',$$

while for every string y ,

$$\sum_{x: f(x; n)=y} D_n(x) \leq q(n) D'_{m(n)}(y). \quad (1.2)$$

Thus the pushforward of D_n under f is pointwise dominated, up to a polynomial factor, by the target distribution at the prescribed target parameter.

Positive example. For the uniform ensemble U_n on $\{0, 1\}^n$, a worst-case polynomial-time decider is immediately an **AvgP** decider: once $T_A(x; n) \leq r(n)$ for every input, choose any $\varepsilon > 0$ and use $p(n) = r(n)^\varepsilon$ in (1.1).

Nonexample. Knowing only that the median running time is polynomial does not imply (1.1). An algorithm may run quickly on two thirds of the mass and take exponential time on the remaining one third; its median is small, but the probability of an exponential running time does not decay polynomially in the time threshold.

Theorem 1.2 (Tail-fractional-moment characterization of average polynomial time). *Let D be an ensemble and let A be an errorless deterministic decider on $\text{Supp}(D_n)$ with finite running time $T_A(x; n) \geq 1$. The following are equivalent.*

1. *There exist $\varepsilon > 0$ and a polynomial p for which (1.1) holds for every n and every $t \geq 1$.*
2. *There exist $\alpha > 0$ and a polynomial q such that*

$$\mathbb{E}_{x \sim D_n} [T_A(x; n)^\alpha] \leq q(n) \quad \text{for every } n. \quad (1.3)$$

Ordinary expected polynomial running time, $\mathbb{E}T_A \leq \text{poly}(n)$, is sufficient for these conditions but is not necessary.

Proof. Assume first that (1.3) holds. For every $t \geq 1$, Markov's inequality applied to the nonnegative random variable T_A^α gives

$$\Pr[T_A \geq t] = \Pr[T_A^\alpha \geq t^\alpha] \leq \frac{\mathbb{E}[T_A^\alpha]}{t^\alpha} \leq \frac{q(n)}{t^\alpha}.$$

Thus the tail condition holds with $\varepsilon = \alpha$.

Conversely suppose (1.1) holds. Choose any $0 < \alpha < \varepsilon$, for instance $\alpha = \varepsilon/2$. Because D_n has finite support and T_A is finite on that support, finite Fubini applied to

$$T_A^\alpha = \alpha \int_0^{T_A} t^{\alpha-1} dt$$

gives the layer-cake identity

$$\mathbb{E}[T_A^\alpha] = \alpha \int_0^\infty t^{\alpha-1} \Pr[T_A > t] dt.$$

Split the integral at 1. The first part is at most 1, while the tail hypothesis yields

$$\alpha \int_1^\infty t^{\alpha-1} \Pr[T_A > t] dt \leq \alpha p(n) \int_1^\infty t^{\alpha-1-\varepsilon} dt = \frac{\alpha}{\varepsilon - \alpha} p(n).$$

Hence (1.3) holds with a polynomial multiple of $1 + p(n)$.

Finally, expected polynomial time is the special case $\alpha = 1$ of the moment condition. To see that it is not necessary, let D_n be uniform on $\{0, 1\}^n$ and consider an always-correct decider whose running time is 2^{2n} on the single input 0^n and n^2 on every other input. Then

$$\mathbb{E}T_A \geq 2^{-n} 2^{2n} = 2^n,$$

which is exponential, whereas for $\alpha = 1/4$,

$$\mathbb{E}[T_A^{1/4}] \leq n^{1/2} + 2^{-n} 2^{n/2} = n^{1/2} + 2^{-n/2}.$$

The fractional moment is polynomially bounded, so the algorithm is polynomial-time on average in the robust sense of the theorem. $\square$

Worked check. If $\Pr[T_A \geq t] \leq n^3/t^2$, choosing $\alpha = 1$ in the second half of the proof gives

$$\mathbb{E}T_A \leq 1 + n^3.$$

If only the weaker tail $n^3/t^{1/2}$ is known, choosing $\alpha = 1/4$ instead gives a polynomial quarter-moment even though the first moment need not be polynomial.

Source and ownership boundary. Levin's average-case convention is the robust polynomial tail/fractional-moment notion, not naive expected polynomial time. The tail formulation and its equivalent positive-moment formulation are reconstructed here with local constants. The separate labels **PComp** and **PSamp** are intentional because the complete-problem theorem for computable ensembles and the stronger samplable-ensemble extension use different encoding arguments.

Dependency checkpoint. The proof uses only the finite-probability convention and Markov inequality from Volume I plus the finite Fubini calculation written explicitly above. It does not use any later fine-grained, meta-complexity, or cryptographic theorem.

1.1 Reader checkpoint and exercises

Synthesis and transition. The robust tail convention makes average running time stable under polynomial distortions, but it does not by itself make reductions safe. Chapter 2 adds the missing measure-transport condition: a reduction must control how much source probability can concentrate on a rare target set.

1. Starting from a tail bound $\Pr[T \geq t] \leq n^4/t^3$, derive an explicit polynomial bound on $\mathbb{E}[T]$ and then repeat the calculation for the fractional moment $\mathbb{E}[T^{1/2}]$.
2. Suppose $\mathbb{E}[T^{1/3}] \leq n^7$. Use Markov's inequality to give a polynomial tail bound of the form required by Definition 1.1.
3. For the positive example in the chapter, compute the ordinary expectation and one positive fractional moment. Explain exactly which of the two is polynomially bounded.
4. Give a simple ensemble whose point probabilities are polynomial-time computable and also polynomial-time samplable. Identify which parts of the algorithms certify **PComp** and which certify **PSamp**.
5. Explain why changing the universal polynomial-tail exponent by a constant factor does not change the qualitative **AvgP** notion, while demanding finite ordinary expectation would change it.
6. Construct a two-point runtime distribution illustrating why a small probability of an enormous running time can satisfy a fractional-moment bound. Quantify both the probability and the exceptional running time.

Chapter 2

Domination-Preserving Average-Case Reductions

A worst-case many-one reduction only has to preserve yes- and no-instances. In average-case complexity it must also prevent a rare target set from receiving a large amount of source probability. Equation (1.2) is exactly the quantitative condition needed for that transfer.

How to read this chapter. The theorem is a measure-transport statement disguised as a reduction theorem. Keep three objects separate while reading it: logical correctness of the map, the time needed to compute the map, and pointwise domination of the pushforward distribution. Only the third controls whether a rare bad set for the target algorithm can acquire substantial probability after pulling it back to the source. The worked domination calculation should be read before the final source-boundary paragraph.

Learning goals. After this chapter, a reader should be able to compute a pushforward distribution; check the polynomial domination inequality point by point; explain why an ordinary worst-case many-one reduction need not preserve average tractability; and reproduce the tail calculation proving closure of AvgP under the selected domination-preserving reduction model.

Theorem 2.1 (Closure of AvgP under domination-preserving reductions). *Suppose*

$$(L, D) \leq_{\text{AvgP}} (L', D')$$

through a reduction f , target-parameter polynomial m , and domination polynomial q as in Definition 1.1. If $(L', D') \in \text{AvgP}$, then $(L, D) \in \text{AvgP}$.

More explicitly, suppose the reduction takes at most $r(n) \geq 1$ steps and a target decider A' satisfies, for constants $\varepsilon > 0$ and a polynomial p ,

$$\Pr_{y \sim D'_N} [T_{A'}(y; N) \geq s] \leq \frac{p(N)}{s^\varepsilon} \quad (N \geq 1, s \geq 1).$$

Then the composed decider has a tail bound

$$\Pr_{x \sim D_n} [T_A(x; n) \geq t] \leq \frac{P(n)}{t^\varepsilon}$$

for a polynomial P determined explicitly by p, q, m, r , and ε .

Proof. Define

$$A(x; n) := A'(f(x; n); m(n)).$$

Correctness follows from the reduction equivalence in Definition 1.1. Absorb fixed simulation costs into $r(n)$ so that

$$T_A(x; n) \leq r(n) + T_{A'}(f(x; n); m(n)).$$

Fix n and $t \geq 1$. First suppose $t > 2r(n)$. Then $T_A(x; n) \geq t$ implies

$$T_{A'}(f(x; n); m(n)) \geq t - r(n) > t/2.$$

Let

$$B_t := \{y : T_{A'}(y; m(n)) \geq t/2\}.$$

Summing the pointwise domination inequality (1.2) over $y \in B_t$ gives

$$\begin{aligned} \Pr_{x \sim D_n} [f(x; n) \in B_t] &= \sum_{y \in B_t} \sum_{x: f(x; n)=y} D_n(x) \\ &\leq q(n) \sum_{y \in B_t} D'_{m(n)}(y) \\ &= q(n) \Pr_{y \sim D'_{m(n)}} [T_{A'}(y; m(n)) \geq t/2] \\ &\leq \frac{2^\varepsilon q(n) p(m(n))}{t^\varepsilon}. \end{aligned}$$

Therefore the same bound holds for the event $T_A \geq t$.

Now suppose $1 \leq t \leq 2r(n)$. The trivial probability bound gives

$$\Pr[T_A \geq t] \leq 1 \leq \frac{(2r(n))^\varepsilon}{t^\varepsilon}.$$

If $k = \lceil \varepsilon \rceil$, then $(2r(n))^\varepsilon \leq (2r(n))^k$ because $r(n) \geq 1$. Hence the polynomial

$$P(n) := (2r(n))^k + 2^\varepsilon q(n) p(m(n))$$

dominates the numerator in both threshold regimes. This proves the asserted AvgP tail bound for the composed decider. $\square$

Worked domination check. Suppose a reduction has constant domination factor four and negligible local overhead, while the target bad-time set obeys

$$\Pr_{D'_N} [T_{A'} \geq s] \leq N^2/s^3.$$

For thresholds above twice the reduction time, the pulled-back bad set has probability at most

$$4 \frac{m(n)^2}{(t/2)^3} = \frac{32m(n)^2}{t^3}.$$

The exceptional probability increases only by the permitted domination factor; it cannot become a constant-density source set unless the target set already had correspondingly large mass.

Why ordinary many-one reduction is insufficient. A correct reduction could map a large fraction of source inputs to one target string that has tiny D' -probability. A target average-case algorithm is allowed to be very slow on such a rare string, so correctness alone would not transfer average tractability. Pointwise pushforward domination is what prevents this concentration of probability mass.

Source and ownership boundary. The reduction model is the deterministic domination-preserving model used in the classical Levin-style theory and in the Bogdanov–Trevisan survey. The proof above is local: it transfers the polynomial running-time tail directly and therefore does not rely on a heuristic-scheme equivalence. Randomized encodings needed for the broader samplable-ensemble completeness theorem are not silently included in this result.

Dependency checkpoint. The theorem depends only on Definition 1.1, finite summation, and the target AvgP tail inequality. It has no dependency on the later complete-problem, worst-case-to-average-case, SETH, Kolmogorov-complexity, or meta-complexity blocks.

2.1 Reader checkpoint and exercises

Synthesis and transition. Logical correctness plus running-time efficiency is not enough for an average-case reduction; pointwise pushforward domination prevents probability mass from collapsing onto a rare hard target. Chapter 3 uses exactly this condition to design a complete distribution whose encoded instances have enough target mass.

1. For a deterministic map f from a four-point source space to a three-point target space, choose explicit source probabilities and compute the pushforward distribution $f(D)$.
2. Construct an ordinary many-one reduction that sends many source points to one target point and explain how it can violate any polynomial domination bound.
3. In the proof of Theorem 2.1, redo the tail calculation when the reduction time is bounded by $m(n)^3$ instead of $m(n)$.
4. Show that multiplying the domination polynomial by a constant does not affect closure, but replacing polynomial domination by an exponential factor can destroy the conclusion.
5. Verify the worked domination example at thresholds t and $2t$ and identify where the reduction-running-time cutoff enters.
6. State in one sentence each the roles of correctness, efficiency, and domination in the theorem; then give an example of a failure caused by dropping each role.

Chapter 3

A Complete Problem for Computable Ensembles

The domination condition makes completeness possible only after the target distribution has been chosen with care. A fixed-length uniform target assigns every individual output exponentially small mass, which is too little to dominate a source input of noticeable probability. The construction below solves this by randomizing the description lengths of a padded bounded-halting instance. The resulting target ensemble has an exactly computable cumulative distribution and gives a string with description lengths a, b probability proportional to 2^{-a-b} , up to a polynomial factor.

How to read this chapter. The construction solves two different problems at once. The target language must simulate an NP verifier, while the target distribution must assign enough mass to the particular encoding of a source input to satisfy the domination inequality from Chapter 2. Follow the proof by tracking payload length, total padded length, and target point mass in parallel. The theorem is only a completeness theorem for PComp ensembles; the final boundary paragraph explains why polynomial-time samplability needs an additional randomized encoding theorem.

Learning goals. After this chapter, a reader should be able to describe the padded bounded-halting language and its length-mixture distribution; prove that the target point probabilities and cumulative probabilities are polynomial-time computable; derive the polynomial domination factor from dyadic coding; and identify precisely which step fails if one tries to replace the source PComp hypothesis by an arbitrary PSamp ensemble.

The padded bounded-halting language. For an integer $N \geq 1$ put $L_N = \lceil \log_2(N+1) \rceil$. If $0 \leq a, b \leq N$, $M \in \{0, 1\}^a$, and $z \in \{0, 1\}^b$, define

$$\text{pad}_N(a, b, M, z) := 1^N 0 \text{bin}_{L_N}(a) \text{bin}_{L_N}(b) M 0^{N-a} z 0^{N-b}. \quad (3.1)$$

Every such string has the same length $3N + 1 + 2L_N$. The first block is a unary copy of the time budget; the next two fixed-width blocks give the lengths of the machine description and payload; the final two N -bit slots contain those strings followed by zero padding.

Let PBH be the language of strings of the form (3.1) for which the Turing machine represented by M has an accepting nondeterministic computation on z of at most N steps. Invalid strings are rejected. As in the machine convention of Volume II, every bit string denotes a machine, with invalid descriptions interpreted as one fixed rejecting machine. A bounded accepting computation is a polynomial-size certificate, so $\text{PBH} \in \text{NP}$.

For each N , define a distribution H_N by the experiment

1. choose a, b independently and uniformly from $\{0, \dots, N\}$;
2. choose $M \in \{0, 1\}^a$ and $z \in \{0, 1\}^b$ independently and uniformly;
3. output $\text{pad}_N(a, b, M, z)$.

Thus every valid padded string has exact point mass

$$H_N(\text{pad}_N(a, b, M, z)) = \frac{2^{-a-b}}{(N+1)^2}. \quad (3.2)$$

Computability of the target ensemble. The ensemble $H = \{H_N\}$ belongs to **PComp**. Indeed, its support has the one fixed length $3N + 1 + 2L_N$. For a query string of a different length the cumulative probability is immediately zero or one according to length-lexicographic order. At the support length, iterate over the $O(N^2)$ pairs (a, b) . The first two fixed-width fields decide the lexicographic order of the corresponding blocks. Within one pair (a, b) , the two padded N -bit slots are monotone in the binary integers represented by M and z , so the number of support strings not exceeding the query is obtained by two integer-rank calculations rather than exponential enumeration. Multiplying those counts by the exact mass (3.2) and summing $O(N^2)$ rational numbers computes F_{H_N} exactly in time polynomial in N .

Theorem 3.1 (Padded bounded halting is complete for $(\text{NP}, \text{PComp})$). *The distributional problem (PBH, H) belongs to $(\text{NP}, \text{PComp})$. Moreover, for every $(L, D) \in (\text{NP}, \text{PComp})$ there is a deterministic domination-preserving average-case reduction*

$$(L, D) \leq_{\text{AvgP}} (\text{PBH}, H).$$

Consequently (PBH, H) is complete for $(\text{NP}, \text{PComp})$ under the reduction convention of Definition 1.1.

Proof. Membership has already been checked above: $\text{PBH} \in \text{NP}$ and $H \in \text{PComp}$. Fix now an arbitrary $(L, D) \in (\text{NP}, \text{PComp})$. We construct the reduction in four steps.

Step 1: a computable short injective code for D_n . For $x \in \text{Supp}(D_n)$ let

$$A_x := \sum_{y \prec x} D_n(y), \quad B_x := A_x + D_n(x), \quad P_x := D_n(x).$$

These quantities are exactly computable in polynomial time from the cumulative-distribution algorithm in Definition 1.1. The half-open intervals $I_x = [A_x, B_x)$ are pairwise disjoint and have length P_x .

Put

$$\ell_x := \left\lceil \log_2 \frac{1}{P_x} \right\rceil + 2, \quad m_x := \frac{A_x + B_x}{2},$$

and let $C_n(x)$ be the first ℓ_x binary digits after the binary point of m_x , using the canonical terminating expansion for dyadic rationals. Equivalently, if $c_x = \lfloor 2^{\ell_x} m_x \rfloor$, then $C_n(x)$ is the length- ℓ_x binary representation of c_x . The dyadic interval

$$J_x := \left[\frac{c_x}{2^{\ell_x}}, \frac{c_x + 1}{2^{\ell_x}} \right)$$

contains m_x and has length

$$2^{-\ell_x} \leq \frac{P_x}{4}.$$

Since m_x is distance $P_x/2$ from either endpoint of I_x , this implies $J_x \subset I_x$. Therefore different source strings have different dyadic cells and hence different codewords $C_n(x)$. The map C_n is injective and polynomial-time computable, and

$$|C_n(x)| \leq \log_2 \frac{1}{D_n(x)} + 3. \quad (3.3)$$

Because the exact cumulative probabilities are output in polynomial time, every positive $D_n(x)$ has a polynomial-bit rational representation; hence the right-hand side of (3.3) is polynomially bounded in n .

Step 2: put the source parameter into the payload. Use a fixed self-delimiting pairing $\langle n, c \rangle$ satisfying, for $|c|$ polynomially bounded in n ,

$$|\langle n, c \rangle| \leq |c| + O(\log(n+2) + \log(|c|+2)). \quad (3.4)$$

For example, prefix the ordinary binary encodings of n and $|c|$ by unary encodings of their binary lengths. Set

$$z_x := \langle n, C_n(x) \rangle.$$

Then $x \mapsto z_x$ is injective for fixed n , and from (3.3)–(3.4) there is a polynomial s and a constant d such that

$$|z_x| \leq s(n), \quad 2^{-|z_x|} \geq \frac{D_n(x)}{n^d} \quad (3.5)$$

for all sufficiently large n ; enlarging the final domination polynomial handles the finitely many smaller parameters.

Step 3: one fixed nondeterministic target machine. Let V be a fixed polynomial-time verifier for L . Construct one nondeterministic machine $M_{L,D}$ which, on a payload z , parses $z = \langle n, c \rangle$, guesses a string x of length at most the support bound for D_n and a witness w of the verifier's polynomially bounded length, computes $C_n(x)$ from the encoded cumulative-distribution algorithm for D , checks $C_n(x) = c$, and accepts exactly when $V(x, w)$ accepts. All these operations take time polynomial in n . Since C_n is injective, for the particular payload z_x we have

$$M_{L,D} \text{ accepts } z_x \iff x \in L. \quad (3.6)$$

Let $a_0 := |\langle M_{L,D} \rangle|$, a constant depending on the fixed distributional problem but not on n or x .

Choose a polynomial target parameter $N = m(n)$ large enough that

$$N \geq a_0, \quad N \geq s(n),$$

and that every accepting computation of $M_{L,D}$ arising from the preceding verification can be completed within N steps. Define

$$f(x; n) := \text{pad}_{m(n)}(a_0, |z_x|, \langle M_{L,D} \rangle, z_x). \quad (3.7)$$

The map is deterministic and polynomial-time computable. Equation (3.6) gives

$$x \in L \iff f(x; n) \in \text{PBH}.$$

Step 4: domination. For fixed n , the reduction is injective because $x \mapsto z_x$ is injective. Hence the pushforward mass of the particular output $f(x; n)$ is exactly $D_n(x)$. By (3.2),

$$H_{m(n)}(f(x; n)) = \frac{2^{-a_0 - |z_x|}}{(m(n) + 1)^2}.$$

Combining this with (3.5) gives

$$D_n(x) \leq 2^{a_0} n^d (m(n) + 1)^2 H_{m(n)}(f(x; n)).$$

The coefficient on the right is polynomial in n . Outputs outside the image have pushforward mass zero, so the same polynomial proves the pointwise domination condition for every target string. Thus f is a domination-preserving average-case reduction, completing the proof. $\square$

Worked probability check. Ignore the constant machine-description factor and suppose a source point has $D_n(x) = 2^{-20}$, while its paired payload has length 26. If the chosen target parameter is $N = 100$, then its target point mass is

$$\frac{2^{-26}}{101^2}.$$

The ratio between source and target masses is $2^6 \cdot 101^2$, a polynomial factor. The purpose of compression is exactly to keep the payload length within an additive logarithmic overhead of $\log_2(1/D_n(x))$; padding the payload to length N before sampling would destroy this property.

Source and ownership boundary. The classical Levin completeness theorem and its modern survey presentation use a bounded-halting problem equipped with a carefully normalized near-uniform distribution on encoded instances. The QRCC proof above uses a locally defined padded length-mixture ensemble H because its point masses and cumulative-distribution computation are transparent. This is a local reconstruction of the computable-ensemble completeness theorem type, not a claim that H is the historical source's literal target distribution.

The theorem is deliberately only for (NP, PComp). Extending completeness to all polynomial-time samplable ensembles requires the separate Impagliazzo–Levin randomized encoding argument and is not a consequence of the deterministic compression used here.

Dependency checkpoint. The proof uses Definition 1.1, the Volume-II verifier characterization of NP at the level of its already fixed witness convention, and elementary dyadic interval coding. It does not use the later hard-core, direct-product, fine-grained, Kolmogorov-complexity, MCSP, or hardness-magnification results.

3.1 Reader checkpoint and exercises

Synthesis and transition. The complete computable ensemble is a coding theorem as much as a complexity theorem: description length must track information content closely enough that the target point mass polynomially dominates the source mass. The deterministic coding argument stops at PComp. Chapter 4 therefore studies hardness amplification separately rather than pretending it supplies the missing samplable-ensemble completeness theorem.

1. For the chapter's length-mixture distribution, compute the probability of one target encoding at two different padded lengths and compare their ratio.
2. Reproduce the dyadic interval argument showing that a source point of probability p can be assigned a description of length within $O(\log n)$ of $\log_2(1/p)$.
3. Explain why padding the payload to the final target length before assigning probability can destroy polynomial domination.

4. Verify that the cumulative target probability up to a lexicographically specified string is polynomial-time computable under the chapter's representation.
5. Trace the verifier simulation for a toy NP language with a one-bit witness and identify the time bound encoded in the bounded-halting instance.
6. Identify the exact sentence in the proof that uses PComp. Explain why a sampler alone does not supply the same operation and what kind of extra randomized encoding theorem is therefore needed.

Chapter 4

Average-Hardness Amplification and Black-Box Barriers

The phrase “hard on average” hides several logically different operations. One may start with a function that already makes every small predictor fail on a noticeable fraction of uniformly random inputs and concentrate that error on a dense hard core. One may then combine independent inputs so that predicting one aggregate bit becomes much harder. Neither operation by itself proves that an NP-complete problem is hard under a polynomial-time samplable distribution. In the opposite direction, a sufficiently black-box worst-case-to-average-case reduction for NP would itself have unexpected complexity consequences.

This chapter separates those interfaces. The first theorem translates the already source-verified Impagliazzo hard-core theorem from Volume IX into a distributional statement and records exactly what computability it does *not* provide. The second proves a conservative independent-input XOR amplification theorem with every density, advantage, and circuit-size loss visible. The final deep input freezes the Bogdanov–Trevisan nonadaptive barrier in the exact approximate-oracle model used by the average-case literature.

How to read this chapter. There are three logically separate arrows. Section 1 turns noticeable predictor failure into a hard-core distribution; Section 2 amplifies independent-bit hardness by XOR; Section 3 records a black-box barrier showing that a broad nonadaptive worst-case-to-average-case route for NP would have an unexpected nonuniform consequence. Do not combine the arrows into one theorem: the first does not supply an efficient sampler, the second assumes independent product inputs, and the third is a limitation theorem for a specified reduction model rather than an impossibility theorem for average-case hardness.

Learning goals. After this chapter, a reader should be able to distinguish a hard set from a samplable hard distribution; compute the domination constant of a hard-core distribution; explain why independent product sampling is essential to the local XOR proof; derive the all-easy-component error term $(1 - \delta)^k$; define a nonadaptive black-box worst-case-to-average-case reduction without hiding the quantifier over approximate oracles; and state precisely what Bogdanov–Trevisan rules out and what it leaves open.

4.1 From a hard-core set to a dominated hard-core distribution

Fix a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. For a distribution D on $\{0, 1\}^n$, a circuit-size bound T , and $0 \leq \alpha \leq 1/2$, say that f is (T, α) -unpredictable under D if every Boolean circuit C of size at most T satisfies

$$\Pr_{x \leftarrow D} [C(x) = f(x)] \leq \frac{1}{2} + \alpha. \quad (4.1)$$

For $0 < \delta \leq 1$, a distribution ν is δ^{-1} -dominated by the uniform distribution when

$$\nu(x) \leq \delta^{-1} 2^{-n} \quad \text{for every } x \in \{0, 1\}^n. \quad (4.2)$$

This is pointwise domination in the sense of Chapter 1. It is a measure-theoretic statement, not an algorithm for evaluating or sampling ν .

Theorem 4.1 (Hard-core distributions from mild uniform average hardness). *Let $0 < \delta < 1/2$, $0 < \eta < 1/2$, and*

$$n \leq S < \frac{2^n}{n}. \quad (4.3)$$

Suppose every Boolean circuit C of size at most S satisfies

$$\Pr_{x \leftarrow U_n} [C(x) = f(x)] \leq 1 - \delta. \quad (4.4)$$

Let $c > 0$ be the universal constant from Theorem 6.2, and put

$$S_{\text{hc}} := \lfloor c\eta^2 \delta^2 S \rfloor. \quad (4.5)$$

Then there is a set $H \subseteq \{0, 1\}^n$ with $|H| \geq \delta 2^n$ such that the uniform distribution $\nu_H := U_H$ has both properties

$$\nu_H(x) \leq \delta^{-1} U_n(x) \quad \text{for every } x, \quad (4.6)$$

and

$$\Pr_{x \leftarrow \nu_H} [C(x) = f(x)] \leq \frac{1}{2} + \eta \quad (4.7)$$

for every circuit C of size at most S_{hc} .

The theorem does not assert that membership in H is polynomial-time decidable, nor that ν_H is a PComp or PSamp ensemble. Such an algorithmic conclusion requires additional structure not supplied by the hard-core theorem alone.

Proof. Theorem 6.2 applied to (4.4) gives a hard-core set H of cardinality at least $\delta 2^n$ on which every circuit of size at most S_{hc} has success at most $1/2 + \eta$. This is exactly (4.7).

For $x \notin H$, $\nu_H(x) = 0$. For $x \in H$,

$$\nu_H(x) = \frac{1}{|H|} \leq \frac{1}{\delta 2^n} = \delta^{-1} U_n(x), \quad (4.8)$$

which proves (4.6). Nothing in this counting argument constructs an efficient membership test or sampler for H , establishing the final scope statement as well. $\square$

Why domination is the right distributional translation. Equation (4.6) says that the hard core cannot concentrate more than a factor $1/\delta$ above uniform mass at any point. When $\delta \geq 1/\text{poly}(n)$, this is polynomial domination, the same quantitative relation that appears in average-case reductions. But domination and efficient samplability are independent properties: an existential hard set can be dense without coming with any efficient way to recognize or sample it.

Worked checkpoint. If $\delta = 1/3$, then $|H| \geq 2^n/3$ and every point of U_H has mass at most $3 \cdot 2^{-n}$. Choosing $\eta = 1/10$ gives hard-core circuit size $\lfloor cS/900 \rfloor$. The density loss and the circuit-size loss are logically different: large support alone does not imply strong prediction hardness.

4.2 Independent-input XOR amplification

For $k \geq 1$, define

$$f^{\oplus k}(x_1, \dots, x_k) := f(x_1) \oplus \dots \oplus f(x_k). \quad (4.9)$$

The input distribution below is the product U_n^k . This is a genuine distributional problem in the sense of Definition 1.1; it should not be confused with a correlated k -tuple distribution or with the vector-valued direct product $(f(x_1), \dots, f(x_k))$.

Theorem 4.2 (Independent-product XOR amplification for a distributional problem). *Under the hypotheses (4.3)–(4.4), let $k \geq 1$ and $0 < \varepsilon < 1/2$ satisfy*

$$\varepsilon > 2(1 - \delta)^k. \quad (4.10)$$

Set

$$T := \left\lfloor \frac{c\varepsilon^2\delta^2S}{16} \right\rfloor. \quad (4.11)$$

If $T \geq 1$, then every Boolean circuit D of size at most $T - 1$ satisfies

$$\Pr_{(x_1, \dots, x_k) \leftarrow U_n^k} [D(x_1, \dots, x_k) = f^{\oplus k}(x_1, \dots, x_k)] \leq \frac{1}{2} + \varepsilon. \quad (4.12)$$

If $T = 0$, the circuit-size conclusion is vacuous.

Proof. Apply Theorem 4.1 with hard-core advantage $\eta = \varepsilon/4$. It gives a set H of density

$$\rho := |H|/2^n \geq \delta \quad (4.13)$$

on which every circuit of size at most T predicts f with success at most $1/2 + \varepsilon/4$.

If $\rho < 1$, let G be uniform on the complement of H ; if $\rho = 1$, give the complementary component weight zero. In either case

$$U_n = \rho U_H + (1 - \rho)G. \quad (4.14)$$

Taking k independent copies expands U_n^k as a convex mixture indexed by subsets $J \subseteq [k]$: coordinates in J come independently from U_H , and the rest from G . The all- G component has weight

$$a = (1 - \rho)^k \leq (1 - \delta)^k < \frac{\varepsilon}{2}. \quad (4.15)$$

Suppose a circuit D of size at most $T - 1$ had success strictly greater than $1/2 + \varepsilon$ in (4.12). Let M be its largest success among mixture components with $J \neq \emptyset$. Since success on the all- G component is at most one,

$$\Pr[D = f^{\oplus k}] \leq a + (1 - a)M. \quad (4.16)$$

If $M \leq 1/2 + \varepsilon/4$, then

$$a + (1 - a)M \leq \frac{1}{2} + \frac{a}{2} + \frac{\varepsilon}{4} < \frac{1}{2} + \frac{\varepsilon}{2}, \quad (4.17)$$

a contradiction. Hence some nonempty component J has success above $1/2 + \varepsilon/4$.

Choose $j \in J$. Averaging over all coordinates except j fixes strings $(x_i)_{i \neq j}$ for which the remaining random input $x_j \leftarrow U_H$ still gives success above $1/2 + \varepsilon/4$. Hardwire those strings and the advice bit

$$b := \bigoplus_{i \neq j} f(x_i). \quad (4.18)$$

Then

$$C(x) := D(x_1, \dots, x_{j-1}, x, x_{j+1}, \dots, x_k) \oplus b \quad (4.19)$$

has size at most T and predicts f on U_H above the hard-core threshold, a contradiction. $\square$

Parameter checkpoint. For $\delta = 1/3$ and $k = 8$,

$$2(1 - \delta)^k = 2(2/3)^8 < 0.079. \quad (4.20)$$

Thus $\varepsilon = 0.1$ satisfies (4.10), and the amplified task is unpredictable to advantage 0.1 against circuits of size roughly $cS/14400$. The theorem uses kn truly independent input bits. It does not claim the constant-factor input-length compression needed in the Impagliazzo–Wigderson E -hardness bridge.

Historical source comparison. The independent-input XOR mechanism is classically attributed to Yao. A sharper standard circuit formulation gives circuit size $\varepsilon^2 S/(400n)$ under the same threshold $\varepsilon > 2(1 - \delta)^k$. Theorem 4.2 deliberately keeps the weaker δ^2 loss inherited from the already frozen Impagliazzo hard-core theorem, because that is the form reconstructed line by line here. The sharper historical constant is source comparison, not a hidden strengthening of the local proof.

4.3 What a black-box worst-case-to-average-case reduction means

The negative result below concerns a specific reduction model. It does not rule out adaptive, non-black-box, algebraically structured, or problem-specific worst-case-to-average-case reductions.

Definition 4.3 (Approximate oracle and nonadaptive black-box worst-case-to-average-case reduction). Let $(L', \mathcal{D})$ be a distributional decision problem. An *approximate oracle with error* $\gamma(m)$ is a function

$$A : \{0, 1\}^* \longrightarrow \{0, 1, \perp\} \quad (4.21)$$

such that for every length m ,

$$\Pr_{y \leftarrow D_m} [A(y) \neq L'(y)] < \gamma(m), \quad (4.22)$$

where $\perp$ counts as failure. The oracle is *errorless* when $A(y) \in \{L'(y), \perp\}$ for every y .

A randomized polynomial-time oracle machine R is a *worst-case-to-average-case reduction* from a language L to $(L', \mathcal{D})$ with error bound γ if, for every approximate oracle A satisfying (4.22), every sufficiently large input length, and every input $x \in \{0, 1\}^n$,

$$\Pr_R [R^A(x) = L(x)] > \frac{2}{3}. \quad (4.23)$$

The reduction is *nonadaptive* when, after fixing x and the internal random coins of R , all oracle queries are determined before any oracle answer is received. Thus all queries may be issued in parallel. The adjective *black-box* emphasizes the universal quantifier over oracles A : the correctness proof may use only oracle answers, not code for an algorithm implementing A .

Theorem 4.4 (Bogdanov–Trevisan nonadaptive black-box barrier; source-owned). *Let L be a language. Suppose there exists a nonadaptive worst-case-to-average-case reduction in the sense of Definition 4.3 from L to some distributional decision problem*

$$(L', \mathcal{D}) \in (\text{NP}, \text{PSamp}). \quad (4.24)$$

Then

$$L \in \text{NP}/\text{poly} \cap \text{coNP}/\text{poly}. \quad (4.25)$$

Consequently, if L is NP-complete under polynomial-time many-one reductions, the standard nonuniform consequences include $\text{coNP} \subseteq \text{NP}/\text{poly}$, and the polynomial hierarchy collapses to its third level.

Source-owned proof architecture. The proof is not reconstructed here. Its key point is precisely the universal black-box quantifier in Definition 4.3. Bogdanov and Trevisan simulate the reduction by a constant-round proof protocol with advice that encodes suitable query-distribution statistics. Nonadaptivity lets all oracle questions be treated simultaneously. The NP witness structure of L' and the samplability of $\mathcal{D}$ let the protocol police the simulated approximate oracle well enough to place L in both nonuniform NP and nonuniform coNP. The source’s argument, not this paragraph, owns those protocol and estimation details.

Primary-source boundary. Andrej Bogdanov and Luca Trevisan, *On Worst-Case to Average-Case Reductions for NP Problems*, *SIAM Journal on Computing* 36(4) (2006), 1119–1159, DOI 10.1137/S0097539705446974. Their theorem is exactly the nonadaptive statement (4.25) for a target in $(\text{NP}, \text{PSamp})$; no restriction is imposed that each query itself have the target distribution. Their survey restates the same theorem as Theorem 7.5 and explicitly contrasts it with adaptive and BPP-class/non-black-box variants.

4.4 What the chapter does not prove

Three boundaries are worth retaining.

First, Theorem 4.1 produces an existential dominated hard distribution, not a polynomial-time samplable ensemble. Thus the theorem is a nonuniform predictor-hardness translation, not an automatic DistNP completeness theorem.

Second, Theorem 4.2 uses the full independent product distribution. A short-seed derandomized direct product is a stronger object; Volume IX keeps the Impagliazzo–Wigderson version source-owned for exactly this reason.

Third, Theorem 4.4 is a barrier to a broad but specific black-box, nonadaptive method. It does not prove that worst-case NP hardness can never imply average-case NP hardness. Structured lattice reductions, adaptive reductions, and non-black-box reductions fall outside its hypotheses.

Samplable-ensemble completeness checkpoint. Chapter 3 proved the selected complete problem only for polynomial-time computable ensembles. Impagliazzo–Levin’s stronger completeness theorem for polynomial-time samplable distributions uses an additional randomized encoding/search-to-decision argument. Nothing in the present hard-core or XOR proof supplies that missing randomized completeness step, so it remains logically separate rather than being smuggled into the notation PSamp.

4.5 Reader checkpoint and exercises

Synthesis and transition. Hard-core concentration and XOR amplification show how mild predictor hardness can be strengthened, while the Bogdanov–Trevisan theorem marks a precise limit of one black-box route from worst-case NP hardness. None of these results changes the exponential-time hypotheses used in fine-grained complexity. Chapter 5 therefore begins a new axis of analysis by fixing ETH, SETH, and reduction budgets from scratch.

1. Starting from the hard-core theorem’s density parameter, compute the normalization of the dominated hard-core distribution and verify its pointwise domination bound.
2. In Theorem 4.1, identify which conclusion is existential and explain why it does not imply efficient sampling from the hard-core distribution.
3. For $k = 3$ independent input bits with individual prediction advantage ε , write the XOR target explicitly and use the theorem’s bound to track the resulting advantage.
4. Explain why sharing randomness among the k coordinates would invalidate the independent-input XOR proof without an additional theorem.
5. State the nonadaptive black-box hypothesis of Theorem 4.4 and name two reduction models explicitly outside its scope.
6. Draw a three-arrow diagram separating hard-core concentration, XOR amplification, and the black-box barrier. Label every arrow as local proof or source-owned input.

Chapter 5

ETH, SETH, Fine-Grained Reductions, and Sparsification

Average-case complexity changes the input measure. Fine-grained complexity changes a different part of the question: instead of asking only whether a problem is polynomial-time solvable, it asks whether the exponent of the best known running time can be improved. This chapter fixes the hypotheses and the reduction bookkeeping before any conditional lower bound is stated.

How to read this chapter. This chapter is a convention layer for the reductions that follow. First freeze the quantifier order in ETH and SETH. Next read a fine-grained reduction as a budget statement: a target exponent saving must translate into some source exponent saving after summing every oracle call and all nonoracle work. Finally treat sparsification as a deep preprocessing input whose bounded-occurrence conclusion is what later keeps Orthogonal Vectors in logarithmic dimension.

Learning goals. After this chapter, a reader should be able to state deterministic ETH and SETH without interchanging their quantifiers; verify the running-time budget in the selected fine-grained reduction definition; explain why fixed polynomial and polylogarithmic factors do not absorb an exponent saving; and state the exact sparsification interface used by the split-and-list reduction.

Definition 5.1 (ETH, SETH, and budgeted fine-grained reductions). For an integer $k \geq 2$, a k -CNF formula is a conjunction of clauses, each clause being a disjunction of at most k literals. The problem k -SAT asks whether such a formula is satisfiable. If F has n variables, define its deterministic exponential-time exponent by

$$s_k := \inf \{ \sigma \geq 0 : k\text{-SAT is solvable in } 2^{\sigma n} \text{poly}(|F|) \text{ time} \}.$$

Because a $(k-1)$ -CNF is also a k -CNF, s_k is nondecreasing in k , and exhaustive search gives $s_k \leq 1$.

The *Exponential Time Hypothesis* (ETH) is the assertion $s_3 > 0$. The *Strong Exponential Time Hypothesis* (SETH) is

$$\lim_{k \rightarrow \infty} s_k = 1.$$

Equivalently, SETH says that for every $\varepsilon > 0$ there is a fixed k for which no deterministic algorithm solves k -SAT in $O(2^{(1-\varepsilon)n} \text{poly}(|F|))$ time. These are deterministic hypotheses in this volume. Randomized ETH/SETH are separately named hypotheses and are not inferred from the deterministic versions.

Let P, Q be problems with reference time scales $T_P, T_Q \geq 1$. A *budgeted deterministic fine-grained reduction* from (P, T_P) to (Q, T_Q) is an oracle reduction with the following property: for every $\varepsilon > 0$ there is $\delta > 0$ such that, on every size- n instance of P , the nonoracle work is at most $T_P(n)^{1-\delta}$ polylog $T_P(n)$ and the oracle queries have sizes $m_1, \dots, m_r$ satisfying

$$\sum_{i=1}^r T_Q(m_i)^{1-\varepsilon} \leq T_P(n)^{1-\delta} \text{polylog } T_P(n).$$

Thus a $T_Q^{1-\varepsilon}$ algorithm for Q yields a $T_P^{1-\delta}$ algorithm for P , up to the displayed polylogarithmic slack. A usual polynomial-time many-one reduction need not satisfy this exponent budget and therefore is not automatically a fine-grained reduction.

Sanity checks. If k -SAT could be solved in $1.99^n \text{poly}(|F|)$ time for every fixed k , then SETH would be false, because $1.99^n = 2^{(1-\eta)n}$ for the fixed constant $\eta = 1 - \log_2 1.99 > 0$. By contrast, an improvement from 2^n to $2^n/n^{100}$ does not by itself refute SETH: it does not save a positive constant in the exponent.

Theorem 5.2 (Sparsification lemma; exact interface used below). *For every fixed $k \geq 2$ and every $\eta > 0$ there is a constant $c = c(k, \eta)$ and an algorithm which, given a k -CNF formula F on n variables, outputs formulas $F_1, \dots, F_t$ such that*

1. $t \leq 2^m$;
2. F is satisfiable if and only if at least one F_i is satisfiable;
3. every F_i is a k -CNF on the same variable set in which every variable occurs in at most c clauses, and hence every F_i has at most cn clauses;
4. the complete list is produced in $2^m \text{poly}(|F|)$ time, where the polynomial may depend on fixed k, η .

Proof status and source boundary. This is the Impagliazzo–Paturi–Zane sparsification lemma and remains a registered Level-C/deep input in the QRCC architecture. The statement above freezes the exact interface needed by the next chapter: the number of disjuncts, bounded occurrence, linear clause count, and total generation time are all explicit. No local proof is claimed in this pass.

Why sparsification is load-bearing. The split-and-list reduction in the next chapter creates one vector coordinate per clause. Without sparsification a k -CNF may have superlinear many clauses, so the resulting vector dimension need not be $O(n) = O(\log N)$. Sparsification is therefore not cosmetic preprocessing; it is exactly what turns the clause-coordinate reduction into the low-dimensional Orthogonal Vectors regime.

5.1 Reader checkpoint and exercises

Synthesis and transition. ETH and SETH specify the hypothesized source-time scales, a fine-grained reduction specifies how an exponent improvement would propagate, and sparsification supplies the bounded-density instances needed for geometric reductions. Chapter 6 combines all three ingredients in the explicit SETH-to-Orthogonal-Vectors chain.

1. Negate ETH and SETH carefully, preserving the order of the quantifiers over k and the saving parameter. Explain why the two negations are not the same statement.

2. Given a target running time $T_Q(m)^{1-\varepsilon}$, write the budget inequality a reduction must satisfy to obtain a constant exponent saving for the source problem.
3. Show that $T^{1-\delta} \log^c T \leq T^{1-\delta/2}$ for all sufficiently large T and fixed positive δ, c .
4. Apply the sparsification theorem symbolically with parameter η and compute the maximum number of sparse branches and clauses per branch.
5. Explain why bounded occurrence implies a linear clause count, and why linear clause count is the property required by the clause-coordinate OV reduction.
6. Identify which result in this chapter is source-owned. List every quantitative feature of its statement that the next chapter actually uses.

Chapter 6

Orthogonal Vectors and the SETH Reduction

We now give the complete split-and-list reduction that turns a sparse CNF formula into a bichromatic Orthogonal Vectors instance. The proof is elementary once the sparsification theorem has supplied only linearly many clauses, but every exponent and dimension conversion matters.

How to read this chapter. The chapter has two jobs. The first half converts sparse SAT into low-dimensional Orthogonal Vectors and then formalizes how fine-grained exponent savings compose. The second half is a complete geometric case study: a three-bit coordinate gadget converts orthogonality into an exact Hamming closest-pair threshold. Keep the symbols n (SAT variables), N (number of vectors per side), and d or D (dimension) separate; most mistakes in this chain come from silently changing one of those scales.

Learning goals. After this chapter, a reader should be able to construct the split-and-list vectors clause by clause; derive $d = O(\log N)$ from sparsification; state the deterministic low-dimensional OVC with its quantifiers; compose two budgeted fine-grained reductions; amplify bounded-error oracle calls without losing the constant exponent saving; and verify the identity turning the three-bit gadget into an affine Hamming-distance statistic.

Bichromatic Orthogonal Vectors. For $d, N \geq 1$, an instance of $\text{OV}_{N,d}$ consists of two sets $A, B \subseteq \{0, 1\}^d$ with $|A|, |B| \leq N$. It is a yes-instance if there are $a \in A$ and $b \in B$ such that their ordinary integer inner product is zero:

$$\langle a, b \rangle = \sum_{j=1}^d a_j b_j = 0.$$

Because the coordinates are Boolean, this is equivalent to saying that no coordinate has $a_j = b_j = 1$.

Theorem 6.1 (Split-and-list reduction from sparse SAT to Orthogonal Vectors). *Let F be a CNF formula on $n \geq 1$ variables with m clauses. In $O(mn2^{n/2})$ deterministic time one can construct sets $A, B \subseteq \{0, 1\}^m$ with*

$$|A| = 2^{\lfloor n/2 \rfloor}, \quad |B| = 2^{\lceil n/2 \rceil}$$

such that

$$F \text{ is satisfiable} \iff \exists a \in A, b \in B : \langle a, b \rangle = 0.$$

In particular, if $m \leq cn$ and $N := 2^{\lceil n/2 \rceil}$, then the construction has dimension

$$d = m \leq 2c \log_2 N.$$

Together with Theorem 5.2, this implies deterministic SETH $\Rightarrow$ the deterministic low-dimensional Orthogonal Vectors Conjecture stated below.

Proof. Partition the variables into disjoint sets X_L, X_R with $|X_L| = \lfloor n/2 \rfloor$ and $|X_R| = \lceil n/2 \rceil$. Write the clauses as $C_1, \dots, C_m$.

For every assignment α to X_L , define a vector $a(\alpha) \in \{0, 1\}^m$ by

$$a(\alpha)_j = \begin{cases} 0, & \text{if some literal of } C_j \text{ whose variable lies in } X_L \text{ is satisfied by } \alpha, \\ 1, & \text{otherwise.} \end{cases}$$

Define $b(\beta)$ analogously for assignments β to X_R . Let A and B contain all such vectors. Enumerating the assignments and scanning all clauses gives the stated $O(mn2^{n/2})$ bound; the extra factor n is a safe bound for inspecting the encoded literals.

Fix assignments α, β . For a clause C_j , the product $a(\alpha)_j b(\beta)_j$ equals one exactly when neither half-assignment satisfies any literal of C_j . Since every variable lies in exactly one half, this happens exactly when the combined assignment $\alpha \cup \beta$ falsifies C_j . Hence

$$\langle a(\alpha), b(\beta) \rangle = 0$$

if and only if no clause is falsified, which is equivalent to $\alpha \cup \beta$ satisfying F . This proves the correctness equivalence.

Now assume $m \leq cn$ and set $N = 2^{\lceil n/2 \rceil}$. Since $\log_2 N = \lceil n/2 \rceil \geq n/2$, we have $n \leq 2 \log_2 N$, and therefore $d = m \leq 2c \log_2 N$. If an exactly- N formulation of OV is desired, duplicate arbitrary existing vectors on the smaller side; duplication cannot create a new inner-product value and hence does not change whether an orthogonal pair exists.

It remains to verify the SETH implication. Suppose the deterministic low-dimensional OVC were false. Negating the conjecture below yields a constant $\varepsilon_0 > 0$ such that for every constant C there is a deterministic algorithm solving OV in dimension at most $C \log_2 N$ in $O(N^{2-\varepsilon_0} \text{poly}(d))$ time. Replacing ε_0 by $\min\{\varepsilon_0, 1\}$ if necessary, assume $0 < \varepsilon_0 \leq 1$ and put $\eta = \varepsilon_0/4$.

Fix any k . Apply sparsification with this k and η . It produces at most 2^m formulas, each with at most $c(k, \eta)n$ clauses. For each sparse formula use the split-and-list construction and then the OV algorithm corresponding to $C = 2c(k, \eta)$. Since $N = 2^{\lceil n/2 \rceil}$,

$$N^{2-\varepsilon_0} = 2^{(1-\varepsilon_0/2)n+O(1)}.$$

Multiplying by the at most 2^m sparse branches gives total oracle time

$$2^{(1-\varepsilon_0/2+\eta)n} \text{poly}(|F|) = 2^{(1-\varepsilon_0/4)n} \text{poly}(|F|).$$

The sparsification and instance-construction work is no larger in exponential order: its exponent is at most $\max\{\eta, 1/2 + \eta\} \leq 3/4$, while $1 - \varepsilon_0/4 \geq 3/4$. Thus, for the same positive constant $\delta = \varepsilon_0/4$, every fixed k admits a $2^{(1-\delta)n} \text{poly}(|F|)$ deterministic SAT algorithm. This is the negation of SETH. Contrapositively, deterministic SETH implies the deterministic low-dimensional OVC. $\square$

Conjecture 6.2 (Orthogonal Vectors conjecture: low-dimensional form). *For every $\varepsilon > 0$ there exists a constant $C = C(\varepsilon) > 0$ such that no deterministic algorithm solves all instances of $\text{OV}_{N,d}$ with $d \leq C \log_2 N$ in $O(N^{2-\varepsilon} \text{poly}(d))$ time.*

Randomized and moderate-dimensional variants. A randomized OVC replaces “deterministic algorithm” by a bounded-error randomized algorithm. The proof above derives only the deterministic conjecture from deterministic SETH; deriving the randomized conjecture requires the correspondingly randomized SETH assumption. A second common formulation concerns $d = \omega(\log N)$ and conjectures $N^{2-o(1)}$ time. The two formulations are recorded separately in this volume; no equivalence is silently asserted.

Worked clause coordinate. Suppose $C_j = (x_1 \vee \neg x_4 \vee x_7)$, with $x_1, x_4 \in X_L$ and $x_7 \in X_R$. If a left assignment sets $x_1 = 0, x_4 = 1$, then it satisfies neither left literal and the j th coordinate of its vector is one. A right assignment with $x_7 = 0$ also has j th coordinate one, so the product in that coordinate is one and the pair cannot be orthogonal. If instead $x_7 = 1$, the right coordinate is zero; this clause contributes zero to the inner product because the combined assignment satisfies it.

Source and ownership boundary. The sparse-CNF interface is owned by the Impagliazzo–Paturi–Zane sparsification lemma. The clause-coordinate split-and-list reduction is proved locally above and matches the Williams SETH-to-OV route. Later geometric, dynamic, string, and data-structure lower bounds should depend on this theorem or on an explicitly stated OV conjecture, rather than citing SETH without carrying the intermediate dimension and exponent bookkeeping.

Theorem 6.3 (Fine-grained reduction calculus). *Assume all reference time scales below are nondecreasing, unbounded, and at least 2 on nontrivial inputs, so fixed powers dominate polylogarithmic factors. For the budgeted fine-grained reductions of Definition 5.1, the following rules hold.*

1. Algorithm transfer. *If (P, T_P) fine-grained reduces to (Q, T_Q) and, for some $\varepsilon > 0$, Q has a deterministic $T_Q(m)^{1-\varepsilon} \text{polylog } T_Q(m)$ algorithm, then there is $\delta > 0$ for which P has a deterministic $T_P(n)^{1-\delta} \text{polylog } T_P(n)$ algorithm.*
2. Composition. *If (P, T_P) fine-grained reduces to (Q, T_Q) and (Q, T_Q) fine-grained reduces to (R, T_R) , then (P, T_P) fine-grained reduces to (R, T_R) .*
3. Bounded-error oracle preservation. *In item 1, if the target algorithm is randomized with two-sided error at most $1/3$, while the reduction itself is deterministic and is correct when all oracle answers are correct, then the transferred source algorithm can be made bounded-error with constant error below $1/3$ without losing the constant exponent saving.*
4. Preprocessing/query bookkeeping. *Suppose an oracle problem has preprocessing scale $P_Q(m)$ and per-query scale $U_Q(m)$. If, for every desired savings $\varepsilon_p, \varepsilon_q > 0$, a source reduction has nonoracle work plus target workload*

$$\sum_j P_Q(m_j)^{1-\varepsilon_p} + \sum_j r_j U_Q(m_j)^{1-\varepsilon_q} \leq T_P(n)^{1-\delta} \text{polylog } T_P(n)$$

for some $\delta > 0$, where database j has size m_j and receives r_j queries, then a target data structure achieving those two improved scales yields the corresponding $T_P^{1-\delta}$ source algorithm. Preprocessing and query exponents may not be silently merged into one cost when the reduction uses many queries per preprocessed instance.

Proof. For item 1, fix the target saving ε and apply the definition of the reduction with that same ε . Its nonoracle work and the sum of the running times of all target calls are each bounded by the displayed source budget. Adding them changes only a constant factor.

For composition, fix a desired saving $\varepsilon > 0$ for the final problem R . Apply the Q -to- R reduction with ε and let its resulting source saving be $\gamma > 0$. Now apply the P -to- Q reduction with requested

target saving $\gamma/2$, obtaining a source saving $\delta > 0$. Consider a Q -oracle query of size m generated by the outer reduction. The complete simulation of that query by the inner reduction has nonoracle work and total R -oracle budget at most

$$T_Q(m)^{1-\gamma} \text{polylog } T_Q(m).$$

For all sufficiently large m this is at most $T_Q(m)^{1-\gamma/2}$; the finitely many smaller sizes contribute only a constant-factor table lookup or fixed overhead. Summing over all outer queries and using the outer budget gives

$$\sum_i T_Q(m_i)^{1-\gamma/2} \leq T_P(n)^{1-\delta} \text{polylog } T_P(n).$$

The outer nonoracle work obeys the same source bound. Any extra polylogarithmic factors introduced by nesting are absorbed by reducing δ by a fixed constant factor. The resulting composed oracle algorithm therefore satisfies Definition 5.1.

For item 3, apply the deterministic reduction with requested target saving $\varepsilon/2$. Let R_n be an a priori upper bound on the number of target calls made on size- n inputs. Since every query contributes at least one unit to the oracle-cost budget, one may take $R_n \leq T_P(n)^{O(1)}$. Amplify each target call independently from error $1/3$ to at most $1/(10(R_n + 1))$ by $O(\log(R_n + 1))$ repetitions and majority vote. A union bound shows that all oracle answers are simultaneously correct with probability at least $9/10$. The amplification factor is $O(\log(R_n + 1)) = \text{polylog } T_P(n)$. Before amplification, the total target time is bounded by the reduction budget because $T_Q(m)^{1-\varepsilon} \leq T_Q(m)^{1-\varepsilon/2}$ for every target query. Multiplying the entire oracle workload by the amplification factor therefore introduces only one additional polylogarithmic factor in the source time; it does not require any lower bound relating an individual target-query size to n . Conditional on all answers being correct the deterministic reduction returns the correct source answer, so the final error is at most $1/10 < 1/3$.

Item 4 is direct substitution, but the two terms must remain separate. Run the improved target preprocessing algorithm once for each database j and then answer its r_j queries with the improved query algorithm. By the assumed two-budget inequality their total cost plus the source reduction's own work is at most $T_P(n)^{1-\delta} \text{polylog } T_P(n)$. This proves the claimed transfer and also explains why charging preprocessing anew for every query, or conversely ignoring repeated query cost, can give a false conditional exponent. $\square$

Worked exponent composition. Suppose P and Q both have quadratic reference scales. An outer reduction can tolerate a Q -solver of cost $M^{2-0.1}$ and in return gives an $N^{2-0.02}$ solver for P . If an inner reduction turns an R saving into a Q saving of 0.2 in the quadratic exponent, then it is more than sufficient for the outer requested saving 0.1. The proof above formalizes this familiar calculation without assuming that a single oracle query dominates all others.

Dependency checkpoint. The calculus uses only Definition 5.1, elementary summation, majority amplification, and the union bound. It is independent of SETH and of the sparsification deep input; those are needed for the concrete OV chain, not for the abstract reduction algebra.

6.1 A complete geometric case study: Hamming closest pair

The abstract reduction calculus becomes useful only after one complete downstream chain is written with the parameters visible. We choose a deliberately elementary geometric target so that no new deep theorem is hidden inside the case study.

Bichromatic Hamming closest pair. For integers $N, D \geq 1$ and a threshold R , an instance of the decision problem $\text{BHCP}_{N,D}$ consists of two sets $X, Y \subseteq \{0, 1\}^D$, each of size at most N . It is a yes-instance when some $(x, y) \in X \times Y$ satisfies $d_H(x, y) \leq R$, where

$$d_H(x, y) := |\{j \in [D] : x_j \neq y_j\}|.$$

The adjective *bichromatic* means that the pair must use one point from each input set.

Theorem 6.4 (OV-to-bichromatic-Hamming-closest-pair reduction). *There is a deterministic reduction taking every bichromatic Orthogonal Vectors instance $A, B \subseteq \{0, 1\}^d$, with $|A|, |B| \leq N$, to a bichromatic Hamming closest-pair instance $X, Y \subseteq \{0, 1\}^{3d}$, with the same two set sizes and threshold $R = d$, such that*

$$\exists a \in A, b \in B : \langle a, b \rangle = 0 \iff \exists x \in X, y \in Y : d_H(x, y) \leq d.$$

The reduction runs in $O(Nd)$ time. Consequently, if for some $\varepsilon > 0$ and every constant $C > 0$ bichromatic Hamming closest pair in dimension $D \leq C \log_2 N$ were solvable in $O(N^{2-\varepsilon} \text{poly}(D))$ deterministic time, then the low-dimensional Orthogonal Vectors Conjecture would be false. In combination with Theorem 6.1, deterministic SETH therefore implies that for every $\varepsilon > 0$ there is a constant $C = C(\varepsilon) > 0$ for which no such $O(N^{2-\varepsilon} \text{poly}(D))$ Hamming algorithm exists on all instances with $D \leq C \log_2 N$.

Proof. Define two coordinate encodings $\alpha, \beta : \{0, 1\} \rightarrow \{0, 1\}^3$ by

$$\alpha(0) = 000, \quad \alpha(1) = 011, \quad \beta(0) = 001, \quad \beta(1) = 100.$$

A direct four-case calculation gives

$d_H(\alpha(u), \beta(v))$	$v = 0$	$v = 1$
$u = 0$	1	1
$u = 1$	1	3

and hence, for all Boolean u, v ,

$$d_H(\alpha(u), \beta(v)) = 1 + 2uv. \tag{6.1}$$

For $a = (a_1, \dots, a_d)$ concatenate the d three-bit blocks

$$\alpha^*(a) := \alpha(a_1)\alpha(a_2) \cdots \alpha(a_d) \in \{0, 1\}^{3d},$$

and define $\beta^*(b)$ analogously using β . Let $X = \{\alpha^*(a) : a \in A\}$ and $Y = \{\beta^*(b) : b \in B\}$. Summing (6.1) over the independent blocks yields the exact identity

$$d_H(\alpha^*(a), \beta^*(b)) = \sum_{j=1}^d (1 + 2a_j b_j) = d + 2\langle a, b \rangle. \tag{6.2}$$

Because $\langle a, b \rangle$ is a nonnegative integer, (6.2) equals d exactly for orthogonal pairs and is at least $d + 2$ for every nonorthogonal pair. Thus the threshold $R = d$ gives the claimed if-and-only-if reduction. Writing the $3d$ output bits for each of at most $2N$ vectors costs $O(Nd)$ time.

For the conditional lower bound, suppose a Hamming algorithm with a fixed saving $\varepsilon > 0$ existed for every constant logarithmic dimension bound. Put $\varepsilon_0 = \min\{\varepsilon, 1/2\} > 0$. The same algorithm is also an $O(N^{2-\varepsilon_0} \text{poly}(D))$ algorithm. If an OV instance has $d \leq C_0 \log_2 N$, the reduction has $D = 3d \leq 3C_0 \log_2 N$, so invoke the Hamming algorithm at constant $3C_0$. Its running time plus the $O(Nd)$ construction cost is

$$O(N^{2-\varepsilon_0} \text{polylog } N + N \log N) = O(N^{2-\varepsilon_0} \text{polylog } N),$$

since $2 - \varepsilon_0 \geq 3/2$. This contradicts Conjecture 6.2. The final SETH consequence follows from Theorem 6.1. $\square$

Why the three-bit gadget is useful. A tempting one-bit encoding cannot distinguish the three allowed coordinate pairs $(0, 0)$, $(0, 1)$, $(1, 0)$ from the forbidden pair $(1, 1)$ by one common Hamming baseline. The two asymmetric three-bit maps above make every allowed pair contribute exactly one unit and make the forbidden pair contribute three. Tensoring by concatenation then turns the Boolean inner product into an affine Hamming-distance statistic with no probabilistic argument and no distortion loss.

Parameter ledger. The reduction preserves the number of points on each side, multiplies dimension by exactly three, uses threshold $R = d$, introduces an additive distance gap of two, and costs only linear time in the output length. Thus no preprocessing, approximation, randomized-error, or hidden polynomial parameter enters this particular conditional lower-bound chain.

6.2 Reader checkpoint and exercises

Synthesis and transition. The chapter exhibits the full fine-grained workflow: sparsify, split, encode, track exponent and dimension, then transport the conditional lower bound through a second exact reduction. Chapter 7 changes viewpoint from running-time exponents to individual-object description length; this shift is intentional and no implication between SETH and Kolmogorov incompressibility is asserted.

1. For a four-clause CNF split over two left and two right variables, construct all left and right vectors explicitly and verify the orthogonality/satisfiability equivalence.
2. Starting from $N = 2^{\lceil n/2 \rceil}$ and $m \leq cn$, derive the constant multiplying $\log_2 N$ in the OV dimension bound.
3. Negate the low-dimensional OVC exactly and use the proof's choice $\eta = \varepsilon_0/4$ to recompute the final SAT exponent.
4. Give a two-step toy fine-grained reduction chain and use Theorem 6.3 to calculate a valid source saving.
5. Amplify a $1/3$ -error oracle to error at most $1/(100q)$ for q oracle calls and estimate the logarithmic repetition overhead.
6. For each Boolean coordinate pair (a_i, b_i) , compute the three-bit gadget distance and verify $d_H(\alpha(a), \beta(b)) = d + 2\langle a, b \rangle$ by summing coordinates.

Chapter 7

Kolmogorov Complexity and the Incompressibility Method

Fine-grained complexity asks how quickly an algorithm can solve a problem. Kolmogorov complexity asks how short a program can describe an individual object. The basic incompressibility method is a counting principle: there are too few short programs to describe every n -bit string. The machine constant is real and must remain visible whenever a numerical threshold is used.

How to read this chapter. Start with the machine conventions before using any counting argument. Plain complexity C , prefix-free complexity K , time-bounded complexity K^t , and Levin Kt answer different questions and are only invariant up to the stated machine-dependent constants. The main theorem is then a counting theorem: short descriptions cover too few strings. Its zero-run application shows how a high-complexity assumption can force a concrete combinatorial property without making any resource-bounded pseudorandomness claim.

Learning goals. After this chapter, a reader should be able to distinguish C , K , K^t , and Kt ; explain the role of optimal machines and additive compiler constants; prove the basic incompressibility count; and turn a hypothetical long zero run into an explicit shorter description with all self-delimiting overhead accounted for.

Definition 7.1 (Plain, prefix-free, time-bounded, and Levin complexity). A *description machine* is a partial computable map $M : \{0, 1\}^* \rightarrow \{0, 1\}^*$. Its plain description complexity is

$$C_M(x) := \min\{|p| : M(p) = x\},$$

with value $+\infty$ if no program outputs x . An *optimal plain machine* U_C is one for which, for every description machine M , there is a constant c_M satisfying $C_{U_C}(x) \leq C_M(x) + c_M$ for every x . We fix one such machine and write $C(x) = C_{U_C}(x)$.

A machine is *prefix-free* if its halting domain is prefix-free. For a prefix-free machine M , define

$$K_M(x) := \min\{|p| : M(p) = x\}.$$

Fix an optimal prefix-free machine U_K and write $K(x) = K_{U_K}(x)$. If U and V are two optimal machines of the same type, optimality in both directions gives a constant $c_{U,V}$ with

$$|C_U(x) - C_V(x)| \leq c_{U,V} \quad \text{or} \quad |K_U(x) - K_V(x)| \leq c_{U,V},$$

respectively. This is the additive *invariance constant*; statements such as “ $K(x) \geq n - 10$ ” therefore depend on the chosen optimal machine up to an additive constant.

Let $t : \mathbb{N} \rightarrow \mathbb{N}$ be a time bound. The *time-bounded prefix-free complexity* relative to the fixed machine U_K is

$$K^t(x) := \min\{|p| : U_K(p) = x \text{ within } t(|x|) \text{ steps}\}.$$

Finally, fixing a universal machine U with an explicit step-count convention, the *Levin Kt complexity* is

$$Kt_U(x) := \min_{p, T} \{|p| + \lceil \log_2 T \rceil : U(p) = x \text{ within } T \geq 1 \text{ steps}\}.$$

Time-bounded quantities retain the displayed machine and simulation convention when quantitative constants matter; unbounded additive invariance alone does not erase simulation-time overhead.

Positive and negative examples. The string 0^n has $C(0^n) \leq O(\log n)$: a fixed program can decode a self-delimiting encoding of n and print n zeros. By contrast, the theorem below shows that most n -bit strings have plain and prefix-free complexity at least $n - O(1)$. “Looks irregular” is not a definition of incompressibility; the inequality against a fixed universal machine is.

Theorem 7.2 (Counting theorem and incompressibility method). *Fix the optimal machines in Definition 7.1. For every $n \geq 1$ and integer $r \geq 0$,*

$$\begin{aligned} |\{x \in \{0, 1\}^n : C(x) < n - r\}| &< 2^{n-r}, \\ |\{x \in \{0, 1\}^n : K(x) < n - r\}| &< 2^{n-r}. \end{aligned}$$

Consequently, if X is uniform on $\{0, 1\}^n$, then

$$\Pr[C(X) \geq n - r] > 1 - 2^{-r}, \quad \Pr[K(X) \geq n - r] > 1 - 2^{-r}.$$

Moreover, there is a constant c_0 depending only on the fixed plain universal machine and the integer coding convention such that every $x \in \{0, 1\}^n$ with $C(x) \geq n - r$ has no run of consecutive zeros longer than

$$r + 4 \lceil \log_2(n + 1) \rceil + c_0.$$

Thus any object violating an efficiently encodable combinatorial regularity by too large a margin would itself yield a shorter description; this is the basic incompressibility proof method.

Proof. There are exactly 2^j binary programs of length j . Hence the total number of programs of length strictly less than $n - r$ is

$$\sum_{j=0}^{n-r-1} 2^j = 2^{n-r} - 1$$

when $n - r \geq 1$, and is zero when $n - r \leq 0$. Each program can output at most one string, so fewer than 2^{n-r} strings of length n can have plain complexity below $n - r$. The same counting argument applies to prefix-free programs; prefix-freeness is not even needed for this finite bound. Dividing by 2^n gives the two probability inequalities.

For the run consequence, let x contain a block of L consecutive zeros beginning at position i , using positions numbered from 0. Put $\ell = \lceil \log_2(n + 1) \rceil$. Encode n by the self-delimiting word

$$1^\ell 0 \text{ bin}_\ell(n),$$

which has length $2\ell + 1$, then encode i and L in exactly ℓ bits each, and finally append the $n - L$ bits obtained from x by deleting the selected zero run. A fixed decoder recovers n , then i, L , and

inserts L zeros at the indicated position. After absorbing the fixed decoder into the chosen optimal machine, this gives

$$C(x) \leq (n - L) + (2\ell + 1) + 2\ell + c'_0 = n - L + 4\ell + c_0$$

for a fixed constant c_0 . If $C(x) \geq n - r$, comparison of the two inequalities yields $L \leq r + 4\ell + c_0$, as claimed. $\square$

Worked count. For $n = 20$ and $r = 5$, fewer than 2^{15} of the 2^{20} strings have $C(x) < 15$. Thus more than $31/32$ of all twenty-bit strings require at least fifteen bits in their shortest plain description under the fixed machine. The exact identity of those strings depends on the machine; the counting fraction does not.

Source and ownership boundary. The counting proof is completely internal. The only universal-machine fact used in the run consequence is the fixed additive compiler constant built into Definition 7.1. Later uses of Kt , MKtP, or circuit-minimization must not replace this unbounded counting theorem by a resource-bounded claim without separately tracking simulation time.

7.1 Reader checkpoint and exercises

Synthesis and transition. Incompressibility is an existence/counting method about individual strings under a fixed description machine. Chapter 8 turns related resource measures into decision problems—MCSP and MKtP—where the input encoding, threshold convention, and computational complexity of deciding the measure become the central questions.

1. Prove directly that fewer than 2^{n-r} strings have plain complexity below $n - r$.
2. Explain why the analogous counting statement for prefix-free K can be sharpened using Kraft's inequality, and state which part of the present chapter does not need that sharpening.
3. For the zero-run encoder, derive the $4\lceil \log_2(n + 1) \rceil + O(1)$ overhead term and identify the data that must be self-delimited.
4. Compare $K^t(x)$ and $Kt(x)$ conceptually: which fixes a time budget externally, and which charges logarithmically for the running time inside the objective?
5. Give an example showing why changing the optimal machine may change the exact shortest program but only by an additive constant in the invariance theorem.
6. Explain why an incompressibility theorem for unbounded C does not automatically imply a lower bound for Kt , MCSP, or any polynomial-time pseudorandomness test.

Chapter 8

Minimum Circuit Size, MKtP, and the First Meta-Complexity Interfaces

Meta-complexity turns a complexity measure itself into the object of computation. This chapter begins with the two basic decision problems needed later: minimum circuit size and a Levin- Kt threshold problem. The input-length convention is crucial. An n -variable Boolean function is given by all $N = 2^n$ truth-table bits, so an algorithm polynomial in N may still take exponential time in the number n of function variables.

How to read this chapter. The chapter changes perspective from measuring a string or function to deciding a statement about that measure. Always distinguish the truth-table input length $N = 2^n$ from the number n of function variables. Read the exact small MCSP examples before the NP-membership proof if the basis convention is unfamiliar, then compare MCSP with the machine-dependent MKtP definition. The final theorem is a one-way implication: an efficient exact MCSP solver produces a large constructive property useful against polynomial-size circuits; it is not itself a circuit lower bound.

Learning goals. After this chapter, a reader should be able to state MCSP and MKtP with their encodings fixed; prove that standard MCSP lies in NP despite a binary threshold; distinguish standard MCSP from gap, search, implicit, and oracle variants; verify the counting argument that makes the MCSP-derived property large; and run or interpret the finite Kolmogorov/MCSP/MKtP laboratory without confusing its finite candidate table with the universal machine.

Definition 8.1 (Minimum Circuit Size Problem). Fix once and for all the bounded-fan-in basis $\mathcal{B} = \{\text{AND}_2, \text{OR}_2, \text{NOT}\}$, and count noninput gates as circuit size. Let $T \in \{0, 1\}^N$ have length $N = 2^n$ and list the values of a Boolean function $f_T : \{0, 1\}^n \rightarrow \{0, 1\}$ in lexicographic input order. For a nonnegative integer threshold s given in binary,

$$(T, s) \in \text{MCSP}_{\mathcal{B}} \iff N = 2^n \text{ for some integer } n \geq 1, \\ \text{size}_{\mathcal{B}}(f_T) \leq s.$$

A truth-table length that is not 2^n for an integer $n \geq 1$ is, by convention, a no-instance; the omitted zero-variable case is trivial and may be restored without changing any complexity statement. A *gap-MCSP* promise problem specifies two thresholds $s_{\text{yes}} < s_{\text{no}}$ and promises that the minimum circuit size is either at most s_{yes} or greater than s_{no} . An *approximation* problem asks for an estimate of the minimum size, while a *search-MCSP* problem asks for an actual small circuit. These are distinct problems and will not be identified without a reduction.

Basis convention. The fixed basis is part of the quantitative definition. Any two fixed functionally complete bounded-fan-in bases simulate one another with constant-factor circuit-size overhead, but exact small-instance values and gap thresholds can change by those constants. Consequently the basis is always displayed when an exact threshold is used.

Definition 8.2 (MKtP in the Levin- Kt convention). Use the universal step-counting machine U fixed in Definition 7.1, so that

$$Kt_U(x) = \min_{p, t \geq 1} \{ |p| + \lceil \log_2 t \rceil : U(p) \text{ outputs } x \text{ within } t \text{ steps} \}.$$

The corresponding threshold problem is

$$\text{MKtP}_U := \{(x, \theta) : Kt_U(x) \leq \theta\},$$

where θ is a nonnegative integer encoded in binary. Gap, approximation, search, and average-case versions are defined only after their two thresholds, approximation factor, required witness, or input ensemble has respectively been fixed.

This book's notation Kt is the displayed program-length-plus-log-time quantity. Some meta-complexity papers use symbols such as KT , Kt , or MKTP for different machine and time-cost conventions. No external theorem about “MKtP” is imported until its convention has been matched to this definition or an explicit reduction has been supplied.

Proposition 8.3 (MCSP belongs to NP). *Under Definition 8.1, $\text{MCSP}_{\mathcal{B}} \in \text{NP}$ when complexity is measured in the ordinary input length $N + O(\log s)$. More explicitly, there is an absolute constant $c > 0$ such that every $N = 2^n$ -bit truth table has a circuit of size at most $cN(n + 1)$. Hence a yes-instance either has $s \geq cN(n + 1)$, in which case a canonical circuit is known deterministically, or admits a circuit certificate whose encoding length and verification time are polynomial in the input length.*

Proof. For every assignment $a \in \{0, 1\}^n$ with $f_T(a) = 1$, form the minterm

$$M_a(x) = \bigwedge_{i=1}^n \begin{cases} x_i, & a_i = 1, \\ \neg x_i, & a_i = 0. \end{cases}$$

using a binary tree of AND gates. Share the n negations among all minterms and OR the at most N minterms whose truth-table entries are one. Constants can be handled by fixed constant-size circuits. The resulting disjunctive-normal-form circuit has at most $cN(n + 1)$ gates for a fixed absolute c .

Now let $S_0 = cN(n + 1)$. If the encoded threshold satisfies $s \geq S_0$, the verifier accepts after checking the input format, because the preceding construction proves that every represented function has a circuit of size at most S_0 . If $s < S_0$, a nondeterministic verifier guesses a DAG description of at most s gates. Each gate records one of the three fixed gate types and at most two predecessors among the $n + s$ available input/gate nodes, so the certificate has $O(s \log(n + s)) = \text{poly}(N)$ bits. The verifier evaluates the guessed circuit on each of the N assignments and compares the resulting output bit with the corresponding entry of T . This costs $O(Ns)$ gate evaluations plus polynomial encoding overhead, hence polynomial time in $N + \log(s + 1)$. The certificate exists exactly when the minimum circuit size is at most s . $\square$

Conjecture 8.4 (Open status of standard MCSP NP-hardness). *As of 2 September 2026, it remains open whether the standard total problem $\text{MCSP}_{\mathcal{B}}$ is NP-hard, and hence NP-complete by Proposition 8.3, under deterministic polynomial-time many-one reductions. This status statement does not assert that gap-MCSP, partial-function MCSP, implicit-MCSP, random-oracle versions, or stronger reduction notions have the same known hardness status.*

Why the status sentence is narrow. Modern results establish substantial hardness evidence for variants and under additional hypotheses. Those results do not convert the preceding open problem into a theorem about standard MCSP. Later chapters therefore attach a dated source table to each variant before using a hardness statement.

Example 8.5 (Exact three-variable MCSP instances). Let the input variables be x_1, x_2, x_3 and use the basis of Definition 8.1. The full truth table has length $N = 8$. With input wires not counted as gates: The following sizes are exact.

1. $\text{size}_{\mathcal{B}}(x_1) = 0$: the output may be the input wire itself.
2. $\text{size}_{\mathcal{B}}(\neg x_1) = 1$: one NOT gate suffices, while a zero-gate circuit can only expose an input wire, not its negation.
3. $\text{size}_{\mathcal{B}}(x_1 \wedge x_2) = 1$: one AND gate suffices, while a zero-gate circuit cannot depend on both variables.
4. $\text{size}_{\mathcal{B}}(x_1 \wedge x_2 \wedge x_3) = 2$: two AND gates suffice, while one bounded-fan-in gate cannot depend on all three distinct inputs.

Thus even these smallest examples distinguish the variable count $n = 3$ from the actual MCSP input length $N = 8$.

Theorem 8.6 (An efficient exact MCSP algorithm yields a natural property). *Assume there is a deterministic algorithm deciding $\text{MCSP}_{\mathcal{B}}$ in time polynomial in its truth-table input length. Then there is a property $\Gamma = (\Gamma_n)_{n \geq 1}$ of n -variable Boolean functions that is polynomial-time constructive from the full truth table, large, and useful against polynomial-size $\mathcal{B}$ -circuits in the sense of the natural-property definition owned by Volume X.*

One explicit choice is

$$\Gamma_n := \{f : \text{size}_{\mathcal{B}}(f) > s(n)\}, \quad s(n) := \left\lceil n^{\log_2(n+2)} \right\rceil. \quad (8.1)$$

For all sufficiently large n , a uniformly random n -variable Boolean function belongs to Γ_n with probability at least $1/2$, while for every fixed k no function in Γ_n has a circuit of size at most n^k once n is sufficiently large.

Proof. Let $N = 2^n$. Given the N -bit truth table of f , compute the integer $s(n)$ from (8.1) and run the assumed exact MCSP algorithm on $(T_f, s(n))$. Accept precisely when MCSP rejects. Since $\log s(n) = O((\log n)^2)$, forming the threshold is polynomial in N , and the assumed algorithm runs in $N^{O(1)}$ time. Thus Γ is constructive in the truth-table input length.

For usefulness, fix $k > 0$. Because $\log_2(n+2) \rightarrow \infty$, eventually $s(n) > n^k$. Every function in Γ_n then has circuit size greater than $s(n)$ and therefore greater than n^k . This holds for every fixed polynomial exponent k .

It remains to prove largeness without appealing to an unproved circuit lower bound. Let $r \geq 1$. A size- r circuit over the fixed basis can be described gate by gate. Each gate chooses one of a constant number of types and at most two predecessors among at most $n + r$ previously available nodes. Hence, for a fixed constant c_1 depending only on the basis, the number of circuits of size exactly r is at most

$$(c_1(n+r)^2)^r.$$

Therefore the total number E_n of functions computable by circuits of size at most $s = s(n)$ is bounded by the number of such circuit descriptions:

$$E_n \leq 1 + \sum_{r=1}^s (c_1(n+s)^2)^r \leq 1 + s(c_1(n+s)^2)^s. \quad (8.2)$$

Taking base-two logarithms and using $s = n^{\Theta(\log n)} = 2^{\Theta((\log n)^2)}$ gives

$$\log_2 E_n = O(s \log(n + s)) = 2^{O((\log n)^2)} = o(2^n) = o(N). \quad (8.3)$$

There are exactly 2^N Boolean functions on n variables. Equations (8.2)–(8.3) imply that, for all sufficiently large n ,

$$\Pr_f[\text{size}_{\mathcal{B}}(f) \leq s(n)] = \frac{E_n}{2^N} \leq \frac{1}{2}.$$

Thus $\Pr_f[f \in \Gamma_n] \geq 1/2$. In particular this is at least 2^{-bn} for any fixed $b > 0$ and all sufficiently large n , so it satisfies the usual natural-property largeness condition. Constructivity, largeness, and usefulness have all been verified separately. $\square$

Interpretation and barrier boundary. The theorem is only an implication from a hypothetical fast exact MCSP algorithm to a natural property. It does not prove that such an algorithm exists, nor does it by itself contradict the natural-proofs barrier: the barrier additionally assumes sufficiently strong pseudorandom functions inside the target circuit class. This separation of hypotheses is essential when meta-complexity is used as evidence about circuit lower bounds.

Dependency checkpoint. The NP-membership proof uses only the fixed bounded-fan-in circuit convention and exhaustive truth-table verification. The natural-property theorem uses the natural-property definition from Volume X plus the elementary circuit-description count proved above. The *Kt* definition depends only on Chapter 7's fixed universal-machine convention. No hardness-magnification theorem and no cryptographic assumption has entered yet.

8.1 An exact finite Kolmogorov/MCSP/MKtP laboratory

The reader laboratory `code/volume_14/kolmogorov_mcsp_lab.py` checks three finite interfaces with exact integer arithmetic. It is deliberately diagnostic: none of the asymptotic statements in this volume use code as a proof substitute.

Incompressibility counting checkpoint. For eight-bit strings and descriptions of length strictly smaller than five, there are only

$$1 + 2 + 4 + 8 + 16 = 31$$

possible descriptions. Any single-valued decoder can therefore describe at most 31 distinct strings with such descriptions, leaving at least

$$2^8 - 31 = 225$$

strings outside the short-description image. The laboratory verifies this exact finite count. It does not attempt to compute the universal-machine Kolmogorov complexity of arbitrary strings.

Exact two-variable MCSP checkpoint. Using the fixed basis from Definition 8.1, the laboratory performs a breadth-first search over finite sets of available truth-table signals. Two input wires cost zero; each NOT, AND, or OR gate costs one. The quotient by available signal sets is exact for this finite reachability problem because future gates depend only on which Boolean signals have already

been constructed. All 16 two-variable Boolean functions are reached, and the search verifies, among other cases,

$$\text{size}(x_1) = 0, \quad \text{size}(\neg x_1) = 1, \quad \text{size}(x_1 \wedge x_2) = 1, \quad \text{size}(x_1 \oplus x_2) = 4. \quad (8.4)$$

It also records a finite promise census for gap thresholds one and three, with nonempty yes, gap, and no regions.

Finite Levin-cost checkpoint. For a displayed finite table of candidate program strings, step bounds, and outputs, the laboratory computes

$$|p| + \lceil \log_2 t \rceil \quad (8.5)$$

exactly and tests threshold monotonicity. This is a finite-table analogue of Definition 8.2; the candidate table is not claimed to be the universal machine fixed in Chapter 7.

Self-test. Executing

```
python code/volume_14/kolmogorov_mcsp_lab.py -self-test
```

prints PASS: QRCC Volume XIV Kolmogorov/MCSP/MKtP laboratory only after all three exact checkpoints succeed.

8.2 Reader checkpoint and exercises

Synthesis and transition. MCSP and MKtP are the first decision problems in the volume whose inputs ask whether a represented object has a short computational description. Their exact encodings explain why apparently weak lower bounds can later be magnified. Chapter 9 therefore keeps the truth-table scale N , variable scale n , promise gap, and computational model visible at every theorem statement.

1. For a three-variable function, write its eight-bit truth table in lexicographic order and verify one exact circuit size from Example 8.5.
2. Give an explicit polynomial bound on the certificate length used in the proof that MCSP lies in NP when $s < cN(n+1)$.
3. Explain why a binary-encoded threshold s does not allow the NP verifier to guess an exponentially larger circuit than the input length can support.
4. Compare the YES/NO promises of exact MCSP, gap-MCSP, search-MCSP, and the approximation problem. State why hardness of one version is not silently transferred to another.
5. Reproduce the counting estimate for circuits of size at most $s(n)$ in Theorem 8.6 and show that its logarithm is $o(2^n)$.
6. Run the Chapter-8 laboratory self-test. Then change one finite candidate Levin-cost entry and predict which assertion should fail before rerunning it.

Chapter 9

Hardness Magnification: Weak Lower Bounds with Strong Consequences

Hardness magnification is a parameter-sensitive implication, not the slogan that “any small lower bound becomes a large one.” The input to MCSP is an entire truth table of length $N = 2^n$; the circuit threshold in the promise problem is naturally a function of the much smaller variable count $n = \log_2 N$; and the lower-bound hypothesis may concern formulas, circuits, a promise gap, or a particular *Kt* convention. Changing any of these coordinates can invalidate the implication.

This chapter freezes two representative magnification theorems at their primary-source regimes. The first turns an almost-linear *formula* lower bound for an approximate MCSP promise problem into a superpolynomial formula lower bound for a language in NP. The second turns a barely superlinear *circuit* lower bound for a gap version of MKtP into $\text{EXP} \not\subseteq \text{SIZE}(\text{poly})$. Their deep reductions remain source-owned; the surrounding definitions, convention transfer, and parameter-sensitivity audit are local.

How to read this chapter. Treat every magnification theorem as a microscope with calibrated settings. Before reading its conclusion, write down the truth-table length N , variable count $n = \log_2 N$, promise gap, computational model, and order of asymptotic quantifiers. The two deep theorems are source-owned at those exact settings; only the convention transfer and parameter-sensitivity proposition are local. The closing figure is a summary of permitted implications and failure zones, not a new theorem.

Learning goals. After this chapter a reader should be able to write a magnification implication with all quantifiers visible; distinguish n from the truth-table input length $N = 2^n$; define the exact MCSP and MKtP promise problems that occur in the selected theorems; explain why formula and circuit lower bounds cannot be interchanged; convert between the source and QRCC *Kt* conventions without losing a hidden constant; and identify which changes in a hypothesis are genuine strengthenings, weakenings, or logically incomparable promise modifications.

9.1 Definition-first magnification bookkeeping

Definition 9.1 (Hardness magnification and the two gap conventions). A *hardness-magnification theorem* is a quantified implication of the form

$$[Q_{\theta,N} \notin \mathcal{C}_{\text{weak}}[t(N)]] \implies [L \notin \mathcal{C}_{\text{strong}}[\text{poly}]], \quad (9.1)$$

where the family of promise problems $Q_{\theta,N}$, every parameter in θ , the weak model $\mathcal{C}_{\text{weak}}$, its size bound $t(N)$, the magnified language/class L , and the order of all asymptotic quantifiers are part of the statement. A theorem with different choices of any of these data is a different magnification theorem.

For the selected MCSP theorem, let $N = 2^n$ and use the fixed circuit basis $\mathcal{B}$ from Definition 8.1. For a threshold $s = s(n)$ and $0 < \delta = \delta(n) < 1$, the promise problem

$$(1, 1 - \delta)\text{-MCSP}_{\mathcal{B}}[s] \quad (9.2)$$

has as YES instances those N -bit truth tables T for which some $\mathcal{B}$ -circuit of size at most $s(n)$ computes f_T exactly, and as NO instances those T for which every such circuit C satisfies

$$\Pr_{x \leftarrow U_n} [C(x) = f_T(x)] < 1 - \delta(n). \quad (9.3)$$

Inputs satisfying neither condition lie outside the promise.

A Boolean *formula* is a fan-out-one rooted tree over a fixed bounded-fan-in De Morgan basis; its size in this chapter is its number of leaves. For a promise problem Q on N -bit inputs, $Q \in \text{Formula}[t(N)]$ means that there is a nonuniform formula family of at most $t(N)$ leaves that is correct on every promised input. The notation $\text{Formula}[N(\log N)^{O(1)}]$ means that one fixed exponent c gives size at most $N(\log N)^c$ for all sufficiently large input lengths. Likewise $\text{SIZE}[t(N)]$ denotes nonuniform Boolean circuits of size at most $t(N)$.

For the selected MKtP theorem, it is useful to display the source convention separately. With the same fixed universal step-counting machine U as in Definition 8.2, put

$$Kt_U^{\log}(x) := \min_{p, t \geq 1} \{ |p| + \log_2 t : U(p) \text{ outputs } x \text{ within } t \text{ steps} \}. \quad (9.4)$$

For real threshold functions $a(N) < b(N)$, the promise problem $\text{GapMKtP}_U^{\log}[a, b]$ has YES instances with $Kt_U^{\log}(x) \leq a(N)$ and NO instances with $Kt_U^{\log}(x) > b(N)$. The QRCC convention of Definition 8.2 instead uses $Kt_U(x) = \min(|p| + \lceil \log_2 t \rceil)$. For every nonempty string x ,

$$Kt_U^{\log}(x) \leq Kt_U(x) < Kt_U^{\log}(x) + 1. \quad (9.5)$$

We write $\text{GapMKtP}_U[a, b]$ for the analogous QRCC-ceiling promise problem.

Proof of the convention comparison (9.5). For each fixed pair (p, t) , $\log_2 t \leq \lceil \log_2 t \rceil < \log_2 t + 1$. Taking the minimum of the left inequality over all valid computations gives $Kt_U^{\log}(x) \leq Kt_U(x)$. If (p, t) minimizes the source quantity, using that same valid computation in the ceiling convention gives $Kt_U(x) < Kt_U^{\log}(x) + 1$. No change of universal machine is involved. $\square$

Why the truth-table scale must stay visible. In (9.2), $s(n) = n^k$ means $s = (\log_2 N)^k$ as a function of the actual input length. Thus a formula lower bound of order $N(\log N)^{O(1)}$ is “barely” above the linear input-reading scale, while the internal MCSP circuit threshold is only polylogarithmic in N . Replacing n by N in one of these two places changes the theorem by an exponential factor.

9.2 MCSP formula magnification

Theorem 9.2 (Oliveira–Santhanam MCSP formula magnification; source-owned). *Fix an integer $k \geq 1$, put*

$$s(n) = n^k, \quad \delta(n) = n^{-k}, \quad N = 2^n. \quad (9.6)$$

If

$$(1, 1 - \delta)\text{-MCSP}_{\mathcal{B}}[s] \notin \text{Formula}[N(\log N)^{O(1)}], \quad (9.7)$$

then there exists a language $L \in \text{NP}$, on ordinary m -bit inputs, such that

$$L \notin \text{Formula}[\text{poly}(m)]. \quad (9.8)$$

Source ownership and exact locator. Theorem 9.2 is Theorem 1(i) of Igor C. Oliveira and Rahul Santhanam, *Hardness Magnification for Natural Problems*, FOCS 2018, pp. 65–76, DOI 10.1109/FOCS.2018.00016. The primary statement uses the same $s(n) = n^k$, $\delta(n) = n^{-k}$, $N = 2^n$, and almost-linear formula hypothesis displayed above. The compression/projection and nonuniform derandomization argument is deep enough that this book keeps the proof source-owned rather than replacing it by a proof sketch counted as local closure.

Proof architecture, not a proof substitute. The source proves the contrapositive. Assuming every NP language has polynomial-size formulas, it samples a small projection of the input truth table and encodes the question “does the sample admit a small exact circuit?” as a compressed language in NP. Exact-small-circuit YES instances survive every projection, while the agreement gap in a NO instance makes a random projection reject all candidate small circuits with high probability. A careful nonuniform fixing of the random choices converts the randomized compression into an almost-linear formula for the original promise problem. The magnification occurs because the compressed input length is controlled by $s(n)/\delta(n) = n^{2k} = \text{poly}(n)$ while the original truth-table length is $N = 2^n$.

Example 9.3 (The microscope at $k = 2$). For $k = 2$, the internal circuit threshold is $s(n) = n^2 = (\log_2 N)^2$ and the agreement gap is $\delta(n) = 1/n^2 = 1/(\log_2 N)^2$. The antecedent asks for a lower bound only against formulas of size $N(\log N)^{O(1)}$ on the N -bit meta-complexity input, yet its consequence is a superpolynomial formula lower bound for some explicit language in NP on ordinary m -bit inputs. Neither replacing the n^2 circuit threshold by N^2 nor deleting the approximation gap is licensed by the theorem.

9.3 MKtP circuit magnification and convention transfer

Theorem 9.4 (Oliveira–Santhanam MKtP circuit magnification; source-owned). *Let N denote the input-string length. In the source convention (9.4), if there exists a fixed $\Delta > 0$ such that for every $\varepsilon > 0$ the promise problem*

$$\text{GapMKtP}_U^{\log}[N^\varepsilon, N^\varepsilon + 5 \log_2 N] \quad (9.9)$$

has no Boolean circuits of size $N^{1+\Delta}$, then

$$\text{EXP} \not\subseteq \text{SIZE}(\text{poly}). \quad (9.10)$$

In the QRCC ceiling convention, the following slightly stronger antecedent is sufficient for the same conclusion: there exists a fixed $\Delta > 0$ such that, for every $\varepsilon > 0$,

$$\text{GapMKtP}_U[N^\varepsilon, N^\varepsilon + 5 \log_2 N + 1] \notin \text{SIZE}[N^{1+\Delta}]. \quad (9.11)$$

Source ownership and exact locator. The source-convention implication (9.9)–(9.10) is Theorem 3 of Oliveira–Santhanam, FOCS 2018, DOI 10.1109/FOCS.2018.00016. The source quantifier order is

$$\exists \Delta > 0 \forall \varepsilon > 0, \quad (9.12)$$

and is part of the theorem. The proof combines the compression idea behind the MCSP theorem with highly efficient error-correcting codes and an EXP compressed language; it remains source-owned here.

Local transfer from the source convention to (9.11). Fix $\varepsilon > 0$ and abbreviate $a = N^\varepsilon$ and $b = N^\varepsilon + 5 \log_2 N$. Suppose, contrapositively, that the source promise $\text{GapMKtP}_U^{\log}[a, b]$ has a circuit C_N of size at most $N^{1+\Delta}$. We claim that the same circuit solves the QRCC promise $\text{GapMKtP}_U[a, b+1]$.

If $Kt_U(x) \leq a$, then (9.5) gives $Kt_U^{\log}(x) \leq a$, so x is a source YES instance. If $Kt_U(x) > b+1$, then the strict upper inequality in (9.5) forces $Kt_U^{\log}(x) > b$, so x is a source NO instance. Hence every circuit for the source promise is also a circuit of the same size for the QRCC promise with the shifted NO threshold. Therefore hardness of every QRCC promise in (9.11) implies the corresponding source hardness in (9.9), and the source-owned theorem yields (9.10). $\square$

Quantifier warning. A lower bound for one convenient value of ε does not satisfy (9.12), nor may the fixed Δ be chosen after ε . The theorem is a uniform family of barely-superlinear lower-bound hypotheses over all constant powers N^ε in the Kt threshold.

9.4 Parameter sensitivity and barrier boundaries

Proposition 9.5 (Parameter sensitivity of the magnification microscope). *The hypotheses of Theorems 9.2 and 9.4 cannot be freely altered along any of the following coordinates: truth-table length, promise gap, computational model, Kt convention, or asymptotic quantifier order. More precisely:*

1. For MCSP, increasing the circuit threshold s enlarges the YES set and shrinks the NO set, while increasing δ shrinks the NO set. Promise problems at different (s, δ) are therefore not interchangeable without an explicit promise-preserving reduction.
2. For every size bound t , $\text{Formula}[t] \subseteq \text{SIZE}[O(t)]$ after a fixed basis conversion. Thus a circuit lower bound at a given scale implies the corresponding formula lower bound up to constants, but a formula lower bound does not by itself imply a circuit lower bound.
3. The two scales satisfy $n = \log_2 N$. Consequently $n^k = (\log_2 N)^k$, whereas $N^{1+\varepsilon} = 2^{(1+\varepsilon)n}$. Replacing one scale by the other changes a magnification premise exponentially.
4. The $Kt^{\log}$ and QRCC-ceiling conventions differ by less than one unit, exactly as in (9.5); the one-unit promise shift in (9.11) is sufficient, whereas silently identifying the two threshold problems is not exact.
5. The MKtP theorem uses $\exists \Delta > 0 \forall \varepsilon > 0$. The logically weaker statement $\forall \varepsilon > 0 \exists \Delta_\varepsilon > 0$ or a lower bound at one isolated ε does not imply its hypothesis.
6. A worst-case exact solver is, in particular, a zero-error average-case solver under any input distribution, but the converse need not hold. Hence a known worst-case lower bound cannot simply be substituted into a magnification theorem whose antecedent demands an average-case lower bound.

Finally, the Razborov–Rudich natural-proofs barrier of Theorem 12.4 is a logically separate conditional obstruction: it constrains large, constructive properties useful against circuit classes containing sufficiently strong pseudorandom functions. It neither proves nor refutes the magnification implications above.

Proof. For item 1, if $s' \geq s$, every exact circuit witnessing the YES condition at threshold s also witnesses it at s' , so the YES set can only grow. The NO condition quantifies over *all* circuits up to the threshold; enlarging that family can only make the NO condition harder to satisfy. If $\delta' \geq \delta$, then $1 - \delta' \leq 1 - \delta$, so demanding that every candidate circuit have agreement below $1 - \delta'$ is stronger and the NO set can only shrink. These changes alter the promise domain in opposite directions, which prevents an automatic hardness transfer.

Item 2 follows by viewing a formula as its underlying circuit DAG: fan-out one is a restriction, not an additional computational resource. A fixed bounded-fan-in basis change costs at most a constant factor per gate. The reverse inclusion fails as a structural statement because circuits may share subcomputations whereas formulas cannot.

Item 3 is the identity $N = 2^n$. The displayed conversions follow by taking base-two logarithms and exponentiating; no complexity assumption is involved.

Item 4 is equation (9.5), and the promise-preserving implication is proved in the local part of Theorem 9.4. Item 5 is elementary predicate logic: allowing Δ to depend on ε changes the order of quantifiers and does not provide one fixed superlinear exponent valid for every threshold exponent.

For item 6, a device correct on every input is correct on every input on which an average-case device is required to answer; therefore worst-case solvability implies the corresponding average-case solvability. Taking contrapositives shows that an average-case lower bound implies a worst-case lower bound, not conversely. The final sentence merely compares hypotheses: Theorem 12.4 assumes a cryptographic pseudorandom-function ensemble and tests a truth-table property, whereas Theorems 9.2–9.4 are conditional implications from explicit meta-complexity lower bounds. No implication between those premises has been asserted. $\square$

microscope	weak antecedent	magnified consequence	failure zone
MCSP / formulas	$(1, 1 - n^{-k})\text{-MCSP}[n^k] \notin \text{Formula}[N(\log N)^{O(1)}]$	Some $L \in \text{NP}$ has no polynomial-size formulas	Changing n to N ; deleting the gap; replacing formulas by circuits without a theorem
MKtP / circuits	For one fixed $\Delta > 0$ and every $\varepsilon > 0$, gap-MKtP near N^ε has no size- $N^{1+\Delta}$ circuits	$\text{EXP} \not\subseteq \text{SIZE}(\text{poly})$	Reversing quantifiers; mixing Kt conventions; proving only one ε
Barrier overlay	Natural-proof constructivity/largeness/usefulness plus a PRF hypothesis	Conditional obstruction to a proof technique, not a magnification consequence	Treating a barrier theorem as a lower bound or as a refutation of magnification

Figure 9.1: The hardness-magnification microscope. The striking implication is meaningful only when its input scale, promise gap, computational model, and quantifier order stay fixed.

A later circuit-scale comparison. Oliveira, Pich, and Santhanam’s 2021 Theory of Computing refinement (DOI 10.4086/toc.2021.v017a011) gives, for a universal constant $c \geq 1$, a different

magnification regime: if there is $\varepsilon > 0$ such that for every sufficiently small $\beta > 0$,

$$\text{GapMCSP} \left[\frac{2^{\beta n}}{cn}, 2^{\beta n} \right] \notin \text{Circuit}[N^{1+\varepsilon}], \quad (9.13)$$

then $\text{NP} \not\subseteq \text{Circuit}[\text{poly}]$. This comparison is recorded to show how magnification thresholds evolved; it is *not* substituted for the formula theorem registered as QRCC-XIV-THM-026.

Dependency checkpoint. The two magnification implications are source-owned at their exact primary-source scopes. All notation needed to read them is local or recalled explicitly. Proposition 9.5 is elementary bookkeeping plus the already-audited natural-proofs barrier. No claimed proof of $\text{P} \neq \text{NP}$, $\text{NP} \not\subseteq \text{P/poly}$, or $\text{EXP} \not\subseteq \text{P/poly}$ appears here; every strong separation remains the consequent of a stated weak-lower-bound hypothesis.

9.5 Reader checkpoint and exercises

Synthesis and transition. Hardness magnification works only when a weak lower bound is measured at the theorem's exact scale. The source-owned MCSP and MKtP implications, together with the local parameter-sensitivity audit, prepare the final chapter: modern learning and cryptographic results use meta-complexity as an interface, but with still different density, sampling, approximation, and quantum-state conventions.

1. Convert between n and $N = 2^n$ for each threshold in the MCSP magnification theorem and explain why replacing n^k by N^k is an exponential change.
2. Draw the YES, promise-gap, and NO regions for $(1, 1 - n^{-k})\text{-MCSP}[n^k]$ using the chapter's agreement convention.
3. State the quantifier order $\exists\Delta\forall\varepsilon$ in the MKtP theorem and write a logically weaker order that would not suffice.
4. Starting from the inequality relating $Kt^{\log}$ to the QRCC ceiling convention, verify the one-unit shift in the NO threshold of equation (9.11).
5. Explain why a formula lower bound need not imply a circuit lower bound, while a circuit lower bound transfers to formulas up to the stated basis-conversion overhead.
6. Give a hypothetical average-case lower bound and a worst-case lower bound and use item 6 of Proposition 9.5 to determine which implication between them is valid.
7. Using Figure 9.1, choose one failure-zone modification and explain precisely which hypothesis of the corresponding source theorem it violates.
8. Compare hardness magnification with the natural-proofs barrier: identify the premise and conclusion of each and explain why neither is a direct proof of a major circuit separation.

Chapter 10

Learning, Cryptography, and the Meta-Complexity Frontier

The final chapter of this volume records three places where meta-complexity acts as an interface rather than as an isolated decision problem. First, a sufficiently strong natural lower-bound property can be converted into a nontrivial learner for the same circuit class. Second, average-case hardness of natural meta-complexity problems can characterize the existence of one-way functions. Third, current work connects meta-complexity to hardness of approximation, quantum cryptography, and explicit construction of random-looking strings. The first two interfaces are deep theorems; the last three are dated frontiers. Their proof ownership is therefore stated explicitly rather than blurred into a survey paragraph.

How to read this chapter. The chapter is an interface map, so theorem ownership matters as much as theorem content. Read the learning theorem as a conversion from a constant-density constructive property to a membership-query learner; read the two cryptographic theorems as exact characterizations or implications under their sampler and entropy hypotheses; then read the final three items only as dated frontier maps. The quantum subsection intentionally names quantum meta-complexity objects without defining their full state model here; no completed classical theorem in Volume XIV depends on those deferred definitions.

Learning goals. After this chapter, a reader should be able to distinguish truth-table constructivity from membership- query learning; state the locally-samplable GapMCSP characterization with the gap and average-case quantifiers visible; compute the entropy-deficiency hypotheses in the SAT/Clique implication; explain which approximation-hardness theorems concern conditional K^t , oracle circuit size, or multivalued MCSP rather than ordinary MCSP; and separate classical meta-complexity from the newer quantum-state and quantum-samplable-distribution interfaces.

10.1 From natural properties to learning

For this section, a circuit class Λ is a family of Boolean circuits over a fixed finite basis. For a size function u , the notation $\Lambda[u]$ means the n -input functions in Λ having size at most $u(n)$. A property R of n -variable Boolean functions is called *P-natural* when membership in R is decidable in time polynomial in the 2^n -bit truth-table length. The source theorem below assumes the stronger

constant-density form of largeness

$$\Pr_{f \leftarrow \mathcal{F}_n} [f \in R] \geq \frac{1}{5}, \quad (10.1)$$

not merely the $2^{-O(n)}$ largeness admitted by Definition 12.2. It is *useful against* $\Lambda[u]$ when every sufficiently large truth table in R avoids $\Lambda[u]$.

Given membership-query access to $f : \{0, 1\}^n \rightarrow \{0, 1\}$, an ε -approximate learner outputs a Boolean circuit C satisfying

$$\Pr_{x \leftarrow U_n} [C(x) \neq f(x)] \leq \varepsilon. \quad (10.2)$$

This is PAC learning over the uniform distribution with membership queries; the output circuit need not itself lie in Λ .

Theorem 10.1 (Carmosino–Impagliazzo–Kabanets–Kolokolova; source-owned). *Let Λ be any circuit class containing $\text{AC}^0[p]$ for some prime p . Let R be a P -natural property of largeness at least $1/5$ that is useful against $\Lambda[u]$, where u is a size function. There is a randomized membership-query learner with the following property. Given oracle access to any n -variable $f \in \Lambda[s_f]$ and an error parameter $0 < \varepsilon < 1$, it outputs a circuit satisfying (10.2) in time*

$$\text{poly}\left(n, \frac{1}{\varepsilon}, 2^{u^{-1}(\text{poly}(n, 1/\varepsilon, s_f))}\right). \quad (10.3)$$

The source’s circuit-builder convention permits the constant success probability to be amplified in the usual way.

Exact source and ownership. This is Theorem 5.1 of Marco L. Carmosino, Russell Impagliazzo, Valentine Kabanets, and Antonina Kolokolova, *Learning Algorithms from Natural Proofs*, CCC 2016, LIPIcs 50, Article 10, DOI 10.4230/LIPIcs.CCC.2016.10. The proof is source-owned here. Its two nontrivial reconstruction layers—black-box hardness amplification within $\text{AC}^0[p]$ and the Nisan–Wigderson distinguisher-to-predictor reconstruction—are precisely the parts that make the implication much stronger than the elementary natural-property construction in Theorem 8.6.

Why the hypotheses matter. The constant $1/5$ in (10.1) supplies a constant distinguishing gap between uniformly random truth tables and the low-complexity truth tables output by the conditional generator. Usefulness guarantees that the generator outputs lie outside R at the selected output length. The inclusion $\text{AC}^0[p] \subseteq \Lambda$ is needed because the amplification and generator circuitry must remain inside the class against which the natural property is useful. Finally, membership queries matter because the reconstruction algorithm evaluates the unknown target at adaptively chosen points; this theorem is not a claim about sample-only PAC learning.

Example 10.2 (Weakly exponential usefulness gives quasipolynomial learning). Suppose $u(m) = 2^{m^\gamma}$ for a fixed $0 < \gamma \leq 1$ and $s_f = n^{O(1)}$. Then $u^{-1}(q) = (\log q)^{1/\gamma}$, so for inverse-polynomial ε the last term in (10.3) is

$$2^{(\log n)^{O(1)}} = n^{(\log n)^{O(1)}}.$$

Thus a natural property useful against weakly exponential size yields quasipolynomial-time learning. This is the regime used in the source to learn $\text{AC}^0[p]$ from the Razborov–Smolensky natural lower bounds.

10.2 One-way functions from average-case meta-complexity

We recall the exact average-case language used in the primary source. For a distribution ensemble $D = \{D_n\}$, a promise problem is *weakly average-case hard on D* when every probabilistic polynomial-time algorithm errs with probability at least $1/n^{O(1)}$ for almost all input lengths. It is *strongly average-case hard on D* when every such algorithm errs with probability at least

$$\frac{1}{2} - \frac{1}{n^{\omega(1)}} \quad (10.4)$$

for almost all lengths. A t -local sampler has random access to its random input and, to compute any designated output bit, runs for at most t steps; hence it reads at most t random-input bits for that output coordinate.

For thresholds $s < c$, $\text{GapMCSP}[s, c]$ distinguishes truth tables of circuit complexity at most s from truth tables of circuit complexity at least c , leaving intermediate complexities outside the promise.

Theorem 10.3 (Ilango–Ren–Santhanam local-samplability characterization; source-owned). *The following are equivalent.*

1. *One-way functions exist.*
2. *For some constant $\delta > 0$ and some $s(n) = \Omega(n^\delta)$, there is an n^δ -locally samplable distribution D such that*

$$\text{GapMCSP}[s, sn^{5\delta}] \quad (10.5)$$

is weakly average-case hard on D .

3. *For every constant $\delta > 0$, there is an n^δ -locally samplable distribution D such that*

$$\text{GapMCSP}\left[n^\delta, o\left(\frac{n}{\log n}\right)\right] \quad (10.6)$$

is strongly average-case hard on D .

Exact source and ownership. This is Theorem 3 of Rahul Ilango, Hanlin Ren, and Rahul Santhanam, *Hardness on Any Samplable Distribution Suffices: New Characterizations of One-Way Functions by Meta-Complexity*, ECCC TR21-082 (2021). The theorem is source-owned. In particular, the nonexistence-of-OWFs direction uses a coding/probability-estimation argument adapted to local samplers; the converse uses pseudorandomness to mix genuinely random truth tables with outputs of small-complexity generators. Neither direction is replaced here by a slogan about MCSP being “cryptographically hard.”

Hypothesis audit. The sampler in (10.5)–(10.6) is not an arbitrary polynomial-time sampler: locality is part of the characterization. The NO threshold in (10.6) is asymptotically below the counting barrier $\Theta(n/\log n)$ for random truth tables. Weak and strong average-case hardness are also not interchangeable; the theorem deliberately states different parameter regimes for the two notions.

For the next theorem, if D is a distribution on m bits, its *Shannon entropy deficiency* is

$$\text{def}(D) := m - H(D),$$

where $H(D)$ is Shannon entropy in bits. Thus smaller deficiency means a distribution closer, in entropy, to the uniform m -bit distribution.

Theorem 10.4 (High-entropy SAT/Clique hardness implies one-way functions; source-owned). *Fix an integer k , let $\Delta \geq 2^{k+3}$ be a sufficiently large integer, and let $t : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $t(n) = \omega(\log n)$. If either of the following holds, then one-way functions exist:*

1. *k -SAT on formulas with Δn clauses is strongly average-case hard with respect to some polynomial-time samplable distribution whose Shannon entropy deficiency is at most*

$$\frac{\Delta n}{2^{k+1}}; \quad (10.7)$$

2. *t -Clique is strongly average-case hard with respect to some polynomial-time samplable distribution whose Shannon entropy deficiency is at most*

$$0.99 \binom{t}{2}. \quad (10.8)$$

If “samplable” is strengthened to “logspace-samplable” in the corresponding hypothesis, then the source concludes that one-way functions computable in NC^0 exist.

Exact source and ownership. This is Theorem 4 of Ilango–Ren–Santhanam, ECCC TR21-082. The parameters in (10.7)–(10.8) are retained literally: the Clique bound is a constant fraction of the $\binom{t}{2}$ edge-description bits, not a bound of order t ; and the SAT clause count is Δn with $\Delta \geq 2^{k+3}$. The source also observes that the Clique assumption follows from the Planted Clique hypothesis and the SAT assumption from pseudorandomness of suitable random local functions. The reduction to the preceding meta-complexity characterization remains source-owned.

10.3 Dated frontier: cryptographic hardness of approximating meta-complexity

Remark 10.5 (Frontier map: approximation hardness, audited 2026-09-02). The strongest currently source-verified approximation-hardness results used by this book do *not* collapse into a theorem that ordinary MCSP or MKtP is NP-hard to approximate. Huang–Ilango–Ren, *NP-Hardness of Approximating Meta-Complexity: A Cryptographic Approach*, SIAM Journal on Computing 54(4), 819–886 (2025), DOI 10.1137/23M1608483, proves three distinct results:

1. assuming subexponentially secure witness encryption, essentially optimal NP-hardness of approximating conditional time-bounded Kolmogorov complexity $K^t(x \mid y)$ in the regime $t \gg |y|$;
2. unconditionally, near-optimal NP-hardness of a gap Minimum *Oracle* Circuit Size Problem, with YES complexity at most $2^{\varepsilon n}$ and NO objects essentially as hard as random truth tables;
3. with probability one over a random oracle O , NP-hardness of approximating a multivalued version mvMCSP^O under quasipolynomial-time reductions with oracle access.

These are different models and reduction regimes. In particular, item 2 is MOCSP rather than ordinary MCSP, and item 3 is a random-oracle statement rather than an unrelativized NP-hardness result.

10.4 Dated frontier: quantum meta-complexity

Remark 10.6 (Frontier map: quantum meta-complexity, audited 2026-09-02). The classical strings of this volume already meet quantum sampling and state-complexity questions, but the underlying quantum-state definitions belong to Volume XV. Three source-verified landmarks fix the current interface.

1. Cavalar–Goldin–Gray–Hall, *A Meta-Complexity Characterization of Quantum Cryptography*, EUROCRYPT 2025, DOI 10.1007/978-3-031-91098-2_4, shows that one-way puzzles exist iff there is a quantum-polynomial-time samplable distribution of classical strings on which approximating Kolmogorov complexity is hard; their equivalence also passes through hardness of probability estimation.
2. Hiroka–Morimae, *Quantum Cryptography and Meta-Complexity*, CRYPTO 2025, gives a GapK characterization of one-way puzzles under quantum-average hardness and relates stronger hardness to quantum pseudorandom generators.
3. Cavalar–Chen–Coladangelo–Gray–Hu–Ji–Li, ECCC TR25-221, *A Meta-Complexity Characterization of Minimal Quantum Cryptography*, characterizes EFI pairs (equivalently, quantum commitments in their setting) via single-copy hardness of a Kolmogorov-like complexity measure on a nonuniformly samplable distribution of pure states, and relates this to nonuniform single-copy-secure pseudorandom-state generators.

No theorem about minimum quantum circuit size is promoted here merely from these results. The state, channel, fidelity, sampling, and adversary conventions required to compare such notions are reserved for Volume XV; this row is a cross-volume source map, not a surrogate quantum definition.

10.5 Dated frontier: pseudorandomness and constructive hard strings

Remark 10.7 (Frontier map: constructive hard strings, audited 2026-09-02). Korten–Santhanam, *How to Construct Random Strings*, CCC 2025, DOI 10.4230/LIPIcs.CCC.2025.35, gives current positive evidence for efficiently constructing strings that are random-like in a time-bounded Kolmogorov sense. Under explicit hierarchy-style hardness assumptions of the form

$$\text{TIME}[T(n)] \not\subseteq \text{TIME}^{\text{NP}}[T(n)^\varepsilon]/2^{\varepsilon n}, \quad (10.9)$$

for appropriately chosen time-constructible T , they construct pseudorandom generators with nearly optimal iterated-logarithmic seed length and derive deterministic constructions of n -bit strings with $K^{\text{poly}(n)}$ complexity $\tilde{\Omega}(n)$. The same framework yields algorithms for Range Avoidance on uniform circuit families with close-to-minimal stretch and gives conditional explicit constructions of objects such as near-optimal rigid matrices and Ramsey graphs.

The source also records two boundaries that matter for this volume. General Range Avoidance can remain hard even when the uniform version becomes tractable under the stated complexity assumptions, and the paper proves a barrier to a direct relativizing reduction from constructing K^{poly} -random strings to constructing hard truth tables. Thus “meta-hardness gives pseudorandomness” is not a single unconditional transformation: uniformity, time bounds, advice, oracle access, and the form of the hardness assumption are all live parameters.

10.6 Volume XIV closure checkpoint

The chapter's six registered items have deliberately different ownership types. The learning theorem and the two one-way-function theorems are deep primary-source results with exact hypotheses frozen above. The final three items are dated frontier maps. None is counted as a local proof merely because its proof architecture is explained. Together with Chapters 1–9, however, they close the registered Volume-XIV agenda: average-case complexity, fine-grained complexity, Kolmogorov and meta-complexity, hardness magnification, learning, cryptographic characterizations, and the current classical/quantum frontier now all have explicit reader-facing ownership boundaries.

10.7 Reader checkpoint and exercises

End-of-volume synthesis. Volume XIV began by changing the input measure, then changed the running-time exponent, then changed the object of computation to description and circuit complexity themselves. The final interfaces show why those distinctions matter: learning needs a constructive large property, cryptographic characterizations need precise sampler and average-case hypotheses, hardness magnification needs exact parameter scales, and quantum frontiers introduce a genuinely new state model. A reader should leave the volume able to say not only what a theorem concludes, but also which distribution, encoding, sampler, model, and ownership convention makes the conclusion valid.

1. For $u(m) = 2^{m^{1/2}}$, compute the inverse u^{-1} used in Theorem 10.1 and simplify the learner running time for polynomial-size targets and inverse-polynomial error.
2. Explain why constant-density largeness is stronger than the weaker natural-property largeness convention recalled from Volume X, and identify where the learning theorem uses the stronger condition.
3. Compare weak and strong average-case hardness in Theorem 10.3. Write the two error lower bounds and explain why their parameter regimes cannot be interchanged.
4. For a distribution on m bits with Shannon entropy $m - r$, compute its entropy deficiency and specialize the calculation to each bound in Theorem 10.4.
5. Make a three-column table separating the approximation-hardness frontier results for conditional K^t , oracle circuit size, and multivalued MCSP. State why none alone settles standard total MCSP NP-hardness.
6. Explain why the quantum frontier is recorded without making Volume XIV depend formally on the still-to-be-built quantum-state definitions of Volume XV.
7. Choose one result from Chapters 1–4 and one from Chapters 8–10 and describe a valid conceptual connection between them without asserting a theorem that the book has not proved.
8. Produce a one-page ownership map for Volume XIV: mark each central item as local proof, verified deep input, open problem, executable example/laboratory, or dated frontier. Check that every arrow you draw follows a registered dependency or an explicitly labelled conceptual transition.

Internalization and Reconstruction Guide: Volume XIV

Purpose. Track how hardness changes when the input distribution, approximation scale, parameter regime, or problem encoding changes; distinguish unconditional theorems from fine-grained and meta-complexity hypotheses.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-X-DEF-001: Boolean circuit and formula models.
2. QRCC-X-DEF-018: Natural property against a circuit class.
3. QRCC-IX-DEF-029: One-way-function interface.
4. QRCC-I-THM-002: Markov inequality.
5. QRCC-II-THM-018: Verifier characterization of NP.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XIV-DEF-001 (Distributional decision problem and AvgP); QRCC-XIV-DEF-007 (ETH, SETH, and fine-grained reductions); QRCC-XIV-DEF-015 (Kolmogorov and time-bounded Kolmogorov complexity); QRCC-XIV-DEF-017 (Minimum Circuit Size Problem); QRCC-XIV-DEF-020 (MKtP and gap meta-complexity problems).

Chapter retrieval map

Ch. 1: Distributional Problems and Average Polynomial Time: enter through QRCC-XIV-DEF-001 (Distributional decision problem and AvgP); exit by being able to state QRCC-XIV-THM-001 (Tail characterization of average polynomial time) with its ownership boundary.

- Ch. 2: Domination-Preserving Average-Case Reductions:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIV-THM-003 (Closure of AvgP under domination-preserving reductions) with its ownership boundary.
- Ch. 3: A Complete Problem for Computable Ensembles:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIV-THM-002 (Padded bounded halting is complete for (NP,PComp)) with its ownership boundary.
- Ch. 4: Average-Hardness Amplification and Black-Box Barriers:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIV-BB-006 (Limitations of black-box worst-case-to-average-case reductions for NP) with its ownership boundary.
- Ch. 5: ETH, SETH, Fine-Grained Reductions, and Sparsification:** enter through QRCC-XIV-DEF-007 (ETH, SETH, and fine-grained reductions); exit by being able to state QRCC-XIV-BB-008 (Sparsification lemma) with its ownership boundary.
- Ch. 6: Orthogonal Vectors and the SETH Reduction:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIV-THM-009 (SETH-to-Orthogonal-Vectors reduction) with its ownership boundary.
- Ch. 7: Kolmogorov Complexity and the Incompressibility Method:** enter through QRCC-XIV-DEF-015 (Kolmogorov and time-bounded Kolmogorov complexity); exit by being able to state QRCC-XIV-THM-016 (Incompressibility method) with its ownership boundary.
- Ch. 8: Minimum Circuit Size, MKtP, and the First Meta-Complexity Interfaces:** enter through QRCC-XIV-DEF-017 (Minimum Circuit Size Problem); exit by being able to state QRCC-XIV-PROP-018 (MCSP belongs to NP) with its ownership boundary.
- Ch. 9: Hardness Magnification: Weak Lower Bounds with Strong Consequences:** enter through QRCC-XIV-DEF-025 (Hardness magnification); exit by being able to state QRCC-XIV-PROP-028 (Parameter sensitivity and magnification barriers) with its ownership boundary.
- Ch. 10: Learning, Cryptography, and the Meta-Complexity Frontier:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XIV-BB-023 (One-way functions from average-case hardness of meta-complexity) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XIV-THM-001 — Tail characterization of average polynomial time. Dependencies: QRCC-XIV-DEF-001, QRCC-I-THM-002. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XIV-THM-002 — Padded bounded halting is complete for (NP,PComp). Dependencies: QRCC-XIV-DEF-001, QRCC-II-THM-018. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

3. QRCC-XIV-BB-008 — Sparsification lemma. Dependencies: QRCC-II-THM-006, QRCC-XIV-DEF-007. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XIV-THM-016 — Incompressibility method. Dependencies: QRCC-XIV-DEF-015. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-XIV-PROP-028 — Parameter sensitivity and magnification barriers. Dependencies: QRCC-XIV-THM-026, QRCC-XIV-THM-027, QRCC-X-THM-019. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-XIV-CONJ-011 — Orthogonal Vectors conjecture. Dependencies: QRCC-XIV-THM-009. *open frontier: separate the conjecture from the strongest proved substitute and never route a later theorem through the conjecture.*

Common failure modes

- moving from worst-case to average-case hardness without specifying a distribution;
- hiding parameter blowups in fine-grained reductions;
- calling a conditional lower bound unconditional;
- mixing decision, search, approximation, promise, and meta-complexity formulations;

Boundary and degenerate cases

- degenerate distributions concentrated on one input;
- parameters so large a fine-grained statement becomes ordinary polynomial time;
- promise gaps collapsing to zero;
- hypothesis statements at the boundary between classical and quantum models;

Retrieval and reconstruction drill

1. State QRCC-XIV-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XIV-THM-001. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XIV-THM-002, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.

5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XIV-CONJ-011, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XV changes the computational model itself to quantum computation and nonlocal interaction.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XV

Quantum Complexity and Nonlocal Computation

Chapter 1

Finite-Dimensional Quantum States and Distinguishability

Quantum complexity theory is not obtained by replacing probabilities by amplitudes inside classical complexity definitions. The basic objects have changed: a state can be mixed, a subsystem is obtained by partial trace, a measurement is a family of positive operators, and a physically allowed transformation is a completely positive trace-preserving map. This chapter fixes those conventions before any quantum circuit, query, verifier, or nonlocal-game model is introduced.

1.1 How to read this chapter

There are three layers. First learn the state/channel language in Definition 1.1. Second learn the two quantitative notions of distinguishability in Definition 1.2: trace distance is operationally tied to binary state discrimination, while fidelity is analytically convenient and behaves naturally under purification. Third read Theorems 1.3–1.6 as a single chain: Schmidt decomposition controls purifications, channels contract trace distance, fidelity and trace distance are quantitatively equivalent, and cloning nonorthogonal states would violate that contraction.

1.2 Learning goals

After this chapter, a reader should be able to distinguish vectors, pure states, and density operators; compute a partial trace; verify that a POVM gives a probability distribution; state complete positivity and trace preservation without invoking an undefined physical slogan; compute trace distance and fidelity for two pure qubit states; prove purification uniqueness with the necessary support/dimension qualification; prove trace-distance data processing; derive the Fuchs–van de Graaf inequalities in the unsquared-fidelity convention; and explain exactly why the no-cloning contradiction fails for orthogonal states.

Literature and convention checkpoint. The operator conventions are calibrated to Watrous, *The Theory of Quantum Information* (Cambridge, 2018), especially Chapters 2–3. Nielsen–Chuang is used as a pedagogical circuit and information-theory cross-check. We use the *unsquared* fidelity $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$, so the Fuchs–van de Graaf inequalities read $1 - F \leq D \leq \sqrt{1 - F^2}$ for $D = \frac{1}{2}\|\rho - \sigma\|_1$. No squared-fidelity convention is silently substituted later.

Definition 1.1 (Finite-dimensional quantum states, measurements, channels, and purification). A *register* X is a finite-dimensional complex Hilbert space $\mathcal{H}_X$. We write $\mathcal{L}(X)$ for its linear operators, I_X for the identity, and A^* for the adjoint of $A \in \mathcal{L}(X)$. An operator H is *Hermitian* when $H = H^*$ and is *positive semidefinite*, written $H \succeq 0$, when $\langle v, Hv \rangle \geq 0$ for every $v \in \mathcal{H}_X$. The trace is denoted by Tr . The *kernel* of an operator A is $\ker A := \{v : Av = 0\}$, its *rank* is the dimension of its image, and for a positive semidefinite operator P its *support* $\text{supp}(P)$ is the orthogonal complement of $\ker P$ (equivalently, the span of the eigenvectors with positive eigenvalue). The orthogonal projection onto this subspace is the *support projection* of P .

A *density operator* or *state* of X is an operator

$$\rho \in \mathcal{D}(X) := \{\rho \in \mathcal{L}(X) : \rho \succeq 0, \text{Tr } \rho = 1\}.$$

A unit vector $|\psi\rangle \in \mathcal{H}_X$ represents the *pure state* $|\psi\rangle\langle\psi|$; vectors differing by a global complex phase represent the same pure state. A density operator is pure exactly when it has rank one. Composite registers use the tensor product: XY has Hilbert space $\mathcal{H}_X \otimes \mathcal{H}_Y$.

For $R \in \mathcal{L}(XY)$, the *partial trace over* Y , denoted $\text{Tr}_Y R \in \mathcal{L}(X)$, is the unique operator satisfying

$$\text{Tr}(A \text{Tr}_Y R) = \text{Tr}((A \otimes I_Y)R) \quad \text{for every } A \in \mathcal{L}(X).$$

If $\{|y_j\rangle\}$ is any orthonormal basis of Y , equivalently

$$\text{Tr}_Y R = \sum_j (I_X \otimes \langle y_j|) R (I_X \otimes |y_j\rangle).$$

A unit vector $|\psi\rangle \in \mathcal{H}_X \otimes \mathcal{H}_Y$ is a *purification* of $\rho \in \mathcal{D}(X)$ when $\text{Tr}_Y |\psi\rangle\langle\psi| = \rho$. The register Y is then called a purifying register. For a purification $|\psi\rangle \in X \otimes Y$, the *Schmidt support in* Y means the support of its reduced state $\text{Tr}_X |\psi\rangle\langle\psi|$. This terminology is used in the purification-uniqueness theorem below.

A finite-outcome *POVM* (positive operator-valued measure) on X with outcome set Ω is a family $\{M_a : a \in \Omega\} \subseteq \mathcal{L}(X)$ such that $M_a \succeq 0$ and $\sum_a M_a = I_X$. Measuring ρ produces outcome a with probability

$$p(a) = \text{Tr}(M_a \rho).$$

This definition specifies outcome probabilities only; a postmeasurement state requires an instrument and is not determined by the POVM alone.

A linear map $\Phi : \mathcal{L}(X) \rightarrow \mathcal{L}(Y)$ is *completely positive* if $\Phi \otimes \text{id}_R$ maps positive semidefinite operators to positive semidefinite operators for every finite register R . It is *trace preserving* if $\text{Tr } \Phi(A) = \text{Tr } A$ for every $A \in \mathcal{L}(X)$. A *quantum channel* is a completely positive trace-preserving (CPTP) map. An isometry $V : \mathcal{H}_X \rightarrow \mathcal{H}_Y$ with $V^*V = I_X$ induces the channel $\rho \mapsto V\rho V^*$; a unitary channel is the equal-dimension special case.

Worked partial trace. For the Bell vector $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$,

$$|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Tracing out the second qubit kills the two off-diagonal tensor factors and gives $\text{Tr}_2 |\Phi^+\rangle\langle\Phi^+| = I_2/2$. Thus a globally pure state may have a mixed marginal; this is the first reason a density-operator model is indispensable.

Definition 1.2 (Trace distance, fidelity, and binary distinguishing advantage). For $A \in \mathcal{L}(X)$ let $|A| := \sqrt{A^*A}$ and define the *trace norm* $\|A\|_1 := \text{Tr } |A|$. For states $\rho, \sigma \in \mathcal{D}(X)$ their *trace distance* is

$$D(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1.$$

For a Hermitian H , write $H = H_+ - H_-$ for its Jordan decomposition, where $H_+, H_- \succeq 0$ have orthogonal supports. Then $\|H\|_1 = \text{Tr } H_+ + \text{Tr } H_-$.

For positive semidefinite operators P, Q , their *fidelity* is

$$F(P, Q) := \|\sqrt{P}\sqrt{Q}\|_1.$$

In particular $0 \leq F(\rho, \sigma) \leq 1$ for states. For pure states $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\varphi\rangle\langle\varphi|$,

$$F(\rho, \sigma) = |\langle\psi|\varphi\rangle|.$$

This is the unsquared fidelity convention throughout Volume XV.

Suppose one of ρ and σ is chosen with prior probability $1/2$ and a binary POVM $\{M, I - M\}$ is used to guess which state was chosen. Its success probability is

$$p_{\text{succ}}(M) = \frac{1}{2} \text{Tr}(M\rho) + \frac{1}{2} \text{Tr}((I - M)\sigma).$$

The *optimal binary distinguishing advantage* is $\max_{0 \preceq M \preceq I} p_{\text{succ}}(M) - 1/2$.

Operational check: the Helstrom variational formula. Let $H = \rho - \sigma$. Because $\text{Tr } H = 0$, its Jordan parts have equal trace and

$$\max_{0 \preceq M \preceq I} \text{Tr}(MH) = \text{Tr } H_+ = \frac{1}{2} \|H\|_1.$$

Indeed $\text{Tr}(MH) \leq \text{Tr}(MH_+) \leq \text{Tr } H_+$, and equality is obtained by taking M to be the projection onto the support of H_+ . Hence

$$\max_M p_{\text{succ}}(M) = \frac{1}{2} (1 + D(\rho, \sigma)), \quad \max_M (p_{\text{succ}}(M) - \frac{1}{2}) = \frac{1}{2} D(\rho, \sigma).$$

Thus trace distance is not merely a matrix norm: it is exactly the equal-prior bias available to the best binary measurement, up to the displayed factor $1/2$.

Theorem 1.3 (Schmidt decomposition and uniqueness of purification). *Let X, Y be finite registers and let $|\psi\rangle \in \mathcal{H}_X \otimes \mathcal{H}_Y$ be a unit vector. There exist an integer $r \leq \min\{\dim X, \dim Y\}$, positive numbers $s_1, \dots, s_r$ with $\sum_i s_i^2 = 1$, and orthonormal families $\{|x_i\rangle\}_{i=1}^r \subseteq \mathcal{H}_X$ and $\{|y_i\rangle\}_{i=1}^r \subseteq \mathcal{H}_Y$ such that*

$$|\psi\rangle = \sum_{i=1}^r s_i |x_i\rangle |y_i\rangle.$$

The nonzero eigenvalues of both reduced states are $s_1^2, \dots, s_r^2$ with multiplicity.

Moreover, if $|\psi\rangle \in X \otimes Y$ and $|\varphi\rangle \in X \otimes Z$ purify the same state $\rho \in \mathcal{D}(X)$, then there is an isometry from the Schmidt support of ψ in Y onto the Schmidt support of φ in Z that maps one purification to the other. If the two purifying spaces are padded to equal dimension, the isometry extends to a unitary on the purifying register.

Proof. Fix orthonormal bases of X and Y and write $|\psi\rangle = \sum_{j,k} a_{jk}|j\rangle|k\rangle$. Regard the coefficient array $A = (a_{jk})$ as a matrix from Y to X . A singular-value decomposition gives $A = U\Sigma V^*$, where the positive diagonal entries of Σ are $s_1, \dots, s_r$. Expanding the matrix identity in the corresponding left and conjugated right singular-vector bases gives exactly $|\psi\rangle = \sum_i s_i |x_i\rangle |y_i\rangle$. Normalization gives $\sum_i s_i^2 = 1$.

Taking partial traces in this expression gives

$$\rho_X = \sum_i s_i^2 |x_i\rangle\langle x_i|, \quad \rho_Y = \sum_i s_i^2 |y_i\rangle\langle y_i|,$$

which proves the reduced-spectrum assertion.

For purification uniqueness, diagonalize the common reduced state as $\rho = \sum_{i=1}^r \lambda_i |x_i\rangle\langle x_i|$ with $\lambda_i > 0$. Define

$$|y_i\rangle := \lambda_i^{-1/2} (\langle x_i| \otimes I_Y) |\psi\rangle, \quad |z_i\rangle := \lambda_i^{-1/2} (\langle x_i| \otimes I_Z) |\varphi\rangle.$$

The identity of the two X marginals implies $\langle y_i | y_j \rangle = \langle z_i | z_j \rangle = \delta_{ij}$, and therefore

$$|\psi\rangle = \sum_i \sqrt{\lambda_i} |x_i\rangle |y_i\rangle, \quad |\varphi\rangle = \sum_i \sqrt{\lambda_i} |x_i\rangle |z_i\rangle.$$

The map $|y_i\rangle \mapsto |z_i\rangle$ is an isometry between the two Schmidt supports and sends $|\psi\rangle$ to $|\varphi\rangle$ when applied on the purifying factor. After padding the two ambient purifying spaces to the same finite dimension, extend orthonormal bases on the complements; the resulting extension is unitary. $\square$

Purification-overlap calculation used below. Fix an orthonormal basis and define $\text{vec}(A) = \sum_{j,k} A_{jk} |j\rangle |k\rangle$. Then $\text{Tr}_2 |\text{vec}(A)\rangle\langle \text{vec}(A)| = AA^*$ and $\langle \text{vec}(A) | \text{vec}(B) \rangle = \text{Tr}(A^*B)$. Consequently $\text{vec}(\sqrt{\rho})$ purifies ρ . If $\sqrt{\rho}\sqrt{\sigma} = W|\sqrt{\rho}\sqrt{\sigma}|$ is a polar decomposition, choose a unitary extension of W^* on the relevant support. Then a purification of σ can be chosen so that its overlap with $\text{vec}(\sqrt{\rho})$ has absolute value $\|\sqrt{\rho}\sqrt{\sigma}\|_1 = F(\rho, \sigma)$. This is the finite-dimensional Uhlmann overlap fact needed in Theorem 1.5; no stronger channel-fidelity theorem is being imported.

Theorem 1.4 (Data processing for trace distance). *For every quantum channel $\Phi : \mathcal{L}(X) \rightarrow \mathcal{L}(Y)$ and all $\rho, \sigma \in \mathcal{D}(X)$,*

$$D(\Phi(\rho), \Phi(\sigma)) \leq D(\rho, \sigma).$$

In particular trace distance cannot increase under a partial trace or under the classical outcome channel of a measurement.

Proof. It is enough to prove contraction of the trace norm on Hermitian operators. Let $H = H_+ - H_-$ be the Jordan decomposition. A completely positive map is positive, so $\Phi(H_+), \Phi(H_-) \succeq 0$. The triangle inequality and trace preservation give

$$\|\Phi(H)\|_1 \leq \|\Phi(H_+)\|_1 + \|\Phi(H_-)\|_1 = \text{Tr } \Phi(H_+) + \text{Tr } \Phi(H_-) = \|H\|_1.$$

Apply this to $H = \rho - \sigma$ and divide by two.

The partial trace is CPTP. A POVM $\{M_a\}$ defines the classical-outcome channel

$$\Phi_M(\rho) = \sum_a \text{Tr}(M_a \rho) |a\rangle\langle a|.$$

This map is trace preserving because $\sum_a M_a = I$. It is completely positive because each positive functional $Z \mapsto \text{Tr}(M_a Z)$ is completely positive and the classical flags $|a\rangle\langle a|$ occupy orthogonal output blocks. Applying the first part proves both stated monotonicity consequences. $\square$

Worked measurement check. For $\rho = |0\rangle\langle 0|$ and $\sigma = |+\rangle\langle +|$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, the pure-state formula derived below gives $D(\rho, \sigma) = 1/\sqrt{2}$. Any fixed measurement can only reduce this distance. Measuring in the computational basis produces distributions $(1, 0)$ and $(1/2, 1/2)$, whose total-variation distance is $1/2 < 1/\sqrt{2}$; the computational-basis measurement is therefore not the optimal discriminator.

Theorem 1.5 (Fuchs–van de Graaf inequalities). *For all states $\rho, \sigma \in \mathcal{D}(X)$,*

$$1 - F(\rho, \sigma) \leq D(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)^2}.$$

If both states are pure, say $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\varphi\rangle\langle\varphi|$, then

$$D(\rho, \sigma) = \sqrt{1 - |\langle\psi|\varphi\rangle|^2}, \quad F(\rho, \sigma) = |\langle\psi|\varphi\rangle|.$$

Thus the upper Fuchs–van de Graaf inequality is an equality for every pure-state pair; the lower inequality is an equality for pure states precisely at the identical-ray and orthogonal endpoints.

Proof. We first prove the lower bound. Put $A = \sqrt{\rho} - \sqrt{\sigma}$ and write its Jordan decomposition $A = Q_0 - Q_1$ with support projections Π_0, Π_1 . The Hermitian trace-norm variational formula implies

$$\|\rho - \sigma\|_1 \geq \text{Tr}((\Pi_0 - \Pi_1)(\rho - \sigma)).$$

Using $\rho - \sigma = \frac{1}{2}[A(\sqrt{\rho} + \sqrt{\sigma}) + (\sqrt{\rho} + \sqrt{\sigma})A]$ and cyclicity of trace, the right side equals

$$\text{Tr}((Q_0 + Q_1)(\sqrt{\rho} + \sqrt{\sigma})).$$

All four factors $Q_0, Q_1, \sqrt{\rho}, \sqrt{\sigma}$ are positive semidefinite, so $\text{Tr}(RS) \geq 0$ for every two of them. Hence

$$\begin{aligned} \text{Tr}((Q_0 + Q_1)(\sqrt{\rho} + \sqrt{\sigma})) &\geq \text{Tr}((Q_0 - Q_1)(\sqrt{\rho} - \sqrt{\sigma})) \\ &= \text{Tr } A^2 = \|\sqrt{\rho} - \sqrt{\sigma}\|_2^2. \end{aligned}$$

Finally

$$\|\sqrt{\rho} - \sqrt{\sigma}\|_2^2 = 2 - 2\text{Tr}(\sqrt{\rho}\sqrt{\sigma}) \geq 2 - 2F(\rho, \sigma),$$

because $\text{Tr}(\sqrt{\rho}\sqrt{\sigma})$ is real nonnegative and no larger than $\|\sqrt{\rho}\sqrt{\sigma}\|_1$. Dividing by two gives $D(\rho, \sigma) \geq 1 - F(\rho, \sigma)$.

For the upper bound, use the purification-overlap calculation after Theorem 1.3 to choose purifications $|u\rangle, |v\rangle$ of ρ, σ with $|\langle u|v\rangle| = F(\rho, \sigma)$. For two unit vectors, the Hermitian operator $|u\rangle\langle u| - |v\rangle\langle v|$ has its nonzero spectrum in their two-dimensional span and has eigenvalues $\pm\sqrt{1 - |\langle u|v\rangle|^2}$. Therefore

$$D(|u\rangle\langle u|, |v\rangle\langle v|) = \sqrt{1 - F(\rho, \sigma)^2}.$$

Partial trace maps these purifications to ρ and σ , so Theorem 1.4 gives the desired upper bound.

Taking $u = \psi$ and $v = \varphi$ proves the displayed pure-state formula. The endpoint equality claims follow by solving $1 - c = \sqrt{1 - c^2}$ for $0 \leq c \leq 1$ and observing that the upper inequality was already exact. $\square$

Normalization check. Some literature defines “fidelity” as $F(\rho, \sigma)^2$. Under that convention the displayed upper bound is written $D \leq \sqrt{1 - F_{\text{sq}}}$. Volume XV never uses that convention without an explicit subscript; this prevents a square from silently moving between complexity parameters later.

Theorem 1.6 (No cloning for nonorthogonal pure states). *Let $|0_B\rangle$ be a fixed blank state. There is no quantum channel $\mathcal{C}$ satisfying*

$$\mathcal{C}(|\psi\rangle\langle\psi| \otimes |0_B\rangle\langle 0_B|) = |\psi\psi\rangle\langle\psi\psi|$$

and the analogous equation for a second pure state $|\varphi\rangle$ whenever $0 < |\langle\psi|\varphi\rangle| < 1$. Orthogonal states are not excluded by this theorem, and a single known state can of course be prepared repeatedly.

Proof. Put $c = |\langle\psi|\varphi\rangle|$. Tensoring with the same blank state does not change the pure state overlap, so Theorem 1.5 gives input trace distance

$$D_{\text{in}} = \sqrt{1 - c^2}.$$

If $\mathcal{C}$ cloned both states, the two outputs would have overlap c^2 and therefore trace distance

$$D_{\text{out}} = \sqrt{1 - c^4}.$$

For $0 < c < 1$ one has $1 - c^4 > 1 - c^2$, hence $D_{\text{out}} > D_{\text{in}}$, contradicting trace-distance contraction under the channel $\mathcal{C}$ from Theorem 1.4.

If $c = 0$, the two states are perfectly distinguishable and a channel can measure in a basis containing them and prepare two copies of the identified state. If $c = 1$, the two vectors describe the same pure state up to global phase, so the two cloning requirements are identical. These are exactly the excluded endpoints. $\square$

Source-owned mixed-state extension. Wootters–Zurek (1982) is the historical pure-state no-cloning source. The stronger no-broadcasting statement for mixed states is due to Barnum–Caves–Fuchs–Jozsa–Schumacher (1996): a finite family of density operators is broadcastable exactly when its members commute pairwise. That stronger result is recorded here as context, not counted as a locally proved theorem in this chapter.

1.3 Entropy, accessible information, and the Holevo bound

Entropy conventions. All logarithms in this section are base two. For a density operator ρ , its *von Neumann entropy* is

$$H(\rho) := -\text{Tr}(\rho \log \rho),$$

with the usual convention $0 \log 0 = 0$. For density operators ρ, σ , their *quantum relative entropy* is

$$D_{\text{q}}(\rho \parallel \sigma) := \text{Tr}(\rho(\log \rho - \log \sigma))$$

when $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$, and is $+\infty$ otherwise. For a bipartite state ω_{AB} , the quantum mutual information is

$$I(A : B)_{\omega} := D_{\text{q}}(\omega_{AB} \parallel \omega_A \otimes \omega_B) = H(\omega_A) + H(\omega_B) - H(\omega_{AB}).$$

For an ordinary finite joint distribution $q(x, y)$, $I(X : Y)$ denotes the classical mutual information, equivalently the classical relative entropy of q from $q_X q_Y$.

An *ensemble* is a finite family $\mathcal{E} = \{p_x, \rho_x\}_{x \in \mathcal{X}}$, where p is a probability distribution and each ρ_x is a state. Its average state is $\bar{\rho} = \sum_x p_x \rho_x$. Introduce a classical register X with distinguished basis $\{|x\rangle\}$ and the classical–quantum state

$$\omega_{XQ} = \sum_x p_x |x\rangle\langle x| \otimes \rho_x.$$

The *Holevo quantity* is

$$\chi(\mathcal{E}) := I(X : Q)_\omega = H(\bar{\rho}) - \sum_x p_x H(\rho_x).$$

If a POVM $\{M_y\}_{y \in \mathcal{Y}}$ is applied to Q , the resulting joint distribution is $q(x, y) = p_x \text{Tr}(M_y \rho_x)$. The *accessible information* $I_{\text{acc}}(\mathcal{E})$ is the supremum of $I(X : Y)$ over all finite POVMs.

Source-owned entropy input. We use the finite-dimensional monotonicity of quantum relative entropy: for every quantum channel Φ and positive states ρ, σ ,

$$D_q(\Phi(\rho) \| \Phi(\sigma)) \leq D_q(\rho \| \sigma).$$

This is the exact data-processing theorem used in Watrous, *The Theory of Quantum Information*, Theorem 5.35. Its proof rests on the joint convexity of quantum relative entropy and is deliberately source-owned here; it is not replaced by the classical data-processing theorem from Volume I.

Theorem 1.7 (Holevo information bound). *Let $\mathcal{E} = \{p_x, \rho_x\}_{x \in \mathcal{X}}$ be a finite ensemble of states on Q , and let $\{M_y\}_{y \in \mathcal{Y}}$ be any finite POVM. If $q(x, y) = p_x \text{Tr}(M_y \rho_x)$, then*

$$I(X : Y)_q \leq \chi(\mathcal{E}) = H\left(\sum_x p_x \rho_x\right) - \sum_x p_x H(\rho_x).$$

Consequently

$$I_{\text{acc}}(\mathcal{E}) \leq \chi(\mathcal{E}) \leq \log \dim Q.$$

Equivalently, if $\omega_{XQ} = \sum_x p_x |x\rangle\langle x| \otimes \rho_x$, every measurement channel $\Phi_M : Q \rightarrow Y$ satisfies

$$I(X : Y)_{(I_X \otimes \Phi_M)(\omega)} \leq I(X : Q)_\omega.$$

Proof. The measurement defines the classical-output channel

$$\Phi_M(\tau) = \sum_y \text{Tr}(M_y \tau) |y\rangle\langle y|.$$

Put $\omega_X = \sum_x p_x |x\rangle\langle x|$ and $\omega_Q = \bar{\rho}$. By definition of quantum mutual information,

$$I(X : Q)_\omega = D_q(\omega_{XQ} \| \omega_X \otimes \omega_Q).$$

Apply the source-owned quantum relative-entropy data-processing theorem to the channel $I_X \otimes \Phi_M$. This gives

$$\begin{aligned} I(X : Q)_\omega &\geq D_q((I_X \otimes \Phi_M)(\omega_{XQ}) \| (I_X \otimes \Phi_M)(\omega_X \otimes \omega_Q)) \\ &= I(X : Y)_q, \end{aligned}$$

because both output operators are diagonal in the classical basis and quantum relative entropy therefore reduces to ordinary relative entropy of the corresponding probability distributions. Taking the supremum over measurements proves $I_{\text{acc}}(\mathcal{E}) \leq \chi(\mathcal{E})$.

It remains only to justify the entropy expression and the dimension bound. The cq state is block diagonal, so its eigenvalues are the numbers $p_x \lambda_{x,j}$, where $\{\lambda_{x,j}\}_j$ are the eigenvalues of ρ_x . Hence

$$H(\omega_{XQ}) = H(p) + \sum_x p_x H(\rho_x), \quad H(\omega_X) = H(p), \quad H(\omega_Q) = H(\bar{\rho}),$$

which yields the displayed formula for χ . Nonnegativity of entropy and the elementary bound $H(\bar{\rho}) \leq \log \dim Q$ then imply $\chi(\mathcal{E}) \leq \log \dim Q$. $\square$

Worked two-state ensemble. Take two equiprobable pure qubit states $|0\rangle$ and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Their average state has eigenvalues

$$\lambda_{\pm} = \frac{1 \pm 2^{-1/2}}{2},$$

so, because the signal states are pure,

$$\chi = h_2(\lambda_+) \approx 0.6009 \text{ bits.}$$

A computational-basis measurement gives output distribution $(3/4, 1/4)$ and conditional entropy $H(Y|X) = 1/2$, hence

$$I(X : Y) = h_2(1/4) - \frac{1}{2} \approx 0.3113 < \chi.$$

Thus the theorem is a genuine information restriction, not merely the trivial one-bit alphabet bound.

1.4 Synthesis and transition

The chapter has established a compact chain of interfaces that will be reused throughout Volume XV. Density operators and CPTP maps replace classical point states and stochastic maps; partial trace formalizes forgetting a subsystem; trace distance measures operational distinguishability; fidelity provides a purification-friendly overlap quantity; and Fuchs–van de Graaf lets us move between those two metrics without losing track of normalization. No cloning is then not a separate physical slogan but an immediate information-theoretic consequence of contractivity. Finally, the Holevo bound explains a different limitation: a measurement may convert a quantum ensemble into classical data, but it cannot increase the mutual information already present in the cq state.

The next chapter must add a *uniform circuit* model before complexity classes are mentioned. In particular, an arbitrary real or complex matrix entry will never count as a free gate description: finite gate sets, approximation precision, ancillas, measurement, and classical control all need explicit complexity accounting.

1.5 Reader checkpoint and exercises

1. Compute both reduced density operators of $(\sqrt{3}|00\rangle + |11\rangle)/2$ and identify its Schmidt coefficients.
2. Let $\rho = \text{diag}(3/4, 1/4)$ and $\sigma = \text{diag}(1/4, 3/4)$. Compute $D(\rho, \sigma)$ and $F(\rho, \sigma)$ exactly and verify both Fuchs–van de Graaf inequalities.
3. Prove directly from the POVM axioms that $a \mapsto \text{Tr}(M_a \rho)$ is a probability distribution. Identify where positivity and the normalization $\sum_a M_a = I$ are used.

4. Show from the defining duality of partial trace that $\text{Tr}_Y(A \otimes B) = A \text{Tr } B$.
5. For two pure states with overlap c , derive the two nonzero eigenvalues of $|\psi\rangle\langle\psi| - |\varphi\rangle\langle\varphi|$ explicitly in a basis of their span and recover $D = \sqrt{1 - |c|^2}$.
6. Give an explicit channel that clones two fixed orthogonal basis states. Explain why the same construction does not clone an arbitrary superposition of them.
7. Let Φ be the completely depolarizing qubit channel $\Phi(\rho) = I_2/2$. Check Theorem 1.4 on two orthogonal pure inputs and interpret the strict loss of distinguishability.
8. Explain why a POVM does not determine a unique postmeasurement state. Give two different physical implementations of the same computational-basis POVM with different postmeasurement behavior on the measured system.
9. For an ensemble of mutually orthogonal pure states with probabilities p_x , show directly that $\chi = H(p)$ and exhibit a measurement attaining this value.
10. Recompute the worked $\{|0\rangle, |+\rangle\}$ ensemble for the Hadamard-basis measurement and compare its mutual information with the computational-basis value and the Holevo upper bound.

Chapter 2

Uniform Quantum Circuits and Approximation

2.1 How to read this chapter

Quantum algorithms become complexity-theoretic objects only after their finite descriptions and approximation costs are fixed. This chapter therefore separates three levels that are often blurred: the abstract unitary implemented by a circuit, the finite gate alphabet used to describe the circuit, and the classical algorithm that generates the n -input circuit. After fixing the model and the basic error ledger, this chapter proves exact continuous universality, separates qualitative finite-set density from quantitative compilation, source-owns the Solovay–Kitaev recursion at its exact hypotheses, and proves deferred measurement for the extended circuit model.

2.2 Learning goals

After this chapter’s foundation section, the reader should be able to distinguish circuit size from depth, uniform from nonuniform circuit families, exact gates from approximated target unitaries, and unitary circuits from circuits with intermediate measurements and classical control. The reader should also be able to propagate operator-norm gate errors through an entire circuit, reconstruct the two-level-unitary universality proof, state precisely what density does and does not imply, use the Solovay–Kitaev polylogarithmic precision bound, and prove deferred measurement by a coherent-record invariant.

2.3 Uniform circuit families

Definition 2.1 (Uniform quantum circuit families). A k -qubit gate is a unitary operator on $(\mathbb{C}^2)^{\otimes k}$; throughout this volume an *elementary gate* acts on one or two qubits unless another arity is explicitly declared. A *finite gate set* $\mathcal{G}$ is a fixed finite collection of such gates. Because $\mathcal{G}$ is fixed once and for all, its matrix entries are part of the computational model and are not free advice supplied with an input.

A quantum circuit C on m wires is a finite acyclic sequence of gate applications, each naming a gate and the wire or pair of wires on which it acts. Its *size* $|C|$ is the number of gate applications. Its *depth* is the minimum number of parallel layers in which gate applications on disjoint wire sets may be scheduled. An n -qubit input may be augmented by $a(n)$ ancilla qubits initialized to $|0\rangle$; unless stated otherwise, ancillas are counted among circuit wires but not among input bits.

A family $\{C_n\}_{n \geq 1}$ is *polynomial-time uniform* if a deterministic Turing machine, on input 1^n , outputs the complete gate-and-wire description of C_n in time polynomial in n . A polynomial-size family additionally satisfies $|C_n| \leq p(n)$ for some polynomial p . When the output is classical, designated qubits are measured in the computational basis at the end.

For an operator A , write

$$\|A\|_{\text{op}} := \sup_{\|v\|_2=1} \|Av\|_2.$$

A circuit unitary V ε -approximates a target unitary U when $\|U - V\|_{\text{op}} \leq \varepsilon$. Approximation precision is never implicit: if ε depends on n , the cost of producing the approximating gate sequence is part of the circuit complexity accounting.

An *extended quantum circuit* may additionally perform computational-basis measurements during the computation, store their outcomes as classical bits, and condition later gate applications on previous outcome bits. Such a measurement record and the corresponding classical control are separate resources from a unitary gate. The deferred-measurement theorem later in this chapter will state precisely when they can be replaced by coherent ancillas and final measurements.

Why the finite-description clause matters. If an arbitrary one-qubit rotation angle of infinite precision were allowed as one free input-specific gate, infinitely many classical bits could be hidden in a single matrix entry. Uniform quantum complexity therefore fixes a finite gate alphabet (or an explicitly equivalent finitely describable model) and charges for approximation.

Operator-norm composition ledger. Let $U_1, \dots, U_m$ and $V_1, \dots, V_m$ be unitaries on the same space, with $\|U_j - V_j\|_{\text{op}} \leq \varepsilon_j$. The telescoping identity

$$U_m \cdots U_1 - V_m \cdots V_1 = \sum_{j=1}^m U_m \cdots U_{j+1} (U_j - V_j) V_{j-1} \cdots V_1$$

and $\|U\|_{\text{op}} = 1$ for every unitary U imply

$$\|U_m \cdots U_1 - V_m \cdots V_1\|_{\text{op}} \leq \sum_{j=1}^m \varepsilon_j.$$

Thus a circuit of m target gates is approximated within total error ε whenever each gate is compiled to operator-norm accuracy at most ε/m . This elementary estimate is the error budget used by the forthcoming finite-gate universality and Solovay–Kitaev statements.

2.4 Exact and approximate universality

The first issue is algebraic rather than asymptotic: can one- and two-qubit gates generate an arbitrary finite-dimensional unitary at all? The answer is yes for arbitrary one-qubit gates together with CNOT. The proof below is constructive and deliberately allows clean ancillas, because later complexity applications care about polynomial overhead and exact restoration of workspace rather than ancilla-free gate-count optimality.

For $d \geq 1$, $U(d)$ denotes the group of $d \times d$ unitary matrices and $SU(d) := \{U \in U(d) : \det U = 1\}$. A finite set $\mathcal{G}_1 \subseteq SU(2)$ is *inverse-closed* when $G^* \in \mathcal{G}_1$ for every $G \in \mathcal{G}_1$. Its generated subgroup is *dense in $SU(2)$* when every element of $SU(2)$ is a limit, in operator norm, of finite words over $\mathcal{G}_1$. These are topological/algebraic conditions; density by itself does not give an efficient word-length bound.

Theorem 2.2 (Universal gate simulation). *Let $\mathcal{U}_{\text{cont}}$ consist of all one-qubit unitaries together with CNOT. For every $n \geq 1$ and every $U \in U(2^n)$, there is a circuit over $\mathcal{U}_{\text{cont}}$, using $O(n)$ clean ancillas that are returned to $|0\rangle$, whose action on the input register is exactly U . One may choose the circuit to have $O(n4^n)$ elementary gates.*

Now fix a finite inverse-closed set $\mathcal{G}_1 \subset SU(2)$ whose generated subgroup is dense in $SU(2)$, and put $\mathcal{G} = \mathcal{G}_1 \cup \{\text{CNOT}\}$. For unitaries on the same space define the phase-insensitive operator distance

$$d_o(U, V) := \min_{|\lambda|=1} \|U - \lambda V\|_{\text{op}}.$$

If an exact $\mathcal{U}_{\text{cont}}$ -circuit for U contains M one-qubit gates, then for every $\varepsilon > 0$ there exists a $\mathcal{G}$ -circuit V with

$$d_o(U, V) \leq \varepsilon.$$

More precisely, it is enough to approximate every one-qubit gate, after removing its scalar phase, within operator-norm error ε/M . This second assertion is qualitative: density alone supplies no efficient bound on the length of the approximating words. The quantitative polylogarithmic bound is Theorem 2.3.

Proof. Write $N = 2^n$. We first record three exact synthesis facts.

One controlled one-qubit gate. Every $W \in U(2)$ has an Euler form

$$W = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta).$$

Set

$$A = R_z(\beta) R_y(\gamma/2), \quad B = R_y(-\gamma/2) R_z(-(\delta + \beta)/2), \quad C = R_z((\delta - \beta)/2).$$

Using $XR_y(\theta)X = R_y(-\theta)$ and $XR_z(\theta)X = R_z(-\theta)$, direct multiplication gives

$$ABC = I, \quad AXBXC = R_z(\beta) R_y(\gamma) R_z(\delta).$$

Thus the temporal sequence $C, \text{CNOT}, B, \text{CNOT}, A$, together with the one-qubit phase $\text{diag}(1, e^{i\alpha})$ on the control, implements controlled- W exactly.

Many controls. Toffoli is exactly synthesizable from one-qubit gates and CNOT. One convenient verification is the Boolean phase identity

$$4abc = a + b + c - (a \oplus b) - (a \oplus c) - (b \oplus c) + (a \oplus b \oplus c) \pmod{8}$$

for $a, b, c \in \{0, 1\}$. Compute the three pairwise parities and the triple parity into clean ancillas using CNOTs. Apply $T = \text{diag}(1, e^{i\pi/4})$ to each of the three data bits and to the triple-parity ancilla, apply T^* to each pair-parity ancilla, and then uncompute the parities. The accumulated phase is $(-1)^{abc}$, hence this is controlled-controlled- Z ; conjugating the target by H gives Toffoli. Consequently the AND of m control bits can be computed into $O(m)$ clean ancillas by a Toffoli chain, a one-control construction can be applied to the resulting flag, and the chain can be uncomputed. Hence every multiply controlled one-qubit unitary has an exact $O(m)$ -size implementation over $\mathcal{U}_{\text{cont}}$ with $O(m)$ clean ancillas. Controls on the value zero are converted to controls on one by surrounding X gates.

Two-level unitaries. A two-level unitary is the identity on all computational basis vectors except two, say $|x\rangle$ and $|y\rangle$. Translate by x using at most n one-qubit X gates, so the distinguished pair becomes 0^n and $d = x \oplus y \neq 0$. Choose a coordinate j with $d_j = 1$. For every other coordinate k with $d_k = 1$, apply CNOT from j to k . This invertible linear basis permutation maps d to e_j . The desired two-level action has therefore become one arbitrary one-qubit unitary on wire j , conditioned on every other wire being zero. The preceding multi-control construction implements it exactly.

Undo the CNOTs and the translation. Thus every two-level unitary costs $O(n)$ elementary gates and $O(n)$ reusable clean ancillas.

We can now synthesize an arbitrary U . Gaussian elimination with complex Givens rotations gives two-level unitaries $G_1, \dots, G_t$, with $t \leq N(N-1)/2$, such that

$$G_t \cdots G_1 U = D$$

for a diagonal unitary D . Hence $U = G_1^* \cdots G_t^* D$. Each G_j^* is handled by the two-level construction. Finally write $D = \text{diag}(e^{i\theta_x})_{x \in \{0,1\}^n}$. For each basis string x , compute the predicate “the input equals x ” into a clean flag using X gates for negative literals and a Toffoli AND chain, apply $\text{diag}(1, e^{i\theta_x})$ to the flag, and uncompute. Multiplying these basis-state phases realizes D . The Givens portion dominates the cost, giving $O(nN^2) = O(n4^n)$ elementary gates; the same $O(n)$ ancillas are recycled and finish in $|0\rangle$.

For the finite-set assertion, decompose U by the exact construction and write every one-qubit factor as $e^{i\alpha_j} S_j$ with $S_j \in SU(2)$. The scalar phases multiply to one overall phase on the entire circuit and therefore disappear in $d_\circ$. Density of $\langle \mathcal{G}_1 \rangle$ gives, for every j , a word V_j with $\|S_j - V_j\|_{\text{op}} \leq \varepsilon/M$. Keep every CNOT exact. The operator-norm composition ledger from Definition 2.1 then gives total phase-insensitive error at most $M(\varepsilon/M) = \varepsilon$. $\square$

Worked two-level synthesis. Suppose a three-qubit two-level unitary acts only on $|000\rangle$ and $|101\rangle$. Here $d = 101$. CNOT from the first wire to the third sends 101 to 100. The problem is now an arbitrary unitary on the first wire conditioned on wires two and three both being zero. Surround those two controls by X gates, use the exact two-control construction above, and reverse the basis change. No other computational basis vector is altered.

Source checkpoint. Barenco et al. prove the same continuous universality statement for arbitrary one-qubit gates plus XOR/CNOT. The proof above internalizes the required synthesis route rather than using that theorem as an unexpanded black box. The finite-set statement is intentionally conditional on a stated density hypothesis; no quantitative word-length estimate is smuggled into the word “dense.”

Theorem 2.3 (Solovay–Kitaev approximation theorem). *Fix a finite inverse-closed gate set $\mathcal{G}_1 \subset SU(2)$ whose generated subgroup is dense in $SU(2)$. There exist constants $C > 0$, $c > 0$, and $\varepsilon_0 > 0$, depending only on $\mathcal{G}_1$, and a classical compilation algorithm such that for every $U \in SU(2)$ and $0 < \varepsilon < \varepsilon_0$, the algorithm outputs a word W over $\mathcal{G}_1$ satisfying*

$$\|U - W\|_{\text{op}} \leq \varepsilon, \quad |W| \leq C \log^c(1/\varepsilon).$$

The classical compilation work is polynomial in $\log(1/\varepsilon)$, apart from the unavoidable cost of emitting the output word. In the Dawson–Nielsen recursion one may take the sequence-length exponent

$$c = \frac{\log 5}{\log(3/2)} \approx 3.97.$$

Consequently, if an exact circuit contains M one-qubit gates, then compiling each such gate to accuracy ε/M gives an overall approximation using

$$O(M \log^c(M/\varepsilon))$$

finite-set one-qubit gates, in addition to the exact CNOTs.

Source-owned proof architecture. The quantitative recursion is retained as a deep input. At a fixed coarse scale, density provides a constant lookup net because $\mathcal{G}_1$ is fixed. A target near the identity is expressed as a balanced group commutator VWV^*W^* with V and W closer to the identity by a square-root scale. Recursive approximations to V and W improve the residual error from scale δ to order $\delta^{3/2}$, while the word-length recurrence multiplies by five. Solving the two recurrences yields $c = \log 5 / \log(3/2)$. Dawson–Nielsen give the explicit pedagogical algorithm and the quoted 3.97 sequence-length exponent. The final circuit-level bound in the theorem is local: apply this one-qubit statement with error ε/M and then invoke the composition ledger.

Why this theorem is separate from universality. Theorem 2.2 says approximation words exist because a subgroup is dense. Solovay–Kitaev says that such words can be found with only polylogarithmic dependence on precision. This distinction is exactly what prevents a complexity argument from treating arbitrary real-valued rotations as free gates.

2.5 Deferred measurement

Theorem 2.4 (Deferred measurement principle). *Consider an extended circuit in the sense of Definition 2.1: every intermediate measurement is in the computational basis, its outcome is stored as a classical bit, and later one- or two-qubit unitary gates may be conditioned on previously stored outcome bits. There is a circuit with no intermediate measurements that uses one fresh record qubit for each measured bit, replaces every classical control by coherent quantum control, and has exactly the same joint probability distribution for all recorded measurement bits and all designated final computational-basis output bits when the record qubits are measured at the end.*

If the original unitary gates come from a fixed finite library and each classical control depends on a single recorded bit, the transformed circuit has linear size overhead and one fresh qubit per intermediate measurement when expressed over the exact continuous library of Theorem 2.2; the constant depends only on the fixed library. If a fixed approximately universal finite gate set is required instead, Theorem 2.3 compiles this continuous circuit to any prescribed total error with polylogarithmic precision overhead. Polynomial-size classical control logic can likewise be made reversible and simulated with polynomial overhead.

Proof. Introduce one fresh record qubit r_j , initialized to $|0\rangle$, for the j -th intermediate measurement. Instead of measuring a data qubit q in the computational basis, apply $\text{CNOT}_{q \rightarrow r_j}$ and leave r_j unmeasured. Whenever the original circuit applies a unitary W conditioned on the stored bit being one, apply the coherent controlled- W with r_j as control. Theorem 2.2 exactly decomposes each fixed controlled gate into the elementary continuous library; because its arity is at most three, a fixed finite gate library incurs only a constant precompilation cost per such control.

We prove equality of outcome statistics by maintaining the diagonal blocks of the record register. After some measurements, let R denote the coherent record and write the transformed density operator as

$$\tilde{\rho}_{RQ} = \sum_{z,z'} |z\rangle\langle z'|_R \otimes X_{z,z'}.$$

The invariant is that the diagonal block $X_{z,z}$ equals the subnormalized branch state $p_z \rho_z$ of the original measured circuit conditioned on classical record z . Before any measurement this is tautological.

An unconditional unitary sends every diagonal block to $UX_{z,z}U^*$, exactly as in every classical

branch. A classically controlled unitary has a branch-dependent unitary U_z ; its coherent replacement

$$\sum_z |z\rangle\langle z|_R \otimes U_z$$

sends the diagonal block to $U_z X_{z,z} U_z^*$, again preserving the invariant.

Now suppose the next measured data qubit has projectors $P_b = |b\rangle\langle b|$ for $b \in \{0, 1\}$. In the measured circuit, branch z splits into the two subnormalized states

$$P_b X_{z,z} P_b, \quad b \in \{0, 1\}.$$

In the coherent circuit, append $|0\rangle$ and apply CNOT from the measured qubit to the fresh record qubit. The diagonal block indexed by the enlarged record (z, b) is exactly $P_b X_{z,z} P_b$. Hence the invariant survives every intermediate measurement as well.

At the end, probabilities obtained by measuring the record register and the designated output qubits depend only on these diagonal blocks. They therefore agree exactly with the branch probabilities of the original extended circuit. This proves semantic equivalence for the claimed outputs.

The resource bounds are direct. Each measurement contributes one record qubit and one CNOT. For a fixed finite library of target unitaries there are only finitely many one-record controlled versions, each acting on at most three qubits, so Theorem 2.2 gives a constant-size exact decomposition over the continuous one-qubit-plus-CNOT library that may be fixed once and for all. If the final circuit must use an approximately universal finite alphabet, Theorem 2.3 and the operator-norm ledger compile the resulting continuous circuit to any chosen total error. If a control predicate is specified by a polynomial-size classical Boolean circuit, standard reversible evaluation using Toffoli and clean workspace computes the predicate coherently, applies the controlled unitary, and uncomputes; the preceding Toffoli construction shows polynomial overhead in the present gate model. $\square$

Worked deferral example. Prepare $|+\rangle_q |0\rangle_t$, measure q , and apply X to t iff the outcome is one. The measured circuit produces $(0, 0)$ and $(1, 1)$, each with probability $1/2$. In the deferred circuit, add a record qubit r , apply CNOT $q \rightarrow r$, then CNOT $r \rightarrow t$. The state is

$$\frac{|000\rangle + |111\rangle}{\sqrt{2}},$$

so measuring r and t at the end gives exactly the same two outcomes and probabilities. The coherence is retained globally, but it cannot change the final classical branch statistics because the record values occupy orthogonal subspaces.

Source checkpoint. The informal deferred-measurement principle is standard in quantum-computation texts. Gurevich and Blass give a precise syntax/semantics treatment for circuits with measurement and classical channels and prove a general deferral theorem. The statement above is intentionally narrower: computational- basis measurements and classical control of later unitaries. Its coherent-record proof is fully internalized here.

2.6 Synthesis and transition

Chapter 2 now separates three logically different facts. Continuous exact universality is an algebraic decomposition theorem. Finite-set universality follows qualitatively from density, while Solovay–Kitaev supplies the quantitative polylogarithmic precision cost. Deferred measurement then shows

that the uniform complexity model may, for classical output statistics, postpone intermediate computational-basis measurements without changing the computation. Chapter 3 can therefore define the quantum query model and analyze phase estimation, amplitude amplification, Simon's algorithm, and query lower bounds without ambiguity about gate precision or intermediate measurements.

2.7 Reader checkpoint and exercises

1. Give an example of a depth-2 circuit having size greater than 2, and explain why size and depth measure different resources.
2. Prove the operator-norm composition estimate directly by induction on the number of gates.
3. Verify the identity $ABC = I$ and $AXBXC = R_z(\beta)R_y(\gamma)R_z(\delta)$ used in the controlled-one-qubit construction.
4. Check the Boolean phase identity for controlled-controlled- Z on all eight triples $(a, b, c) \in \{0, 1\}^3$.
5. For the basis strings $x = 0110$ and $y = 1101$, explicitly give a sequence of X and CNOT basis permutations that sends the pair to $0^4, e_j$ for some j .
6. Suppose an exact circuit has $M = n^3$ one-qubit gates and final operator-norm error at most 2^{-n} is required. What per-gate accuracy suffices, and what Solovay–Kitaev gate-count bound follows in terms of n ?
7. Explain why density of a gate-generated subgroup by itself does not imply a polynomial-time quantum circuit model. Which extra quantitative statement does Solovay–Kitaev supply?
8. Reprove deferred measurement for one computational-basis measurement followed by two different classically controlled gates, using the diagonal-block invariant from the theorem.
9. Give an example of a polynomial-size circuit family that is not polynomial-time uniform, and identify exactly where nonuniform information enters its descriptions.
10. Why does the finite-set approximation theorem use phase-insensitive distance? Explain why a scalar phase on a one-qubit gate becomes only a scalar phase on the entire multi-qubit circuit.

Chapter 3

Quantum Queries, Search, and Hidden Structure

3.1 How to read this chapter

Quantum speedups are cleanest when the expensive information source is isolated as an oracle. The query model deliberately ignores the cost of oracle-independent unitary work and counts only uses of that oracle. This makes it possible to prove both upper and lower bounds in the same model, but it also makes model discipline essential: a standard reversible function oracle, a phase oracle, and a controlled oracle are related interfaces, not automatically identical black boxes.

The chapter first fixes those interfaces. It then develops four basic patterns. Phase estimation turns an eigenphase into a Fourier sample and records exactly what controlled powers cost. Grover search and amplitude amplification reduce an algorithm to a two-dimensional rotation. A hybrid argument proves that the resulting square-root speedup for unstructured search is optimal in the bounded-error query model. Finally, Simon's algorithm shows a qualitatively different phenomenon: interference reveals a hidden linear constraint and produces an exponential oracle separation.

3.2 Learning goals

After this chapter the reader should be able to define bounded-error quantum query complexity without smuggling oracle-dependent information into the inter-query unitaries; convert a Boolean bit oracle to a phase oracle; state when controlled-oracle access is an additional resource; derive the exact phase-estimation output distribution; prove the Grover rotation formula; reconstruct the hybrid $\Omega(\sqrt{N})$ lower bound; and derive both the quantum and classical query bounds for Simon's hidden-XOR promise problem.

3.3 The quantum query model

Definition 3.1 (Quantum query model). Let $n, m \geq 1$, let $X = \{0, 1\}^n$, and let $f : X \rightarrow \{0, 1\}^m$. The *standard reversible function oracle* for f is the unitary

$$O_f |x\rangle|z\rangle = |x\rangle|z \oplus f(x)\rangle, \quad x \in X, z \in \{0, 1\}^m,$$

where $\oplus$ denotes bitwise addition modulo two. A use of O_f or $O_f^* = O_f$ is one *standard query*. The first register is the *query register* and the second is the *answer register*.

For a Boolean function $\chi : X \rightarrow \{0, 1\}$, its *phase oracle* is

$$P_\chi|x\rangle = (-1)^{\chi(x)}|x\rangle.$$

One standard query implements one phase query: prepare an answer qubit in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ and use phase kickback,

$$O_\chi(|x\rangle|-\rangle) = (-1)^{\chi(x)}|x\rangle|-\rangle.$$

A *controlled standard oracle* is the separately specified unitary

$$CO_f|c\rangle|x\rangle|z\rangle = |c\rangle|x\rangle|z \oplus cf(x)\rangle, \quad c \in \{0, 1\},$$

where $cf(x)$ is 0^m for $c = 0$ and $f(x)$ for $c = 1$. Controlled access to an arbitrary unknown unitary is not free by definition. For a classical function oracle O_f , however, CO_f can be simulated with two standard queries: compute $f(x)$ into a clean m -qubit ancilla, add it into z controlled by c , and query O_f again to uncompute the ancilla.

A T -query quantum algorithm for f is a circuit of the form

$$U_T O_f U_{T-1} O_f \cdots O_f U_0,$$

possibly with additional ancillas and a final measurement, where every U_j is independent of the particular oracle f . If another oracle interface is being counted, such as P_χ or CO_f , that interface is stated explicitly. *Query complexity* counts only oracle calls; ordinary gate complexity, workspace, and the cost of synthesizing the U_j are separate resources.

A *promise decision problem* is a disjoint pair $(\mathcal{F}_{\text{yes}}, \mathcal{F}_{\text{no}})$ of allowed oracles. For $0 \leq \delta < 1/2$, its quantum query complexity Q_δ is the least T for which a T -query algorithm outputs the correct yes/no answer with probability at least $1 - \delta$ for every oracle in the promise. The convention $Q := Q_{1/3}$ is called *bounded-error quantum query complexity*. For a function or relation problem, the same notation means that the required output is produced with probability at least $1 - \delta$ on every promised oracle. The case $\delta = 0$ is *exact query complexity*.

Model checkpoint. The standard and phase oracles are interchangeable in the direction needed by Grover search because one standard Boolean query gives one phase query. The reverse direction requires a controlled-phase interface and is therefore not silently assumed. Lower bounds below are proved directly for the standard reversible oracle, so no oracle-equivalence shortcut is needed.

3.4 Phase estimation

For real numbers a, b , define their distance modulo one by

$$\|a - b\|_{\mathbb{T}} := \min_{k \in \mathbb{Z}} |a - b - k|.$$

For $M \geq 1$, the quantum Fourier transform over $\mathbb{Z}_M$ is the unitary

$$F_M|k\rangle = \frac{1}{\sqrt{M}} \sum_{y=0}^{M-1} e^{2\pi i ky/M} |y\rangle.$$

When $M = 2^t$, the usual binary factorization of the Fourier kernel gives an exact circuit over arbitrary one-qubit phase rotations and CNOT with $O(t^2)$ elementary gates, including the final bit reversal. Chapter 2 explains how such a continuous-gate circuit is compiled to a fixed finite gate alphabet with an explicit total approximation budget.

Algorithm 3.2 (Quantum phase estimation). *Let U be a unitary, let $|u\rangle$ be a normalized eigenvector satisfying*

$$U|u\rangle = e^{2\pi i\varphi}|u\rangle, \quad \varphi \in [0, 1),$$

and let $t \geq 1$ with $M = 2^t$. Suppose that controlled applications of $U^{2^0}, U^{2^1}, \dots, U^{2^{t-1}}$ are available. Prepare t control qubits in $|0^t\rangle$, apply Hadamards, apply the controlled powers, apply F_M^{-1} to the control register, and measure it as an integer $Y \in \{0, \dots, M-1\}$. Then

$$\Pr[Y = y] = \left| \frac{1}{M} \sum_{k=0}^{M-1} e^{2\pi i k(\varphi - y/M)} \right|^2.$$

If y_0 is a nearest integer to $M\varphi$ modulo M , then

$$\Pr[Y = y_0] \geq \frac{4}{\pi^2}, \quad \left\| \frac{y_0}{M} - \varphi \right\|_{\mathbb{T}} \leq \frac{1}{2M}.$$

More generally, let $p \geq 1$ and $0 < \eta < 1$. Set

$$s = \left\lceil \log_2 \left(1 + \frac{1}{\eta} \right) \right\rceil, \quad t = p + s.$$

A single run with these t control qubits satisfies

$$\Pr \left[\left\| \frac{Y}{M} - \varphi \right\|_{\mathbb{T}} < 2^{-p} \right] \geq 1 - \eta.$$

The algorithm uses exactly t controlled-power calls. If only controlled- U itself is available and U^{2^j} is implemented by literal repetition, those calls expand to $1 + 2 + \dots + 2^{t-1} = 2^t - 1$ controlled- U uses. Any faster problem-specific implementation of the powers must be charged according to its actual circuit cost; phase estimation does not make powered unitaries free.

Proof. After the Hadamards, the control register is $M^{-1/2} \sum_{k=0}^{M-1} |k\rangle$. Writing k in binary and applying the corresponding controlled powers multiplies the $|k\rangle|u\rangle$ term by $e^{2\pi i k\varphi}$. Hence immediately before the inverse Fourier transform the joint state is

$$\frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} e^{2\pi i k\varphi} |k\rangle |u\rangle.$$

The definition of F_M^{-1} gives the displayed output amplitude and probability.

Choose a nearest integer y_0 to $M\varphi$ modulo M and write $\Delta = M\varphi - y_0$ with $|\Delta| \leq 1/2$. If $\Delta = 0$, the outcome is y_0 with probability one. Otherwise the geometric-series identity gives

$$\left| \frac{1}{M} \sum_{k=0}^{M-1} e^{2\pi i k\Delta/M} \right| = \frac{|\sin(\pi\Delta)|}{M|\sin(\pi\Delta/M)|}.$$

Because $0 \leq \pi|\Delta| \leq \pi/2$, the inequalities $\sin x \geq 2x/\pi$ and $\sin x \leq x$ on this interval imply

$$\frac{|\sin(\pi\Delta)|}{M|\sin(\pi\Delta/M)|} \geq \frac{2}{\pi}.$$

Squaring proves the $4/\pi^2$ bound.

For the high-probability estimate, keep y_0 and let $d_M(y, y_0)$ denote cyclic integer distance in $\mathbb{Z}_M$. If $d_M(y, y_0) = k \geq 1$, choose the representative $x = M\varphi - y$ with $|x| \leq M/2$. Then $|x| \geq k - 1/2$. The same geometric-series formula, now using $|\sin(\pi x)| \leq 1$ and $|\sin(\pi x/M)| \geq 2|x|/M$, yields

$$\Pr[Y = y] \leq \frac{1}{4(k - 1/2)^2}.$$

There are at most two outcomes at each cyclic distance k . Put $R = 2^s \geq 2$. If $d_M(Y, y_0) \leq R - 1$, then

$$\left\| \frac{Y}{M} - \varphi \right\|_{\mathbb{T}} \leq \frac{R - 1 + 1/2}{M} < \frac{R}{M} = 2^{-p}.$$

Therefore

$$\Pr \left[\left\| \frac{Y}{M} - \varphi \right\|_{\mathbb{T}} \geq 2^{-p} \right] \leq \sum_{k=R}^{\infty} \frac{1}{2(k - 1/2)^2} \leq \frac{1}{R - 1} \leq \eta.$$

For the penultimate inequality, bound $(k - 1/2)^{-2} \leq (k - 1)^{-2}$ and compare the decreasing tail of $\sum_{j \geq R-1} j^{-2}$ with its first term plus an integral. The resource accounting follows from the controlled powers used in the preparation of the Fourier phase state. $\square$

Worked phase-estimation example. If $\varphi = 5/8$ and $t = 3$, then $M\varphi = 5$ is an integer. The pre-QFT control state is exactly $F_8|5\rangle$, so the inverse QFT returns $Y = 5$ with probability one. This exact-grid case is useful for separating the Fourier transform itself from the approximation tail that appears for a generic eigenphase.

Source checkpoint. Kitaev's eigenvalue-measurement procedure is the primary historical source for phase estimation. The QFT-form circuit used here is the standard textbook realization. The probability and tail bounds above are proved locally, and the controlled-power accounting is stated explicitly because later order-finding applications implement powers by arithmetic circuits rather than by literal repeated application of one black-box unitary.

3.5 Grover rotation and amplitude amplification

For a subspace $\mathcal{G} \subseteq \mathcal{H}$, $\Pi_{\mathcal{G}}$ denotes its orthogonal projector: the Hermitian operator satisfying $\Pi_{\mathcal{G}}^2 = \Pi_{\mathcal{G}}$ with image $\mathcal{G}$. The associated *reflection* is $I - 2\Pi_{\mathcal{G}}$ (or its negative, which differs only by a global phase when used alone). Measuring the *good/bad decomposition* means the two-outcome projective measurement $\{\Pi_{\mathcal{G}}, I - \Pi_{\mathcal{G}}\}$.

Algorithm 3.3 (Grover search and amplitude amplification). *Let $\mathcal{H}$ be a finite-dimensional Hilbert space and let $\Pi_{\mathcal{G}}$ be the orthogonal projector onto a designated good subspace. Let A be a unitary and suppose*

$$|\psi\rangle := A|0\rangle = \sqrt{a}|\psi_{\mathcal{G}}\rangle + \sqrt{1-a}|\psi_B\rangle, \quad 0 < a < 1,$$

where $|\psi_{\mathcal{G}}\rangle$ and $|\psi_B\rangle$ are normalized vectors in the good and orthogonal bad subspaces. Define $\theta \in (0, \pi/2)$ by $\sin^2 \theta = a$, and define the two reflections

$$S_G = I - 2\Pi_{\mathcal{G}}, \quad S_{\psi} = 2|\psi\rangle\langle\psi| - I = -A(I - 2|0\rangle\langle 0|)A^*.$$

Then the amplitude-amplification iterate $Q = S_{\psi}S_G$ satisfies, for every integer $r \geq 0$,

$$Q^r|\psi\rangle = \sin((2r+1)\theta)|\psi_{\mathcal{G}}\rangle + \cos((2r+1)\theta)|\psi_B\rangle,$$

so measurement of the good/bad decomposition succeeds with exact probability $\sin^2((2r+1)\theta)$.

If $a \leq 1/2$ is known and r is an integer nearest to $\pi/(4\theta) - 1/2$, then

$$r = O(a^{-1/2}), \quad \Pr[\text{good}] \geq 1 - a.$$

If a is unknown but positive, choosing a uniformly random number of iterations from successively larger ranges, increasing the range geometrically until it is of order $1/\sqrt{a}$, finds a good outcome with constant probability using expected $O(a^{-1/2})$ applications of S_G , A , and A^* . Constant success can be amplified to $1 - \delta$ by independent restarts at an additional factor $O(\log(1/\delta))$.

For unstructured search, let $f : [N] \rightarrow \{0, 1\}$ have K marked inputs, let $|\psi\rangle = N^{-1/2} \sum_{x=0}^{N-1} |x\rangle$, and let $S_G = P_f$ be the phase oracle for the marked set. Then $a = K/N$. In particular, when $K = 1$, Grover search uses $O(\sqrt{N})$ standard Boolean queries and can achieve success at least $1 - 1/N$ when $N \geq 2$.

Proof. The span of $|\psi_G\rangle, |\psi_B\rangle$ is invariant under both reflections. In the ordered basis $(|\psi_G\rangle, |\psi_B\rangle)$,

$$S_G = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S_\psi = \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}.$$

Thus

$$Q = S_\psi S_G = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}.$$

Applying this rotation to $(\sin \theta, \cos \theta)^T$ gives $(\sin 3\theta, \cos 3\theta)^T$, and induction gives the formula for $Q^r |\psi\rangle$.

If r is nearest to $\pi/(4\theta) - 1/2$, then $|(2r+1)\theta - \pi/2| \leq \theta$. Hence the success probability is at least $\cos^2 \theta = 1 - a$. Also $\sin \theta \leq \theta$, so $\theta \geq \sqrt{a}$. Therefore $r \leq \pi/(4\theta) + 1/2 = O(a^{-1/2})$.

For completeness, the unknown- a mechanism can be quantified without knowing θ . If j is uniform on $\{0, \dots, m-1\}$, then the rotation formula and the finite cosine sum give

$$\frac{1}{m} \sum_{j=0}^{m-1} \sin^2((2j+1)\theta) = \frac{1}{2} - \frac{\sin(4m\theta)}{4m \sin(2\theta)}.$$

Consequently, whenever $m \geq 1/\sin(2\theta)$, the average success probability is at least $1/4$. For $a \leq 1/2$, $\sin(2\theta) = 2\sqrt{a(1-a)} \geq \sqrt{2a}$, so a geometrically increasing schedule reaches such an m after total work $O(a^{-1/2})$; the geometric sum of all earlier ranges has the same order. A measured candidate is verified by one ordinary query. If $a > 1/2$, direct preparation and measurement already has success probability greater than $1/2$, so the same asymptotic conclusion is trivial.

In unstructured search, the good component is the normalized uniform superposition of marked inputs, and its squared amplitude in the uniform starting state is K/N . Phase kickback from Definition 3.1 implements S_G with one standard Boolean query. Taking $K = 1$ gives the claimed $O(\sqrt{N})$ query bound and, from the known- a estimate, success at least $1 - 1/N$. $\square$

Worked Grover example. For $N = 4$ and one marked input, $a = 1/4$ and $\theta = \pi/6$. One Grover iterate changes the good amplitude from $\sin(\pi/6) = 1/2$ to $\sin(3\pi/6) = 1$, so the marked item is found with certainty. This is the smallest nontrivial case in which the geometric rotation is visible without any asymptotic approximation.

Source checkpoint. Grover introduced the square-root search algorithm. Boyer–Brassard–Høyer–Tapp supplied the tight success formula and an unknown-number-of-solutions search procedure, while Brassard–Høyer–Mosca–Tapp formulated amplitude amplification at the general initial-success amplitude a . The two-dimensional proof above is local and fixes the reflection signs explicitly.

3.6 Why unstructured search cannot be faster

Theorem 3.4 (Quantum search lower bound). *For $N \geq 1$, consider Boolean oracles $f_0, f_1, \dots, f_N$ on $[N] = \{1, \dots, N\}$, where $f_0(x) = 0$ for every x and $f_i(x) = 1$ exactly when $x = i$. Any quantum algorithm that, using the standard reversible oracle of Definition 3.1, distinguishes f_0 from every f_i with success probability at least $2/3$ requires*

$$T \geq \frac{\sqrt{N}}{12}$$

queries. Hence bounded-error quantum query complexity of OR and of deciding whether an unstructured search instance has zero or one marked item is $\Omega(\sqrt{N})$. Any algorithm that locates the unique marked item with probability at least $2/3$ also uses $\Omega(\sqrt{N})$ queries.

Proof. By adding workspace, postpone all measurements to the end. Run the algorithm against the all-zero oracle f_0 . Immediately before query $t \in \{0, \dots, T-1\}$, write its normalized state as

$$|\phi_t\rangle = \sum_{j=1}^N |j\rangle |\eta_{t,j}\rangle, \quad \sum_{j=1}^N \|\eta_{t,j}\|_2^2 = 1,$$

where the second vector contains the answer register and all remaining workspace.

Fix i . Replace the all-zero queries by the f_i query one at a time, starting from the last query and telescoping the resulting final states. At a step where the pre-query state is the all-zero-run state, the two oracles differ only on query-register basis state $|i\rangle$. Since the answer-qubit operators are I and X and $\|X - I\|_{\text{op}} = 2$,

$$\|(O_{f_i} - O_{f_0})|\phi_t\rangle\|_2 \leq 2\|\eta_{t,i}\|_2.$$

All intervening transformations are unitary, so the hybrid triangle inequality gives, for the final pure states $|\Psi_i\rangle$ and $|\Psi_0\rangle$,

$$d_i := \|\Psi_i - \Psi_0\|_2 \leq 2 \sum_{t=0}^{T-1} \|\eta_{t,i}\|_2.$$

Cauchy–Schwarz and normalization now imply

$$\sum_{i=1}^N d_i^2 \leq 4T \sum_{t=0}^{T-1} \sum_{i=1}^N \|\eta_{t,i}\|_2^2 = 4T^2.$$

Let E be the final measurement effect corresponding to the answer “a marked item exists.” The success assumption gives $\langle \Psi_i | E | \Psi_i \rangle \geq 2/3$ and $\langle \Psi_0 | E | \Psi_0 \rangle \leq 1/3$. For any two unit vectors $|v\rangle, |w\rangle$ and any $0 \leq E \leq I$,

$$|\langle v | E | v \rangle - \langle w | E | w \rangle| \leq 2\|v - w\|_2,$$

by adding and subtracting $\langle w | E | v \rangle$ and using Cauchy–Schwarz. Therefore every $d_i \geq 1/6$. Combining with the preceding sum gives

$$\frac{N}{36} \leq 4T^2,$$

which is the stated $T \geq \sqrt{N}/12$.

Finally, if an algorithm locates the unique marked item using T queries, run it on the promise “zero or one marked item” and verify its proposed location with one additional standard query. On f_0 verification always rejects; on f_i it accepts with the locator’s success probability. Thus the decision lower bound applies to $T + 1$, giving the same $\Omega(\sqrt{N})$ asymptotic bound. $\square$

Interpretation. The proof does not argue that a quantum query “looks at only one address.” During a query the address register may be in a superposition of all N inputs. Instead, it compares the marked-oracle computation with the zero-oracle computation and shows that the total squared distinguishability that T queries can accumulate over all possible marked locations is only $O(T^2)$. Grover’s algorithm matches exactly this quadratic information-growth scale.

Source checkpoint. The hybrid method is the Bennett–Bernstein–Brassard–Vazirani route to the quantum search lower bound. The theorem above is a deliberately finite standard-oracle version with an explicit constant; it is proved locally rather than importing the stronger random-oracle consequences of the original paper.

3.7 Simon’s hidden-XOR algorithm

For $x, y \in \{0, 1\}^n$, write $x \cdot y = \sum_{j=1}^n x_j y_j \pmod{2}$. For nonzero $s \in \{0, 1\}^n$, define

$$s^\perp := \{y \in \{0, 1\}^n : s \cdot y = 0\},$$

which is an $(n - 1)$ -dimensional vector subspace over $\mathbb{F}_2$.

Algorithm 3.5 (Simon’s algorithm). *Let $n \geq 2$. Suppose an oracle $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ satisfies the Simon promise: there is an unknown nonzero string $s \in \{0, 1\}^n$ such that*

$$f(x) = f(x') \iff x' = x \text{ or } x' = x \oplus s.$$

For any integer $r \geq 1$, Simon’s quantum algorithm uses

$$q = n - 1 + r$$

standard queries and returns s with probability at least $1 - 2^{-r}$. Its oracle-independent quantum gates and classical postprocessing are polynomial in $n + r$.

By contrast, every bounded-error randomized classical query algorithm for recovering s under the same promise requires $\Omega(2^{n/2})$ queries. Thus the promise problem exhibits an exponential separation between quantum and randomized classical query complexity.

Proof. For one quantum sample, begin with $|0^n\rangle|0^n\rangle$, apply $H^{\otimes n}$ to the first register, and query O_f :

$$\frac{1}{2^{n/2}} \sum_x |x\rangle |f(x)\rangle.$$

Measure the second register. If its value is $f(x_0)$, the Simon promise says that the first register collapses to

$$\frac{|x_0\rangle + |x_0 \oplus s\rangle}{\sqrt{2}}.$$

Applying $H^{\otimes n}$ gives amplitude at $y \in \{0, 1\}^n$ proportional to

$$(-1)^{x_0 \cdot y} + (-1)^{(x_0 \oplus s) \cdot y} = (-1)^{x_0 \cdot y} (1 + (-1)^{s \cdot y}).$$

It vanishes when $s \cdot y = 1$ and has equal magnitude for every $y \in s^\perp$. Hence one query produces an independent uniform sample from $s^\perp$. The intermediate measurement can be deferred by Chapter 2 if a fully unitary presentation is preferred.

Let $d = n - 1 = \dim s^\perp$, and take $q = d + r$ independent samples. They fail to span $s^\perp$ only if there exists a nonzero linear functional on $s^\perp$ that vanishes on every sampled vector. There are $2^d - 1$ nonzero linear functionals, and any fixed one vanishes on one uniform sample with probability $1/2$. By the union bound,

$$\Pr[\text{samples do not span } s^\perp] \leq (2^d - 1)2^{-q} < 2^{d-q} = 2^{-r}.$$

When the samples span $s^\perp$, Gaussian elimination over $\mathbb{F}_2$ finds the one-dimensional orthogonal complement $(s^\perp)^\perp = \{0, s\}$ and hence recovers the unique nonzero s .

It remains to justify the classical lower bound. By Yao's minimax principle it is enough to exhibit a distribution on promised inputs for which every deterministic algorithm using $q = o(2^{n/2})$ queries has success bounded away from $2/3$. Choose s uniformly from the $2^n - 1$ nonzero strings. Then assign independent uniformly random distinct labels in $\{0, 1\}^n$ to the 2^{n-1} unordered pairs $\{x, x \oplus s\}$ and let f be the resulting 2-to-1 function.

Before the deterministic algorithm observes a collision, all returned labels are distinct. By symmetry, conditioned on the queried points $x_1, \dots, x_k$ and on seeing no collision, the secret is uniform over the nonzero strings not equal to any difference $x_i \oplus x_j$ with $i < j$. There are at most $\binom{k}{2}$ excluded differences. Therefore, as long as $\binom{q}{2} \leq (2^n - 1)/2$, the conditional probability that query $k + 1$ creates the first collision is at most

$$\frac{k}{2^n - 1 - \binom{q}{2}}.$$

Summing over $k < q$ gives

$$\Pr[\text{a collision occurs}] \leq \frac{\binom{q}{2}}{2^n - 1 - \binom{q}{2}}.$$

If no collision occurs, the transcript is compatible with at least $2^n - 1 - \binom{q}{2}$ secrets, all with equal posterior probability, so even the best final guess has conditional success probability at most the reciprocal of this number. For $q \leq 2^{n/2-3}$ and all sufficiently large n , the sum of these two success contributions is less than $1/3$. Hence any classical randomized algorithm with worst-case success at least $2/3$ must use $q = \Omega(2^{n/2})$ queries. Finitely many small n affect only the implicit constant. $\square$

Worked Simon example. For $n = 3$ and secret $s = 101$, every quantum sample satisfies $y_1 \oplus y_3 = 0$. Thus

$$s^\perp = \{000, 010, 101, 111\}.$$

If two independent samples are, for example, 010 and 101, they span this two-dimensional subspace. Solving the two equations $010 \cdot s = 0$ and $101 \cdot s = 0$ leaves the one-dimensional solution space $\{000, 101\}$, recovering the promised nonzero secret.

Source checkpoint. Simon's original oracle separation is the primary source. The modern hidden-XOR promise formulation used here makes the sampling subspace and postprocessing especially transparent. The quantum sampling proof, the rank-success estimate, and the birthday-style randomized classical lower bound are all written locally.

3.8 What these algorithms are really exploiting

Phase estimation and Simon's algorithm are both interference procedures, but their structure is different from Grover search. Grover starts with a one-dimensional success predicate and repeatedly rotates amplitude toward it; the hybrid lower bound shows that no black-box method can improve its square-root scale for unstructured search. Simon's promise, by contrast, supplies algebraic structure: each query can be converted into a uniformly random linear equation orthogonal to one fixed hidden vector. The exponential speedup comes from extracting that global invariant rather than from amplifying the probability of one marked basis state.

Phase estimation supplies the bridge to the next chapter. Once a computational problem constructs a unitary whose eigenphase encodes arithmetic information and implements the powers U^{2^j} efficiently, the inverse Fourier transform converts that phase into a classical approximation. The important complexity lesson is that the efficiency lies jointly in the Fourier readout and in the problem-specific implementation of those powered unitaries.

3.9 Reader checkpoint and exercises

1. Prove explicitly that the standard Boolean oracle O_f is self-inverse and that phase kickback uses exactly one standard query. Then write the two-query compute–control–uncompute circuit for CO_f when f has an m -bit output.
2. In phase estimation, derive the geometric-series expression for the amplitude of outcome y and check separately the case in which $M\varphi - y$ is an integer.
3. For $\varphi = 1/3$ and $t = 3$, compute all eight phase-estimation probabilities exactly in terms of trigonometric functions and verify that the two outcomes nearest to $8/3$ carry most of the mass.
4. Derive the binary product formula for $F_{2^t}|x\rangle$ and use it to justify the $O(t^2)$ exact continuous-gate QFT circuit, including bit reversal.
5. Starting from the two reflection matrices in the Grover proof, verify the exact identity $Q^r|\psi\rangle = \sin((2r+1)\theta)|\psi_G\rangle + \cos((2r+1)\theta)|\psi_B\rangle$ without using a geometric picture.
6. Prove the finite-sum identity

$$\frac{1}{m} \sum_{j=0}^{m-1} \sin^2((2j+1)\theta) = \frac{1}{2} - \frac{\sin(4m\theta)}{4m \sin(2\theta)}$$

and identify exactly where the case $\sin(2\theta) = 0$ must be separated.

7. Rework the hybrid search lower bound with an arbitrary error parameter $0 < \delta < 1/2$. Determine an explicit constant $c_\delta > 0$ such that $T \geq c_\delta \sqrt{N}$.
8. In Simon's algorithm, prove directly that $s^\perp$ has exactly 2^{n-1} elements and that $(s^\perp)^\perp = \{0, s\}$ without invoking any finite-field classification theorem.
9. Improve the union-bound estimate for the probability that $n-1+r$ random vectors span an $(n-1)$ -dimensional binary vector space by computing the exact full-rank probability $\prod_{j=0}^{n-2} (1 - 2^{j-(n-1+r)})$.

10. Explain why Simon's classical lower bound is a birthday phenomenon. For a fixed query budget q , compare the probability scale of seeing a collision with the $\Theta(q^2/2^n)$ birthday estimate and identify the role of adaptivity in the conditional-symmetry argument.

Chapter 4

Quantum Fourier Sampling, Order Finding, and Shor's Algorithms

4.1 How to read this chapter

The previous chapter separated the query interface from the ordinary circuit cost. That distinction becomes decisive here. Shor's speedup does not come from pretending that the powers $U, U^2, U^4, \dots$ are free oracle calls. It comes from an arithmetic unitary whose powers have a special form: after repeated squaring has computed the classical constants $a, a^2, a^4, \dots \pmod{N}$, each controlled power is just reversible multiplication by one known constant. Phase estimation can therefore read an eigenphase while the complete arithmetic circuit still has size polynomial in the input bit length.

The chapter has three layers. First we close the classical continued-fraction interface and use phase estimation to recover an exact multiplicative order with constant success per pair of Fourier samples. Second we state the finite Abelian hidden subgroup theorem with an explicit source-owned boundary. Third we separate Shor's algorithms into their quantum cores and classical reductions: factoring uses order finding and a nontrivial square root of one, while cyclic discrete logarithm is presented as a two-dimensional Abelian hidden-subgroup problem.

4.2 Learning goals

After this chapter the reader should be able to define multiplicative order and the continued-fraction convergents used by Shor postprocessing; prove the $1/(2q^2)$ rational-recovery criterion; diagonalize modular multiplication; account for every controlled modular power by repeated squaring; prove a constant-probability exact order-finding algorithm without assuming a lucky coprime Fourier numerator; state precisely the finite Abelian hidden subgroup interface and its annihilator; prove the classical reduction from order finding to factoring, including the failure probability; and reduce cyclic-group discrete logarithm to an Abelian hidden subgroup without importing generic classical DLP algorithms into the dependency chain.

4.3 Arithmetic and continued fractions

For an integer $N \geq 2$, write

$$(\mathbb{Z}/N\mathbb{Z})^\times := \{\bar{a} \in \mathbb{Z}/N\mathbb{Z} : \gcd(a, N) = 1\}$$

for the multiplicative group of invertible residue classes. If $\gcd(a, N) = 1$, the *multiplicative order of a modulo N* is

$$\text{ord}_N(a) := \min\{r \geq 1 : a^r \equiv 1 \pmod{N}\}.$$

For a prime p and a positive integer m , $v_p(m)$ is the largest $e \geq 0$ for which $p^e \mid m$. In particular, $v_2(m)$ is the *two-adic valuation* of m .

A *simple continued fraction* is an expression

$$[a_0; a_1, a_2, \dots] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}},$$

where $a_0 \in \mathbb{Z}$ and $a_j \in \mathbb{Z}_{\geq 1}$ for $j \geq 1$. The truncations $p_j/q_j = [a_0; \dots, a_j]$, written in lowest terms with $q_j > 0$, are its *convergents*. They obey

$$p_j = a_j p_{j-1} + p_{j-2}, \quad q_j = a_j q_{j-1} + q_{j-2},$$

with $p_{-2} = 0, p_{-1} = 1, q_{-2} = 1, q_{-1} = 0$, and hence

$$p_j q_{j-1} - p_{j-1} q_j = (-1)^{j-1}. \quad (4.1)$$

A rational number has a finite simple continued fraction. If its last partial quotient is at least 2, replacing the tail a_n by $a_n - 1, 1$ gives the second finite expansion; these two expansions have opposite parity.

Lemma 4.1 (Continued-fraction rational recovery). *Let $x \in \mathbb{R}$ and let p/q be a reduced rational number with $q \geq 1$. If*

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2}, \quad (4.2)$$

then p/q is a convergent of the simple continued fraction of x .

Proof. If $x = p/q$, the conclusion is immediate. If $q = 1$, then p is an integer and $|x - p| < 1/2$. For $x > p$, one has $\lfloor x \rfloor = p$, so $p/1$ is the initial convergent. For $x < p$, one has $\lfloor x \rfloor = p - 1$ and the fractional part of x lies in $(1/2, 1)$, so the next partial quotient is 1 and $\lfloor p - 1; 1 \rfloor = p$. Thus assume $q \geq 2$ and $x \neq p/q$. Choose one of the two finite continued-fraction expansions

$$\frac{p}{q} = [a_0; \dots, a_n] = \frac{p_n}{q_n}$$

so that, with $\varepsilon := (-1)^{n-1}$,

$$x = \frac{p}{q} - \varepsilon \frac{\alpha}{q^2}, \quad 0 < \alpha < \frac{1}{2}. \quad (4.3)$$

The parity choice is possible because the two finite expansions have opposite values of $(-1)^{n-1}$. Let p_{n-1}/q_{n-1} be the preceding convergent. Equation (4.1) gives $p q_{n-1} - p_{n-1} q = \varepsilon$. Define

$$\omega := \frac{1}{\alpha} - \frac{q_{n-1}}{q}.$$

Since $q_{n-1} < q$ and $\alpha < 1/2$, one has $\omega > 1$. Direct substitution using (4.3) and the determinant identity yields

$$x = \frac{\omega p + p_{n-1}}{\omega q + q_{n-1}} = [a_0; \dots, a_n, \omega].$$

Because $\omega > 1$, applying the ordinary continued-fraction algorithm to the remaining complete quotient ω extends the displayed finite prefix. Thus $p/q = p_n/q_n$ occurs as a convergent of x . $\square$

Why the q^2 scale matters. The theorem is not merely a numerical approximation statement. It turns a real approximation into an exact discrete candidate. If one knows in advance that the unknown denominator is below N , an approximation error below $1/(2N^2)$ is enough to force the reduced fraction to appear among the continued-fraction convergents. Shor postprocessing therefore needs quadratically fine phase resolution, but only logarithmically many phase-estimation control qubits.

4.4 Order finding by phase estimation

Let $N \geq 3$, let $n = \lceil \log_2 N \rceil$, and let a satisfy $1 \leq a < N$ and $\gcd(a, N) = 1$. On the 2^n computational basis states define the *modular multiplication unitary*

$$U_a|z\rangle = \begin{cases} |az \bmod N\rangle, & 0 \leq z < N, \\ |z\rangle, & N \leq z < 2^n. \end{cases} \quad (4.4)$$

Multiplication by a permutes the valid residue classes because a is invertible modulo N ; the identity extension therefore makes (4.4) a unitary on the full n -qubit space.

Algorithm 4.2 (Order finding by quantum Fourier sampling). *Let $N \geq 3$, $1 \leq a < N$, $\gcd(a, N) = 1$, and let $r = \text{ord}_N(a)$. For every $0 < \delta < 1$ there is a quantum algorithm that outputs the exact integer r with probability at least $1 - \delta$ using*

$$O\left(\log \frac{1}{\delta}\right)$$

independent pairs of phase-estimation samples and a number of quantum gates and classical bit operations polynomial in $\log N$ and $\log(1/\delta)$.

For one phase sample, set

$$t = \lceil \log_2(2N^2) \rceil, \quad M = 2^t, \quad (4.5)$$

run Algorithm 3.2 on U_a with target state $|1\rangle$, and measure $Y \in \{0, \dots, M-1\}$. The controlled powers are not black-box powers: compute

$$b_j = a^{2^j} \bmod N$$

classically by repeated squaring, and implement controlled- $U_a^{2^j}$ as controlled reversible multiplication by the known constant b_j modulo N , with the identity action on basis states $z \geq N$.

For each observed rational Y/M , compute its continued-fraction convergents and retain their positive denominators below N . From two independent samples form all least common multiples of one retained denominator from each sample. Repeat this two-sample experiment

$$K = \left\lceil \frac{\log(1/\delta)}{-\log(1 - 4/\pi^4)} \right\rceil \quad (4.6)$$

times, pool all candidate least common multiples $R < N$, and test them by modular exponentiation. Return the least positive candidate satisfying $a^R \equiv 1 \pmod{N}$. If there is no such candidate, return FAIL.

Proof. The orbit of $|1\rangle$ under U_a consists of $|1\rangle, |a\rangle, \dots, |a^{r-1}\rangle$. For $s \in \{0, \dots, r-1\}$ define

$$|u_s\rangle := \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle. \quad (4.7)$$

These vectors are orthonormal, and shifting the orbit index gives

$$U_a |u_s\rangle = e^{2\pi i s / r} |u_s\rangle. \quad (4.8)$$

Fourier orthogonality also gives

$$|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle. \quad (4.9)$$

Thus phase estimation acts on an equal superposition of eigenphases s/r ; after measuring the control register, s is effectively uniform in $\mathbb{Z}_r$.

Fix one such s . By Algorithm 3.2, with probability at least $4/\pi^2$ the measured Y is a nearest grid point to Ms/r and

$$\left| \frac{Y}{M} - \frac{s}{r} \right| \leq \frac{1}{2M}. \quad (4.10)$$

For $s = 0$ this favorable outcome is $Y = 0$. Since r is the order of an element of $(\mathbb{Z}/N\mathbb{Z})^\times$, one has $r \leq \varphi(N) \leq N - 1$. If $1 \leq s < r$, then $\min(s/r, 1 - s/r) \geq 1/r > 1/N$, while $1/(2M) < 1/N$; hence the nearest phase-estimation outcome in (4.10) lies on the ordinary interval side of both endpoints and can be interpreted with ordinary, not wrapped, distance in $[0, 1]$.

Reduce s/r to p_s/q_s . Then

$$q_s = \frac{r}{\gcd(s, r)} \leq r < N. \quad (4.11)$$

Because $M \geq 2N^2$,

$$\left| \frac{Y}{M} - \frac{p_s}{q_s} \right| \leq \frac{1}{2M} \leq \frac{1}{4N^2} < \frac{1}{2q_s^2}. \quad (4.12)$$

Lemma 4.1 therefore places p_s/q_s among the convergents of Y/M . On the nearest-grid event, the retained denominator list consequently contains q_s .

Now take two independent uniform values $s_1, s_2 \in \mathbb{Z}_r$. For a prime $p \mid r$, the full p -power dividing r appears in $\text{lcm}(q_{s_1}, q_{s_2})$ unless both s_1 and s_2 are divisible by p . The latter event has probability $1/p^2$. Hence, by a union bound over the distinct prime divisors of r ,

$$\Pr[\text{lcm}(q_{s_1}, q_{s_2}) \neq r] \leq \sum_{p \mid r} \frac{1}{p^2} \leq \sum_{m=2}^{\infty} \frac{1}{m^2} \leq \frac{1}{4} + \int_2^{\infty} \frac{dx}{x^2} = \frac{3}{4}. \quad (4.13)$$

Thus the denominator pair has least common multiple exactly r with probability at least $1/4$. Combining this with the two independent nearest-grid events, one two-sample experiment places the exact order r in its tested candidate set with probability at least

$$\frac{1}{4} \left(\frac{4}{\pi^2} \right)^2 = \frac{4}{\pi^4}. \quad (4.14)$$

If r is present, every candidate R satisfying $a^R \equiv 1 \pmod{N}$ is a multiple of r by the definition of multiplicative order. Therefore the least valid pooled candidate is exactly r . The K independent experiments in (4.6) miss the favorable event with probability at most $(1 - 4/\pi^4)^K \leq \delta$.

It remains to charge the circuit cost. Equation (4.4) gives $U_a^{2^j} = U_{a^{2^j \bmod N}}$ on the valid residue subspace. The constants b_j are obtained by $t = O(\log N)$ classical modular squarings. Standard reversible add–multiply–reduce circuits implement multiplication by each known b_j with size polynomial in $n = O(\log N)$ and clean their workspace by compute–copy–uncompute. Continued fractions have $O(\log M) = O(\log N)$ convergents, so enumerating denominator pairs, least common multiples, and modular-exponentiation checks is also polynomial. Repetition contributes only the factor $O(\log(1/\delta))$. $\square$

Worked order-finding example. Take $N = 15$ and $a = 2$. The order is $r = 4$ because $2, 4, 8, 1$ is the multiplicative orbit modulo 15. Here (4.5) gives $M = 512$, which is divisible by 4, so the four eigenphases $0, 1/4, 1/2, 3/4$ lie exactly on the phase-estimation grid. Samples with $s = 1$ or 3 contribute denominator 4, while $s = 2$ contributes denominator 2 and $s = 0$ contributes denominator 1. Two samples recover 4 by their denominator least common multiple unless their numerators jointly miss the full factor 2^2 ; the general proof above is the prime-by-prime version of this calculation.

Circuit-cost checkpoint. The phrase “apply $U_a^{2^j}$ ” abbreviates a compiled arithmetic circuit, not 2^j literal uses of an unknown oracle. Repeated squaring changes the classical multiplier, so the j th controlled power can be synthesized with polynomially many elementary gates. This is the complexity interface that makes the order-finding theorem polynomial-time rather than merely query-efficient.

Source checkpoint. Shor’s expanded primary treatment explicitly separates reversible modular exponentiation, the Fourier step, order finding, factoring, and discrete logarithm. It constructs modular powers by repeated squaring and recovers rational phase information with continued fractions. The phase-estimation presentation above is an equivalent eigenvalue form of the Fourier-sampling step. The two-sample least-common-multiple success bound (4.13)–(4.14) and the complete continued-fraction lemma are written locally; no coprimality of a single sampled numerator is assumed.

4.5 The finite Abelian hidden subgroup interface

Let

$$G = \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k} \quad (4.15)$$

be an explicitly represented finite Abelian group with componentwise addition. Its character group $\hat{G}$ and the normalized finite-group Fourier transform were defined in Volume I. For a subgroup $H \leq G$, define the *annihilator*

$$H^\perp := \{\chi \in \hat{G} : \chi(h) = 1 \text{ for every } h \in H\}. \quad (4.16)$$

Let S be a finite set whose elements have unique binary encodings of length polynomial in the input length. A function $f : G \rightarrow S$ *hides* H when

$$f(x) = f(y) \iff x - y \in H. \quad (4.17)$$

Reversible oracle access means a unitary $O_f|x, z\rangle = |x, z \oplus \text{enc}(f(x))\rangle$ on a sufficiently large answer register, with one occurrence charged as one oracle call. The *finite Abelian hidden subgroup problem*

is: given the integers $m_1, \dots, m_k$ in binary and reversible oracle access to such an f , output a finite generating set for the hidden subgroup H . The input-length parameter is

$$L := k + \sum_{j=1}^k \lceil \log_2 m_j \rceil, \quad (4.18)$$

plus the fixed circuit cost assigned to one oracle call.

Theorem 4.3 (Finite Abelian hidden subgroup theorem — verified deep input). *For the explicit product group (4.15), every oracle satisfying (4.17) admits a bounded-error quantum algorithm that, for every $0 < \delta < 1$, outputs generators for H with probability at least $1 - \delta$ using a number of oracle calls, quantum gates, and classical postprocessing bit operations polynomial in L and $\log(1/\delta)$.*

At the Fourier-sampling interface, a coset state of H transforms to a state supported on the annihilator $H^\perp$. The source-owned reconstruction layer turns sufficiently many such character constraints into generators for H using polynomial-time integer linear algebra. QRCC treats that general reconstruction theorem, including its finite-precision and arbitrary-component bookkeeping, as the deep input in this statement.

Exact source boundary. Kitaev's Abelian stabilizer formulation gives a polynomial-time quantum procedure based on eigenvalue measurement and Fourier analysis and contains factoring and discrete logarithm as special cases. In the finite explicit-product model used here, the standard Abelian-HSP formulation (4.17) is the reader-facing interface. QRCC does *not* claim that the short annihilator discussion below is a replacement proof of the general theorem: the polynomial reconstruction of an arbitrary hidden subgroup from encoded character data remains source-owned.

Why the annihilator appears. For one left coset $x + H$, its normalized state is

$$|x + H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |x + h\rangle.$$

Applying the finite-group Fourier transform gives the character amplitude

$$\frac{1}{\sqrt{|G||H|}} \sum_{h \in H} \overline{\chi(x + h)} = \frac{\overline{\chi(x)}}{\sqrt{|G||H|}} \sum_{h \in H} \overline{\chi(h)}. \quad (4.19)$$

The subgroup-character sum is $|H|$ for $\chi \in H^\perp$ and 0 otherwise by character orthogonality. Thus the quantum sample really is a linear constraint on the hidden subgroup. What is deep in Theorem 4.3 is the uniform polynomial-time machinery that converts those constraints into generators over arbitrary products of cyclic groups.

Worked hidden-subgroup example. Let $G = \mathbb{Z}_8$ and $H = \{0, 4\}$. Writing $\chi_y(x) = e^{2\pi i xy/8}$, the condition $\chi_y(4) = 1$ is $(-1)^y = 1$. Hence

$$H^\perp = \{\chi_0, \chi_2, \chi_4, \chi_6\}.$$

Fourier sampling therefore never outputs an odd character index. In this one-dimensional example, the visible parity constraint already identifies the order-two hidden subgroup; the deep input is needed for the general multidimensional reconstruction problem, not for this calculation.

4.6 Shor factoring and discrete logarithm

We now use the two quantum interfaces for two different classical reductions. The factoring branch needs only exact order finding. The discrete-logarithm branch is cleaner as a two-dimensional hidden subgroup problem and therefore uses Theorem 4.3.

Algorithm 4.4 (Shor factoring and cyclic-group discrete logarithm). *The following two search problems have bounded-error quantum algorithms with running time polynomial in the input bit length and $\log(1/\delta)$.*

1. **Factoring.** *Given a composite integer $N \geq 4$ and $0 < \delta < 1$, output a nontrivial factor of N with probability at least $1 - \delta$. Even inputs return the factor 2. Perfect powers are detected classically by testing integer e th roots for $2 \leq e \leq \lfloor \log_2 N \rfloor$; if $N = b^e$ with $1 < b < N$, return b . In the remaining case N is odd and has at least two distinct prime divisors. Repeatedly choose a uniform $a \in \{1, \dots, N-1\}$. If $1 < \gcd(a, N) < N$, return that gcd. Otherwise use Algorithm 4.2 to estimate the exact order $r = \text{ord}_N(a)$ with internal error at most $1/12$. When r is even, set $x = a^{r/2} \bmod N$ and $d = \gcd(x-1, N)$; return d only when $1 < d < N$. Repeating this outer trial $O(\log(1/\delta))$ times gives the claimed success probability.*
2. **Cyclic discrete logarithm.** *Let $G = \langle g \rangle$ be a cyclic group of known order m with unique encodings (each group element has exactly one valid binary representation), reversible multiplication and inversion circuits of size $\text{poly}(\log m)$, and let $h \in G$. Under the promise $h = g^x$, output the unique $x \in \mathbb{Z}_m$ with probability at least $1 - \delta$ in quantum time $\text{poly}(\log m, \log(1/\delta))$. Apply Theorem 4.3 to $A = \mathbb{Z}_m \times \mathbb{Z}_m$ and*

$$f(a, b) := g^a h^{-b}. \quad (4.20)$$

From generators returned for the hidden subgroup, recover x by extended Euclidean arithmetic on their second coordinates.

Proof. We prove the two reductions separately.

Factoring. After the even and perfect-power cases have been removed, write

$$N = \prod_{i=1}^k p_i^{e_i}, \quad k \geq 2, \quad (4.21)$$

with distinct odd primes p_i . A uniform $a \in \{1, \dots, N-1\}$ either has a nontrivial gcd with N , which immediately succeeds, or is a uniform element of $(\mathbb{Z}/N\mathbb{Z})^\times$ conditioned on being a unit. It is therefore enough to show that at least half of the units have the order pattern needed by the reduction.

For a unit a , let r_i be its order modulo $p_i^{e_i}$ and put $T_i = v_2(r_i)$. The Chinese remainder theorem makes the residue components, and hence the random variables T_i , independent for uniform a . The reduction map

$$(\mathbb{Z}/p_i^{e_i}\mathbb{Z})^\times \longrightarrow \mathbb{F}_{p_i}^\times$$

has kernel of odd order $p_i^{e_i-1}$. Consequently its Sylow two-subgroup injects into, and has the same order as, the Sylow two-subgroup of $\mathbb{F}_{p_i}^\times$. By the cyclicity theorem for finite-field multiplicative groups from Volume III, that two-subgroup is cyclic.

If its order is 2^{s_i} , a uniform element of the cyclic two-group has

$$\Pr[T_i = 0] = 2^{-s_i}, \quad \Pr[T_i = j] = 2^{j-1-s_i} \quad (1 \leq j \leq s_i). \quad (4.22)$$

Every atom in (4.22) is therefore at most $1/2$. Conditioning on T_1 and using independence,

$$\Pr[T_1 = T_2 = \dots = T_k] \leq 2^{-(k-1)} \leq \frac{1}{2}. \quad (4.23)$$

Let $r = \text{ord}_N(a) = \text{lcm}(r_1, \dots, r_k)$ and $T = \max_i T_i = v_2(r)$. If the values T_i are not all equal, then $T \geq 1$, so r is even. For a component with $T_i < T$, $r/2$ is a multiple of r_i and hence $a^{r/2} \equiv 1 \pmod{p_i^{e_i}}$. For a component with $T_i = T$, the element $a^{r_i/2}$ is the unique element of order two in the cyclic Sylow two-subgroup and is therefore -1 modulo $p_i^{e_i}$; because r/r_i is odd in its two-part, $a^{r/2} \equiv -1 \pmod{p_i^{e_i}}$. Thus

$$x = a^{r/2} \not\equiv \pm 1 \pmod{N}, \quad x^2 \equiv 1 \pmod{N}. \quad (4.24)$$

It follows that $\gcd(x-1, N)$ is strictly between 1 and N .

By (4.23), conditional on a being a unit, the arithmetic good event has probability at least $1/2$. Algorithm 4.2 returns the exact order with probability at least $11/12$ for every fixed unit a . Therefore one outer trial succeeds with conditional probability at least $11/24$ on the unit branch, while a nonunit branch succeeds immediately. Hence every outer trial has success probability at least $11/24$, and K independent trials fail with probability at most

$$\left(\frac{13}{24}\right)^K. \quad (4.25)$$

Taking $K \geq \log(1/\delta)/\log(24/13)$ proves the success claim. Integer gcds, perfect-power tests, modular exponentiation, and the order-finding circuit all have polynomial bit complexity in $\log N$ and $\log(1/\delta)$.

Cyclic discrete logarithm. The problem is the cyclic-group DLP defined in Volume III. Write $h = g^x$. For $(a, b), (a', b') \in A = \mathbb{Z}_m^2$, equation (4.20) gives

$$f(a, b) = f(a', b') \iff a - a' - x(b - b') \equiv 0 \pmod{m}. \quad (4.26)$$

Thus f hides the subgroup

$$H = \{(xb, b) : b \in \mathbb{Z}_m\} = \langle (x, 1) \rangle \leq \mathbb{Z}_m^2. \quad (4.27)$$

The oracle for (4.20) is implemented reversibly using repeated squaring of the fixed group elements g and h , so its gate cost is polynomial in $\log m$ under the stated group-operation model.

Run Theorem 4.3 with error parameter δ and suppose it returns generators $(a_1, b_1), \dots, (a_s, b_s)$ for H . Projection of (4.27) onto the second coordinate is all of $\mathbb{Z}_m$, so the residues $b_1, \dots, b_s$ generate $\mathbb{Z}_m$. Equivalently,

$$\gcd(b_1, \dots, b_s, m) = 1. \quad (4.28)$$

Extended Euclid therefore finds integers $c_0, c_1, \dots, c_s$ satisfying

$$c_0 m + \sum_{j=1}^s c_j b_j = 1. \quad (4.29)$$

Because H is a subgroup, the same linear combination of its generators belongs to H and has second coordinate 1. By (4.27), the unique element of H with second coordinate 1 is $(x, 1)$. Hence

$$x \equiv \sum_{j=1}^s c_j a_j \pmod{m}, \quad (4.30)$$

which reconstructs the discrete logarithm in polynomial classical time. The only failure event is the bounded-error hidden-subgroup call, so the final success probability is at least $1 - \delta$. $\square$

Worked factoring example. For $N = 21$ choose $a = 2$. One finds $2^6 \equiv 1 \pmod{21}$ and no smaller positive exponent returns 1, so $r = 6$. Since r is even,

$$x = 2^{r/2} = 2^3 = 8, \quad x^2 \equiv 1 \pmod{21},$$

and $8 \not\equiv \pm 1 \pmod{21}$. The gcd step gives $\gcd(8 - 1, 21) = 7$. The quantum subroutine is used only to find the order; the extraction of the factor from a good even order is entirely classical.

Worked discrete-logarithm example. In $G = \mathbb{F}_7^\times = \langle 3 \rangle$ of order 6, let $h = 5 = 3^5$. The function $f(a, b) = 3^a 5^{-b}$ on $\mathbb{Z}_6^2$ hides $H = \langle (5, 1) \rangle$. A generating set for this subgroup determines a combination with second coordinate 1, and the matching first coordinate is 5 (mod 6). The hidden subgroup encodes the unknown logarithm as a slope.

Hypothesis checkpoint. The discrete-logarithm theorem assumes a cyclic group of known order, unique encodings, and efficient reversible group arithmetic. These are model assumptions, not consequences of the phrase “finite Abelian group.” The theorem therefore does not silently claim an algorithm for every possible black-box encoding. Likewise, the factoring proof uses the exact order returned by Algorithm 4.2; approximate phase data is converted to an exact checked integer before the classical square-root argument begins.

Source checkpoint. Shor’s primary paper supplies the polynomial-time reductions for factoring and discrete logarithm and explicitly charges reversible modular exponentiation. The probability calculation in the factoring proof above is reconstructed locally in two-adic language. For discrete logarithm, QRCC uses the clean Abelian-HSP kernel (4.27), aligned with Kitaev’s stabilizer formulation, rather than importing Pollard rho or baby-step–giant-step into the quantum proof architecture.

4.7 What the speedup is really using

The three quantum mechanisms in Chapters 3–4 should now be kept conceptually separate. Grover search accelerates an unstructured predicate by a two-dimensional amplitude rotation, and the hybrid lower bound shows that the quadratic scale is optimal in the oracle model. Simon’s algorithm turns every oracle call into a random linear equation over $\mathbb{F}_2$. Shor order finding instead constructs a unitary with arithmetic eigenphases and uses Fourier resolution plus continued fractions to convert those phases into an exact period.

The arithmetic circuit is as important as the Fourier transform. Phase estimation alone would be exponentially expensive if $U_a^{2^j}$ were implemented by 2^j literal multiplications. Modular arithmetic supplies a compressed representation of those powers through the classically known constants $a^{2^j} \bmod N$. The algorithmic speedup therefore comes from the compatibility of three layers:

$$\text{efficient reversible arithmetic} \longrightarrow \text{Fourier/phase sampling} \longrightarrow \text{exact classical reconstruction.} \quad (4.31)$$

Dropping any one of these layers obscures where the polynomial-time claim is actually proved.

4.8 Reader checkpoint and exercises

1. Prove the recurrence and determinant identity (4.1) for continued-fraction convergents from the matrix product

$$\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_j & 1 \\ 1 & 0 \end{pmatrix}.$$

Then identify exactly why the strict constant $1/2$ in Lemma 4.1 is sufficient for the parity argument.

2. For $N = 21$ and $a = 2$, write the orbit eigenvectors (4.7) explicitly and verify their fourier orthogonality and eigenvalues. Determine the reduced denominator q_s for every $s \in \mathbb{Z}_6$.
3. Prove directly that $q_s = r / \gcd(s, r)$. Then, for a fixed prime power $p^e \parallel r$, prove that the p^e factor is missing from q_s exactly when $p \mid s$.
4. Replace the crude bound in (4.13) by $\sum_p p^{-2} < 0.453$ over primes. Compute the resulting improved constant success probability for a pair of phase samples. Explain why the coarse $1/4$ denominator-success bound is nevertheless convenient for a theorem statement.
5. Give a reversible-circuit sketch for multiplying an n -bit residue by a fixed constant $b \pmod{N}$. Identify the ancilla registers, where modular reduction occurs, and how compute-copy-uncompute returns the workspace to zero.
6. For $G = \mathbb{Z}_{12}$ and $H = \{0, 4, 8\}$, compute $H^\perp$ explicitly under the characters $\chi_y(x) = e^{2\pi i xy/12}$. Verify equation (4.19) by direct summation.
7. Fill in the group-theoretic detail in the factoring proof: prove that the kernel of $(\mathbb{Z}/p^e\mathbb{Z})^\times \rightarrow \mathbb{F}_p^\times$ has order p^{e-1} and hence odd order for odd p . Deduce that the Sylow two-subgroup of the unit group is cyclic from the cyclicity of $\mathbb{F}_p^\times$.
8. In a cyclic group of order 2^s , count the elements whose order has two-adic valuation exactly j and rederive (4.22). Show that the probability in (4.23) can be as large as $1/2$ when there are exactly two prime-power components.
9. In the DLP reduction, prove that any generating set (a_j, b_j) of the subgroup (4.27) satisfies (4.28). Conversely, show that if a collection of points in H has second coordinates with gcd 1 together with m , then those points generate an element $(x, 1)$ even if they do not generate all of H .
10. Compare the order-finding presentation in this chapter with direct period finding on the state $M^{-1/2} \sum_k |k\rangle |a^k \bmod N\rangle$. Derive the same rational-approximation condition and explain precisely where the phase-estimation eigenbasis (4.7) is hidden in the direct Fourier-sampling calculation.

4.9 Transition to quantum complexity classes

Chapters 1–4 have supplied states and measurements, circuit uniformity and compilation, oracle algorithms and lower bounds, and now arithmetic quantum algorithms. The next chapter changes scale: rather than analyzing one algorithm, it defines the class BQP and the reduction conventions needed to compare quantum computation with classical randomized and deterministic complexity classes. The amplification and uniformity clauses in that definition will inherit the resource discipline fixed in the present chapters.

Chapter 5

BQP, Counting Bounds, and the PH Frontier

5.1 How to read this chapter

The first four chapters treated particular quantum algorithms. This chapter changes level: the object is now an entire complexity class. Three bookkeeping choices become load-bearing. First, a BQP machine is a *uniform finite-description* computation, not a circuit whose real-valued gates can hide input-specific advice. Second, bounded error is part of the class definition but the numerical constants $2/3$ and $1/3$ are not: repetition can drive the error exponentially low with polynomial overhead. Third, upper bounds such as $\text{BQP} \subseteq \text{PP}$ are statements about the sign of an exactly defined classical counting quantity. They therefore require a precision normalization before quantum amplitudes are converted into signed path counts.

The final section is deliberately different. BQP versus the polynomial hierarchy is a dated frontier, not a local theorem. The Raz–Tal oracle separation is a theorem about relativized/black-box computation; it is strong evidence that quantum computation can escape the polynomial hierarchy, but it is not an unrelativized proof that $\text{BQP} \not\subseteq \text{PH}$.

5.2 Learning goals

After this chapter the reader should be able to define BQP and Promise-BQP in the uniform circuit model; state classical-oracle and bounded-error quantum reduction conventions without confusing them; amplify any inverse-polynomial BQP gap to exponentially small error by a coherent majority construction; simulate a BPP computation quantumly; convert a rationalized quantum path sum into a GapP sign test; prove $\text{BQP} \subseteq \text{PP} \subseteq \text{PSPACE}$; and distinguish the unrelativized BQP–PH question from oracle separations and Forrelation evidence.

5.3 BQP, promise problems, and reduction conventions

Definition 5.1 (BQP and quantum polynomial-time reductions). Fix once and for all a finite gate set of the kind admitted by Definition 2.1, together with its polynomial-time uniform compiler. Gate-set changes among the standard efficiently universal finite models are understood through Theorems 2.2–2.3; an approximation tolerance is part of the resource accounting rather than an uncharged real parameter.

A language $L \subseteq \{0, 1\}^*$ belongs to BQP if there are polynomials s, a and a polynomial-time uniform family of circuits $\{C_n\}$ with at most $s(n)$ gates and $a(n)$ clean ancillas such that, on input

$$|x\rangle|0^{a(n)}\rangle, \quad |x| = n,$$

a designated output qubit measured in the computational basis satisfies

$$x \in L \implies \Pr[C_n(x) = 1] \geq \frac{2}{3}, \quad x \notin L \implies \Pr[C_n(x) = 1] \leq \frac{1}{3}. \quad (5.1)$$

The circuit family itself is the finite object generated from 1^n ; the input string is supplied only through its computational-basis input wires. Intermediate computational-basis measurements and polynomial-time classical control give the same class by deferred measurement, but the definition above uses a final classical output bit.

A *promise problem* is a pair $\Pi = (\Pi_{\text{yes}}, \Pi_{\text{no}})$ of disjoint sets. It belongs to Promise-BQP if one uniform polynomial-size family obeys the inequalities in (5.1) on the two promised sets; behavior outside $\Pi_{\text{yes}} \cup \Pi_{\text{no}}$ is unrestricted. More generally, a family has *error at most* $\varepsilon(n)$ if each promised input of length n is answered incorrectly with probability at most $\varepsilon(n)$.

For a total language A , the class BQP^A augments the circuit model by the standard reversible membership oracle

$$O_A|z, b\rangle = |z, b \oplus A(z)\rangle. \quad (5.2)$$

Each occurrence of O_A is one oracle call, the circuit may make only polynomially many calls, and all nonoracle gates still obey the ordinary uniform finite-gate cost model. This is the oracle convention used by the BQP–PH frontier below.

Finally, let Π and Γ be promise problems. A *bounded-error quantum many-one reduction* from Π to Γ is a polynomial-time uniform polynomial-size quantum transducer—that is, a uniform quantum circuit family with a designated computational-basis output register—that, on classical input x , outputs a classical string Y such that for some fixed $\eta < 1/2$,

$$x \in \Pi_{\text{yes}} \implies \Pr[Y \in \Gamma_{\text{yes}}] \geq 1 - \eta, \quad x \in \Pi_{\text{no}} \implies \Pr[Y \in \Gamma_{\text{no}}] \geq 1 - \eta. \quad (5.3)$$

An output outside the promise of Γ counts as reduction failure. Deterministic polynomial-time many-one reductions remain the classical notion from Volume II. The two notions are not silently identified.

Why the reduction convention is asymmetric. A randomized or quantum map can fail before the target problem is even invoked. Equation (5.3) therefore charges outputs outside the target promise as failures and keeps the reduction error separate from the target solver error. If a reduction has error η and the target solver has error δ , sequential composition has error at most $\eta + \delta$ by the union bound. Chapter 3 amplification then lets either error budget be made negligible when the underlying gap is bounded away from $1/2$.

Classical input and output are conventions, not physical restrictions. The state during the computation may of course be an arbitrary superposition or entangled state. “Classical input” means only that the input register starts in a computational-basis state $|x\rangle$; “classical output” means that the decision is the outcome of a specified final computational-basis measurement. This is exactly the interface needed to compare a quantum circuit family with the language classes of Volume II.

5.4 Amplification without changing the class

Theorem 5.2 (BQP amplification). *Let Π be a promise problem. Suppose a polynomial-time uniform polynomial-size quantum circuit family accepts every $x \in \Pi_{\text{yes}}$ with probability at least $1/2 + \gamma(n)$ and every $x \in \Pi_{\text{no}}$ with probability at most $1/2 - \gamma(n)$, where $\gamma(n) \geq 1/p(n)$ for some polynomial p .*

For every polynomially bounded integer function $q(n) \geq 1$, there is a polynomial-time uniform polynomial-size family for Π with error at most $2^{-q(n)}$. It is enough to use an odd number

$$m(n) \geq \frac{2(q(n) + 1) \ln 2}{\gamma(n)^2} \quad (5.4)$$

of independent copies and accept exactly when a strict majority of their output bits are 1. In particular, the constants $2/3$ and $1/3$ in the definition of BQP may be replaced by any inverse-polynomial gap around $1/2$, and BQP admits exponentially small error with polynomial overhead.

Proof. Fix an input of length n and write $\gamma = \gamma(n)$. Run m copies on disjoint work registers. The joint state is a tensor product of the m individual output/work states. Hence if the m output qubits were measured separately, their bits $X_1, \dots, X_m$ would be independent Bernoulli variables with a common success parameter bounded on the appropriate side of $1/2$. Rather than measure them early, compute their strict-majority bit reversibly into one clean ancilla using an ordinary polynomial-size classical Boolean circuit made reversible by compute–copy–uncompute. Measuring only that final ancilla produces exactly the same majority distribution.

For a yes-instance let $p \geq 1/2 + \gamma$ and $X = \sum_i X_i$, so $\mu = mp$. Failure means $X \leq m/2$. Write

$$\frac{m}{2} = (1 - \delta)\mu, \quad \delta = \frac{p - 1/2}{p}.$$

The lower-tail Chernoff bound from Theorem 2.4 gives

$$\Pr[X \leq m/2] \leq \exp\left(-\frac{\mu\delta^2}{2}\right) = \exp\left(-\frac{m(p - 1/2)^2}{2p}\right) \leq e^{-m\gamma^2/2}. \quad (5.5)$$

For a no-instance, $p \leq 1/2 - \gamma$. If $p = 0$ the majority accepts with probability zero. Otherwise set $\delta = (1/2 - p)/p$ and use the general upper-tail form of the same theorem:

$$\Pr[X \geq m/2] \leq \exp\left(-\frac{\mu\delta^2}{2 + \delta}\right) = \exp\left(-\frac{m(1/2 - p)^2}{p + 1/2}\right) \leq e^{-m\gamma^2}. \quad (5.6)$$

Thus both error probabilities are at most $e^{-m\gamma^2/2}$. The choice (5.4) makes this at most $2^{-(q+1)} < 2^{-q}$.

The circuit-size overhead is m copies plus a polynomial-size reversible majority circuit, hence polynomial because $m = O(qp^2)$. A uniform generator first emits the m disjoint copies and then emits the fixed reversible-majority construction, so uniformity is preserved. The defining BQP family already uses a fixed finite gate alphabet. If one instead begins with an ideal continuous-gate description, compile the base circuit only accurately enough to retain, say, half of the inverse-polynomial gap and then amplify that fixed compiled family; exponentially accurate gate synthesis is not required merely because the final decision error is exponentially small. $\square$

Worked error budget. For the standard BQP gap $2/3$ versus $1/3$, one has $\gamma = 1/6$. Taking $m \geq 72(q + 1) \ln 2$ odd copies is already sufficient for error at most 2^{-q} . The numerical constant is not optimized; the important point is linear growth in q for a constant base gap.

Hypothesis checkpoint. Independence comes from using disjoint copies with fresh clean ancillas. Reusing a work register without uncomputing it can correlate the trials and invalidates the Bernoulli-product calculation. Likewise, “repeat until confident” is not a uniform polynomial-time construction unless the repetition count is fixed as a polynomial function of the input length and requested error exponent.

Source checkpoint. Bernstein–Vazirani’s quantum complexity framework establishes bounded-error quantum polynomial time as a robust class and explicitly supports exponentially small error and subroutine composition. The finite-circuit Chernoff calculation above is local: no separate quantum concentration theorem is being imported.

5.5 Signed counting and rational quantum path sums

The next theorem compares a quantum acceptance probability with a PP majority. Two elementary classical interfaces are needed first.

A function $g : \{0, 1\}^* \rightarrow \mathbb{Z}$ is a *GapP function* if

$$g = f_+ - f_- \quad (5.7)$$

for two $\#P$ functions f_+, f_- . By the closure properties proved in Theorem 9.2, GapP is closed under addition, subtraction, polynomially many products, and sums over polynomial-length strings. For example, $(f - g)(h - k) = fh + gk - fk - gh$ is again a difference of $\#P$ functions.

Lemma 5.3 (PP as the sign of a GapP function). *A language L belongs to PP if and only if there is a GapP function g such that*

$$x \in L \implies g(x) > 0, \quad x \notin L \implies g(x) \leq 0. \quad (5.8)$$

Proof. If a PP machine has exactly $2^{r(n)}$ binary branches and $A(x)$ of them accept, then $g(x) = 2A(x) - 2^{r(n)}$ is GapP and has exactly the sign required by (5.8).

Conversely write $g = f - h$ with $f, h \in \#P$. Pad their witnessing nondeterministic machines so that each uses exactly $r(n)$ binary choices. A new machine uses one leading choice bit. On leading bit 0 it simulates the f -machine and accepts exactly on its accepting paths. On leading bit 1 it simulates the h -machine and accepts exactly on its *rejecting* paths. Its total number of accepting branches is

$$f(x) + (2^{r(n)} - h(x)),$$

out of $2^{r(n)+1}$ branches, so it accepts on a strict majority exactly when $f(x) > h(x)$. $\square$

Lemma 5.4 (Gaussian-rational unitary approximation). *Fix a constant dimension d and a unitary $U \in U(d)$ whose entries can be approximated to b binary bits in time polynomial in b . For every rational $\varepsilon > 0$, one can compute in time polynomial in $\log(1/\varepsilon)$ an exactly unitary matrix $V \in U(d)$ whose entries lie in $\mathbb{Q}(i)$ and a phase $|\lambda| = 1$ such that*

$$\|V - \lambda U\|_{\text{op}} \leq \varepsilon. \quad (5.9)$$

The numerators and denominators of the entries of V have bit length polynomial in $\log(1/\varepsilon)$.

Proof. Choose a computable phase λ for which -1 is not an eigenvalue of λU ; only finitely many phases are forbidden. Its Cayley transform

$$A = (I - \lambda U)(I + \lambda U)^{-1} \quad (5.10)$$

is skew-Hermitian. Approximate the upper-triangular entries of A by Gaussian rationals, choose rational purely imaginary diagonal entries, and impose $A'_{ji} = -\overline{A'_{ij}}$. Thus A' is exactly skew-Hermitian with entries in $\mathbb{Q}(i)$ and can be made arbitrarily close to A . Put

$$V = (I - A')(I + A')^{-1}. \quad (5.11)$$

Because A' is skew-Hermitian, direct multiplication gives $V^*V = I$. Fixed-dimensional rational matrix inversion keeps all entries in $\mathbb{Q}(i)$, with polynomial bit complexity. The inverse Cayley map is continuous at A , so choosing the rational approximation to sufficiently many $O(\log(1/\varepsilon))$ bits gives (5.9). All operations occur in constant dimension, hence the claimed uniform running time and bit-length bound. $\square$

Why exact rational unitarity matters. Entrywise rounding of a unitary matrix need not remain unitary. Lemma 5.4 first approximates the skew-Hermitian Cayley parameter and then maps back to the unitary group exactly. The counting proof can therefore use exact rational arithmetic without changing the circuit into a nonphysical linear map.

5.6 Classical and counting upper bounds

We recall the classical class names at the point where they enter the statement. A language is in BPP when a polynomial-time randomized machine accepts yes-instances with probability at least $2/3$ and no-instances with probability at most $1/3$; ordinary repetition makes the particular constants inessential. A language is in PP when a polynomial-time balanced nondeterministic machine has a strict majority of accepting branches exactly on yes-instances, as equivalently characterized by Lemma 5.3. Finally, PSPACE is the class decidable by deterministic Turing machines using polynomial work space, with no polynomial-time requirement. These are the Volume-II/IX conventions recalled here so that every class name in the next theorem is defined before use.

Theorem 5.5 (Classical and counting upper bounds for BQP). *The standard uniform classes satisfy*

$$\text{BPP} \subseteq \text{BQP} \subseteq \text{PP} \subseteq \text{PSPACE}. \quad (5.12)$$

The middle containment remains valid in the finite uniform circuit model of Definition 5.1; it does not assume exact access to arbitrary real or complex gate entries.

Proof. Step 1: BPP $\subseteq$ BQP. Let a BPP machine on n -bit inputs use exactly $r(n)$ fair random bits and compute the deterministic predicate $M(x, z)$. Prepare $r(n)$ qubits in the uniform superposition

$$2^{-r(n)/2} \sum_{z \in \{0,1\}^{r(n)}} |z\rangle$$

using Hadamard gates, reversibly compute $M(x, z)$ into an output qubit, and measure that qubit. Its acceptance probability is exactly the BPP acceptance probability. Reversible evaluation of a polynomial-time classical computation has polynomial overhead by the standard compute-copy-uncompute construction. If the chosen finite BQP alphabet does not contain these ideal gates exactly,

the compilation theorem from Chapter 2 approximates the complete simulation to a sufficiently small constant error, preserving the $2/3$ versus $1/3$ gap.

Step 2: $\text{BQP} \subseteq \text{PP}$. Let $L \in \text{BQP}$ and let C_n be its uniform finite-gate family, with $T(n)$ gates after padding. For a fixed input length, apply Lemma 5.4 to each elementary gate type at accuracy

$$\varepsilon_j = \frac{1}{96T(n)}. \quad (5.13)$$

Replacing each gate by its exactly unitary Gaussian-rational approximation changes the overall circuit, up to an irrelevant product of global phases, by operator norm at most $1/96$. If $|\psi\rangle$ and $|\phi\rangle$ are the two final normalized states, then

$$|\langle\psi|P|\psi\rangle - \langle\phi|P|\phi\rangle| \leq 2\|\psi - \phi\|_2 \leq \frac{1}{48} \quad (5.14)$$

for the accepting projector P . Thus the rationalized circuit has acceptance probability at least $2/3 - 1/48 > 1/2$ on yes-instances and at most $1/3 + 1/48 < 1/2$ on no-instances.

For gate j , choose a positive common denominator d_j for all entries of its rationalized local matrix and write every entry as

$$\frac{a + ib}{d_j}, \quad a, b \in \mathbb{Z}.$$

Put $D = \prod_{j=1}^T d_j$. The uniform rationalization procedure and fixed local dimension imply that D has polynomially many bits and is computable in polynomial time from 1^n .

Fix a final computational-basis string y . Expanding the circuit by inserting computational-basis resolutions of the identity between gates gives

$$\langle y|C'_n|x, 0^a\rangle = \frac{A_y(x)}{D}, \quad (5.15)$$

where $A_y(x)$ is a sum, over exponentially many computational paths, of products of polynomial-bit Gaussian integers. Its real and imaginary parts are GapP functions of (x, y) . To see this directly, a nondeterministic machine guesses one computational path, computes its signed integer real and imaginary weights in polynomial time, and realizes each nonnegative weight $w < 2^{b(n)}$ as exactly w accepting branches by guessing a $b(n)$ -bit integer u and testing $u < w$. Positive and negative path weights are counted by separate $\#P$ machines, and their differences give the two GapP components.

Let Y_1 be the set of final basis strings whose designated output bit is 1. By GapP closure under products and exponentially indexed sums over polynomial-length strings,

$$F(x) := \sum_{y \in Y_1} ((\Re A_y(x))^2 + (\Im A_y(x))^2) \quad (5.16)$$

is GapP. Equation (5.15) and the Born rule give the *exact* rationalized acceptance probability

$$p'(x) = \frac{F(x)}{D^2}. \quad (5.17)$$

The nonnegative polynomial-time computable integer D^2 is itself a $\#P$ function: guess a sufficiently long binary integer u and accept exactly when $u < D^2$. Hence

$$H(x) := 2F(x) - D^2 \quad (5.18)$$

is GapP. By (5.14), $H(x) > 0$ for $x \in L$ and $H(x) < 0$ for $x \notin L$. Lemma 5.3 therefore gives $L \in \text{PP}$.

Step 3: $PP \subseteq PSPACE$. Let a PP machine use exactly $r(n)$ binary nondeterministic choices and run in polynomial time. A deterministic machine enumerates all $2^{r(n)}$ choice strings in lexicographic order, simulates the PP machine on each, and maintains an $(r(n) + 1)$ -bit counter of accepting branches. The enumeration may take exponential time, but it uses only the choice string, one polynomial-size simulation workspace, and the polynomial-size counter. Comparing the final count with $2^{r(n)-1}$ therefore decides the PP language in polynomial space. $\square$

What the counting proof did not assume. The proof never asks a PP machine to manipulate an arbitrary real amplitude exactly. The constant BQP gap first permits a rational *unitary* approximation with polynomially many bits. Only then are amplitudes expanded as finite signed counts. This order is essential: replacing a quantum gate by an arbitrary rounded matrix and then counting its paths would not prove anything about the original unitary computation.

Relation to stronger known upper bounds. Adleman–DeMarrais–Huang established the classical upper bound $BQP \subseteq PP$ using a rational-amplitude quantum model. Fortnow–Rogers sharpened the counting viewpoint to $BQP \subseteq AWPP$ and consequently obtained further lowness results for PP. The present chapter proves only the registered chain (5.12); AWPP is useful source context but is not inserted as an undeclared extra class in the theorem statement.

Worked path-counting example. For the one-qubit circuit H^2 on $|0\rangle$, the two computational paths to output 1 have amplitudes $+1/2$ and $-1/2$ and cancel exactly. A signed count records this cancellation as a zero numerator, whereas an ordinary nonnegative path count would incorrectly regard the two paths as reinforcing. GapP is the classical device that preserves precisely this interference sign information.

5.7 Dated frontier: BQP versus the polynomial hierarchy

Remark 5.6 (BQP versus PH, audited 2026-09-03). The unrelativized relation between BQP and the classical polynomial hierarchy PH remains unresolved at this checkpoint. In particular, this book does *not* assert an unrelativized separation $BQP \not\subseteq PH$.

There is, however, a sharp black-box theorem. Raz–Tal, first at STOC 2019 and in the expanded JACM 2022 version, construct an explicit distributional/Forrelation-type black-box problem for which a quantum algorithm makes one query and runs in $O(\log N)$ time while bounded-depth quasipolynomial-size classical circuits have vanishing distinguishing advantage. Their reductions imply

$$\text{Promise-BQP} \not\subseteq \text{Promise-PH} \quad \text{in the black-box model,} \quad (5.19)$$

and hence there exists an oracle O such that

$$BQP^O \not\subseteq PH^O. \quad (5.20)$$

This settles the long-standing oracle-separation problem, not the unrelativized class separation.

Historically, Aaronson’s BQP-versus-PH program introduced Fourier Checking/Forrelation as the central candidate and obtained an oracle *relation* separation together with nonoracle quantum-logtime versus AC^0 consequences. Raz–Tal supplied the later decision/oracle separation. The logical boundary is therefore:

$$\text{Forrelation evidence} \implies \text{black-box/oracle separation} \not\Rightarrow \text{unrelativized BQP-PH separation.} \quad (5.21)$$

An oracle separation rules out a proof of $\text{BQP} \subseteq \text{PH}$ that relativizes uniformly to all oracles, but it does not exclude a nonrelativizing unrelativized containment proof. Conversely, it is not itself a language in the ordinary unrelativized world separating the two classes.

Why this frontier belongs immediately after the PP upper bound. Theorem 5.5 places BQP inside PP, a powerful counting class, but that upper bound does not locate BQP inside PH. The Raz–Tal theorem shows that any such containment, if true unrelativized, would have to use genuinely nonrelativizing structure. Thus the frontier is not a decorative open problem: it marks a concrete proof barrier between the classical hierarchy developed in Volume II and the quantum class defined here.

5.8 Reader checkpoint and exercises

1. Show that a $3/4$ versus $1/4$ acceptance convention defines the same BQP class as (5.1). Give an explicit odd repetition count that converts one convention to the other.
2. Suppose a Promise-BQP solver has error δ and a bounded-error quantum many-one reduction has error η . Prove the $\eta + \delta$ composition bound, carefully counting an output outside the target promise as reduction failure.
3. In Theorem 5.2, derive the no-instance exponent (5.6) directly from the general upper-tail Chernoff bound, including the case in which the multiplicative deviation parameter exceeds 1.
4. Give a reversible majority circuit using an accumulator that stores the Hamming weight of m output bits. Bound its size and number of clean ancillas polynomially in m , and explain how all scratch space is uncomputed after the majority bit is copied out.
5. Prove the GapP sign characterization in Lemma 5.3 using only the definition of PP from Volume II and balanced binary branching. Identify where strict majority is used.
6. Verify algebraically that if $A'^* = -A'$, then the Cayley inverse $(I - A')(I + A')^{-1}$ is unitary whenever $I + A'$ is invertible. Show that -1 cannot be an eigenvalue of a skew-Hermitian matrix and hence $I + A'$ is always invertible.
7. For a two-gate circuit with Gaussian-rational local matrices, write out the computational-path expansion explicitly and identify the common denominator D . Show how positive and negative real path weights can be represented as the difference of two $\#P$ counts.
8. Explain why the function in (5.16) is GapP even though the sum ranges over exponentially many accepting basis strings. Which closure operation from counting complexity is being used?
9. Give the polynomial-space enumeration algorithm proving $\text{PP} \subseteq \text{PSPACE}$ and write an explicit space bound in terms of the PP machine's running time and number of binary choices.
10. State in your own words the difference among: (a) an unrelativized separation $\text{BQP} \not\subseteq \text{PH}$; (b) an oracle O with $\text{BQP}^O \not\subseteq \text{PH}^O$; and (c) a promise black-box query separation. Explain exactly which of these is supplied by the Raz–Tal theorem and which remains open here.

5.9 Transition to quantum verification

BQP contains computations whose only certificate is the circuit's own interference pattern. The next chapter adds an explicit quantum witness and asks what a polynomial-time quantum verifier can certify. That move introduces QMA, its amplification problem, the local-Hamiltonian completeness interface, and a new proof-ownership boundary: unlike the BQP amplification above, witness-preserving QMA amplification cannot be replaced by naive independent repetition without discussing witness structure.

Chapter 6

QMA, Amplification, and Local Hamiltonians

6.1 How to read this chapter

BQP gives the verifier no untrusted quantum data. QMA changes that interface: a polynomial-time quantum verifier receives a polynomial-size quantum witness whose state may be chosen adversarially. This creates three distinctions that matter throughout the chapter. First, completeness is an existential statement about one witness whereas soundness quantifies over *every* witness. Second, ordinary repetition is not an innocent operation, because Merlin is not required to provide independent copies of a witness. Third, the local-Hamiltonian problem converts verification into ground-state energy; a short history-state slogan is not by itself a proof of the QMA-hardness promise gap.

The chapter therefore separates what is reconstructed locally from what remains source-owned. We prove an in-place QMA amplification theorem with no witness-length increase, prove membership of fixed-locality Hamiltonian in QMA, record the circuit-to-Hamiltonian hardness theorem at an explicit deep-input boundary, and then prove $\text{QMA} \subseteq \text{PP}$ using the finite-gate counting normalization already built in Chapter 5. The final section is a dated frontier: quantum PCP, QMA versus QCMA, and QMA(2) are related questions, but none is silently substituted for another.

6.2 Learning goals

After this chapter the reader should be able to define QMA, QCMA, QMA(2), local Hamiltonian, frustration freeness, and stoquasticity; turn a verifier into its positive-semidefinite acceptance operator; explain why copying an unknown QMA witness is not a valid amplification primitive; derive the two-projection rotation behind in-place amplification; state exactly which part of Local Hamiltonian completeness is source-owned; and prove $\text{QMA} \subseteq \text{PP}$ by tracing an amplified verifier over the witness space.

6.3 Witnessed quantum verification and Hamiltonian promises

Definition 6.1 (QMA, QCMA, QMA(2), and local Hamiltonians). Fix the polynomial-time uniform finite-gate circuit model of Definition 2.1. A *QMA verifier* for inputs of length n is a uniform polynomial-size unitary circuit V_x acting on a witness register W of $w(n)$ qubits and an ancilla

register A of $a(n)$ qubits initialized to $|0^{a(n)}\rangle$, followed by measurement of one designated output qubit. Write Π_{acc} for the orthogonal projector onto output value 1. The associated *acceptance operator on the witness space* is

$$M_x = (I_W \otimes \langle 0^a |) V_x^* \Pi_{\text{acc}} V_x (I_W \otimes |0^a\rangle). \quad (6.1)$$

It is positive semidefinite and satisfies $0 \leq M_x \leq I_W$. For every witness density operator ρ , its acceptance probability is $\text{Tr}(M_x \rho)$; hence the maximum acceptance probability over all mixed witnesses is $\lambda_{\max}(M_x)$ and is attained by a pure eigenstate.

Let $c, s : [0, \infty) \cap \mathbb{N} \rightarrow [0, 1]$ be polynomial-time computable functions with $c(n) - s(n) \geq 1/p(n)$ for some polynomial p . A promise problem $\Pi = (\Pi_{\text{yes}}, \Pi_{\text{no}})$ is in $\text{QMA}(c, s)$ if there is such a verifier satisfying

$$\begin{aligned} x \in \Pi_{\text{yes}} &\implies \exists \rho \quad \text{Tr}(M_x \rho) \geq c(|x|), \\ x \in \Pi_{\text{no}} &\implies \forall \rho \quad \text{Tr}(M_x \rho) \leq s(|x|). \end{aligned} \quad (6.2)$$

The class QMA means $\text{QMA}(2/3, 1/3)$; Theorem 6.3 below shows that any inverse-polynomial gap gives the same class. The class QCMA is defined identically except that Merlin's witness is required to be a computational-basis state $|z\rangle$ for a classical string z of polynomial length.

The class QMA(2) is defined by replacing the one-witness condition in (6.2) with two witness registers W_1, W_2 of polynomial size. Completeness permits one product witness $|\psi\rangle \otimes |\phi\rangle$ that is accepted with the stated probability, while soundness quantifies over every product state on $W_1 \otimes W_2$. Because acceptance is linear, the convex hull of product states—called the set of *separable* mixed states—has the same maximum acceptance as the pure product states. It is therefore equivalent to quantify over separable mixed witnesses. No equality among QMA, QCMA, and QMA(2) is built into these definitions.

Fix an integer $k \geq 1$. A *k-local Hamiltonian instance* consists of n qubits and

$$H = \sum_{j=1}^m H_j, \quad (6.3)$$

where $m \geq 1$ is polynomial in n , each H_j acts nontrivially on at most k specified qubits, $0 \leq H_j \leq I$, and the matrix entries of each local term are rational complex numbers of polynomial bit length in the computational basis. The instance also contains rational thresholds $a < b$ of polynomial bit length with $b - a \geq 1/p(n)$ for some polynomial p . The promise problem is

$$\Pi_{\text{yes}} : \lambda_{\min}(H) \leq a, \quad \Pi_{\text{no}} : \lambda_{\min}(H) \geq b. \quad (6.4)$$

Unless a different reduction is stated explicitly, *QMA-complete* means complete for promise-QMA under deterministic polynomial-time many-one reductions in the sense recalled in Chapter 5.

For positive-semidefinite terms, the Hamiltonian is *frustration free* if there is a state annihilated by every term H_j ; equivalently, every term and their sum have simultaneous ground energy zero. A local term, and hence a Hamiltonian term-by-term, is *stoquastic in the computational basis* if all off-diagonal matrix entries are real and nonpositive. Frustration-free and stoquastic Local Hamiltonian are separate promise families; the unrestricted completeness theorem below must not be transferred to those variants without an additional theorem.

Why the acceptance operator is the right abstraction. Equation (6.1) removes the verifier's initialized workspace while retaining exactly the quadratic form seen by Merlin. The verifier may entangle witness and ancilla arbitrarily, but all of its optimization over witnesses is encoded by one positive-semidefinite contraction M_x . This is the bridge to both amplification and the trace argument in $\text{QMA} \subseteq \text{PP}$.

A warning about repetition. For BQP, Chapter 5 simply prepared independent input/work registers and ran independent copies. In QMA, Merlin supplies only one witness unless the verifier asks for more qubits, and an unknown quantum state cannot be cloned. Even if the protocol requested several registers, a malicious Merlin need not supply independent copies unless the proof verifies the required structure. Marriott–Watrous amplification avoids this issue by recycling the original witness register.

6.4 Two projections and the verifier rotation

The spectral mechanism behind in-place amplification is elementary once the correct two projections are written down.

Lemma 6.2 (Two-projection phase blocks). *Let Δ_0, Δ_1 be orthogonal projectors on a finite-dimensional Hilbert space and put $A = \Delta_0 \Delta_1 \Delta_0$ on $\text{Ran}(\Delta_0)$. If a normalized $u \in \text{Ran}(\Delta_0)$ satisfies $Au = \lambda u$ with $0 < \lambda < 1$, define*

$$v = \frac{\Delta_1 u - \lambda u}{\sqrt{\lambda(1-\lambda)}}. \quad (6.5)$$

Then u, v are orthonormal, $\Delta_0 v = 0$, the plane $\text{span}\{u, v\}$ is invariant under both projectors, and in the ordered basis (u, v) ,

$$\Delta_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Delta_1 = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}. \quad (6.6)$$

For the reflections $R_j = 2\Delta_j - I$, the product $R_1 R_0$ has eigenvalues $e^{\pm 2i\theta}$ where $\lambda = \cos^2 \theta$ and $\theta \in (0, \pi/2)$. The endpoint cases $\lambda = 0, 1$ are the corresponding one-dimensional limiting blocks.

Proof. Since $\Delta_0 u = u$,

$$\langle u, \Delta_1 u \rangle = \langle u, \Delta_0 \Delta_1 \Delta_0 u \rangle = \lambda.$$

Also $\|\Delta_1 u\|^2 = \langle u, \Delta_1 u \rangle = \lambda$. Therefore $\Delta_1 u - \lambda u$ is orthogonal to u and has squared norm $\lambda - \lambda^2 = \lambda(1-\lambda)$, proving that v is normalized and orthogonal to u . Moreover,

$$\Delta_0 v = \frac{\Delta_0 \Delta_1 u - \lambda u}{\sqrt{\lambda(1-\lambda)}} = 0.$$

The definition rearranges to

$$\Delta_1 u = \lambda u + \sqrt{\lambda(1-\lambda)} v.$$

Applying Δ_1 once more and using $\Delta_1^2 u = \Delta_1 u$ determines

$$\Delta_1 v = \sqrt{\lambda(1-\lambda)} u + (1-\lambda)v,$$

which proves (6.6) and invariance.

Write $\lambda = \cos^2 \theta$. From (6.6),

$$R_1 R_0 = \begin{pmatrix} 2\lambda - 1 & -2\sqrt{\lambda(1-\lambda)} \\ 2\sqrt{\lambda(1-\lambda)} & 2\lambda - 1 \end{pmatrix} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}. \quad (6.7)$$

This is a planar rotation with eigenvalues $e^{\pm 2i\theta}$. At $\lambda = 0$ or 1 the same statement is read in the invariant endpoint subspaces, where the phase is respectively π or 0 . $\square$

Interpretation. A verifier acceptance probability λ becomes a rotation phase. Large acceptance means small $\theta = \arccos \sqrt{\lambda}$, and small acceptance means large θ . The amplification problem is thus converted into a phase-separation problem without manufacturing copies of the witness.

6.5 In-place QMA amplification

Theorem 6.3 (Marriott–Watrous QMA amplification). *Let Π have a polynomial-time uniform QMA verifier with witness length $w(n)$, completeness $c(n)$, soundness $s(n)$, and gap*

$$\gamma(n) := c(n) - s(n) \geq \frac{1}{p(n)} \quad (6.8)$$

for some polynomial p . For every polynomial-time computable, polynomially bounded integer function $q(n) \geq 1$, there is a polynomial-time uniform verifier for the same promise problem using the same $w(n)$ witness qubits and only polynomially many additional workspace qubits and gates such that

$$c'(n) \geq 1 - 2^{-q(n)}, \quad s'(n) \leq 2^{-q(n)}. \quad (6.9)$$

Thus the class is unchanged if the constants $2/3, 1/3$ are replaced by any inverse-polynomial completeness–soundness gap, or if exponentially small error is required.

Proof. Fix an input x , suppress its length from the notation, and let $V = V_x$. On witness plus ancilla define

$$\Delta_0 = I_W \otimes |0^a\rangle\langle 0^a|, \quad \Delta_1 = V^* \Pi_{\text{acc}} V, \quad R_j = 2\Delta_j - I. \quad (6.10)$$

Restricted to $\text{Ran}(\Delta_0)$, the operator $\Delta_0 \Delta_1 \Delta_0$ is exactly the acceptance operator M_x with its ancilla reattached. Hence Lemma 6.2 decomposes the relevant space into invariant blocks whose rotation $R := R_1 R_0$ has eigenphases $\pm 2\theta$ with

$$\lambda = \cos^2 \theta, \quad \theta(\lambda) = \arccos \sqrt{\lambda} \in [0, \pi/2]. \quad (6.11)$$

Put $\theta_c = \theta(c)$ and $\theta_s = \theta(s)$. Since

$$\left| \frac{d\theta}{d\lambda} \right| = \frac{1}{2\sqrt{\lambda}\sqrt{1-\lambda}} \geq 1 \quad (0 < \lambda < 1), \quad (6.12)$$

the mean-value theorem, with endpoint values understood by continuity, gives

$$\theta_s - \theta_c \geq c - s = \gamma. \quad (6.13)$$

Run phase estimation from Algorithm 3.2 on R . An eigenvalue $e^{\pm 2i\theta}$ has phase $\varphi = \pm\theta/\pi$ modulo 1, so its torus distance from 0 is θ/π . Choose a precision parameter p_0 with

$$2^{-p_0} \leq \frac{\gamma}{8\pi} \quad (6.14)$$

and failure parameter $\eta = 1/16$. The algorithm uses $t = p_0 + \lceil \log_2(17) \rceil$ control qubits and, with probability at least $15/16$, returns an estimate whose torus distance from the true phase is below 2^{-p_0} . Classify the block as accepting exactly when the estimated torus distance from 0 is below

$$\tau := \frac{\theta_c + \theta_s}{2\pi}. \quad (6.15)$$

By (6.13)–(6.14), every block with $\lambda \geq c$ lies on the accepting side with margin larger than the phase-estimation error, and every block with $\lambda \leq s$ lies on the rejecting side with the same property. Thus one spectral test has error at most $1/16$ on the relevant side of the promise. For completeness, Merlin may choose a top eigenvector of M_x ; for soundness, every spectral block has $\lambda \leq s$, so the same bound holds for an arbitrary witness by spectral decomposition.

No powered unitary is being declared free. If R^{2^j} is obtained by literal repetition, one phase test uses at most $2^t - 1 = O(1/\gamma)$ controlled applications of R . The reflection R_0 is a phase flip conditioned on all ancillas being zero. The reflection $R_1 = V^*(2\Pi_{\text{acc}} - I)V$ uses the original verifier, one output phase flip, and V^* . Controlled versions of these known polynomial-size circuits have polynomial overhead; Chapter 2 compiles any fixed auxiliary gates to the finite library. Because $1/\gamma$ is polynomially bounded, one spectral test has polynomial cost.

Repeat this phase test an odd number m of times with fresh control/work registers but the *same* original witness/system register. This is legitimate because each eigenspace of R is invariant under every controlled power of R . An intermediate phase readout may collapse a superposition of spectral blocks, but it cannot move amplitude from a yes-phase block into a no-phase block or conversely. Conditioned on any history, the next test therefore still has error probability at most $1/16$ on the applicable promise side. If $E_i \in \{0, 1\}$ records an erroneous classification, then for every $u > 0$,

$$\mathbb{E}[e^{uE_i} \mid E_1, \dots, E_{i-1}] \leq \frac{15}{16} + \frac{1}{16}e^u.$$

Iterating conditional expectation and taking $u = \ln 15$ yields

$$\Pr\left[\sum_{i=1}^m E_i \geq \frac{m}{2}\right] \leq \left(\frac{30}{16\sqrt{15}}\right)^m < e^{-m/2}. \quad (6.16)$$

Choose odd $m = O(q)$ so that the last quantity is at most $2^{-(q+2)}$. A reversible majority circuit on the m classification bits decides the amplified verifier. The deferred-measurement theorem (Theorem 2.4) lets the intermediate phase-estimation measurements and classical controls be postponed without changing the output distribution, so this remains a standard final-measurement QMA verifier and still uses only the original $w(n)$ witness qubits.

Finally compile every added fixed-dimensional gate so that the operator-norm error of the entire amplified circuit is at most $2^{-(q+4)}$. The telescoping error ledger of Chapter 2 makes this possible with polynomial overhead because the logarithm of the required precision is polynomial. Acceptance probability changes by at most twice the final-state norm error, which is below the slack between $2^{-(q+2)}$ and 2^{-q} . This proves (6.9). $\square$

Source checkpoint. Marriott and Watrous proved strong QMA error reduction with exponentially small completeness and soundness errors while preserving Merlin’s witness length. The proof above reconstructs the same resource-level conclusion through the two-projection spectral blocks and Chapter 3 phase estimation. It is not a claim that their original in-place circuit is being reproduced line for line; the source gate records the theorem parity, while the local argument certifies the statement used by QRCC.

Worked amplification scale. Suppose the base verifier has $c = 2/3$ and $s = 1/3$. Then $\gamma = 1/3$, so the literal controlled- R cost of one constant-error phase test is already constant. To drive the error below 2^{-100} , only $O(100)$ sequential phase tests are needed. The witness register is not replicated 100 times: all additional qubits are verifier workspace.

6.6 Local Hamiltonian as a verification problem

Theorem 6.4 (Fixed-locality Local Hamiltonian completeness — verified deep input). *For every fixed integer $k \geq 2$, the k -local Hamiltonian promise problem of Definition 6.1, with $0 \leq H_j \leq I$ and inverse-polynomial threshold gap $b - a$, is QMA-complete under deterministic polynomial-time many-one reductions. Membership in QMA is proved locally below. QMA-hardness is source-owned at the circuit-to-Hamiltonian spectral-gap layer: Kitaev’s construction gives a 5-local Hamiltonian with inverse-polynomial promise gap, and the Kempe–Kitaev–Regev locality reduction yields 2-local QMA-completeness; the case of any fixed $k > 2$ follows because a 2-local instance is also k -local.*

Local proof of membership in QMA. Let $H = \sum_{j=1}^m H_j$ and put $r = \lceil \log_2 m \rceil$, $M = 2^r < 2m$. Merlin supplies an n -qubit state ρ . Arthur prepares r fair Hadamard bits, interpreted as $j \in \{0, \dots, M-1\}$. If $j \geq m$, Arthur accepts. If $j < m$, Arthur performs the two-outcome POVM

$$\{I - H_{j+1}, H_{j+1}\}, \quad (6.17)$$

accepting on the first outcome. The ideal acceptance probability is exactly

$$p_{\text{acc}}(\rho) = 1 - \frac{\text{Tr}(H\rho)}{M}. \quad (6.18)$$

Because k is fixed and every local matrix has polynomial-bit entries, a classical uniform procedure can approximate a diagonalization or unitary dilation of the constant-dimensional effect H_{j+1} to inverse-polynomial operator norm and compile the resulting local measurement using Chapter 2. Choose the resulting acceptance-probability error below $(b - a)/(10M)$.

On a yes-instance, a ground state with energy at most a is accepted with probability at least $1 - a/M - (b - a)/(10M)$. On a no-instance, every witness has energy at least b and is accepted with probability at most $1 - b/M + (b - a)/(10M)$. The resulting completeness–soundness gap is at least $4(b - a)/(5M) > 2(b - a)/(5m)$, still inverse polynomial. Theorem 6.3 normalizes this verifier to the standard QMA constants. Hence fixed- k Local Hamiltonian belongs to QMA. $\square$

The source-owned hardness architecture. Let an amplified QMA verifier be written as a sequence $U_T \cdots U_1$. Kitaev adjoins a clock and forms

$$H_{\text{hist}} = H_{\text{in}} + H_{\text{prop}} + H_{\text{out}} + H_{\text{clock}}. \quad (6.19)$$

For an initial work state $|\psi_0\rangle$, the corresponding legal history state is

$$|\eta\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^T U_t \cdots U_1 |\psi_0\rangle \otimes |t\rangle, \quad U_0 = I. \quad (6.20)$$

A unary clock makes each propagation check local; together with the data gate touched at that time, the standard construction is 5-local. H_{clock} penalizes illegal unary clock strings, H_{in} checks clean input ancillas, H_{prop} enforces adjacent-time propagation, and H_{out} penalizes rejection at the final time. An accepting computation therefore gives a low-energy history state.

The converse direction is the deep part. One must prove that every low-energy state is sufficiently close to the legal propagation/history subspace and then transfer verifier soundness into an inverse-polynomial lower bound on the ground energy. This uses the spectral gap of the path-graph propagation Hamiltonian together with a geometric/projection estimate. Kempe–Kitaev–Regev then replace the 5-local interactions by 2-local ones while preserving an inverse-polynomial promise gap. QRCC records these steps as the exact source-owned boundary rather than compressing them into a paragraph labeled “proof.”

A three-time history example. For two gates U_1, U_2 , the legal history state is

$$\frac{1}{\sqrt{3}}(|\psi_0\rangle|0\rangle + U_1|\psi_0\rangle|1\rangle + U_2U_1|\psi_0\rangle|2\rangle). \quad (6.21)$$

The propagation terms do not ask whether any one snapshot is globally correct in isolation; they compare neighboring clock sectors so that zero propagation energy forces the entire sequence to fit together. The spectral gap of this clock-line consistency operator is why a state that is globally inconsistent cannot hide all of its error in a tiny number of time slices.

6.7 From one quantum witness to a PP majority

Theorem 6.5 (QMA is contained in PP). *In the uniform finite-gate model fixed in this volume,*

$$\text{QMA} \subseteq \text{PP}. \quad (6.22)$$

The containment also holds for promise-QMA in the usual promise-class sense: there is a PP language whose answers agree with the QMA promise on every promised input.

Proof. Let V_x be a QMA verifier with $w = w(|x|)$ witness qubits. Apply Theorem 6.3 with

$$\varepsilon = 2^{-(w+6)} \quad (6.23)$$

so that the amplified acceptance operator M_x satisfies

$$\begin{aligned} x \in \Pi_{\text{yes}} &\implies \lambda_{\max}(M_x) \geq 1 - \varepsilon, \\ x \in \Pi_{\text{no}} &\implies M_x \leq \varepsilon I. \end{aligned} \quad (6.24)$$

Feed the amplified verifier the maximally mixed witness $I/2^w$. It can be produced within a unitary circuit by preparing w Bell pairs and leaving one half of every pair unmeasured by the verifier. The acceptance probability is

$$p(x) = \frac{\text{Tr } M_x}{2^w}. \quad (6.25)$$

On a yes-instance, one eigenvalue contributes at least $1 - \varepsilon$, so $p(x) \geq (1 - \varepsilon)2^{-w}$. On a no-instance all 2^w eigenvalues are at most ε , so $p(x) \leq \varepsilon$. With

$$t = 2^{-(w+1)}, \quad (6.26)$$

we therefore have $p(x) > t$ on yes-instances and $p(x) < t$ on no-instances, with distance from the threshold at least $2^{-(w+6)}$.

Construct a witness-free quantum circuit B_x as follows. Flip one fair quantum coin. On the first branch, run the amplified verifier on the maximally mixed witness. On the second branch, generate $w + 1$ fair bits and accept unless all are zero; that branch has exact acceptance probability $1 - t$. Thus

$$\Pr[B_x \text{ accepts}] = \frac{1}{2}p(x) + \frac{1}{2}(1 - t) = \frac{1}{2} + \frac{p(x) - t}{2}. \quad (6.27)$$

Its acceptance probability is therefore strictly above $1/2$ on yes-instances and strictly below $1/2$ on no-instances, with a known inverse-exponential bias of at least $2^{-(w+7)}$.

It remains to turn this strict quantum sign into a PP sign without hiding arbitrary real constants. Apply the Gaussian-rational unitary approximation lemma from Chapter 5 to every fixed-dimensional gate of B_x at sufficient precision that the total acceptance-probability perturbation is below $2^{-(w+9)}$.

The circuit has polynomial size and w is polynomial, so the number of precision bits and the finite-gate compilation cost are polynomial. The strict sign around $1/2$ survives.

The path-sum construction in the proof of Theorem 5.5 now applies verbatim to the exactly Gaussian-rational circuit: after clearing polynomial-bit denominators, twice the accepting numerator minus the total denominator squared is a GapP function whose sign is positive on yes-instances and negative on no-instances. Lemma 5.3 converts that sign into a PP language. Its behavior outside the original QMA promise may be chosen arbitrarily, which is exactly the promise-class containment claim. $\square$

Why amplification is essential here. If we simply averaged an ordinary QMA verifier over the maximally mixed witness, a unique good witness could be diluted by 2^{-w} while no-instances still had constant acceptance. Strong in-place amplification first pushes every no-instance eigenvalue exponentially below that dilution scale, without increasing w . The trace of M_x then separates the two cases.

Source checkpoint. Marriott–Watrous use strong amplification to derive $\text{QMA} \subseteq \text{PP}$. QRCC’s local proof keeps the same witness-dilution idea but routes the final counting conversion through Chapter 5’s finite-gate Gaussian-rational normalization so that the PP majority is an exact integer sign computation.

6.8 Dated frontiers: quantum PCP, QCMA, and unentangled proofs

The following checkpoint is audited as of **2026-09-04**. It is a frontier record, not a theorem claiming that the questions have been resolved.

Hamiltonian quantum PCP. One standard Hamiltonian formulation asks whether there are constants $k \geq 2$ and $\varepsilon > 0$ such that it remains QMA-hard, for normalized fixed- k local terms $0 \leq H_j \leq I$, to distinguish low ground energy from ground energy larger by an *extensive* gap εm . The inverse-polynomial-gap completeness theorem above does not imply such a constant relative gap. The Hamiltonian quantum-PCP conjecture remains open at this checkpoint.

Games versions are not interchangeable with the Hamiltonian conjecture. Multiprover nonlocal games provide a different quantum-PCP-style interface. Modern results around $\text{MIP}^* = \text{RE}$ changed the landscape dramatically, and the games and Hamiltonian formulations cannot be identified by slogan. Natarajan–Nirkhe’s 2024 status analysis in particular records a flaw in an earlier claimed QMA energy-amplification route for the games version. QRCC therefore keeps the two formulations separate unless a stated reduction supplies the bridge.

QMA versus QCMA. The inclusion $\text{QCMA} \subseteq \text{QMA}$ follows immediately from Definition 6.1: a quantum verifier can receive a computational-basis witness. The reverse inclusion, and hence unrelativized equality versus separation, remains unresolved at this checkpoint. Oracle and distribution-testing separations provide evidence about possible distinctions, but they are not an unrelativized separation theorem. In particular, the classical-oracle separation of Bostanci–Haferkamp–Nirkhe–Zhandry now has a STOC 2026 proceedings reference; its relativized nature is unchanged.

QMA(2). By allowing Merlin to split a proof across two promised-unentangled registers, $\text{QMA} \subseteq \text{QMA}(2)$, while a direct exponential-time description/search argument gives the coarse upper bound $\text{QMA}(2) \subseteq \text{NEXP}$, where NEXP denotes nondeterministic exponential time. The exact power of standard $\text{QMA}(2)$ remains unresolved. Recent work gives strong amplification for $\text{QMA}(2)$ and shows that certain restricted or modified unentangled-proof models can behave very differently; those results do not by themselves settle $\text{QMA}(2) = \text{QMA}$ or $\text{QMA}(2) = \text{NEXP}$ for the standard class defined above.

No automatic implication matrix. Quantum PCP asks about robustness of local-Hamiltonian verification, QMA versus QCMA asks whether a quantum witness is more powerful than a classical one, and $\text{QMA}(2)$ asks what is gained by a separability promise across multiple witnesses. A result about one axis may motivate another, but QRCC records no implication among these frontier questions unless an explicit theorem proves it.

6.9 Reader checkpoint and exercises

1. Starting from (6.1), prove directly that $0 \leq M_x \leq I$ and that $\max_{\rho} \text{Tr}(M_x \rho) = \lambda_{\max}(M_x)$. Explain why a pure witness suffices for completeness.
2. Give an explicit two-dimensional example of Lemma 6.2 with $\lambda = 3/4$. Write $R_1 R_0$ numerically and identify its two eigenphases.
3. In Theorem 6.3, verify the derivative bound (6.12) and show carefully that phase-estimation accuracy (6.14) leaves a nonzero classification margin around (6.15).
4. Why does requesting m copies of a QMA witness change the verifier interface? Contrast “Merlin sends m registers” with “Arthur has m independent copies of the same unknown state.”
5. For $H = \sum_{j=1}^m H_j$, derive (6.18). If the promise gap is $b - a = n^{-3}$ and $m = n^2$, what asymptotic inverse-polynomial verification gap is obtained before amplification, ignoring the small implementation slack and using $m \leq M < 2m$?
6. For the three-time history state (6.21), write an idealized propagation penalty whose kernel contains exactly vectors satisfying the two transition constraints. Explain why a soundness proof needs a spectral gap estimate rather than only this kernel calculation.
7. Recheck every inequality in (6.24)–(6.27) for $w = 5$. Determine an explicit lower bound on the PP bias after the fair-branch construction.
8. Explain why the proof of $\text{QMA} \subseteq \text{PP}$ would be invalid if exponentially precise gate constants were simply assumed as free advice. Identify the two places where Chapter 5’s uniform rationalization prevents this loophole.
9. Give one sentence each distinguishing the Hamiltonian quantum-PCP conjecture, the games quantum-PCP program, QMA versus QCMA, and $\text{QMA}(2)$. For each pair, state why a result about one does not automatically settle the other.
10. Consider a diagonal 2-local Hamiltonian. Show that its off-diagonal entries are zero and hence it is stoquastic. Explain why this observation does not imply that arbitrary stoquastic Local Hamiltonian has the same complexity as unrestricted Local Hamiltonian.

Transition. The next chapter changes the communication interface rather than the witness interface. Quantum communication complexity asks how much information must cross a partition when local computation is cheap; the state/channel conventions from Chapter 1 and the query/complexity bookkeeping developed in Chapters 3–6 will be reused there.

Chapter 7

Quantum Communication Complexity

7.1 How to read this chapter

Communication complexity asks a deliberately different question from circuit or query complexity. Alice and Bob each hold part of an input; local computation and local quantum memory are free, while every system that crosses the cut is charged. This makes the resource model part of every theorem statement. A shared Bell pair is not a transmitted qubit, a classical bit is not a free side channel in the ordinary quantum-communication model, and a simultaneous-message protocol is not an interactive protocol with the messages merely written in a different order.

The chapter begins by fixing those conventions. Teleportation and superdense coding then provide exact resource identities that are used as model-conversion checks, not as slogans saying that classical and quantum communication are “the same.” Quantum fingerprinting gives a first exponential separation in the uncorrelated simultaneous-message model. The final theorem records the tight $\Theta(\sqrt{n})$ bounded-error quantum communication complexity of set disjointness, while preserving the Razborov lower bound and the Aaronson–Ambainis upper bound as explicit source-owned deep inputs.

7.2 Learning goals

After this chapter the reader should be able to distinguish unassisted and prior-entanglement quantum communication, one-way, interactive, and simultaneous-message protocols; compute worst-case qubit cost; define exact and bounded-error complexity; write the distributional quantum information cost of an interactive protocol; verify teleportation and superdense coding by Bell-basis identities; derive the swap-test error in quantum fingerprinting from code distance; and state exactly which part of the disjointness $\Theta(\sqrt{n})$ theorem is source-owned.

7.3 Communication models and information cost

Definition 7.1 (Quantum communication complexity). Let $f : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}$ be a finite two-party function. Alice receives a classical input $x \in \mathcal{X}$ and Bob receives $y \in \mathcal{Y}$. A *two-party quantum communication protocol* is a finite alternating sequence of local quantum channels and transmitted finite-dimensional message registers. Local computation, local ancillas, and local memory are free. If message C_i has dimension 2^{q_i} , transmitting it costs q_i qubits; the protocol’s *quantum communication*

cost is

$$\text{QCC}(\Pi) := \sum_i q_i. \quad (7.1)$$

A classical bit sent through the communication channel may be encoded in orthogonal qubit states and therefore costs one qubit in this metric; there is no free classical side channel unless a theorem explicitly introduces a mixed classical/quantum resource ledger.

In the *unassisted* model, the initial private state is a product state independent of (x, y) . In the *prior-entanglement* model, denoted by a superscript $*$, Alice and Bob may initially share an arbitrary finite entangled state $\eta_{T_A T_B}$ independent of (x, y) ; the preparation of that state is not charged to (7.1). Prior entanglement alone carries no input-dependent signal because its reduced state at either party is fixed before the inputs are supplied.

A protocol is *one-way from Alice to Bob* if only Alice sends a message and Bob produces the output. It is *r-message interactive* if at most r nonempty messages cross the cut, in alternating directions, and one designated party produces the output after the final message. In the *simultaneous-message passing* (SMP) model, Alice and Bob each send one message to a referee who has neither input and then measures the joint messages. Unless stated otherwise, QRCC's quantum-SMP model gives Alice and Bob neither shared randomness nor shared entanglement with one another or with the referee.

For $0 \leq \varepsilon < 1/2$, a protocol *computes f with error at most ε* if for every (x, y) its output equals $f(x, y)$ with probability at least $1 - \varepsilon$. Define

$$Q_\varepsilon(f) := \min_{\Pi} \text{QCC}(\Pi), \quad Q_\varepsilon^*(f) := \min_{\Pi \text{ with prior entanglement}} \text{QCC}(\Pi), \quad (7.2)$$

with analogous superscripts 1 and $\parallel$ for the one-way and SMP restrictions. The exact quantities use $\varepsilon = 0$. For a family f_n , asymptotic communication complexity refers to the worst-case cost as a function of n .

There is also a distributional information measure for an interactive protocol. Let μ be a distribution on (X, Y) and purify the classical input state as

$$|\Psi_\mu\rangle_{XYR} = \sum_{x,y} \sqrt{\mu(x,y)} |x\rangle_X |y\rangle_Y |x,y\rangle_R. \quad (7.3)$$

Run the protocol coherently on this input and on any input-independent prior entanglement. Immediately before message C_i is received, let B_{i-1} denote Bob's entire local register when Alice sends, and let A_{i-1} denote Alice's entire local register when Bob sends. For any tripartite state ρ_{UVW} define the *conditional quantum mutual information*

$$I(U : V \mid W)_\rho := H(UW)_\rho + H(VW)_\rho - H(W)_\rho - H(UVW)_\rho. \quad (7.4)$$

The *quantum information cost* of Π on μ is

$$\text{QIC}(\Pi, \mu) := \frac{1}{2} \sum_{i: A \rightarrow B} I(C_i : R \mid B_{i-1}) + \frac{1}{2} \sum_{i: B \rightarrow A} I(C_i : R \mid A_{i-1}). \quad (7.5)$$

This is a distributional quantity and is not the definition of worst-case communication cost. The factor $1/2$ is the standard fully quantum normalization: since $I(C_i : R \mid B_{i-1}) \leq 2H(C_i) \leq 2q_i$ and similarly in the other direction,

$$\text{QIC}(\Pi, \mu) \leq \text{QCC}(\Pi). \quad (7.6)$$

Interface with Volume XII. Volume XII's classical communication model charges transmitted classical bits. Equation (7.1) instead charges transmitted qubits. The two metrics can be compared only after a simulation theorem specifies all extra resources. Teleportation and superdense coding below are exactly such resource identities.

Why the reference register appears in information cost. The reference R remembers a purification of the input distribution. A message can be locally mixed while still carrying correlations about the input, and conditional mutual information measures those correlations relative to what the receiver already holds. The inequality (7.6) is immediate from finite-dimensional entropy bounds, but the deeper operational theory of quantum information complexity is not needed for the remaining proofs in this chapter.

7.4 Bell pairs as exact communication resources

Write

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\beta_{ab}\rangle = (I \otimes X^b Z^a)|\Phi^+\rangle, \quad a, b \in \{0, 1\}. \quad (7.7)$$

The four $|\beta_{ab}\rangle$ are an orthonormal basis of two qubits, the *Bell basis*. One shared copy of $|\Phi^+\rangle$ is called one *ebit*. An ebit is a pre-shared resource, not a transmitted qubit.

Theorem 7.2 (Teleportation and superdense coding). *The following exact resource conversions hold for a noiseless channel.*

1. **Teleportation.** *One arbitrary input qubit at Alice, one ebit shared by Alice and Bob, and two classical bits sent from Alice to Bob suffice to transfer the input state exactly to Bob. The protocol destroys Alice's original copy and sends no qubit carrying the unknown state across the cut.*
2. **Superdense coding.** *One ebit shared by Alice and Bob and one qubit sent from Alice to Bob suffice to transmit two classical bits exactly.*

Consequently a q -qubit protocol can be simulated exactly by a protocol using q fresh shared ebits and $2q$ transmitted classical bits, by teleporting each communicated qubit in its original direction. This consequence is a mixed-resource simulation; it does not say that the standard qubit communication cost of the original protocol is zero.

Proof. For teleportation, let Alice's unknown input be $|\psi\rangle_Q = \alpha|0\rangle + \beta|1\rangle$ and let A, B be the two halves of a shared $|\Phi^+\rangle_{AB}$. Alice applies $\text{CNOT}_{Q \rightarrow A}$ and then a Hadamard gate to Q . A direct expansion gives

$$(H_Q \otimes I_{AB}) \text{CNOT}_{Q \rightarrow A} |\psi\rangle_Q |\Phi^+\rangle_{AB} = \frac{1}{2} \sum_{a,b \in \{0,1\}} |a, b\rangle_{QA} X^b Z^a |\psi\rangle_B. \quad (7.8)$$

Alice measures Q, A in the computational basis, obtaining two classical bits (a, b) , and sends those bits to Bob. Conditioned on the outcome, Bob holds $X^b Z^a |\psi\rangle$. Applying $Z^a X^b$ returns $|\psi\rangle$ up to the irrelevant global phase $(-1)^{ab}$. Each pair (a, b) occurs with probability $1/4$, independent of $|\psi\rangle$, so the classical message does not reveal the unknown state itself. The measurement irreversibly removes Alice's coherent copy, consistent with the no-cloning theorem.

For superdense coding, Alice and Bob again share $|\Phi^+\rangle_{AB}$. To transmit (a, b) , Alice applies $X^b Z^a$ to her half and sends that one qubit to Bob. Up to a global phase, the joint state becomes

one of the four Bell states $|\beta_{ab}\rangle$. Because the Bell states are orthonormal, Bob can perform the Bell-basis projective measurement and recover (a, b) with certainty. Exactly one qubit crossed the cut and exactly one ebit was consumed.

Finally, teleport each message qubit of an arbitrary q -qubit interactive protocol in the same chronological order and direction. Local operations are unchanged. Each simulated transmission consumes one ebit and two classical bits, so the total resource ledger is $(q \text{ ebits}, 2q \text{ classical bits})$, with identical input-output statistics. No classical communication is being declared free in the definition (7.1). $\square$

No-signalling check. Before Alice's two teleportation bits arrive, Bob's reduced state is $I/2$, independent of $|\psi\rangle$. Similarly, acting on one half of an ebit cannot change the other half's reduced state. The protocols exploit correlations plus an actual transmitted message; they do not permit signalling using entanglement alone.

Worked basis check. If $|\psi\rangle = |1\rangle$ and Alice's teleportation measurement returns $(a, b) = (1, 0)$, Bob holds $Z|1\rangle = -|1\rangle$ and his correction Z restores $|1\rangle$. If the result is $(0, 1)$, he holds $X|1\rangle = |0\rangle$ and applies X . These are state-independent correction rules, not a tomography procedure.

7.5 Quantum fingerprints for equality

For $x, y \in \{0, 1\}^n$, define

$$\text{EQ}_n(x, y) = 1 \iff x = y. \quad (7.9)$$

The simultaneous-message setting is crucial: Alice and Bob each send a single short state to a referee and do not share randomness or entanglement.

Theorem 7.3 (Quantum fingerprinting for equality). *There is an absolute constant C such that for every $n \geq 1$ and every $0 < \varepsilon < 1/2$ there is a quantum SMP protocol for EQ_n with no shared randomness or prior entanglement, perfect completeness, error at most ε on unequal inputs, and total communication*

$$Q_\varepsilon^\parallel(\text{EQ}_n) \leq C \log(n+1) \log(1/\varepsilon). \quad (7.10)$$

In particular, constant-error equality has an $O(\log n)$ -qubit quantum fingerprinting protocol.

Proof. Fix a binary code

$$E : \{0, 1\}^n \longrightarrow \{0, 1\}^m \quad (7.11)$$

with $m \leq C_0 n$ and relative minimum distance at least a fixed $\delta > 0$; for definiteness one may take any constant $0 < \delta < 1/2$ and then choose C_0 from the Gilbert–Varshamov theorem recorded as QRCC-III-THM-009. The only coding property used below is

$$x \neq y \implies d_H(E(x), E(y)) \geq \delta m. \quad (7.12)$$

Alice prepares the normalized fingerprint

$$|F_x\rangle := \frac{1}{\sqrt{m}} \sum_{j=1}^m |j\rangle |E(x)_j\rangle, \quad |F_y\rangle := \frac{1}{\sqrt{m}} \sum_{j=1}^m |j\rangle |E(y)_j\rangle, \quad (7.13)$$

and each sends the state to the referee. A fingerprint uses $\lceil \log_2 m \rceil + 1 = O(\log(n+1))$ qubits. Their overlap is exactly

$$\langle F_x | F_y \rangle = \frac{1}{m} \# \{j : E(x)_j = E(y)_j\}. \quad (7.14)$$

Thus equal inputs give overlap 1, whereas unequal inputs satisfy $0 \leq \langle F_x | F_y \rangle \leq 1 - \delta$.

The referee performs the swap test. For arbitrary pure states $|u\rangle, |v\rangle$, the standard Hadamard–controlled-SWAP–Hadamard circuit outputs the “equal” outcome with probability

$$p_{\text{sw}}(u, v) = \frac{1 + |\langle u | v \rangle|^2}{2}. \quad (7.15)$$

Indeed, immediately before measuring the control qubit the probability of outcome 0 is the squared norm of $(|u\rangle|v\rangle + |v\rangle|u\rangle)/2$, which is (7.15). Hence if $x = y$ the referee always accepts, while if $x \neq y$ then

$$p_{\text{sw}}(F_x, F_y) \leq \alpha_\delta := \frac{1 + (1 - \delta)^2}{2} < 1. \quad (7.16)$$

To reduce the unequal-input error, Alice and Bob independently prepare t copies of their own fingerprints and send all of them to the referee. No shared correlation between the senders is required. The referee runs t swap tests on paired copies and accepts only if all accept. Equal inputs still have acceptance probability one, and unequal inputs have probability at most α_δ^t . Taking

$$t = \left\lceil \frac{\log(1/\varepsilon)}{-\log \alpha_\delta} \right\rceil \quad (7.17)$$

gives error at most ε . The two senders together transmit $2t(\lceil \log_2 m \rceil + 1)$ qubits, yielding (7.10). $\square$

Why the extra code-value qubit is useful. One often sees phase fingerprints of the form $m^{-1/2} \sum_j (-1)^{E(x)_j} |j\rangle$. Formula (7.13) uses one additional qubit but makes the overlap equal to the fraction of code coordinates that agree, so a lower bound on Hamming distance immediately becomes a nonnegative upper bound on overlap. This avoids silently needing an upper as well as a lower distance bound.

A concrete scale. If the chosen code has relative distance $\delta = 1/4$, then unequal fingerprints have overlap at most $3/4$ and one swap test accepts them with probability at most $25/32$. Ten paired copies reduce that false-equality probability below $(25/32)^{10} < 0.085$, while the per-sender message remains only a constant multiple of $\log n$ qubits.

7.6 Set disjointness: the source-owned lower-bound frontier

Define

$$\text{DISJ}_n(x, y) = 1 \iff \forall i \in [n], x_i y_i = 0. \quad (7.18)$$

Unlike equality in SMP, disjointness permits full interaction here. The quantum speedup is quadratic rather than exponential.

Theorem 7.4 (Quantum communication complexity of disjointness — deep input). *For bounded error $1/3$,*

$$Q_{1/3}(\text{DISJ}_n) = \Theta(\sqrt{n}), \quad Q_{1/3}^*(\text{DISJ}_n) = \Theta(\sqrt{n}). \quad (7.19)$$

In particular, every bounded-error interactive quantum protocol for disjointness communicates $\Omega(\sqrt{n})$ qubits, even when Alice and Bob have arbitrary input-independent prior entanglement.

Ownership and proof architecture. The lower bound in Theorem 7.4 is retained as a verified deep input from Razborov’s symmetric-predicate communication theorem. Its key mechanism is not the elementary query polynomial method alone. One restricts to fixed Hamming-weight layers and symmetrizes the protocol’s acceptance matrix so that acceptance depends only on the intersection size $|x \cap y|$. Low communication controls the relevant matrix spectrum and yields a low-degree univariate approximation on the intersection parameter. A discrete approximation lower bound then shows that a polynomial bounded on the allowed intersection sizes cannot separate $|x \cap y| = 0$ from $|x \cap y| \geq 1$ with constant error at degree $o(\sqrt{n})$. This is the point at which the approximate-degree language of Theorem 7.4 is conceptually relevant, but Razborov’s communication-to-polynomial reduction remains source-owned rather than being replaced by a query lower-bound slogan. The source theorem explicitly allows prior entanglement.

The matching $O(\sqrt{n})$ upper bound is also source-owned at this checkpoint, via Aaronson–Ambainis. The earlier Buhrman–Cleve–Wigderson distributed-Grover simulation already gives the instructive near-tight bound $O(\sqrt{n} \log n)$: a Grover-style search is performed for an index with $x_i = y_i = 1$, and simulating one coherent distributed predicate query requires sending an $O(\log n)$ -qubit index/work register across the cut. The later protocol removes the logarithmic overhead. QRCC records the sharp theorem but does not compress that communication-efficient search construction into a paragraph and call it a local proof.

Why prior entanglement does not erase the lower bound. Teleportation can trade transmitted qubits for classical bits only by consuming entanglement, but it still requires actual communication. Razborov’s lower bound is formulated directly in the entanglement-assisted quantum model, so the $\Omega(\sqrt{n})$ statement is not inferred by an invalid reduction from the unassisted case.

Four-bit example. Let $x = 1010$ and $y = 0101$. Their coordinatewise AND is 0000, so $\text{DISJ}_4(x, y) = 1$. If y changes to 0110, then the third coordinate intersects and the answer becomes 0. A communication lower bound concerns the worst-case amount of information that must cross the cut to distinguish such cases for arbitrary n ; it is not a statement that either party cannot evaluate its own bits locally.

7.7 Model-comparison checkpoints

Entanglement-assisted versus unassisted complexity. Because an unassisted protocol is a special case of an entanglement-assisted one, $Q_\varepsilon^*(f) \leq Q_\varepsilon(f)$. No reverse inequality follows for free. Teleportation compares a qubit channel to a *classical* channel plus ebits, while superdense coding compares a classical message to a qubit channel plus ebits; neither identity removes communication from the problem.

SMP versus interaction. The fingerprinting theorem is powerful precisely because the referee receives two short uncorrelated quantum states. An interactive protocol has a different information flow and may adapt later messages to earlier ones. Lower bounds proved for SMP cannot be imported as interactive lower bounds without a theorem.

Query versus communication cost. A query algorithm acts on an oracle locally; a communication protocol must distribute the data needed to realize that oracle action. The BCW simulation exposes an $O(\log n)$ communication overhead per ordinary index query, while the sharp disjointness

upper bound requires a more delicate construction. Thus $Q(\text{OR}_n) = \Theta(\sqrt{n})$ in the query model is not, by itself, a proof of $Q(\text{DISJ}_n) = O(\sqrt{n})$ in communication complexity.

7.8 Reader checkpoint and exercises

1. In Definition 7.1, explain why allowing a free classical channel would make teleportation invalidate the intended meaning of qubit communication cost. Give a separate mixed-resource ledger instead.
2. Prove the entropy estimate $I(C : R \mid B) \leq 2H(C) \leq 2\log_2 \dim C$ and derive (7.6) message by message. Which part of the argument uses finite-dimensionality?
3. Expand both sides of (7.8) for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and verify every one of the four conditional Bob states. Track the possible global phase in the correction.
4. Starting from (7.7), prove that the four Bell states are orthonormal. Show explicitly how a CNOT and a Hadamard turn Bell-basis measurement into computational-basis measurement.
5. Suppose Alice sends three classical bits to Bob and they have unlimited prior ebits. Use superdense coding to give an explicit two-qubit transmission protocol. Why does the answer involve a ceiling rather than exactly $3/2$ qubits?
6. Use the finite Gilbert–Varshamov theorem to choose constants C_0 and $\delta > 0$ for (7.11)–(7.12). Which part of the fingerprint proof would fail if the code map were not injective?
7. Derive the swap-test probability (7.15) directly from the symmetric and antisymmetric projectors $(I \pm \text{SWAP})/2$.
8. For $\delta = 1/4$, determine the minimum number of paired fingerprints needed to push the unequal-input error below 10^{-3} . State the total qubit cost in terms of m .
9. Explain carefully why Theorem 7.4 alone is not a proof of the disjointness communication lower bound. Identify the missing communication-matrix/symmetrization interface owned by Razborov’s theorem.
10. The BCW simulation gives $O(\sqrt{n} \log n)$ communication from distributed Grover search, while the sharp theorem is $O(\sqrt{n})$. Explain why deleting the logarithm in the theorem statement without supplying a new communication construction would be a proof gap rather than a cosmetic asymptotic simplification.

Transition. The next chapter keeps the two-party quantum viewpoint but allows the verifier and prover to exchange a sequence of messages. Quantum interactive proofs therefore reuse the communication-register discipline of this chapter together with the verifier and amplification machinery developed in Chapters 5–6.

Chapter 8

Quantum Interactive Proofs and $\text{QIP} = \text{PSPACE}$

8.1 How to read this chapter

A quantum interactive proof has only one computationally bounded participant. The verifier must be a uniform polynomial-time quantum computation, but the prover is an information-theoretic adversary with unbounded computation and private quantum memory. This makes the model different from both QMA and ordinary quantum communication complexity. A QMA witness is sent once and then checked; a QIP prover may adapt later messages to the quantum systems returned by the verifier. Conversely, communication is not the resource being minimized: polynomially many polynomial-size messages are permitted, while verifier time remains polynomial.

The central equality

$$\text{QIP} = \text{PSPACE}$$

has one easy containment and one deep containment. The easy direction is local: classical interactive proofs are a special case of quantum interactive proofs, and Volume XIII already proved $\text{IP} = \text{PSPACE}$. The reverse direction is not replaced here by a slogan about “solving an SDP.” The relevant semidefinite programs have exponential matrix dimension, so a generic polynomial-time SDP algorithm would at best suggest an exponential-time simulation. The decisive theorem of Jain–Ji–Upadhyay–Watrous uses a special parallelizable matrix-multiplicative-weights approximation, after a three-message/public-coin normalization of quantum interactive proofs. Those two mechanisms remain explicit source-owned deep inputs.

8.2 Learning goals

After this chapter the reader should be able to define QIP with all prover/verifier quantifiers visible; distinguish the number of transmitted messages from verifier circuit size; define the quantum public-coin (QMAM) normal form; write the primal/dual conventions for a finite-dimensional semidefinite program; explain why prover actions impose linear reduced-state consistency constraints; solve the one-message acceptance SDP; prove $\text{PSPACE} \subseteq \text{QIP}$ from the classical theorem; and identify exactly where parallelization/public-coin normalization and matrix multiplicative weights enter the source-owned proof of $\text{QIP} \subseteq \text{PSPACE}$.

8.3 Quantum interactive-proof and SDP conventions

Definition 8.1 (Quantum interactive proofs, public coins, and SDP interface). Fix the finite-dimensional quantum-state conventions of Definition 1.1, the polynomial-time uniform finite-gate circuit convention of Definition 2.1, and the classical interactive-proof quantifier convention of Definition 1.1.

A *quantum interactive verifier* for an input $x \in \{0,1\}^*$ has a polynomial-size private register V , a polynomial-size message register M , and a polynomial-time uniform sequence of quantum circuits that alternately acts on $V \otimes M$ and transfers the message register to or from a prover. The verifier's private/work registers and any message register it initializes begin in prescribed computational-basis states. At the end the verifier performs a two-outcome measurement $\{\Pi_{\text{acc}}, I - \Pi_{\text{acc}}\}$ and accepts on the first outcome.

A *prover strategy* has a private register P of unrestricted size and, whenever it holds the message register, may apply an arbitrary quantum channel to $P \otimes M$ before returning M . There is no computational bound on this channel. By purifying the prover's private randomness and dilating its channels, one may equivalently optimize over isometric prover actions on a sufficiently large finite private space; this is a normal-form convenience, not a charge on the prover.

A *message* is one nonempty transmission of a quantum register across the prover–verifier cut. Thus an m -message protocol has at most m such transmissions; this convention counts individual transmissions, not verifier–prover pairs. Message registers, verifier work registers, verifier circuit size, and m are all bounded by polynomials in $|x|$ unless a theorem states a more restrictive bound. Prover computation and prover private memory are not charged.

For fixed verifier V_x and prover strategy P , let $\Pr[V_x^P \text{ accepts}]$ be the final measurement probability. A promise problem $\Pi = (\Pi_{\text{yes}}, \Pi_{\text{no}})$ is in $\text{QIP}(m, c, s)$ if there is a polynomial-time uniform verifier using at most $m(n)$ messages such that

$$\begin{aligned} x \in \Pi_{\text{yes}} &\implies \exists P \quad \Pr[V_x^P \text{ accepts}] \geq c(|x|), \\ x \in \Pi_{\text{no}} &\implies \forall P^* \quad \Pr[V_x^{P^*} \text{ accepts}] \leq s(|x|). \end{aligned} \tag{8.1}$$

The soundness quantifier ranges over every information-theoretically allowed quantum prover strategy. The class QIP permits polynomially many messages and uses the bounded-error normalization $c \geq 2/3$, $s \leq 1/3$. As usual, any alternative completeness–soundness pair invoked later must be stated explicitly rather than silently identified with this normalization.

A *quantum public-coin* or *quantum Arthur–Merlin* verifier sends only fresh uniformly random *classical* strings in its messages to the prover. The random string is revealed to the prover when sent; it is not a hidden quantum register or a coherent superposition of challenges. In the three-message *QMAM* normal form, Merlin sends a quantum message, Arthur samples and sends a uniformly random classical challenge, Merlin returns a second quantum message, and Arthur performs a polynomial-time quantum verification measurement. This public-coin restriction concerns verifier-to-prover messages; Merlin's messages may remain quantum.

For later optimization statements, let $\text{Herm}(d)$ denote the Hermitian $d \times d$ matrices, write $X \succeq 0$ for positive semidefiniteness, and use the Hilbert–Schmidt pairing $\langle A, B \rangle = \text{Tr}(AB)$ on Hermitian matrices. A finite-dimensional *semidefinite program* (SDP) in QRCC means a primal problem

$$\text{maximize } \langle C, X \rangle \quad \text{subject to } \Phi(X) = B, \quad X \succeq 0, \tag{8.2}$$

where Φ is a real-linear map between Hermitian matrix spaces, together with its dual

$$\text{minimize } \langle B, Y \rangle \quad \text{subject to } \Phi^*(Y) \succeq C. \tag{8.3}$$

Here Φ^* is defined by $\langle Y, \Phi(X) \rangle = \langle \Phi^*(Y), X \rangle$. A feasible primal value never exceeds a feasible dual value; this is *weak duality*. When the matrices have dimension $2^{\text{poly}(n)}$ but their entries and constraint maps are described by polynomial-size circuits or polynomial-space index procedures, we call the SDP *succinct*. An *additive δ -approximation* to its value is a number differing from the optimum by at most δ .

The SDP encoding of an interactive prover uses *transcript-consistency constraints*: an operation carried out only by the prover cannot change the reduced density operator on registers retained by the verifier. Such partial-trace equalities are linear constraints on transcript density matrices. Conversely, realizing every consistent sequence of reduced states by a prover strategy requires a purification/isometry argument; that strategy–SDP equivalence is part of the source-owned architecture of Theorem 8.2, not an extra theorem hidden inside this definition.

Three costs that must not be conflated. An m -message QIP protocol may transmit polynomially many qubits in each message. This does not assert a small communication-complexity bound: communication is simply not the resource classifying QIP. Likewise, the prover is unbounded even though the verifier is polynomial time. Finally, an exponential-dimensional SDP can have a polynomial-length *succinct description*; succinctness does not make an ordinary polynomial-time matrix algorithm run in polynomial time on the original input.

8.4 A one-message SDP that can be solved exactly

The QMA acceptance operator from Chapter 6 gives the smallest nontrivial example of the optimization interface. Let M be a positive semidefinite contraction on a d -dimensional witness space. The best one-message prover acceptance probability is

$$\omega(M) = \max\{\text{Tr}(M\rho) : \rho \succeq 0, \text{Tr } \rho = 1\}. \quad (8.4)$$

This is an SDP. Its scalar dual is

$$\min\{t : tI \succeq M\}. \quad (8.5)$$

The primal optimum is at most $\lambda_{\max}(M)$ because $M \preceq \lambda_{\max}(M)I$, and equality is attained by a unit eigenvector of the top eigenvalue. The dual has the same value because $tI \succeq M$ exactly when $t \geq \lambda_{\max}(M)$. Thus

$$\omega(M) = \lambda_{\max}(M). \quad (8.6)$$

The example is intentionally elementary: it shows how an unbounded prover becomes a convex optimization over density operators. In a multi-message QIP protocol, the variable must also encode compatibility of successive transcript states, which is where the larger strategy SDP appears.

Why exponential dimension is not yet a contradiction. If the witness has $w(n)$ qubits, then M is a $2^{w(n)} \times 2^{w(n)}$ matrix. Writing it down entry by entry already costs exponential time. The same issue persists for QIP transcript SDPs. The equality $\text{QIP} = \text{PSPACE}$ therefore needs an algorithm whose use of the succinctly represented matrices can itself be carried out in polynomial space; generic “SDP is polynomial time” reasoning is insufficient.

8.5 The complexity of quantum interactive proofs

Theorem 8.2 (Jain–Ji–Upadhyay–Watrous: QIP = PSPACE — verified deep input). *For promise problems under the bounded-error conventions of Definition 8.1,*

$$\text{QIP} = \text{PSPACE}. \quad (8.7)$$

The containment $\text{PSPACE} \subseteq \text{QIP}$ is proved locally below from Theorem 4.1. The reverse containment is source-owned at two decisive layers:

1. *reducing arbitrary polynomial-message quantum interactive proofs to a constant-message/public-coin normal form without losing a usable completeness–soundness gap; and*
2. *approximating the value of the resulting exponential-dimensional succinct SDP in polynomial space by a parallelized matrix-multiplicative-weights method, with all positivity, conditioning, and precision losses controlled inside the original bounded-error gap.*

Local proof of $\text{PSPACE} \subseteq \text{QIP}$. Volume XIII proved $\text{IP} = \text{PSPACE}$ in Theorem 4.1. Let $L \in \text{PSPACE}$ and take the classical polynomial-time interactive verifier supplied there. A quantum verifier simulates every classical probabilistic step by computational-basis gates and measurements, and sends every classical message as a computational-basis quantum state. An honest classical prover is therefore an allowed quantum prover, so completeness is unchanged.

For soundness, fix any quantum prover strategy against this basis-measuring verifier. Immediately before each classical message is read, the verifier measures the message register in the computational basis. Conditioned on the classical transcript produced so far, the quantum prover therefore induces an ordinary probability distribution over the next classical response. Such randomized behavior is a convex combination of classical response choices, and the information-theoretic soundness condition of the classical protocol already bounds every such strategy. Hence the same completeness and soundness parameters hold against arbitrary quantum provers. Thus $L \in \text{QIP}$. $\square$

Deep-input architecture for $\text{QIP} \subseteq \text{PSPACE}$. The reverse containment is not reproduced as a compressed proof. The literature-calibrated route has the following interfaces.

1. **Three-message and public-coin normalization.** Kitaev–Watrous proved that polynomial-round quantum interactive proofs can be parallelized to three messages with strong error control. Marriott–Watrous subsequently proved that three-or-more-message quantum Arthur–Merlin games have the full power of ordinary QIP; in particular $\text{QMAM} = \text{QIP}$, with Arthur’s message consisting only of public random classical bits. Jain–Ji–Upadhyay–Watrous use this public-coin characterization because it gives a particularly tractable optimization problem.
2. **Strategy value as an SDP.** For a fixed verifier and a fixed public challenge, the prover’s possible messages are represented by positive semidefinite transcript operators. Partial traces enforce that a prover action cannot change Arthur’s retained state. The acceptance probability is linear in these operators, so maximizing cheating acceptance becomes a succinct SDP. Purification/isometry arguments establish that the SDP does not introduce fictitious prover strategies.
3. **Penalty/conditioning step.** The exact consistency equalities are converted to a form suitable for multiplicative weights. This step is quantitative: violating a transcript constraint must incur enough penalty that an approximate optimizer cannot cross the completeness–soundness threshold. Inverse and square-root operations on positive operators must therefore be controlled rather than treated as formal matrix notation.

4. **Parallel matrix multiplicative weights.** A matrix-multiplicative-weights iteration approximately solves the structured SDP. The iteration count and required matrix precision are polynomial in the input length because only inverse-polynomial/additive accuracy inside a constant bounded-error gap is needed. The matrix operations used by each iteration—including matrix exponentials and spectral thresholding to the required precision—have parallel algorithms of polynomial depth on exponentially large but succinct matrices. Polynomial depth in the logarithm of the matrix dimension corresponds to polynomial space for the original input, yielding $\text{QIP} \subseteq \text{PSPACE}$.

What the theorem does *not* say. The equality does not mean a QIP prover can be simulated efficiently, that an optimal prover strategy has a short classical description, or that the verifier can explicitly materialize the exponential-dimensional SDP. It says that the *decision problem* of whether the optimal acceptance probability lies on the yes side or the no side of a bounded gap can be solved in polynomial space.

8.6 A finite verifier thought experiment

Suppose a three-message public-coin verifier receives a first prover register A , chooses a uniformly random bit $r \in \{0, 1\}$, sends r , receives a second register B , and finally measures a projector Π_r on $A \otimes B$ together with its private workspace. For each r , let $X_r \succeq 0$ denote the post-response transcript state seen by the verifier. The first message was sent before r was chosen, so the part of X_0 and X_1 corresponding to the verifier's retained copy of the first-message interface must have the same reduced density operator. That equality is linear in X_0, X_1 , while the average acceptance

$$\frac{1}{2}\langle \Pi_0, X_0 \rangle + \frac{1}{2}\langle \Pi_1, X_1 \rangle \quad (8.8)$$

is linear as well. This is the geometric reason public-coin normalization is useful: adaptive prover behavior is converted into positive matrices tied together by linear consistency constraints. Equation (8.8) is an illustration, not the complete Jain–Ji–Upadhyay–Watrous SDP.

8.7 Reader checkpoint and exercises

1. Give a precise simulation showing that every BQP computation is a zero-prover-message special case of a QIP verifier. Then compare this with the one-prover-message QMA interface of Chapter 6.
2. Starting from (8.4), derive the scalar dual (8.5) directly from weak duality and prove again that both optima equal $\lambda_{\max}(M)$.
3. Explain why a prover-local channel cannot change the verifier's reduced density operator. Which direction of the converse requires a purification/isometry theorem?
4. A transcript SDP has matrix dimension $D = 2^{p(n)}$. Explain why an algorithm polynomial in D is an exponential-time algorithm in n , while an algorithm implementable by circuits of depth $\text{poly}(\log D) = \text{poly}(n)$ can still be simulated in polynomial space.
5. In the local proof of $\text{PSPACE} \subseteq \text{QIP}$, identify exactly where the classical verifier's basis measurement prevents coherent quantum messages from violating classical soundness.

6. Compare private-coin QIP with QMAM. State which messages may be quantum in QMAM and which verifier message is required to be uniformly random classical data.
7. Why would an additive approximation error of 0.4 be useless for distinguishing completeness $2/3$ from soundness $1/3$? Give a sufficient constant additive accuracy and explain why the deep proof must budget every penalty and numerical-approximation error below this slack.

Chapter 9

Nonlocal Games, CHSH, and Entangled-Prover Complexity

9.1 How to read this chapter

A nonlocal game is simultaneously a finite optimization problem, a Bell experiment, and the one-round core of many multi-prover proof systems. The dangerous point is that the phrase “quantum correlation” is not a single mathematical model. Finite-dimensional tensor-product strategies, arbitrary spatial tensor-product strategies, the closure of finite-dimensional correlations, and commuting-operator strategies are related but are not interchangeable. Later complexity and operator-algebra consequences depend precisely on which set is being optimized over.

This chapter therefore fixes the correlation models before proving anything about CHSH. The CHSH game then serves as a completely solvable laboratory: the classical value is $3/4$, the tensor-product quantum value is $\cos^2(\pi/8)$, and the upper bound is proved locally by an exact operator-norm argument. The final section records the $\text{pre-MIP}^* = \text{RE}$ complexity milestones that made recursive compression possible. Those milestones are source-owned because their multilinearity tests, rigidity/self-testing, introspection, PCP interfaces, and compression soundness are far too substantial to replace responsibly by a few pages of prose.

9.2 Learning goals

After this chapter the reader should be able to define a finite nonlocal game and its classical, finite-dimensional tensor-product, spatial, closure-valued, and commuting-operator correlations; explain why a game value is a supremum rather than necessarily an attained maximum; distinguish $\mathcal{C}_q$ from $\mathcal{C}_{qa}$ and $\mathcal{C}_{qc}$; define synchronous games and synchronous correlations; convert a binary POVM into a Hermitian contraction and then into a reflection; prove Tsirelson’s CHSH bound; compute the optimal Bell-pair strategy entry by entry; and state the exact role of the Ito–Vidick, Ji, Natarajan–Wright, and Fitzsimons–Ji–Vidick–Yuen milestones in the road to the next chapter.

9.3 Nonlocal games and correlation models

Definition 9.1 (Nonlocal games and quantum correlation models). *A two-player nonlocal game*

$$G = (X, Y, A, B, \mu, V) \tag{9.1}$$

consists of finite question sets X, Y , finite answer sets A, B , a probability distribution μ on $X \times Y$, and a predicate $V : A \times B \times X \times Y \rightarrow \{0, 1\}$. The verifier samples $(x, y) \sim \mu$, sends x to Alice and y to Bob, receives answers a, b , and accepts exactly when $V(a, b \mid x, y) = 1$. After receiving their questions the players may not communicate.

A *correlation* is a family

$$p = (p(a, b \mid x, y))_{a,b,x,y} \quad (9.2)$$

of conditional probability distributions. The following sets use the same finite alphabets.

1. A *classical* or *local* correlation belongs to $\mathcal{C}_{\text{loc}}$ if there is a finite shared random variable λ with distribution ν and local response distributions $p_A(a \mid x, \lambda)$, $p_B(b \mid y, \lambda)$ such that

$$p(a, b \mid x, y) = \sum_{\lambda} \nu(\lambda) p_A(a \mid x, \lambda) p_B(b \mid y, \lambda). \quad (9.3)$$

Equivalently, $\mathcal{C}_{\text{loc}}$ is the convex hull of deterministic response tables.

2. A *finite-dimensional tensor-product quantum correlation* belongs to $\mathcal{C}_q$ if there are finite-dimensional Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, a density operator ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, and POVMs $\{A_a^x\}_{a \in A}$ and $\{B_b^y\}_{b \in B}$ such that

$$p(a, b \mid x, y) = \text{Tr}(\rho(A_a^x \otimes B_b^y)). \quad (9.4)$$

The local dimensions are finite but are not bounded as a function of the alphabet sizes unless a theorem states such a bound.

3. A *spatial tensor-product quantum correlation* belongs to $\mathcal{C}_{qs}$ if the same formula is allowed on arbitrary separable Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$. This is an exact tensor-product model; it is not, by definition, a closure operation.
4. The *closure-valued quantum correlation set* is

$$\mathcal{C}_{qa} := \overline{\mathcal{C}_q} \quad (9.5)$$

inside the finite-dimensional Euclidean space of conditional probability tables. Membership in $\mathcal{C}_{qa}$ therefore means that the correlation can be approximated arbitrarily well by finite-dimensional tensor-product strategies; it need not be attained by any one such strategy.

5. A *commuting-operator correlation* belongs to $\mathcal{C}_{qc}$ if there are a Hilbert space $\mathcal{H}$, a unit vector $|\psi\rangle \in \mathcal{H}$, POVMs $\{A_a^x\}_a$, $\{B_b^y\}_b$ on $\mathcal{H}$, and

$$[A_a^x, B_b^y] = 0 \quad \text{for every } a, b, x, y, \quad (9.6)$$

such that

$$p(a, b \mid x, y) = \langle \psi | A_a^x B_b^y | \psi \rangle. \quad (9.7)$$

The commuting condition replaces an explicit tensor-factor decomposition.

The elementary inclusions are

$$\mathcal{C}_{\text{loc}} \subseteq \mathcal{C}_q \subseteq \mathcal{C}_{qs} \subseteq \mathcal{C}_{qa} \subseteq \mathcal{C}_{qc}. \quad (9.8)$$

No equality in this display is to be inferred from notation alone. In particular, closure and commuting-operator questions become theorem-level issues later in Volume XV.

For any correlation class $\mathcal{C}$, define its value in G by the linear functional

$$\omega_{\mathcal{C}}(G) := \sup_{p \in \mathcal{C}} \sum_{x,y} \mu(x,y) \sum_{a,b} V(a,b \mid x,y) p(a,b \mid x,y). \quad (9.9)$$

Write $\omega(G)$ for the classical value, $\omega^*(G)$ for the finite-dimensional entangled value $\omega_{\mathcal{C}_q}(G)$, and $\omega^{\text{co}}(G)$ for the commuting-operator value. Since the payoff is a continuous linear functional, taking the Euclidean closure does not change the supremum:

$$\omega^*(G) = \omega_{\mathcal{C}_{qa}}(G). \quad (9.10)$$

Equation (9.10) does *not* say an optimal finite-dimensional strategy exists.

A game is *synchronous* when $X = Y$, $A = B$, and equal questions require equal answers: $V(a,b \mid x,x) = 0$ whenever $a \neq b$. A correlation on matching question/answer sets is *synchronous* if

$$p(a,b \mid x,x) = 0 \quad (a \neq b). \quad (9.11)$$

When a theorem concerns synchronous games it must specify whether the strategy is additionally required to be synchronous, rather than silently identifying the two notions.

Why four quantum models are carried explicitly. For a fixed finite game, replacing $\mathcal{C}_q$ by its closure does not change the number $\omega^*(G)$, but it can change whether value 1 is actually attained. Likewise, a commuting strategy may exist without an approximating finite-dimensional tensor strategy. These distinctions are invisible in CHSH because a 2-qubit strategy already attains the optimum; they become decisive for the undecidability and operator-algebra consequences in Chapter 10.

Binary answers and observables. For a binary POVM $\{M_0, M_1\}$ define its signed observable

$$M := M_0 - M_1 = 2M_0 - I. \quad (9.12)$$

Then $M = M^\dagger$ and $\|M\| \leq 1$. Conversely $M_0 = (I + M)/2$ and $M_1 = (I - M)/2$. If $M^2 = I$, the measurement is projective and M is a *reflection*. Any Hermitian contraction can be dilated to the reflection

$$\widehat{M} = \begin{pmatrix} M & \sqrt{I - M^2} \\ \sqrt{I - M^2} & -M \end{pmatrix} \quad (9.13)$$

on $\mathcal{H} \oplus \mathcal{H}$. Because M commutes with $\sqrt{I - M^2}$, one has $\widehat{M}^2 = I$; embedding a state in the first summand preserves its expectation of M . Thus, for the CHSH upper bound, binary POVMs may be replaced by reflections without changing the four correlators.

9.4 CHSH and Tsirelson's bound

The CHSH game has $X = Y = A = B = \{0, 1\}$, uniform questions, and winning predicate

$$a \oplus b = xy. \quad (9.14)$$

For binary observables A_0, A_1, B_0, B_1 , put

$$E_{xy} := \langle A_x \otimes B_y \rangle, \quad S := E_{00} + E_{01} + E_{10} - E_{11}. \quad (9.15)$$

Since $(-1)^{a \oplus b} = (-1)^{a+b}$, averaging the four winning indicators gives

$$\Pr[\text{win}] = \frac{1}{2} + \frac{S}{8}. \quad (9.16)$$

Theorem 9.2 (Tsirelson bound for CHSH). *For the CHSH game,*

$$\omega(\text{CHSH}) = \frac{3}{4}, \quad \omega^*(\text{CHSH}) = \cos^2\left(\frac{\pi}{8}\right) = \frac{1}{2} + \frac{1}{2\sqrt{2}}. \quad (9.17)$$

The entangled optimum is attained by a pair of qubits in a Bell state and projective binary measurements.

Proof. Classical value. By convexity it suffices to consider deterministic answers. Encode Alice's two answers by signs $\alpha_0, \alpha_1 \in \{\pm 1\}$ and Bob's by $\beta_0, \beta_1 \in \{\pm 1\}$. Their CHSH expression is

$$S_{\text{cl}} = \alpha_0(\beta_0 + \beta_1) + \alpha_1(\beta_0 - \beta_1). \quad (9.18)$$

Exactly one of the two parentheses is zero and the other is ± 2 , so $|S_{\text{cl}}| \leq 2$. Equation (9.16) therefore gives success probability at most $3/4$. The deterministic strategy $a = b = 0$ wins on the three question pairs other than $(1, 1)$, so the bound is attained.

Quantum upper bound. By the reflection dilation (9.13), assume without loss of correlators that $A_0^2 = A_1^2 = B_0^2 = B_1^2 = I$. On the joint Hilbert space define the CHSH operator

$$\mathcal{B} := A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1). \quad (9.19)$$

A direct expansion using the reflection identities gives

$$\mathcal{B}^2 = 4I - [A_0, A_1] \otimes [B_0, B_1]. \quad (9.20)$$

For contractions R, S , $\|[R, S]\| \leq 2\|R\|\|S\|$, hence each commutator in (9.20) has norm at most 2. Therefore

$$\|\mathcal{B}\|^2 = \|\mathcal{B}^2\| \leq 4 + \|[A_0, A_1]\| \|[B_0, B_1]\| \leq 8, \quad (9.21)$$

so $|S| = |\langle \mathcal{B} \rangle| \leq 2\sqrt{2}$. Substitution into (9.16) yields $\Pr[\text{win}] \leq \frac{1}{2} + \frac{1}{2\sqrt{2}}$.

Matching strategy. Let

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad A_0 = Z, \quad A_1 = X, \quad (9.22)$$

and

$$B_0 = \frac{Z + X}{\sqrt{2}}, \quad B_1 = \frac{Z - X}{\sqrt{2}}. \quad (9.23)$$

For $|\Phi^+\rangle$, $\langle Z \otimes Z \rangle = \langle X \otimes X \rangle = 1$ and $\langle Z \otimes X \rangle = \langle X \otimes Z \rangle = 0$. Hence

$$E_{00} = E_{01} = E_{10} = \frac{1}{\sqrt{2}}, \quad E_{11} = -\frac{1}{\sqrt{2}}, \quad (9.24)$$

so $S = 2\sqrt{2}$ and the upper bound is attained. $\square$

What the proof actually used. The upper bound did not assume qubits, pure states, or projective measurements at the start. Binary POVMs were converted to Hermitian contractions and then dilated to reflections. The dimension-independent estimate is therefore an operator inequality, not a special calculation in $\mathbb{C}^2 \otimes \mathbb{C}^2$. The qubit Bell pair appears only in the matching lower-bound strategy.

9.5 Worked CHSH density-matrix calculation

The optimal state from (9.22) has density matrix, in the ordered basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\rho_{\Phi^+} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}. \quad (9.25)$$

Using

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (9.26)$$

one checks directly that

$$\begin{aligned} \text{Tr}(\rho_{\Phi^+} Z \otimes Z) &= 1, & \text{Tr}(\rho_{\Phi^+} X \otimes X) &= 1, \\ \text{Tr}(\rho_{\Phi^+} Z \otimes X) &= 0, & \text{Tr}(\rho_{\Phi^+} X \otimes Z) &= 0. \end{aligned} \quad (9.27)$$

Substituting (9.23) therefore gives all four correlators in (9.24). The reduced state of either qubit is $I/2$, so every individual output is unbiased. For outcomes encoded by $(-1)^a$ and $(-1)^b$ this gives the full joint distribution

$$p(a, b \mid x, y) = \frac{1}{4} \left(1 + (-1)^{a+b} E_{xy} \right). \quad (9.28)$$

For each question pair, the winning parity has probability

$$\frac{1}{2} \left(1 + (-1)^{xy} E_{xy} \right) = \frac{1}{2} + \frac{1}{2\sqrt{2}} = \cos^2\left(\frac{\pi}{8}\right). \quad (9.29)$$

Thus the Bell-pair strategy does not merely achieve the correct average: all four question pairs have the same winning probability. By contrast, the deterministic classical strategy $a = b = 0$ wins three pairs with probability 1 and loses the pair $(1, 1)$ with probability 1.

9.6 Before $\text{MIP}^* = \text{RE}$: the compression staircase

The next chapter states the 2020 theorem that completely changes the complexity landscape. To understand why that theorem was plausible, one needs the sequence of increasingly strong entangled-prover lower bounds and compression results that preceded it. The following item is intentionally a source-frozen milestone bundle, not a claim that one short proof establishes all of these results.

Theorem 9.3 (Selected pre- $\text{MIP}^* = \text{RE}$ milestones — verified deep input). *Under the finite-dimensional entangled-prover convention of Definition 9.1, the pre-2020 literature establishes the following staircase.*

1. **Entanglement does not destroy classical MIP power.** *Ito–Vidick proved*

$$\text{NEXP} \subseteq \text{MIP}^*. \quad (9.30)$$

Their soundness analysis shows that the multilinearity architecture behind classical $\text{MIP} = \text{NEXP}$ can be made robust against entangled provers.

2. **Protocol compression makes tiny nonlocal games computationally hard.** *Ji’s compression theorem transforms an r -prover quantum interactive proof, for polynomially bounded r , into an $(r + 8)$ -player one-round nonlocal game whose questions have logarithmic length*

and whose answers have constant length, with perfect completeness and an inverse-polynomial completeness–soundness gap. In particular, inverse-polynomial approximation of the entangled value of such games is QMIP*-complete and hence NEXP-hard.

3. **The bounded-gap class already exceeds classical MIP.** Natarajan–Wright proved

$$\text{NEXP} \subseteq \text{MIP}^*, \quad (9.31)$$

where $\text{NEXP} = \text{NTIME}(2^{2^{\text{poly}(n)}})$. Since $\text{MIP} = \text{NEXP}$, this gives an unconditional exponential separation in computational power.

4. **Recursive compression reaches arbitrary iterated-exponential time with a shrinking gap.** Fitzsimons–Ji–Vidick–Yuen showed that, for an arbitrary time-constructible number of exponential iterations, the corresponding nondeterministic iterated-exponential-time language has a constant-prover polynomial-time entangled-prover proof with perfect completeness and a quantitatively controlled but generally vanishing completeness–soundness gap. Their recursive compression framework also yields an independent route to uncomputability phenomena for entangled game values and identifies the missing ingredients that the $\text{MIP}^* = \text{RE}$ theorem supplies.

The multilinearity/low-degree tests, rigidity and introspection mechanisms, PCP encodings, compression theorem, and all quantitative soundness bookkeeping in these statements remain source-owned.

Why the gap qualifier in the fourth item matters. A proof system with soundness $1 - 2^{-2^{\cdot}}$ is not automatically a standard bounded-gap MIP^* protocol. The 2019 iterated-compression theorem is therefore evidence about the power and self-reference of the model, but it must not be silently rewritten as a constant-gap class equality. The Natarajan–Wright inclusion (9.31), by contrast, is a standard bounded-gap MIP^* lower bound.

Where correlation-model topology enters. Slofstra proved that finite-dimensional tensor-product correlations are not closed: there are nonlocal games whose value can approach 1 arbitrarily closely with finite-dimensional strategies although no such strategy attains value 1. This is exactly why Definition 9.1 keeps $\mathcal{C}_q$ and its closure $\mathcal{C}_{qa}$ separate even though equation (9.10) gives the same game-value supremum. Chapter 10 will go further and compare $\mathcal{C}_{qa}$ with $\mathcal{C}_{qc}$ using the consequences of $\text{MIP}^* = \text{RE}$.

9.7 Executable finite laboratory

The reader-facing script `code/volume_15/quantum_circuit_lab.py` contains five finite diagnostics used across Chapters 3–9: phase-estimation sampling for an exactly representable eigenphase, the two-dimensional Grover rotation, Simon orthogonality sampling over $\mathbb{F}_2^n$, exact pure-state trace distance, and the CHSH correlator/value calculation above. These checks use only small state vectors and exact or high-precision finite arithmetic. They are regression tests for formulas and conventions, not substitutes for the asymptotic proofs.

9.8 Reader checkpoint and exercises

1. Prove directly from (9.3) that $\mathcal{C}_{\text{loc}}$ is the convex hull of deterministic response tables. Why is the set compact for fixed finite alphabets?

2. Show that every classical correlation is in $\mathcal{C}_q$ by encoding the shared random variable in a diagonal separable state. Then show that every finite-dimensional tensor-product correlation is commuting by letting Alice's and Bob's operators act on separate tensor factors.
3. Verify explicitly that the block operator (9.13) is Hermitian and squares to the identity. Explain why the first-block embedding preserves the expectation of M .
4. Expand (9.19) and derive (9.20) without skipping the signs. Which step would fail if Alice's operators did not commute with Bob's?
5. Starting from (9.25)–(9.27), calculate the four matrices $A_x \otimes B_y$ and verify every correlator in (9.24) by an explicit trace.
6. A sequence of finite-dimensional strategies has winning probabilities $1 - 2^{-d}$ as the local dimension parameter d grows. What can be concluded about $\omega^*(G)$? What cannot be concluded about existence of a perfect finite-dimensional strategy?
7. Explain the distinction between the standard bounded-gap inclusion $\text{NEEXP} \subseteq \text{MIP}^*$ and a recursive-compression theorem whose soundness approaches 1 as the simulated time scale increases.
8. In one paragraph each, identify the proof role of multilinearity/low-degree testing, rigidity or self-testing, introspection, and protocol compression. Do not call any of these ingredients “gap amplification” unless the cited theorem actually supplies a constant-gap transformation.

Chapter 10

Uncomputable Nonlocality and Quantum LDPC

10.1 How to read this chapter

This final chapter of Volume XV closes two arcs that at first look unrelated. The first begins with nonlocal games: the finite-dimensional tensor-product model reaches the recursively enumerable languages, while the commuting-operator model reaches the complementary class. The second begins with quantum error correction: sparse stabilizer checks can simultaneously have constant rate and linear distance, and suitable code Hamiltonians force every sufficiently low-energy state to have large circuit depth. In both arcs, the point is the same: finite local descriptions can encode global phenomena that are invisible to a naive finite-dimensional or shallow-circuit approximation.

The source policy is intentionally asymmetric. The computability reductions, compression machinery, operator-algebra implications, lifted-product code construction, and NLTS clustering argument are deep primary-source inputs and remain source-owned. By contrast, all interfaces used to state those results are defined here, and the elementary semidecision implications are proved locally. This prevents a reader from mistaking a theorem-level boundary for a slogan.

10.2 Learning goals

After this chapter, the reader should be able to define the normalized classes MIP^* and MIP^{co} without conflating tensor-product and commuting-operator correlations; explain why $\text{MIP}^* \subseteq \text{RE}$ is elementary once one can enumerate finite-dimensional strategies; state the strong game-value form of $\text{MIP}^* = \text{RE}$ and its undecidability consequences; distinguish the now-resolved theorem $\text{MIP}^{\text{co}} = \text{coRE}$ from the older frontier status recorded by its stable ID; define binary CSS stabilizer codes and quantum LDPC families; derive the corresponding commuting frustration-free code Hamiltonian; distinguish asymptotic goodness, local testability, NLTS, and the quantum PCP conjecture; and state precisely what the good-qLDPC and NLTS theorems do and do not prove.

10.3 Recognizable languages and the tensor-product multiprover class

Recall from Volume II that a language $L \subseteq \{0,1\}^*$ is *recursively enumerable* or *recognizable*, written $L \in \text{RE}$, if a deterministic Turing machine halts and accepts every $x \in L$ and never accepts any

$x \notin L$; on a no-instance it may run forever. The class **coRE** consists of complements of languages in **RE**. The halting language witnesses $\text{RE} \neq \text{coRE}$.

For the remainder of this section, a *uniform two-prover one-round game family* is a polynomial-time classical algorithm that, on input x , outputs a finite nonlocal game G_x in the sense of Definition 9.1. We use the following constant-gap normalization:

$$\begin{aligned} x \in L &\implies \omega^*(G_x) = 1, \\ x \notin L &\implies \omega^*(G_x) \leq \frac{1}{2}. \end{aligned} \tag{10.1}$$

The resulting class is denoted **MIP***. Fixed constants separated from one another by a positive gap lead to the same standard class after the usual completeness/soundness normalization; we use (10.1) because it is the form in which the undecidability consequence is cleanest.

Theorem 10.1 (**MIP* = RE** — verified deep input). *Under the finite-dimensional tensor-product convention of Definition 9.1,*

$$\text{MIP}^* = \text{RE}. \tag{10.2}$$

More concretely, there is an effective reduction that maps an instance x of a recognizable language to a two-player nonlocal game G_x such that

$$x \in L \implies \omega^*(G_x) = 1, \quad x \notin L \implies \omega^*(G_x) \leq \frac{1}{2}. \tag{10.3}$$

The hard inclusion $\text{RE} \subseteq \text{MIP}^$, including the quantum low-degree/Pauli tests, introspection, recursive compression, fixed-point construction, and constant-gap soundness analysis, remains source-owned.*

Local proof of the easy inclusion $\text{MIP}^ \subseteq \text{RE}$.* Fix a game G with rational verifier data. For each local dimension $d = 1, 2, \dots$, finite-dimensional states and POVMs can be approximated arbitrarily closely by matrices whose real and imaginary parts are rational. Enumerate such rational approximations, discard tuples that fail increasingly accurate positivity/normalization checks, and compute rigorous rational lower bounds on their winning probabilities. Dovetail this search over dimensions and precisions.

If $\omega^*(G) = 1$, some finite-dimensional strategy has value greater than $3/4$, and sufficiently accurate rational data eventually certify a value greater than $2/3$. If $\omega^*(G) \leq 1/2$, no such certificate can occur. Applying this procedure to G_x therefore recognizes the yes-instances of any language in the normalized class (10.1). This proves $\text{MIP}^* \subseteq \text{RE}$. The reverse inclusion is the deep content of Theorem 10.1. $\square$

What “infinite computational power” does and does not mean. The verifier remains an ordinary polynomial-time classical machine and each game is a finite object. The unbounded power comes from allowing arbitrarily large finite-dimensional shared entanglement across the family of strategies. Equation (10.2) does not say that one fixed finite-dimensional device can solve the halting problem, and it does not replace the supremum in ω^* by a maximum.

10.4 Undecidability and the correlation-model separation

Corollary 10.2 (Undecidability and operator-algebra consequences — verified deep input). *There is no algorithm that, on every promised finite two-player nonlocal game G , decides which of*

$$\omega^*(G) = 1, \quad \text{or} \quad \omega^*(G) \leq \frac{1}{2} \tag{10.4}$$

holds. Consequently there is no algorithm that uniformly approximates $\omega^*(G)$ to additive error strictly smaller than $1/4$ for all finite games. Moreover,

$$\mathcal{C}_{qa} \subsetneq \mathcal{C}_{qc}, \quad (10.5)$$

and, through the standard operator-algebra equivalence/implication chain relating Tsirelson’s problem to Connes embedding, the Connes embedding conjecture is false.

The elementary undecidability implication. If an algorithm decided the promise (10.4), composing it with the effective reduction (10.3) would decide the halting problem. Likewise, an additive- ε approximation with $\varepsilon < 1/4$ separates the intervals $\{1\}$ and $[0, 1/2]$ by thresholding at $3/4$. These deductions are local. The correlation-set separation and the operator-algebra consequence use the deep source-owned bridge described next.

Why closure versus commuting operators becomes algorithmic. The entangled value $\omega^*(G)$ is lower semicomputable: finite-dimensional strategies can be enumerated from below as in the preceding proof. The Navascués–Pironio–Acín semidefinite hierarchy provides a source-owned decreasing sequence of efficiently describable upper bounds converging to the commuting value $\omega^{\text{co}}(G)$. If $\mathcal{C}_{qa} = \mathcal{C}_{qc}$, then $\omega^*(G) = \omega^{\text{co}}(G)$ for every finite game, so matching lower and upper semicomputations would approximate the common value to any prescribed constant accuracy. This contradicts (10.4), forcing (10.5). The equivalence of this correlation question with the relevant operator-algebraic form of Connes embedding is not reproved here.

10.5 The commuting-operator class: a formerly open frontier now resolved

For a uniform game family $x \mapsto G_x$, replace the tensor-product value in (10.1) by the commuting value from Definition 9.1:

$$\begin{aligned} x \in L &\implies \omega^{\text{co}}(G_x) = 1, \\ x \notin L &\implies \omega^{\text{co}}(G_x) \leq \tfrac{1}{2}. \end{aligned} \quad (10.6)$$

The corresponding class is MIP^{co} . Here the superscript “co” abbreviates *commuting operator*; in coRE the prefix “co” means *complement*. They are unrelated pieces of notation.

Theorem 10.3 ($\text{MIP}^{\text{co}} = \text{coRE}$ — resolved frontier, verified source). *As of the 2026 literature calibration of this volume,*

$$\text{MIP}^{\text{co}} = \text{coRE}. \quad (10.7)$$

The result, due to Junqiao Lin, appeared in the STOC 2026 proceedings after its 2025 preprint. The stable identifier QRCC-XV-F-040 is retained because this item entered the book as a frontier gate before the theorem was publication-calibrated.

Local proof of the easy inclusion $\text{MIP}^{\text{co}} \subseteq \text{coRE}$. For a promised no-instance of (10.6), the NPA hierarchy supplies successively tighter certified upper bounds on $\omega^{\text{co}}(G_x)$ that converge to a number at most $1/2$. Eventually one of these bounds drops below $3/4$, providing a terminating certificate that $x \notin L$. Thus the complement of L is recognizable, so $L \in \text{coRE}$. The reverse inclusion $\text{coRE} \subseteq \text{MIP}^{\text{co}}$ is the deep theorem: it adapts compression to commuting strategies through tracially embeddable strategies and a weak-compressibility framework, together with the needed testing and soundness machinery. Those ingredients remain source-owned. $\square$

Tensor and commuting models genuinely differ. Combining (10.2), (10.7), and $\text{RE} \neq \text{coRE}$ gives $\text{MIP}^* \neq \text{MIP}^{\text{co}}$. This is not a notational curiosity: the distinction between $\mathcal{C}_{qa}$ and $\mathcal{C}_{qc}$ changes the computational class represented by nonlocal games.

10.6 Quantum LDPC codes and code Hamiltonians

We next switch from interactive proofs to sparse quantum constraints. The bridge is conceptual rather than a reduction: both subjects ask what globally complicated behavior can be forced by locally specified tests.

Definition 10.4 (CSS quantum LDPC codes, code Hamiltonians, and NLTS). Let

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (10.8)$$

For $u = (u_1, \dots, u_n) \in \mathbb{F}_2^n$, set

$$X(u) := \bigotimes_{i=1}^n X^{u_i}, \quad Z(u) := \bigotimes_{i=1}^n Z^{u_i}. \quad (10.9)$$

The *Pauli weight* of a tensor-product Pauli operator is the number of tensor factors different from the identity.

A binary *CSS stabilizer code presentation* consists of matrices $H_X \in \mathbb{F}_2^{m_X \times n}$ and $H_Z \in \mathbb{F}_2^{m_Z \times n}$ satisfying

$$H_X H_Z^\top = 0. \quad (10.10)$$

Rows u of H_X specify commuting checks $X(u)$ and rows v of H_Z specify checks $Z(v)$. Condition (10.10) is exactly the parity condition that every X -check commutes with every Z -check. The *codespace* $\mathcal{Q}$ is their simultaneous $+1$ eigenspace. If $\dim \mathcal{Q} = 2^k$ and the minimum Pauli weight of an operator that preserves $\mathcal{Q}$ but acts nontrivially on it is d , the code has parameters $[[n, k, d]]_2$.

A family of such presentations is a *quantum low-density parity-check (qLDPC) family* if there are constants w, Δ independent of n such that every row of H_X, H_Z has Hamming weight at most w and every column participates in at most Δ rows in total. It is *asymptotically good* if there are constants $r, \delta > 0$ for which

$$k_n \geq rn, \quad d_n \geq \delta n \quad (10.11)$$

for all sufficiently large block lengths in the family. An *explicit family* here means that, from the family index, a deterministic algorithm outputs the two check matrices in time polynomial in the total size of their written descriptions; no efficient decoding theorem is implied by this word.

Choose commuting independent stabilizer generators $S_1, \dots, S_m$ for a CSS code. Its *code Hamiltonian* is

$$H_{\mathcal{Q}} := \sum_{j=1}^m \Pi_j, \quad \Pi_j := \frac{I - S_j}{2}. \quad (10.12)$$

Each Π_j is a projector, the projectors commute, and the Hamiltonian is *frustration free*: its minimum energy is 0 and its ground space is exactly the codespace. For a qLDPC presentation, each term acts on $O(1)$ qubits and each qubit occurs in $O(1)$ terms.

An n -qubit state ρ is *depth- D trivial* if there are $m \geq n$ qubits and a depth- D circuit of bounded-fanin quantum gates that maps $|0\rangle^{\otimes m}$ to a pure state whose reduced state on n designated output qubits is ρ ; gates may connect arbitrary pairs of qubits unless geometric locality is explicitly

imposed. The *preparation depth* of ρ is the minimum such D . A family of local Hamiltonians $H_n = \sum_{j=1}^{m_n} h_{n,j}$ with $0 \preceq h_{n,j} \preceq I$ has the *no low-energy trivial states (NLTS) property* if there is a constant $\varepsilon > 0$ such that every family of states $\{\rho_n\}$ with

$$\mathrm{Tr}(H_n \rho_n) \leq \varepsilon m_n \quad (10.13)$$

requires preparation depth tending to infinity with n ; a theorem may state a quantitative lower bound such as $\Omega(\log n)$.

The *constant-gap quantum PCP conjecture* asks, in one standard formulation, whether there is a constant $\gamma > 0$ for which approximating the ground energy of bounded-degree $O(1)$ -local Hamiltonians to additive error γm remains QMA-hard. NLTS is a structural obstacle expected of such hard Hamiltonians, but the NLTS theorem by itself does not prove quantum PCP hardness. Likewise, qLDPC means sparse checks; it does not by definition mean quantum local testability.

A four-qubit check example. Take the commuting stabilizers $S_X = X^{\otimes 4}$ and $S_Z = Z^{\otimes 4}$. Their supports overlap in four positions, hence they commute. The Hamiltonian

$$H = \frac{I - X^{\otimes 4}}{2} + \frac{I - Z^{\otimes 4}}{2} \quad (10.14)$$

is a sum of two commuting 4-local projectors. A simultaneous $+1$ eigenstate has energy 0; a simultaneous eigenstate with signs $(+1, -1)$ has energy 1. This finite example explains the constraint-to-Hamiltonian dictionary, but it is not an asymptotically good code and does not imply NLTS.

10.7 Asymptotically good quantum LDPC codes

Theorem 10.5 (Panteleev–Kalachev good qLDPC theorem — verified deep input). *For every rate target $R \in (0, 1)$ there is an explicit binary qLDPC family with parameters*

$$[[n, k_n, d_n]]_2, \quad k_n \geq Rn, \quad d_n = \Theta(n) \quad (10.15)$$

as $n \rightarrow \infty$. *Equivalently, the family has simultaneously constant check weight/degree, rate bounded away from zero, and relative distance bounded away from zero.*

Proof ownership and what is not claimed. The primary theorem is stronger in alphabet generality: for every finite field $\mathbb{F}_q$ and every $R \in (0, 1)$ it constructs qLDPC codes $[[n, k \geq Rn, d = \Theta(n)]]_q$. Equation (10.15) is the $q = 2$ specialization needed for this qubit volume. The lifted-product construction over non-Abelian groups, the local-minimality estimates, expansion input, and distance/rate proof remain source-owned. The theorem resolves the asymptotically good qLDPC existence problem; it does *not* state that the quantum codes are locally testable, and it does not supply an efficient decoder merely from the displayed parameters.

10.8 NLTS from good quantum-code Hamiltonians

Theorem 10.6 (Anshu–Breuckmann–Nirkhe NLTS theorem — verified deep input). *There exist a constant $\varepsilon > 0$ and an explicit family of n -qubit Hamiltonians*

$$H^{(n)} = \sum_{i=1}^{m_n} h_i^{(n)}, \quad m_n = \Theta(n), \quad (10.16)$$

that are frustration free, commuting, and $O(1)$ -local, such that every family of states $\{\rho_n\}$ satisfying

$$\mathrm{Tr}(H^{(n)}\rho_n) < \varepsilon n \quad (10.17)$$

has circuit complexity at least $\Omega(\log n)$. In particular these Hamiltonians satisfy NLTS.

Why this theorem is not formally derived from Theorem 10.5. The NLTS proof uses a particular good quantum-code family with an additional clustering property; in the primary proof this property is established for the Leverrier–Zémor quantum Tanner code construction. The Panteleev–Kalachev theorem and the NLTS theorem therefore solve adjacent historic problems, but the latter is not a black-box consequence of *every* good qLDPC code. The clustering theorem, uncertainty argument, robustness from violated checks to separated distributions, and light-cone/circuit-depth lower bound remain source-owned.

NLTS versus quantum PCP. Equation (10.17) says that low energy cannot be witnessed by a shallow quantum circuit for this Hamiltonian family. Quantum PCP asks for a complexity-hardness statement about estimating ground energy at constant extensive precision. The first is a structural circuit lower bound; the second is an approximation-complexity conjecture. A reader should therefore treat NLTS as a major necessary structural milestone toward the quantum-PCP landscape, not as the quantum PCP theorem itself.

10.9 Quantum complexity and correlation-model map

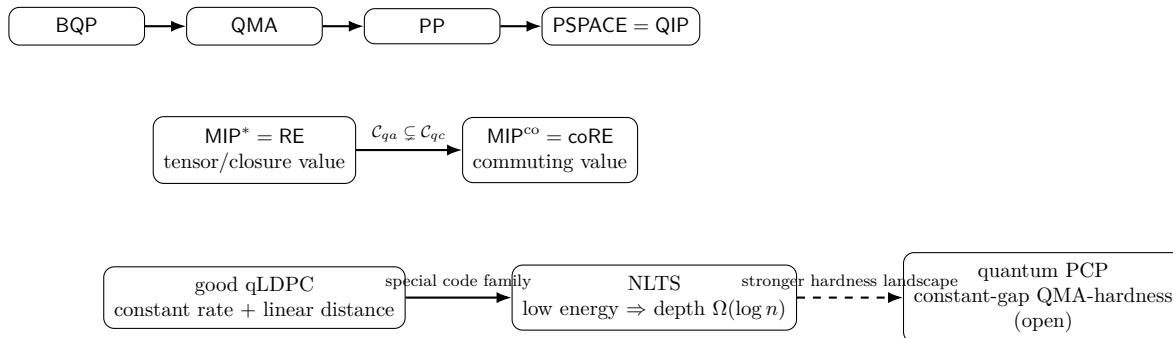


Figure 10.1: Volume-XV synthesis map. Solid arrows/links record proved interfaces used in this volume. The dashed link is contextual: NLTS is not a proof of the quantum PCP conjecture.

How to read the middle row. The horizontal link between MIP^* and $\mathrm{MIP}^{\mathrm{co}}$ is not a class inclusion. It records that changing the correlation model changes the value problem and, ultimately, the computability class. The strict set inclusion $\mathcal{C}_{qa} \subsetneq \mathcal{C}_{qc}$ belongs to the correlation models, not to the language classes.

10.10 Volume XV synthesis

The ten chapters of this volume form four proof arcs.

1. **Quantum information foundations.** States, channels, distinguishability, uniform circuits, and compilation supply a finite-description model in which later complexity statements are meaningful.
2. **Algorithms and single-prover complexity.** Query algorithms, Fourier sampling, Shor's reductions, BQP, QMA, Local Hamiltonian, and QIP connect quantum amplitudes to classical counting and space complexity while keeping amplification and approximation costs explicit.
3. **Communication and nonlocality.** Quantum communication protocols lead to nonlocal games; CHSH exhibits a finite-dimensional separation from classical correlations, while $\text{MIP}^* = \text{RE}$ and $\text{MIP}^{\text{co}} = \text{coRE}$ show that the choice of quantum correlation model eventually controls undecidability itself.
4. **Sparse quantum constraints.** Quantum LDPC codes and NLTS turn sparse commuting checks into Hamiltonians whose low-energy state space has unavoidable circuit complexity. The quantum PCP problem remains beyond this volume as a clearly labelled frontier rather than an implied corollary.

The conceptual lesson is that “quantum complexity” is not one resource. Dimension, approximation precision, interaction, communication, entanglement model, witness size, check sparsity, energy density, and circuit depth are different axes. The volume's definitions and dependency ledger are designed to prevent movement along one axis from being silently charged to another.

10.11 Reader checkpoint and exercises

1. Give a dovetailing algorithm that recognizes the yes-side of (10.4) from below. Identify exactly where finite-dimensionality of the tensor-product strategies is used.
2. Suppose an oracle returns a number $v(G)$ with $|v(G) - \omega^*(G)| \leq 0.24$ for every finite game. Show explicitly how to decide the promise (10.4). Why would error 0.26 not suffice by the same threshold argument?
3. Explain why the chain $\mathcal{C}_q \subseteq \mathcal{C}_{qa} \subsetneq \mathcal{C}_{qc}$ concerns correlation sets whereas $\text{MIP}^* = \text{RE}$ and $\text{MIP}^{\text{co}} = \text{coRE}$ concern language classes. Give one sentence explaining why neither pair should be identified.
4. For binary vectors $u, v \in \mathbb{F}_2^n$, prove $X(u)Z(v) = (-1)^{u \cdot v} Z(v)X(u)$. Deduce the CSS commutation condition (10.10).
5. Let $S_1, \dots, S_m$ be commuting Pauli stabilizers. Prove directly that every $\Pi_j = (I - S_j)/2$ is a projector and that the ground space of (10.12) is the simultaneous $+1$ eigenspace.
6. A family has $k_n = \lfloor 0.1n \rfloor$, $d_n = \lfloor n/20 \rfloor$, check weight at most 12, but one qubit participates in $\sqrt{n}$ checks. Which part of Definition 10.4 fails? Why do rate and distance alone not imply the LDPC property?
7. State Theorem 10.5 with $R = 1/3$ and $q = 2$. List three properties the theorem does *not* assert: choose among efficient decoding, quantum local testability, geometric locality, and constant relative distance.
8. Explain why Theorem 10.6 is not formally obtained by combining only Definition 10.4 and Theorem 10.5. Name the additional source-owned structural ingredient used by the NLTS construction.

9. Compare NLTS with the constant-gap quantum PCP conjecture. Which is a circuit-complexity property of low-energy states, and which is a QMA-hardness statement about energy approximation?
10. Using Figure 10.1, write a one-page dependency narrative from uniform quantum circuits to one of the two end points: uncomputable nonlocal-game values or NLTS Hamiltonians. Every arrow in your narrative must name the resource that changes.

Internalization and Reconstruction Guide: Volume XV

Purpose. Internalize the quantum model before using its speedups or lower bounds: amplitudes, measurements, query access, Fourier sampling, proof systems, communication, nonlocal games, and complexity-class containments all depend on precise operational conventions.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-II-DEF-002: Computability and many-one reduction.
2. QRCC-III-DEF-010: Cyclic-group discrete logarithm and generic algorithms.
3. QRCC-III-THM-002: Cyclicity of finite-field multiplicative groups.
4. QRCC-I-DEF-006: Normalized Fourier transform on a finite abelian group.
5. QRCC-II-THM-011: Polynomial-hierarchy quantifier characterization.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XV-DEF-001 (Quantum states, measurements, and channels); QRCC-XV-DEF-002 (Trace distance and fidelity); QRCC-XV-DEF-008 (Uniform quantum circuit families); QRCC-XV-DEF-012 (Quantum query model); QRCC-XV-DEF-020 (BQP and quantum reductions).

Chapter retrieval map

- Ch. 1: Finite-Dimensional Quantum States and Distinguishability:** enter through QRCC-XV-DEF-001 (Quantum states, measurements, and channels); exit by being able to state QRCC-XV-THM-003 (Schmidt decomposition and purification uniqueness) with its ownership boundary.
- Ch. 2: Uniform Quantum Circuits and Approximation:** enter through QRCC-XV-DEF-008 (Uniform quantum circuit families); exit by being able to state QRCC-XV-THM-009 (Universal gate simulation) with its ownership boundary.

- Ch. 3: Quantum Queries, Search, and Hidden Structure:** enter through QRCC-XV-DEF-012 (Quantum query model); exit by being able to state QRCC-XV-ALG-013 (Quantum phase estimation) with its ownership boundary.
- Ch. 4: Quantum Fourier Sampling, Order Finding, and Shor’s Algorithms:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XV-ALG-017 (Order finding by quantum Fourier sampling) with its ownership boundary.
- Ch. 5: BQP, Counting Bounds, and the PH Frontier:** enter through QRCC-XV-DEF-020 (BQP and quantum reductions); exit by being able to state QRCC-XV-THM-021 (BQP amplification) with its ownership boundary.
- Ch. 6: QMA, Amplification, and Local Hamiltonians:** enter through QRCC-XV-DEF-024 (QMA and local Hamiltonians); exit by being able to state QRCC-XV-BB-026 (k-local Hamiltonian is QMA-complete) with its ownership boundary.
- Ch. 7: Quantum Communication Complexity:** enter through QRCC-XV-DEF-029 (Quantum communication complexity); exit by being able to state QRCC-XV-BB-032 (Quantum communication lower bounds for disjointness) with its ownership boundary.
- Ch. 8: Quantum Interactive Proofs and QIP :** enter through QRCC-XV-DEF-033 (Quantum interactive proofs); exit by being able to state QRCC-XV-BB-034 (QIP equals PSPACE) with its ownership boundary.
- Ch. 9: Nonlocal Games, CHSH, and Entangled-Prover Complexity:** enter through QRCC-XV-DEF-035 (Nonlocal games and correlation models); exit by being able to state QRCC-XV-BB-037 (Entangled-prover complexity before $\mathsf{MIP}^* = \mathsf{RE}$) with its ownership boundary.
- Ch. 10: Uncomputable Nonlocality and Quantum LDPC:** enter through QRCC-XV-DEF-041 (Quantum LDPC and NLTS interfaces); exit by being able to state QRCC-XV-BB-038 ($\mathsf{MIP}^* = \mathsf{RE}$) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XV-THM-003 — Schmidt decomposition and purification uniqueness. Dependencies: QRCC-XV-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XV-ALG-013 — Quantum phase estimation. Dependencies: QRCC-XV-DEF-008, QRCC-XV-THM-010. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-XV-THM-021 — BQP amplification. Dependencies: QRCC-I-THM-004, QRCC-XV-DEF-020. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-XV-BB-032 — Quantum communication lower bounds for disjointness. Dependencies:

QRCC-XII-THM-019, QRCC-XV-DEF-029. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

5. QRCC-XV-BB-037 — Entangled-prover complexity before $\text{MIP}^* = \text{RE}$. Dependencies: QRCC-XV-DEF-035, QRCC-XIII-BB-007. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XV-F-023 — BQP versus the polynomial hierarchy. Dependencies: QRCC-XV-DEF-020, QRCC-II-THM-011. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- squaring amplitudes too early and losing interference;
- confusing query access with a physically specified QRAM implementation;
- ignoring completeness/soundness gaps or amplification costs in quantum proof systems;
- treating tensor-product, commuting-operator, and finite-dimensional nonlocal models as interchangeable;

Boundary and degenerate cases

- global phase and zero amplitudes;
- one-dimensional Hilbert spaces and deterministic measurements;
- exact versus bounded-error algorithms;
- finite-dimensional versus commuting-operator nonlocal models;

Retrieval and reconstruction drill

1. State QRCC-XV-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XV-THM-003. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XV-ALG-013, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.

7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XV-F-023, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XVI uses complexity assumptions to define and prove cryptographic security in a reductionist framework.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XVI

Foundations of Modern Cryptography

Chapter 1

Security Parameters, Games, and Reductions

Modern cryptography starts with a quantified statement rather than with a cipher. One fixes a security parameter, an adversarial resource model, an experiment, and an advantage. A construction is secure only after every efficient adversary has been converted into an efficient solver for a stated assumption with an explicit loss ledger. This chapter fixes that grammar for all of Volume XVI.

1.1 How to read this chapter

Read Definition 1.1 as the type system for the volume. Theorem 1.2 then explains which negligible error terms may be accumulated safely. Theorem 1.3 is the accounting rule used in every later hybrid proof. Figure 1.1 is a map of the same proof pattern.

1.2 Learning goals

A reader should be able to distinguish uniform PPT algorithms from nonuniform circuit adversaries; translate a bit-output security game into an advantage; prove that polynomially many negligible losses remain negligible; audit a reduction for running time and oracle-query overhead; and identify when an apparently harmless reparameterization or union bound ceases to be polynomially controlled.

Literature and convention checkpoint. The experiment-and-reduction discipline is calibrated to Goldreich’s *Foundations of Cryptography*, Katz–Lindell’s modern treatment, Bellare–Rogaway’s concrete-security notes, and Victor Shoup’s *Sequences of Games*. The present volume uses uniform PPT adversaries by default, but records nonuniform circuit security separately whenever a theorem inherited from earlier volumes is stated in that model. A classical theorem is never silently upgraded to superposition-query or quantum-adversary security.

Definition 1.1 (Security parameters, adversaries, games, and concrete advantage). The *security parameter* is an integer $\lambda \geq 1$, presented to every efficient algorithm in unary as 1^λ unless an equivalent uniform convention is explicitly stated. An *ensemble* is a family $X = \{X_\lambda\}_{\lambda \geq 1}$ of random variables whose sample spaces may depend on λ .

A randomized algorithm A is *probabilistic polynomial time* (PPT) if one Turing machine computes $A(1^\lambda, \cdot)$ in time polynomial in λ and the explicit input length. A *nonuniform polynomial-*

size adversary is a family of Boolean or arithmetic circuits $\{C_\lambda\}$ of size $\text{poly}(\lambda)$. These models are not identified: a theorem proved for one model states that model.

A function $\mu : \mathbb{N} \rightarrow [0, \infty)$ is *negligible* if for every polynomial p with positive values for all sufficiently large λ , there exists λ_0 such that

$$\mu(\lambda) < \frac{1}{p(\lambda)} \quad (\lambda \geq \lambda_0).$$

It is *noticeable* if $\mu(\lambda) \geq 1/p(\lambda)$ eventually for some positive polynomial p . A function may be neither negligible nor noticeable if it oscillates; asymptotic statements below are therefore quantified explicitly.

A *security experiment* $G^A(1^\lambda)$ is a probabilistic interactive computation between a challenger and adversary A , possibly with named stateful oracles. It terminates with a bit. For two experiments G_0, G_1 the distinguishing advantage is

$$\text{Adv}_{G_0, G_1}(A, \lambda) := \left| \Pr[G_0^A(1^\lambda) = 1] - \Pr[G_1^A(1^\lambda) = 1] \right|.$$

For a single challenge-bit game one may instead normalize by subtracting the baseline success probability; every theorem states which normalization is in force.

An *oracle algorithm* receives access to explicitly typed query interfaces. In this classical volume, an oracle query sends and returns classical strings unless a later statement explicitly grants quantum superposition access. The *query complexity* $q_A(\lambda)$ is the maximum or expected number of calls, as stated by the theorem.

A *reduction* from breaking construction Π to problem or assumption $\mathcal{H}$ is a uniform algorithm B^A that uses an adversary A as a subroutine. A concrete reduction record consists of functions T_B, Q_B, L such that an adversary with time t , query budget q , and advantage ε produces a solver with time at most $T_B(\lambda, t, q)$, query budget at most $Q_B(\lambda, t, q)$, and advantage at least $L(\lambda, \varepsilon, t, q)$. *Asymptotic security* says the resulting advantage is negligible for every PPT adversary; *concrete security* retains the displayed resource and advantage functions.

Example: distinguishing a biased coin. If G_0 outputs a Bernoulli random variable of parameter $1/2$ and G_1 one of parameter $1/2 + 2^{-\lambda}$, even an unbounded observer of one sample has advantage exactly $2^{-\lambda}$, which is negligible. Repeating the experiment 2^λ times would make the bias visible, but that family of experiments is not polynomially bounded. The difference between “small per trial” and “secure under polynomial resources” is the first recurring quantifier lesson.

Theorem 1.2 (Closure calculus for negligible functions). *Let $\mu, \mu_1, \mu_2, \dots$ be nonnegative functions.*

- (i) *If $\mu_1, \dots, \mu_k$ are negligible for fixed k , then $\sum_{i=1}^k \mu_i$ is negligible. More generally, if $k(\lambda) \leq p(\lambda)$ is polynomially bounded and there is one negligible ν such that $\mu_i(\lambda) \leq \nu(\lambda)$ for every $i \leq k(\lambda)$, then $\sum_{i \leq k(\lambda)} \mu_i(\lambda)$ is negligible.*
- (ii) *If $a(\lambda) \leq p(\lambda)$ is polynomially bounded and μ is negligible, then $a\mu$ is negligible.*
- (iii) *If $q : \mathbb{N} \rightarrow \mathbb{N}$ is polynomially bounded below in the sense that $q(\lambda) \geq \lambda^c$ eventually for some constant $c > 0$, and μ is negligible, then $\mu(q(\lambda))$ is negligible.*
- (iv) *If events $E_1, \dots, E_{k(\lambda)}$ satisfy $\Pr(E_i) \leq \nu(\lambda)$ with k polynomially bounded and ν negligible, then $\Pr(\bigcup_i E_i)$ is negligible.*

None of these conclusions remains valid after replacing the polynomial bound by an arbitrary superpolynomial one.

Proof. For (i), fix a positive polynomial r . In the polynomial-family case choose the negligible bound for ν against the polynomial pr : eventually $\nu(\lambda) < 1/(p(\lambda)r(\lambda))$, whence

$$\sum_{i \leq k(\lambda)} \mu_i(\lambda) \leq p(\lambda)\nu(\lambda) < \frac{1}{r(\lambda)}.$$

The fixed- k statement is the same argument with $p = k$.

Part (ii) is identical: test μ against pr . For (iii), let r have degree d and coefficients bounded in absolute value by C for sufficiently large arguments. There is a constant C' with $r(\lambda) \leq C'\lambda^{d'}$ for some d' . Since $q(\lambda) \geq \lambda^c$, one has $\lambda^{d'} \leq q(\lambda)^{d'/c}$. Negligibility of μ against a polynomial of degree exceeding d'/c gives $\mu(q(\lambda)) < 1/r(\lambda)$ eventually. Part (iv) is the union bound followed by (i).

For the failure mode, let $\mu(\lambda) = 2^{-\lambda}$ and take $k(\lambda) = 2^\lambda$. Then $k(\lambda)\mu(\lambda) = 1$. Thus the word “polynomial” is a hypothesis, not stylistic bookkeeping. $\square$

Theorem 1.3 (Game-hopping and reduction accounting). *Let $G_0, \dots, G_t$ be bit-output games for the same adversary A , with $t = t(\lambda)$ polynomially bounded. Suppose that for each $i < t$ there are nonnegative functions δ_i, β_i and an assumption solver B_i^A such that*

$$\left| \Pr[G_i^A = 1] - \Pr[G_{i+1}^A = 1] \right| \leq \delta_i(\lambda) \leq \alpha_i(\lambda) \text{Adv}_{\mathcal{H}_i}(B_i^A, \lambda) + \beta_i(\lambda). \quad (1.1)$$

Then

$$\text{Adv}_{G_0, G_t}(A, \lambda) \leq \sum_{i=0}^{t-1} \alpha_i \text{Adv}_{\mathcal{H}_i}(B_i^A, \lambda) + \sum_{i=0}^{t-1} \beta_i. \quad (1.2)$$

If a hop changes the game only when an event Bad_i occurs, then it is valid to take $\delta_i \leq \Pr[\text{Bad}_i]$ provided the two games can be coupled to evolve identically until that event. If B_i simulates A with at most s_i extra elementary operations and r_i extra oracle queries, its resource ledger is

$$T_{B_i} \leq T_A + s_i, \quad Q_{B_i} \leq Q_A + r_i, \quad (1.3)$$

up to the explicitly stated cost of answering simulated queries.

Proof. Write $p_i = \Pr[G_i^A = 1]$. The triangle inequality gives $|p_0 - p_t| \leq \sum_i |p_i - p_{i+1}|$; substituting (1.1) proves (1.2). For the bad-event clause, couple the random coins of the two adjacent games. On $\neg \text{Bad}_i$ their output bits are identical, so the probability that the coupled outputs differ is at most $\Pr[\text{Bad}_i]$. For Bernoulli variables, total variation equals the absolute difference of their success probabilities, hence the same upper bound applies. Equation (1.3) is literal accounting of the wrapper computation. No asymptotic step has been used; if the number of hops and every loss multiplier are polynomial and the assumption advantages and additive errors are negligible, Theorem 1.2 converts (1.2) to negligible total advantage. $\square$

Abort conditioning warning. A proof sometimes analyzes a game conditioned on $\neg \text{Bad}$. One may not simply delete the conditioning. If two games agree conditioned on $\neg \text{Bad}$, the unconditional statistical gap is bounded by the probability of the bad event (or by the sum of the two bad probabilities when the events are defined separately). This term must remain in the ledger.

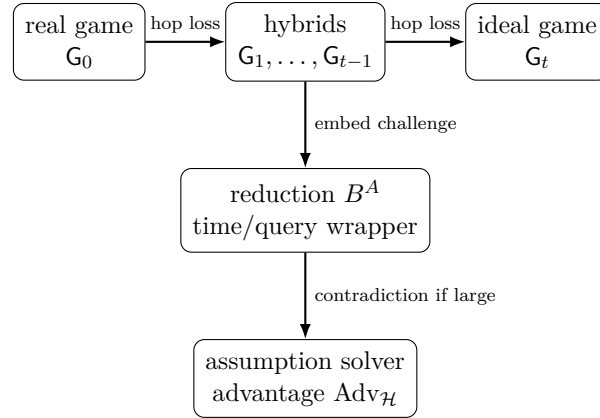


Figure 1.1: Security-game and reduction anatomy. Every arrow carries an advantage, running-time, and query-cost statement. A proof is incomplete if any of these interfaces is only implicit.

1.3 Reader checkpoint

1. Give an example of a function that is neither negligible nor noticeable.
2. Prove that the maximum of finitely many negligible functions is negligible.
3. Find the weakest lower-growth condition on q you need in Theorem 1.2(iii).
4. Show by example that $\mu(\log \lambda)$ need not be negligible when μ is negligible.
5. Derive (1.2) when each hop is signed rather than absolute-valued.
6. Suppose a reduction guesses one of q adversarial queries uniformly. Record its exact advantage loss.
7. Explain why an expected-polynomial-time simulator requires a separate runtime argument before it is a PPT reduction.
8. Construct a pair of games that are identical until a collision event and bound their gap by a birthday term.
9. Explain why classical oracle access does not define quantum superposition-query security.
10. Audit any later proof in this volume by listing challenger state, adversary interface, output event, and loss ledger.

Chapter 2

Perfect Secrecy and the Cost of Information-Theoretic Encryption

Perfect secrecy is the zero-computation baseline against which computational security should be read. It says not that decryption is difficult without the key, but that the ciphertext distribution reveals no information about the message at all. The price of that guarantee is equally exact: fresh secret-key entropy must be at least the message entropy.

2.1 How to read this chapter

Definition 2.1 freezes correctness and perfect/statistical secrecy before any construction is analyzed. Theorem 2.2 gives the group-valued one-time pad; Theorem 2.3 explains why its key length is not an accident.

2.2 Learning goals

A reader should be able to translate perfect secrecy among posterior, independence, and equal-ciphertext- distribution formulations; prove the one-time-pad theorem on an arbitrary finite abelian group; diagnose key reuse algebraically; and prove Shannon’s entropy lower bound without assuming uniform messages.

Literature checkpoint. The definitions and lower bound are calibrated directly to Claude Shannon’s 1949 *Communication Theory of Secrecy Systems*. We state them in modern random-variable language and use the entropy chain rule already proved in Volume I.

Definition 2.1 (Correct encryption, perfect secrecy, and statistical secrecy). A one-time symmetric encryption system consists of finite sets $\mathcal{K}, \mathcal{M}, \mathcal{C}$, a key random variable K independent of the message M , an encryption algorithm $\text{Enc} : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{C}$ (possibly randomized), and a decryption algorithm $\text{Dec} : \mathcal{K} \times \mathcal{C} \rightarrow \mathcal{M} \cup \{\perp\}$. It has *perfect correctness* if $\Pr[\text{Dec}_K(\text{Enc}_K(m)) = m] = 1$ for every key/message pair in the support of key generation, where the probability is over encryption randomness.

It has *perfect secrecy* if for every distribution of M independent of K , the ciphertext $C := \text{Enc}_K(M)$ is independent of M . Equivalently, whenever the conditioning event has positive probability,

$$\Pr[M = m \mid C = c] = \Pr[M = m]. \quad (2.1)$$

For deterministic encryption this is also equivalent to the distribution of $\text{Enc}_K(m)$ being the same for every $m \in \mathcal{M}$.

For $\varepsilon \geq 0$, the system has ε -statistical secrecy for a specified message class if the joint distribution (M, C) has total-variation distance at most ε from $M \times C$, the product of its marginals. Perfect secrecy is the case $\varepsilon = 0$.

Theorem 2.2 (One-time pad over a finite abelian group). *Let $(G, +)$ be a finite abelian group. Take $\mathcal{K} = \mathcal{M} = \mathcal{C} = G$, let K be uniform on G , and define*

$$\text{Enc}_k(m) = m + k, \quad \text{Dec}_k(c) = c - k. \quad (2.2)$$

Then the scheme has perfect correctness and perfect secrecy for every message distribution independent of K . If the same key encrypts two messages, however, the pair of ciphertexts reveals the exact group difference

$$C_1 - C_2 = M_1 - M_2. \quad (2.3)$$

Proof. Correctness is $(m + k) - k = m$. For secrecy, fix $m, c \in G$. Exactly one key, namely $k = c - m$, satisfies $m + k = c$, and the uniform key therefore gives

$$\Pr[C = c \mid M = m] = \frac{1}{|G|},$$

independent of m . Thus C is uniform and independent of M , proving perfect secrecy by Definition 2.1. Equation (2.3) follows by cancellation: $(M_1 + K) - (M_2 + K) = M_1 - M_2$. $\square$

Worked binary example. For $G = \mathbb{F}_2^n$, (2.2) is XOR. With a reused key, two English-like plaintexts leak their XOR; known structure in one message can expose structure in the other. This is not a computational attack on the pad. It is a deterministic identity caused by violating the one-time hypothesis.

Theorem 2.3 (Shannon key-entropy lower bound). *Let M, K, C describe any perfectly correct, perfectly secret one-time encryption system, with K independent of M . Then*

$$H(K) \geq H(M). \quad (2.4)$$

Consequently, if M is uniform on a message set $\mathcal{M}$ and K is supported on $\mathcal{K}$, then

$$|\mathcal{K}| \geq |\mathcal{M}|. \quad (2.5)$$

If in addition K is uniform, its bit length is at least $\log_2 |\mathcal{M}|$.

Proof. Perfect secrecy gives $H(M \mid C) = H(M)$. Perfect correctness says M is determined by (K, C) , hence $H(M \mid K, C) = 0$. Using the conditional chain rule in two orders,

$$\begin{aligned} H(M \mid C) &\leq H(M, K \mid C) \\ &= H(K \mid C) + H(M \mid K, C) \\ &= H(K \mid C) \\ &\leq H(K). \end{aligned}$$

This proves (2.4). If M is uniform, $H(M) = \log_2 |\mathcal{M}|$; every variable supported on $\mathcal{K}$ has entropy at most $\log_2 |\mathcal{K}|$. Combining these inequalities gives (2.5). $\square$

What the lower bound does not say. The theorem is one-time and information-theoretic. Computational encryption deliberately leaves the perfect-secrecy regime, allowing a short key to protect polynomially many long messages against bounded adversaries. The rest of Volume XVI develops the assumptions and reductions that pay for that escape.

2.3 Reader checkpoint

1. Prove directly from Bayes' rule that (2.1) is equivalent to independence of M and C .
2. For deterministic encryption, prove equality of all conditional ciphertext distributions is equivalent to perfect secrecy.
3. Compute the posterior distribution of M in a one-time pad when the key is biased rather than uniform.
4. Give a two-message example where key reuse reveals only a relation, not either plaintext individually.
5. Prove $H(M | C) \leq H(K | C)$ directly from recoverability of M from (K, C) .
6. Determine when equality holds in (2.4).
7. Show that randomized encryption does not invalidate the entropy proof as long as decryption is perfectly correct.
8. Formulate an ε -secrecy analogue in mutual information and compare it with total variation.
9. Explain why computational PRG expansion does not contradict Shannon's lower bound.
10. Generalize the OTP proof to any finite group, identifying where commutativity is and is not needed.

Chapter 3

Computational Indistinguishability and Hybrid Arguments

Computational indistinguishability replaces equality of distributions by equality as seen through a resource-bounded observer. This is the bridge from information theory to modern cryptography: once the observer is restricted, pseudorandomness can stand in for randomness and short secret keys can protect long computations.

3.1 How to read this chapter

Definition 3.1 distinguishes plain, auxiliary-input, and oracle versions of the notion. Theorem 3.2 is the closure rule that will be invoked in PRG, PRF, encryption, MAC, and CCA proofs.

3.2 Learning goals

A reader should be able to convert a distinguisher into an explicit advantage function, prove closure under efficient postprocessing, apply a polynomial hybrid argument with the correct $1/t$ loss, and explain why auxiliary information must be included in the definition rather than waved away as “public context.”

Definition 3.1 (Computational indistinguishability). Two ensembles $X = \{X_\lambda\}$ and $Y = \{Y_\lambda\}$ over the same efficiently representable sample spaces are *computationally indistinguishable*, written $X \approx_c Y$, if for every uniform PPT bit-output distinguisher D the function

$$\left| \Pr[D(1^\lambda, X_\lambda) = 1] - \Pr[D(1^\lambda, Y_\lambda) = 1] \right| \quad (3.1)$$

is negligible. Nonuniform indistinguishability replaces D by polynomial-size circuit families.

For joint ensembles (X, Z) and (Y, Z') , *auxiliary-input indistinguishability* means the same condition applied to the full joint samples. In security proofs Z may contain public keys, leakage, challenge metadata, or prior transcripts; it is part of the distribution, not an informal side channel.

Two oracle ensembles $\mathcal{O}_0, \mathcal{O}_1$ are *classically oracle-indistinguishable* if every PPT oracle machine making polynomially many classical queries has negligible distinguishing advantage. This does not grant superposition queries. A later post-quantum statement must specify whether the adversary is quantum with classical queries or has coherent oracle access.

Statistical closeness implies computational closeness. If the total-variation distance between X_λ and Y_λ is at most $\mu(\lambda)$, then every (possibly unbounded) test has distinguishing gap at most $\mu(\lambda)$. Hence negligible statistical distance is stronger than computational indistinguishability. The converse is precisely what cryptographic assumptions allow to fail.

Theorem 3.2 (Efficient closure and polynomial-hybrid transitivity). *The following hold in either the uniform-PPT or polynomial-size-circuit model.*

- (i) *If $X \approx_c Y$ and T is a probabilistic polynomial-time transformation using fresh coins independent of its input, then $T(X) \approx_c T(Y)$. The same holds jointly with efficiently transformed auxiliary information.*
- (ii) *Let $H_0, \dots, H_t$ be efficiently samplable ensembles with $t = t(\lambda)$ polynomially bounded. If every adjacent pair satisfies $H_i \approx_c H_{i+1}$ with a common negligible upper bound on the relevant distinguishing gaps, then $H_0 \approx_c H_t$.*
- (iii) *Quantitatively, if some distinguisher separates H_0 from H_t with advantage ε , then for that distinguisher there exists $i < t$ with adjacent gap at least ε/t . If the index can be sampled and the adjacent experiment simulated uniformly, a reduction that guesses i loses at most a factor t in advantage.*

Proof. For (i), if D distinguishes $T(X)$ and $T(Y)$, define $D'(z)$ to sample the internal coins of T , compute $T(z)$, and run D . Its runtime is polynomial and its advantage is identical, contradicting $X \approx_c Y$.

For (ii)–(iii), fix a distinguisher D and write $p_i = \Pr[D(H_i) = 1]$. Then

$$|p_0 - p_t| \leq \sum_{i=0}^{t-1} |p_i - p_{i+1}|. \quad (3.2)$$

A polynomial sum of negligible functions is negligible by Theorem 1.2. Conversely, if the left side is ε , at least one summand has absolute value at least ε/t . If the adjacent experiments admit one uniform simulator, a reduction may choose $I \leftarrow \{0, \dots, t-1\}$ and embed its challenge in hop I : before taking absolute values, the signed adjacent gaps telescope, so the averaged signed gap is exactly $(p_0 - p_t)/t$. Choosing the orientation (equivalently, using D or its output complement) that agrees with the sign of $p_0 - p_t$ therefore gives advantage ε/t . This is an existential reduction choice; randomly complementing the output would instead cancel the gap. $\square$

Worked three-hybrid pattern. Suppose a real transcript uses a PRF value, an intermediate game replaces that PRF by a random function, and a final game conditions on no nonce collision. The total distinguishing advantage is at most

$$\text{Adv}_F^{\text{prf}}(B) + \Pr[\text{NonceCollision}]. \quad (3.3)$$

There is no extra mystical “hybrid security” term: every difference must be either a reduction term or an explicitly bounded probability/statistical term.

3.3 Reader checkpoint

1. Prove that computational indistinguishability is symmetric and reflexive.
2. Use Theorem 3.2 to prove transitivity for three ensembles.

3. Give an example showing that an inefficient transformation need not preserve computational indistinguishability.
4. Explain why revealing one hard-core bit as auxiliary information can destroy an otherwise useful indistinguishability statement.
5. Derive the exact hybrid loss for $t = 100$ hops and endpoint advantage 0.1.
6. Show how to turn a signed adjacent gap into an absolute gap using one advice bit in the nonuniform model.
7. State the oracle version of the closure theorem for adaptive classical queries.
8. Explain why polynomially many samples from an ensemble require a fresh hybrid argument rather than automatic one-sample security.
9. Compare negligible statistical distance with computational indistinguishability on a pair of distributions supported on polynomial-size sets.
10. Audit equation (3.3) by naming the two adjacent games explicitly.

Chapter 4

One-Way Functions, Amplification, and the Minicrypt Boundary

A one-way function is the weakest standard computational primitive from which much of private-key cryptography can be built. The gap between “noticeably hard to invert” and “negligibly invertible” is itself nontrivial, and the gap between one-way functions and public-key cryptography is deeper still. This chapter separates those two questions.

4.1 How to read this chapter

Definition 4.1 fixes strong and weak one-wayness. Theorem 4.2 is deliberately source-owned because its direct-product proof is a genuine hardness-amplification argument, not an independence heuristic. The dated frontier following it records what arbitrary one-way functions are *not* presently known to imply.

4.2 Learning goals

A reader should be able to state inversion success so that any valid preimage counts; distinguish weak from strong one-wayness by quantifier order; explain why naive multiplication of inversion probabilities is invalid for a joint inverter; and distinguish a black-box/oracle barrier from an unconditional impossibility theorem in the standard model.

Definition 4.1 (Strong and weak one-way functions). A length-regular family $f = \{f_\lambda : \{0, 1\}^{n(\lambda)} \rightarrow \{0, 1\}^{m(\lambda)}\}$ is efficiently computable if one deterministic polynomial-time algorithm evaluates it and n, m are polynomially related to λ . For an inverter A , define

$$\text{Inv}_f(A, \lambda) := \Pr_{x \leftarrow U_{n(\lambda)}, A} [f_\lambda(A(1^\lambda, f_\lambda(x))) = f_\lambda(x)]. \quad (4.1)$$

Any preimage is accepted; injectivity is not assumed.

The family is a *strong one-way function* if (4.1) is negligible for every PPT A . It is a *weak one-way function* if there exists a polynomial p such that for every PPT A and all sufficiently large λ ,

$$\text{Inv}_f(A, \lambda) \leq 1 - \frac{1}{p(\lambda)}. \quad (4.2)$$

A *one-way permutation* is a strong one-way function whose domain and range coincide and for which each f_λ is a permutation.

A function is *r-regular* if every image point has exactly r preimages; regular families permit r to depend efficiently on the parameter. *Distributional one-wayness* compares a sampler for a random input conditioned on its image with any efficient reconstruction distribution; it is a different notion and is named explicitly when used.

A *fully black-box construction* of primitive P from primitive Q gives the construction oracle access to any implementation of Q and gives a black-box reduction from every breaker of P to a breaker of Q . Relativizing and non-black-box constructions are weaker/different proof-method classes; a black-box separation is not an unconditional separation of the primitives themselves.

Why the weak definition is genuinely weak. Condition (4.2) permits an inverter to succeed on $1 - 1/p$ of all random inputs. Cryptographically that is useless: almost every ciphertext-like image might be invertible. The amplification theorem says the existential assumption nevertheless suffices to obtain a strong OWF after a polynomial input expansion.

Theorem 4.2 (Weak one-way functions amplify to strong one-way functions — verified deep input). *If weak one-way functions exist, then strong one-way functions exist. More concretely, let f satisfy (4.2) for polynomial p . There is a polynomially bounded repetition parameter $t(\lambda)$ and an efficient direct-product transformation*

$$F(x_1, \dots, x_t) = (f(x_1), \dots, f(x_t)) \quad (4.3)$$

(possibly after standard length regularization) such that every PPT inverter for F has negligible success. The input length grows only polynomially.

Proof ownership and the non-independence trap. The theorem is the classical weak-to-strong OWF amplification associated with Yao and developed in full in Goldreich’s *Foundations of Cryptography*, with later security-preserving refinements by Goldreich–Impagliazzo–Levin–Venkatesan. The proof remains source-owned here. The tempting calculation $(1 - 1/p)^t$ is *not* a proof: a joint inverter for (4.3) need not invert coordinates independently. The canonical proof converts any noticeably successful joint inverter into a single-instance inverter that violates (4.2), using a careful embedding/conditioning argument. This is precisely the mechanism that the source gate owns.

Quantitative intuition. Taking t on the order of $\lambda p(\lambda)$ would make $(1 - 1/p)^t \leq e^{-\lambda}$ if independence were available. The real theorem recovers the same qualitative amplification without assuming independence of the adversary’s coordinate successes, at polynomial reduction cost.

4.3 Dated frontier: minimal assumptions for public-key cryptography

Status as of 2026-09-03. It remains unknown whether the mere existence of arbitrary one-way functions implies standard-model key agreement or public-key encryption. Impagliazzo–Rudich established a foundational black-box/oracle barrier: relative to an oracle, one-way permutations can exist while secret-key agreement fails, ruling out relativizing proofs and strongly constraining fully black-box constructions. This does *not* prove that no non-black-box standard-model construction exists.

Recent work continues to refine nearby boundaries—for example bounded-storage variants and fine-grained public-key notions under stronger hardness—without turning the general implication “OWF $\Rightarrow$ key agreement/PKE” into a theorem. Consequently this row remains a dated frontier,

not a negative theorem. Quantum-adversary and quantum-random-oracle variants form a separate frontier and are discussed again in Chapter 10.

4.4 Reader checkpoint

1. Rewrite strong one-wayness with explicit quantifiers over adversaries and polynomials.
2. Give a weak one-way function candidate that would be useless if left unamplified.
3. Explain why inversion accepts any preimage for a noninjective function.
4. Show that a one-way permutation is automatically length preserving.
5. Identify the invalid independence step in the naive proof of (4.3).
6. Explain the difference between a direct-product construction and an XOR hardness amplifier.
7. Define what a relativizing proof of “OWF implies key agreement” would have to mean.
8. Explain why an oracle separation cannot by itself rule out a nonrelativizing construction.
9. Compare the roles of Minicrypt and Cryptomania in Impagliazzo’s five-world picture.
10. State one additional assumption, beyond arbitrary OWFs, under which public-key encryption is known.

Chapter 5

Hard-Core Predicates and the Goldreich–Levin Theorem

A one-way function hides an entire preimage only in the sense that inversion is hard. It does not say that every individual bit of the preimage is unpredictable. Hard-core predicates extract a bit whose prediction would be strong enough to reconstruct a preimage. Goldreich–Levin supplies such a predicate uniformly for every one-way function after adjoining a random inner-product seed.

5.1 How to read this chapter

Definition 5.1 fixes the prediction experiment. Theorem 5.2 is retained as a verified deep theorem because its efficient Hadamard list-decoding reconstruction is a substantial algorithmic proof. The reader-facing text nevertheless derives the Fourier correlation that connects prediction advantage to the list-decoding problem.

5.2 Learning goals

A reader should be able to distinguish inversion hardness from bit unpredictability; derive the Goldreich–Levin correlation identity; explain why a large Fourier coefficient identifies a short list of candidate preimages; and state the theorem with the auxiliary random string r included in the one-way image.

Definition 5.1 (Hard-core predicates). Let $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ be an efficiently computable family. A polynomial-time computable bit $b(x)$ is a *hard-core predicate for f* if every PPT predictor P satisfies

$$\left| \Pr_{x \leftarrow U_n} [P(1^n, f(x)) = b(x)] - \frac{1}{2} \right| \leq \text{negl}(n). \quad (5.1)$$

For a transformed function $F(x, r)$, the predictor receives the full public image $F(x, r)$, including any auxiliary seed r intentionally exposed by the construction.

A polynomial-length map $h(x) \in \{0, 1\}^{\ell(n)}$ is a *hard-core function* if $(f(x), h(x))$ is computationally indistinguishable from $(f(x), U_{\ell(n)})$. A hard-core predicate is the one-bit case, but repeated bits require a separate hybrid or next-bit argument.

Theorem 5.2 (Goldreich–Levin hard-core theorem — verified deep input). *Let $f = \{f_n : \{0, 1\}^n \rightarrow \{0, 1\}^{m(n)}\}$ be a strong one-way function. Define*

$$F_n(x, r) := (f_n(x), r), \quad b(x, r) := \langle x, r \rangle \pmod{2}, \quad (5.2)$$

for independent uniform $x, r \in \{0, 1\}^n$. Then F is one-way and b is a hard-core predicate for F . Quantitatively, a predictor with advantage ε yields, in time $\text{poly}(n, 1/\varepsilon)$ and with inverse-polynomial success, a short list containing a preimage x' satisfying $f(x') = f(x)$; checking the list with f turns this into an inverter.

The local Fourier reduction. Fix an image y and encode a predictor by $g_y(r) := (-1)^{P(y, r)}$. For a candidate x , its normalized Walsh coefficient is

$$\hat{g}_y(x) := \mathbb{E}_{r \leftarrow U_n} g_y(r) (-1)^{\langle x, r \rangle}. \quad (5.3)$$

If P predicts $\langle x, r \rangle$ with conditional success $1/2 + \eta_x$, then $\hat{g}_{f(x)}(x) = 2\eta_x$. Thus a nonnegligible global prediction advantage forces a nonnegligible set of inputs for which the hidden x is a heavy Walsh coefficient of the efficiently queryable Boolean function $g_{f(x)}$.

Why heavy coefficients form a short list. Parseval’s identity gives

$$\sum_{z \in \{0, 1\}^n} \hat{g}_y(z)^2 = 1. \quad (5.4)$$

Hence at most $1/\tau^2$ strings can satisfy $|\hat{g}_y(z)| \geq \tau$. The difficult algorithmic step is not this counting fact but *finding* all such strings in time polynomial in n and $1/\tau$ while accessing g_y as a circuit/oracle. The Goldreich–Levin list-decoding algorithm performs that task. This list-reconstruction mechanism, including its sampling and success analysis, remains source-owned.

One-wayness of the transformed function. An inverter for $F(x, r) = (f(x), r)$ immediately inverts f : on challenge $f(x)$ sample independent uniform r , invoke the F -inverter on $(f(x), r)$, and output its first component. The extra public seed therefore does not weaken one-wayness.

A two-bit worked check. For $n = 2$ and hidden $x = (1, 0)$, the target bit is the first bit of r . A predictor agreeing on three of the four seeds has success $3/4$, advantage $1/4$, and Walsh coefficient $1/2$ at frequency x . Equation (5.4) permits at most four coefficients of magnitude $1/2$; the list-decoding theorem is the scalable version of this finite observation.

5.3 Reader checkpoint

1. Give a one-way-function thought experiment where a particular input bit is trivially revealed.
2. Derive equation (5.3) from the prediction-success probability.
3. Prove the $1/\tau^2$ list-size bound from Parseval.
4. Explain why enumerating all 2^n Walsh coefficients would not prove the cryptographic theorem.
5. Show formally that adjoining r preserves one-wayness.

6. Translate a predictor of advantage $1/n^3$ into the threshold used by a Goldreich–Levin list decoder.
7. Explain why hard-core security is conditioned on seeing $f(x)$ rather than on seeing no auxiliary information.
8. Compare a hard-core predicate with the inner-product two-source extractor from Volume VIII.
9. Explain why one hard-core bit does not automatically give polynomial stretch.
10. State the role of preimage verification in turning list reconstruction into inversion.

Chapter 6

Pseudorandom Generators from One-Wayness

Pseudorandom generators convert short uniform seeds into longer strings that no efficient observer can distinguish from uniform. The first construction in this chapter is elementary once a one-way permutation and a hard-core predicate are available. The second is the deeper HILL theorem: arbitrary one-way functions already suffice.

6.1 How to read this chapter

Definition 6.1 separates stretch from security. Theorem 6.2 is proved locally for one-way permutations. Theorem 6.3 is source-owned because the regularization and pseudoentropy machinery that removes the permutation hypothesis is substantial.

6.2 Learning goals

A reader should be able to define a cryptographic PRG as a deterministic efficient map with positive stretch; prove one-bit stretch from a hard-core predicate; extend one-bit stretch to polynomial stretch by iteration/hybrids; and explain why “OWF implies PRG” is strictly deeper than “OWP plus hard-core bit implies PRG.”

Definition 6.1 (Cryptographic pseudorandom generators). A *pseudorandom generator* (PRG) is a deterministic polynomial-time family

$$G_\lambda : \{0, 1\}^{s(\lambda)} \rightarrow \{0, 1\}^{\ell(\lambda)}, \quad \ell(\lambda) > s(\lambda), \quad (6.1)$$

such that $G_\lambda(U_{s(\lambda)}) \approx_c U_{\ell(\lambda)}$. The difference $\ell - s$ is the *stretch*. A family has *polynomial stretch* if every desired polynomial output length can be obtained with polynomial seed length/runtime by a specified iteration construction.

A generator satisfies *next-bit unpredictability* if, for every PPT predictor and every output position, the next output bit cannot be predicted from the preceding output prefix with more than negligible advantage. Volume IX proved Yao’s next-bit characterization, which equates this condition with pseudorandomness for polynomial output length under the stated adversary model.

Multi-sample security concerns polynomially many independently seeded outputs and follows by an explicit sample-by-sample hybrid from one-sample security. A *post-quantum PRG* replaces the

classical distinguisher by a quantum polynomial-time distinguisher; that stronger adjective is never inferred automatically from a classical theorem.

Theorem 6.2 (One-way permutations with a hard-core predicate yield PRGs). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a one-way permutation and let b be a hard-core predicate for f . Then*

$$G(x) := f(x) \| b(x) \quad (6.2)$$

is a one-bit-stretch PRG. Moreover, for every polynomial $L(n) \geq n$, the iterated output

$$G_L(x) := f^t(x) \| b(f^{t-1}(x)) \| \cdots \| b(f(x)) \| b(x), \quad t = L(n) - n, \quad (6.3)$$

is pseudorandom and computable in polynomial time.

Proof. Because f is a permutation, $f(U_n) = U_n$. If a distinguisher D separates $(f(x), b(x))$ from (U_n, U_1) with advantage ε , define the predictor $P(f(x))$ by comparing $D(f(x), 0)$ and $D(f(x), 1)$ using the standard one-bit prediction-from-distinguishing reduction (or equivalently invoke the next-bit direction of QRCC-IX-THM-007). It predicts $b(x)$ with advantage at least a constant multiple of ε , contradicting the hard-core property. Thus (6.2) is pseudorandom.

For (6.3), it is safer not to invent a backwards-state simulation. Put

$$S_j(x) := f^j(x) \| b(f^{j-1}(x)) \| \cdots \| b(x) \quad (j \geq 0), \quad (6.4)$$

with $S_0(x) = x$. We prove by induction that $S_j(U_n) \approx_c U_{n+j}$ for every polynomially bounded j . The base $j = 1$ is the one-bit generator (6.2). For the induction step define the efficient map

$$T_j(y, u_1, \dots, u_j) := (f(y), b(y), u_1, \dots, u_j).$$

Then $T_j(S_j(x)) = S_{j+1}(x)$, so efficient closure preserves the induction hypothesis. On a uniform input, however,

$$T_j(U_n, U_j) = (G(U_n), U_j) \approx_c U_{n+1+j}$$

by the already proved one-bit pseudorandomness, with the j extra uniform bits carried as independent auxiliary randomness. A triangle inequality gives $S_{j+1}(U_n) \approx_c U_{n+j+1}$. Iterating this argument polynomially many times incurs only a polynomial sum of negligible losses. Since $S_t(x) = G_L(x)$, the desired polynomial-stretch claim follows. $\square$

Direction of the iteration matters. Writing the final state $f^t(x)$ first makes every bit replacement expose exactly the hard-core prediction interface “predict $b(z)$ from $f(z)$.” A proof that prints the bits in a different order must recheck what information the predictor receives; pseudorandomness does not follow from a diagram alone.

Theorem 6.3 (Håstad–Impagliazzo–Levin–Luby — PRGs from arbitrary OWFs; verified deep input). *Pseudorandom generators exist if and only if one-way functions exist. In particular, from any strong one-way function one can construct a deterministic polynomial-time PRG with one-bit stretch, and hence by standard iteration a PRG with arbitrary polynomial stretch.*

Proof ownership. The easy direction is local: if $G : \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ were efficiently invertible on a random output from its image, membership in the sparse image of G could be distinguished from uniform with large advantage, contradicting pseudorandomness after the usual verification step. The difficult OWF $\Rightarrow$ PRG direction is the HILL theorem. Its regularization, entropy-smoothing, and pseudoentropy conversion mechanisms remain source-owned. This boundary is intentional: Goldreich–Levin alone gives a hard-core predicate, but arbitrary OWFs are not permutations, so the simple state-iteration proof of Theorem 6.2 does not apply.

The cryptographic dependency chain so far. The volume has now established the conceptual sequence

$$\text{weak OWF} \implies \text{strong OWF} \implies \text{hard-core bit} \implies \text{PRG}, \quad (6.5)$$

with the first, second, and arbitrary-OWF-to-PRG transformations source-owned at their genuinely deep steps. Chapter 7 turns a PRG into an oracle-like pseudorandom function.

6.3 Reader checkpoint

1. Prove that the image of any one-bit-stretch deterministic generator has uniform measure at most $1/2$ inside the output space.
2. Explain why that sparsity does not make the image efficiently recognizable.
3. Derive a predictor for $b(x)$ from a distinguisher for (6.2).
4. Verify that $f^j(U_n) = U_n$ uses the permutation property.
5. Work out the three-bit-stretch instance of (6.3) and its hybrids.
6. Explain why an arbitrary one-way function cannot simply be iterated as a state permutation.
7. Use Yao's next-bit theorem to give an alternate proof of polynomial stretch.
8. Prove multi-sample security for $q(\lambda)$ independent PRG seeds by a q -hybrid.
9. State the change required in Definition 6.1 for a quantum distinguisher.
10. Explain exactly which arrow in (6.5) is HILL rather than Goldreich–Levin.

Chapter 7

Pseudorandom Functions and Feistel Permutations

A pseudorandom generator produces a long pseudorandom string from a short seed. A pseudorandom function (PRF) is stronger: one short key determines an exponentially large table that remains indistinguishable from a truly random function even under adaptive oracle queries. The GGM tree provides this upgrade. Feistel networks then turn PRFs into efficiently invertible pseudorandom permutations.

7.1 How to read this chapter

Definition 7.1 fixes the oracle experiments. Theorem 7.2 is proved locally by a lazy-tree hybrid. Theorem 7.3 records the exact classical Luby–Rackoff boundary as a verified deep input, because its collision transcript analysis is a substantial theorem. The executable laboratory at the end is diagnostic, not evidence of security.

7.2 Learning goals

A reader should be able to define PRF, PRP, and strong-PRP security as oracle games; evaluate the GGM tree; explain why adaptively queried leaves require consistent lazy sampling; and distinguish the three-round forward-only Feistel guarantee from the four-round chosen-direction guarantee.

Definition 7.1 (Pseudorandom functions and permutations). A keyed family $F = \{F_k : \{0, 1\}^m \rightarrow \{0, 1\}^n\}_{k \in \{0, 1\}^s}$ is a *pseudorandom function* if no PPT adversary making polynomially many adaptive classical oracle queries can distinguish $F_K(\cdot)$ for uniform K from a uniformly random function $R : \{0, 1\}^m \rightarrow \{0, 1\}^n$ sampled lazily and consistently.

A *weak PRF* restricts the adversary to random or otherwise specified nonadaptive inputs; it is not the default notion. *Multi-user PRF security* gives the adversary oracle access to independently keyed instances and requires the number of users in the advantage ledger.

A keyed permutation family $P_k : \{0, 1\}^N \rightarrow \{0, 1\}^N$ is a *PRP* if forward-oracle access to P_K is indistinguishable from a uniformly random permutation. It is a *strong PRP* (SPRP) if the adversary may query both forward and inverse directions, subject to consistency. These are classical query notions. Quantum superposition-query PRFs/PRPs are separate definitions.

GGM construction. Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a length-doubling PRG and write $G(s) = G_0(s) \| G_1(s)$ with $G_b(s) \in \{0, 1\}^n$. For a key $s \in \{0, 1\}^n$ and input $x = x_1 \cdots x_m$, define

$$F_s(x) := G_{x_m}(G_{x_{m-1}}(\cdots G_{x_1}(s) \cdots)). \quad (7.1)$$

Thus a key labels the root of a binary tree and the two halves of G label the children of every expanded node.

Theorem 7.2 (Goldreich–Goldwasser–Micali PRF construction). *If G is a secure length-doubling PRG, then the family (7.1), for every polynomial input length $m(n)$, is a PRF. Quantitatively, an adversary making at most q oracle queries and distinguishing the GGM family from a random function with advantage ε yields a PRG distinguisher with advantage at least $\varepsilon/(qm)$, up to a constant-factor and bookkeeping term from the exact node-selection simulation.*

Proof. Run the adversary with a lazily generated binary tree. Pad its interaction to exactly q oracle queries and regard each length- m query as contributing m chronological expansion *slots*; a slot that revisits an already expanded node is a dummy slot. Thus there are exactly qm slots. In a real expansion the two child labels are $G_0(s_v), G_1(s_v)$, while in an ideal expansion they are two independent uniform n -bit strings. Labels already assigned to a node are never resampled.

For $j = 0, \dots, qm$, let H_j be the experiment in which every genuine expansion whose first-use slot is at most j receives ideal uniform children and every later genuine expansion uses G . Dummy slots make $H_{j-1} = H_j$. The adaptive order is determined by the simulated transcript, so these hybrids are well-defined. If p_j is the adversary's acceptance probability in H_j , then

$$|p_0 - p_{qm}| = \varepsilon, \quad p_0 - p_{qm} = \sum_{j=1}^{qm} (p_{j-1} - p_j). \quad (7.2)$$

Build a PRG distinguisher by choosing J uniformly from $[qm]$ and simulating the hybrid boundary at slot J : genuine expansions before J get independent uniform children, genuine expansions after J use G , and at slot J (if it is a genuine first expansion) the $2n$ -bit PRG challenge supplies the two children. If slot J is dummy, the two challenge worlds induce the same transcript. At a genuine slot, the parent label is uniform: it is either the uniform root or a child created by an earlier ideal expansion. Therefore the two challenge worlds realize H_{J-1} and H_J exactly. Averaging over J , the signed acceptance-probability gap of this distinguisher is

$$\frac{1}{qm} \sum_{j=1}^{qm} (p_{j-1} - p_j) = \frac{p_0 - p_{qm}}{qm},$$

so its absolute PRG advantage is $\varepsilon/(qm)$. This establishes the stated quantitative loss without an additional guess of a “critical” node.

When every genuine expansion is ideal, distinct queried leaves receive consistent independent uniform labels, exactly the lazy-sampling distribution of a random function on the adversary's queried inputs. $\square$

Feistel transform. For a function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$, one Feistel round maps

$$(L, R) \mapsto (R, L \oplus f(R)). \quad (7.2)$$

It is a permutation for every f , with inverse $(L', R') \mapsto (R' \oplus f(L'), L')$. Thus invertibility does not require f itself to be invertible.

Theorem 7.3 (Luby–Rackoff pseudorandom permutations — verified deep input). *Let F_1, F_2, F_3, F_4 be independent secure PRFs on n -bit blocks.*

- (i) *The three-round Feistel permutation $\Psi(F_1, F_2, F_3)$ is a PRP on $2n$ bits against polynomially many forward queries.*
- (ii) *The four-round Feistel permutation $\Psi(F_1, F_2, F_3, F_4)$ is a strong PRP against polynomially many adaptive forward and inverse queries.*

After replacing the PRFs by independent random functions, the information-theoretic distinguishing loss is of birthday order $O(q^2/2^n)$ for q oracle queries, with constants depending on the exact transcript formulation.

Proof ownership and round-count warning. The PRF-to-random-function replacement is an ordinary hybrid using Theorem 7.2. The deep Luby–Rackoff step analyzes collisions among internal Feistel variables and proves that, outside a birthday-probability bad event, the transcript has the same law as a random permutation transcript. That collision argument remains source-owned. Three rounds do not give the same inverse-oracle guarantee as four rounds; “PRP” and “SPRP” are therefore not interchangeable.

7.3 Executable PRG/PRF/Feistel laboratory

The companion program `code/volume_16/crypto_primitives_lab.py` exhaustively checks, on deliberately tiny domains, four invariants: deterministic PRG image size versus uniform output space; GGM tree consistency; Feistel invertibility for arbitrary round functions; and empirical distinguishing advantage of selected small tests. Its output labels empirical frequencies as *experiments*. They do not certify asymptotic pseudorandomness.

7.4 Reader checkpoint

1. Evaluate the GGM tree by hand for a two-bit input.
2. Explain why two equal oracle queries must return the same random-function answer.
3. Prove that at most qm internal nodes are expanded by q length- m queries.
4. Write the PRG reduction for a guessed expansion index explicitly.
5. Show directly that one Feistel round is invertible.
6. Construct a simple distinguisher showing that one Feistel round is not a secure PRP.
7. Explain why three-round security does not automatically survive inverse queries.
8. State a multi-user PRF experiment with u users and q total queries.
9. Explain why exhaustive tests on a 4-bit domain cannot establish a cryptographic PRF theorem.
10. State what must change to define a quantum-superposition-query PRF.

Chapter 8

Symmetric Encryption: Semantic Security and IND-CPA

Computational encryption formalizes the promise that a ciphertext reveals no efficiently usable information about its plaintext. Two languages for that promise became standard: semantic security, which talks about computing functions of a message, and indistinguishability, which asks whether two chosen messages can be told apart. They are equivalent under the conventions fixed here, while indistinguishability is usually cleaner for reductions.

8.1 How to read this chapter

Definition 8.1 separates eavesdropping, chosen-plaintext, nonce, and stateful models. Theorem 8.2 proves semantic/indistinguishability equivalence in the fixed-length private-key setting. Theorem 8.3 then gives a complete PRF-stream construction and loss ledger; the following worked example expands every hybrid.

8.2 Learning goals

A reader should be able to write the IND-CPA experiment, identify legal challenge restrictions, explain why encryption must be randomized/stateful for multi-message security, prove the semantic-security equivalence, and separate PRF replacement from nonce-collision loss.

Definition 8.1 (Symmetric encryption and security notions). A symmetric encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has PPT key generation, possibly randomized/stateful encryption, deterministic or randomized decryption, and correctness $\Pr[\text{Dec}_K(\text{Enc}_K(m)) = m] \geq 1 - \text{negl}(\lambda)$ for every admissible message.

In the *IND-EAV* game, an adversary outputs equal-length messages (m_0, m_1) , receives $c^* \leftarrow \text{Enc}_K(m_b)$ for uniform b , and guesses b . Its normalized advantage is $2|\Pr[b' = b] - 1/2|$. In *IND-CPA*, the adversary additionally receives an encryption oracle before and, unless a theorem says otherwise, after the challenge. Equal challenge lengths prevent trivial length leakage.

Semantic security means that whatever efficiently computable target $h(M)$ an adversary can predict from a ciphertext together with efficiently samplable auxiliary information Z , a PPT simulator given only the permitted public data $(1^\lambda, Z)$ can predict with negligibly smaller success. The message distribution and target family are required to be efficiently samplable/computable.

A *nonce-based* scheme takes an explicit nonce N . Security may require nonces to be unique, random, or adversary chosen; the requirement is part of the theorem. A *stateful* scheme maintains sender state to enforce a property such as counter uniqueness. Reusing a nonce in a scheme proved only for unique/random nonces is misuse, not an attack contradicting the theorem.

Theorem 8.2 (Semantic security equals indistinguishability). *For polynomial-time private-key encryption of fixed-length messages, under the efficient-sampler and auxiliary-information conventions of Definition 8.1, semantic security is equivalent to indistinguishability under the corresponding attack model (EAV or CPA). The reductions incur only polynomial overhead and constant-factor changes in normalized advantage.*

Proof. We prove the EAV form; oracle simulation gives the CPA form.

Assume indistinguishability. Let (M, Z) be an efficiently samplable message/auxiliary distribution and let A compute a target from $(Z, \text{Enc}_K(M))$. Fix a canonical dummy message 0^ℓ of the same declared length and define simulator $S(Z)$ to generate an independent key, encrypt 0^ℓ , and run A on that dummy ciphertext together with Z . The simulator therefore needs no ability to sample the conditional distribution $M \mid Z$. If replacing $\text{Enc}_K(M)$ by an independent encryption of 0^ℓ changed the success probability of predicting $h(M)$ noticeably, a reduction samples (M, Z) itself and uses $(M, 0^\ell)$ as its equal-length IND challenge pair while evaluating the same success predicate. This yields an IND distinguisher with the same noticeable gap. Thus ciphertext access gives no noticeably greater target-prediction success than the simulator's permitted information.

Conversely assume semantic security and suppose an IND adversary distinguishes two efficiently generated equal-length messages m_0, m_1 with nonnegligible advantage. Let $M = m_B$ for uniform bit B and choose target $h(M) = B$ (duplicate messages can be removed or indexed in the sampler so that the target is well defined). The IND adversary predicts B from the ciphertext with success $1/2 + \varepsilon/2$. A simulator with no ciphertext sees auxiliary data independent of B and succeeds with probability at most $1/2$ (up to any explicitly allowed prior bias). This contradicts semantic security. The oracle version runs the same argument while forwarding legal encryption queries. $\square$

PRF stream encryption. Let $F_k : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ be a PRF. To encrypt $m \in \{0, 1\}^\ell$, sample $r \leftarrow \{0, 1\}^n$ and output

$$c = (r, m \oplus F_k(r)). \quad (8.1)$$

Decryption recomputes the pad and XORs it away.

Theorem 8.3 (PRF-based randomized encryption is IND-CPA secure). *For the scheme (8.1), let A make at most q encryption-oracle queries in addition to one challenge. There is a PRF distinguisher B with comparable running time and at most $q + 1$ oracle queries such that*

$$\text{Adv}_{\Pi}^{\text{ind-cpa}}(A) \leq \text{Adv}_F^{\text{prf}}(B) + \frac{q}{2^n}. \quad (8.2)$$

A more conservative proof that aborts on any nonce collision among all $q + 1$ encryption samples may replace the last term by $\binom{q+1}{2}/2^n$.

Proof. Game 0 is the real IND-CPA experiment. Game 1 replaces F_k by a lazy random function R ; the gap is exactly the PRF advantage of a wrapper B that implements every encryption query using its oracle.

In Game 1 let r^* be the challenge nonce and let **Bad** be the event that some other queried encryption uses the same nonce. Each of at most q independently sampled nonces equals r^* with

probability 2^{-n} , so $\Pr[\text{Bad}] \leq q/2^n$. Conditioned on $\neg\text{Bad}$, the random function value $R(r^*)$ has never been exposed. It is therefore a uniform independent ℓ -bit pad, so $m_b \oplus R(r^*)$ is uniform and independent of b . The challenge gives zero distinguishing advantage in that conditioned game. The bad-event coupling lemma from Theorem 1.3 yields (8.2). $\square$

8.3 Worked hybrid: four encryption queries

Take $q = 4$. The proof ledger is

$$\text{Real PRF} \xrightarrow{\text{Adv}_F^{\text{prf}}(B)} \text{Random function} \xrightarrow{\Pr[r^* \in \{r_1, r_2, r_3, r_4\}]} \text{Fresh challenge pad} \xrightarrow{0} \text{Hidden challenge bit.} \quad (8.3)$$

The collision term is at most $4/2^n$. If one instead aborts whenever any of the five nonces collide, the union bound gives $\binom{5}{2}/2^n = 10/2^n$. The latter is weaker but sometimes simpler when several parts of a protocol depend on global nonce uniqueness. The example illustrates a general proof-design choice: use the smallest bad event that makes the conditioned games literally identical.

Stateful counter variant. If sender state guarantees a never-repeated n -bit counter during its supported lifetime, the collision term disappears. Security then depends on enforcing the state invariant. Resetting the sender and reusing counters is outside that theorem.

8.4 Reader checkpoint

1. Write the complete IND-EAV experiment as pseudocode.
2. Explain the equal-message-length restriction in IND games.
3. Give a deterministic encryption scheme and show why multi-message IND-CPA fails.
4. Fill in the conditioning step in the semantic-security $\Rightarrow$ indistinguishability reduction.
5. Derive the $q/2^n$ challenge-collision bound in (8.2).
6. Compare that bound with the all-pairs birthday bound.
7. Explain why a repeated challenge nonce reveals the one-time pad to a chosen-plaintext adversary.
8. Modify (8.1) to use a counter and state the new misuse assumption.
9. State the PRF oracle budget of the reduction B exactly.
10. Explain why post-quantum security of (8.1) requires a quantum-secure PRF theorem rather than only a classical PRF theorem.

Chapter 9

Authentication, MACs, and Digital Signatures

Confidentiality and authenticity are different goals. Encryption tries to hide a message; a message authentication code prevents an active attacker from creating a new valid message/tag pair; a digital signature makes verification public. This chapter develops information-theoretic authentication, computational MACs, one-time signatures, and the deep theorem that arbitrary one-way functions already suffice for full digital signatures.

9.1 How to read this chapter

Definition 9.1 separates MAC and signature experiments before any construction is stated. Theorems 9.2–9.4 are local reductions. The Rompel theorem is source-owned because its universal-one-way-hash and authentication-tree machinery is substantial.

9.2 Learning goals

A reader should be able to define EUF-CMA without confusing secret-key and public-key verification; prove a Wegman–Carter forgery bound; reduce a PRF MAC forgery to PRF distinguishing plus random-tag guessing; and reduce a Lamport one-time forgery to inversion of one randomly embedded one-way image.

Definition 9.1 (Authentication and signature security). A MAC consists of Gen , Tag , Vrfy sharing a secret key. In the *EUFCMA* experiment an adversary receives a tagging oracle and wins by outputting a pair (m^*, t^*) that verifies and whose message m^* was never submitted to the oracle. *Strong unforgeability* additionally treats a new valid tag for a previously queried message as a forgery.

A digital signature scheme has $\text{Gen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$, signing $\text{Sign}_{\text{sk}}(m)$, and public verification $\text{Vrfy}_{\text{pk}}(m, \sigma)$. EUF-CMA gives the adversary the public key and an adaptive signing oracle and requires a valid signature on a never-signed message. A *one-time signature* permits at most one signing query. Multi-user security explicitly counts independently generated keys; it is not silently implied without a hybrid/union-bound argument.

An information-theoretic authentication code may permit an unbounded adversary and measures forgery probability directly over secret keys. Computational MAC/signature security instead quantifies over the chosen efficient adversary model.

Theorem 9.2 (Wegman–Carter one-time authentication). *Let $\mathcal{H}$ be a strongly universal₂ family from message set $\mathcal{M}$ to a finite abelian tag group T : for every distinct m, m' and every $a, b \in T$,*

$$\Pr_{h \leftarrow \mathcal{H}}[h(m) = a \wedge h(m') = b] = |T|^{-2}. \quad (9.1)$$

Choose $h \leftarrow \mathcal{H}$ and a uniform one-time mask $S \leftarrow T$, and authenticate one message m by

$$t = h(m) + S. \quad (9.2)$$

After seeing one valid pair (m, t) , any unbounded adversary that outputs $m' \neq m$ and tag t' succeeds with probability exactly $1/|T|$ (or at most that value for an almost-strongly-universal family with the corresponding parameter).

Proof. Fix the observed (m, t) and candidate forgery (m', t') with $m' \neq m$. For each hash key h , the observed tag determines the mask uniquely as $S = t - h(m)$. The forgery verifies exactly when

$$h(m') - h(m) = t' - t. \quad (9.3)$$

Under strong universality, conditioned on $h(m) = a$, the value $h(m')$ is uniform on T . Equivalently the difference in (9.3) is uniform, so the probability of the required value is $1/|T|$. No computational restriction was used. $\square$

Hash-key reuse. The theorem is one-time because the mask S is one-time. A long-lived universal-hash key can be reused with independent secret masks under the corresponding authentication theorem, but key recycling after accept/reject feedback requires a separate leakage analysis. We do not infer that stronger statement from the one-message proof.

Theorem 9.3 (PRF-based MAC security). *Let $F_k : \mathcal{M} \rightarrow \{0, 1\}^\tau$ be a PRF and define $\text{Tag}_k(m) = F_k(m)$ with verification by equality. For an EUF-CMA adversary making at most q tagging queries, there is a PRF distinguisher B such that*

$$\Pr[\text{Forge}_A] \leq \text{Adv}_F^{\text{prf}}(B) + 2^{-\tau}. \quad (9.4)$$

For variable-length messages, the same theorem applies after an injective, prefix-free, efficiently computable encoding into the PRF domain (within the declared maximum message length); merely concatenating variable-length blocks without domain separation is not covered.

Proof. Replace F_k by a lazy random function R . The game gap is the PRF advantage of the natural wrapper. In the random-function game, the forgery message m^* is new by the EUF-CMA winning condition, so $R(m^*)$ has never been returned and is a uniform independent τ -bit string even after adaptive prior queries. The adversary's proposed t^* matches it with probability exactly $2^{-\tau}$. Summing the hybrid gap and ideal-world success proves (9.4). $\square$

Theorem 9.4 (Lamport one-time signatures from one-way functions). *Let $f : \{0, 1\}^s \rightarrow \{0, 1\}^m$ be a one-way function and let messages be ℓ -bit strings. Generate independent secrets $x_{i,b} \leftarrow \{0, 1\}^s$ for $i \in [\ell]$, $b \in \{0, 1\}$, publish $y_{i,b} = f(x_{i,b})$, and sign $m = m_1 \cdots m_\ell$ by revealing $(x_{1,m_1}, \dots, x_{\ell,m_\ell})$. Then the scheme is one-time EUF-CMA secure. More precisely, a forger succeeding with probability ε yields an inverter for f with success at least $\varepsilon/(2\ell)$, up to the negligible correctness/format terms of the exact family model.*

Proof. Given a random inversion challenge $y = f(x)$, choose a random coordinate $I \in [\ell]$ and random bit $B \in \{0, 1\}$. Put y in public-key position (I, B) and generate every other secret/public pair honestly. When the adversary makes its sole signing query m , the reduction can answer unless $m_I = B$; if equality holds it aborts. If it does not abort and the adversary later produces a valid forgery on $m' \neq m$, then whenever $m'_I = B$ the forged signature reveals a string x' with $f(x') = y$.

Condition on a successful forgery. The set $D = \{i : m_i \neq m'_i\}$ is nonempty. The random I lies in D with probability at least $1/\ell$. Given such an I , exactly one choice of B equals m'_I and is opposite m_I , so with probability $1/2$ the reduction both answers the signing query and extracts a preimage from the forgery. Thus the inversion success is at least $\varepsilon/(2\ell)$. Because the one-way definition accepts any preimage, injectivity of f is unnecessary. $\square$

Signing long messages. If an L -bit document is first compressed to an ℓ -bit digest, the resulting theorem must separately assume that the adversary cannot find a different message with the same signed digest under the chosen hashing interface. One-wayness of f alone does not provide collision resistance of an arbitrary message-compression hash.

Theorem 9.5 (Rompel — digital signatures from one-way functions; verified deep input). *If one-way functions exist, then polynomial-time digital signature schemes secure against adaptive chosen-message attacks exist.*

Proof ownership. Rompel's STOC 1990 theorem closes the generic Minicrypt signature chain. The construction passes through universal one-way hash functions and authentication-tree machinery that upgrades one-time signatures to many-message public verification. Those mechanisms remain source-owned. The local Lamport proof above isolates the simple one-time core and makes clear why the many-message theorem is not obtained by merely reusing the same Lamport key.

9.3 Reader checkpoint

1. Distinguish MAC EUF-CMA from signature EUF-CMA in one sentence each.
2. Prove equation (9.3).
3. Generalize Theorem 9.2 to an ε -almost strongly universal family.
4. Explain why fresh one-time masks can hide repeated use of a universal-hash key.
5. Reproduce the random-function proof of (9.4).
6. Give a variable-length encoding that is injective and prefix-free.
7. Work out the Lamport reduction when $\ell = 3$.
8. Explain why the reduction loses a factor 2ℓ rather than merely ℓ .
9. State what additional property is needed to sign arbitrary long documents through hashing.
10. Explain why Rompel's theorem does not imply public-key encryption from OWFs.

Chapter 10

Chosen-Ciphertext Security, Hybrid Encryption, and Post-Quantum Proof Discipline

Chosen-ciphertext security is where composition mistakes become costly. A decryption oracle forces the proof to account for malformed ciphertexts, replay exclusions, authentication failures, and proof-system simulation. This chapter closes the classical foundations volume with three composition paradigms and a dated warning about transporting classical reductions into quantum-adversary models.

10.1 How to read this chapter

Definitions 10.1–10.3 fix CCA, KEM/DEM, simulation-sound proof, and DDH interfaces before any theorem uses them. Encrypt-then-MAC and KEM–DEM are proved locally. Naor–Yung is proved at the generic hybrid level assuming its simulation-sound NIZK interface. The concrete Cramer–Shoup algebraic construction remains source-owned.

10.2 Learning goals

A reader should be able to write the IND-CCA2 experiment with its challenge-ciphertext exclusion; explain why ciphertext integrity neutralizes decryption-oracle attacks on Encrypt-then-MAC; state the exact simulation-soundness role in Naor–Yung; prove a KEM–DEM composition hybrid; and identify why quantum adversaries can invalidate classical oracle-programming or rewinding arguments even when the underlying hardness assumption remains quantum resistant.

Definition 10.1 (Public-key CCA security, authenticated encryption, and KEM/DEM). A public-key encryption scheme has $\text{Gen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$, randomized $\text{Enc}_{\text{pk}}(m)$, and $\text{Dec}_{\text{sk}}(c) \in \mathcal{M} \cup \{\perp\}$. IND-CPA gives the adversary pk and a challenge encryption of one of two equal-length messages. In *IND-CCA2*, it additionally receives a decryption oracle before and after the challenge, except that the exact challenge ciphertext c^* may not be queried. IND-CCA1 allows decryption queries only before the challenge. *Non-malleability* is a related relation-based notion and is not identified with IND-CCA without the hypotheses of a stated equivalence theorem.

For symmetric encryption, *ciphertext integrity* (INT-CTXT) requires that an adversary with an encryption oracle cannot produce a new ciphertext accepted by decryption. An *authenticated*

encryption scheme supplies both confidentiality and integrity under its declared nonce/associated-data model. Associated data (AD) is authenticated but not encrypted and is an explicit input to both algorithms.

A *key encapsulation mechanism* (KEM) consists of key generation, encapsulation $\text{Encap}_{\text{pk}} \rightarrow (c_K, K)$, and decapsulation $\text{Decap}_{\text{sk}}(c_K) \rightarrow K$ or $\perp$. IND-CCA KEM security asks whether the challenge key paired with c_K^* is the real encapsulated key or an independent uniform key, while granting a decapsulation oracle on every $c_K \neq c_K^*$. A *data encapsulation mechanism* (DEM) is a one-time symmetric encryption/authenticated-encryption primitive under a uniformly random session key. A KEM–DEM ciphertext is a pair (c_K, c_D) ; the theorem must state how c_K is bound into the DEM, usually as associated data.

Definition 10.2 (Simulation-sound noninteractive-proof interface). A CRS-model noninteractive proof/argument system for relation R consists of $\text{Setup}(1^\lambda) \rightarrow \text{crs}$, proving $\text{Prove}_{\text{crs}}(x, w)$, and verification $\text{Vrfy}_{\text{crs}}(x, \pi)$. Completeness requires accepting true statements with valid witnesses. Computational soundness forbids PPT production of an accepting proof for $x \notin L_R$. Zero knowledge provides a simulator producing a simulated CRS and proofs computationally indistinguishable from real ones for true statements.

The system is *simulation sound* for the interface used here if, even after receiving polynomially many simulated proofs under the simulated CRS, no PPT adversary can produce a new accepting proof of a false statement outside the explicitly excluded simulated instances. The exact theorem below assumes this multi-theorem adaptive form. Setup trust and CRS generation are part of the model.

Definition 10.3 (Abstract DDH group family). A DDH group family is a PPT sampler that on 1^λ outputs a description of a cyclic group G of prime order q , a generator g , canonical encodings, and efficient group operations/exponentiation, with $\log q = \text{poly}(\lambda)$. The *decisional Diffie–Hellman (DDH) assumption* says that for uniform $a, b, c \in \mathbb{Z}_q$, the ensembles

$$(G, q, g, g^a, g^b, g^{ab}) \quad \text{and} \quad (G, q, g, g^a, g^b, g^c) \quad (10.1)$$

are computationally indistinguishable in the declared adversary model. This abstract interface is used only to state standard-model constructions such as Cramer–Shoup; no concrete elliptic-curve or finite-field group is assumed secure merely by fitting the syntax.

Theorem 10.4 (Encrypt-then-MAC gives authenticated encryption). *Let E be an IND-CPA secure symmetric encryption scheme and M a MAC providing ciphertext integrity against chosen-message attacks on the ciphertext domain (for example a secure PRF MAC with appropriate encoding). With independent keys define*

$$c \leftarrow E_{k_E}(m), \quad t \leftarrow M_{k_M}(c), \quad C = (c, t), \quad (10.2)$$

and decrypt only after verifying t . Then Encrypt-then-MAC is IND-CCA secure and INT-CTXT secure. A concrete proof bounds CCA advantage by an IND-CPA term for E plus a MAC/ciphertext-integrity forgery term, with the exact query overhead of the simulation.

Proof. Consider an IND-CCA adversary. Couple the real game with a game that rejects every decryption query (c, t) not previously produced by the encryption simulation. If the games ever differ, the adversary has produced a new valid ciphertext/tag pair, yielding a MAC/ciphertext-integrity

forgery. Conditioned on no such event, decryption queries reveal nothing beyond plaintexts of ciphertexts already generated by the simulator, which it can answer from its own encryption records. In particular the challenge cannot be decrypted because the exact challenge pair is excluded by the CCA game.

The remaining experiment is therefore an IND-CPA encryption game with tags that the wrapper can compute under its independently sampled MAC key. Any residual challenge-bit advantage yields an IND-CPA breaker for E . The game-hopping theorem adds the two losses. The same forgery game directly proves INT-CTXT. The proof relies on authenticating the *ciphertext*; it does not prove the analogous claim for MAC-then-encrypt or encrypt-and-MAC. $\square$

Theorem 10.5 (Naor–Yung/Sahai double encryption with simulation-sound proofs). *Let E_0, E_1 be independent public-key encryption schemes that are IND-CPA secure. Suppose there is a CRS-model simulation-sound NIZK proof system for the relation asserting that two ciphertexts encrypt the same plaintext under the two public keys. Encrypt m by*

$$c_0 \leftarrow E_{0, \text{pk}_0}(m), \quad c_1 \leftarrow E_{1, \text{pk}_1}(m), \quad \pi \leftarrow \text{Prove}(c_0, c_1; \text{coins/message witness}), \quad (10.3)$$

and decrypt only if π verifies, then decrypt either component. Under the stated proof-system interface, the resulting public-key encryption is IND-CCA secure.

Proof. The reduction uses the zero-knowledge simulator to generate the CRS and to simulate the challenge proof, so it need not know witnesses for both challenge ciphertexts during hybrids. First replace the challenge plaintext inside c_0 using IND-CPA security of E_0 , while simulating the proof. Then replace it inside c_1 using IND-CPA security of E_1 . At the endpoint the challenge pair contains encryptions whose joint distribution is independent of the challenge bit, up to the two IND-CPA losses and the proof simulation gap.

It remains to simulate the decryption oracle without owning both secret keys in each hybrid. For any adversarial ciphertext with an accepting proof, simulation soundness implies—except with negligible probability—that the two components encrypt the same plaintext, unless the tuple is one of the explicitly simulated challenge instances. Therefore the reduction may decrypt using the secret key it knows; consistency with the unknown component follows from soundness. A new accepting inconsistent pair would itself violate simulation soundness. The exact challenge tuple is excluded by IND-CCA2. Summing the proof-simulation, two IND-CPA, and simulation-soundness losses yields negligible total advantage. $\square$

Why ordinary zero knowledge is insufficient, and source boundary. The CCA adversary sees a simulated challenge proof and can then submit new proof-bearing ciphertexts to the decryption oracle. Ordinary soundness is defined under an honestly generated CRS with no simulated proof exposure. Simulation soundness is the property that closes precisely this adaptive gap. The original Naor–Yung 1990 theorem is the historical double-encryption/NIZK paradigm in the preprocessing (“lunchtime”) chosen-ciphertext model. The adaptive CCA2 interface used in the theorem above is calibrated to the later Naor–Yung/Sahai line with a strengthened noninteractive proof property; the two security models are not identified.

Theorem 10.6 (Cramer–Shoup standard-model adaptive-CCA encryption — verified deep input). *Under the DDH assumption in an appropriate efficiently samplable prime-order group family and the specified universal one-way/hash interface, the Cramer–Shoup public-key encryption scheme is secure against adaptive chosen-ciphertext attack in the standard model.*

Proof ownership. The primary Cramer–Shoup construction and reduction remain source-owned. The reader-facing statement keeps the group model, DDH assumption, hash requirement, decryption validity equation, and adversary interface explicit; the algebraic case split that turns a successful malformed-ciphertext attack into a DDH distinguisher is not compressed into a pseudo-proof.

Theorem 10.7 (KEM–DEM composition). *Let K be an IND-CCA secure KEM and let D be a one-time authenticated DEM secure under a uniform session key. Construct hybrid encryption by*

$$(c_K, K_s) \leftarrow \text{Encap}_{\text{pk}}, \quad c_D \leftarrow \text{DEM.Enc}_{K_s}(m; \text{ad} = c_K), \quad C = (c_K, c_D). \quad (10.4)$$

Decryption decapsulates c_K and then verifies/decrypts c_D using associated data c_K . Then the hybrid scheme is IND-CCA secure. Its advantage is bounded by the KEM IND-CCA advantage plus the one-time DEM authenticated-encryption advantage, with matching decapsulation/decryption query overhead.

Proof. Game 0 is the real hybrid CCA game. In Game 1 replace the challenge session key associated with the challenge encapsulation c_K^* by an independent uniform key. The KEM IND-CCA game gives exactly this replacement while allowing the reduction to decapsulate every $c_K \neq c_K^*$. Hence the gap is bounded by KEM advantage.

Now condition on the random challenge session key. For any decryption query with $c_K \neq c_K^*$, the reduction decapsulates normally and answers. Queries with $c_K = c_K^*$ but modified c_D are exactly one-time DEM decryption/forgery queries under the challenge session key and associated data c_K^* ; the exact challenge pair is excluded. Thus the remaining challenge-bit advantage is bounded by the DEM’s one-time CCA/authenticated-encryption advantage. Binding c_K as associated data prevents an attacker from moving the DEM component to a different encapsulation without authentication failure. Summing the two hybrid gaps proves the theorem. $\square$

10.3 Dated frontier: when do classical reductions survive quantum adversaries?

Status as of 2026-09-03. There is no blanket theorem saying that a classical construction remains secure against quantum adversaries whenever its underlying hardness assumption is “post-quantum.” At least three adversary interfaces must be separated:

- (a) a quantum polynomial-time adversary that receives only classical inputs and makes only classical oracle queries;
- (b) a quantum adversary making coherent superposition queries to a classical function oracle;
- (c) protocols whose classical proof uses rewinding, oracle programming, or extraction from an adversary’s internal state.

Zhandry’s quantum-PRF work shows that classical-query security and superposition-query security can separate, while Boneh–Zhandry formulate quantum chosen-message/chosen-ciphertext models in which oracle queries themselves may be coherent. Watrous and Unruh developed quantum rewinding techniques for specific proof structures, but classical rewinding is not generically valid in the presence of unknown quantum state.

The preservation rule for this book is therefore syntactic and theorem-driven: a classical black-box reduction may often lift to a QPT adversary with *classical* interfaces if the simulator can run that adversary coherently as an opaque subroutine and the assumption itself is quantum hard,

but superposition oracles, measurement, cloning, and rewinding require a dedicated theorem. No result in Chapters 1–10 is labelled post-quantum solely because its classical assumption has a quantum-resistant candidate.

Volume XVI closure map. The volume’s classical dependency spine is

$$\begin{aligned} \text{OWF} \implies \text{hard-core bits / PRGs} \implies \text{PRFs} \implies \begin{cases} \text{IND-CPA encryption,} \\ \text{MACs,} \\ \text{Feistel PRPs,} \end{cases} & (10.5) \\ \text{OWF} \implies \text{signatures,} & \quad \text{CCA compilers/KEM-DEM} \implies \text{active-channel encryption.} \end{aligned}$$

The public-key arrow from arbitrary OWFs remains absent; Chapter 4 records that frontier rather than drawing a conjectural implication.

10.4 Reader checkpoint

1. Write the exact IND-CCA2 challenge-ciphertext exclusion.
2. Explain why Encrypt-then-MAC decryption verifies before decrypting.
3. Identify the event that converts a new accepted EtM ciphertext into a MAC forgery.
4. State why the Naor–Yung proof needs simulation soundness after a simulated challenge proof is shown.
5. Distinguish a DDH group *interface* from a claim that a named concrete group is secure.
6. Reproduce the two hybrids in the KEM–DEM proof.
7. Explain why c_K is included as DEM associated data in (10.4).
8. Give an example of a classical reduction step that is problematic under quantum rewinding.
9. Distinguish a QPT adversary with classical PRF queries from a superposition-query adversary.
10. Using (10.5), identify which arrows are local proofs and which rely on deep source-owned theorems.

Internalization and Reconstruction Guide: Volume XVI

Purpose. Learn cryptography as quantified reduction statements: correctness plus an adversarial game, a computational assumption, and a reduction whose loss and oracle access are explicit; composition is part of the theorem, not an afterthought.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-008: Entropy chain rule.
2. QRCC-IX-THM-007: Yao next-bit characterization.
3. QRCC-XV-DEF-020: BQP and quantum reductions.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XVI-DEF-001 (Security parameters, adversaries, and games); QRCC-XVI-DEF-004 (Perfect secrecy); QRCC-XVI-DEF-007 (Computational indistinguishability); QRCC-XVI-DEF-009 (One-way functions and permutations); QRCC-XVI-DEF-012 (Hard-core predicates).

Chapter retrieval map

- Ch. 1: Security Parameters, Games, and Reductions:** enter through QRCC-XVI-DEF-001 (Security parameters, adversaries, and games); exit by being able to state QRCC-XVI-THM-002 (Closure calculus for negligible functions) with its ownership boundary.
- Ch. 2: Perfect Secrecy and the Cost of Information-Theoretic Encryption:** enter through QRCC-XVI-DEF-004 (Perfect secrecy); exit by being able to state QRCC-XVI-THM-005 (One-time pad perfect secrecy) with its ownership boundary.
- Ch. 3: Computational Indistinguishability and Hybrid Arguments:** enter through QRCC-XVI-DEF-007 (Computational indistinguishability); exit by being able to state

QRCC-XVI-THM-008 (Hybrid transitivity of computational indistinguishability) with its ownership boundary.

- Ch. 4: One-Way Functions, Amplification, and the Minicrypt Boundary:** enter through QRCC-XVI-DEF-009 (One-way functions and permutations); exit by being able to state QRCC-XVI-THM-010 (Weak one-way functions amplify to strong one-way functions) with its ownership boundary.
- Ch. 5: Hard-Core Predicates and the Goldreich–Levin Theorem:** enter through QRCC-XVI-DEF-012 (Hard-core predicates); exit by being able to state QRCC-XVI-THM-013 (Goldreich–Levin hard-core theorem) with its ownership boundary.
- Ch. 6: Pseudorandom Generators from One-Wayness:** enter through QRCC-XVI-DEF-014 (Cryptographic pseudorandom generators); exit by being able to state QRCC-XVI-BB-016 (One-way functions imply pseudorandom generators) with its ownership boundary.
- Ch. 7: Pseudorandom Functions and Feistel Permutations:** enter through QRCC-XVI-DEF-017 (Pseudorandom functions and permutations); exit by being able to state QRCC-XVI-THM-018 (Goldreich–Goldwasser–Micali PRF construction) with its ownership boundary.
- Ch. 8: Symmetric Encryption: Semantic Security and IND-CPA:** enter through QRCC-XVI-DEF-020 (Symmetric-encryption security); exit by being able to state QRCC-XVI-THM-021 (Semantic security equals indistinguishability) with its ownership boundary.
- Ch. 9: Authentication, MACs, and Digital Signatures:** enter through QRCC-XVI-DEF-023 (Authentication and signature security); exit by being able to state QRCC-XVI-BB-027 (Digital signatures from one-way functions) with its ownership boundary.
- Ch. 10: Chosen-Ciphertext Security, Hybrid Encryption, and Post-Quantum Proof Discipline:** enter through QRCC-XVI-DEF-028 (Chosen-ciphertext security and KEM/DEM composition); exit by being able to state QRCC-XVI-BB-031 (Cramer–Shoup standard-model CCA encryption) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XVI-THM-002 — Closure calculus for negligible functions. Dependencies: QRCC-XVI-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XVI-THM-008 — Hybrid transitivity of computational indistinguishability. Dependencies: QRCC-XVI-DEF-007, QRCC-XVI-THM-003. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
3. QRCC-XVI-THM-013 — Goldreich–Levin hard-core theorem. Dependencies: QRCC-XVI-DEF-012. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

4. QRCC-XVI-THM-018 — Goldreich–Goldwasser–Micali PRF construction. Dependencies: QRCC-XVI-DEF-014, QRCC-XVI-DEF-017. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
5. QRCC-XVI-BB-027 — Digital signatures from one-way functions. Dependencies: QRCC-XVI-THM-026, QRCC-XVI-BB-016, QRCC-XVI-DEF-023. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XVI-F-011 — Minimal assumptions for public-key cryptography. Dependencies: QRCC-XVI-DEF-009. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- putting the security parameter or adversary quantifier in the wrong order;
- assuming negligible functions remain negligible under superpolynomially many uses;
- forgetting state, nonce, or multi-user conditions when composing primitives;
- claiming CCA/authentication security from an IND-CPA argument without the required compiler;

Boundary and degenerate cases

- empty messages and deterministic keys;
- security games with zero queries;
- nonce reuse/state reset;
- polynomial versus superpolynomially many composed users/sessions;

Retrieval and reconstruction drill

1. State QRCC-XVI-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XVI-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XVI-THM-008, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.

6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XVI-F-011, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XVII instantiates those definitions with number-theoretic groups, RSA, elliptic curves, and pairings.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XVII

Number-Theoretic and Elliptic-Curve Cryptography

Chapter 1

Computational Number Theory: Arithmetic and Primality

Number-theoretic cryptography lives simultaneously in algebra and in bit complexity. Congruences that are one-line identities mathematically can hide a nontrivial computational cost, while a fast arithmetic routine is useless cryptographically unless its input distribution and failure probability are also explicit. This chapter fixes those conventions before any hardness assumption is introduced.

1.1 How to read this chapter

Definition 1.1 is the type system for Volume XVII. Theorem 1.2 collects the constructive arithmetic used by RSA, finite-field DLP, and elliptic-curve implementations. Algorithm 1.3 then separates a practical one-sided randomized primality test from the existence of deterministic polynomial-time primality testing.

1.2 Learning goals

A reader should be able to translate arithmetic cost into input-bit length; use extended Euclid and CRT constructively; audit repeated-squaring costs; distinguish a probable-prime test from a proof of primality; and recognize exactly which part of the Miller–Rabin error theorem is a counting theorem about strong liars.

Literature and convention checkpoint. The arithmetic conventions are calibrated against the basic-algorithms chapters of Galbraith and the companion Algebra/algorithmic-number-theory sources. The strong probable-prime theorem is calibrated to Miller and Rabin, while deterministic polynomial-time primality is recorded at the Agrawal–Kayal–Saxena endpoint. No heuristic factoring statement is used to justify primality.

Definition 1.1 (Computational number-theory model). Integers are encoded in signed binary; the *bit length* of a nonzero integer n is $\ell(n) = \lfloor \log_2 |n| \rfloor + 1$. Arithmetic complexity is measured in bit operations unless a theorem explicitly counts ring, field, or group operations. A randomized algorithm has an explicit internal random tape and a stated success probability over that tape and over any declared input distribution.

For $N > e$, $0 \leq \alpha \leq 1$, and $c > 0$, write

$$L_N[\alpha, c] := \exp\left((c + o(1))(\log N)^\alpha (\log \log N)^{1-\alpha}\right).$$

An integer is *B-smooth* if every prime divisor is at most B . A cyclic group instance consists of an encoding set, algorithms for multiplication/inversion/equality, its order when known, and one named generator when cyclicity is part of the interface. A *generic-group algorithm* may manipulate only opaque encodings through those group operations; a representation-specific algorithm may additionally use coordinates such as residues, field elements, or elliptic-curve coordinates.

A complexity law labelled *heuristic* is not an unconditional worst-case theorem: every smoothness, random-map, or independence heuristic used in deriving it remains part of the statement. Security parameters are inherited from Definition 1.1; cryptographic advantage is always measured against that explicit computational model.

Theorem 1.2 (Constructive CRT and fast modular arithmetic). *Let $m_1, \dots, m_t \geq 2$ be pairwise coprime and let $M = \prod_i m_i$.*

- (i) *For residues $a_i \pmod{m_i}$ there is a unique $x \pmod{M}$ satisfying all congruences. If $M_i = M/m_i$ and $u_i M_i \equiv 1 \pmod{m_i}$, then*

$$x \equiv \sum_{i=1}^t a_i u_i M_i \pmod{M}. \quad (1.1)$$

- (ii) *The extended Euclidean algorithm computes $g = \gcd(a, b)$ and integers u, v with $ua + vb = g$; when $g = 1$, $u \pmod{b}$ is the inverse of $a \pmod{b}$.*
- (iii) *Binary repeated squaring computes $a^e \pmod{m}$ using at most $2\lceil \log_2 e \rceil + 1$ modular multiplications. Hence CRT, inversion, and modular exponentiation are polynomial-time in the binary input length using schoolbook arithmetic, and remain so with faster integer multiplication.*

Proof. For (i), the i th summand in (1.1) is a_i modulo m_i and every other summand is 0 modulo m_i . If x, y satisfy all congruences, each m_i divides $x - y$; pairwise coprimality implies $M \mid (x - y)$. For (ii), Euclid repeatedly replaces (a, b) by $(b, a - qb)$; back substitution expresses the final gcd as an integer linear combination. The residues strictly decrease, so the number and bit cost of divisions are polynomial in the input length. For (iii), scan the binary expansion of e , squaring the running base once per bit and multiplying it into the accumulator when the bit is 1. Reduction modulo m after every operation keeps all intermediate residues below m . There are at most one square and one multiply per bit, and each elementary integer operation has polynomial bit cost. $\square$

Worked CRT example. The system $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$ has $M = 105$ and reconstructs $x \equiv 23 \pmod{105}$. The point of the formula is not merely existence: RSA CRT decryption later uses exactly the same reconstruction to replace one exponentiation modulo $N = pq$ by two smaller exponentiations.

Algorithm 1.3 (Miller–Rabin and deterministic primality interfaces — verified deep input). *Let $n > 2$ be odd and write $n - 1 = 2^s d$ with d odd. For a base $a \in \{2, \dots, n - 2\}$, the strong probable-prime test accepts when either*

$$a^d \equiv 1 \pmod{n}$$

or $a^{2^r d} \equiv -1 \pmod{n}$ for some $0 \leq r < s$; otherwise a is a Miller–Rabin witness and the test rejects n as composite. Every prime passes every base. Rabin’s strong-liar theorem implies that for every odd composite n , a base drawn uniformly according to the standard Miller–Rabin experiment is a strong liar with probability at most $1/4$; a sampled nonunit can instead reveal compositeness

immediately by a gcd. Consequently k independent rounds leave a composite integer accepted with probability at most 4^{-k} . Separately, deterministic primality belongs to polynomial time by the AKS theorem.

What is local and what is source-owned. Prime correctness is immediate from the cyclic structure of $\mathbb{F}_p^*$: if p is prime and $a^d \not\equiv 1$, repeatedly squaring a^d until reaching $a^{p-1} = 1$ forces the preceding nontrivial square root of 1 to be -1 . The 4^{-k} amplification is then just independence. The global *strong-liar counting theorem* for every odd composite, including prime powers and products with unequal two-adic structure, is the nontrivial Rabin/Miller source-owned step; AKS is also source-owned. The reader should therefore not mistake the short prime-side argument for a proof of the one-quarter bound.

1.3 Reader checkpoint

1. Run extended Euclid on (65537, 3120) and recover the corresponding inverse modulo 3120.
2. Prove uniqueness in CRT without using the explicit reconstruction formula.
3. Derive the bound $2\lceil \log_2 e \rceil + 1$ for square-and-multiply.
4. Explain why reducing after every multiplication controls bit length.
5. For $n = 21$, write $n - 1 = 2^s d$ and test bases 2 and 8 by hand.
6. Prove that a prime has no nontrivial square root of 1 modulo p .
7. Explain why a gcd discovered while choosing a Miller–Rabin base is a certificate of compositeness.
8. Show that $k = \lceil \lambda/2 \rceil$ independent rounds make the error at most $2^{-\lambda}$.
9. Compare “probable prime” with “primality is in P” as logical statements.
10. Identify every place Definition 1.1 distinguishes bit complexity from group-operation complexity.

Chapter 2

Integer Factoring: Random Walks and Sieves

Factoring is the first hardness problem in the volume whose best practical algorithms mix rigorous algebra with probabilistic modeling of arithmetic data. That boundary matters: Pollard’s rho method has a clean birthday-paradox core, while the familiar asymptotic laws for the quadratic and number-field sieves rely on smoothness and independence heuristics at specific steps.

2.1 How to read this chapter

Definition 2.1 fixes the factoring distributions. Algorithm 2.2 proves the exact collision-to-factor mechanism and isolates the random-map model. The black-box result 2.3 records sieve complexity with the word “heuristic” kept inside the theorem.

2.2 Learning goals

A reader should be able to turn a collision modulo an unknown prime factor into a gcd; derive the birthday scale $\sqrt{p}$ in an idealized random-map model; distinguish this model from the deterministic polynomial iteration actually used by Pollard rho; and read $L_N[\alpha, c]$ notation without converting a heuristic law into an unconditional theorem.

Definition 2.1 (Factoring problems and distributions). The *integer-factorization problem* maps a composite integer $N > 1$ to its prime-power factorization. A weaker *nontrivial-factor problem* asks for d with $1 < d < N$ and $d \mid N$. An *RSA modulus* is sampled as $N = pq$ from a stated distribution of distinct odd primes; any bit-length balance condition is part of the distribution, not part of the definition of compositeness.

For a randomized factoring algorithm A , the success probability is taken over its internal randomness and, when declared, the sampling of N . A statement about a fixed N and a random auxiliary map is different from an average-case statement over RSA moduli. Smoothness uses Definition 1.1.

Algorithm 2.2 (Pollard rho factoring in the random-map model). *Let N be composite and let p be a prime divisor. Generate an iteration $x_{i+1} = f(x_i) \bmod N$ and use Floyd or Brent cycle detection while computing gcds of differences. If for some $i \neq j$*

$$x_i \equiv x_j \pmod{p} \quad \text{but} \quad x_i \not\equiv x_j \pmod{N}, \quad (2.1)$$

then $d = \gcd(x_i - x_j, N)$ is a nontrivial factor whenever the difference is not simultaneously 0 modulo every prime divisor of N .

In the idealized model in which the residues modulo p encountered before the first collision behave like independent uniform samples, the expected collision scale is $\Theta(\sqrt{p})$ and a constant success probability is obtained after $O(\sqrt{p})$ samples. This birthday statement is rigorous for independent uniform samples; modeling the deterministic polynomial iteration (often $f(x) = x^2 + c$) as a random mapping is a heuristic about Pollard rho on a particular integer.

Proof of the rigorous part. Condition (2.1) says $p \mid (x_i - x_j)$, so the gcd is greater than 1. If $N \nmid (x_i - x_j)$, the gcd is strictly less than N . For independent samples $X_1, \dots, X_t$ from a p -element set, the probability of no collision is

$$\prod_{j=0}^{t-1} \left(1 - \frac{j}{p}\right) \leq \exp\left(-\frac{t(t-1)}{2p}\right). \quad (2.2)$$

Thus $t = c\sqrt{p}$ gives collision probability at least $1 - e^{-c^2/2+o(1)}$. Floyd's tortoise-and-hare method detects a cycle with constant memory once the iteration enters one. Equation (2.2) does not prove that a fixed polynomial f generates independent samples; that is precisely the heuristic boundary stated in the algorithm. $\square$

Failure modes are informative. A gcd equal to 1 means the tested pair has not collided modulo any prime factor. A gcd equal to N means the pair collided modulo all factors simultaneously and is unhelpful. Restarting with a new seed or new polynomial changes the walk. The algorithm never receives p ; p appears only in the analysis.

Theorem 2.3 (Quadratic-sieve and number-field-sieve complexity — verified heuristic input). *For general integer factorization, the standard heuristic analyses predict*

$$T_{\text{QS}}(N) = L_N[1/2, 1], \quad T_{\text{GNFS}}(N) = L_N\left[1/3, (64/9)^{1/3}\right], \quad (2.3)$$

up to lower-order terms and implementation-dependent polynomial factors. These laws depend on smoothness probabilities and on modeling collected relations as sufficiently random/independent. They are not stated here as unconditional worst-case running-time theorems for QS or GNFS. Rigorous subexponential factoring bounds exist for other algorithms, but they do not license replacing the heuristic qualifiers in (2.3).

Why sieves beat rho asymptotically. Pollard rho attacks a factor p at a square-root scale. Sieves instead search for many arithmetic values that factor over a small factor base, then solve a sparse linear relation that produces a congruence of squares. NFS moves the relation search into two linked number-ring representations, changing the smoothness balance and producing the $L_N[1/3, c]$ regime. The algebraic relation is exact; the asymptotic yield of usable smooth relations is where the heuristic analysis enters.

2.3 Reader checkpoint

1. Prove the collision-to-gcd implication in (2.1).
2. Derive inequality (2.2) from $1 - x \leq e^{-x}$.

3. Find c giving idealized collision probability at least 0.9.
4. Explain why Pollard rho can use constant memory although the birthday proof imagines stored samples.
5. Give an example where $\gcd(x_i - x_j, N) = N$.
6. Distinguish random input, random seed, and random-map assumptions.
7. Compare $L_N[1/2, 1]$ with $N^{1/4}$ on a logarithmic scale.
8. Verify that $L_N[1/3, c]$ is subexponential but superpolynomial.
9. Explain the role of a factor base in a congruence-of-squares factorization method.
10. State precisely why equation (2.3) is not a proof that GNFS factors every N within that time.

Chapter 3

RSA, Rabin, and Provable Encodings

RSA and Rabin are often introduced as exponentiation formulas. Cryptographically, the formulas are only the beginning: one must specify the sampling domain, distinguish inversion from factoring, handle nonunits, and add an encoding whose security theorem matches the attack model. This chapter separates those layers.

3.1 How to read this chapter

Definition 3.1 separates the assumptions. Theorem 3.2 proves RSA correctness on all residues for the declared key relation. Theorem 3.3 gives the factoring equivalence for Rabin square-root inversion. Theorem 3.4 is source-owned because OAEP and FDH security reductions contain substantial random-oracle programming.

3.2 Learning goals

A reader should be able to prove RSA correctness without assuming the plaintext is a unit; reconstruct CRT decryption; distinguish the RSA assumption from the factoring assumption; carry out the Rabin root-to-factor reduction with its $1/2$ extraction probability; and state why textbook RSA is not IND-CPA encryption or an EUF-CMA signature scheme.

Definition 3.1 (RSA and Rabin assumptions). Let $N = pq$ for distinct odd primes and let $\lambda(N) = \text{lcm}(p-1, q-1)$. An *RSA key* chooses e with $\gcd(e, \lambda(N)) = 1$ and $d \equiv e^{-1} \pmod{\lambda(N)}$. The public permutation on $\mathbb{Z}_N^*$ is $x \mapsto x^e$; the *RSA inversion assumption* states that inverting a sampled public instance on a random unit is hard in the declared key distribution. This is not defined to be equivalent to factoring.

The *strong-RSA problem* gives a random unit u and asks for any exponent $e > 1$ and v with $v^e = u \pmod{N}$; it is a different assumption. The *Rabin map* is $x \mapsto x^2 \pmod{N}$. For Blum moduli $p \equiv q \equiv 3 \pmod{4}$, square roots have especially simple exponent formulas, but the factoring equivalence below only needs distinct odd primes and access to square roots of random quadratic residues. *Quadratic residuosity* asks, under an explicitly sampled Jacobi-+1 distribution, whether a residue is a square; it is again distinct from both factoring and RSA inversion.

Theorem 3.2 (RSA correctness and CRT decryption). *With $N = pq$ and $ed \equiv 1 \pmod{\lambda(N)}$, every $m \in \mathbb{Z}_N$ satisfies*

$$(m^e)^d \equiv m \pmod{N}. \quad (3.1)$$

The decryption can be computed from $d_p = d \bmod (p-1)$ and $d_q = d \bmod (q-1)$ by exponentiating separately modulo p and q and reconstructing with Theorem 1.2. Correctness fails to be guaranteed if e is not invertible modulo $\lambda(N)$ or if the public modulus is malformed relative to the declared key-generation algorithm.

Proof. Write $ed = 1 + k\lambda(N)$. Modulo p , if $p \mid m$ then both sides of (3.1) are 0. Otherwise Fermat's theorem gives $m^{p-1} = 1$, and $p-1 \mid \lambda(N)$ implies $m^{ed} = m(m^{p-1})^{k\lambda(N)/(p-1)} \equiv m \pmod{p}$. The same argument holds modulo q . CRT then gives (3.1) modulo N . Reducing d modulo $p-1$ and $q-1$ preserves the two prime-modulus powers, after which constructive CRT reconstructs the unique residue modulo N . $\square$

Why the nonunit case matters. A proof only on $\mathbb{Z}_N^*$ is enough to discuss the RSA permutation assumption but is not enough for the literal arithmetic claim on every encoded residue. The two-case prime-modulus proof above is the small repair that prevents that common gap.

Theorem 3.3 (Rabin inversion is equivalent to factoring). *For $N = pq$ with distinct odd primes, knowing the factorization computes all square roots modulo N in randomized polynomial time (and by the elementary exponent formula when N is a Blum modulus). Conversely, suppose an algorithm A receives a uniformly random quadratic residue $y = x^2 \bmod N$ generated from uniform $x \in \mathbb{Z}_N^*$ and, with probability ε , returns some square root z of y . Then a randomized reduction factors N with probability at least $\varepsilon/2$ per successful-oracle experiment, using polynomial additional work.*

Proof. Given p, q , work separately in $\mathbb{F}_p$ and $\mathbb{F}_q$. Because y is a quadratic residue, $X^2 - y$ splits into linear factors in each field. Apply the locally proved Cantor–Zassenhaus algorithm QRCC-III-ALG-006 to factor these degree-two polynomials in Las Vegas time polynomial in $\log p$ and $\log q$; the two linear factors give the two roots modulo each prime. Constructive CRT QRCC-XVII-THM-002 combines the four sign choices, yielding all four roots modulo N . For $p \equiv q \equiv 3 \pmod{4}$, a root of a quadratic residue y modulo either prime is simply $y^{(p+1)/4}$ or $y^{(q+1)/4}$, so this direction becomes elementary and deterministic.

For the converse, sample uniform $x \in \mathbb{Z}_N^*$, set $y = x^2 \bmod N$, and query $A(y)$. Conditional on this fixed y , the sampled x is uniform among its four roots. Whatever root z the algorithm returns, exactly two of the four choices satisfy $x \equiv \pm z \pmod{N}$; for the other two, x and z agree in sign modulo one prime and disagree modulo the other. Hence with conditional probability $1/2$,

$$1 < \gcd(x - z, N) < N, \quad (3.2)$$

which exposes one prime factor. Multiplying by the oracle success probability gives at least $\varepsilon/2$. $\square$

Theorem 3.4 (RSA-OAEP and RSA-FDH — verified deep inputs). *In the random-oracle model, the Fujisaki–Okamoto–Pointcheval–Stern analysis proves adaptive chosen-ciphertext security for RSA-OAEP under the RSA assumption through the partial-domain-one-wayness route, which is equivalent to RSA one-wayness for the RSA function; the reduction is not tight. Separately, RSA Full-Domain Hash signatures are EUF-CMA secure in the random-oracle model under RSA inversion, with a quantitative loss depending on hash/signing queries (and tighter variants due to later work such as Coron). These are theorems about specified transforms in the random-oracle model, not proofs that arbitrary standardized padding or arbitrary hash-then-RSA encodings are secure.*

Textbook warning. Deterministic textbook RSA leaks equality and is multiplicatively malleable, so it is not IND-CPA. Raw RSA signatures are likewise algebraically forgeable. The padding/encoding theorem is therefore not an implementation detail appended after the mathematics; it is part of the cryptographic construction.

3.3 Reader checkpoint

1. Prove RSA correctness modulo $p^k q^\ell$ under an appropriate Carmichael-exponent condition.
2. Explain why $ed \equiv 1 \pmod{\varphi(N)}$ is sufficient but not necessary when $\lambda(N)$ is used.
3. Work out CRT decryption for a toy RSA modulus.
4. Give an explicit multiplicative malleability identity for textbook RSA.
5. Enumerate the four square roots of a quadratic residue modulo $N = 77$.
6. Prove the conditional $1/2$ extraction probability in the Rabin reduction.
7. Explain why a deterministic root oracle does not defeat that reduction.
8. Distinguish factoring, RSA inversion, strong RSA, and quadratic residuosity.
9. State the random-oracle assumption used by FDH in one precise sentence.
10. Explain the historical OAEP proof-gap lesson: why a proof for a generic trapdoor permutation need not transfer unchanged to RSA.

Chapter 4

Discrete Logarithms: Generic and Representation-Specific Algorithms

Discrete logarithms illustrate why a complexity statement must name the representation. Generic attacks see only group operations and meet a square-root barrier; finite-field encodings expose multiplicative and polynomial structure to index calculus. Elliptic curves are valuable partly because known representation-specific attacks behave very differently there.

4.1 How to read this chapter

Definition 4.1 separates generic and representation-specific models. Algorithm 4.2 gives the local Pollard-rho extraction algebra. Theorems 4.3 and 4.4 are deliberately source-owned because they encode, respectively, a generic lower-bound argument and a field-regime-dependent algorithmic literature.

4.2 Learning goals

A reader should be able to recover a logarithm from a useful rho collision; explain the prime-order assumption in that division step; state Shoup's lower bound without claiming it for algorithms that inspect field coordinates; and distinguish prime-, medium-, and small-characteristic finite-field DLP regimes.

Definition 4.1 (Discrete logarithms and generic groups). Let $G = \langle g \rangle$ be a finite cyclic group of order r . The *discrete logarithm problem* (DLP) receives (G, g, h) with $h \in G$ and asks for $x \in \mathbb{Z}_r$ satisfying $h = g^x$. A *generic-group algorithm* receives random-looking encodings and oracle access to group operations and equality only. A *representation-specific* algorithm may use the encoding, for example the fact that $G \subset \mathbb{F}_q^*$ or that elements are curve points.

If $r = \prod_i \ell_i^{e_i}$ is smooth, Pohlig–Hellman reduces DLP to prime-power-order subgroups; thus cryptographic generic hardness is controlled by a large prime divisor of r . *Index calculus* means a representation-specific relation-collection method over a factor base followed by linear algebra and a descent. Its complexity must be stated for the exact field regime.

Algorithm 4.2 (Pollard rho for discrete logarithms). *Let $G = \langle g \rangle$ have prime order q and let $h = g^x$. Maintain a pseudorandom walk whose state has the tracked representation*

$$X_i = g^{a_i} h^{b_i}. \tag{4.1}$$

If cycle detection finds $X_i = X_j$ with $i \neq j$, then

$$a_i - a_j \equiv x(b_j - b_i) \pmod{q}. \quad (4.2)$$

Whenever $b_j \not\equiv b_i \pmod{q}$, the logarithm is

$$x \equiv (a_i - a_j)(b_j - b_i)^{-1} \pmod{q}. \quad (4.3)$$

In the idealized random-partition/random-walk model, a useful collision occurs after $\Theta(\sqrt{q})$ group steps with constant probability and constant-memory cycle detection. For composite order, equation (4.2) may have a noninvertible coefficient; one instead works in large prime-order factors or solves the resulting congruence with its gcd information.

Proof of the extraction and idealized collision claim. Equality in (4.1) gives $g^{a_i+xb_i} = g^{a_j+xb_j}$, hence (4.2). Prime order makes every nonzero $b_j - b_i$ invertible, giving (4.3). Under the idealized uniform-sample model, the birthday calculation of (2.2) with p replaced by q gives the $\Theta(\sqrt{q})$ scale. As in integer rho, applying that model to a particular deterministic partition function is a modeling assumption, not an identity. $\square$

Theorem 4.3 (Shoup generic-group lower bound — verified deep input). *In Shoup’s generic-group model, for a cyclic group whose order has a largest prime divisor p , a generic algorithm using m oracle operations has DLP success probability at most $O(m^2/p)$, with an explicit constant in the primary theorem. Thus constant success requires $\Omega(\sqrt{p})$ generic operations. The theorem does not apply to algorithms exploiting special encodings, such as finite-field index calculus, and therefore does not imply a universal $\sqrt{|G|}$ lower bound for every concrete representation.*

Theorem 4.4 (Finite-field index-calculus regimes — verified deep input). *The finite-field DLP has no single representation-independent asymptotic law. In large-prime fields, number-field-sieve variants give subexponential heuristic algorithms in an $L_q[1/3, c]$ regime. In extension fields, the constants and even the asymptotic regime depend on the characteristic, extension degree, and representation; in suitable small-characteristic families, quasi-polynomial algorithms are known. Any cryptographic parameter claim must therefore name the field family and best applicable attack, rather than cite “index calculus” as one universal complexity theorem.*

Generic does not mean weak. Shoup’s theorem is powerful precisely because it isolates what can be proved without representation structure. Its limitation is equally important: a finite field supplies addition, multiplication, low-degree polynomials, norms, and smoothness notions far beyond an opaque group oracle. The absence of an analogous known index-calculus attack on generic elliptic-curve groups is a fact about present algorithms, not a theorem that such an attack can never exist.

4.3 Reader checkpoint

1. Derive equation (4.2) from a rho collision.
2. Explain exactly where prime order is used in equation (4.3).
3. Give a composite-order example where $b_j - b_i$ is not invertible.
4. Re-derive the birthday scale for a q -element group.

5. Explain why constant memory does not contradict the need to detect a collision.
6. State the model restriction in Shoup's theorem.
7. Give one piece of structure visible in $\mathbb{F}_p^*$ but hidden in a generic group.
8. Explain how Pohlig–Hellman changes the relevant group-size parameter.
9. Compare a generic lower bound with a heuristic upper bound as logical statements.
10. Explain why “ECC has no index calculus” should be read as an algorithmic-status statement, not an impossibility theorem.

Chapter 5

Diffie–Hellman Assumptions and Random Self-Reduction

The same algebraic tuple supports several distinct assumptions. Computing g^{ab} , distinguishing it from random, and deriving a symmetric key from it are different tasks. This chapter fixes their types and proves the randomization identities that let reductions move between worst-looking and uniformly shifted instances in a prime-order cyclic group.

5.1 How to read this chapter

Definition 5.1 names CDH, DDH, gap-DH, and hashed variants separately. Theorem 5.2 proves only what the key-agreement algebra plus the declared KDF assumption actually yields. Theorem 5.3 gives exact random self-reductions.

5.2 Learning goals

A reader should be able to distinguish CDH from DDH; prove the Diffie–Hellman shared-secret identity; state why a raw group element is not automatically a uniformly distributed symmetric key; randomize DLP, CDH, and DDH challenges algebraically; and identify subgroup/cofactor caveats.

Definition 5.1 (Diffie–Hellman assumptions). Let $G = \langle g \rangle$ be a sampled cyclic group, normally of prime order q in the theorems below. The *computational Diffie–Hellman* (CDH) problem maps (g, g^a, g^b) to g^{ab} . The *decisional Diffie–Hellman* (DDH) problem distinguishes (g, g^a, g^b, g^{ab}) from (g, g^a, g^b, g^c) for independent uniform exponents. A *gap-DH* group supports an efficient DDH test while CDH remains assumed hard.

A *hashed-DH* statement additionally names a key-derivation function K and asserts pseudo-randomness of $K(g^{ab}, \text{ctx})$ under a specified model; this does not follow formally from CDH alone. Static, ephemeral, one-pass, authenticated, and multi-user variants are separate experiments. Every theorem below names the variant it uses.

Theorem 5.2 (Diffie–Hellman correctness and passive-security interface). *Alice samples $a \in \mathbb{Z}_q$ and sends $A = g^a$; Bob samples $b \in \mathbb{Z}_q$ and sends $B = g^b$. Then*

$$B^a = A^b = g^{ab}. \tag{5.1}$$

If the session key is $K(g^{ab}, A, B, \text{ctx})$ and the declared hashed-DH/KDF assumption says this value is computationally indistinguishable from a fresh key given the public transcript, then the passive key-indistinguishability game reduces identically to that assumption. Algebraic correctness alone does not imply authentication, active security, or pseudorandomness of the raw encoding of g^{ab} .

Proof. Equation (5.1) is $B^a = (g^b)^a = g^{ab} = (g^a)^b = A^b$. A passive distinguisher seeing the transcript and session-key challenge can be used unchanged in the hashed-DH experiment because the distributions are the same by definition. No statement about an active adversary follows: an unauthenticated attacker can replace A and B and establish separate keys with each party. $\square$

Theorem 5.3 (Random self-reduction of DLP, CDH, and DDH). *Let $G = \langle g \rangle$ have prime order q .*

(i) *From a DLP challenge $h = g^x$, choose uniform r and give $hg^r = g^{x+r}$ to a solver. Its exponent is uniform in $\mathbb{Z}_q$, and subtracting r recovers x .*

(ii) *From a CDH challenge (g^a, g^b) , choose uniform r, s and ask a solver on (g^{a+r}, g^{b+s}) . From $g^{(a+r)(b+s)}$ recover*

$$g^{ab} = g^{(a+r)(b+s)} (g^a)^{-s} (g^b)^{-r} g^{-rs}. \quad (5.2)$$

(iii) *Transform a DDH tuple (g^a, g^b, T) to*

$$(g^{a+r}, g^{b+s}, T(g^a)^s (g^b)^r g^{rs}). \quad (5.3)$$

If $T = g^{ab}$ the transformed fourth component is the correct DH value; if T is uniform independent, so is the transformed fourth component. The first two exponents are uniform.

Thus solvers whose advantage is guaranteed on uniform prime-order instances extend to arbitrary fixed instances of the same group through these exact randomizations.

Proof. Addition by an independent uniform element is a bijection of $\mathbb{Z}_q$, proving uniformity in (i) and of the shifted exponents in (ii)–(iii). Expanding $(a+r)(b+s) = ab + as + br + rs$ gives (5.2). Multiplying a uniform independent group element by a fixed element preserves uniformity, giving the random branch of (iii). $\square$

Scope warning. Prime order removes subgroup leakage and makes the later Schnorr extraction denominator invertible whenever the challenges differ. Some randomization identities remain algebraically valid in composite-order groups, but protocol security can change dramatically when cofactors, invalid-subgroup inputs, or pairing-friendly structure expose extra tests.

5.3 Reader checkpoint

1. Prove equation (5.1) in any abelian cyclic group.
2. Give a man-in-the-middle transcript showing unauthenticated DH is not actively secure.
3. Explain why CDH hardness does not syntactically imply DDH hardness.
4. Prove uniformity of $x + r \pmod{q}$ for fixed x .
5. Verify equation (5.2) exponent by exponent.
6. Prove that the random branch of (5.3) remains uniform.

7. Explain what a gap-DH group provides and why pairings can create such a setting.
8. State one difference between static and ephemeral DH assumptions.
9. Explain why a KDF theorem must name its context/domain separation.
10. Identify one composite-order subgroup attack that the prime-order interface rules out abstractly.

Chapter 6

ElGamal, Schnorr Identification, and Fiat–Shamir Boundaries

This chapter turns the group assumptions into concrete public-key mechanisms. ElGamal is a direct DDH hybrid. Schnorr identification is a three-move proof whose completeness, special soundness, and honest-verifier zero knowledge are elementary algebra. Turning that protocol into a signature by hashing the challenge is substantially more delicate and remains source-owned.

6.1 How to read this chapter

Definitions 6.1–6.3 introduce the encryption, Sigma-protocol, and random-oracle/forking interfaces in that order. Theorems 6.4 and 6.5 are fully local. The Schnorr-signature theorem 6.6 is not inferred merely by replacing a verifier challenge with a hash.

6.2 Learning goals

A reader should be able to write an exact DDH reduction for ElGamal; extract a Schnorr witness from two accepting transcripts with the same commitment; simulate the honest-verifier transcript perfectly; and state what rewinding/programming assumptions a Fiat–Shamir security theorem adds.

Definition 6.1 (ElGamal and Schnorr security models). Fix a prime-order group $G = \langle g \rangle$ of order q . ElGamal key generation samples x and publishes $y = g^x$; its message space for the theorem below is G . Schnorr uses the same relation $y = g^x$ as a proof-of-knowledge relation. Every nonce is sampled freshly from $\mathbb{Z}_q$; nonce reuse is outside the security theorem and can expose the secret key.

Encryption security is the IND-CPA game of Volume XVI instantiated on group-valued messages. Identification security is defined separately below. A Fiat–Shamir signature computes its challenge by a domain-separated hash of the statement, commitment, message, and declared context; the random-oracle model is not silently identified with a fixed deployed hash function.

Definition 6.2 (Three-move identification interface). A public-coin three-move protocol has messages (t, c, z) : prover commitment t , uniformly sampled verifier challenge c from a declared challenge space, and response z . *Completeness* means an honest prover with a valid witness is accepted. *Special soundness* means two accepting transcripts with the same t and distinct challenges efficiently determine a witness. *Honest-verifier zero knowledge* (HVZK) means there is an efficient

simulator producing the same (perfect HVZK) or computationally indistinguishable transcript distribution as an honest execution with a uniform honest challenge.

Definition 6.3 (Random-oracle and forking interface). A random oracle H is a consistent uniformly sampled function exposed by query access. A classical programmable-random-oracle reduction may answer previously unseen queries by lazy sampling and, when the proof theorem permits, program selected fresh inputs. A *forking experiment* reruns an adversary with related internal randomness/transcript prefix but changes a selected fresh oracle answer to obtain two accepting transcripts with the same commitment and different challenges. Its theorem must state the hash-query budget, signing-query budget, nonce model, abort conditions, and success loss. Classical rewinding is not automatically a theorem about quantum-superposition oracle queries.

Theorem 6.4 (ElGamal IND-CPA security under DDH). *Let $y = g^x$. Encrypt $m \in G$ by sampling $r \leftarrow \mathbb{Z}_q$ and outputting*

$$(c_1, c_2) = (g^r, my^r). \quad (6.1)$$

Decrypt by $m = c_2/(c_1)^x$. For the single-user IND-CPA game, every adversary A gives a DDH distinguisher B with exactly the same challenge advantage, up to simulation overhead. In a u -user reduction that must guess which independently generated user becomes the challenge target, the elementary guessing reduction loses at most a factor u .

Proof. Correctness is $my^r/(g^r)^x = mg^{xr}/g^{rx} = m$. Given a DDH challenge (g, g^x, g^r, T) , use g^x as the public key and return challenge ciphertext $(g^r, m_b T)$. If $T = g^{xr}$ this is real ElGamal. If T is uniform in G , multiplication by fixed m_b leaves the second component uniform and independent of b . Thus the IND-CPA distinguishing gap is the DDH gap. The multi-user statement follows by choosing a uniform target index and embedding the DDH public key only there. $\square$

Theorem 6.5 (Schnorr identification: completeness, special soundness, and perfect HVZK). *With public key $y = g^x$, the prover samples $r \leftarrow \mathbb{Z}_q$ and sends $t = g^r$; the verifier sends $c \leftarrow \mathbb{Z}_q$; the prover responds $z = r + cx \pmod{q}$; and the verifier accepts iff*

$$g^z = ty^c. \quad (6.2)$$

The protocol has perfect completeness. From two accepting transcripts (t, c, z) and (t, c', z') with $c \neq c'$, one extracts

$$x = (z - z')(c - c')^{-1} \pmod{q}. \quad (6.3)$$

It also has perfect HVZK: sample independent uniform c, z and set $t = g^z y^{-c}$.

Proof. For an honest prover, $g^z = g^{r+cx} = ty^c$. Dividing two accepting equations with equal t gives $g^{z-z'} = y^{c-c'} = g^{x(c-c')}$; prime order and $c \neq c'$ make the denominator invertible, proving (6.3). In a real transcript, r, c are independent uniform and $z = r + cx$; for every fixed c , addition by cx maps uniform r bijectively to uniform z , with $t = g^{z-cx} = g^z y^{-c}$. This is exactly the simulator distribution. $\square$

Nonce-reuse catastrophe. If two responses use the same r under distinct challenges, equation (6.3) recovers x even without a rewinding reduction. Implementations therefore need strong nonce generation or a separately analyzed deterministic-nonce derivation.

Theorem 6.6 (Schnorr signatures in the random-oracle model — verified deep input). *The Fiat–Shamir Schnorr signature obtained by setting the challenge to a domain-separated random-oracle value admits classical-ROM EUF-CMA reductions to discrete-log hardness under the exact nonce and oracle conditions of the selected forking-lemma analysis. The standard proof is non-tight and its success depends on hash/signing-query counts; later analyses change tightness and modeling details. This volume therefore keeps the forking/reprogramming mechanism source-owned rather than presenting Theorem 6.5 as if it alone proved signature security.*

6.3 Reader checkpoint

1. Verify ElGamal decryption algebraically.
2. Construct the exact DDH challenge ciphertext used in the IND-CPA reduction.
3. Explain why the random DDH branch hides the message bit perfectly.
4. Derive Schnorr special soundness from two accepting equations.
5. Prove the HVZK simulator distribution is exact, not merely computationally close.
6. Show algebraically how nonce reuse reveals x .
7. Distinguish HVZK from malicious-verifier zero knowledge.
8. Explain why Fiat–Shamir changes the proof model even though the verification equation is unchanged.
9. State why classical rewinding is not automatically valid against quantum random-oracle queries.
10. Identify the source of a factor- u loss in a simple multi-user ElGamal reduction.

Chapter 7

Elliptic Curves over Finite Fields

Elliptic-curve cryptography is not obtained by replacing multiplication signs with point-addition signs. The group law rests on nonsingular projective geometry, the group order is controlled by Frobenius, and secure parameter generation requires knowing that order. This chapter exposes those mathematical interfaces before any curve-based cryptosystem is instantiated.

7.1 How to read this chapter

Definition 7.1 fixes the curve model and explicitly restricts the displayed short Weierstrass formulas to characteristic other than 2, 3. The group-law theorem 7.2 remains source-owned at associativity/projective exceptional cases. Hasse and Schoof are source-owned deep inputs because compressing either proof would defeat the volume's proof-ownership discipline.

7.2 Learning goals

A reader should be able to test nonsingularity of a short Weierstrass model; use the chord–tangent formulas with the point at infinity; state Hasse's bound and the Frobenius trace; explain why point counting is part of cryptographic group selection; and describe the modular-torsion architecture of Schoof's algorithm.

Definition 7.1 (Elliptic curves over finite fields). For a field K of characteristic not 2 or 3, a short Weierstrass curve is

$$E : y^2 = x^3 + Ax + B, \tag{7.1}$$

with discriminant $\Delta = -16(4A^3 + 27B^2) \neq 0$. Its K -points are the affine solutions together with the projective point at infinity O . In characteristics 2 and 3 one uses an appropriate general Weierstrass equation; the short formulas below are not silently reused there.

For $E/\mathbb{F}_q$, write $\#E(\mathbb{F}_q)$ for the number of rational points and let $\pi : (x, y) \mapsto (x^q, y^q)$ be Frobenius. The n -torsion $E[n]$ means points killed by multiplication by n over an algebraic closure, with the usual characteristic caveat when n is divisible by $\text{char } K$. A quadratic twist is a curve becoming isomorphic over a quadratic extension but not necessarily over the base field. Endomorphisms are algebraic group maps $E \rightarrow E$ fixing O .

Theorem 7.2 (Elliptic-curve group law — verified deep input). *For a nonsingular curve (7.1), set $-(x, y) = (x, -y)$ and $P + O = P$. If $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are not inverses, define*

$$\lambda = \begin{cases} (y_2 - y_1)/(x_2 - x_1), & P \neq Q, \\ (3x_1^2 + A)/(2y_1), & P = Q \neq -P, \end{cases}$$

then

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1, \quad (7.2)$$

and set $P + Q = (x_3, y_3)$. Together with $P + (-P) = O$, these rules make $E(K)$ an abelian group. The full proof of associativity and the projective exceptional cases is source-owned and calibrated to the standard projective/divisor construction in canonical elliptic-curve references; the formulas and all domain conditions needed later are explicit here.

Local formula check. Substituting the line $y = y_1 + \lambda(x - x_1)$ into (7.1) yields a cubic whose three roots are x_1, x_2, x_3 . Comparing the x^2 coefficient gives $x_1 + x_2 + x_3 = \lambda^2$, hence the first formula in (7.2); reflecting the third line intersection gives the second. This verifies closure/formulas but is not being mislabeled as a full associativity proof.

Theorem 7.3 (Hasse bound and Frobenius polynomial — verified deep input). *For $E/\mathbb{F}_q$ there is an integer t such that*

$$\#E(\mathbb{F}_q) = q + 1 - t, \quad |t| \leq 2\sqrt{q}. \quad (7.3)$$

On every prime-to-characteristic Tate module, Frobenius satisfies

$$\pi^2 - t\pi + q = 0, \quad (7.4)$$

so the characteristic polynomial is $T^2 - tT + q$; its roots have complex absolute value $\sqrt{q}$. The proof is source-owned because it uses the arithmetic geometry of isogenies/Tate modules (or an equivalent form of the Riemann hypothesis for genus one), not merely the affine addition formulas.

Theorem 7.4 (Schoof deterministic point counting — verified deep input). *There is a deterministic algorithm polynomial in $\log q$ that computes $\#E(\mathbb{F}_q)$ for an elliptic curve over a finite field. In Schoof's original characteristic- > 3 presentation, one determines the trace t modulo enough small primes $\ell \neq \text{char } \mathbb{F}_q$ by evaluating the Frobenius relation (7.4) on $E[\ell]$ through division-polynomial arithmetic, then reconstructs the unique integer t in Hasse's interval by CRT. Modern variants improve constants/exponents but do not change this logical architecture.*

Why Hasse makes CRT finite. Once the product of the tested small primes exceeds $4\sqrt{q}$, there is at most one integer in $[-2\sqrt{q}, 2\sqrt{q}]$ with the accumulated residues. Hasse's theorem therefore converts modular information about Frobenius into the exact group order.

7.3 Reader checkpoint

1. Compute the discriminant of $y^2 = x^3 + 2x + 2$ over $\mathbb{F}_{17}$.
2. Derive the x_3 formula in (7.2) from the line-cubic intersection.
3. Explain why a vertical line gives $P + (-P) = O$.

4. Identify why the displayed doubling denominator fails when $y_1 = 0$ and interpret the group result.
5. State what changes in characteristic 2 or 3.
6. For $q = 101$, give the Hasse interval for $\#E(\mathbb{F}_q)$.
7. Derive $\#E(\mathbb{F}_q) = q + 1 - t$ from the fixed-point interpretation of Frobenius at a conceptual level.
8. Explain why knowing only an approximate curve order is insufficient for subgroup selection.
9. Determine how large a product of small ℓ values suffices to reconstruct t uniquely.
10. Explain which part of Schoof is algebraic manipulation and which part depends on Hasse's deep bound.

Chapter 8

Elliptic-Curve Cryptographic Groups

Once an elliptic curve has been counted, cryptography still needs a prime-order subgroup, encoding and validation rules, cofactor handling, and a statement about twists. This chapter transfers the generic Diffie–Hellman, ElGamal, and Schnorr reductions to that interface and works one complete small example.

8.1 How to read this chapter

Definition 8.1 is deliberately implementation-aware: “a curve” is not yet a cryptographic group. Theorem 8.2 is a transfer theorem, not a new hardness proof. Example 8.3 shows the entire arithmetic chain on a curve too small to be secure.

8.2 Learning goals

A reader should be able to select and validate a prime-order subgroup; explain cofactor clearing and twist security; instantiate ECDH and EC-ElGamal reductions without changing their algebra; distinguish EC-Schnorr from ECDSA; and identify invalid-point failures in a toy computation.

Definition 8.1 (Elliptic-curve cryptographic groups). A curve-based group instance specifies $E/\mathbb{F}_q$, a prime r , a point $P \in E(\mathbb{F}_q)$ of order r , and cofactor $h = \#E(\mathbb{F}_q)/r$. Public inputs are accepted only after the declared encoding is parsed, the curve equation is checked, the point is not O when forbidden, and the subgroup condition $[r]Q = O$ (or an equivalent safe cofactor-clearing rule) is enforced. *Twist security* requires that malformed or alternative-curve inputs not land in exploitable small subgroups under the implementation’s formulas.

The *ECDLP* is DLP in $\langle P \rangle$. ECDH/CDH and EC-DDH are the corresponding group problems. Scalar multiplication means $[k]P$ via a declared double-and-add/window algorithm. Hash-to-curve is a separate encoding map with its own indistinguishability/uniformity theorem; “hash bytes and interpret as x ” is not the default definition.

Theorem 8.2 (ECDH, EC-ElGamal, and EC-Schnorr transfer). *Let $\langle P \rangle \subseteq E(\mathbb{F}_q)$ have prime order r and suppose every adversarial point is validated into that subgroup.*

- (i) *ECDH with public points $A = [a]P$, $B = [b]P$ has shared point $[a]B = [b]A = [ab]P$ and inherits the declared hashed-EC-DH passive-security interface.*
- (ii) *EC-ElGamal on point messages, $(R, S) = ([k]P, M + [k]Q)$ for $Q = [x]P$, is IND-CPA under EC-DDH by the same reduction as Theorem 6.4.*

- (iii) Schnorr identification with additive notation, commitment $T = [r]P$ and verification $[z]P = T + [c]Q$, has the same perfect completeness, special soundness, and perfect HVZK as Theorem 6.5; the classical-ROM signature theorem transfers only under the source-stated EC-DLP/random-oracle assumptions.

This theorem concerns EC-Schnorr. It does not claim an ECDSA security theorem.

Proof. All three proofs use only the prime-order cyclic-group axioms and equality of encodings after validation. Replace multiplicative g^a by additive $[a]P$, products by sums, and inverses by negatives in the proofs of Theorems 5.2, 6.4, and 6.5. Validation ensures the adversary has not changed the group in which those proofs operate. $\square$

Example 8.3 (Complete toy elliptic-curve cryptosystem). Consider

$$E/\mathbb{F}_{17} : y^2 = x^3 + 2x + 2.$$

Direct enumeration gives $\#E(\mathbb{F}_{17}) = 19$, so every nonidentity point has prime order 19. Take $P = (5, 1)$. Repeated addition yields

$$[5]P = (9, 16), \quad [7]P = (0, 6), \quad [16]P = (10, 11), \quad [19]P = O.$$

If Alice uses $a = 5$ and Bob $b = 7$, both ECDH computations give $[5]([7]P) = [7]([5]P) = [35]P = [16]P = (10, 11)$.

For toy EC-ElGamal, let Alice's public key be $Q = [5]P$, encode the message as $M = [9]P = (7, 6)$, and choose $k = 3$. Then

$$C_1 = [3]P = (10, 6), \quad C_2 = M + [3]Q = [9 + 15]P = [5]P = (9, 16).$$

Decryption computes $C_2 - [5]C_1 = [5 - 15]P = [9]P = M$ modulo the order 19. The byte pair $(1, 1)$ is not on the curve because $1^2 \not\equiv 1^3 + 2 + 2 \pmod{17}$; accepting it as a point would violate the group interface before any cryptographic assumption is reached.

The example is not a parameter recommendation. A nineteen-element group is exhaustively breakable. Its purpose is to make point counting, subgroup order, scalar arithmetic, protocol correctness, and validation failure visible in one finite calculation.

8.3 Reader checkpoint

1. Enumerate all points of the toy curve and verify the count 19.
2. Compute $[2]P, [3]P, [4]P$ from the chord-tangent formulas.
3. Verify the displayed ECDH shared point independently from both sides.
4. Verify the EC-ElGamal decryption numerically.
5. Explain why prime group order makes cofactor clearing trivial in the example.
6. Give the validation checklist for a received compressed curve point.
7. Explain an invalid-curve/small-subgroup attack at the interface level.
8. Translate the multiplicative ElGamal DDH proof into additive notation explicitly.
9. Explain why ECDSA is not covered by the Schnorr transfer statement.
10. State why a standardized hash-to-curve map requires more than reducing a hash modulo q .

Chapter 9

Pairings, Miller’s Algorithm, and ECDLP Transfer Attacks

Pairings add a new operation that is simultaneously constructive and cryptanalytic. Bilinearity enables new protocols, while the same map can move an elliptic-curve logarithm into a finite field where index calculus applies. The embedding degree and the exact source/target groups therefore become security parameters rather than background algebra.

9.1 How to read this chapter

Definition 9.1 fixes a cryptographic pairing interface. The Weil-pairing theorem 9.2 remains source-owned at divisor theory. Algorithm 9.3 internalizes Miller’s addition-chain recurrence. The MOV/Frey–Rück theorem is source-owned because the full reduction needs torsion rationality and pairing nondegeneracy conditions beyond the loop calculation.

9.2 Learning goals

A reader should be able to state bilinearity/nondegeneracy and embedding degree; derive Miller’s f_{m+n} recurrence from divisors; explain final exponentiation for the reduced Tate pairing; state the MOV attack condition; and distinguish a fast pairing computation from a proof that a pairing-friendly curve is secure.

Definition 9.1 (Bilinear pairings). Let r be prime. A cryptographic bilinear pairing instance has cyclic groups G_1, G_2, G_T of order r and an efficiently computable map

$$e : G_1 \times G_2 \rightarrow G_T$$

such that $e([a]P, [b]Q) = e(P, Q)^{ab}$ and nondegeneracy means suitable nonidentity inputs have nonidentity pairing. Symmetric/type-1 and asymmetric/type-3 settings are not identified.

For $E/\mathbb{F}_q$ with $r \mid \#E(\mathbb{F}_q)$ and $\gcd(r, q) = 1$, the *embedding degree* is the smallest $k \geq 1$ with $r \mid (q^k - 1)$. Weil and Tate pairings arise on suitable r -torsion/subquotient domains over extension fields. The reduced Tate pairing includes final exponentiation by $(q^k - 1)/r$ to remove the target quotient ambiguity. Distortion maps, twists, subgroup choice, and cofactor clearing are additional typed interfaces when used.

Theorem 9.2 (Weil pairing properties — verified deep input). *For r prime to the characteristic and the full r -torsion $E[r]$ over a suitable extension, the Weil pairing*

$$e_r : E[r] \times E[r] \rightarrow \mu_r$$

is bilinear, alternating ($e_r(P, P) = 1$), nondegenerate, and Galois-compatible. In particular, $e_r([a]P, [b]Q) = e_r(P, Q)^{ab}$. The precise divisor-function definition and proof of independence from auxiliary divisors, bilinearity, and nondegeneracy remain source-owned in the standard divisor-theoretic references; these properties are the exact interface used below.

Algorithm 9.3 (Miller pairing algorithm). *Fix $P \in E$ and for an integer $m \geq 1$ let f_m be a rational function with divisor*

$$\operatorname{div}(f_m) = m(P) - ([m]P) - (m-1)(O). \quad (9.1)$$

Let $\ell_{R,S}$ denote the line through R, S (tangent if $R = S$) and v_T the vertical line through T . Then, up to a nonzero scalar irrelevant to pairing evaluation,

$$f_{m+n} = f_m f_n \frac{\ell_{[m]P, [n]P}}{v_{[m+n]P}}. \quad (9.2)$$

A binary addition chain therefore evaluates $f_r(Q)$ using $O(\log r)$ doubling/addition line-function steps. For a reduced Tate pairing one subsequently raises the result to $(q^k - 1)/r$. Denominator elimination is an optimization only under field/subgroup conditions that make the omitted denominator lie in a subfield killed by final exponentiation; it is not a general algebraic identity.

Proof. The divisor of the line $\ell_{R,S}$ is $(R) + (S) + (-(R+S)) - 3(O)$, while the vertical line through $R+S$ has divisor $(R+S) + (-(R+S)) - 2(O)$. Hence the quotient has divisor

$$(R) + (S) - (R+S) - (O). \quad (9.3)$$

Adding (9.3), with $R = [m]P, S = [n]P$, to the divisors (9.1) for f_m and f_n gives exactly the divisor (9.1) for f_{m+n} . Rational functions with the same divisor differ by a nonzero scalar, proving (9.2) at the pairing-evaluation level. Repeated doubling and optional addition follow the binary expansion of r , so the loop has $O(\log r)$ group/line steps. $\square$

Theorem 9.4 (MOV/Frey–Rück pairing reduction — verified deep input). *Let a prime-order subgroup $\langle P \rangle$ of an elliptic curve have embedding degree k with respect to r . Under the torsion-rationality/nondegeneracy hypotheses of the MOV or Frey–Rück pairing construction, an ECDLP instance in that subgroup can be mapped in polynomial time to a DLP in an order- r subgroup of $\mathbb{F}_{q^k}^*$. Consequently curve selection must make the best finite-field DLP attack at the embedding degree at least as hard as the intended curve-group attack. The theorem does not say that every curve has small embedding degree.*

Algorithm 9.5 (Finite-field and elliptic-curve arithmetic laboratory). *The executable file `code/volume_17/number_theory_ecc_lab.py` checks extended Euclid and CRT, Miller–Rabin on an exhaustive small range, deterministic Pollard-rho factor recovery on selected composites, the complete toy curve of Example 8.3, scalar multiplication, point validation, a Miller line-function recurrence checkpoint, and an abstract prime-order bilinear-map identity. The lab is a finite invariant check, not evidence for cryptographic parameter security.*

9.3 Reader checkpoint

1. Prove bilinearity implies $e([a]P, Q) = e(P, Q)^a$ by repeated addition.
2. Compute the embedding degree when $q = 5$ and $r = 3$.
3. Verify divisor equation (9.3).
4. Derive the doubling specialization of Miller recurrence (9.2).
5. Count Miller-loop iterations from the binary length of r .
6. Explain the purpose of final exponentiation in a reduced Tate pairing.
7. State why denominator elimination needs an additional subfield hypothesis.
8. Explain how a nondegenerate pairing transfers a logarithm equation to G_T .
9. State the curve-selection consequence of a small embedding degree.
10. Distinguish the Weil pairing theorem from the computational Miller algorithm.

Chapter 10

Pairing-Based Cryptography and the Assumption Map

Pairings were first a cryptanalytic warning and then became a design tool. This final chapter keeps those two roles visible at once: Joux’s three-party identity is elementary bilinearity, whereas identity-based encryption and short-signature security depend on carefully programmed random-oracle reductions and pairing-specific assumptions. Concrete curve security is explicitly dated because finite-field attacks continue to change parameter estimates.

10.1 How to read this chapter

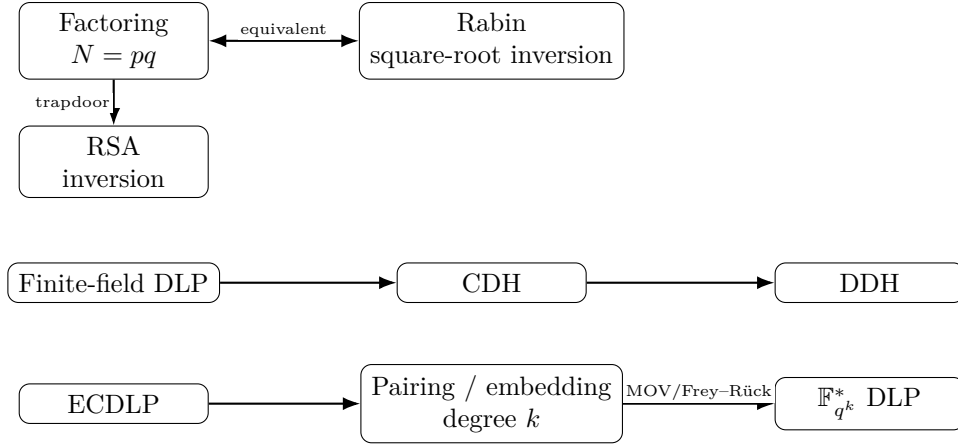
The frontier statement 10.1 comes first so that no later construction is read as a parameter recommendation. Figure 10.1 separates assumptions and reductions. Boneh–Franklin and BLS are source-owned; Theorem 10.4 proves only the Joux shared-key identity plus its assumption interface.

10.2 Learning goals

A reader should be able to distinguish BDH/CDH/DDH-style pairing assumptions; state the security level of a construction only relative to its exact group/hash/oracle model; explain why TNFS changes target-field security estimates; derive Joux’s one-round three-party key; and read the reduction map without reversing one-way implications.

Theorem 10.1 (Pairing assumptions and concrete-security audit — dated frontier, 2026-09-03). *Pairing-based parameters must be evaluated simultaneously in the source elliptic-curve groups and the target finite field $\mathbb{F}_{q^k}^*$. Tower/extended-tower number-field-sieve developments changed historical security estimates for several pairing-friendly families, so embedding degree and field size cannot be translated to security bits by an obsolete generic formula. Current protocol specifications must also bind subgroup checks, cofactors, twists, serialization, and a standardized hash-to-curve interface; RFC 9380, for example, specifies suites including BLS12-381 but is not by itself a theorem that every protocol using that curve meets a chosen security target.*

This row is a dated methodology frontier, not a fixed list of approved parameters. Deployment choices must be refreshed against current finite-field DLP records/attacks, protocol standards, and implementation side-channel requirements.



Arrows are known implication/reduction directions in the stated models, not equivalences unless marked. Knowing the factors gives the RSA trapdoor, but a polynomial-time RSA inverter is not known in general to factor. Likewise CDH and DDH can separate in pairing/gap-DH settings.

Figure 10.1: Number-theoretic assumption and reduction map.

Theorem 10.2 (Boneh–Franklin identity-based encryption — verified deep input). *In the random-oracle model, the Boneh–Franklin pairing construction gives a fully functional identity-based encryption system. In the CRYPTO 2001 presentation, BasicIdent is proved one-way identity-based secure (ID-OWE) under the stated Weil Diffie–Hellman (WDH) assumption, with adaptive private-key extraction queries away from the challenge identity. Applying the Fujisaki–Okamoto-style transform yields FullIdent, which is proved adaptively chosen-ciphertext secure (IND-ID-CCA) in the same random-oracle/WDH framework. The reductions program identity/hash queries and account explicitly for extraction and, for FullIdent, decryption queries. Those mechanisms remain source-owned; this theorem is not a claim that BasicIdent is itself IND-ID-CPA, nor a claim about arbitrary deterministic string-to-point maps.*

Theorem 10.3 (Boneh–Lynn–Shacham short signatures — verified deep input). *For the source-specified pairing group and random-oracle hash-to-group model, BLS signatures verify by one bilinear equation and satisfy the source theorem’s existential-unforgeability guarantee under its CDH/co-CDH or gap-group assumption (depending on the symmetric/asymmetric formulation). A modern instantiation must additionally specify subgroup validation and a hash-to-curve suite; the abstract theorem does not certify an arbitrary encoding or aggregation protocol.*

Theorem 10.4 (Joux one-round tripartite Diffie–Hellman). *Let $G = \langle P \rangle$ have prime order r and let $e : G \times G \rightarrow G_T$ be a symmetric nondegenerate bilinear pairing. Alice, Bob, and Carol broadcast $A = [a]P$, $B = [b]P$, and $C = [c]P$. Without another round they compute, respectively,*

$$e(B, C)^a = e(A, C)^b = e(A, B)^c = e(P, P)^{abc}. \quad (10.1)$$

Thus the protocol has perfect algebraic agreement. Passive key secrecy requires a separately stated bilinear Diffie–Hellman-type assumption and a KDF; authentication is not provided by (10.1).

Proof. Bilinearity gives $e([b]P, [c]P)^a = e(P, P)^{abc}$, and cyclically the same exponent is obtained by the other two parties. The security qualification follows for the same reason as ordinary DH: equality of honest keys is an algebraic statement, while pseudorandomness of the value to an observer is a computational assumption. $\square$

Volume closure. The number-theoretic chain now has three explicit layers. Arithmetic algorithms explain what honest parties can compute; attack algorithms delimit which algebraic problems are plausible assumptions in each representation; security reductions then use those assumptions without reversing their logical direction. Elliptic curves and pairings do not erase that hierarchy—they make it more important because one object can expose several linked groups with different best attacks.

10.3 Reader checkpoint

1. Explain why pairing security must consider both G_1/G_2 and G_T attacks.
2. State what embedding degree contributes to target-field size.
3. Explain why the pairing-security row is dated rather than certified permanently.
4. Read every arrow in Figure 10.1 in the correct direction.
5. Explain why factoring breaking RSA does not prove RSA inversion is equivalent to factoring.
6. State one difference between BasicIdent and FullIdent security goals.
7. Explain why BLS needs a typed hash-to-group interface.
8. Derive all three equalities in (10.1).
9. Give a man-in-the-middle reason Joux agreement is not authenticated by bilinearity alone.
10. Summarize, in one paragraph, the difference between an algebraic correctness theorem, a hardness assumption, and a reduction theorem.

Internalization and Reconstruction Guide: Volume XVII

Purpose. Connect number-theoretic algorithms and algebraic groups to public-key constructions while keeping algebraic correctness, computational assumptions, encodings, padding, and random-oracle or pairing models distinct.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-XVI-DEF-023: Authentication and signature security.
2. QRCC-XVI-DEF-001: Security parameters, adversaries, and games.
3. QRCC-XVI-DEF-028: Chosen-ciphertext security and KEM/DEM composition.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XVII-DEF-001 (Computational number-theory model); QRCC-XVII-DEF-004 (Factoring problems and distributions); QRCC-XVII-DEF-007 (RSA and Rabin assumptions); QRCC-XVII-DEF-011 (Discrete logarithm and generic groups); QRCC-XVII-DEF-015 (Diffie–Hellman assumptions).

Chapter retrieval map

- Ch. 1: Computational Number Theory: Arithmetic and Primality:** enter through QRCC-XVII-DEF-001 (Computational number-theory model); exit by being able to state QRCC-XVII-ALG-003 (Miller–Rabin and deterministic primality interfaces) with its ownership boundary.
- Ch. 2: Integer Factoring: Random Walks and Sieves:** enter through QRCC-XVII-DEF-004 (Factoring problems and distributions); exit by being able to state QRCC-XVII-ALG-005 (Pollard rho factoring analysis) with its ownership boundary.
- Ch. 3: RSA, Rabin, and Provable Encodings:** enter through QRCC-XVII-DEF-007 (RSA and Rabin assumptions); exit by being able to state QRCC-XVII-BB-010 (RSA padding and

full-domain-hash signatures) with its ownership boundary.

- Ch. 4: Discrete Logarithms: Generic and Representation-Specific Algorithms:** enter through QRCC-XVII-DEF-011 (Discrete logarithm and generic groups); exit by being able to state QRCC-XVII-ALG-012 (Pollard rho for discrete logarithms) with its ownership boundary.
- Ch. 5: Diffie–Hellman Assumptions and Random Self-Reduction:** enter through QRCC-XVII-DEF-015 (Diffie–Hellman assumptions); exit by being able to state QRCC-XVII-THM-016 (Diffie–Hellman key agreement correctness) with its ownership boundary.
- Ch. 6: ElGamal, Schnorr Identification, and Fiat–Shamir Boundaries:** enter through QRCC-XVII-DEF-018 (ElGamal and Schnorr security models); exit by being able to state QRCC-XVII-BB-021 (Schnorr signature security in the random-oracle model) with its ownership boundary.
- Ch. 7: Elliptic Curves over Finite Fields:** enter through QRCC-XVII-DEF-022 (Elliptic curves over finite fields); exit by being able to state QRCC-XVII-BB-024 (Hasse bound and Frobenius characteristic polynomial) with its ownership boundary.
- Ch. 8: Elliptic-Curve Cryptographic Groups:** enter through QRCC-XVII-DEF-026 (Elliptic-curve cryptographic groups); exit by being able to state QRCC-XVII-THM-027 (Elliptic-curve Diffie–Hellman and signatures) with its ownership boundary.
- Ch. 9: Pairings, Miller’s Algorithm, and ECDLP Transfer Attacks:** enter through QRCC-XVII-DEF-030 (Bilinear pairings); exit by being able to state QRCC-XVII-ALG-032 (Miller pairing algorithm) with its ownership boundary.
- Ch. 10: Pairing-Based Cryptography and the Assumption Map:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XVII-THM-033 (Boneh–Franklin identity-based encryption) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XVII-ALG-003 — Miller–Rabin and deterministic primality interfaces. Dependencies: QRCC-XVII-DEF-001, QRCC-XVII-THM-002. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
2. QRCC-XVII-BB-010 — RSA padding and full-domain-hash signatures. Dependencies: QRCC-XVII-THM-008, QRCC-XVI-DEF-023. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-XVII-THM-016 — Diffie–Hellman key agreement correctness. Dependencies: QRCC-XVII-DEF-015, QRCC-XVI-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
4. QRCC-XVII-BB-024 — Hasse bound and Frobenius characteristic polynomial. Dependencies: QRCC-XVII-DEF-022. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

5. QRCC-XVII-ALG-032 — Miller pairing algorithm. Dependencies: QRCC-XVII-THM-031, QRCC-XVII-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
6. QRCC-XVII-F-036 — Pairing assumptions and concrete-security audit. Dependencies: QRCC-XVII-THM-028, QRCC-XVII-ALG-032. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- using a group operation before validating subgroup/order conditions;
- treating RSA correctness as a security proof or discrete-log hardness as DDH hardness;
- ignoring encoding, padding, hash-to-curve, or random-oracle interfaces;
- forgetting exceptional elliptic-curve points or pairing embedding-degree conditions;

Boundary and degenerate cases

- noninvertible residues modulo composite moduli;
- small or anomalous subgroups;
- the elliptic-curve point at infinity and exceptional denominators;
- pairing-friendly versus ordinary curves;

Retrieval and reconstruction drill

1. State QRCC-XVII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XVII-ALG-003. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XVII-BB-010, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.

8. Starting from QRCC-XVII-F-036, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XVIII turns from construction to cryptanalysis, beginning with hash functions and the Wang differential methodology.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XVIII

Hash Functions and the Wang Cryptanalytic Revolution

Chapter 1

Hash Functions: Security Notions and Generic Bounds

A hash function compresses a large message space to a short digest, so collisions exist by counting. Cryptographic security therefore cannot mean injectivity; it means that a specified computational experiment makes the relevant witness hard to find. This chapter fixes those experiments and separates classical from quantum query access before any concrete construction is discussed.

1.1 How to read this chapter

Definition 1.1 is the type system for the entire volume. Theorem 1.2 then gives the exact random-function bookkeeping behind the familiar $2^{n/2}$ collision and 2^n preimage scales, with the quantum collision endpoint explicitly source-owned. Theorem 1.3 explains why security properties must be transported through a domain-extension construction rather than merely asserted for the compression function.

1.2 Learning goals

A reader should be able to distinguish collision, second-preimage, preimage, target-collision, multicollision, and chosen-prefix experiments; compute exact generic success probabilities; state the classical and quantum query models; and audit which security property a domain-extension theorem actually transfers.

Literature and convention checkpoint. Classical random-function bounds are elementary and proved locally. Quantum preimage search reuses the audited Grover/BBBV interface from Volume XV. The tight quantum collision scale is calibrated to the Brassard–Høyer–Tapp upper bound and the Aaronson–Shi/Ambainis lower-bound line and remains source-owned. Domain extension is calibrated to the Merkle–Damgård theorem and later indistinguishability literature; no compression-function assumption is silently promoted to a random-oracle claim.

Definition 1.1 (Hash-function security notions). A *compression function* is a map $f : \{0, 1\}^c \times \{0, 1\}^b \rightarrow \{0, 1\}^c$; a *variable-length hash family* is an efficiently computable family $H_\lambda : \{0, 1\}^* \rightarrow \{0, 1\}^{n(\lambda)}$ with a specified padding/encoding rule and security parameter λ . For a fixed public hash H , the standard witness problems are:

- a *collision*: $M \neq M'$ with $H(M) = H(M')$;

- a *preimage* of a target digest y : an M with $H(M) = y$;
- a *second preimage* of a given message M : an $M' \neq M$ with $H(M') = H(M)$;
- a *t -multicollision*: distinct $M_1, \dots, M_t$ having one common digest;
- a *chosen-prefix collision*: for two externally fixed prefixes P, P' , suffixes S, S' with $H(P\|S) = H(P'\|S')$.

A *target-collision* experiment fixes or samples the first message before the adversary chooses its colliding partner; its quantifier order must therefore be stated rather than identified with ordinary collision resistance.

A *classical query algorithm* learns oracle values $F(x)$ adaptively. A *quantum query algorithm* accesses F through the reversible oracle $|x, z\rangle \mapsto |x, z \oplus F(x)\rangle$ and query complexity counts such calls, with all nonquery computation stated separately. A *random oracle* is a uniformly sampled function with the specified domain/range. *Indifferentiability* compares an ideal primitive with a construction plus simulator; it is stronger and structurally different from collision resistance.

Theorem 1.2 (Generic collision and preimage query bounds). *Let $F : D \rightarrow R$ be uniformly random with $|R| = N = 2^n$, and suppose $q \leq |D|$ distinct inputs are queried.*

(i) *The probability of no collision among the q answers is*

$$\prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right),$$

so

$$1 - e^{-q(q-1)/(2N)} \leq \Pr[a \text{ collision among the answers}] \leq \frac{q(q-1)}{2N}.$$

In particular constant classical collision probability requires and is achieved with $q = \Theta(N^{1/2})$ queries.

- (ii) *For a fixed target $y \in R$, the success probability of q distinct classical queries is $1 - (1 - 1/N)^q \leq q/N$; constant success therefore needs $\Theta(N)$ classical queries.*
- (iii) *In the reversible black-box model, preimage search has tight query complexity $\Theta(N^{1/2})$. Collision search has tight query complexity $\Theta(N^{1/3})$ under the standard promised collision formulation. The preimage statement follows from the Volume-XV Grover/search lower-bound interface; the collision statement is a verified deep input.*

Proof. Conditioned on the first i answers being distinct, the next uniformly random answer avoids them with probability $1 - i/N$. Multiplying gives the exact no-collision probability. Since $1 - x \leq e^{-x}$, the product is at most $e^{-q(q-1)/(2N)}$, giving the lower bound on collision probability. The union bound over the $\binom{q}{2}$ pairs gives the upper bound. Taking $q = c\sqrt{N}$ gives a constant lower bound for any fixed $c > 0$, while the union bound forces $q = \Omega(\sqrt{N})$ for constant success.

For a fixed y , every fresh query misses independently with probability $1 - 1/N$, so the exact formula and the union bound follow. The quantum preimage clause is exactly unstructured search. The $N^{1/3}$ collision theorem uses the Brassard–Høyer–Tapp algorithm and the matching polynomial-method lower bounds and is intentionally not compressed into a local proof here. $\square$

Worked scale check. For a 256-bit ideal digest, $q = 2^{64}$ classical queries give collision probability at most about 2^{-129} , whereas $q \approx 2^{128}$ reaches constant collision probability. This says nothing about a concrete hash whose structure admits a differential attack: the random-function benchmark is the baseline that concrete cryptanalysis tries to beat.

Theorem 1.3 (Domain-extension security bookkeeping). *Let T map every message M to a finite block sequence $E(M) = (B_1, \dots, B_\ell)$ together with a fixed initial state h_0 , and let $h_i = f(h_{i-1}, B_i)$. Suppose the encoding E is injective and prefix-free. If two distinct messages satisfy $T(M) = T(M')$, then at the first position where their encoded computation histories merge into a common chaining state there is either*

- (a) *a collision in f on two distinct state/block inputs, or*
- (b) *an encoding violation (noninjective or non-prefix-free representation).*

Consequently a collision-resistance reduction for an iterated construction must name the encoding property and the exact compression-function collision it extracts. The theorem does not imply preimage resistance, second-preimage resistance, pseudorandomness, or random-oracle indistinguishability.

Proof. Write the two state/block transcripts from h_0 to their common final state. If the encoded block strings were identical, injectivity would force $M = M'$, contrary to hypothesis. Traverse backward from the common final state until the predecessor transcript entries first differ. The two distinct predecessor state/block pairs map under f to the same next state, yielding a compression collision. If this backward comparison cannot be made because one encoded block string is a strict prefix of the other, prefix-freeness has failed. The final sentence is logical: none of those other security experiments is reduced by the collision-extraction argument. $\square$

1.3 Reader checkpoint

1. Derive the exact no-collision probability for $q = 3$ and range size N .
2. Use the union bound to determine a sufficient q for collision probability at most 2^{-40} for $n = 256$.
3. Show that $1 - (1 - 1/N)^q \leq q/N$ without using calculus.
4. Explain why second-preimage resistance has a different quantifier order from collision resistance.
5. Give a chosen-prefix collision experiment in which the adversary does not choose either prefix.
6. State precisely the reversible oracle used by a quantum hash-query algorithm.
7. Explain why the random-function birthday bound is not a security proof for MD5 or SHA-1.
8. Construct a non-prefix-free encoding for which appending blocks creates an ambiguity.
9. In Theorem 1.3, identify the exact point at which injectivity is used.
10. List four security properties that Theorem 1.3 deliberately does not transfer.

Chapter 2

Iterated Hashing, Length Extension, and Sponge Security

Most deployed hashes are not arbitrary functions: they extend a fixed-width primitive to arbitrary messages. The extension rule is therefore part of the cryptographic object. This chapter contrasts strengthened Merkle–Damgård iteration, Davies–Meyer compression, and the sponge construction, and separates collision preservation from indistinguishability.

2.1 How to read this chapter

Definition 2.1 fixes the constructions. Theorem 2.2 proves the classical collision-preservation theorem, Proposition 2.3 isolates plain Merkle–Damgård length extension, and Black Box 2.4 states the sponge indistinguishability interface at its source-owned level.

2.2 Learning goals

A reader should be able to reconstruct a Merkle–Damgård collision reduction, explain why strengthening matters, compute a length extension from a chaining value, distinguish Davies–Meyer from a hash mode, and explain the roles of sponge rate and capacity.

Literature and convention checkpoint. The collision theorem is calibrated to Merkle and Damgård. Length extension is derived directly from the public iteration rule and is not confused with a generic attack on HMAC. The sponge statement is calibrated to Bertoni, Daemen, Peeters, and Van Assche; its simulator proof remains source-owned.

Definition 2.1 (Merkle–Damgård, Davies–Meyer, and sponge constructions). Let $f : \{0, 1\}^c \times \{0, 1\}^b \rightarrow \{0, 1\}^c$. A *Merkle–Damgård iteration* pads/encodes a message as blocks $B_1, \dots, B_\ell$, fixes $h_0 = IV$, and sets $h_i = f(h_{i-1}, B_i)$, outputting h_ℓ . *MD strengthening* appends a fixed-format terminal encoding of the original bit length. The encoding is self-delimiting/prefix-free, and different original lengths give different terminal length-bearing blocks (equivalently, the strengthening is suffix-disambiguating at the final compression step).

Given a block cipher $E_K(X)$, *Davies–Meyer compression* is $f(h, m) = E_m(h) \oplus h$; this defines a compression function, not by itself a variable-length hash.

A *sponge* uses a fixed permutation P on $b = r + c$ bits. After injective domain-separated padding into r -bit blocks, absorption xors each block into the first r state bits and applies P ; squeezing

reads r output bits and applies P again as needed. The parameters r and c are the *rate* and *capacity*. A wide-pipe design uses an internal chaining state wider than the final digest; HAIFA additionally makes iteration position/salt-like metadata explicit. Keyed sponge modes must name their key-injection/domain-separation rule rather than inheriting unkeyed claims automatically.

Theorem 2.2 (Merkle–Damgård collision preservation). *For strengthened Merkle–Damgård with fixed IV , any collision in the variable-length hash yields a collision in f with at most linear overhead in the two padded message lengths. Hence collision resistance of f implies collision resistance of the strengthened iteration under the same computational model.*

Proof. Let $B_1, \dots, B_s$ and $B'_1, \dots, B'_t$ be the two strengthened padded block strings and let h_i, h'_j be their chaining states, with $h_0 = h'_0 = IV$ and $h_s = h'_t$.

If the original message lengths differ, the terminal length-bearing blocks are different by the strengthening convention. Hence the two final compression inputs (h_{s-1}, B_s) and (h'_{t-1}, B'_t) are distinct unless both predecessor states and terminal blocks coincide; the latter is impossible because the terminal blocks differ. Since their outputs are the common final digest, the final step already gives a collision in f .

It remains to consider equal original lengths. Then the strengthened padded strings have the same number of blocks. They are distinct because the underlying messages are distinct. Starting from the common initial state, let j be the first index at which either the block or the next chaining state differs. Because the final states agree, there is a least $k > j$ at which the two state sequences merge again. At that merger, $h_{k-1} \neq h'_{k-1}$ or $B_k \neq B'_k$, while

$$f(h_{k-1}, B_k) = h_k = h'_k = f(h'_{k-1}, B'_k).$$

Thus the two distinct compression inputs form a collision. Computing both transcripts and locating the final-step or first-merger witness takes time linear in the padded message lengths. $\square$

Proposition 2.3 (Length extension for plain Merkle–Damgård). *Suppose a Merkle–Damgård digest exposes the final chaining value and the attacker knows the encoded length of a message M . From $H(M)$ and any extension X , one can compute*

$$H(M \parallel \text{pad}(M) \parallel X)$$

without knowing M : initialize the public compression iteration at state $H(M)$ and process the blocks of X and the final padding corresponding to the extended total length. Therefore a naive secret-prefix MAC $H(K \parallel M)$ is generically length-extendable when the key length is known or guessable. This proposition does not apply unchanged to HMAC.

Proof. The digest $H(M)$ is exactly the chaining state after the padded representation of M . Every subsequent compression call is public and depends only on the current state and the next block. Knowing the original encoded length determines the padding already absorbed and the length field needed at the end of the extension. Replaying those public calls computes the extended digest. HMAC nests keyed applications and does not expose the inner state as its final public digest, so the displayed continuation argument is not an HMAC forgery proof. $\square$

Theorem 2.4 (Sponge indistinguishability — verified deep input). *For the sponge construction built from an ideal random permutation on $b = r + c$ bits with injective padding, there is a simulator for the permutation interface such that, for a bounded number q of construction/permutation queries, the*

joint system is indifferentiable from a random oracle up to a birthday-type advantage governed by the capacity c (with the exact theorem's query/output-length terms as in Bertoni–Daemen–Peeters–Van Assche). In particular, the meaningful generic security scale is controlled by capacity, not merely by the visible output length.

Ownership boundary. The local text defines the simulator game and explains the capacity-collision intuition: bad events arise when hidden c -bit portions of internal states collide or become consistently constrained by the adversary. The complete lazy sampling/simulator consistency proof and its exact constants are source-owned. No claim is made that every concrete permutation inherits ideal-permutation security.

2.3 Reader checkpoint

1. Explain why MD strengthening makes padded encodings prefix-free.
2. Reconstruct the backward first-merge argument in Theorem 2.2.
3. Give an example showing why an unstrengthened iteration can make the proof awkward when one block string extends another.
4. Derive Davies–Meyer compression from a block-cipher call and feed-forward.
5. For a 1600-bit sponge with rate 1088, compute the capacity.
6. Explain why increasing output length without increasing capacity need not raise every generic security level.
7. Write the state from which a length-extension computation begins.
8. Explain why knowing the original message length matters in Proposition 2.3.
9. State why the same continuation argument does not directly forge HMAC.
10. Distinguish collision preservation from random-oracle indifferentiability in one paragraph.

Chapter 3

Birthday, Multicollision, and Long-Message Second-Preimage Attacks

Even when the compression function behaves ideally, iteration creates combinatorial structure. Birthday search is only the first layer: multicollisions can be assembled recursively, and very long messages expose expandable-message second-preimage tradeoffs. These attacks are generic in a strong sense—they do not need the bit-level weaknesses that drive the Wang revolution later in the volume.

3.1 How to read this chapter

Theorem 3.1 sharpens the birthday baseline. Theorem 3.2 proves Joux’s multicollision recursion locally. Black Box 3.3 then states the Kelsey–Schneier long-message second-preimage theorem with its length dependence explicit.

3.2 Learning goals

A reader should be able to compute collision probabilities with constants, build a 2^k multicollision from k compression collisions, explain why concatenating hash outputs does not always multiply security as naively expected, and identify the expandable-message mechanism behind long-message second-preimage attacks.

Literature and convention checkpoint. The multicollision theorem is calibrated to Joux (CRYPTO 2004). The long-message second-preimage theorem is calibrated to Kelsey–Schneier (EUROCRYPT 2005). Generic attack costs count compression calls and memory separately; no concrete MD4/MD5 differential shortcut is used in this chapter.

Theorem 3.1 (Birthday collision theorem with constants). *For independent uniform samples $Y_1, Y_2, \dots$ from an N -element set, let T be the first index at which a collision appears. Then*

$$\Pr[T > q] = \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right), \quad \Pr[T \leq q] \geq 1 - e^{-q(q-1)/(2N)}.$$

Moreover

$$\mathbb{E}T = \sum_{q \geq 0} \Pr[T > q] = \sqrt{\frac{\pi N}{2}} + O(1).$$

Thus the expected first collision occurs at approximately $1.2533\sqrt{N}$ samples.

Proof. The exact product and exponential bound were proved in Theorem 1.2. Write

$$S_q := \Pr[T > q], \quad a_q := \frac{q(q-1)}{2N}.$$

For $q \leq N/2$, the elementary inequality $0 \leq -\log(1-x) - x \leq 2x^2$ for $0 \leq x \leq 1/2$ gives

$$0 \leq -\log S_q - a_q \leq \frac{2}{N^2} \sum_{i < q} i^2 \leq \frac{2q^3}{3N^2}.$$

Hence, using $1 - e^{-u} \leq u$,

$$0 \leq e^{-a_q} - S_q \leq e^{-a_q} \frac{2q^3}{3N^2}. \quad (3.2)$$

For $q \geq 2$, $a_q \geq q^2/(4N) - 1$, so summing (3.2) and comparing with the integral of $x^3 e^{-x^2/(4N)}$ gives

$$\sum_{q \leq N/2} (e^{-a_q} - S_q) = O(1). \quad (3.3)$$

For $q > N/2$, the exponential upper bound gives a total tail $O(Ne^{-cN})$. Therefore

$$\mathbb{E}T = \sum_{q \geq 0} e^{-q(q-1)/(2N)} + O(1). \quad (3.4)$$

Moreover

$$\left| e^{-q(q-1)/(2N)} - e^{-q^2/(2N)} \right| \leq \frac{q}{2N} e^{-q(q-1)/(2N)},$$

and the sum of the right-hand side is $O(1)$ by the same integral comparison. Finally, since $x \mapsto e^{-x^2/(2N)}$ is decreasing,

$$\sum_{q \geq 0} e^{-q^2/(2N)} = \int_0^\infty e^{-x^2/(2N)} dx + O(1) = \sqrt{\frac{\pi N}{2}} + O(1).$$

Combining this with (3.4) proves the displayed expectation with the claimed $O(1)$ remainder. $\square$

Theorem 3.2 (Joux multicollision attack on iterated hashes). *Let an iterated n -bit hash expose an n -bit chaining state and suppose a collision from any prescribed chaining state can be found in expected $C 2^{n/2}$ compression calls. Then for every $k \geq 1$ one can construct 2^k distinct equal-length messages with a common final hash state using expected $kC 2^{n/2}$ compression calls and storage linear in k collision pairs (besides the collision-search workspace).*

Proof. Starting from $h_0 = IV$, find two one-block strings B_1^0, B_1^1 with $f(h_0, B_1^0) = f(h_0, B_1^1) = h_1$. From h_1 find a new pair B_2^0, B_2^1 colliding to h_2 , and continue for k levels. For every bit string $\sigma \in \{0, 1\}^k$, the concatenation $B_1^{\sigma_1} \cdots B_k^{\sigma_k}$ follows the same state sequence $h_1, \dots, h_k$. Hence all 2^k messages collide. Linearity of expectation gives the work bound. Equal block count ensures identical final strengthening length fields when the construction includes a final length block. $\square$

Concatenated-hash warning. If two iterated hashes process the same message, a multicollision cheaply supplies a large candidate set already equal under the first component. One can then search that structured set for collisions in the second component. The security of $H_1(M)\|H_2(M)$ therefore cannot in general be obtained by simply adding the nominal collision-bit strengths of H_1 and H_2 .

Theorem 3.3 (Kelsey–Schneier second preimages on long iterated messages — verified deep input). *For an n -bit Damgård–Merkle iterated hash with n -bit intermediate states and a target message of 2^k blocks, Kelsey and Schneier construct expandable messages and obtain a generic second-preimage attack requiring about*

$$k 2^{n/2+1} + 2^{n-k+1}$$

compression-function work, up to the theorem’s lower-order and memory terms. The gain is specific to sufficiently long targets; it does not state a comparable shortcut for an arbitrary short message.

Mechanism map. An expandable message is a family of prefixes of many different block lengths that all reach one common chaining state. Once such a family exists, the attacker searches for a bridge from that state into one of the many internal states of the long target, then chooses the expandable-prefix length so that the bridge lands at the correct absolute position. Constructing the family repeatedly uses birthday collisions; the bridge benefits from the target’s many available internal states. The exact multi-level construction and constants remain source-owned.

3.3 Reader checkpoint

1. Compute the approximate number of samples for 50% collision probability in a 128-bit range.
2. Derive the tail-sum identity $\mathbb{E}T = \sum_{q \geq 0} \Pr[T > q]$ for integer-valued $T \geq 1$.
3. Evaluate $\int_0^\infty e^{-x^2/2} dx$ from the Gaussian integral.
4. Construct explicitly the four messages in the $k = 2$ Joux multicollision tree.
5. Explain why all Joux messages should have equal block length under strengthening.
6. Compare the work for a 2^{20} -multicollision with naive independent collision search.
7. Explain how multicollisions undermine naive security addition for concatenated iterated hashes.
8. Define an expandable message in your own words without referring to a concrete hash.
9. For $n = 160$ and $k = 60$, compare the two terms in the Kelsey–Schneier work formula.
10. Explain why the long-message attack is not a contradiction of ordinary 2^n second-preimage intuition for short messages.

Chapter 4

Differential Paths, Bit Conditions, and Message Modification

The Wang cryptanalytic revolution replaced generic birthday reasoning by a local analysis of how modular additions, Boolean functions, rotations, and message schedules transport carefully engineered differences. The central object is not merely an XOR difference but a path of state differences together with bit conditions that make low-probability carries and Boolean transitions behave as desired.

4.1 How to read this chapter

Definition 4.1 separates exact path conditions from heuristic search concepts. Theorem 4.2 gives an exact conditional-probability calculus. Algorithm 4.3 formalizes early-round message modification, and Example 4.4 gives a complete toy ARX collision.

4.2 Learning goals

A reader should be able to propagate XOR and modular differences, compute carries exactly, use the probability chain rule rather than multiply unjustified marginals, invert an ARX step to enforce a target state word, and distinguish proved message correction from heuristic neutral-bit or tunnel search.

Literature and convention checkpoint. The terminology is calibrated to the Wang-era MD4/MD5/SHA attack papers. Later differential-path literature is used for the modern search vocabulary. Whenever the book says that a path has probability 2^{-w} , the conditioning model and independence claim are stated. Search heuristics are not promoted to theorems merely because they work well in implementations.

Definition 4.1 (Differential paths and bit conditions). For w -bit words X, X' , the *XOR difference* is $\Delta_{\oplus}X = X \oplus X'$ and the *modular difference* is $\Delta_{+}X = X' - X \pmod{2^w}$. A *signed bit difference* records, at each bit, whether the transition is $0 \rightarrow 1$, $1 \rightarrow 0$, or unchanged rather than only recording XOR support.

A *differential path* specifies differences of selected message/state words through successive steps. A *bit condition* is an equality or relation among concrete bits sufficient to make one local transition follow the path. A *local collision* is a nonzero disturbance that is canceled after a bounded number

of steps. A *disturbance vector* records where such disturbances are injected. *Message modification* changes still free message words to force early bit conditions. A *neutral bit* is a message degree of freedom that can be changed while preserving already enforced conditions with useful probability. A *tunnel* or advanced modification is a multi-step freedom used to reach later conditions. Those latter notions are algorithmic search interfaces, not independence assumptions.

Theorem 4.2 (Differential-path probability calculus). *Fix a finite word-level step computation and a distribution on its unconstrained input bits. Let $C_1, \dots, C_t$ be the successive bit-condition events required by a differential path. Then exactly*

$$\Pr[C_1 \cap \dots \cap C_t] = \prod_{i=1}^t \Pr[C_i \mid C_1 \cap \dots \cap C_{i-1}],$$

with the convention that factors are evaluated only after conditioning on a positive-probability prefix. For modular addition $z = x + y \pmod{2^w}$, writing $c_0 = 0$ for the input carry,

$$z_i = x_i \oplus y_i \oplus c_i, \quad c_{i+1} = \text{maj}(x_i, y_i, c_i),$$

so every conditional transition probability for a frozen finite set of bit constraints can be computed exactly by a finite recurrence over carry states. Multiplying unconditional per-step probabilities is valid only when the needed conditional-independence statement has separately been proved.

Proof. The first identity is the probability chain rule, obtained by repeatedly using $\Pr[A \cap B] = \Pr[A] \Pr[B \mid A]$. The binary addition identities are the truth table for one full adder: parity gives the sum bit and majority gives the outgoing carry. Conditioning on any fixed pattern of message/state bits leaves a finite distribution on the remaining bits and the two possible carry states. Propagating their probabilities from least significant to most significant bit is therefore exact dynamic programming. The final warning follows because conditional probabilities can change after earlier path conditions restrict the same carries or Boolean inputs. $\square$

Algorithm 4.3 (Wang-style message modification framework). *Consider an ARX step of the form*

$$Q_{i+1} = Q_i + \text{ROL}_{s_i}(Q_{i-3} + \Phi_i(Q_i, Q_{i-1}, Q_{i-2}) + M_{\pi(i)} + K_i) \pmod{2^w}.$$

If the message word $M_{\pi(i)}$ has not yet been constrained elsewhere, any desired target value Q_{i+1}^ can be enforced exactly by setting*

$$M_{\pi(i)} = \text{ROR}_{s_i}(Q_{i+1}^* - Q_i) - Q_{i-3} - \Phi_i(Q_i, Q_{i-1}, Q_{i-2}) - K_i \pmod{2^w}. \quad (4.1)$$

Thus early-round bit conditions that can be represented by admissible target state words can be corrected deterministically. When message words are reused by later rounds, preserving old conditions while changing later ones becomes a constrained search problem; neutral-bit, tunnel, and advanced message-modification claims must then be verified for the particular primitive/path rather than inferred from (4.1).

Proof. Subtract Q_i , apply the inverse rotation ROR_{s_i} , and solve the remaining modular linear equation for $M_{\pi(i)}$. Rotation and addition by a fixed word are permutations of $\mathbb{Z}/2^w\mathbb{Z}$, so the solution is unique. Once a message word participates in another already-fixed step, changing it can disturb that step; no unconditional freedom remains, which is exactly why advanced modification is primitive-specific. $\square$

Example 4.4 (A complete local collision in an 8-bit ARX toy hash). Define the one-block toy compression

$$F(h, m) = \text{ROL}_1(h + m \bmod 256) \oplus m.$$

For $h = 0$, take $m = 7\mathbf{f}$ and $m' = 80$. Then

$$\text{ROL}_1(7\mathbf{f}) = \mathbf{fe}, \quad \text{ROL}_1(80) = 01,$$

and hence

$$F(0, 7\mathbf{f}) = \mathbf{fe} \oplus 7\mathbf{f} = 81 = 01 \oplus 80 = F(0, 80).$$

The collision is deterministic, not probabilistic: it exhibits how a carry across the most significant boundary, followed by rotation and feed-forward XOR, can cancel a large visible input difference. The example is intentionally small enough that every bit can be audited by hand; it is not presented as a model of MD5 attack complexity.

4.3 Reader checkpoint

1. Compute both XOR and modular differences between $7\mathbf{f}$ and 80 .
2. Derive the full-adder carry recurrence from its eight-row truth table.
3. Give two dependent events for which multiplying marginal probabilities is wrong.
4. Explain why the chain-rule product in Theorem 4.2 is always valid.
5. Invert a simplified step $Q' = \text{ROL}_5(Q + M)$ for M .
6. Verify Equation (4.1) by substituting it back into the ARX step.
7. Explain why a reused message word is not automatically a free correction variable.
8. Distinguish a bit condition from a differential path.
9. Verify the toy collision bit by bit, including the rotation wraparound.
10. State one reason a successful automated path search does not by itself prove an independence-based probability estimate.

Chapter 5

MD4, RIPEMD, and the First Wang Collision Paths

MD4 established the design lineage that later led to MD5, SHA-0, and SHA-1. Its deliberately simple ARX structure also made it the natural laboratory in which differential paths, bit conditions, and message modification became concrete cryptanalytic tools. This chapter fixes the exact primitive interfaces and records the EUROCRYPT 2005 breaks at source-owned strength.

5.1 How to read this chapter

Definition 5.1 gives a bit-exact MD4 interface and distinguishes the original RIPEMD family member analyzed by Wang et al. from later RIPEMD-128/160 variants. Black Boxes 5.2 and 5.3 record the collision and second-preimage results without turning paper-specific path tables into universal theorems.

5.2 Learning goals

A reader should be able to implement MD4 from the definition, identify where message words are reused, explain why message modification is unusually effective in early rounds, distinguish a collision attack from a weak-message second-preimage result, and state exactly which RIPEMD version a cited attack addresses.

Literature and convention checkpoint. The MD4 specification is calibrated to RFC 1320. The collision theorem is calibrated to Wang, Lai, Feng, Chen, and Yu (EUROCRYPT 2005). The second-preimage theorem is calibrated to Yu, Wang, Zhang, and Wang (CANS 2005). The word “RIPEMD” is not used as a synonym for RIPEMD-160.

Definition 5.1 (MD4 and RIPEMD step interfaces). MD4 operates on four 32-bit words (A, B, C, D) , 512-bit message blocks parsed as little-endian words $X_0, \dots, X_{15}$, and

$$F(x, y, z) = (x \wedge y) \vee (\neg x \wedge z), \quad G(x, y, z) = (x \wedge y) \vee (x \wedge z) \vee (y \wedge z), \quad H(x, y, z) = x \oplus y \oplus z.$$

Each step replaces one register by a left rotation of its sum with one Boolean-function value, one message word, and a round constant modulo 2^{32} , while the active register cycles through A, D, C, B .

Round 1 uses F , zero constant, message order $0, 1, \dots, 15$, and rotation pattern $(3, 7, 11, 19)$ repeated. Round 2 uses G , constant `5a827999`, rotations $(3, 5, 9, 13)$ repeated, and message order

$0, 4, 8, 12, 1, 5, 9, 13, 2, 6, 10, 14, 3, 7, 11, 15.$

Round 3 uses H , constant `6ed9eba1`, rotations $(3, 9, 11, 15)$ repeated, and message order

$0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15.$

The final working words are added componentwise to the incoming chaining value. MD4 padding appends a 1 bit, zeros to $448 \bmod 512$, and the original length as a 64-bit little-endian integer.

In this volume, *RIPEMD* in Black Box 5.2 means the original 128-bit RIPEMD design analyzed in the cited paper. RIPEMD-128 and RIPEMD-160 are distinct later designs and receive no attack claim merely from that name overlap.

Theorem 5.2 (Wang–Lai–Feng–Chen–Yu attacks on MD4 and RIPEMD — verified deep input). *The EUROCRYPT 2005 Wang–Lai–Feng–Chen–Yu paper gives differential collision attacks against full MD4 and the specified RIPEMD variant using signed modular differences, sufficient bit conditions, and message modification. The reader-facing theorem in this volume records the primitive version, attack goal, published differential-path conditions, and claimed work factors from that paper; the large path-condition verification remains source-owned and is not generalized to RIPEMD-128 or RIPEMD-160.*

Why this was a methodological break. The attack does not wait for a random birthday collision. It first chooses a disturbance pattern, derives a path of state differences that should cancel, translates that path into bit conditions, then modifies message words so many early conditions hold deterministically. The remaining search therefore pays only for the residual conditions. That sequence–path, sufficient conditions, message modification, residual search—is the reusable conceptual contribution.

Theorem 5.3 (Second-preimage attack on MD4 weak messages — verified deep input). *Yu, Wang, Zhang, and Wang give a second-preimage attack for MD4 based on special weak-message/differential conditions. Their CANS 2005 analysis reports that a uniformly random message satisfies the relevant weak-message property with a small specified probability and gives message-modification procedures that can move a message toward that weak class at substantially less than exhaustive-search cost. This is a structured weak-message result; it is not a theorem that arbitrary MD4 second preimages can be found at the same cost.*

Scope audit. Collision resistance asks the attacker to choose both messages. Second-preimage resistance fixes the first message. A collision differential can therefore be very powerful without automatically yielding an arbitrary-target second-preimage attack. The CANS result succeeds by identifying messages with extra differential structure, so its quantifier order and probability of landing in the weak class are part of the theorem statement.

5.3 Reader checkpoint

1. Write the MD4 round-1 step that updates A from (A, B, C, D, X_0) .
2. Explain why little-endian parsing is part of a bit-exact cryptanalytic statement.

3. Identify the four occurrences of message word X_0 across the three MD4 rounds.
4. Verify that $F(x, y, z)$ selects y when $x = 1$ and z when $x = 0$ bitwise.
5. Compare the Boolean functions F, G, H in terms of nonlinearity and symmetry.
6. Explain why feed-forward prevents a compression collision from being described solely by equal final working registers.
7. State the versioning warning attached to the word RIPEMD in this chapter.
8. Explain why early message modification can be deterministic while later conditions remain probabilistic.
9. Contrast the quantifier order of Black Boxes 5.2 and 5.3.
10. Explain why a weak-message second-preimage theorem must report the density of weak messages.

Chapter 6

MD5: Full Collisions, Chosen Prefixes, and the Rogue-CA Lesson

MD5 added a fourth round and more irregular scheduling to MD4, yet the same differential philosophy eventually produced practical full collisions and later chosen-prefix collisions. The historical lesson is not merely that MD5 was broken: it is that collision resistance is a protocol assumption whenever a signature authenticates a digest.

6.1 How to read this chapter

Definition 6.1 is sufficient for a clean-room MD5 implementation. Black Box 6.2 records the Wang–Yu full collision method; Black Box 6.3 separates later chosen-prefix techniques and the rogue-CA case study from the original identical-prefix collision attack.

6.2 Learning goals

A reader should be able to implement MD5, distinguish identical-prefix and chosen-prefix collisions, explain why protocol parsing/prediction constraints matter, and state what the rogue-CA demonstration did and did not prove.

Literature and convention checkpoint. The primitive specification is RFC 1321. The full collision attack is calibrated to Wang and Yu, EUROCRYPT 2005. Chosen-prefix methodology is calibrated to Stevens and collaborators, with the CRYPTO 2009 rogue-CA construction used only as a protocol case study. RFC 6151 supplies later operational security context.

Definition 6.1 (MD5 compression function). MD5 parses each 512-bit block as little-endian words $M_0, \dots, M_{15}$ and maintains four 32-bit words (A, B, C, D) . Let

$$F(x, y, z) = (x \wedge y) \vee (\neg x \wedge z),$$

$$G(x, y, z) = (x \wedge z) \vee (y \wedge \neg z),$$

$$H(x, y, z) = x \oplus y \oplus z,$$

$$I(x, y, z) = y \oplus (x \vee \neg z).$$

For step $i = 0, \dots, 63$, define $K_i = \lfloor 2^{32} |\sin(i+1)| \rfloor$ and choose (Φ_i, g_i) by

$$(F, i), \quad (G, (5i+1) \bmod 16), \quad (H, (3i+5) \bmod 16), \quad (I, 7i \bmod 16)$$

in the four successive 16-step rounds. With shift amounts

$$(7, 12, 17, 22), (5, 9, 14, 20), (4, 11, 16, 23), (6, 10, 15, 21)$$

repeated four times within their rounds, one step is

$$(A, B, C, D) \leftarrow (D, B + \text{ROL}_{s_i}(A + \Phi_i(B, C, D) + M_{g_i} + K_i), B, C) \pmod{2^{32}}.$$

After 64 steps the working words are added to the incoming chaining value componentwise. Padding appends 1, then zeros to length $448 \bmod 512$, then the original bit length modulo 2^{64} in little-endian order; the digest serializes the four chaining words little-endian.

Theorem 6.2 (Wang–Yu full MD5 collision attack — verified deep input). *Wang and Yu’s EUROCRYPT 2005 method constructs collisions for full 64-step MD5 by combining a modular-difference path, sufficient bit conditions, and message modification. The source reports practical collision generation on then-current hardware and supplies explicit colliding message blocks. In this volume the complete published path and its sufficient-condition audit remain source-owned; the executable laboratory independently verifies a canonical published collision pair against the local MD5 implementation.*

Verification is not reconstruction. Checking that a published pair hashes to the same digest verifies the primitive implementation and the concrete collision. It does not by itself verify that every sufficient condition in the attack paper is necessary, that the search complexity estimate follows, or that an implementation of the path generator is correct. Those are distinct audit layers.

Theorem 6.3 (Chosen-prefix MD5 collisions and the rogue-CA case study — verified deep input). *Later chosen-prefix methods allow two externally selected prefixes P and P' to be extended by computed suffixes S, S' satisfying $\text{MD5}(P\|S) = \text{MD5}(P'\|S')$. Stevens and collaborators refined this method sufficiently to construct a rogue certification-authority certificate in the historical MD5/X.509 setting, with the published construction reporting a work factor around 2^{49} MD5 compression calls for the relevant collision generation. The case study depends additionally on certificate-format, serial-number, validity-time, and signature-workflow conditions; those operational assumptions are not part of the abstract chosen-prefix theorem.*

Protocol lesson. If a signer signs only $H(M)$ and an adversary can prepare two semantically different valid objects with the same hash, a signature issued for one may validate the other. But turning that logical observation into an actual certificate attack required precise control of both encodings and the CA issuance process. Cryptanalytic capability and protocol exploitability are related but not identical claims.

6.3 Reader checkpoint

1. Compute g_i for steps 0, 16, 32, 48 in the four MD5 rounds.
2. Explain why the constants K_i can be generated from the specification rather than stored as an independent theorem.
3. Compare MD4 and MD5 in number of rounds and Boolean functions.
4. Explain why a published collision pair is a useful regression test for an MD5 implementation.

5. Distinguish collision-pair verification from verification of an attack's complexity analysis.
6. Define identical-prefix and chosen-prefix collisions side by side.
7. Explain why chosen-prefix collisions are more relevant to structured signed objects.
8. List three non-hash conditions required by the rogue-CA demonstration.
9. Explain why the rogue-CA case study is not an attack on RSA's number-theoretic hardness.
10. State the modern design lesson about signing raw legacy hash outputs.

Chapter 7

SHA-0: Disturbance Vectors and Efficient Full Collisions

SHA-0 occupies a crucial historical position: it is close enough to SHA-1 that their comparison isolates the effect of one message-expansion rotation, yet weak enough that full collisions became practical before SHA-1. The Wang–Yu–Yin attack made disturbance vectors and local correction chains especially transparent.

7.1 How to read this chapter

Definition 7.1 gives the full SHA-0 computation and identifies the single schedule operation that SHA-1 later changes. Black Box 7.2 records the CRYPTO 2005 full attack and its published work factor; the executable lab verifies a published SHA-0 collision independently.

7.2 Learning goals

A reader should be able to implement SHA-0, derive its 80-word schedule, describe a six-step local-collision correction pattern at a conceptual level, and explain why changing the message expansion can radically alter differential propagation even when the round function is unchanged.

Literature and convention checkpoint. The attack statement is calibrated to Wang, Yu, and Yin, CRYPTO 2005, which reports full 80-step collisions below 2^{39} hash operations. The book separates that result from the earlier Joux/Biham–Chen collision line and from later SHA-1 attacks.

Definition 7.1 (SHA-0 compression function). SHA-0 parses each 512-bit block as sixteen 32-bit big-endian words $W_0, \dots, W_{15}$ and expands them by

$$W_t = W_{t-3} \oplus W_{t-8} \oplus W_{t-14} \oplus W_{t-16}, \quad 16 \leq t < 80.$$

It maintains five 32-bit words (A, B, C, D, E) . At step t ,

$$T = \text{ROL}_5(A) + f_t(B, C, D) + E + W_t + K_t \pmod{2^{32}},$$

then

$$(A, B, C, D, E) \leftarrow (T, A, \text{ROL}_{30}(B), C, D).$$

For steps 0–19, $f_t = (B \wedge C) \vee (\neg B \wedge D)$ and $K_t = 5a827999$; for 20–39 and 60–79, $f_t = B \oplus C \oplus D$ with constants `6ed9eba1` and `ca62c1d6` respectively; for 40–59, $f_t = (B \wedge C) \vee (B \wedge D) \vee (C \wedge D)$ and $K_t = 8f1bbcdc$. The five final working words are added to the incoming state. Padding is the SHA big-endian length-encoding convention.

Theorem 7.2 (Wang–Yu–Yin efficient collision attack on SHA-0 — verified deep input). *Wang, Yu, and Yin give a differential attack on full 80-step SHA-0 using low-weight disturbance vectors, local collision correction, and message modification, with reported complexity below 2^{39} SHA-0 evaluations. Their paper supplies full-collision data and a path-construction methodology. The path search and full sufficient-condition proof are source-owned; a published collision is independently verified in the Volume-XVIII executable laboratory.*

Disturbance-correction picture. A difference inserted in one schedule word perturbs several subsequent state words. Carefully chosen later schedule differences can cancel that disturbance after a short window, creating a local collision. The global attack selects a disturbance vector whose expanded pattern can be decomposed into compatible local corrections, then uses message modification to satisfy early nonlinear bit conditions. This is more structured than a generic birthday search and more constrained than choosing arbitrary differences at every round.

Why the schedule matters. SHA-0’s expansion is linear over XOR and contains no one-bit rotation of the newly formed schedule word. SHA-1 adds that rotation. The change does not make the schedule nonlinear, but it spreads bit positions and makes low-weight correction patterns harder to maintain. The next chapter shows that the same philosophy still broke SHA-1 below its ideal 2^{80} collision scale, but at vastly greater cost.

7.3 Reader checkpoint

1. Compute W_{16} and W_{17} symbolically from the sixteen SHA-0 input words.
2. State the endian convention difference between MD5 and SHA-0.
3. Write the state update from (A, B, C, D, E) to the next five words.
4. Identify the two XOR-function round ranges in SHA-0.
5. Explain why the message expansion is linear over $\mathbb{F}_2$.
6. Define a disturbance vector independently of any particular attack paper.
7. Explain the role of a local collision in building a full differential path.
8. Compare the 2^{39} reported attack scale with the ideal 160-bit birthday scale.
9. Explain why collision verification and disturbance-path reconstruction are separate tasks.
10. Predict qualitatively what a one-bit rotation in the schedule expansion does to a sparse bit disturbance.

Chapter 8

SHA-1: From the Wang Path to SHAttered and Chosen Prefixes

SHA-1 survived much longer than MD5, but its cryptanalytic history provides the clearest case study of theory becoming practice. A 2005 differential method pushed full SHA-1 below the birthday bound; in 2017 the first public full collision was computed; by 2020 chosen-prefix collisions were practical. These are distinct milestones and must not be collapsed into one claim.

8.1 How to read this chapter

Definition 8.1 isolates SHA-1’s single compression-level change from SHA-0. Black Boxes 8.2–8.4 separate the theoretical, practical identical-prefix, and practical chosen-prefix milestones. Figure 8.1 summarizes the disturbance-correction genealogy without implying that one paper contains all later techniques.

8.2 Learning goals

A reader should be able to implement SHA-1, explain the role of its rotated message expansion, distinguish predicted attack complexity from an actually computed collision, and understand why chosen-prefix control raises the protocol risk beyond identical-prefix collisions.

Literature and convention checkpoint. The theoretical attack is Wang–Yin–Yu, CRYPTO 2005. The first public full collision is Stevens, Bursztein, Karpman, Albertini, and Markov (2017). The first practical chosen-prefix collision is Leurent–Peyrin (USENIX Security 2020). The book preserves that chronology and does not attribute SHAttered’s concrete collision pair to the 2005 paper.

Definition 8.1 (SHA-1 compression function). SHA-1 uses the SHA-0 state update, round functions, constants, padding, and five-word feed-forward of Definition 7.1. Its message expansion differs for $16 \leq t < 80$ by

$$W_t = \text{ROL}_1(W_{t-3} \oplus W_{t-8} \oplus W_{t-14} \oplus W_{t-16}).$$

Thus SHA-0 and SHA-1 have the same 160-bit digest size and 80-step nonlinear state recurrence but different linear message-expansion operators. In this chapter a *full SHA-1 collision* means all 80 steps with the standard IV, padding, and digest; a freestart or reduced-step collision must be labelled as such.

Theorem 8.2 (Wang–Yin–Yu theoretical collision attack on full SHA-1 — verified deep input). *The CRYPTO 2005 Wang–Yin–Yu paper gives a differential collision-search method for full 80-step SHA-1 with a claimed complexity below 2^{69} hash operations, beating the 2^{80} ideal birthday scale. It develops low-weight disturbance vectors, adjusted first-round paths, and local-collision corrections. The statement is a theoretical attack method and complexity estimate; it is not a claim that the paper published the later SHattered collision pair.*

Theorem 8.3 (SHattered: first practical full SHA-1 collision — verified deep input). *Stevens, Bursztein, Karpman, Albertini, and Markov reported the first publicly computed full SHA-1 collision in 2017, using a PDF-format identical-prefix construction and a large CPU/GPU computation. The two published files have the same SHA-1 digest while differing as byte strings. This establishes practical full collision generation; it does not by itself provide chosen-prefix control.*

Theorem 8.4 (Practical chosen-prefix collision for SHA-1 — verified deep input). *Leurent and Peyrin give the first practical chosen-prefix collision for SHA-1 and demonstrate it in a PGP/Web-of-Trust setting. Their 2020 report gives GPU-normalized complexity estimates around $2^{63.4}$ SHA-1 equivalents for the chosen-prefix attack and documents the concrete computational campaign. The theorem is about chosen-prefix collision capability and its demonstrated protocol consequence, not a universal signature forgery theorem.*

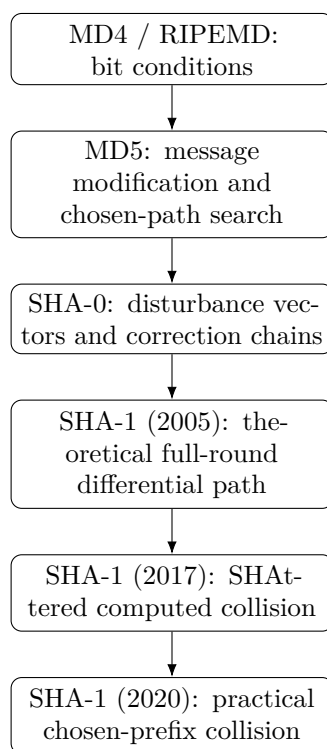


Figure 8.1: MD4-to-SHA-1 disturbance-correction genealogy. Arrows denote methodological inheritance, not theorem implication.

Chronology is part of correctness. A theoretical path can show that the ideal security exponent is wrong years before engineering makes a collision practical. A practical identical-prefix collision

can in turn precede a chosen-prefix capability. Recording those steps separately prevents hindsight from turning a progression of cryptanalytic achievements into one undifferentiated “SHA-1 was broken” statement.

8.3 Reader checkpoint

1. Write the single formula that distinguishes SHA-1’s schedule from SHA-0’s.
2. Explain why that schedule change preserves linearity over XOR.
3. Compare the ideal 160-bit collision exponent with the 2005 claimed SHA-1 attack exponent.
4. State why Black Box 8.2 is not the SHAttered result.
5. Distinguish identical-prefix and chosen-prefix collision control.
6. Explain why chosen-prefix control matters for structured signed objects.
7. Interpret every arrow in Figure 8.1 as methodological rather than logical implication.
8. Explain why a practical collision need not imply a practical preimage attack.
9. State one protocol assumption needed to turn a hash collision into a signature-level attack.
10. Give a three-stage chronology of SHA-1 cryptanalysis using 2005, 2017, and 2020.

Chapter 9

SM3, SHA-3, and the Modern Hash-Design Boundary

The Wang-era attacks transformed hash design. Modern standards do not simply enlarge digests: they alter message expansion, state width, domain separation, and in SHA-3’s case the entire sponge architecture. SM3 remains an iterated 256-bit design with a SHA-2-like state structure, while SHA-3 is permutation/sponge based. Their security records must therefore be stated by primitive, round count, and attack goal.

9.1 How to read this chapter

Definition 9.1 gives a bit-exact SM3 compression interface and recalls the distinct SHA-3 sponge interface. Theorem 9.2 is a standards/specification record, Black Box 9.3 freezes representative reduced-round SM3 cryptanalysis, Frontier 9.4 dates NIST’s SHA-3 revision process, and the executable lab verifies all five legacy/SM3 implementations against independent vectors and collision fixtures.

9.2 Learning goals

A reader should be able to implement SM3, distinguish full-primitive security from reduced-round cryptanalysis, compare Merkle–Damgård and sponge design interfaces, and interpret a standards revision without inferring a cryptanalytic break.

Literature and convention checkpoint. SM3 status is calibrated to GB/T 32905-2016 and ISO/IEC 10118-3:2018. Representative reduced-round records use Wang–Shen preimage/pseudo-collision results and Kircanski–Shen–Wang–Youssef/Bai et al. boomerang analyses. SHA-3 revision status is taken from NIST’s official 2025 decision to update FIPS 202 and revise SP 800-185; as of the 2026-09-03 volume freeze, that editorial/streaming revision process is not recorded as a break of full SHA-3.

Definition 9.1 (SM3 and modern hash-design interfaces). SM3 maps 512-bit blocks to a 256-bit chaining state of eight 32-bit words. Parse a block as big-endian $W_0, \dots, W_{15}$ and for $16 \leq j < 68$ set

$$W_j = P_1(W_{j-16} \oplus W_{j-9} \oplus \text{ROL}_{15}(W_{j-3})) \oplus \text{ROL}_7(W_{j-13}) \oplus W_{j-6},$$

with $W'_j = W_j \oplus W_{j+4}$ for $0 \leq j < 64$, where

$$P_0(x) = x \oplus \text{ROL}_9(x) \oplus \text{ROL}_{17}(x), \quad P_1(x) = x \oplus \text{ROL}_{15}(x) \oplus \text{ROL}_{23}(x).$$

For $j < 16$, $FF_j(x, y, z) = GG_j(x, y, z) = x \oplus y \oplus z$; for $j \geq 16$,

$$FF_j = (x \wedge y) \vee (x \wedge z) \vee (y \wedge z), \quad GG_j = (x \wedge y) \vee (\neg x \wedge z).$$

Let $T_j = 79\text{cc}4519$ for $j < 16$ and $7\text{a}879\text{d}8\text{a}$ otherwise. With working state (A, B, C, D, E, F, G, H) ,

$$\begin{aligned} SS_1 &= \text{ROL}_7(\text{ROL}_{12}(A) + E + \text{ROL}_j(T_j)), \\ SS_2 &= SS_1 \oplus \text{ROL}_{12}(A), \\ TT_1 &= FF_j(A, B, C) + D + SS_2 + W'_j, \\ TT_2 &= GG_j(E, F, G) + H + SS_1 + W_j, \end{aligned} \quad (\text{mod } 2^{32}),$$

then

$$(A, B, C, D, E, F, G, H) \leftarrow (TT_1, A, \text{ROL}_9(B), C, P_0(TT_2), E, \text{ROL}_{19}(F), G).$$

After 64 steps the working state is XORed with the incoming state. Padding appends 1, zeros to $448 \bmod 512$, and the original length as a 64-bit big-endian integer.

By contrast, SHA-3/SHAKE use the Keccak-f permutation inside the sponge interface of Definition 2.1. A claim about SM3 iteration therefore does not automatically transfer to SHA-3, and vice versa.

Theorem 9.2 (SM3 specification and standard-status record — verified official input). *The bit-level algorithm of Definition 9.1 is the SM3 hash algorithm standardized in GB/T 32905-2016, effective in China from 2017-03-01, and included in the published ISO/IEC 10118-3:2018 dedicated hash-function standard, which was confirmed in 2024. Standards adoption is a specification/interoperability fact; it is not a theorem that full SM3 has ideal collision or preimage security.*

Theorem 9.3 (Representative reduced-round SM3 cryptanalysis — verified deep input). *Published cryptanalysis breaks reduced versions of SM3 without establishing a full-round collision or preimage break. Representative records include preimage attacks on 29/30-step SM3 and pseudo-preimage/pseudo-collision attacks through 31/32 steps in the Wang–Shen line, together with boomerang distinguishers on reduced compression functions extending into the mid-30-step range. Every such record is indexed by attack goal, target (hash or compression function), round/step count, and work factor; none is rewritten as an attack on full 64-step SM3.*

Remark 9.4 (SHA-3 and derived-function revision status — dated frontier, 2026-09-03). NIST announced on 2025-03-12 that it would update FIPS 202 and revise SP 800-185 after public review. The FIPS 202 update was described primarily as improving editorial quality and updating references to withdrawn techniques; the SP 800-185 revision was planned to add streaming specifications for SHAKE128/SHAKE256 and consider other technical comments. At the 2026-09-03 freeze of this volume, the project records this as an active standards-revision process, not as evidence of a cryptanalytic break of the full SHA-3 permutation/sponge designs.

Algorithm 9.5 (Bit-exact hash and collision-verification laboratory). *The executable source code `volume_18/hash_collision_lab.py` implements MD4, MD5, SHA-0, SHA-1, and SM3 from the reader-facing definitions. It checks independent known-answer vectors; verifies the canonical Wang-era 128-byte MD5 collision pair and the published 2048-bit SHA-0 collision pair; confirms that the colliding byte strings are distinct; and emits compression-level difference checkpoints. SHA-1/SHattered verification is represented by official digest metadata rather than bundling copyrighted PDF fixtures into the canonical source archive.*

9.3 Reader checkpoint

1. Compute W'_{17} from the expanded SM3 words.
2. Explain the different roles of P_0 and P_1 in SM3.
3. State the two round ranges in which SM3 changes its Boolean functions and constants.
4. Compare SM3 feed-forward with SHA-1 feed-forward.
5. Explain why a 32-step preimage result is not a preimage attack on full 64-step SM3.
6. Distinguish a compression-function distinguisher from a hash-function collision.
7. State the Chinese and ISO standards status recorded in Theorem 9.2.
8. Explain why SHA-3's sponge capacity has no direct analogue in the 256-bit SM3 chaining state.
9. Summarize the 2025 NIST SHA-3 revision decision without calling it a security break.
10. Explain why self-contained test vectors are preferable to bundling third-party SHattered PDF files into the source tree.

Chapter 10

MAC Cryptanalysis, Collision Composition, and Automated Path Search

A collision attack becomes cryptographically consequential only through a surrounding construction. Conversely, a secure construction can remain sound even when its underlying hash has lost ordinary collision resistance. This terminal chapter separates the HMAC/NMAC-MD5 cryptanalysis record from naive “collision implies MAC break” claims, proves an abstract chosen-prefix composition lemma, and dates the increasingly automated search for differential paths.

10.1 How to read this chapter

Black Box 10.1 records the exact HMAC/NMAC-MD5 and MD5-MAC cryptanalytic scope. Theorem 10.2 is a local composition theorem explaining when chosen-prefix collisions transfer through a parser/signature layer. Frontier 10.3 separates formally verified path constraints from heuristic optimizer performance.

10.2 Learning goals

A reader should be able to explain why an MD5 collision is not automatically an HMAC-MD5 forgery, state an abstract signature-transfer condition for chosen-prefix collisions, distinguish parser validity from hash equality, and audit SAT/MILP/SMT-assisted differential search without confusing solver soundness with cryptanalytic completeness.

Literature and convention checkpoint. HMAC/NMAC-MD5 scope is calibrated to Wang, Yu, Wang, Zhang, and Zhan (EUROCRYPT 2009) and the later RFC 6151 security summary. The chosen-prefix composition theorem is local and protocol-agnostic. Automated-search status is calibrated to the HashClash/Stevens lineage and later constraint-based differential-search literature, but remains a dated frontier because tool capabilities and primitive coverage continue to change.

Theorem 10.1 (Cryptanalysis of HMAC/NMAC-MD5 and MD5-MAC — verified deep input). *The EUROCRYPT 2009 Wang–Yu–Wang–Zhang–Zhan paper gives cryptanalytic results against constructions based on MD5, including a distinguishing attack on HMAC/NMAC-MD5 without related keys and attacks on MD5-MAC variants, under the paper’s explicit chosen-message/data/time models.*

In particular, the widely cited HMAC-MD5 distinguisher is not a claim of a practical existential forgery from an ordinary MD5 collision. Construction, keying model, oracle access, and attack goal are part of every result in this row.

Why HMAC survives the naive length-extension argument. HMAC computes an outer keyed hash of an inner keyed digest. The public tag is therefore not the raw chaining state of $H(K\|M)$, and the continuation attack of Proposition 2.3 does not expose the internal state needed to append blocks. This is a structural statement, not a claim that every use of HMAC-MD5 should be considered modern or recommended.

Theorem 10.2 (Chosen-prefix collision composition lemma). *Let H be a hash, let $\mathcal{P}$ be a deterministic parser, and let $\text{Verify}(pk, M, \sigma)$ accept exactly when a signature verification algorithm accepts σ on digest $H(M)$ together with any fixed protocol/domain-separation metadata. Suppose two byte-string builders produce valid, semantically distinct objects*

$$M_0 = P_0\|S_0, \quad M_1 = P_1\|S_1,$$

where P_0, P_1 satisfy all externally fixed prefix constraints, S_0, S_1 satisfy all parser/length/checksum fields not authenticated independently of H , and $H(M_0) = H(M_1)$. If the signed verification input is identical for the two objects except through $H(M_i)$, then every signature σ accepted for M_0 is also accepted for M_1 .

The conclusion can fail in any of the following cases: parsing changes the signed byte representation; a length, checksum, or identity field is authenticated outside H ; randomized or domain-separated hashing changes the two digest inputs; or the signer checks semantic constraints before issuing the signature.

Proof. By hypothesis the parser accepts both byte strings and the verification algorithm receives the same fixed metadata. The only message-dependent verification input is therefore $H(M_i)$, and these digests are equal. Hence the verifier executes on identical cryptographic inputs and has the same acceptance result for both messages. Each listed failure mode violates one of those hypotheses, so it is not covered by the implication. $\square$

Remark 10.3 (Automated differential-path search — dated frontier, 2026-09-03). Modern differential cryptanalysis increasingly encodes bit conditions, modular-addition carries, message schedules, and path constraints into SAT, SMT, MILP, constraint-programming, or specialized search engines. A correct encoding can give machine-checkable soundness for any returned path; optimizer heuristics, symmetry breaking, neutral-bit ranking, and search completeness are separate questions. The Volume-XVIII frontier therefore records, for each tool and primitive version, (i) the exact constraint semantics, (ii) whether every valid path is representable, (iii) whether UNSAT is a proof of nonexistence in the encoded model, and (iv) which performance claims are empirical. The neighboring Volume XIX registry owns the general automated-cryptanalysis methodology.

Volume closure. Hash cryptanalysis now has three layers that must be kept distinct. Generic combinatorics explains birthday, multicollision, and long-message phenomena even for idealized compression. Differential cryptanalysis exploits the specific Boolean/ARX/message-schedule structure of MD4-family designs. Protocol cryptanalysis asks whether the resulting capability crosses a parser, MAC, signature, certificate, or application boundary. A rigorous security claim names the layer at which it lives.

10.3 Reader checkpoint

1. Explain why an MD5 collision is not automatically an HMAC-MD5 forgery.
2. State the message-dependent input to the verifier in Theorem 10.2.
3. Give one parser condition that could invalidate a chosen-prefix composition attempt.
4. Give one independently authenticated metadata field that could invalidate the theorem's hypothesis.
5. Explain why randomized hashing can defeat direct signature transfer even when raw MD5 collisions exist.
6. Distinguish a distinguisher from a forgery attack on a MAC.
7. Explain what SAT-solver UNSAT proves when the cryptanalytic encoding is sound but incomplete.
8. Explain what SAT-solver SAT proves when the encoding is sound.
9. Name the three layers of hash cryptanalysis summarized in the volume closure.
10. Give an example of a statement that would improperly conflate generic, differential, and protocol cryptanalysis.

Internalization and Reconstruction Guide: Volume XVIII

Purpose. Read hash cryptanalysis at three layers: generic query bounds, structural properties of domain extension, and concrete differential attacks; a practical collision result never automatically implies a generic break of every derived construction.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-XVI-DEF-023: Authentication and signature security.
2. QRCC-XV-ALG-014: Grover search and amplitude amplification.
3. QRCC-XV-THM-015: Quantum search lower bound.
4. QRCC-XVI-THM-003: Game-hopping and reduction accounting lemma.
5. QRCC-I-DEF-005: Total variation and statistical tests.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XVIII-DEF-001 (Hash-function security notions); QRCC-XVIII-DEF-004 (Merkle–Damgard, Davies–Meyer, and sponge constructions); QRCC-XVIII-DEF-011 (Differential paths and bit conditions); QRCC-XVIII-DEF-014 (MD4 and RIPEMD step functions); QRCC-XVIII-DEF-017 (MD5 compression function).

Chapter retrieval map

- Ch. 1: Hash Functions: Security Notions and Generic Bounds:** enter through QRCC-XVIII-DEF-001 (Hash-function security notions); exit by being able to state QRCC-XVIII-THM-002 (Generic collision and preimage query bounds) with its ownership boundary.
- Ch. 2: Iterated Hashing, Length Extension, and Sponge Security:** enter through QRCC-XVIII-DEF-004 (Merkle–Damgard, Davies–Meyer, and sponge constructions); exit

by being able to state QRCC-XVIII-BB-007 (Sponge indistinguishability theorem) with its ownership boundary.

- Ch. 3: Birthday, Multicollision, and Long-Message Second-Preimage Attacks:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XVIII-BB-010 (Kelsey–Schneier second-preimage attack on long messages) with its ownership boundary.
- Ch. 4: Differential Paths, Bit Conditions, and Message Modification:** enter through QRCC-XVIII-DEF-011 (Differential paths and bit conditions); exit by being able to state QRCC-XVIII-ALG-013 (Wang-style message modification framework) with its ownership boundary.
- Ch. 5: MD4, RIPEMD, and the First Wang Collision Paths:** enter through QRCC-XVIII-DEF-014 (MD4 and RIPEMD step functions); exit by being able to state QRCC-XVIII-BB-015 (Wang–Lai–Feng–Chen–Yu attacks on MD4 and RIPEMD) with its ownership boundary.
- Ch. 6: MD5: Full Collisions, Chosen Prefixes, and the Rogue-CA Lesson:** enter through QRCC-XVIII-DEF-017 (MD5 compression function); exit by being able to state QRCC-XVIII-BB-018 (Wang–Yu full MD5 collision attack) with its ownership boundary.
- Ch. 7: SHA-0: Disturbance Vectors and Efficient Full Collisions:** enter through QRCC-XVIII-DEF-020 (SHA-0 compression function); exit by being able to state QRCC-XVIII-BB-021 (Wang–Yu–Yin efficient collision attack on SHA-0) with its ownership boundary.
- Ch. 8: SHA-1: From the Wang Path to SHattered and Chosen Prefixes:** enter through QRCC-XVIII-DEF-022 (SHA-1 compression function); exit by being able to state QRCC-XVIII-BB-023 (Wang–Yin–Yu theoretical collision attack on full SHA-1) with its ownership boundary.
- Ch. 9: SM3, SHA-3, and the Modern Hash-Design Boundary:** enter through QRCC-XVIII-DEF-026 (SM3 and modern hash-design interfaces); exit by being able to state QRCC-XVIII-BB-028 (Differential and structural cryptanalysis of reduced SM3) with its ownership boundary.
- Ch. 10: MAC Cryptanalysis, Collision Composition, and Automated Path Search:** enter through the inherited definitions listed at the start of the chapter; exit by being able to state QRCC-XVIII-BB-030 (Cryptanalysis of HMAC/NMAC-MD5 and MD5-MAC) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XVIII-THM-002 — Generic collision and preimage query bounds. Dependencies: QRCC-XVIII-DEF-001, QRCC-XV-ALG-014, QRCC-XV-THM-015. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

2. QRCC-XVIII-BB-010 — Kelsey–Schneier second-preimage attack on long messages. Dependencies: QRCC-XVIII-DEF-004, QRCC-XVIII-THM-009. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-XVIII-BB-015 — Wang–Lai–Feng–Chen–Yu attacks on MD4 and RIPEMD. Dependencies: QRCC-XVIII-ALG-013, QRCC-XVIII-DEF-014. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XVIII-BB-021 — Wang–Yu–Yin efficient collision attack on SHA-0. Dependencies: QRCC-XVIII-ALG-013, QRCC-XVIII-DEF-020. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XVIII-BB-028 — Differential and structural cryptanalysis of reduced SM3. Dependencies: QRCC-XVIII-DEF-026, QRCC-XVIII-THM-012. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XVIII-F-029 — SHA-3 and derived-function revision status. Dependencies: QRCC-XVIII-DEF-026. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- equating collision resistance with preimage or second-preimage resistance;
- assuming Merkle–Damgård domain extension preserves every security notion;
- multiplying differential step probabilities as independent without a justified model;
- confusing reduced-round/theoretical attacks with practical full-standard breaks;

Boundary and degenerate cases

- one-block versus very long messages;
- identical-prefix versus chosen-prefix collisions;
- full versus reduced rounds;
- the difference between a standard being deprecated and a primitive being mathematically broken;

Retrieval and reconstruction drill

1. State QRCC-XVIII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XVIII-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XVIII-BB-010, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?

4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XVIII-F-029, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XIX generalizes the attack viewpoint to block ciphers, permutations, automated trail search, faults, and quantum-assisted attacks.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XIX

Symmetric Cryptanalysis and Automated Attacks

Chapter 1

Block-Cipher Structure, S-box Spectra, and Feistel Networks

Symmetric cryptanalysis begins with a precise state model. Before probabilities, biases, trails, or solver objectives can be trusted, the reader must know what the state variables are, where the key enters, and which transformations are permutations. This chapter establishes the algebraic bookkeeping used throughout the volume and proves the elementary S-box identities that later automated attacks must encode exactly.

1.1 How to read this chapter

Definition 1.1 fixes the block-cipher, SPN, Feistel, round-key, and S-box conventions. Theorem 1.2 proves that Feistel invertibility does not require an invertible round function. Proposition 1.3 then collects exact DDT, Walsh/LAT, and algebraic-degree identities.

1.2 Learning goals

A reader should be able to write a round function with all key additions visible, invert an arbitrary Feistel round, compute a difference distribution table and Walsh spectrum, convert Walsh coefficients to linear correlations and biases, and explain which affine transformations preserve S-box differential uniformity and algebraic degree.

Literature checkpoint. The conventions are calibrated to the standard differential/linear cryptanalysis literature and to Daemen–Rijmen style S-box analysis. The results below are elementary and proved locally; no concrete-cipher security claim is inferred from a good DDT or LAT alone.

Definition 1.1 (Block ciphers and design structures). A *block cipher* is a family of permutations $E_K : \{0, 1\}^n \rightarrow \{0, 1\}^n$ indexed by a key K . An r -round iterated cipher writes

$$X_i = R_i(X_{i-1}, K_i), \quad 1 \leq i \leq r,$$

where the *round keys* K_i are derived from the master key by a specified key schedule. A *substitution-permutation network* (SPN) alternates parallel nonlinear S-boxes with linear/affine diffusion and key addition. An $m \times n$ *S-box* is a map $S : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$. A *Feistel round* on two m -bit branches has the form

$$\Phi_{F,k}(L, R) = (R, L \oplus F(R, k)),$$

with no invertibility assumption on F . XOR difference is $x \oplus x'$. For masks a, b , the dot products $a \cdot x$ and $b \cdot S(x)$ are taken over $\mathbb{F}_2$.

Theorem 1.2 (Feistel invertibility and round-function independence). *For every function $F : \{0, 1\}^m \times \mathcal{K} \rightarrow \{0, 1\}^m$ and every fixed round key k , the Feistel map $\Phi_{F,k}$ is a permutation with inverse*

$$\Phi_{F,k}^{-1}(L', R') = (R' \oplus F(L', k), L').$$

Consequently any composition of Feistel rounds is invertible even when every round function is many-to-one.

Proof. If $(L', R') = (R, L \oplus F(R, k))$, then $R = L'$ and $L = R' \oplus F(L', k)$. Substitution gives the displayed inverse. A composition of bijections is a bijection. $\square$

Proposition 1.3 (S-box DDT, Walsh/LAT, and algebraic-degree identities). *Let $S : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$. Define*

$$\text{DDT}_S(a, b) = |\{x : S(x) \oplus S(x \oplus a) = b\}|,$$

and

$$W_S(u, v) = \sum_{x \in \mathbb{F}_2^m} (-1)^{u \cdot x \oplus v \cdot S(x)}, \quad C_S(u, v) = 2^{-m} W_S(u, v).$$

Then:

- (i) $\sum_b \text{DDT}_S(a, b) = 2^m$ for every a ; for $a \neq 0$, every DDT entry is even.
- (ii) If S is a permutation, $\text{DDT}_S(a, 0) = 0$ for $a \neq 0$ and $W_S(0, v) = 0$ for $v \neq 0$.
- (iii) For every fixed v , Parseval gives $\sum_u W_S(u, v)^2 = 2^{2m}$.
- (iv) Precomposition by an invertible affine map permutes DDT input/output labels and Walsh input masks; postcomposition by an invertible affine output map permutes output differences and output masks. Thus differential uniformity and the maximum absolute nontrivial correlation are invariant under affine equivalence.
- (v) If $m = n$ and A, B are invertible affine maps, the vectorial algebraic degree of $B \circ S \circ A$ equals that of S .

Proof. For (i), every x contributes to exactly one output difference. If $a \neq 0$, x and $x \oplus a$ are distinct and produce the same derivative value, so solutions occur in pairs. For a permutation, $S(x) = S(x \oplus a)$ implies $a = 0$, proving the first claim in (ii); the second is balance of each nonzero linear combination $v \cdot S(x)$. For (iii), apply Walsh Parseval to the Boolean function $x \mapsto v \cdot S(x)$. Invertible affine changes of variables are bijections of the input or output space, so they only relabel differences/masks and can introduce at most a sign in Walsh coefficients. Finally, invertible affine substitutions preserve total degree, while an invertible linear recombination of output coordinates cannot lower the maximum coordinate degree in both directions; applying the same argument to the inverse affine equivalence proves equality. $\square$

Worked micro-example. For the permutation $S = (0, 2, 3, 1)$ on two bits, the nonzero input difference 01₂ produces only even DDT counts. Computing all four inputs gives output differences 10, 01, 01, 10, hence counts two and two. This tiny example is the same invariant later encoded by the automated trail laboratory.

1.3 Reader checkpoint

1. Invert two Feistel rounds symbolically without assuming anything about the round functions.
2. Prove directly that $\text{DDT}_S(a, b)$ is even for $a \neq 0$.
3. Compute the full DDT of the two-bit S-box in the worked example.
4. Compute $W_S(u, v)$ for all masks of that S-box.
5. Convert correlation $C = 1/4$ to a linear-approximation success probability and bias.
6. Show that XORing a constant to every S-box output does not change DDT counts.
7. Show that an invertible input linear map preserves differential uniformity.
8. Give a noninvertible output map that can reduce vectorial algebraic degree.
9. Explain why a low maximum DDT entry is not by itself a proof of full-cipher resistance.
10. Identify the exact theorem property an automated S-box model must preserve to certify trail weights.

Chapter 2

Differential Cryptanalysis and Markov-Cipher Bookkeeping

Differential cryptanalysis studies how input differences propagate through a keyed permutation. The central mathematical issue is conditional probability: multiplying local DDT entries is justified only under hypotheses that make each fresh round input appropriately distributed. This chapter states those hypotheses explicitly and separates an elementary characteristic-probability theorem from the source-owned full-DES attack.

2.1 How to read this chapter

Definition 2.1 distinguishes a characteristic from a differential. Theorem 2.2 proves the product rule in the independent-round-key Markov model. Black Box 2.3 records the Biham–Shamir full-DES result without replacing its key-recovery statistics by a toy argument.

2.2 Learning goals

A reader should be able to convert DDT entries to one-round transition probabilities, distinguish a single trail from a differential hull, prove when trail probabilities multiply, and audit the data/time/key-ranking quantities in a concrete differential attack.

Definition 2.1 (Differential characteristics and differentials). For two encryptions under the same key write $\Delta X_i = X_i \oplus X'_i$. A *differential characteristic* (or trail) is a prescribed sequence $(\Delta X_0, \dots, \Delta X_r)$. Its probability is the probability that all intermediate differences equal the prescribed values. A *differential* fixes only selected endpoint differences and sums the probabilities of all compatible internal trails, including their dependence. A *right pair* obeys the selected trail or differential event; a *wrong pair* does not. Data complexity counts queried/chosen plaintexts or pairs, time complexity counts the stated primitive operations, and success probability is separate from both.

Theorem 2.2 (Characteristic probability with independent round keys). *Consider r rounds*

$$X_i = G_i(X_{i-1} \oplus K_i),$$

where $K_1, \dots, K_r$ are mutually independent uniform n -bit round keys and independent of the chosen plaintext pair. For differences $\delta_0, \dots, \delta_r$, define

$$p_i(\delta_{i-1}, \delta_i) = 2^{-n} |\{x : G_i(x) \oplus G_i(x \oplus \delta_{i-1}) = \delta_i\}|.$$

Then

$$\Pr[\Delta X_1 = \delta_1, \dots, \Delta X_r = \delta_r \mid \Delta X_0 = \delta_0] = \prod_{i=1}^r p_i(\delta_{i-1}, \delta_i).$$

The theorem does not assert the same factorization for a deterministic correlated key schedule.

Proof. Condition on the complete history through round $i - 1$. The value X_{i-1} may then be arbitrary, but $X_{i-1} \oplus K_i$ is uniform because K_i is fresh uniform and independent of that history. The paired round input is displaced from it by exactly δ_{i-1} . Hence the conditional probability of obtaining δ_i is $p_i(\delta_{i-1}, \delta_i)$, independent of earlier events. The chain rule multiplies these conditional probabilities. A real key schedule can correlate round keys with the history, so the conditioning argument need not survive unchanged. $\square$

Theorem 2.3 (Biham–Shamir differential cryptanalysis of full DES — verified deep input). *Biham and Shamir developed differential cryptanalysis into a key-recovery attack on the full 16-round DES faster than exhaustive key search, using a large chosen-plaintext structure, a high-probability differential architecture through most rounds, filtering, and statistical ranking of last-round subkey candidates. Their CRYPTO’92/full monograph analysis reports about 2^{47} chosen plaintexts, about 2^{36} usable ciphertext pairs after filtering, and roughly 2^{37} data-analysis work in the stated model. This concrete DES result is source-owned: S-box tables, DES expansion and permutation structure, key-schedule effects, filtering, and signal-to-noise analysis are not replaced by the independent-round-key theorem above.*

Audit warning. A product of local DDT probabilities is a trail calculation, not automatically the probability of the endpoint differential. Distinct trails can contribute to the same endpoints, and a deterministic key schedule can alter the independence assumptions used in Theorem 2.2.

2.3 Reader checkpoint

1. Convert a DDT entry 6 for a 4-bit S-box to a one-round transition probability.
2. Give two distinct internal trails with the same endpoint differential.
3. Prove the chain-rule factorization in Theorem 2.2 for two rounds explicitly.
4. Identify where fresh round-key independence enters that proof.
5. Give a correlated round-key example for which the argument cannot be reused verbatim.
6. Explain the difference between a right pair and a right key guess.
7. Why can filtering reduce analysis time without reducing initial data complexity?
8. Separate the DES attack’s chosen-plaintext, usable-pair, and time figures.
9. Explain why the theorem does not prove security of DES against differential cryptanalysis.
10. State what an automated differential tool must report besides its best trail probability.

Chapter 3

Linear Cryptanalysis, Piling Up, and Hulls

Linear cryptanalysis replaces differences by correlations between parity masks. Its elementary core is a Fourier identity: the correlation of an XOR of independent Boolean variables is the product of their signed correlations. The difficult part of a real attack is not that lemma but choosing approximations, accounting for hulls and key masks, collecting enough known plaintexts, and ranking candidate keys.

3.1 How to read this chapter

Definition 3.1 fixes masks, correlation, bias, trails, and hulls. Theorem 3.2 proves the piling-up lemma. Black Box 3.3 records Matsui's DES attack. Example 3.4 combines differential and linear calculations on one toy SPN.

3.2 Learning goals

A reader should be able to translate between bias and signed correlation, prove the piling-up lemma from character multiplicativity, distinguish a single linear trail from a linear hull, and explain which statistical and key-schedule assumptions separate Matsui's concrete DES attack from the elementary correlation identity.

Definition 3.1 (Linear approximations and hulls). For Boolean Z , define its signed correlation $\text{cor}(Z) = \mathbb{E}[(-1)^Z]$ and bias $\epsilon(Z) = \Pr[Z = 0] - 1/2 = \text{cor}(Z)/2$. For a keyed mapping, a *linear approximation* compares an input parity $u \cdot X$, an output parity $v \cdot Y$, and possibly a key parity. A *linear trail* specifies intermediate masks. A *linear hull* fixes endpoints and combines all compatible trails with their signs and key dependence; absolute trail correlations must not simply be added.

Theorem 3.2 (Piling-up lemma). *Let $Z_1, \dots, Z_t$ be independent Boolean random variables with biases ϵ_i . Then*

$$\text{cor}(Z_1 \oplus \dots \oplus Z_t) = \prod_{i=1}^t \text{cor}(Z_i), \quad \epsilon(Z_1 \oplus \dots \oplus Z_t) = 2^{t-1} \prod_{i=1}^t \epsilon_i.$$

If the Z_i are not independent, the formula need not hold.

Proof. Because $(-1)^{z_1 \oplus \dots \oplus z_t} = \prod_i (-1)^{z_i}$, independence gives

$$\mathbb{E}[(-1)^{\oplus_i Z_i}] = \prod_i \mathbb{E}[(-1)^{Z_i}].$$

Since each signed correlation equals twice the bias, dividing the first identity by two yields the second. $\square$

Theorem 3.3 (Matsui linear cryptanalysis of DES — verified deep input). *Matsui’s EURO-CRYPT’93 method uses biased linear relations across DES, partial encryption/decryption under candidate subkeys, and statistical ranking to obtain a known-plaintext attack. The original paper reports an attack on full 16-round DES requiring about 2^{47} known plaintexts in its stated model; the subsequent CRYPTO’94 experimental work validated linear cryptanalysis on DES-scale implementations. The exact DES masks, key-bit relations, Algorithm 1/2 ranking logic, and empirical success calculations remain source-owned.*

Example 3.4 (Full differential and linear analysis of a toy SPN). Take the 4-bit permutation

$$S = [0, E, 4, D, 7, 1, A, F, C, 3, 9, 8, 5, B, 6, 2]$$

and a two-nibble SPN whose diffusion swaps the two output nibbles before the next S-box layer. A complete analysis proceeds in four reproducible steps: (1) enumerate the DDT and record nonzero transitions; (2) enumerate the Walsh/LAT and record signed correlations; (3) search two-round differential trails by multiplying exact transition probabilities; (4) search two-round linear trails by multiplying signed correlations. The executable laboratory in Chapter 9 uses the same S-box and independently verifies the best trail certificate. The point is methodological: a tiny cipher can be exhaustively checked, so probability arithmetic and solver encodings can be tested before they are trusted on a large primitive.

3.3 Reader checkpoint

1. Prove the piling-up lemma for two variables by a four-row probability table.
2. Translate biases $1/8$ and $-1/4$ into signed correlations and combine them.
3. Give dependent bits for which piling up fails.
4. Explain why a linear hull can exhibit cancellation.
5. Compute one nonzero Walsh coefficient of the toy S-box.
6. Explain why Matsui’s attack is known-plaintext rather than chosen-plaintext in its basic form.
7. Distinguish an endpoint approximation from a specified linear trail.
8. State the role of partial decryption in last-round key ranking.
9. Why must a solver preserve correlation signs, not only absolute weights, when reasoning about hulls?
10. Compare the statistical roles of a differential right-pair count and a linear parity count.

Chapter 4

Differential-Linear, Boomerang, and Rectangle Cryptanalysis

Two long weak effects can sometimes be joined more effectively than one searches for a single end-to-end trail. Differential-linear attacks join a differential prefix to a linear suffix; boomerang and rectangle attacks join two differentials through quartet structures. Their convenient product formulas are useful only when the assumptions behind them are visible.

4.1 How to read this chapter

Definition 4.1 fixes the experiment types. Theorem 4.2 proves the standard idealized differential-linear bias composition. Theorem 4.3 proves the idealized p^2q^2 boomerang estimate. Black Box 4.4 records the modern key-schedule-aware rectangle line.

4.2 Learning goals

A reader should be able to state differential-linear, boomerang, rectangle, and BCT experiments without conflating them; derive the idealized differential-linear and p^2q^2 formulas from explicit conditional-independence assumptions; and identify why key-schedule constraints and middle-round dependence require more refined analysis.

Definition 4.1 (Differential-linear, boomerang, and rectangle attacks). Write a cipher as $E = E_1 \circ E_0$. A *differential-linear attack* chooses an input difference for E_0 and a linear mask through E_1 , testing a parity relation on paired ciphertexts. A *boomerang attack* chooses a differential $\alpha \rightarrow \beta$ through E_0 and a differential $\gamma \rightarrow \delta$ through E_1^{-1} , creates a pair with difference α , perturbs ciphertexts by γ , decrypts, and tests whether the returned pair has difference α . A *rectangle attack* is a chosen/known-plaintext quartet formulation that amplifies related boomerang structures. A *boomerang connectivity table* records S-box compatibility of upper and lower differences; it captures dependence that the naive product model ignores.

Theorem 4.2 (Idealized differential-linear bias composition). *Suppose an input pair reaches a prescribed middle difference with probability p . Conditioned on that event, suppose the same suffix linear approximation, with signed correlation c , applies independently to the two members of the pair; conditioned on failure of the middle differential, suppose the tested parity difference is unbiased.*

Then the final pair test has success probability

$$\frac{1}{2} + \frac{pc^2}{2}.$$

Equivalently, if the one-text linear bias is $\epsilon = c/2$, the pair-test bias is $2p\epsilon^2$.

Proof. Let Z, Z' be the two suffix approximation error bits. Under the successful middle differential they are independent with $\text{cor}(Z) = \text{cor}(Z') = c$. By the piling-up lemma, $\text{cor}(Z \oplus Z') = c^2$, so the desired parity equality holds with probability $(1 + c^2)/2$. On the complementary event it holds with probability $1/2$ by hypothesis. Averaging gives $p(1 + c^2)/2 + (1 - p)/2 = 1/2 + pc^2/2$. $\square$

Theorem 4.3 (Idealized boomerang quartet probability). *Let the upper differential through E_0 have probability p and the lower inverse differential through E_1^{-1} have probability q . If the two upper differential events for the quartet are independent, the two lower events are independent, and upper/lower compatibility introduces no additional dependence, then the boomerang returns with probability p^2q^2 .*

Proof. A successful quartet requires two prescribed upper transitions and two prescribed lower transitions. Under the three explicit independence/compatibility assumptions, the joint probability is the product $ppqq$. $\square$

Why the theorem is only an idealized baseline, and source boundary. Real ciphers violate the independence assumptions through S-box compatibility, clustering, key schedules, and shared internal values. Wagner’s 1999 boomerang paper supplies the historical attack framework and idealized product analysis. The Boomerang Connectivity Table is a later tool of Cid–Huang–Peyrin–Sasaki–Song (EUROCRYPT 2018) designed to expose S-box compatibility that the naive product model misses. The two source layers are therefore recorded separately; BCT is not retroactively part of Wagner’s original theorem.

Theorem 4.4 (Rectangle attacks with key-schedule-aware guessing — verified deep input). *The Biham–Dunkelman–Keller rectangle framework and later Wang-group work refine boomerang quartets into practical key-recovery attacks. In particular, Dong, Qin, Sun, and Xiaoyun Wang (EUROCRYPT 2022) show that linear key schedules can impose nonlinear relations on quartets and develop key-guessing strategies that avoid generating quartets which cannot suggest a valid key candidate. The primitive-specific quartet structures, key-schedule relations, and attack complexities remain source-owned.*

4.3 Reader checkpoint

1. Derive the $pc^2/2$ differential-linear bias from signed correlations.
2. State exactly which independence assumptions the derivation uses.
3. Give a numerical example with $p = 2^{-4}$ and $c = 2^{-2}$.
4. Derive the idealized p^2q^2 boomerang estimate.
5. Explain why an S-box BCT can invalidate a naive product estimate.
6. Distinguish a boomerang distinguisher from a rectangle key-recovery attack.

7. Explain how a key schedule can make some quartets impossible for every master key.
8. Why can clustering increase an observed quartet probability?
9. State what must be measured empirically before turning a quartet probability into data complexity.
10. Explain why the two local theorems are baselines rather than security theorems for a concrete cipher.

Chapter 5

Integral, Impossible-Differential, and Rebound Methods

Integral cryptanalysis propagates sets or multisets rather than individual differences. Rebound cryptanalysis uses available degrees of freedom to solve a low-probability middle differential efficiently and pays probability mainly in outbound phases. Both methods exploit structure that is invisible in a single-trail probability.

5.1 How to read this chapter

Definition 5.1 separates integral, impossible, and rebound vocabulary. Theorem 5.2 proves the elementary propagation invariants. The rebound attack and its automated classical/quantum descendants remain source-owned in Black Boxes 5.3 and 5.4.

5.2 Learning goals

A reader should be able to propagate balanced multisets through affine and bijective layers, distinguish integral and impossible-differential reasoning from rebound matching, and explain the inbound/outbound split and the additional resource assumptions introduced by automated or quantum rebound attacks.

Definition 5.1 (Integral, impossible-differential, and rebound methods). An *integral structure* is a multiset of states in which selected coordinates range over a prescribed set while others may be constant, balanced, or unknown. A coordinate is *balanced* if the XOR of its multiset values is zero. An *impossible differential* is an input/output difference pair proved to have probability zero, often by contradicting forward and backward propagation. A *rebound attack* splits a differential path into an *inbound* phase solved by matching/degree-of-freedom techniques and one or more *outbound* phases whose probabilities are paid probabilistically. These notions describe attack structure, not automatically key recovery.

Theorem 5.2 (Integral propagation through affine and bijective layers). *Let $\mathcal{M}$ be an even-cardinality multiset of vectors over $\mathbb{F}_2^n$.*

(i) *For an affine map $A(x) = Lx \oplus c$,*

$$\bigoplus_{x \in \mathcal{M}} A(x) = L \left(\bigoplus_{x \in \mathcal{M}} x \right).$$

Thus a zero XOR-sum is preserved by every affine layer.

- (ii) *If a coordinate of $\mathcal{M}$ takes every value of $\mathbb{F}_2^m$ with the same multiplicity and S is a bijection of $\mathbb{F}_2^m$, then after applying S that coordinate still takes every value with the same multiplicity. For $m \geq 2$ it is therefore balanced.*

Proof. XOR distributes through the linear map. The constant c appears an even number of times and cancels, proving (i). A bijection merely permutes the elements of $\mathbb{F}_2^m$, proving (ii). The XOR of all m -bit words is zero for $m \geq 2$ because each coordinate contains equally many zeros and ones. $\square$

Theorem 5.3 (The rebound attack on AES-like hashing — verified deep input). *Mendel, Rechberger, Schläffer, and Thomsen introduced the rebound attack at FSE 2009 for reduced Whirlpool and Grøstl. The method uses a match-in-the-middle inbound phase to satisfy a dense part of an AES-like differential path with available degrees of freedom, followed by probabilistic outbound propagation. Their concrete reduced-round collision results and the detailed inbound matching equations remain source-owned.*

Theorem 5.4 (Automatic classical and quantum rebound attacks — verified deep input). *Dong, Zhang, Sun, Wei, Xiaoyun Wang, and Hu (ASIACRYPT 2021) automate classical and quantum rebound-style search for AES-like hashing using related-key differentials, with explicit classical/quantum resource models. The paper’s primitive instances, optimization models, and quantum subroutines are retained as source-owned deep input. A quantum speedup is not represented as a free square root of every classical time exponent: QRAM, query, gate, and memory assumptions must be stated separately.*

5.3 Reader checkpoint

1. Prove part (i) for a multiset of four vectors.
2. Why is the even-cardinality hypothesis needed for an affine constant to cancel?
3. Show that a bijection preserves the all-values property.
4. Explain why “balanced” is weaker than “all values equally often.”
5. Construct a one-bit edge case for the XOR of the complete domain.
6. Explain the miss-in-the-middle logic of an impossible differential.
7. Separate inbound work from outbound probability in a rebound attack.
8. Why is rebound especially natural for AES-like permutations with strong local diffusion?
9. List four resource quantities that must be reported for a quantum rebound claim.
10. Explain why Theorem 5.2 alone does not yield a practical integral attack.

Chapter 6

Meet-in-the-Middle, Biclique, and Automated Matching Attacks

Meet-in-the-middle (MITM) attacks trade memory for avoiding a full Cartesian search. Modern variants choose matching states, partial key guesses, neutral words, and recomputation boundaries algorithmically. Biclique cryptanalysis is a related full-keyspace technique whose real advantage must be measured against the exact work model, not merely against an exponent written beside brute force.

6.1 How to read this chapter

Definition 6.1 fixes MITM, partial matching, and biclique terminology. Theorem 6.2 proves a generic batching tradeoff; the full-AES biclique and automated MITM frameworks remain source-owned in Black Boxes 6.3 and 6.4.

6.2 Learning goals

A reader should be able to formulate a meet-in-the-middle equality, derive the batching memory–time tradeoff including verification cost, distinguish partial matching from exact matching, and interpret biclique or automated MITM claims using their stated time, memory, data, and recomputation models.

Definition 6.1 (Meet-in-the-middle and biclique attacks). A *meet-in-the-middle decomposition* expresses a target condition as $L(k_L) = R(k_R)$ for two candidate sets K_L, K_R . A *matching state* is the value compared by the two partial computations. A *partial matching* compares only selected bits/bytes and creates false positives that require verification. A *biclique* is a complete bipartite structure of related internal computations used to amortize parts of key search; its dimension and recomputation strategy are part of the attack specification. Time, memory, data, precomputation, and verification cost are recorded separately.

Theorem 6.2 (Generic MITM memory–time tradeoff). *Suppose a candidate is a pair $(k_L, k_R) \in K_L \times K_R$, $|K_L| = N_L$, $|K_R| = N_R$, and a correct pair satisfies $L(k_L) = R(k_R)$ for efficiently computable values. With memory for M left values, $1 \leq M \leq N_L$, an exact batching algorithm uses*

$$O\left(N_L + \left\lceil \frac{N_L}{M} \right\rceil N_R\right)$$

map evaluations, $O(M)$ stored entries, plus verification of matching candidates. With $M = N_L$, this becomes $O(N_L + N_R)$ time and $O(N_L)$ memory.

Proof. Partition K_L into batches of at most M . For each batch, compute its L -values and store them in a lookup table, then scan all N_R right candidates and test whether $R(k_R)$ occurs in the table. Across all batches, each left candidate is evaluated once, giving N_L left evaluations, while the right side is scanned $\lceil N_L/M \rceil$ times. Every exact match is emitted for final verification, so no solution is lost. $\square$

Theorem 6.3 (Biclique cryptanalysis of full AES — verified deep input). *Bogdanov, Khovratovich, and Rechberger (ASIACRYPT 2011) introduced biclique cryptanalysis for full AES and reported key-recovery complexities below exhaustive search for AES-128/192/256, including about $2^{126.1}$ computations for AES-128 in their stated work model. The biclique construction, related-key differentials, recomputation accounting, and practical interpretation of the modest exponent reduction remain source-owned.*

Theorem 6.4 (Automated meet-in-the-middle attacks — verified deep input). *Recent Wang-group work and collaborators formulate MITM cryptanalysis as structured optimization: automatic search of MITM preimage attacks on AES-like hashing, MITM attacks on sponge-based hashing, and automated MITM attacks on Feistel networks choose neutral variables, matching constraints, and recomputation boundaries algorithmically. The concrete primitive models and optimized parameters remain source-owned; this row records the methodology and exact publication scope rather than claiming a universal MITM optimizer.*

6.3 Reader checkpoint

1. Apply Theorem 6.2 with $N_L = N_R = 2^{32}$ and $M = 2^{16}$.
2. Explain the additional cost caused by partial rather than full matching.
3. Why can a wrong matching state create many false positives?
4. Distinguish data complexity from MITM table memory.
5. Explain how recomputation trades time for memory.
6. What is amortized by a biclique?
7. Why is $2^{126.1}$ versus 2^{128} a mathematically real but operationally modest improvement?
8. Give a reason an automated search might choose a non-obvious middle state.
9. State what an optimization model must encode to prove it did not omit a class of matches.
10. Explain why the generic theorem does not certify any AES biclique parameter.

Chapter 7

Algebraic Degree, Higher Derivatives, and Cube Attacks

Cube attacks use higher-order derivatives of Boolean polynomials to eliminate monomials and expose lower-degree “superpolys” in secret variables. The key algebra is exact and simple; the cryptanalytic art is choosing public variables and conditions so that the surviving polynomial is useful.

7.1 How to read this chapter

Definition 7.1 fixes ANF, derivatives, cubes, and superpolys. Theorem 7.2 proves the exact cube-sum derivative identity. The conditional-cube attack on reduced-round Keccak keyed modes remains source-owned in Black Box 7.3.

7.2 Learning goals

A reader should be able to express Boolean functions in ANF, prove the cube-sum/higher-derivative identity monomial by monomial, derive degree reduction under cube summation, and separate this exact algebra from the heuristic or primitive-specific task of finding useful conditional cubes.

Definition 7.1 (Algebraic and cube attacks). Every Boolean function has a unique algebraic normal form (ANF), a multilinear polynomial over $\mathbb{F}_2$. For a public variable x_i , define the derivative $\Delta_i f(x) = f(x) \oplus f(x \oplus e_i)$. For a set I , write $\Delta_I = \prod_{i \in I} \Delta_i$. A *cube sum* fixes all non-cube public variables and XORs the output over all $2^{|I|}$ assignments to the cube variables. The resulting function of secret variables is the *superpoly*. A cube attack has an offline phase that identifies useful cubes/superpolys and an online phase that queries the keyed primitive and solves the resulting secret relations.

Theorem 7.2 (Cube-sum derivative theorem). *Let $f(x_1, \dots, x_m, z)$ be a Boolean polynomial, where z denotes all non-cube variables, and let $I \subseteq \{1, \dots, m\}$. For any fixed base assignment with cube coordinates set to zero,*

$$\bigoplus_{a \in \mathbb{F}_2^I} f(x_I = a, z) = \Delta_I f(0, z).$$

Equivalently, in the ANF of f , every monomial not containing all variables $\prod_{i \in I} x_i$ cancels, while a monomial $(\prod_{i \in I} x_i)m(z)$ contributes exactly $m(z)$. Hence the cube sum has degree at most $\deg(f) - |I|$ in the remaining variables.

Proof. It suffices to check a monomial. If a monomial omits some cube variable x_j , then as the cube is summed, assignments pair by toggling x_j and the two monomial values are identical, so they cancel over $\mathbb{F}_2$. If the monomial contains every cube variable, its value is nonzero only when all included cube variables equal one, leaving precisely the residual factor in non-cube variables. Linearity over the ANF proves the identity. Each derivative removes at least one cube variable from every surviving monomial, giving the degree bound. $\square$

Theorem 7.3 (Conditional cube attacks on reduced-round Keccak keyed modes — verified deep input). *Huang, Xiaoyun Wang, Xu, Meiqin Wang, and Zhao (EUROCRYPT 2017) introduced conditional cube testers for reduced-round Keccak sponge/keyed modes. Key-dependent bit conditions reduce diffusion of selected cube variables; later work uses MILP to search such variables and conditions. The reported attacks are on specified reduced-round Keccak-MAC/Keyak settings and do not threaten the full-round SHA-3 security margin. Primitive-specific conditions and complexities remain source-owned.*

7.3 Reader checkpoint

1. Compute $\Delta_1(x_1x_2 \oplus x_3)$.
2. Compute the two-variable cube sum of $x_1x_2z \oplus x_1z \oplus 1$.
3. Prove that a first derivative lowers algebraic degree by at least one when nonzero.
4. Explain why a cube larger than the total degree gives zero.
5. Distinguish cube dimension from the degree of the resulting superpoly.
6. Why are public/tweakable variables essential to an online cube attack?
7. Separate the offline and online phases.
8. Explain how key-dependent conditions can alter effective diffusion.
9. Why does a reduced-round Keccak-MAC attack not imply a SHA3-256 collision attack?
10. State what a MILP model for conditional cubes must preserve to be cryptanalytically meaningful.

Chapter 8

Differential Fault Analysis and Automated Fault Search

Fault attacks change the computational model: the adversary observes not only correct encryption but outputs produced under an injected physical or logical fault. Algebraically, this creates paired executions with a constrained internal difference. Security claims are therefore meaningless unless the fault location, timing, granularity, persistence, and observability model are explicit.

8.1 How to read this chapter

Definition 8.1 makes the fault interface explicit before any security claim. Theorem 8.2 derives the local last-round key equations, while Black Box 8.3 records the automated MITM-DFA line with its concrete fault model.

8.2 Learning goals

A reader should be able to state a fault model before deriving equations, obtain last-round DFA key constraints by inverting the final S-box, distinguish location/timing/value assumptions, and explain what an automated fault-search framework certifies relative to its encoded model.

Definition 8.1 (Fault and side-channel-assisted cryptanalysis). A *fault model* specifies where and when a computation may be perturbed, the allowed fault values/distributions, whether the attacker knows the fault, and which correct/faulty outputs are observable. *Differential fault analysis* (DFA) compares correct and faulty executions to derive key-dependent equations. A *persistent fault* remains across executions; an *ineffective fault* is injected but leaves an observed value unchanged and can itself leak information. Side-channel information and faults are additional interfaces beyond black-box chosen-plaintext access and must be costed separately.

Theorem 8.2 (Last-round differential-fault key equations for an SPN). *Suppose a bitwise final round is*

$$C = S(X) \oplus K,$$

where S is an invertible byte S-box applied componentwise. If a faulty computation enters that S-box layer with $X' = X \oplus \Delta$, producing C' , then every candidate last-round key K consistent with the fault satisfies

$$S^{-1}(C \oplus K) \oplus S^{-1}(C' \oplus K) = \Delta.$$

If the fault was injected one linear layer earlier and $\Delta = L(e_j\eta)$ for a specified fault position j and an unknown nonzero byte value η , then the candidate key must satisfy the component equations above simultaneously for some allowed η . Intersecting candidate sets from independent faulty ciphertexts cannot discard the true key under the stated fault model.

Proof. From $C = S(X) \oplus K$ and invertibility, $X = S^{-1}(C \oplus K)$; similarly $X' = S^{-1}(C' \oplus K)$. XORing gives the first equation. A fault $e_j\eta$ before an invertible linear diffusion layer becomes $L(e_j\eta)$ by linearity. The true key and true fault value satisfy all resulting component equations, so intersecting the candidate supersets over several observations retains the true key. $\square$

Theorem 8.3 (Automated meet-in-the-middle DFA on AES-like ciphers — verified deep input). *Yu, Dong, Qin, Kang, Jia, Xiaoyun Wang, and Zhang (TCHES 2023) formulate search for MITM differential-fault attacks on AES-like ciphers as constrained optimization, choosing fault locations, matching points, and independent computation paths. Their models yield attacks on specified lightweight/SPN designs with faults injected earlier than in prior DFA work. The MILP formulations, primitive-specific fault assumptions, and reported parameters remain source-owned.*

8.3 Reader checkpoint

1. Derive the last-round DFA equation from the correct and faulty ciphertext equations.
2. Why must the S-box be invertible for that exact form?
3. Propagate a one-byte fault through a linear diffusion matrix symbolically.
4. Explain why the true key survives candidate-set intersection.
5. Give a fault-model change that invalidates the theorem.
6. Distinguish random-value, stuck-at, and known-value faults.
7. Explain why a fault attack is not a black-box chosen-ciphertext attack.
8. What additional experimental uncertainty appears in physical fault injection?
9. Why can MITM extend a DFA farther from the final round?
10. List the fault-model metadata required for a reproducible cryptanalytic claim.

Chapter 9

SAT, MILP, SMT, and Constraint-Programmed Cryptanalysis

Automated cryptanalysis is only as trustworthy as its encoding. A solver can prove optimality or unsatisfiability inside a mathematical model, but it cannot prove that the model represents the cipher unless that connection is established independently. This chapter formalizes that interface and provides an executable toy laboratory with an independent certificate checker.

9.1 How to read this chapter

Definition 9.1 separates cipher relations from their solver encoding. Theorem 9.2 proves the exact encoding interface, Black Box 9.3 calibrates modern machine-assisted search, and Code Laboratory 9.4 supplies an independently checked toy certificate.

9.2 Learning goals

A reader should be able to define sound, complete, and objective-faithful trail encodings; prove local-to-global correctness of a composed SAT/MILP/SMT/CP model; read a solver certificate independently of the optimizer that produced it; and separate model correctness from heuristic search performance.

Definition 9.1 (SAT, MILP, SMT, and CP cryptanalytic encodings). A *trail-search instance* consists of state variables $x_0, \dots, x_r$, local transition relations $R_i \subseteq X_{i-1} \times X_i$, endpoint constraints, and optionally an additive cost $w = \sum_i w_i(x_{i-1}, x_i)$. A SAT/MILP/SMT/CP *encoding* introduces solver variables and constraints Φ_i intended to represent each R_i . It is *sound* if every satisfying solver assignment decodes to a valid trail, and *complete* if every valid trail has a satisfying encoding. An objective is *faithful* if its solver value equals the declared trail cost after decoding. Solver optimality is meaningful only after these three properties are established.

Theorem 9.2 (Soundness and completeness of exact local-relation encodings). *For each round i , suppose Φ_i is an exact encoding of R_i in the following sense: an assignment to the visible state variables (x_{i-1}, x_i) extends to the auxiliary variables of Φ_i if and only if $(x_{i-1}, x_i) \in R_i$. Let*

$$\Phi = \Phi_{\text{in}} \wedge \Phi_1 \wedge \dots \wedge \Phi_r \wedge \Phi_{\text{out}},$$

where the endpoint encodings are exact. Then Φ is satisfiable if and only if a valid trail satisfying the endpoint conditions exists. If additionally each encoded local cost equals w_i on every satisfying assignment, minimizing the sum of encoded costs returns a globally minimum-cost trail.

Proof. For soundness, restrict any satisfying assignment of Φ to its visible states. Exactness of every Φ_i implies $(x_{i-1}, x_i) \in R_i$, and exact endpoint constraints give a valid trail. For completeness, begin with any valid trail. Exactness supplies auxiliary assignments satisfying every Φ_i ; because adjacent relations share the same visible state variables, these assignments combine into a satisfying assignment of Φ . If local encoded costs are faithful, their sum equals the trail cost for every feasible assignment, so an optimizer's minimum equals the minimum over valid trails. $\square$

Theorem 9.3 (MILP/SAT-assisted cryptanalytic search — verified deep input). *Constraint-based cryptanalysis has developed from SAT encodings of hash-function problems and MILP models for active S-box bounds into compact differential/linear/division-property models and primitive-specific automated attack frameworks. Mouha, Wang, Gu, and Preneel (2011) are a canonical MILP reference for differential/linear bounds; later work substantially extends the modeling toolkit. Individual solver formulations, symmetry breaking, optimizer performance, and primitive-specific completeness claims remain source-owned and must be audited per model.*

Example 9.4 (Solver-certified trail-search laboratory). The executable file `code/volume_19/trail_search_lab.py` computes the toy S-box DDT, builds the exact nonzero transition graph, searches a two-round differential trail by dynamic programming, emits a certificate containing every transition and exact numerator/denominator probability, and then runs an independent verifier which checks each DDT entry and the product. It also exhaustively confirms that no competing two-round trail with the same endpoints has larger probability. The laboratory therefore tests the same soundness/completeness split as Theorem 9.2 without depending on a proprietary solver.

9.3 Reader checkpoint

1. Distinguish solver soundness from encoding soundness.
2. Give an underconstrained encoding that is satisfiable but unsound.
3. Give an overconstrained encoding that is sound but incomplete.
4. Prove the theorem for a two-round relation chain.
5. Why must auxiliary variables be existentially extendable in both directions?
6. Explain why minimizing active S-boxes need not minimize exact differential weight.
7. What does an UNSAT result prove after exactness has been established?
8. Why should a cryptanalytic artifact include its solver version and parameters?
9. Explain the role of the independent certificate checker in the laboratory.
10. List three reasons two correct encodings can have very different solving times.

Chapter 10

Quantum Resource Accounting and Reproducible Automated Attacks

The final chapter puts the attack families into one resource language. Quantum algorithms can change query exponents, but a cryptanalytic claim also depends on reversible-circuit cost, qubits, classical memory, QRAM assumptions, data access, and success amplification. Machine-assisted cryptanalysis adds a second obligation: the search artifact must be sufficiently specified that another team can distinguish a theorem from an optimizer run.

10.1 How to read this chapter

Definition 10.1 fixes quantum resource accounting. Theorem 10.2 records the Grover and generic-collision baselines, Black Box 10.3 gives the low-QRAM refinement, and the final frontier/figure separate reproducibility requirements from attack-family taxonomy.

10.2 Learning goals

A reader should be able to distinguish quantum query complexity from reversible gate, qubit, memory, and QRAM costs; state Grover and generic collision baselines in the correct oracle model; and specify the artifacts, versions, resource accounting, and independent verification required for a reproducible machine-assisted cryptanalytic claim.

Definition 10.1 (Quantum cryptanalytic resource models). A quantum cryptanalytic resource statement records at least: oracle/query complexity; logical gate count and depth; logical qubits; classical memory; coherent quantum memory; whether QRAM is assumed and its access model; classical or quantum data access; precomputation; success probability and amplification; and the cost of reversibly implementing the primitive. A *black-box key-search oracle* marks candidate keys satisfying a verification predicate. A *collision oracle* provides reversible access to the function in the standard quantum query model. Query bounds alone are not wall-clock or fault-tolerant resource estimates.

Theorem 10.2 (Grover key search and quantum collision baselines). *For an ideal k -bit key space with a unique key recognized by an efficiently reversible verification predicate, quantum key search uses $\Theta(2^{k/2})$ oracle queries for constant success, up to the standard amplification factors. For a random-function collision problem with range size $N = 2^n$, the standard quantum query complexity*

is $\Theta(N^{1/3}) = \Theta(2^{n/3})$. These are query baselines: the reversible cost of one key verification or function query and the memory/QRAM architecture multiply or otherwise refine the physical resource estimate.

Proof. The key-search statement is the unstructured-search theorem already established in Volume XV: Grover amplitude amplification gives the upper bound and the quantum search lower bound gives matching $\Omega(2^{k/2})$ queries. The collision statement is the generic quantum-collision theorem already source-verified in Volume XVIII. Replacing an abstract oracle by a concrete cipher/hash circuit requires implementing that oracle reversibly, so query counts do not erase gate, depth, qubit, or memory costs. $\square$

Theorem 10.3 (Low-QRAM quantum collision attacks on AES-like hashing — verified deep input). *Dong, Sun, Shi, Gao, Xiaoyun Wang, and Hu (ASIACRYPT 2020) develop dedicated quantum collision attacks on AES-like hashing under low quantum-random-access-memory regimes, refining the resource tradeoffs of generic or earlier quantum rebound approaches. The algorithms depend on primitive-specific differential structures and explicit quantum memory models; their exponents are therefore not presented as generic consequences of the $2^{n/3}$ collision baseline.*

Remark 10.4 (Machine-assisted cryptanalysis reproducibility standard — dated frontier, 2026-09-03). A machine-assisted cryptanalytic claim should publish, when legally and practically possible, the exact primitive version and parameters; the formal constraints/objective; tool and solver versions; seeds and nondeterministic settings; hardware/software environment; input instances; raw certificates or trails; independent verification code; and scripts that regenerate headline tables. “Artifact available”, “artifact functional”, and “results reproduced” are distinct levels of evidence. Security-conference artifact processes, including CHES’s evolving badge practice, show that community standards continue to change; this row is therefore a dated methodological frontier rather than a mathematical theorem.

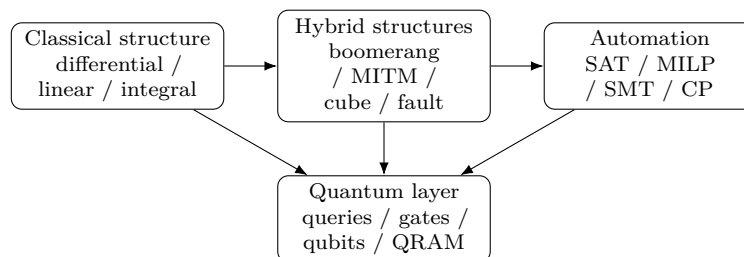


Figure 10.1: Attack taxonomy and resource cube. Automation is a search methodology, not a separate mathematical attack notion; quantumization changes the resource model and may reuse classical differential/algebraic structure.

Volume closure. Volume XIX separates four statements that are frequently blurred together: an algebraic/probabilistic attack principle, a concrete primitive attack, an automated search result, and a physical resource estimate. Local theorems prove the reusable algebra. Deep primitive attacks retain source ownership. Solver results require exact encodings and certificates. Quantum results require explicit memory and reversible-circuit models.

10.3 Reader checkpoint

1. Translate a $k = 128$ ideal key space into the Grover query scale.
2. Translate a 256-bit random-function output into the generic quantum collision-query scale.
3. Explain why those two exponents refer to different problems.
4. Give three costs hidden by an abstract key-search oracle.
5. Distinguish coherent memory from classical RAM.
6. Explain why low-QRAM cryptanalysis is not characterized by query complexity alone.
7. List the minimum files you would require to review a solver-found differential trail.
8. Distinguish artifact availability, functionality, and reproduced-result evidence.
9. Use Figure 10.1 to classify a quantum rebound attack.
10. Give a checklist for deciding whether a headline “quantum speedup” is comparable to the classical attack it cites.

Internalization and Reconstruction Guide: Volume XIX

Purpose. Treat symmetric cryptanalysis as a modeling discipline: distinguish characteristics from differentials, correlations from probabilities, exact encodings from heuristic independence, and classical complexity from quantum resource models.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-XVIII-DEF-004: Merkle–Damgard, Davies–Meyer, and sponge constructions.
2. QRCC-XVIII-THM-002: Generic collision and preimage query bounds.
3. QRCC-I-THM-016: Parseval/Plancherel theorem.
4. QRCC-XV-DEF-012: Quantum query model.
5. QRCC-XV-ALG-014: Grover search and amplitude amplification.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XIX-DEF-001 (Block ciphers and design structures); QRCC-XIX-DEF-004 (Differential characteristics and differentials); QRCC-XIX-DEF-007 (Linear approximations and hulls); QRCC-XIX-DEF-010 (Differential-linear, boomerang, and rectangle attacks); QRCC-XIX-DEF-014 (Integral, impossible, and rebound methods).

Chapter retrieval map

- Ch. 1: Block-Cipher Structure, S-box Spectra, and Feistel Networks:** enter through QRCC-XIX-DEF-001 (Block ciphers and design structures); exit by being able to state QRCC-XIX-PROP-003 (S-box DDT, LAT, and algebraic-degree identities) with its ownership boundary.
- Ch. 2: Differential Cryptanalysis and Markov-Cipher Bookkeeping:** enter through QRCC-XIX-DEF-004 (Differential characteristics and differentials); exit by being able to

state QRCC-XIX-BB-006 (Biham–Shamir differential cryptanalysis of DES) with its ownership boundary.

- Ch. 3: Linear Cryptanalysis, Piling Up, and Hulls:** enter through QRCC-XIX-DEF-007 (Linear approximations and hulls); exit by being able to state QRCC-XIX-BB-009 (Matsui linear cryptanalysis of DES) with its ownership boundary.
- Ch. 4: Differential-Linear, Boomerang, and Rectangle Cryptanalysis:** enter through QRCC-XIX-DEF-010 (Differential-linear, boomerang, and rectangle attacks); exit by being able to state QRCC-XIX-BB-013 (Wang-group rectangle and key-schedule attacks) with its ownership boundary.
- Ch. 5: Integral, Impossible-Differential, and Rebound Methods:** enter through QRCC-XIX-DEF-014 (Integral, impossible, and rebound methods); exit by being able to state QRCC-XIX-BB-016 (Rebound attacks on AES-like hashing) with its ownership boundary.
- Ch. 6: Meet-in-the-Middle, Biclique, and Automated Matching Attacks:** enter through QRCC-XIX-DEF-018 (Meet-in-the-middle and biclique attacks); exit by being able to state QRCC-XIX-BB-020 (Biclique cryptanalysis of full AES) with its ownership boundary.
- Ch. 7: Algebraic Degree, Higher Derivatives, and Cube Attacks:** enter through QRCC-XIX-DEF-022 (Algebraic and cube attacks); exit by being able to state QRCC-XIX-BB-024 (Conditional cube attacks on Keccak keyed modes) with its ownership boundary.
- Ch. 8: Differential Fault Analysis and Automated Fault Search:** enter through QRCC-XIX-DEF-025 (Fault and side-channel-assisted cryptanalysis); exit by being able to state QRCC-XIX-BB-027 (Automated meet-in-the-middle differential fault analysis) with its ownership boundary.
- Ch. 9: SAT, MILP, SMT, and Constraint-Programmed Cryptanalysis:** enter through QRCC-XIX-DEF-028 (SAT, MILP, SMT, and CP encodings); exit by being able to state QRCC-XIX-BB-030 (MILP/SAT-assisted cryptanalytic search frontier) with its ownership boundary.
- Ch. 10: Quantum Resource Accounting and Reproducible Automated Attacks:** enter through QRCC-XIX-DEF-031 (Quantum cryptanalytic resource models); exit by being able to state QRCC-XIX-BB-033 (Wang-group low-QRAM quantum collision attacks) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XIX-PROP-003 — S-box DDT, LAT, and algebraic-degree identities. Dependencies: QRCC-XIX-DEF-001, QRCC-I-THM-016. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XIX-BB-009 — Matsui linear cryptanalysis of DES. Dependencies: QRCC-XIX-THM-008. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

3. QRCC-XIX-BB-016 — Rebound attacks on AES-like hashing. Dependencies: QRCC-XIX-DEF-014, QRCC-XVIII-DEF-004. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XIX-BB-024 — Conditional cube attacks on Keccak keyed modes. Dependencies: QRCC-XIX-THM-023, QRCC-XVIII-DEF-004. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XIX-BB-030 — MILP/SAT-assisted cryptanalytic search frontier. Dependencies: QRCC-XIX-THM-029. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XIX-F-034 — Machine-assisted cryptanalysis reproducibility standard. Dependencies: QRCC-XIX-DEF-028, QRCC-XIX-DEF-031. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- treating one characteristic probability as the full differential probability;
- losing sign and normalization in linear-correlation composition;
- calling a SAT/MILP encoding complete without proving both soundness and completeness;
- comparing quantum query counts while suppressing reversible-circuit, memory, or QRAM costs;

Boundary and degenerate cases

- zero-correlation or probability-one trails;
- independent-round-key idealization versus real key schedules;
- single-fault versus multi-fault models;
- classical memory versus QRAM assumptions;

Retrieval and reconstruction drill

1. State QRCC-XIX-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XIX-PROP-003. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XIX-BB-009, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.

5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XIX-F-034, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XX moves to post-quantum assumptions and standards, with lattice/coding geometry as the central hardness source.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XX

Lattices and Post-Quantum Cryptography

Chapter 1

Lattice Geometry, Duality, and the Shortest-Vector Landscape

Lattice cryptography uses two kinds of statements that must not be conflated: exact geometric theorems about an arbitrary lattice, and algorithmic cost models whose strongest practical forms are often heuristic. This chapter fixes the Euclidean normalization, records the transference interface used later by Gaussian arguments, and separates provable SVP algorithms from heuristic concrete-security estimates.

1.1 How to read this chapter

Definition 1.1 fixes the lattice and approximation-problem conventions. Theorem 1.2 is a verified deep input: its constants are the Euclidean Banaszczyk transference constants. Black Boxes 1.3 and 1.4 then distinguish provable exact-SVP algorithms from heuristic sieving claims.

1.2 Learning goals

A reader should be able to compute a dual lattice and determinant, distinguish λ_1 from the full sequence of successive minima, state GapSVP/SIVP/CVP/BDD with an approximation factor, use duality without losing norm conventions, and explain why a heuristic exponent is not a theorem about worst-case running time.

Definition 1.1 (Lattice geometry and computational problems). A full-rank *lattice* $L \subset \mathbb{R}^n$ is $B\mathbb{Z}^n$ for an invertible real matrix B . Its determinant is $\det L := |\det B|$ and its Euclidean dual is

$$L^* := \{y \in \mathbb{R}^n : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in L\} = B^{-T}\mathbb{Z}^n.$$

The i th *successive minimum* $\lambda_i(L)$ is the least r for which $L \cap rB_2^n$ spans an i -dimensional subspace. The *covering radius* is $\mu(L) := \sup_{t \in \mathbb{R}^n} \text{dist}(t, L)$.

SVP $_\gamma$ asks for a nonzero lattice vector of norm at most $\gamma\lambda_1(L)$; GapSVP $_\gamma$ distinguishes $\lambda_1(L) \leq d$ from $\lambda_1(L) > \gamma d$ on promised inputs; SIVP $_\gamma$ asks for n independent vectors of maximum norm at most $\gamma\lambda_n(L)$; CVP $_\gamma$ asks, given t , for $x \in L$ with $\|t - x\|_2 \leq \gamma \text{dist}(t, L)$. For $\alpha < 1/2$, BDD $_\alpha$ is the promised CVP problem with $\text{dist}(t, L) < \alpha\lambda_1(L)$. A *unique-SVP gap* means that every nonparallel lattice vector is longer than a declared multiple of a shortest vector. Classical, randomized, and quantum reductions are always named explicitly.

Theorem 1.2 (Euclidean transference bounds — verified deep input). *For every full-rank $L \subset \mathbb{R}^n$ and $1 \leq i \leq n$, Banaszczyk’s Euclidean transference theorem gives*

$$1 \leq \lambda_i(L) \lambda_{n-i+1}(L^*) \leq n.$$

For the covering radius,

$$1 \leq 2\lambda_1(L^*)\mu(L) \leq n.$$

Also $\det(L^) = (\det L)^{-1}$. These statements use the Euclidean norm and the dual defined above; later smoothing arguments use the upper bounds, not an unspecified “duality principle.”*

Source boundary. The determinant identity follows immediately from B^{-T} , but the dimension-linear upper transference bounds are retained as source-owned geometry. They are bound to Banaszczyk’s 1993 theorem rather than to an unresolved companion pointer.

Theorem 1.3 (Provable and heuristic SVP algorithms — verified deep input). *Exact Euclidean SVP has randomized single-exponential algorithms. As a historical provable anchor, the Micciancio–Voulgaris list-sieve analysis gives an exact-SVP algorithm with time $2^{3.199n+o(n)}$ and space $2^{1.325n+o(n)}$; subsequent provable work improves asymptotic exponents. Practical lattice-security estimates also use enumeration, BKZ, and heuristic sieves whose costs depend on pruning, block size, preprocessing, memory, nearest-neighbor models, and sometimes quantum assumptions. A heuristic concrete-security exponent is therefore not promoted to a worst-case theorem in this volume.*

Theorem 1.4 (Wang–Liu–Tian–Bi two-level heuristic sieve — verified deep input). *The 2011 two-level refinement of the Nguyen–Vidick heuristic sieve reports, in its heuristic model, time $2^{0.3836n+o(n)}$ polynomial operations and space $2^{0.2557n+o(n)}$. The theorem label records the paper’s claimed heuristic complexity; it is not a proof that every n -dimensional lattice can be solved within those bounds.*

Worked micro-example. For $B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, $\det L = 6$ and $B^{-T} = \begin{pmatrix} 1/2 & 0 \\ -1/6 & 1/3 \end{pmatrix}$, so $\det L^* = 1/6$. The product identity is exact even though shortest-vector computation may be difficult in high dimension.

1.3 Reader checkpoint

1. Compute the dual basis of a diagonal lattice $\text{diag}(a_1, \dots, a_n)\mathbb{Z}^n$.
2. Prove $\det(L^*) = (\det L)^{-1}$ from a basis matrix.
3. Distinguish $\lambda_1(L)$ from $\lambda_n(L)$ on a rectangular two-dimensional lattice.
4. State the promise in GapSVP_γ .
5. Explain why BDD requires a radius below half the minimum distance for uniqueness.
6. Apply the $i = 1$ transference inequality to bound $\lambda_n(L^*)$ from $\lambda_1(L)$.
7. Explain which part of Theorem 1.2 is nontrivial.
8. Classify an enumeration cost model as proved, empirical, or heuristic.
9. Explain why memory must be reported with a sieving exponent.
10. State precisely what the Wang two-level exponent does and does not claim.

Chapter 2

Discrete Gaussians, Smoothing, and Lattice Sampling

Discrete Gaussians are the analytic bridge between lattice geometry and cryptographic reductions. The normalization matters: changing whether s or $s/\sqrt{2\pi}$ is called the Gaussian width changes every displayed threshold. We therefore fix one convention and keep the strongest smoothing and sampler claims source-owned.

2.1 How to read this chapter

Definition 2.1 fixes ρ_s , $D_{L+c,s}$, statistical distance, and the smoothing parameter. Theorem 2.2 records the standard Poisson/smoothing consequences in that normalization. Black Box 2.3 is the GPV sampling interface; Black Box 2.4 records a restricted transference theorem for lattices having a shortest-vector gap.

2.2 Learning goals

A reader should be able to translate Gaussian conventions, compute a one-dimensional smoothing bound, explain why dual short vectors control modular nonuniformity, distinguish a distributional sampler theorem from a heuristic sampler, and identify the gap hypothesis in a restricted transference statement.

Definition 2.1 (Discrete Gaussians and smoothing). For $s > 0$, set $\rho_s(x) := \exp(-\pi\|x\|_2^2/s^2)$ and $\rho_s(S) := \sum_{x \in S} \rho_s(x)$ for a countable S . The discrete Gaussian on a coset $L + c$ is

$$\Pr[D_{L+c,s} = x] = \frac{\rho_s(x)}{\rho_s(L+c)} \quad (x \in L+c).$$

For $0 < \varepsilon < 1$, the *smoothing parameter* is

$$\eta_\varepsilon(L) := \inf\{s > 0 : \rho_{1/s}(L^* \setminus \{0\}) \leq \varepsilon\}.$$

The statistical distance of distributions P, Q on a countable space is $\Delta(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$. A real random variable X is σ -subgaussian when $\mathbb{E}e^{t(X-\mathbb{E}X)} \leq e^{\sigma^2 t^2/2}$ for every real t . All Gaussian widths in this volume use the $e^{-\pi\|x\|^2/s^2}$ convention.

Theorem 2.2 (Smoothing and Gaussian tail interface — verified deep input). *Let $L \subset \mathbb{R}^n$ be full rank, $0 < \varepsilon < 1$, and $s \geq \eta_\varepsilon(L)$. Poisson summation gives, for every $c \in \mathbb{R}^n$,*

$$(1 - \varepsilon) \frac{s^n}{\det L} \leq \rho_s(L + c) \leq (1 + \varepsilon) \frac{s^n}{\det L}.$$

Consequently Gaussian mass is nearly independent of the coset, reduction modulo L after adding sufficiently wide continuous Gaussian noise is close to uniform, and discrete-Gaussian projections have Gaussian tails up to the $(1 + \varepsilon)/(1 - \varepsilon)$ smoothing loss. These consequences are understood in the displayed normalization; the Fourier/Poisson proof and its sharper variants are source-owned to Micciancio–Regev.

Theorem 2.3 (GPV discrete-Gaussian sampling — verified deep input). *Given a lattice basis B and Gaussian width above the appropriate Gram–Schmidt threshold, the GPV/Klein-type sampler runs efficiently and outputs a distribution statistically close to the target discrete Gaussian over the specified coset. The required width is governed essentially by the longest Gram–Schmidt vector of the supplied basis, and the output distribution is designed to hide the detailed geometry of the trapdoor basis. The exact negligible-distance parameter is kept source-owned.*

Theorem 2.4 (Gap-lattice transference of Wei–Tian–Wang — verified deep input). *Wei, Tian, and Xiaoyun Wang prove strengthened transference relations for lattices whose shortest vector is n^ε -unique, including relations involving successive minima, covering radius, dual geometry, and a smoothing- parameter application. The gap hypothesis is part of the theorem: this row is not a replacement for the unrestricted Banaszczyk bounds of Theorem 1.2.*

Worked micro-example. For $L = \mathbb{Z}$, $L^* = \mathbb{Z}$ and $\rho_{1/s}(\mathbb{Z} \setminus \{0\}) = 2 \sum_{k \geq 1} e^{-\pi s^2 k^2}$. Any s making this sum at most ε is a valid upper bound on $\eta_\varepsilon(\mathbb{Z})$. This directly displays why a wider primal Gaussian suppresses nonzero dual frequencies.

2.3 Reader checkpoint

1. Translate $\rho_s(x)$ to a normal density variance convention.
2. Compute the first two terms in the smoothing sum for $L = \mathbb{Z}$.
3. Explain why the dual lattice occurs in the definition of η_ε .
4. State the difference between continuous and discrete Gaussian sampling.
5. Explain the role of statistical distance in a trapdoor sampler theorem.
6. Identify the basis quality parameter used by GPV sampling.
7. Explain why smoothing is useful for reducing information about a lattice coset.
8. Give the exact gap hypothesis attached to Black Box 2.4.
9. Explain why that result cannot be quoted for an arbitrary lattice.
10. Describe how Theorem 1.2 and smoothing interact without conflating their sources.

Chapter 3

Worst-Case to Average-Case Reductions

The conceptual origin of lattice cryptography is Ajtai’s discovery that solving a suitable random modular problem can be at least as hard as solving geometric lattice problems on every input. The logical direction is easy to misstate: a reduction from worst case to average case converts an average-case solver into a worst-case solver.

3.1 How to read this chapter

Definition 3.1 fixes distributions, parameter maps, and reduction directions. Black Box 3.2 records Ajtai’s seminal theorem with its historical scope and its published correction boundary visible.

3.2 Learning goals

A reader should be able to draw the reduction arrow correctly, separate distributional hardness from uniform sampling, track dimension/modulus/norm parameters through a reduction, distinguish classical from quantum reductions, and explain why a worst-case connection does not by itself give a concrete security level.

Definition 3.1 (Worst-case/average-case lattice reductions). A parameterized average-case problem is a family $(\mathcal{D}_n, R_n)$ in which the instance is sampled from a specified distribution $\mathcal{D}_n$ and $R_n(x, y)$ defines valid outputs. A randomized polynomial-time *worst-case to average-case reduction* from problem W to this family is an algorithm that, given an arbitrary promised instance x of W and oracle access to a solver that succeeds on the declared average distribution, outputs a valid solution to x with the declared success probability. The reduction record includes the output dimension, modulus, norm/error bound, number of oracle samples, approximation factor, failure probability, and whether the reduction is classical or quantum. A concrete-security estimate additionally needs the cost of the best known attacks at the target parameters; it is not implied by the asymptotic reduction alone.

Theorem 3.2 (Ajtai’s worst-case/average-case breakthrough — verified deep input). *Ajtai’s 1996 construction gives a random modular subset-sum/lattice problem such that an efficient algorithm succeeding with nonnegligible constant probability on the random instances yields randomized*

polynomial-time algorithms for specified worst-case lattice approximation tasks, including approximating shortest-vector length, unique shortest-vector variants, and short independent/basis problems within polynomial factors. The theorem is a worst-case-to-average-case hardness connection, not an assertion that every later SIS/LWE parameter set is covered unchanged.

Historical proof audit. The ECCC record for Ajtai’s report includes a corrected proof of Lemma 4. The volume therefore cites the corrected record rather than silently treating every line of the first public draft as immutable. Later SIS reductions use cleaner Gaussian machinery and receive their own source ownership in Chapter 4.

Worked reduction logic. Suppose an average-case oracle succeeds with probability at least $1/2$ on $\mathcal{D}_n$, and reduction R uses it to solve every n -dimensional instance of W . Then an efficient average-case oracle would contradict worst-case hardness of W . The converse does not follow: failure to solve the average distribution is not a proof that every worst-case instance is hard.

3.3 Reader checkpoint

1. Draw the oracle-reduction arrow from a worst-case input to average instances.
2. Explain the contrapositive cryptographic hardness inference.
3. Distinguish an average distribution from “a typical lattice” in informal language.
4. List five parameters that a lattice hardness reduction must track.
5. Explain why a quantum reduction and a classical reduction are different security statements.
6. State what success probability an average-case oracle is promised to achieve in a typical reduction theorem.
7. Explain why asymptotic worst-case hardness does not determine a NIST security category.
8. Identify the historical correction noted for Ajtai’s report.
9. Explain why later SIS/LWE theorems are not merely restatements of Ajtai 1996.
10. Give one invalid converse inference from a worst-case/average-case theorem.

Chapter 4

Short Integer Solutions and Collision-Resistant Hashing

The Short Integer Solution problem turns a random modular linear map into a cryptographic hardness assumption. Its cleanest pedagogical application is collision resistance: two distinct short-domain inputs with the same image differ by a nonzero short kernel vector.

4.1 How to read this chapter

Definition 4.1 fixes SIS and the q -ary kernel lattice. Theorem 4.2 is proved locally and gives the exact collision-to-SIS reduction. Black Box 4.3 then records the much deeper worst-case hardness theorem.

4.2 Learning goals

A reader should be able to translate a modular matrix equation into a q -ary lattice, prove the collision reduction, check the compression condition $2^m > q^n$, and distinguish the easy pigeonhole existence of collisions from the computational hardness of finding one.

Definition 4.1 (SIS and q -ary lattices). For $A \in \mathbb{Z}_q^{n \times m}$, define the integer q -ary kernel lattice

$$\Lambda_q^\perp(A) := \{z \in \mathbb{Z}^m : Az \equiv 0 \pmod{q}\}.$$

The Euclidean $\text{SIS}_{n,m,q,\beta}$ problem samples A uniformly and asks for $0 \neq z \in \mathbb{Z}^m$ with $Az \equiv 0 \pmod{q}$ and $\|z\|_2 \leq \beta$. ISIS instead asks for a short preimage of a specified syndrome. Ring-SIS and Module-SIS replace scalar coordinates by a declared quotient ring/module; they are different average-case assumptions and are not identified with ordinary SIS without a reduction. A hash family is collision resistant when no efficient adversary, for a random public key, finds distinct domain points with the same hash except with negligible probability.

Theorem 4.2 (SIS matrix hashing is collision resistant under SIS). *Let $A \leftarrow \mathbb{Z}_q^{n \times m}$ and define $h_A : \{0,1\}^m \rightarrow \mathbb{Z}_q^n$ by $h_A(x) = Ax \bmod q$. Every collision $x \neq x'$ yields*

$$z := x - x' \in \{-1, 0, 1\}^m \setminus \{0\}, \quad Az \equiv 0 \pmod{q}, \quad \|z\|_2 \leq \sqrt{m}.$$

Hence an adversary finding hash collisions with probability δ gives an equally successful solver for $\text{SIS}_{n,m,q,\sqrt{m}}$. If $2^m > q^n$, the function compresses and collisions exist by pigeonhole; their existence does not provide an efficient method to find them.

Proof. From $Ax = Ax' \pmod{q}$ we get $A(x - x') = 0 \pmod{q}$. Distinctness makes $z = x - x'$ nonzero, each coordinate of z is in $\{-1, 0, 1\}$, and therefore $\|z\|_2^2 \leq m$. The reduction simply runs the collision adversary and outputs z , so its success probability and essentially its running time are unchanged. Finally the domain has 2^m elements while the range has q^n elements; if $2^m > q^n$, pigeonhole guarantees some collision. $\square$

Theorem 4.3 (Worst-case hardness of random modular short solutions — verified deep input). *Micciancio–Regev’s Gaussian-measure framework shows that finding sufficiently short nonzero solutions to random modular linear equations is at least as hard, under explicit polynomial-time reductions and parameter maps, as approximating standard worst-case lattice problems within an almost-linear $n \log^{O(1)} n$ factor in the relevant theorem regimes. The source-owned reduction supplies the geometric hardness behind SIS; Theorem 4.2 supplies only the elementary cryptographic collision interface.*

Worked micro-example. Take $q = 5$, $n = 1$, $m = 3$, and $A = (1, 2, 4)$. The inputs $x = (1, 0, 1)$ and $x' = (0, 0, 0)$ do not collide because the image is $0 \pmod{5}$ only if $1 + 4 = 0 \pmod{5}$ —in fact they *do* collide, so $z = (1, 0, 1)$ is an SIS solution with norm $\sqrt{2}$. The example is intentionally tiny and has no security meaning.

4.3 Reader checkpoint

1. Verify that $\Lambda_q^\perp(A)$ is an additive subgroup of $\mathbb{Z}^m$.
2. Prove the norm bound $\|x - x'\|_2 \leq \sqrt{m}$ for binary inputs.
3. State the compression condition for h_A .
4. Explain why collision existence and collision finding are different claims.
5. Work out all collisions for the one-row toy matrix in the example.
6. Replace the Euclidean norm by ℓ_∞ and derive the corresponding collision bound.
7. Explain what changes in ISIS.
8. Explain why Module-SIS is not definitionally the same problem as SIS.
9. Identify which theorem is locally proved and which hardness theorem is source-owned.
10. Give the exact reduction output produced from a collision adversary.

Chapter 5

Learning With Errors: Reductions, Switching, and Toy Geometry

Learning With Errors replaces exact modular equations by noisy ones. This small perturbation supports public-key cryptography but also makes parameter bookkeeping central: the secret distribution, error distribution, modulus, sample count, and representation convention are part of the problem.

5.1 How to read this chapter

Definition 5.1 fixes search/decision LWE. Black Box 5.2 records Regev’s quantum worst-case reduction; Theorem 5.3 proves an elementary prime-modulus search-to-decision reduction; Black Box 5.4 records later classical worst-case reductions; and Theorem 5.5 proves deterministic rounding/noise accounting. Example 5.6 keeps all arithmetic small enough to inspect by hand.

5.2 Learning goals

A reader should be able to define the LWE distribution precisely, recover a uniform secret using a decision oracle in the prime-modulus model, propagate deterministic modulus-switching errors, and distinguish theorem-level worst-case reductions from concrete primal/dual attacks.

Definition 5.1 (Learning with errors). Fix modulus $q \geq 2$, dimension n , sample count m , a secret distribution S on $\mathbb{Z}_q^n$, and an integer error distribution χ reduced modulo q . An LWE sample with secret $s \leftarrow S$ is

$$(a, b), \quad a \leftarrow \mathbb{Z}_q^n, \quad e \leftarrow \chi, \quad b = \langle a, s \rangle + e \pmod{q}.$$

Search-LWE asks to recover s from independent samples. Decision-LWE asks to distinguish those samples from independent uniform pairs in $\mathbb{Z}_q^n \times \mathbb{Z}_q$. The *noise rate* is a declared scale such as $\alpha = \sigma/q$ when χ has width σ . A *normal form* changes the secret/error/sample representation through a proved reduction; it is not an informal license to replace a uniform secret by a binary one. *Primal* and *dual* attacks refer to lattice embeddings or short-dual-vector distinguishers attached to the declared parameters.

Theorem 5.2 (Regev’s quantum worst-case reduction to LWE — verified deep input). *In Regev’s standard prime-modulus/discretized-Gaussian formulation, when the noise and modulus satisfy the*

theorem's parameter inequalities (in particular the noise magnitude is large enough relative to $\sqrt{n}/q$), an efficient LWE solver yields a quantum polynomial-time algorithm for worst-case GapSVP and SIVP within $\tilde{O}(n/\alpha)$ -type approximation factors. The iterative discrete-Gaussian sampling argument and exact parameter losses remain source-owned.

Theorem 5.3 (Prime-modulus search-to-decision and secret randomization). *Let q be prime and let the secret be uniform in $\mathbb{Z}_q^n$. Suppose a decision algorithm on m samples has acceptance-probability gap at least $\delta > 0$ between LWE and uniform. Then, for any target failure probability $\eta > 0$, the secret can be recovered coordinate by coordinate using $O(nq\delta^{-2}\log(nq/\eta))$ independent decision experiments, each on a fresh m -sample batch. Moreover, for any known $t \in \mathbb{Z}_q^n$, replacing b by $b + \langle a, t \rangle$ changes secret s to $s + t$ without changing the error. Stronger binary/small-secret normal forms require additional reductions and are not claimed by this local proof.*

Proof. Fix coordinate i and candidate $g \in \mathbb{Z}_q$. For each input sample (a, b) choose independent uniform $r \in \mathbb{Z}_q$ and output

$$a' = a + re_i, \quad b' = b + rg.$$

If $g = s_i$, then $b' = \langle a', s \rangle + e$, so the transformed samples are LWE. If $g \neq s_i$, then $b' = \langle a', s \rangle + e + r(g - s_i)$. Conditional on a' , the value r remains uniform because $(a_i, r) \mapsto (a_i + r, r)$ is bijective; since q is prime, multiplication by nonzero $g - s_i$ permutes $\mathbb{Z}_q$. Thus b' is uniform and independent of a' . To make the statistical decision explicit, first estimate the algorithm's acceptance probability on independent uniform m -sample batches, and for each candidate g estimate its acceptance probability on independently transformed m -sample batches. The correct candidate differs from the uniform baseline by at least δ in absolute value, while every wrong candidate has exactly the uniform distribution. By Hoeffding's inequality, $O(\delta^{-2}\log(nq/\eta))$ repetitions per candidate estimate all nq tests to additive error below $\delta/3$ simultaneously except with probability at most η . The unique separated candidate is then s_i , and repeating over coordinates recovers s with the stated experiment bound.

For the secret-shift interface, use one common public shift t across every sample in a single LWE batch: $b + \langle a, t \rangle = \langle a, s + t \rangle + e$. If a reduction needs a freshly uniform secret from an arbitrary fixed secret, choose a fresh uniform t for that oracle invocation; then $s + t$ is exactly uniform. Independent amplified oracle trials use fresh shifts and fresh sample batches. $\square$

Theorem 5.4 (Classical worst-case reductions for LWE — verified deep input). *Peikert's 2009 work gave classical worst-case connections for appropriate LWE/shortest-vector variants, and Brakerski–Langlois–Peikert–Regev–Stehlé later established classical hardness of standard LWE with polynomial modulus under explicit parameter transformations. These theorems differ from Regev's quantum reduction in the lattice problem, modulus/noise regime, dimension tradeoff, and reduction architecture; the distinctions are retained rather than collapsed into “LWE is classically hard.”*

Theorem 5.5 (Deterministic modulus switching and rounding error). *Represent $a \in \mathbb{Z}_q^n$ and $b \in \mathbb{Z}_q$ by centered integer lifts and suppose $b = \langle a, s \rangle + e$ over the integers before reduction modulo q . For $p \geq 2$, set $a' = \lfloor (p/q)a \rfloor$ and $b' = \lfloor (p/q)b \rfloor$, coordinatewise for a . Write $\delta_a = a' - (p/q)a$ and $\delta_b = b' - (p/q)b$, so every coordinate of δ_a and δ_b has magnitude at most $1/2$. Then*

$$b' - \langle a', s \rangle = \frac{p}{q}e + \delta_b - \langle \delta_a, s \rangle,$$

and hence

$$|b' - \langle a', s \rangle| \leq \frac{p}{q}|e| + \frac{1 + \|s\|_1}{2}.$$

This is a correctness/noise identity; distribution-preserving LWE modulus-switching reductions require additional randomization and hypotheses.

Proof. Substitute $a' = (p/q)a + \delta_a$ and $b' = (p/q)b + \delta_b$, use $b - \langle a, s \rangle = e$, and apply the triangle inequality together with $|\delta_b| \leq 1/2$ and $|\langle \delta_a, s \rangle| \leq \|\delta_a\|_\infty \|s\|_1 \leq \|s\|_1/2$. $\square$

Example 5.6 (Complete toy LWE encryption and exhaustive attack geometry). Take $q = 17$, $n = 2$, secret $s = (3, 5)$, and samples

$$(a_1, b_1) = ((2, 7), 8), \quad (a_2, b_2) = ((6, 1), 7), \quad (a_3, b_3) = ((4, 9), 5),$$

which correspond to centered errors $(1, 1, -1)$. Exhaustive scoring over all 17^2 secrets by the sum of squared centered residuals identifies $(3, 5)$ as a best tiny-noise explanation. A toy Regev-style encryption can select a subset of samples, sum their a_i values into u , sum b_i values and add 0 or $\lfloor q/2 \rfloor$ into v , and decrypt by checking whether $v - \langle u, s \rangle$ is nearer 0 or $q/2$. The associated primal embedding and dual distinguisher are finite lattices that can be enumerated at these dimensions. None of these toy parameters has cryptographic security.

5.3 Reader checkpoint

1. Write the exact joint distribution of one LWE sample.
2. Verify the three toy errors in Example 5.6.
3. Prove that the candidate-randomization map in Theorem 5.3 preserves uniform a' .
4. Identify where primality of q is used in that proof.
5. Derive the secret-shift identity.
6. Explain why binary-secret LWE needs a separate reduction.
7. Re-derive the modulus-switching error identity symbolically.
8. Bound the rounding term when $s \in \{-1, 0, 1\}^n$.
9. Explain why deterministic rounding alone is not a distributional hardness reduction.
10. Contrast Regev's quantum reduction with the later classical reduction line.

Chapter 6

Ring-LWE, Module-LWE, and Structured-Lattice Encryption

Structure makes lattice cryptography compact and fast, but structure also changes the hardness question. Ring-LWE and Module-LWE are not ordinary LWE with faster multiplication: they expose algebraic symmetries that must be included in the assumption and attack model.

6.1 How to read this chapter

Definition 6.1 fixes a power-of-two cyclotomic coefficient model and module notation. Black Box 6.2 owns the deeper Ring-LWE worst-case reduction. Theorem 6.3 proves correctness and an explicit two-hybrid IND-CPA interface for a generic Module-LWE encryption core. Frontier 6.4 explains why structured attack evidence must remain dated.

6.2 Learning goals

A reader should be able to distinguish LWE, Ring-LWE, and Module-LWE instance spaces, expand a module-LWE decryption error exactly, run an IND-CPA hybrid under declared pseudorandomness assumptions, and explain why algebraic symmetry is both an efficiency source and an attack surface.

Definition 6.1 (Ring-LWE and Module-LWE in a fixed cyclotomic model). Let N be a power of two, $R := \mathbb{Z}[X]/(X^N + 1)$, and $R_q := R/qR$. Elements of R_q are represented by N coefficients modulo q , with multiplication reduced by $X^N = -1$. For module rank k , R_q^k is a free module and $R_q^{k \times k}$ denotes matrices over R_q . A Module-LWE sample has

$$A \leftarrow R_q^{k \times k}, \quad t = As + e \in R_q^k,$$

where s, e are sampled from declared short coefficient distributions; the decision problem distinguishes (A, t) from uniform. Rank $k = 1$ gives a ring-structured case. Ring-LWE worst-case theorems are naturally formulated using the canonical embedding, ideals, and the codifferent of the underlying number field: those algebraic objects are recalled inside the source-owned theorem when needed. An *algebraic attack surface* is any exploitable property arising from ring/module structure, automorphisms, subfields, special ideals, or representation choices.

Theorem 6.2 (Lyubashevsky–Peikert–Regev Ring-LWE reduction — verified deep input). *The LPR Ring-LWE theorem gives worst-case-to-average-case hardness reductions from approximate problems on ideal lattices in suitable number fields to the corresponding Ring-LWE distribution, with the canonical embedding/error distribution and algebraic dual/codifferent conventions explicit. Its worst-case problem is restricted to structured ideal lattices; it is not the unrestricted Euclidean-lattice theorem of ordinary LWE.*

Theorem 6.3 (Generic Module-LWE PKE correctness and IND-CPA hybrid). *Work in R_q above and encode each message coefficient $m_j \in \{0, 1\}$ as $\text{Enc}(m)_j = m_j \lfloor q/2 \rfloor$. Generate $A \leftarrow R_q^{k \times k}$, short s, e , and $t = As + e$. To encrypt m , sample independent short r, e_1, e_2 and set*

$$u = A^T r + e_1, \quad v = t^T r + e_2 + \text{Enc}(m).$$

Decrypt coefficientwise from the centered representative of $v - s^T u$. The error polynomial is exactly

$$\nu = e^T r + e_2 - s^T e_1.$$

If every coefficient of ν has magnitude $< q/4$ (with the obvious margin adjustment when q is odd), decryption is correct. If (H1) $(A, As + e)$ is computationally indistinguishable from (A, U) and (H2), for uniform t , the pair $(A^T r + e_1, t^T r + e_2)$ is computationally indistinguishable from uniform over $R_q^k \times R_q$, then the scheme is IND-CPA secure with advantage at most the sum of the two distinguishing advantages.

Proof. Expand $v - s^T u = (As + e)^T r + e_2 + \text{Enc}(m) - s^T (A^T r + e_1)$; commutativity of R_q cancels $s^T A^T r$, leaving $\text{Enc}(m) + \nu$. The $q/4$ coefficient margin makes nearest-half-modulus decoding unambiguous. For security, Game 0 is the real IND-CPA game. In Game 1 use H1 to replace public t by uniform, changing the adversary’s success probability by at most ϵ_1 . In Game 2 use H2 to replace the message-independent pair $(u, v - \text{Enc}(m))$ by uniform, changing it by at most ϵ_2 . Adding a fixed encoding to a uniform v leaves it uniform, so Game 2 contains no information about the challenge bit. The triangle inequality gives advantage at most $\epsilon_1 + \epsilon_2$. $\square$

Remark 6.4 (Structured-lattice hardness and symmetry audit — dated frontier, 2026-09-03). Ring/module structure reduces keys and accelerates arithmetic, but it creates algebraic symmetries absent from a random unstructured lattice. A current security review must therefore record the exact ring/module family, automorphism and subfield structure, representation, attack model, parameter set, and whether an attack is asymptotic, heuristic, or implemented. Hardness of unrestricted LWE/SIS is not silently transferred to a structured family; conversely, the absence of a known structural attack is not a reduction. This methodological frontier is frozen on the displayed date.

Worked micro-example. In $R_{17} = \mathbb{Z}_{17}[X]/(X^4 + 1)$, multiplication by X maps $a_0 + a_1X + a_2X^2 + a_3X^3$ to $-a_3 + a_0X + a_1X^2 + a_2X^3$. The operation is a signed cyclic rotation, illustrating the symmetry that yields fast structured arithmetic and must also be visible to cryptanalysis.

6.3 Reader checkpoint

1. Multiply two degree-three polynomials modulo $X^4 + 1$.
2. Explain how Module-LWE interpolates between ring and less-structured module settings.

3. Expand the decryption error ν in Theorem 6.3.
4. State the coefficientwise correctness margin.
5. Identify the two computational assumptions used in the IND-CPA hybrid.
6. Explain why the hybrid proof does not establish CCA security.
7. Distinguish unrestricted lattice hardness from ideal/module lattice hardness.
8. Give two examples of algebraic attack surfaces.
9. Explain why an automorphism can be useful both for implementation and for attacks.
10. State the epistemic rule enforced by Frontier 6.4.

Chapter 7

Trapdoors, Rejection Sampling, and Standardized Lattice Signatures

Lattice signatures follow two influential architectures. GPV-style hash-and-sign uses a trapdoor to sample short preimages, while Fiat–Shamir-with-aborts signatures use rejection sampling so that public transcripts reveal little about a short secret. Modern standards inherit ideas from these lines without being identical to the academic schemes.

7.1 How to read this chapter

Definition 7.1 fixes trapdoors, preimage sampling, rejection sampling, and signature security. Black Box 7.2 records GPV; Theorem 7.3 proves the exact rejection-sampling identity underlying abort techniques; the remaining black boxes separate ML-DSA, FN-DSA, and Wang-group compact-gadget research. The executable laboratory is referenced at the end.

7.2 Learning goals

A reader should be able to state what a lattice trapdoor enables, prove exact rejection sampling, explain why aborts can remove secret dependence from a proposal distribution, and distinguish a final FIPS standard from a selected algorithm whose FIPS is still in development.

Definition 7.1 (Lattice trapdoors and signatures). For a public linear map f_A such as $x \mapsto Ax \bmod q$, a *trapdoor* is secret auxiliary information allowing efficient inversion or short-preimage sampling while the public map remains one-way under the declared assumption. $\text{SamplePre}(A, T, y)$ denotes a trapdoor algorithm outputting a short x with $f_A(x) = y$ according to a declared near-Gaussian distribution. *Hash-and-sign* hashes a message to a target and uses the trapdoor to sample a short preimage. *Rejection sampling* proposes $X \sim \mu$ and accepts with a declared probability so that the conditional accepted distribution equals a target ν . *Fiat–Shamir with aborts* applies this idea to a commitment/challenge/response transcript. EUF-CMA and strong unforgeability use the definitions of Volume XVI; random-oracle and quantum-random-oracle models are always distinguished.

Theorem 7.2 (GPV trapdoor sampling — deep input). *Gentry–Peikert–Vaikuntanathan construct lattice trapdoor tools with preimage sampling and derive hash-and-sign signatures whose security is tied to worst-case lattice hardness in the random-oracle model. A core theorem samples a discrete*

Gaussian above a basis-dependent Gram–Schmidt threshold while making the output distribution essentially independent of the particular trapdoor geometry. The exact trapdoor generation and reduction parameters remain source-owned.

Theorem 7.3 (Exact rejection sampling and the abort principle). *Let μ, ν be distributions on a countable set Ω with $\nu(x) \leq M\mu(x)$ for every x and some $M \geq 1$. Propose $X \leftarrow \mu$ and accept $X = x$ with probability $\nu(x)/(M\mu(x))$ whenever $\mu(x) > 0$. Then the acceptance probability is exactly $1/M$ and the distribution of X conditioned on acceptance is exactly ν . Consequently, if a lattice-signature proposal distribution depending on a secret can be pointwise dominated and the accepted target ν is secret-independent, rejection sampling removes that dependence from the accepted sample. The full Fiat–Shamir/forking security reduction is a separate source-owned theorem.*

Proof. For every x ,

$$\Pr[X = x \text{ and accept}] = \mu(x) \frac{\nu(x)}{M\mu(x)} = \frac{\nu(x)}{M}.$$

Summing over x gives $\Pr[\text{accept}] = 1/M$. Dividing the joint probability by $1/M$ gives $\Pr[X = x \mid \text{accept}] = \nu(x)$. The cryptographic corollary is exactly this identity after verifying the domination bound for the concrete proposal/target pair. $\square$

Theorem 7.4 (ML-DSA/FIPS 204 interface — verified official standard). *FIPS 204, published 13 August 2024, specifies ML-DSA, a module-lattice digital-signature standard derived from the CRYSTALS-Dilithium design line. This volume treats the FIPS algorithms, parameter sets, validation conditions, and current errata as an official specification interface; it does not identify the standard byte-level algorithms with the abstract rejection-sampling theorem above.*

Theorem 7.5 (FN-DSA/Falcon status and trapdoor-sampling interface — verified official status). *Falcon was selected by NIST for standardization as FN-DSA, but as of 4 September 2026 FIPS 206 remains in development. Its design uses NTRU-lattice trapdoors and Gaussian sampling with implementation-sensitive numerical requirements. Accordingly this row records both the mathematical interface and the dated standardization state; it does not call an unpublished final FIPS “final.”*

Theorem 7.6 (Compact lattice gadget and hash-and-sign signatures — verified deep input). *Yu, Jia, and Xiaoyun Wang’s 2023 compact lattice-gadget work develops a smaller gadget/trapdoor interface and applications to hash-and-sign signatures. The concrete construction, norm bounds, and security reduction are retained as a primary- source deep input and are not inferred from GPV by analogy.*

Remark 7.7 (Lattice and PQC parameter laboratory). The executable file `code/volume_20/1 attice_pqc_lab.py` performs exact rational Gram–Schmidt checks, a toy SIS collision search, the toy LWE residual attack from Chapter 5, modulus-switching identity checks, finite rejection-sampling verification, and parsing of synthetic .rsp-style test-vector records. It deliberately does not reimplement ML-KEM, ML-DSA, FN-DSA, or any production cryptography.

7.3 Reader checkpoint

1. Explain what information a lattice trapdoor adds to a public linear map.
2. State the distributional goal of GPV preimage sampling.
3. Prove the acceptance probability in Theorem 7.3.
4. Explain why pointwise domination is necessary for the stated sampler.
5. Distinguish rejection sampling from simply discarding “large” signatures.
6. Explain which part of Fiat–Shamir security is not proved by Theorem 7.3.
7. State the publication date and status class of FIPS 204.
8. State the 4 September 2026 status of FIPS 206.
9. Explain why Falcon’s numerical Gaussian sampler is implementation-sensitive.
10. Run the laboratory and identify which checks are exact rather than Monte Carlo.

Chapter 8

Code-Based KEMs, Decoding Hardness, and Decryption Failures

Code-based post-quantum cryptography has a different hardness substrate from lattices. Its canonical public-key forms are generator-matrix (McEliece) and parity-check/syndrome (Niederreiter) systems, while modern compact proposals introduce quasi-cyclic structure and decryption-failure phenomena that require their own analysis.

8.1 How to read this chapter

Definition 8.1 fixes decoding and KEM terminology. Black Box 8.2 records the classical NP-completeness theorem; Theorem 8.3 proves the elementary McEliece/Niederreiter correctness and syndrome duality; the final two black boxes separate HQC’s NIST selection status from a concrete BIKE weak-key/decryption-failure attack.

8.2 Learning goals

A reader should be able to derive a syndrome equation, explain the algebraic duality of McEliece and Niederreiter representations, distinguish generic syndrome-decoding hardness from structural-code security, and explain why a tiny decryption-failure probability can become a cryptanalytic oracle.

Definition 8.1 (Code-based KEMs and decoding assumptions). For a binary linear $[n, k, d]$ code $C \subseteq \mathbb{F}_2^n$, a generator matrix $G \in \mathbb{F}_2^{k \times n}$ has row space C , and a parity-check matrix $H \in \mathbb{F}_2^{(n-k) \times n}$ satisfies $HG^T = 0$. The *syndrome* of y is Hy^T . Syndrome decoding asks, given (H, s, t) , for an error e of Hamming weight at most t with $He^T = s$. Information-set decoding is the broad attack family that guesses coordinates expected to contain sufficiently few errors and solves the induced linear system. Binary Goppa, QC-MDPC, and QC-LDPC name specific structured code families and are not interchangeable. A *decryption-failure rate* (DFR) is the probability, under a declared key and ciphertext distribution, that legitimate decapsulation fails. A *weak key* is a key class with atypical security or failure behavior. KEM security uses the definitions of Volume XVI.

Theorem 8.2 (Syndrome decoding hardness — verified deep input). *Berlekamp, McEliece, and van Tilborg proved in 1978 that the general decoding problem for linear codes and the associated weight problem are NP-complete. This is a generic worst-case complexity theorem. It does not prove that*

every structured public code family or every chosen parameter set has a specified concrete security level.

Theorem 8.3 (McEliece/Niederreiter correctness and syndrome duality). *Let C have generator G , parity check H , and an efficient decoder correcting every error of weight at most t . In the idealized McEliece form, encryption sends message $m \in \mathbb{F}_2^k$ to $c = mG + e$ with $\text{wt}(e) \leq t$; decoding c recovers mG and hence m when G has rank k . In the idealized Niederreiter form, an error e of weight at most t is represented by its syndrome $y = He^T$ and recovered by syndrome decoding. For a McEliece ciphertext,*

$$Hc^T = H(mG + e)^T = He^T.$$

Thus the error component of the generator-matrix form maps exactly to the syndrome form. This algebraic identity does not by itself prove indistinguishability of a disguised structured code from a random code.

Proof. The decoder corrects $c = mG + e$ to the unique codeword mG in the declared radius; full row rank of G then determines m . Niederreiter correctness is precisely the promise that the syndrome decoder returns the unique declared error. Finally $HG^T = 0$, so $H(mG)^T = 0$ and the displayed syndrome equality follows. Structural hiding and CCA/KEM transforms are separate security layers. $\square$

Theorem 8.4 (HQC design and NIST selection status — verified official status). *HQC is a code-based KEM using quasi-cyclic structure and decoding/noise assumptions. NIST selected HQC for standardization in March 2025. As of 4 September 2026 it is selected and standardization work is underway, but it is not one of the final FIPS 203/204/205 standards. The scheme’s complete security proof and parameter analysis remain source/standardization owned.*

Theorem 8.5 (Wang-group BIKE weak keys and decryption-failure attacks — verified deep input). *Tianrui Wang, Anyu Wang, and Xiaoyun Wang identify a new class of weak keys for BIKE and exploit decryption-failure behavior in key-recovery attacks. The result is tied to the BIKE construction, parameter/failure model, and oracle conditions analyzed in the primary paper; it is not a generic break of code-based cryptography or of HQC.*

Worked micro-example. For $G = (I_k \mid P)$ and $H = (P^T \mid I_{n-k})$ over $\mathbb{F}_2$, direct multiplication gives $HG^T = 0$ because the two copies of P^T cancel. Adding an error e to a codeword changes the syndrome from zero to He^T and nothing else.

8.3 Reader checkpoint

1. Verify $HG^T = 0$ for the systematic matrices in the example.
2. Derive the syndrome of a McEliece ciphertext.
3. State what a unique decoder guarantees at radius t .
4. Distinguish decoding NP-completeness from average-case structured-code hardness.
5. Explain the idea of information-set decoding.
6. Define a decryption-failure rate with its probability space.

7. Explain how repeated failure observations can leak key information.
8. State HQC's status as of 4 September 2026.
9. Explain why the BIKE attack does not imply an HQC break.
10. Identify which layers still require a KEM transform beyond the algebraic theorem.

Chapter 9

Hash-Based Signatures and the Merkle Composition Principle

Hash-based signatures avoid lattice and code assumptions, but they require careful composition of one-time/few-time signatures, authentication trees, and domain-separated hashes. The Merkle tree is the core reusable proof idea; modern stateless hypertree designs add many layers beyond it.

9.1 How to read this chapter

Definition 9.1 fixes one-time, stateful, stateless, and Merkle terminology. Theorem 9.2 proves a bounded-use composition theorem. Black Box 9.3 records the SLH-DSA/FIPS 205 interface, and Frontier 9.4 freezes the separate NIST additional-signature process at its current Round-3 state.

9.2 Learning goals

A reader should be able to verify a Merkle authentication path, reduce a forged tree signature to either an underlying one-time forgery or a hash collision, distinguish stateful and stateless designs, and distinguish the nine candidates initially advanced by NIST from the eight still active in Round 3 Round-3 signature candidates without confusing them with final standards.

Definition 9.1 (Hash-based signatures). A one-time signature (OTS) key pair is intended to sign at most one message; a few-time signature permits a bounded number. A *Merkle tree* places verification-key commitments at leaves and labels each internal node by a collision-resistant hash of its two ordered children with domain separation. An *authentication path* is the sibling label at each level needed to recompute the root from one leaf. A *stateful* hash-based scheme must prevent reuse of a leaf/OTS state; XMSS and LMS are canonical standardized examples. A *stateless* design chooses paths without persistent signer state and may use hypertrees and few-time components such as FORS. *Multi-target security* accounts for an adversary attacking many keys/nodes/targets simultaneously rather than silently applying a single-target bound.

Theorem 9.2 (Merkle-tree signature composition). *Let a height- h binary Merkle tree commit to 2^h OTS verification keys. Suppose each leaf is used at most once, leaf/internal-node hashing is domain separated, the OTS is one-time EUF-CMA secure, and the tree hash is collision resistant. Any successful forgery against the bounded-use Merkle signature either (i) forges the OTS under a leaf key whose signing right was not validly used for that forged message, or (ii) yields two different*

ordered child inputs with the same internal-node hash. Quantitatively, for reductions with matching time up to tree simulation,

$$\text{Adv}_{\text{Merkle}} \leq 2^h \text{Adv}_{\text{OTS}} + \text{Adv}_{\text{CR}}. \quad (9.1)$$

The factor 2^h is only the random-leaf embedding loss; the hash-collision branch requires no additional internal-node guess.

Proof. Partition a successful forgery by its attacked leaf j . For the OTS branch, embed the one-time-signature challenge verification key at a uniformly chosen leaf $J \in [2^h]$ and generate all other leaves honestly. If the signing algorithm ever legitimately uses leaf J , forward that single message to the OTS signing oracle; otherwise make no signing query for the challenge key. Conditioned on a successful forgery whose authentication path contains the honest verification key at leaf j , the event $J = j$ has probability 2^{-h} , and the forged message/signature pair is a valid OTS forgery. Thus the total probability of this branch is at most $2^h \text{Adv}_{\text{OTS}}$.

In the other branch, compute the entire honest path for the forged leaf and the path supplied by the forgery. Their leaf/path data differ but their final root is the same. Move upward from the first differing value and choose the first level at which the two path computations merge. At that level two distinct domain-separated ordered child encodings hash to the same parent. This is a collision extracted *deterministically* from the forgery, so no random guess of a level or internal node is needed. Hence this branch is bounded directly by Adv_{CR} . Adding the disjoint branches proves (9.1). $\square$

Theorem 9.3 (SLH-DSA/FIPS 205 interface — verified official standard). *FIPS 205, published 13 August 2024, specifies SLH-DSA, derived from SPHINCS+, as a stateless hash-based digital-signature standard. Its hypertree, WOTS+, FORS, address/domain-separation rules, parameter sets, and byte-level algorithms are an official-standard interface. Theorem 9.2 supplies only the elementary Merkle composition idea, not a replacement proof of the complete standard.*

Remark 9.4 (NIST additional post-quantum signatures — dated frontier, 2026-09-04). NIST IR 8610 advanced nine additional-signature candidates to Round 3 on 14 May 2026: FAEST, HAWK, MAYO, MQOM, QR-UOV, SDitH, SNOVA, SQIsign, and UOV. On 29 July 2026 the HAWK team withdrew HAWK from consideration following disclosure of a mathematical vulnerability. Thus, as of 4 September 2026, eight candidates remain active in Round 3: FAEST, MAYO, MQOM, QR-UOV, SDitH, SNOVA, SQIsign, and UOV. All are evaluation candidates, not final FIPS standards. The historical nine-candidate selection and the current eight-candidate ledger are deliberately kept distinct.

Worked micro-example. In a height-two Merkle tree, four leaf commitments produce two level-one hashes and one root. A signature for leaf 2 contains its OTS signature, the sibling leaf-3 commitment, and the hash of leaves 0–1. Recomputing exactly two internal hashes reaches the root.

9.3 Reader checkpoint

1. Draw a height-three Merkle authentication path.
2. Count the number of sibling hashes in a height- h path.
3. Identify the two bad events in Theorem 9.2.
4. Explain why leaf reuse violates the theorem’s premise.

5. Explain the role of domain separation in tree hashing.
6. Distinguish a stateful XMSS/LMS deployment from stateless SLH-DSA.
7. Explain why the elementary Merkle proof is not the full SPHINCS+/SLH-DSA proof.
8. List the nine candidates initially advanced to Round 3, identify HAWK as withdrawn, and list the eight active candidates as of 4 September 2026.
9. Explain why “Round 3” is not “standardized.”
10. Give one place where multi-target loss enters a hash-based security reduction.

Chapter 10

PQC Standards, Transform Layers, and Cryptanalytic Status

A post-quantum scheme lives simultaneously in several layers: a mathematical assumption, a security reduction, a concrete parameter set, an implementation, and a standardization state. The final chapter makes these layers explicit, then records ML-KEM, the Fujisaki–Okamoto transform interface, the SIKE break, a lattice-symmetry case study, and the current NIST status ledger.

10.1 How to read this chapter

Definition 10.1 defines the status vocabulary before any dated claim is made. Theorem 10.2 owns the final FIPS 203 interface; Theorem 10.3 owns the deep FO transform; Black Boxes 10.4–10.5 are cryptanalytic case studies; and Frontier 10.6 is frozen to 4 September 2026. Figure 10.1 summarizes the assumption-to-standard map without drawing false reduction arrows.

10.2 Learning goals

A reader should be able to separate scheme design from a final standard, state the purpose of an FO-style transform, explain why the SIKE break is an isogeny case study rather than a lattice break, and read a standards map without confusing selection, draft, final publication, and migration guidance.

Definition 10.1 (PQC standards and cryptanalytic status). A *design* is an algorithmic proposal; a *proof* is a theorem in a declared security model; a *concrete parameter claim* maps attacks and reductions to a claimed work factor; an *implementation* realizes algorithms in software/hardware and adds side-channel/fault concerns; *selection* is an official decision to standardize or advance an algorithm; a *draft standard* is a published draft open to revision; a *final standard* is a final issued document such as a FIPS; *certification/validation* concerns conformance under a designated program; and *migration guidance* concerns deployment policy. A *cryptanalytic break* states exactly which security goal, parameter/version, oracle model, complexity, and implementation assumptions are violated. Status claims in this chapter carry an explicit date.

Theorem 10.2 (ML-KEM/FIPS 203 specification interface — verified official standard). *FIPS 203, final on 13 August 2024, specifies ML-KEM with parameter sets ML-KEM-512, ML-KEM-768,*

and ML-KEM-1024. The standard fixes byte encodings, key generation, encapsulation, decapsulation, validation conditions, and the complete algorithm interface. NIST subsequently published an errata/planning note indicating issues to be corrected in a future revision. This row therefore treats the current FIPS plus official errata as the normative interface rather than reconstructing production ML-KEM from the generic PKE theorem of Chapter 6.

Theorem 10.3 (Fujisaki–Okamoto-style KEM transform — verified deep input). *Fujisaki–Okamoto-style transforms convert an appropriate public-key encryption core into a CCA-secure encryption/KEM interface in a random-oracle setting by deriving coins/keys from hashed inputs and checking ciphertext consistency by re-encryption or an equivalent validity mechanism. Modern modular analyses quantify the required one-way/IND-CPA assumptions, random-oracle queries, multi-user losses, and the effect of decryption failures; implicit-rejection variants return a pseudo-random fallback key rather than exposing a validity bit. The complete reduction, including failure terms, is source-owned rather than compressed into a one-paragraph proof.*

Theorem 10.4 (SIKE/SIDH key-recovery break — verified deep input). *Castryck and Decru’s 2022/2023 attack gives an efficient key-recovery attack on SIDH and thereby invalidates the SIDH/SIKE hardness basis targeted by the attack. The construction uses deep isogeny/arithmetic-geometry structure and is retained as a source-owned post-quantum case study. No companion-algebraic-geometry theorem is needed as a load-bearing QRCC claim: the reader-facing conclusion is only the exact cryptanalytic status of the affected isogeny design.*

Theorem 10.5 (Lattice-automorphism randomization reductions — verified deep input). *Jiang, Anyu Wang, Luo, Liu, Yu, and Xiaoyun Wang study randomization and the automorphism problem arising from symmetry of $\mathbb{Z}^n$, giving reductions/algorithms that clarify how lattice automorphisms can change algorithmic search spaces. The exact reduction statements and parameter regimes remain source-owned. The result is included to reinforce the Volume-XX rule that algebraic symmetry is a cryptanalytic object, not merely an implementation convenience.*

Remark 10.6 (NIST PQC status ledger — dated frontier, 2026-09-04). On 4 September 2026, NIST lists FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) as final standards from 13 August 2024. FN-DSA/FIPS 206 remains in development. HQC has been selected for standardization but is not a final FIPS. NIST initially advanced nine additional-signature candidates to Round 3 in May 2026, but HAWK was withdrawn on 29 July 2026; eight candidates therefore remain active, as recorded in Frontier 9.4. These are status facts, not mathematical reductions, and the row must be refreshed at the next canonical release if NIST changes any of them.

Volume closure. Volume XX now has one discipline across geometry, reductions, schemes, attacks, and standards: every arrow has a type. Worst-case-to-average-case reductions are theorem arrows; scheme security proofs are game reductions; concrete estimates are attack-model calculations; implementation claims are engineering evidence; and NIST labels are dated administrative states. Keeping those layers separate is the main defense against overclaiming in post-quantum cryptography.

10.3 Reader checkpoint

1. Distinguish selection, draft standard, and final standard.

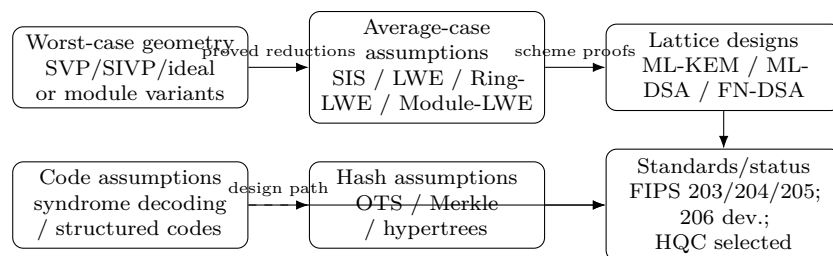


Figure 10.1: Post-quantum assumption and standardization map. Solid arrows marked “proved reductions” are theorem-level interfaces; arrows into the standards box denote design/selection lineage, not reductions between hardness assumptions. The SIKE/SIDH break is deliberately omitted from the surviving-standard arrows and discussed as a cryptanalytic case study.

2. Name the three final NIST PQC FIPS documents in this volume.
3. List the three ML-KEM parameter-set names.
4. Explain the role of re-encryption or validity checking in an FO-style transform.
5. Explain why decryption failures must appear in a modern FO analysis.
6. State exactly what the Castryck–Decru case study breaks.
7. Explain why that break is not a lattice-cryptography break.
8. State the 4 September 2026 status of FN-DSA and HQC.
9. Explain the difference between a reduction arrow and a standardization-lineage arrow in Figure 10.1.
10. Give a checklist for updating Frontier 10.6 at the next release.

Internalization and Reconstruction Guide: Volume XX

Purpose. Connect lattice geometry to average-case assumptions and post-quantum schemes while tracking norms, discrete-Gaussian parameters, algebraic structure, decryption failure, reductions, attacks, and standardization status separately.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-XVI-THM-003: Game-hopping and reduction accounting lemma.
2. QRCC-XVI-DEF-028: Chosen-ciphertext security and KEM/DEM composition.
3. QRCC-III-ALG-017: LLL basis-reduction algorithm.
4. QRCC-XVI-DEF-023: Authentication and signature security.
5. QRCC-III-THM-001: Classification of finite fields.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XX-DEF-001 (Lattice geometry and computational problems); QRCC-XX-DEF-005 (Discrete Gaussian and smoothing parameter); QRCC-XX-DEF-009 (Worst-case/average-case lattice reductions); QRCC-XX-DEF-011 (SIS and q-ary lattices); QRCC-XX-DEF-014 (Learning with errors).

Chapter retrieval map

- Ch. 1: Lattice Geometry, Duality, and the Shortest-Vector Landscape:** enter through QRCC-XX-DEF-001 (Lattice geometry and computational problems); exit by being able to state QRCC-XX-BB-003 (Classical and heuristic SVP algorithms) with its ownership boundary.
- Ch. 2: Discrete Gaussians, Smoothing, and Lattice Sampling:** enter through QRCC-XX-DEF-005 (Discrete Gaussian and smoothing parameter); exit by being able to state

QRCC-XX-BB-007 (Efficient discrete Gaussian sampling over lattices) with its ownership boundary.

- Ch. 3: Worst-Case to Average-Case Reductions:** enter through QRCC-XX-DEF-009 (Worst-case/average-case lattice reductions); exit by being able to state QRCC-XX-BB-010 (Ajtai worst-case to average-case lattice reduction) with its ownership boundary.
- Ch. 4: Short Integer Solutions and Collision-Resistant Hashing:** enter through QRCC-XX-DEF-011 (SIS and q-ary lattices); exit by being able to state QRCC-XX-BB-013 (Worst-case hardness of SIS) with its ownership boundary.
- Ch. 5: Learning With Errors: Reductions, Switching, and Toy Geometry:** enter through QRCC-XX-DEF-014 (Learning with errors); exit by being able to state QRCC-XX-BB-015 (Regev quantum reduction to LWE) with its ownership boundary.
- Ch. 6: Ring-LWE, Module-LWE, and Structured-Lattice Encryption:** enter through QRCC-XX-DEF-019 (Ring-LWE and Module-LWE); exit by being able to state QRCC-XX-BB-020 (Ring-LWE worst-case reduction) with its ownership boundary.
- Ch. 7: Trapdoors, Rejection Sampling, and Standardized Lattice Signatures:** enter through QRCC-XX-DEF-023 (Lattice trapdoors and signatures); exit by being able to state QRCC-XX-BB-024 (GPV trapdoors and hash-and-sign signatures) with its ownership boundary.
- Ch. 8: Code-Based KEMs, Decoding Hardness, and Decryption Failures:** enter through QRCC-XX-DEF-029 (Code-based KEMs and decoding assumptions); exit by being able to state QRCC-XX-BB-030 (Syndrome decoding hardness) with its ownership boundary.
- Ch. 9: Hash-Based Signatures and the Merkle Composition Principle:** enter through QRCC-XX-DEF-034 (Hash-based signatures); exit by being able to state QRCC-XX-BB-036 (SLH-DSA/SPHINCS+ security interface) with its ownership boundary.
- Ch. 10: PQC Standards, Transform Layers, and Cryptanalytic Status:** enter through QRCC-XX-DEF-038 (PQC standards and cryptanalytic status); exit by being able to state QRCC-XX-BB-041 (SIKE key-recovery break as a post-quantum case study) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XX-BB-003 — Classical and heuristic SVP algorithms. Dependencies: QRCC-XX-DEF-001, QRCC-III-ALG-017. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
2. QRCC-XX-BB-010 — Ajtai worst-case to average-case lattice reduction. Dependencies: QRCC-XX-DEF-009. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-XX-BB-015 — Regev quantum reduction to LWE. Dependencies: QRCC-XX-DEF-014, QRCC-XX-THM-006. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

4. QRCC-XX-BB-024 — GPV trapdoors and hash-and-sign signatures. Dependencies: QRCC-XX-BB-007, QRCC-XX-DEF-023. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XX-BB-036 — SLH-DSA/SPHINCS+ security interface. Dependencies: QRCC-XX-THM-035, QRCC-XVIII-DEF-001. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XX-F-022 — Structured-lattice hardness and symmetry audit. Dependencies: QRCC-XX-DEF-019. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- mixing Euclidean and other lattice norms inside a reduction;
- forgetting the smoothing/Gram–Schmidt threshold in discrete-Gaussian sampling;
- treating plain LWE, Ring-LWE, and Module-LWE assumptions as automatically equivalent;
- confusing standardization status, parameter security, and worst-case reduction guarantees;

Boundary and degenerate cases

- rank one and determinant-one toy lattices;
- error width near decryption-failure thresholds;
- prime versus composite modulus;
- heuristic lattice attack costs versus reduction-theoretic hardness;

Retrieval and reconstruction drill

1. State QRCC-XX-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XX-BB-003. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XX-BB-010, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.

6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XX-F-022, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XXI layers interactive proof, zero-knowledge, MPC, and succinct-proof machinery on top of the cryptographic foundations.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XXI

Zero Knowledge, Secure Computation, and Succinct Proofs

Chapter 1

Commitments: Hiding, Binding, and Setup

Commitment schemes are the first place where “hiding” and “binding” must be treated as distinct quantified experiments. This chapter fixes that distinction and gives two canonical constructions with very different setup assumptions.

1.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

1.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 1.1 (Commitment schemes and their security). A bit-commitment scheme has a commit phase producing a transcript c and an opening phase revealing a bit b and decommitment data d . Correctness means honest openings verify. Hiding means the commit-phase views for $b = 0$ and $b = 1$ are identical, statistically close, or computationally indistinguishable, according to the declared level. Binding means that no unbounded or polynomial-time committer, according to the declared level, can produce one commitment admitting accepting openings to both bits except with the stated probability. Interactive commitments include the receiver’s coins in the transcript. Equivocation and extraction are simulator capabilities and are not part of ordinary hiding/binding unless stated. Any common-reference-string or group-generation assumption is part of the scheme definition.

Theorem 1.2 (Naor-style commitment from a pseudorandom generator). *Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ be a pseudorandom generator. In the protocol where the receiver samples uniform $R \in \{0, 1\}^{3n}$, the committer samples $s \in \{0, 1\}^n$ and sends $C = G(s)$ for $b = 0$ and $C = G(s) \oplus R$ for $b = 1$, the commitment is computationally hiding and, except with probability at most 2^{-n} over R , perfectly binding. Consequently a one-way permutation with a hard-core predicate yields such a commitment after the standard iterated hard-core PRG construction.*

Proof. For hiding, replace $G(s)$ by a uniform $U \in \{0, 1\}^{3n}$. Both U and $U \oplus R$ are uniform, so any distinguisher between the two commitments distinguishes $G(s)$ from uniform with the same advantage. For binding, a double opening exists exactly when $R = G(s_0) \oplus G(s_1)$ for some pair (s_0, s_1) . There are at most 2^{2n} such differences among 2^{3n} possible R , hence a uniform receiver string is bad with probability at most 2^{-n} . Conditioned on a good R , no commitment has openings to both bits. The final sentence uses the earlier QRCC hard-core-bit interface to obtain a length-expanding PRG from a one-way permutation. $\square$

Theorem 1.3 (Pedersen commitment). *Let $G = \langle g \rangle$ be a prime-order group of order q , let $h = g^x$ for unknown $x \in \mathbb{Z}_q$, and define $\text{Com}(m; r) = g^m h^r$. For $m, r \in \mathbb{Z}_q$ the scheme is perfectly hiding. A polynomial-time adversary producing distinct openings $(m, r) \neq (m', r')$ to the same commitment computes $\log_g h$ unless $r = r'$, in which case equality forces $m = m'$.*

Proof. For fixed m , the map $r \mapsto m + xr$ permutes $\mathbb{Z}_q$, so g^{m+xr} is uniform in G . For two distinct openings, $g^{m-m'} = h^{r'-r} = g^{x(r'-r)}$. If $r' \neq r$, division in $\mathbb{Z}_q$ gives $x = (m - m')(r' - r)^{-1}$. If $r' = r$, prime order gives $m = m'$, contrary to distinct messages. Thus binding is computational under discrete-log hardness and fails information-theoretically for a party knowing x . $\square$

1.3 Reader checkpoint

1. Write the hiding and binding experiments separately.
2. Repeat the Naor binding count for stretch $2n + k$.
3. Explain why the receiver must choose R after the PRG is fixed.
4. Show that Naor hiding does not require R to be secret.
5. Prove Pedersen perfect hiding by an explicit bijection on randomness.
6. Extract x from two Pedersen openings.
7. Explain why a trapdoor setup party can equivocate Pedersen commitments.
8. State the subgroup-validation condition needed in a non-prime ambient group.
9. Compare statistical binding with computational binding.
10. Give one protocol use where equivocation is useful to a simulator.

Chapter 2

Sigma Protocols and Partial Knowledge

Three-move public-coin protocols provide a compact algebraic interface for identification, proofs of knowledge, and later Fiat–Shamir transforms.

2.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

2.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 2.1 (Sigma protocols). For relation R , a Sigma protocol has prover message a , uniformly sampled public challenge e from a finite challenge set E , response z , and deterministic verification predicate $V(x, a, e, z)$. It has completeness if honest transcripts accept, special soundness if two accepting transcripts with the same a and distinct challenges efficiently yield a witness, and special honest-verifier zero knowledge if, for each prescribed challenge, a simulator samples an accepting transcript with the honest conditional distribution. The knowledge error is the maximum success probability that can be explained without enabling extraction under the selected extractor theorem.

Theorem 2.2 (Schnorr Sigma protocol). Let $G = \langle g \rangle$ have prime order q , statement $y = g^w$, challenge $e \in E \subseteq \mathbb{Z}_q$, and protocol $a = g^r$, $z = r + ew \pmod{q}$, accepting when $g^z = ay^e$. The protocol has perfect completeness, special soundness, and perfect special honest-verifier zero knowledge. For distinct e, e' , extraction returns $w = (z - z')(e - e')^{-1} \pmod{q}$.

Proof. Completeness is $g^{r+ew} = g^r(g^w)^e$. From two accepting transcripts, division gives $g^{z-z'} = y^{e-e'} = g^{w(e-e')}$, and invertibility of $e - e'$ modulo prime q gives the formula. To simulate a prescribed e , choose uniform $z \in \mathbb{Z}_q$ and set $a = g^z y^{-e}$. This is exactly the honest distribution because $r = z - ew$ is uniform. $\square$

Theorem 2.3 (Selected AND/OR composition). *For Sigma protocols with a common finite-field challenge space, parallel AND composition preserves perfect completeness and special honest-verifier zero knowledge, and has special soundness by componentwise extraction. The standard two-branch OR composition proves knowledge of a witness for at least one branch: simulate the unknown branch at a chosen subchallenge, run the known branch honestly, and choose subchallenges summing to the verifier’s global challenge. Two accepting transcripts with the same first message and distinct global challenges yield a witness for at least one branch.*

Proof. The AND claims follow by products of honest distributions and component extractors. For OR, write $e = e_0 + e_1$. A prover knowing branch 0 chooses e_1 first and simulates branch 1, commits honestly in branch 0, receives e , sets $e_0 = e - e_1$, and answers branch 0. The simulated transcript is distributed as an honest-verifier transcript because one subchallenge is uniform and determines the other. For two global challenges, at least one branch subchallenge differs; the corresponding repeated first message and distinct challenges trigger that branch’s special-soundness extractor. $\square$

2.3 Reader checkpoint

1. Verify Schnorr completeness algebraically.
2. Derive the Schnorr extractor.
3. Write the exact Schnorr simulator.
4. Explain why prime order matters for extraction.
5. Prove AND-composition completeness.
6. Construct the two-branch OR transcript distribution.
7. Show why one branch challenge must differ across two distinct global challenges.
8. Generalize OR composition to k branches.
9. Distinguish witness indistinguishability from zero knowledge.
10. Explain why Fiat–Shamir has not yet been invoked.

Chapter 3

Zero Knowledge and Simulation

Zero knowledge is a statement about distributions of verifier views, not about an informal absence of leaked facts. The simulator and its running-time convention must therefore be part of every theorem.

3.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

3.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 3.1 (Zero knowledge and simulation). An interactive proof for language L is perfect, statistical, or computational zero knowledge if for every probabilistic polynomial-time verifier V^* and every polynomial-length auxiliary input, there is an expected-polynomial-time simulator whose output distribution on yes instances is respectively identical, statistically close, or computationally indistinguishable from the complete view of V^* . The view contains its input, auxiliary input, random tape, and received messages. Sequential and concurrent executions are different models and are named explicitly.

Theorem 3.2 (Perfect zero knowledge for graph isomorphism). *Given isomorphic graphs G_0, G_1 on the same vertex set and an isomorphism $\phi : G_0 \rightarrow G_1$, the protocol that sends a random relabeling $H = \pi(G_0)$, receives challenge $b \in \{0, 1\}$, and answers with an isomorphism $G_b \rightarrow H$ has perfect completeness, soundness error $1/2$ against nonisomorphic inputs, and perfect zero knowledge against the honest verifier.*

Proof. Completeness is immediate. If G_0, G_1 are nonisomorphic, one fixed H cannot be isomorphic to both, so at most one challenge can be answered. For honest-verifier zero knowledge, choose uniform b and uniform isomorphism σ from G_b to a random relabeling $H = \sigma(G_b)$, and output (H, b, σ) . Because G_0 and G_1 are isomorphic, this equals the honest transcript distribution exactly. $\square$

Theorem 3.3 (GMW: NP has computational zero knowledge from one-way functions — verified deep input). *Assuming one-way functions, every language in NP has a computational zero-knowledge interactive proof with negligible soundness after polynomial repetition. The claim is auxiliary-input computational zero knowledge in the classical interactive model; the generic commitment/coloring compiler and its full malicious-verifier simulation remain source-owned.*

Theorem 3.4 (Sequential composition). *A polynomial number of sequential executions of computational zero-knowledge protocols remains computational zero knowledge, provided each invocation's simulator is valid with arbitrary polynomial auxiliary input and expected polynomial time; the composed simulator runs the component simulators in order, passing the evolving simulated view as auxiliary state.*

Proof. Consider hybrids H_j in which the first j sessions are simulated and the remaining sessions are real. If adjacent hybrids were distinguishable with non-negligible advantage, fixing the preceding transcript as auxiliary input would distinguish a single real session from its simulator. Summing a polynomial number of negligible adjacent gaps remains negligible. Expected running times add over a polynomial number of sessions. The argument is inherently sequential: in a concurrent schedule the auxiliary state may depend on messages from overlapping unfinished sessions, so this proof does not establish concurrent composition. $\square$

3.3 Reader checkpoint

1. Define the verifier view explicitly.
2. Prove the graph-isomorphism soundness bound.
3. Show equality of the honest transcript and simulator distributions.
4. Explain why this theorem is HVZK as stated.
5. State the GMW assumption exactly.
6. Describe the role of commitments in the GMW architecture.
7. Construct the sequential-composition hybrids.
8. Prove that a polynomial sum of negligible functions is negligible.
9. Explain the auxiliary-input requirement.
10. Give a concurrent schedule that breaks the sequential proof architecture.

Chapter 4

Knowledge Extraction and Repetition

Knowledge soundness strengthens acceptance soundness by requiring an extractor. Rewinding is an algorithmic resource, and its success probability must be accounted for.

4.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

4.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 4.1 (Proofs of knowledge and extraction). For relation R , a proof of knowledge has an extractor that, given black-box access to a prover convincing the verifier with probability ε above a declared knowledge error κ , outputs w with $(x, w) \in R$ in expected polynomial time in the security parameter and in an appropriate inverse success gap. Rewinding means restoring the prover to an earlier state while changing verifier randomness. Simulation extractability is a stronger noninteractive notion and is not implied by ordinary proof of knowledge.

Theorem 4.2 (Special soundness gives knowledge extraction). *Let a Sigma protocol have challenge set E of size N and deterministic special-soundness extractor. If a prover, after its first message and internal state are fixed, answers a uniform challenge successfully on an average fraction ε of challenges, then the standard rewind-and-rechallenge extractor obtains two distinct accepting challenges with expected $O((\varepsilon(\varepsilon - 1/N))^{-1})$ trials whenever $\varepsilon > 1/N$. Thus the natural knowledge error is at most $1/N$.*

Proof. For a fixed first-message/internal state let K be the number of answerable challenges. Two independent uniform challenges are both answerable and distinct with probability

$$\frac{K(K-1)}{N^2}. \quad (4.1)$$

If the first-message state is itself random, take expectation over it. The one-run acceptance probability is $\varepsilon = \mathbb{E}[K]/N$, and Jensen's inequality gives $\mathbb{E}[K^2] \geq (\mathbb{E}K)^2$. Therefore

$$\Pr[\text{two accepting distinct challenges}] = \frac{\mathbb{E}[K(K-1)]}{N^2} \geq \varepsilon \left(\varepsilon - \frac{1}{N} \right). \quad (4.2)$$

When the theorem is read after conditioning on a fixed first-message state, this is simply the identity $(K/N)((K-1)/N)$. Repeating the rewind experiment until success therefore has expected $O((\varepsilon(\varepsilon - 1/N))^{-1})$ trials whenever $\varepsilon > 1/N$. Special soundness applied to the two accepting transcripts returns the witness. At the boundary $K = 1$, acceptance probability $1/N$ may reveal no witness, which explains the knowledge error. $\square$

Theorem 4.3 (Parallel repetition of special-sound protocols). *For t parallel copies using independent challenges from E , the challenge space has size N^t . If the product protocol retains special soundness in the sense that two accepting global transcripts with the same first message and distinct challenge vectors yield a witness from a coordinate where the challenges differ, then its knowledge error is at most N^{-t} and the rewind extractor of Theorem 4.2 applies to the global challenge space.*

Proof. Two distinct challenge vectors differ in some coordinate. That coordinate contains identical first messages and two distinct accepting subchallenges, hence yields a witness by component special soundness. The only success pattern not forcing two answerable global challenges has size one, so its probability under a uniform global challenge is N^{-t} . $\square$

4.3 Reader checkpoint

1. Define knowledge error.
2. Compute the two-challenge extraction probability for k answerable challenges.
3. Explain the $1/N$ boundary.
4. Apply the extractor to Schnorr.
5. Show that two distinct challenge vectors differ in a coordinate.
6. Derive the N^{-t} repeated knowledge error.
7. Compare this theorem with general interactive-proof parallel repetition.
8. Explain why rewinding a quantum state is not automatic.
9. Distinguish black-box from non-black-box extraction.
10. State one stronger property needed for simulation-extractable NIZKs.

Chapter 5

Fiat–Shamir: Classical ROM and Quantum Boundaries

Fiat–Shamir turns public-coin interaction into a hash-derived challenge, but its security is model-sensitive. Classical forking arguments and QROM measure-and-reprogram arguments are therefore separated.

5.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

5.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 5.1 (Fiat–Shamir and oracle models). For a public-coin protocol with first message a , the Fiat–Shamir transform sets the challenge to $e = H(\text{dst}||x||a||\text{ctx})$, with explicit domain-separation string dst and context. In the classical random-oracle model (ROM), adversaries make classical adaptive oracle queries and reductions may program selected points. In the quantum ROM (QROM), adversaries may query the oracle in superposition, so classical transcript-forking and point-programming arguments do not transfer automatically.

Theorem 5.2 (Classical forking lemma — verified deep input). *A Pointcheval–Stern style classical forking lemma converts a successful ROM forger whose accepting transcript contains a hash-derived challenge into, with an explicit loss depending on the forging probability and number of oracle queries, two accepting transcripts sharing the relevant prefix but using distinct oracle answers. For Schnorr-type signatures these transcripts expose the signing witness through special soundness. The exact probability bound and replay schedule are source-owned here.*

Theorem 5.3 (Fiat–Shamir in the QROM — dated frontier, 2026-09-04). *There is no blanket theorem saying that every classically secure Fiat–Shamir instantiation remains secure against quantum random-oracle queries. Positive QROM results depend on protocol properties and reduction frameworks such as measure-and-reprogram; multi-round variants retain explicit polynomial/query losses rather than a free classical-to-quantum transfer. The known positive theorems and negative/separation phenomena make protocol-class hypotheses essential. Every downstream QRCC QROM claim must therefore identify the exact transform theorem it uses.*

5.3 Reader checkpoint

1. Write a domain-separated Fiat–Shamir challenge.
2. Explain why the transcript prefix is part of the hash input.
3. State what a classical fork attempts to reproduce.
4. Show how two Schnorr responses reveal the secret.
5. Explain why programming one classical query is conceptually simpler than one quantum query.
6. Define a superposition oracle query.
7. State the purpose of measure-and-reprogram.
8. Explain why a QROM loss must appear in concrete security accounting.
9. Give an example of a statement that would overclaim QROM security.
10. List the exact model fields that a Fiat–Shamir theorem should record.

Chapter 6

Oblivious Transfer and Completeness

Oblivious transfer is a small functionality with a large completeness role. This chapter carefully avoids the false slogan that arbitrary public-key encryption automatically gives OT.

6.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

6.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 6.1 (Oblivious transfer and correlated randomness). In one-out-of-two OT, a sender holds (m_0, m_1) and a receiver holds $b \in \{0, 1\}$; correctness gives the receiver m_b , receiver privacy hides b from the sender, and sender privacy hides m_{1-b} from the receiver. Security may be semi-honest or malicious and is defined by real/ideal simulation. Correlated OT and OT extension change the input distribution or amortization but not the underlying ideal functionality.

Theorem 6.2 (Dual-mode encryption interface gives semi-honest OT). *Suppose a public-key system has two computationally indistinguishable public-key distributions: valid keys with known secret keys, and oblivious keys for which ciphertexts computationally hide the plaintext even from the key sampler; furthermore one can combine a receiver-generated pair so that exactly one position is valid while the sender cannot tell which. Then encrypting m_0, m_1 under the two keys gives semi-honest one-out-of-two OT.*

Proof. The receiver generates the pair with the valid key in position b , decrypts the corresponding ciphertext, and correctness gives m_b . Indistinguishability of the two key-pair distributions hides b from the sender. The unchosen position is an oblivious key, so its ciphertext hides m_{1-b} from the receiver. These simulations compose directly in the static semi-honest model. The theorem deliberately assumes the dual-mode/oblivious-key interface; ordinary IND-CPA encryption alone does not imply it. This interface is calibrated to the Peikert–Vaikuntanathan–Waters dual-mode

framework; stronger malicious or composable conclusions require the corresponding stronger protocol theorem. $\square$

Theorem 6.3 (OT completeness — verified deep input). *In standard simulation-based models, secure oblivious transfer is complete for general secure two-party computation: Boolean-circuit evaluation can be compiled from local gates, symmetric-key garbling, and OT for transferring the receiver’s input-wire labels. Exact malicious-security, fairness, and setup claims depend on the chosen compiler and remain source-owned.*

6.3 Reader checkpoint

1. Write the ideal OT functionality.
2. Separate sender and receiver privacy.
3. Explain the dual-key condition.
4. Prove receiver correctness in the abstract OT construction.
5. Write the hybrid for receiver privacy.
6. Write the hybrid for sender privacy.
7. Explain why arbitrary IND-CPA PKE is insufficient.
8. Describe correlated OT.
9. Describe what OT extension amortizes.
10. Identify the role of OT in a garbled-circuit compiler.

Chapter 7

Garbled Circuits and Two-Party Computation

Yao’s paradigm separates circuit correctness from the cryptographic mechanism hiding wire values. Malicious security requires further enforcement and is source-owned.

7.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

7.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 7.1 (Secure two-party computation). A two-party protocol securely realizes functionality $\mathcal{F}$ if every allowed real-world adversary has an ideal-world simulator producing an indistinguishable joint output distribution. The model declares semi-honest or malicious behavior, static or adaptive corruption, abort/fairness/guaranteed-output semantics, setup assumptions, and the circuit representation of $\mathcal{F}$. Selective-failure leakage is treated as an attack, not as ordinary abort.

Theorem 7.2 (Selected Yao garbling interface). *Fix a public Boolean circuit C and a static evaluator input before the garbling is generated. Assign independent random labels K_w^0, K_w^1 to each wire and, for every binary gate g , publish four authenticated encryptions mapping the two input labels to the correct output label. Under a dual-key cipher/PRF interface ensuring that one valid input-label pair reveals exactly one table entry while the remaining entries are pseudorandom, evaluation is correct and the evaluator-side garbling view is simulatable in the semi-honest model from $(1^\kappa, C, C(x))$ together with the evaluator’s own prescribed input information. Thus the leakage function in this selected theorem reveals the full circuit C ; adaptive-input privacy, authenticity, obliviousness, malicious security, and hidden-circuit privacy are separate notions.*

Proof. Correctness is inductive over a topological ordering: valid labels for the input wires select the unique row corresponding to their semantic bits and decrypt to the label for $g(b_0, b_1)$. For

privacy, replace inaccessible table entries gate by gate with uniform strings under the dual-key security assumption. The remaining accessible row can be generated from the actual path of wire labels, while unused semantic labels are never exposed. A simulator therefore, given the public circuit and prescribed output, samples random labels along the evaluation path and programs only the output decoding consistent with the ideal functionality; the evaluator's own input labels are treated as its prescribed input information rather than hidden simulator output. The privacy hybrid is the selected static/full-circuit-leakage garbling notion calibrated to the formal simulation framework of Bellare–Hoang–Rogaway. This is not a proof of adaptive input privacy, authenticity, obliviousness, hidden-circuit privacy, or malicious two-party security; malicious-input consistency and cut-and-choose are not hidden inside the proof. $\square$

Theorem 7.3 (Malicious two-party computation from OT and commitments — verified deep input). *Classical compilers combine OT, commitments, zero-knowledge proofs, consistency checks, and/or cut-and-choose to obtain maliciously secure two-party computation under explicit setup and abort/fairness conventions. This volume records the compiler architecture but source-owns the complete simulation proof and concrete overhead.*

7.3 Reader checkpoint

1. Define the ideal/real experiment.
2. Explain point-and-permute.
3. Prove gate-by-gate garbling correctness.
4. Identify the inaccessible garbled rows.
5. Write one privacy hybrid replacing an inaccessible row.
6. Explain why output decoding leaks the intended output.
7. Describe selective failure.
8. Explain why malicious security requires input consistency.
9. Compare abort security with fairness.
10. Identify where commitments enter a malicious compiler.

Chapter 8

Secret Sharing and Multiparty Computation

Information-theoretic MPC makes the corruption thresholds visible in algebra. Shamir sharing and degree reduction explain the passive $t < n/2$ threshold; active security needs verifiable sharing and the stricter $t < n/3$ threshold.

8.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

8.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 8.1 (Multiparty computation and secret sharing). A $(t+1)$ -threshold Shamir sharing of $s \in \mathbb{F}$ among distinct nonzero points $\alpha_1, \dots, \alpha_n$ chooses a uniform polynomial p of degree at most t with $p(0) = s$ and gives share $p(\alpha_i)$ to party i . Linear secret sharing means authorized linear combinations of secrets correspond to the same combinations of shares. Passive adversaries follow the protocol; active adversaries may deviate. Honest-majority, broadcast, verifiable secret sharing (VSS), preprocessing, and arithmetic-circuit assumptions are always explicit.

Theorem 8.2 (Shamir secret sharing). *For n distinct nonzero evaluation points in a field $\mathbb{F}$ with $|\mathbb{F}| > n$, Shamir degree- t sharing reconstructs from any $t+1$ shares, reveals no information about the secret to any set of at most t shares, is linear, and turns pointwise multiplication of two degree- t sharings into evaluations of a degree-at-most- $2t$ polynomial whose constant term is the product.*

Proof. Reconstruction is Lagrange interpolation at zero. For privacy, fix any $k \leq t$ observed shares and any candidate secret s . The affine system specifying $p(0) = s$ and those k evaluations has exactly $|\mathbb{F}|^{t-k}$ degree- t solutions, independent of s , so the observed share vector has the same distribution for every secret. Linearity is coefficientwise. Products of degree- t polynomials have degree at most $2t$ and constant term equal to the product of the secrets. $\square$

Theorem 8.3 (BGW passive MPC at honest majority). *Let $t < n/2$ and let $\mathbb{F}$ contain at least $n + 1$ elements. Arithmetic circuits over $\mathbb{F}$ can be evaluated with perfect passive security against any coalition of at most t parties using Shamir sharing. Addition is local; after multiplication, the degree- $2t$ sharing can be linearly reduced to degree t because $2t < n$ permits interpolation of the product secret from all evaluation points followed by resharing through fixed linear recombination.*

Proof. Addition follows from linearity. For multiplication, parties locally obtain evaluations $h(\alpha_i)$ of $h = pq$, degree at most $2t$. Since $2t < n$, there are public coefficients λ_i with $h(0) = \sum_i \lambda_i h(\alpha_i)$. Each party i Shamir-shares $\lambda_i h(\alpha_i)$ with fresh degree- t randomness; summing these sharings yields a fresh degree- t sharing of $h(0)$. A passive coalition of size at most t sees, at every refresh, at most t evaluations of fresh degree- t polynomials. By Theorem 8.2, those views can be sampled from the coalition's inputs, outputs, and prescribed public openings, giving a perfect simulator by induction over gates. $\square$

Theorem 8.4 (BGW active MPC and VSS — verified deep input). *With authenticated broadcast and the classical synchronous complete-network model, unconditional active security for general computation is achievable for $t < n/3$ corruptions; VSS and consistency mechanisms enforce well-defined shared values. The threshold is tight in the BGW model. The complete active-security compiler is source-owned.*

Theorem 8.5 (Beaver multiplication triples). *Suppose parties hold linear secret sharings of x, y and a preprocessed random triple (a, b, c) with $c = ab$. Opening $d = x - a$ and $e = y - b$ permits a sharing of*

$$xy = c + db + ea + de.$$

If a, b are uniform and hidden from the adversary before opening, then d, e reveal no information about x, y beyond what follows from the selected corruption model.

Proof. Expanding $(a + d)(b + e)$ proves correctness. Because a and b are independent uniform masks, $d = x - a$ and $e = y - b$ are uniform from the view of any adversary lacking the masks, even conditioned on x, y . Linearity of the sharing scheme computes the displayed expression; the public scalar term de may be added to one designated share or through the public-sharing convention. $\square$

Example 8.6 (Complete secret-shared multiplication). Over $\mathbb{F}_{17}$ with threshold $t = 1$, share $x = 4$ by $p(X) = 4 + 3X$ and $y = 7$ by $q(X) = 7 + 5X$ at $1, 2, 3$. Let the triple be $a = 6, b = 2, c = 12$. After sharing the triple, opening gives $d = x - a = -2 = 15$ and $e = y - b = 5$. The reconstructed product is $12 + 15 \cdot 2 + 5 \cdot 6 + 15 \cdot 5 = 147 \equiv 11 \pmod{17}$, matching $4 \cdot 7 = 28 \equiv 11$. Each opened difference is one-time masked by the corresponding random triple component.

8.3 Reader checkpoint

1. Compute Lagrange coefficients for three evaluation points.
2. Prove privacy of one Shamir share when $t = 1$.
3. Show linearity of Shamir sharing.
4. Explain degree doubling under multiplication.
5. Derive the condition $2t < n$ for passive degree reduction.

6. Describe the passive BGW simulator gate by gate.
7. State the active BGW threshold.
8. Verify the Beaver identity.
9. Repeat the worked multiplication example with new values.
10. Explain why Beaver triples move multiplication cost into preprocessing.

Chapter 9

Polynomial Commitments and Interactive Oracle Proofs

Modern succinct arguments separate an algebraic proof protocol from a commitment layer. This chapter records KZG, transparent inner-product folding, and FRI as distinct interfaces.

9.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

9.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 9.1 (Polynomial commitments and IOPs). A polynomial commitment binds a prover to a bounded-degree polynomial and later supports evaluation openings. Security may include evaluation binding, hiding, and extractability and may require a structured reference string. An interactive oracle proof (IOP) lets the prover send oracle strings that the verifier samples at a small number of positions. A proximity test distinguishes words near a code such as Reed–Solomon from words far from it. A polynomial IOP represents oracle words as evaluations of low-degree polynomials.

Theorem 9.2 (KZG polynomial commitments — verified deep input). *In bilinear groups with an SRS containing powers of a hidden scalar τ , KZG commits to a degree- d polynomial by a single group element and opens an evaluation $f(z) = y$ using a quotient witness for $(f(X) - y)/(X - z)$. Verification is one pairing equation. Binding/extractability uses the selected pairing/algebraic assumption and the degree bound; knowledge of the toxic waste τ destroys the intended binding guarantee. The exact reduction is source-owned.*

Theorem 9.3 (Inner-product logarithmic folding invariant). *Let $\mathbf{a}, \mathbf{b} \in \mathbb{F}^{2m}$ split as $(\mathbf{a}_L, \mathbf{a}_R)$ and $(\mathbf{b}_L, \mathbf{b}_R)$. For nonzero challenge x , define*

$$\mathbf{a}' = x\mathbf{a}_L + x^{-1}\mathbf{a}_R, \quad \mathbf{b}' = x^{-1}\mathbf{b}_L + x\mathbf{b}_R.$$

Then

$$\langle \mathbf{a}', \mathbf{b}' \rangle = \langle \mathbf{a}_L, \mathbf{b}_L \rangle + \langle \mathbf{a}_R, \mathbf{b}_R \rangle + x^2 \langle \mathbf{a}_L, \mathbf{b}_R \rangle + x^{-2} \langle \mathbf{a}_R, \mathbf{b}_L \rangle.$$

If the two cross terms are separately committed in the verifier equation, one round halves the vector dimension while preserving an equivalent verification relation; repeating gives $\log_2 n$ rounds for n a power of two.

Proof. Expand the bilinear inner product of the two folded vectors. The four resulting terms have coefficients $1, x^2, x^{-2}, 1$ exactly as displayed. A commitment relation that moves the two cross terms into challenge-weighted group elements therefore has the same truth value before and after folding. Each round halves dimension, so after $\log_2 n$ rounds only scalars remain. This theorem is the algebraic folding invariant; discrete-log binding and Fiat–Shamir noninteractivity are separate assumptions. $\square$

Theorem 9.4 (FRI low-degree proximity testing — verified deep input). *FRI gives an interactive-oracle proof of proximity for Reed–Solomon type evaluation codes by repeatedly folding a claimed evaluation function to a smaller domain using verifier challenges, committing each layer, and checking a small number of consistency paths plus a final low-degree condition. Completeness, soundness as a function of rate/distance and query count, and the precise folding domain hypotheses are retained at the primary-source level rather than replaced by a slogan that “FRI proves low degree.”*

Example 9.5 (Executable laboratory contract). The Volume-XXI laboratory reproduces Schnorr extraction, Shamir interpolation/privacy checks, Beaver multiplication, Merkle authentication, polynomial interpolation, a toy sum-check identity, and one algebraic FRI-style folding step over a finite field. It is pedagogical code and carries no production-security claim.

9.3 Reader checkpoint

1. Define evaluation binding.
2. Write the KZG quotient polynomial identity.
3. Explain toxic waste.
4. Expand the inner-product fold by hand.
5. Count rounds when $n = 1024$.
6. Explain which part of the inner-product argument uses discrete-log binding.
7. Describe one FRI folding step conceptually.
8. Distinguish low degree from proximity to low degree.
9. Explain why Merkle commitments add a hash assumption to an IOP.
10. State which laboratory checks are algebraic rather than cryptographic security tests.

Chapter 10

SNARKs, STARKs, Recursion, and Post-Quantum Frontiers

Succinct proof systems combine several layers: arithmetization, an interactive or oracle proof, commitments, and often a Fiat–Shamir transform. Security claims are meaningful only when all layers and setup assumptions are named.

10.1 How to read this chapter

The stable identifiers in this chapter are part of the project-wide theorem registry. Security notions are interpreted only in the model explicitly stated here; source-owned results are not silently strengthened.

10.2 Learning goals

After this chapter, a reader should be able to state the relevant syntax, reproduce the local algebraic or simulation argument, identify the reduction assumption, and separate classical, random-oracle, setup, and post-quantum claims.

Definition 10.1 (Succinct arguments and proof composition). A succinct noninteractive argument (SNARG) has proofs and/or verifier work sublinear in a declared computation measure; a SNARK additionally carries a declared knowledge-soundness notion. Setup may be circuit-specific, universal/updatable, or transparent. Recursive composition verifies proofs inside proofs. An accumulation scheme reduces several verification claims to one accumulated claim, while a folding scheme reduces multiple structured satisfiability instances to one; neither term by itself implies zero knowledge, noninteractivity, or an argument of knowledge.

Theorem 10.2 (Groth16 interface — verified deep input). *Groth’s 2016 pairing-based construction compiles an arithmetic circuit through a QAP and, after circuit-dependent structured setup, gives constant-size proofs consisting of three group elements with a verifier using a constant number of pairings plus work linear in the public input. Completeness and zero knowledge are perfect in the source construction; knowledge soundness is proved against polynomially bounded adversaries restricted to generic bilinear-group operations. Trusted setup and toxic-waste erasure are part of the interface.*

Theorem 10.3 (Transparent STARK composition interface — verified deep input). *A STARK-style transparent argument composes an algebraic intermediate representation, a polynomial IOP with low-degree/proximity testing such as FRI, collision-resistant Merkle commitments, and normally Fiat–Shamir for noninteractivity. Its total soundness error is an explicit budget across algebraic testing, oracle sampling, commitment binding, and the Fiat–Shamir model. “Transparent” removes a secret structured setup; it does not remove cryptographic assumptions or automatically prove QROM security.*

Theorem 10.4 (Recursive arguments, accumulation, and folding — dated frontier, 2026-09-04). *Halo-style accumulation, Nova-style relaxed-R1CS folding, HyperNova-style customizable-constraint folding, cycle-of-curves recursion, lookup arguments, and proof-carrying-data systems occupy distinct interfaces. In particular, a Nova-family folding primitive reduces structured instances but is weaker than a SNARK or argument of knowledge by itself; additional cryptographic layers turn folding into recursive arguments. QRCC records construction-specific setup, zero-knowledge, soundness/knowledge, and verifier-cost claims rather than treating “folding” as a synonym for recursive SNARK.*

Theorem 10.5 (Post-quantum succinct arguments — dated frontier, 2026-09-04). *Post-quantum succinctness is construction-specific. Kilian/PCP-based succinct arguments have standard-model post-quantum security under collapsing-hash assumptions, later IOP-based lines extend this under collapse-binding vector commitments, and lattice/SIS-based succinct arguments form a separate line with different proof sizes, verifier costs, extraction models, and assumptions. These theorems do not automatically make a Fiat–Shamir-transformed transparent proof QROM-secure, while pairing-based KZG/Groth16 systems are not post-quantum under their standard discrete-log/pairing assumptions. No construction receives a blanket “post-quantum” label without a construction-specific quantum security theorem.*

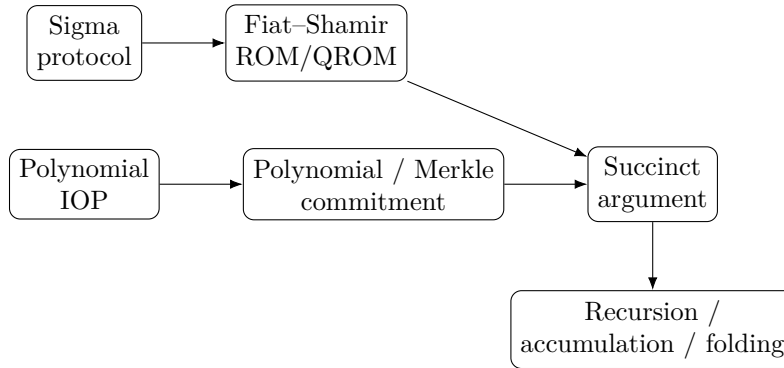


Figure 10.1: Transformation map. Every arrow changes assumptions or security models; it is not an implication without the corresponding theorem.

10.3 Reader checkpoint

1. Define SNARG and SNARK separately.
2. List Groth16 setup assumptions.

3. Explain why three group elements does not mean three field elements.
4. List the layers of a STARK-style composition.
5. Explain transparent setup.
6. Distinguish accumulation from folding.
7. State the relaxed-R1CS role in Nova.
8. Explain why KZG/Groth16 are not post-quantum under standard assumptions.
9. State what must be checked for a hash-based proof in the QROM.
10. Trace one path through Figure 10.1 and list every added assumption.

Internalization and Reconstruction Guide: Volume XXI

Purpose. Unify commitments, Sigma protocols, zero knowledge, extraction, oblivious transfer, MPC, polynomial commitments, and succinct proofs through the simulator/extractor viewpoint and explicit setup/knowledge assumptions.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-XVII-DEF-011: Discrete logarithm and generic groups.
2. QRCC-XVI-THM-003: Game-hopping and reduction accounting lemma.
3. QRCC-XVII-DEF-030: Bilinear pairings.
4. QRCC-XX-THM-035: Merkle-tree signature composition.
5. QRCC-XVI-THM-013: Goldreich–Levin hard-core theorem.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XXI-DEF-001 (Commitment schemes and their security); QRCC-XXI-DEF-004 (Sigma protocols); QRCC-XXI-DEF-007 (Zero knowledge and simulation); QRCC-XXI-DEF-011 (Proofs of knowledge and extraction); QRCC-XXI-DEF-014 (Fiat–Shamir transform and oracle models).

Chapter retrieval map

- Ch. 1: Commitments: Hiding, Binding, and Setup:** enter through QRCC-XXI-DEF-001 (Commitment schemes and their security); exit by being able to state QRCC-XXI-THM-002 (Commitments from one-way permutations) with its ownership boundary.
- Ch. 2: Sigma Protocols and Partial Knowledge:** enter through QRCC-XXI-DEF-004 (Sigma protocols); exit by being able to state QRCC-XXI-THM-005 (Schnorr Sigma protocol) with its ownership boundary.

- Ch. 3: Zero Knowledge and Simulation:** enter through QRCC-XXI-DEF-007 (Zero knowledge and simulation); exit by being able to state QRCC-XXI-BB-009 (Every NP language has a computational zero-knowledge proof under one-way functions) with its ownership boundary.
- Ch. 4: Knowledge Extraction and Repetition:** enter through QRCC-XXI-DEF-011 (Proofs of knowledge and extraction); exit by being able to state QRCC-XXI-THM-012 (Sigma protocols are proofs of knowledge) with its ownership boundary.
- Ch. 5: Fiat-Shamir: Classical ROM and Quantum Boundaries:** enter through QRCC-XXI-DEF-014 (Fiat-Shamir transform and oracle models); exit by being able to state QRCC-XXI-THM-015 (Classical forking lemma) with its ownership boundary.
- Ch. 6: Oblivious Transfer and Completeness:** enter through QRCC-XXI-DEF-017 (Oblivious transfer and correlated randomness); exit by being able to state QRCC-XXI-BB-019 (Oblivious transfer completeness) with its ownership boundary.
- Ch. 7: Garbled Circuits and Two-Party Computation:** enter through QRCC-XXI-DEF-020 (Secure two-party computation); exit by being able to state QRCC-XXI-BB-022 (Maliciously secure two-party computation from OT and commitments) with its ownership boundary.
- Ch. 8: Secret Sharing and Multiparty Computation:** enter through QRCC-XXI-DEF-023 (Multiparty computation and secret sharing); exit by being able to state QRCC-XXI-BB-026 (BGW active-security and verifiable-secret-sharing theorem) with its ownership boundary.
- Ch. 9: Polynomial Commitments and Interactive Oracle Proofs:** enter through QRCC-XXI-DEF-028 (Polynomial commitments and interactive oracle proofs); exit by being able to state QRCC-XXI-BB-029 (KZG polynomial commitment) with its ownership boundary.
- Ch. 10: SNARKs, STARKs, Recursion, and Post-Quantum Frontiers:** enter through QRCC-XXI-DEF-032 (Succinct arguments and proof composition); exit by being able to state QRCC-XXI-BB-033 (Groth16 succinct argument interface) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XXI-THM-002 — Commitments from one-way permutations. Dependencies: QRCC-XVI-THM-013, QRCC-XXI-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XXI-BB-009 — Every NP language has a computational zero-knowledge proof under one-way functions. Dependencies: QRCC-II-THM-005, QRCC-XVI-DEF-009, QRCC-XXI-DEF-001, QRCC-XXI-DEF-007. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-XXI-THM-015 — Classical forking lemma. Dependencies: QRCC-XXI-DEF-014, QRCC-XXI-THM-005, QRCC-XVI-THM-003. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XXI-BB-022 — Maliciously secure two-party computation from OT and commitments. Dependencies: QRCC-XXI-DEF-001, QRCC-XXI-BB-019, QRCC-XXI-DEF-020. *source-owned*

interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.

5. QRCC-XXI-BB-029 — KZG polynomial commitment. Dependencies: QRCC-XVII-DEF-030, QRCC-XXI-DEF-028. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XXI-F-016 — Fiat–Shamir in the quantum random-oracle model. Dependencies: QRCC-XV-DEF-020, QRCC-XXI-DEF-014. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- confusing honest-verifier zero knowledge with general zero knowledge;
- using special soundness as a full proof-of-knowledge theorem without challenge/knowledge-error accounting;
- moving from semi-honest to malicious MPC without authentication/consistency machinery;
- calling every succinct proof a SNARK while suppressing setup, transparency, or knowledge assumptions;

Boundary and degenerate cases

- challenge space of size one;
- perfect/statistical/computational simulation boundaries;
- standalone versus concurrent composition;
- trusted setup, transparent setup, and random-oracle transformations;

Retrieval and reconstruction drill

1. State QRCC-XXI-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XXI-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XXI-BB-009, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.

6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XXI-F-016, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XXII asks what can be computed directly on hidden data or hidden programs, separating homomorphism, functionality, and obfuscation.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XXII

Homomorphic, Functional, and Program Cryptography

Chapter 1

Homomorphic Encryption Before FHE

Partially homomorphic encryption already displays the algebraic phenomenon on which FHE is built. The point of this chapter is to separate correctness and homomorphism, which are elementary algebra, from semantic security, which is an assumption-backed cryptographic claim.

1.1 How to read this chapter

All group domains and randomness spaces are explicit. Malleability is a feature here, not an authentication guarantee.

1.2 Learning goals

After this chapter, a reader should be able to state compact homomorphic-evaluation syntax, prove Paillier addition and ElGamal multiplication/rerandomization, and explain why IND-CPA security does not imply ciphertext integrity.

Definition 1.1 (Homomorphic-encryption syntax and security). A public-key homomorphic-encryption scheme consists of probabilistic polynomial-time algorithms KeyGen , Enc , Dec together with an evaluation algorithm Eval . Correctness requires decryption of an evaluated ciphertext to equal the prescribed plaintext computation except with the declared failure probability. A leveled scheme supports every circuit up to a public depth bound; a fully homomorphic scheme supports arbitrary polynomial-size circuits while remaining *compact*, meaning that decryption cost and ciphertext size after evaluation are bounded independently of the evaluated circuit size. An evaluation key is public auxiliary information consumed by Eval . IND-CPA security is the usual chosen-plaintext indistinguishability experiment. Circuit privacy, threshold or multi-key decryption, decryption-failure bounds, and any noise budget are additional properties and must be stated separately.

Theorem 1.2 (Paillier additive homomorphism). *Let $N = pq$ for distinct odd primes and $\lambda = \text{lcm}(p-1, q-1)$. Choose $g \in \mathbb{Z}_{N^2}^*$ such that $\mu = L(g^\lambda \bmod N^2)^{-1} \bmod N$ exists, where $L(1+uN) = u \bmod N$. For $m \in \mathbb{Z}_N$ and $r \in \mathbb{Z}_N^*$ set*

$$\text{Enc}(m; r) = g^m r^N \pmod{N^2}.$$

Then $\text{Dec}(c) = L(c^\lambda \bmod N^2) \mu \bmod N$ decrypts correctly and

$$\text{Enc}(m_1; r_1) \text{Enc}(m_2; r_2) = \text{Enc}(m_1 + m_2; r_1 r_2).$$

Multiplication by ρ^N publicly rerandomizes a ciphertext. The semantic-security theorem under Paillier's decisional composite-residuosity assumption is source-owned and is not inferred from the algebraic identities alone.

Proof. We first justify the randomizer cancellation. Regard $r \in \mathbb{Z}_N^*$ as any integer lift coprime to N . Then r is a unit modulo both p^2 and q^2 . The two unit groups have orders $\varphi(p^2) = p(p-1)$ and $\varphi(q^2) = q(q-1)$. Since $N\lambda = pq \operatorname{lcm}(p-1, q-1)$ is divisible by both orders, Euler's theorem gives

$$r^{N\lambda} \equiv 1 \pmod{p^2}, \quad r^{N\lambda} \equiv 1 \pmod{q^2}.$$

The moduli p^2, q^2 are coprime, so CRT yields $r^{N\lambda} \equiv 1 \pmod{p^2 q^2} = \pmod{N^2}$.

Thus $c^\lambda \equiv g^{m\lambda} \pmod{N^2}$. Because μ is defined, g^λ lies in the subgroup $1 + N\mathbb{Z}_{N^2}$; write $g^\lambda \equiv 1 + N\alpha \pmod{N^2}$ with $\alpha = L(g^\lambda)$ invertible modulo N . The binomial theorem gives $g^{m\lambda} \equiv (1 + N\alpha)^m \equiv 1 + mN\alpha \pmod{N^2}$, since every term of degree at least two is divisible by N^2 . Hence $L(c^\lambda) = m\alpha$ and multiplication by $\mu = \alpha^{-1}$ recovers m modulo N .

Finally,

$$\operatorname{Enc}(m_1; r_1) \operatorname{Enc}(m_2; r_2) = g^{m_1 + m_2} (r_1 r_2)^N = \operatorname{Enc}(m_1 + m_2; r_1 r_2),$$

and $c\rho^N$ changes the encryption randomness from r to $r\rho$ without changing the plaintext. $\square$

Theorem 1.3 (ElGamal multiplicative homomorphism and rerandomization). *Let $G = \langle g \rangle$ be a finite cyclic group, let the secret key be x , and let $h = g^x$. Encrypt $m \in G$ with randomness r as $\operatorname{Enc}(m; r) = (g^r, mh^r)$. Componentwise multiplication of two ciphertexts encrypts $m_1 m_2$ with randomness $r_1 + r_2$, and multiplication by an encryption of 1_G publicly rerandomizes. Under the declared DDH group family, the usual ElGamal IND-CPA theorem is source-owned.*

Proof. For ciphertexts $(u_i, v_i) = (g^{r_i}, m_i h^{r_i})$,

$$(u_1 u_2, v_1 v_2) = (g^{r_1 + r_2}, m_1 m_2 h^{r_1 + r_2}),$$

which is exactly a fresh encryption with summed randomness. Taking $m_2 = 1_G$ proves public rerandomization. Decryption is $v/u^x = m$. $\square$

1.3 Reader checkpoint

1. Verify the Paillier decryption formula when $g = 1 + N$.
2. Show directly that $r^{N\lambda} \equiv 1 \pmod{N^2}$ for $r \in \mathbb{Z}_N^*$ lifted to $\mathbb{Z}_{N^2}^*$.
3. Compute a Paillier encryption for $N = 15$ using admissible parameters.
4. Prove that Paillier scalar multiplication by k is ciphertext exponentiation by k .
5. Explain why Paillier rerandomization does not authenticate the ciphertext.
6. Derive ElGamal decryption from the Diffie–Hellman relation.
7. Prove ElGamal rerandomization using an encryption of 1_G .
8. Explain why ordinary ElGamal is multiplicatively rather than additively homomorphic.
9. State precisely where DDH enters the ElGamal security theorem.
10. Give one example where homomorphic malleability is useful and one where it is dangerous.

Chapter 2

Noise, Relinearization, and Modulus Switching

Lattice-based homomorphic encryption turns correctness into a quantitative invariant: a ciphertext carries a plaintext together with structured error, and evaluation must keep the error inside a decryption region.

2.1 How to read this chapter

The scalar presentation below is deliberately pedagogical. It isolates phase algebra that is inherited coefficientwise by ring variants; it is not a substitute for an RLWE security proof or production parameter analysis.

2.2 Learning goals

After this chapter, a reader should be able to track a ciphertext phase, derive addition and one multiplication with relinearization, and prove a deterministic rounded modulus-switching bound.

Definition 2.1 (Noise-bearing lattice encryption). Fix integers $p \geq 2$ and $q \gg p$. For secret $s \in \mathbb{Z}_q$, a two-component ciphertext $c = (c_0, c_1) \in \mathbb{Z}_q^2$ has *phase* $\phi_s(c) = c_0 + c_1 s \bmod q$. In the exact-message phase convention, a valid ciphertext for $m \in \mathbb{Z}_p$ has a canonical integer lift

$$\phi_s(c) = m + pe,$$

with error e small enough that no ambiguous wraparound occurs. A degree-two ciphertext (d_0, d_1, d_2) has phase $d_0 + d_1 s + d_2 s^2$. A gadget basis $g_0, \dots, g_{\ell-1}$ permits a digit decomposition $u = \sum_j u_j g_j$. An evaluation key for relinearization consists of ciphertext-like pairs whose phases are $g_j s^2 + p\eta_j$. Modulus switching maps a ciphertext from modulus q to q' by coefficientwise scaled rounding; its correctness is a phase-approximation statement, not a security theorem.

Theorem 2.2 (Exact-message phase invariant under addition, multiplication, and relinearization). Suppose ciphertexts c, c' have lifted phases $m + pe$ and $m' + pe'$ with $m, m' \in \mathbb{Z}_p$. Addition has phase $(m + m') + p(e + e')$. Their tensor product has degree-two phase

$$mm' + p(me' + m'e + pee').$$

If the quadratic coefficient is decomposed as $d_2 = \sum_j u_j g_j$ and the relinearization key has phases $g_j s^2 + p\eta_j$, relinearization preserves plaintext $mm' \bmod p$ and changes the error to

$$e_{\text{mult}} = me' + m'e + pee' + \sum_j u_j \eta_j.$$

Whenever the resulting lifted phase remains inside $(-q/2, q/2)$, reduction modulo p decrypts the intended exact plaintext.

Proof. Addition is immediate from linearity of ϕ_s . For multiplication, multiply the two lifted phases:

$$(m + pe)(m' + pe') = mm' + p(me' + m'e + pee').$$

The tensor-product coefficients give exactly the same polynomial evaluated at s . To relinearize, replace $d_2 s^2$ by the sum of the evaluation-key phases weighted by the gadget digits u_j . The $g_j s^2$ terms reconstruct $d_2 s^2$ and the only change is the added multiple $p \sum_j u_j \eta_j$. Thus the plaintext modulo p is unchanged. The no-wrap condition ensures that the canonical lift used for reduction is the intended integer representative. $\square$

Theorem 2.3 (Rounded modulus-switching lemma). *Let $\hat{s} = (1, s_1, \dots, s_n) \in \mathbb{Z}^{n+1}$ and let $c \in \mathbb{Z}^{n+1}$ be fixed integer lifts of a ciphertext modulo q . For $0 < q' < q$, define $c'_i = \lfloor (q'/q)c_i \rfloor$. Then*

$$\left| \langle c', \hat{s} \rangle - \frac{q'}{q} \langle c, \hat{s} \rangle \right| \leq \frac{1}{2} \|\hat{s}\|_1.$$

Consequently, if $\langle c, \hat{s} \rangle = \Delta m + e$ and $\Delta' = (q'/q)\Delta$, then $\langle c', \hat{s} \rangle = \Delta' m + e'$ with

$$|e'| \leq \frac{q'}{q} |e| + \frac{1}{2} \|\hat{s}\|_1.$$

If decoding rounds by the scale and the right-hand side is below $\Delta'/2$, the decoded message is unchanged. Production BGV modulus reduction additionally exploits plaintext-modulus congruences and ring structure and remains source-owned.

Proof. Write $c'_i = (q'/q)c_i + \rho_i$ with $|\rho_i| \leq 1/2$. Taking the inner product with $\hat{s}$ gives the first bound by the triangle inequality. Substituting the scaled-message representation gives the displayed formula for e' and its bound. The final decoding claim is nearest-integer uniqueness inside half a scale. $\square$

2.3 Reader checkpoint

1. Verify the exact-message addition invariant.
2. Derive the multiplication error term before relinearization.
3. Explain why the evaluation key encrypts a gadget multiple of s^2 .
4. Bound $\sum_j |u_j \eta_j|$ for balanced base- B digits.
5. State the no-wrap condition needed for exact decryption.
6. Reprove modulus switching using rounding errors ρ_i .

7. Compare an ℓ_1 secret bound with an ℓ_2 bound.
8. Explain why smaller modulus reduces the old error but adds rounding error.
9. Give a parameter example where switching preserves nearest-scale decoding.
10. Explain why this lemma alone is not a complete BGV security or correctness theorem.

Chapter 3

Bootstrapping: Turning Shallow Evaluation into FHE

Bootstrapping is a compiler theorem. The cryptographic construction and the compiler must be kept conceptually separate: one provides a scheme that can evaluate enough of decryption, and the other converts that capability into unbounded refresh.

3.1 How to read this chapter

The local theorem proves only the generic compiler under an explicit encrypted-secret-key interface. Gentry’s ideal-lattice construction, squashing step, and hardness assumptions remain source-owned.

3.2 Learning goals

After this chapter, a reader should be able to define bootstrappability, locate the circular/KDM-security hypothesis, and explain how refresh prevents noise from growing with total circuit depth.

Definition 3.1 (Bootstrapping and circular-security interfaces). Let D_{sk} denote the Boolean or arithmetic circuit computing decryption under secret key sk . A scheme is *bootstrappable* for a class of ciphertexts if its evaluation procedure can correctly evaluate the required augmented decryption circuit on encryptions of the secret-key bits together with an encryption of the target ciphertext bits, producing a fresh ciphertext of the same plaintext. The public collection $\text{Enc}_{pk}(sk_i)$ is an encrypted-secret-key or bootstrapping key. Circular security means that publishing encryptions of secret-key information under a key related cyclically to that same secret does not destroy the declared security notion; KDM security is the more general security notion allowing encryptions of key-dependent messages. A key-chain construction replaces a cycle by encryptions from one key to a different next key and therefore gives a leveled interface without assuming a literal key cycle.

Theorem 3.2 (Generic bootstrapping compiler). *Let Π be a compact IND-CPA encryption scheme whose evaluator can correctly evaluate its augmented decryption circuit on the relevant ciphertext class. Suppose the public evaluation material contains encryptions of the required secret-key bits and that the resulting key-dependent-message exposure is secure under the explicitly declared circular/KDM assumption. Then the refresh map*

$$\text{Refresh}(c) = \text{Eval}_{pk}(D, \text{Enc}_{pk}(sk_1), \dots, \text{Enc}_{pk}(sk_t), c)$$

returns a ciphertext of the same plaintext with the post-evaluation noise profile of a fresh evaluable ciphertext. Interleaving bounded-depth evaluation with refresh yields fully homomorphic evaluation of arbitrary polynomial-size circuits. If one instead uses a sufficiently long acyclic chain of keys, the same argument yields leveled FHE without a circular-security hypothesis.

Proof. Correctness of Eval applied to the decryption circuit implies that the plaintext encrypted by Refresh(c) is $D_{sk}(c)$, hence the original message. Bootstrappability places the output back inside the scheme's admissible ciphertext class, so another bounded block of gates can be evaluated before the next refresh. Repeating this process a polynomial number of times evaluates any polynomial-size circuit while compactness keeps ciphertext and decryption complexity independent of the circuit size. The only security information added by the compiler beyond ordinary public parameters is the encrypted-key material. Thus the security reduction requires exactly the stated circular/KDM interface. Replacing the self-encryption by encryptions of sk_i under pk_{i+1} removes the cycle for an a priori bounded number of refreshes. $\square$

Theorem 3.3 (Gentry's first ideal-lattice FHE construction — verified deep input). *Gentry's STOC 2009 construction gives the first fully homomorphic public-key encryption scheme. The source architecture begins with a somewhat homomorphic ideal-lattice scheme, applies a squashing transformation that lowers decryption-circuit complexity under an additional sparse-subset-sum-style assumption, and then applies the bootstrapping theorem. The original ideal-lattice security and squashing assumptions, parameter regime, and decryption-circuit analysis are source-owned; this book does not replace them by a generic LWE claim.*

3.3 Reader checkpoint

1. Draw the data flow of the refresh circuit.
2. Explain why evaluating decryption does not reveal the plaintext to the evaluator.
3. Identify the exact public information responsible for circular-security concerns.
4. Show how a key chain removes a literal cycle.
5. Explain why a finite key chain gives leveled rather than unbounded FHE.
6. Distinguish bootstrappability from semantic security.
7. Explain why compactness is part of the FHE definition.
8. State what squashing changes in Gentry's original architecture.
9. Explain why the ideal-lattice theorem is not replaced by the local phase model of Chapter 2.
10. Give a circuit-depth schedule with two refresh operations.

Chapter 4

BGV, BFV, and Exact Homomorphic Arithmetic

Modern exact-arithmetic FHE separates several mechanisms that were entangled in the first construction: RLWE encryption, key switching, modulus management, SIMD packing, and optional bootstrapping.

4.1 How to read this chapter

The local BFV recurrence is an explicit phase model. BGV’s asymptotic leveled theorem and concrete bootstrapping algorithms are source-owned.

4.2 Learning goals

After this chapter, a reader should be able to distinguish BGV from BFV scaling conventions, derive a transparent multiplication-error recurrence, and identify which practical acceleration claims are algorithmic benchmarks rather than asymptotic security theorems.

Definition 4.1 (BGV and BFV exact-arithmetic families). Let $R = \mathbb{Z}[X]/(\Phi(X))$, $R_q = R/qR$, and $R_t = R/tR$. RLWE-based exact-arithmetic schemes encrypt plaintexts in R_t using ciphertexts in one or more rings R_q . BGV-style presentations keep plaintext separated from error by a factor of t in the phase and use modulus reduction to control error relative to q . BFV-style presentations commonly encode a plaintext at scale $\Delta \approx q/t$ and restore the scale after multiplication. Relinearization/key switching maps ciphertexts under higher-degree secret monomials or old keys back to the chosen key using evaluation-key encryptions. A modulus chain is a decreasing sequence of ciphertext moduli. Batching or SIMD packing uses a Chinese-remainder decomposition of the plaintext ring when the algebraic conditions permit. Residue-number-system implementations represent large moduli by smaller coprime factors; this is an implementation representation, not a distinct security assumption.

Theorem 4.2 (BGV leveled FHE — verified deep input). *Brakerski–Gentry–Vaikuntanathan construct leveled FHE from LWE and ring-LWE without invoking bootstrapping inside the leveled evaluator. The construction uses dimension/modulus management, key switching and modulus reduction to support an a priori polynomial circuit depth while preserving compact decryption. The exact approximation factors, dimension growth, modulus chain, and LWE/RLWE reduction parameters are retained as source-owned theorem data rather than compressed into the scalar invariant of Chapter 2.*

Theorem 4.3 (Simplified BFV plaintext scaling and multiplication-error recurrence). *Work in one coefficient of a BFV-style phase model with scale $\Delta > 0$. Suppose two ciphertexts have lifted phases*

$$z_i = \Delta m_i + e_i \quad (i = 1, 2).$$

Form the degree-two tensor product, divide its coefficients by Δ with coefficientwise nearest-integer rounding, and then apply a relinearization map whose phase error is η_{ks} . If ρ denotes the total phase error introduced by the coefficientwise scaling/rounding step, the resulting phase is

$$\Delta m_1 m_2 + e_{\text{mult}}, \quad e_{\text{mult}} = m_1 e_2 + m_2 e_1 + \frac{e_1 e_2}{\Delta} + \rho + \eta_{\text{ks}}.$$

Hence

$$|e_{\text{mult}}| \leq |m_1||e_2| + |m_2||e_1| + \frac{|e_1 e_2|}{\Delta} + |\rho| + |\eta_{\text{ks}}|.$$

Nearest-scale decoding is correct whenever $|e_{\text{mult}}| < \Delta/2$. The full polynomial-ring BFV bounds, ciphertext rounding map, and RLWE security reduction remain source-owned.

Proof. The raw product phase is

$$z_1 z_2 = \Delta^2 m_1 m_2 + \Delta(m_1 e_2 + m_2 e_1) + e_1 e_2.$$

Scaling by $1/\Delta$ gives the desired main term $\Delta m_1 m_2$ and the first three error terms. Coefficientwise rounding changes the evaluated phase by ρ , and relinearization changes it by η_{ks} by definition of the key-switching error. The triangle inequality proves the bound, and nearest-scale decoding is unique inside half the scale. $\square$

Theorem 4.4 (Large-plaintext BGV bootstrapping of Ma–Huang–Wang–Wang — verified algorithmic input). *Ma, Huang, Anyu Wang and Xiaoyun Wang (EUROCRYPT 2024) accelerate the digit-removal stage of BGV bootstrapping for large plaintext prime p using low-degree null polynomials over $\mathbb{Z}_{p^e}$. Their analyzed digit-extraction cost improves the prior square-root-in- p dependence to a bound controlled by a small support parameter B , and their HELib experiments include $p \in \{17, 127, 257, 8191, 65537\}$. The paper reports bootstrapping-throughput speedups ranging from about 1.38 to 151 over the compared baseline configurations and, for $p = 65537$, an approximately 80-fold digit-removal speedup with total bootstrapping reduced from more than twelve hours to under fourteen minutes. These are construction- and implementation-specific claims, not a replacement for the BGV security theorem.*

4.3 Reader checkpoint

1. Compare the plaintext placement conventions of BGV and BFV.
2. Derive the raw BFV product phase.
3. Explain the purpose of dividing by Δ after multiplication.
4. Bound the local recurrence for $|m_i| \leq M$ and $|e_i| \leq E$.
5. Explain why relinearization changes dimension but should preserve plaintext.
6. Describe how an RNS representation differs from modulus switching.
7. State one algebraic condition enabling batching.

8. Explain why the BGV leveled theorem is stronger than the scalar local invariant.
9. Separate the asymptotic and empirical claims in the large- p bootstrapping result.
10. Explain why a faster bootstrap does not alter the underlying RLWE assumption by itself.

Chapter 5

CKKS and Approximate Homomorphic Arithmetic

CKKS changes the meaning of correctness: a decrypted value is expected to approximate the desired numerical answer, and cryptographic noise becomes one component of a total numerical error budget.

5.1 How to read this chapter

The theorem proves a simplified scaled-number invariant. The canonical embedding, slot structure, RLWE reduction, and production rescaling algorithms are source-owned.

5.2 Learning goals

After this chapter, a reader should be able to separate encoding error from encryption error, propagate an approximate product bound, and explain why successful numerical experiments are not by themselves cryptographic correctness or security proofs.

Definition 5.1 (Approximate homomorphic arithmetic). In CKKS, a vector of complex slots is represented through the canonical embedding of a cyclotomic number ring and scaled by a positive factor Δ before rounding to a ring element. Decoding divides by the current scale and applies the inverse embedding, thereby returning an approximation. The total error includes encoding/rounding error, encryption noise, homomorphic-operation error and rescaling error. Rescaling removes a modulus factor and approximately divides both plaintext scale and error by that factor. Slot rotations and conjugation are automorphisms implemented with evaluation keys. A numerical-correctness claim must state an error norm and tolerance; it is distinct from IND-CPA security and from a bound on decryption-failure probability.

Theorem 5.2 (Scaled approximate-arithmetic invariant). *In a simplified one-slot CKKS model, suppose a ciphertext at scale Δ decrypts to a lifted phase $z = \Delta(x + \varepsilon)$, where $x \in \mathbb{C}$ is the intended value and ε is total current error. Addition of two equal-scale ciphertexts represents $x_1 + x_2$ with error $\varepsilon_1 + \varepsilon_2$. Multiplication produces scale Δ^2 and represents*

$$x_1x_2 + x_1\varepsilon_2 + x_2\varepsilon_1 + \varepsilon_1\varepsilon_2.$$

If rescaling back to scale Δ adds decoded rounding error δ_{rs} , the post-rescale error is

$$\varepsilon_{\text{mult}} = x_1\varepsilon_2 + x_2\varepsilon_1 + \varepsilon_1\varepsilon_2 + \delta_{\text{rs}}.$$

Thus $|\varepsilon_{\text{mult}}| \leq |x_1||\varepsilon_2| + |x_2||\varepsilon_1| + |\varepsilon_1\varepsilon_2| + |\delta_{\text{rs}}|$. The same algebra applies slotwise after the source-owned canonical-embedding machinery.

Proof. Addition is linear. For multiplication, multiply $(x_1 + \varepsilon_1)(x_2 + \varepsilon_2)$ and separate the desired product from the three error terms. Rescaling changes the represented scale and introduces the declared rounding term δ_{rs} . The displayed bound is the triangle inequality. $\square$

Remark 5.3 (Verified-numerics and CKKS certification frontier — dated 3 September 2026). The engineering frontier is not summarized by a single “precision bit” number. Security-guideline work now separates concrete LWE/RLWE security from functional parameters, while compiler and library work tracks level/scale scheduling, bootstrapping precision and active-decryption hazards. In particular, CKKS deployments must account for the Li–Micciancio chosen-plaintext-with-decryption-output attack model when decryption results or partial decryptions are exposed. The frontier ledger therefore records error certificates, scale scheduling, decryption interfaces and security models separately and is frozen to 2026-09-03.

Example 5.4 (End-to-end encrypted polynomial evaluation). For the polynomial $f(x) = x^2 + 2x + 1$ and an encoded value $x = 1.25$, suppose the initial CKKS decoded error satisfies $|\varepsilon| \leq 2^{-20}$ and the rescale contributes at most 2^{-19} . The local invariant bounds the squared-term error by $2|x|2^{-20} + 2^{-40} + 2^{-19} < 2^{-17}$; adding the linear term contributes at most 2^{-19} more before any key-switching allowance. A BFV evaluation of the same polynomial over $\mathbb{Z}_t$ instead aims at the exact residue class and uses a half-scale decryption inequality. Numerical agreement in either toy calculation verifies arithmetic bookkeeping only; it does not establish RLWE hardness or production parameter safety.

5.3 Reader checkpoint

1. Separate CKKS encoding error from encryption noise.
2. Derive the addition error formula.
3. Derive the multiplication error formula.
4. Evaluate the bound when $|x_i| \leq 2$ and $|\varepsilon_i| \leq 10^{-6}$.
5. Explain why rescaling changes both modulus and scale.
6. Explain why slot rotations require evaluation keys.
7. Compare exact BFV correctness with approximate CKKS correctness.
8. Repeat the worked polynomial example at $x = -0.75$.
9. Explain why empirical output accuracy is not an IND-CPA proof.
10. Give one application-specific numerical tolerance and translate it into a required error budget.

Chapter 6

FHEW, TFHE, and Functional Bootstrapping

Gate-oriented FHE changes the computational granularity: rather than preserving a large arithmetic plaintext through a deep circuit, a bootstrap can itself implement a Boolean gate or a lookup table.

6.1 How to read this chapter

Blind rotation, external products and torus discretization are source-owned algorithms. The chapter emphasizes their interfaces and the boundary between ordinary programmable bootstrapping and full-domain lookup.

6.2 Learning goals

After this chapter, a reader should be able to define blind rotation and programmable bootstrapping, state the negacyclic lookup restriction, and compare one-bootstrap gate computation with leveled ring arithmetic.

Definition 6.1 (Gate and functional bootstrapping). A torus-LWE ciphertext represents a phase in $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with small error. A gadget decomposition writes a coefficient in a public radix so that a GSW/RLWE-style ciphertext can act through an external product. *Blind rotation* homomorphically rotates an encrypted test polynomial by an amount encoded in an LWE phase. Sample extraction converts a selected rotated coefficient back to an LWE ciphertext, usually followed by key switching. Programmable or functional bootstrapping chooses the test polynomial so that refresh simultaneously evaluates a lookup-table function. In the standard negacyclic ring, the lookup table must respect the induced negacyclic relation or the message domain must be restricted; *full-domain functional bootstrapping* removes or circumvents that restriction by an additional construction.

Theorem 6.2 (FHEW/TFHE programmable-bootstrapping interface — verified deep input). *FHEW and TFHE realize fast gate bootstrapping by combining LWE ciphertexts with GSW/RLWE-style external products, blind rotation, sample extraction and key switching. TFHE’s torus formulation supports bootstrapped Boolean gates and, with a test vector encoding an admissible lookup table, programmable bootstrapping that refreshes while evaluating that function. Correctness depends on phase separation and an explicit output-noise bound; security depends on the LWE/RLWE assumptions and on the distribution of the bootstrapping/evaluation keys. The full algorithmic proof and parameter analysis are source-owned.*

Theorem 6.3 (Efficient full-domain functional bootstrap of Ma–Huang–Wang–Zhou–Wang — verified deep input). *Ma, Huang, Anyu Wang, Zhou and Xiaoyun Wang, in TCHES 2024(1), give several full-domain functional-bootstrapping algorithms and homomorphic digit-decomposition algorithms for FHEW/TFHE-style computation. Their constructions remove the ordinary negacyclic lookup restriction over the intended full message domain, reduce output-noise growth relative to compared prior methods, and report runtime reductions up to 39.2% for the tested FDFB configurations; the two reported digit-decomposition algorithms run about $2.0\times$ and $1.5\times$ as fast as the compared predecessors. The exact parameter sets, OpenFHE implementation choices and security analysis are source-owned.*

Remark 6.4 (Homomorphic-noise and bootstrapping laboratory). The executable laboratory code `/volume_22/homomorphic_program_crypto_lab.py` verifies Paillier and ElGamal homomorphism, exact phase/noise algebra, deterministic rounded modulus switching, scaled approximate arithmetic, the matrix PIR correctness calculation of Chapter 8, and threshold ElGamal recombination from Chapter 10. It deliberately does not claim to implement production FHE or to validate cryptographic parameter security.

6.3 Reader checkpoint

1. Define the torus $\mathbb{T}$ and explain why a phase can encode a bit.
2. Describe the role of gadget decomposition in an external product.
3. Explain blind rotation without using implementation jargon.
4. State the ordinary negacyclic lookup restriction.
5. Explain why programmable bootstrapping refreshes and evaluates simultaneously.
6. Compare FHEW/TFHE gate bootstrapping with CKKS rescaling.
7. Separate output-noise analysis from security of a bootstrapping key.
8. Explain what “full domain” changes in functional bootstrapping.
9. Distinguish a benchmark speedup from an asymptotic theorem.
10. Run the Volume-XXII laboratory and identify which tests are algebraic rather than cryptographic.

Chapter 7

Functional Encryption

Functional encryption changes the authorization object: a secret key no longer decrypts an entire plaintext, but instead reveals a prescribed function of it.

7.1 How to read this chapter

Security notions for FE are unusually sensitive to the permitted function family, number of keys, challenge restrictions and whether security is simulation- or indistinguishability-based. No theorem here silently upgrades one notion to another.

7.2 Learning goals

After this chapter, a reader should be able to state FE syntax, distinguish collusion resistance from function privacy, and interpret inner-product FE as a restricted but concrete functionality rather than general-purpose FE.

Definition 7.1 (Functional encryption). For a function family $\mathcal{F}$, functional encryption has algorithms Setup , KeyGen , Enc , Dec such that a key sk_f decrypts a ciphertext of x to $f(x)$ and should reveal no additional information permitted by the declared security game. *Selective* security fixes challenge information before setup; *adaptive* security allows later choice. Indistinguishability and simulation security are different notions and can diverge for expressive functionalities. Collusion resistance addresses adversaries holding many functional keys. Function privacy additionally hides information about f . Inner-product FE reveals $\langle x, y \rangle$ for a key indexed by y ; predicate encryption reveals only a predicate outcome. Message and function domains, number of keys, leakage and setup assumptions are part of the theorem statement.

Theorem 7.2 (Inner-product functional encryption — verified deep input). *Abdalla, Bourse, De Caro and Pointcheval give simple functional-encryption schemes for inner products in cyclic-group settings, including a construction whose security is based on DDH. A functional key indexed by a vector y allows recovery of $\langle x, y \rangle$ from an encryption of x over the declared bounded algebraic domain. The exact challenge restrictions, admissible message/function spaces, collusion model and reduction loss are source-owned; the theorem is not promoted to function-hiding or general-circuit FE.*

Remark 7.3 (General-purpose functional-encryption assumptions — dated 3 September 2026). The FE frontier remains a hierarchy rather than one primitive. Boneh–Sahai–Waters show why naive simulation definitions fail for sufficiently expressive functionalities; efficient restricted FE such as inner products can rest on conventional group assumptions, while general-circuit FE typically uses much stronger tools such as iO, multilinear-style assumptions, or specialized lattice constructions with bounded-functionality/collusion tradeoffs. Recent work also studies compact lattice FE lower bounds and registered FE/ABE models. Every frontier entry is therefore indexed by functionality, selective/adaptive security, collusion bound, setup and assumption family; this ledger is frozen to 2026-09-03.

7.3 Reader checkpoint

1. Write the four FE algorithms and their inputs.
2. Explain why ordinary public-key encryption is a special functionality pattern.
3. Distinguish selective from adaptive FE security.
4. Distinguish simulation from indistinguishability security.
5. Define collusion resistance in words.
6. Give an inner-product application that reveals less than the full vector.
7. Explain why inner-product FE is not general-circuit FE.
8. State what additional property function privacy would require.
9. Explain why bounded message spaces can matter to an inner-product construction.
10. Build a table whose columns are functionality, setup, security game and assumption.

Chapter 8

Attribute-Based Encryption and Private Retrieval

Attribute-based encryption attaches decryption authority to access policies; private information retrieval hides which database location a client asks for. Both can be expressed using linear algebra, but their security goals are different.

8.1 How to read this chapter

The Waters CP-ABE theorem is source-owned. The PIR construction is elementary and proves an explicit $O(\sqrt{n})$ communication tradeoff from additive homomorphism.

8.2 Learning goals

After this chapter, a reader should be able to distinguish KP-ABE from CP-ABE, explain LSSS reconstruction, and derive correctness/privacy/communication for a matrix-based homomorphic PIR protocol.

Definition 8.1 (Attribute-based encryption and private retrieval). In ciphertext-policy attribute-based encryption (CP-ABE), a ciphertext carries an access policy and a user key carries attributes; decryption succeeds when the attribute set satisfies the policy. Key-policy ABE reverses these roles. A monotone access structure can be represented by a linear secret-sharing scheme (LSSS) matrix whose authorized rows reconstruct the shared secret by public linear coefficients. Collusion resistance means that combining unauthorized user keys does not create an authorized key. Private information retrieval (PIR) lets a client retrieve database item x_j while hiding j from the server. Computational PIR permits computational privacy. A leakage profile specifies any revealed sizes, access patterns or result metadata; ordinary PIR hides the query index, not necessarily the server's work or all application leakage.

Theorem 8.2 (Waters ciphertext-policy ABE from bilinear maps — verified deep input). *Waters' PKC 2011 construction realizes expressive CP-ABE for LSSS/monotone access policies in the standard model. In its most efficient presentation the ciphertext and main encryption/decryption work grow linearly with the access-formula representation, and selective CPA security is proved under the decisional q -parallel BDHE assumption; the paper also gives tradeoffs under weaker bilinear assumptions. The pairing equations, exact selective-security game and dual-system-style proof details are source-owned. No adaptive-security claim is inferred from the selectively secure theorem.*

Theorem 8.3 (Matrix homomorphic single-server PIR). *Let an $n = rc$ symbol database over $\mathbb{Z}_N$ be arranged as a matrix $X = (x_{ab}) \in \mathbb{Z}_N^{r \times c}$, and let E be an IND-CPA additively homomorphic encryption scheme over $\mathbb{Z}_N$. To retrieve x_{ij} , the client sends c ciphertexts $q_b = E(1)$ for $b = j$ and $q_b = E(0)$ otherwise. The server returns, for every row a ,*

$$C_a = \prod_{b=1}^c q_b^{x_{ab}}.$$

Then C_a encrypts x_{aj} , so decrypting response C_i returns x_{ij} . Query privacy reduces by a hybrid to IND-CPA security of E . The protocol uses c query ciphertexts and r response ciphertexts with $O(rc) = O(n)$ server exponentiations; choosing $r, c \asymp \sqrt{n}$ gives $O(\sqrt{n})$ ciphertext communication.

Proof. Additive homomorphism gives

$$C_a = E\left(\sum_b x_{ab} \mathbf{1}[b = j]\right) = E(x_{aj}),$$

proving correctness. For privacy, queries for two indices differ only in which positions encrypt zero or one. Replace the ciphertexts one at a time; an adversary distinguishing the two query distributions across the hybrid sequence yields an IND-CPA distinguisher for one component. The cost follows by counting ciphertexts and matrix entries processed by the server. $\square$

8.3 Reader checkpoint

1. Distinguish KP-ABE and CP-ABE.
2. Explain how an LSSS reconstructs a secret for an authorized set.
3. State the selective-security limitation of the Waters theorem used here.
4. Explain collusion resistance in an ABE system.
5. Derive the PIR response formula row by row.
6. Prove query privacy with an explicit hybrid sequence.
7. Minimize $r + c$ subject to $rc \geq n$.
8. Compute the server work for a 100×100 database matrix.
9. Explain why hiding the query index does not hide response length in general.
10. Compare this PIR protocol with downloading the entire database.

Chapter 9

Program Obfuscation and Witness Encryption

Obfuscation is a warning against conflating desirable intuition with realizable definitions. General virtual-black-box obfuscation is impossible, whereas indistinguishability obfuscation asks only that equivalent circuits of the same size have indistinguishable obfuscations.

9.1 How to read this chapter

The Barak et al. impossibility and Jain–Lin–Sahai construction are source-owned. The elementary implication $\text{iO} \Rightarrow \text{witness encryption}$ is proved locally because it follows directly from the definitions.

9.2 Learning goals

After this chapter, a reader should be able to state VBB and iO, explain why the impossibility theorem does not rule out every restricted obfuscator, and prove the circuit compiler from iO to witness encryption.

Definition 9.1 (Program obfuscation and witness encryption). A virtual-black-box (VBB) obfuscator preserves functionality and requires that whatever an efficient adversary learns from the obfuscated program can be simulated with oracle access to the program, up to the declared auxiliary-input model. An indistinguishability obfuscator (iO) preserves functionality and requires computationally indistinguishable obfuscations for every pair of same-size circuits computing the same function. Polynomial slowdown bounds the size and running-time expansion. Witness encryption for an NP relation $R(x, w)$ encrypts a message to an instance x : any valid witness decrypts, while for x with no witness, encryptions of equal-length messages are computationally indistinguishable. Null-iO and differing-inputs obfuscation are distinct weaker/alternative notions and are not identified with general iO.

Theorem 9.2 (Impossibility of general virtual-black-box obfuscation — verified deep input). *Barak, Goldreich, Impagliazzo, Rudich, Sahai, Vadhan and Yang construct efficiently computable circuit families possessing a property that can be recovered from code implementing a member but cannot be recovered from black-box oracle access in the required simulator sense. Consequently a general-purpose VBB obfuscator for all circuits cannot satisfy the proposed black-box definition. The unobfuscatable-family construction uses self-reference and auxiliary circuit interactions; its full separation proof is source-owned. The theorem does not rule out obfuscation for every restricted function class.*

Theorem 9.3 (Indistinguishability obfuscation from well-founded assumptions — verified deep input). *Jain, Lin and Sahai construct iO for all polynomial-size circuits assuming subexponential security of four explicitly stated assumption families: $SXDH$ in asymmetric prime-order bilinear groups, LWE with their subexponential modulus-to-noise regime, an LPN assumption over a prime field with inverse-polynomial error rate and polynomially many samples, and an NC^0 Boolean pseudorandom generator with mildly superlinear stretch. This is a conditional construction, not unconditional existence of iO . The parameter coupling and proof are source-owned. The STOC 2021 result was subsequently published in Journal of the ACM in 2026.*

Theorem 9.4 (iO implies witness encryption). *Assume iO for polynomial-size circuits. Then every NP language L has witness encryption for one-bit messages. For a polynomial-time verifier V and instance x , define*

$$F_{x,b}(w) = \begin{cases} b, & V(x, w) = 1, \\ 0, & V(x, w) = 0. \end{cases}$$

After padding $F_{x,0}$ and $F_{x,1}$ to the same size, set $\text{Enc}(x, b) = iO(F_{x,b})$ and $\text{Dec}(C, w) = C(w)$. If $x \in L$, every valid witness recovers b ; if $x \notin L$, the two circuits compute the same constant-zero function and iO security makes the ciphertext distributions indistinguishable.

Proof. Correctness is iO functionality preservation: for $V(x, w) = 1$, the obfuscated circuit returns b . If $x \notin L$, no witness is accepted, so $F_{x,0}$ and $F_{x,1}$ are both the constant-zero function. Padding makes their sizes equal, and the definition of iO gives computational indistinguishability of their obfuscations. No public-key primitive is needed for this implication. $\square$

Remark 9.5 (iO assumptions and efficiency frontier — dated 3 September 2026). The frontier separates classical iO from quantum-circuit obfuscation and tracks assumption simplification separately from efficiency. The Jain–Lin–Sahai route remains a theorem from explicit subexponential assumptions, now with a 2026 JACM publication, while general quantum iO from standard falsifiable assumptions remains unresolved and recent 2026 work gives black-box barriers for related QMA witness-encryption/null-qIO routes under restricted reductions. No such quantum barrier is retroactively stated as a classical- iO impossibility. The ledger is frozen to 2026-09-03.

9.3 Reader checkpoint

1. State the VBB simulator intuition precisely enough to see what oracle access means.
2. State iO for same-size equivalent circuits.
3. Explain why the VBB impossibility does not logically imply iO impossibility.
4. List the four assumption families in the Jain–Lin–Sahai theorem.
5. Explain why “well-founded assumptions” does not mean unconditional security.
6. Prove correctness of the iO -to-witness-encryption construction.
7. Prove its security on false instances.
8. Explain the need to pad the two circuits to equal size.
9. Extend the one-bit construction to fixed-length messages.
10. Distinguish the classical iO frontier from the quantum- iO frontier.

Chapter 10

Threshold, Multi-Key, and Verifiable Homomorphic Computation

Distributed homomorphic computation adds a second security surface: not only must evaluated ciphertexts remain secure, but key generation, partial decryption and result release must withstand malformed shares and chosen-ciphertext-style interactions.

10.1 How to read this chapter

Threshold ElGamal supplies a fully transparent local example. Modern threshold/multi-key FHE and verifiable evaluation remain construction-specific frontiers.

10.2 Learning goals

After this chapter, a reader should be able to distinguish threshold from multi-key FHE, prove Shamir-combined ElGamal decryption, and explain why proofs of valid partial decryption matter against active participants.

Definition 10.1 (Threshold and multi-key homomorphic computation). Threshold encryption distributes one logical secret key among parties so that an authorized subset can decrypt while an unauthorized subset cannot. Distributed key generation creates such shares without a single dealer learning the full secret. A partial decryption is a party's local contribution to final decryption; malicious-security variants require proofs that partial decryptions are well formed. Multi-key FHE instead evaluates ciphertexts created under different public keys and outputs a ciphertext requiring a prescribed joint decryption. Key aggregation compresses or combines key material when the construction permits. Verifiable homomorphic evaluation adds a proof that a ciphertext is the result of the claimed computation. Circuit privacy hides information about the evaluated circuit beyond its output and declared leakage. These properties are independent unless a theorem links them.

Theorem 10.2 (Threshold ElGamal decryption from Shamir sharing). *Let $G = \langle g \rangle$ have prime order q , let an ElGamal secret exponent $x \in \mathbb{Z}_q$ be Shamir-shared by a degree- t polynomial f with $f(0) = x$, and let party i hold $x_i = f(\alpha_i)$. For ciphertext $(u, v) = (g^r, m(g^x)^r)$, any set S of $t + 1$ parties computes partial decryptions $d_i = u^{x_i}$ and public Lagrange coefficients λ_i satisfying $\sum_{i \in S} \lambda_i x_i = x$. Then*

$$d = \prod_{i \in S} d_i^{\lambda_i} = u^x, \quad m = v/d.$$

Any coalition of at most t shares has perfect information-theoretic privacy about x by Shamir sharing. In an active protocol, each party can attach a Chaum–Pedersen/Sigma proof that $\log_g(g^{x_i}) = \log_u(d_i)$; special soundness and zero knowledge give the usual validity interface, while robustness against abort and full malicious threshold-security composition remain construction-owned.

Proof. By Lagrange interpolation at zero, $\sum_{i \in S} \lambda_i x_i = x$ in $\mathbb{Z}_q$. Therefore

$$\prod_{i \in S} d_i^{\lambda_i} = u^{\sum_i \lambda_i x_i} = u^x,$$

and ElGamal decryption gives $v/u^x = m$. Privacy of at most t key shares is exactly the Shamir theorem. For validity, the equality-of-discrete-logs Sigma protocol proves knowledge of a single exponent linking (g, g^{x_i}) and (u, d_i) ; its two accepting transcripts with different challenges extract that exponent by the same algebra as Schnorr special soundness. $\square$

Remark 10.3 (Verifiable, threshold, and multi-key FHE frontier — dated 3 September 2026). Current distributed-FHE work must be indexed by adversary model, setup, interaction, decryption-output exposure, proof system and key count. In particular, approximate and exact FHE can require additional protections when partial decryption outputs are exposed, and smudging or stronger decryption-security notions cannot be omitted from threshold claims. Multi-key and threshold TFHE/BGV/BFV/CKKS variants, low-communication threshold protocols, and verifiable FHE continue to trade ciphertext growth, interaction and proof overhead against stronger malicious-security guarantees. This frontier is frozen to 2026-09-03.

Remark 10.4 (Homomorphic and program-cryptography assumption map). The reader-facing map for this volume has four horizontal layers: *algebraic homomorphism* (Paillier/ElGamal); *noise-bearing lattice evaluation* (BGV/BFV/CKKS/FHEW/TFHE); *function- or policy-restricted decryption* (FE/ABE/PIR); and *program cryptography* (VBB/iO/witness encryption). Vertical annotations mark bootstrapping/evaluation keys, LWE/RLWE or group assumptions, setup and circular/KDM exposure, while side arrows connect MPC/zero knowledge to threshold decryption and verifiable evaluation. The map explicitly separates correctness, IND–CPA security, numerical accuracy, circuit privacy and malicious robustness.

10.3 Reader checkpoint

1. Distinguish threshold encryption from multi-key FHE.
2. Compute Lagrange coefficients for three Shamir points with threshold one.
3. Derive the threshold ElGamal recombination equation.
4. Explain why t Shamir shares reveal no information about x .
5. State the equality-of-discrete-logs statement proved by a partial-decryption proof.
6. Explain why a malformed partial decryption can break robustness without a validity proof.
7. Distinguish abort robustness from confidentiality.
8. Explain why decryption-output exposure changes the threat model for threshold FHE.
9. Trace one path through the assumption map from RLWE to threshold CKKS.
10. Trace one path from iO to witness encryption and explain which implication is proved locally.

Internalization and Reconstruction Guide: Volume XXII

Purpose. Understand computation on encrypted or hidden data by tracking a correctness invariant and a security interface: noise growth, rescaling, bootstrapping, functionality leakage, and program hiding are distinct axes.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-XX-DEF-019: Ring-LWE and Module-LWE.
2. QRCC-XVII-DEF-030: Bilinear pairings.
3. QRCC-XVII-THM-002: Chinese remainder and fast modular arithmetic.
4. QRCC-XVII-THM-019: ElGamal IND-CPA security under DDH.
5. QRCC-XX-THM-016: LWE search-to-decision and normal-form reductions.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XXII-DEF-001 (Homomorphic-encryption syntax and security); QRCC-XXII-DEF-004 (Noise-bearing lattice encryption); QRCC-XXII-DEF-007 (Bootstrapping and circular-security interfaces); QRCC-XXII-DEF-010 (BGV and BFV exact-arithmetic families); QRCC-XXII-DEF-014 (Approximate homomorphic arithmetic).

Chapter retrieval map

- Ch. 1: Homomorphic Encryption Before FHE:** enter through QRCC-XXII-DEF-001 (Homomorphic-encryption syntax and security); exit by being able to state QRCC-XXII-THM-002 (Paillier additive homomorphism) with its ownership boundary.
- Ch. 2: Noise, Relinearization, and Modulus Switching:** enter through QRCC-XXII-DEF-004 (Noise-bearing lattice encryption); exit by being able to state QRCC-XXII-THM-005 (LWE encryption addition and multiplication invariant) with its ownership boundary.

- Ch. 3: Bootstrapping: Turning Shallow Evaluation into FHE:** enter through QRCC-XXII-DEF-007 (Bootstrapping and circular-security interfaces); exit by being able to state QRCC-XXII-BB-009 (First ideal-lattice FHE construction) with its ownership boundary.
- Ch. 4: BGV, BFV, and Exact Homomorphic Arithmetic:** enter through QRCC-XXII-DEF-010 (BGV and BFV exact-arithmetic families); exit by being able to state QRCC-XXII-BB-011 (BGV leveled-FHE theorem) with its ownership boundary.
- Ch. 5: CKKS and Approximate Homomorphic Arithmetic:** enter through QRCC-XXII-DEF-014 (Approximate homomorphic arithmetic); exit by being able to state QRCC-XXII-THM-015 (CKKS approximate-arithmetic invariant) with its ownership boundary.
- Ch. 6: FHEW, TFHE, and Functional Bootstrapping:** enter through QRCC-XXII-DEF-017 (Gate bootstrapping and functional bootstrapping); exit by being able to state QRCC-XXII-BB-018 (TFHE programmable-bootstrapping interface) with its ownership boundary.
- Ch. 7: Functional Encryption:** enter through QRCC-XXII-DEF-020 (Functional encryption); exit by being able to state QRCC-XXII-THM-021 (Inner-product functional encryption in a bilinear-group model) with its ownership boundary.
- Ch. 8: Attribute-Based Encryption and Private Retrieval:** enter through QRCC-XXII-DEF-023 (Attribute-based encryption and private retrieval); exit by being able to state QRCC-XXII-BB-024 (Ciphertext-policy ABE from bilinear maps) with its ownership boundary.
- Ch. 9: Program Obfuscation and Witness Encryption:** enter through QRCC-XXII-DEF-026 (Program obfuscation and witness encryption); exit by being able to state QRCC-XXII-BB-028 (Indistinguishability obfuscation from well-founded assumptions) with its ownership boundary.
- Ch. 10: Threshold, Multi-Key, and Verifiable Homomorphic Computation:** enter through QRCC-XXII-DEF-031 (Threshold and multi-key homomorphic computation); exit by being able to state QRCC-XXII-THM-032 (Threshold ElGamal decryption from Shamir sharing) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XXII-THM-002 — Paillier additive homomorphism. Dependencies: QRCC-XVII-THM-002, QRCC-XXII-DEF-001. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*
2. QRCC-XXII-BB-009 — First ideal-lattice FHE construction. Dependencies: QRCC-XXII-THM-008, QRCC-XX-DEF-019. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-XXII-THM-015 — CKKS approximate-arithmetic invariant. Dependencies: QRCC-XXII-DEF-014, QRCC-XXII-THM-006. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

4. QRCC-XXII-THM-021 — Inner-product functional encryption in a bilinear-group model. Dependencies: QRCC-XVII-DEF-030, QRCC-XXII-DEF-020. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XXII-BB-028 — Indistinguishability obfuscation from well-founded assumptions. Dependencies: QRCC-XXII-DEF-026, QRCC-XX-DEF-019. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XXII-F-016 — Verified numerics and error certification for CKKS. Dependencies: QRCC-XXII-THM-015. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- mixing ciphertext modulus, plaintext scale, and approximation error;
- treating leveled evaluation as bootstrapped FHE without an evaluable decryption circuit;
- confusing functional-encryption leakage with obfuscation security;
- using VBB impossibility as though it ruled out indistinguishability obfuscation;

Boundary and degenerate cases

- zero multiplicative depth and no-bootstrapping cases;
- rounding exactly on a modulus-switch boundary;
- exact BFV/BGV arithmetic versus CKKS approximate arithmetic;
- function privacy, message privacy, and program privacy as separate notions;

Retrieval and reconstruction drill

1. State QRCC-XXII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XXII-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XXII-BB-009, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.

6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XXII-F-016, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XXIII returns to finite fields at a deeper arithmetic-geometric level and connects zeta algorithms with coding and decoding complexity.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XXIII

Finite-Field Arithmetic, Zeta Functions, Coding, and the Wan Program

Chapter 1

Characters, Gauss Sums, and Square-Root Cancellation

Finite-field character sums are the first bridge from elementary arithmetic to the cohomological estimates used later in the volume. This chapter fixes conventions carefully, proves the elementary Fourier identities locally, and then records the Weil and Deligne inputs with their hypotheses visible.

1.1 How to read this chapter

The Gauss-sum theorem is proved completely. The Weil and Deligne bounds are source-owned deep inputs: the reader should learn their exact hypotheses and cancellation scale, not mistake the one-line statements for replacement proofs of the underlying curve or ℓ -adic theory.

1.2 Learning goals

After this chapter, a reader should be able to normalize additive and multiplicative characters, prove both orthogonality relations, compute the magnitude of a nontrivial Gauss sum, and check when a quoted square-root bound is invalid because an exceptional power or singular leading form is present.

Definition 1.1 (Additive and multiplicative character sums). Let $\mathbb{F}_q$ be the field with $q = p^a$ elements. An *additive character* is a homomorphism $\psi : (\mathbb{F}_q, +) \rightarrow \mathbb{C}^\times$; it is nontrivial if it is not identically 1. A *multiplicative character* is a homomorphism $\chi : \mathbb{F}_q^\times \rightarrow \mathbb{C}^\times$; for a nontrivial χ we extend it by $\chi(0) = 0$. Its order is its order in the character group of $\mathbb{F}_q^\times$.

For nontrivial χ, ψ , the *Gauss sum* is

$$G(\chi, \psi) = \sum_{x \in \mathbb{F}_q} \chi(x) \psi(x).$$

For a polynomial or rational function f on a specified domain, a character sum means a finite sum such as $\sum_x \psi(f(x))$ or $\sum_x \chi(f(x))$, with poles omitted and the convention at zeros stated. *Square-root cancellation* means a bound of order $q^{1/2}$ in one free variable, or $q^{n/2}$ in n free variables, up to a complexity factor independent of q .

Theorem 1.2 (Orthogonality and Gauss-sum magnitude). *Let ψ be a nontrivial additive character and χ a nontrivial multiplicative character of $\mathbb{F}_q$. Then*

$$\sum_{x \in \mathbb{F}_q} \psi(ax) = \begin{cases} q, & a = 0, \\ 0, & a \neq 0, \end{cases} \quad \sum_{x \in \mathbb{F}_q^\times} \chi(x)^j = \begin{cases} q-1, & \chi^j = 1, \\ 0, & \chi^j \neq 1. \end{cases}$$

Moreover $|G(\chi, \psi)| = \sqrt{q}$.

Proof. If $a \neq 0$, multiplication by a permutes $\mathbb{F}_q$. Writing $S = \sum_x \psi(x)$ and choosing b with $\psi(b) \neq 1$ gives $S = \sum_x \psi(x+b) = \psi(b)S$, hence $S = 0$. The multiplicative relation is identical: if a character θ of $\mathbb{F}_q^\times$ is nontrivial, choose b with $\theta(b) \neq 1$ and compare $\sum_x \theta(x)$ with the same sum after $x \mapsto bx$.

Because character values have absolute value one on $\mathbb{F}_q^\times$,

$$|G|^2 = \sum_{x, y \in \mathbb{F}_q^\times} \chi(x/y) \psi(x-y) = \sum_{t \in \mathbb{F}_q^\times} \chi(t) \sum_{y \neq 0} \psi((t-1)y).$$

For $t = 1$ the inner sum is $q-1$; for $t \neq 1$ it is -1 . Since $\sum_{t \neq 0} \chi(t) = 0$, we have $\sum_{t \neq 1} \chi(t) = -1$. Thus $|G|^2 = (q-1) - (-1) = q$. $\square$

Theorem 1.3 (Weil bound for a selected one-variable multiplicative character sum — verified deep input). *Let χ be a nontrivial multiplicative character of order m on $\mathbb{F}_q^\times$. Let $f \in \mathbb{F}_q[X]$ have $r \geq 1$ distinct zeros over $\overline{\mathbb{F}}_q$, and suppose that f is not a constant times an m th power in $\overline{\mathbb{F}}_q[X]$. With $\chi(0) = 0$,*

$$\left| \sum_{x \in \mathbb{F}_q} \chi(f(x)) \right| \leq (r-1)\sqrt{q}.$$

This is the polynomial Kummer-curve form used in this volume. Rational-function and mixed additive-multiplicative variants require conductor/pole corrections and are not silently substituted for this statement.

Theorem 1.4 (Deligne's smooth-leading-form exponential-sum bound — verified deep input). *Let $f \in \mathbb{F}_q[X_1, \dots, X_n]$ have degree d with $p \nmid d$. Assume that its degree- d homogeneous part defines a smooth hypersurface in $\mathbb{P}_{\mathbb{F}_q}^{n-1}$. For every nontrivial additive character ψ of $\mathbb{F}_q$,*

$$\left| \sum_{x \in \mathbb{F}_q^n} \psi(f(x)) \right| \leq (d-1)nq^{n/2}.$$

The smoothness and prime-to- p hypotheses are part of the theorem. Singular leading forms require different estimates.

1.3 Worked hypothesis audit

For a quadratic polynomial $f(x) = ax^2 + bx + c$ with $a \neq 0$ and odd q , Theorem 1.4 with $n = 1, d = 2$ gives the sharp scale $\sqrt{q}$. By contrast, if $f = g^m$ in the multiplicative setting, then $\chi(f(x))$ can be identically 1 away from zeros, so a square-root bound would be false; this is why the exceptional-power clause in Theorem 1.3 is structural rather than cosmetic.

1.4 Reader checkpoint

1. Prove additive orthogonality without using Fourier terminology.
2. Prove multiplicative orthogonality for every character power χ^j .
3. Recompute $|G(\chi, \psi)|^2$ by the change of variables $x = ty$.
4. Explain why the convention $\chi(0) = 0$ is harmless for nontrivial χ .
5. Give an example showing failure of the Weil bound when f is an m th power.
6. Check the Weil bound for $f(X) = X(X - 1)$ and a quadratic character.
7. State exactly the smooth-leading-form hypothesis in the Deligne theorem.
8. Explain why $p \nmid d$ appears in the selected Deligne statement.
9. Compare the cancellation scales $q^{1/2}$ and $q^{n/2}$ with the trivial bounds.
10. Identify which results in this chapter are locally proved and which are source-owned.

Chapter 2

Zeta Functions and the Weil Package

Point counts over all finite extensions can be compressed into one generating function. The resulting zeta function is elementary to define but unexpectedly rigid: Dwork proves rationality for every finite-field variety, and the smooth-projective case carries the full Weil package.

2.1 How to read this chapter

First master the logarithmic generating function and the elementary affine/projective calculations. Dwork rationality and the cohomological Weil conjectures are source-owned; the small elliptic-curve example is included to make the abstract reciprocal-root language concrete.

2.2 Learning goals

A reader should be able to move between extension-field point counts and $Z(X, T)$, compute Z for affine and projective spaces, read Frobenius eigenvalue information from a curve numerator, and separate Dwork's rationality theorem from Grothendieck's cohomological formalism and Deligne's weight theorem.

Definition 2.1 (Hasse–Weil zeta function). Let X be a separated scheme of finite type over $\mathbb{F}_q$ and put $N_r = |X(\mathbb{F}_{q^r})|$. Its *Hasse–Weil zeta function* is the formal power series

$$Z(X, T) = \exp \left(\sum_{r \geq 1} N_r \frac{T^r}{r} \right).$$

When $Z(X, T)$ is rational, its zeros and poles may be described through reciprocal roots: a factor $1 - \alpha T$ has reciprocal root α . A reciprocal root has *weight* w if every complex embedding has absolute value $q^{w/2}$. For smooth projective X , the *Betti-number degree* b_i means the dimension of the i th ℓ -adic cohomology group for a prime $\ell \neq p$. The Euler product $\prod_{x \in |X|} (1 - T^{\deg x})^{-1}$, when interpreted formally, is equivalent to the exponential definition.

Theorem 2.2 (Zeta functions of affine and projective space). *For every $n \geq 0$,*

$$Z(\mathbb{A}_{\mathbb{F}_q}^n, T) = \frac{1}{1 - q^n T}, \quad Z(\mathbb{P}_{\mathbb{F}_q}^n, T) = \prod_{i=0}^n \frac{1}{1 - q^i T}.$$

Moreover $Z(\mathbb{G}_m^n, T)$ is determined by $|\mathbb{G}_m^n(\mathbb{F}_{q^r})| = (q^r - 1)^n$ and hence equals

$$\prod_{j=0}^n (1 - q^j T)^{-(-1)^{n-j} \binom{n}{j}}.$$

Proof. We have $|\mathbb{A}^n(\mathbb{F}_{q^r})| = q^{nr}$. Since $-\log(1 - aT) = \sum_{r \geq 1} a^r T^r / r$, the affine formula follows. Also $|\mathbb{P}^n(\mathbb{F}_{q^r})| = 1 + q^r + \cdots + q^{nr}$; adding the corresponding logarithms yields the product formula. Finally $(q^r - 1)^n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} q^{jr}$, and exponentiating term by term gives the torus formula. $\square$

Theorem 2.3 (Dwork rationality theorem — verified deep input). *For every separated scheme X of finite type over $\mathbb{F}_q$, the formal series $Z(X, T)$ is a rational function of T . Dwork’s proof constructs p -adic analytic trace operators and Fredholm determinants; the operator-theoretic normalization needed for that proof is developed as a source-owned interface in Chapter 4.*

Theorem 2.4 (Weil conjectures for smooth projective varieties — verified deep input). *Let $X/\mathbb{F}_q$ be smooth, projective, and geometrically connected of dimension d . Then*

$$Z(X, T) = \prod_{i=0}^{2d} P_i(T)^{(-1)^{i+1}},$$

where $P_i(T) \in \mathbb{Z}[T]$, $P_0 = 1 - T$, $P_{2d} = 1 - q^d T$, and $\deg P_i = b_i$. Every reciprocal root of P_i has complex absolute value $q^{i/2}$. Poincaré duality yields the usual functional equation after the Euler characteristic and power of q are inserted. QRCC attributes rationality historically to Dwork, the cohomological factorization/functional-equation framework to Grothendieck, and the Riemann-hypothesis weight assertion to Deligne rather than compressing these contributions into one anonymous theorem.

Example 2.5 (A small elliptic curve and its Frobenius polynomial). Consider $E/\mathbb{F}_5$ given by $y^2 = x^3 + x + 1$. Direct counting gives $|E(\mathbb{F}_5)| = 9$, so the Frobenius trace is $a = 5 + 1 - 9 = -3$. Thus

$$Z(E, T) = \frac{1 + 3T + 5T^2}{(1 - T)(1 - 5T)}.$$

If α, β are the reciprocal roots of the numerator, then $\alpha + \beta = -3$ and $\alpha\beta = 5$, hence

$$|E(\mathbb{F}_{25})| = 25 + 1 - (\alpha^2 + \beta^2) = 26 - ((-3)^2 - 10) = 27.$$

The laboratory in Chapter 8 verifies the $\mathbb{F}_{25}$ count directly using $\mathbb{F}_5[u]/(u^2 - 2)$. The Weil bound is visible as $|a| = 3 \leq 2\sqrt{5}$.

2.3 Reader checkpoint

1. Derive the affine-space zeta formula directly from the logarithmic definition.
2. Derive the projective-space formula from its cell decomposition.
3. Expand the torus formula for $n = 1$ and $n = 2$.
4. Prove the formal identity between the Euler product and extension-field point counts.

5. For the displayed elliptic curve, verify the nine $\mathbb{F}_5$ points by hand.
6. Compute N_2 from a and q using $\alpha^2 + \beta^2$.
7. Explain what “weight i ” says about the roots of P_i .
8. Distinguish rationality from the Riemann-hypothesis part of the Weil conjectures.
9. Explain why smoothness and projectivity matter in the stated Weil package.
10. Identify which part of the small-curve computation uses a deep theorem and which part is elementary.

Chapter 3

Newton Polygons and Nondegenerate Exponential Sums

Newton polygons turn valuations of Frobenius eigenvalues into geometry. Their elementary calculus is simple enough to prove locally; identifying those slopes for exponential-sum L -functions is the deep part.

3.1 How to read this chapter

Treat the polygon calculus as exact algebra. Then read the Adolphson–Sperber theorem as a source-owned bridge from a Laurent polynomial’s Newton polytope to the degree and cohomological size of its exponential-sum L -function.

3.2 Learning goals

A reader should be able to construct a Newton polygon from coefficients, recover slopes from reciprocal roots over a splitting field, prove the product rule, understand why specialization can move a polygon upward, and state the nondegeneracy condition used by Adolphson and Sperber.

Definition 3.1 (L -functions, Newton polygons, and nondegeneracy). Let (K, v) be a complete discretely valued field, normalized by $v(\pi) = 1$. For $P(T) = 1 + a_1T + \cdots + a_mT^m \in K[T]$ with $a_m \neq 0$, the *Newton polygon* $\text{NP}(P)$ is the lower convex hull of $(i, v(a_i))$, $0 \leq i \leq m$, with $a_0 = 1$. If $P(T) = \prod_i (1 - \alpha_i T)$ in an algebraic closure, its slope multiset is the multiset $\{v(\alpha_i)\}$ with the standard normalization.

For a Laurent polynomial $f = \sum_u a_u X^u$ in n variables, its Newton polytope at infinity is $\Delta(f) = \text{conv}(\{0\} \cup \{u : a_u \neq 0\}) \subset \mathbb{R}^n$. For a face τ not containing 0, write f_τ for the sum of monomials with exponent in τ . The polynomial is *nondegenerate with respect to* $\Delta(f)$ if the equations $X_i \partial f_\tau / \partial X_i = 0$ for $1 \leq i \leq n$ have no common solution in $(\overline{\mathbb{F}}_q^\times)^n$ for every such face. The exponential-sum L -function is $L^*(f, T) = \exp(\sum_{r \geq 1} S_r^*(f) T^r / r)$, where $S_r^*(f)$ is the additive-character sum over $(\mathbb{F}_{q^r}^\times)^n$ with the character lifted by the field trace.

Theorem 3.2 (Newton-polygon product and specialization calculus). *Let $P, Q \in 1 + TK[T]$.*

1. *The slope multiset of PQ is the disjoint union of the slope multisets of P and Q ; equivalently, $\text{NP}(PQ)$ is obtained by concatenating the slopes of $\text{NP}(P)$ and $\text{NP}(Q)$ and sorting them increasingly.*

2. Let $\varphi : K \rightarrow K'$ be a homomorphism into another valued field satisfying $v'(\varphi(a)) \geq v(a)$ for every coefficient under consideration. If P^φ has the same degree as P , then $\text{NP}(P^\varphi)$ lies on or above the lower convex hull determined by the coefficient valuations of P . In particular, coefficient specialization cannot move a vertex below its pre-specialization valuation bound.

Proof. Over a common splitting field write $P = \prod_i (1 - \alpha_i T)$ and $Q = \prod_j (1 - \beta_j T)$. Their product has precisely the combined reciprocal-root multiset, and the Newton polygon theorem identifies polygon slopes with the corresponding valuations; this proves (1). For (2), every coefficient point $(i, v'(\varphi(a_i)))$ lies on or above $(i, v(a_i))$. Taking lower convex hulls preserves this pointwise lower bound. The same-degree hypothesis excludes loss of the terminal point through a zero leading coefficient. $\square$

Theorem 3.3 (Adolphson–Sperber nondegenerate Newton-polyhedron theorem — verified deep input). *Let f be a Laurent polynomial over $\mathbb{F}_q$ that is nondegenerate with respect to an n -dimensional Newton polytope $\Delta(f)$. Then the toric exponential-sum L -function has the expected cohomological concentration; in particular*

$$L^*(f, T)^{(-1)^{n-1}}$$

is a polynomial of degree $n! \text{Vol}(\Delta(f))$, with Euclidean volume normalized so that a fundamental lattice simplex has volume $1/n!$. The stronger purity and Newton-versus-Hodge assertions require the additional hypotheses stated in the Adolphson–Sperber papers and remain source-owned rather than being bundled into this degree statement.

3.3 A two-slope example

If $P(T) = (1 - \pi T)(1 - u\pi^3 T)$ with u a unit, then the slopes are 1 and 3. Multiplying by $Q(T) = 1 - vT$ with v a unit adds slope 0, so the product slopes are 0, 1, 3. This simple bookkeeping is the local algebra behind much more sophisticated slope decompositions.

3.4 Reader checkpoint

1. Draw the Newton polygon of $1 + \pi T + \pi^4 T^2$.
2. Verify the product rule for three explicit linear factors.
3. Explain why a coefficient whose valuation increases can move a polygon upward.
4. Give an example where degree loss would invalidate the specialization conclusion.
5. Compute $\Delta(f)$ for $f = X + Y + (XY)^{-1}$.
6. State the face-by-face nondegeneracy test for that polynomial.
7. Explain the normalization of $n! \text{Vol}(\Delta)$.
8. Distinguish the degree conclusion from purity and from a Newton–Hodge inequality.
9. Explain why the Adolphson–Sperber theorem is a natural input to p -adic algorithms.
10. Identify the exact local theorem proved in this chapter.

Chapter 4

Dwork Operators, Trace Formulas, and Cohomological Determinants

Dwork’s method converts point counts into traces of p -adic Frobenius operators and then traces into Fredholm determinants. The analytic construction is deep; the purpose here is to expose the complete interface used by later arithmetic and algorithmic chapters.

4.1 How to read this chapter

This is an interface chapter. The definitions explain what overconvergence, complete continuity, and a Fredholm determinant mean in the selected model. The two theorems are source-owned; no finite-dimensional analogy is presented as a proof of Dwork’s analytic convergence arguments.

4.2 Learning goals

A reader should be able to recognize the three stages point count $\rightarrow$ trace $\rightarrow$ determinant, explain why an infinite-dimensional determinant needs nuclearity/completely-continuous hypotheses, and read an alternating cohomological determinant without confusing it with an ordinary finite matrix identity.

Definition 4.1 (Selected Dwork-cohomology interface). Fix a finite extension $K/\mathbb{Q}_p$ with valuation ring $\mathcal{O}_K$. An *overconvergent power series* on a chosen affinoid model is a power series whose coefficients satisfy a growth condition strong enough to converge on a strictly larger p -adic domain than the closed unit domain. A continuous endomorphism U of a p -adic Banach space is *completely continuous* (or nuclear in the Dwork setting used here) when it admits finite-rank approximations strong enough for traces $\mathrm{Tr}(U^r)$ and the Fredholm determinant

$$\det(1 - TU) = \exp \left(- \sum_{r \geq 1} \mathrm{Tr}(U^r) \frac{T^r}{r} \right)$$

to be defined on a nonzero p -adic disk. A *Dwork complex* $C_f^\bullet$ is the selected overconvergent cochain complex attached to f , together with Frobenius operators F_i on its terms or cohomology. The splitting function and Frobenius lift are the source-specific analytic data used to build those F_i .

Theorem 4.2 (Dwork trace formula in the selected hypersurface/torus model — verified deep input). *For a polynomial or Laurent polynomial f in the Dwork model fixed by the primary source, there is an overconvergent Banach space and a completely continuous Frobenius operator α_f such that the extension-field exponential sums are recovered from traces of iterates of α_f after the explicit normalization factors prescribed by the model. Equivalently, the exponential generating series of those traces is a finite quotient or alternating product of Fredholm determinants. The normalization by powers of q , the auxiliary ψ -operators, and the passage between affine and toric point counts are part of the source-owned theorem data and are not suppressed when the theorem is applied.*

Theorem 4.3 (Cohomological determinant formula for zeta and L -functions — verified deep input). *Let $C_f^\bullet$ be a Dwork cohomological complex for which Frobenius F_i acts completely continuously and for which the trace formula of Theorem 4.2 applies. Then the associated zeta or exponential-sum L -function has the alternating determinant form*

$$L_f(T) = \prod_i \det(1 - TF_i \mid H^i(C_f^\bullet))^{(-1)^{i+1}},$$

with the precise shift and normalization chosen according to the affine, toric, or hypersurface model. When the relevant cohomology is finite-dimensional, this expression is rational. In the original Dwork proof, rationality may also emerge from cancellation between Fredholm determinants before finite-dimensional cohomology is isolated.

4.3 Finite-dimensional sanity model

For a finite matrix A , the identity

$$\det(1 - TA) = \exp \left(- \sum_{r \geq 1} \text{Tr}(A^r) T^r / r \right)$$

follows formally from eigenvalues. Dwork's achievement is not this identity; it is the construction of an infinite-dimensional p -adic operator for which the traces encode arithmetic and the Fredholm determinant converges. Keeping this distinction prevents the sanity model from being mistaken for the proof.

4.4 Reader checkpoint

1. Derive the trace–determinant identity for a diagonal finite matrix.
2. Explain why complete continuity is needed in infinite dimension.
3. Distinguish a Frobenius lift from the geometric Frobenius endomorphism on points.
4. Explain what overconvergence buys analytically.
5. Describe the point-count $\rightarrow$ trace step in one sentence.
6. Describe the trace $\rightarrow$ determinant step in one sentence.
7. Explain why an alternating product appears after passing to a complex.
8. State a condition under which the determinant expression is visibly rational.

9. Identify which normalizations this chapter deliberately leaves source-owned.
10. Explain how Chapter 3's Newton polygons can be read from Frobenius determinants.

Chapter 5

Unit Roots, Dwork's Conjecture, and the Wan Program

Wan's work makes p -adic variation itself an object of arithmetic study. Unit-root pieces of Frobenius need not give rational functions, but Dwork conjectured a strong p -adic meromorphy property. Wan proved the conjecture and developed higher-rank and variation theories around it.

5.1 How to read this chapter

The first two theorem rows are source-owned and should be read as landmarks in a program rather than as isolated black boxes. The final row is a dated frontier: it records what Wan–Zhang had actually published by the PASS105 certification date and what remains open.

5.2 Learning goals

A reader should be able to define slope-zero/unit-root parts, distinguish rationality from p -adic meromorphy, explain why ordinarity stabilizes a slope decomposition in families, and identify the publication status of the current rigid-cohomology/Betti-bound frontier.

Definition 5.1 (Unit roots and p -adic meromorphy). For a Frobenius operator over a p -adic coefficient field, a reciprocal eigenvalue is a *unit root* if its p -adic valuation is 0; more generally, the *slope- s part* is the generalized Frobenius subspace with valuation slope s . A family is *ordinary* in the selected sense when its Newton polygon attains the prescribed Hodge lower bound so that the relevant slope pieces have the expected ranks. A power series or quotient is *p -adic meromorphic* on a domain if it is locally a quotient of convergent p -adic analytic functions there. Dwork's unit-root conjecture asks for meromorphic continuation of unit-root L -functions arising from ordinary geometric families; it does not assert that those functions are rational.

Theorem 5.2 (Wan's proof of Dwork's unit-root conjecture — verified deep input). *For the ordinary geometric families and unit-root zeta/ L -functions covered by Wan's 1999 Annals theorem, the unit-root function admits the conjectured p -adic meromorphic continuation. The proof rewrites the unit-root contribution as a p -adic limit of nuclear σ -module constructions and controls the resulting Fredholm determinants. The exact family, ordinarity, coefficient-field, and slope hypotheses are retained from the primary theorem rather than generalized here to arbitrary F -isocrystals.*

Theorem 5.3 (Wan's higher-rank and Newton-variation theorems — verified deep inputs). *Wan's two JAMS papers of 2000 extend the unit-root meromorphy program through rank-one and higher-rank overconvergent σ -modules, with the higher-rank case reduced to the rank-one theory in the source's precise category. His 2004 Asian Journal of Mathematics paper develops variation of p -adic Newton polygons for exponential-sum L -functions, including decomposition and generic/specialized polygon comparisons under the nondegeneracy and family hypotheses stated there. These statements are kept as separately source-gated components; no claim is made that every p -adic family has uniform slopes or a constant Newton polygon.*

Theorem 5.4 (Rigid cohomology and explicit Betti bounds — dated PASS105 frontier). *As verified on 2026-09-03, Wan–Zhang's rigid-cohomology work is no longer only a preprint frontier. Revisiting Dwork cohomology: visibility and divisibility of Frobenius eigenvalues in rigid cohomology appeared in Compositio Mathematica 161 (2025), 1215–1249. Betti number bounds for varieties and exponential sums appeared in Advances in Mathematics 490 (2026), article 110852. The latter gives new explicit compact-support Betti-number bounds for affine varieties defined by bounded-degree equations and asymptotically sharp behavior in important complete-intersection regimes. The general problem of asymptotically optimal bounds in all parameter regimes remains a frontier and is not promoted to a closed theorem here.*

5.3 Reader checkpoint

1. Define a unit root in valuation language.
2. Explain why a unit-root L -function need not be rational.
3. Distinguish ordinary from merely smooth.
4. Explain the role of a slope decomposition in a family.
5. State the conclusion of Wan's 1999 theorem without overgeneralizing its hypotheses.
6. Explain why higher rank is a genuinely new problem.
7. Describe what it means for Newton polygons to vary in a family.
8. Give the publication year and venue of each Wan–Zhang result recorded here.
9. State one conclusion that is published and one question that remains frontier-level.
10. Explain why this chapter uses dated frontier language.

Chapter 6

Point Counting as an Algorithmic Problem

The same Frobenius data that controls zeta functions can sometimes be computed efficiently. Three algorithmic paradigms illustrate different parameter regimes: torsion arithmetic for elliptic curves, Monsky–Washnitzer cohomology for hyperelliptic curves, and Dwork/trace-formula methods for fixed-dimensional varieties in small characteristic.

6.1 How to read this chapter

Keep existence, algorithm, and bit complexity separate. Theorems below are source-owned because their precision management and complexity analyses are substantial. The reader-facing goal is to know exactly which parameters are polynomial and which are fixed or restricted.

6.2 Learning goals

A reader should be able to define the input-size parameters for point counting, explain Schoof’s CRT strategy, state Kedlaya’s fixed-characteristic complexity regime, and state the characteristic and fixed-dimension hypotheses in Lauder–Wan.

Definition 6.1 (Point-counting algorithm model). A *point-counting algorithm* receives equations over $\mathbb{F}_q$ together with an explicit encoding of $\mathbb{F}_q$ and outputs one or more counts $|X(\mathbb{F}_{q^r})|$ or the rational zeta function. *Bit complexity* counts bit operations in the encoded input; it must be distinguished from arithmetic-operation complexity over $\mathbb{F}_q$ or a p -adic ring. A p -adic algorithm also specifies working precision, precision loss under reduction, and the cost of Frobenius lifting. A theorem polynomial in $\log q$ with genus or dimension fixed is not automatically polynomial when those geometric parameters are part of the input.

Theorem 6.2 (Schoof point counting for elliptic curves — verified deep algorithmic input). *In the original characteristic- $p > 3$ setting, Schoof gives a deterministic algorithm that computes $\#E(\mathbb{F}_q)$ for an elliptic curve $E/\mathbb{F}_q$ in time polynomial in $\log q$; the original paper gives an $O((\log q)^9)$ bound in its bit-operation model. Writing*

$$\#E(\mathbb{F}_q) = q + 1 - t, \quad |t| \leq 2\sqrt{q},$$

the algorithm determines $t \bmod \ell$ for enough small primes $\ell \neq p$ by testing the Frobenius relation on $E[\ell]$, then reconstructs the unique integer t in the Hasse interval by the Chinese remainder theorem.

Theorem 6.3 (Kedlaya’s hyperelliptic p -adic algorithm — verified deep input). *For an arbitrary hyperelliptic curve of genus g over $\mathbb{F}_{p^n}$ with odd p , Kedlaya computes the characteristic polynomial of Frobenius using Monsky–Washnitzer cohomology and p -adic approximation. For fixed p , the original asymptotic running-time statement is $O(g^{4+\varepsilon}n^{3+\varepsilon})$ for every fixed $\varepsilon > 0$, with precision and memory governed by the cohomological reduction. Later improvements are not retroactively attributed to the 2001 theorem.*

Theorem 6.4 (Lauder–Wan zeta algorithms in small characteristic — verified deep input). *Lauder and Wan give a deterministic polynomial-time method for computing the zeta function of an arbitrary variety of fixed dimension over a finite field in their suitably-small-characteristic regime. The method makes Dwork’s trace formula algorithmic, combines it with degree bounds and semilinear reduction, and keeps the characteristic restriction explicit. It therefore does not assert a single uniform polynomial-time theorem simultaneously in unrestricted characteristic, dimension, degree, and coefficient size.*

6.3 Three different polynomialities

Schoof is polynomial in $\log q$ for elliptic curves. Kedlaya’s original bound is especially effective when p is fixed and genus is moderate. Lauder–Wan fixes dimension and restricts the characteristic regime. These are complementary statements, not contradictory complexity estimates for the same input model.

6.4 Reader checkpoint

1. Explain why the Hasse bound makes CRT reconstruction possible in Schoof’s algorithm.
2. Determine how large the product of auxiliary primes must be to recover t uniquely.
3. Explain why one works modulo division polynomials rather than enumerating $E[\ell]$ points.
4. State the characteristic restriction in Schoof’s original paper.
5. State Kedlaya’s field and curve hypotheses.
6. Identify the parameters in $O(g^{4+\varepsilon}n^{3+\varepsilon})$.
7. Explain why “fixed p ” is a material complexity qualifier.
8. State the fixed-dimension, small-characteristic regime of Lauder–Wan.
9. Distinguish computing one point count from computing a full zeta function.
10. Give one reason precision loss matters in a p -adic point-counting implementation.

Chapter 7

Subset Sums and the Li–Wan Distinct-Coordinate Sieve

Many finite-field counting problems ask for tuples satisfying an equation and also require all coordinates to be distinct. Classical inclusion–exclusion over all pairwise equalities is enormous. Li and Wan reorganize it by permutation cycle type.

7.1 How to read this chapter

The distinct-coordinate identity is proved locally and is the reusable combinatorial mechanism. The quantitative finite-field subset-sum estimates are source-owned because their strength comes from combining this sieve with character-sum bounds.

7.2 Learning goals

A reader should be able to formulate fixed-cardinality subset sum as a distinct-coordinate problem, prove the Li–Wan sieve by signed permutations, group the sieve by cycle type in the symmetric case, and understand where Weil-type estimates enter the quantitative theory.

Definition 7.1 (Subset-sum counts and distinct-coordinate restrictions). Let $D \subseteq \mathbb{F}_q$ and $b \in \mathbb{F}_q$. Write $N_D(k, b)$ for the number of k -element subsets $S \subseteq D$ with $\sum_{x \in S} x = b$. For $X \subseteq D^k$, write $\bar{X}$ for the elements of X whose coordinates are pairwise distinct. For a permutation $\tau \in S_k$ with cycle decomposition, X_τ is the subset of X on which coordinates belonging to the same cycle of τ are equal, and $c(\tau)$ is the number of cycles. If X is invariant under coordinate permutations, then $|X_\tau|$ depends only on the cycle type of τ , permitting a partition-indexed form of the sieve.

Theorem 7.2 (Li–Wan distinct-coordinate sieve). *For every finite set D and every $X \subseteq D^k$,*

$$|\bar{X}| = \sum_{\tau \in S_k} (-1)^{k-c(\tau)} |X_\tau|.$$

If X is symmetric under coordinate permutations, the terms may be collected by cycle type, reducing the distinct-coordinate calculation from pairwise inclusion–exclusion to a sum indexed by partitions of k .

Proof. Sum the right-hand side pointwise over $x = (x_1, \dots, x_k) \in X$. Let the equality classes of the coordinates of x have sizes $b_1, \dots, b_s$. A permutation τ contributes to $x \in X_\tau$ exactly when every cycle of τ lies inside one equality class. Since $(-1)^{k-c(\tau)}$ is the sign of τ , the total contribution of x factors as

$$\prod_{j=1}^s \sum_{\sigma \in S_{b_j}} \text{sgn}(\sigma).$$

The inner signed sum equals 1 for $b_j = 1$ and 0 for $b_j \geq 2$, because S_{b_j} then has equally many even and odd permutations. Therefore the point contributes 1 exactly when all coordinates are distinct and 0 otherwise. Summing over x proves the formula. Symmetry makes $|X_\tau|$ constant on conjugacy classes, which are indexed by partitions of k . $\square$

Theorem 7.3 (Li–Wan finite-field subset-sum estimates — verified deep input). *For $D = \mathbb{F}_q$, $D = \mathbb{F}_q^\times$, and further structured finite-field sets treated in their papers, Li and Wan combine the distinct-coordinate sieve with character-sum estimates to obtain explicit formulas or error bounds for $N_D(k, b)$, with main term of the expected scale $\binom{|D|}{k}/q$ in the uniform regimes and with the exceptional characteristic/divisibility cases recorded separately. The theorem family is source-owned because the error term depends on D, k, p and on the relevant character sums; QRCC does not replace it by a fictitious universal error bound.*

7.3 Elementary uniformity baseline

If $D = \mathbb{F}_q$ and $p \nmid k$, translation by c sends a k -subset of sum b to one of sum $b + kc$. Hence all q sum classes have equal size and

$$N_{\mathbb{F}_q}(k, b) = \frac{1}{q} \binom{q}{k}.$$

The difficult cases begin when translation does not give exact uniformity or D is restricted.

7.4 Reader checkpoint

1. Rewrite $N_D(k, b)$ as a count of ordered distinct tuples divided by $k!$.
2. Prove that $(-1)^{k-c(\tau)}$ is the sign of τ .
3. Verify the sieve explicitly for $k = 2$.
4. Verify it for $k = 3$ by grouping permutations by cycle type.
5. Explain why symmetry reduces the number of terms to the partition number $p(k)$.
6. Prove the translation-uniformity baseline when $p \nmid k$.
7. Give an example where $p \mid k$ and translation does not move the sum.
8. Explain where character orthogonality can enforce the equation $\sum x_i = b$.
9. Explain why square-root character bounds improve the subset-sum error term.
10. Distinguish the locally proved sieve identity from the source-owned quantitative estimates.

Chapter 8

Reed–Solomon Geometry and Deep Holes

Reed–Solomon decoding converts polynomial agreement into Hamming geometry. The elementary covering radius is exact, but classifying all words attaining that radius leads quickly to finite-field subset sums and character-sum estimates.

8.1 How to read this chapter

First prove the covering-radius theorem from interpolation and root counting. Then read the Li–Wan deep-hole theorem as a genuinely stronger arithmetic statement for the standard full evaluation set. The executable laboratory checks only small finite examples, not the asymptotic deep-hole theorem.

8.2 Learning goals

A reader should be able to pass between a received vector and its interpolation polynomial, compute distance from polynomial agreement, prove that degree- k words are deep holes, and state one parameter-exact Li–Wan non-deep-hole theorem.

Definition 8.1 (Reed–Solomon geometry and deep holes). Let $D = \{\alpha_1, \dots, \alpha_n\} \subseteq \mathbb{F}_q$ consist of distinct points and $1 \leq k < n$. The evaluation Reed–Solomon code is

$$RS(D, k) = \{(f(\alpha_1), \dots, f(\alpha_n)) : f \in \mathbb{F}_q[X], \deg f < k\}.$$

The Hamming distance $d(u, v)$ is the number of unequal coordinates and $d(u, C) = \min_{c \in C} d(u, c)$. The covering radius is $\rho(C) = \max_u d(u, C)$; a word at distance $\rho(C)$ is a *deep hole*. Every received word has a unique interpolation polynomial $u(X)$ of degree $< n$. For u represented by a polynomial, agreement with a codeword represented by g occurs exactly at roots in D of $u - g$.

Theorem 8.2 (Covering radius and elementary Reed–Solomon deep holes). *For every evaluation set D of size n and $1 \leq k < n$,*

$$\rho(RS(D, k)) = n - k.$$

Moreover, if a received word has interpolation polynomial u of degree exactly k , then $d(u, RS(D, k)) = n - k$ and the word is a deep hole.

Proof. Given any received word, choose any k evaluation positions. There is a unique polynomial g of degree $< k$ agreeing with the word on those positions, so every word is within distance at most $n - k$ of the code. Now let $\deg u = k$. For every g with $\deg g < k$, the nonzero polynomial $u - g$ has degree k and hence at most k roots in D . Thus u agrees with any codeword in at most k positions, so its distance is at least $n - k$. Together with the upper bound this gives equality and proves both assertions. $\square$

Theorem 8.3 (Li–Wan non-deep-hole regime for the standard Reed–Solomon code — verified deep input). *Let $C = RS(\mathbb{F}_q, k)$ and let a received word have interpolation polynomial of degree $k + d$, where $1 \leq d \leq q - 1 - k$. Li and Wan prove, for every fixed $\varepsilon > 0$, that if*

$$q > \max\{(k + 1)^2, d^{2+\varepsilon}\}, \quad k > \left(\frac{2}{\varepsilon} + 1\right)d + \frac{8}{\varepsilon} + 2,$$

then the word is not a deep hole: $d(u, C) < q - k$. Their stronger “ordinary word” theorem gives the exact distance $q - (k + d)$ under a related explicit parameter regime. These are character-sum theorems for the standard full evaluation set, not a classification for arbitrary D .

Example 8.4 (Executable finite-field zeta and coding laboratory). The executable file `code/volume_23/finite_field_zeta_coding_lab.py` checks: prime-field character orthogonality and a quadratic Gauss sum; the $\mathbb{F}_5$ and $\mathbb{F}_{25}$ counts of Example 2.5; Newton-polygon slope bookkeeping; the Li–Wan distinct-coordinate identity on finite sets; and Reed–Solomon covering-radius/deep-hole instances. It is a sanity laboratory, not a verifier for Deligne, Dwork, Wan, or asymptotic decoding-hardness theorems.

8.3 Reader checkpoint

1. Prove the upper bound $\rho \leq n - k$ by interpolation.
2. Prove the lower bound using a degree- k polynomial.
3. Explain why the argument works for every evaluation set of distinct points.
4. Show that $n - k + 1$ is the minimum distance of $RS(D, k)$.
5. For $D = \mathbb{F}_7$ and $k = 2$, construct an explicit deep hole.
6. Explain why degree $k + 1$ already introduces a subset-sum condition.
7. State both inequalities in the displayed Li–Wan parameter regime.
8. Explain why that theorem does not classify all deep holes.
9. Distinguish the standard full evaluation set from arbitrary generalized evaluation sets.
10. Run the laboratory and identify which checks are exact exhaustive finite tests.

Chapter 9

Distance Distributions and Interpolation-Based List Decoding

Unique decoding asks for one nearest codeword; distance distributions ask how many codewords lie on every Hamming shell. List decoding lies between these viewpoints and becomes algebraic when agreement constraints are encoded by a bivariate interpolation polynomial.

9.1 How to read this chapter

The factor-extraction lemma is proved locally and is the algebraic core common to Sudan/Guruswami–Sudan style interpolation. Multiplicity choices, optimal degree budgets, and final decoding radii belong to the full source-owned algorithms and are not hidden inside the local theorem.

9.2 Learning goals

A reader should be able to define distance distributions and agreement, prove the weighted-degree factor-extraction criterion, understand how interpolation converts many agreements into a factor $Y - f(X)$, and state the scope of Li–Wan’s distance-distribution theorem family.

Definition 9.1 (Distance distributions and decoding promises). For a code $C \subseteq \mathbb{F}_q^n$ and received word u , the distance distribution is $A_r(u) = |\{c \in C : d(u, c) = r\}|$. A bounded-distance decoder is promised that a codeword lies within a specified radius and returns one; a list decoder returns all codewords within the radius. For an evaluation code, the *agreement* of a polynomial f with received pairs (α_i, y_i) is the number of indices with $f(\alpha_i) = y_i$. For $Q(X, Y) = \sum q_{ij} X^i Y^j$, the $(1, k-1)$ weighted degree is $\max\{i + (k-1)j : q_{ij} \neq 0\}$.

Theorem 9.2 (Interpolation factor-extraction certificate). *Let $\alpha_1, \dots, \alpha_n$ be distinct and let $Q(X, Y) \in \mathbb{F}_q[X, Y]$ be nonzero with $(1, k-1)$ weighted degree at most D . Suppose $Q(\alpha_i, y_i) = 0$ for every i . If $f \in \mathbb{F}_q[X]$ has $\deg f < k$ and agrees with the received word in more than D positions, then*

$$Q(X, f(X)) \equiv 0,$$

and therefore $Y - f(X)$ divides $Q(X, Y)$ in $\mathbb{F}_q[X, Y]$.

Proof. Every monomial $X^i Y^j$ of Q becomes $X^i f(X)^j$ of degree at most $i + (k-1)j \leq D$, so $R(X) = Q(X, f(X))$ has degree at most D . At each agreement position, $R(\alpha_i) = Q(\alpha_i, y_i) = 0$.

More than D distinct roots force $R = 0$. Divide Q by the monic polynomial $Y - f(X)$ in the polynomial ring $\mathbb{F}_q[X][Y]$. The remainder is independent of Y and equals $Q(X, f(X)) = 0$, proving divisibility. $\square$

Theorem 9.3 (Li–Wan distance-distribution bounds for standard Reed–Solomon codes — verified deep input). *Let $f \in \mathbb{F}_q[X]$ have degree $k + m$. Li and Wan study, for each $r \geq 0$, the number of $g \in \mathbb{F}_q[X]$ with $\deg g \leq k - 1$ for which $f + g$ has exactly r distinct roots in $\mathbb{F}_q$. Their 2020 theorem family gives explicit error bounds and, in stated asymptotic regimes, formulas for the distance/list distribution of the standard $[q, k, q - k + 1]_q$ Reed–Solomon code. The dependence on p, q, k, m, r is retained in the source; QRCC does not replace it by a universal Poisson or binomial heuristic.*

9.3 From certificate to algorithm

The local theorem says what happens once a suitable Q exists. Sudan and Guruswami–Sudan choose a monomial space (and, in the latter case, multiplicities) large enough that interpolation has a nonzero solution while keeping weighted degree low enough for factor extraction. Solving that optimization yields the decoding radius; it is deliberately not folded into Theorem 9.2.

9.4 Reader checkpoint

1. Translate Hamming distance into polynomial agreement for Reed–Solomon codes.
2. Prove the weighted-degree substitution bound used in the theorem.
3. Explain why distinct evaluation points are necessary for the root count.
4. Prove the factor theorem over $\mathbb{F}_q[X][Y]$.
5. Construct a small Q for three received points and $k = 2$.
6. Explain how increasing interpolation multiplicity changes the constraint count.
7. Distinguish factor extraction from the existence of a useful interpolation polynomial.
8. Define $A_r(u)$ and relate it to the number of roots of $f - g$.
9. State the polynomial family counted by Li–Wan in their distance-distribution work.
10. Explain why an asymptotic distance distribution is stronger information than one decoding radius.

Chapter 10

Decoding Hardness and the Arithmetic–Computation Bridge

Structured codes sit at an unusual boundary: their algebra gives fast encoding and powerful decoding algorithms, yet natural decoding problems remain hard in important parameter regimes. Precise reduction language is essential because unconditional NP-hardness, conditional cryptographic hardness, and approximation hardness are different statements.

10.1 How to read this chapter

Read every hardness theorem together with its code family, input representation, reduction type, and radius/rate regime. The dated frontier records the strongest statements verified in PASS105 without extrapolating them to all primitive Reed–Solomon or post-quantum code families.

10.2 Learning goals

A reader should be able to distinguish maximum-likelihood, bounded-distance, list, and minimum-distance problems; explain the Guruswami–Vardy evaluation-set model; state Cheng–Wan’s conditional positive-rate primitive result; and keep Gap Minimum Distance approximation hardness separate from Reed–Solomon maximum-likelihood decoding.

Definition 10.1 (Decoding-complexity and reduction taxonomy). *Maximum-likelihood decoding* asks for a codeword minimizing Hamming distance to a received word. Its decision form asks whether that minimum is at most a threshold. *Bounded-distance decoding* fixes a radius and asks for a nearby codeword under a promise or decision formulation; *list decoding* asks for all nearby codewords. The *minimum-distance problem* is a property of the code itself and asks for the least nonzero codeword weight; its gap version asks to distinguish two promised distance ranges. An unconditional NP-hardness reduction is logically different from a reduction saying that an efficient decoder would yield an efficient discrete-logarithm algorithm. A positive-rate family has k/n bounded away from zero; a primitive Reed–Solomon code uses essentially all nonzero field elements as evaluation points, whereas an evaluation-set/generalized instance may include $D \subseteq \mathbb{F}_q$ as part of the input.

Theorem 10.2 (Guruswami–Vardy maximum-likelihood decoding hardness — verified deep input). *The decision version of maximum-likelihood decoding is NP-complete for the Reed–Solomon family in the Guruswami–Vardy representation, in which the field/evaluation-set data and the received word*

form the instance; the hardness persists even with unlimited preprocessing of the code. The construction uses algebraically structured Reed–Solomon codes, but its freedom to choose the evaluation set is material. This theorem is therefore not silently restated as unconditional NP-hardness for every fixed primitive positive-rate Reed–Solomon family.

Theorem 10.3 (Cheng–Wan positive-rate primitive Reed–Solomon hardness — verified conditional input). *Cheng and Wan show that maximum-likelihood decoding of primitive/extended-primitive Reed–Solomon codes at any fixed positive information rate is at least as hard as specified discrete-logarithm problems over finite-field extensions. In the 2010 full version, for polynomial-time computable $k(q)$ with*

$$\sqrt{q} \leq k(q) \leq q - \sqrt{q},$$

a randomized polynomial-time maximum-likelihood decoder for the family $RS_q[q, k(q)]$ would yield randomized polynomial-time algorithms for the finite-field discrete-logarithm instances appearing in their reduction. They also derive conditional bounded-distance hardness beyond a $2/3 + \varepsilon$ fraction of the minimum distance under explicit extension-degree assumptions. The conclusion is conditional cryptographic hardness, not an unconditional NP-hardness theorem.

Theorem 10.4 (Cheng–Wan deterministic Gap Minimum Distance reduction — verified deep input). *For the minimum-distance problem on general linear codes, Cheng and Wan derandomize the earlier gap reduction. Consequently, no deterministic polynomial-time algorithm approximates the minimum distance within any fixed constant factor unless $P = NP$; for every fixed $\varepsilon > 0$ they also obtain inapproximability within $2^{(\log n)^{1-\varepsilon}}$ under the stronger assumption $NP \not\subseteq DTIME(2^{\text{polylog } n})$. Reed–Solomon/Hadamard structure is a technical ingredient in the reduction, but the theorem’s target is Gap Minimum Distance for linear codes, not primitive Reed–Solomon maximum-likelihood decoding.*

Theorem 10.5 (Structured-code decoding complexity — dated PASS105 frontier). *As of the PASS105 source audit on 2026-09-03, the verified landscape contains: Guruswami–Vardy’s unconditional NP-hardness for the evaluation-set Reed–Solomon model; Gandikota–Ghazi–Grigorescu’s later NP-hardness for bounded-distance Reed–Solomon decoding at radii $N - K - c \log N / \log \log N$ for $K = \Theta(N)$; and Cheng–Wan’s conditional discrete-logarithm hardness for primitive positive-rate families. These do not collapse to a single theorem asserting unconditional NP-hardness of maximum-likelihood decoding for every standard primitive positive-rate family. Complexity for highly structured algebraic-geometry, alternant, and code-based post-quantum families is tracked separately because changing the representation or parameter regime changes the problem.*

10.3 Reader checkpoint

1. Distinguish maximum-likelihood decoding from minimum-distance computation.
2. Explain why allowing the evaluation set as input matters in Guruswami–Vardy.
3. State the rate window $\sqrt{q} \leq k(q) \leq q - \sqrt{q}$ in Cheng–Wan.
4. Explain why a reduction from discrete logarithm gives conditional rather than NP-hardness.
5. State what target problem appears in the Cheng–Wan Gap Minimum Distance theorem.

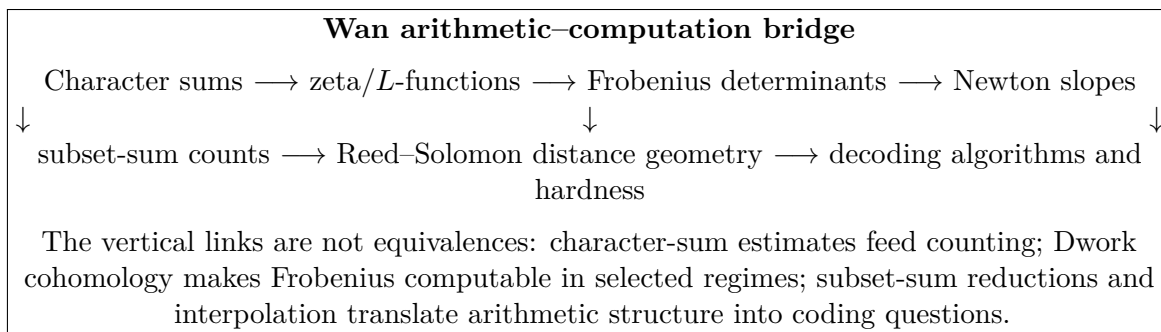


Figure 10.1: Arithmetic and computational interfaces used throughout Volume XXIII.

6. Explain the meaning of a constant-factor inapproximability result.
7. Compare maximum-likelihood and bounded-distance radii in the verified Reed–Solomon results.
8. Explain why no theorem in this chapter proves all primitive Reed–Solomon decoding NP-hard.
9. Trace one path through Figure 10.1 from character sums to decoding complexity.
10. Give one future source-calibration question that must keep code representation and rate explicit.

Internalization and Reconstruction Guide: Volume XXIII

Purpose. Move between character sums, zeta functions, p-adic/cohomological structure, point-counting algorithms, finite-field sieves, Reed–Solomon geometry, and decoding complexity without conflating structural and algorithmic statements.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-015: Fourier inversion on finite abelian groups.
2. QRCC-XVII-DEF-022: Elliptic curves over finite fields.
3. QRCC-III-ALG-004: Euclidean polynomial arithmetic.
4. QRCC-III-THM-010: Reed–Solomon distance theorem.
5. QRCC-III-ALG-013: Sudan list-decoding theorem.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XXIII-DEF-001 (Additive and multiplicative character sums); QRCC-XXIII-DEF-005 (Zeta functions of varieties over finite fields); QRCC-XXIII-DEF-009 (L-functions, Newton polygons, and slopes); QRCC-XXIII-DEF-012 (Dwork cohomology and nuclear operators); QRCC-XXIII-DEF-015 (Unit-root zeta functions and p-adic meromorphy).

Chapter retrieval map

Ch. 1: Characters, Gauss Sums, and Square-Root Cancellation: enter through QRCC-XXIII-DEF-001 (Additive and multiplicative character sums); exit by being able to state QRCC-XXIII-BB-003 (Weil bound for one-variable character sums) with its ownership boundary.

- Ch. 2: Zeta Functions and the Weil Package:** enter through QRCC-XXIII-DEF-005 (Zeta functions of varieties over finite fields); exit by being able to state QRCC-XXIII-BB-007 (Dwork rationality theorem) with its ownership boundary.
- Ch. 3: Newton Polygons and Nondegenerate Exponential Sums:** enter through QRCC-XXIII-DEF-009 (L-functions, Newton polygons, and slopes); exit by being able to state QRCC-XXIII-BB-011 (Adolphson–Sperber nondegenerate Newton-polyhedron theorem) with its ownership boundary.
- Ch. 4: Dwork Operators, Trace Formulas, and Cohomological Determinants:** enter through QRCC-XXIII-DEF-012 (Dwork cohomology and nuclear operators); exit by being able to state QRCC-XXIII-THM-013 (Dwork trace formula in a basic hypersurface model) with its ownership boundary.
- Ch. 5: Unit Roots, Dwork’s Conjecture, and the Wan Program:** enter through QRCC-XXIII-DEF-015 (Unit-root zeta functions and p-adic meromorphy); exit by being able to state QRCC-XXIII-BB-016 (Wan’s proof of Dwork’s conjecture) with its ownership boundary.
- Ch. 6: Point Counting as an Algorithmic Problem:** enter through QRCC-XXIII-DEF-019 (Algorithms for zeta functions and point counting); exit by being able to state QRCC-XXIII-BB-021 (Kedlaya p-adic point-counting algorithm) with its ownership boundary.
- Ch. 7: Subset Sums and the Li–Wan Distinct-Coordinate Sieve:** enter through QRCC-XXIII-DEF-023 (Finite-field subset-sum and distinct-coordinate sieves); exit by being able to state QRCC-XXIII-BB-025 (Wan finite-field subset-sum estimates) with its ownership boundary.
- Ch. 8: Reed–Solomon Geometry and Deep Holes:** enter through QRCC-XXIII-DEF-026 (Reed–Solomon geometry and deep holes); exit by being able to state QRCC-XXIII-BB-028 (Wan program on Reed–Solomon deep holes) with its ownership boundary.
- Ch. 9: Distance Distributions and Interpolation-Based List Decoding:** enter through QRCC-XXIII-DEF-029 (Distance distributions and list/bounded-distance decoding); exit by being able to state QRCC-XXIII-BB-031 (Wan distance-distribution results for Reed–Solomon codes) with its ownership boundary.
- Ch. 10: Decoding Hardness and the Arithmetic–Computation Bridge:** enter through QRCC-XXIII-DEF-032 (Decoding complexity and reduction taxonomy); exit by being able to state QRCC-XXIII-BB-033 (NP-hardness of maximum-likelihood decoding for generalized Reed–Solomon codes) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XXIII-BB-003 — Weil bound for one-variable character sums. Dependencies: QRCC-XXIII-DEF-001. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*

2. QRCC-XXIII-BB-011 — Adolphson–Sperber nondegenerate Newton-polyhedron theorem. Dependencies: QRCC-XXIII-DEF-009. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
3. QRCC-XXIII-BB-016 — Wan’s proof of Dwork’s conjecture. Dependencies: QRCC-XXIII-DEF-015. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XXIII-BB-025 — Wan finite-field subset-sum estimates. Dependencies: QRCC-XXIII-THM-024, QRCC-XXIII-BB-003. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XXIII-BB-031 — Wan distance-distribution results for Reed–Solomon codes. Dependencies: QRCC-XXIII-DEF-029. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XXIII-F-018 — Rigid cohomology and explicit Betti bounds. Dependencies: QRCC-XXIII-DEF-012. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- using Weil/Deligne bounds without checking smoothness or nondegeneracy hypotheses;
- treating a Newton polygon identity as a cohomological existence theorem;
- mixing arithmetic-operation complexity with bit complexity in point counting;
- conflating generalized Reed–Solomon MLD hardness, primitive-code conditional hardness, and general linear-code Gap Minimum Distance;

Boundary and degenerate cases

- affine/projective spaces of dimension zero;
- singular versus smooth/nondegenerate varieties;
- very small characteristic;
- Reed–Solomon words already inside the code versus genuine deep holes;

Retrieval and reconstruction drill

1. State QRCC-XXIII-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XXIII-BB-003. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XXIII-BB-011, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?

4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.
5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XXIII-F-018, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

Volume XXIV closes the series by interpreting number-theoretic cancellation and dynamics as deterministic pseudorandomness against structured tests.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Part XXIV

Arithmetic and Dynamical
Pseudorandomness

Chapter 1

Möbius Cancellation and Random Benchmarks

Arithmetic pseudorandomness begins with a distinction that will be enforced throughout this volume: a random model supplies a benchmark, whereas cancellation of the Möbius or Liouville function is a theorem only when an arithmetic argument proves it. The prime number theorem gives the first basic cancellation statement; an independent-sign model supplies the comparison scale.

1.1 How to read this chapter

The prime-number-theorem equivalence is a source-owned analytic-number-theory interface. The random benchmark is proved locally and is never used as evidence that the Möbius function is actually random.

1.2 Learning goals

The reader should be able to normalize Möbius/Liouville correlations, distinguish Cesàro and logarithmic averages, state the summatory-Möbius form of the prime number theorem, and compute expectation and variance in an independent-sign comparison model.

Definition 1.1 (Arithmetic pseudorandomness and test classes). The *Möbius function* is $\mu(1) = 1$, $\mu(n) = 0$ when a square prime divides n , and $\mu(n) = (-1)^r$ when n is a product of r distinct primes. The *Liouville function* is $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ counts prime factors with multiplicity. A function $f : \mathbb{N} \rightarrow \mathbb{C}$ is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $(m, n) = 1$, and *1-bounded* if $|f(n)| \leq 1$.

For bounded sequences a_n, b_n , their Cesàro correlation up to N is $N^{-1} \sum_{n \leq N} a_n \overline{b_n}$. Their logarithmic correlation on $[1, N]$ is $H_N^{-1} \sum_{n \leq N} a_n \overline{b_n} / n$, where $H_N = \sum_{n \leq N} 1/n$. We say that a is *orthogonal* to a test class $\mathcal{T}$ if the relevant correlations tend to 0 for every $b \in \mathcal{T}$. A *quantitative orthogonality bound* records an explicit decay rate. Fourier phases, prime-dilate tests, nilsequences, short-interval tests, and deterministic dynamical sequences are different test classes; success against one class is not silently promoted to another.

Theorem 1.2 (Prime number theorem as Möbius cancellation — verified deep input). *Let $M(x) = \sum_{n \leq x} \mu(n)$. The prime number theorem is equivalent to*

$$M(x) = o(x) \quad (x \rightarrow \infty).$$

Equivalently, $x^{-1} \sum_{n \leq x} \mu(n) \rightarrow 0$. In this volume the analytic equivalence is source-owned: the required Dirichlet-series identities, zero-free statement at $s = 1$, and Tauberian/partial-summation bridges are not reconstructed from a random-model heuristic.

Theorem 1.3 (Independent-sign benchmark). *Let $\varepsilon_1, \dots, \varepsilon_N$ be independent random variables taking values ± 1 with equal probability, and let $a_1, \dots, a_N \in \mathbb{C}$ satisfy $|a_n| \leq 1$. Put*

$$S_N = \frac{1}{N} \sum_{n \leq N} \varepsilon_n a_n.$$

Then $\mathbb{E}S_N = 0$ and

$$\mathbb{E}|S_N|^2 = \frac{1}{N^2} \sum_{n \leq N} |a_n|^2 \leq \frac{1}{N}.$$

Consequently $\mathbb{P}(|S_N| \geq t) \leq (Nt^2)^{-1}$ for every $t > 0$.

Proof. Independence and centering give $\mathbb{E}\varepsilon_n = 0$ and $\mathbb{E}(\varepsilon_m \varepsilon_n) = 0$ for $m \neq n$, while $\varepsilon_n^2 = 1$. Expanding $|S_N|^2$ therefore kills every off-diagonal term and leaves $N^{-2} \sum |a_n|^2$. Markov's inequality applied to the nonnegative variable $|S_N|^2$ gives the final claim. $\square$

1.3 Benchmark audit

The theorem predicts root- N fluctuations for independent signs. Möbius values are neither independent nor identically distributed: squareful integers force zeros and multiplicativity couples values at different integers. Thus $N^{-1/2}$ is a useful comparison scale, not a theorem about $M(N)$.

1.4 Reader checkpoint

1. Compute $\mu(n)$ and $\lambda(n)$ for $1 \leq n \leq 12$.
2. Give a multiplicative function that is not completely multiplicative.
3. Rewrite $M(N) = o(N)$ as a Cesàro-correlation statement.
4. Explain why the independent-sign theorem does not prove the prime number theorem.
5. Prove the variance identity when $|a_n|$ is unrestricted.
6. Derive the displayed probability bound directly from Chebyshev/Markov.
7. Compare Cesàro and logarithmic weights on the interval $[N/2, N]$.
8. Explain why zeros of μ matter when comparing it with Rademacher signs.
9. Name two later test classes that are stronger than fixed periodic tests.
10. Identify the locally proved and source-owned results in this chapter.

Chapter 2

Fourier Tests and Davenport Cancellation

Fourier phases are the first structured deterministic tests against which Möbius exhibits cancellation far stronger than the prime number theorem alone. Davenport’s uniform estimate is the source-owned analytic engine; its periodic and linear-phase consequences are elementary once that engine is available.

2.1 How to read this chapter

The uniform exponential-sum estimate is not reproved. The corollary for arbitrary fixed periodic sequences is proved by finite Fourier expansion, making exact use of the uniformity in the frequency.

Definition 2.1 (Exponential and Fourier tests). Write $e(t) = e^{2\pi it}$. For a bounded arithmetic function f , its interval exponential sum is

$$S_f(N; \alpha) = \sum_{n \leq N} f(n)e(\alpha n), \quad \alpha \in \mathbb{R}/\mathbb{Z}.$$

A *linear phase* is $n \mapsto e(\alpha n)$. A sequence b_n is *q-periodic* if $b_{n+q} = b_n$ for all n . Fourier uniformity at the linear level means a bound on $\sup_{\alpha} |S_f(N; \alpha)|$. Rational frequencies, major/minor-arc decompositions, and cyclic-group Fourier transforms are bookkeeping devices; this chapter uses only the interval normalization above.

Theorem 2.2 (Davenport’s Möbius exponential-sum bound — verified deep input). *For every fixed $A > 0$ there is a constant C_A such that, for all $N \geq 2$,*

$$\sup_{\alpha \in \mathbb{R}/\mathbb{Z}} \left| \sum_{n \leq N} \mu(n)e(\alpha n) \right| \leq C_A N(\log N)^{-A}.$$

The theorem is source-owned to Davenport’s 1937 part-II paper and its standard modern formulation. The statement here uses only the uniform logarithmic-power decay; it makes no unverified claim about an optimal constant.

Corollary 2.3 (Periodic and linear-phase Möbius orthogonality). *For every $A > 0$ and every $\alpha \in \mathbb{R}/\mathbb{Z}$,*

$$\sum_{n \leq N} \mu(n) e(\alpha n) = O_A(N(\log N)^{-A})$$

uniformly in α . If b_n is a fixed bounded q -periodic sequence, then

$$\sum_{n \leq N} \mu(n) b_n = O_{A,b}(N(\log N)^{-A}).$$

Proof. The first statement is Theorem 2.2. For the second, Fourier inversion on $\mathbb{Z}/q\mathbb{Z}$ gives constants $\widehat{b}(r)$ such that

$$b_n = \sum_{r=0}^{q-1} \widehat{b}(r) e(rn/q).$$

Insert this finite expansion into the correlation and apply the uniform Davenport bound to each of the q rational frequencies. The finite sum of Fourier coefficients is absorbed into the constant depending on b . $\square$

2.2 Worked comparison

The parity sequence $(-1)^n = e(n/2)$ is a single rational Fourier mode. A general fixed periodic sequence is only finitely more complicated. Neither conclusion controls a quadratic phase $e(\alpha n^2)$ by itself; later Gowers and nilsequence inputs address stronger structured tests.

2.3 Reader checkpoint

1. Verify that $e(t)$ depends only on $t \bmod 1$.
2. Compute the Fourier expansion of $(-1)^n$.
3. Write the period-three indicator $1_{n \equiv 1 \pmod{3}}$ as three Fourier modes.
4. Explain where uniformity in α is used in the periodic corollary.
5. Show that the trivial bound for the Davenport sum is N .
6. Compare logarithmic-power saving with a hypothetical $N^{1/2}$ bound.
7. Explain why a fixed periodic sequence is a finite-dimensional Fourier object.
8. Give a polynomial phase not covered by the corollary.
9. State the exact DOI correction made for Davenport part II.
10. Explain which analytic work remains source-owned.

Chapter 3

Prime Dilates: Kátai and Bourgain–Sarnak–Ziegler

Kátai’s criterion is a reusable bridge: to rule out correlation with every bounded multiplicative function, it can suffice to show that the test sequence decorrelates from itself along distinct prime dilates. This chapter proves a qualitative form completely and then records the dynamical BSZ specialization.

3.1 How to read this chapter

The local proof exposes all truncation terms. It uses only divergence of the sum of reciprocal primes, elementary divisor counting, multiplicativity away from the exceptional set $p \mid m$, and Cauchy–Schwarz.

Definition 3.1 (Prime-dilate correlations and the KBSZ hypothesis). For a 1-bounded sequence $a = (a_n)$ and distinct primes p, q , its (p, q) *prime-dilate correlation* is

$$C_{p,q}(M) = \frac{1}{M} \sum_{m \leq M} a_{pm} \overline{a_{qm}}.$$

The *qualitative KBSZ hypothesis* is $C_{p,q}(M) \rightarrow 0$ as $M \rightarrow \infty$ for every distinct sufficiently large primes p, q . A *bounded multiplicative weight* is a multiplicative $\nu : \mathbb{N} \rightarrow \mathbb{C}$ with $|\nu(n)| \leq 1$.

Theorem 3.2 (Kátai orthogonality criterion). *Let $|a_n| \leq 1$. Assume that $C_{p,q}(M) \rightarrow 0$ for every pair of distinct sufficiently large primes p, q . Then, for every bounded multiplicative function ν ,*

$$\frac{1}{N} \sum_{n \leq N} \nu(n) a_n \longrightarrow 0.$$

Proof. Fix a large W and a finite set P of primes $p \geq W$. Put

$$A = \sum_{p \in P} \frac{1}{p}, \quad \omega_P(n) = \sum_{p \in P} 1_{p \mid n}.$$

For fixed P , elementary divisor counting gives

$$\sum_{n \leq N} |\omega_P(n) - A|^2 \leq 2AN + o_P(N).$$

Hence, with $S_N = \sum_{n \leq N} \nu(n)a_n$, Cauchy–Schwarz gives

$$AS_N = \sum_{p \in P} \sum_{\substack{n \leq N \\ p|n}} \nu(n)a_n + O\left(N\sqrt{A} + o_P(N)\right).$$

Write $n = pm$. Except when $p \mid m$, multiplicativity gives $\nu(pm) = \nu(p)\nu(m)$; the exceptional terms contribute at most

$$2N \sum_{p \in P} p^{-2} \leq \frac{2AN}{W}.$$

Thus it remains to bound

$$T = \sum_{p \in P} \nu(p) \sum_{m \leq N/p} \nu(m)a_{pm}.$$

After extending missing summands by zero, Cauchy–Schwarz yields

$$|T|^2 \leq N \sum_{m \leq N} \left| \sum_{\substack{p \in P \\ pm \leq N}} \nu(p)a_{pm} \right|^2.$$

The diagonal contribution is $O(AN^2)$. Since P is fixed, every off-diagonal term is $o_P(N^2)$ by the assumed prime-dilate correlations. Therefore $|T| \leq N\sqrt{A + o_P(1)}$. Dividing by AN gives

$$\limsup_{N \rightarrow \infty} \frac{|S_N|}{N} \ll A^{-1/2} + W^{-1}.$$

Choose W beyond the finitely many excluded primes and then choose a finite $P \subset [W, \infty)$ with A arbitrarily large, using divergence of $\sum_p 1/p$. The limsup is therefore zero. $\square$

Theorem 3.3 (Bourgain–Sarnak–Ziegler dynamical specialization). *Let (X, T) be a compact topological dynamical system, $F \in C(X)$ with $\|F\|_\infty \leq 1$, and $x \in X$. Put $a_n = F(T^n x)$. If for every distinct sufficiently large primes p, q ,*

$$\frac{1}{M} \sum_{m \leq M} F(T^{pm} x) \overline{F(T^{qm} x)} \rightarrow 0,$$

then $N^{-1} \sum_{n \leq N} \nu(n)F(T^n x) \rightarrow 0$ for every bounded multiplicative ν , in particular for μ .

Proof. The displayed assumption is exactly the prime-dilate hypothesis of Theorem 3.2 for the bounded sequence $a_n = F(T^n x)$. Apply that theorem. $\square$

3.2 Hypothesis audit

The criterion is one-way. Pair decorrelation is a sufficient mechanism, not a characterization of Möbius disjointness. Quantitative versions require quantitative uniformity in the prime set and are source-owned to Kátai/BSZ refinements rather than inferred from the qualitative proof.

3.3 Reader checkpoint

1. Expand the first two moments of $\omega_P(n)$.
2. Show $\sum_{p \in P} p^{-2} \leq A/W$.
3. Locate exactly where multiplicativity is used.
4. Explain why the terms with $p \mid m$ must be removed.
5. Explain why P is fixed before $N \rightarrow \infty$.
6. Explain why divergence of $\sum_p 1/p$ is enough at the final step.
7. State a quantitative hypothesis stronger than the qualitative one.
8. Apply the theorem to a sequence known to have zero prime-dilate correlations.
9. Explain why the BSZ statement is a corollary, not a new analytic input here.
10. Identify the theorem source used to literature-calibrate the criterion.

Chapter 4

Gowers Uniformity of Arithmetic Weights

Linear Fourier tests see only first-order structure. Gowers norms test higher-order additive patterns and are the natural interface between multiplicative functions and additive-combinatorial counting lemmas.

Definition 4.1 (Interval Gowers uniformity). For a finite interval $I = [N] = \{1, \dots, N\}$ and $k \geq 1$, extend $f : I \rightarrow \mathbb{C}$ by zero outside I . The normalized interval Gowers norm $\|f\|_{U^k[N]}$ is the standard local-cube average

$$\|f\|_{U^k[N]}^{2^k} = \frac{\sum_{x, h_1, \dots, h_k} \prod_{\omega \in \{0,1\}^k} \mathcal{C}^{|\omega|} f(x + \omega \cdot h)}{\sum_{x, h_1, \dots, h_k} \prod_{\omega \in \{0,1\}^k} 1_{[N]}(x + \omega \cdot h)},$$

where the sums range over integer tuples for which the denominator is nonzero and $\mathcal{C}$ denotes complex conjugation. A function is U^k -uniform when this quantity is small. A *majorant* is a nonnegative function ν with $|f| \leq \nu$; relative counting requires pseudorandomness assumptions on ν in addition to smallness of a Gowers norm.

Theorem 4.2 (Quantitative Gowers uniformity of Möbius and von Mangoldt weights — verified deep input). *For every fixed $k \geq 2$, Tao–Teräväinen prove a constant $c = c(k) > 0$ for which*

$$\|\mu\|_{U^k[N]} \ll (\log \log N)^{-c}.$$

They also prove the corresponding quantitative statement for the von Mangoldt function after subtraction of the appropriate Cramér/ W -tricked model, with the auxiliary W -parameter and error terms kept as in the source. The result is all fixed orders k , not merely the earlier qualitative fixed-order theory.

Theorem 4.3 (Generalized von Neumann control — verified companion import). *Let $f_1, \dots, f_t$ be 1-bounded functions on a finite abelian group and let $L_1, \dots, L_t$ be a finite-complexity system of affine-linear forms in which one selected form has Cauchy–Schwarz complexity at most s . In the bounded setting, the normalized multilinear average is controlled by the U^{s+1} norm of the selected function, up to the boundary/ambient-group translation recorded in the Volume V companion theorem ‘QRCC-V-THM-003’. Relative/majorized versions use the separately registered dense-model interface and are not inferred from this bounded statement.*

4.1 Interface audit

The 2025 quantitative theorem strengthens the arithmetic input; it does not remove the need for a counting lemma. Conversely, generalized von Neumann control is formal additive combinatorics and says nothing about Möbius until an arithmetic theorem supplies a small Gowers norm.

4.2 Reader checkpoint

1. Check that $\|f\|_{U^1}$ is controlled by the mean under the chosen convention.
2. Write the four factors in the U^2 cube.
3. Explain why zero extension creates boundary bookkeeping.
4. State the difference between bounded and majorized counting.
5. Explain what “all fixed orders” means in the Tao–Teräväinen theorem.
6. Why is $(\log \log N)^{-c}$ still a genuine quantitative decay?
7. Identify the exact Volume V companion theorem imported here.
8. Give a three-term-progression average controlled by U^2 .
9. Explain why a Gowers estimate is not itself a prime-counting theorem.
10. State which result in this chapter changed materially because of the 2025 publication.

Chapter 5

Nilsequences and Möbius Orthogonality

Nilsequences are the structured objects detected by inverse Gowers theory. Green and Tao’s strong orthogonality theorem therefore sits at the meeting point of arithmetic cancellation, higher-order Fourier analysis, and dynamics.

Definition 5.1 (Filtered nilmanifolds and nilsequences). A *nilmanifold* is a compact homogeneous space G/Γ with G a nilpotent Lie group and Γ a discrete cocompact subgroup. A *filtration* is a descending sequence $G = G_0 \supseteq G_1 \supseteq \cdots$ with $[G_i, G_j] \subseteq G_{i+j}$. A sequence $g : \mathbb{Z} \rightarrow G$ is *polynomial* relative to the filtration when its iterated discrete derivatives fall into the corresponding levels. A *polynomial nilsequence* is $n \mapsto F(g(n)\Gamma)$ for a Lipschitz $F : G/\Gamma \rightarrow \mathbb{C}$. Dimension, filtration degree, a rational Mal’cev description, and $\|F\|_{\text{Lip}}$ are part of its quantitative complexity.

Theorem 5.2 (Green–Tao strong Möbius orthogonality to nilsequences — verified deep input). *Fix a filtered nilmanifold of bounded quantitative complexity and a bounded Lipschitz polynomial nilsequence $F(g(n)\Gamma)$. For every $A > 0$,*

$$\left| \frac{1}{N} \sum_{n \leq N} \mu(n) F(g(n)\Gamma) \right| \ll (\log N)^{-A},$$

with the implied constant depending on the fixed nilmanifold/filtration, rationality data, Lipschitz norm and A as in Green–Tao. Uniformity is therefore over nilsequences of controlled complexity, not over arbitrary nilmanifolds with parameters growing unchecked with N .

Corollary 5.3 (Möbius disjointness for nilsystems). *Let $X = G/\Gamma$ be a fixed compact nilmanifold, let $T(x) = gx$ be a nilrotation, let $x \in X$, and let $F \in C(X)$. Then*

$$\frac{1}{N} \sum_{n \leq N} \mu(n) F(T^n x) \longrightarrow 0.$$

Proof. If F is Lipschitz, $F(T^n x) = F(g^n x)$ is a degree-one polynomial nilsequence, so Theorem 5.2 applies. For general continuous F , compactness of X gives uniform approximation by a Lipschitz function F_ε . Since $|\mu(n)| \leq 1$, replacing F by F_ε changes the normalized correlation by at most $\|F - F_\varepsilon\|_\infty$. First let $N \rightarrow \infty$, then $\varepsilon \rightarrow 0$. $\square$

5.1 Boundary audit

The corollary is qualitative for every continuous observable on a fixed nilsystem. A quantitative rate for an arbitrary continuous F requires quantitative approximation information and is not claimed here.

5.2 Reader checkpoint

1. Define a one-step nilmanifold and give a torus example.
2. Show that $n \mapsto g^n$ is polynomial for the lower-central-series filtration.
3. Explain why a rationality parameter is needed in a quantitative theorem.
4. Distinguish a basic nilsequence from a uniform limit of nilsequences.
5. Locate the Lipschitz hypothesis in the Green–Tao theorem.
6. Prove the uniform-approximation step in the corollary.
7. Explain why compactness is used.
8. Relate linear phases to one-step nilsequences.
9. Explain why the corollary does not produce a universal rate for all continuous functions.
10. Identify the source-owned and locally proved statements.

Chapter 6

Transference and Linear Patterns in the Primes

The primes are sparse, so dense additive-combinatorial theorems cannot be applied to their indicator without a transference layer. This chapter records the exact architecture rather than compressing the sieve, dense-model and inverse-theory components into one black box.

Definition 6.1 (Majorants, the W -trick, and relative counting). A *pseudorandom majorant* for a nonnegative weight f is a nonnegative weight ν with $0 \leq f \leq \nu$ and normalized mean near 1, together with specified linear-forms/correlation estimates. The W -trick fixes W as a product of small primes and studies residue classes $Wn + b$ with $(b, W) = 1$, removing small-prime local irregularities. A *relative counting theorem* transfers a dense counting statement to $f \leq \nu$ once the dense-model and relative-uniformity hypotheses on ν are verified. Local factors encode congruence obstructions of an affine-linear system.

Theorem 6.2 (Green–Tao transference for progressions in primes — verified deep input). *For every $k \geq 1$, the primes contain a k -term arithmetic progression. In the source architecture, this follows by W -tricking the primes, majorizing the resulting weight by a Goldston–Yıldırım/enveloping-sieve pseudorandom measure, applying a relative Szemerédi theorem/dense-model transference principle, and using positive relative density. Each ingredient has its own quantitative hypotheses; “the primes look random” is not a substitute for them.*

Theorem 6.3 (Finite-complexity systems of linear equations in primes — verified deep input). *For a fixed finite-complexity system of affine-linear forms satisfying the nondegeneracy and local-solubility hypotheses in Green–Tao–Ziegler, the number of prime solutions in a growing convex region has the predicted product of archimedean volume and local factors, with an $o(1)$ relative error in the theorem’s normalization. The statement is restricted to the source’s finite-complexity regime; pairwise proportional linear parts and singular local systems require separate treatment. Tao–Teräväinen’s 2025 Gowers-uniformity estimates give quantitative refinements for finite-complexity prime systems, but they do not erase these local hypotheses.*

6.1 Architecture checklist

Sparse majorant $\rightarrow$ dense model $\rightarrow$ Gowers/counting control $\rightarrow$ local factors is a chain of implications. Reversing an arrow, or omitting the local-solubility check, changes the theorem.

6.2 Reader checkpoint

1. Explain why the prime indicator has density tending to zero.
2. Define the role of W in the W -trick.
3. Give a local congruence obstruction for a simple affine system.
4. Explain why a majorant needs more than a pointwise upper bound.
5. Identify the dense-model interface already owned by Volume V.
6. State what a relative counting lemma adds to generalized von Neumann.
7. Explain why local factors appear in the final asymptotic.
8. Distinguish Green–Tao progressions from the full linear-equations theorem.
9. Explain how the 2025 quantitative Gowers result refines but does not replace GTZ.
10. List the hypotheses one must audit before applying the source-owned theorem.

Chapter 7

Chowla, Elliott, and Logarithmic Correlations

Chowla-type conjectures ask for cancellation in products of shifted multiplicative functions. Logarithmic averaging has allowed decisive progress, but the resulting theorems must not be conflated with the original unweighted conjectures.

Definition 7.1 (Chowla/Elliott correlations). For distinct integer shifts $h_1, \dots, h_k$, the order- k Liouville correlation is

$$\frac{1}{N} \sum_{n \leq N} \prod_{j=1}^k \lambda(n + h_j).$$

The *Cesàro Chowla conjecture* predicts that this tends to 0 for every $k \geq 1$. A *logarithmically averaged* version replaces the average by a normalized sum with weight $1/n$, possibly over a moving interval $[x/\omega(x), x]$. Elliott-type statements replace λ by bounded multiplicative functions and must exclude *pretentious* functions, namely functions that remain at finite/small pretentious distance from a character times n^{it} . Odd- and even-order logarithmic statements are tracked separately.

Theorem 7.2 (Two-point logarithmically averaged Chowla/Elliott — verified deep input). *Tao proves the logarithmically averaged two-point Chowla theorem: for distinct affine shifts (equivalently nonzero determinant in the two affine forms) the normalized logarithmic two-point Liouville correlation tends to zero in the moving-scale regime of the source. The same paper proves the corresponding two-point Elliott statement for 1-bounded multiplicative functions under the required nonpretentiousness hypothesis. This is not the unweighted two-point Chowla theorem.*

Theorem 7.3 (Odd-order logarithmically averaged Chowla — verified deep input). *For every odd k and every k distinct fixed shifts, Tao–Teräväinen prove the logarithmically averaged order- k Chowla correlation tends to zero. The theorem is odd-order and logarithmically averaged; it does not close the even orders $k \geq 4$ or the full Cesàro conjecture.*

Theorem 7.4 (Chowla/Elliott status ledger — dated PASS106 frontier). *As verified on 2026-09-03, the two-point logarithmic theorem and all odd-order logarithmic Chowla cases are established; higher-uniformity short-interval/averaged variants have also advanced substantially. The general unweighted Cesàro Chowla conjecture remains open, and the full even-order logarithmic problem is not silently promoted from the known odd-order and two-point cases. Pretentious Elliott variants require their own obstruction hypotheses. Every later revision of this ledger must record averaging, order, shift and exceptional-set regimes separately.*

7.1 Non-equivalence warning

“Logarithmic,” “almost all scales,” “short intervals,” and “Cesàro” are mathematically different quantifiers. Progress in one cell of the ledger is not evidence that a neighboring cell is solved.

7.2 Reader checkpoint

1. Write the order-two Liouville correlation with shifts 0 and 1.
2. Explain the difference between Cesàro and logarithmic weighting.
3. State why distinct shifts are required.
4. Give an example of a pretentious multiplicative function.
5. Explain what is known for odd logarithmic orders.
6. Explain why order four is not covered by the odd-order theorem.
7. State the main distinction between Chowla and Elliott formulations.
8. Explain what a moving logarithmic interval changes.
9. Name one short-interval result that is not the full Chowla conjecture.
10. Summarize the open cells in the dated frontier.

Chapter 8

Short Intervals and Local Arithmetic Randomness

Global cancellation can hide long local irregularities. Matomäki–Radziwiłł theory shows that bounded multiplicative functions nevertheless exhibit strong cancellation on almost all short intervals, and later work upgrades this to higher-order local uniformity on average.

Definition 8.1 (Short-interval averages and exceptional sets). For $x \asymp X$ and $1 \leq H \leq X$, the short-interval mean of f is

$$A_f(x; H) = \frac{1}{H} \sum_{x < n \leq x+H} f(n).$$

A property holds for *almost all* $x \in [X, 2X]$ if the set of exceptions has size $o(X)$. An *averaged short-interval Gowers estimate* means an integral or average over base points x of a local Gowers norm; it is not a uniform assertion for every interval. Pretentious and nonpretentious multiplicative functions are separated because their local means have different main terms.

Theorem 8.2 (Matomäki–Radziwiłł short-interval cancellation — verified deep input). *A selected consequence of Matomäki–Radziwiłł is the following. If $H = H(X) \rightarrow \infty$ with $H \leq X$, then for every $\varepsilon > 0$ the proportion of $x \in [X, 2X]$ for which*

$$\left| \frac{1}{H} \sum_{x < n \leq x+H} \mu(n) \right| > \varepsilon$$

tends to zero as $X \rightarrow \infty$. The underlying theorem applies more generally to bounded multiplicative functions with the appropriate global/pretentious mean term.

Theorem 8.3 (Higher short-interval uniformity on average — verified deep input). *Fix $k \geq 1$ and $0 < \theta < 1$. Matomäki–Radziwiłł–Tao–Teräväinen–Ziegler prove, in particular, that for $X^\theta \leq H \leq X$,*

$$\int_X^{2X} \|\lambda\|_{U^{k+1}([x, x+H])} dx = o(X),$$

with corresponding polynomial-phase and nilsequence formulations and applications to averaged Chowla-type patterns. The integral over x is part of the theorem: it is not a pointwise uniform bound for every short interval.

Theorem 8.4 (Arithmetic-pseudorandomness laboratory — completed artifact). *The executable laboratory ‘code/volume_24/arithmetic_pseudorandomness_lab.py’ computes μ by a sieve, checks the independent-sign second-moment identity by exhaustive enumeration at a small size, verifies the finite Kátai divisor second-moment inequality, checks the cyclic U^2 Fourier identity numerically, and reports short-interval Möbius diagnostics. Every numerical output is labeled finite-sample evidence rather than an asymptotic theorem.*

8.1 Interpretation

The almost-all theorem allows exceptional intervals. The higher-uniformity theorem averages local norms over x . Neither may be rewritten as a uniform-in- x square-root law.

8.2 Reader checkpoint

1. Define “almost all” base points precisely.
2. Explain why $H(X) \rightarrow \infty$ can be much smaller than any fixed power of X .
3. State the selected Möbius consequence of Matomäki–Radziwiłł.
4. Explain the role of a pretentious main term in the general theorem.
5. Distinguish an averaged local norm from a supremum over intervals.
6. State the $X^\theta \leq H \leq X$ range in the higher-uniformity theorem.
7. Explain why the laboratory cannot verify an asymptotic theorem.
8. Run the lab with a larger cutoff and compare the reported short-interval statistic.
9. Relate local U^{k+1} uniformity to polynomial-phase tests.
10. Identify which short-interval claims remain source-owned.

Chapter 9

Topological Dynamics and Möbius Disjointness

Sarnak’s program asks whether zero-entropy deterministic dynamics is asymptotically orthogonal to Möbius. Many structured classes are known, but the full conjecture remains open. We prove a transparent continuous-eigenfunction case and then record stronger source-owned criteria.

Definition 9.1 (Möbius disjointness and uniform discrete spectrum). A *topological dynamical system* is a compact metric space X with a continuous map $T : X \rightarrow X$. It is *Möbius disjoint* if for every $x \in X$ and $F \in C(X)$,

$$\frac{1}{N} \sum_{n \leq N} \mu(n) F(T^n x) \rightarrow 0.$$

A nonzero $\phi \in C(X)$ is a *continuous eigenfunction* if $\phi \circ T = \zeta \phi$ for some $|\zeta| = 1$. We say the system has *uniform discrete spectrum* if the complex linear span of its continuous eigenfunctions is dense in $C(X)$ in the uniform norm. This is a deliberately topological class; it is stronger than merely saying that each invariant measure has measure-theoretic discrete spectrum.

Theorem 9.2 (Möbius disjointness for uniform discrete spectrum). *Every topological dynamical system with uniform discrete spectrum is Möbius disjoint. In particular, compact abelian group rotations with their usual character eigenfunctions satisfy the conclusion.*

Proof. For a continuous eigenfunction ϕ with eigenvalue $\zeta = e(\alpha)$,

$$\phi(T^n x) = \zeta^n \phi(x) = e(\alpha n) \phi(x).$$

Corollary 2.3 therefore gives Möbius orthogonality to every continuous eigenfunction and hence to every finite linear combination of them. Given $F \in C(X)$ and $\varepsilon > 0$, choose such a combination G with $\|F - G\|_\infty < \varepsilon$. Because $|\mu| \leq 1$, the normalized correlations for F and G differ by at most ε . Let $N \rightarrow \infty$ and then $\varepsilon \rightarrow 0$. $\square$

Theorem 9.3 (Selected Möbius-disjointness criteria beyond nilsystems — verified deep inputs). *Huang–Wang–Ye prove Möbius disjointness when every invariant Borel probability measure has sub-polynomial measure complexity; their applications include discrete-spectrum regimes beyond the elementary continuous-eigenfunction theorem above. Kanigowski–Lemańczyk–Radziwiłł prove complementary rigidity criteria based on bounded-prime-volume rigidity or polynomial-rate rigidity on a linearly dense class of observables, yielding further zero-entropy examples. These theorems*

retain their measure-complexity/rigidity hypotheses; zero topological entropy alone is not substituted for them.

Theorem 9.4 (Structured-sequence correlation audit — completed example). *The Volume XXIV laboratory compares finite Möbius correlations against a periodic sign, an irrational linear phase, a quadratic phase, and a bracket-type nilsequence. The periodic and linear tests are governed by Corollary 2.3; the fixed nilsequence is governed by Theorem 5.2; a finite quadratic-phase value is reported only as a diagnostic unless it is embedded in an explicitly calibrated higher-order theorem. No finite table is interpreted as proof of a limiting rate.*

Theorem 9.5 (Sarnak conjecture and deterministic-sequence frontier — dated PASS112 frontier). *As verified on 2026-09-04, Sarnak’s Möbius-disjointness conjecture for all zero-topological-entropy systems remains open. Strong class theorems include sub-polynomial measure complexity and rigidity criteria, and 2026 work continues to add specific classes and averaging regimes: examples include an infinite-dimensional-torus theorem, a short-interval/Furstenberg-flow result, and a singular-spectrum/system-class criterion. These are class- or regime-specific advances, not a proof of the general conjecture. The frontier ledger therefore records the system class, observable regularity and averaging mode for every claimed advance.*

9.1 Reader checkpoint

1. Prove the eigenfunction orbit formula used in Theorem 27.
2. Explain why uniform density of eigenfunctions is stronger than measure-theoretic discrete spectrum.
3. Give the continuous eigenfunctions of an irrational circle rotation.
4. Explain the uniform-approximation step in the proof.
5. State the measure-complexity condition in qualitative form.
6. State one rigidity mechanism from the source-owned criterion.
7. Explain why zero entropy alone cannot replace those hypotheses in a theorem statement.
8. Match each structured-sequence example to the theorem that actually controls it.
9. Explain why a 2026 special-class theorem does not solve Sarnak globally.
10. List the quantifiers that must be recorded in the frontier ledger.

Chapter 10

Homogeneous Dynamics as Deterministic Randomness

Unipotent dynamics is deterministic yet can exhibit strikingly rigid statistical behavior. Ratner's theorems classify invariant measures and orbit closures, while effective results in selected settings quantify deviations. The lesson is not that deterministic orbits are independent random samples, but that algebraic structure can force precise statistical laws.

Definition 10.1 (Homogeneous dynamics and quantitative equidistribution). Let G be a Lie group and $\Gamma \leq G$ a lattice; $X = G/\Gamma$ is a finite-volume homogeneous space. A one-parameter subgroup $U = \{u_t\}$ is *Ad-unipotent* if $\text{Ad}(u_t)$ is unipotent for every t . A probability measure is *homogeneous* if it is the invariant probability measure on a closed orbit Hx of a closed subgroup $H \leq G$. *Polynomial divergence* means that relative displacement under $u_{-t}gu_t$ is polynomial in t in suitable Lie coordinates; *additional invariance* is invariance of a limiting measure under a subgroup larger than the original flow. *Quantitative equidistribution* is an explicit error estimate comparing an orbit average of a regular observable with the invariant spatial average, with Sobolev norms, spectral data and cusp/orbit-height parameters stated.

Theorem 10.2 (Ratner measure classification and orbit closure — verified deep inputs). *Let G/Γ be a finite-volume homogeneous space and let U be generated by Ad-unipotent one-parameter subgroups in the setting of Ratner's theorems. Every ergodic U -invariant probability measure is homogeneous: it is the invariant probability measure on a closed orbit Hx for a closed subgroup H containing the acting unipotent directions. Correspondingly, orbit closures of unipotent trajectories are homogeneous. The exact standing hypotheses and subgroup conclusions are source-owned to Ratner's 1991 measure-classification and orbit-closure theorems.*

Theorem 10.3 (Polynomial divergence and additional invariance — verified source-owned mechanism). *In Ratner's proof architecture, if two recurrent/generic unipotent trajectories start at a small transverse displacement g , then conjugation $u_{-t}gu_t$ varies polynomially because the adjoint unipotent action is nilpotent. Choosing times at which this polynomial displacement has a controlled nontrivial limit and combining recurrence, genericity and nondivergence can force a limiting invariant measure to acquire additional invariance. This shearing mechanism is a source-owned step in the classification argument; it is not asserted here as an assumption-free standalone classification theorem.*

Theorem 10.4 (Selected quantitative homogeneous equidistribution input — verified deep input). *For the horocycle flow on the unit tangent bundle of a noncompact finite-area hyperbolic surface,*

Strömbergsson gives effective bounds for deviations of ergodic averages. The explicit rate depends on small Laplace eigenvalues and on cusp excursion of the associated geodesic. This selected theorem is an effective rank-one homogeneous-dynamics input; it is not a universal “effective Ratner theorem” for arbitrary G/Γ .

Theorem 10.5 (Arithmetic dynamics and higher-order pseudorandomness — dated PASS112 frontier). *As verified on 2026-09-04, effective equidistribution is highly setting-dependent. Strong results exist for horocycle flows and many arithmetic homogeneous actions, while general effective analogues of Ratner’s qualitative classification, Möbius disjointness for broader skew/homogeneous systems, sparse-orbit questions and higher-rank interactions remain active. March/April/July 2026 Sarnak-program advances add further specific classes or averaging regimes without resolving the full conjecture. This ledger records the ambient group, lattice, regularity norm, spectral gap/nondivergence assumptions and averaging scale for each quantitative theorem.*

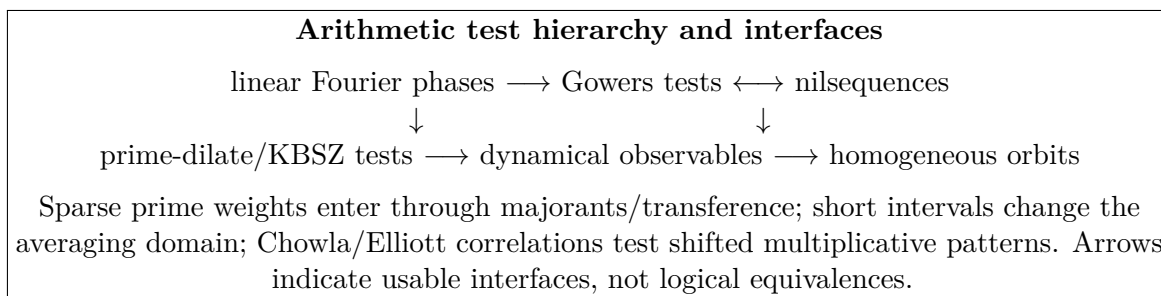


Figure 10.1: The hierarchy of structured tests used in Volume XXIV.

10.1 Closing perspective

Volume XXIV ends the current twenty-four-volume architecture by showing the same discipline at three scales: elementary Fourier tests, higher-order arithmetic structure, and homogeneous dynamics. “Pseudorandom” always means pseudorandom relative to an explicitly named test class and theorem.

10.2 Reader checkpoint

1. Define an Ad-unipotent one-parameter subgroup.
2. Explain what a homogeneous probability measure is.
3. State the qualitative content of Ratner measure classification.
4. Distinguish measure classification from orbit closure.
5. Compute why conjugation by a unipotent matrix produces polynomial coordinates in a simple 2×2 example.
6. Explain the phrase “additional invariance.”
7. State the setting of Strömbergsson’s selected effective theorem.
8. List two parameters on which its effective error can depend.

9. Explain why the figure arrows are not equivalences.
10. State one global closure task that remains after the twenty-four-volume architecture is drafted.

Internalization and Reconstruction Guide: Volume XXIV

Purpose. See arithmetic pseudorandomness as cancellation against increasingly structured deterministic tests: phases, bilinear criteria, Gowers norms, nilsequences, transference, correlations, short intervals, and dynamical systems.

Study contract. On a second pass, do not read linearly. Start from the dependency map below, reconstruct one theorem family from memory, test the boundary cases, and only then return to the chapter proof or source interface. A locally audited theorem should be reproducible from the listed prerequisites; a source-owned theorem should be reproducible at the level of its exact statement, hypotheses, conclusion, and role in the later argument, not by inventing a shortened proof.

Prerequisite ladder

The most frequently reused earlier-volume interfaces are:

1. QRCC-I-THM-004: Chernoff bounds.
2. QRCC-I-THM-016: Parseval/Plancherel theorem.
3. QRCC-V-THM-003: Generalized von Neumann inequality.
4. QRCC-V-THM-010: Dense model theorem.
5. QRCC-V-THM-011: Relative generalized von Neumann theorem.

Before attempting the reconstruction drill, be able to state from memory these local conventions: QRCC-XXIV-DEF-001 (Arithmetic pseudorandomness and test classes); QRCC-XXIV-DEF-004 (Exponential and Fourier tests for multiplicative functions); QRCC-XXIV-DEF-007 (Kátai-Bourgain-Sarnak-Ziegler criteria); QRCC-XXIV-DEF-010 (Gowers uniformity of arithmetic functions); QRCC-XXIV-DEF-013 (Nilsequences and polynomial orbits).

Chapter retrieval map

- Ch. 1: Möbius Cancellation and Random Benchmarks:** enter through QRCC-XXIV-DEF-001 (Arithmetic pseudorandomness and test classes); exit by being able to state QRCC-XXIV-THM-002 (Prime number theorem as Möbius cancellation) with its ownership boundary.
- Ch. 2: Fourier Tests and Davenport Cancellation:** enter through QRCC-XXIV-DEF-004 (Exponential and Fourier tests for multiplicative functions); exit by being able to state

QRCC-XXIV-BB-005 (Davenport bound for Möbius exponential sums) with its ownership boundary.

- Ch. 3: Prime Dilates: Kátaı and Bourgain–Sarnak–Ziegler:** enter through QRCC-XXIV-DEF-007 (Kátaı–Bourgain–Sarnak–Ziegler criteria); exit by being able to state QRCC-XXIV-THM-008 (Kátaı orthogonality criterion) with its ownership boundary.
- Ch. 4: Gowers Uniformity of Arithmetic Weights:** enter through QRCC-XXIV-DEF-010 (Gowers uniformity of arithmetic functions); exit by being able to state QRCC-XXIV-BB-012 (Gowers-uniformity results for Möbius and von Mangoldt weights) with its ownership boundary.
- Ch. 5: Nilsequences and Möbius Orthogonality:** enter through QRCC-XXIV-DEF-013 (Nilsequences and polynomial orbits); exit by being able to state QRCC-XXIV-BB-014 (Green–Tao Möbius–nilsequence theorem) with its ownership boundary.
- Ch. 6: Transference and Linear Patterns in the Primes:** enter through QRCC-XXIV-DEF-016 (Pseudorandom majorants and transference for primes); exit by being able to state QRCC-XXIV-BB-017 (Green–Tao transference architecture for arithmetic progressions in primes) with its ownership boundary.
- Ch. 7: Chowla, Elliott, and Logarithmic Correlations:** enter through QRCC-XXIV-DEF-019 (Chowla, Elliott, and logarithmic correlations); exit by being able to state QRCC-XXIV-BB-020 (Logarithmically averaged two-point Chowla/Elliott theorem) with its ownership boundary.
- Ch. 8: Short Intervals and Local Arithmetic Randomness:** enter through QRCC-XXIV-DEF-023 (Multiplicative functions in short intervals); exit by being able to state QRCC-XXIV-BB-024 (Matomäki–Radziwiłł short-interval theorem) with its ownership boundary.
- Ch. 9: Topological Dynamics and Möbius Disjointness:** enter through QRCC-XXIV-DEF-026 (Möbius disjointness in topological dynamics); exit by being able to state QRCC-XXIV-BB-028 (Selected Möbius-disjointness criteria beyond nilsystems) with its ownership boundary.
- Ch. 10: Homogeneous Dynamics as Deterministic Randomness:** enter through QRCC-XXIV-DEF-030 (Homogeneous dynamics as deterministic randomness); exit by being able to state QRCC-XXIV-BB-031 (Ratner measure classification and orbit-closure theorems) with its ownership boundary.

Major theorem-family reconstruction map

For each anchor below, first write the statement and dependency arrows without looking at the text. Then use the status-specific reconstruction rule.

1. QRCC-XXIV-THM-002 — Prime number theorem as Möbius cancellation. Dependencies: QRCC-XXIV-DEF-001. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
2. QRCC-XXIV-THM-008 — Kátaı orthogonality criterion. Dependencies: QRCC-XXIV-DEF-007, QRCC-I-THM-016. *local proof: reconstruct the proof from the hypotheses and named dependencies before rereading the certified argument.*

3. QRCC-XXIV-BB-014 — Green–Tao Möbius–nilsequence theorem. Dependencies: QRCC-XXIV-BB-005, QRCC-XXIV-DEF-013. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
4. QRCC-XXIV-BB-020 — Logarithmically averaged two-point Chowla/Elliott theorem. Dependencies: QRCC-XXIV-DEF-019. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
5. QRCC-XXIV-BB-028 — Selected Möbius-disjointness criteria beyond nilsystems. Dependencies: QRCC-XXIV-THM-009, QRCC-XXIV-DEF-026. *source-owned interface: reconstruct the exact statement, model, and dependency role; do not replace the cited deep proof by a slogan.*
6. QRCC-XXIV-F-022 — Chowla and Elliott status ledger. Dependencies: QRCC-XXIV-DEF-019, QRCC-XXIV-BB-020, QRCC-XXIV-BB-021. *dated frontier: state what is proved, what remains open, and the date/model boundary before using the item as context.*

Common failure modes

- mixing Cesaro, logarithmic, and short-interval averages;
- treating multiplicativity as independence;
- using a nilsequence theorem as though it proved the full Sarnak conjecture;
- calling a qualitative homogeneous-dynamics theorem effective without a quantitative source;

Boundary and degenerate cases

- periodic sequences and pure phases as baseline tests;
- pretentious multiplicative functions with obvious structured correlations;
- interval length comparable to 1 or to the whole range;
- nilsystems/discrete-spectrum systems versus the unresolved full zero-entropy frontier;

Retrieval and reconstruction drill

1. State QRCC-XXIV-DEF-001 from memory, including every normalization and quantified parameter. Give one example and one non-example.
2. Reconstruct the dependency path ending at QRCC-XXIV-THM-002. For each arrow, say exactly what output of the earlier item is consumed by the later one.
3. For QRCC-XXIV-THM-008, separate the theorem statement from proof technique, algorithmic consequence, and historical attribution. Which of these four pieces is actually certified here?
4. Choose one local proof in the volume and write a five-line proof skeleton containing only invariants, reductions, or inequalities. Expand each line only after the skeleton is logically complete.

5. Choose one source-owned or frontier result and write an interface card: hypotheses, conclusion, parameter dependence, model, source owner, and the first later theorem that uses it. Do not reproduce an uncited proof.
6. Exhibit one of the listed failure modes concretely by constructing a smallest or simplest example where the tempting but invalid inference fails.
7. Audit two of the boundary cases above. State whether the theorem remains true, becomes vacuous, changes normalization, or requires an excluded hypothesis.
8. Starting from QRCC-XXIV-F-022, explain the conceptual handoff to the next volume in at most six sentences and identify one dependency that must not be reversed.

Transition

After this volume there is no Volume XXV: the conceptual transition is to the whole-series synthesis, where probability, additive structure, complexity, cryptography, and arithmetic dynamics are compared as different notions of distinguishability and structure.

Gate-D certification rule. A reader has internalized this volume when they can reproduce the prerequisite ladder, distinguish local proofs from source-owned interfaces, recover the major theorem-family arrows, diagnose the listed failure modes, and solve the reconstruction drill without using the table of contents as a substitute for memory.

Registry-only publication interfaces

The following registered rows did not carry a stable PDF anchor in the certified chapter sources at the Gate-E cross-reference audit. Rather than silently editing audited theorem chapters, this edition publishes their exact registry interfaces here. The stable ID therefore resolves to this publication-interface page.

QRCC-IV-CODE-023: Quasirandom graph diagnostic laboratory

Volume/chapter: 4/8. **Kind/owner/status:** CODE/A/CODE_COMPLETE_PASS107.

Registered scope. Code brief comparing cut discrepancy, nonprincipal spectrum, C4 density, sampled subgraph counts and graphon step approximations for random and structured graph families.

Source/provenance key. INTERNAL_PEDAGOGY; INTERNAL:QRCC-PASS5.

QRCC-IV-THM-012: Compactness of graphon space modulo weak isomorphism

Volume/chapter: 4/8. **Kind/owner/status:** THM/C/DEEP_INPUT_VERIFIED_PASS107.

Registered scope. State that graphons modulo measure-preserving weak isomorphism form a compact metric space in cut distance, and give the finite weak-regularity approximation/proof architecture used to obtain sequential compactness.

Source/provenance key. GRAPHON_LIMITS_FOUNDATIONS; <https://doi.org/10.1016/j.jc.2006.05.002>; <https://doi.org/10.1016/j.aim.2008.07.008>.

QRCC-IX-CONJ-024: BPP equals P

Volume/chapter: 9/10. **Kind/owner/status:** CONJ/F/OPEN_PROBLEM.

Registered scope. State $BPP=P$ and its uniform/promise variants separately; record all conditional implications proved in this volume.

Source/provenance key. BPP_EQUALS_P; https://doi.org/10.1007/978-3-642-22670-0_20.

QRCC-IX-CONJ-025: Promise-BPP equals Promise-P

Volume/chapter: 9/10. **Kind/owner/status:** CONJ/F/OPEN_PROBLEM.

Registered scope. State the promise-class derandomization problem and explain why it is formally stronger than the language equality without silently identifying them.

Source/provenance key. PROMISE_BPP_EQUALS_P; <https://doi.org/10.1016/j.ip1.2019.105841>.

QRCC-X-CODE-029: Small-circuit enumeration and restriction laboratory

Volume/chapter: 10/5. **Kind/owner/status:** CODE/A/CODE_COMPLETE_PASS38.

Registered scope. Executable exact-rational restriction laboratory: enumerate truth-table-distinct tiny width-two DNFs, apply the QRCC product restriction exactly, compute minimum residual decision-tree depth by exhaustive dynamic programming, reproduce the Chapter-4 equality-DNF tails, and compare finite tails with the local and source-owned switching bounds. Fourier/LMN scope is deliberately removed.

Source/provenance key. INTERNAL_CODE; code/.

QRCC-X-CONJ-023: NP not subset P/poly

Volume/chapter: 10/12. **Kind/owner/status:** CONJ/F/OPEN_PROBLEM.

Registered scope. State the nonuniform circuit lower-bound conjecture and its consequences for P versus NP, polynomial hierarchy, cryptography, and natural proofs.

Source/provenance key. NP_NOT_PPOLY; <https://doi.org/10.1145/800141.804678>.

QRCC-X-CONJ-024: NEXP not subset P/poly

Volume/chapter: 10/12. **Kind/owner/status:** CONJ/F/OPEN_PROBLEM.

Registered scope. State the exponential-time nonuniform lower-bound problem separately from known restricted-class results and from uniform hierarchy theorems.

Source/provenance key. NEXP_NOT_PPOLY; <https://doi.org/10.1142/S0219061324500235>.

QRCC-XIV-CONJ-012: 3SUM conjecture

Volume/chapter: 14/6. **Kind/owner/status:** CONJ/F/OPEN_PROBLEM.

Registered scope. State the real/integer and randomized/deterministic 3SUM hypotheses used in conditional lower bounds, without conflating them with SETH.

Source/provenance key. THREESUM_CONJECTURE; <https://doi.org/10.4230/LIPIcs.SoCG.2021.2>.

QRCC-XIV-CONJ-013: APSP conjecture

Volume/chapter: 14/6. **Kind/owner/status:** CONJ/F/OPEN_PROBLEM.

Registered scope. State the weighted directed APSP fine-grained hypothesis and distinguish combinatorial from algebraic algorithm versions.

Source/provenance key. APSP_CONJECTURE; <https://doi.org/10.1145/3186893>.

Index of registered results, constructions, and frontiers

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